SUNAPI

SUNAPI Complete API Reference Guide

Hanwha

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

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www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

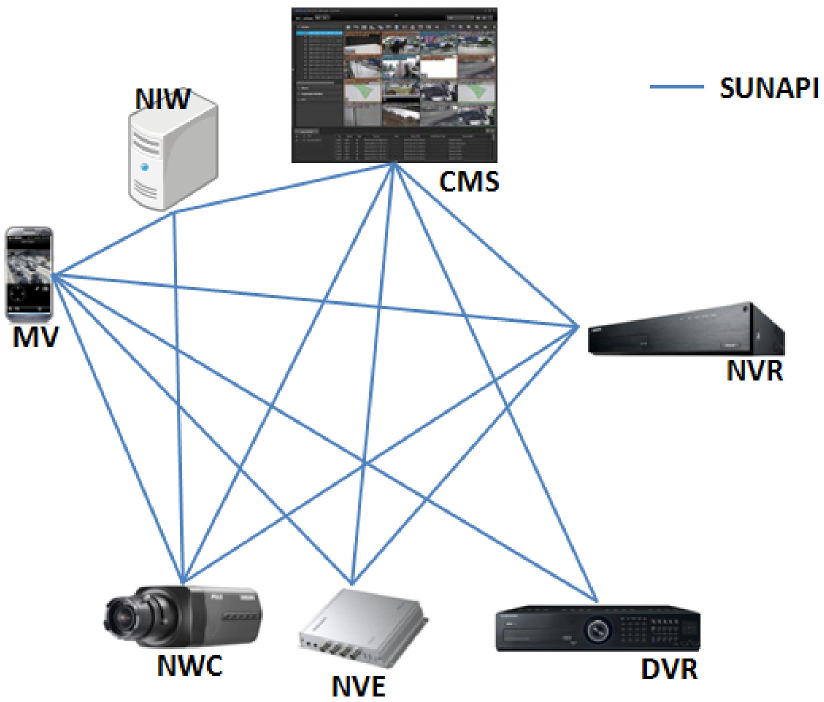
Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 1. Introduction to SUNAPI

1.1. Overview

The Smart Unified Network API (SUNAPI) provides a single, seamless interface for controlling and configuring the various products which make up a networked video security system. It can integrate products such as network cameras, storage devices (NVR and DVR), encoders and central monitoring software (CMS), including both Hanwha Vision products and products from other third-party vendors.

SUNAPI allows you to access product features by simply entering standard HTTP URLs. The URLs pass variables to SUNAPI’s CGI, which interfaces with the specific product. This simplified interface system makes it possible for central monitoring software to access the features of a diverse set of products in a standardized way. This makes SUNAPI a valuable tool for developers of central monitoring software and other network video applications.



SUNAPI provides CGIs to perform the following functions:

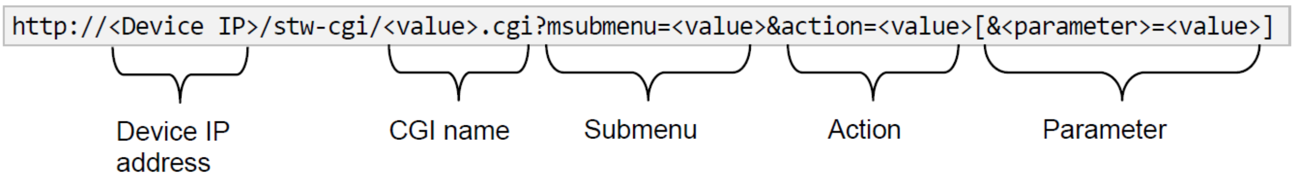
- System configuration: system.cgi configures general system settings and performs factory resets to restore the system. It can also view system logs, event logs and access logs.
- Network management: network.cgi sets up the network which connects the video security products. It configures IP, DNS, DDNS and other settings.
- Security management: security.cgi sets up security features such as RTSP, SSL and IP address filtering (which allows only authorized addresses to connect). It also sets up systems users and configures their access permissions.
- Video streaming: video.cgi requests http retrieval of JPG images or MJPEG stream, and media.cgi configures profiles, which include video/audio codec settings, audio input/output settings, etc.

- Image adjustment: image.cgi configures camera features such as BLC, privacy, white balance, exposure and OSD.
- Server settings: transfer.cgi configures the FTP and SMTP server settings for transferring event video and images.
- Event monitoring and handling: eventsource.cgi, eventrules.cgi, and eventstatus.cgi manage events such as motion detection, face detection and tampering.
- PTZ control: ptzcontrol.cgi and ptzconfig.cgi manage the camera’s pan, tilt and zoom and detail configuration.
- Searching stored data: recording.cgi configures recording schedules and related settings. It also provides an interface to search recorded videos.
- Open SDK installation: opensdk.cgi installs and manages the Open SDK application.

1.2. SUNAPI Command Structure

SUNAPI commands are HTTP URLs. Each URL specifies the target device’s IP address and the CGI which provides the command. This is followed by a query string which specifies the command’s submenu, action and parameters.

1.2.1. Syntax



The following is an example of a device information request:

```
http://192.168.0.100/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view
```

The text in the URL command is case-sensitive. For example, if you type **msubmenu=DeviceInfo** instead of **msubmenu=deviceinfo**, an error message will be generated.

To set a port number, add a colon after the IP address and enter the number in the format of<Device IP>:<Port Number> as in the example below.

```
http://192.168.0.100:20/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view
```

NOTE

For NVR, a channel number is mandatory wherever it is applicable.
If the channel parameter is not provided in the view action, a response for all channels is given. If it is not provided in the set action, the change will be applied to channel **0**.

1.2.2. Submenus

Each CGI is divided into submenus which perform specific functions. For example, **system.cgi** has submenus such as **deviceinfo**, which sets up product information, and **date**, which sets the date. In the SUNAPI syntax, the submenu is specified in the query string by assigning a value to the 'msubmenu' variable. For example, msubmenu=deviceinfo.

1.2.3. Actions

An action must be specified for each SUNAPI command. SUNAPI offers the following action types:

- view: Requests the current setting information.
  Generally, the **view** action does not take parameters, and returns only response parameters and values, which are marked as **RES** on the parameter table in the SUNAPI reference document. In some cases, the view action does take request parameters.
- set: Specifies new settings with new parameters.
- control: Controls data settings.
- update: Updates existing parameters with new values. Updates the current data. The **update** and **add** actions are used instead of the **set** action to add or modify list data such as the user list data or the IP filtering list data.
- add: Adds new parameters.
- install: Installs data.
- remove: Deletes the existing parameters.

1.3. Response Format

A SUNAPI response is in text format for all the commands, except for the attributes response, which is in XML format. From SUNAPI version 2.5 onwards two types of response format (Text and JSON) are supported; by default, the response format will be text and if the request has special header (Accept: application/json) the response will be in JSON format.

1.4. Examples

Below are Examples for getting and setting information.

1.4.1. Example of the View Action

To get device information, the following request is used:

REQUEST

```
http://192.168.0.100/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view
```

ACTUAL REQUEST MESSAGE

For the default request as shown below, the response will be in text format.

```
GET /stw-cgi/system.cgi?msubmenu=deviceinfo&action=view HTTP/1.1
User-Agent: xxxxx
Host: 192.168.0.100
Accept: */*
```

For a request in the format below, the response will be in JSON format from SUNAPI 2.5 onwards.

JSON REQUEST MESSAGE

```
GET /stw-cgi/system.cgi?msubmenu=deviceinfo&action=view HTTP/1.1
User-Agent: XXXXXX
Host: 192.168.0.100
Accept: application/json
```

The following response will be sent:

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Model=XNF-9010RV
SerialNumber=ZLNA70GKA0001JZ
FirmwareVersion=2.01.00_20200721_R201
BuildDate=2020.07.21
WebURL=http://www.hanwhavision.com/
DeviceType=NWC
ConnectedMACAddress=00:09:18:5B:BD:3A
ISPVersion=1.00_200721
BootloaderVersion=ver=U-Boot 2020.07-svn2693 (Jul 21 2020 - 18:09:00
CGIVersion=2.5.7
ONVIFVersion=19.12
DeviceName=Camera
DeviceLocation=Location
DeviceDescription=Description
Memo=Memo
Language=English
PasswordStrength=Strong
OpenSDKVersion=4.00_200702
```

```
FirmwareGroup=XNF-9010RV
```

JSON RESPONSE


```
{
    "Model": "XNF-9010RV",
    "SerialNumber": "ZLNA70GKA0001JZ",
    "FirmwareVersion": "2.01.00_20200721_R201",
    "BuildDate": "2020.07.21",
    "WebURL": "http://www.hanwhavision.com/",
    "DeviceType": "NWC",
    "ConnectedMACAddress": "00:09:18:5B:BD:3A",
    "ISPVersion": "1.00_200721",
    "BootloaderVersion": "ver=U-Boot 2020.07-svn2693 (Jul 06 2020 - 18:09:00
",
    "CGIVersion": "2.5.7",
    "ONVIFVersion": "19.12",
    "DeviceName": "Camera",
    "DeviceLocation": "Location",
    "DeviceDescription": "Description",
    "Memo": "Memo",
    "Language": "English",
    "PasswordStrength": "Strong",
    "OpenSDKVersion": "4.00_200702",
    "FirmwareGroup": "XNF-9010RV"
}
```


1.4.2. Example of the Set Action

To set the language to English, the following request should be used:

REQUEST

```
http://192.168.0.100/stw-
cgi/system.cgi?msubmenu=deviceinfo&action=set&Language=English
```

The following response is sent:

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain_
```

```
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type:application/json_
<Body>
```


```
{
    "Response": "Success"
}
```


When an error occurs, an error message will be returned, as shown below:

REQUEST

```
http://192.168.0.100/stw-
cgi/system.cgi?msubmenu=deviceinfo&action=set&Language=
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain_
<Body>
```

```
NG
Error Code: 604
Error Details:
Invalid Input Value(s)
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type:application/json_
<Body>
```

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Input Value(s)"
    }
}
```

1.4.3. Note on Sample Requests & Responses

The SUNAPI reference provides code samples in each Chapter. The samples in SUNAPI documentation are meant for reference only. The returned Data in the samples can differ from the actual data depending on the product model and its configuration.

1.5. Access level

SUNAPI has 4 types of access for CGI requests, Guests, Users, and Admin, which are based on the SUNAPI action level.

If a user doesn't have the permission to perform the particular action, then all the CGI parameters corresponding to the action will not be accessed. For example, an Access Level specified as "Admin" will be given only with the admin login, and other user logins will not have access to the corresponding features.

Access Level is stated in the SUNAPI reference documents.

1.6. Authentication

For Hanwha Vision products, the Digest Authentication is used for secure authentication.

NOTE	It is strongly recommended to always use HTTPS for all SUNAPI communications to ensure secure data transmission and protect credentials.

1.7. Initial Password Setup

Starting from SUNAPI version 2.5.5, devices in factory default state require initial password configuration before any SUNAPI or ONVIF operations can be performed. The initial password can be set using either pw_init.cgi (a hidden CGI requiring no authentication) or the IP Installer Protocol.

1.7.1. Checking Password Initialization Status

To verify if the device password has been initialized:

REQUEST

```
http://<ip>/init-cgi/pw_init.cgi?msubmenu=statuscheck&action=view
```

If the device is not initialized, the response will include an RSA public key for password encryption:

RESPONSE


```
{
    "Initialized": false,
    "Language": "English",
    "MaxChannel": 2,
    "SpecialType": "none",
    "NewPasswordPolicy": true,
    "MaxPasswordLength": 64,
    "SupportedPublicKeyFormats": ["PKCS1", "X509"],
    "PublicKey": "<RSA Public Key>",
    "Manufacturer": "Hanwha Vision"
}
```


1.7.2. Setting the Initial Password

The initial password can only be set once. Subsequent attempts will fail.

To set the password with encryption (recommended):

REQUEST (POST)

```
http://<DeviceIP>/init-
cgi/pw_init.cgi?msubmenu=setinitpassword&action=set&IsPasswordEncrypted=True
```

POST Payload

```
<base64 encoded encrypted password>
```

1.7.3. Password Encryption Procedure

To encrypt the password for secure transmission:

Step 1: Obtain the RSA Public Key

The public key is included in the statuscheck response when the device is not initialized.

REQUEST

```
http://<ip>/init-
cgi/pw_init.cgi?msubmenu=statuscheck&action=view&PublicKeyFormat=X509
```

Alternatively, for password changes on initialized devices, retrieve the public key using the **rsa** submenu in **security.cgi** and update the password using the **users** submenu in the same cgi.

REQUEST

```
http://<ip>/stw-cgi/security.cgi?msubmenu=rsa&action=view
```

RESPONSE


```
{
    "PublicKey": "-----BEGIN RSA PUBLIC KEY-----\n<key content>\n-----END
RSA PUBLIC KEY-----\n"
}
```


Step 2: Encrypt the Password

Encrypt the password using the RSA public key with RSA_PKCS1_PADDING padding scheme.

Step 3: Encode and Send

Base64 encode the encrypted binary data and send it as the POST message body with the IsPasswordEncrypted parameter set to True.

Example:

```
http://<DeviceIP>/init-
cgi/pw_init.cgi?msubmenu=setinitpassword&action=set&IsPasswordEncrypted=True
```

POST Body:


```
Vvg2Ku93HlReI+3RseQgfYQoxUFkh9P2L5RvjZ+bLovLTGMQ23OFT+BDaIwMcgvs...
```


NOTE

- Until the initial password is set, all SUNAPI and ONVIF CGIs will be disabled.
- Plain text password setting is deprecated and should not be used.
- After initial password setup, ONVIF services are enabled by default along with SUNAPI services.
- This encryption method applies to all SUNAPI submenus where password is a parameter.

1.8. Index numbers

All index numbers start from **1** in SUNAPI, except channel number, deviceid(POS) and the snmptrap submenu in network.cgi, which all start from **0**. In this reference document, the hash sign (#) represents an index number.

1.9. Multipart Boundary

The multipart boundary string is not officially defined in SUNAPI. Although "SamsungTechwin" has been commonly used as the boundary string in multipart responses, clients should not rely on a fixed value and should always get the multipart boundary from the HTTP header.

```
Content-Type: multipart/xxxxxxx; boundary=*"boundarystring"*
```

1.10. Error codes

When an error occurs during a **set** action command, the following error codes can be returned:

Error Code	Define	Response(Text)	Response(JSON)	Description

600	Invalid submenu	NG
		Error Code: 600
		Error Details:
		Submenu Not Found
			{
			"Response": "Fail",
			"Error": {
			"Code": 600,
			"Details": "Submenu Not Found"
			}
			}
				An unsupported menu was requested

601	Invalid action	NG
		Error Code: 601
		Error Details:
		Action Not Found
			{
			"Response": "Fail",
			"Error": {
			"Code": 601,
			"Details": "Action Not Found "
			}
			}
				An unsupported action was requested

602	Invalid parameter	NG
		Error Code: 602
		Error Details:
		Invalid Parameter(s)
			{
			"Response": "Fail",
			"Error": {
			"Code": 602,
			"Details": "Invalid Parameter(s) "
			}
			}
				The parameters don’t match the existing settings in the request.
				
				e.g., requesting IPv6Address when IPv6Type=Auto

Error Code	Define	Response(Text)	Response(JSON)	Description

603	Missing Required Parameter	NG
		Error Code: 603
		Error Details:
		Missing Required Parameter(s)
			{
			"Response": "Fail",
			"Error": {
			"Code": 603,
			"Details": "Missing Required Parameter(s) "
			}
			}
				The user request is missing a required parameter.

604	Invalid Value	NG
		Error Code: 604
		Error Details:
		Invalid Input Value(s)
			{
			"Response": "Fail",
			"Error": {
			"Code": 604,
			"Details": "Invalid Input Value(s) "
			}
			}
				The user request has an invalid parameter value.

605	List is full	NG
		Error Code: 605
		Error Details:
		List Full
			{
			"Response": "Fail",
			"Error": {
			"Code": 605,
			"Details": "List Full"
			}
			}
				The user has attempted to add data to a list that is full.

606	Duplicate value	NG
		Error Code: 606
		Error Details:
		Duplicate Value
			{
			"Response": "Fail",
			"Error": {
			"Code": 606,
			"Details": "Duplicate Value"
			}
			}
				The user has attempted to set a value that is already being used in another field.

607	Unknown error	NG
		Error Code: 607
		Error Details:
		Unknown Error
			{
			"Response": "Fail",
			"Error": {
			"Code": 607,
			"Details": "Unknown Error"
			}
			}
				System-related error

Error Code	Define	Response(Text)	Response(JSON)	Description

608	Feature not implemented OR Not supported	NG
		Error Code: 608
		Error Details:
		Feature(s) Not Implemented OR Not Supported
			{
			"Response": "Fail",
			"Error": {
			"Code": 608,
			"Details": "Feature(s) Not Implemented OR Not Supported"
			}
			}
				When an unsupported service or feature is requested.

609	Not Authorized	NG
		Error Code: 609
		Error Details:
		Not Authorized
			{
			"Response": "Fail",
			"Error": {
			"Code": 609,
			"Details": "Not Authorized"
			}
			}
				A login error has occurred.

610	Cannot Access Resource	NG
		Error Code: 610
		Error Details:
		Cannot Access Resource
			{
			"Response": "Fail",
			"Error": {
			"Code": 610,
			"Details": "Cannot Access Resource"
			}
			}
				The other session is using the resource.

611	Invalid File	NG
		Error Code: 611
		Error Details:
		Invalid File
			{
			"Response": "Fail",
			"Error": {
			"Code": 611,
			"Details": "Invalid File"
			}
			}
				File is invalid.

612	Configuration Not Found	NG
		Error Code: 612
		Error Details:
		Configuration Not Found
			{
			"Response": "Fail",
			"Error": {
			"Code": 612,
			"Details": "Configuration Not Found"
			}
			}
				A configuration is not found.
				
				e.g., no response for camera-related CGI requests if the camera is not registered in the NVR

Error Code	Define	Response(Text)	Response(JSON)	Description

613	Cannot Set Simultaneously	NG
		Error Code: 613
		Error Details:
		Tried to set values at once that can’t be set simultaneously
			{
			"Response": "Fail",
			"Error": {
			"Code": 613,
			"Details": "Tried to set values at once that can’t be set simultaneously"
			}
			}
				When some features are not available to be set at the same time

614	Service Not Ready	NG
		Error Code: 614
		Error Details:
		Service not ready
			{
			"Response": "Fail",
			"Error": {
			"Code": 614,
			"Details": "Service not ready"
			}
			}
				When some features are not ready to service

615	Configuration Cannot Applied	NG
		Error Code: 615
		Error Details:
		The configuration cannot be applied
			{
			"Response": "Fail",
			"Error": {
			"Code": 615,
			"Details": "The configuration cannot be applied"
			}
			}
				When some configurations cannot be set

616	Configuration Incompatible	NG
		Error Code: 616
		Error Details:
		The configuration is not compatible
			{
			"Response": "Fail",
			"Error": {
			"Code": 616,
			"Details": "The configuration is not compatible"
			}
			}
				When some configurations are not compatible

Error Code	Define	Response(Text)	Response(JSON)	Description

700	Requested parameter is not found	NG
		Error Code: 700
		Error Details:
		<Tag Name>: Not Found
			{
			"Response": "Fail",
			"Error": {
			"Code": 700,
			"Details": "<Tag Name>: Not Found "
			}
			}
				attributes.cgi error
				
				The user request contains an unsupported tag

701	Error in XML file	NG
		Error Code: 701
		Error Details:
		Invalid XML
			{
			"Response": "Fail",
			"Error": {
			"Code": 701,
			"Details": "Invalid XML"
			}
			}
				attributes.cgi error
				
				The XML format is incorrect

702	System Menu Used	NG
		Error Code: 702
		Error Details:
		System Menu Used
			{
			"Response": "Fail",
			"Error": {
			"Code": 702,
			"Details": " System Menu Used "
			}
			}
				Settings are not adjustable in the web browser/protocol when the system menu is active in the NVR SET UI.
				NVR ONLY

1.11. Special HTTP Error codes

The following special error codes will be sent when additional password is required or when the device is locked on 5 successive login attempts with wrong password.

Error Code	Status Message	Title	Body
490	Status: 490 AccountBlocked	490 - Account Blocked	You have exceeded the maximum number of login attempts, Please try after some time.
491	Status: 491 AdditionalPasswordRequired	491 - Additional Password Required	Additional Passwords is required for this action

1.12. Style Conventions on SUNAPI Reference

The SUNAPI reference document is for developers of central monitoring software (CMS) as well as developers of applications for viewing, streaming and webcasting video.

1.12.1. Syntax

Text in the angle brackets is replaceable. In the example below, the text <Device IP> can be replaced with the actual IP address of the device, such as 192.168.0.100.

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=<value>&action=<value>&<parameter>=<value>
```

1.12.2. Parameters

Parameter information is displayed in tables like the one below:

Action	Parameter	Request/ Response	Type/ Value	Description
view set control	Parameter name	REQ RES	<int> <float> <bool> <string> <enum> <csv>	Brief parameter description

- The Action column shows the type of action performed on the parameter.
- The Parameter column shows the name of the parameter.
- The Request/Response column shows if the parameter is a request parameter (REQ) or a response parameter (RES).
- The Type/
  Value column shows the value types as well as the allowed values and ranges. Value types are as follows:
  - <enum> indicates enumeration values.
  - <int> indicates integer values.
  - <float> indicates floating point values
  - <string> indicates string data.
  - <bool> indicates boolean values.
  - <csv> indicates comma-separated values. Set multiple values separated by commas.
- The Description column offers a brief description of the parameter.

1.13. Support

Visit http://step.hanwhavision.com for the latest documents and technical support.

Chapter 2. Revision History

2.1. Version 2.6.8

Date: 2026-03-25

System

- storageinfo
  - Storages parameter added to control action for multi-storage format operations
  - Storage.#.SlotIndex.# parameter pattern added
- powermode
  - ConstraintOption parameter added (2SD_NoIR, None, 1SD_50PercentIR, FPS20_IR10m)
- powermodeoptions submenu added
  - Provides available power mode constraint options for different PoE modes

Security

- users
  - OpenAPPAccess parameter added for user access control

Media

- videosourceoptions
  - AvailableSensorCaptureFrameRateForAISupport parameter added

Image

- camera
  - BCRHDREnable parameter added for Barcode Reader HDR support
  - SequenceGain parameter added for BCR cameras
  - LensModel enum updated to include SLA-T10XX/SLA-T16XX
- whiteled
  - DwellTime parameter description updated and clarified

EventRules

- ledpreset
  - LightMode enum extended with On, Off, Blink, Rotating, FadeIn, FadeOut, FadeInAndOut, Strobe
  - LightModeAfterDuration enum extended with same values as LightMode
  - Custom value added to Color and ColorAfterDuration enums
  - HexColorCode parameter added (format: #RRGGBB)
  - HexColorCodeAfterDuration parameter added (format: #RRGGBB)
  - Brightness parameter now supported for both BCR cameras and network cameras

  - BrightnessAfterDuration parameter now supported for both BCR cameras and network cameras
  - EventActivePeriod parameter added - when true, disables post-duration actions
- dynamicrules
  - AccessGranted and AccessDenied event source types added for access control
  - Rule.#.EventSource.#.CredentialHolderTokens parameter added to filter events by credential holder tokens
  - Rule.#.EventSource.#.AccessGroupTokens parameter added to filter events by access group tokens
- handover2
  - CA certificate validation support added

EventSources

- earlyfiredetection
  - Rule.#.MinTemperatureCoordinates parameter added to check action
  - Rule.#.MaxTemperatureCoordinates parameter added to check action

Eventstatus

- New events added (for SPS-A100M Audio Beacon)
  - SIPCall - for SIP call events
  - AudioClipPlayback - for audio playback events
  - TwoWayTalk - for two-way audio communication events

Transfer

- smtp
  - CA certificate validation support added

PTZ

- digitalrtz
  - Mode parameter added for QNF-C9010 camera support
  - view action enhanced with additional parameters:
    - Zoom, ZoomPulse, ActualZoomPulse parameters added
    - SubViewIndex and ViewModeIndex changed from RES to REQ type

Attributes

- System/Support
  - DebugLogBackup capability attribute added
- Eventsource/Support
  - EarlyFireDetection.MinMaxCoordinates attribute added
- Image/Support

  - BCRHDREnable attribute added
  - SequenceGain attribute added for BCR cameras
- Privacy
  - MaskPattern attributes added

2.2. Version 2.6.7

Date: 2025-08-27

System

- sdcardstatus submenu added
- sdcardinfo
  - control action added, with format mode support.

Eventrules

- audiooutfiles
  - MP3 file format support added.
- handover2
  - install action added for certificate
  - CACertificateInstalled parameter added to view action
  - CertificateType parameter added to remove action

Network

- snmp
  - ShareCommunityStringWithV1 parameter added.

EventSources

- temperaturechangedetection
  - TemperatureChange.ROI.#.PolygonCoordinates parameter added.
  - TemperatureChange.ROI.#.DetectionType parameter added.

opensdk

- appmetadatacount submenu added.

Image

- autoimagealignment
  - HorizontalAdjustmentEnable parameter added.
  - HorizontalAdjustmentPosition parameter added.

Transfer

- smtp

- install action added for certificate
- CACertificateInstalled parameter added to view action
- CertificateType parameter added to remove action

2.3. Version 2.6.6

Date: 2025-02-26

Introduction

- New Error code 702 defined.

IpInstaller

- New device type **AccessController** defined.

Accesscontrol

- capabilities submenu added.
- door submenu added.
- configuration submenu added.
- credentialreader submenu added.
- area submenu added.
- securitylevel submenu added.
- authenticationprofile submenu added.
- accesspoint submenu added.
- accessschedule submenu added.
- accessprofile submenu added.
- accessgroup submenu added.
- credentialholder submenu added.
- credentialholderimage submenu added.
- credentialmethod submenu added.
- accessrule submenu added.

System

- geoorientation submenu added.
- aux submenu added.
- storageinfo
  - Storage.#.SlotIndex.#.Health parameter added.
  - Storage.#.SlotIndex.#.TotalSpace parameter added.
  - Storage.#.SlotIndex.#.UsedSpace parameter added.

- Storage.#.SlotIndex.#.Status parameter added.
- Storage.#.SlotIndex parameter added.

Network

- portconf
  - WebSessionTimeout parameter added.

Security

- 802Dot1x
  - MACsecMode parameter added.
  - ConnectivityAssociationKey parameter added.
  - ConnectivityAssociationKeyName parameter added.
- users
  - LogAccess parameter added.
- apisecurity submenu added.

Media

- videoprofile
  - IsDPMProfile parameter added.
- audiooutput
  - EQPreset parameter added.
- audioinputoptions submenu added.
- videocodecinfo
  - <EncodingType>.DPM.<Width>X<Height>.Width parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.Height parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MaxFPS parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MinFPS parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.DefaultFPS parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MaxCBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MinCBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.DefaultCBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MaxVBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.MinVBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.DefaultVBRTargetBitrate parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.IsTruncated parameter added.
  - <EncodingType>.DPM.<Width>X<Height>.FrameLockMaxFPS parameter added.

Image

- camera
  - ImagePresetMode.#.WDRMotionArtifactReduction parameter added.
  - WDRMotionArtifactReduction parameter added.
  - TemperatureOffsetEnable parameter added.
  - TemperatureOffset parameter added.
- imageoptions
  - TemperatureOffsetOffset.Celcius.Min parameter added.
  - TemperatureOffsetOffset.Celcius.Max parameter added.
  - TemperatureOffsetOffset.Fahrenheit.Min parameter added.
  - TemperatureOffsetOffset.Fahrenheit.Max parameter added.
- imageenhancements2
  - ContrastAdaptiveLevel parameter added.
  - ImagePresetMode.#.ContrastAdaptiveLevel parameter added.
- ssdr
  - ImagePresetMode.#.HighLightLevel parameter added.
  - HighLightLevel parameter added.
- dynamicprivacymask submenu added.

Transfer

- ftp
  - DirectoryDepth parameter added

EventRules

- dynamicrules
  - Language parameter added.
  - Rule.#.EventSource.#.EventName_<Language> parameter added.
  - EarlyFireDetection, TemperatureDifference, CaseTampering enum values added to Rule.#.EventSource.#.Type parameter.
- dynamicrulesoptions
  - EventSource policy enum updated.
  - MonitorAlert,LEDPreset enum values added to EventSource.#.ActionTypes and AppEventSource.#.ActionTypes.
  - EarlyFireDetection, TemperatureDifference, CaseTampering enum values added to EventSource.#.Type parameter.
- ledpreset
  - add and remove action added.

- Duration parameter added.
- ColorAfterDuration parameter added.
- LightModeAfterDuration parameter added.
- audiooutfiles
  - loopcount parameter added in set action.

EventSources

- earlyfiredetection submenu added.
- earlyfiredetectionoptions submenu added.
- temperaturedifference submenu added.
- temperaturedifferenceoptions submenu added.
- tankleveldetection submenu added.
- casetampering submenu added.
- boxtemperaturedetection
  - ROI.#.DetectionDistance parameter added.
  - ROI.#.TemperatureOffsetEnable parameter added.
  - ROI.#.TemperatureOffset parameter added.
- boxtemperaturedetectionoptions
  - TemperatureOffset.Celcius.Min parameter added.
  - TemperatureOffset.Celcius.Max parameter added.
  - TemperatureOffset.Fahrenheit.Min parameter added.
  - TemperatureOffset.Fahrenheit.Max parameter added.

2.4. Version 2.6.5

Date: 2024-09-02

System

- logserver
  - Format parameter added
  - ConnectionType parameter added

Network

- interface
  - SpeedDuplex parameter added
- gpuportmap submenu added

Image

- multiimageosd
  - CustomFontSize parameter added.
- imagealignment
  - SensorResolution parameter added.

Media

- Metaimagetransfer
  - Base64ObjectImageEnable parameter added.

Transfer

- jetsonflashingserver submenu added.

EventSources

- casetampering submenu added.
- metaimagetransfer
  - Base64ObjectImageEnable parameter added.

Security

- macfilter submenu added.

Opensdk

- apps
  - <AppID>.Status enum list updated.
  - <AppID>.ChannelType parameter added.
  - ActivatedChannel parameter added.

2.5. Version 2.6.4

Date: 2024-02-01

System

- deviceinfo
  - DeviceType list extended with EmergencyBell.
  - SpeakerType enum list updated

Network

- sipaccount
  - MediaEncryptionMode parameter added
- interface
  - IPv4DHCPVendorIdentifier parameter added

Media

- speakervolume
  - Volume parameter added for INDEPENDENT mode speaker

Opensdk

- apps
  - AppName parameter added.
    - This mainly added for helping users to distinguish original app name with AppID generated by a device.

Bypass

- bypass
  - Method parameter added

Image

- camera
  - AFLKShutterSpeed parameter added
  - ManualShutterSpeed parameter added
  - TemperatureRangeMode parameter added
- whiteled
  - SupplementLightDNSwitchTime parameter added
  - SupplementLightDurationMode parameter added
  - SupplementLightColorToBWThreshold parameter added
  - SupplementLightBWToColorThreshold parameter added
- imageoptions
  - AFLKMode.#.ShutterSpeed parameter added
  - SupplementLightDurationMode.#.BWToColor parameter added
  - SupplementLightDurationMode.#.ColorToBW parameter added
  - MinManualShutterSpeed parameter added
  - MaxManualShutterSpeed parameter added
- focus
  - FocusFeedbackEnable parameter added
  - ResetMaxFocus parameter added

Display

- pvmsetup, pvmpowerschedule and bannerimage submenus added

EventRules

- dynamicrules

- MonitorAlert enum value added to EventAction.#.Type parameter
- Rule.#.EventAction.#.MonitorAlertMode parameter added
- EventAction.#.MonitorAlertMode parameter added

- dynamicrulesoptions
  - EventSource policy enum updated
  - MonitorAlert enum value added to EventSource.#.ActionTypes and AppEventSource.#.ActionTypes
- rules
  - Information on deprecation has been included
- ledpreset
  - Brightness parameter added
  - AntiFlashing parameter added
  - PrechargeTime parameter added
- audiooutfiles
  - Conditions for INDEPENDENT mode speaker added
- ttsfiles
  - Conditions for INDEPENDENT mode speaker added

EventSources

- Objectdetection
  - VideoOutOverlay.<ObjectType>.LineColor parameter added
- Sourceoptions
  - VideoOutOverlayObjectTypes parameters added
- boxtemperaturedetectionoptions
  - TemperatureRangeMode parameters added
- boxtemperaturedetection
  - AreaName parameters added

EventActions

- complexaction
  - Information on deprecation has been included

EventStatus

- eventstatus
  - Details regarding the minimum SUNAPI version necessary for SchemaBased events have been incorporated

OpenSdk

- apps

    - <AppID>.ModelName parameter added

2.6. Version 2.6.3

Date: 2023-09-11

System

- deviceinfo
    - WisenetPlatformVersion parameter added
- geolocation
    - Mode enum updated with Off
- registeredsubdevices
    - Device.#.Type enum updated
    - Device.#.AudioSourceType parameter added

Security

- users
    - VideoProfileSettingAccess
    - ImageSettingAccess parameter added
    - VideoSetupAccess parameter added
    - FocusSetupAccess parameter added
    - PlaybackAccess parameter added
    - Check action added with CurrentUserCount and MaxUserCount parameter.

Media

- speakervolume submenu added
- backchannelinfo submenu added
- speakerstatus submenu added
- networkaudioinput submenu added

Image

- whiteled submenu added
- autofocuszone submenu added
- wiseautofocus submenu added
- overlay
    - PresetNamePositionX and PresetNamePositionY parameter added
    - QuickZoomAndFocusEnable parameter added

Transfer

- ftp
    - SFTP support added

Network

- sipaudiocodec submenu added

EventSources

- defocusdetection
    - OneShotAF parameter added
- callrequest
    - MicrophoneActivationMode parameter added
- autotracking
    - LostModeTimer parameter added
    - Redetection parameter added
    - RedectionTimer parameter added
- boxtemperaturedetection
    - ROI.#.AreaName
    - ROI.#.PolygonCoordinates
    - ROI.#.AreaColor
    - ROI.#.Condition.#.Duration
    - ROI.#.Condition.#.DetectionType
    - ROI.#.Condition.#.ThresholdTemperature
    - ROI.#.Condition.#.TemperatureType
    - ROI.#.Condition.#.Enable
    - EntireAreaAvgTemperatureOverlay
    - EntireAreaMinTemperatureOverlay
    - EntireAreaMaxTemperatureOverlay
    - check action added
- vehiclecount submenu added

EventRules

- dynamicrules
    - LightAlarm enum value added to EventAction.#.Type parameter
- dynamicrulesoptions
    - EventSource policy enum updated
    - LightAlarm enum value added to EventSource.#.ActionTypes and AppEventSource.#.ActionTypes

- audiooutfiles
  - DeviceID parameter added for install action
  - Name parameter added for remove action
  - Volume parameter added for control action
  - LoopCount parameter added for control action
- ttsfiles
  - Name parameter added for install action
  - DeviceID parameter added for install action
  - Language parameter added for install action
  - Gender parameter added for install action
  - Pitch parameter added for install action
  - Speed parameter added for install action
  - DelayBetweenSentence parameter added for install action
  - DelayBetweenComma parameter added for install action
  - Volume parameter added for control action
  - LoopCount parameter added for control action
  - Name parameter added for remove action

PTZControl

- movestatus submenu added

PTZConfig

- group
  - Name parameter added
- tour
  - TourMode
  - EnhancedSequence
  - SequenceType
  - PresetIndex
  - GroupIndex
  - TraceIndex
  - SwingMode
- trace
  - view action added
  - Trace.#.Enable parameter added
  - Trace.#.Name parameter added

- autorun
  - AutorunDuringSequence parameter added
- presetvideoanalysis2
  - DefinedArea.#.RuleName
  - DefinedArea.#.ObjectTypeFilter
  - DefinedArea.#.ObjectTypeFilterDetails
  - Line.#.RuleName
  - Line.#.ObjectTypeFilter
  - Line.#.ObjectTypeFilterDetails
- presetobjectdetection submenu added
- ptzsettings
  - QuickZoomEnable parameter added

Recoding

- vehiclecountsearch submenu added

2.7. Version 2.6.2

Date: 2022-01-30

System

- New device class IPAS (IP Audio System) added
- systemlog
  - MQTTConnection enum value added to Type parameter
  - SSDManagement enum value added to Type parameter
- deviceinfo
  - DeviceType list extended with NetworkSpeaker and NetworkMic.
  - SpeakerType parameter added
- registeredsubdevices submenu added
- speakergroups submenu added
- ssdstorage submenu added
- localvms submenu added

Network

- sipsetup
  - CallStopEnable parameter added
- sipaccount
  - SIPProxyAddress parameter added

- sipcall
  - TestCallState parameter added
  - CallRequest parameter added
  - Ringing enum value added to CallState parameter
  - StopCallRequest parameter deprecated (Use CallRequest parameter instead)
- mqttclient submenu added

Recording

- eventsearch submenu added

Image

- imageoptions
  - OSDType.Title.SupportedLanguages parameter added
- focus
  - SimplefocusSchedule added

Media

- videoprofile
  - IsDPMProfile parameter added
- speakervolume submenu added

Eventsources

- proximitysensor submenu added
- dtmf
  - HandoverIndex parameter added
- mqttpublication, mqttsubscription submenu added

Eventstatus

- New Events Added
  - ProximitySensor
  - MQTTSubscription
- eventstatus
  - MQTTSubscription parameter added

Eventrules

- dynamicrules
  - Rule.#.EventAction.#.MQTTMessageIndex parameter added
  - MQTTPublication enum value added to EventAction.#.Type parameter
  - MQTTSubscription enum value added to EventSource.#.Type parameter

- dynamicrulesoptions
  - MQTTPublication enum value added to EventSource.#.ActionTypes and AppEventSource.#.ActionTypes parameter
- audiooutfiles
  - Name parameter added under control operation
  - SpeakerID parameter added under control operation
  - GroupID parameter adder under control operation
- ttsfiles submenu added

AI

- metaattributesearch
  - async search support added
- ocrsearch
  - async search support added
- facerecognitionsearch
  - async search support added

2.8. Version 2.6.1

Date: 2022-08-01

System

- powermode submenu added
- eventlog
  - Supported Type extended with DynamicRule

Media

- metadatashare submenu added
- cameradiscovery
  - ANALOG enum value added to Protocol parameter

PTZConfig

- ptzsettings
  - MountPosition parameter added

Image

- camera
  - ThermalVariationSensitivity parameter added
- ptr
  - PanAngle parameter added to view response

- TiltAngle parameter added to view response
- privacy
  - AdjustMode parameter added
- imagealignment
  - AlphaBlendingLevel parameter added
- imageoptions
  - LCEMode.#.Top parameter added
  - LCEMode.#.Bottom parameter added
  - LCEMode.#.Left parameter added
  - LCEMode.#.Right parameter added
- thermalnuc submenu added
- stereosensorcalibration
  - ZoomScale parameter added to view response

EventRules

- dynamicrules
  - Rule.#.Status parameter added
  - Rule.#.EventSource.#.Type supported values updated
  - Rule.#.EventSource.#.AppName parameter added
  - Rule.#.EventSource.#.RuleIndexType parameter added
  - Rule.#.EventSource.#.RuleIndex parameter added
  - Rule.#.EventSource.#.Channel parameter added
  - Rule.#.EventSource.#.State parameter added
  - Rule.#.EventAction.#.Type parameter added
  - Rule.#.EventAction.#.AudioClipIndex parameter added
  - Rule.#.EventAction.#.HandoverIndex parameter added
- dynamicruleoptions submenu added
- schedulelist
  - Schedule.#.IsFixed parameter added
  - EveryDay Parameter added
  - EveryDay<h> parameter added
  - <dddh>.FromTo parameter added
  - EveryDay<h>.FromTo parameter added

Application programmers guide

- pw_init cgi statuscheck submenu

    - MaxPasswordLength parameter added

IP Installer

- RSA Key Response updated to include maximum password length supported
  Device Type 0x07: LEDBox newly added

2.9. Version 2.6.0

Date: 2022-01-24

System

- systemimage submenu added.
- localvms
    - LicenseBackup and LicenseRestore support added

Network

- tutk submenu deprecated.
- p2p submenu added.
- interface.
    - ICMPEnable parameter added.

Security

- users
    - Optional current password verification support added.
    - New user authority option for Privacy area access added.

Image

- multiimageosd
    - New parameters added to support the changing image position.
- privacy
    - New parameters added to support the reordering mask index.

Media

- videoinput submenu added.
- videocodecinfo
    - MinFPS parameter added.

Event

- ledindicator
    - EnableFlickeringAlarm parameter added
- eventstatusschema submenu modified.

PTZ

- exclusiveptzcontrol submenu added.

2.10. Version 2.5.9

Date: 2021-08-09

Network

- Sipsetup submenu added.
- Nattraversal submenu added.
- Sipaccount submenu added.
- Siprecipients submenu added.
- Sipcall submenu added.

Security

- 802Dot1x
  - ClientCertificateInUse parameter added.
  - CACertificateInUse parameter added.
- ssl
  - Certificate.#.Issuer parameter added.
  - Certificate.#.SerialNumber parameter added.
  - Certificate.#.Signature parameter added.
  - Certificate.#.Thumbprint parameter added.
  - Certificate.#.IsEncrypted parameter added.
  - Certificate.#.Version parameter added.
- rsa
  - PublicKeyFormat parameter added.
- Cacertificate submenu added.

Media

- Wisestream
  - AISupportEnable parameter added.
- Videoprofile
  - IsEvenNumberFrameRateProfile parameter added.
  - IsVoIPProfile parameter added.
- Videocodecinfo
  - VoIP profile resolution and bitrate parameters added.
- Camera

  - SubShutterSpeedRatio parameter added.
  - PreferShutterAISupportEnable parameter added.
- Imageoptions
  - MaxDISSensorFrameRate parameter added.
- eqsettings
  - Removed because this submenu has been deprecated.

Transfer

- ftp
  - ReportyFileType parameter added.
  - PreEventDuration and PostEventDuration parameter added.

Eventrules

- handover2
  - Action parameter added.
  - Query parameter added.

Eventsources

- Videoanalysis2
  - DefinedArea.#.ObjectTypeFilterDetails parameter added.
  - Line.#.ObjectTypeFilterDetails parameter added.
- Heatmap
  - AutoReference parameter added.
  - ManualModeEnable parameter added.
  - ManualReference parameter added.
- Callrequest submenu added.
- Dtmf submenu added.
- Tamperswitch submenu added.
- Objectdetection
  - ObjectTypeDetails parameter added.
  - HandoverIndex parameter added.
- Metaimagetransfer
  - ImageQuality parameter added.
- Socialdistancingviolation submenu added.
- Indicationpass submenu added.
- Ledindicator submenu added.

- Parkingdetection submenu added.

Ptzcontrol

- Digitalrtz submenu added.
- Supportedptzactions submenu added.

Opensdk

- Apps
    - Update action based on AppID added.
- Metaframeschema submenu added.
- Metaframecapability submenu added.

2.11. Version 2.5.8

Date: 2021-02-18

System

- Deviceinfo
    - New Device type "IOBox" added
    - CameraRegistrationMode parameter added for recorder.
- Geolocation submenu added
- Clientregister submenu added

Network

- Onvifdiscovery submenu added
- Ssl submenu updated for self-signed certificate creation.
- Camerausers
    - IsInitPasswordSet parameter added to view

Video

- Slideshow submenu added

Media

- Videooutput
    - Quadview type is added
- Videoprofile
    - RTPMulticastType and RTPMulticastAddressIPv6 parameters added for configuring Ipv6 multicast address to profile.

Image

- Irled

    - LEDMaxPowerLevel and Zone.#.Level parameter added
- Imagealignment
    - FovAngle parameter added
- Imageoptions
    - LensModel.#.SupportLDCMode parameter added
- Thermalpalettesetting
    - ThermalVariationSensistivity parameter added
- Focuspreset submenu added
- Stereosensorcalibration submenu added
- Blackbodyconfig submenu added
- Blackbodyconfigoptions submenu added
- Radiometrysettings submenu added
- Radiometrysettingsoptions submenu added

Eventsources

- Thermaldetectionmode submenu added
- Bodytemperaturedetection submenu added
- Temperaturemeasurementregion submenu added
- Maskdetection submenu added
- Cellmotion submenu added
- Autotracking
    - ObjectFilterEnable and ObjectTypeFilter parameter added

Eventrules

- Ioboxregister submenu added

Ptzcontrol

- Areazoom
    - Profile parameter added
- Aux
    - Activate status parameter added

Ptzconfig

- Ptzsettings
    - ProportionalPTSpeed parameter added
- Ptcorrection submenu added

Ai

- Aiengine
  - Facerecognitionagreementtime and Agreementstatus parameter added

2.12. Version 2.5.7

Date:2020-07-16

System

- Deviceinfo
  - GUIVersion parameter added for Recoder
- Date
  - DateFormat and Timeformat parameters added for Recorder
- Firmwareupdate
  - OnlineUpgrade parameter added for Recorder
- Monitorout
  - Optimalresolution parameter added for Recorder
- Peerconnectioninfo submenu added

Network

- Interface
  - Interfacelabel parameter added
- Snmptrap
  - UseCommunity parameter added
- Dhcpserver and poestatus submenus added

Security

- SSL
  - Clientcertificateauthentication and updatehostname parameters added
- Users
  - Admin access privilege option added for normal users
- Camerausers
  - Password encryption support added
- Cameravalidationstatus, clienthttpsstatus, and tlsversion submenus added

Video

- Thumbnail submenu added

Media

- Videoprofile

- Showall parameter added to show non-video profiles
- Videocodecinfo
  - Compression level parameters added
- Videoencoderinstances submenu added

Image

- Camera submenu updated
- Whitebalance
  - Whitebalance mode parameter added
- Imageenhancements, imageenhacement2
  - LDCmode,xce,Disfocallength and gammacontrol parameters added
- Focus
  - Focuscontinuos and Zoomcontinuos parameters added
- Iamgeoptions submenu updated

Eventsources

- Videoanalysis2
  - Rulename and objectfilter parameters added
- Tamperingdetection
  - Rulename parameter added
- Defocusdetection
  - Rulename parameter added
- Sourceoptions
  - Exclude area min and max index parameters added
  - Minimumareasizeinpixel parameter added

Eventrules

- Dynamicrule and schedulelist submenus added
- Handover, Handover2
  - Connectionmode parameter added for http/https

Io

- Alarmoutput
  - IOPortindex parameter added for notifying the physical port index
- Ioport submenu added for configurable IO

Ptzconfig

- Digitalautotracking

◦ Objecttype filter parameter added

Recording

- General
    ◦ Substream recording parameters added
- Timeline
    ◦ BkID parameter added for recorder (bookmark)
- Bookmark and diskutility submenus added

Opensdk

- Opensdkeventinfo
    ◦ Type parameter added

Ai

- Metaattributesearch, ocrsearch, objectdetectfromimage, imagelibrary, facerecognitionsearch, aiengine, and aitimeline submenus added

2.13. Version 2.5.6

Date:2020-02-20

System

- Deviceinfo
    ◦ ActualDeviceTypeparameter added
- Stratocast, Stratocastregister submenu added
- Serial
    ◦ DeviceId parameter added
- Systemlog
    ◦ Type parameter added
- Storageinfo
    ◦ IsSDCardEncrypted parameter added
    ◦ NewDASPassword parameter added
    ◦ IsNewDASPasswordEncrypted parameter added
    ◦ IsDASEncryptable parameter added
    ◦ DASPassword parameter added
    ◦ IsDASPasswordEncrypted parameter added

Network

- Interface
    ◦ MTUSize parameter added

- Ethstatus submenu added

Security

- Camerausers submenu added

Media

- Videosource
  - VideoType parameter added
  - Add/Remove actions added
- Videooutput
  - RebootRequired parameter added
- Streamurl
  - RecordStreamType parameter added
- Videocodecinfo
  - <EncodingType>.General.<Width>X<Height>.FrameLockMaxFPS added
  - <EncodingType>.Record.<Width>X<Height>.FrameLockMaxFPS added
- Cameraregister
  - IsBypassSupported parameter added
  - VideoState, AudioState parameter added

Image

- Camera
  - ImagePresetMode.#.BLCAreaCoordinates
  - ImagePresetMode.#.NormalizedBLC added
  - BLCAreaCoordinates parameter added
  - NormalizedBLCLevel parameter added
- Ptr
  - Mode parameter added
- Ptrpreset submenu added
- Imageoptions
  - BLCAreaCoordinates-based parameter added
  - OSDType.Title.SupportedSpecialCharacters parameter added
  - BacklightType-based parameter added
  - MaxWDRSensorFrameRate parameter added
- directionindicator, noisereduction submenus added

Eventsources

- Overspeed submenu added
- Tamperingdetection
    - ChannelIDList parameter added
- Temperaturechangedetection
    - TemperatureChange.ROI.#.HandoverIndex parameter added
- Boxtemperaturedetection
    - ROI.#.HandoverIndex parameter added

Eventrules

- Rules
    - Support for PeopleCount added

Eventstatus

- Support for OpenSDK event added
- Eventstatus
    - Support for EventFilter monitoridff added
- Metadataschema
    - EventName parameter added

Ptzcontrol

- Osdmenu, Rs485Command submenus added

Ptzconfig

- Panzeroposition submenu added
- Ptzprotocol
    - Status parameta added

Opensdk

- Apps, Debug, Appstatus, Manifest
    - Channel parameter added
- Opensdkeventinfo submenu added

Display

- videooutlayout and spoutout submenus added

2.14. Version 2.5.5

Date:2019-01-20

System

- Deviceinfo

- ONVIFVersion parameter added
- Serial
  - Serialinterface parameter is added
- Storageinfo
  - DasPassword support is added
- Usbconfig
  - Enable parameter added

Network

- Rtspovertls submenu added

Media

- Videosource
  - VideoMode & Sensorcapturesize support list extended
- Videosourceoptions
  - Sensorcapturesize support extended
- Videoprofile
  - Isfixedframerateprofile parameter added
- Cameraregister
  - Camchannel parameter added
- Cameraconnection
  - Camchannel parameter added

Image

- Camera
  - ImagePresetMode based settings added
- Whitebalance
  - ImagePresetMode based settings added
- Imageenhancements2 submenu added
- Ssdr
  - Imagepresetmode based settings added
- Focus
  - Imagepresetmode based settings added
- Overlay
  - Imagepresetmode based settings added
- Multilineosd

- Imagepresetmode based settings added
- Imageoptions
  - AgcMode and Imagepresetmode parameter added
- Autoimagealignment submenu added
- Imagepreset2 submenu added
- Imagepresetschedule submenu added
- Thermalpalletesetting submenu added
- Thermalpalletesettingoptions submenu added
- Spottemperaturereading submenu added

Transfer

- Dataserver submenu added

Eventsources

- Videoanalysis2
  - Roi based Duration support added
  - Defined area intrusion duration added
  - DetectionResult overlay support added
- Wiperhousing submenu added
- SoureOptions
  - Eventsource expanded for BoxTemperature detection
  - Eventaction expanded for AudioClip playback
- Boxtemperaturedetection submenu added
- Boxtemperaturedetectionoptions submenu added

Eventrules

- Rules
  - Support for boxtemperaturedeteciton and audioclip added
- Audiooutfiles submenu added
- Internalhandovercalibration submenu added
- Audiooutfileschedule submenu added

Eventstatus

- Eventstatus
  - Support for BoxTemperatureDetection,HousingTampering, WaterLevelWarning added
- Metadataschema submenu added
- Eventstatusschema submenu added

Ptzconfig

- Presetimageconfig
  - OpticalDefogFilterEnable parameter added
- Presetvideoanalysis2
  - ROI duration and Defined area instrusionduration support added
- Ptzsettings
  - Imagepresetmode support added

Recording

- Timeline
  - Boxtemperaturedetection event support added

2.15. Version 2.5.4

Date:2018-08-07

System

- Deviceinfo
  - DeviceReady and DeviceMode parameters added
- Date
  - LastSyncTime, Week, DSTStart, DSTEnd, TimeZone and ActivateServer parameters added
- Storageinfo
  - SDCardEncryption parameter added
- Added holday, hddalarm, raid, monitorin and monitorout submenus

Network

- Added dhcpclients submenu

Media

- Videosource
  - RemoteName, VideoType and VideoMode parameters added
- Added videosourceoptions and eqsettings submenus
- Videoprofile
  - H264.MinDynamicFPS parameter added
- Audioinput
  - NoiseReduction and NoiseReductionSensitivity parameters added
- Videocodecinfo
  - AllProfiles parameter added
- Added Cameraregister, Autoregister and Cameradiscovery submenus

- Camerapasswordchange
    - OldPassword and IsPasswordEncrypted parameters added
- Added Cameraconnection submenu

Image

- Camera
    - LensModel, TemperatureUnit, ThermalColorPalette, ExposureControlSpeed parameters added
    - DayNightMode extended with ScheduleBW
- Focus
    - AutoFocusRange parameter added
- Imageoptions
    - CompensationMode.#.SensorCaptureFrameRate.#.DefaultMaxShutterSpeed
    - and CompensationMode.#.SensorCaptureFrameRate.#.DefaultMinShutterSpeed parameters added

Eventsources

- Videoanalysis2
    - ROI.#.HandoverIndex, DefinedArea.#.HandoverIndex and Line.#.HandoverIndex parameters added
- Audioanalysis
    - ConfigurationToken and HandoverIndex parameters added
- Facedetection
    - DynamicArea parameter added
- Heatmap
    - Resolution parameter added
- Tamperingdetection
    - HandoverIndex parameter added
- Sourceoptions
    - ROIIncludeMinIndex, ROIIncludeMaxIndex,
    - ROIExcludeMinIndex, ROIExcludeMaxIndex, DefinedAreaIncludeMinIndex,
    - DefinedAreaIncludeMaxIndex, DefinedAreaExcludeMinIndex and
    - DefinedAreaExcludeMaxIndex parameters added
- Added temperaturechangedetection, temperaturechangedetectionoptions and shockdetection submenus

Eventactions

- Complexaction parameter added

Eventrules

- Rules
    - EventSource enum extended with ShockDetection
- Added handover2 submenu

Eventstatus

- Eventstatus
    - Channel.#.EventType parameter extended with ShockDetection, TemperatureChangeDetection

Ptzcontrol

- Continuous
    - ViewModeType parameter added
- Stop
    - ViewModeType parameter added

Ptzconfig

- Presetimageconfig
    - AutoFocusRange parameter added
- presetvideoanalysis2
    - ROI.#.HandoverIndex, DefinedArea.#.HandoverIndex and Line.#.HandoverIndex parameters added
- Ptzsettings
    - SpeedType parameter added
- Ptlimits
    - DaysAfterReboot and StartTime parameters added
- Ptzprotocol
    - ConnectionPortType parameter added

Recording

- Timeline
    - Type and Channel.#.Result.#.Type parameters extended withShockDetection, TemperatureChangeDetection

Display

- Added layout, decoderboardinfo, wall, sequence and favorite submenus

2.16. Version 2.5.3

Date: 2017-09-15

System

- Added power, iscsidiscovery and sdcardinfo submenus
- Deviceinfo
  - RequestedClientIPAddress parameter added
- Date
  - SyncType enum extended with GPS
- Serial
  - Signaltermination parameter added
- Systemlog
  - GsensorEvent and GPSDisconnect event added
- Eventlog
  - QueueEvent added
- Profileaccesinfo
  - Channel-based profilenamelist and currentbitrate parameters added
- Databasereset
  - Datatype extended with QueueEvents
- Storageinfo:
  - IsNASPasswordEncrypted and IsCHAPPasswordEncrypted added
- digitalsignage:
  - IsFTPPasswordEncrypted added

Network

- Added Mts and wifi submenus
- Interface
  - InterfaceType extended with WIFI
  - IsPPoEPAsswordEncrypted parameter added
- Dynamicdns
  - IsPublicPasswordEncrypted parameter added
- Snmp
  - IsPasswordEncrypted parameter added

Security

- Added rsa submenu
- 802Dot1x
  - IsEAPOLPasswordEncrypted parameter added
- Additionalpassword

    - IsPasswordEncrypted parameter added
- Users
    - IsPasswordEncrypted parameter added
- Usergroups
    - EventMenuAccess extended with Gsensor
    - NetworkMenuAccess extended with MTS

Media

- Added setsynchronizationpoint submenu
- Videosource
    - Videosourcetoken parameter added
- Videooutput
    - Channel parameter added
- Videoprofile
    - Viewmodetype extended with QuadView.# parameter.
- Multicast
    - Channel parameter added
- Mediaoptions
    - Viewmodetype extended with QuadView.# parameter.
- Videocodecinfo
    - Viewmode extended with QuadView.#

Image

- Added ptr, ptrzusage, scheduler and flangeback submenus
- Camera
    - LensModel and IrisMode parameters added
- Focus
    - ImagePreview parameter added
    - IRShift parameter added
    - Temperature compensation enable added
- Fisheyelens
    - RPLnumber parameter added
    - LensModel parameter enum extended with BoowonOpticals
- Fisheyesetup
    - LensModel parameter enum extended with BoowinOpticals
    - ViewModeType parameter enum extended with QuadView.#

- Multilineosd
    - OSDBlink parameter is added
    - ViewModeType parameter enum extended with QuadView.#
- Viewmodes
    - ViewMode.#.Type parameter enum extended with QuadView.#
- Imageoptions
    - ViewModeType parameter enum extended with QuadView.#

Transfer

- ftp
    - IsPasswordEncrypted parameter added
- Smtp
    - IsPasswordEncrypted parameter added
- http
    - IsPasswordEncrypted parameter added
    - IsProxyPasswordEncrypted parameter added

Eventsources

- Added Peoplecount, Heatmap and QueueManagement and gsensor submenus
- Facedetection
    - Overlaycolor parameter is added
- Sourceoptions
    - Extended the enum with peoplecount, heatmap and queuemanagement.

Eventactions

- Added Complexaction submenu
- Smtp
    - Supported enum in Systemevent extended with EmergencyTrigger and gsensor

Eventrules

- Handover
    - IsPasswordEncrypted parameter added

Eventstatus

- Added Eventscheme submenu
- QueueEvent support added to Channel.#.Eventype
- EmergencyTrigger, InternalHDDWarmup, GSensorEvent, GPSDisconnect, WiFiSignalChanged added to Systemevent

Ptzconfig

- Presetimageconfig
  - Gamma, Control, LDCEnable, LDCMode, LDCLevel, SSNRMode, SSNR2DLevel, SSNR3DLevel parameters added
- Ptzsettings
  - Imagepreview parameter added
- Ptlimits
  - Channel, TiltRange parameter added

Recording

- Added queuesearch submenu added
- General
  - Recordoverlap extended support for FogDetection, AudioAnalysis, EmergencyTrigger, GSensorEvent
- Timeline
  - Type parameter extended for QueueEvent, Videoloss, EmergencyTrigger, InternalHDDWarmup, GSensorEvent

Opensdk

- Added Debug submenu

2.17. Version 2.5.2

Date: 2017-05-15

System

- Added logserver and sessioninfo submenu
- systemlog
  - Added DatabaseRemove, USBWIFIConnect, and USBWIFIDisconnect to Type
- eventlog
  - Added FogDetection and AudioAnalysis to Type
- firmwareupdate
  - Added check action

Network

- Added tutk submenu

Media

- Added mediaoption submenu
- videoprofile

    - Added CropRatio
    - Added MJPEG.PriorityType
    - Added H264.DynamicFPSEnable
    - Added H265.DynamicFPSEnable
- videocodecinfo
    - Added <EncodingType>.General.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.Record.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.Email.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.DigitalPTZ.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.Record.<ViewMode>.<Width>X<Height>.DefaultFPS
    - Added <EncodingType>.Email.<ViewMode>.<Width>X<Height>.DefaultFPS

Image

- camera
    - Added On and Off values to HLCMode
    - Added Black, Blue, Red, Cyan, and Magenta values to HLCMaskColor
    - Added HLCDim
    - Added HLCAreaTop
    - Added HLCAreaBottom
    - Added HLCAreaLeft
    - Added HLCAreaRight
    - Added PreferedShutterSpeed
    - Added SSNR2DLevel
    - Added SSNR3DLevel
    - Added WDRSeamlessTransition
    - Added WDRLowLight
    - Added WDRIRLEDEnable
- imageenhancements
    - Added control action
    - Added Contrast
    - Added LDCEnable
    - Added LDCMode
    - Added LDCLevel

- imageoptions
  - Added LeftHalfView and RightHalfView values to ViewModeType
  - Added TopLeftCoordinates
  - Added BottomRightCoordinates
  - Added CompensationMode.#.DefaultAutoShortShutterSpeed
  - Added CompensationMode.#.DefaultAutoLongShutterSpeed
  - Added CompensationMode.#.SensorCaptureFrameRate.#.DefaultPreferShutterSpeed
  - Added CompensationMode.#.SensorCaptureFrameRate.#.AutoShortShutterSpee
  - Added CompensationMode.#.SensorCaptureFrameRate.#.AutoLongShutterSpeed
  - Added CompensationMode.#.SensorCaptureFrameRate.#.PreferShutterSpeed

Recording

- Timeline
  - Added FogDetection and AudioAnalysis to Type
  - Added DefocusDetection, FogDetection, and AudioAnalysis to Channel.#.Result.#.Type

PTZ

- Added ptzmode submenu

Event

- Added videoanalysis2 submenu under eventsources.cgi
- Added audioanalysis submenu under eventsources.cgi
- Added fogdetection submenu under eventsources.cgi
- Added samples submenu under eventsources.cgi
- Added pushnotification under eventstatus.cgi
- facedetection
  - Added DetectionArea.#.Mode
- tamperingdetection
  - Added DarknessDetection, Duration, SensitivityLevel, and ThresholdLevel parameters
- defocusdetection
  - Added Duration, ThresholdLevel, and AutoSimpleFocus parameters
- heatmap
  - Added the check action
- eventrules.cgi
  - Added FogDetection,and AudioAnalysis values to EventSource
- eventstatus.cgi

    - FogDetection, SDFormat, SDFail, SDFull, SDInsert, AudioAnalytics, and USBWIFIConnect to Channel.#.EventType

OpenSDK

- Changed ApplicationSessionID to ApplicationSessionId

2.18. Version 2.5.1

Date: 2016-09-30

System

- Added databasereset submenu
- Systemlog
    - New Type (databasefull) added

Media

- Videoprofilepolicy
    - Added Channel
- streamuri
    - DeviceID added
    - Profiletoken added
- Videoprofile
    - Added IsDigitalPTZProfile
    - Added Bitrate
    - Added H264.BitrateControlType
    - Added H265.BitrateControlType
    - Added MPEG4.BitrateControlType
    - Added ViewModeIndex
    - Added ViewModeType
- Videocodecinfo
    - ChannelIDList added
    - ViewMode added
    - MaxUnknownTargetBitrate added
    - MinUnknownTargetBitrate added
    - <EncodingType>.IsDigitalPTZSupported
    - Viewmode based resolution and fps added

Network

- Portconf

    - ProtocolType added
- Added Standbydeviceinfo submenu

Eventsources

- Added Heatmap submenu
- Added peoplecount submenu
- Added autotracking submenu

Recording

- Added Posconf submenu
- Added poseventconf submenu
- Added posdata submenu
- Added poscalendar submenu
- Timeline
    - PrimaryDeviceIPAddress added
- Added metadata submenu
- Calendarsearch
    - PrimaryDeviceIPAddress added
    - IgnoreChannelBasedResults added
    - Result added
- Searchrecordingperiod
    - ResultsInUTC added
- Storage
    - Channel added
- Added smartsearch submenu
- Heatmapsearch
    - ResultImageType added
    - ResultAsImage added
- Added peoplecountsearch submenu

Security

- Usergroups
    - DeviceMenuAccess added
- Added Additionalpassword submenu

PtzControl

- SubViewIndex parameter added for all ptz operations

EventStatus

- Eventstatus
    - Pos EventType added

Image

- Privacy
    - Channel Added
    - Mode added
    - Mask Index added
    - Mask Coordinates added
- Viewmodes
    - ViewMode Type added
- Focus
    - Mode added
- Fisheyelens
    - LensModel added
- Added Fisheyesetup submenu
- Added Imagealignment submenu
- Added imageoptions submenu

Eventrules

- Handover
    - PresetIndex added
- Added scheduler submenu

Ptzconfig

- Preset
    - SubViewIndex added
- PtzSettings
    - RememberLastPosition and RememberLastPositionduration added

2.19. Version 2.5.0

Date: 2016-06-20

NOTE	JSON response support added

System

- Device Information

    - Added Memo
- Firmwareupdate
    - Added IgnoreMultipartResponse
- OnvifFeature
    - Added FocusControl

Network

- Added portconf submenu

Media

- Videoprofile
    - Added IsFixedProfile
    - Added H264.DynamicGOVLength
    - Added H265.DynamicGOVLength
- Videocodecinfo
    - Added <encodingtype>.Record.<width>X<height>.DefaultCBRTarget
    - Added <encodingtype>.Record.<width>X<height>.DefaultVBRTarget
    - Added <encodingtype>.DigitalPTZ.<width>X<height>.Width
    - Added <encodingtype>.DigitalPTZ.<width>X<height>.Height
    - <EncodingType>.DigitalPTZ.<Width>X<Height>.MaxFPS
    - <EncodingType>.DigitalPTZ.<Width>X<Height>.MinCBRTargetBitrate
    - <EncodingType>.DigitalPTZ.<Width>X<Height>.MaxCBRTargetBitrate
    - <EncodingType>.DigitalPTZ.<Width>X<Height>.DefaultCBRTargetBitrate

Image

- Camera
    - Added SSNRMode
    - Added ImagePreview
- Whitebalance
    - Added ImagePreview
- Imageenhancements
    - Added ImagePreview
- Irled
    - Added Mode
    - Added ImagePreview
- SSDR

    - Added Imagepreview
- Focus
    - Added FastAutoFocus
- Overlay
    - Added ImagePreview
- Privacy
    - Added CommonMaskColor
- Added fisheyesetup submenu
- Added multiimageosd submenu
- Added multilineosd submenu
- Added imageoptions submenu
- Imagepreset
    - Added imagepreview
    - Added Mode
    - Added Schedule.#.Mode

Eventsources

- Added defocusdetection submenu
- Added Sourceoptions submenu
- Audiodetection
    - Added check action
- Rules
    - Added eventsource

Eventrules

- Added handover submenu

Eventstatus

- Eventstatus
    - Profile.#.DigitalAutoTracking added

Ptz

- Continuous
    - Added IgnoreIfBusy
    - Added digitalautotracking submenu

2.20. Version 2.4.3

Date: 2016-05-15

System

- Eventlog
  - Type: added Defocusdetecion, TrackingStart and TrackingEnd.

Media

- Videoprofile
  - IsDigitalPTZProfile added

Image

- Focus
  - Channel added for NVR
  - Mode added for NVR

EventStatus

- Eventstatus
  - New eventtypes added (DefocusDetection and Tracking added)

Recording

- General
  - RecordOverlap (DefocusDetection and Tracking added)
- Timeline
  - Type(DefocusDetection and Tracking added)
- Added Smartsearch submenu

2.21. Version 2.4.2

Date: 2016-01-30

System

- Added vehicleinformation submenu

Network

- DNS
  - Interfacename added
- Dynamicdns
  - Interfacename added

Media

- Streamuri
  - Eventsearchtype added

Eventstatus

- Eventstatus
  - New eventtypes added (SDFail and SDFull added)

Eventactions

- Smtp
  - SystemEvent added
  - RecipientGroupID.#.SystemEvent added

Ptzcontrol

- Added Aux submenu

Recording

- Storage
  - Channel added
- Timeline
  - EventSearchType added

2.22. Version 2.4.1

Date: 2015-03-28

System

- Added gps submenu
- Added autobackup submenu
- Added digitalsignage submenu

Network

- Interface
  - Added IPv4SubnetMask
- Dns
  - Added IPType

Security

- Usergroups
  - DeviceMenuAccess Added

Media

- Videoprofilepolicy

- LiveMode added

- Videoprofile
  - H265.BitrateControlType added
  - H265GoveLength added
  - H265.Profile added
  - H265.EntropyCoding added
  - H265.SmartCodecEnable added
- Added camerapasswordchange submenu

Image

- Imageenhancements
  - LDCEnable added
  - LDCLevel added
- Flip
  - Rotate added

Transfer

- ftp
  - Status added
- Smtp
  - Status added

2.23. Version 2.4.0

Date: 2014-02-28

System

- Device Information
  - Added MicomVersion
- Date and Time
  - Added NTPStatus and NTPLastUpdatedTime
- Factory Reset
  - Added the ‘None’ value to ExcludeSettings
- Storage Information
  - Added SlotNumber, Storage.#.Usage, Storage.#.Model, and Storage.#.Temperature, TargetIP, Port, TargetIQN, CHAPUserID, and CHAPPassword

Network

- Added svp submenu (15 SVP)

- Added bandwidth submenu (16 Bandwidth)
- Network Interface
    - Added IsDefaultGateway, and HostName
- DNS
    - Added Type
- DDNS
    - Added SamsungQuickConnectStatus
- SNMP Trap Settings
    - Added Trap.#.LinkDown and Trap.#.WarmStart
- RTSP
    - Added MobilePort

Security

- Added usergroups submenu (8 User Group Configuration)
- Added authority submenu (9 Authority)
- IP Address Filtering
    - Added IPType for the set action
- 802.1x
    - Added InterfaceName for the install and remove actions
- User Configuration
    - Added UserName, ViewerAccess, GroupID, and UserLevel

Video/Audio

- Added sessionkey submenu (2.10 Session Key)
- Video Source
    - Added Name, and State
- Video-Profile Policy
    - Added NetworkProfile and LiveProfile
- Stream URI
    - Added ClientType

Event

- Added networkalarminput submenu (2.9 Network Alarm Input)
- Added smtp submenu of eventactions.cgi (3.1 Email Sending)
- Added complexaction submenu of eventactions.cgi (3.2 Complex Action)
- Event Status

- Added SystemEvent for the check action

Transfer

- Added smtpusers submenu (4 SMTP Users)
- Added smtpgroups submenu (5 SMTP User Groups)

PTZ

- Continuous PTZ Operation Control
  - Added Iris
- Stop Control
  - Added ‘Pan’, ‘Tilt’ and ‘Zoom’ values to OperationType

I/O

- Added alarmreset submenu (5 Alarm Reset)
- Alarm Output
  - Added AlarmOutput.#.<dddh>

Recording

- Added manualrecording submenu(6 Manual Recording)
- Added searchrecordingperiod submenu (9 Recording Period)
- Added heatmapsearch submenu (10 Heat Map Search)
- Storage
  - Added DiskEndBeep
- General
  - Added FullFrameBandwidth, FullFrameRate, KeyFrameBandWidth, KeyFrameRate, Codec, RecordOverlap, SourceProfile, Resolution, FrameRate, CompressionLevel, AudioEnable, and BitrateLimit
- Recording Schedule
  - Added the ‘enum’ to value type <ddd>, EveryDay, <dddh>, and EveryDay<h> for NVR
- Timeline
  - Added the ‘UserInput’ value to Channel.#.Result.#.Type

2.24. Version 2.3.2

Date: 2014-12-12

System

- Device Information
  - Added OpenSDKVersion
- Storage Information

- Added Storage.#.Status

Network

- SNMP Trap Settings
  - Added p.#.AlarmInput.#, Trap.#.AlarmOutput.#, and Trap.#.TamperingDetection

Security

- 802.1x Setup
  - Added EAPOLType

Video/Audio

- Stream URI
  - Added OverlappedID

Image

- Camera
  - Added the 'Off' and' On' values to AFLKMode
  - Added the 'P-Iris-SLAM2890PN' value to IrisMode
  - Added AGCLevel
- White Balance
  - Added the 'Mercury' and 'Sodium' values to WhiteBalanceMode
- Focus
  - Added Channel and FocusAreaCoordinate for the Control action
- Overlay
  - Added AzimuthEnable
- Privacy
  - Added MaskIndex, MaskPattern, and ZoomThresholdEnable
- Image Preset
  - Added the 'UserPreset ' value to Mode
  - Added the 'Off' and' UserPreset' values to Schedule.#.Mode

Event

- Video Analytics Setup
  - Added the 'Off' value to EntireAreaMode
  - Added DetectionType for the view action
- Event Status
  - Added Aux for check, monitor, and monitordiff actions

PTZ

- Requesting Camera's Position Information
  - Added ZoomPulse and Iris
- Preset Image Configuration
  - Added AfterActionTrackingTime and AFLKMode
  - Added the 'Off' and 'VideoAnalytics' values to AfterAction
  - Added the 'Mercury' and 'Sodium' values to WhiteBalanceMode
- Preset Video Analysis Setup
  - Added ROI.#.Coordinate and ROIMode
- PTZ Settings
  - Added DigitalPTZEnable, and NorthDirection

Recording

- Added overlapped submenu (Chapter 5 Overlapped Recording)
- Storage
  - Added AutoDeleteEnable and AutoDeleteDays
- Timeline
  - Added the 'UserInput'value to Type and Channel.#.Result.#.Type
  - Added OverlappedID

2.25. Version 2.3.1

NOTE Internal release

Video/Audio

- Video Profile
  - Added ViewModeIndex
- Video Codec Info
  - Added ViewMode
  - Added <EncodingType>.General.<ViewMode>.<Width>X<Height>.Width, <EncodingType>.General.<ViewMode>.<Width>X<Height>.Height, <EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxFPS, <EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxCBRTargetBitrate, <EncodingType>.General.<ViewMode>.<Width>X<Height>.MinCBRTargetBitrate, <EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultCBRTargetBitrate, <EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxVBRTargetBitrate, <EncodingType>.General.<ViewMode>.<Width>X<Height>.MinVBRTargetBitrate, <EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultVBRTargetBitrate

Image

- Added the fisheyelens submenu
- Added the viewmodes submenu

Event

- Event Status
    - Added IncludeTimestamp and Timestamp for check, monitor, and monitordiff actions

PTZ

- Absolute Position Control
    - Added ViewModeIndex and SubViewIndex
- Relative Position Control
    - Added ViewModeIndex and SubViewIndex
- Continuous PTZ Operation Control
    - Added ViewModeIndex
- Requesting Camera’s Position Information
    - Added ViewModeIndex and SubViewIndex
- Moving to Preset Position
    - Added ViewModeIndex
- Group Control
    - Added ViewModeIndex
- Moving to Home Position
    - Added ViewModeIndex
- Stop Control
    - Added ViewModeIndex
- Movement Control
    - Added ViewModeIndex
- Group Setup
    - Added ViewModeIndex
- Home Position Setup
    - Added ViewModeIndex
- Preset Configuration
    - Added ViewModeIndex

2.26. Version 2.3.0

NOTE	Internal release

System

- Device Information
    - Added PasswordStrength
- Event Log
    - Added the ‘OpenSDK’ and ‘PTZMotion’ values to Type
- Profile Access Information
    - Added the ‘Optimized’ value to User.#.ClientNetworkConnectionStatus

Video/Audio

- Video Profile
    - Added ATCTrigger and ATCEventType

Event

- Auto Tracking
    - Added the TrackingAreas parameter to the view action
    - Added the TargetLockCoordinate parameter to the control action
    - Added the TrackingAreaEnable parameter to the set action
    - Added Channel, TrackingAreaID, Coordinate parameters to the add action
    - Added Channel and TrackingAreaID to the remove action
- Alarm Input
    - Changed Alarminput.#.Enable to AlarmInput.#.Enable
- Event Rules
    - Added the ‘OpenSDK’ and ‘UserInput’ values to EventSource
- Event Status
    - Added the 'PTZMotion', 'UserInput' values to Channel.#.EventType for the check, monitor, and monitordiff actions

Image

- Camera
    - Deleted the ‘Off’ value from AFLKMode

PTZ

- Auto Run Setup
    - Added the ‘Schedule’ value to Mode of the set action
    - Added the SwingMode parameter to the set action
    - Added the update action including the Channel, ScheduleMode, FromTo, Preset, Group, Trace, AutoPanSpeed, AutoPanTiltAngle, Tour, and SwingMode parameters
- Preset Image Configuration

- Added the DefogMode and DefogLevel parameters
- Changed the 'OneShot' value of FocusMode to 'OneShotAutoFocus'

- Preset Video Analysis Setup
  - Added MinimumObjectSizeInPixels, MaximumObjectSizeInPixels, IVRuleType, DetectionResultOverlay, and DisplayRules parameters
- PTZ Settings
  - Added the ProportionalPTSpeedMode parameter

I/O

- Added the UserInput submenu (4 User Input State)

Recording

- Timeline
  - Added the 'UserInput' value to Type

Open SDK

- Added opensdk.cgi with the apps, appstatus, and manifest submenus

2.27. Version 2.2.1

NOTE	Internal release

Date: 2014-01-17

System

- 2 Device Information
  - Added PTZBoardVersion, InterfaceBoardVersion, and TrackingVersion
- 3 Date and Time
  - Added NTPURLList
- 5 System Log
  - Added the 'NASFormat', 'NASFail', 'NASFull', 'NASConnect, and 'NASDisconnect' values to Type
- 7 Event Log
  - Added the 'Tracking' value to Type
- 11 Firmware Update
  - Added Status, FirmwareModule, and Progress
- 14 Storage Information
  - Added Storage.#.UsedSpace, Storage.#.TotalSpace, Storage.#.Type, Storage, Enable, DefaultFolder, NASIP, NASUserID, NASPassword, Status, and Mode parameters

Network

- Removed submenus - arping, upnpnat
- 2 Network Interface
  - Removed AutoIPv6Address, IsDefaultGateway, HostName, DomainName, and IPv6RouterAdvertisement
  - Added ‘Default’ value to IPv6Type
  - Removed the InterfaceType value
  - Removed value ranges for IPv4PrefixLength and IPv6PrefixLength
- 3 DNS
  - Removed Type and SearchDomainList
- 4 DDNS
  - Removed ‘www.changeip.org’ value from PublicServiceEntry
  - Eliminated maximum value length for SamsungProductID, PublicHostName, PublicUserName, and PublicPassword
- 5 Bonjour
  - Eliminated maximum value length for FriendlyName
- 6 UPnP Discovery
  - Eliminated maximum value length for FriendlyName
- 8 SNMP Setup
  - Eliminated maximum value length for ReadCommunity, WriteCommunity, and UserPassword
- 10 QoS Setup
  - Removed value ranges for Index, PrefixLength, and DSCP
- 13 RTSP
  - Changed values ‘0’ and ‘60’ to ‘0s’ and ‘60s’ for Timeout

Security

- 2 IP Address Filtering
  - Removed value ranges for IPIndex and Mask
- 3 802.1x Setup
  - Removed value of Status
  - Eliminated maximum value length for EAPOLId and EAPOLPassword
- 7 Configuring Users
  - Eliminated maximum value length for UserID, Index, and Password

Video/Audio

- 2.1 Snapshot
  - Changed ProfileID to Profile

- Removed value range for Profile

- 2.2 Stream
  - Added the Resolution parameter
  - Changed ProfileID to Profile
  - Removed value ranges for Profile, FrameRate, and CompressionLevel
- 2.3 Video Source
  - Added value ‘20’ to SensorCaptureFrameRate
- 2.5 Video-Profile Policy
  - Removed value ranges for DefaultProfile, EventProfile, and RecordProfile
- 2.6 Video Profile
  - Added MPEG4.BitrateControlType, MPEG4.GOVLength MPEG4.PriorityType, and H264.SmartCodecEnable
  - Removed the value for Resolution
  - Removed value ranges for Profile, Name, ATCLimit, SVNPMulticastTTL, RTPMulticastTTL, CropAreaCoordinate, FrameRate, CompressionLevel, Bitrate, and H264.GOVLength
- 2.7 Audio Input
  - Added the value to SampleRate
- 2.8 Audio Output
  - Added the value to SampleRate
- 2.9 Stream URI
  - Removed OverlappedID
  - Removed value range for Profile
- 2.10 Multicast
  - Removed value range for Profile
- 2.11 Video Codec Info
  - Added Profile parameter

Image

- Added the following submenus
  - irled (Chapter 5 IR LED)
  - smartcodec (Chapter 10 Smart Codec)
  - fisheyelens (Chapter 12 Fisheye Lens)
  - imagepreset (Chapter 13 Image Preset)
- 2 Camera
  - Added HLCMode, HLCLevel, HLCMaskTone, HLCMaskColor, WDRLimit, WDRBlackLevel, WDRWhiteLevel, ShutterMode, ManualShutterSpeed, DayNightModeSchedule.EveryDay,

DayNightModeSchedule.EveryDay.FromTo, DayNightSwitchingTimeColorToBW, DayNightSwitchingBrightnessColorToBW, DayNightSwitchingTimeBWToColor, DayNightSwitchingBrightnessBWToColor, NegativeModeEnable, SensUpMode, and SensUpLevel parameters

    - Removed the values from AutoLongShutterSpeed, AutoShortShutterSpeed, DayNightSwitchingTime and IrisFno
    - Removed value ranges for BLCAreaTop, BLCAreaBottom, BLCAreaLeft, BLCAreaRight, and SSNRLevel
    - Added values 'P-Iris-SLAM3180PN' and 'P-Iris-M13VP288IR' to IrisMode
- 3 White Balance
    - Added values 'ATW1', 'ATW2', '3200K', and '5600K' to WhiteBalanceMode
    - Removed value ranges for WhiteBalanceManualRedLevel and WhiteBalanceManualBlueLevel
- 4 Image Enhancement
    - Removed value ranges for SharpnessLevel, Brightness, Gamma, Saturation, and DefogLevel
    - Added CAR
- 7 SSDR
    - Removed value range for Level
- 8 Focus
    - Added Mode, ZoomTrackingMode, ZoomTrackingSpeed, and LensResetSchedule of set action and Zoom of the control action
    - Removed value from Focus
- 9 Overlay
    - Added PTZPositionEnable, CameraIDEnable, PresetNameEnable, and OSDColor
    - Removed value ranges for Title, TitlePositionX, and TitlePositionY
- 11 Privacy
    - Changed the value range for MaskIndex and MaskName

Transfer

- FTP
    - Changed maximum value length of Host
- SMTP
    - Changed maximum value length of Host

Event

- Added following submenus
    - videoloss (Chapter 2.5 Video Loss)
    - autotracking (Chapter 2.6 Auto Tracking)

- Video Analytics Setup
  - Added ObjectSize, MinimumObjectSizeInPixels, and MaximumObjectSizeInPixels
  - Removed value from ROIIndex
  - Removed the value range for MinimumObjectSize and MaximumObjectSize
  - Changed the parameter name from DefinedArea.#.Coordinates to DefinedArea.#.Coordinate and from Line.#.Coordinates to Line.#.Coordinate.
  - Changed the value type of EntireAreaMode from enum to csv
- Face Detection Setup
  - Changed the parameter name from DetectionArea.#.Coordinates to DetectionArea.#.Coordinate
  - Removed the value from DetectionAreaIndex
  - Removed the value range for Sensitivity
- Audio Detection
  - Removed the value range for InputThresholdLevel
- Event Rules
  - Added EveryDay, EveryDay<h>, and EveryDay<h>.FromTo
  - Removed the value range for PresetNumber
- Event Status
  - Removed the values for AlarmInput and AlarmOutput
  - Removed ChangedConfigURI from the check action
  - Added SystemEvent to the monitor and monitordiff actions
  - Added value ‘Tracking’ to Channel.#.EventType

I/O

- Alarm Output
  - Removed value from AlarmOutput.#.ManualDuration and AlarmOutput.#.ManualDuration

PTZ

- Added following submenus of ptzcontrol.cgi
  - absolute (Chapter 2.1 Absolute Position Control)
  - relative (Chapter 2.2 Relative Position Control)
  - query (Chapter 2.4 Requesting Current PTZ Info)
  - swing (Chapter 2.6 Swing Control)
  - group (Chapter 2.7 Group Control)
  - tour (Chapter 2.8 Tour Control)
  - trace (Chapter 2.9 Trace Control)
  - home (Chapter 2.10 Moving to Home Position)

- areazoom (Chapter 2.11 Zoom Area)

- Added following submenus of ptzconfig.cgi
  - swing (Chapter 3.1 Swing Setup)
  - group (Chapter 3.2 Group Setup)
  - tour (Chapter 3.3 Tour Setup)
  - trace (Chapter 3.4 Trace Setup)
  - autorun (Chapter 3.5 Auto Run Setup)
  - home (Chapter 3.6 Home Position Setup)
  - presetimageconfig (Chapter 3.8 Preset Image configuration)
  - presetvideoanalysis (Chapter 3.9 Preset Video Analysis Setup)
  - ptzsettings (Chapter 3.10 PTZ Settings)
  - ptlimits (Chapter 3.11 PT Operation Limits)
- 2.3 Continuous PTZ Operation Control
  - Added the NormalizedSpeed parameter
  - Removed value ranges for Pan, Tilt, Zoom, and Duration
- 2.12 Stop Control
  - Added values ‘Focus’ and ‘Iris’ to OperationType
- 2.13 Movement Control
  - Removed value ranges for MoveSpeed

Recording

- General
  - Removed values for PreEventDuration and PostEventDuration

2.28. Version 2.0.0

Date: 2013-03-29

NOTE	Internal release

-First edition

Attributes

SUNAPI

v2.6.8
2026-04-09

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Contact Information

Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 3. Overview

3.1. attributes.cgi

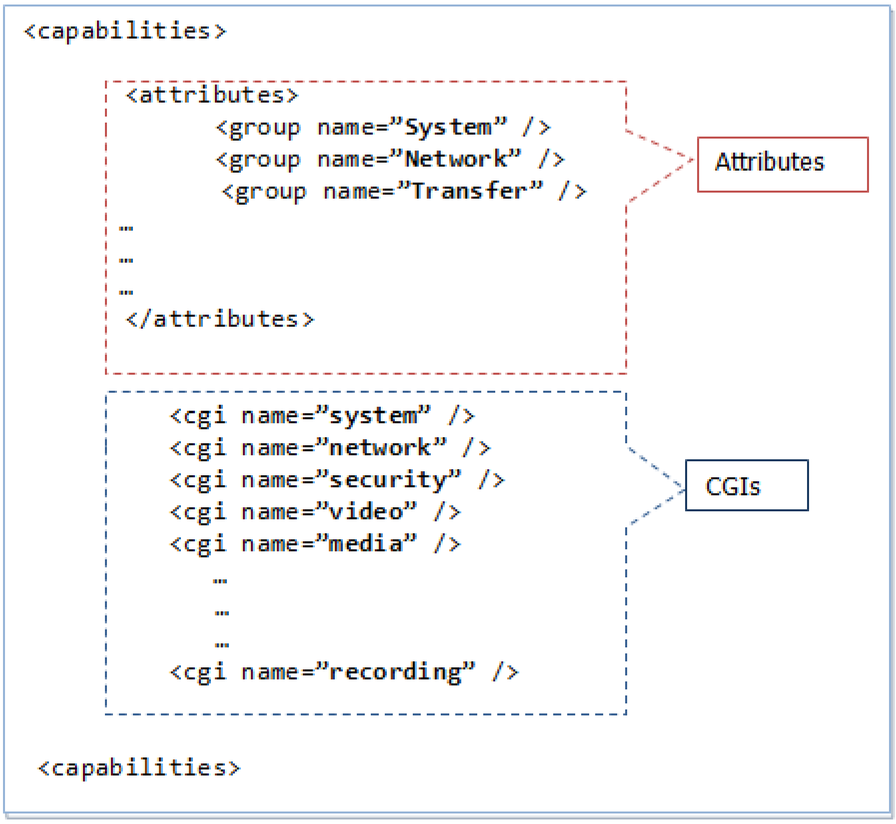
The attributes and capabilities of Hanwha Vision video surveillance devices are requested by using **attributes.cgi**.

The following CGI command returns device attributes, including parameters and allowed value ranges, in XML format.

```
http://<Device IP>/stw-cgi/attributes.cgi
```

The XML response contains two sections.

- attributes
- cgi



Each section can be requested separately. Details are provided in Chapter 2 and Chapter 3.

Chapter 4. Attributes

4.1. Overview

The attributes section of the XML contains a few high-level attributes of the features that the device supports.

4.2. Attribute request

The following command returns only the attributes data.

```
http://<Device IP>/stw-cgi/attributes.cgi/attributes
```

4.3. Group and category

Attributes are grouped by feature and the corresponding CGI, with group names such as **System**, **Network**, **Transfer**, etc. Each group is further divided into the following categories:

- **Property**: General information about the device and its CGI (typically fixed for the device and firmware).
- **Support**: Indicates whether a particular feature is supported.
- **Limit**: Specifies the device's limitations for each feature (e.g., number of video channels or maximum number of users).

4.4. Example data

Below is the **attributes** section of the XML file.

The **attributes** section consists of groups such as System, Network, IO, etc. The System group includes three categories: Property, Support, and Limit. For example, the Limit category shows the NIC count and maximum number of channels, while the Support category indicates whether system configuration or firmware updates are supported.

```
<attributes>

<group name="System">
 <category name="Property">
  <attribute name="ISPVersion" type="bool" value="True" accesslevel="guest" />
  <attribute name="PTZBoardVersion" type="bool" value="False"
       accesslevel="guest" />
  <attribute name="InterfaceBoardVersion" type="bool" value="False"
       accesslevel="guest" />
  …
 </category>
 <category name="Support">
      <attribute name="ConfigRestore" type="bool" value="True" accesslevel="admin"/>
  <attribute name="ConfigBackup" type="bool" value="True" accesslevel="admin" />
  <attribute name="FWUpdate" type="bool" value="True" accesslevel="admin" />
  …
  …
 </category>
 <category name="Limit">
  <attribute name="NICCount" type="int" value="1" accesslevel="guest" />
  <attribute name="MaxChannel" type="int" value="1" accesslevel="guest" />
  …
  …
 </category>
</group>
```

```
…
<group name="Network">
   <category name="Limit">
    <attribute name="MaxIPv4Filter" type="int" value="10" accesslevel="guest" />
…
    …
   </category>
</group>
…
<group name="IO">
   <category name="Limit">
    <attribute name="MaxAlarmOutput" type="int" value="1" accesslevel="suser" />
..
   </category>
   <category name="Support">
    <attribute name="AlarmOutput" type="bool" value="True" accesslevel="suser" />
..
   </category>
</group>
…
…
…
</attributes>
```

4.5. Attributes

The following is an attribute table.

4.5.1. System Group

Category	Attribute	Type	Description
Property	MicomVersion	<bool>	Indicates whether the device can provide the current version of the microcontroller firmware.
	ISPVersion	<bool>	Indicates whether the device can provide the current version of the Image Signal Processor (ISP) firmware.
	PTZBoardVersion	<bool>	Indicates whether the device can provide the current version of the PTZ (Pan-Tilt-Zoom) board firmware.
	InterfaceBoardVersion	<bool>	Indicates whether the device can provide the current version of the interface board firmware.

Category	Attribute	Type	Description
	TrackingVersion	<bool>	Indicates whether the device can provide the current version of the tracking firmware.
	BootLoaderVersion	<bool>	Indicates whether the device can provide the current version of the bootloader firmware.
	ModelName	<string>	Model name of the device.
	ONVIFVersion	<string>	Current version of the ONVIF (Open Network Video interface Forum) protocol in the device.
	SUNAPIVersion	<string>	Current version of the SUNAPI (Smart Unified Network API) protocol in the device.
	FirmwareVersion	<string>	Current version of the firmware in the device.
	ModelType	<enum> Box, Bullet, Dome, Vandal, PTZ, IOBox, LEDBox, Speaker, Mic	Indicates model type of the device.
	WisenetPlatformVersion	<bool>	Indicates whether the device can provide the current version of the Wisenet platform.
Support	GlobalConfiguration	<bool>	Applicable only to NVR models.
	RS422	<bool>	Indicates whether the device supports RS422 communication protocol.
	ConfigRestore	<bool>	Indicates whether the device supports configuration restoration.
	ConfigRestore.ExcludeGroups	<enum> Network, Camera, Authority	Specifies the features that can be excluded during configuration restoration.
	ConfigRestore.ExcludeGroups.Network.Selectable	<bool>	Indicates if network groups can be selectively excluded during configuration restoration.
	ConfigRestore.ExcludeGroups.Camera.Selectable	<bool>	Indicates if camera groups can be selectively excluded during configuration restoration.
	ConfigRestore.ExcludeGroups.Authority.Selectable	<bool>	Indicates if authority groups can be selectively excluded during configuration restoration.

Category	Attribute	Type	Description
	ConfigBackup	<bool>	Indicates whether the device supports configuration backup.
	FWUpdate	<bool>	Indicates whether the device supports firmware updates.
	Shutdown	<bool>	Indicates whether the device supports a shutdown feature.
	Restart	<bool>	Indicates whether the device supports a restart feature.
	GPS	<bool>	Indicates whether the device supports GPS functionality.
	AutoBackup	<bool>	Indicates whether the device supports automatic backups.
	DigitalSignage	<bool>	Indicates whether the device supports digital signage features.
	OpenSDK	<bool>	Indicates whether the device supports OpenSDK (when supported apps can be installed.)
	FreeStyleSplitMode	<bool>	Indicates whether the device supports free style split mode. NVR ONLY
	DisplayMerge	<csv> Device, Monitor, Tile	Indicates whether the device supports display merge functionality.
	SDCardEncryption	<bool>	Indicates whether the device supports SD card encryption.
	FullDuplex	<bool>	Indicates whether the device supports full duplex communication in RS485 NVR ONLY
	OneOpenAppPerChannel [**deprecated**]	<bool>	Indicates whether the device supports one open application per channel (deprecated feature).
	SupportChannelExpansionFeature	<bool>	Indicates whether the device supports channel expansion features.
	LogServer	<bool>	Indicates whether the device supports log server functionality.
	AIFeatures	<bool>	Indicates whether the device supports AI features.
	AIEngine	<bool>	Indicates whether the device includes an AI engine.

Category	Attribute	Type	Description
	Stratocast	<bool>	Indicates whether the device supports Stratocast Cloud.
	HybridThermal	<bool>	Indicates whether the device supports hybrid thermal imaging.
	ManufacturerChange	<bool>	Indicates whether the device supports changing the manufacturer name to legacy names such as HanwhaTechwin and SamsungTechwin for backward compatibility.
	SSDStorage	<bool>	Indicates whether the device supports SSD storage.
	LocalVMS	<bool>	Indicates whether the device supports a local Video Management System (VMS).
	VirtualCropChannel	<bool>	Indicates whether the device supports virtual crop channels.
	MultiLanguageEventSchema	<bool>	Indicates whether the device supports multiple language in event schema.
	LiveEventDisplay	<bool>	Indicates whether the device supports live event display.
	PowerMode	<bool>	Indicates whether the device supports different power modes.
	SubmonitorMaxResolution	<enum> 4K, HD	Indicates the maximum resolution supported by sub-monitors.
	ChannelBasedOpenApp	<bool>	Indicates whether the device supports channel-based open applications, allowing multiple instances of the same application to run, one per channel
	PVM	<bool>	Indicates whether the device supports Public View Monitor (PVM).
	DebugLogBackup	<bool>	Indicates whether the serial log backup function is supported. Backed up logs can be decrypted using a decryption tool.
Limit	MaxHDMIOut	<int>	Maximum number of HDMI outputs supported.
	NICCount	<int>	Number of Network Interface Cards (NICs) supported.
	MaxChannel	<int>	Maximum number of channels supported.
	MaxPOS	<int>	Maximum number of Point of Sale (POS) systems supported.

Category	Attribute	Type	Description
	MaxVGAOut	<int>	Maximum number of VGA outputs supported.
	MaxHDMIOut	<int>	Maximum number of HDMI outputs supported.
	MaxHDMIIn	<int>	Maximum number of HDMI inputs supported.
	MaxLiveSession	<int>	Maximum number of live sessions supported.
	MaxSearchSession	<int>	Maximum number of search sessions supported.
	MaxBackupSession	<int>	Maximum number of backup sessions supported.
	OpenSDK.MaxApps	<int>	Maximum number of applications supported by OpenSDK.
	MaxSSDStorage	<int>	Maximum SSD storage capacity supported.
	MaxPartitionPerSSD	<int>	Maximum number of partitions per SSD supported.
	NTPURLLength	<int>	Maximum length of the NTP (Network Time Protocol) URL supported.
	MaxAnalog	<int>	Maximum number of analog inputs supported.
	MaxVGASpotCount	<int>	Maximum number of VGA spot outputs supported.
	MaxAnalogSpotCount	<int>	Maximum number of analog spot outputs supported.
	ChannelExpansionLimit	<int>	Maximum limit for channel expansion.
	MaxAudioSubDevices	<int>	Maximum number of sub devices supported. IPAS ONLY
	MaxSpeakerGroups	<int>	Maximum number of speaker groups supported IPAS ONLY

4.5.2. Network Group

Category	Attribute	Type	Description
Limit	MaxIPv4Filter	<int>	Maximum number of IPv4 filters supported.
	MaxIPv6Filter	<int>	Maximum number of IPv6 filters supported.
	MaxIPv4QoS	<int>	Maximum number of IPv4 Quality of Service (QoS) rules supported.
	MaxIPv6QoS	<int>	Maximum number of IPv6 Quality of Service (QoS) rules supported.

Category	Attribute	Type	Description
	MaxNetCam	<int>	Maximum number of network cameras supported.
	MaxSIPAccountCount	<int>	Maximum number of SIP accounts supported.
	MaxRecipientCount	<int>	Maximum number of recipients supported in SIP.
	MaxRecipientInGroup	<int>	Maximum number of recipients in a group supported in SIP.
Support	BandwidthControl	<bool>	Capability to control bandwidth.
	POE	<bool>	Capability to support Power over Ethernet (PoE).
	WiFi	<bool>	Capability to support WiFi.
	TUTK	<bool>	Capability to support TUTK protocol.
	POEExtender	<bool>	Capability to support PoE extender.
	MTUSize	<bool>	Capability to configure Maximum Transmission Unit (MTU) size.
	SIP	<bool>	Capability to support SIP protocol.
	DDNSP2P	<bool>	Capability to support Dynamic DNS and Peer-to-Peer (P2P).
	MQTT	<bool>	Capability to support MQTT protocol.
	NetworkAudioIn	<bool>	Capability to support network audio input. IPAS ONLY

4.5.3. Transfer Group

Category	Attribute	Type	Description
Support	SMTP	<bool>	Capability If SMTP notification is supported.
	HTTP	<bool>	Capability if HTTP based event notification is supported.
	FTP	<bool>	Capability if FTP based image/video upload is supported.
	DataServer	<bool>	Capability if Dataserver connection is supported. CAMERA ONLY
Limit	SMTP.MaxRecipients	<int>	Maximum number of recipients supported in SMTP.

4.5.4. Security Group

Category	Attribute	Type	Description
Limit	MaxAdditionalPasswords	<int>	Maximum number of additional passwords supported. NVR ONLY
	MaxUser	<int>	Maximum number of users supported.
	MaxGroup	<int>	Maximum number of groups supported. NVR ONLY
	MaxUserPerGroup	<int>	Maximum number of users per group supported. NVR ONLY
	MaxSMTPGroup	<int>	Maximum number of SMTP groups supported.
	MaxSMTPUser	<int>	Maximum number of SMTP users supported.
	MaxSMTPUserPerGroup	<int>	Maximum number of SMTP users per group supported.
Support	AdminIDChangeable	<bool>	Capability to change the administrator ID.
	ClientCertificateAuthentication	<bool>	Capability to authenticate using client certificates.
	AdminAccess	<bool>	Capability if admin access level is supported for normal user.
	MaxSelfSignedCertificates	<int>	Maximum number of self-signed certificates supported.
	NewPasswordPolicy	<bool>	Capability to enforce a new password policy.
	CurrentPasswordVerification	<enum> Optional, Mandatory	Capability to verify the current password. On devices with Mandatory value,user password change will not be allowed without presenting current password.
	SupportCertificateFormat	<enum> PKCS1, PKCS8, PKCS12	Certificate types can be installed to the device.

4.5.5. Media Group

Category	Channel	Attribute	Type	Description
Limit	Channel.#	MaxAudioInput	<int>	Maximum number of audio inputs supported per channel.
		MaxAudioOutput	<int>	Maximum number of audio outputs supported per channel.
		MaxProfile	<int>	Maximum number of profiles supported per channel.
		MaxResolution	<enum>	Maximum resolution supported per channel.
		MaxBannerImage	<int>	Maximum number of banner images supported per channel.
	StreamingMetadata		<csv> POS, GPS	Metadata supported for streaming, in CSV format. NVR ONLY
	StreamingProfiles		<csv> Network, Live, Recording, All	Profiles supported for streaming, in CSV format. NVR ONLY
	MaxVideoOutput		<int>	Maximum number of video outputs supported.
	MaxAudioInput		<int>	Maximum number of audio inputs supported.
	GOVLengthMultiplierFactor		<int>	fps * GOVLengthMultiplierFactor = GOV Length.
Support	Channel.#	WiseStream	<bool>	Indicates whether the device supports WiseStream technology.
		DynamicGOV	<bool>	Indicates whether the device supports Dynamic Group of Video (GOV) technology.
		DynamicFPS	<bool>	Indicates whether the device supports Dynamic Frames Per Second (FPS) adjustment.
		Protocol.SUNAPI	<bool>	Indicates whether the device supports the SUNAPI protocol.
		ChannelAudioOutput	<bool>	Indicates whether the device supports audio output on a per-channel basis.

Category	Channel	Attribute	Type	Description
		DeviceAudioOutput	<bool>	Indicates whether the device supports audio output for the entire device.
		MultiAudioOutput	<bool>	Indicates whether the device supports multiple audio outputs simultaneously.
		AudioIn	<bool>	Indicates whether the device supports audio input.
		VideoOut	<bool>	Indicates whether the device supports video output.
		ATC	<bool>	Indicates whether the device supports Automatic Target Calibration (ATC).
		Crop	<bool>	Indicates whether the device supports video cropping.
		VideoEncodingType	<enum>	Indicates the types of video encoding supported by the device.
		AudioEncodingType	<enum>	Indicates the types of audio encoding supported by the device.
		VideoSetting	<bool>	Indicates whether the device supports customizable video settings.
		ExactTimeStamp	<bool>	Indicates whether the device supports exact time stamping of video frames.
		Live	<bool>	Indicates whether the device supports live streaming.
		FixedProfileCodecChange	<bool>	Indicates whether the device supports changing codecs for fixed profiles.
		ShowNonVideoProfile	<bool>	Indicates whether the device can display non-video profiles.
		ProfileAddByIndex	<bool>	Indicates whether the device supports adding profiles by index.
		Stream.UDP	<bool>	Indicates whether the device supports streaming over UDP.
		Stream.TCP	<bool>	Indicates whether the device supports streaming over TCP.

Category	Channel	Attribute	Type	Description
		Stream.Multicast	<bool>	Indicates whether the device supports multicast streaming.
		Stream.MulticastIPV6	<bool>	Indicates whether the device supports multicast streaming over IPv6.
		Stream.RTPOverRTSPOverTCP	<bool>	Indicates whether the device supports RTP over RTSP over TCP streaming.
		Stream.RTSPOverHTTP	<bool>	Indicates whether the device supports RTSP over HTTP streaming.
		Stream.RTSPOverHTTPS	<bool>	Indicates whether the device supports RTSP over HTTPS streaming.
		Stream.MultiProfiles	<bool>	Indicates whether the device supports multiple streaming profiles.
		Stream.Metadata	<bool>	Indicates whether the device supports streaming metadata.
		Stream.SRTP.UDP	<bool>	Indicates whether the device supports secure RTP (SRTP) over UDP.
		Stream.SRTP.TCP	<bool>	Indicates whether the device supports secure RTP (SRTP) over TCP.
		Stream.SRTP.Multicast	<bool>	Indicates whether the device supports secure RTP (SRTP) multicast streaming.
		Stream.SRTP.RTSPOverRTSPOverTCP	<bool>	Indicates whether the device supports secure RTP (SRTP) over RTSP over TCP streaming.
		Stream.SRTP.RTSPOverHTTP	<bool>	Indicates whether the device supports secure RTP (SRTP) over RTSP over HTTP streaming.
		Stream.SRTP.RTSPOverHTTPS	<bool>	Indicates whether the device supports secure RTP (SRTP) over RTSP over HTTPS streaming.
		Stream.RTSPBlocksize	<bool>	Indicates whether the device supports RTSP block size configuration.

Category	Channel	Attribute	Type	Description
		Stream.BackchannelOverPost	<bool>	Indicates whether the device supports backchannel over POST for streaming.
		Metadata.ImageType	<csv> Object, Scene, None	Specifies the types of images supported for metadata, in CSV format.
		Metadata.ImageTransfer	<bool>	Indicates whether the device supports image transfer for metadata.
		Metadata.ClassTypes	<bool>	Indicates whether the device supports classification types for metadata.
		Metadata.ClassTypeDetails	<bool>	Indicates whether the device supports detailed classification types for metadata.
		Metadata.Base64ObjectImage	<bool>	Indicates whether the device supports Base64 encoding for object images in metadata.
		SensorCaptureMode	<bool>	Indicates whether the device supports sensor capture mode.
		DynamicPrivacyMask	<bool>	Indicates whether the device supports dynamic privacy masking.
		NoRecordProfileLimitation	<csv> GOVLength,EntropyCoding,DynamicGOV,DynamicFPS	Specifies features that have no limitations in profiles used for the camera’s internal recording.
		WiseStreamVersion	<csv> 1,2,3	Specifies the versions of WiseStream supported, in CSV format.
	GlobalSensorMode		<bool>	Indicates whether changing the sensor mode of one channel also changes the sensor mode of other channels, as seen in some panoramic cameras.
	MetadataShare		<bool>	Indicates whether the device supports the configuration to share metadata between channels.

Category	Channel	Attribute	Type	Description
	DynamicPrivacyMask		<bool>	Indicates whether the device supports dynamic privacy masking.
	SlideShow.ImageFormat		<csv> BMP, JPG, PNG	Specifies the image formats supported for slideshows, in CSV format.
	BannerImage.ImageFormat		<csv> BMP, JPG, PNG	Specifies the image formats supported for banner images, in CSV format.

4.5.6. Image Group

Category	Channel	Attribute	Type	Description
Limit	Channel.#	MaxSmartCodecArea	<int>	Maximum number of smart codec areas supported in the channel.
		MaxPrivacyMask	<int>	Maximum number of privacy masks supported in the channel.
		MaxPrivacyMask.Rectangle	<int>	Maximum number of rectangular privacy masks supported in the channel.
		MaxPrivacyMask.Polygon	<int>	Maximum number of polygon privacy masks supported in the channel.
		MaxOSDTitles	<int>	Maximum number of On-Screen Display (OSD) titles supported in the channel.
		MaxOSDDates	<int>	Maximum number of On-Screen Display (OSD) dates supported in the channel.
		ViewModes	<int>	Number of view modes supported in the channel.
		ViewModes.MaxActive	<int>	Maximum number of active view modes supported in the channel.
		MaxIRZoneCountInCeilingMode	<int>	Maximum number of IR zones supported in ceiling mode in the channel.
		MaxIRZoneCountInWallMode	<int>	Maximum number of IR zones supported in wall mode in the channel.
		MaxFocusPreset	<int>	Maximum number of focus presets supported in the channel.

Category	Channel	Attribute	Type	Description
		MaxIRZoneCount	<int>	Maximum number of IR zones supported in the channel.
		MaxAutoFocusZoneCount	<int>	Maximum number of auto focus zones supported in the channel.
Support	Channel.#	Defog	<bool>	Indicates whether the device supports defogging in the channel.
		ProfileBasedDewarpedView	<bool>	Indicates whether the device supports profile-based dewarped views in the channel.
		MultiImager	<bool>	Indicates whether the channel is a stitched channel (i.e., composed of multiple image sensors).
		OSDLanguages	<csv> English, French, German, Spanish, Italian, Chinese, Russian, Korean, Polish, Japanese, Dutch, Portugese, Turkish, Czech, Danish, Swedish, Thai, Romania, Serbian, Croatian, Hungarian, Greek, Finnish, Norsk, TraditionalChinese	Specifies the On-Screen Display (OSD) languages supported in the channel, in CSV format.

Category	Channel	Attribute	Type	Description
		Gamma	<bool>	Indicates whether the device supports gamma adjustment in the channel.
		Saturation	<bool>	Indicates whether the device supports saturation adjustment in the channel.
		Sharpness	<bool>	Indicates whether the device supports sharpness adjustment in the channel.
		Brightness	<bool>	Indicates whether the device supports brightness adjustment in the channel.
		ResetFocus	<bool>	Indicates whether the device supports focus reset in the channel.
		SimpleFocus	<bool>	Indicates whether the device supports simple focus functionality in the channel.
		SimpleFocus.FocusArea	<bool>	Indicates whether the device supports defining focus areas for simple focus in the channel.
		PrivacyMask.ZoomThreshold	<bool>	Indicates whether the device supports privacy mask zoom threshold in the channel.
		Privacy.MaskColor.Global	<bool>	Indicates whether the device supports global privacy mask color, meaning that if the color of one privacy mask is changed, it changes the color of other privacy masks as well.
		Privacy.MaskPattern	<bool>	Indicates whether the device supports a mask pattern int the channel.
		FocusAdjust	<bool>	Indicates whether the device supports focus adjustment in the channel.
		ZoomAdjust	<bool>	Indicates whether the device supports zoom adjustment in the channel.
		IRLED	<bool>	Indicates whether the device supports IR LED in the channel.
		ZoneBasedIRLED	<bool>	Indicates whether the device supports zone-based IR LED in the channel.
		DIS	<bool>	Indicates whether the device supports Digital Image Stabilization (DIS) in the channel.
		P-Iris	<bool>	Indicates whether the device supports P-Iris in the channel.

Category	Channel	Attribute	Type	Description
		DayNight	<bool>	Indicates whether the device supports day/night mode in the channel.
		FisheyeLens	<bool>	Indicates whether the device supports fisheye lens in the channel.
		DewarpedView	<bool>	Indicates whether the device supports dewarped view in the channel.
		ThermalFeatures	<bool>	Indicates whether the device supports thermal features in the channel.
		PTRZ	<bool>	Indicates whether the device supports Pan, Tilt, Rotate, and Zoom (PTRZ) in the channel.
		ViewModes.360Panorama	<bool>	Indicates whether the device supports 360-degree panorama view mode in the channel.
		ViewModes.360Panorama.SubViews	<int>	Number of sub-views supported in 360-degree panorama view mode in the channel.
		ViewModes.360Panorama.PTZ	<bool>	Indicates whether the device supports PTZ (Pan-Tilt-Zoom) in 360-degree panorama view mode in the channel.
		ViewModes.360Panorama.PTZ.Features	<csv>	Specifies the PTZ features supported in 360-degree panorama view mode in the channel, in CSV format.
		ViewModes.OneOverviewAndTripleView	<bool>	Indicates whether the device supports one overview and triple view mode in the channel.
		ViewModes.OneOverviewAndTripleView.SubViews	<int>	Number of sub-views supported in one overview and triple view mode in the channel.
		ViewModes.OneOverviewAndTripleView.PTZ	<bool>	Indicates whether the device supports PTZ in one overview and triple view mode in the channel.
		ViewModes.OneOverviewAndTripleView.PTZ.Features	<csv>	Specifies the PTZ features supported in one overview and triple view mode in the channel, in CSV format.
		ViewModes.OneOverviewAndOctaView	<bool>	Indicates whether the device supports one overview and octa view mode in the channel.

Category	Channel	Attribute	Type	Description
		ViewModes.OneOverviewAndOctaView.SubViews	<int>	Number of sub-views supported in one overview and octa view mode in the channel.
		ViewModes.OneOverviewAndOctaView.PTZ	<bool>	Indicates whether the device supports PTZ in one overview and octa view mode in the channel.
		ViewModes.OneOverviewAndOctaView.PTZ.Features	<csv>	Specifies the PTZ features supported in one overview and octa view mode in the channel, in CSV format.
		ViewModes.Overview	<bool>	Indicates whether the device supports overview mode in the channel.
		ViewModes.Overview.SubViews	<int>	Number of sub-views supported in overview mode in the channel.
		ViewModes.Overview.PTZ	<bool>	Indicates whether the device supports PTZ in overview mode in the channel.
		ViewModes.Overview.PTZ.Features	<csv>	Specifies the PTZ features supported in overview mode in the channel, in CSV format.
		ViewModes.LeftHalfView	<bool>	Indicates whether the device supports left half view mode in the channel.
		ViewModes.LeftHalfView.SubViews	<int>	Number of sub-views supported in left half view mode in the channel.
		ViewModes.LeftHalfView.PTZ	<bool>	Indicates whether the device supports PTZ in left half view mode in the channel.
		ViewModes.LeftHalfView.PTZ.Features	<csv>	Specifies the PTZ features supported in left half view mode in the channel, in CSV format.
		ViewModes.RightHalfView	<bool>	Indicates whether the device supports right half view mode in the channel.
		ViewModes.RightHalfView.SubViews	<int>	Number of sub-views supported in right half view mode in the channel.
		ViewModes.RightHalfView.PTZ	<bool>	Indicates whether the device supports PTZ in right half view mode in the channel.
		ViewModes.RightHalfView.PTZ.Features	<csv>	Specifies the PTZ features supported in right half view mode in the channel, in CSV format.

Category	Channel	Attribute	Type	Description
		ViewModes.SingleView	<bool>	Indicates whether the device supports single view mode in the channel.
		ViewModes.SingleView.SubViews	<int>	Number of sub-views supported in single view mode in the channel.
		ViewModes.SingleView.PTZ	<bool>	Indicates whether the device supports PTZ in single view mode in the channel.
		ViewModes.SingleView.PTZ.Features	<csv>	Specifies the PTZ features supported in single view mode in the channel, in CSV format.
		ViewModes.QuadView	<bool>	Indicates whether the device supports quad view mode in the channel.
		ViewModes.QuadView.SubViews	<int>	Number of sub-views supported in quad view mode in the channel.
		ViewModes.QuadView.PTZ	<bool>	Indicates whether the device supports PTZ in quad view mode in the channel.
		ViewModes.QuadView.PTZ.Features	<csv>	Specifies the PTZ features supported in quad view mode in the channel, in CSV format.
		ViewModes.QuadView.#	<bool>	Indicates whether the device supports a specific quad view mode in the channel.
		ViewModes.QuadView..#.SubViews	<int>	Number of sub-views supported in a specific quad view mode in the channel.
		ViewModes.QuadView..#.PTZ	<bool>	Indicates whether the device supports PTZ in a specific quad view mode in the channel.
		ViewModes.QuadView..#.PTZ.Features	<csv>	Specifies the PTZ features supported in a specific quad view mode in the channel, in CSV format.
		ViewModes.Panorama	<bool>	Indicates whether the device supports panorama view mode in the channel.
		ViewModes.Panorama.SubViews	<int>	Number of sub-views supported in panorama view mode in the channel.
		ViewModes.Panorama.PTZ	<bool>	Indicates whether the device supports PTZ in panorama view mode in the channel.
		ViewModes.Panorama.PTZ.Features	<csv>	Specifies the PTZ features supported in panorama view mode in the channel, in CSV format.

Category	Channel	Attribute	Type	Description
		ViewModes.DoublePanorama	<bool>	Indicates whether the device supports double panorama view mode in the channel.
		ViewModes.DoublePanorama.SubViews	<int>	Number of sub-views supported in double panorama view mode in the channel.
		ViewModes.DoublePanorama.PTZ	<bool>	Indicates whether the device supports PTZ in double panorama view mode in the channel.
		ViewModes.DoublePanorama.PTZ.Features	<csv>	Specifies the PTZ features supported in double panorama view mode in the channel, in CSV format.
		ViewModes.CropView	<bool>	Indicates whether the device supports crop view mode in the channel.
		ViewModes.CropView.SubViews	<int>	Number of sub-views supported in crop view mode in the channel.
		ViewModes.CropView.PTZ	<bool>	Indicates whether the device supports PTZ in crop view mode in the channel.
		ViewModes.CropView.PTZ.Features	<csv>	Specifies the PTZ features supported in crop view mode in the
		NormalizedOSDRange	<bool>	Indicates whether the device supports a normalized range for On-Screen Display (OSD) settings in the channel.
		CustomOSDFontSize	<bool>	Indicates whether the device supports custom font sizes for OSD in the channel.
		AutoImageAlignmentSupport	<bool>	Indicates whether the device supports automatic image alignment in the channel.
		ImageSettings	<bool>	Indicates whether the device supports configurable image settings in the channel.
		Iris	<bool>	Indicates whether the device supports iris control in the channel.
		Iris-Fno	<bool>	Indicates whether the device supports F-number settings for the iris in the channel.

Categor y	Channel	Attribute	Type	Description
		LDC	<bool>	Indicates whether the device supports Lens Distortion Correction (LDC) in the channel.
		Crop	<bool>	Indicates whether the device supports image cropping in the channel.
		DynamicArea	<bool>	Indicates whether the device supports dynamic area settings in the channel.
		AutoFocus	<bool>	Indicates whether the device supports auto-focus in the channel.
		DirectionIndicator	<bool>	Indicates whether the device supports direction indicators in the channel.
		OSDArrow	<bool>	Indicates whether the device supports arrows in the OSD settings in the channel.
		Contrast	<bool>	Indicates whether the device supports contrast adjustment in the channel.
		WhiteBalance	<bool>	Indicates whether the device supports white balance adjustment in the channel.
		BackLight	<bool>	Indicates whether the device supports backlight control in the channel.
		DayNightSwitchThresholdAdjust	<bool>	Indicates whether the device supports adjusting the threshold for day/night switching in the channel.
		XCE	<bool>	Indicates whether the device supports Extreme Contrast Enhancement (XCE) in the channel.
		FocusContinuousAdjust	<bool>	Indicates whether the device supports continuous focus adjustment in the channel.
		ZoomContinuousAdjust	<bool>	Indicates whether the device supports continuous zoom adjustment in the channel.
		AntiFlicker	<bool>	Indicates whether the device supports anti-flicker settings in the channel.
		AGC	<bool>	Indicates whether the device supports Automatic Gain Control (AGC) in the channel.

Category	Channel	Attribute	Type	Description
		ShutterControl	<bool>	Indicates whether the device supports shutter control in the channel.
		ThermalVariationSensitivity	<bool>	Indicates whether the device supports thermal variation sensitivity adjustments in the channel.
		FocusPreset	<bool>	Indicates whether the device supports focus presets in the channel.
		HorizontalFlip	<bool>	Indicates whether the device supports horizontal flip in the channel.
		VerticalFlip	<bool>	Indicates whether the device supports vertical flip in the channel.
		Rotate	<enum>	Specifies the rotation options supported in the channel.
		UnifiedFlip	<bool>	Indicates whether the device supports unified flip functionality in the channel, meaning that when horizontal flip is applied, vertical flip will also be applied.
		RotateAndLDCIncompatible	<bool>	Indicates whether the device's rotate feature is incompatible with Lens Distortion Correction (LDC) in the channel.
		RotateAndVideoOutIncompatible	<bool>	Indicates whether the device's rotate feature is incompatible with video output in the channel.
		PrivacyIndexReorder	<bool>	Indicates whether the device supports reordering the privacy mask index in the channel.
		ImageAlignment	<csv>	Specifies the image alignment options supported in the channel, in CSV format.
		SSNR2DLevel	<bool>	Indicates whether the device supports 2D Smart Signal-to-Noise Ratio (SSNR) adjustment in the channel.
		SSNR3DLevel	<bool>	Indicates whether the device supports 3D Smart Signal-to-Noise Ratio (SSNR) adjustment in the channel.
		ThermalNUC	<bool>	Indicates whether the device supports Thermal Non-Uniformity Correction (NUC) in the channel.

Category	Channel	Attribute	Type	Description
		PrivacyAndDISIncompatible	<bool>	Indicates whether the device’s privacy features are incompatible with Digital Image Stabilization (DIS) in the channel.
		AutoFocusZone	<bool>	Indicates whether the device supports auto-focus zones in the channel.
		WiseAutoFocus	<bool>	Indicates whether the device supports WiseAutoFocus in the channel.
		WhiteLED	<bool>	Indicates whether the device supports white LED in the channel.
		WhiteLEDSupplementLightModeAndIRLEDIncompatible	<bool>	Indicates whether the device’s white LED supplement light mode is incompatible with IR LED in the channel.
		WhiteLEDSupplementLightModeAndDayNightIncompatible	<bool>	Indicates whether the device’s white LED supplement light mode is incompatible with day/night mode in the channel.
		AntiflickerShutter	<bool>	Indicates whether the device supports anti-flicker shutter settings in the channel.
		WDRAndDefogIncompatible	<bool>	Indicates whether the device’s defog mode setting is incompatible with WDR in the channel. If WDR is enabled, Defog mode will be disabled and cannot be controlled.
		HDR	<bool>	Indicates whether the device supports HDR (High Dynamic Range) in the channel.
	GlobalRotateView		<bool>	Indicates whether the device supports global rotate view.
	GlobalLDCMode		<bool>	Indicates whether the device supports global Lens Distortion Correction (LDC) mode.
	GlobalMaskPattern		<bool>	Indicates whether the device supports a global mask pattern.
	GlobalMultiImageOSD		<bool>	When true, the OSD image will be applied to all the channels.

4.5.7. PTZSupport Group

Category	Channel	Attribute	Type	Description
Support	Channel.#	Preset	<bool>	Indicates whether the device supports preset configuration in the channel.
		Swing	<bool>	Indicates whether the device supports swing functionality in the channel.
		Group	<bool>	Indicates whether the device supports group settings in the channel.
		DigitalZoom	<bool>	Indicates whether the device supports digital zoom in the channel.
		Trace	<bool>	Indicates whether the device supports trace functionality in the channel.
		AutoRun	<bool>	Indicates whether the device supports auto-run feature in the channel.
		Home	<bool>	Indicates whether the device supports home position setting in the channel.
		Tour	<bool>	Indicates whether the device supports tour functionality in the channel.
		PresetRename	<bool>	Indicates whether the device supports renaming presets in the channel.
		PTZLimit	<bool>	Indicates whether the device supports PTZ (Pan-Tilt-Zoom) limit settings in the channel.
		AreaZoom	<bool>	Indicates whether the device supports area zoom in the channel.
		Query.Pan	<bool>	Indicates whether the device supports querying pan position in the channel.
		Query.Tilt	<bool>	Indicates whether the device supports querying tilt position in the channel.
		Query.Zoom	<bool>	Indicates whether the device supports querying zoom level in the channel.
		Query.Focus	<bool>	Indicates whether the device supports querying focus in the channel.
		Query.Iris	<bool>	Indicates whether the device supports querying iris settings in the channel.
		OSDMenu.On	<bool>	Indicates whether the device supports turning on the OSD menu in the channel.

Category	Channel	Attribute	Type	Description
		OSDMenu.Off	<bool>	Indicates whether the device supports turning off the OSD menu in the channel.
		OSDMenu.Up	<bool>	Indicates whether the device supports navigating up in the OSD menu in the channel.
		OSDMenu.Down	<bool>	Indicates whether the device supports navigating down in the OSD menu in the channel.
		OSDMenu.Right	<bool>	Indicates whether the device supports navigating right in the OSD menu in the channel.
		OSDMenu.Left	<bool>	Indicates whether the device supports navigating left in the OSD menu in the channel.
		OSDMenu.Select	<bool>	Indicates whether the device supports selecting items in the OSD menu in the channel.
		OSDMenu.Return	<bool>	Indicates whether the device supports returning to the previous menu in the OSD in the channel.
		Absolute.Pan	<bool>	Indicates whether the device supports absolute pan position control in the channel.
		Absolute.Tilt	<bool>	Indicates whether the device supports absolute tilt position control in the channel.
		Absolute.Zoom	<bool>	Indicates whether the device supports absolute zoom level control in the channel.
		Absolute.Focus	<bool>	Indicates whether the device supports absolute focus control in the channel.
		Absolute.Iris	<bool>	Indicates whether the device supports absolute iris control in the channel.
		Absolute.PanSpeed	<bool>	Indicates whether the device supports setting the pan speed in the channel.
		Absolute.TiltSpeed	<bool>	Indicates whether the device supports setting the tilt speed in the channel.

Category	Channel	Attribute	Type	Description
		Absolute.ZoomSpeed	<bool>	Indicates whether the device supports setting the zoom speed in the channel.
		Continuous.Pan	<bool>	Indicates whether the device supports continuous pan control in the channel.
		Continuous.Tilt	<bool>	Indicates whether the device supports continuous tilt control in the channel.
		Continuous.Zoom	<bool>	Indicates whether the device supports continuous zoom control in the channel.
		Continuous.Focus	<bool>	Indicates whether the device supports continuous focus control in the channel.
		Continuous.Iris	<bool>	Indicates whether the device supports continuous iris control in the channel.
		Continuous.PanSpeed	<bool>	Indicates whether the device supports continuous pan speed adjustment in the channel.
		Continuous.TiltSpeed	<bool>	Indicates whether the device supports continuous tilt speed adjustment in the channel.
		Continuous.ZoomSpeed	<bool>	Indicates whether the device supports continuous zoom speed adjustment in the channel.
		Relative.Pan	<bool>	Indicates whether the device supports relative pan control in the channel.
		Relative.Tilt	<bool>	Indicates whether the device supports relative tilt control in the channel.
		Relative.Zoom	<bool>	Indicates whether the device supports relative zoom control in the channel.
		Relative.Focus	<bool>	Indicates whether the device supports relative focus control in the channel.
		Relative.Iris	<bool>	Indicates whether the device supports relative iris control in the channel.
		3AxisPTZ	<bool>	Indicates whether the device supports 3-axis PTZ (Pan-Tilt-Zoom) control in the channel.
		Azimuth	<bool>	Indicates whether the device supports azimuth control in the channel.
		ProfileBasedDigitalPTZ	<bool>	Indicates whether the device supports profile-based digital PTZ in the channel.

Category	Channel	Attribute	Type	Description
		DigitalAutoTracking	<bool>	Indicates whether the device supports digital auto-tracking in the channel.
		DigitalPTZ	<bool>	Indicates whether the device supports digital PTZ in the channel.
		ExternalPTZ	<bool>	Indicates whether the device supports external PTZ control in the channel.
		RealPTZ	<bool>	Indicates whether the device supports real PTZ control in the channel.
		ZoomOnly	<bool>	Indicates whether the device supports zoom-only control in the channel.
		AuxCommands	<csv>	Specifies the auxiliary commands supported in the channel, in CSV format.
		PanTiltOnly	<bool>	Indicates whether the device supports pan and tilt control only in the channel.
		PanZeroPosition	<bool>	Indicates whether the device supports setting the pan zero position in the channel.
		SmartZoom	<bool>	Indicates whether the device supports smart zoom in the channel.
		PTCorrection	<bool>	Indicates whether the device supports PT correction in the channel.
		AIAutoTracking	<bool>	Indicates whether the device supports AI auto-tracking in the channel.
		PresetImageConfig	<bool>	Indicates whether the device supports preset based image configuration in the channel.
		PresetImageConfig.XCE	<bool>	Indicates whether the device supports Extreme Contrast Enhancement (XCE) in preset image configuration in the channel.
		PresetVideoAnalysis	<bool>	Indicates whether the device supports preset based video analysis in the channel.
		MountPosition	<bool>	Indicates whether the device supports mount position settings in the channel.
		PresetObjectDetection	<bool>	Indicates whether the device supports preset based object detection in the channel.

Category	Channel	Attribute	Type	Description
		QuickZoom	<bool>	Indicates whether the device supports quick zoom in the channel.
	GlobalPTZMode		<bool>	Indicates that the PTZ mode selection is global and not channel specific.
	DigitalRTZOnlyInDefaultSettings		<bool>	Indicates whether the device supports digital RTZ (Real Time Zoom). For fisheye cameras, digital RTZ is supported only in ceiling mode.
Limit	Channel.#	MaxPreset	<int>	Maximum number of presets supported in the channel.
		MaxGroupCount	<int>	Maximum number of groups supported in the channel.
		MaxTourCount	<int>	Maximum number of tours supported in the channel.
		MaxTraceCount	<int>	Maximum number of traces supported in the channel.
		MaxPresetCountPerGroup	<int>	Maximum number of presets supported per group in the channel.
		MaxGroupCountPerTour	<int>	Maximum number of groups supported per tour in the channel.

4.5.8. Recording Group

Category	Channel or Attribute	Attribute	Type	Description
Support	Channel.#	QueueManagement	<bool>	Indicates whether the device supports queue management on a per-channel basis.
		SearchVideoSummary	<bool>	Indicates whether the device supports searching video summaries.
		Backup	<bool>	Indicates whether the device supports backup functionality.
		SearchCalendar	<bool>	Indicates whether the device supports calendar-based search functionality.

Category	Channel or Attribute	Attribute	Type	Description
		SearchTimeline	<bool>	Indicates whether the device supports timeline-based search functionality.
		SearchEvent	<bool>	Indicates whether the device supports event-based search functionality.
		SearchMotionGrid	<bool>	Indicates whether the device supports motion grid search functionality.
		PeopleCountSearch	<bool>	Indicates whether the device supports people count search functionality.
		SmartSearch	<bool>	Indicates whether the device supports smart search functionality.
		SearchHeatMap	<bool>	Indicates whether the device supports heat map search functionality.
		AISearchTypes	<csv> PersonSearch, FRSearch, FaceSearch, VehicleSearch, OCRSearch	Specifies the types of AI search supported, in CSV format.
		SearchBookmark	<bool>	Indicates whether the device supports bookmark search functionality.
	SearchMetadata		<bool>	Indicates whether the device supports searching metadata.
	ContinuousPlayNVR		<bool>	Indicates whether the device supports continuous playback on NVR.

Category	Channel or Attribute	Attribute	Type	Description
	RecordStreamLimitation		<bool>	Indicates whether the device has recording stream limitations.
	RAID		<bool>	Indicates whether the device supports RAID storage configuration.
	FailOverRecording		<bool>	Indicates whether the device supports failover recording functionality.
	SearchPOS		<bool>	Indicates whether the device supports POS (Point of Sale) search functionality.
	SearchPeriod		<bool>	Indicates whether the device supports period-based search functionality.
	Recording		<bool>	Indicates whether the device supports recording functionality.
	ManualRecordingStart		<bool>	Indicates whether the device supports manual recording start functionality.
	ManualRecordingStop		<bool>	Indicates whether the device supports manual recording stop functionality.
	RecordingStatus		<bool>	Indicates whether the device supports recording status reporting.
	Overlapped		<bool>	Indicates whether the device supports overlapped recordings.

Category	Channel or Attribute	Attribute	Type	Description
	PlaybackType		<enum> Pause, Seek, ForwardStep, BackwardStep, Slow	Specifies the types of playback
	PlaybackSpeed		<enum> -64x, -32x, -16x, -x8, -4x, -2x, -1x, 1x, 2x, 4x, 8x, 16x, 32x, 64x, -0.125x, -0.25x, -0.5x, 0.125x, 0.25x, 0.5x	Supported playback speeds.
	NAS		<bool>	Indicates whether the device supports NAS (Network Attached Storage).
	iSCSI		<bool>	Indicates whether the device supports iSCSI (Internet Small Computer System Interface) storage.
	SearchByUTCTime		<bool>	Indicates whether the device supports searches by UTC time.
	HeatMapGridArea		<string>	Specifies the grid areas for heat maps in WidthxHeight.

Category	Channel or Attribute	Attribute	Type	Description
	RecordPriorityOrder		<csv> FogDetection, AudioAnalysis, EmergencyTrigger, GSensorEvent, AudioDetection, DefocusDetection, Tracking, VideoAnalysis, AlarmInput, Normal, MotionDetection, Manual	Specifies the recording priority order, in CSV format.
	DualTrackRecording		<bool>	Indicates whether the device supports dual track recording.
	DiskUtility		<csv> SMART	Specifies the disk utility features supported, in CSV format.
	DistributedRecording		<bool>	Indicates whether the device supports distributed recording.
	GlobalRecordFileType		<bool>	Indicates whether the device supports global record file types.
	SimplePlaybackPage		<bool>	Indicates whether the device supports a simple playback page.
Limit	MaxPlaybackChannels		<int>	Maximum number of playback channels supported.
	MaxSmartSearchDays		<int>	Maximum number of days supported for smart searches.

Category	Channel or Attribute	Attribute	Type	Description
	MaxMetadataSearchDays		<int>	Maximum number of days supported for metadata searches.
	MaxSmartSearchIncludeAreas		<int>	Maximum number of include areas supported for smart searches.
	MaxSmartSearchExcludeAreas		<int>	Maximum number of exclude areas supported for smart searches.
	MaxSmartSearchLines		<int>	Maximum number of lines supported for smart searches.
	NumberOfStorageDevices		<int>	Number of storage devices supported.
	PreferRecordingCodec		<enum> MJPEG, H264, H265	Specifies the preferred recording codecs.
	RecordingGOVLengthMultiplierFactor		<float>	Specifies the multiplier factor for the recording GOP length.
	OpenSDK.CloudEdgeRecording		<bool>	Indicates whether the camera supports the cloud edge recording OpenSDK API.

4.5.9. EventSource Group

Category	Channel or Attribute	Attribute	Type	Description
Limit	Channel.#	MaxROI.Include	<int>	Maximum number of Regions of Interest (ROI) that can be included in the channel.
		MaxROI.Exclude	<int>	Maximum number of Regions of Interest (ROI) that can be excluded in the channel.
		MaxIVRule.Area.Include	<int>	Maximum number of Intelligent Videoanalytics (IV) rule areas that can be included in the channel.

Category	Channel or Attribute	Attribute	Type	Description
		MaxIVRule.Area.Exclude	<int>	Maximum number of Intelligent Videoanalytics (IV) rule areas that can be excluded in the channel.
		MaxIVRule.Line	<int>	Maximum number of Intelligent Videoanalytics (IV) rule lines that can be set in the channel.
		MaxFaceDetection.Area.Include	<int>	Maximum number of face detection areas that can be included in the channel.
		MaxFaceDetection.Area.Exclude	<int>	Maximum number of face detection areas that can be excluded in the channel.
		MaxParkingArea	<int>	Maximum number of parking areas that can be defined in the channel.
		MaxExcludeParkingArea	<int>	Maximum number of excluded parking areas that can be defined in the channel.
		MaxPeopleCountRule	<int>	Maximum number of people count rules that can be defined in the channel.
		MaxVehicleCountRule	<int>	Maximum number of vehicle count rules that can be defined in the channel.
		MaxTemperatureDifferenceRule	<int>	Maximum number of temperature difference rules that can be defined in the channel.
		MaxEarlyFireDetectionArea	<int>	Maximum number of early fire detection areas that can be defined in the channel.

Category	Channel or Attribute	Attribute	Type	Description
		MaxTankLevelDetectionArea	<int>	Maximum number of tank level detection areas that can be defined in the channel.
	MaxQueues		<int>	Maximum number of queues supported.
	MaxPeopleCountRule		<int>	Maximum number of people count rules supported.
	MaxVehicleCountRule		<int>	Maximum number of vehicle count rules supported.
	MaxHeatMap.Area.Exclude		<int>	Maximum number of excluded areas in heat maps supported.
	MaxROI		<int>	Maximum number of Regions of Interest (ROI) supported.
	MaxFaceDetectionArea		<int>	Maximum number of face detection areas supported.
	MaxTamperingDetectionArea		<int>	Maximum number of tampering detection areas supported.
	MaxTrackingArea		<int>	Maximum number of tracking areas supported.
	MaxAlarmInput		<int>	Maximum number of alarm inputs supported.
	MaxNetworkAlarmInput		<int>	Maximum number of network alarm inputs supported.
	MaxIVRule		<int>	Maximum number of Intelligent Videoanalytics (IV) rules supported.
	MaxIVRule.Line		<int>	Maximum number of Intelligent Videoanalytics (IV) rule lines supported.

Category	Channel or Attribute	Attribute	Type	Description
	MaxIVRule.Area		<int>	Maximum number of Intelligent Videoanalytics (IV) rule areas supported.
	ROICoordinate.MinX		<int>	Minimum X-coordinate for ROI supported.
	ROICoordinate.MinY		<int>	Minimum Y-coordinate for ROI supported.
	ROICoordinate.MaxX		<int>	Maximum X-coordinate for ROI supported.
	ROICoordinate.MaxY		<int>	Maximum Y-coordinate for ROI supported.
	MaxIVRule.Area.Include		<int>	Maximum number of Intelligent Videoanalytics (IV) rule areas that can be included.
	MaxIVRule.Area.Exclude		<int>	Maximum number of Intelligent Videoanalytics (IV) rule areas that can be excluded.
	MaxROI.Include		<int>	Maximum number of Regions of Interest (ROI) that can be included.
	MaxROI.Exclude		<int>	Maximum number of Regions of Interest (ROI) that can be excluded.
	MaxCoordinate.DefinedArea.AppearDisappear		<int>	Maximum number of defined areas for appear/disappear events supported.
	MaxCoordinate.DefinedArea.Entering		<int>	Maximum number of defined areas for entering events supported.
	MaxCoordinate.DefinedArea.Exiting		<int>	Maximum number of defined areas for exiting events supported.
	MaxHeatMapLevel		<int>	Maximum number of heat map levels supported.

Category	Channel or Attribute	Attribute	Type	Description
	MaxBoxTemperatureDetectionArea		<int>	Maximum number of box temperature detection areas supported.
	MaxBodyTemperatureDetectionArea		<int>	Maximum number of body temperature detection areas supported.
	MaxDTMFCodeCount		<int>	Maximum number of DTMF code counts supported.
	MaxDynamicRule		<int>	Maximum number of dynamic rules supported.
	MaxDynamicRule.EventSource		<int>	Maximum number of event sources for dynamic rules supported.
	MaxScheduleCount		<int>	Maximum number of schedules supported.
	MaxMQTTMessageCount		<int>	Maximum number of MQTT messages supported.
	MaxMQTTSubscriptionCount		<int>	Maximum number of MQTT subscriptions supported.
	MaxAudioFiles		<int>	Maximum number of audio files supported in the channel. IPAS ONLY
	MaxTTSFiles		<int>	Maximum number of text-to-speech (TTS) files supported in the channel. IPAS ONLY

Category	Channel or Attribute	Attribute	Type	Description
Support	Channel.#	VideoAnalytics	<csv> Passing, Entering, Exiting, Appearing, Disappearing, Intrusion, Loitering	List of video analytics features supported in the channel.
		AudioAnalytics	<csv> Scream, Gunshot, Explosion, GlassBreak	List of audio analytics features supported in the channel.
		ThermalGridArea	<string>	Thermal grid area as WidthxHeight in the channel.
		IVRule	<bool>	Indicates if Intelligent Videoanalytics (IV) rules are supported.
		GetIVRule	<bool>	Indicates if retrieval of Intelligent Videoanalytics (IV) rules is supported.
		OverlayIVRule	<bool>	Indicates if overlay of Intelligent Videoanalytics (IV) rules is supported.
		AdjustMDIVRuleOnFlipMirror	<bool>	Indicates whether motion detection Intelligent Videoanalytics (IV) rules need to be adjusted when the Flip or Mirror settings are changed.
		ObjectDetection	<bool>	Indicates if object detection is supported.
		ObjectTypes	<csv> Person, Vehicle, License Plate, Face	List of object types supported for detection.

Category	Channel or Attribute	Attribute	Type	Description
		ObjectDetection.EventActions	<csv>	List of event actions supported for object detection.
		DefocusDetection	<bool>	Indicates if defocus detection is supported.
		DefocusDetection.OneShotAF	<bool>	Indicates if one-shot autofocus for defocus detection is supported.
		FaceDetection	<bool>	Indicates if face detection is supported.
		HeadDetection	<bool>	Indicates if head detection is supported.
		FaceDetection.EventActions	<csv>	List of event actions supported for face detection.
		Tracking	<bool>	Indicates if tracking is supported.
		Tracking.EventActions	<csv>	List of event actions supported for tracking.
		Metadata	<bool>	Indicates if metadata is supported.
		AdvancedMotionDetection	<bool>	Indicates if advanced motion detection is supported.
		TemperatureChangeDetection	<bool>	Indicates if temperature change detection is supported.
		TemperatureChangeDetection.AreaType	<enum> Rectangle, Polygon	Indicates the supported area types for temperature change detection.
		BodyTemperatureDetection	<bool>	Indicates if body temperature detection is supported.
		BodyTemperatureDetection.EventActions	<csv>	List of event actions supported for body temperature detection.

Category	Channel or Attribute	Attribute	Type	Description
		BoxTemperatureDetection	<bool>	Indicates if box temperature detection is supported.
		BoxTemperatureDetection.AreaType	<enum> Rectangle, Polygon	Indicates the supported area types for box temperature detection.
		BoxTemperatureDetection.EntireArea	<bool>	Indicates whether box temperature detection supports the entire area.
		BoxTemperatureDetection.MinMaxCoordinates	<bool>	Indicates whether box temperature detection supports specifying minimum and maximum temperature coordinates.
		TamperingDetection.DarknessDetection	<bool>	Indicates whether darkness detection is supported in tampering detection.
		EarlyFireDetection	<bool>	Indicates whether early fire detection is supported.
		EarlyFireDetection.AreaType	<enum> Rectangle, Polygon	Indicates the area types supported for early fire detection.
		EarlyFireDetection.MinMaxCoordinates	<bool>	Indicates whether early fire detection supports specifying minimum and maximum temperature coordinates.
		TemperatureDifference	<bool>	Indicates if temperature difference is supported.
		TankLevelDetection	<bool>	Indicates if tank level detection is supported.
		TankLevelDetection.EventActions	<bool>	List of event actions supported for tank level detection.
		MaskDetection	<bool>	Indicates if mask detection is supported.

Category	Channel or Attribute	Attribute	Type	Description
		MaskDetection.EventActions	<csv>	List of event actions supported for mask detection.
		ParkingDetection	<bool>	Indicates if parking detection is supported.
		ParkingDetection.EventActions	<csv>	List of event actions supported for parking detection.
		LEDIndicator	<bool>	Indicates if LED indicator is supported.
		LEDIndicator.SourceChannel	<csv>	List of source channels for LED indicator.
		SocialDistancingViolation	<bool>	Indicates if social distancing violation detection is supported.
		SocialDistancingViolation.EventActions	<csv>	List of event actions supported for social distancing violation.
		CallRequest	<bool>	Indicates if call request feature is supported.
		CallRequest.EventActions	<csv>	List of event actions supported for call request.
		TamperingSwitch	<bool>	Indicates if tampering switch is supported.
		TamperingSwitch.EventActions	<csv>	List of event actions supported for tampering switch.
		DTMF	<bool>	Indicates if Dual-tone multi-frequency (DTMF) signaling is supported.
		DTMF.EventActions	<csv>	List of event actions supported for DTMF signaling.
		ProximitySensor	<bool>	Indicates if proximity sensor is supported.
		ProximitySensor.EventActions	<csv>	List of event actions supported for proximity sensor.

Category	Channel or Attribute	Attribute	Type	Description
		VA.ObjectTypeFilter	<bool>	Indicates if object type filter is supported for video analytics.
		VA.MotionDetection.ObjectTypeFilter	<bool>	Indicates if object type filter is supported for motion detection in video analytics.
		VA.MetadataInExcludeArea	<bool>	Indicates if metadata in exclude areas is supported for video analytics.
		VA.Overlay	<bool>	Indicates if overlay is supported for video analytics.
	ROIType		<enum>	Indicates the type of ROI (Region of Interest).
	TamperingDetection		<bool>	Indicates if tampering detection is supported.
	NetworkAlarmInput		<bool>	Indicates if network alarm input is supported.
	AlarmInput		<bool>	Indicates if alarm input is supported.
	AlarmOutput		<bool>	Indicates if alarm output is supported.
	AudioDetection		<bool>	Indicates if audio detection is supported.
	UserInput		<bool>	Indicates if user input is supported.
	UserInput.EventActions		<csv>	List of event actions supported for user input.
	NetworkDisconnect		<bool>	Indicates if network disconnect detection is supported.
	VideoLoss		<bool>	Indicates if video loss detection is supported.
	VA.Passing		<bool>	Indicates if passing detection is supported in video analytics.

Category	Channel or Attribute	Attribute	Type	Description
	VA.Enter		<bool>	Indicates if enter detection is supported in video analytics.
	VA.Exit		<bool>	Indicates if exit detection is supported in video analytics.
	VA.Appear		<bool>	Indicates if appear detection is supported in video analytics.
	VA.Disappear		<bool>	Indicates if disappear detection is supported in video analytics.
	VA.MotionDetection		<bool>	Indicates if motion detection is supported in video analytics.
	VA.MotionDetection.Overlay		<bool>	Indicates if overlay is supported for motion detection in video analytics.
	VA.MotionDetection.ObjectSize		<bool>	Indicates if object size detection is supported in motion detection for video analytics.
	VA.AppearDisappearType		<enum> Rectangle, Polygon	Indicates the type of appear/disappear events in video analytics.
	VA.<int>rusion		<bool>	Indicates if intrusion detection is supported in video analytics.
	VA.Loitering		<bool>	Indicates if loitering detection is supported in video analytics.
	TamperingDetection.EventActions		<csv>	List of event actions supported for tampering detection.
	AlarmInput.EventActions		<csv>	List of event actions supported for alarm input.

Category	Channel or Attribute	Attribute	Type	Description
	AudioDetection.EventActions		<csv>	List of event actions supported for audio detection.
	NetworkDisconnect.EventActions		<csv>	List of event actions supported for network disconnect detection.
	VideoLoss.EventActions		<csv>	List of event actions supported for video loss detection.
	VA.MotionDetection.EventActions		<csv>	List of event actions supported for motion detection in video analytics.
	Timer		<bool>	Indicates if timer functionality is supported.
	Timer.EventActions		<csv>	List of event actions supported for timer.
	DefocusDetection		<bool>	Indicates if defocus detection is supported.
	DefocusDetection.EventActions		<csv>	List of event actions supported for defocus detection.
	OpenSDK		<bool>	Indicates if OpenSDK is supported.
	OpenSDK.EventActions		<csv>	List of event actions supported for OpenSDK.
	FogDetection		<bool>	Indicates if fog detection is supported.
	FogDetection.EventActions		<csv>	List of event actions supported for fog detection.
	TemperatureChangeDetection		<bool>	Indicates if temperature change detection is supported.
	TemperatureChangeDetection.EventActions		<csv>	List of event actions supported for temperature change detection.

Category	Channel or Attribute	Attribute	Type	Description
	ShockDetection		<bool>	Indicates if shock detection is supported.
	ShockDetection.EventActions		<csv>	List of event actions supported for shock detection.
	GlobalShockDetection		<bool>	Indicates if shock detection is not a channel-based event but a device-wise event.
	AudioAnalysis		<bool>	Indicates if audio analysis is supported.
	WaterLevelWarning		<bool>	Indicates if water level warning is supported.
	HousingTampering		<bool>	Indicates if housing tampering detection is supported.
	BoxTemperatureDetection.EventActions		<csv>	List of event actions supported for box temperature detection.
	BodyTemperatureDetection		<bool>	Indicates if body temperature detection is supported.
	BodyTemperatureDetection.EventActions		<csv>	List of event actions supported for body temperature detection.
	TankLevelDetection		<bool>	Indicates if tank level detection is supported.
	TankLevelDetection.EventActions		<bool>	List of event actions supported for tank level detection.
	TamperingDetection.DarknessDetection		<bool>	Indicates if tampering detection in darkness is supported.
	ChannelBasedROICoordinate		<bool>	Indicates whether channel-based ROI (Region of Interest) coordinates are supported.
	GlobalAudioDetection		<bool>	Indicates if global audio detection is supported.

Category	Channel or Attribute	Attribute	Type	Description
	GlobalAudioAnalysis		<bool>	Indicates if global audio analysis is supported.
	CallRequest		<bool>	Indicates if call request feature is supported.
	CallRequest.EventActions		<csv>	List of event actions supported for call request.
	TamperingSwitch		<bool>	Indicates if tampering switch is supported.
	TamperingSwitch.EventActions		<csv>	List of event actions supported for tampering switch.
	DTMF		<bool>	Indicates if Dual-tone multi-frequency (DTMF) signaling is supported.
	DTMF.EventActions		<csv>	List of event actions supported for DTMF signaling.
	ProximitySensor		<bool>	Indicates if proximity sensor is supported.
	ProximitySensor.EventActions		<csv>	List of event actions supported for proximity sensor.
	DynamicRule		<bool>	Indicates if dynamic rule functionality is supported.

4.5.10. IO Group

Category	Attribute	Type	Description
Limit	MaxAlarmOutput	<int>	Maximum number of alarm outputs supported.
	MaxAux	<int>	Maximum number of auxiliary pins supported.
	MaxConfigurableIO	<bool>	Maximum number of configurable I/O pins supported.
Support	AlarmOutput	<bool>	If alarm output is supported, this capability will be set to true.
	Aux	<bool>	Capability to notify if auxiliary is supported.

Category	Attribute	Type	Description
	RS485	<bool>	Capability to notify if RS485 is supported.
	AlarmReset	<bool>	Capability if alarm reset feature is supported.
	PowerRelayIndices	<int>	Capability if power relay is supported.
	ConfigurableIO	<bool>	Capability if configurable I/O is supported.
	ExternalConfigurableIO	<bool>	Indicates the number of configurable I/O pins that can be linked externally, such as with an IOBox.

4.5.11. Display Group

NOTE

Supported only on decoder devices.

Category	Attribute	Type	Description
Limit	MaxHDMISplit	<int>	Maximum number of HDMI splits supported.
	MaxVGASplit	<int>	Maximum number of VGA splits supported.
	MaxBNCSplit	<bool>	Maximum number of BNC splits supported.

Chapter 5. CGI

5.1. Overview

Each CGI section of the XML represents the submenus, actions, and parameters available in the corresponding CGI. The XML data includes the parameter data type, allowed ranges or values, and whether the parameter is for requests or responses.

The following command returns only the CGI part of the XML.

```
http://<Device IP>/stw-cgi/attributes.cgi/<cgi name>
```

5.2. Examples

The following example shows a request for only the system-related submenus and parameters of the device.

REQUEST

```
http://<Device IP>/stw-cgi/attributes.cgi/system
```

RESPONSE

```
HTTP Code: 200 OK
Content-Type: text/xml
<Body>
```

```
<cgi name="system">
<submenu name="deviceinfo">
 <action name="view" accesslevel="guest">
   <parameter name="Model" request="false" response="true">
  <dataType>
          <string maxlen="31" />
  </dataType>
   </parameter>
    <parameter name="FirmwareVersion" request="false" response="true">
  <dataType>
      <string maxlen="15" />
  </dataType>
   </parameter>
     …
 </action>
 <action name="set" accesslevel="admin">
    <parameter name="DeviceName" request="true" response="true">
       <dataType>
          <string maxlen="8" />
  </dataType>
    </parameter>
    <parameter name="Language" request="true" response="true">
  <dataType>
      <enum>
          <entry value="English" />
          <entry value="Korean" />
          …
      </enum>
  </dataType>
    </parameter>
 </action>
…
…
</submenu>
```

```
<submenu name="date">
  <action name="view" accesslevel="guest">
    <parameter name="LocalTime" request="false" response="true">
   <dataType>
          <string format="YYYY-MM-DD hh:mm:ss"/>
   </dataType>
    </parameter>
      …
  </action>
  <action name="set" accesslevel="admin">
      <parameter name="SyncType" request="true" response="true">
   <dataType>
       <enum>
          <entry value="NTP"/>
          <entry value="Manual"/>
       </enum>
   </dataType>
      </parameter>
      <parameter name="NTPURLList" request="true" response="true">
        <dataType>
       <csv>
          <entry value="MaxEntries=5"/>
       </csv>
          </dataType>
      </parameter>
      …
  </action>
…
</submenu>
<submenu name="systemlog">
…
</submenu>
…
…
</cgi>
```

System

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 6. Overview

6.1. Description

system.cgi configures general system settings for Hanwha Vision video surveillance products. These settings include device information, system date and time, and serial port settings. **system.cgi** can also perform important system functions like factory resets, firmware updates and accessing logs.

The following submenus are used to configure system settings:

- **deviceinfo**: Sets and requests device information.
- **date**: Sets and requests the system date, time and time zone.
- **serial**: Configures the RS-485 settings.
- **systemlog**: Reads the system log.
- **accesslog**: Reads the HTTP client access log.
- **eventlog**: Reads the event log.
- **profileaccessinfo**: Requests information on the profile currently being used and on the users accessing the current profile.
- **factoryreset**: Resets the system back to factory default settings. It offers the option to exclude the network configuration, camera mapping information or user access level information.
- **power**: Resets the system power.
- **firmwareupdate**: Updates the device firmware.
- **configbackup**: Makes a copy of all the system settings for backup.
- **configrestore**: Restores the system configuration by using the backup.
- **storageinfo**: Requests storage device information.
- **gps**: Requests the Global Positioning System (GPS) information of the NVR.
- **autobackup**: Makes a backup of the videos recorded on NVR automatically on the server when the NVR is connected to the server.
- **digitalsignage**: Displays advertisements from the FTP server if no input or event occurs within the specified time.
- **vehicleinformation**: Stores the vehicle-related information if mobile NVR is used.
- **onviffeature**: Enables some features in the ONVIF protocol.
- **databasereset**: Resets the database content in the camera.
- **logserver**: Configure clients to receive log information from device.
- **sessioninfo**: Gets the current session information of NVR.
- **sdcardinfo**: Gets the details of the current SD card.
- **iscsidiscovery**: Used to get the ISCSI targets.
- **holiday**: Configures the holiday settings for the device.

- **hddalarm**: Configures the hddalarm settings for the device.
- **monitorin**: Configures the monitor input settings for the device.
- **monitorout**: Configures the monitor output settings for the device.
- **usbconfig**: Configures the USB port on the device.
- **stratocast**: Configures the stratocast service URL and communication interval.
- **stratocastregister**: Reads the current status regarding the stratocast service registration.
- **peerconnectioninfo**: Reads the current status of the peer connection information.
- **clientregister**: Client can connect to IO Box and checks connection status.
- **geolocation**: Gets and configures the geolocation information of the device.
- **systemimage**: Gets the image files used by the NVR device to the client.
- **powermode**: Used to configure the power mode of the device.
- **registeredsubdevices**: Used to manage the network speaker and mic in IP Audio System (IPAS).
- **speakergroups**: Used to manage speaker groups in IP Audio System (IPAS).
- **ssdstorage**: Used to manage the SSD storage in cameras that support SSD storage.
- **localvms**: Used to install and manage the VMS application running inside camera.
- **geoorientation**: Used to configure the orientation of the device.
- **aux**: Controls the auxiliary PTZ operations.
- **sdcardstatus**: Gets the current status of sd cards for each slot

Chapter 7. Device Information

7.1. Description

The **deviceinfo** submenu requests and sets device information.

Access level

Action	Camera	Encoder	NVR	Decoder	IPAS
view	Guest	Guest	User	User	Guest
set	Admin	Admin	User	User	Admin

7.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
deviceinfo&action=<value>[&<parameter>=<value>]
```

7.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the device information settings.
	DeviceReady	RES	<bool>	Gives device ready status DECODER ONLY
	Model	RES	<string>	Model name
	SerialNumber	RES	<string>	Device serial number CAMERA ONLY ENCODER ONLY IPAS ONLY
	FirmwareVersion	RES	<string>	Current firmware version
	BuildDate	RES	<string>	Date the firmware was built
	WebURL	RES	<string>	Web address

Action	Parameters	Request/ Response	Type/ Value	Description
	DeviceType	RES	<enum> NWC, NVR, DVR, Encoder, Decoder, Hybrid, IOBox, NetworkSpeaker, NetworkMic, AudioBridge, EmergencyBell	Device type
	SpeakerType	RES	<enum> MASTER ,SLAVE, INDEPENDENT, SERVER	When DeviceType is NetworkSpeaker, speaker type is provided. IPAS ONLY
	ConnectedMACAddress	RES	<string>	MAC address
	ISPVersion	RES	<string>	ISP version CAMERA ONLY
	WisenetPlatformVersion	RES	<string>	Provided only when the device is based on WisenetPlatform.
	PTZBoardVersion	RES	<string>	PTZ board version CAMERA ONLY
	InterfaceBoardVersion	RES	<string>	Interface board version CAMERA ONLY
	TrackingVersion	RES	<string>	Tracking version CAMERA ONLY
	BootloaderVersion	RES	<string>	Boot loader version CAMERA ONLY ENCODER ONLY
	CGIVersion	RES	<string>	CGI version

Action	Parameters	Request/ Response	Type/ Value	Description
	GUIVersion	RES	<string>	Recorder's GUI version NVR ONLY
	MicomVersion	RES	<string>	Micom version NVR ONLY
	PasswordStrength	RES	<enum> Weak, Strong	Strength of the password as per the password policy CAMERA ONLY ENCODER ONLY
	FirmwareGroup	RES	<string>	Shows firmware group name
	OpenSDKVersion	RES	<string>	Open SDK application version CAMERA ONLY
	ONVIFVersion	RES	<string>	ONVIF transmitter version supported by device CAMERA ONLY
	RequestedClientIPAddress	RES	<string>	Client's address as seen by device. NVR ONLY
	ActualDeviceType	RES	<enum> Encoder	When the device is an encoder and **DeviceType** is changed to the network camera, this parameter shows up.
set	DeviceName	REQ, RES	<string>	Device name
	DeviceLocation	REQ, RES	<string>	Location information CAMERA ONLY ENCODER ONLY
	Memo	REQ, RES	<string>	Additional information about the device CAMERA ONLY ENCODER ONLY
	DeviceDescription	REQ, RES	<string>	Detailed information about the device CAMERA ONLY ENCODER ONLY
	Language	REQ, RES	<enum>	Language of the interface

Action	Parameters	Request/ Response	Type/ Value	Description
	CameraRegistrationMode	REQ,RES	<enum> Normal, PnP	Camera registration mode in PoE-supported NVR. NVR ONLY
	DeviceMode	REQ, RES	<enum> StandAlone, VMS	Mode of the device DECODER ONLY

7.4. Examples

7.4.1. Getting device information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Model=XND-8080R
SerialNumber=ZECA6V2HB00008M
FirmwareVersion=1.29.99_20190125
BuildDate=2019.01.25
WebURL=http://www.hanwhavision.com/
DeviceType=NWC
ConnectedMACAddress=00:16:6C:F9:1F:EE
ISPVersion=1.46_180907
BootloaderVersion=ver=U-Boot 2016.01-svn2152 (Nov 29 2017 - 20:20:02
CGIVersion=2.5.5
ONVIFVersion=18.6
DeviceName=Camera
DeviceLocation=Location
DeviceDescription=Description
Memo=Memo
Language=English
PasswordStrength=Strong
OpenSDKVersion=3.50_181011
```

```
FirmwareGroup=XND-8080R
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Model": "XND-8080R",
    "SerialNumber": "ZECA6V2HB00008M",
    "FirmwareVersion": "1.29.99_20190125",
    "BuildDate": "2019.01.25",
    "WebURL": "http://www.hanwhavision.com/",
    "DeviceType": "NWC",
    "ConnectedMACAddress": "00:16:6C:F9:1F:EE",
    "ISPVersion": "1.46_180907",
    "BootloaderVersion": "ver=U-Boot 2016.01-svn2152 (Nov 29 2017 - 20:20:02
",
    "CGIVersion": "2.5.5",
    "ONVIFVersion": "18.6",
    "DeviceName": "Camera",
    "DeviceLocation": "Location",
    "DeviceDescription": "Description",
    "Memo": "Memo",
    "Language": "English",
    "PasswordStrength": "Strong",
    "OpenSDKVersion": "3.50_181011",
    "FirmwareGroup": "XND-8080R"
}
```


7.4.2. Setting the language as English

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=deviceinfo&action=set&Language=English
```

Chapter 8. Date and Time

8.1. Description

The **date** submenu requests and sets the system date, time and time zone.

Access level

Action	Camera	Encoder	NVR	Decoder	IPAS
view	Guest	Guest	User	User	Guest
set	Admin	Admin	User	User	Admin

8.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
date&action=<value>[&<parameter>=<value>...]
```

8.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Read system date and time settings
	TimeZoneList	REQ		Time zone list The response contains a list of time zones (with indexes) supported by the device. The time zone index should be used to set TimeZoneIndex. CAMERA ONLY ENCODER ONLY IPAS ONLY
	LocalTime	RES	<string>	Local time The time is defined in the <YYYY-MM-DD hh:mm:ss> format.
	UTCTime	RES	<string>	UTC time The time is defined in the <YYYY-MM-DD hh:mm:ss> format.

Action	Parameters	Request/ Response	Type/ Value	Description
	NTPStatus	RES	<enum> Success, Fail	NTP status This parameter is valid only when **SyncType** is set to NTP. NVR ONLY
	NTPLastUpdatedTime	RES	<string>	NTP last updated time This parameter is valid only when **SyncType** is set to NTP. NVR ONLY
	Week	RES	<enum> First, Second, Third, Fourth, Last	Week of the current month NVR ONLY DECODER ONLY
	DateFormat	RES	<enum> YYYY-MM-DD,DD-MM-YYYY,MM-DD-YYYY	The current system's date display format NVR ONLY
	TimeFormat	RES	<enum> HMS24, HMS12	The current system's hour display format NVR ONLY
set	SyncType	REQ, RES	<enum> NTP, Manual, GPS	Time synchronization types • NTP: Time sync with NTP • Manual: Manual time synchronization (**Year**, **Month**, **Day**, **Hour**, **Minute**, and **Second** parameters should be set together.) • GPS: Use GPS as the source (NVR Only) **Note** Either **SyncType** or **TimeZoneIndex** must be sent for the **set** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	NTPURLList	REQ, RES	<csv>	NTP URL list This parameter is valid only when **SyncType** is set to NTP.
	DSTEnable	REQ, RES	<bool> True, False	Enables or disables daylight saving time adjustments. This parameter needs to be configured in both scenarios, regardless of whether the timezone is set using **TimeZoneIndex** or **POSIXTimeZone** parameter. **Note** Not supported in IPAS devices.
	TimeZoneIndex	REQ, RES	<int>	Index number of time zone **Note** Either **SyncType** or **TimeZoneIndex** must be sent for the **set** action. CAMERA ONLY ENCODER ONLY IPAS ONLY
	POSIXTimeZone	REQ, RES	<string>	Time zone with or without DST (daylight saving time). To enable DST, **DSTEnable** parameter should be sent along with this.
	Year	REQ	<int>	Year of the system time This parameter is valid only when **SyncType** is set to Manual.
	Month	REQ	<int>	Month of the system time The values must be within the range of 1 to 12. This parameter is valid only when **SyncType** is set to Manual.

Action	Parameters	Request/ Response	Type/ Value	Description
	Day	REQ	<int>	Day of the system time The values must be within the range of 1 to 31. This parameter is valid only when **SyncType** is set to Manual.
	Hour	REQ	<int>	Hour of the system time The values must be within the range of 0 to 23. This parameter is valid only when **SyncType** is set to Manual.
	Minute	REQ	<int>	Minute of the system time The values must be within the range of 0 to 59. This parameter is valid only when **SyncType** is set to Manual.
	Second	REQ	<int>	Second of the system time The values must be within the range of 0 to 59. This parameter is valid only when **SyncType** is set to Manual.
	TimeZone	REQ, RES	<enum>	TimeZone information based on GMT NVR ONLY DECODER ONLY
	DateFormat	REQ, RES	<enum> YYYY-MM-DD,DD-MM-YYYY,MM-DD-YYYY	Sets the current system's date display format NVR ONLY
	TimeFormat	REQ, RES	<enum> HMS24, HMS12	Sets the current system's hour display format NVR ONLY

8.4. Examples

8.4.1. Getting system date/time information

The date and time information are provided in both local time and UTC time.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=date&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
NTPURLList=pool.ntp.org,asia.pool.ntp.org,europe.pool.ntp.org,north-
america.pool.ntp.org,time.nist.gov
LocalTime=2015-06-29 07:26:27
UTCTime=2015-06-29 07:26:27
SyncType=Manual
DSTEnable=False
TimeZoneIndex=33
POSIXTimeZone=STWT0STWST,M3.5.0/1,M10.5.0/2:0:0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "NTPURLList": [
        "pool.ntp.org",
        "asia.pool.ntp.org",
        "europe.pool.ntp.org",
        "north-america.pool.ntp.org",
        "time.nist.gov"
    ],
    "LocalTime": "2015-06-29 07:26:27",
    "UTCTime": "2015-06-29 07:26:27",
```

```
    "SyncType": "Manual",
    "DSTEnable": false,
    "TimeZoneIndex": 33,
    "POSIXTimeZone": "STWT0STWST,M3.5.0/1,M10.5.0/2:0:0"
}
```

The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
NTPURLList=pool.ntp.org,asia.pool.ntp.org,europe.pool.ntp.org,north-
america.pool.ntp.org,time.nist.gov
LocalTime=2015-06-29 07:26:27
UTCTime=2015-06-29 07:26:27
SyncType=Manual
DSTEnable=False
TimeZoneIndex=33
POSIXTimeZone=STWT0STWST,M3.5.0/1,M10.5.0/2:0:0
DateFormat=YYYY-MM-DD
TimeFormat=HMS24
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "NTPLastUpdatedTime": "2015-06-29 16:32:24",
    "LocalTime": "2015-06-29 16:32:24",
    "UTCTime": "2015-06-29 07:32:24",
    "SyncType": "Manual",
    "DSTEnable": false,
    "POSIXTimeZone": "STWT-9STWST,M3.5.0/1:00:00,M10.5.0/1:00:00",
    "DateFormat": "YYYY-MM-DD",
    "TimeFormat": "HMS24"
```

```
}
```

8.4.2. Time zones

Getting the time zone list

The **TimeZoneList** returns a list of available time zones and their index numbers, which **TimeZoneIndex** refers to for setting time zones.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=date&action=view&TimeZoneList
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
00:(GMT-12:00) International Date Line West
01:(GMT-11:00) Coordinated Universal Time-11
02:(GMT-10:00) Hawaii
03:(GMT-09:00) Alaska[March.2nd.Sun/02:00:00,November.1st.Sun/02:00:00]
04:(GMT-08:00) Pacific Time (US &
Canada)[March.2nd.Sun/02:00:00,November.1st.Sun/02:00:00]
05:(GMT-08:00) Baja
California[April.1st.Sun/02:00:00,October.last.Sun/02:00:00]
06:(GMT-07:00) Chihuahua, La Paz,
Mazatlan[April.1st.Sun/02:00:00,October.last.Sun/02:00:00]
07:(GMT-07:00) Mountain Time (US &
Canada)[March.2nd.Sun/02:00:00,November.1st.Sun/02:00:00]
08:(GMT-07:00) Arizona
09:(GMT-06:00) Saskatchewan
10:(GMT-06:00) Central America
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TimeZones": [
        {
            "TimeZoneIndex": 0,
            "TimeZone": "(GMT-12:00) International Date Line West"
        },
        {
            "TimeZoneIndex": 1,
            "TimeZone": "(GMT-11:00) Coordinated Universal Time-11"
        },
        {
            "TimeZoneIndex": 2,
            "TimeZone": "(GMT-10:00) Hawaii"
        },
        {
            "TimeZoneIndex": 3,
            "TimeZone": "(GMT-09:00) Alaska",
            "StartTime": "March.2nd.Sun/02:00:00",
            "EndTime": "November.1st.Sun/02:00:00"
        },
        {
            "TimeZoneIndex": 4,
            "TimeZone": "(GMT-08:00) Pacific Time (US & Canada)",
            "StartTime": "March.2nd.Sun/02:00:00",
            "EndTime": "November.1st.Sun/02:00:00"
        },
        {
            "TimeZoneIndex": 5,
            "TimeZone": "(GMT-08:00) Baja California",
            "StartTime": "April.1st.Sun/02:00:00",
            "EndTime": "October.last.Sun/02:00:00"
        },
        {
            "TimeZoneIndex": 6,
            "TimeZone": "(GMT-07:00) Chihuahua, La Paz, Mazatlan",
            "StartTime": "April.1st.Sun/02:00:00",
            "EndTime": "October.last.Sun/02:00:00"
        },
        {
            "TimeZoneIndex": 7,
            "TimeZone": "(GMT-07:00) Mountain Time (US & Canada)",
```

```
            "StartTime": "March.2nd.Sun/02:00:00",
            "EndTime": "November.1st.Sun/02:00:00"
        },
        {
            "TimeZoneIndex": 8,
            "TimeZone": "(GMT-07:00) Arizona"
        },
        {
            "TimeZoneIndex": 9,
            "TimeZone": "(GMT-06:00) Saskatchewan"
        },
        {
            "TimeZoneIndex": 10,
            "TimeZone": "(GMT-06:00) Central America"
        }
    ]
}
```

This chapter is for network cameras only.

The time zone can be changed annually.

Setting the time zone for Hawaii

To set the time zone for Hawaii, **TimeZoneIndex** should be set to 02. (Refer to the data returned by **TimeZoneList**.)

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=date&action=set&TimeZoneIndex=02
```

8.4.3. Daylight saving time

To enable daylight saving time, **DSTEnable** should be set together with **TimeZoneIndex**.

Setting the system time to October 10, 2012, 00:10:00 in the Alaska time zone with DST

Setting the Alaska time zone with DST will change the system time 1 hour ahead of standard Alaska local time.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=date&action=set&SyncType=Manual&Year=2012&Month=10&D
```

```
ay=10&Hour=0&Minute=10&Second=0&TimeZoneIndex=03&DSTEnable=True
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=date&action=set&SyncType=Manual&Year=2012&Month=10&D
ay=10&Hour=0&Minute=10&Second=0&DSTEnable=True&POSIXTimeZone=STWT+9
```

8.4.4. POSIX TimeZone

POSIXTimeZone requests the time zone with or without DST. The following example is for Istanbul.

```
POSIXTimeZone=STWT-2STWST,M3.5.0/3,M10.5.0/4:0:0
```

It follows the syntax of POSIX Time zone;

```
std offset dst[offset],start[/time],end[/time]
STWT -2 STWST ,M3.5.0/3 ,M10.5.0/4:0:0
```

STWT-2STWST indicates that it is 2 hours ahead of GMT. M3.5.0/3,M10.5.0/4:0:0 indicates that daylight saving time starts on Sunday (0), the last week (5) of March (M3), at 3:00 AM, and it ends on Sunday (0), the last week (5) of October (M10), at 4:00 AM.

8.4.5. System time sync

Using Manual Time Sync and changing the date and time to January 1, 2000, 00:10:31

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=date&action=set&SyncType=Manual&Year=2000&Month=1&Da
y=1&Hour=0&Minute=10&Second=31
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 9. RS-485

9.1. Description

The **serial** submenu configures the RS-485 settings.

Access level

Action	Camera	Encoder	NVR
view	Admin	Admin	User
set	Admin	Admin	(Not supported)

9.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
serial&action=<value>[&<parameter>=<value>...]
```

9.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the RS-485/RS-422 settings.
set	SerialInterface	REQ, RES	RS422, RS485	Chooses the serial mode. CAMERA ONLY
	BaudRate	REQ, RES	<enum> 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200	Baud rate (read only for NVR)
	ParityBit	REQ, RES	<enum> None, Even, Odd	Parity bit mode (read only for NVR) • None: Off • Even: Even parity • Odd: Odd parity
	StopBits	REQ, RES	<enum> 1, 2	Stop bit (read only for NVR)

Action	Parameters	Request/ Response	Type/ Value	Description
	DataBits	REQ, RES	<enum> 7, 8	Data bits (read only for NVR)
	SignalTermination	REQ, RES	<bool> True, False	Sets the signal ending status CAMERA ONLY ENCODER ONLY
	DeviceId	REQ, RES	<int>	Hybrid NVR ID NVR ONLY

9.4. Examples

9.4.1. Getting serial port settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=serial&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
BaudRate=115200
ParityBit=None
StopBits=1
DataBits=8
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "BaudRate": "115200",
    "ParityBit": "None",
    "StopBits": "1",
```

```
    "DataBits": "8"
}
```

9.4.2. Configuring serial port settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=serial&action=set&BaudRate=9600&ParityBit=None&StopBits=2&DataBits=8
```

Chapter 10. System Log

10.1. Description

The **systemlog** submenu reads the system log. Each line of a system log response consists of the following:

- Date
- Time
- Type
- Description

Access level

Action	Camera	Encoder	NVR
view	User	Admin	User

10.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
systemlog&action=<value>[&<parameter>=<value>...]
```

10.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Level	REQ, RES	<enum> All	Log level CAMERA ONLY ENCODER ONLY
	Type	REQ, RES	<csv> Refer to **SystemLog Types** block below.	Detailed log type
	FromDate	REQ	<string>	Search start date The date is specified in the format of <YYYY-MM-DD>.
	ToDate	REQ	<string>	Search end date The date is specified in the format of <YYYY-MM-DD>.

SystemLog Types

```
 PowerOn, PowerOff, PowerRebootConfigChange, Backup, FWUpdate, FactoryReset,
HDDFull, HDDFail, HDDNone, FanError, SDFormat, SDFail, SDFull, SDInsert,
SDRemove, Network, TimeChange, Record, ConfigurationBackup,
ConfigurationRestore, NASFormat, NASFail, NASFull, NASConnect,
NASDisconnect, LocalSetupStart, LocalSetupEnd, RemoteSetup, PlaybackStart,
PlaybackEnd, CodecError, SystemUpgrade, DiskFull, DiskFail,
RecoverFromPowerFail, BackupStart, BackupEnd, BackupStop, BackupFail,
BackupOverwrite, InternalHDDErase, ExternalHDDErase, USBHDDErase,
USBMemoryErase, StreamCorrupt, ManualRecordStart, ManualRecordEnd,
OverwriteDecoding, DecoderRestart, HDDError, PTZModeIn, PTZModeOut,
RecordingError, NetworkBackupStart, NetworkBackupEnd, NetworkBackupStop,
AutoDeleteStart, AutoDeleteEnd, PasswordChange, RebootExternalHDD,
FrameFanFail, FrameFanRepair, CPUFanFail, CPUFanRepair, LeftFanFail,
LeftFanRepair, RightFanFail, RightFanRepair, EmergencyReset,
NetCamTrafficOverflow, NetCamTrafficRelease, DualSMPSFail, DualSMPSRepair,
iSCSIAttach, iSCSIDetach, iSCSIConnect, iSCSIDisconnect, RecordFrameDrop,
AlarmOut1, AlarmOut2, AlarmOut3, AlarmOut4, AlarmOutBeep, Net1Connect,
Net1Disconnect, Net2Connect, Net2Disconnect, Net3Connect, Net3Disconnect,
Net4Connect, Net4Disconnect, USBHDDConnect, USBHDDDisconnect,
DSPVASystemStart, DSPVASystemFault, DSPVAAMDLoadFail, DSPVAAMDStart,
DSPVAAMDReset, DSPDisplayStart, DSPDisplayFail, BatteryFailRecover,
NetCamConnect, NetCamDisconnect, NetProfileReplace, NetProfileRestore,
VideoLossRecordProfileReplace, VideoLossRecordProfileRestore, RecordStart,
RecordEnd, RAIDEnable, RAIDDisable, RAIDSetup, RAIDBuildStart, RAIDBuildEnd,
RAIDBuildCancel, RAIDBuildFail, RAIDDegrade, RAIDRebuildStart,
RAIDRebuildEnd, RAIDRebuildFail, RAIDFail, InternalHDDConnect,
InternalHDDDisconnect, InternalHDDWarmup, DatabaseFull, DatabaseRemove,
USBWIFIConnect, USBWIFIDisconnect, GSensorEvent, GPSDisconnect, HWSelfTest,
SWSelfTest, FanStopped, GPSInfo, EmergencyTrigger, MQTTConnection
```

10.4. Examples

10.4.1. Getting all system logs

Using the **view** action without specifying the **type** parameter returns logs of all types.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=systemlog&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=1000
[2015-06-30 13:18:29] [Network] Physical network is connected
[2015-06-30 13:18:21] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-30 13:18:14] [PowerOn] Camera System Power On
[2015-06-26 17:01:33] [Network] Physical network is connected
[2015-06-26 17:01:25] [Network] System gets an IPv4 address: 192.168.10.54
[2015-06-26 17:01:18] [PowerOn] Camera System Power On
[2015-06-26 17:00:21] [FWUpdate] System Firmware Update Complete
[2015-06-26 16:36:34] [Network] Physical network is connected
[2015-06-26 16:36:26] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-26 16:36:19] [PowerOn] Camera System Power On
[2015-06-26 16:35:26] [FWUpdate] System Firmware Update Complete
[2015-06-26 12:42:59] [Network] Physical network is connected
[2015-06-26 12:42:51] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-26 12:42:44] [PowerOn] Camera System Power On
[2015-06-25 14:41:41] [ConfigChange] Profile 5 RTP Multicast Port: 0 =>
47806
[2015-06-25 14:41:41] [ConfigChange] Profile 5 RTP Multicast IPv4 Address:
=> 224.16.17.52
[2015-06-25 14:41:41] [ConfigChange] Profile 5 RTP Multicast: off => on
[2015-06-25 14:41:41] [FanStopped] [Right] Fan failure
[2015-06-25 14:41:42] [FanStopped] [Left] Fan failure
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SystemLog": [
        {
            "Date": "2015-06-30 13:18:29",
```

```
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-30 13:18:21",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-30 13:18:14",
            "Type": "PowerOn",
            "Description": "Camera System Power On"
        },
        {
            "Date": "2015-06-26 17:01:33",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-26 17:01:25",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.10.54"
        },
        {
            "Date": "2015-06-26 17:01:18",
            "Type": "PowerOn",
            "Description": "Camera System Power On"
        },
        {
            "Date": "2015-06-26 17:00:21",
            "Type": "FWUpdate",
            "Description": "System Firmware Update Complete"
        },
        {
            "Date": "2015-06-26 16:36:34",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-26 16:36:26",
            "Type": "Network",
```

```
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-26 16:36:19",
            "Type": "PowerOn",
            "Description": "Camera System Power On"
        },
        {
            "Date": "2015-06-26 16:35:26",
            "Type": "FWUpdate",
            "Description": "System Firmware Update Complete"
        },
        {
            "Date": "2015-06-26 12:42:59",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-26 12:42:51",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-26 12:42:44",
            "Type": "PowerOn",
            "Description": "Camera System Power On"
        },
        {
            "Date": "2015-06-25 14:41:41",
            "Type": "ConfigChange",
            "Description": "Profile 5 RTP Multicast Port: 0 => 47806"
        },
        {
            "Date": "2015-06-25 14:41:41",
            "Type": "ConfigChange",
            "Description": "Profile 5 RTP Multicast IPv4 Address: =>
224.16.17.52"
        },
        {
            "Date": "2015-06-25 14:41:41",
            "Type": "ConfigChange",
```

```
            "Description": "Profile 5 RTP Multicast: off => on"
        },
        {
            "Date": "2015-06-25 14:41:41",
            "Type": "FanStopped",
            "Description": "[Right] Fan failure"
        },
        {
            "Date": "2015-06-25 14:41:42",
            "Type": "FanStopped",
            "Description": "[Left] Fan failure"
        }
        ...
    ]
}
```

10.4.2. Getting the time change log and the network log

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=systemlog&action=view&Type=TimeChange,Network
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=85
[2015-06-30 13:18:29] [Network] Physical network is connected
[2015-06-30 13:18:21] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-26 17:01:33] [Network] Physical network is connected
[2015-06-26 17:01:25] [Network] System gets an IPv4 address: 192.168.10.54
[2015-06-26 16:36:34] [Network] Physical network is connected
[2015-06-26 16:36:26] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-26 12:42:59] [Network] Physical network is connected
[2015-06-26 12:42:51] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-25 03:32:40] [Network] Physical network is connected
[2015-06-25 03:32:31] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-23 16:23:27] [TimeChange] Time Change: 2015-03-23 16:23:18 => 2015-
```

```
06-23 16:23:27
[2015-03-23 16:19:04] [TimeChange] Time Change: 2015-06-23 16:19:11 => 2015-
03-23 16:19:04
[2015-06-22 16:21:59] [Network] Physical network is connected
[2015-06-22 16:21:45] [Network] System gets an IPv4 address: 192.168.75.51
[2015-06-22 16:09:04] [Network] Physical network is connected
[2015-06-22 16:08:50] [Network] System gets an IPv4 address: 192.168.75.51
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SystemLog": [
        {
            "Date": "2015-06-30 13:18:29",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-30 13:18:21",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-26 17:01:33",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-26 17:01:25",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.10.54"
        },
        {
            "Date": "2015-06-26 16:36:34",
            "Type": "Network",
            "Description": "Physical network is connected"
```

```
        },
        {
            "Date": "2015-06-26 16:36:26",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-26 12:42:59",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-26 12:42:51",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-25 03:32:40",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-25 03:32:31",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-23 16:23:27",
            "Type": "TimeChange",
            "Description": "Time Change: 2015-03-23 16:23:18 => 2015-06-23
16:23:27"
        },
        {
            "Date": "2015-03-23 16:19:04",
            "Type": "TimeChange",
            "Description": "Time Change: 2015-06-23 16:19:11 => 2015-03-23
16:19:04"
        },
        {
            "Date": "2015-06-22 16:21:59",
            "Type": "Network",
```

```
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-22 16:21:45",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        },
        {
            "Date": "2015-06-22 16:09:04",
            "Type": "Network",
            "Description": "Physical network is connected"
        },
        {
            "Date": "2015-06-22 16:08:50",
            "Type": "Network",
            "Description": "System gets an IPv4 address: 192.168.75.51"
        }
    ]
}
```


10.4.3. Getting the time change log

This example reads time changes between January 1, 2015 and June 30, 2015

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=systemlog&action=view&Type=TimeChange&FromDate=2015-01-01&ToDate=2015-06-30
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=2
[2015-06-23 16:23:27] [TimeChange] Time Change: 2015-03-23 16:23:18 => 2015-06-23 16:23:27
[2015-03-23 16:19:04] [TimeChange] Time Change: 2015-06-23 16:19:11 => 2015-03-23 16:19:04
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SystemLog": [
        {
            "Date": "2015-06-23 16:23:27",
            "Type": "TimeChange",
            "Description": "Time Change: 2015-03-23 16:23:18 => 2015-06-23
16:23:27"
        },
        {
            "Date": "2015-03-23 16:19:04",
            "Type": "TimeChange",
            "Description": "Time Change: 2015-06-23 16:19:11 => 2015-03-23
16:19:04"
        }
    ]
}
```

Chapter 11. Access Log

11.1. Description

The **accesslog** submenu reads the HTTP client access log.

Access level

Action	Camera	Encoder	NVR
view	User	Admin	User

11.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
accesslog&action=<value>[&<parameter>=<value>...]
```

11.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ, RES	<csv> AdminLogin, AdminLogout, UserLogin, UserLogout, GuestLogin, GuestLogout, Login, SessionTimeout	Log type If **Type** is not sent, the response will contain all user-level data.
	FromDate	REQ	<string>	Search start date The date is specified in the format of <YYYY-MM-DD>.
	ToDate	REQ	<string>	Search end date The date is specified in the format of <YYYY-MM-DD>.

11.4. Examples

11.4.1. Getting all access logs

Using the **view** action without the **Type** parameter returns all response parameters and values.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=accesslog&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=402
[2015-06-30 13:30:49] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-30 13:22:19] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:37:06] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:19:13] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:19:04] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:18:23] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:18:19] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:02:53] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:53:44] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:43:59] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 16:43:57] [AdminLogin] Admin Log In Success: 192.168.75.137
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AccessLog": [
        {
            "Date": "2015-06-30 13:30:49",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-30 13:22:19",
            "Type": "AdminLogin",
```

```
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:37:06",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:19:13",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:19:04",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:18:23",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:18:19",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:02:53",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:53:44",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:43:59",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
```

```
        },
        {
            "Date": "2015-06-26 16:43:57",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        }
        ...
    ]
}
```

11.4.2. Getting the log data on admin logins and logouts

This example reads the log information on admin logins and logouts between June 1, 2015 and June 30, 2015.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=accesslog&action=view&Type=AdminLogin,AdminLogout&FromDate=2015-06-01&ToDate=2015-06-30
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=402
[2015-06-30 13:30:49] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-30 13:22:19] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:37:06] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:19:13] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:19:04] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:18:23] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 17:18:19] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 17:02:53] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:53:44] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:43:59] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 16:43:57] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:28:52] [AdminLogin] Admin Log In Success: 192.168.75.137
[2015-06-26 16:28:35] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 16:28:19] [AdminLogin] Admin Log In Success: 192.168.75.137
```

```
[2015-06-26 16:26:16] [AdminLogout] Admin Log Out: 192.168.75.137
[2015-06-26 16:26:06] [AdminLogin] Admin Log In Success: 192.168.75.137
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AccessLog": [
        {
            "Date": "2015-06-30 13:30:49",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-30 13:22:19",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:37:06",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:19:13",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:19:04",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:18:23",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
```

```
        },
        {
            "Date": "2015-06-26 17:18:19",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 17:02:53",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:53:44",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:43:59",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:43:57",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:28:52",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:28:35",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:28:19",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        },
```

```
        {
            "Date": "2015-06-26 16:26:16",
            "Type": "AdminLogout",
            "Description": "Admin Log Out: 192.168.75.137"
        },
        {
            "Date": "2015-06-26 16:26:06",
            "Type": "AdminLogin",
            "Description": "Admin Log In Success: 192.168.75.137"
        }
        ...
    ]
}
```

Chapter 12. Event Log

12.1. Description

The **eventlog** submenu reads the event log. Each line of an event log response consists of the following:

- Date
- Time
- Type
- Description

Access level

Action	Camera	Encoder	NVR
view	User	Admin	User

12.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
eventlog&action=<value>[&<parameter>=<value>...]
```

12.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ, RES	<csv> Refer to **EventLog Types** block below.	Event Type **Note** The actual event logs may differ from model to model, depending on the functions supported by the product. For example, if a model does not support Face Detection, no "Face Detection" log will be available. Please check the events the device supports using attributes.cgi.
	FromDate	REQ	<string>	Search start date The date is specified in the format of <YYYY-MM-DD>.

Action	Parameters	Request/ Response	Type/ Value	Description
	ToDate	REQ	<string>	Search end date The date is specified in the format of <YYYY-MM-DD>.
check	Type	RES	<csv> Refer to **EventLog Types** block below.	Event type

EventLog Types

```
VideoAnalysis, Passing, Entering, Exiting, Appearing, Disappearing,
ScheduledEvent, MotionDetection, NetworkDisconnect, FaceDetection,
TamperingDetection, AlarmOutput, AlarmInput, AudioDetection, GotoPreset,
Aux, Videoloss, Tracking, OpenSDK, PTZMotion, UserInput, AlarmDetect,
SceneChange, CameraAlarm, CameraAppearDisappear, VideoLossRelease, AMDStart,
AMDStop, TamperMask, TamperRotate, TamperImage, LowFpsStart, LowFpsEnd,
DefocusDetection, TrackingStart, TrackingEnd, FogDetection, AudioAnalysis,
QueueEvent, ShockDetection, TemperatureChangeDetection,
BoxTemperatureDetection,HousingTampering, WaterLevelWarning,
ObjectDetection, BodyTemperatureDetection, MaskDetection,
SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived,
ProximitySensor, ParkingDetection, DynamicRule, MQTTSubscription, LEDPreset,
SIPCall, AudioClipPlayback, TwoWayTalk, CaseTampering, SIPCall,
AudioClipPlayback, TwoWayTalk, CaseTampering
```

12.4. Examples

12.4.1. Getting all event logs

Using the **view** action without specifying the **Type** parameter returns all data.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=eventlog&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=598
[2015-06-25 14:34:29] [PTZMotion] PTZ STOP : 192.168.75.130
[2015-06-25 14:34:27] [PTZMotion] PTZ UI ZOOM OUT Click : 192.168.75.130
[2015-06-25 14:34:25] [PTZMotion] PTZ STOP : 192.168.75.130
...
[2015-06-17 02:08:31] [TamperingDetection] Tampering Event Detected
[2015-06-17 02:08:13] [TamperingDetection] Tampering Event Detected
[2015-06-17 02:07:18] [TamperingDetection] Tampering Event Detected
[2015-06-17 02:06:54] [TamperingDetection] Tampering Event Detected
...
[2015-06-12 10:08:01] [FaceDetection] Face Detection Start
[2015-06-12 10:07:59] [FaceDetection] Face Detection End
[2015-06-12 10:07:55] [FaceDetection] Face Detection Start
[2015-06-12 10:06:57] [FaceDetection] Face Detection End
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ChannelEventLog": [
        {
            "Channel": 0,
            "EventLog": [
                {
                    "Date": "2015-06-25 14:34:29",
                    "Type": "PTZMotion",
                    "Description": "PTZ STOP : 192.168.75.130"
                },
                {
                    "Date": "2015-06-25 14:34:27",
                    "Type": "PTZMotion",
                    "Description": "PTZ UI ZOOM OUT Click : 192.168.75.130"
                },
                {
                    "Date": "2015-06-25 14:34:25",
                    "Type": "PTZMotion",
                    "Description": "PTZ STOP : 192.168.75.130"
```

```
                },
                {
                    "Date": "2015-06-17 02:08:31",
                    "Type": "TamperingDetection",
                    "Description": "Tampering Event Detected"
                },
                {
                    "Date": "2015-06-17 02:08:13",
                    "Type": "TamperingDetection",
                    "Description": "Tampering Event Detected"
                },
                {
                    "Date": "2015-06-17 02:07:18",
                    "Type": "TamperingDetection",
                    "Description": "Tampering Event Detected"
                },
                {
                    "Date": "2015-06-17 02:06:54",
                    "Type": "TamperingDetection",
                    "Description": "Tampering Event Detected"
                },
                {
                    "Date": "2015-06-12 10:08:01",
                    "Type": "FaceDetection",
                    "Description": "Face Detection Start"
                },
                {
                    "Date": "2015-06-12 10:07:59",
                    "Type": "FaceDetection",
                    "Description": "Face Detection End"
                },
                {
                    "Date": "2015-06-12 10:07:55",
                    "Type": "FaceDetection",
                    "Description": "Face Detection Start"
                },
                {
                    "Date": "2015-06-12 10:06:57",
                    "Type": "FaceDetection",
                    "Description": "Face Detection End"
                }
```

```
            ]
        }
    ]
}
```

12.4.2. Getting motion detection and face detection events

This example reads the motion detection and face detection event logs from June 1, 2015 to July 30, 2015.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=eventlog&action=view&Type=MotionDetection,FaceDetect
ion&FromDate=2015-06-01&ToDate=2015-07-30
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Total=6
[2015-06-12 10:08:01] [FaceDetection] Face Detection Start
[2015-06-12 10:07:59] [FaceDetection] Face Detection End
[2015-06-12 10:07:55] [FaceDetection] Face Detection Start
[2015-06-12 10:06:57] [FaceDetection] Face Detection End
[2015-07-06 19:14:10] [MotionDetection] Motion Detection Start
[2015-07-06 19:14:13] [MotionDetection] Motion Detection End
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ChannelEventLog": [
        {
            "Channel": 0,
            "EventLog": [
                {
```

```
                    "Date": "2015-06-12 10:08:01",
                    "Type": "FaceDetection",
                    "Description": "Face Detection Start"
                },
                {
                    "Date": "2015-06-12 10:07:59",
                    "Type": "FaceDetection",
                    "Description": "Face Detection End"
                },
                {
                    "Date": "2015-06-12 10:07:55",
                    "Type": "FaceDetection",
                    "Description": "Face Detection Start"
                },
                {
                    "Date": "2015-06-12 10:06:57",
                    "Type": "FaceDetection",
                    "Description": "Face Detection End"
                },
                {
                    "Date": "2015-07-06 19:14:10",
                    "Type": "MotionDetection",
                    "Description": "Motion Detection Start"
                },
                {
                    "Date": "2015-07-06 19:14:13",
                    "Type": "MotionDetection",
                    "Description": "Motion Detection End"
                }
            ]
        }
    ]
}
```

Chapter 13. Profile Access Information

13.1. Description

The **profileaccessinfo** submenu requests information about the currently used profile and the users accessing it.

NOTE This chapter is for network cameras only. Attribute to check for max channels: "attributes/System/Limit/**MaxChannel**"

Access level

Action	Camera	Encoder
view	Admin	Admin

13.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
profileaccessinfo&action=<value>[&<parameter>=<value>...]
```

13.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ViewGroup	REQ	<enum> Profile, User, All	Type of target information. • Profile: Profile info • User: User info • All: Both profile and user info
	Channel.#.Profile.#.CurrentBitrate	RES	<int>	The profile's current bit rate(in kbps) This parameter is valid only when **ViewGroup** is set to Profile or All.
	Channel.#.Profile.#.TotalBitrate	RES	<int>	The profile's total bit rate(in kbps) This parameter is valid only when **ViewGroup** is set to Profile or All.
	Channel.#.Profile.#.CurrentFPS	RES	<int>	The profile's current frame rate This parameter is valid only when **ViewGroup** is set to Profile or All.

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel.#.Profile.#.TotalFPS	RES	<int>	The profile's total frame rate This parameter is valid only when **ViewGroup** is set to Profile or All.
	Channel.#.Profile.#.ATC	RES	<int>	The profile's ATC This parameter is valid only when **ViewGroup** is set to Profile or All.
	Channel.#.Profile.#.ConcurrentUserCount	RES	<int>	The number of users using the profile This parameter is valid only when **ViewGroup** is set to Profile or All.
	User.#.ProfileNameList	RES	<csv>	The profile name list This parameter is valid only when **ViewGroup** is set to User or All.
	User.#.ClientIPAddress	RES	<string>	The user's IP address. This parameter is valid only when **ViewGroup** is set to User or All.
	User.#.CurrentBitrate	RES	<int>	Bit rate transferred to the user This parameter is valid only when **ViewGroup** is set to User or All.
	User.#.ClientNetworkConnectionStatus	RES	<enum> Good, Bad, Optimized	User and camera's connection status This parameter is valid only when **ViewGroup** is set to User or All.
	User.#.Channel.#.ProfileNameList	RES	<csv>	Channel-based profile name list (Applicable for Multi directional camera)
	User.#.Channel.#.CurrentBitrate	RES	<int>	Channel-based current bitrate (Applicable for Multi directional camera)

NOTE # represents the index number of the channel, profile or user.

13.4. Examples

13.4.1. Getting the profile access information

Specifying the **ViewGroup** as Profile returns profile access information only and does not return user access information.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=profileaccessinfo&action=view&ViewGroup=Profile
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.0.CurrentBitrate=0
Channel.0.Profile.0.TotalBitrate=6144
Channel.0.Profile.0.CurrentFPS=0
Channel.0.Profile.0.TotalFPS=5
Channel.0.Profile.0.ATC=0
Channel.0.Profile.0.ConcurrentUserCount=0
Channel.0.Profile.1.CurrentBitrate=0
Channel.0.Profile.1.TotalBitrate=512
Channel.0.Profile.1.CurrentFPS=0
Channel.0.Profile.1.TotalFPS=30
Channel.0.Profile.1.ATC=0
Channel.0.Profile.1.ConcurrentUserCount=0
Channel.0.Profile.9.CurrentBitrate=0
Channel.0.Profile.9.TotalBitrate=300
Channel.0.Profile.9.CurrentFPS=0
Channel.0.Profile.9.TotalFPS=3
Channel.0.Profile.9.ATC=0
Channel.0.Profile.9.ConcurrentUserCount=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ProfileAccessInfo": {
        "ProfileInfo": [
            {
                "Channel": 0,
                "Profiles": [
                    {
                        "Profile": 0,
                        "CurrentBitrate": 0,
                        "TotalBitrate": 6144,
                        "CurrentFPS": 0,
                        "TotalFPS": 5,
                        "ATC": 0,
                        "ConcurrentUserCount": 0
                    },
                    {
                        "Profile": 1,
                        "CurrentBitrate": 0,
                        "TotalBitrate": 512,
                        "CurrentFPS": 0,
                        "TotalFPS": 30,
                        "ATC": 0,
                        "ConcurrentUserCount": 0
                    },
                    {
                        "Profile": 9,
                        "CurrentBitrate": 0,
                        "TotalBitrate": 300,
                        "CurrentFPS": 0,
                        "TotalFPS": 3,
                        "ATC": 0,
                        "ConcurrentUserCount": 0
                    }
                ]
            }
        ]
    }
}
```

13.4.2. Getting the user information

This example reads profile information which includes the number of users accessing simultaneously.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=profileaccessinfo&action=view&ViewGroup=User
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
User.1.ProfileNameList=H.264
User.1.ClientIPAddress=192.168.75.137
User.1.ClientBitrate=195
User.1.ClientNetworkConnectionStatus=Good
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ProfileAccessInfo": {
        "Users": [
            {
                "User": 1,
                "ProfileNameList": [
                    "H.264"
                ],
                "ClientIPAddress": "192.168.75.137",
                "ClientBitrate": 195,
                "ClientNetworkConnectionStatus": "Good"
            }
        ]
    }
```

```
}
```

Chapter 14. Factory Reset

14.1. Description

The **factoryreset** submenu resets the system to default settings.

The device can restart after a factory reset.

Access level

Action	Camera	Encoder	NVR	IPAS
control	Admin	Admin	Admin	Admin

14.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
factoryreset&action=<value>[&<parameter>=<value>...]
```

14.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	ExcludeSettings	REQ	<csv> Authority, Network, Camera, None	Selects a group of settings to be excluded from being set to defaults during a factory reset • Authority: Resets the system to default settings except for the user permission configuration. • Network: Resets the system to default settings except for the network configuration. • Camera: Resets the system to default settings except for the camera mapping information.

14.4. Examples

14.4.1. Resetting the system except for the network configuration

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/system.cgi?msubmenu=factoryreset&action=control&ExcludeSettings=Network
```

Chapter 15. Control System Power

15.1. Description

The **power** submenu controls the system power.

Access level

Action	Camera	Encoder	NVR
view	User	N/A	User
set	Admin	N/A	User
check	Admin	N/A	User
control	Admin	Admin	User

15.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
power&action=<value>[&<parameter>=<value>...]
```

15.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the power reset settings
set	DelayBooting	REQ, RES	<enum> Off, 5s, 10s, 15s, 30s, 1m, 2m, 3m, 5m, 10m	Based on this setting device, booting time after restart can be controlled(for NVR only)
	EnableAlarmOutput	REQ, RES	<bool> True, False	Option to enable/disable alarmout (for NVR only)
	ShutdownTime	REQ, RES	<int>	Based on this setting, shutdown time can be controlled in seconds (for NVR only)
	PowerCalculationPeriod	REQ, RES	<enum> DAY, WEEK, MONTH	
	EnableECOMode	REQ, RES	<bool> True, False	
	ECOMode.LimitedIR	REQ, RES	<bool> True, False	It operates only when EnableECOMode is set to True.
	ECOMode.HeaterOff	REQ, RES	<bool> True, False	It operates only when EnableECOMode is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
check	Type	REQ	<enum> Monitoring	default Monitoring
	PoEClass	RES	<int>	
	PowerRequested	RES	<float>	
	PowerConsumption.Current	RES	<float>	
	PowerConsumption.Max	RES	<float>	Value calculated during "PowerCalculationPeriod"
	PowerConsumption.Avg	RES	<float>	Value calculated during "PowerCalculationPeriod"
control	Type	REQ	<enum> Restart, Shutdown	Reset type • Restart: Restarts the system.Shutdown: Shutdown the system (for NVR only). **Note** **Type** must be sent together with the **control** action.

15.4. Examples

15.4.1. Restarting the system

To restart the system with the **control** action, the **Type** parameter must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=power&action=control&Type=Restart
```

15.4.2. Getting the power configuration

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=power&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
  "PowerCalculationPeriod": "DAY",
  "EnableECOMode": false
}
```

15.4.3. checking the power

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=power&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
  "PoEClass": 3,
  "PowerRequested": 12.95,
  "PowerConsumption": {
    "Current": 6.15,
    "Max": 9.74,
    "Avg": 6.55
  }
}
```

Chapter 16. Diagnoses the system

16.1. Description

The **diagnostics** submenu diagnoses the system.

Access level

Action	Camera	Encoder	NVR
view	Admin	N/A	N/A
control	Admin	N/A	N/A

16.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
diagnostics&action=<value>[&<parameter>=<value>...]
```

16.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the system diagnostics
	CPULoad	RES	<float>	
	Power	RES	<float>	
	LifeTime	RES	<string>	<Num>W<Num>D<Num>H<Num>M<Num>S
	UpTime	RES	<float>	<Num>W<Num>D<Num>H<Num>M<Num>S
	CloudConnectivityStatus	RES	<enum> Connected Disconnected	
	ConnectionStatus.User.#.ProfileNameList	RES	<csv>	
	ConnectionStatus.User.#.ClientIPAddress	RES	<string>	
	ConnectionStatus.User.#.ClientBitrate	RES	<int>	
	ConnectionStatus.User.#.ClientNetworkConnectionStatus	RES	<enum> Good, Bad, Optimized	

Action	Parameters	Request/ Response	Type/ Value	Description
	PanMotorRev	RES	<string>	
	TiltMotorRev	RES	<string>	
	FirmwareVersion	RES	<string>	
	WiperMovementCount	RES	<int>	
	Temperature.Celsius.Body	RES	<float>	
	Temperature.Celsius.Head	RES	<float>	
	Temperature.Celsius.Window	RES	<float>	
	Temperature.Fahrenheit.Body	RES	<float>	
	Temperature.Fahrenheit.Head	RES	<float>	
	Temperature.Fahrenheit.Window	RES	<float>	
	Humidity.Body	RES	<float>	
	Humidity.Head	RES	<float>	
	Humidity.Window	RES	<float>	
	RecordingStatus.Storage.#.Type	RES	<enum> DAS, NAS	
	RecordingStatus.Storage.#.Enabled	RES	<bool> True, False	
	RecordingStatus.Storage.#.Status	RES	<enum> NASRecording, NASReady, NASConnect, NASDisconnect, NASFail, NASFormat, NASFull, SDRecording, SDReady, SDFail, SDFormat, SDFull, SDInsert, SDRemove	
	AppStatus.AppID.#.AppName	RES	<string>	

Action	Parameters	Request/ Response	Type/ Value	Description
	AppStatus.AppID.#.Status	RES	<enum> UnInstalling, Installed, Installing, StartedNotRunning, Running, Stopped, Starting, Stopping	
control	Mode	RES	<enum> Reset	
	Type	RES	<enum> WIPER_MOVEMENT_COUNT	

16.4. Examples

16.4.1. Diagnoses the system

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=diagnostics&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
  "FirmwareVersion": "99.99.99_20230305",
  "CPULoad": 21.61,
  "Power": 21.27,
  "LifeTime": "0000W00D15H23M37S",
  "UpTime": "0000W00D14H28M22S",
  "CloudConnectivityStatus": "Disconnected",
  "ConnectionStatus": {
    "Users": [
      {
        "User": 1,
        "ProfileNameList": [
          "H.264"
        ],
```

```
        "ClientIPAddress": "192.168.75.84",
        "ClientBitrate": 0,
        "ClientNetworkConnectionStatus": "Good"
      }
    ]
  },
  "PanMotorRev": "293rev",
  "TiltMotorRev": "700rev",
  "WiperMovementCount": 39,
  "Temperature": {
    "Celsius": {
      "Body": 38.21,
      "Head": 38.13,
      "Window": 31.26
    },
    "Fahrenheit": {
      "Body": 100.79,
      "Head": 100.63,
      "Window": 88.27
    }
  },
  "Humidity": {
    "Body": 25.59,
    "Head": 28.13,
    "Window": 44.22
  },
  "RecordingStatus": [
    {
      "Storage": 1,
      "Type": "DAS",
      "Enabled": true,
      "Status": "SDInsert"
    },
    {
      "Storage": 2,
      "Type": "NAS",
      "Enabled": false,
      "Status": "NASDisconnect"
    }
  ],
  "AppStatus": [
```

```
    {
      "AppName": "WiseAI",
      "AppID": "WiseAI",
      "Status": "Running"
    }
  ]
}
```

16.4.2. Control related to diagnostics

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=diagnostics&action=control&Mode=Reset&Type=WIPER_MOVEMENT_COUNT
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 17. Firmware Update

17.1. Description

The **firmwareupdate** submenu updates the device firmware.

The device can restart after a firmware update.

Access level

Action	Camera	Encoder	NVR	IPAS
control	Admin	Admin	Admin	Admin
check			Admin	
set			Admin	
view			Admin	

NOTE Attribute to check for Firmware Update: "attributes/System/Support/**FWUpdate**"

17.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
firmwareupdate&action=<value>[&<parameter>=<value>...]
```

17.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
check	FWVersionAvailableInServer	RES	<string>	Firmware version available in server. NVR ONLY
control	DownloadFromServer	REQ	<bool> True, False	If true, the firmware is downloaded from the server. NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Type	REQ	<enum> Normal, FactoryDefault	Firmware update type • Normal: Only updates the firmware. • FactoryDefault: Updates the firmware and resets the system to factory defaults. **Note** **Type** must be sent together with the **control** action. In **IPAS** FactoryDefault type is not supported.
	IgnoreMultipartResponse	REQ	<bool>	If true, intermediate events will not be sent from the device. This is used to ignore multipart messages
	Status	RES	<enum> DownloadAck, DownloadOK, DownloadFail, Start, End, OK, Skip, Fail, UpdatingISP, Alive	Status of firmware update • DownloadAck: Successfully received the firmware data • DownloadOK: Finished receiving firmware data • DownloadFail: Failed to download the firmware data • Start: Starting firmware update • End: Completed firmware update • OK: Completed each firmware module update • Skip: Skips update for the firmware module using the latest version • Fail: Failed to update firmware • Alive: Reports periodically that the device is alive.

Action	Parameters	Request/ Response	Type/ Value	Description
	FirmwareModule	RES	<enum> None, Kernel, App, Web, ISP	Firmware module to update CAMERA ONLY ENCODER ONLY
	Progress	RES	<int>	Progress of firmware update The value must be in the range of 0 to 100; it indicates the progress in percent. 0 means that 0% of update has been completed and 100 means 100% has been completed.
set	OnlineUpgrade	REQ	<bool> True, False	To enable online upgrade of NVR NVR ONLY
view	OnlineUpgrade	RES	<bool> True, False	To check the current online upgrade settings NVR ONLY

17.4. Examples

17.4.1. Normal type firmware updates

REQUEST

The file content is provided in the HTTP body as below according to the format given in RFC 1867.

```
POST /stw-cgi/system.cgi?msubmenu=firmwareupdate&action=control&Type=Normal
HTTP/1.1

Content-Length: <content length>
Content-Type: multipart/form-data; boundary=<boundary>

--<boundary>
Content-Disposition: form-data; name="UploadedFile";
filename="<file name>"
Content-Type: application/octet-stream

<firmware file content>

--<boundary>--
```

TEXT RESPONSE

The status of firmware update and the progress in percent.

```
HTTP/1.1 200 OK
```

```
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
--<MultiPartBoundary>
Content-Type: text/plain
--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=0%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=4%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=8%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=13%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
```

```
FirmwareModule=None
Progress=17%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=21%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=26%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=30%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=34%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=39%

--<MultiPartBoundary>
Content-Type: text/plain
```

```
Status=DownloadAck
FirmwareModule=None
Progress=43%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=47%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=52%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=56%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=60%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=65%

--<MultiPartBoundary>
Content-Type: text/plain
```

```
Status=DownloadAck
FirmwareModule=None
Progress=69%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=73%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=78%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=82%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=86%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=91%

--<MultiPartBoundary>
```

```
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=95%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=99%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadOK
FirmwareModule=None
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive
```

```
--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=OK
FirmwareModule=Kernel
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=OK
FirmwareModule=App
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=OK
FirmwareModule=Web
Progress=100%
```

```
--<MultiPartBoundary>
Content-Type: text/plain

Status=End
FirmwareModule=None
Progress=100%

--<MultiPartBoundary>
Content-type:text/plain

OK
```

JSON RESPONSE

Status of the firmware update, and the progress as a percentage.

```
HTTP/1.1 200 OK
```

```
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
--<MultiPartBoundary>
Content-Type: application/json

--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "DownloadAck"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 4
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 8
}
```

```
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 13
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 17
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 21
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 26
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 30
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 34
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 39
```

```
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 43
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 47
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 52
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 56
}

--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 60
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 65
}
--<MultiPartBoundary>
Content-Type: application/json
```

```
{
    "Progress": 69
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 73
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 78
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 82
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 86
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 91
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 95
}
--<MultiPartBoundary>
Content-Type: application/json
```

```
{
    "Progress": 99
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 100
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "DownloadOK"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Start"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
```

```
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "OK",
    "FirmwareModule": "Kernel",
    "Progress": 100
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "OK",
    "FirmwareModule": "App",
    "Progress": 100
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json
```

```
{
    "Status": "OK",
    "FirmwareModule": "Web",
    "Progress": 100
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "End"
}
--<MultiPartBoundary>
Content-type:application/json

{
    "Response": "Success"
}
```

CURL command

A firmware update can be tested with CURL as below. To learn about CURL, please refer to http://curl.haxx.se.

NOTE For getting JSON response, add the -H ted with CURL as below. To header to the request.

```
curl --digest -u <userid>:<password> -F
UploadedFile=@snb5004_Series_1.13_131218.img "http://<Device IP>/stw-
cgi/system.cgi?msubmenu=firmwareupdate&action=control&Type=Normal" -H
"Expect:"
```

The above command will produce a request to the device as below:

```
POST /stw-cgi/system.cgi?msubmenu=firmwareupdate&action=control&Type=Normal
HTTP/1.1
Content-Length: 27920272
Content-Type: multipart/form-data; boundary=<boundary>
```

```
--<boundary>
Content-Disposition: form-data; name="UploadedFile";
```

```
filename="snb5004_Series_1.13_131218.img"
Content-Type: application/octet-stream

<firmware file content>

--<boundary>--
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=0%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=4%

--<MultiPartBoundary>
Content-Type: text/plain

Status=DownloadAck
FirmwareModule=None
Progress=8%

--<MultiPartBoundary>
Content-Type: text/plain

...

--<MultiPartBoundary>
Content-Type: text/plain
```

```
Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=OK
FirmwareModule=App
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=Alive

--<MultiPartBoundary>
Content-Type: text/plain

Status=OK
FirmwareModule=Web
Progress=100%

--<MultiPartBoundary>
Content-Type: text/plain

Status=End
FirmwareModule=None
Progress=100%

--<MultiPartBoundary>
Content-type:text/plain

OK
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
```

```
--<MultiPartBoundary>
Content-Type: application/json
```

```
{
    "Status": "DownloadAck"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 4
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Progress": 8
}
...

--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "Alive"
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "OK",
    "FirmwareModule": "Web",
    "Progress": 100
}
--<MultiPartBoundary>
Content-Type: application/json

{
    "Status": "End"
}
--<MultiPartBoundary>
Content-type:application/json

{
```

```
    "Response": "Success"
}
```

Chapter 18. Configuration Backup

18.1. Description

The **configbackup** submenu makes a copy of all system settings for backup.

The format of the configuration is device-dependent.

The device can restart after making a copy of the system settings for backup.

Access level

Action	Camera	Encoder	NVR
control	Admin	Admin	User

> **NOTE** Attribute to check for Configuration Backup: "attributes/System/Support/**ConfigBackup**"

18.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=configbackup&action=<value>
```

CURL command The configuration backup can be tested with CURL as below. To learn about CURL, please refer to http://curl.haxx.se.

```
curl -v --digest -u admin:isv13579! "http://<Device IP>/stw-
cgi/system.cgi?msubmenu=configbackup&action=control" > config.bin
```

Chapter 19. Configuration Restore

19.1. Description

The **configrestore** submenu restores the system configuration by using the backup.

You can reset multiple cameras to the same configuration by using **configbackup** and **configrestore**.

The format of the configuration is device-dependent.

The device can restart after a configuration restore.

Access level

Action	Camera	Encoder	NVR
control	Admin	Admin	Admin

NOTE Attribute to check for Configuration Restore: "attributes/System/Support/**ConfigRestore**"

19.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
configrestore&action=<value>[&<parameter>=<value>...]
```

19.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	ExcludeSettings	REQ	<csv> Authority, Network, Camera, None	Selects a group of settings to be excluded from system configuration restore • Authority: Restores the system configuration except for the user permission settings. • Network: Restores the system configuration except for the network settings. • Camera: Restores the system configuration except for the camera mapping information assigned to each channel. • None: Restores all settings.

19.4. Examples

19.4.1. Restoring the system configuration except for current network settings

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=configrestore&action=control&ExcludeSettings=Network
```

File content

The file content is provided in the HTTP body as below in URL-encoded format after being encoded in base64 format.

```
<Body>
```

```
POST /stw-
cgi/system.cgi?msubmenu=configrestore&action=control&ExcludeSettings=Network
HTTP/1.1
Content-type: application/x-www-form-urlencoded; charset=utf-8
Content-Length: <content length>

<config file content>
```

CURL command

The configuration backup and configuration restore can be tested with CURL as below. To learn about CURL, please refer to http://curl.haxx.se.

CURL command for configuration backup looks like below;

```
curl -v --digest -u admin:isv13579! "http://<Device IP>/stw-
cgi/system.cgi?msubmenu=configbackup&action=control" > config.bin
```

The configuration file must be base64 encoded before sending the POST request.

```
openssl base64 -in config.bin -out encoded.bin
```

The CURL command for configuration restore is as below;

```
curl -v --digest -u <userid>:<password> --data-urlencode @encoded.bin
"http://<Device IP>/stw-
cgi/system.cgi?msubmenu=configrestore&action=control&ExcludeSettings=Network
" -H "Expect:"
```

Chapter 20. Storage Information

20.1. Description

The **storageinfo** submenu requests storage device information.

NOTE	Attributes to check if the device supports DAS encryption: **"attributes/System/Support/SDCardEncryption"**

Access level

Action	Camera	NVR
view	User	User
set	Admin	User
control	Admin	User

20.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
storageinfo&action=<value>[&<parameter>=<value>...]
```

20.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the storage information settings.
	SlotNumber	REQ, RES	<int>	Slot number NVR ONLY
	Storage.#.Usage	RES	<enum> Internal, External	Usage of corresponding storage NVR ONLY
	Storage.#.Model	RES	<string>	Model of corresponding storage NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	UsedSpace	RES	<string>	Amount of storage device space currently in use (in megabytes) UsedSpace means the sum of the used space for all storage devices; Sum of Storage.1.UsedSpace, Storage.2.UsedSPace, and so forth.
	TotalSpace	RES	<string>	Total storage device space (in megabytes) TotalSpace means the sum of total space of all storage devices; Sum of Storage.1.TotalSpace, Storage.2.TotalSPace, and so forth.
	Storage.#.Temperature	RES	<string>	Temperature of storage NVR ONLY
	Storage.#.UsedSpace	RES	<string>	Amount of storage device space currently in use (in megabytes) in corresponding storage
	Storage.#.TotalSpace	RES	<string>	Total storage device space (in megabytes) in corresponding storage
	Storage.#.Type	RES	<enum> DAS, NAS	Storage type of the corresponding storage **Note** Attribute to check for NAS Support: "attributes/Recording/Support/NAS"
	Storage.#.FileSystem	RES	<enum>	File system of the corresponding storage This parameter is valid only when **Storage.#.Type** is set to DAS. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Storage.#.Status	RES	<enum> Normal, Error, Active, Formatting, Lock, Check, Error. InvalidSlot, Error. UnknownRAID, Full, PWError	Status of the corresponding storage CAMERA ONLY
	Storage	REQ	<int>	Storage number
	IsSDCardEncrypted	RES	<bool> True, False	Status of storage encryption
	Storage.#.SlotIndex	RES	<int>	A storage slot index
	Storage.#.SlotIndex.#. Health	RES	<int>	lifetime of slot storage(0~100%)
	Storage.#.SlotIndex.#. TotalSpace	RES	<string> MB	Total space of this slot storage
	Storage.#.SlotIndex.#. UsedSpace	RES	<string> MB	Used space of this slot storage
	Storage.#.SlotIndex.#. Status	RES	<enum>	Status of this slot storage
set	Storage	REQ	<int>	Storage number (read-only for NVR) **Note** **Storage** and **Enable** must be sent together for the **set** action.
	Enable	REQ, RES	<bool> True, False	Enables or disables the storage **Note** **Storage** and **Enable** must be sent together for the **set** action. CAMERA ONLY
	DefaultFolder	REQ, RES	<string>	Path on the NAS for recording This parameter is valid only when Type is set to NAS. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	NASIP	REQ, RES	<string>	NAS IP address This parameter is valid only when Type is set to NAS. CAMERA ONLY
	NASUserID	REQ, RES	<string>	NAS user ID This parameter is valid only when Type is set to NAS. CAMERA ONLY
	NASPassword	REQ, RES	<string>	NAS password This parameter is valid only when **Type** is set to NAS. CAMERA ONLY
	IsNASPasswordEncrypted	REQ	<bool> True, False	When this is set as true, the password is encrypted using the public key obtained using the **rsa** submenu of security.cgi, and sent as payload content for the POST command. CAMERA ONLY
	FileSystem	REQ, RES	<enum> VFAT, ext4	File system This parameter is valid only when **Storage.#.Type** is set to DAS. CAMERA ONLY
	TargetIP	REQ, RES	<string>	Target IP address (read-only for NVR)
	IsDASEncryptEnable	REQ, RES	<bool> True, False	Enables or disables disk array storage encryption CAMERA ONLY
	NewDASPassword	REQ, RES	<string>	New DAS password for password change

Action	Parameters	Request/ Response	Type/ Value	Description
	IsNewDASPasswordEncrypted	REQ, RES	<bool> True, False	When this is set to true, the new DAS password is encrypted using the public key obtained from the **rsa** submenu of security.cgi, and sent as payload content for the POST command. CAMERA ONLY
	DASPassword	REQ, RES	<string>	DAS password used for encryption CAMERA ONLY
	IsDASPasswordEncrypted	REQ	<bool> True, False	When this is set as true, the DAS password is encrypted using the public key obtained using the **rsa** submenu of security.cgi and sent as payload content for the POST command. CAMERA ONLY
	Port	REQ, RES	<string>	Port number (read-only for NVR)
	TargetIQN	REQ, RES	<string>	Target IQN (read-only for NVR)
	CHAPUserID	REQ, RES	<string>	CHAP (Challenge Handshake Authentication Protocol) user ID (read-only for NVR)
	CHAPPassword	REQ	<string>	CHAP (Challenge Handshake Authentication Protocol) password (Read-only for NVR)
	IsCHAPPasswordEncrypted	REQ	<bool> True, False	Returns true if password sent is encrypted Encrypted password should be sent as a post message
	SMBVersion	REQ	<enum> v1.0, v2.0, v3.0	Configure SMB Version s
control	Status	RES	<enum> Fail, Success	Connection status This parameter is valid only when **Mode** is set to NASTest. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Storage	REQ	<int>	Storage number **Note** To use the **control** action, **Storage** and **Mode** must be sent together.
	Mode	REQ	<enum> Format, NASTest, Connect, Disconnect	Storage mode NASTest is available when **Storage.#.Type** is set to NAS and Storage.# is not in an active state (i.e., already connected and enabled for recording). **Note** To use the **control** action, **Mode** must be sent together with exactly one of **Storage** or **Storages**. Do not include both **Storage** and **Storages** in the same request.
	Storages	REQ	<csv> Storage.#.SlotIndex.#, Storage.#	Storages detail to control

20.4. Examples

20.4.1. Getting the current storage info when the device supports SD card encryption

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=storageinfo&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
UsedSpace=105
```

```
TotalSpace=30185

Storage.1.Type=DAS
Storage.1.UsedSpace=105
Storage.1.TotalSpace=30185
Storage.1.FileSystem=VFAT
Storage.1.Enable=True
Storage.1.Status=Active
Storage.1.IsDASEncrypteEnable=False
Storage.1.DASPassword=
Storage.1.IsSDCardEncrypted=False

Storage.2.Type=NAS
Storage.2.UsedSpace=0
Storage.2.TotalSpace=0
Storage.2.Enable=False
Storage.2.Status=
Storage.2.DefaultFolder=SNB6004_test
Storage.2.NASIP=192.168.75.180
Storage.2.NASUserID=admin
Storage.2.NASPassword=admin4321
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "UsedSpace": "105",
    "TotalSpace": "30185",
    "Storages": [
        {
            "Storage": 1,
            "Type": "DAS",
            "UsedSpace": "105",
            "TotalSpace": "30185",
            "FileSystem": "VFAT",
            "Enable": true,
            "Status": "Active",
            "DASConfig": {
```

```
                "IsDASEncryptEnable": false,
                "DASPassword": "",
                "IsSDCardEncrypted": false
            },
            "SlotInformation": [
                {
                    "SlotIndex": 1,
                    "Health": -1,
                    "TotalSpace": "0",
                    "UsedSpace": "0",
                    "Status": ""
                },
                {
                    "SlotIndex": 2,
                    "Health": -1,
                    "TotalSpace": "0",
                    "UsedSpace": "0",
                    "Status": ""
                }
            ]
        },
        {
            "Storage": 2,
            "Type": "NAS",
            "UsedSpace": "0",
            "TotalSpace": "0",
            "Enable": false,
            "Status": "Active",
            "NASConfig": {
                "DefaultFolder": "SNB6004_test",
                "NASIP": "192.168.75.180",
                "NASUserID": "admin",
                "NASPassword": "admin4321"
            }
        }
    ]
}
```


20.4.2. Enabling storage 1

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Storage=1&Enable=True
```

20.4.3. Setting storage mode to NASTest

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=control&Storage=2&Mode=NASTest
```

20.4.4. Initially set new SD card password

REQUEST Get

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&NewDASPassword={Password}
```

REQUEST Post

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&IsNewDASPasswordEncrypted=True
```

```
<SDEncryption>
    <NewDASPassword>{RSA encrypted pw}</NewDASPassword>
</SDEncryption>
```

- {RSA encrypted pw} should be rsa encrypted & base64 encoded.
- whole message should be URL encoded.

20.4.5. Set SD card's password to decrypt SD card(In case of SD Card was encrypted by other camera device, user want to use this sd card in this camera device.)

REQUEST Get

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&DASPassword={Password}
```

REQUEST Post

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&IsDASPasswordEncrypted=True
```

```
<SDEncryption>
    <DASPassword>{RSA encrypted pw}</DASPassword>
</SDEncryption>
```

- {RSA encrypted pw} should be rsa encrypted & base64 encoded.
- whole message should be URL encoded.

20.4.6. Change SD card password

REQUEST Get

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&DASPassword={Password}&NewDASPassword={Password}
```

REQUEST Post

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=set&Enable=True&Storage=1&IsDASEn
cryptEnable=True&IsDASPasswordEncrypted=True&IsNewDASPasswordEncrypted=True
```

```
<SDEncryption>
    <DASPassword>{RSA encrypted pw}</DASPassword>
    <NewDASPassword>{RSA encrypted pw}</NewDASPassword>
</SDEncryption>
```

- {RSA encrypted pw} should be rsa encrypted & base64 encoded.
- whole message should be URL encoded.

20.4.7. Format NAS and slot 1,2 SD

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=storageinfo&action=control&Storages=Storage.1.SlotIn
```

```
dex.1,Storage.1.SlotIndex.2,Storage.2&Mode=Format
```

Chapter 21. GPS

21.1. Description

The **gps** submenu requests the GPS (global positioning system) information of the NVR.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

21.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
gps&action=<value>[&<parameter>=<value>...]
```

21.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Check	REQ	<enum> Once, Periodically	Requests the GPS data only once or periodically
	Periodicity	REQ	<int>	Interval to request the GPS data The range is from 1 to 300.
	GPSData	RES	<string>	GPS data $GPRMC sentence is only available.

21.4. Examples

21.4.1. Getting the GPS data only one time

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=gps&action=view&Check=Once
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
GPSData=
$GPRMC,084142.00,A,3729.03548,N,12653.80696,E,11.581,119.28,290610,,,A*51
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "GPSData":
"$GPRMC,084142.00,A,3729.03548,N,12653.80696,E,11.581,119.28,290610,,,A*51"
}
```

21.4.2. Requesting the GPS data every 5 seconds

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=gps&action=view&Check=Periodically&Periodicity=5
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GPSData=
$GPRMC,102122.00,A,3521.03548,N,12843.80696,E,12.381,118.17,290610,,,A*51
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "GPSData":
"$GPRMC,102122.00,A,3521.03548,N,12843.80696,E,12.381,118.17,290610,,,A*51"
}
```

Chapter 22. Automatic Backup

22.1. Description

The **autobackup** submenu automatically backs up videos recorded on the NVR to the server when the NVR is connected.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

22.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
autobackup&action=<value>[&<parameter>=<value>...]
```

22.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the automatic backup settings.
	Status	RES	<enum>	Backup status
set	ConnectionType	REQ, RES	<enum> WiFi, Ethernet	Connection type
	ServerSSID	REQ, RES	<string>	Server SSID
	PollingFrequency	REQ, RES	<enum> Off, 5s, 10s, 20s, 30s, 1m, 5m, 10m	Polling frequency interval

22.4. Examples

22.4.1. Getting the current auto backup settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=autobackup&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=processing
ConnectionType=Wifi
ServerSSID=stw_post
PollingFrequency=30s
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "processing",
    "ConnectionType": "Wifi",
    "ServerSSID": "stw_post",
    "PollingFrequency": "30s"
}
```

22.4.2. Setting to make backups through the WiFi connection

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=autobackup&action=set&ConnectionType=WiFi&ServerSSID
=stw_post&PollingFrequency=5s
```

22.4.3. Setting to make backups through the Ethernet connection

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/system.cgi?msubmenu=autobackup&action=set&ConnectionType=Ethernet
```

Chapter 23. Digital Signage

23.1. Description

The **digitalsignage** submenu is to display the advertisements from the FTP server if a certain input or event does not occur for the specified time.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

23.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
digitalsignage&action=<value>[&<parameter>=<value>...]
```

23.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the settings for the digital signage function.
set	AutoStart	REQ, RES	<bool> True, False	Enables or disables the digital signage function.
	AutoStartDuration	REQ, RES	<int> 10s, 20s, 30s, 1m, 2m, 3m, 4m, 5m, 10m, 20m, 30m, 1h	Auto start duration **AutoStartDuration** is valid only when **AutoStart** is set to True.
	FTPSync	REQ, RES	<bool> True, False	Enables or disables the sync in the FTP server to get advertisements **FTPSync** is valid only when **AutoStart** is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
	FTPServer	REQ, RES	<string>	FTP server **FTPServer** is valid only when **AutoStart** and **FTPSync** are both set to True.
	FTPUserName	REQ, RES	<string>	FTP user name **FTPUserName** is valid only when **AutoStart** and **FTPSync** are both set to True.
	FTPPassword	REQ, RES	<string>	FTP password **FTPPassword** is valid only when **AutoStart** and **FTPSync** are both set to True.
	IsFTPPasswordEncrypted	REQ	<bool> True, False	When this is set as true, password is encrypted using the public key obtained using the **rsa** submenu of security.cgi and sent as payload content for the POST command.
	FTPFilename	REQ, RES	<string>	FTP file name **FTPFilename** is valid only when **AutoStart** and **FTPSync** are both set to True.

23.4. Examples

23.4.1. Getting the current digital signage settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=digitalsignage&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AutoStart=True
```

```
AutoStartDuration=1m
FTPSync=True
FTPServer=ftpserver
FTPUserName=anonymous
FTPPassword={password}
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AutoStart": true,
    "AutoStartDuration": "1m",
    "FTPSync": true,
    "FTPServer": "ftpserver",
    "FTPUserName": "anonymous",
    "FTPPassword": {password}
}
```

23.4.2. Setting to use the advertisements from the FTP server

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=digitalsignage&action=set&AutoStart=Ture&AutoStartDu
ration=1m&FTPSync=True&FTPServer=stw_signage_server&FTPUserName=admin&FTPPas
sword={password}
```

23.4.3. Setting to use the advertisements from USB

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=digitalsignage&action=set&AutoStart=True&AutoStartDu
ration=30s&FTPSync=False
```

Chapter 24. Vehicle Information

The **vehicleinformation** submenu specifies the vehicle-related information on the device.

> **NOTE** This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

24.1. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
vehicleinformation&action=<value>[&<parameter>=<value>...]
```

24.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the vehicle information configured on the device
set	Company	REQ, RES	<string>	Company information of the vehicle
	Maker	REQ, RES	<string>	Vehicle brand name
	Model	REQ, RES	<string>	Vehicle model name
	RegistrationNumber	REQ, RES	<string>	Registration number of the vehicle (Number plate information)
	License	REQ, RES	<string>	Driver License information
	DriverName	REQ, RES	<string>	Driver name
	DelayUnits	REQ, RES	<enum> Seconds, Minutes, Hours	Shutdown delay unit
	ShutdownDelay	REQ, RES	<int>	Shutdown delay duration

24.3. Examples

Gets the vehicle information stored in the device.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=vehicleinformation&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Company=ABC Corporation
Maker=Hyundai
Model=Sonata
RegistrationNumber=AZ1000
License=AZ10001001
DriverName=Thompson
DelayUnits=Seconds
ShutdownDelay=30
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Company": "ABC Corporation",
    "Maker": "Hyundai",
    "Model": "Sonata",
    "RegistrationNumber": "AZ1000",
    "License": "AZ10001001",
    "DriverName": "Thompson",
    "DelayUnits": "Seconds",
    "ShutdownDelay": 30
}
```

Chapter 25. ONVIF Feature

The **onviffeature** submenu is used to enable or disable some features exposed in the ONVIF protocol.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin
set	Admin

25.1. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
onviffeature&action=<value>[&<parameter>=<value>...]
```

25.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the ONVIF feature that can be enabled or disabled
set	FocusControl	REQ, RES	<bool>	Enables or disables focus control feature using ONVIF image service.

25.3. Examples

Get the current FocusControl status for onvif

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=onviffeature&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
FocusControl=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "FocusControl": false
}
```

Chapter 26. Database Reset

This **databasereset** submenu is used to reset the database entries.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
control	Admin

26.1. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
databasereset&action=<value>[&<parameter>=<value>...]
```

26.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	IncludeDataType	REQ	<csv> PeopleCount, VehicleCount, HeatMap, QueueEvents, All	Based on the parameter, the passed database will be reset.

26.3. Examples

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=databasereset&action=control&IncludeDataType=All
```

Chapter 27. Log Server

This **logserver** submenu is used to configure clients to receive log messages.

Access level

Action	Camera	NVR
view	User	User
add/update	User	User
remove	User	User

27.1. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
logserver&action=<value>[&<parameter>=<value>...]
```

27.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	RES	<int>	Index of the logserver
add/update	Enable	REQ, RES	<bool> True, False	Enable or disable sending
	IPType	REQ, RES	<enum> IPv4	IP type selection
	IPAddress	REQ, RES	<string>	IP address
	Port	REQ, RES	<int>	Port number
	ConnectionType	REQ, RES	<enum> UDP,TCP	Transport layer protocol type
	Format	REQ, RES	<enum> RFC3164, RFC5424	Log server format
remove	Index	REQ	<int>	Index of the logserver

27.3. Examples

27.3.1. Getting the current logserver settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=logserver&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "LogServer": [
        {
            "Index": 1,
            "Enable": true,
            "IPType": "IPv4",
            "IPAddress": "192.168.1.100",
            "Port": 501,
            "ConnectionType": "UDP"
        }
    ]
}
```


27.3.2. Add a new client

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=logserver&action=add&Enable=True&IPType=IPv4&IPAddre
ss=192.168.1.103&Port=503&ConnectionType=UDP
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

27.3.3. Remove client using index

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=logserver&action=remove&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 28. Session Info

This **sessioninfo** submenu is used to get the current sessions in use.

NOTE	This chapter applies to NVR only.

Access level

Action	NVR
view	User

28.1. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
sessioninfo&action=<value>[&<parameter>=<value>...]
```

28.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Live	RES	<int>	Current live session count
	Search	RES	<int>	Current search session count
	Backup	RES	<int>	Current backup session count

28.3. Examples

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=sessioninfo&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Live=2
Search=0
Backup=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Live": 2,
    "Search": 0,
    "Backup": 0
}
```

Chapter 29. SD card information

29.1. Description

The **sdcardinfo** submenu is used to get the details of the current SD card.

NOTE This chapter applies to multi directional cameras only.

Access level

Action	Camera
view	User

29.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
sdcardinfo&action=<value>[&<parameter>=<value>...]
```

29.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	
	Channel.#.UsedSpace	RES	<string>	SD card space used
	Channel.#.TotalSpace	RES	<string>	Total SD card capacity
	Channel.#.FileSystem	RES	<enum> EXT4, VFAT	Filesystem used in SD card
	Channel.#.Status	RES	<enum> Normal, Error, Active, Formatting, Lock, Full	Current state of SD card
control	Channel	REQ	<csv>	channel number
	Mode	REQ	<enum> Format	Format the SD card of specified channel **Note** Channel-based format is supported only in certain multi-sensor cameras

29.4. Examples

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=sdcardinfo&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.UsedSpace=15067
Channel.0.TotalSpace=30420
Channel.0.FileSystem=VFAT
Channel.0.Status=Active
Channel.1.UsedSpace=2269
Channel.1.TotalSpace=30420
Channel.1.FileSystem=VFAT
Channel.1.Status=Active
Channel.2.UsedSpace=0
Channel.2.TotalSpace=0
Channel.2.FileSystem=VFAT
Channel.2.Status=
Channel.3.UsedSpace=0
Channel.3.TotalSpace=0
Channel.3.FileSystem=VFAT
Channel.3.Status=
Channel.3.Status=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
[
    {
        "Channel": 0,
        "UsedSpace": "15067",
```

```
        "TotalSpace": "30420",
        "FileSystem": "VFAT",
        "Status": "Active"
    },
    {
        "Channel": 1,
        "UsedSpace": "2269",
        "TotalSpace": "30420",
        "FileSystem": "VFAT",
        "Status": "Active"
    },
    {
        "Channel": 2,
        "UsedSpace": "0",
        "TotalSpace": "0",
        "FileSystem": "VFAT",
        "Status": ""
    },
    {
        "Channel": 3,
        "UsedSpace": "0",
        "TotalSpace": "0",
        "FileSystem": "VFAT",
        "Status": ""
    }
]
```

29.4.1. Formatting the SD card in channel 1

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=sdcardinfo&action=control&Channel=1&Mode=Format
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
  "Response": "Success"
}
```

Chapter 30. ISCSI Discovery

30.1. Description

This **iscsidiscovery** submenu is used to get the ISCSI targets.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
control	ADMIN

30.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
iscsidiscovery&action=<value>[&<parameter>=<value>...]
```

30.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Mode	REQ	<enum> Discover	
	PortalIP	REQ	<string>	Storage group IP address
	Port	REQ	<int>	Port number of storage group
	CHAPUserID	REQ	<string>	User name
	CHAPPassword	REQ	<string>	Password
	IsCHAPPasswordEncrypted	REQ	<bool> True, False	When set to true, the password is encrypted using the public key provided by the **rsa** submenu of security.cgi and sent as a post payload.
	AvailableTargets	RES	<csv>	List of available targets

30.4. Examples

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=iscsidiscovery&action=control&Mode=Discover&PortalIp
```

```
=192.168.17.1&Port=3260&CHAPUserId=Martin&CHAPPassword={password}
```

TEXT RESPONSE

```
AvailableTargets=Target1,Target2,Target3
```

JSON RESPONSE

```
{
    "AvailableTargets": [
        "Target1",
        "Target2",
        "Target3"
    ]
}
```

Chapter 31. Holiday

31.1. Description

The **holiday** submenu configures the holiday settings for the device.

Access level

Action	NVR	Decoder
view	User	User
set	User	User
remove	User	User

31.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
holiday&action=<value>[&<parameter>=<value>...]
```

31.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the holiday settings.
	Year	REQ	<int>	Year for the holidays to be searched The values must be within the range of 2000 to 2037.
	Month	REQ	<csv> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12	Month(s) for the holiday to be searched
	Holiday	RES	<csv> 1, 2, 3, 4, , , , 31	Holiday number in a year and in a month(s)
set	Year	REQ, RES	<int>	Year for the holiday The values must be within the range of 2000 to 2037.
	Month	REQ, RES	<int>	Month for the holiday

Action	Parameters	Request/ Response	Type/ Value	Description
	Day	REQ, RES	<csv> 1, 2, 3, 4, , , , 31	Day for the holiday
	Week	REQ, RES	<enum> First, Second, Third, Fourth, Last	Week for the holiday
	<ddd>	REQ, RES	<bool> True, False	Day of week for the holiday <ddd> represents day of the week, and should be specified in the short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g.) ‘SUN=True’ indicates every Sunday of the month is set as the holiday.
remove	Year	REQ	<int>	Holiday year
	Month	REQ	<int>	Holiday month
	Holiday	REQ	<csv> 1, 2, 3, 4, , , , 31	Holiday number

31.4. Examples

31.4.1. Getting holiday settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=holiday&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Year.2018.Month.1.Holiday=
Year.2018.Month.1.Day=
```

```
Year.2018.Month.1.Week.First.WeekDay=
Year.2018.Month.1.Week.Second.WeekDay=
Year.2018.Month.1.Week.Third.WeekDay=
Year.2018.Month.1.Week.Fourth.WeekDay=
Year.2018.Month.1.Week.Last.WeekDay=
Year.2018.Month.2.Holiday=
Year.2018.Month.2.Day=
Year.2018.Month.2.Week.First.WeekDay=
Year.2018.Month.2.Week.Second.WeekDay=
Year.2018.Month.2.Week.Third.WeekDay=
Year.2018.Month.2.Week.Fourth.WeekDay=
Year.2018.Month.2.Week.Last.WeekDay=
Year.2018.Month.3.Holiday=
Year.2018.Month.3.Day=
Year.2018.Month.3.Week.First.WeekDay=
Year.2018.Month.3.Week.Second.WeekDay=
Year.2018.Month.3.Week.Third.WeekDay=
Year.2018.Month.3.Week.Fourth.WeekDay=
Year.2018.Month.3.Week.Last.WeekDay=
Year.2018.Month.4.Holiday=7,8,22,27
Year.2018.Month.4.Day=7,8,22,27
Year.2018.Month.4.Week.First.WeekDay=SAT
Year.2018.Month.4.Week.Second.WeekDay=SUN
Year.2018.Month.4.Week.Third.WeekDay=
Year.2018.Month.4.Week.Fourth.WeekDay=
Year.2018.Month.4.Week.Last.WeekDay=
Year.2018.Month.5.Holiday=1,7
Year.2018.Month.5.Day=1,7
Year.2018.Month.5.Week.First.WeekDay=
Year.2018.Month.5.Week.Second.WeekDay=
Year.2018.Month.5.Week.Third.WeekDay=
Year.2018.Month.5.Week.Fourth.WeekDay=
Year.2018.Month.5.Week.Last.WeekDay=
Year.2018.Month.6.Holiday=
Year.2018.Month.6.Day=
Year.2018.Month.6.Week.First.WeekDay=
Year.2018.Month.6.Week.Second.WeekDay=
Year.2018.Month.6.Week.Third.WeekDay=
Year.2018.Month.6.Week.Fourth.WeekDay=
Year.2018.Month.6.Week.Last.WeekDay=
Year.2018.Month.7.Holiday=
```

```
Year.2018.Month.7.Day=
Year.2018.Month.7.Week.First.WeekDay=
Year.2018.Month.7.Week.Second.WeekDay=
Year.2018.Month.7.Week.Third.WeekDay=
Year.2018.Month.7.Week.Fourth.WeekDay=
Year.2018.Month.7.Week.Last.WeekDay=
Year.2018.Month.8.Holiday=
Year.2018.Month.8.Day=
Year.2018.Month.8.Week.First.WeekDay=
Year.2018.Month.8.Week.Second.WeekDay=
Year.2018.Month.8.Week.Third.WeekDay=
Year.2018.Month.8.Week.Fourth.WeekDay=
Year.2018.Month.8.Week.Last.WeekDay=
Year.2018.Month.9.Holiday=
Year.2018.Month.9.Day=
Year.2018.Month.9.Week.First.WeekDay=
Year.2018.Month.9.Week.Second.WeekDay=
Year.2018.Month.9.Week.Third.WeekDay=
Year.2018.Month.9.Week.Fourth.WeekDay=
Year.2018.Month.9.Week.Last.WeekDay=
Year.2018.Month.10.Holiday=
Year.2018.Month.10.Day=
Year.2018.Month.10.Week.First.WeekDay=
Year.2018.Month.10.Week.Second.WeekDay=
Year.2018.Month.10.Week.Third.WeekDay=
Year.2018.Month.10.Week.Fourth.WeekDay=
Year.2018.Month.10.Week.Last.WeekDay=
Year.2018.Month.11.Holiday=
Year.2018.Month.11.Day=
Year.2018.Month.11.Week.First.WeekDay=
Year.2018.Month.11.Week.Second.WeekDay=
Year.2018.Month.11.Week.Third.WeekDay=
Year.2018.Month.11.Week.Fourth.WeekDay=
Year.2018.Month.11.Week.Last.WeekDay=
Year.2018.Month.12.Holiday=
Year.2018.Month.12.Day=
Year.2018.Month.12.Week.First.WeekDay=
Year.2018.Month.12.Week.Second.WeekDay=
Year.2018.Month.12.Week.Third.WeekDay=
Year.2018.Month.12.Week.Fourth.WeekDay=
Year.2018.Month.12.Week.Last.WeekDay=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Year": 2018,
    "MonthWiseHolidays": [
        {
            "Month": 1,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": []
                },
                {
                    "Week": "Second",
                    "WeekDay": []
                },
                {
                    "Week": "Third",
                    "WeekDay": []
                },
                {
                    "Week": "Fourth",
                    "WeekDay": []
                },
                {
                    "Week": "Last",
                    "WeekDay": []
                }
            ]
        },
        {
            "Month": 2,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
```

```
            {
                "Week": "First",
                "WeekDay": []
            },
            {
                "Week": "Second",
                "WeekDay": []
            },
            {
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
        "Month": 3,
        "Holiday": [],
        "Day": [],
        "WeekwiseHolidays": [
            {
                "Week": "First",
                "WeekDay": []
            },
            {
                "Week": "Second",
                "WeekDay": []
            },
            {
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
```

```
                    "WeekDay": []
                },
                {
                    "Week": "Last",
                    "WeekDay": []
                }
            ]
        },
        {
            "Month": 4,
            "Holiday": [
                "7",
                "8",
                "22",
                "27"
            ],
            "Day": [
                "7",
                "8",
                "22",
                "27"
            ],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": [
                        "SAT"
                    ]
                },
                {
                    "Week": "Second",
                    "WeekDay": [
                        "SUN"
                    ]
                },
                {
                    "Week": "Third",
                    "WeekDay": []
                },
                {
                    "Week": "Fourth",
```

```
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
        "Month": 5,
        "Holiday": [
            "1",
            "7"
        ],
        "Day": [
            "1",
            "7"
        ],
        "WeekwiseHolidays": [
            {
                "Week": "First",
                "WeekDay": []
            },
            {
                "Week": "Second",
                "WeekDay": []
            },
            {
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
```

```
        {
            "Month": 6,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": []
                },
                {
                    "Week": "Second",
                    "WeekDay": []
                },
                {
                    "Week": "Third",
                    "WeekDay": []
                },
                {
                    "Week": "Fourth",
                    "WeekDay": []
                },
                {
                    "Week": "Last",
                    "WeekDay": []
                }
            ]
        },
        {
            "Month": 7,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": []
                },
                {
                    "Week": "Second",
                    "WeekDay": []
                },
                {
```

```
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
        "Month": 8,
        "Holiday": [],
        "Day": [],
        "WeekwiseHolidays": [
            {
                "Week": "First",
                "WeekDay": []
            },
            {
                "Week": "Second",
                "WeekDay": []
            },
            {
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
```

```
            "Month": 9,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": []
                },
                {
                    "Week": "Second",
                    "WeekDay": []
                },
                {
                    "Week": "Third",
                    "WeekDay": []
                },
                {
                    "Week": "Fourth",
                    "WeekDay": []
                },
                {
                    "Week": "Last",
                    "WeekDay": []
                }
            ]
        },
        {
            "Month": 10,
            "Holiday": [],
            "Day": [],
            "WeekwiseHolidays": [
                {
                    "Week": "First",
                    "WeekDay": []
                },
                {
                    "Week": "Second",
                    "WeekDay": []
                },
                {
                    "Week": "Third",
```

```
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
        "Month": 11,
        "Holiday": [],
        "Day": [],
        "WeekwiseHolidays": [
            {
                "Week": "First",
                "WeekDay": []
            },
            {
                "Week": "Second",
                "WeekDay": []
            },
            {
                "Week": "Third",
                "WeekDay": []
            },
            {
                "Week": "Fourth",
                "WeekDay": []
            },
            {
                "Week": "Last",
                "WeekDay": []
            }
        ]
    },
    {
        "Month": 12,
```

```
                "Holiday": [],
                "Day": [],
                "WeekwiseHolidays": [
                    {
                        "Week": "First",
                        "WeekDay": []
                    },
                    {
                        "Week": "Second",
                        "WeekDay": []
                    },
                    {
                        "Week": "Third",
                        "WeekDay": []
                    },
                    {
                        "Week": "Fourth",
                        "WeekDay": []
                    },
                    {
                        "Week": "Last",
                        "WeekDay": []
                    }
                ]
            }
        ]
}
```

31.4.2. Setting June 2018 as the holiday

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=holiday&action=set&Year=2018&Month=6&Day=2,3,9,10,16,17,23,24
```

31.4.3. Deselecting April 2018 from the holidays

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=holiday&action=remove&Year=2018&Month=4
```

Chapter 32. HDD Alarm

32.1. Description

The **hddalarm** submenu configures the HDD alarm settings for the device.

Access level

Action	NVR	Decoder
view	User	User
set	User	User

32.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
hddalarm&action=<value>[&<parameter>=<value>...]
```

32.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads HDD Alarm settings.
	Type	REQ	<csv> Check, Replace, iSCSI	Alarm output terminal type Check status means that the HDD is operating but it has problems that require technical examination. Replace status means that the HDD has defect and requires immediate replacement.
set	Type	REQ, RES	<csv> Check, Replace, iSCSI	Alarm output terminal type
	AlarmOutput	REQ, RES	<csv> 1, 2, 3, 4, Beep, None	Alarm output number If Beep was selected, a beep will sound. Alarm signal will output through the alarm out port on the rear side when select <1>, <2>, <3> and <4>.

Action	Parameters	Request/ Response	Type/ Value	Description
	Duration	REQ, RES	<enum> None, 5s, 10s, 20s, 30s, Always	Alarm duration for the alarm signal and beep sound

32.4. Examples

32.4.1. Getting HDD alarm settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=hddalarm&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Type.Check.AlarmOutput=Beep
Type.Check.Duration=Always
Type.Replace.AlarmOutput=Beep
Type.Replace.Duration=Always
Type.Replace.AlarmOutput=Beep
Type.Replace.Duration=Always
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "HDDAlarm": [
        {
            "Type": "Check",
            "AlarmOutput": [
                "Beep"
            ],
```

```
            "Duration": "Always"
        },
        {
            "Type": "Replace",
            "AlarmOutput": [
                "Beep"
            ],
            "Duration": "Always"
        },
        {
            "Type": "Replace",
            "AlarmOutput": [
                "Beep"
            ],
            "Duration": "Always"
        }
    ]
}
```


32.4.2. Setting HDD alarm

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=hddalarm&action=set&Type=Check&AlarmOutput=1&Duration=10s
```

Chapter 33. Monitor Input

33.1. Description

The **monitorin** submenu configures the monitor input settings for the device.

Access level

Action	Decoder
view	User
set	User

33.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
monitorin&action=<value>[&<parameter>=<value>...]
```

33.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view		REQ	<csv>	Reads monitor in settings for the device.
	Index	REQ	<csv>	Index number
	Index.#.VideoOutput	RES	<enum> HDMI, VGA	Video output format of monitor in device
	Index.#.ConnectedVGAIndex	RES	<int>	Connected VGA index number
	Index.#.Use	RES	<bool> True, False	Enable or disable status of monitor in device
set	Index.#.Resolution	REQ, RES	<enum> 1280x720_HDMI, 1920x1080_HDMI, 1280x720_DVI, 1920x1080_DVI	Resolution of monitor in device

33.4. Examples

33.4.1. Getting monitor input settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=monitorin&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Index.1.Use=True
Index.1.ConnectedVGAIndex=1
Index.1.VideoOutput=HDMI
Index.1.Resolution=1920x1080_HDMI
Index.2.Use=True
Index.2.ConnectedVGAIndex=2
Index.2.VideoOutput=HDMI
Index.2.Resolution=1920x1080_HDMI
Index.3.Use=False
Index.3.ConnectedVGAIndex=3
Index.3.VideoOutput=HDMI
Index.3.Resolution=1920x1080_HDMI
Index.4.Use=False
Index.4.ConnectedVGAIndex=4
Index.4.VideoOutput=HDMI
Index.4.Resolution=1920x1080_HDMI
Index.5.Use=False
Index.5.ConnectedVGAIndex=5
Index.5.VideoOutput=HDMI
Index.5.Resolution=1920x1080_HDMI
Index.6.Use=False
Index.6.ConnectedVGAIndex=6
Index.6.VideoOutput=HDMI
Index.6.Resolution=1920x1080_HDMI
Index.7.Use=False
Index.7.ConnectedVGAIndex=7
Index.7.VideoOutput=HDMI
Index.7.Resolution=1920x1080_HDMI
```

```
Index.8.Use=False
Index.8.ConnectedVGAIndex=8
Index.8.VideoOutput=HDMI
Index.8.Resolution=1920x1080_HDMI
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MonitorIn": [
        {
            "Index": 1,
            "Use": true,
            "ConnectedVGAIndex": 1,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 2,
            "Use": true,
            "ConnectedVGAIndex": 2,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 3,
            "Use": false,
            "ConnectedVGAIndex": 3,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 4,
            "Use": false,
            "ConnectedVGAIndex": 4,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
```

```
        {
            "Index": 5,
            "Use": false,
            "ConnectedVGAIndex": 5,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 6,
            "Use": false,
            "ConnectedVGAIndex": 6,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 7,
            "Use": false,
            "ConnectedVGAIndex": 7,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        },
        {
            "Index": 8,
            "Use": false,
            "ConnectedVGAIndex": 8,
            "VideoOutput": "HDMI",
            "Resolution": "1920x1080_HDMI"
        }
    ]
}
```


33.4.2. Setting monitor input resolution as 1280X720_HDMI at Index 6

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=monitorin&action=set&Index.6.Resolution=1280x720_HDM
I
```

Chapter 34. Monitor Out

34.1. Description

The **monitorout** submenu configures the monitor output settings for the device.

Access level

Action	Decoder	NVR
view	User	User
set	User	User

34.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
monitorout&action=<value>[&<parameter>=<value>...]
```

34.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads monitor out settings.
	Index	REQ	<csv>	Index number
	Index.#.VideoOutput	RES	<enum> HDMI, VGA	Video output format of monitor out device
	Index.#.ConnectedVGAIndex	RES	<int>	Connected VGA index number
	Index.#.Use	RES	<bool> True, False	Enable or disable status of monitor out device
	Index.#.OptimalResolution	RES	<string> widthxheight	Optimal video output resolution (e.g. 1920x1080)
set	Index.#.Resolution	REQ, RES	<enum> 858x480, 1280x1024, 1280x720, 1920x1080, 2560x1440, 3840x2160	Resolution of monitor out device

Action	Parameters	Request/ Response	Type/ Value	Description
	Display	REQ, RES	<csv> ChannelName, IPAddress, Date, Time, Icon, Resolution, FPS, MonitorIndex, None	Display output

34.4. Examples

34.4.1. Getting monitor out settings

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=monitorout&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Display=None
Index.1.Use=True
Index.1.ConnectedVGAIndex=1
Index.1.VideoOutput=HDMI
Index.1.Resolution=1280x1024
Index.2.Use=True
Index.2.ConnectedVGAIndex=1
Index.2.VideoOutput=HDMI
Index.2.Resolution=1280x1024
Index.3.Use=True
Index.3.ConnectedVGAIndex=2
Index.3.VideoOutput=HDMI
Index.3.Resolution=1280x1024
Index.4.Use=True
Index.4.ConnectedVGAIndex=2
```

```
Index.4.VideoOutput=HDMI
Index.4.Resolution=1280x1024
Index.5.Use=False
Index.5.ConnectedVGAIndex=3
Index.5.VideoOutput=HDMI
Index.5.Resolution=1280x1024
Index.6.Use=False
Index.6.ConnectedVGAIndex=3
Index.6.VideoOutput=HDMI
Index.6.Resolution=1280x1024
Index.7.Use=False
Index.7.ConnectedVGAIndex=4
Index.7.VideoOutput=HDMI
Index.7.Resolution=1280x1024
Index.8.Use=False
Index.8.ConnectedVGAIndex=4
Index.8.VideoOutput=HDMI
Index.8.Resolution=1280x1024
Index.9.Use=False
Index.9.ConnectedVGAIndex=5
Index.9.VideoOutput=HDMI
Index.9.Resolution=1280x1024
Index.10.Use=False
Index.10.ConnectedVGAIndex=5
Index.10.VideoOutput=HDMI
Index.10.Resolution=1280x1024
Index.11.Use=False
Index.11.ConnectedVGAIndex=6
Index.11.VideoOutput=HDMI
Index.11.Resolution=1280x1024
Index.12.Use=False
Index.12.ConnectedVGAIndex=6
Index.12.VideoOutput=HDMI
Index.12.Resolution=1280x1024
Index.13.Use=False
Index.13.ConnectedVGAIndex=7
Index.13.VideoOutput=HDMI
Index.13.Resolution=1280x1024
Index.14.Use=False
Index.14.ConnectedVGAIndex=7
Index.14.VideoOutput=HDMI
```

```
Index.14.Resolution=1280x1024
Index.15.Use=False
Index.15.ConnectedVGAIndex=8
Index.15.VideoOutput=HDMI
Index.15.Resolution=1280x1024
Index.16.Use=False
Index.16.ConnectedVGAIndex=8
Index.16.VideoOutput=HDMI
Index.16.Resolution=1280x1024
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Display": [
        "None"
    ],
    "MonitorOut": [
        {
            "Index": 1,
            "Use": true,
            "ConnectedVGAIndex": 1,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 2,
            "Use": true,
            "ConnectedVGAIndex": 1,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 3,
            "Use": true,
            "ConnectedVGAIndex": 2,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
```

```
        },
        {
            "Index": 4,
            "Use": true,
            "ConnectedVGAIndex": 2,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 5,
            "Use": false,
            "ConnectedVGAIndex": 3,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 6,
            "Use": false,
            "ConnectedVGAIndex": 3,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 7,
            "Use": false,
            "ConnectedVGAIndex": 4,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 8,
            "Use": false,
            "ConnectedVGAIndex": 4,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 9,
            "Use": false,
            "ConnectedVGAIndex": 5,
            "VideoOutput": "HDMI",
```

```
            "Resolution": "1280x1024"
        },
        {
            "Index": 10,
            "Use": false,
            "ConnectedVGAIndex": 5,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 11,
            "Use": false,
            "ConnectedVGAIndex": 6,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 12,
            "Use": false,
            "ConnectedVGAIndex": 6,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 13,
            "Use": false,
            "ConnectedVGAIndex": 7,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 14,
            "Use": false,
            "ConnectedVGAIndex": 7,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 15,
            "Use": false,
            "ConnectedVGAIndex": 8,
```

```
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        },
        {
            "Index": 16,
            "Use": false,
            "ConnectedVGAIndex": 8,
            "VideoOutput": "HDMI",
            "Resolution": "1280x1024"
        }
    ]
}
```

34.4.2. Setting monitor out resolution as 1280x720 at Index 6

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=monitorout&action=set&Index.6.Resolution=1280x720
```

Chapter 35. USB Configuration

35.1. Description

The **usbconfig** submenu configures the USB port on the camera.

NOTE This chapter applies to camera only.

Access level

Action	Camera
view	User
set	User

35.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
usbconfig&action=<value>[&<parameter>=<value>...]
```

35.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Enable	REQ, RES	<bool> True, False	Enables or disables the USB port.

35.4. Examples

35.4.1. Getting USB configuration usbconfig

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=usbconfig&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Enable": true
}
```


35.4.2. Setting to enable the USB port

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=usbconfig&action=set&Enable=True
```

Chapter 36. Stratocast Service Configuration

36.1. Description

The **stratocast** submenu provides URLs to enroll devices to Stratocast web cloud service provided by Genetec.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin
set	Admin

36.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
stratocast&action=<value>[&<parameter>=<value>...]
```

36.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	ServiceURL	REQ, RES	<string>	URL of Stratocast web cloud service in EN
	DeviceEntryURL	REQ, RES	<string>	Device entry point When **SimplifiedEnrollmentEnable** is set to true, the device sends its own information to **DeviceEntryURL**. Stratocast collects and takes this information to the queue for registration. When user reads the QR code of the device with Stratocast mobile application, the device will finally be activated.

Action	Parameters	Request/ Response	Type/ Value	Description
	ProbeServiceURL	REQ, RES	<string>	URL of Stratocast probe server The device sends its own information such as CPU/memory usage and logs to the probe server. The probe server can send commands to the camera (e.g. reboot)
	CameraProbeEnable	REQ, RES	<bool> True, False	Enables or disables sending device information to the probe server
	ProbeInterval	REQ, RES	<int>	Interval between probes The range is from 30 to 300.
	SimplifiedEnrollmentEnable	REQ, RES	<bool> True, False	Enables or disables simplified enrollment When simplified enrollment is enabled, users can activate their cameras with a QR code using Stratocast mobile application without accessing any web viewers

36.4. Examples

36.4.1. Getting the current configurations

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=stratocast&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ServiceURL=app.stratcast.com
DeviceEntryURL=
ProbeServiceURL=app.stratocast.com
CameraProbeEnable=True
ProbeInterval=30
```

```
SimplifiedEnrollmentEnable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ServiceURL": "app.startocast.com",
    "DeviceEntryURL": "",
    "ProbeServieURL": "app.stratocast.com",
    "CamerProbeEnable": true,
    "ProbeInterval": 30,
    "SimplifiedEnrollmentEnable": false
}
```


36.4.2. Enabling the transfer of the camera information to the probe server

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=stratocast&action=set&CameraProbeEnable=True&ProbeIn
terval=100
```

Chapter 37. Status of Stratocast Service

37.1. Description

The **stratocastregister** submenu provides the current status of registration process on Stratocast web cloud services.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin
set	Admin
check	Admin

37.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
stratocastregister&action=<value>[&<parameter>=<value>...]
```

37.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	ActivationCode	REQ, RES	<string>	Unique identification code to register the device with Stratocast service **ActivationCode** must be issued from Stratocast server.
check	RegistrationPhase	RES	<enum> ActivationReady, GetEnrollmentInfo, GetNTPServerInfo, RegistrationDevice , GetLoadBalancerInfo, GetSSHServerInfo, ConnectSSHSession, ActivationDone	Current status when registration process is ongoing

Action	Parameters	Request/ Response	Type/ Value	Description
	Detail	RES	<string>	With **RegistrationPhase**, an additional description is provided

37.4. Examples

37.4.1. Getting the activation code

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=stratocastregister&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ActivationCode=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ActivationCode": ""
}
```

37.4.2. Setting the activation code issued by the Stratocast service

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=stratocastregister&action=set&ActivationCode=0123456
7890
```

37.4.3. Checks the current status of registration process

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=stratocastregister&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
RegistrationPhase=ActivationDone
Detail=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "RegistrationPhase": "ActivationDone",
    "Detail": ""
}
```

Chapter 38. Peer Connection Information

38.1. Description

The **peerconnectioninfo** submenu provides the session status of the currently connected client.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin

38.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
peerconnectioninfo&action=<value>[&<parameter>=<value>...]
```

38.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
	Client.#.IPAddress	RES	<string>	IP address IP address of the connected user

Action	Parameters	Request/ Response	Type/ Value	Description
	Client.#.ClientHttpsStatus	RES	<enum> NO_HTTPS, HTTPS_WITHOUT_CLIENT_CERT, HTTPS_WITH_INVALID_CLIENT_CERT, HTTPS_WITH_VALID_CLIENT_CERT	Client HTTPS status HTTPS status of the connected user • NO_HTTPS: Client is connected using HTTP • HTTPS_WITHOUT_CLIENT_CERT: Client is connected using HTTPS without client certificate • HTTPS_WITH_INVALID_CLIENT_CERT: Client is connected using HTTPS with client certificate • HTTPS_WITH_VALID_CLIENT_CERT: Client connected with a valid certificate.

38.4. Examples

38.4.1. Getting peer connection information

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=peerconnectioninfo&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Client.1.IPAddress=192.168.71.93
Client.1.ClientHttpsStatus=NO_HTTPS
Client.2.IPAddress=192.168.71.93
Client.2.ClientHttpsStatus=HTTPS_WITHOUT_CLIENT_CERT
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Clients": [
        {
            "Index": 1,
            "IPAddress": "192.168.71.93",
            "ClientHttpsStatus": "NO_HTTPS"
        },
        {
            "Index": 2,
            "IPAddress": "192.168.71.93",
            "ClientHttpsStatus": "HTTPS_WITHOUT_CLIENT_CERT"
        }
    ]
}
```

Chapter 39. IOBox connection

39.1. Description

The **clientregister** submenu is used for IO Box. It allows clients (cameras) to connect to IO Box and check the connection status. This submenu is related to **ioboxregister** submenu (camera submenu), which is used to configure connection information for IO Box.

> **NOTE**
> This chapter applies to IO Box only.
> Currently, the only supported client is camera, but other VMS or devices may be supported in the future.

Access level

Action	IOBox
check	Admin
control	Admin

39.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
clientregister&action=<value>[&<parameter>=<value>...]
```

39.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
check	ConnectionStatus	RES	<enum> "NotConnected", "ConnectedByMe", "ConnectedByOther"	• NotConnected: IO Box is not connected to client. Please check **ioboxregistrer** submenu's configuration and request **ConnectionRequest=True**. • ConnectedByMe: IO Box is connected to client which requests check action. • ConnectedByOther: IO Box is already connected to another client.

Action	Parameters	Request/ Response	Type/ Value	Description
control	ConnectionRequest	REQ	<bool>	Request to connect to IO Box based on **ioboxregister** submenu's configuration. Only 1 to 1 connection is possible.

39.4. Examples

39.4.1. Getting IOBox connection information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=clientregister&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ConnectionStatus": "NotConnected"
}
```

Chapter 40. Geolocation

40.1. Description

The **geolocation** submenu provides how to set and get the geolocation information of the device.

> **NOTE** This chapter applies to cameras only.

Access level

Action	Camera
view	User
set	Admin

40.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
geolocation&action=<value>[&<parameter>=<value>...]
```

40.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Mode	REQ, RES	<enum> Off, Static, Auto	When GPS sensor is supported, Mode is set to Auto
	Latitude	REQ, RES	<float>	Latitude of device
	Longitude	REQ, RES	<float>	Longitude of device
	Elevation	REQ, RES	<float>	Elevation of device in meters

40.4. Examples

40.4.1. Getting geolocation information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=geolocation&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Mode=Static
Longitude=30.3333
Latitude=12.0
Elevation=40.44
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Mode": "Static",
    "Longitude": 30.3333,
    "Latitude": 12,
    "Elevation": 40.44
}
```

40.4.2. Setting longitude of device

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=geolocation&action=set&Longitude=24.99
```

Chapter 41. SystemImage

41.1. Description

The **systemimage** submenu transmits image files from the device to the client.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

41.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
systemimage&action=<value>[&<parameter>=<value>...]
```

41.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ImageType	RES	<enum> HDDMAP, P2PQRCODE, QRHELP, MOBILEIOS, MOBILEAND	

41.4. Examples

41.4.1. Retrieving the p2pqrcode image

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=systemimage&action=view&ImageType=P2PQRCODE
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: image/png or jpeg
```

```
<Body>
```

```
<PNG image data>
```

Chapter 42. PowerMode

42.1. Description

The **powermode** submenu is used to configure the input power mode of the device.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Guest
set	Admin

42.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
powermode&action=<value>[&<parameter>=<value>...]
```

42.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Mode	REQ, RES	<enum> PoE+, PoE	Input power mode
	ConstraintOption	REQ, RES	<enum> 2SD_NoIR, None, 1SD_50PercentIR, FPS20_IR10m	Power constraint option

42.4. Examples

42.4.1. Getting the current power mode

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=powermode&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Mode=PoE+
ConstraintOption=None
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Mode": "PoE+",
    "ConstraintOption": "None"
}
```

42.4.2. Changing the power mode

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=powermode&action=set&Mode=PoE
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```json
{
    "Response": "Success"
}
```

Chapter 43. PowerMode Options

43.1. Description

The **powermodeoptions** submenu is used to view available power mode constraint options.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin

43.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
powermodeoptions&action=<value>[&<parameter>=<value>...]
```

43.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ConstraintOption	RES	<enum> 2SD_NoIR, None, 1SD_50PercentIR, FPS20_IR10m	Available power constraint options

43.4. Examples

43.4.1. Getting the available power mode constraint options

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=powermodeoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```



```
{
    "PowerModeOptions": [
        {
            "ConstraintOption": [
                "1SD_50PercentIR",
                "2SD_NoIR"
            ],
            "Mode": "PoE"
        },
        {
            "ConstraintOption": [
                "None"
            ],
            "Mode": "PoE+"
        }
    ]
}
```

Chapter 44. Registered Subdevices

44.1. Description

The **registeredsubdevices** submenu can be used to get the list of registered subdevices. A subdevice can be either a speaker or mic.

> **NOTE**
> This chapter applies to only **IPAS**.
> Only when SpeakerType is MASTER/SERVER this submenu is supported. To ensure the device appears in this submenu after registration, it must be associated to a zone or source.

Access level

Action	IPAS
view	Admin

44.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
registeredsubdevices&action=<value>[&<parameter>=<value>...]
```

44.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Device.#.IP	RES	<string>	IP V4 address format or IPV6 address format
	Device.#.ID	RES	<int>	Unique id of the device
	Device.#.Type	RES	<enum> SPEAKER, SPEAKER_WITH_AUDIO_SOURCE, AUDIO_SOURCE	Can be one of the enum values.
	Device.#.AudioSourceType	RES	<csv> TTS, BGM	Source type if device type has audio source support.
	Device.#.Name	RES	<string>	Optional name of the device

44.4. Examples

44.4.1. Getting sub devices information

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=registeredsubdevices&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Device": [
        {
            "IP": "192.168.71.22",
            "ID": 1221,
            "Name": "Speaker1stFloor",
            "Type": "SPEAKER",
            "AudioSourceType": ""
        },
        {
            "IP": "192.168.71.23",
            "ID": 1223,
            "Name": "Speaker2ndFloor",
            "Type": "SPEAKER",
            "AudioSourceType": ""
        },
        {
            "IP": "192.168.71.24",
            "ID": 1224,
            "Name": "Speaker3ndFloor",
            "Type": "SPEAKER_WITH_AUDIO_SOURCE",
            "AudioSourceType": "BGM"
        },
        {
            "IP": "192.168.71.20",
            "ID": 1225,
            "Name": "sourceRoom1",
```

```
            "Type": "AUDIO_SOURCE",
            "AudioSourceType": "TTS"
        },
        {
            "IP": "192.168.71.21",
            "ID": 1226,
            "Name": "sourceRoom2",
            "Type": "AUDIO_SOURCE",
            "AudioSourceType": "BGM"
        }
    ]
}
```

Chapter 45. Speaker groups

45.1. Description

The **speakergroups** submenu can be used to get speaker groups configured on the MASTER or SERVER speaker.

NOTE	This chapter applies to only IPAS.

Access level

Action	IPAS
view	Admin

45.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
speakergroups&action=<value>[&<parameter>=<value>]
```

45.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Group.#.ID	RES	<int>	Unique id of the device
	Group.#.DeviceIDs	RES	<csv>	Array of device ids in the group, only device type speaker can be part of group.
	Group.#.Name	RES	<string>	Optional name of the group

45.4. Examples

45.4.1. Getting group information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=speakergroups&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
  "Group":[
    {
       "ID": 1,
       "Name": "Speakerslistin1stand2nd",
       "DeviceIDs": [ 1221,1223]
    },
    {
       "ID": 2,
       "Name": "Speakerslistin1stand3rd",
        "DeviceIDs": [ 1221,1224]
    }
    ]
}
```

Chapter 46. SSDStorage

46.1. Description

The **ssdstorage** submenu is used to manage the SSD storage installed on the device.

> **NOTE**
> This chapter applies to cameras only
> Attribute to check: "attributes/System/Support/**SSDStorage**"

Access level

Action	Camera
view	Admin
set	Admin
update	Admin
control	Admin

46.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
ssdstorage&action=<value>[&<parameter>=<value>...]
```

46.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Storage	REQ	<int>	Storage Index, if passed only requested storage information is provided in response. **Note** For max supported SSD Storage count check below attributes "attributes/System/Limit/**MaxSSD Storage**"
	Storage.#.Enable	RES	<bool>	Storage enable state.

Action	Parameters	Request/ Response	Type/ Value	Description
	Storage.#.Status	RES	<enum> None,Error, Uninitialized, Initializing, Formatting, Wait,InUse	Storage status
	Storage.#.TotalSpace	RES	<string>	Total Space in MB provided as string.
	Storage.#.Partition.#.Device	RES	<string>	Device path eg: /dev/sda1
	Storage.#.Partition.#.MountPoint	RES	<string>	mount path eg: /mnt/sda1
	Storage.#.Partition.#.FileSystem	RES	<enum> ext4	Total Space in MB provided as string.
	Storage.#.Partition.#.Size	RES	<string>	Parition size in MB provided as string.
	Storage.#.Partition.#.FreeSize	RES	<string>	Free Space in MB provided as string.
	Storage.#.Partition.#.Status	RES	<enum> None, Formatting, InUse	Total Space in MB provided as string.
set	Storage.#.Enable	REQ	<bool>	Enable or Disable a storage.
update	Storage	<int>	REQ	Storage Index
	TotalPartitionCount	<int>	REQ	Total partition count **Note** For max parition count supported check below attributes "attributes/System/Limit/**MaxPartitionPerSSD**"
	Partition.#.SizeInMB	<int>	REQ	Partition size in MB.
control	Mode	REQ	<enum> Format	control operation supported on the SSD .
	Storage.#.Partition	REQ	<int>	Parition Index on which the control operation is applicable.
check	Storage	REQ	<int>	SSD Storage Index.

Action	Parameters	Request/ Response	Type/ Value	Description
	Storage.#.Health.WriteSpeed	<RES>	<int>	Write speed in MB/s.
	Storage.#.Health.ReadSpeed	<RES>	<int>	Read speed in MB/s.
	Storage.#.Health.PowerOnHours	<RES>	<int>	Total power on hours.
	Storage.#.Health.TemperatureInCelsius	<RES>	<int>	Temperature of SSD in celcius.
	Storage.#.Health.ValidSpareBlock	<RES>	<int>	Total number of valid spareblock.
	Storage.#.Health.SpareBlockAlarm	<RES>	<enum> Good,Normal,Poor,Bad	Status of SSD.
	Storage.#.Health.RemainingLifeInPercentage	<RES>	<int>	Remining lifetime in percentage.
	Storage.#.Health.LifetimeAlarm	<RES>	<enum> Good, Normal, Poor, Bad	Health state of SSD.
	Storage.#.SMART.Model	<RES>	<string>	Model name.
	Storage.#.SMART.SerialNumber	<RES>	<string>	Serial number of SSD.
	Storage.#.SMART.FirmwareVersion	<RES>	<string>	Firmware version of SSD.
	Storage.#.SMART.Attributes.#.Id	<RES>	<string>	SMART attribute id.
	Storage.#.SMART.Attributes.#.Name	<RES>	<string>	SMART attribute name.
	Storage.#.SMART.Attributes.#.Value	<RES>	<int>	SMART attribute value.

46.4. Examples

46.4.1. View Storage and Partition Information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=ssdstorage&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Storage.1.Enable=True
Storage.1.Status=InUse
Storage.1.TotalSpace=976762
Storage.1.Partition.1.Device=/dev/sda1
Storage.1.Partition.1.MountPoint=/mnt/sda1
Storage.1.Partition.1.FileSystem=ext4
Storage.1.Partition.1.Size=93356
Storage.1.Partition.1.FreeSize=87764
Storage.1.Partition.1.Status=InUse
Storage.1.Partition.2.Device=/dev/sda2
Storage.1.Partition.2.MountPoint=/mnt/sda2
Storage.1.Partition.2.FileSystem=ext4
Storage.1.Partition.2.Size=93356
Storage.1.Partition.2.FreeSize=88512
Storage.1.Partition.2.Status=InUse
Storage.1.Partition.3.Device=/dev/sda3
Storage.1.Partition.3.MountPoint=/mnt/sda3
Storage.1.Partition.3.FileSystem=ext4
Storage.1.Partition.3.Size=93356
Storage.1.Partition.3.FreeSize=88512
Storage.1.Partition.3.Status=InUse
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Storage": [
        {
            "Index": 1,
            "Enable": true,
            "Status": "InUse",
```

```
            "TotalSpace": "976762",
            "Partition": [
                {
                    "Index": 1,
                    "Device": "/dev/sda1",
                    "MountPoint": "/mnt/sda1",
                    "FileSystem": "ext4",
                    "Size": "93356",
                    "FreeSize": "87764",
                    "Status": "InUse"
                },
                {
                    "Index": 2,
                    "Device": "/dev/sda2",
                    "MountPoint": "/mnt/sda2",
                    "FileSystem": "ext4",
                    "Size": "93356",
                    "FreeSize": "88512",
                    "Status": "InUse"
                },
                {
                    "Index": 3,
                    "Device": "/dev/sda3",
                    "MountPoint": "/mnt/sda3",
                    "FileSystem": "ext4",
                    "Size": "93356",
                    "FreeSize": "88512",
                    "Status": "InUse"
                }
            ]
        }
    ]
}
```


46.4.2. Enable SSD Storage

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=ssdstorage&action=set&Storage.1.Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

46.4.3. Create Partitions in SSD

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=ssdstorage&action=update&Storage=1&TotalPartitionCou
nt=3&Partition.1.SizeInMB=100000&Partition.2.SizeInMB=100000&Partition.3.Siz
eInMB=100000
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

46.4.4. Format parition

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=ssdstorage&action=control&Mode=Format&Storage.1.Part
ition=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

46.4.5. Check the Health status

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=ssdstorage&action=check
```

JSON RESPONSE (Recommended over TEXT response)

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Storage": [
        {
            "Index": 1,
            "Health": {
                "WriteSpeed": 0,
                "ReadSpeed": 0,
                "PowerOnHours": 770,
                "TemperatureInCelsius": 59,
                "ValidSpareBlock": 100,
                "SpareBlockAlarm": "Good",
                "RemainingLifeInPercentage": 100,
                "LifetimeAlarm": "Good"
            },
            "SMART": {
                "Model": "TS2TMTS952T2",
                "SerialNumber": "H512150083",
                "FirmwareVersion": "02J0T6OC\n",
                "Attributes": [
                    {
                        "Index": 1,
                        "Id": "01",
                        "Name": "Read Error Rate",
                        "Value": 0
                    },
                    {
                        "Index": 2,
                        "Id": "05",
                        "Name": "Reallocated Sectors Count",
                        "Value": 0
                    },
                    {
                        "Index": 3,
                        "Id": "09",
                        "Name": "Power-On Hours",
```

```
                    "Value": 770
                },
                {
                    "Index": 4,
                    "Id": "0C",
                    "Name": "Power Cycle Count",
                    "Value": 36
                },
                {
                    "Index": 5,
                    "Id": "94",
                    "Name": "SLC Total Erase Count",
                    "Value": 148
                },
                {
                    "Index": 6,
                    "Id": "95",
                    "Name": "SLC Maximum Erase Count",
                    "Value": 13
                },
                {
                    "Index": 7,
                    "Id": "96",
                    "Name": "SLC Minimum Erase Count",
                    "Value": 0
                },
                {
                    "Index": 8,
                    "Id": "97",
                    "Name": "SLC Average Erase Count",
                    "Value": 1
                },
                {
                    "Index": 9,
                    "Id": "9F",
                    "Name": "DRAM one bit error count",
                    "Value": 0
                },
                {
                    "Index": 10,
                    "Id": "A0",
```

```
                        "Name": "Uncorrectable sectors count when
read/write",
                        "Value": 0
                    },
                    {
                        "Index": 11,
                        "Id": "A1",
                        "Name": "Number of Valid Spare Blocks",
                        "Value": 114
                    },
                    {
                        "Index": 12,
                        "Id": "A3",
                        "Name": "Number of Initial Invalid Blocks",
                        "Value": 34
                    },
                    {
                        "Index": 13,
                        "Id": "A4",
                        "Name": "TLC Total Erase Count",
                        "Value": 100
                    },
                    {
                        "Index": 14,
                        "Id": "A5",
                        "Name": "TLC Maximum Erase Count",
                        "Value": 4
                    },
                    {
                        "Index": 15,
                        "Id": "A6",
                        "Name": "TLC Minimum Erase Count",
                        "Value": 0
                    },
                    {
                        "Index": 16,
                        "Id": "A7",
                        "Name": "TLC Average Erase Count",
                        "Value": 0
                    },
                    {
```

```
                "Index": 17,
                "Id": "A8",
                "Name": "Max Erase Count of Spec",
                "Value": 3000
            },
            {
                "Index": 18,
                "Id": "A9",
                "Name": "Remain Life (percentage)",
                "Value": 100
            },
            {
                "Index": 19,
                "Id": "B1",
                "Name": "Total Wear Level Count",
                "Value": 0
            },
            {
                "Index": 20,
                "Id": "B5",
                "Name": "Total Program Fail Count",
                "Value": 0
            },
            {
                "Index": 21,
                "Id": "B6",
                "Name": "Total Erase Fail Count",
                "Value": 0
            },
            {
                "Index": 22,
                "Id": "C0",
                "Name": "Power-Off Retract Count",
                "Value": 28
            },
            {
                "Index": 23,
                "Id": "C2",
                "Name": "Temperature",
                "Value": 59
            },
```

```
                    {
                        "Index": 24,
                        "Id": "C3",
                        "Name": "Total Correctable Count",
                        "Value": 0
                    },
                    {
                        "Index": 25,
                        "Id": "C4",
                        "Name": "Reallocation Event Count",
                        "Value": 0
                    },
                    {
                        "Index": 26,
                        "Id": "C7",
                        "Name": "Ultra DMA CRC Error Count",
                        "Value": 0
                    },
                    {
                        "Index": 27,
                        "Id": "E8",
                        "Name": "Available Reserved Space",
                        "Value": 100
                    },
                    {
                        "Index": 28,
                        "Id": "F1",
                        "Name": "Total LBA Written (each write unit=32MB)",
                        "Value": 1723
                    },
                    {
                        "Index": 29,
                        "Id": "F2",
                        "Name": "Total LBA Read (each read unit=32MB)",
                        "Value": 1021
                    },
                    {
                        "Index": 30,
                        "Id": "F5",
                        "Name": "Flash Write Sector Count",
                        "Value": 5376
```

```
                    }
                ]
            }
        }
    ]
}
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Storage.1.Health.WriteSpeed=8
Storage.1.Health.ReadSpeed=1276
Storage.1.Health.PowerOnHours=3705
Storage.1.Health.TemperatueInCelsius=59
Storage.1.Health.ValidSpareBlock=100
Storage.1.Health.SpareBlockAlarm=Good
Storage.1.Health.RemainingLifeInPercentage=100
Storage.1.Health.LifetimeAlarm=Good
Storage.1.SMART.Model=TS2TMTS952T2
Storage.1.SMART.SerialNumber=H512150083
Storage.1.SMART.FirmwareVersion=02J0T6OC

Storage.1.SMART.Attributes.1.Id=01
Storage.1.SMART.Attributes.1.Name=Read Error Rate
Storage.1.SMART.Attributes.1.Value=0
Storage.1.SMART.Attributes.2.Id=05
Storage.1.SMART.Attributes.2.Name=Reallocated Sectors Count
Storage.1.SMART.Attributes.2.Value=0
Storage.1.SMART.Attributes.3.Id=09
Storage.1.SMART.Attributes.3.Name=Power-On Hours
Storage.1.SMART.Attributes.3.Value=3705
Storage.1.SMART.Attributes.4.Id=0C
Storage.1.SMART.Attributes.4.Name=Power Cycle Count
Storage.1.SMART.Attributes.4.Value=112
Storage.1.SMART.Attributes.5.Id=94
Storage.1.SMART.Attributes.5.Name=SLC Total Erase Count
Storage.1.SMART.Attributes.5.Value=208
Storage.1.SMART.Attributes.6.Id=95
```

```
Storage.1.SMART.Attributes.6.Name=SLC Maximum Erase Count
Storage.1.SMART.Attributes.6.Value=72
Storage.1.SMART.Attributes.7.Id=96
Storage.1.SMART.Attributes.7.Name=SLC Minimum Erase Count
Storage.1.SMART.Attributes.7.Value=0
Storage.1.SMART.Attributes.8.Id=97
Storage.1.SMART.Attributes.8.Name=SLC Average Erase Count
Storage.1.SMART.Attributes.8.Value=2
Storage.1.SMART.Attributes.9.Id=9F
Storage.1.SMART.Attributes.9.Name=DRAM one bit error count
Storage.1.SMART.Attributes.9.Value=0
Storage.1.SMART.Attributes.10.Id=A0
Storage.1.SMART.Attributes.10.Name=Uncorrectable sectors count when
read/write
Storage.1.SMART.Attributes.10.Value=0
Storage.1.SMART.Attributes.11.Id=A1
Storage.1.SMART.Attributes.11.Name=Number of Valid Spare Blocks
Storage.1.SMART.Attributes.11.Value=114
Storage.1.SMART.Attributes.12.Id=A3
Storage.1.SMART.Attributes.12.Name=Number of Initial Invalid Blocks
Storage.1.SMART.Attributes.12.Value=34
Storage.1.SMART.Attributes.13.Id=A4
Storage.1.SMART.Attributes.13.Name=TLC Total Erase Count
Storage.1.SMART.Attributes.13.Value=339
Storage.1.SMART.Attributes.14.Id=A5
Storage.1.SMART.Attributes.14.Name=TLC Maximum Erase Count
Storage.1.SMART.Attributes.14.Value=10
Storage.1.SMART.Attributes.15.Id=A6
Storage.1.SMART.Attributes.15.Name=TLC Minimum Erase Count
Storage.1.SMART.Attributes.15.Value=0
Storage.1.SMART.Attributes.16.Id=A7
Storage.1.SMART.Attributes.16.Name=TLC Average Erase Count
Storage.1.SMART.Attributes.16.Value=0
Storage.1.SMART.Attributes.17.Id=A8
Storage.1.SMART.Attributes.17.Name=Max Erase Count of Spec
Storage.1.SMART.Attributes.17.Value=3000
Storage.1.SMART.Attributes.18.Id=A9
Storage.1.SMART.Attributes.18.Name=Remain Life (percentage)
Storage.1.SMART.Attributes.18.Value=100
Storage.1.SMART.Attributes.19.Id=B1
Storage.1.SMART.Attributes.19.Name=Total Wear Level Count
```

```
Storage.1.SMART.Attributes.19.Value=0
Storage.1.SMART.Attributes.20.Id=B5
Storage.1.SMART.Attributes.20.Name=Total Program Fail Count
Storage.1.SMART.Attributes.20.Value=0
Storage.1.SMART.Attributes.21.Id=B6
Storage.1.SMART.Attributes.21.Name=Total Erase Fail Count
Storage.1.SMART.Attributes.21.Value=0
Storage.1.SMART.Attributes.22.Id=C0
Storage.1.SMART.Attributes.22.Name=Power-Off Retract Count
Storage.1.SMART.Attributes.22.Value=102
Storage.1.SMART.Attributes.23.Id=C2
Storage.1.SMART.Attributes.23.Name=Temperature
Storage.1.SMART.Attributes.23.Value=59
Storage.1.SMART.Attributes.24.Id=C3
Storage.1.SMART.Attributes.24.Name=Total Correctable Count
Storage.1.SMART.Attributes.24.Value=0
Storage.1.SMART.Attributes.25.Id=C4
Storage.1.SMART.Attributes.25.Name=Reallocation Event Count
Storage.1.SMART.Attributes.25.Value=0
Storage.1.SMART.Attributes.26.Id=C7
Storage.1.SMART.Attributes.26.Name=Ultra DMA CRC Error Count
Storage.1.SMART.Attributes.26.Value=0
Storage.1.SMART.Attributes.27.Id=E8
Storage.1.SMART.Attributes.27.Name=Available Reserved Space
Storage.1.SMART.Attributes.27.Value=100
Storage.1.SMART.Attributes.28.Id=F1
Storage.1.SMART.Attributes.28.Name=Total LBA Written (each write unit=32MB)
Storage.1.SMART.Attributes.28.Value=1723
Storage.1.SMART.Attributes.29.Id=F2
Storage.1.SMART.Attributes.29.Name=Total LBA Read (each read unit=32MB)
Storage.1.SMART.Attributes.29.Value=1025
Storage.1.SMART.Attributes.30.Id=F5
Storage.1.SMART.Attributes.30.Name=Flash Write Sector Count
Storage.1.SMART.Attributes.30.Value=14700
```

Chapter 47. LocalVMS

47.1. Description

The **localvms** submenu used to manage the vms application installed on the device.

NOTE

This chapter applies to cameras only
Attribute to check: "attributes/System/Support/**LocalVMS**"

Access level

Action	Camera
view	Admin
control	Admin

47.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
localvms&action=<value>[&<parameter>=<value>...]
```

47.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	VmsName	RES	<enum> None, Wave	VMS application name, if not installed *None" is returned
	Version	RES	<string>	Version of the VMS application
	Status	RES	<enum> None, NotInstalled, Ready, Running, Installing, Stopped, Error	Status of the VMS application
	Location	RES	<string>	Parition where the VMS application is installed

Action	Parameters	Request/ Response	Type/ Value	Description
control	Mode	REQ	<enum> Install, Uninstall, Start, Stop, LicenseBackup, LicenseRestore	Operation to perform on the VMS **Note** For **Install** operation HTTP POST is used to send the installation file as binary octet stream.
	VmsName	REQ	<enum> Wave	Name of the vms application to be installed
	Location	REQ	<string>	Optional partition location where the application has to be installed. eg: /mnt/sda1

47.4. Examples

47.4.1. View the status of the installed application

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=localvms&action=view
```

TEXT RESPONSE

```
VmsName=Wave
Version=1.2XXX
Status=Ready
Location=/mnt/sda1
```

JSON RESPONSE

```
{
    "VmsName": "Wave",
    "Version": "1.2XXX",
    "Status": "Ready",
    "Location": "/mnt/sda1"
}
```

47.4.2. Install VMS Application

REQUEST [POST]

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&Mode=Install&VmsName=Wave&Lo
cation=/mnt/sda1
```

POST BODY

```
Content-Length: 180324956
Content-Type: application/octet-stream
```

```
Binary Data
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

47.4.2.1. Using Curl to install VMS

REQUEST

```
curl -v --digest -u user:password -H "Content-Type:application/octet-stream"
--data-binary "@wave-server-5.0.0.34745-linux_arm64-beta.deb" -H "Expect:"
"http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&Mode=Install&VmsName=Wave&Lo
```

```
cation=/mnt/sda1"
```

47.4.3. Uninstall VSM Application

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&Mode=Uninstall
```

47.4.4. Start VMS Application

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&&Mode=Start
```

47.4.5. Stop VMS Application

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&&Mode=Stop
```

47.4.6. Take License Backup in Camera Storage

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&Mode=LicenseBackup
```

47.4.7. Restore License from Camera Storage

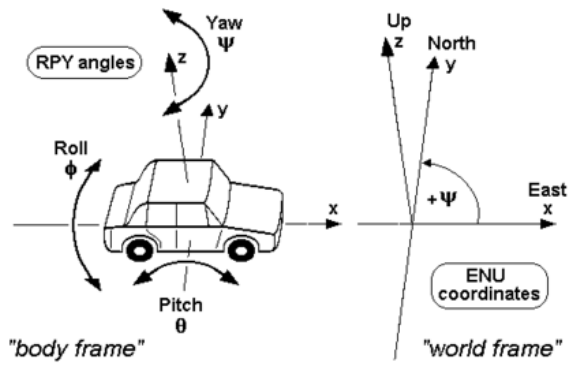
REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=localvms&action=control&Mode=LicenseRestore
```

Chapter 48. GeoOrientation

48.1. Description

The **geoorientation** submenu explains how to set and retrieve the geoorientation information of the device. It follows the ENU (East-North-Up) reference frame.



source: https://commons.wikimedia.org/wiki/File:RPY_angles_of_cars.png

Access level

Action	Camera
view	User
set	Admin

48.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
geoorientation&action=<value>[&<parameter>=<value>...]
```

48.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Mode	REQ, RES	<enum> Off, Static	Mode can be set to off or static mode.
	NorthOffset	REQ, RES	<float> 0.0 to 360.0	Angle between magnetic north and device. 0.0 degree is aligned with North.

Action	Parameters	Request/ Response	Type/ Value	Description
	Yaw	RES	<float> -180.0 to 180.0	When yaw angle is 0.0 its aligned with East. 90 Degree is aligned with North, considering East as the reference in ENU. Basically, Yaw = 90-NorthOffset (if Yaw is less than -180 then yaw = yaw + 360 to keep value in range) **READ ONLY**
	Pitch	RES	<float> -90.0 to 90.0	Angle between the device and the horizon, when 0.0 its aligned with Horizon. **READ ONLY**
	Roll	RES	<float> -180.0 to 180.0	Angle between the device and the ground, when 0.0 its aligned with ground. **READ ONLY**
	PitchRollPreset	REQ,RES	<enum> CEILING , WALL_VERTICAL_MIC_TOP, WALL_VERTICAL_MIC_BOTTOM, WALL_HORIZONTAL_MIC_LEFT, WALL_HORIZONTAL_MIC_RIGHT	Based on mount position preset, Pitch and Roll values will be automatically configured.

Table 1. Preset and Pitch-Roll angle mapping

PitchRollPreset	Pitch	Roll
CEILING	0.0	0.0
WALL_VERTICAL_MIC_TOP	90.0	0.0
WALL_VERTICAL_MIC_BOTTOM	90.0	180.0
WALL_HORIZONTAL_MIC_LEFT	90.0	-90.0
WALL_HORIZONTAL_MIC_RIGHT	90.0	90.0

48.4. Examples

48.4.1. Getting geoorientation information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=geoorientation&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Mode=Off
NorthOffset=90.00000
Yaw=0.000000
Pitch=0.000000
Roll=0.000000
PitchRollPreset=CEILING
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Mode": "Off",
    "NorthOffset": 90.0,
    "Yaw": 0.0,
    "Pitch": 0.0,
    "Roll": 0.0,
    "PitchRollPreset":"CEILING"
}
```

48.4.2. Setting North Offset and Mount Position

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/system.cgi?msubmenu=geoorientation&action=set&Mode=Static&NorthOffset=30
&PitchRollPreset=CEILING
```

Chapter 49. Aux control

49.1. Description:

The **aux** submenu in system cgi provides the exactly same functionalities with the one in ptzcontrol cgi. This dupliceted set has been defined because to provide auxilary commands even for devices don’t have PTZ. The supported auxiliary commands are listed in the attribute xml (refer to "AuxCommands" in "System/Support").

Access level

Action	Camera
control	Suser
check	Suser

49.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=
aux&action=<value>[&<parameter>=<value>]
```

49.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Command	REQ	<string> HeaterOn, FanOn, HeaterOff, WiperOn, VibrationOn, PanDriveHeaterOn, PanDriveHeaterOff	Refer to "AuxCommands" in attributes.cgi for supported commands.
check	Channel	REQ	<int>	Channel ID
	AuxType	REQ	<enum> Heater	Refer to "AuxType" in the attributes.cgi/cgis section for supported values.
	Activate	RES	<bool> True, False	Checks if a specific aux is activated

49.4. Examples

49.4.1. Control auxiliary camera functions

This example controls an auxiliary camera function: the heater.

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=aux&action=control&Command=HeaterOn
```

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=aux&action=control&Command=HeaterOff
```

49.4.2. Check whether an auxiliary function is activated or not

REQUEST

```
http://<Device IP>/stw-
cgi/system.cgi?msubmenu=aux&action=check&Channel=0&AuxType=Heater
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Channel": 0,
    "Activate": true
}
```

Chapter 50. SDCardStatus

50.1. Description

The **sdcardstatus** submenu shows the status of the SD card in each slot.

Access level

Action	Camera
view	Admin
set	Admin

50.2. Syntax

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=sdcardstatus&action=<value>
```

50.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index.#.Status	RES	<enum> "Normal", "Error", "Active", "Formatting", "Lock", "Full", "Wait", "NoProfile", "StreamingLimitExceed", "PWError"	Shows the status

50.4. Examples

50.4.1. Getting the information

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=sdcardstatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
  "SdcardStatus": [
    {
      "Index": 0,
      "Status": "Error"
    },
    {
      "Index": 1,
      "Status": "Active"
    }
  ]
}
```

Network

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 51. Overview

51.1. Description

This document explains network.cgi.

network.cgi manages the network settings of video surveillance products.

The following submenus of network.cgi are used to configure network settings:

- **interface**: Requests and configures the network interface settings.
- **dns**: Requests and configures the DNS server settings.
- **dynamicdns** : Requests and configures the DDNS server settings.
- **bonjour**: Requests and configures Bonjour discovery.
- **upnpdiscovery**: Requests and configures UPnP discovery.
- **zeroconf**: Requests and configures the zero configuration mechanism.
- **snmp** : Requests and configures the SNMP settings.
- **snmptrap**: Requests and configures the settings for SNMP trap messages.
- **qos**: Requests and configures the QoS (Quality of Service) settings
- **http**: Requests and configures the HTTP port number.
- **https**: Requests and configures the HTTPS port number.
- **rtsp**: Requests and configures the RTSP server settings.
- **svnp**: Requests and configures the SVNP (Samsung Video Network Protocol) settings.
- **svp**: Requests and configures the SVP settings.
- **bandwidth**: Requests and configures the bandwidth.
- **multicastsetup**: Requests and configures multicast settings.
- **gpuportmap**: Requests and configures all related to internal/external Port Mapping settings in a device.
- **portconf**: Requests and configures all port-related settings in a device.
- **standbydeviceinfo**: Requests and configures the standby device IP address.
- **mts**: Requests and configures the Mobile Tracking System (MTS) configuration in NVR.
- **Wifi**: Requests and configures the WiFi configuration and WiFi scanning device.
- **dhcpclients**: Requests and configures the DHCP client configuration.
- **poestatus**: Requests and configures the PoE ports’ configuration supported by the device.
- **dhcpserver**: Used to configure DHCP server on the NVR device.
- **onvifdiscovery**: Requests and configures ONVIF Discovery-related settings.
- **sipsetup**: Requests and configures SIP general settings.

- **sipaccount**: Requests and configures SIP account settings.
- **siprecipients**: Requests and configures SIP recipient settings.
- **sipcall**: Requests and controls SIP call state.
- **sipaudiocodec**: Requests and configures SIP audio codec settings.
- **nattraversal**: Requests and configures NAT traversal settings.
- **p2p**: Sets and retrieves the p2p activation state on the NVR device.
- **ethstatus**: Gets the inbound and outbound traffic information in real time.
- **rtspovertls**: Supports RTSP over TLS (RTSPS) settings.
- **mqttclient**: Requests and configures the MQTT client settings.

Chapter 52. Network Interface

52.1. Description

The **interface** submenu configures the network interface settings.

NOTE IPV6 is not supported in **IPAS**.

Access level

Action	Camera	NVR	Encoder	Decoder	IPAS
view	Admin	User	Admin	User	Admin
set	Admin	User	Admin	User	Admin

52.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
interface&action=<value>[&<parameter>=<value>...]
```

52.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the network interface settings
	InterfaceName	REQ	<csv>	Interface name **Note** Not supported in IPAS devices.
	MACAddress	RES	<string>	MAC address (read-only)
	LinkStatus	RES	<enum> Connected, Disconnected	Current link status (read-only)
	InterfaceType	RES	<enum> Ethernet, WiFi	Interface type (read-only)
	BroadcastAddress	RES	<string>	Broadcast address (read-only)

Action	Parameter	Request/ Response	Type/ Value	Description
	IPv6DefaultAddress	RES	<string>	Default IPv6 address when IPv6 is enabled CAMERA ONLY ENCODER ONLY
	InterfaceLabel	RES	<enum> All, Camera, Viewer, iSCSI, Wi-Fi	Interface type in NVR NVR ONLY
set	IPv4Type	REQ, RES	<enum> Manual, DHCP, PPPoE	IP Type • Manual: Static IP. Manually set the IP, gateway, subnet mask and DNS. • DHCP: DHCP mode. Auto-configure the IP, gateway, and subnet mask. • PPPoE: PPPoE mode.
	IPv4Address	REQ, RES	<string>	IPv4 address **IPv4Address** is valid only when **IPv4Type** is set to Manual.
	IPv4PrefixLength	REQ, RES	<int>	Prefix length **IPv4PrefixLength** is valid only when **IPv4Type** is set to Manual. CAMERA ONLY ENCODER ONLY
	IPv4Gateway	REQ, RES	<string>	Gateway IP address **IPv4Gateway** is valid only when **IPv4Type** is set to Manual.
	IPv4DHCPVendorIdentifier	REQ, RES	<string>	VendorClassIdentifier or Option60 in DHCP IPv4, used by device to identify themselves to DHCP server. CAMERA ONLY
	IsDefaultGateway	REQ, RES	<bool> True, False	Default gateway NVR ONLY DECODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	HostName	REQ, RES	<string>	Host name If **IPv4Type** is not set to Manual, **HostName** will be read-only. **Note** This parameter is read-only parameter in **IPAS**.
	IPv4SubnetMask	REQ, RES	<string>	IPv4 subnet mask. This parameter is valid only when **IPv4Type** is set to Manual.
	PPPoEUserName	REQ, RES	<string>	User name **PPPoEUserName** is valid only when **IPv4Type** is set to PPPoE.
	PPPoEPassword	REQ, RES	<string>	**PPPoEPassword** is valid only when **IPv4Type** is set to PPPoE.
	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras and IPAS)
	Enable	REQ, RES	<bool> True, False	Enable or Disable Network Interface NVR ONLY DECODER ONLY
	IPv6Enable	REQ, RES	<bool> True, False	Whether to use IPv6
	IPv6Type	REQ, RES	<enum> Auto, Manual, Default	IPv6 type • Auto: IP address auto configuration. • Manual: Manual configuration. • Default: default IP configuration
	IPv6Address	REQ, RES	<string>	IPv6 address **IPv6Address** is valid only when **IPv6Type** is set to Manual.
	IPv6PrefixLength	REQ, RES	<int>	IPv6 prefix length **IPv6PrefixLength** is valid only when **IPv6Type** is set to Manual.

Action	Parameter	Request/ Response	Type/ Value	Description
	IPv6DefaultGateway	REQ, RES	<string>	IPv6 default gateway. **IPv6DefaultGateway** is valid only when **IPv6Type** is set to Manual.
	IsPPPoEPasswordEncrypted	REQ	<bool> True, False	When this is set as true, password is encrypted using the public key obtained through the **rsa** submenu of security.cgi, and sent as payload content for the POST command.
	MTUSize	REQ, RES	<int>	Maximum Transmission Unit size of the network interface in bytes. The range is from 1280 to 1500.
	ICMPEnable	REQ, RES	<bool> True, False	Sets ICMP enable or not. CAMERA ONLY
	SpeedDuplex	REQ, RES	<enum> AutoNegotiate, 10M_HalfDuplex, 10M_FullDuplex, 100M_HalfDuplex, 100M_FullDuplex, 1000M_HalfDuplex, 1000M_FullDuplex	SpeedDuplex is an option for setting the speed and duplex communication mode of a network adapter.

52.4. Examples

52.4.1. Getting the system network interface settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=interface&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```


```
InterfaceName=NetworkInterface1
MACAddress=00:09:18:69:B8:81
HostName=XNV-C7083R-00091869B881
LinkStatus=Connected
InterfaceType=Ethernet
BroadcastAddress=192.168.75.255
IPv4Type=Manual
IPv4Address=192.168.75.91
IPv4PrefixLength=24
IPv4SubnetMask=255.255.255.0
IPv4Gateway=192.168.75.1
IPv4DHCPVendorIdentifier=cam1
IPv6Enable=True
IPv6Type=Default
IPv6Address=fe80::209:18ff:fe69:b881
IPv6PrefixLength=64
IPv6DefaultGateway=
IPv6DefaultAddress=fe80::209:18ff:fe69:b881
MTUSize=1500
ICMPEnable=True
```


JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NetworkInterfaces": [
        {
            "InterfaceName": "NetworkInterface1",
            "MACAddress": "00:09:18:69:B8:81",
            "HostName": "XNV-C7083R-00091869B881",
            "LinkStatus": "Connected",
            "InterfaceType": "Ethernet",
            "BroadcastAddress": "192.168.75.255",
            "IPv4Type": "Manual",
```

```
                "IPv4Address": "192.168.75.91",
                "IPv4PrefixLength": 24,
                "IPv4SubnetMask": "255.255.255.0",
                "IPv4Gateway": "192.168.75.1",
                "IPv4DHCPVendorIdentifier": "cam1",
                "IPv6Enable": true,
                "IPv6Type": "Default",
                "IPv6Address": "fe80::209:18ff:fe69:b881",
                "IPv6PrefixLength": 64,
                "IPv6DefaultGateway": "",
                "IPv6DefaultAddress": "fe80::209:18ff:fe69:b881",
                "MTUSize": 1500,
                "ICMPEnable": true
            }
        ]
}
```


The following sample response is for NVR.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
HostName=PRN-4011
InterfaceName=Network1
MACAddress=00:09:15:05:06:26
LinkStatus=Connected
InterfaceType=Ethernet
BroadcastAddress=192.168.255.255
IPv4Type=Manual
IPv4Address=192.168.90.147
IPv4Gateway=192.168.90.1
IPv4SubnetMask=255.255.255.0
IsDefaultGateway=False
PPPoEUserName=
PPPoEPassword=
IPv6Enable=True
IPv6Type=Default
IPv6Address=fe80::209:15ff:fe05:626
```

```
IPv6PrefixLength=64
IPv6DefaultGateway=ff02::c
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "HostName": "PRN-4011",
    "NetworkInterfaces": [
        {
            "InterfaceName": "Network1",
            "MACAddress": "00:09:15:05:06:26",
            "LinkStatus": "Connected",
            "InterfaceType": "Ethernet",
            "BroadcastAddress": "192.168.255.255",
            "IPv4Type": "Manual",
            "IPv4Address": "192.168.90.147",
            "IPv4Gateway": "192.168.90.1",
            "IPv4SubnetMask": "255.255.255.0",
            "IsDefaultGateway": false,
            "PPPoEUserName": "",
            "PPPoEPassword": "",
            "IPv6Enable": true,
            "IPv6Type": "Default",
            "IPv6Address": "fe80::209:15ff:fe05:626",
            "IPv6PrefixLength": 64,
            "IPv6DefaultGateway": "ff02::c"
        }
    ]
}
```


52.4.2. Setting a static IP

Assigning a static IP manually.

To set **IPv4Type** to Manual, **IPv4Address**, **IPv4PrefixLength**, and **IPv4Gateway**must be set together.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=interface&action=set&IPv4Type=Manual&IPv4Address=19
2.168.0.101&IPv4PrefixLength=24&IPv4Gateway=192.168.0.1
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=interface&action=set&InterfaceName=Network1&IPv4Typ
e=Manual&IPv4Address=192.168.0.101&IsDefaultGateway=True
```

52.4.3. Setting DHCP mode

If **IPv4Type** is set to DHCP, an available IP address on the same network is assigned to the Hanwha Vision video surveillance product.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=interface&action=set&IPv4Type=DHCP
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=interface&action=set&InterfaceName=Network1&IPv4Typ
e=DHCP
```

52.4.4. Setting the PPPoEPassword with encrypted password

The PPPoEPassword can be configured as shown in the command below. PPPoEPassword should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi to obtain an RSA key). Base 64 encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=interface&action=set&IPv4Type=PPPoE&PPPoEUserName=U
ser1&IsPPPoEPasswordEncrypted=True
```

Chapter 53. DNS

53.1. Description

The **dns** submenu configures the DNS (Domain Name System) server settings.

Access level

Action	Camera	NVR	Encoder	Decoder	IPAS
view	Admin	User	Admin	User	Admin
set	Admin	User	Admin	User	Admin

53.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
dns&action=<value>[&<parameter>=<value>...]
```

53.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the DNS settings.
	InterfaceName	REQ	<string>	For filtering the response based on interface name **Note** Not supported in IPAS devices.
set	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras and IPAS)
	Type	REQ, RES	<enum> Manual, DHCP	Type
	IPType	REQ, RES	<enum> IPv4, IPv6	IP Type NVR ONLY DECODER ONLY
	PrimaryDNS	REQ, RES	<string>	Primary DNS server IP address The address can be either IPv4 or IPv6.

Action	Parameter	Request/ Response	Type/ Value	Description
	SecondaryDNS	REQ, RES	<string>	Secondary DNS server IP address The address can be either IPv4 or IPv6. CAMERA ONLY ENCODER ONLY IPAS ONLY

53.4. Examples

53.4.1. Getting the current DNS settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dns&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=Network1
IPType=IPv4
Type=Manual
PrimaryDNS=168.126.63.1
IPType=IPv6
Type=DHCP
PrimaryDNS=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "DNS": [
```

```
        {
            "InterfaceName": "Network1",
            "DNSByIPType": [
                {
                    "IPType": "IPv4",
                    "Type": "Manual",
                    "PrimaryDNS": "168.126.63.1"
                },
                {
                    "IPType": "IPv6",
                    "Type": "DHCP",
                    "PrimaryDNS": ""
                }
            ]
        }
    ]
}
```


53.4.2. Setting primary and secondary DNS server addresses

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dns&action=set&PrimaryDNS=10.1.20.12&SecondaryDNS=192.1.168.8
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dns&action=set&InterfaceName=Network1&PrimaryDNS=1.1.1.2&Type=Manual
```

Chapter 54. DDNS

54.1. Description

The **dynamicdns** submenu configures the DDNS (Dynamic DNS) server settings.

Access level

Action	Camera	NVR	Encoder
view	Admin	User	Admin
set	Admin	User	Admin

54.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
dynamicdns&action=<value>[&<parameter>=<value>...]
```

54.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the DDNS settings.
	InterfaceName	REQ	<string>	For filtering the response based on interface name.
	SamsungServerName	RES	<string>	Samsung DDNS server URL (read-only) **SamsungServerName** is valid only when **Type** is set to SamsungDDNS.
	Status	RES	<string>	DDNS status (read-only)

Action	Parameter	Request/ Response	Type/ Value	Description
	UpnpStatus	RES	<string> "Success", "Fail[Need Initialization]", "Fail[Invalid Configuration]", "Fail[Router UPnP Disabled]", "Fail[Router Not Found]", "Fail[Need To Restart Router]", "Fail[UPnP Port clash]", "Fail[Quick Connect Not Supported]" "Fail[Unknown Error]"	UpnP Status (read-only) **Note** Success : Quick Connect Success Fail[Need Initialization]: Connection Failed. Fail[Invalid Configuration]: Invalid Network Configuration Fail[Router UPnP Disabled] : UpnP disable state NVR ONLY
	UpnpPortClashInfo	RES	<csv>	ClashPort List NVR ONLY
set	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras)
	Type	REQ, RES	<enum> Off, SamsungDDNS, PublicDDNS	Dynamic DNS type **Note** The **Type** parameter must be sent together with the **set** action.
	SamsungProductID	REQ, RES	<string>	Product ID for Samsung DDNS **SamsungProductID** is valid only when **Type** is set to SamsungDDNS.

Action	Parameter	Request/ Response	Type/ Value	Description
	SamsungQuickConnect	REQ, RES	<bool> True, False	Enables or disables Samsung DDNS Quick Connect **SamsungQuickConnect** is valid only when **Type** is set to SamsungDDNS.
	PublicServiceEntry	REQ, RES	<enum> www.dyndns.org, www.no-ip.org	Public DDNS Service **PublicServiceEntry** is valid only when **Type** is set to PublicDDNS.
	PublicHostName	REQ, RES	<string>	Public DDNS Hostname **PublicHostName** is valid only when **Type** is set to PublicDDNS.
	PublicUserName	REQ, RES	<string>	Public DDNS user ID **PublicUserName** is valid only when **Type** is set to PublicDDNS.
	PublicPassword	REQ, RES	<string>	Public DDNS user password **PublicPassword** is valid only when **Type** is set to PublicDDNS.
	IsPublicPasswordEncrypted	REQ,	<bool> True, False	When this is set to true, password is encrypted using the public key obtained using the **rsa** submenu of security.cgi, and sent as payload content for the POST command.

54.4. Examples

54.4.1. Getting the current DDNS configuration settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dynamicdns&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=1a5a97d2-464a-4222-91c6-140ff36b82b6
Status=Trying
Type=SamsungDDNS
SamsungServerName=ddns.hanwha-security.com
SamsungProductID=snb6004gt
SamsungQuickConnect=False
PublicServiceEntry=www.dyndns.org
PublicHostName=host
PublicUserName=user
PublicPassword=pw
UpnpStatus=Fail[Router Not Found]
UpnpPortClashInfo=1024,1025,1026,1027,1028,1029,1030,1031,1032,1033
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "DynamicDNS": [
        {
            "InterfaceName": "1a5a97d2-464a-4222-91c6-140ff36b82b6",
            "Status": "Trying",
            "Type": "SamsungDDNS",
            "SamsungServerName": "ddns.hanwha-security.com",
            "SamsungProductID": "snb6004gt",
            "SamsungQuickConnect": false,
            "PublicServiceEntry": "www.dyndns.org",
            "PublicHostName": "host",
            "PublicUserName": "user",
            "PublicPassword": "pw",
            "UpnpStatus":   "Fail[Router Not Found]",
            "UpnpPortClashInfo":    [1024, 1025, 1026, 1027, 1028, 1029,
1030, 1031, 1032, 1033]

        }
    ]
}
```

54.4.2. Setting the DDNS type

The accompanying parameters differ depending on the **Type**.

Using the Samsung DDNS service

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dynamicdns&action=set&Type=SamsungDDNS&SamsungProductID=PRODUCTID&SamsungQuickConnect=True
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dynamicdns&action=set&InterfaceName=Network1&Type=SamsungDDNS&SamsungProductId=1111&SamsungQuickConnect=True
```

Using the public DDNS service

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dynamicdns&action=set&Type=PublicDDNS&PublicServiceEntry=www.no-ip.org&PublicHostName=host_name&PublicUserName=user_name&PublicPassword=password
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi&msubmenu=dynamicdns&action=set&InterfaceName=Network1&Type=PublicDDNS&PublicServiceEntry=www.changeip.com&PublicHostname=hostname&PublicUsername=user&PublicPassword=password
```

54.4.3. Setting the PublicPassword with encrypted password

The PublicPassword can be configured as shown in the command below. PublicPassword should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi to obtain an RSA key). Base 64 Encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=dynamicdns&action=set&Type=PublicDDNS&PublicService
Entry=www.no-
ip.org&PublicHostName=host_name&PublicUserName=user_name&IsPublicPasswordEnc
rypted=True
```

Chapter 55. Bonjour

55.1. Description

The **bonjour** submenu enables or disables the Bonjour discovery mechanism and sets the name for it.

The network camera that is connected to the local area network can be automatically searched on the PC with Bonjour service.

NOTE This chapter applies to network cameras and encoders only.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

55.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
bonjour&action=<value>[&<parameter>=<value>]
```

55.3. Parameters

Action	Parameter	Request Response	Type/ Value	Description
view				Reads the Bonjour settings.
set	Enable	REQ, RES	<bool> True, False	Whether to use Bonjour. **Note** **Enable** must be sent together with the **set** action.
	FriendlyName	REQ, RES	<string>	Device name or short description

55.4. Examples

55.4.1. Getting Bonjour settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=bonjour&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
FriendlyName=WISENET-QNO-6082R-E4302266BDEE
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Enable": true,
    "FriendlyName": "WISENET-QNO-6082R-E4302266BDEE"
}
```


55.4.2. Enabling Bonjour and setting the friendly name as 'QNO6082R'

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=bonjour&action=set&Enable=True&FriendlyName=QNO6082
R
```

Chapter 56. UPnP Discovery

56.1. Description

The **upnpdiscovery** submenu enables or disables UPnP discovery and sets the name for it.

> **NOTE** This chapter applies to network cameras and encoders only.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

56.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
upnpdiscovery&action=<value>[&<parameter>=<value>]
```

56.3. Parameters

Action	Parameter	Request/Response	Type/Value	Description
view				Reads the UPnP discovery settings.
set	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras)
	Enable	REQ, RES	<bool> True, False	Enables or disables UPnP discovery. If this function is enabled, cameras can be automatically searched in the client and operating system that supports the UPnP protocol. **Note** **Enable** must be sent together with the **set** action.
	FriendlyName	REQ, RES	<string>	Device name or short description

56.4. Examples

56.4.1. Getting UPnP discovery settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=upnpdiscovery&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=1a5a97d2-464a-4222-91c6-140ff36b82b6
Enable=True
FriendlyName=WISENET-QNO-6082R-E4302266BDEE
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "UpnpDiscovery": [
        {
            "InterfaceName": "1a5a97d2-464a-4222-91c6-140ff36b82b6",
            "Enable": true,
            "FriendlyName": "WISENET-QNO-6082R-E4302266BDEE"
        }
    ]
}
```

56.4.2. Enabling UPnP discovery and setting the friendly name as ‘QNO6082R’

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=upnpdiscovery&action=set&Enable=True&FriendlyName=Q
NO6082R
```

Chapter 57. Zeroconf

57.1. Description

The **zeroconf** submenu enables or disables Zeroconf (Zero configuration networking), which supports automatic configuration of link-local IP addresses.

NOTE This chapter applies to network cameras and encoders only.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

57.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
zeroconf&action=<value>[&<parameter>=<value>]
```

57.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the zero configuration settings.
	IPAddress	RES	<string>	IP address assigned by the Zeroconf protocol (read-only) IPv4 format is allowed.
	SubnetMask	RES	<string>	Subnet mask assigned by the Zeroconf protocol (read-only) IPv4 format is allowed.
set	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras)
	Enable	REQ, RES	<bool> True, False	Enables or disables Zeroconf. **Note** **Enable** must be sent together with the **set** action.

57.4. Examples

57.4.1. Getting current Zeroconf settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=zeroconf&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=1a5a97d2-464a-4222-91c6-140ff36b82b6
IPAddress=169.254.4.236
SubnetMask=16
Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ZeroConf": [
        {
            "InterfaceName": "1a5a97d2-464a-4222-91c6-140ff36b82b6",
            "IPAddress": "169.254.4.236",
            "SubnetMask": "16",
            "Enable": true
        }
    ]
}
```


57.4.2. Enabling Zeroconf

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/network.cgi?msubmenu=zeroconf&action=set&Enable=True
```

Chapter 58. SNMP Setup

58.1. Description

The **snmp** submenu configures the SNMP (Simple Network Management Protocol) settings.

NOTE SNMP Version 1 and Version 2 not supported in IPAS

Access level

Action	Camera	NVR	IPAS
view	Admin	User	Admin
set	Admin	User	Admin

58.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
snmp&action=<value>[&<parameter>=<value>...]
```

58.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SNMP settings.
set	Version1	REQ, RES	<bool> True, False	Enables or disables SNMP version 1.
	Version2	REQ, RES	<bool> True, False	Enables or disables SNMP version 2.
	Version3	REQ, RES	<bool> True, False	Enables or disables SNMP version 3.
	ShareCommunityStringWithV1	REQ, RES	<bool> True, False	When set to True, SNMPv1 and SNMPv2c share the same community strings. When set to False, SNMPv1 uses its default community strings: "public" for read-only access and "private" for write access.
	ReadCommunity	REQ, RES	<string>	SNMP read community name **ReadCommunity** is valid only when **Version2** is set to True.

Action	Parameter	Request/ Response	Type/ Value	Description
	WriteCommunity	REQ, RES	<string>	SNMP write community name **WriteCommunity** is valid only when **Version2** is set to True.
	UserPassword	REQ, RES	<string>	User password **UserPassword** is valid only when **Version3** is set to True.
	IsPasswordEncrypted	REQ	<bool> True, False	When this is set to true, password is encrypted using the public key obtained through the **rsa** submenu of security.cgi, and sent as payload content for the POST command.

58.4. Examples

58.4.1. Getting current SNMP settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=snmp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Version1=False
Version2=True
Version3=False
ShareCommunityStringWithV1=False
ReadCommunity=public
WriteCommunity=write
UserPassword={password}
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Version1": false,
    "Version2": true,
    "Version3": false,
    "ShareCommunityStringWithV1": false,
    "ReadCommunity": "public",
    "WriteCommunity": "write",
    "UserPassword": "{password}"
}
```

58.4.2. Setting SNMP

ReadCommunity and **WriteCommunity** values can be set only when **Version2** is set to True, as in the example below.

REQUEST

```
http://<DeviceIP>/stw-
cgi/network.cgi?msubmenu=snmp&action=set&Version2=True&ReadCommunity=public&
WriteCommunity=write
```

58.4.3. Setting the Password with encrypted password

The Password can be configured as shown in the command below. Password should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi to obtain an RSA key). Base 64 Encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=snmp&action=set&Version1=False&Version2=False&Versi
on3=True& IsPasswordEncrypted=True
```

Chapter 59. SNMP Trap Settings

59.1. Description

The **snmptrap** submenu configures the trap settings of SNMP (Simple Network Management Protocol).

Access level

Action	Camera	NVR	Encoder	Decoder
view	Admin	User	Admin	User
set	Admin	User	Admin	User

59.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
snmptrap&action=<value>[&<parameter>=<value>...]
```

59.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SNMP trap settings.
set	Enable	REQ, RES	<bool> True, False	Enables or disables trap message transmission.
	Trap.#.Address	REQ, RES	<string>	IP address IPv4 format is allowed. **Trap.#.Address** is valid only when **Enable** is set to True.
	Trap.#.Community	REQ, RES	<string>	SNMP community **Trap.#.Community** is valid only when **Enable** is set to True. CAMERA ONLY ENCODER ONLY
	Trap.#.AuthenticationFailure	REQ, RES	<bool> True, False	Enables or disables trap notification in the event of an authentication failure (read-only for NVR)
	Trap.#.ColdStart	REQ, RES	<bool> True, False	Enables or disables ColdStart.

Action	Parameter	Request/ Response	Type/ Value	Description
	Trap.#.LinkUp	REQ, RES	<bool> True, False	Enables or disables linkUp trap notification. (read-only for NVR)
	Trap.#.LinkDown	REQ, RES	<bool> True, False	Enables or disables linkDown trap notification. (read-only for NVR) NVR ONLY DECODER ONLY
	Trap.#.WarmStart	REQ, RES	<bool> True, False	Enables or disables WarmStart. (read-only for NVR) NVR ONLY DECODER ONLY
	Trap.#.AlarmInput.#	REQ, RES	<bool> True, False	Enables or disables alarm inputs CAMERA ONLY ENCODER ONLY
	Trap.#.AlarmOutput.#	REQ, RES	<bool> True, False	Enables or disables alarm outputs CAMERA ONLY ENCODER ONLY
	Trap.#.TamperingDetection	REQ, RES	<bool> True, False	Enables or disables tampering detection CAMERA ONLY ENCODER ONLY
	Trap.#.UseCommunity	REQ, RES	<bool> True, False	Enables or disables Community. To disable Community feature, user can set this parameter as False. Then Community field would be erased. If user set Community filed, this parameter would be set as True automatically. CAMERA ONLY

NOTE # represents the index number of the trap. Available trap numbers may vary depending on the device. Please check the device attributes using attributes.cgi.

59.4. Examples

59.4.1. Getting current SNMP Trap settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=snmptrap&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
Trap.0.Address=0.0.0.0
Trap.0.AuthenticationFailure=False
Trap.0.ColdStart=False
Trap.0.WarmStart=False
Trap.0.LinkUp=False
Trap.0.LinkDown=False
Trap.0.UseCommunity=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false,
    "SNMPTrap": [
        {
            "Trap": 0,
            "Address": "0.0.0.0",
            "AuthenticationFailure": false,
            "ColdStart": false,
            "WarmStart": false,
            "LinkUp": false,
            "LinkDown": false,
            "UseCommunity": false
        }
    ]
```

```
}
```

59.4.2. Setting an SNMP trap

REQUEST

```
http://<DeviceIP>/stw-
cgi/network.cgi?msubmenu=snmptrap&action=set&Enable=True&Trap.0.Address=127.
0.0.1&Trap.0.Community=community&Trap.0.AuthenticationFailure=False&Trap.0.L
inkUp=False
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=snmptrap&action=set&Enable=True&Trap.0.Address=127.
0.0.1
```

59.4.3. Setting community not to use

REQUEST

```
http://<DeviceIP>/stw-
cgi/network.cgi?msubmenu=snmptrap&action=set&Trap.0.UseCommunity=False
```

The following request example is for NVR only.

Chapter 60. QoS Setup

60.1. Description

The **qos** submenu configures the QoS (Quality of Service) settings. It specifies the priority to secure a stable transfer rate for a specific IP.

> **NOTE**
> This chapter applies to network cameras and encoder only.
> Attribute to check for maximum IPV4QoS Addresses:
> "attributes/Network/Limit/**MaxIPv4QoS**"
> Attribute to check for maximum IPV6QoS Addresses:
> "attributes/Network/Limit/**MaxIPv6QoS**"

Access level

Action	Camera	Encoder
view	Admin	Admin
add, update	Admin	Admin
remove	Admin	Admin

60.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
qos&action=<value>[&<parameter>=<value>...]
```

60.3. Parameters

Acton	Parameter	Request/ Response	Type/ Value	Description
view				Reads the QoS settings.
add, update	Enable	REQ, RES	<bool> True, False	Enables or disables QoS for registered IPv4/IPv6 addresses. **Note** **IPType**, **IP Address** and **Enable** must be sent together with the **add** action.
	Index	REQ, RES	<int>	Index of the registered QoS setting **Note** **IPType** and **Index** must be sent together with the **update** action.

Acton	Parameter	Request/ Response	Type/ Value	Description
	IPType	REQ, RES	<enum> IPv4, IPv6	QoS IP type
	IPAddress	REQ, RES	<string>	QoS IP address (IPv4 or IPv6 address)
	PrefixLength	REQ, RES	<int>	Prefix length for IPv4 and IPv6
	DSCP	REQ, RES	<int>	Priority of QoS DSCP (Differentiated Services Code Point)
remove	Index	REQ	<int>	Index that is to be deleted **Note** **IPType** and **Index** must be sent together with the **remove** action.
	IPType	REQ	<enum> IPv4, IPv6	IP type that is to be deleted **Note** **IPType** and **Index** must be sent together with the **remove** action.

60.4. Examples

60.4.1. Getting current QoS settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=qos&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
qos.Index.IPType=Enable/IPAddress/PrefixLength/DSCP
qos.1.IPv4=True/192.168.75.137/32/63
qos.2.IPv4=True/192.168.75.135/32/63
qos.1.IPv6=True/2001:1:1:1:1:1:1:1/128/63
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "QOS": [
        {
            "IPType": "IPv4",
            "IPList": [
                {
                    "Index": 1,
                    "IPAddress": "192.168.75.137",
                    "PrefixLength": 32,
                    "Enable": true,
                    "DSCP": 63
                },
                {
                    "Index": 2,
                    "IPAddress": "192.168.75.135",
                    "PrefixLength": 32,
                    "Enable": true,
                    "DSCP": 63
                }
            ]
        },
        {
            "IPType": "IPv6",
            "IPList": [
                {
                    "Index": 1,
                    "IPAddress": "2001:1:1:1:1:1:1:1",
                    "PrefixLength": 128,
                    "Enable": true,
                    "DSCP": 63
                }
            ]
        }
    ]
}
```

60.4.2. Adding a QoS-enabled IPv4 Address

To add a QoS IP with the **add** action, the **IPType**, **IP Address**, and **Enable** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=qos&action=add&IPType=IPv4&IPAddress=10.10.10.10&DS
CP=1&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

60.4.3. Updating a QoS-enabled IPv4 Address

To edit the QoS IP/DSCP settings with the **update** action, the **IPType** and **Index** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=qos&action=update&Index=1&Enable=False&IPType=IPv4&
IPAddress=10.10.10.10&PrefixLength=5
```

60.4.4. Deleting a QoS IPv6 Address

To remove the QoS IP/DSCP settings with the **remove** action, the **IPType** and **Index** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=qos&action=remove&Index=2&IPType=IPv6
```

Chapter 61. HTTP Port

61.1. Description

The **http** submenu configures the HTTP port number.

Access level

Action	Camera	NVR	Encoder	Decoder
view	Admin	User	Admin	User
set	Admin	User	Admin	User

61.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
http&action=<value>[&<parameter>=<value>...]
```

61.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the HTTP port settings
set	Port	REQ, RES	<int>	HTTP port number **Note** **Port** must be sent together with the **set** action.

61.4. Getting the HTTP port number

61.4.1. Getting the current HTTP port number

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=http&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Port=80
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
"Port": 80
}
```


61.5. Setting the HTTP port

61.5.1. Setting the HTTP port to ‘8080’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=http&action=set&Port=8080
```

Chapter 62. HTTPS Port

62.1. Description

The **https** submenu configures the HTTPS (Secure Hypertext Transfer Protocol) port number.

Access level

Action	Camera	NVR	Encoder	Decoder
view	Admin	User	Admin	User
set	Admin	(Not supported)	Admin	(Not supported)

62.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
https&action=<value>[&<parameter>=<value>...]
```

62.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the HTTPS port settings.
set	Port	REQ, RES	<int>	HTTPS port number (read-only for NVR) **Note** **Port** must be sent together with the **set** action.

62.4. Examples

62.4.1. Getting the current HTTPS port number

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=https&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Port=443
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
"Port": 443
}
```


62.4.2. Setting the HTTPS port to ‘8080’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=https&action=set&Port=8080
```

Chapter 63. RTSP

63.1. Description

The **rtsp** submenu configures the RTSP (Real Time Streaming Protocol) server settings.

Access level

Action	Camera	NVR	Encoder
view	Admin	User	Admin
set	Admin	User	Admin

63.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
rtsp&action=<value>[&<parameter>=<value>...]
```

63.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the RTSP settings.
	MobilePort	RES	<int>	Mobile port NVR ONLY
set	Port	REQ, RES	<int>	RTSP port number
	Timeout	REQ, RES	<enum> 0s, 60s	RTSP timeout CAMERA ONLY ENCODER ONLY
	ProfileSessionPolicy	REQ, RES	<enum> Continue, Disconnect	Keep connection when profile setting is changed CAMERA ONLY ENCODER ONLY

63.4. Examples

63.4.1. Getting current RTSP settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=rtsp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Port=554
Timeout=60s
ProfileSessionPolicy=Continue
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
"Port": 554,
"Timeout": "60s",
"ProfileSessionPolicy": "Continue"
}
```


63.4.2. Setting the RTSP port to ‘1024’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=rtsp&action=set&Port=1024
```

Chapter 64. SVNP

64.1. Description

The **svnp** submenu configures the SVNP (Samsung Video Network Protocol) settings. SVNP is Samsung's own video network protocol for integrating Hanwha network cameras with 3rd party applications.

> **NOTE** This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

64.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
svnp&action=<value>[&<parameter>=<value>...]
```

64.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SVNP settings.
set	Port	REQ, RES	<int>	SVNP port number

64.4. Examples

64.4.1. Getting current SVNP settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=svnp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Port=4520
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Port": 4520
}
```

64.4.2. Setting the SVNP port to ‘6002’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=svnp&action=set&Port=6002
```

Chapter 65. SVP

65.1. Description

The **svp** submenu configures the SVP (Smart Viewer Protocol) port number.

NOTE	This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

65.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
svp&action=<value>[&<parameter>=<value>...]
```

65.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SVP (Smart Viewer Protocol) settings
set	Port	REQ, RES	<int>	SVP port number

65.4. Examples

65.4.1. Getting current SVP settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=svp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Port=554
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Port": 554
}
```

65.4.2. Setting the SVP port to ‘555’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=svp&action=set&Port=555
```

Chapter 66. Bandwidth

66.1. Description

The **bandwidth** submenu configures the bandwidth.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

66.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
bandwidth&action=<value>[&<parameter>=<value>...]
```

66.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the bandwidth settings
	InterfaceName	REQ	<csv>	Interface name
set	InterfaceName	REQ, RES	<string>	Interface name
	Bandwidth	REQ, RES	<int>	Bandwidth

66.4. Examples

66.4.1. Getting the current bandwidth settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=bandwidth&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=Network1
Bandwidth=400
InterfaceName=Network2
Bandwidth=0
InterfaceName=Network3
Bandwidth=0
InterfaceName=Network4
Bandwidth=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NetworkBandwidths": [
        {
            "InterfaceName": "Network1",
            "Bandwidth": 400
        },
        {
            "InterfaceName": "Network2",
            "Bandwidth": 0
        },
        {
            "InterfaceName": "Network3",
            "Bandwidth": 0
        },
        {
            "InterfaceName": "Network4",
            "Bandwidth": 0
        }
    ]
}
```


66.4.2. Getting the bandwidth settings of the interface name ‘Network 1’ and ‘Network 2’

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=bandwidth&action=view&InterfaceName=Network1,Networ
k2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=Network1
Bandwidth=400
InterfaceName=Network2
Bandwidth=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "NetworkBandwidths": [
        {
            "InterfaceName": "Network1",
            "Bandwidth": 400
        },
        {
            "InterfaceName": "Network2",
            "Bandwidth": 0
        }
    ]
}
```

66.4.3. Setting the bandwidth for the interface name ‘Network3’

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=bandwidth&action=set&InterfaceName=Network3&Bandwidth=200
```

Chapter 67. Multicast Setup

67.1. Description

The **multicastsetup** submenu can be used to configure the multicast configuration of a streaming server. Using this interface client, the multicast IP, port and TTL values can be set.

> **NOTE** This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	Admin

67.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
multicastsetup&action=<value>[&<parameter>=<value>...]
```

67.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the multicast settings
set	TTL	REQ, RES	<int>	TTL setting in seconds
	IPAddress	REQ, RES	<string>	Multicast IP address used for streaming
	PortRange	REQ, RES	<enum> 8000~8159, 8160~8319, 8320~8479, 8480~8639, 8640~8799, 8800~8959, 8960~9119, 9120~9279, 9280~9439, 9440~9599, 9600~9759, 9760~9919	Port range

Action	Parameter	Request/ Response	Type/ Value	Description
	PortStart	REQ, RES	<int>	Starting this port number any port would be used for multicast streaming

67.4. Examples

67.4.1. Getting the current multicast settings

The following command gets the current multicast settings.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=multicastsetup&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
PortRange=8000~8159
TTL=5
IPAddress=224.126.63.1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PortRange": "8000~8159",
    "TTL": 5,
    "IPAddress": "224.126.63.1"
}
```

67.4.2. Setting the different ipaddress

The following command sets a new multicast address.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=multicastsetup&action=set&IPAddress=224.126.64.122
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 68. GPU Port Mapping Configuration

68.1. Description

The **gpuportmap** submenu can be used to view and set the port map settings of the device.

Access level

Action	Camera
view	User
set	Admin
add	Admin
update	Admin
remove	Admin

68.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
gpuportmap&action=<value>[&<parameter>=<value>...]
```

68.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the current settings
set	Bridge.Begin	REQ, RES	<int>	Begin value of Port Bridge
	Bridge.End	REQ, RES	<int>	End value of Port Bridge
add/update	PortMap.Index	REQ, RES	<int>	If Index is not specified when adding, it is allocated from Device.
	PortMap.External	REQ, RES	<int>	External Port of Port Map
	PortMap.Internal	REQ, RES	<int>	Port to be Port Mapped internally
remove	PortMap.Index	REQ	<int>	Delete PortMap settings for the specified Index

68.4. Examples

68.4.1. Getting the port configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=gpuportmap&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Bridge.Begin=1024
Bridge.End=65535
PortMap.1.External=8080
PortMap.1.Internal=80
PortMap.2.External=2222
PortMap.2.Internal=22
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Bridge": {
        "Begin": 1024,
        "End": 65535
    },
    "PortMap": [
        {
            "External": 8080,
            "Index": 1,
            "Internal": 80
        },
        {
            "External": 2222,
            "Index": 2,
            "Internal": 22
        }
    ]
```

```
}
```

68.4.2. Set Begin and End port of Bridge

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=gpuportmap&action=set&Bridge.Begin=1024&Bridge.End=
65535
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

68.4.3. Add PortMap setting

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=gpuportmap&action=add&PortMap.External=8080&PortMap
.Internal=80
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Index": 1
}
```

Chapter 69. Port Configuration

69.1. Description

The **portconf** submenu can be used to view and set the port settings of the device.

Access level

Action	Camera	NVR	Encoder	Decoder	IPAS
view	User	User	User	User	User
set	Admin	Admin	Admin	Admin	Admin

69.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
portconf&action=<value>[&<parameter>=<value>...]
```

69.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	FixedPorts	RES	<csv>	List of fixed ports used by the device
	UsedPorts	RES	<csv>	List of ports used by the device
	RouterUsedPorts	RES	<csv>	Provides a list of ports used by the router when using UPNP NVR ONLY
set	ProtocolType	REQ, RES	<enum> TCP, UDP-UNICAST, UDP-MULTICAST	Transmission mode used by SVP server NVR ONLY DECODER ONLY
	WebSessionTimeout	REQ, RES	<int>	Digest Nonce expire time in minute
	HTTPPort	REQ, RES	<int>	HTTP port used by the device **Note** **set** action not supported in IPAS devices.

Action	Parameters	Request/ Response	Type/ Value	Description
	HTTPSPort	REQ, RES	<int>	HTTPs port used in the device. **Note** **set** action not supported in IPAS devices.
	RTSPPort	REQ, RES	<int>	RTSP server port
	DevicePort	REQ, RES	<int>	Used to configure SVNP port for camera and SVP port for NVR **Note** This parameter is deprecated.
	RTSPTimeout	REQ, RES	<enum> 0s, 60s	RTSP timeout setting CAMERA ONLY ENCODER ONLY

69.4. Examples

69.4.1. Getting the port configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=portconf&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
FixedPorts=22,25,110,111,123,443,3260,3702,4520,9923
UsedPorts=80,554,555,556,557,558,7001,7002,7003,7004,7005,7006,7007,7008,700
9,7010,7011,7012,7013,7014,7015,7016,7017,7018,7019,7020,7021,7022,7023,7024
,7025,7026,7027,7028,7029,7030,7031,7032,7033,7034,7035,7036,7037,7038,7039,
7040,7041,7042,7043,7044,7045,7046,7047,7048,7049,7050,7051,7052,7053,7054,7
055,7056,7057,7058,7059,7060,7061,7062,7063,7064,10001,10002,10003,25000
RouterUsedPorts=22,25,110,111,123,443,3260,3702,4520,9923
ProtocolType=UDP-UNICAST
HTTPPort=80
HTTPSPort=443
```

```
RTSPPort=558
DevicePort=554
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "FixedPorts": [
        22,
        25,
        110,
        111,
        123,
        443,
        3260,
        3702,
        4520,
        9923
    ],
    "UsedPorts": [
        80,
        554,
        555,
        556,
        557,
        558,
        7001,
        7002,
        7003,
        7004,
        7005,
        7006,
        7007,
        7008,
        7009,
        7010,
        7011,
        7012,
```

```
        7013,
        7014,
        7015,
        7016,
        7017,
        7018,
        7019,
        7020,
        7021,
        7022,
        7023,
        7024,
        7025,
        7026,
        7027,
        7028,
        7029,
        7030,
        7031,
        7032,
        7033,
        7034,
        7035,
        7036,
        7037,
        7038,
        7039,
        7040,
        7041,
        7042,
        7043,
        7044,
        7045,
        7046,
        7047,
        7048,
        7049,
        7050,
        7051,
        7052,
        7053,
```

```
        7054,
        7055,
        7056,
        7057,
        7058,
        7059,
        7060,
        7061,
        7062,
        7063,
        7064,
        10001,
        10002,
        10003,
        25000
    ],
    "RouterUsedPorts":  [
        22,
        25,
        110,
        111,
        123,
        443,
        3260,
        3702,
        4520,
        9923
    ],
    "ProtocolType": "UDP-UNICAST",
    "HTTPPort": 80,
    "HTTPSPort": 443,
    "RTSPPort": 558,
    "DevicePort": 554
}
```

Chapter 70. Standby Device Setting

70.1. Description

The **standbydeviceinfo** submenu is used to set the standby device IP address.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	Admin

70.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
standbydeviceinfo&action=<value>[&<parameter>=<value>...]
```

70.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the standby device IP address
set	IPAddress	REQ, RES	<string>	Sets the standby device IP address

70.4. Examples

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=standbydeviceinfo&action=set&IPAddress=192.168.90.1
22
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 71. MTS (mobile tracking system) settings

71.1. Description

The **mts** submenu is used to configure the MTS (mobile tracking system) in NVR

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

71.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
mts&action=<value>[&<parameter>=<value>...]
```

71.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	BaseURL	RES	<string>	URL to access the server
set	Enable	REQ, RES	<bool> True, False	Enable MTS
	Version	REQ, RES	<string>	MTS version
	Periodicity	REQ, RES	<enum>	Periodicity of updating the GPS location
	EnableEventNotification	REQ, RES	<bool> True, False	Enable Event Notification to server.
	EnableJPEGPush	REQ, RES	<bool> True, False	Enable JPEG snapshot transmission.
	IPType	REQ, RES	<enum>	Server IP type
	IPAddress	REQ, RES	<string>	Server IP address.
	Port	REQ, RES	<int>	Server port.

71.4. Examples

71.4.1. Getting the MTS configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=mts&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
Version=v1
Periodicity=5s
EnableEventNotification=False
EnableJPEGPush=False
IPType=IPv4
IPAddress=129.123.123.12
Port=3000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Enable": false,
    "Version": "v1",
    "Periodicity": "5s",
    "EnableEventNotification": false,
    "EnableJPEGPush": false,
    "IPType": "IPv4",
    "IPAddress": "129.123.123.12",
    "Port": 3000
}
```

71.4.2. Setting the MTS configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=mts&action=set&Enable=True&Version=V2&Periodicity=10s&EnableEventNotification=True&EnableJPEGPush=True&IPType=IPv4&IPAddress=255.255.255.255&Port=2555
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 72. WiFi settings

72.1. Description

The **wifi** submenu is used to set and get WiFi connection configuration

NOTE This chapter applies to NVR only.

Access level

Action	Camera	NVR
view	User	User
set	Admin	User
control		User
add/update	Admin	
remove	Admin	

72.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
wifi&action=<value>[&<parameter>=<value>...]
```

72.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	ConnectedSSID	RES	<string>	Reads the SSID on the device
	SignalStrength	RES	<int>	Reads the Signal Strength
	WPSEnable	RES	<bool> True, False	Reads the WPS (WiFi Protected Setup).
	WiFiMode	REQ	<enum> Station, Direct	Reads the WiFi Mode.
set	Enable	REQ, RES	<bool> True, False	Enabling/Disabling WiFi feature.
control	Mode	REQ	<enum> Scan, Connect	Mode to scan for available WiFi devices, or to connect to the WiFi device

Action	Parameter	Request/ Response	Type/ Value	Description
	WiFiMode	REQ	<enum> Station, Direct	Station mode
	SSID	REQ	<string>	WiFi device ID The **SSID** parameter is valid only when **Mode** is set to Connect. The **SSID** and **Password** parameters should be sent along with the **SecurityMode** parameter. The maximum length of the string should be less than 32 characters.
	Password	REQ	<string>	WiFi device password The **Password** parameter is valid only when **Mode** is set to Connect, and **SecurityMode** is not None. The **SSID** and **Password** parameters should be sent along with the **SecurityMode** parameter.
	IsPasswordEncrypted	REQ	<bool> True, False	True if password sent is encrypted Encrypted password should be sent as post message.
	SecurityMode	REQ	<enum> None, WEP, PSK, Dot1X, Extended	Encoding type The **SecurityMode** parameter is valid only when **Mode** is set to Connect.
	AvailableNetworks.#.SSID	RES	<string>	WiFi device ID The maximum length of the string should be less than 32 characters.
	AvailableNetworks.#.SecurityMode	RES	<enum> None, WEP, WPA/WPA2 PSK	Encoding type
	AvailableNetworks.#.SignalStrength	RES	<int>	Signal strength of the WiFi connection The range should be 0 to 100.

72.4. Examples

72.4.1. Getting the WiFi configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=wifi&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ConnectedSSID=
SignalStrength=0
WiFiMode=
WPSEnable=False
Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ConnectedSSID": "",
    "SignalStrength": 0,
    "WiFiMode": "",
    "WPSEnable": false,
    "Enable": true
}
```

72.4.2. Scanning nearby WiFi device information

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=wifi&action=control&Mode=Scan&WiFiMode=Station
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AvailableNetworks.1.SSID=iptime_pky
AvailableNetworks.1.SecurityMode=WPA/WPA2PSK
AvailableNetworks.1.SignalStrength=5
AvailableNetworks.2.SSID=kuhyeon_iptime
AvailableNetworks.2.SecurityMode=WPA/WPA2PSK
AvailableNetworks.2.SignalStrength=4
AvailableNetworks.3.SSID=iptime5G_pky
AvailableNetworks.3.SecurityMode=None
AvailableNetworks.3.SignalStrength=5
AvailableNetworks.4.SSID=iMacT
AvailableNetworks.4.SecurityMode=WPA/WPA2PSK
AvailableNetworks.4.SignalStrength=3
AvailableNetworks.5.SSID=TP-LINK_A3AF00
AvailableNetworks.5.SecurityMode=WPA/WPA2PSK
AvailableNetworks.5.SignalStrength=3
AvailableNetworks.6.SSID=SYG IPT 2g
AvailableNetworks.6.SecurityMode=WEP
AvailableNetworks.6.SignalStrength=3
AvailableNetworks.7.SSID=iptime
AvailableNetworks.7.SecurityMode=None
AvailableNetworks.7.SignalStrength=4
AvailableNetworks.8.SSID=NBAndroidHotspot2608
AvailableNetworks.8.SecurityMode=WPA/WPA2PSK
AvailableNetworks.8.SignalStrength=3
AvailableNetworks.9.SSID=B2C_PART4-5G
AvailableNetworks.9.SecurityMode=WPA/WPA2PSK
AvailableNetworks.9.SignalStrength=1
AvailableNetworks.10.SSID=B2C_PART1-5G
AvailableNetworks.10.SecurityMode=WPA/WPA2PSK
AvailableNetworks.10.SignalStrength=1
AvailableNetworks.11.SSID=NETGEAR27
AvailableNetworks.11.SecurityMode=WPA/WPA2PSK
AvailableNetworks.11.SignalStrength=3
AvailableNetworks.12.SSID=iptime_br
AvailableNetworks.12.SecurityMode=WPA/WPA2PSK
```

```
AvailableNetworks.12.SignalStrength=2
AvailableNetworks.13.SSID=KJK
AvailableNetworks.13.SecurityMode=WPA/WPA2PSK
AvailableNetworks.13.SignalStrength=3
AvailableNetworks.14.SSID=B2C_PART2-5G
AvailableNetworks.14.SecurityMode=WPA/WPA2PSK
AvailableNetworks.14.SignalStrength=1
AvailableNetworks.15.SSID=iptime_neighbor
AvailableNetworks.15.SecurityMode=None
AvailableNetworks.15.SignalStrength=4
AvailableNetworks.16.SSID=Mobile_Guest
AvailableNetworks.16.SecurityMode=WPA/WPA2PSK
AvailableNetworks.16.SignalStrength=2
AvailableNetworks.17.SSID=Mobile88_Guest
AvailableNetworks.17.SecurityMode=WPA/WPA2PSK
AvailableNetworks.17.SignalStrength=2
AvailableNetworks.18.SSID=MobileASUS_88
AvailableNetworks.18.SecurityMode=WPA/WPA2PSK
AvailableNetworks.18.SignalStrength=2
AvailableNetworks.19.SSID=iptime_sooj
AvailableNetworks.19.SecurityMode=WPA/WPA2PSK
AvailableNetworks.19.SignalStrength=2
AvailableNetworks.20.SSID=iptime_hwan
AvailableNetworks.20.SecurityMode=WPA/WPA2PSK
AvailableNetworks.20.SignalStrength=2
AvailableNetworks.21.SSID=UCOMM_IPTIME(Smart_Cam)
AvailableNetworks.21.SecurityMode=WPA/WPA2PSK
AvailableNetworks.21.SignalStrength=2
AvailableNetworks.22.SSID=MVipTime
AvailableNetworks.22.SecurityMode=WPA/WPA2PSK
AvailableNetworks.22.SignalStrength=2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AvailableNetworks": [
        {
```

```
            "SSID": "iptime_pky",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 5
        },
        {
            "SSID": "kuhyeon_iptime",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 4
        },
        {
            "SSID": "iptime5G_pky",
            "SecurityMode": "None",
            "SignalStrength": 5
        },
        {
            "SSID": "iMacT",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 3
        },
        {
            "SSID": "TP-LINK_A3AF00",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 3
        },
        {
            "SSID": "SYG IPT 2g",
            "SecurityMode": "WEP",
            "SignalStrength": 3
        },
        {
            "SSID": "iptime",
            "SecurityMode": "None",
            "SignalStrength": 4
        },
        {
            "SSID": "NBAndroidHotspot2608",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 3
        },
        {
            "SSID": "B2C_PART4-5G",
```

```
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 1
        },
        {
            "SSID": "B2C_PART1-5G",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 1
        },
        {
            "SSID": "NETGEAR27",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 3
        },
        {
            "SSID": "iptime_br",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "KJK",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "B2C_PART2-5G",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 1
        },
        {
            "SSID": "iptime_neighbor",
            "SecurityMode": "None",
            "SignalStrength": 4
        },
        {
            "SSID": "Mobile_Guest",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "Mobile88_Guest",
            "SecurityMode": "WPA/WPA2PSK",
```

```
            "SignalStrength": 2
        },
        {
            "SSID": "MobileASUS_88",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "iptime_sooj",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "iptime_hwan",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "UCOMM_IPTIME(Smart_Cam)",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "MVipTime",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 2
        },
        {
            "SSID": "iptime_yjcho",
            "SecurityMode": "WPA/WPA2PSK",
            "SignalStrength": 3
        }
    ]
}
```


72.4.3. Connecting WiFi device

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=wifi&action=control&Mode=Connect&SSID=Samsung_5G&Se
```

```
curityMode=WEP&Password={password}
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=wifi&action=control&Mode=Connect&SSID=Samsung_5G&Se
curityMode=None
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

72.4.4. Enabling WiFi feature

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=wifi&action=set&Enable=False
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 73. DHCP clients

73.1. Description

The **dhcpclients** submenu is used to get DHCP clients' information.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

73.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
dhcpclients&action=view[&<parameter>=<value>...]
```

73.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	InterfaceName	REQ, RES	<csv>	Requests information about specific interface name
	IPType	REQ, RES	<enum> IPv4, IPv6	Requests information about specific IP type
	Index	RES	<int>	Shows index of the response list.
	IPAddress	RES	<string>	Shows IP address of the client
	MacAddess	RES	<string>	Shows Mac address of the client

73.4. Examples

73.4.1. Getting DHCP client list

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dhcpclients&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Index.1.InterfaceName=Network1
Index.1.IPType=IPv4
Index.1.IPAddress=192.168.111.18
Index.1.MACAddress=e1:6c:d6:ae:52:90
Index.2.InterfaceName=Network1
Index.2.IPType=IPv4
Index.2.IPAddress=192.168.111.19
Index.2.MACAddress=00:09:18:ff:ff:ff
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "DHCPClients": [
        {
            "Index": 1,
            "InterfaceName": "Network1",
            "IPType": "IPv4",
            "IPAddress": "192.168.111.18",
            "MACAddress": "e1:6c:d6:ae:52:90"
        },
        {
            "Index": 2,
            "InterfaceName": "Network1",
            "IPType": "IPv4",
            "IPAddress": "192.168.111.19",
            "MACAddress": "00:09:18:ff:ff:ff"
        }
    ]
}
```

Chapter 74. RTSP over TLS settings

74.1. Description

The **rtspovertls** submenu supports RTSP over TLS (RTSPS).

NOTE This chapter applies to NWC only.

Access level

Action	Camera
view	User
set	User

74.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
rtspovertls&action=<value>[&<parameter>=<value>...]
```

74.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				
set	Enable	REQ, RES	<bool> True, False	It is rtsp over tls
	Port	REQ, RES	<int>	It is rtsp over tls

74.4. Examples

74.4.1. Getting the RTSP over TLS configuration information

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=rtspovertls&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
Port=1024
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Enable": true,
    "Port": 1024
}
```


74.4.2. Setting the RTSP over TLS configuration information

REQUEST

```
http://<Device IP>/stw-cgi
/network.cgi?msubmenu=rtspovertls&action=set&Enable=True&Port=1024
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 75. EthStatus

75.1. Description

The **ethstatus** submenu is used to get the inbound and outbound traffic info in real time.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

75.2. Syntax

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=ethstatus&action=<value>[&<parameter>=<value>...]
```

75.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	EthIndex	RES	<int>	Ethernet Index
	InBound	RES	<float>	Inbound data rate in MBs
	OutBound	RES	<float>	Outbound rate in MBs

75.4. Examples

75.4.1. Getting the current inbound and outbound rates

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=ethstatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EthernetStaus": [
        {
            "EthIndex": 0,
            "InBound": 8.181055,
            "OutBound": 1.010742
        }
    ]
}
```

Chapter 76. PoeStatus

76.1. Description

The **poestatus** submenu is used to get or change the settings of POE ports that device supports.

For camera users, they can find out whether their cameras support POE ports or not by referring to **POEExtender** parameter in attributes (attributes/network/support).

Action	Camera	NVR
view	User	User
set	Admin	Admin

76.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
poestatus&action=<value>[&<parameter>=<value>...]
```

76.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Port	REQ, RES	<csv>	PoE port number CAMERA ONLY
	Total	RES	<float>	Shows each port's activation status.
	Port.#.Enable	RES	<bool>	Shows power consumption of each port. CAMERA ONLY
	Port.#.PowerConsumption	RES	<float>	Shows power consumption of each port.
	Port.#.Summary	RES	<enum> NoFault, VoltageFault, ThermalShutdownFault, LoadDisconnect, OverloadTotalPower	Shows each port's status. NVR ONLY
set	Enable	REQ, RES	<bool>	Configures or sees activation status CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	Port	RES	<int>	Configures specified port's settings. CAMERA ONLY
	Port.#.Enable	REQ	<bool>	Configures each port's activation status NVR ONLY

76.4. Examples

76.4.1. Getting the current PoE ports' status (NVR)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=poestatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Total": 0,
    "Ports": [
        {
            "Port": 0,
            "PowerConsumption": 0,
            "Enable": true,
            "Summary": "NoFault"
        },
        {
            "Port": 1,
            "PowerConsumption": 0,
            "Enable": true,
            "Summary": "NoFault"
        },
        {
            "Port": 2,
            "PowerConsumption": 0,
```

```
            "Enable": true,
            "Summary": "NoFault"
        },
        {
            "Port": 3,
            "PowerConsumption": 0,
            "Enable": true,
            "Summary": "NoFault"
        },
        {
            "Port": 4,
            "PowerConsumption": 0,
            "Enable": true,
            "Summary": "NoFault"
        }
    ]
}
```

76.4.2. Getting the current PoE ports’ status (camera)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=poestatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Total": 0,
    "Ports": [
        {
            "Port": 1,
            "Enable": true,
            "PowerConsumption": 0
        }
    ]
}
```

76.4.3. Setting the PoE port to activate (camera)

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=poestatus&action=set&Port=1&Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

76.4.4. Setting the PoE port to activate (NVR)

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=poestatus&action=set&Port.0.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 77. DHCP Server

77.1. Description

The **dhcpserver** submenu is used to configure DHCP server on device for each network interface.

NOTE	This chapter applies to NVR only and IPv6 DHCP server can be configured only after enabling IPv6 on the interface.

Access level

Action	NVR
view	User
set	User

77.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
dhcpserver&action=<value>[&<parameter>=<value>...]
```

77.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	InterfaceName	REQ	<enum> Network1, Network2, Network3	Network interface name
	CheckServer	REQ	<bool> True, False	Check if another DHCP is running on the network **Note** Response can take some time in this case.
	ExternalDhcpServer	RES	<bool> True, False	If True, another DHCP server is already running on the same network. If checkserver parameter is delivered through the request, this parameter is added to the response.

Action	Parameter	Request/ Response	Type/ Value	Description
set	InterfaceName	REQ, RES	<enum> Network1, Network2, Network3	Network interface name used as reference for set operation (Fixed value cannot be changed)
	Enable	REQ, RES	<bool> True, False	Enable DHCP server
	IPType	REQ, RES	<enum> IPv4, IPv6	IPType Selection **Note** Both IPv4 and IPv6 DHCP servers can be enabled on the same interface.
	IPRangeFrom	REQ, RES	<string>	IP Allocation starting range
	IPRangeTo	REQ, RES	<string>	IP Allocation limit
	IPLeaseTime	REQ, RES	<int> secs	Lease duration in seconds (valid range : 3600 - 172800)

77.4. Examples

77.4.1. Getting the current DHCP server settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=dhcpserver&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "DHCPServer": [
        {
            "InterfaceName": "Network1",
            "IPType": "IPv4",
            "Enable": true,
            "IPRangeFrom": "192.168.75.2",
            "IPRangeTo": "192.168.75.254",
```

```
            "IPLeaseTime": 86400
        },
        {
            "InterfaceName": "Network1",
            "IPType": "IPv6",
            "Enable": false,
            "IPRangeFrom": "",
            "IPRangeTo": "",
            "IPLeaseTime": 86400
        },
        {
            "InterfaceName": "Network2",
            "IPType": "IPv4",
            "Enable": false,
            "IPRangeFrom": "192.168.2.2",
            "IPRangeTo": "192.168.2.254",
            "IPLeaseTime": 86400
        },
        {
            "InterfaceName": "Network2",
            "IPType": "IPv6",
            "Enable": false,
            "IPRangeFrom": "",
            "IPRangeTo": "",
            "IPLeaseTime": 86400
        },
        {
            "InterfaceName": "Network3",
            "IPType": "IPv4",
            "Enable": false,
            "IPRangeFrom": "192.168.3.2",
            "IPRangeTo": "192.168.3.254",
            "IPLeaseTime": 86400
        },
        {
            "InterfaceName": "Network3",
            "IPType": "IPv6",
            "Enable": false,
            "IPRangeFrom": "",
            "IPRangeTo": "",
            "IPLeaseTime": 86400
```

```
        }
    ]
}
```

77.4.2. Setting DHCP server settings

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=dhcpserver&action=set&Enable=True&InterfaceName=Net
work2&IPType=IPv4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Response":"Success"
}
```

Chapter 78. ONVIF Discovery

78.1. Description

The **onvifdiscovery** submenu enables or disables ONVIF Discovery mechanism. In factory default, ONVIF discovery is enabled. When it is enabled, camera can be discovered using the ONVIF ws-discovery protocol.

NOTE This chapter applies to network cameras and NVR only.

Access level

Action	Camera	NVR
view	Admin	Admin
set	Admin	Admin

78.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
onvifdiscovery&action=<value>[&<parameter>=<value>]
```

78.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the onvifdiscovery settings.
	InterfaceName	REQ	<string>	Requests information about specific interface name
set	InterfaceName	REQ, RES	<string>	Interface name
	Enable	REQ, RES	<bool> True, False	Enables or disables ONVIF discovery service **Note** **Enable** must be sent together with the **set** action.

78.4. Examples

78.4.1. Getting the current onvif discovery settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=onvifdiscovery&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=NetworkInterface1
Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "OnvifDiscovery": [
        {
            "InterfaceName": "NetworkInterface1",
            "Enable": true
        }
    ]
}
```

78.4.2. Enabling ONVIF discovery service

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=onvifdiscovery&action=set&Enable=True
```

Chapter 79. SIP Setup

79.1. Description

The **sipsetup** submenu is used to configure SIP general settings.

> **NOTE**
>
> This chapter is applicable to intercom cameras.
>
> Attribute to check for feature support: "attributes/Network/Support/SIP"

Access level

Action	Camera
view	Admin
set	Admin

79.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
sipsetup&action=<value>[&<parameter>=<value>]
```

79.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the SIP settings.
set	SIPEnable	REQ, RES	<bool> True, False	Whether to use SIP
	AllowIncomingCalls	REQ, RES	<bool> True, False	Whether to allow incoming calls
	SIPPort	REQ, RES	<int>	Port number to be used by SIP
	SIPTLSPort	REQ, RES	<int>	Port number to be used by SIPoverTLS
	RTSPStartPort	REQ, RES	<int>	Initial port number to be used by RTSP
	TransportAutoSwitchEnable	REQ, RES	<bool> True, False	Whether to use transport auto switch
	AllowContactRewrite	REQ, RES	<bool> True, False	Whether to use contact rewrite
	CallDurationLimitEnable	REQ, RES	<bool> True, False	Whether to use call duration limit
	CallDurationLimit	REQ, RES	<int>	Call duration limit (minutes)

Action	Parameters	Request/ Response	Type/ Value	Description
	RegistrationInterval	REQ, RES	<int>	Registration interval (seconds)
	AudioDirection	REQ, RES	<enum> ReceiveOnly, SendOnly, SendAndReceive	Audio direction
	CallStopEnable	REQ, RES	<bool> True, False	If enabled, user can stop the call with the call button
	TouchlessSensorRange	REQ, RES	<enum> 15cm, 20cm, 25cm, 30cm, 35cm, 40cm	The maximum distance from touchless sensor

79.4. Examples

79.4.1. Getting the current sip setting values from cameras (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=sipsetup&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SIPEnable": true,
    "AllowIncomingCalls": false,
    "SIPPort": 5060,
    "SIPTLSPort": 5061,
    "RTPStartPort": 4000,
    "TransportAutoSwitchEnable": true,
    "AllowContactRewrite": true,
    "CallDurationLimitEnable": false,
```

```
    "CallDurationLimit": 60,
    "RegistrationInterval": 300,
    "AudioDirection": "SendAndReceive",
    "CallStopEnable": true,
    "TouchlessSensorRange" : "30cm"
}
```

79.4.2. Setting SIP configurations

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=sipsetup&action=set&
SIPEnable=True&SIPPort=4521&SIPTLSPort=5001&AllowIncomingCalss=False&RTSPSta
rtPort=4000&TransportAutoSwitchEnable=True&AllowContactRewrite=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 80. SIP Account Settings

80.1. Description

The **sipaccount** submenu configures credential and connection settings of the SIP registrar

NOTE

This chapter is applicable to intercom cameras.

Attributes to check for feature support: "Attributes/Network/Support/SIP"

Access level

Action	Camera
view	Admin
add/update	Admin
remove	Admin

80.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
sipaccount&action=<value>[&<parameter>=<value>]
```

80.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	REQ	<int>	Account index
	Accounts.#.Status	RES	<enum> Unavailable , Available	Account status
	Accounts.#.Type	RES	<enum> PeerToPeer , Registrar	Account type
	Accounts.#.IsFixed	RES	<bool> True, False	Whether the default account is fixed

Action	Parameters	Request/ Response	Type/ Value	Description
	Accounts.#.ConnectionTypeInUse	RES	<enum> Domain, BackupDomain, Registrar, BackupRegistrar, PeerToPeer	Connection the account is using
add/update	Index	REQ, RES	<int>	Account index
	Name	REQ, RES	<string>	Account name
	Enable	REQ, RES	<bool> True, False	Whether to use account
	UserID	REQ, RES	<string>	SIP user ID
	AuthenticationID	REQ, RES	<string>	SIP authentication ID
	Password	REQ, RES	<string>	SIP password
	CallerID	REQ, RES	<string>	SIP Caller ID
	DomainName	REQ, RES	<string>	SIP Domain name
	RegistrarAddress	REQ, RES	<string>	SIP Registrar address
	BackupDomainName	REQ, RES	<string>	SIP backup domain name
	BackupRegistrarAddress	REQ, RES	<string>	SIP backup registrar address
	SIPProxyAddress	REQ, RES	<string>	SIP Proxy address
	TransportMode	REQ, RES	<enum> UDP, TCP, TLS	SIP transport mode
	MediaEncryptionMode	REQ, RES	<enum> None, Prefer, Force	SIP Media encryption mode **Note** This parameter is valid only when **TransportMode** set to **TLS**
	IsPasswordEncrypted	REQ	<bool> True, False	When this is set to true, the password is encrypted using the public key obtained through the **rsa** submenu of security and cgi, and it is sent as payload content for the POST command.
remove	Index	REQ	<csv>	Account index

80.4. Examples

80.4.1. Getting current SIP account settings (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=sipaccount&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Accounts": [
        {
            "Index": 0,
            "Status": "Available",
            "Name": "peer-to-peer",
            "Enable": true,
            "Type": "PeerToPeer",
            "UserID": "peer-to-peer",
            "AuthenticationID": "",
            "Password": "",
            "CallerID": "",
            "DomainName": "192.168.71.90",
            "RegistrarAddress": "192.168.71.90",
            "BackupDomainName": "",
            "BackupRegistrarAddress": "",
            "SIPProxyAddress": "",
            "TransportMode": "UDP",
            "MediaEncryptionMode": "None",
            "IsFixed": true,
            "ConnectionTypeInUse": "PeerToPeer"
        },
        {
            "Index": 1,
            "Status": "Unavailable",
            "Name": "account1",
            "Enable": true,
```

```
                "Type": "Registrar",
                "UserID": "admin",
                "AuthenticationID": "0001",
                "Password": "",
                "CallerID": "00001",
                "DomainName": "192.168.125.213",
                "RegistrarAddress": "192.168.125.213",
                "BackupDomainName": "192.168.125.213",
                "BackupRegistrarAddress": "1",
                "SIPProxyAddress": "",
                "TransportMode": "UDP",
                "MediaEncryptionMode": "None",
                "IsFixed": false,
                "ConnectionTypeInUse": "Domain"
            }
        ]
}
```

80.4.2. Adding a SIP account

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=sipaccount&action=add&Name=test&Enable=True&UserID=
test&AuthenticationID=1234&Password={password}&CallerID=0011&DomainName=doma
inName&RegistrarAddress=addr&TransportMode=TLS
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success",
    "Index": 1
}
```

80.4.3. Updating account settings

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
sipaccount&action=update&Index=1&Name=name&Enable=True&UserID=admin&Password
={password}&DomainName=192.168.125.213&TransportMode=UDP&AuthenticationID=00
01&CallerID=0001&BackupDomainName=192.168.125.213&RegistrarAddress=192.168.1
25.213&BackupRegistrarAddress=192.168.125.214
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

80.4.4. Remove SIP account

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=sipaccount&action=remove&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 81. SIP Recipient Settings

81.1. Description

The **siprecipients** submenu is used to manage SIP recipients. It can add, update, or delete SIP recipients, and you can choose the type of call requests.

NOTE

This chapter is applicable to intercom cameras.

Attributes to check for feature support: "Attributes/Network/Support/SIP"

Access level

Action	Camera
view	Admin
set	Admin
add	Admin
update	Admin
remove	Admin

81.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
siprecipients&action=<value>[&<parameter>=<value>]
```

81.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	REQ	<int>	Recipient index
	Group	REQ	<enum> None, 1~5	Group number for filter of view
set	CallRequestType	REQ, RES	<enum> Single, Multiple	Recipient calling type when a CallRequest event occurs
	RecipientInSingleMode	REQ, RES	<enum> None, #	Recipient when the **CalRequestType** is in Single mode It can set the recipient index

Action	Parameters	Request/ Response	Type/ Value	Description
	RecipientInMultipleMode	REQ, RES	<enum> None, 1~5	Recipient when the **CallRequestType** is in Multiple mode It can set the group number
add	Name	REQ	<string>	Recipient name
	TransportMode	REQ	<enum> Registrar, PeerToPeer	Recipient transport mode
	Address	REQ	<string>	Recipient address
	Group	REQ	<enum> None, 1~5	Recipient group
update	Recipients.#.Name	REQ, RES	<string>	Recipient name
	Recipients.#.TransportMode	REQ, RES	<enum> Registrar, PeerToPeer	Recipient transport mode
	Recipients.#.Address	REQ, RES	<string>	Recipient address
	Recipients.#.Group	REQ, RES	<enum> None, 1~5	Recipient group
remove	Index	REQ	<csv> All, #	Recipient index

81.4. Examples

81.4.1. Getting SIP recipient settings (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=siprecipients&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "CallRequestType": "Single",
    "RecipientInSingleMode": "None",
```

```
    "RecipientInMultipleMode": "None",
    "Recipients": [
        {
            "Index": 1,
            "Name": "recipient1",
            "TransportMode": "Registrar",
            "Address": "192.168.125.123",
            "Group": [
                "1",
                "2",
                "3"
            ]
        }
    ]
}
```


81.4.2. Adding a recipient by assigning them to groups 1, 2, or 3

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=siprecipients&action=add&Name=recipient1&TransportM
ode=Registrar&Address=192.168.125.123&Group=1,2,3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success",
    "Index": 1
}
```


81.4.3. Updating a recipient for Index 1

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=siprecipients&action=update&Recipients.1.Name=recip
```

```
ient2&Recipients.1.TransportMode=PeerToPeer&Recipients.1.Address=192.168.125
.123&Recipients.1.Group=1,2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


81.4.4. Selecting a target recipient and setting the Call request type to Single mode

The group you are setting up must contain at least one recipient.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=siprecipients&action=set&CallRequestType=Single&Rec
ipientInSingleMode=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


81.4.5. Selecting a target recipient group and setting the Call request type to Multiple mode

You can set a group index of previously registered recipients to a group.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=siprecipients&action=set&CallRequestType=Multiple&R
ecipientInMultipleMode=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


81.4.6. Removing all recipients

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=siprecipients&action=remove&Index=All
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 82. SIP Call

82.1. Description

The **sipcall** submenu is used for getting SIP call states and controlling SIP calls.

NOTE	This chapter is applicable to intercom cameras. Attributes to check for feature support: "Attributes/Network/Support/SIP"

Access level

Action	Camera
check	Admin
control	Admin

82.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
sipcall&action=<value>[&<parameter>=<value>]
```

82.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
check	CallState	RES	<enum> Idle, Call, Ringing	SIP call state
	TestCallState	RES	<enum> Idle, Trying, Success, Failure	State for test call control commands
control	StopCallRequest	REQ	<bool> True, False	SIP call stop request **Note** Deprecated parameter, Use **CallRequest** instead of this parameter

Action	Parameters	Request/ Response	Type/ Value	Description
	CallRequest	REQ	<enum> StopVMS, StartVMS, StopCall, TestCall	control the callrequest • StopVMS: Stop VMS call session • StartVMS: Stop CallRequest event to connect from VMS, return error if SIP is already connected • StopCall: Stop all call session • TestCall: Start test call

82.4. Examples

82.4.1. Getting the current SIP call state (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=sipcall&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "CallState": "Idle",
    "TestCallState": "Idle"
}
```

82.4.2. Stopping a call request

This command is used to mean that the VMS has accepted the CallRequest. If you request this command, CallRequest and SIP Calling will be aborted.

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=sipcall&action=control&CallRequest=StartVMS
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 83. SIP Audio Codec

83.1. Description

The **sipaudiocodec** submenu is used to get a list of audio codecs used in SIP and set which codecs to use and their priority.

> **NOTE**
> This chapter is applicable to intercom cameras.
> Attributes to check for feature support: "Attributes/Network/Support/SIP"

Access level

Action	Camera
view	Admin
set	Admin

83.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
sipaudiocodec&action=<value>[&<parameter>=<value>]
```

83.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Codec.#.Index	RES	<int>	Audio codec index
	Codec.#.Name	RES	<string>	Audio codec name
set	Codec.#.Enable	REQ	<bool> True, False	Whether to use this codec
	Codec.#.Priority	REQ	<int> 1 ~ 254	Audio codec priority The higher the number, the higher the priority

83.4. Examples

83.4.1. Getting the current SIP audio codec list (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=sipaudiocodec&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Codecs": [
        {
            "Index": 1,
            "Name": "PCMU/8000/1",
            "Enable": true,
            "Priority": 254
        },
        {
            "Index": 2,
            "Name": "PCMA/8000/1",
            "Enable": true,
            "Priority": 253
        },
        {
            "Index": 3,
            "Name": "opus/48000/2",
            "Enable": true,
            "Priority": 252
        },
        {
            "Index": 4,
            "Name": "G722/16000/1",
            "Enable": true,
            "Priority": 251
        },
        {
            "Index": 5,
            "Name": "L16/44100/1",
            "Enable": true,
            "Priority": 250
        }
    ]
}
```

83.4.2. Changing the priority of the audio codec whose index is 1 and enable it

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=sipaudiocodec&action=set&Codec.1.Priority=240&Codec
.1.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Response": "Success"
}
```

Chapter 84. NAT Traversal Settings

84.1. Description

The **nattraversal** submenu provides settings for the NAT Traversal features.

> **NOTE**
> This chapter is applicable to intercom cameras.
> Attributes to check for feature support: "Attributes/Network/Support/SIP"

Access level

Action	Camera
view	Admin
set	Admin

84.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
nattraversal&action=<value>[&<parameter>=<value>]
```

84.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	ICEEnable	REQ, RES	<bool> True, False	Whether to use ICE
	STUNEnable	REQ, RES	<bool> True, False	Whether to use STUN
	STUNAddress	REQ, RES	<string>	STUN address
	TURNEnable	REQ, RES	<bool> True, False	Whether to use TURN
	TURNAddress	REQ, RES	<string>	TURN address
	TURNUserID	REQ, RES	<string>	TURN user ID
	TURNUserPassword	REQ, RES	<string>	TURN user password

Action	Parameters	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<bool> True, False	When this is set to true, the password is encrypted using the public key obtained through the **rsa** submenu of security and cgi, and is sent as payload content for the POST command.

84.4. Examples

84.4.1. Getting current NAT traversal settings (this submenu only supports JSON responses)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=nattraversal&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ICEEnable": true,
    "STUNEnable": true,
    "STUNAddress": "192.168.71.27",
    "TURNEnable": true,
    "TURNAddress": "192.168.71.27",
    "TURNUserID": "user1",
    "TURNUserPassword": ""
}
```


84.4.2. Changing the NAT traversal settings

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=nattraversal&action=set&ICEEnable=True&STUNEnable=T
rue&STUNAddress=192.168.71.27&TURNEnable=True&TURNAddress=192.168.71.27&TURN
UserID=user1&TURNUserPassword={password}
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 85. P2P

85.1. Description

The **p2p** submenu is used to set and retrieve the p2p activation state.

NOTE This chapter applies to NVR only

Access level

Action	NVR
view	User
set	User

85.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
p2p&action=<value>[&<parameter>=<value>]
```

85.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Status	RES	<enum> Success, NoCertificate, NetworkUnavailable,, DeviceKeyIDFail,, ConfigDownLoadFail,, ConnectionFail, Unknown	Success - Succeeded NoCertificate - No public certificate installed. DeviceKeyIDFail - Failed to create device ID and device key. ConfigDownLoadFail - Failed to download P2P setting information. ConnectionFail - Failed to connect server. NetworkUnavailable - Network settings are wrong. Unknown - Unknown Errors
set	Enable	REQ, RES	<bool> True, False	P2P use, not used.

85.4. Examples

85.4.1. Retrieving the current P2P state.

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=p2p&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
Status=Success
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "P2P": {
            "Enable":true,
            "Status": "Unknown"
    }
}
```

Chapter 86. MQTT client settings

86.1. Description

The **mqttclient** submenu is used to configure the MQTT client

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin
check	Admin

86.2. Syntax

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=
mqttclient&action=<value>[&<parameter>=<value>...]
```

86.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Enable	REQ, RES	<bool> True, False	Configures whether to enable the MQTT client.
	Username	REQ, RES	<string>	Username to use to connect to the MQTT broker
	Password	REQ, RES	<string>	Password to use to connect to the MQTT broker
	Address	REQ, RES	<string>	Address of the MQTT broker
	Port	REQ, RES	<int>	Port of the MQTT broker
	Transport	REQ, RES	<enum> TCP, TLS, WebSocket, WebSocket Secure	Transport type to connect to the MQTT broker

Action	Parameters	Request/ Response	Type/ Value	Description
	BasePath	REQ, RES	<string>	BasePath for WebSocket **Note** This parameter is valid only when **Transport** set to **WebSocket** or **WebSocketSecure**
	CleanSessionEnable	REQ, RES	<bool> True, False	Whether the CleanSession flag is enabled when connecting to an MQTT broker
	CustomClientIDEnable	REQ, RES	<bool> True, False	Whether to use a custom client ID when connecting to an MQTT broker
	CustomClientID	REQ, RES	<string>	Custom client ID **Note** This parameter is only valid when **CustomClientIDEnable** is set to **True**, and if it is an empty string value, it is used an auto-generated id.
	KeepAliveInterval	REQ, RES	<int>	Keep alive interval
	ConnectionTimeout	REQ, RES	<int>	Connection timeout
	CACertificateName	REQ, RES	<string>	The name of the CA certificate installed in the device **Note** For a list of CA certificates, check the **cacertificate** submenu of security.cgi.
	ClientCertificateName	REQ, RES	<string>	The name of the client certificate installed in the device **Note** For a list of client certificates, check the **ssl** submenu of security.cgi.
	VerifyServerCertification	REQ, RES	<bool> True, False	Whether to verify the MQTT broker's certificate
	ALPN	REQ, RES	<string>	ALPN (Application Layer Protocol Negotiation) protocol ID

Action	Parameters	Request/ Response	Type/ Value	Description
	AutoReconnectEnable	REQ, RES	<bool> True, False	Whether to automatically reconnect to the MQTT broker
	DefaultTopicPrefix	REQ, RES	<string>	Default prefix for MQTT topics
	ConnectionMessageIndex	REQ, RES	<int>	Index of the message to be sent when connected to the MQTT broker. If the value is set to 0, it is unspecified. **Note** To set this item, **MQTT Publication** message must be set in advance
	LWTMessageIndex	REQ, RES	<int>	Message index to be sent when MQTT broker abnormal disconnection. If the value is set to 0, it is unspecified. **Note** To set this item, **MQTT Publication** message must be set in advance
	IsPasswordEncrypted	REQ, RES	<bool> True, False	When this is set to true, the password is encrypted using the public key obtained through the **rsa** submenu of security and cgi, and is sent as payload content for the POST command.
check	Status	RES	<enum> Disconnected, Connected	Current connection status of MQTT client

86.4. Examples

86.4.1. Getting the MQTT client configuration information (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-cgi/network.cgi?msubmenu=mqttclient&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "Enable": true,
    "Username": "user",
    "Password": "",
    "Address": "192.168.0.2",
    "Port": 1883,
    "Transport": "TCP",
    "BasePath": "",
    "CleanSessionEnable": false,
    "CustomClientIDEnable": false,
    "CustomClientID": "00:09:18:FF:FF:FF",
    "KeepAliveInterval": 30,
    "ConnectionTimeout": 60,
    "CACertificateName": "HTW_rootca",
    "VerifyServerCertification": false,
    "ClientCertificateName": "HTW_default",
    "ALPN": "",
    "AutoReconnectEnable": false,
    "DefaultTopicPrefix": "TestPrefix",
    "ConnectionMessageIndex": 0,
    "LWTMessageIndex": 0
}
```


86.4.2. Setting the MQTT client configuration information

REQUEST

```
http://<Device IP>/stw-
cgi/network.cgi?msubmenu=msubmenu=mqttclient&action=set&Enable=True&Username
=user&Password={password}&Address=192.168.0.2&Port=1883&Transport=TCP&CleanS
essionEnable=False&CustomClientIDEnable=False&KeepAliveInterval=30&Connectio
nTimeout=30&CACertificateName=HTW_rootca&VerifyServerCertification=False&Cli
entCertificateName=HTW_default&AutoReconnectEnable=False&DefaultTopicPrefix=
TestPrefix
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Security

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 87. Overview

87.1. Description

Hanwha Vision provides network security and authentication methods for secure data transfers. **security.cgi** is used to configure the general security settings for Hanwha Vision video surveillance devices.

The following submenus of security.cgi are used for network security settings:

- **ipfilter**: Requests and configures IP address filtering to block or allow certain IP addresses.
- **802Dot1x**: Requests and configures the parameters required for accessing an 802.1x protected network.
- **rtsp**: Requests and selects the RTSP authentication method.
- **ssl**: Requests and configures the SSL (Secure Socket Layer) settings.
- **guest**: Requests and enables/disables guest logins.
- **users**: Requests, adds and deletes system users and sets access permissions.
- **usergroups**: Requests, adds and deletes system user groups and sets access permissions for the user groups.
- **authority**: Sets access permissions and auto logout times.
- **additionalpassword**: Sets multiple passwords for users.
- **rsa**: Requests public RSA key.
- **camerausers**: Configures default user credentials used by the NVR to connect to cameras during discovery or connection testing.
- **clienthttpsstatus**: Requests the mutual authentication status of the client.
- **tlsversion:** Requests and configures the TLS (Transport Layer Security) settings.
- **cameravalidationstatus**: Requests the camera validation status from the NVR.
- **cacertificate**: Configures and handles the CA certificate used in a device.
- **macfilter**: Requests and configures MAC address filtering to block or allow certain MAC addresses.

NOTE	It is highly recommended to avoid using plain text passwords in the submenus. Password encryption should be used. Refer to the Application Programming Guide Document for details.

Chapter 88. IP Address Filtering

88.1. Description

The **ipfilter** submenu configures IP address filtering. This submenu controls device access based on IP filter settings. The feature provides both Allow and Deny filtering types. Deny filtering allows access from all IP addresses except those listed, while Allow filtering blocks access from all IP addresses except those listed.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User
add, update	Admin	User
remove	Admin	User

88.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
ipfilter&action=<value>[&<parameter>=<value>...]
```

88.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the IP address filtering settings.
set	AccessType	REQ, RES	<enum> Allow, Deny	Access type • Allow: Allows access to listed IP addresses • Deny: Blocks access to listed IP addresses **Note** **AccessType** must be sent together with the **set** action.
	IPType	REQ, RES	<enum>	IP type

Action	Parameter	Request/ Response	Type/ Value	Description
add, update	IPIndex	REQ, RES	<int>	Position of an IP address in the list **Note** **IPType**, **IPIndex**, and **Enable** must be sent together for the **update** action.
	IPType	REQ, RES	<enum> IPv4, IPv6	IP type The IP type is either IPv4 or IPv6. **Note** **IPType**, **Address**, and **Enable** parameters must be sent together for the **add** action.
	Address	REQ, RES	<string>	IP address to be configured for device access IPv4 and IPv6 are both allowed.
	Mask	REQ, RES	<int>	Netmask for the IP address
	Enable	REQ, RES	<bool> True, False	Whether to apply IPv4 or IPv6 address filtering
remove	IPType	REQ	<enum> IPv4, IPv6	IP type that is to be removed. **Note** **IPType** and **IPIndex** must be sent together for the **remove** action.
	IPIndex	REQ	<int>	IP Index that is to be removed.

88.4. Examples

88.4.1. Getting the current IP filtering settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=ipfilter&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
"ipfilter.IPType.IPIndex=Address/Mask/Enable"
AccessType=Deny
ipfilter.IPv4.1=192.168.75.135/32/False
ipfilter.IPv6.1=2001:1:1:1:1:1:1:1/128/True
```

JSON RESPONSE (For Camera)

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AccessType": "Deny",
    "IPFilters": [
        {
            "IPType": "IPv4",
            "Filters": [
                {
                    "IPIndex": 1,
                    "Address": "192.168.75.135",
                    "Mask": 32,
                    "Enable": false
                }
            ]
        },
        {
            "IPType": "IPv6",
            "Filters": [
                {
                    "IPIndex": 1,
                    "Address": "2001:1:1:1:1:1:1:1",
                    "Mask": 128,
                    "Enable": true
                }
            ]
        }
    ]
}
```

JSON RESPONSE (For NVR)

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "IPFilters": [
        {
            "IPType": "IPv4",
            "AccessType": "Deny",
            "Filters": [
                {
                    "IPIndex": 1,
                    "Address": "192.168.75.135",
                    "Mask": 32,
                    "Enable": false
                }
            ]
        },
        {
            "IPType": "IPv6",
            "AccessType": "Allow",
            "Filters": [
                {
                    "IPIndex": 1,
                    "Address": "2001:1:1:1:1:1:1:1",
                    "Mask": 128,
                    "Enable": true
                }
            ]
        }
    ]
}
```


88.4.2. Setting an IPv6 address and enabling filtering

To add the IP address to be filtered with the **add** action, the **IPType**, **Address**, and **Enable** parameters must be set.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=ipfilter&action=add&IPType=IPv6&Address=fe80::2091:18ff:fe71:1111&Mask=32&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
IPIndex=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

88.4.3. Updating an IPv6 address and disabling filtering

To modify a filtered address via the **update** action, the **IPType**, **IPIndex**, and **Enable** parameters must be set.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=ipfilter&action=update&IPIndex=1&IPType=IPv6&Address=fe80::2091:18ff:fe71:1111&Mask=32&Enable=False
```

88.4.4. Deleting an IPv4 address from IP filtering list

To remove a filtered address with the **remove** action, the **IPType** and **IPIndex** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=ipfilter&action=remove&IPIndex=1&IPType=IPv4
```

Chapter 89. 802.1x Setup

89.1. Description

The **802Dot1x** submenu requests and configures the parameters required for accessing an 802.1x protected network.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User
install	Admin	User
remove	Admin	User

89.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
802Dot1x&action=<value>[&<parameter>=<value>...]
```

89.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the 802.1x settings
	Status	RES	<enum> Unauthorized, Authorized, Stopped	Status (read-only)
	CACertificateInstalled	RES	<bool> True, False	Whether or not a CA certificate is installed
	ClientCertificateInstalled	RES	<bool> True, False	Whether or not a public certificate containing the client certificate key is installed
	ClientPrivateKeyInstalled	RES	<bool> True, False	Whether or not a public certificate containing the client private key is installed
	IsPasswordSet	RES	<bool> True, False	Whether or not the EAPOL password is set
set	InterfaceName	REQ, RES	<string>	Interface name (read-only for network cameras)

Action	Parameter	Request/ Response	Type/ Value	Description
	Enable	REQ, RES	<bool> True, False	Whether to activate 802.1x mode
	EAPOLVersion	REQ, RES	<enum> 1, 2	EAPOL (Extensible Authentication Protocol over LAN) version
	EAPOLId	REQ, RES	<string>	EAPOL ID
	EAPOLPassword	REQ, RES	<string>	EAPOL password
	IsEAPOLPasswordEncrypted	REQ	<bool> True, False	If set to true, EAPOLPassword is encrypted using the public key provided by the **rsa** submenu of **security**.cgi and sent as post payload. Refer to the Application Programmer's Guide.
	ClientCertificateInUse	REQ, RES	<string>	Set or get the client certificate to (in) use
	CACertificateInUse	REQ,RES	<string>	Set or get the CA certificate to (in) use
	EAPOLType	REQ, RES	<enum> EAP-TLS, LEAP	EAPOL type
	MACsecMode	REQ, RES	<enum> StaticCAK, DynamicCAK, None	Change the MACsec mode configuration.
	ConnectivityAssociationKey	REQ, RES	<string>	Configure connectivity association key
	ConnectivityAssociationKeyName	REQ, RES	<string>	Configure connectivity association keyname
install	InterfaceName	REQ	<string>	Interface name NVR ONLY
remove	CertificateType	REQ	<enum> CACertificate, ClientCertificate, ClientPrivateKey	Certificate type **Note** **CertificateType** must be sent together with the **remove** action
	InterfaceName	REQ	<string>	Interface name NVR ONLY

89.4. Examples

89.4.1. Getting the current 802.1x settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=802Dot1x&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
InterfaceName=1a5a97d2-464a-4222-91c6-140ff36b82b6
Enable=False
EAPOLType=EAP-TLS
Status=Stopped
CACertificateInstalled=False
ClientCertificateInstalled=False
ClientPrivateKeyInstalled=False
ClientCertificateInUse=
CACertificateInUse=
EAPOLVersion=1
EAPOLId=test
EAPOLPassword=
IsPasswordSet=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "IEEE802Dot1x": [
        {
            "InterfaceName": "NetworkInterface1",
            "Enable": false,
            "Status": "Stopped",
            "CACertificateInstalled": false,
```

```
            "ClientCertificateInstalled": false,
            "ClientPrivateKeyInstalled": false,
            "ClientCertificateInUse": "",
            "CACertificateInUse": "",
            "EAPOLVersion": "1",
            "EAPOLId": "",
            "EAPOLPassword": "",
            "EAPOLType": "EAP-TLS",
            "IsPasswordSet": false
        }
    ]
}
```

89.4.2. Setting the EAPOL

Setting the EAPOL version to ‘1’, the EAPOL ID to ‘test’ and the EAPOL password to ‘123’

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=802Dot1x&action=set&EAPOLVersion=1&EAPOLId=test&EAPOLPassword=123
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=802Dot1x&action=set&InterfaceName=Network2&EAPOLVersion=1&EAPOLId=test&EAPOLPassword=123
```

89.4.3. Installing an 802.1x certificate

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=802Dot1x&action=install
```

When requesting installation of an 802.1x certificate, data should be sent via the POST method in the following format.

The certtype value is the CACertificate, ClientCertificate or ClientPrivateKey. The certlength value is the data size of the certificate. The certdata value is certificate data.

```
<SetData802Dot1x>
    <PublicCertType>certtype</PublicCertType>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
</SetData802Dot1x>
```

The following XML is for camera models supporting PKCS12 certificate installation but **cacertificate** submenu is not supported. (Check first attribute value **Security/Support/SupportCertificateFormat**.) When the **PublicCertType** is set to ClientCertificatePKCS12, then one more tag **PassPhrase** should be sent together. **PassPhrase** is used to decrypt the certificate file.

```
<SetData802Dot1x>
    <PublicCertType>certtype</PublicCertType>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
    <PassPhrase>passphrase</PassPhrase>
</SetData802Dot1x>
```

(The following example is for NVR only.)

```
<SetData802Dot1x>
    <CertificateName>certname</CertificateName>
    <PublicCertType>certtype</PublicCertType>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
</SetData802Dot1x>
```

NOTE Now ssl and cacertificate handles all the certificates used in a device. So we recommend using those submenus to install new certificates and just use **ClientCertificateInUse** and **CACertificateInUse** parameter to set the certificates that you want to use.

CURL command

802.1x certificate install can be tested with CURL as below. To learn about CURL, please refer to http://curl.haxx.se.

NOTE To get a JSON response add the -H with CURL as shown below in the header of the request.

```
curl -v --digest -u <userid>:<password> --data-urlencode @802dot1x.xml
"http://<Device IP>/stw-cgi/security.cgi?msubmenu=802Dot1x&action=install"
```

```
-H "Expect:"
```

(The following example is for NVR only.)

```
curl --digest -u <userid>:<password> "http://<Device IP>/stw-
cgi/security.cgi?msubmenu=802Dot1x&action=install&InterfaceName=Network2" -H
"Expect:" --data-urlencode @802dot1x.xml
```

The above command will produce a request to the device as below:

```
POST /stw-cgi/security.cgi?msubmenu=802Dot1x&action=install HTTP/1.1
Content-Length: 1985
Content-Type: application/x-www-form-urlencoded
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
{
    "Response": "Success"
}
```

89.4.4. Deleting a certificate

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=802Dot1x&action=remove&CertificateType=CACertifica
te
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=802Dot1x&action=remove&InterfaceName=Network2&Cert
ificateType=CACertificate
```

Chapter 90. RTSP Authentication

90.1. Description

The **rtsp** submenu selects the RTSP authentication method.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User

90.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
rtsp&action=<value>[&<parameter>=<value>...]
```

90.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the RTSP authentication settings.
	RTSPAuthentication	RES	<enum> Anonymous, Protected	Whether to allow anonymous access without login for RTSP URL requests
set	RTSPAuthentication	REQ, RES	<enum> Anonymous, Protected	Whether to allow anonymous access without logging in for RTSP URL requests (read-only for NVR) **Note** **RTSPAuthentication** must be sent together with the **set** action.

90.4. Examples

90.4.1. Getting the current RTSP authentication settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=rtsp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
RTSPAuthentication=Protected
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "RTSPAuthentication": "Protected"
}
```

90.4.2. Enabling anonymous RTSP authentication

To use the **set** action, the **RTSPAuthentication** parameter must be set at the same time.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=rtsp&action=set&RTSPAuthentication=Anonymous
```

Chapter 91. SSL (HTTPS) Settings

91.1. Description

The **ssl** submenu configures the SSL (Secure Sockets Layer) settings.

Access level

Action	Camera	NVR	IPAS
view	Admin	User	Admin
set	Admin	User	
install	Admin	User	
remove	Admin	User	

91.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
ssl&action=<value>[&<parameter>=<value>...]
```

91.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SSL (HTTPS) settings
	PublicCertificateInstalled	RES	<bool> True, False	Whether or not a public certificate is installed (read-only)
	PublicCertificateName	RES	<csv>	Public certificate name
	UpdateDeviceHostname	RES	<bool>	Whether device hostname is updated
	CertificateInUse	RES	<string>	Name of certificate currently in use
	Certificate.#.CertificateName	RES	<string>	Certificate name
	Certificate.#.Type	RES	<enum> Unique, SelfSigned, Public	Certificate type
	Certificate.#.Subject	RES	<string>	Subject of the certificate
	Certificate.#.SubjectAlternativeName	RES	<string>	Subject alternative name (SAN)

Action	Parameter	Request/ Response	Type/ Value	Description
	Certificate.#.Issuer	RES	<string>	Issuer
	Certificate.#.IssueDate	RES	<string>	Issued date
	Certificate.#.ExpiryDate	RES	<string>	Expiry date
	Certificate.#.Version	RES	<string>	Version
	Certificate.#.SerialNumber	RES	<string>	Serial number
	Certificate.#.Signature	RES	<string>	Signature
	Certificate.#.Thumbprint	RES	<string>	Thumbprint
	Certificate.#.IsRemovable	RES	<bool>	Whether the certificate can be deleted
	Certificate.#.IsEncrypted	RES	<bool>	Whether the certificate is encrypted
add	CertificateName	REQ	<string>	The name of the certificate
	Type	REQ	<enum> SelfSigned	The type of the certificate • SelfSigned: Access using a built-in certificate from the device
	CommonName	REQ	<string>	Common name (CN)
	SubjectAlternativeName	REQ	<string>	Subject alternative name (SAN)
	ExpiryDate	REQ	<string>	Certificate expiry date
	Country	REQ	<string>	Country © **Note** ISO-3166-1 alpha-2 codes
	Province	REQ	<string>	Province (ST)
	Location	REQ	<string>	Location (L)
	Organization	REQ	<string>	Organization (O)
	Division	REQ	<string>	Division (OU)

Action	Parameter	Request/ Response	Type/ Value	Description
	EmailID	REQ	<string>	E-mail address **Note** Multiple e-mails are separated by commas
set	Policy	REQ, RES	<enum> HTTP, HTTPSProprietary, HTTPSPublic, HTTPandHTTPSProprietary, HTTPandHTTPSPublic	Select the SSL method: • HTTP: Access using only HTTP • HTTPSProprietary: Access using a built-in certificate in the device • HTTPSPublic: Access using a certificate that the user has installed (HTTPSPublic mode is valid only when the certificate is installed) • HTTPandHTTPSProprietary: HTTP and HTTPSProprietary mode • HTTPandHTTPSPublic: HTTP and HTTPSPublic mode
	UpdateDeviceHostname	REQ	<bool> True, False	Whether the device hostname is updated
	CertificateInUse	REQ	<string>	Name of certificate currently in use
	ClientCertificateAuthenticationEnable	REQ, RES	<bool> True, False	Whether mutual authentication is enabled

Action	Parameter	Request/ Response	Type/ Value	Description
	ClientCertificateAuthenticationMode	REQ, RES	<enum> ALLOW_ALL_CERT, ALLOW_CERT_FROM_KNOWN_CA, ALLOW_CERT_FROM_VALID_CLIENT_AND_KNOWN_CA	Select the mode of certificate authentication • ALLOW_ALL_CERT: Allows all connections • ALLOW_CERT_FROM_KNOWN_CA: Allows only mutually authenticated connections • ALLOW_CERT_FROM_VALID_CLIENT_AND_KNOWN_CA: Allows only mutually authenticated connections (including Device ID authentication) **Note** Only works if **Policy** is HTTPSProprietary.
remove	CertificateName	REQ	<string>	Certificate name
install				POST method

91.4. Examples

91.4.1. Getting the current SSL settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=ssl&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Policy=HTTPandHTTPSProprietary
PublicCertificateInstalled=False
SelfSignedCertificateInstalled=True
PublicCertificateName=
UpdateDeviceHostname=False
CertificateInUse=HTW_default
```

```
ClientCertificateAuthenticationEnable=False
ClientCertificateAuthenticationMode=ALLOW_ALL_CERT
Certificate.1.CertificateName=HTW_default
Certificate.1.Type=Unique
Certificate.1.Subject=/C=KR/O=Hanwha Vision/OU=Security
Device/CN=00091867F9BC.hanwhavision.com/serialNumber=5f55694d-f741-41a6-
9c6f-ceb4b1ee124c
Certificate.1.SubjectAlternativeName=00091867F9BC.hanwhavision.com
Certificate.1.Issuer=/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2
Certificate.1.IssueDate=Dec 1 10:05:06 2020 GMT
Certificate.1.ExpiryDate=Nov 24 10:05:06 2050 GMT
Certificate.1.Version=V3
Certificate.1.SerialNumber=00 12 35 8F
Certificate.1.Signature=sha256WithRSAEncryption
Certificate.1.Thumbprint=e268cf387b1a19c5ea3c04be2f24a2053069d8742b9741f1d03
4937872ce8538
Certificate.1.IsRemovable=False
Certificate.1.IsEncrypted=False
Certificate.2.CertificateName=test
Certificate.2.Type=Public
Certificate.2.Subject=/C=FR/ST=Radius/O=Example
Inc./CN=user@example.org/emailAddress=user@example.org
Certificate.2.SubjectAlternativeName=-
Certificate.2.Issuer=/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority
Certificate.2.IssueDate=Jul 1 04:36:15 2021 GMT
Certificate.2.ExpiryDate=May 10 04:36:15 2031 GMT
Certificate.2.Version=V3
Certificate.2.SerialNumber=00
Certificate.2.Signature=sha256WithRSAEncryption
Certificate.2.Thumbprint=2b58473f4655282053630336ad25a84eae3e53155030a6fa10e
dc3576feec51d
Certificate.2.IsRemovable=True
Certificate.2.IsEncrypted=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Policy": "HTTPandHTTPSProprietary",
    "PublicCertificateInstalled": false,
    "SelfSignedCertificateInstalled": true,
    "PublicCertificateName": "",
    "UpdateDeviceHostname": false,
    "CertificateInUse": "HTW_default",
    "ClientCertificateAuthenticationEnable": false,
    "ClientCertificateAuthenticationMode": "ALLOW_ALL_CERT",
    "Certificates": [
        {
            "CertificateName": "HTW_default",
            "Type": "Unique",
            "Subject": "/C=KR/O=Hanwha Vision/OU=Security
Device/CN=00091867F9BC.hanwhavision.com/serialNumber=5f55694d-f741-41a6-
9c6f-ceb4b1ee124c",
            "SubjectAlternativeName": "00091867F9BC.hanwhavision.com",
            "Issuer": "/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2",
            "IssueDate": "Dec 1 10:05:06 2020 GMT",
            "ExpiryDate": "Nov 24 10:05:06 2050 GMT",
            "Version": "V3",
            "SerialNumber": "00 12 35 8F ",
            "Signature": "sha256WithRSAEncryption",
            "Thumbprint":
"e268cf387b1a19c5ea3c04be2f24a2053069d8742b9741f1d034937872ce8538",
            "IsRemovable": false,
            "IsEncrypted": false
        },
        {
            "CertificateName": "test",
            "Type": "Public",
            "Subject": "/C=FR/ST=Radius/O=Example
Inc./CN=user@example.org/emailAddress=user@example.org",
            "SubjectAlternativeName": "-",
            "Issuer": "/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority",
            "IssueDate": "Jul 1 04:36:15 2021 GMT",
            "ExpiryDate": "May 10 04:36:15 2031 GMT",
            "Version": "V3",
            "SerialNumber": "00",
```

```
                "Signature": "sha256WithRSAEncryption",
                "Thumbprint":
"2b58473f4655282053630336ad25a84eae3e53155030a6fa10edc3576feec51d",
                "IsRemovable": true,
                "IsEncrypted": true
            }
        ]
}
```

91.4.2. Enabling HTTPS

To use the **set** action, the **Policy** parameter must be set at the same time.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=ssl&action=set&Policy=HTTPSPublic
```

91.4.3. Installing a certificate

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=ssl&action=install
```

When requesting a certificate to be installed, data should be sent via the POST method in the following format.

The certname value is the certificate name. The certlength value is the certificate data size. The certdata value is the certificate data. The keylength is the key data size. The keydata value is the key data.

```
<SetHTTPSData>
    <PublicCertName>certname</PublicCertName>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
    <KeyLength>keylength</KeyLength>
    <KeyData>keydata</KeyData>
</SetHTTPSData>
```

The following XML is for camera models supporting PKCS12 certificate installation. Check first attribute value **Security/Support/SupportCertificateFormat**. When the **Type** tag is not included, the certificate installation request is treated as PKCS1,PCKS8(original request). **FilePassword** is used to decrypt the certificate file and **KeyPassword** is used to encrypt the certificate when it is stored to the device. If **AllPasswordEncrypted** is set to False, passwords are sent in simple plaintext. If it is True, passwords are encrypted using the public key provided by the **rsa** submenu of **security**.cgi.

```
<SetHTTPSData>
    <Type>PKCS12</Type>
    <PublicCertName>certname</PublicCertName>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
    <FilePassword>password</FilePassword>
    <KeyPassword>password</KeyPassword>
    <AllPasswordEncrypted>True</AllPasswordEncrypted>
    <KeyLength>keylength</KeyLength>
    <KeyData>keydata</KeyData>
</SetHTTPSData>
```

CURL command

Certificate installation can be tested with CURL as shown below. To learn about CURL, please refer to http://curl.haxx.se.

NOTE To get a JSON response, add the -H "Accept: application/json" header to the request.

```
curl -v --digest -u <userid>:<password> --data-urlencode @ssl.xml
"http://<Device IP>/stw-cgi/security.cgi?msubmenu=ssl&action=install" -H
"Expect:"
```

The above command will produce a request to the device that looks like below:

```
POST /stw-cgi/security.cgi?msubmenu=ssl&action=install HTTP/1.1
Content-Length: 3775
Content-Type: application/x-www-form-urlencoded
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
{
    "Response": "Success"
}
```

91.4.4. Add a self-signed certificate

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=ssl&action=add&CertificateName=certName&Type=SelfS
igned&CommonName=192.168.75.123&SubjectAlternativeName=domain.com,testdom.co
m&ExpiryDate=2021-09-
09&Country=KR&Province=Gyeonggi&Location=Bundang&Organization=Hanwha&Divisio
n=RSDT&EmailID=test@hanwha.com
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE


```
{
    "Response": "Success"
}
```


91.4.5. Use a specific certificate

To use a specific certificate, the **CertificateInUse** parameter must be set along with the **Policy** parameter.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=ssl&action=remove&CertificateInUse=certname&Policy
=HTTPandHTTPSProprietary
```

91.4.6. Deleting a certificate

To delete a certificate with the **remove** action, the **CertificateName** parameter must be set at the same time.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=ssl&action=remove&CertificateName=certname
```

Chapter 92. Guest User Login

92.1. Description

The **guest** submenu enables or disables guest logins.

NOTE This chapter applies to the network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

92.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
guest&action=<value>[&<parameter>=<value>...]
```

92.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the guest user login settings.
set	Enable	REQ, RES	<bool> True, False	Enables or disables guest logins. **Note** **Enable** must be sent together with the **set** action.

92.4. Examples

92.4.1. Getting the current guest login setting

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=guest&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false
}
```

92.4.2. Enabling guest login

To use the **set** action, the **Enable** parameter must be set at the same time.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=guest&action=set&Enable=True
```

Chapter 93. User Configuration

93.1. Description

The **users** submenu adds and deletes system users and sets access permissions.

Access level

Action	Camera	NVR	IPAS
view	User	User	User
add	Admin	Admin	
update	Admin	Admin	Admin
remove	Admin	Admin	
check	Admin		

93.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
users&action=<value>[&<parameter>=<value>...]
```

93.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	UserID	REQ	<string>	User ID 'admin' cannot be used as the **UserID** value.
add, update	Index	REQ, RES	<int>	Index of the registered user account. **Note** **Index**, **UserID**, and **Password** must be sent together for the **update** action.

Action	Parameter	Request/ Response	Type/ Value	Description
	UserID	REQ, RES	<string>	User ID **Note** **UserID** and **Password** must be sent together with the **add** action. In **IPAS** devices, UserID is fixed and cannot be changed. Only following userids are supported admin,setup, user and guest.
	UserName	REQ, RES	<string>	User name This parameter is for users only; the admin user name cannot be changed. NVR ONLY
	Enable	REQ, RES	<bool> True, False	Whether to activate or deactivate the user account. CAMERA ONLY
	VideoProfileAccess	REQ, RES	<bool> True, False	Whether to grant video profile permission. If **VideoProfileAccess** is set to False, only the default profile will be allowed when accessing video. If **VideoProfileAccess** is set to True, all profiles are allowed when accessing video. This parameter can't be false, if "VideoProfileSettingAccess" is set to true. CAMERA ONLY
	PTZAccess	REQ, RES	<bool> True, False	Whether to grant PTZ control permission. For PTZ models,the values for **PTZAccess**, **AlarmOutputAccess**, and **AudioOutAccess** should be the same. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	AudioInAccess	REQ, RES	<bool> True, False	Whether to grant audio input control permission. CAMERA ONLY IPAS ONLY
	AudioOutAccess	REQ, RES	<bool> True, False	Whether to grant audio output control permission. For PTZ models, the values for **PTZAccess**, **AlarmOutputAccess**, and **AudioOutAccess** should be the same. CAMERA ONLY IPAS ONLY
	AlarmOutputAccess	REQ, RES	<bool> True, False	Whether to grant alarm output control permission. For PTZ models, values for **PTZAccess**, **AlarmOutputAccess**, and **AudioOutAccess** should be the same. CAMERA ONLY IPAS ONLY
	ViewerAccess	REQ, RES	<bool> True, False	Whether to grant access permission to a viewer **ViewerAccess** is valid only when UserID is NOT set to Admin. NVR ONLY
	AdminAccess	REQ, RES	<bool> True, False	Whether or not to grant admin permission. If this parameter is True, it means that user has full access rights (e.g. administrator). CAMERA ONLY IPAS ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	PrivacyAreaAccess	REQ, RES	<bool> True, False	Whether to grant access permission to create/remove privacy areas If this parameter is True, the user can create a new privacy area or configure one. PTZAccess will be enbled automatically when the device supports PTZ. This parameter can’t be false, if "VideoSetupAccess" is set to true. CAMERA ONLY
	Password	REQ, RES	<string>	User password
	IsPasswordEncrypted	REQ	<bool> True, False	When set to true, password is encrypted using the public key provided by the **rsa** submenu of **security**.cgi, and sent as post payload. Refer to the Application Programmer’s Guide.
	GroupID	REQ, RES	<string>	Group ID **GroupID** is invalid, if **UserLevel** is set to Admin, NVR ONLY
	UserLevel	RES	<enum> Guest, Admin, User	User level If **UserLevel** is set to Admin for NVR, **GroupID** is invalid. NVR ONLY
	VideoProfileSettingAccess	REQ, RES	<bool> True,False	It allows user to set/add/update/control/remove below submenus. "videosource","videoprofile","videoprofilepolicy". And it also allows "VideoProfileAccess". If this parameter is set to true, "VideoProfileAccess" automatically becomes true.

Action	Parameter	Request/ Response	Type/ Value	Description
	ImageSettingAccess	REQ, RES	<bool> True,False	It allows user to set/add/update/control/remove below submenus. "camera","presetimageconfig","videosource","ssdr","imagepreset2","whitebalance","imagepresetschedule","imageenhancements","imageenhancements2","lpcapturesetup","imagecgi-scheduler","irled","overlay","multiimageosd","multilineosd","ptzcontrolcgi-aux"
	VideoSetupAccess	REQ, RES	<bool> True,False	It allows user to set/add/update/control/remove below submenus. "flip","usbconfig","videooutput","videoinput","fisheyesetup" And it also allows "PTZAccess", "PrivacyAreaAccess". If this parameter is set to true, "PrivacyAreaAccess", "PTZAccess" automatically becomes true.
	FocusSetupAccess	REQ, RES	<bool> True,False	It allows user to set/add/update/control/remove below submenus. "focuspreset","focus"
	PlayBackAccess	REQ, RES	<bool> True,False	It allows user to set/add/update/control/remove below submenus view action. storageinfo(view),sdcardinfo(view),overlapped(view),timeline(view),calendarsearch(view)
	LogAccess	REQ, RES	<bool> True,False	It allows user to use below submenu. systemlog(view), eventlog(view), accesslog(view), logserver(view, update, add, remove)
	OpenAPPAccess	REQ, RES	<bool> True,False	It allows user to use below submenu. apps(view), appstatus, dockerstatus, appmetadatacount

Action	Parameter	Request/ Response	Type/ Value	Description
remove	Index	REQ	<int>	Index number of the user that is to be removed **Note** Either **Index** or **UserID** must be sent together with the **remove** action. CAMERA ONLY
	UserID	REQ	<string>	User ID that is to be removed CAMERA ONLY
check	CurrentUserCount	RES	<int>	It shows how many current users are there. CAMERA ONLY
	MaxUserCount	RES	<int>	It indicates maximum user count that can be created. CAMERA ONLY

> **Note**
> To find out the max users supported by the device, refer to the Attributes/Security/Limit/MaxUser attribute in the device attributes section.

93.4. Examples

93.4.1. Getting current users settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
"Users.Index=UserID/Password/Enable/VideoProfileAccess/PTZAccess/AudioInAcce
ss/AudioOutAccess/AlarmOutputAccess/AdminAccess/PrivacyAreaAccess/OpenAPPAcc
ess"
Users.0=admin//True/True/True/True/True/True/True/True/True
Users.1=user1//False/False/False/False/False/False/False/False/False
Users.2=user2//False/False/False/False/False/False/False/False/False
```

```
Users.3=user3//False/False/False/False/False/False/False/False/False
Users.4=user4//False/False/False/False/False/False/False/False/False
Users.5=user5//False/False/False/False/False/False/False/False/False
Users.6=user6//False/False/False/False/False/False/False/False/False
Users.7=user7//False/False/False/False/False/False/False/False/False
Users.8=user8//False/False/False/False/False/False/False/False/False
Users.9=user9//False/False/False/False/False/False/False/False/False
Users.10=user10//False/False/False/False/False/False/False/False/False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Users": [
        {
            "Index": 0,
            "UserID": "admin",
            "Password": "",
            "Enable": true,
            "VideoProfileAccess": true,
            "PTZAccess": true,
            "AudioInAccess": true,
            "AudioOutAccess": true,
            "AlarmOutputAccess": true,
            "AdminAccess": true,
            "PrivacyAreaAccess": true,
            "OpenAPPAccess": true
        },
        {
            "Index": 1,
            "UserID": "user1",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
```

```
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 2,
            "UserID": "user2",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 3,
            "UserID": "user3",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 4,
            "UserID": "user4",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
```

```
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 5,
            "UserID": "user5",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 6,
            "UserID": "user6",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 7,
            "UserID": "user7",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
```

```
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 8,
            "UserID": "user8",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 9,
            "UserID": "user9",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        },
        {
            "Index": 10,
            "UserID": "user10",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
```

```
                "AudioInAccess": false,
                "AudioOutAccess": false,
                "AlarmOutputAccess": false,
                "AdminAccess": false,
                "PrivacyAreaAccess": false,
                "OpenAPPAccess": false
            }
        ]
}
```

The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
"Users.Index=UserID/UserName/Password/GroupID/UserLevel/ViewerAccess"
Users.0=admin/admin///Admin/True
Users.1=tarak/tarak//Group 1/User/True
Users.2=vmsuser/vmsuser//Group 2/User/True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Users": [
        {
            "Index": 0,
            "UserID": "admin",
            "UserName": "admin",
            "Password": "",
            "GroupID": "",
            "ViewerAccess": true,
            "UserLevel": "Admin"
        },
```

```
        {
            "Index": 1,
            "UserID": "tarak",
            "UserName": "tarak",
            "Password": "",
            "GroupID": "Group 1",
            "ViewerAccess": true,
            "UserLevel": "User"
        },
        {
            "Index": 2,
            "UserID": "vmsuser",
            "UserName": "vmsuser",
            "Password": "",
            "GroupID": "Group 2",
            "ViewerAccess": true,
            "UserLevel": "User"
        }
    ]
}
```

93.4.2. Getting ‘user1’

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=users&action=view&UserID=user1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
"Users.Index=UserID/Password/Enable/VideoProfileAccess/PTZAccess/AudioInAcce
ss/AudioOutAccess/AlarmOutputAccess/AdminAccess/PrivacyAreaAccess/OpenAPPAcc
ess"
Users.1=user1//False/False/False/False/False/False/False/False/False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Users": [
        {
            "Index": 1,
            "UserID": "user1",
            "Password": "",
            "Enable": false,
            "VideoProfileAccess": false,
            "PTZAccess": false,
            "AudioInAccess": false,
            "AudioOutAccess": false,
            "AlarmOutputAccess": false,
            "AdminAccess": false,
            "PrivacyAreaAccess": false,
            "OpenAPPAccess": false
        }
    ]
}
```

93.4.3. Adding a user

This is an example in which a user is added that has permission to access all video profiles but without PTZ control.

To add a user, the **UserID** and **Password** parameters must be set together.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=add&UserID=user2&Password={password}&VideoProfileAccess=True&PTZAccess=False
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
OK
Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

The following example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=add&UserId=user1&UserName=username1&Password={password}&GroupId=group1&UserLevel=User&ViewerAccess=False
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "Response": "Success"
}
```

93.4.4. Updating 'user2' to give permission for PTZ control

To update user information for the cameras, the **UserID**, **Index**, and **Password** parameters must be sent together.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=update&UserID=user2&Index=2&Password={password}&VideoProfileAccess=True&PTZAccess=True
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=update&UserId=admin&Password={password}
```

93.4.5. Updating password

When a user (including admin users) wants to change their current password, it's recommended to use an encrypted password using the RSA PublicKey retrievable from the **rsa** submenu.

if below attribute is supported, current password verification is supported. Value can be 'Optional' or 'Mandatory', if its optional current password verification is optional, if its mandatory only below JSON format can be used for changing password.

```
<attribute name="CurrentPasswordVerification" type="enum" value="Optional" accesslevel="suser"/>
```

JSON FORMAT (This should be encrypted using the RSA public key and base64 encoding.)

```
{
"CurrentPassword":"q1w2e3r4t5!!",
"NewPassword":"qwerty99!!"
}
```

The post body is sent in the below format,
Base64(RSAEncrypt(JsonString))

If above attribute is not supported, only new password can be encrypted using the RSA public key, (example if new password is "qwerty99!!" then only this password needs to be encrypted)

The post body is sent in the below format,
Base64(RSAEncrypt(newpassword))

REQUEST

```
http://192.168.75.196/stw-cgi/security.cgi?msubmenu=users&action=update&UserID=user1&Index=1&IsPasswordEncrypted=True
```

POST BODY


```
gH3EsB2O9IfcNY02fQLyOMYA//3ES1nGDT2UUInzN51i0Bw849tuJsdASRWrJt5P+oC1mH05vt2W7VqICpTZSK4bg578nonpBbv3uTtsTzMyqDrK71hNsblcISZSDTYLa61IeawC88X1//0UTsIHUtT4XKaiEEAyMFXo7E12dj3itf1ySj//Emo7zq321bRqE3ELOxKQnNFJi7DQf92gHcCN3Tr1WosK3vIAuZNiOD2txsEIBPxWdS+4RGZNAKzDy62yrJaoIe1lvU3xLA94KH7mGgtsebdXx1X6xay0wTyEtBf8/TSt2in/PHKx8ZGJxqTq0eSZDsf40nFXkl5K7w==
```


If an admin is changing the password of a normal user, verifying the current password is not required. Only the password string can be encrypted using the RSA public key and be sent in the POST body.

> **NOTE** Plain text password updates will soon be deprecated.

93.4.6. Deleting ‘user2’

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=remove&UserID=user2
```

Deleting ‘user2’ via the user’s index number

Assuming the 'user2' index is '3', user2 can be deleted by using the index number.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=remove&Index=3
```

93.4.7. Getting current users count & maximum users count

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "CurrentUserCount": 10,
    "MaxUserCount": 10
}
```

Chapter 94. User Group Configuration

94.1. Description

The **usergroups** submenu adds and deletes system user groups and sets access permissions for the user groups.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
add, update	Admin
remove	Admin

94.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
usergroups&action=<value>[&<parameter>=<value>...]
```

94.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	UserGroupID	REQ	<string>	User group ID
add, update	Index	REQ, RES	<int>	User group index **Note** **Index** must be sent for the **update** action.
	UserGroupID	REQ, RES	<string>	User group ID
	LiveChannel	REQ, RES	<csv> #, None	Live channel • None: Disables all channels
	SearchChannel	REQ, RES	<csv> #, None	Search channel • None: Disables all channels
	BackupChannel	REQ, RES	<csv> #, None	Backup channel • None: Disables all channels

Action	Parameter	Request/ Response	Type/ Value	Description
	PTZAccess	REQ, RES	<bool> True, False	Whether to grant permission for PTZ control **PTZAccess** is valid only when **LiveChannel** is enabled.
	RecordStartAccess	REQ, RES	<bool> True, False	Whether to grant access permission to start recording
	RecordStopAccess	REQ, RES	<bool> True, False	Whether to grant access permission to stop recording
	AlarmOutputAccess	REQ, RES	<bool> True, False	Whether to grant permission to access the alarm output
	ShutdownAccess	REQ, RES	<bool> True, False	Whether to grant permission to access shutdown
	MenuAccess	REQ, RES	<csv> System, Device, Record, Event, Network, None	Accessible menu
	SystemMenuAccess	REQ, RES	<csv> DateTimeLanguage, SystemManagement, SystemLog, EventLog, BackupLog, Holiday, None	Accessible system menu
	DeviceMenuAccess	REQ, RES	<csv> CameraRegistration, CameraSetup, LiveSetup, ChannelSetup, DeviceFormat, iSCSI, RAID, HDDAlarm, Monitor, POSConf, POSEventConf, None	Accessible device menu
	RecordMenuAccess	REQ, RES	<csv> RecordingSchedule, NvrRecordSetup, NetCamRecordSetup, RecordOption, None	Accessible record menu

Action	Parameter	Request/ Response	Type/ Value	Description
	EventMenuAccess	REQ, RES	<csv> NvrSensorDetection, NetCamSensorDetection, NvrEventDetection, NetCamEventDetection, VideoLossDetection, AlarmSchedule, Gsensor, None	Accessible event menu
	NetworkMenuAccess	REQ, RES	<csv> NetworkInterface, NetworkPort, DDNS, IPFilter, SSL, 802.1x, LiveStreaming, SMTP, EventMail, GroupAndRecipientEmail, SNMP, DHCPServer,MTS, None	Accessible network menu
remove	Index	REQ	<int>	User group index **Note** Either **Index** or **UserGroupID** must be sent together with the **remove** action.
	UserGroupID	REQ	<string>	User group ID that is to be removed

94.4. Examples

94.4.1. Getting current user group settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=usergroups&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Index.1.UserGroupID=Group 1
Index.1.LiveChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,
22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,4
7,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.SearchChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,2
1,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46
,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.BackupChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,2
1,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46
,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.MenuAccess=System,Device,Record,Event,Network
Index.1.SystemMenuAccess=DateTimeLanguage,SystemManagement,SystemLog,BackupL
og,EventLog
Index.1.DeviceMenuAccess=CameraRegistration,CameraSetup,LiveSetup,ChannelSet
up,DeviceFormat,HDDAlarm,iSCSI,RAID,Monitor
Index.1.RecordMenuAccess=RecordingSchedule,NvrRecordSetup,NetCamRecordSetup,
RecordOption
Index.1.EventMenuAccess=NvrSensorDetection,NetCamSensorDetection,NvrEventDet
ection,NetCamEventDetection,VideoLossDetection,AlarmSchedule
Index.1.NetworkMenuAccess=NetworkInterface,NetworkPort,DDNS,IPFilter,SSL,802
.1x,LiveStreaming,SMTP,EventMail,GroupAndRecipientEmail,SNMP,DHCPServer
Index.1.RecordStartAccess=True
Index.1.RecordStopAccess=True
Index.1.PTZAccess=True
Index.1.AlarmOutputAccess=True
Index.1.ShutdownAccess=True
Index.2.UserGroupID=Group 2
Index.2.LiveChannel=None
Index.2.SearchChannel=None
Index.2.BackupChannel=None
Index.2.MenuAccess=None
Index.2.SystemMenuAccess=None
Index.2.DeviceMenuAccess=None
Index.2.RecordMenuAccess=None
Index.2.EventMenuAccess=None
Index.2.NetworkMenuAccess=None
Index.2.RecordStartAccess=False
Index.2.RecordStopAccess=False
Index.2.PTZAccess=False
Index.2.AlarmOutputAccess=False
```

```
Index.2.ShutdownAccess=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "UserGroups": [
        {
            "Index": 1,
            "UserGroupID": "Group 1",
            "LiveChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
                "10", "11", "12", "13", "14", "15", "16", "17", "18", "19",
                "20", "21", "22", "23", "24", "25", "26", "27", "28", "29",
                "30", "31", "32", "33", "34", "35", "36", "37", "38", "39",
                "40", "41", "42", "43", "44", "45", "46", "47", "48", "49",
                "50", "51", "52", "53", "54", "55", "56", "57", "58", "59",
                "60", "61", "62", "63"
            ],
            "SearchChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
                "10", "11", "12", "13", "14", "15", "16", "17", "18", "19",
                "20", "21", "22", "23", "24", "25", "26", "27", "28", "29",
                "30", "31", "32", "33", "34", "35", "36", "37", "38", "39",
                "40", "41", "42", "43", "44", "45", "46", "47", "48", "49",
                "50", "51", "52", "53", "54", "55", "56", "57", "58", "59",
                "60", "61", "62", "63"
            ],
            "BackupChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
                "10", "11", "12", "13", "14", "15", "16", "17", "18", "19",
                "20", "21", "22", "23", "24", "25", "26", "27", "28", "29",
                "30", "31", "32", "33", "34", "35", "36", "37", "38", "39",
                "40", "41", "42", "43", "44", "45", "46", "47", "48", "49",
                "50", "51", "52", "53", "54", "55", "56", "57", "58", "59",
                "60", "61", "62", "63"
            ],
            "MenuAccess": [
```

```
                "System",
                "Device",
                "Record",
                "Event",
                "Network"
            ],
            "SystemMenuAccess": [
                "DateTimeLanguage",
                "SystemManagement",
                "SystemLog",
                "BackupLog",
                "EventLog"
            ],
            "DeviceMenuAccess": [
                "CameraRegistration",
                "CameraSetup",
                "LiveSetup",
                "ChannelSetup",
                "DeviceFormat",
                "HDDAlarm",
                "iSCSI",
                "RAID",
                "Monitor"
            ],
            "RecordMenuAccess": [
                "RecordingSchedule",
                "NvrRecordSetup",
                "NetCamRecordSetup",
                "RecordOption"
            ],
            "EventMenuAccess": [
                "NvrSensorDetection",
                "NetCamSensorDetection",
                "NvrEventDetection",
                "NetCamEventDetection",
                "VideoLossDetection",
                "AlarmSchedule"
            ],
            "NetworkMenuAccess": [
                "NetworkInterface",
                "NetworkPort",
```

```
                "DDNS",
                "IPFilter",
                "SSL",
                "802.1x",
                "LiveStreaming",
                "SMTP",
                "EventMail",
                "GroupAndRecipientEmail",
                "SNMP",
                "DHCPServer"
            ],
            "RecordStartAccess": true,
            "RecordStopAccess": true,
            "PTZAccess": true,
            "AlarmOutputAccess": true,
            "ShutdownAccess": true
        },
        {
            "Index": 2,
            "UserGroupID": "Group 2",
            "LiveChannel": [
                "None"
            ],
            "SearchChannel": [
                "None"
            ],
            "BackupChannel": [
                "None"
            ],
            "MenuAccess": [
                "None"
            ],
            "SystemMenuAccess": [
                "None"
            ],
            "DeviceMenuAccess": [
                "None"
            ],
            "RecordMenuAccess": [
                "None"
            ],
```

```
            "EventMenuAccess": [
                "None"
            ],
            "NetworkMenuAccess": [
                "None"
            ],
            "RecordStartAccess": false,
            "RecordStopAccess": false,
            "PTZAccess": false,
            "AlarmOutputAccess": false,
            "ShutdownAccess": false
        }
    ]
}
```

94.4.2. Getting ‘Group 1’ user group settings

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=usergroups&action=view&UserGroupID=Group 1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Index.1.UserGroupID=Group 1
Index.1.LiveChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,
22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,4
7,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.SearchChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,2
1,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46
,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.BackupChannel=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,2
1,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46
,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Index.1.MenuAccess=System,Device,Record,Event,Network
Index.1.SystemMenuAccess=DateTimeLanguage,SystemManagement,SystemLog,BackupL
og,EventLog
```

```
Index.1.DeviceMenuAccess=CameraRegistration,CameraSetup,LiveSetup,ChannelSet
up,DeviceFormat,HDDAlarm,iSCSI,RAID,Monitor
Index.1.RecordMenuAccess=RecordingSchedule,NvrRecordSetup,NetCamRecordSetup,
RecordOption
Index.1.EventMenuAccess=NvrSensorDetection,NetCamSensorDetection,NvrEventDet
ection,NetCamEventDetection,VideoLossDetection,AlarmSchedule
Index.1.NetworkMenuAccess=NetworkInterface,NetworkPort,DDNS,IPFilter,SSL,802
.1x,LiveStreaming,SMTP,EventMail,GroupAndRecipientEmail,SNMP,DHCPServer
Index.1.RecordStartAccess=True
Index.1.RecordStopAccess=True
Index.1.PTZAccess=True
Index.1.AlarmOutputAccess=True
Index.1.ShutdownAccess=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "UserGroups": [
        {
            "Index": 1,
            "UserGroupID": "Group 1",
            "LiveChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23",
                "24", "25", "26", "27", "28", "29", "30", "31", "32", "33",
"34",
                "35", "36", "37", "38", "39", "40", "41", "42", "43", "44",
"45",
                "46", "47","48", "49", "50", "51", "52", "53", "54", "55",
"56",
                "57", "58", "59", "60", "61", "62", "63"
            ],
            "SearchChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
```

```
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23",
                "24", "25", "26", "27", "28", "29", "30", "31", "32", "33",
"34",
                "35", "36", "37", "38", "39", "40", "41", "42", "43", "44",
"45",
                "46", "47","48", "49", "50", "51", "52", "53", "54", "55",
"56",
                "57", "58", "59", "60", "61", "62", "63"
            ],
            "BackupChannel": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23",
                "24", "25", "26", "27", "28", "29", "30", "31", "32", "33",
"34",
                "35", "36", "37", "38", "39", "40", "41", "42", "43", "44",
"45",
                "46", "47","48", "49", "50", "51", "52", "53", "54", "55",
"56",
                "57", "58", "59", "60", "61", "62", "63"
            ],
            "MenuAccess": [
                "System",
                "Device",
                "Record",
                "Event",
                "Network"
            ],
            "SystemMenuAccess": [
                "DateTimeLanguage",
                "SystemManagement",
                "SystemLog",
                "BackupLog",
                "EventLog"
            ],
            "DeviceMenuAccess": [
                "CameraRegistration",
                "CameraSetup",
                "LiveSetup",
```

```
            "ChannelSetup",
            "DeviceFormat",
            "HDDAlarm",
            "iSCSI",
            "RAID",
            "Monitor"
        ],
        "RecordMenuAccess": [
            "RecordingSchedule",
            "NvrRecordSetup",
            "NetCamRecordSetup",
            "RecordOption"
        ],
        "EventMenuAccess": [
            "NvrSensorDetection",
            "NetCamSensorDetection",
            "NvrEventDetection",
            "NetCamEventDetection",
            "VideoLossDetection",
            "AlarmSchedule"
        ],
        "NetworkMenuAccess": [
            "NetworkInterface",
            "NetworkPort",
            "DDNS",
            "IPFilter",
            "SSL",
            "802.1x",
            "LiveStreaming",
            "SMTP",
            "EventMail",
            "GroupAndRecipientEmail",
            "SNMP",
            "DHCPServer"
        ],
        "RecordStartAccess": true,
        "RecordStopAccess": true,
        "PTZAccess": true,
        "AlarmOutputAccess": true,
        "ShutdownAccess": true
    }
```

```
    ]
}
```

94.4.3. Adding a user group

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=usergroups&action=add&UserGroupID=Group3&LiveChann
el=1,11,21&SearchChannel=2,12,22&BackupChannel=3,13,33
```

94.4.4. Updating a user group

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=usergroups&action=update&UserGroupID=Group3&Backup
Channel=13,21,3
```

94.4.5. Removing a user group

A user group can be deleted with the index number or the user group ID.

REQUEST

```
http://<Device IP>/ stw-
cgi/security.cgi?msubmenu=usergroups&action=remove&Index=3
```

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=usergroups&action=remove&UserGroupID=Group3
```

Chapter 95. Authority

95.1. Description

The **authority** submenu sets the access permissions.

NOTE	This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	Admin

95.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
authority&action=<value>[&<parameter>=<value>...]
```

95.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the authority settings
set	LiveAccess	REQ, RES	<bool> True, False	Whether to grant permission for live access
	PTZAccess	REQ, RES	<bool> True, False	Whether to grant permission for PTZ control
	RemoteAlarmOutput	REQ, RES	<bool> True, False	Whether to grant permission for remote alarm output control
	ShutDown	REQ, RES	<bool> True, False	Whether to grant permission to shut down
	RecordStartAccess	REQ, RES	<bool> True, False	Whether to grant permission for record start control
	RecordStopAccess	REQ, RES	<bool> True, False	Whether to grant permission for record stop control
	SearchAccess	REQ, RES	<bool> True, False	Whether to grant permission for search control
	BackupAccess	REQ, RES	<bool> True, False	Whether to grant permission for backup access

Action	Parameter	Request/ Response	Type/ Value	Description
	NetworkAccess	REQ, RES	<bool> True, False	Whether to grant permission for network access
	WebviewerAccess	REQ, RES	<bool> True, False	Whether to grant permission for Web viewer access
	AutoLogoutTime	REQ, RES	<enum> Off, 1m, 2m, 3m, 5m, 10m	Auto logout time
	IDManualInput	REQ, RES	<bool> True, False	Whether to grant permission to manually input the ID

95.4. Examples

95.4.1. Getting current permission settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=authority&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
LiveAccess=True
PTZAccess=False
RemoteAlarmOutput=False
ShutDown=False
RecordStopAccess=False
SearchAccess=False
BackupAccess=False
RecordStartAccess=False
NetworkAccess=True
WebviewerAccess=True
IDManualInput=False
AutoLogoutTime=OFF
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "LiveAccess": true,
    "PTZAccess": false,
    "RemoteAlarmOutput": false,
    "ShutDown": false,
    "RecordStopAccess": false,
    "SearchAccess": false,
    "BackupAccess": false,
    "RecordStartAccess": false,
    "NetworkAccess": true,
    "WebviewerAccess": true,
    "IDManualInput": false,
    "AutoLogoutTime": "OFF"
}
```


95.4.2. Setting the access permission

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=authority&action=set&LiveAccess=False&PTZAccess=False&RemoteAlarmOutput=False&ShutDown=False&RecordStopAccess=False&SearchAccess=False&BackupAccess=false
```

Chapter 96. Additional Password

96.1. Description

The **additionalpassword** submenu sets multiple passwords for the user.

> **NOTE** This chapter applies to NVR only.

Access level

Action	NVR
view	User
add, update	Admin
remove	Admin

96.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
additionalpassword&action=<value>[&<parameter>=<value>...]
```

96.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	UserID	REQ	<string>	User ID
add/update	UserID	REQ, RES	<string>	User ID
	Enable	REQ, RES	<bool> True, False	To enable a multi password rule
	PasswordIndex	REQ, RES	<int>	Index for password
	Password	REQ, RES	<string>	Additional password
	IsPasswordEncrypted	REQ	<bool> True, False	If set to true, **Password** is encrypted using the public key provided by the **rsa** submenu of **security.cgi**, and sent as post payload. Refer to the Application Programmer's Guide.
remove	UserID	REQ	<string>	User ID

Action	Parameter	Request/ Response	Type/ Value	Description
	PasswordIndex	REQ	<csv>	Password Index that is to be removed **Note** All passwords will be removed for the corresponding user if PasswordIndex is not provided.

96.4. Examples

96.4.1. Getting additional password settings for all users

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=additionalpassword&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
UserID.User1.GroupID=Group1
UserID.User1.Enable=True
UserID.User2.GroupID=Group2
UserID.User2.Enable=False
UserID.User3.GroupID=Group2
UserID.User3.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AdditionalPasswords": [
        {
            "UserID": "User1",
```

```
            "GroupID": "Group1",
            "Enable": true
        },
        {
            "UserID": "User2",
            "GroupID": "Group2",
            "Enable": false
        },
        {
            "UserID": "User3",
            "GroupID": "Group2",
            "Enable": true
        }
    ]
}
```

96.4.2. Getting additional password settings for user "User1"

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=additionalpassword&action=view&UserID=User1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
UserID.User1.GroupID=Group1
UserID.User1.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AdditionalPasswords": [
```

```
        {
            "UserID": "User1",
            "GroupID": "Group1",
            "Enable": true
        }
    ]
}
```

96.4.3. Adding a Password for a User

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?
msubmenu=additionalpassword&action=add&UserID=User1&Password={password}
```

96.4.4. Updating Password

REQUEST

```
http://<Device IP/stw-
cgi/security.cgi?msubmenu=additionalpassword&action=update&UserID=User1&Pass
wordindex=1&Password={password}
```

96.4.5. Removing all Passwords for a User

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=additionalpassword&action=remove&UserID=User1
```

96.4.6. Removing Password Index 1,2,3

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=additionalpassword&action=remove&UserID=User1&Pass
wordIndex=1,2,3
```

Chapter 97. Getting public key

97.1. Description

The **rsa** submenu is used to get the public key from the device. This can be used to encrypt the password information sent to the device.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera	NVR	IPAS
view	Admin	Admin	Admin

97.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
rsa&action=<value>[&<parameter>=<value>...]
```

97.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	PublicKeyFormat	REQ	<enum> PKCS1, X509	Optional request parameter to specify the RSA public key format. If not specified, the PKCS1 format is served. CAMERA ONLY
	PublicKey	RES	<string>	RSA public key

97.4. Examples

NOTE Please refer to the application programming guide for information on how to use this public key to encrypt passwords.

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=rsa&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
PublicKey=-----BEGIN RSA PUBLIC KEY-----
MIIBCgKCAQEAugtcj1ql+mvkzK60Hph3an0Z/rdtZ/NF84m0TAsQuiDheGnN7dYJ
nZfRit5PcdugQO7XAkVq9DBy6kWrgrMqlzS9PwwEN7cBgFmyU/yJvpnZNrxlDLFB
ELlXgEYVih5yTSSoa6uWy8cSnGrnY1Ywymh8JGvuk0xZFc09eBCnIogQqydQb1AP
OnUqTx5JaCdnitYekHRWdyh1XY3wJnV6Ykb8hfnwzhrbz4P2bOCTW/ISE5hl2qvP
WrSBk+EEH2Wfcwfu2785iyu6mDzoCT64XcyOgscLkLhkg2IAXhoel+TNDdN0zzOX
dIpDg2Vi4s3ObPG4KSLnStXJJYIDx3CrRwIDAQAB
-----END RSA PUBLIC KEY-----
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PublicKey": "-----BEGIN RSA PUBLIC KEY-----
\nMIIBCgKCAQEAugtcj1ql+mvkzK60Hph3an0Z/rdtZ/NF84m0TAsQuiDheGnN7dYJ\nnZfRit5P
cdugQO7XAkVq9DBy6kWrgrMqlzS9PwwEN7cBgFmyU/yJvpnZNrxlDLFB\nELlXgEYVih5yTSSoa6
uWy8cSnGrnY1Ywymh8JGvuk0xZFc09eBCnIogQqydQb1AP\nOnUqTx5JaCdnitYekHRWdyh1XY3w
JnV6Ykb8hfnwzhrbz4P2bOCTW/ISE5hl2qvP\nWrSBk+EEH2Wfcwfu2785iyu6mDzoCT64XcyOgs
cLkLhkg2IAXhoel+TNDdN0zzOX\ndIpDg2Vi4s3ObPG4KSLnStXJJYIDx3CrRwIDAQAB\n-----
END RSA PUBLIC KEY-----\n"
}
```

Example of when key format is specified.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=rsa&action=view&PublicKeyFormat=X509
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
PublicKey=-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAxp5XPQay2FVUJZXCV2K2
7Wv7uLtZ3vIwsSiAHVJZwSmAQV7H23ElmBCNEWCk96mdjonZnHpQ0mWj3hsDkO48
qGnELbsrqfTuUF5Ulze7+f34aX/Mg9pwbOruZE3CRbcsxc2JTTbm0sLoVnSV7pPn
Lg/r4dzp7ll3fL4WfKere/sXmRdeZ+2ugVzrCGSovOX4madkAtwCEszOZedIWe85
AkDN42Aw11sknn66EkDZAMVrpI5g0nfrdUYTKxh/e+LAVOfMSHdFaMht4rSTaXN7
z+RxPh5Ro0UN5Ha9buNtiXUB4VkjV440/Be13njHt+d5HZduGh0WhggIjgyTJGsN
7wIDAQAB
-----END PUBLIC KEY-----
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PublicKey": "-----BEGIN PUBLIC KEY-----
\nMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAxp5XPQay2FVUJZXCV2K2\n7Wv7uLtZ
3vIwsSiAHVJZwSmAQV7H23ElmBCNEWCk96mdjonZnHpQ0mWj3hsDkO48\nqGnELbsrqfTuUF5Ulz
e7+f34aX/Mg9pwbOruZE3CRbcsxc2JTTbm0sLoVnSV7pPn\nLg/r4dzp7ll3fL4WfKere/sXmRde
Z+2ugVzrCGSovOX4madkAtwCEszOZedIWe85\nAkDN42Aw11sknn66EkDZAMVrpI5g0nfrdUYTKx
h/e+LAVOfMSHdFaMht4rSTaXN7\nz+RxPh5Ro0UN5Ha9buNtiXUB4VkjV440/Be13njHt+d5HZdu
Gh0WhggIjgyTJGsN\n7wIDAQAB\n-----END PUBLIC KEY-----\n"
}
```

Chapter 98. Configure default camera user credentials in NVR

98.1. Description

The **camerausers** submenu is used to configure camera's default set of username and password in NVR. If NVR discovers some cameras or does connection test, it automatically tries to connect with camera with these configured user credentials.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
add/update	Admin
remove	Admin

98.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
camerausers&action=<value>[&<parameter>=<value>...]
```

98.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	UserID	REQ	<string>	
	IsInitPasswordSet	RES	<bool> True, False	Initial password status whether it is set or not
add/update	Index	REQ, RES	<int>	Index to be added
	UserID	REQ, RES	<string>	ID to try connecting with its camera
	Password	REQ, RES	<string>	Password to try connecting with its camera
	IsPasswordEncrypted	REQ	<bool> True, False	When this is set to true, password is encrypted using the public key obtained from the **rsa** submenu of security.cgi, and sent as payload content for the POST command.

Action	Parameter	Request/ Response	Type/ Value	Description
set	InitPassword	REQ	<string>	Initializes camera with this password
	IsPasswordEncrypted	REQ	<bool> True, False	When this is set to True, InitPassword parameter is not sent and instead the password is encrypted using the public key obtained from the **rsa** submenu of security.cgi, and sent as payload content for the POST command.
remove	Index	REQ	<int>	Index to be removed

98.4. Examples

98.4.1. Viewing a user

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=camerausers&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Users": [
        {
            "Index": 0,
            "UserID": "admin"
        },
        {
            "Index": 1,
            "UserID": "admin"
        },
        {
            "Index": 2,
            "UserID": "root"
        }
    ]
```

```
}
```

NOTE Please refer to the application programming guide for information on how to use this public key to encrypt passwords.

98.4.2. Adding a user

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=camerausers&action=add&UserId=admin&password={pass
word}&IsPasswordEncrypted=false
```

98.4.3. Updating a user

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=camerausers&action=update&Index=0&UserId=admin&pas
sword={password}&IsPasswordEncrypted=False
```

98.4.4. Removing a user

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=camerausers&action=remove&Index=1
```

Chapter 99. Getting Client Mutual Authenticate Status

99.1. Description

The **clienthttpsstatus** submenu is used to get the client's mutual authentication status from the device.

NOTE This chapter applies to network cameras only.
Attribute that checks for the client certificate authentication supports: "attributes/Security/Support/ClientCertificateAuthentication"

Access level

Action	Camera
view	User

99.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
clienthttpsstatus&action=<value>[&<parameter>=<value>...]
```

99.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Read client's mutual certificate status

Action	Parameter	Request/ Response	Type/ Value	Description
	ClientHttpsStatus	RES	<enum> NO_HTTPS, HTTPS_WITHOUT_CLIENT_CERT, HTTPS_WITH_INVALID_CLIENT_CERT, HTTPS_WITH_VALID_CLIENT_CERT	Current client's HTTPS connection status • NO_HTTPS: The client uses no HTTPS • HTTPS_WITHOUT_CLIENT_CERT: The client uses HTTPS, but the device does not use mutual authentication • HTTPS_WITH_INVALID_CLIENT_CERT: The device has mutual authentication enabled, but the client's certificate is invalid • HTTPS_WITH_VALID_CLIENT_CERT: The device has mutual authentication enabled and the client's certificate is valid

99.4. Examples

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=clienthttpsstatus&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

ClientHttpsStatus=NO_HTTPS

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

{

"ClientHttpsStatus": "NO_HTTPS"

}

Chapter 100. Getting TLS Configuration

100.1. Description

The **tlsversion** submenu is used to get the TLS configuration from the device.

> **NOTE**
> This chapter applies to network cameras and IPAS.
> In IPAS only view operation is supported.

Access level

Action	Camera
view	Admin
set	

100.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
tlsversion&action=<value>[&<parameter>=<value>...]
```

100.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads TLS configuration.
	Version	REQ	<csv> TLSv1_0, TLSv1_1, TLSv1_2, TLSv1_3	TLS Version Getting the configuration of specific TLS version
	Version.#.SupportedCipherMode	RES	<csv> Compatible, Secure	Supported cipher mode • Compatible: Supports a compatible cipher suite • Secure: Supports a secure cipher suite
set	Version.#.Enable	REQ, RES	<bool> True, False	Enables or disables TLS version If a specific TLS version to be enabled does not support Secure mode, **CipherMode** will automatically change to Compatible mode

Action	Parameter	Request/ Response	Type/ Value	Description
	CipherMode	REQ, RES	<enum> Compatible, Secure	Cipher mode • Compatible: Device uses a compatible cipher suite • Secure: Device uses a more secure cipher suite To use a specific mode, all versions must support the mode.

100.4. Examples

100.4.1. Getting all versions of the TLS configurations

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=tlsversion&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Version.TLSv1_0.Enable=False
Version.TLSv1_0.SupportedCipherModes=Compatible
Version.TLSv1_1.Enable=False
Version.TLSv1_1.SupportedCipherModes=Compatible
Version.TLSv1_2.Enable=True
Version.TLSv1_2.SupportedCipherModes=Compatible,Secure
Version.TLSv1_3.Enable=True
Version.TLSv1_3.SupportedCipherModes=Compatible,Secure
CipherMode=Secure
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Versions": [
        {
            "Version": "TLSv1_0",
            "Enable": false,
            "SupportedCipherModes": [
                "Compatible"
            ]
        },
        {
            "Version": "TLSv1_1",
            "Enable": false,
            "SupportedCipherModes": [
                "Compatible"
            ]
        },
        {
            "Version": "TLSv1_2",
            "Enable": true,
            "SupportedCipherModes": [
                "Compatible",
                "Secure"
            ]
        },
        {
            "Version": "TLSv1_3",
            "Enable": true,
            "SupportedCipherModes": [
                "Compatible",
                "Secure"
            ]
        }
    ],
    "CipherMode": "Secure"
}
```


100.4.2. Getting ‘TLSv1_3’ configuration

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/security.cgi?msubmenu=tlsversion&action=view&Version=TLSv1_3
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Version.TLSv1_3.Enable=True
Version.TLSv1_3.SupportedCipherModes=Compatible,Secure
CipherMode=Secure
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Versions": [
        {
            "Version": "TLSv1_3",
            "Enable": true,
            "SupportedCipherModes": [
                "Compatible",
                "Secure"
            ]
        }
    ],
    "CipherMode": "Secure"
}
```


100.4.3. Enabling TLS v1.2 and v1.3

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=tlsversion&action=set&Version.TLSv1_2.Enable=True&
Version.TLSv1_3.Enable=True
```

100.4.4. Setting cipher mode to compatible mode

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=tlsversion&action=set&CipherMode=Compatible
```

Chapter 101. Getting Camera’s validation status from NVR

101.1. Description

The **cameravalidationstatus** submenu is used to get the camera’s validation status from NVR.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

101.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
cameravalidationstatus&action=<value>[&<parameter>=<value>...]
```

101.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads camera validation status
	Channel.#.Connected	RES	<bool> True, False	Camera connection status
	Channel.#.CameraValidationStatus	RES	<enum> UNKNOWN, HTTP, OTHER_CERT, CHANGED_CERT, INVALID_DEVICE_CERT, VALID_DEVICE_CERT	Current channels connection status **Note** Valid status is only supported for cameras connected using SUNAPI in NVR

101.4. Examples

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=cameravalidationstatus&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Connected=True
Channel.0.CameraValidationStatus=HTTP
Channel.1.Connected=True
Channel.1.CameraValidationStatus=HTTP
Channel.2.Connected=True
Channel.2.CameraValidationStatus=HTTP
....
Channel.3.Connected=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "cameravalidationstatus": [
        {
            "Channel": 0,
            "Connected": true,
            "CameraValidationStatus": "HTTP"
        },
        {
            "Channel": 1,
            "Connected": true,
            "CameraValidationStatus": "HTTP"
        },
        {
            "Channel": 2,
            "Connected": true,
            "CameraValidationStatus": "HTTP"
        },
        ...
    ]
```

```
}
```

Chapter 102. CA Certificate Settings

102.1. Description

The **cacertificate** submenu handles CA certificates.

Access level

Action	Camera
view	Admin
install	Admin
remove	Admin

102.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
cacertificate&action=<value>[&<parameter>=<value>...]
```

102.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the SSL (HTTPS) settings
	Certificate.#.CertificateName	RES	<string>	Certificate name
	Certificate.#.Type	RES	<enum> Unique, Public, SelfSigned	Type of the certificate
	Certificate.#.Subject	RES	<string>	Subject of the certificate
	Certificate.#.SubjectAlternativeName	RES	<string>	Subject alternative name (SAN)
	Certificate.#.Issuer	RES	<string>	Issuer
	Certificate.#.IssueDate	RES	<string>	Issued date
	Certificate.#.ExpiryDate	RES	<string>	Expiry date
	Certificate.#.Version	RES	<string>	Version
	Certificate.#.SerialNumber	RES	<string>	Serial number
	Certificate.#.Signature	RES	<string>	Signature

Action	Parameter	Request/ Response	Type/ Value	Description
	Certificate.#.Thumbprint	RES	<string>	Thumbprint
	Certificate.#.IsRemovable	RES	<bool>	Whether the certificate can be deleted
remove	CertificateName	REQ	<string>	Certificate name
install				POST method

102.4. Examples

102.4.1. Getting the CA certificates

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=cacertificate&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Certificate.1.CertificateName=HTW_rootca
Certificate.1.Type=Unique
Certificate.1.Subject=/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2
Certificate.1.SubjectAlternativeName=-
Certificate.1.Issuer=/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2
Certificate.1.IssueDate=Feb 17 04:15:52 2020 GMT
Certificate.1.ExpiryDate=Feb 2 04:15:52 2080 GMT
Certificate.1.Version=V3
Certificate.1.SerialNumber=00 85 BE B7 49 3A 83 3A E7
Certificate.1.Signature=sha256WithRSAEncryption
Certificate.1.Thumbprint=3676ae9bd6ebb4f3543b00c08898d17b7fa96b7e61726a4bd5f
60c09062c4fce
Certificate.1.IsRemovable=False
Certificate.2.CertificateName=helloCA
Certificate.2.Type=Public
Certificate.2.Subject=/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority
```

```
Certificate.2.SubjectAlternativeName=-
Certificate.2.Issuer=/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority
Certificate.2.IssueDate=Mar 9 01:46:25 2020 GMT
Certificate.2.ExpiryDate=Jan 16 01:46:25 2030 GMT
Certificate.2.Version=V3
Certificate.2.SerialNumber=00 62 0A 2B 24 D9 7E DD 53 B2 B5 F6 71 DF 72 92
08 F0 7F 25 9A
Certificate.2.Signature=sha256WithRSAEncryption
Certificate.2.Thumbprint=68b8a185ae17600815b25cd1c42ecbbd62f2b962a62787376fc
53f9efb211391
Certificate.2.IsRemovable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Certificates": [
        {
            "CertificateName": "HTW_rootca",
            "Type": "Unique",
            "Subject": "/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2",
            "SubjectAlternativeName": "-",
            "Issuer": "/C=KR/O=Hanwha Vision/OU=Security Solution/CN=Hanwha
Vision Private Root CA 2",
            "IssueDate": "Feb 17 04:15:52 2020 GMT",
            "ExpiryDate": "Feb 2 04:15:52 2080 GMT",
            "Version": "V3",
            "SerialNumber": "00 85 BE B7 49 3A 83 3A E7 ",
            "Signature": "sha256WithRSAEncryption",
            "Thumbprint":
"3676ae9bd6ebb4f3543b00c08898d17b7fa96b7e61726a4bd5f60c09062c4fce",
            "IsRemovable": false
        },
        {
            "CertificateName": "helloCA",
            "Type": "Public",
```

```
            "Subject": "/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority",
            "SubjectAlternativeName": "-",
            "Issuer": "/C=FR/ST=Radius/L=Somewhere/O=Example
Inc./emailAddress=admin@example.org/CN=Example Certificate Authority",
            "IssueDate": "Mar 9 01:46:25 2020 GMT",
            "ExpiryDate": "Jan 16 01:46:25 2030 GMT",
            "Version": "V3",
            "SerialNumber": "00 62 0A 2B 24 D9 7E DD 53 B2 B5 F6 71 DF 72 92
08 F0 7F 25 9A ",
            "Signature": "sha256WithRSAEncryption",
            "Thumbprint":
"68b8a185ae17600815b25cd1c42ecbbd62f2b962a62787376fc53f9efb211391",
            "IsRemovable": true
        }
    ]
}
```

102.4.2. Installing CA certificate

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=cacertificate&action=install
```

When requesting a certificate to be installed, data should be sent via the POST method in the following format.

The certname value is the certificate name. The certlength value is the certificate data size. The certdata value is the certificate data. The keylength is the key data size. The keydata value is the key data.

```
<SetHTTPSData>
    <PublicCertName>certname</PublicCertName>
    <CertLength>certlength</CertLength>
    <CertData>certdata</CertData>
</SetHTTPSData>
```

CURL command

The certificate installation can be tested with CURL as below. To learn about CURL, please refer to http://curl.haxx.se.

NOTE To get a JSON response, add the -H "Accept: application/json" header to the request.

```
curl -v --digest -u <userid>:<password> --data-urlencode @cert.xml
"http://<Device IP>/stw-
cgi/security.cgi?msubmenu=cacertificate&action=install" -H "Expect:"
```

The above command will produce a request to the device that appears as below:

```
POST /stw-cgi/security.cgi?msubmenu=ssl&action=install HTTP/1.1
Content-Length: 3775
Content-Type: application/x-www-form-urlencoded
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
{
    "Response": "Success"
}
```

102.4.3. Deleting a certificate

To delete a certificate with the **remove** action, the **CertificateName** parameter must be set at the same time.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=cacertificate&action=remove&CertificateName=certna
me
```

Chapter 103. MAC Address Filtering

103.1. Description

The **macfilter** submenu configures MAC address filtering. This submenu controls the device access based MAC filter settings. The feature provides both Allow and Deny filtering types. Deny filtering allows all MAC addresses except the listed addresses, while Allow filtering blocks all MAC addresses except the listed addresses.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User
add, update	Admin	User
remove	Admin	User

103.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
macfilter&action=<value>[&<parameter>=<value>...]
```

103.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the MAC address filtering settings.
set	AccessType	REQ, RES	<enum> Allow, Deny	Access type • Allow: Allows access to listed MAC addresses • Deny: Blocks access to listed MAC addresses **Note** **AccessType** must be sent together with the **set** action.

Action	Parameter	Request/ Response	Type/ Value	Description
add, update	Index	REQ, RES	<int>	Position of a MAC address in the list **Note** **Index**, and **Enable** must be sent together for the **update** action.
	Address	REQ, RES	<string>	MAC address to be configured for device access
	Enable	REQ, RES	<bool> True, False	Whether to apply MAC address filtering
remove	Index	REQ	<int>	MAC address Index that is to be removed.

103.4. Examples

103.4.1. Getting the current MAC filtering settings

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=macfilter&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AccessType": "Allow",
    "Filters": [
        {
            "Index": 1,
            "Address": "00:00:00:00:00:00",
            "Enable": false
        },
        {
            "Index": 2,
            "Address": "11:11:11:11:11:11",
            "Enable": true
        }
```

```
    ]
}
```

103.4.2. Setting an MAC address and enabling filtering

To add the MAC address to be filtered with the **add** action, **Address** and **Enable** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=macfilter&action=add&Address=00:00:00:00:00:00&Ena
ble=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

103.4.3. Updating an MAC address and disabling filtering

To modify a filtered address via the **update** action, **Index** and **Enable** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=macfilter&action=update&Index=1&Address=11:11:11:1
1:11:11&Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
```

```
}
```

103.4.4. Deleting an MAC address from MAC filtering list

To remove a filtered address with the **remove** action, **Index** parameters must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/security.cgi?msubmenu=macfilter&action=remove&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Response": "Success"
}
```

Chapter 104. API Security configuration

104.1. Description

The **apisecurity** submenu supports security-related settings for the API. For example, web server-related settings are possible.

Access level

Action	Camera	view
Admin	set	Admin

104.2. Syntax

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=
apisecurity&action=<value>[&<parameter>=<value>...]
```

104.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads API security related settings
set	CORSMode	REQ, RES	<enum> AllowAll, Restricted	CORSMode • AllowAll: In API response, allow all origins regardless of origin and respond with cors header • Restricted: If the origin is different in the API response, the cors header is not responded.

104.4. Examples

104.4.1. Getting the current corsmode

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=apisecurity&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "CORSMode": "AllowAll"
}
```

104.4.2. Setting corsmode to Restricted

REQUEST

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=apisecurity&action=set&CORSMode=Restricted
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Video/Audio

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 105. Overview

105.1. Description

This document describes how to request and configure settings for HTTP-based video streaming using SUNAPI. It also explains the RTSP (Real Time Streaming Protocol) API for video streaming.

SUNAPI consists of **video.cgi** and **media.cgi** for video streaming over HTTP; video.cgi requests JPEG image or MJPEG video, and media.cgi configures the video and audio settings.

video.cgi uses the following submenus to request images and video:

- **snapshot**: Requests JPEG images with the specified settings.
- **stream**: Requests MJPEG video with the specified settings.
- **thumbnail**: Requests the thumbnail snapshot of a recorded data for the requested duration in NVR.
- **slideshow**: Displays an image slideshow whenever MotionDetection is not happening.

media.cgi uses the following submenus to configure video and audio:

- **videosource**: Configures the video source settings.
- **videooutput**: Configures the video output settings.
- **videoprofilepolicy**: Configures the policy settings of the video profile.
- **videoprofile**: Configures the video profile settings.
- **audioinput**: Configures the input settings of the audio source.
- **audiooutput**: Configures the output settings of the audio source.
- **streamuri**: Requests the streaming URI for the profile.
- **sessionkey**: Request the session key settings.
- **multicast**: Controls the multicast operation.
- **videocodecinfo**: Requests the video codec capabilities.
- **wisestream**: Configures the WiseStream settings.
- **camerapasswordchange**: Applies the new password for all connected cameras.
- **mediaoptions**: Requests the media options available in the device.
- **Setsynchronizationpoint**: Requests the intra frame for the given profile.
- **videosourceoptions**: Provides different source settings that are possible for a video source
- **cameraregister**: Allows camera registration in NVR and decoder
- **autoregister**: Allows multiple cameras at the same time
- **cameradiscovery**: Allows the discovery of cameras in the network
- **cameraconnection**: Allows a test of camera registration in NVR and decoder
- **videoencoderinstances**: Requests the maximum streaming counts for a device

- **videoinput**: Configures the video input format in the encoder.
- **metadatashare**: Configures the metadata sharing settings.
- **speakervolume**: Configures the speaker volume in an IP Audio System (IPAS).
- **backchannelinfo**: Used to get the backchannel streaming urls of network speaker (IPAS).
- **speakerstatus**: Used to get the current speaker status (IPAS).
- **networkaudioinput**: Used to configure the port and multicast settings in Speaker for receiving audio over network (IPAS).

Chapter 106. Video/Audio Setting

106.1. Snapshot

106.1.1. Description

The **snapshot** submenu requests JPEG images.

Access level

Action	Camera	NVR	Encoder
view	Guest	User	Guest

106.1.2. Syntax

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=
snapshot&action=<value>[&<parameter>=<value>]
```

106.1.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID. Starts from 0.
	Profile	REQ	<int>	Selects an MJPEG profile if multiple exist. If omitted or not MJPEG, the lowest-numbered MJPEG profile is used. CAMERA ONLY ENCODER ONLY

106.1.4. Examples

106.1.4.1. Requesting JPEG image

REQUEST

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=snapshot&action=view
```

REQUEST

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=snapshot&action=view&Channel=5
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: image/jpeg
Content-length: <image size>
<Body>
```

```
<JPEG image data>
```

106.2. Stream

106.2.1. Description

The **stream** submenu requests real-time video using the HTTP protocol.

NOTE This chapter applies to network camera and encoder only.

Access level

Action	Camera	Encoder
view	Guest	Guest

106.2.2. Syntax

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=
stream&action=<value>[&<parameter>=<value>]
```

106.2.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Profile	REQ	<int>	Profile number Only MJPEG profiles can be requested.
	CodecType	REQ	<enum> MJPEG	Video codec type. This is an optional parameter.
	Resolution	REQ	<enum>	Video resolution
	FrameRate	REQ	<int>	Video encoding frame rate

Action	Parameters	Request/ Response	Type/ Value	Description
	CompressionLevel	REQ	<int>	Video encoding compression level Higher values increase compression rate and reduces video quality. **Note** If the value is out of the camera's supported range, it will be automatically adjusted.
	Channel	REQ	<int>	Channel ID

106.2.4. Examples

106.2.4.1. Requesting MJPEG stream

Getting MJPEG stream

REQUEST

```
http://<Device IP>/stw-
cgi/video.cgi?msubmenu=stream&action=view&Profile=1&CodecType=MJPEG
```

106.3. Thumbnail

106.3.1. Description

The **thumbnail** submenu is used to get the thumbnail image from the recorded data at regular intervals as a single image.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

106.3.2. Syntax

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=
thumbnail&action=<value>[&<parameter>=<value>]
```

106.3.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel for which a thumbnail is requested
	FromDate	REQ	<string>	From date in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	To date in UTC format YYYY-MM-DDTHH:MM:SSZ
	Width	REQ	<int>	Width of each thumbnail in the grid
	Height	REQ	<int>	Height of each thumbnail in the grid
	Column	REQ	<int>	Number of columns in the thumbnail image grid
	Row	REQ	<int>	Number of rows in the thumbnail image grid
	OverlappedID	REQ	<int>	Recording overlapped ID

106.3.4. Examples

106.3.4.1. Requesting a thumbnail

Getting a thumbnail with each grid's image size 160x120 and total grid size of 10x10

REQUEST

```
http://<Device IP>/stw-
cgi/video.cgi?msubmenu=thumbnail&action=view&Channel=0&FromDate=2020-07-
07T08:00:00Z&ToDate=2020-07-
07T16:00:00Z&Width=160&Height=120&OverlappedID=100&Row=10&Column=10
```

106.4. Slideshow

106.4.1. Description

The **slideshow** submenu is used to show images one by one at a set interval. If motion detection happens, the camera HDMI video streams live for a set period of time. After that period, if there is no more motion, it shows images uploaded in the SD card. This feature is usually used in a retail environment.

NOTE	This chapter applies to cameras only.

Access level

Action	Camera
view	Admin

106.4.2. Syntax

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=
slideshow&action=<value>[&<parameter>=<value>]
```

106.4.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ImageList	RES	<csv>	Displays list of images uploaded in SD card
	Status	RES	<enum> Idle, Uploading	If the camera is uploading (installing) BMP images to the SD card, Status is "Uploading"; otherwise, it is "Idle".
install	ImageName	REQ	<string> e.g., 01_slide.bmp	This command must include a POST message with a BMP image file. Only .bmp files are supported and will be uploaded to the SD card.
remove	Mode	REQ	<enum> All, Selected	• All: remove all saved images. • Selected: remove a particular image. "ImageName" should be used together.
	ImageName	REQ	<csv>	Name of the image to be deleted.
set	Enable	REQ, RES	<bool>	Enables the slideshow feature. If Motion is detected, it streams live video through HDMI output. If no Motion is detected, it shows images in the list one by one with dwell time.
	ImageDwellTime	REQ, RES	<int>	Sets the display time for an image
	LiveDwellTime	REQ, RES	<enum> 5, 10, 30, 60, 120	Sets the duration for live video display

106.4.4. Examples

106.4.4.1. Requesting view action

Getting a configuration for slideshow

REQUEST

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=slideshow&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Enable": false,
    "ImageDwellTime": 1,
    "LiveDwellTime": 5,
    "ImageList": "sky.bmp",
    "Status": "Idle"
}
```


106.5. Video Source

106.5.1. Description

The **videosource** submenu configures and views video source settings.

Access level

Action	Camera	NVR	Encoder	Decoder
view	Guest	User	Guest	User
set	User	User	Admin	User
add	User	-	-	-
remove	User	-	-	-

106.5.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videosource&action=<value>[&<parameter>=<value>]
```

106.5.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video source settings.
	Channel	REQ	<csv>	Channel ID list
	VideoSourceToken	RES	<string>	Video source token CAMERA ONLY
	Type	RES	<enum> NTSC, PAL	Video source type
	State	RES	<enum> On, Deactive	State (read-only for encoder) • On: Camera is connected, • Deactive: Camera is not connected. **Note** Applicable only to encoders and multi-sensor cameras with changeable lenses.
	Fixed	RES	<bool>	Returns True if the target channel can't be modified. Some camera models that support 'SupportChannelExpansionFeatrue' also support channel settings like adding, deleting, or changing view type. Channels that support those features present Fixed parameter as False. CAMERA ONLY
	IsCroppedChannel	RES	<bool>	Returns True if the target channel is cropped and created from the original video channel. CAMERA ONLY
set	Channel	REQ, RES	<int>	Channel ID
	SensorCaptureSize	REQ, RES	<enum> 2M, 3M, 2M_double_ shutter	Capture size (read-only for NVR) The value may vary depending on the model. Please check the device attributes using attributes.cgi.

Action	Parameters	Request/ Response	Type/ Value	Description
	VideoType	REQ, RES	<enum> NTSC, PAL, AHD, TVI, CVI, CVBS	Video Type The value may vary depending on the model. Please check the device attributes using attributes.cgi. ENCODER ONLY
	VideoMode	REQ, RES	<enum> Auto, Manual, JPMode	Video Mode The value may vary depending on the model. Please check the device attributes using attributes.cgi. ENCODER ONLY
	SensorCaptureFrameRate	REQ, RES	<enum> 20, 25, 30, 50, 60	Capture frame rate The number of frames that a camera sensor captures in one second. CAMERA ONLY
	Name	REQ, RES	<string>	Name NVR ONLY
	State	REQ, RES	<enum> On, Off, Covert1, Covert2, Deactive	State NVR ONLY
	ViewType	REQ	<enum> QuadView, DoublePanorama, Panorama, QuadView.1, QuadView.2, QuadView.3, QuadView.4	Tries to change target channel's view type. If this command works, profiles for that channel would be re-generated according to changed ViewType. This parameter is applicable only for Not Fixed Channel. FISHEYE ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Ratio	REQ, RES	<enum> 16:9, 4:3	Tries to change target channel's ratio. This parameter should be requested with Channel parameter. This parameter is applicable only for Not Fixed Channel. CAMERA ONLY
	CroppedCoordinates	REQ, RES	<string>	Tries to change the coordinates to be cut from the original source. This parameter should be requested with Channel parameter. This parameter is applicable only for Not Fixed Channel. Coordinates should be in the format of <x1, y1, x2, y2> with even numbers only. CAMERA ONLY
add	ViewType	REQ	<csv> QuadView, DoublePanorama, Panorama, QuadView.1, QuadView.2, QuadView.3, QuadView.4, DPTZView	Tries to make a new channel with selected view type. User can make multiple channels at once using csv values. If this command works, profiles for new channel would be generated according to selected ViewType. This parameter is applicable only for Not Fixed Channel.
	Ratio	REQ, RES	<enum> 16:9, 4:3	Tries to make a new channel with requested ratio. This parameter should be requested with 'CroppedCoordinates' parameter. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	CroppedCoordinates	REQ, RES	<string>	Tries to make a new channels with requested the coordinates to be cut from the original source. This parameter should be requested with 'Ratio' parameter. Coordinates should be in the format of <x1, y1, x2, y2> with even numbers only. CAMERA ONLY **Note** Cameras Supporting Virtual Channel Addition
remove	Channel	REQ	<csv> All	Tries to remove all Not Fixed Channel created before. Some models may support deleting individual channels. **Note** Cameras Supporting Virtual Channel Addition

106.5.4. Examples

106.5.4.1. Getting video source settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videosource&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.SensorCaptureSize=2M
Channel.0.SensorCaptureFrameRate=30
Channel.0.VideoSourceToken=VideoSourceToken-01
Channel.0.VideoType=CVI
Channel.0.VideoMode=Auto
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoSources": [
        {
            "Channel": 0,
            "SensorCaptureSize": "2M",
            "SensorCaptureFrameRate": "30",
            "VideoSourceToken": "VideoSourceToken-01",
            "VideoType": "CVI",
            "VideoMode": "Auto"
        }
    ]
}
```


The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Type=NTSC
Channel.0.SensorCaptureSize=Unknown
Channel.0.Name=Cam 01
Channel.0.State=On
Channel.1.Type=NTSC
Channel.1.SensorCaptureSize=Unknown
Channel.1.Name=Cam 02
Channel.1.State=On
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "VideoSources": [
        {
            "Channel": 0,
            "Type": "NTSC",
            "SensorCaptureSize": "Unknown",
            "Name": "CAM 01",
            "State": "On"
        },
        {
            "Channel": 1,
            "Type": "NTSC",
            "SensorCaptureSize": "Unknown",
            "Name": "CAM 02",
            "State": "On"
        }
    ]
}
```


106.5.4.2. Setting a sensor to capture 25 frames per second

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=set&SensorCaptureFrameRate=25
```

106.5.4.3. Setting sensor capture size to 12M

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=set&SensorCaptureSize=12M
```

106.5.4.4. Setting a video type to CVI and video mode to Auto

The following example is for encoder only.

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/media.cgi?msubmenu=videosource&action=set&Channel=0&VideoType=CVI&VideoM
ode=Auto
```

106.5.4.5. Changing a view type from QuadView to SinglePanorama to Channel 2.

The following example is for fisheye only.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=set&Channel=2&ViewMode=Panorama
```

106.5.4.6. Adding multiple channels

The following example is for fisheye only.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=add&ViewType=Panorama,QuadView
```

106.5.4.7. Adding a new cropped channel

The following example is for camera only that supports ChannelExpansionFeature.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=add&Ratio=16:9&CroppedCoordinates=
720,465,2640,1545
```

106.5.4.8. Removing all generated channels

The following example is for fisheye only.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videosource&action=remove&Channel=All
```

106.6. Video Output

106.6.1. Description

The **videooutput** submenu configures and views the video output settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

106.6.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videooutput&action=<value>[&<parameter>=<value>]
```

106.6.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video output settings.
	Channel	REQ	<csv>	Channel ID list MULTI DIRECTION CAMERA ONLY
	RebootRequired	RES	<bool>	Returns True if a device should reboot after video output setting has been changed. CAMERA ONLY
set	Channel	REQ, RES	<int>	Channel ID MULTI DIRECTION CAMERA ONLY
	Type	REQ, RES	<enum> NTSC, PAL, HDMI_1080p, HDMI_720p, HDMI_QuadView	Video output type
	Enable	REQ, RES	<bool> True, False	Enables or disables video output

106.6.4. Examples

106.6.4.1. Getting video output settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videooutput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Type=NTSC
Channel.0.Enable=True
Channel.0.RebootRequired=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoOutputs": [
        {
            "Channel": 0,
            "Type": "NTSC",
            "Enable": true,
            "RebootRequired": true
        }
    ]
}
```

106.6.4.2. Setting video output type to NTSC

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videooutput&action=set&Type=NTSC&Channel=0
```

106.7. Video-Profile Policy

106.7.1. Description

The **videoprofilepolicy** submenu configures and views the default, record and email profile settings.

Access level

Action	Camera	NVR	Encoder
view	Guest	User	Admin
set	User	User	Admin

106.7.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videoprofilepolicy&action=<value>[&<parameter>=<value>]
```

106.7.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video-profile policy settings
	Channel	REQ	<csv>	Channel ID NVR ONLY MULTI DIRECTION CAMERA ONLY
set	Channel	REQ, RES	<int>	Channel ID
	LiveMode	REQ, RES	<enum> Auto, Manual, Record	Live mode setting NVR ONLY
	DefaultProfile	REQ, RES	<int>	Default profile index CAMERA ONLY ENCODER ONLY
	EventProfile	REQ, RES	<int>	Event profile index Only the MJPEG codec is supported for the Event profile. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	RecordProfile	REQ, RES	<int>	Recording profile index MPEG4 is not supported for the Record profile.
	NetworkProfile	REQ, RES	<int>	Network profile number NVR ONLY
	LiveProfile	REQ, RES	<int>	Live profile number NVR ONLY

106.7.4. Examples

106.7.4.1. Getting video profile policy settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofilepolicy&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.DefaultProfile=2
Channel.0.EventProfile=1
Channel.0.RecordProfile=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoProfilePolicies": [
        {
            "Channel": 0,
            "DefaultProfile": 2,
```

```
            "EventProfile": 1,
            "RecordProfile": 1
        }
    ]
}
```

The following response example is for NVR only.

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.NetworkProfile=4
Channel.0.LiveProfile=4
Channel.0.RecordProfile=2
Channel.0.LiveMode=Auto
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoProfilePolicies": [
        {
            "Channel": 0,
            "NetworkProfile": 4,
            "LiveProfile": 4,
            "RecordProfile": 2,
            "LiveMode": "Auto"
        }
    ]
    ...
}
```

106.7.4.2. Setting the profile policy

Setting the default profile to 1, event profile to 1 and record profile to 2

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofilepolicy&action=set&DefaultProfile=1&EventProfile=1&RecordProfile=2
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofilepolicy&action=set&Channel=3&LiveProfile=3&NetworkProfile=1&RecordProfile=2
```

106.8. Video Profile

106.8.1. Description

The **videoprofile** submenu configures and views the video profile settings.

Access level

Action	Camera	NVR	Encoder
view	Guest	User	Guest
add, update	User	User	Admin
remove	User	User	Admin

NOTE

The default profile cannot be deleted or renamed, and its encoding type cannot be altered.
For devices that support MPEG and H.264, Profile 1 and 2 are the default profiles; and Profile 1, 2 and 3 are the default profiles when MPEG, H.264, and MPEG4 are supported.
Attribute to check for maximum profiles: "attributes/Media/Limit/**MaxProfile**"
Attribute to check for maximum resolution: "attributes/Media/Limit/**MaxResolution**"
Attribute to check for ATC: "attributes/Media/Support/**ATC**"
Attribute to check for Crop: "attributes/Media/Support/**Crop**"
Attribute to check for supported VideoEncodings: "attributes/Media/Support/**VideoEncodingType**"

106.8.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
```

```
videoprofile&action=<value>[&<parameter>=<value>]
```

106.8.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video profile settings
	Channel	REQ, RES	<csv>	Channel ID
	Channel.#.Profile	REQ	<csv>	Profile ID
	IsFixedProfile	RES	<bool> True, False	Gives information on whether profile is fixed profile or not.
	ProfileToken	RES	<string>	Profile Token
	ShowAll	REQ	<bool> True, False	Shows all profiles including non-video profiles. (When this parameter is not delivered, audio only or meta only profile will not be shown.) CAMERA ONLY
	MaxGOVLength	RES	<int>	Maximum length of GOV (Group of Video) This parameter is valid only when **EncodingType** is set to H264 or MPEG4. The maximum length of GOV should be less than eight times of the framerate. CAMERA ONLY ENCODER ONLY
	MinGOVLength	RES	<int>	Minimum length of GOV (Group of Video) This parameter is valid only when **EncodingType** is set to H264 or MPEG4. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	IsEvenNumberFrameRateProfile	RES	<bool>	If this param is true, framerate should always be an even number. Even if framerate is set to an odd number, it is autoadjusted. CAMERA ONLY
	IsDPMProfile	RES	<bool>	This parameter indicates whether this profile provides Dynamic Privacy Masking (DPM). CAMERA ONLY
add, update	Channel	REQ, RES	<int>	Channel ID
	Profile	REQ, RES	<int>	Profile number **Note** **Profile** must be sent together with the **update** action. For the **add** action, the **Profile** parameter is generally not supported. Only if the **ProfileAddByIndex** attribute is true, the **Profile** parameter can be used for the **add** action.
	Name	REQ, RES	<string>	Profile name A profile name should be unique. **Note** **Name** and **EncodingType** must be sent together with the **add** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	ATCMode	REQ, RES	<enum> Disabled, AdjustFramerate, AdjustCompressionlevel	ATC (Auto Transmit Control) mode This adjusts the video properties according to the variance in the network bandwidth, controlling the bit rate. ATC mode is used to set the way to adjust the bit rate. • Disabled: Turns off ATC. • AdjustFramerate: Controls the frame rate if the network bandwidth drops off. • AdjustCompressionlevel: Controls the compression rate if the network bandwidth drops off. If **ATCTrigger** is set to Event, only AdjustFramerate for **ATCMode** is valid; if **ATCTrigger** is set to Network, AdjustFramerate and AdjustCompressionlevel for **ATCMode** are valid. CAMERA ONLY ENCODER ONLY
	ATCSensitivity	REQ, RES	<enum> VeryLow, Low, Medium, High, VeryHigh	ATC sensitivity This parameter is valid only when **ATCMode** is set to AdjustFramerate or AdjustCompressionlevel, and **ATCTrigger** is set to Network. CAMERA ONLY ENCODER ONLY
	ATCLimit	REQ, RES	<int>	ATC level limit This parameter is valid only when **ATCMode** is set to AdjustFramerate or AdjustCompressionlevel, and **ATCTrigger** is set to Network. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	ATCTrigger	REQ, RES	<enum> Network, Event	ATC (Auto Transmit Control) trigger This parameter is valid only when **ATCMode** is set to AdjustFramerate or AdjustCompressionlevel. If **ATCTrigger** is set to Event, only AdjustFramerate for **ATCMode** is valid; if **ATCTrigger** is set to Network, AdjustFramerate and AdjustCompressionlevel for **ATCMode** are valid. CAMERA ONLY ENCODER ONLY
	ATCEventType	REQ, RES	<csv> MotionDetection	ATC event type • MotionDetection: Sends only I-Frames normally, but sends videos with full-frames for motion detection events. **ATCEventType** is valid only when **ATCTrigger** is set to Event. CAMERA ONLY ENCODER ONLY
	SVNPMulticastEnable	REQ, RES	<bool> True, False	Enables or disables SVNP (Samsung Video Network Protocol) multicasting. If the **SVNPMulticastEnable** parameter is set to True, the **SVNPMulticastAddress** and **SVNPMulticastPort** parameter should be set together. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	SVNPMulticastAddress	REQ, RES	<string>	SVNP (Samsung Video Network Protocol) multicasting address The value must be in the format of an IPv4 address, and within the range of 224.0.0.1 to 239.255.255.254. **SVNPMulticastAddress** is valid only when **SVNPMulticastEnable** is set to True. CAMERA ONLY ENCODER ONLY
	SVNPMulticastPort	REQ, RES	<int>	SVNP multicasting port. **SVNPMulticastPort** is valid only when **SVNPMulticastEnable** is set to True. CAMERA ONLY ENCODER ONLY
	SVNPMulticastTTL	REQ, RES	<int>	SVNP (Samsung Video Network Protocol) TTL CAMERA ONLY ENCODER ONLY
	RTPMulticastEnable	REQ, RES	<bool> True, False	Enables or disables RTP (Real-time Transport Protocol) multicasting If the RTPMulticastEnable parameter is set to True, the **RTPMulticastAddress** and **RTPMulticastPort** parameter must be sent together. CAMERA ONLY ENCODER ONLY
	RTPMulticastType	REQ, RES	<enum> IPV4, IPv6	Based on this selection, IPV6 or IPV4 multicast address will be applicable for the profile. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	RTPMulticastAddress	REQ, RES	<string>	RTP multicasting address The value must be in the format of an IPv4 address, and within the range of 224.0.0.1 to 239.255.255.254. **RTPMulticastAddress** is valid only when **RTPMulticastEnable** is set to True. CAMERA ONLY ENCODER ONLY
	RTPMulticastAddressIPv6	REQ, RES	<string>	RTP multicasting address The value must be in the format of an IPv4 address, and within the range of ff0x::/32 **RTPMulticastAddress** is valid only when **RTPMulticastEnable** is set to True and RTPMulticastType is set to IPV6 CAMERA ONLY ENCODER ONLY
	RTPMulticastPort	REQ, RES	<int>	RTP multicasting port. **RTPMulticastPort** is valid only when **RTPMulticastEnable** is set to True. CAMERA ONLY ENCODER ONLY
	RTPMulticastTTL	REQ, RES	<int>	RTP TTL CAMERA ONLY ENCODER ONLY
	EncodingType	REQ, RES	<enum> MJPEG, H264, MPEG4, H265	Encoding type (read-only for NVR) **Note** **Name** and **EncodingType** must be sent together with the **add** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	Resolution	REQ, RES	<enum>	List of supported resolutions The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	CropRatio	REQ, RES	<enum> Manual, 16:9, 4:3	Crop ratio setting If **CropEncodingEnable** is set to True, then **CropRation** must be set together with **CropEncodingEnable**. CAMERA ONLY ENCODER ONLY
	CropEncodingEnable	REQ, RES	<bool> True, False	Enables or disables the crop encoding This can transmit multiple cropped images from the whole area, as well as a full overview image simultaneously. If **CropEncodingEnable** is set to True, then **CropAreaCoordinate** must be set together with **CropEncodingEnable**. CAMERA ONLY ENCODER ONLY
	CropAreaCoordinate	REQ, RES	<string>	Crop area coordinates (vertices of the crop area) **CropAreaCoordinate** is valid only when **CropEncodingEnable** is set to True. Coordinates should be in the format of <x1, y1, x2, y2> with even numbers only. CAMERA ONLY ENCODER ONLY
	FrameRate	REQ, RES	<int>	Video encoding frame rate

Action	Parameters	Request/ Response	Type/ Value	Description
	CompressionLevel	REQ, RES	<int>	Video encoding compression level The higher the level, the lower the quality.
	Bitrate	REQ, RES	<int>	Video bit rate CAMERA ONLY ENCODER ONLY
	IsDigitalPTZProfile	REQ, RES	<bool> True, False	Enables or disables Digital PTZ for a profile **Note** **Not all profiles support Digital PTZ. Use the videocodecinfo** submenu to learn whether a particular profile supports Digital PTZ or not.
	IsDPMProfile	REQ, RES	<bool>	Set which profile to be used in Dynamic privacy mask.
	MPEG4.BitrateControlType	REQ, RES	<enum> CBR, VBR, Unknown	MPEG4 video bit rate control type • CBR: Constant Bit Rate. Varies the video quality and fixes network transfer bit rate. • VBR: Variable Bit Rate. This varies the network transfer bit rate, to optimize quality. **Note** Unknown value is supported only for NVR.
	MPEG4.GOVLength	REQ, RES	<int>	MPEG4 GOV length CAMERA ONLY ENCODER ONLY
	MPEG4.PriorityType	REQ, RES	<enum> FrameRate, CompressionLevel	MPEG4 video encoding priority type CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	MJPEG.PriorityType	REQ, RES	<enum> FrameRate, Bitrate	MJPEG.video encoding priority type CAMERA ONLY ENCODER ONLY
	H264.BitrateControlType	REQ, RES	<enum> CBR, VBR, Unknown	H264 video bit rate control type (supports the camera only) • CBR: Constant Bit Rate • VBR: Variable Bit Rate **Note** Unknown value is supported only for NVR.
	H264.GOVLength	REQ, RES	<int>	H264 GOV length The GOV length is 8*FrameRate. However if FrameRate is 1, the GOV length is 1 as an exception. CAMERA ONLY ENCODER ONLY
	H264.PriorityType	REQ, RES	<enum> FrameRate, CompressionLevel	H264 video encoding priority type This parameter is valid only when **ATCMode** is NOT set to Disabled. CAMERA ONLY ENCODER ONLY
	H264.Profile	REQ, RES	<enum> BaseLine, Main, High	H264 profile CAMERA ONLY ENCODER ONLY
	H264.EntropyCoding	REQ, RES	<enum> CAVLC, CABAC	H264 video entropy encoding CAMERA ONLY ENCODER ONLY
	H264.DynamicGOVEnable	REQ, RES	<bool> True, False	Enables or disables H264 Dynamic GOV CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	H264.DynamicGOVLength	REQ, RES	<int>	H264 Dynamic GOV length This parameter is valid only when **H264.DynamicGOVEnable** is NOT set to False. CAMERA ONLY ENCODER ONLY
	H264.SmartCodecEnable	REQ, RES	<bool> True, False	Enables or disables H264 smart codec **H264.SmartCodecEnable** is valid only when **H264.BitrateControlType** is set to CBR. CAMERA ONLY ENCODER ONLY
	H264.DynamicFPSEnable	REQ, RES	<bool> True, False	Enables or disables H264 Dynamic FPS CAMERA ONLY ENCODER ONLY
	H264.MinDynamicFPS	REQ, RES	<int>	Minimum value for h264 dynamic fps. CAMERA ONLY ENCODER ONLY
	H265.BitrateControlType	REQ, RES	<enum> CBR, VBR, Unknown	H.265 video bit rate control type (supports the camera only) • CBR: Constant Bit Rate • VBR: Variable Bit Rate **Note** Unknown value is supported only for NVR.
	H265.GOVLength	REQ, RES	<int>	H.265 GOV length The GOV length is 8*FrameRate. However if FrameRate is 1, the GOV length is 1 as an exception. CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	H265.PriorityType	REQ, RES	<enum> FrameRate, CompressionLevel	H.265 video encoding priority type This parameter is valid only when **ATCMode** is NOT set to Disabled. CAMERA ONLY ENCODER ONLY
	H265.Profile	REQ, RES	<enum> Main, Main10, MainStillPicture	H.265 profile
	H265.EntropyCoding	REQ, RES	<enum> CAVLC, CABAC	H.265 video entropy encoding CAMERA ONLY ENCODER ONLY
	H265.SmartCodecEnable	REQ, RES	<bool> True, False	Enables or disables H.265 smart codec **H265.SmartCodecEnable** is valid only when **H265.BitrateControlType** is set to CBR. CAMERA ONLY ENCODER ONLY
	H265.DynamicGOVEnable	REQ, RES	<bool> True, False	Enables or disables H.265 Dynamic GOV CAMERA ONLY ENCODER ONLY
	H264.DynamicGOVLength	REQ, RES	<int>	H.265 Dynamic GOV length This parameter is valid only when H.265.DynamicGOV is NOT set to Disabled. CAMERA ONLY ENCODER ONLY
	H265.DynamicFPSEnable	REQ, RES	<bool> True, False	Enables or disables H.265 Dynamic FPS CAMERA ONLY ENCODER ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	H265.MinDynamicFPS	REQ, RES	<int>	Minimum value for H.265 dynamic fps.
	AudioInputEnable	REQ, RES	<bool> True, False	Enables or disables the audio input.
	ViewModeIndex	REQ, RES	<int>	View mode of index number. If a user wants to set a profile as dewarped view, this should be set as 1. This should set with ViewModeType.
	ViewModeType	REQ, RES	<enum> LeftHalfView, RightHalfView, Overview, 360Panorama, Panorama, DoublePanorama, QuadView, SingleView, OneOverviewAndTripleView, OneOverviewAndOctaView, CropView, QuadView.#, Bright, Dark, SubregionView.1, SubregionView.2	View mode type If SensorCaptureSize is 2M_double_shutter, Overview is used in interleaved (double shutter) mode.
	IsFixedFrameRateProfile	REQ, RES	<bool> True, False	Applicable only for H.264/H.265 profiles. Only one profile of the 10 profiles can be configured as FixedFrameRateProfile. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	IsVoIPProfile	REQ, RES	<bool> True, False	If it is enabled, this profile is used in SIP communication. **Note** Use the **videocodecinfo** submenu to refer to detailed profile setting dependency.
remove	Channel	REQ	<int>	Channel ID
	Profile	REQ	<int>	Profile index **Note** **Profile** must be sent together with the **remove** action.

106.8.4. Examples

106.8.4.1. Getting video profile settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.ATCTrigger=Network
Channel.0.Profile.1.ATCEventType=MotionDetection
Channel.0.Profile.1.ATCMode=Disabled
Channel.0.Profile.1.ATCSensitivity=VeryHigh
Channel.0.Profile.1.ATCLimit=50
Channel.0.Profile.1.RTPMulticastEnable=False
Channel.0.Profile.1.RTPMulticastType=IPV4
Channel.0.Profile.1.RTPMulticastAddress=
Channel.0.Profile.1.RTPMulticastAddressIPv6=
Channel.0.Profile.1.RTPMulticastPort=0
```

```
Channel.0.Profile.1.RTPMulticastTTL=5
Channel.0.Profile.1.CropEncodingEnable=False
Channel.0.Profile.1.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.1.CropRatio=Manual
Channel.0.Profile.1.Resolution=1920x1080
Channel.0.Profile.1.FrameRate=2
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.Bitrate=6144
Channel.0.Profile.1.MJPEG.PriorityType=Bitrate
Channel.0.Profile.1.AudioInputEnable=True
Channel.0.Profile.1.ViewModeType=Overview
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.ProfileToken=0bfe0cc8-6a06-4973-87dc-dc695a3224b0
Channel.0.Profile.2.Name=H.264
Channel.0.Profile.2.EncodingType=H264
Channel.0.Profile.2.ATCTrigger=Network
Channel.0.Profile.2.ATCEventType=MotionDetection
Channel.0.Profile.2.ATCMode=Disabled
Channel.0.Profile.2.ATCSensitivity=VeryHigh
Channel.0.Profile.2.ATCLimit=50
Channel.0.Profile.2.RTPMulticastEnable=False
Channel.0.Profile.2.RTPMulticastType=IPV4
Channel.0.Profile.2.RTPMulticastAddress=
Channel.0.Profile.2.RTPMulticastAddressIPv6=
Channel.0.Profile.2.RTPMulticastPort=0
Channel.0.Profile.2.RTPMulticastTTL=5
Channel.0.Profile.2.CropEncodingEnable=False
Channel.0.Profile.2.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.2.CropRatio=Manual
Channel.0.Profile.2.Resolution=1920x1080
Channel.0.Profile.2.FrameRate=30
Channel.0.Profile.2.CompressionLevel=10
Channel.0.Profile.2.Bitrate=30720
Channel.0.Profile.2.H264.BitrateControlType=VBR
Channel.0.Profile.2.H264.PriorityType=FrameRate
Channel.0.Profile.2.H264.GOVLength=1
Channel.0.Profile.2.H264.Profile=BaseLine
Channel.0.Profile.2.H264.EntropyCoding=CAVLC
Channel.0.Profile.2.H264.SmartCodecEnable=False
Channel.0.Profile.2.H264.MaxGOVLength=240
Channel.0.Profile.2.H264.MinGOVLength=1
```

```
Channel.0.Profile.2.H264.DynamicFPSEnable=False
Channel.0.Profile.2.H264.MinDynamicFPS=1
Channel.0.Profile.2.H264.DynamicGOVEnable=False
Channel.0.Profile.2.H264.DynamicGOVLength=200
Channel.0.Profile.2.H264.MaxDynamicGOVLength=480
Channel.0.Profile.2.AudioInputEnable=False
Channel.0.Profile.2.ViewModeType=Overview
Channel.0.Profile.2.IsFixedProfile=True
Channel.0.Profile.2.ProfileToken=7c3efdcd-f995-4fd4-9c8c-766044665c23
Channel.0.Profile.3.Name=H.265
Channel.0.Profile.3.EncodingType=H265
Channel.0.Profile.3.ATCTrigger=Network
Channel.0.Profile.3.ATCEventType=MotionDetection
Channel.0.Profile.3.ATCMode=Disabled
Channel.0.Profile.3.ATCSensitivity=VeryHigh
Channel.0.Profile.3.ATCLimit=50
Channel.0.Profile.3.RTPMulticastEnable=False
Channel.0.Profile.3.RTPMulticastType=IPV4
Channel.0.Profile.3.RTPMulticastAddress=
Channel.0.Profile.3.RTPMulticastAddressIPv6=
Channel.0.Profile.3.RTPMulticastPort=0
Channel.0.Profile.3.RTPMulticastTTL=5
Channel.0.Profile.3.CropEncodingEnable=False
Channel.0.Profile.3.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.3.CropRatio=Manual
Channel.0.Profile.3.Resolution=1920x1080
Channel.0.Profile.3.FrameRate=25
Channel.0.Profile.3.CompressionLevel=10
Channel.0.Profile.3.Bitrate=2048
Channel.0.Profile.3.H265.BitrateControlType=VBR
Channel.0.Profile.3.H265.PriorityType=FrameRate
Channel.0.Profile.3.H265.GOVLength=50
Channel.0.Profile.3.H265.Profile=Main
Channel.0.Profile.3.H265.EntropyCoding=CABAC
Channel.0.Profile.3.H265.SmartCodecEnable=False
Channel.0.Profile.3.H265.MaxGOVLength=200
Channel.0.Profile.3.H265.MinGOVLength=1
Channel.0.Profile.3.H265.DynamicFPSEnable=False
Channel.0.Profile.3.H265.MinDynamicFPS=1
Channel.0.Profile.3.H265.DynamicGOVEnable=False
Channel.0.Profile.3.H265.DynamicGOVLength=200
```

```
Channel.0.Profile.3.H265.MaxDynamicGOVLength=480
Channel.0.Profile.3.AudioInputEnable=False
Channel.0.Profile.3.ViewModeType=Overview
Channel.0.Profile.3.IsFixedProfile=True
Channel.0.Profile.3.ProfileToken=c848c619-4653-4483-ba3e-c1ca0030056e
Channel.0.Profile.4.Name=PLUGINFREE
Channel.0.Profile.4.EncodingType=H264
Channel.0.Profile.4.ATCTrigger=Network
Channel.0.Profile.4.ATCEventType=MotionDetection
Channel.0.Profile.4.ATCMode=Disabled
Channel.0.Profile.4.ATCSensitivity=VeryHigh
Channel.0.Profile.4.ATCLimit=50
Channel.0.Profile.4.RTPMulticastEnable=False
Channel.0.Profile.4.RTPMulticastType=IPV4
Channel.0.Profile.4.RTPMulticastAddress=
Channel.0.Profile.4.RTPMulticastAddressIPv6=
Channel.0.Profile.4.RTPMulticastPort=0
Channel.0.Profile.4.RTPMulticastTTL=5
Channel.0.Profile.4.CropEncodingEnable=False
Channel.0.Profile.4.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.4.CropRatio=Manual
Channel.0.Profile.4.Resolution=1280x720
Channel.0.Profile.4.FrameRate=20
Channel.0.Profile.4.CompressionLevel=10
Channel.0.Profile.4.Bitrate=2048
Channel.0.Profile.4.H264.BitrateControlType=VBR
Channel.0.Profile.4.H264.PriorityType=FrameRate
Channel.0.Profile.4.H264.GOVLength=20
Channel.0.Profile.4.H264.Profile=Main
Channel.0.Profile.4.H264.EntropyCoding=CABAC
Channel.0.Profile.4.H264.SmartCodecEnable=False
Channel.0.Profile.4.H264.MaxGOVLength=160
Channel.0.Profile.4.H264.MinGOVLength=1
Channel.0.Profile.4.H264.DynamicFPSEnable=False
Channel.0.Profile.4.H264.MinDynamicFPS=1
Channel.0.Profile.4.H264.DynamicGOVEnable=False
Channel.0.Profile.4.H264.DynamicGOVLength=200
Channel.0.Profile.4.H264.MaxDynamicGOVLength=480
Channel.0.Profile.4.AudioInputEnable=False
Channel.0.Profile.4.ViewModeType=Overview
Channel.0.Profile.4.IsFixedProfile=False
```

```
Channel.0.Profile.4.ProfileToken=77db8fa8-a4fd-4061-b365-30af89242f5a
Channel.0.Profile.10.Name=MOBILE
Channel.0.Profile.10.EncodingType=H264
Channel.0.Profile.10.ATCTrigger=Network
Channel.0.Profile.10.ATCEventType=MotionDetection
Channel.0.Profile.10.ATCMode=Disabled
Channel.0.Profile.10.ATCSensitivity=VeryHigh
Channel.0.Profile.10.ATCLimit=50
Channel.0.Profile.10.RTPMulticastEnable=False
Channel.0.Profile.10.RTPMulticastType=IPV4
Channel.0.Profile.10.RTPMulticastAddress=
Channel.0.Profile.10.RTPMulticastAddressIPv6=
Channel.0.Profile.10.RTPMulticastPort=0
Channel.0.Profile.10.RTPMulticastTTL=5
Channel.0.Profile.10.CropEncodingEnable=False
Channel.0.Profile.10.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.10.CropRatio=Manual
Channel.0.Profile.10.Resolution=1920x1080
Channel.0.Profile.10.FrameRate=30
Channel.0.Profile.10.CompressionLevel=10
Channel.0.Profile.10.Bitrate=30720
Channel.0.Profile.10.H264.BitrateControlType=VBR
Channel.0.Profile.10.H264.PriorityType=FrameRate
Channel.0.Profile.10.H264.GOVLength=1
Channel.0.Profile.10.H264.Profile=BaseLine
Channel.0.Profile.10.H264.EntropyCoding=CAVLC
Channel.0.Profile.10.H264.SmartCodecEnable=False
Channel.0.Profile.10.H264.MaxGOVLength=240
Channel.0.Profile.10.H264.MinGOVLength=1
Channel.0.Profile.10.H264.DynamicFPSEnable=False
Channel.0.Profile.10.H264.MinDynamicFPS=1
Channel.0.Profile.10.H264.DynamicGOVEnable=False
Channel.0.Profile.10.H264.DynamicGOVLength=200
Channel.0.Profile.10.H264.MaxDynamicGOVLength=480
Channel.0.Profile.10.AudioInputEnable=False
Channel.0.Profile.10.ViewModeType=Overview
Channel.0.Profile.10.IsFixedProfile=False
Channel.0.Profile.10.ProfileToken=c2ecc6cd-457a-429b-8e59-793cef09e759
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoProfiles": [
        {
            "Channel": 0,
            "Profiles": [
                {
                    "Profile": 1,
                    "Name": "MJPEG",
                    "EncodingType": "MJPEG",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastType": "IPV4",
                    "RTPMulticastAddress": "",
                    "RTPMulticastAddressIPv6": "",
                    "RTPMulticastPort": 0,
                    "RTPMulticastTTL": 5,
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1920x1080",
                    "FrameRate": 2,
                    "CompressionLevel": 10,
                    "Bitrate": 6144,
                    "PriorityType": "Bitrate",
                    "AudioInputEnable": true,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": true,
                    "ProfileToken": "0bfe0cc8-6a06-4973-87dc-dc695a3224b0"
                },
                {
                    "Profile": 2,
```

```
                "Name": "H.264",
                "EncodingType": "H264",
                "ATCTrigger": "Network",
                "ATCEventType": "MotionDetection",
                "ATCMode": "Disabled",
                "ATCSensitivity": "VeryHigh",
                "ATCLimit": 50,
                "RTPMulticastEnable": false,
                "RTPMulticastType": "IPV4",
                "RTPMulticastAddress": "",
                "RTPMulticastAddressIPv6": "",
                "RTPMulticastPort": 0,
                "RTPMulticastTTL": 5,
                "CropEncodingEnable": false,
                "CropAreaCoordinate": "320,28,1600,1052",
                "CropRatio": "Manual",
                "Resolution": "1920x1080",
                "FrameRate": 30,
                "CompressionLevel": 10,
                "Bitrate": 30720,
                "H264": {
                    "BitrateControlType": "VBR",
                    "PriorityType": "FrameRate",
                    "GOVLength": 1,
                    "Profile": "BaseLine",
                    "EntropyCoding": "CAVLC",
                    "SmartCodecEnable": false,
                    "MaxGOVLength": 240,
                    "MinGOVLength": 1,
                    "DynamicFPSEnable": false,
                    "MinDynamicFPS": 1,
                    "DynamicGOVEnable": false,
                    "DynamicGOVLength": 200,
                    "MaxDynamicGOVLength": 480
                },
                "AudioInputEnable": false,
                "ViewModeType": "Overview",
                "IsFixedProfile": true,
                "ProfileToken": "7c3efdcd-f995-4fd4-9c8c-766044665c23"
            },
            {
```

```
            "Profile": 3,
            "Name": "H.265",
            "EncodingType": "H265",
            "ATCTrigger": "Network",
            "ATCEventType": "MotionDetection",
            "ATCMode": "Disabled",
            "ATCSensitivity": "VeryHigh",
            "ATCLimit": 50,
            "RTPMulticastEnable": false,
            "RTPMulticastType": "IPV4",
            "RTPMulticastAddress": "",
            "RTPMulticastAddressIPv6": "",
            "RTPMulticastPort": 0,
            "RTPMulticastTTL": 5,
            "CropEncodingEnable": false,
            "CropAreaCoordinate": "320,28,1600,1052",
            "CropRatio": "Manual",
            "Resolution": "1920x1080",
            "FrameRate": 25,
            "CompressionLevel": 10,
            "Bitrate": 2048,
            "H265": {
                "BitrateControlType": "VBR",
                "PriorityType": "FrameRate",
                "GOVLength": 50,
                "Profile": "Main",
                "EntropyCoding": "CABAC",
                "SmartCodecEnable": false,
                "MaxGOVLength": 200,
                "MinGOVLength": 1,
                "DynamicFPSEnable": false,
                "MinDynamicFPS": 1,
                "DynamicGOVEnable": false,
                "DynamicGOVLength": 200,
                "MaxDynamicGOVLength": 480
            },
            "AudioInputEnable": false,
            "ViewModeType": "Overview",
            "IsFixedProfile": true,
            "ProfileToken": "c848c619-4653-4483-ba3e-c1ca0030056e"
        },
```

```
                {
                    "Profile": 4,
                    "Name": "PLUGINFREE",
                    "EncodingType": "H264",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastType": "IPV4",
                    "RTPMulticastAddress": "",
                    "RTPMulticastAddressIPv6": "",
                    "RTPMulticastPort": 0,
                    "RTPMulticastTTL": 5,
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1280x720",
                    "FrameRate": 20,
                    "CompressionLevel": 10,
                    "Bitrate": 2048,
                    "H264": {
                        "BitrateControlType": "VBR",
                        "PriorityType": "FrameRate",
                        "GOVLength": 20,
                        "Profile": "Main",
                        "EntropyCoding": "CABAC",
                        "SmartCodecEnable": false,
                        "MaxGOVLength": 160,
                        "MinGOVLength": 1,
                        "DynamicFPSEnable": false,
                        "MinDynamicFPS": 1,
                        "DynamicGOVEnable": false,
                        "DynamicGOVLength": 200,
                        "MaxDynamicGOVLength": 480
                    },
                    "AudioInputEnable": false,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": false,
                    "ProfileToken": "77db8fa8-a4fd-4061-b365-30af89242f5a"
```

```
                },
                {
                    "Profile": 10,
                    "Name": "MOBILE",
                    "EncodingType": "H264",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastType": "IPV4",
                    "RTPMulticastAddress": "",
                    "RTPMulticastAddressIPv6": "",
                    "RTPMulticastPort": 0,
                    "RTPMulticastTTL": 5,
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1920x1080",
                    "FrameRate": 30,
                    "CompressionLevel": 10,
                    "Bitrate": 30720,
                    "H264": {
                        "BitrateControlType": "VBR",
                        "PriorityType": "FrameRate",
                        "GOVLength": 1,
                        "Profile": "BaseLine",
                        "EntropyCoding": "CAVLC",
                        "SmartCodecEnable": false,
                        "MaxGOVLength": 240,
                        "MinGOVLength": 1,
                        "DynamicFPSEnable": false,
                        "MinDynamicFPS": 1,
                        "DynamicGOVEnable": false,
                        "DynamicGOVLength": 200,
                        "MaxDynamicGOVLength": 480
                    },
                    "AudioInputEnable": false,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": false,
```

```
                    "ProfileToken": "c2ecc6cd-457a-429b-8e59-793cef09e759"
                }
            ]
        }
    ]
}
```


106.8.4.2. Getting Profile 1 information

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videoprofile&action=view&Profile=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.ATCTrigger=Network
Channel.0.Profile.1.ATCEventType=MotionDetection
Channel.0.Profile.1.ATCMode=Disabled
Channel.0.Profile.1.ATCSensitivity=VeryHigh
Channel.0.Profile.1.ATCLimit=50
Channel.0.Profile.1.RTPMulticastEnable=False
Channel.0.Profile.1.RTPMulticastAddress=
Channel.0.Profile.1.RTPMulticastPort=0
Channel.0.Profile.1.RTPMulticastTTL=5
Channel.0.Profile.1.CropEncodingEnable=False
Channel.0.Profile.1.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.1.CropRatio=Manual
Channel.0.Profile.1.Resolution=1920x1080
Channel.0.Profile.1.FrameRate=2
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.Bitrate=6144
Channel.0.Profile.1.MJPEG.PriorityType=Bitrate
Channel.0.Profile.1.AudioInputEnable=True
Channel.0.Profile.1.ViewModeType=Overview
```

```
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.ProfileToken=0bfe0cc8-6a06-4973-87dc-dc695a3224b0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoProfiles": [
        {
            "Channel": 0,
            "Profiles": [
                {
                    "Profile": 1,
                    "Name": "MJPEG",
                    "EncodingType": "MJPEG",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastAddress": "",
                    "RTPMulticastPort": 0,
                    "RTPMulticastTTL": 5,
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1920x1080",
                    "FrameRate": 2,
                    "CompressionLevel": 10,
                    "Bitrate": 6144,
                    "PriorityType": "Bitrate",
                    "AudioInputEnable": true,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": true,
                    "ProfileToken": "0bfe0cc8-6a06-4973-87dc-dc695a3224b0"
                }
            ]
```

```
        }
    ]
}
```

106.8.4.3. Getting all profiles including audio only and meta only profiles

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videoprofile&action=view&ShowAll=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.ATCTrigger=Network
Channel.0.Profile.1.ATCEventType=MotionDetection
Channel.0.Profile.1.ATCMode=Disabled
Channel.0.Profile.1.ATCSensitivity=VeryHigh
Channel.0.Profile.1.ATCLimit=50
Channel.0.Profile.1.RTPMulticastEnable=False
Channel.0.Profile.1.RTPMulticastAddress=
Channel.0.Profile.1.RTPMulticastPort=0
Channel.0.Profile.1.RTPMulticastTTL=5
Channel.0.Profile.1.CropEncodingEnable=False
Channel.0.Profile.1.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.1.CropRatio=Manual
Channel.0.Profile.1.Resolution=1920x1080
Channel.0.Profile.1.FrameRate=2
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.Bitrate=6144
Channel.0.Profile.1.MJPEG.PriorityType=Bitrate
Channel.0.Profile.1.AudioInputEnable=True
Channel.0.Profile.1.ViewModeType=Overview
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.ProfileToken=0bfe0cc8-6a06-4973-87dc-dc695a3224b0
Channel.0.Profile.2.Name=H.264
```

```
Channel.0.Profile.2.EncodingType=H264
Channel.0.Profile.2.ATCTrigger=Network
Channel.0.Profile.2.ATCEventType=MotionDetection
Channel.0.Profile.2.ATCMode=Disabled
Channel.0.Profile.2.ATCSensitivity=VeryHigh
Channel.0.Profile.2.ATCLimit=50
Channel.0.Profile.2.RTPMulticastEnable=False
Channel.0.Profile.2.RTPMulticastAddress=
Channel.0.Profile.2.RTPMulticastPort=0
Channel.0.Profile.2.RTPMulticastTTL=5
Channel.0.Profile.2.CropEncodingEnable=False
Channel.0.Profile.2.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.2.CropRatio=Manual
Channel.0.Profile.2.Resolution=1920x1080
Channel.0.Profile.2.FrameRate=30
Channel.0.Profile.2.CompressionLevel=10
Channel.0.Profile.2.Bitrate=30720
Channel.0.Profile.2.H264.BitrateControlType=VBR
Channel.0.Profile.2.H264.PriorityType=FrameRate
Channel.0.Profile.2.H264.GOVLength=1
Channel.0.Profile.2.H264.Profile=BaseLine
Channel.0.Profile.2.H264.EntropyCoding=CAVLC
Channel.0.Profile.2.H264.SmartCodecEnable=False
Channel.0.Profile.2.H264.MaxGOVLength=240
Channel.0.Profile.2.H264.MinGOVLength=1
Channel.0.Profile.2.H264.DynamicFPSEnable=False
Channel.0.Profile.2.H264.MinDynamicFPS=1
Channel.0.Profile.2.H264.DynamicGOVEnable=False
Channel.0.Profile.2.H264.DynamicGOVLength=200
Channel.0.Profile.2.H264.MaxDynamicGOVLength=480
Channel.0.Profile.2.AudioInputEnable=False
Channel.0.Profile.2.ViewModeType=Overview
Channel.0.Profile.2.IsFixedProfile=True
Channel.0.Profile.2.ProfileToken=7c3efdcd-f995-4fd4-9c8c-766044665c23
Channel.0.Profile.3.Name=H.265
Channel.0.Profile.3.EncodingType=H265
Channel.0.Profile.3.ATCTrigger=Network
Channel.0.Profile.3.ATCEventType=MotionDetection
Channel.0.Profile.3.ATCMode=Disabled
Channel.0.Profile.3.ATCSensitivity=VeryHigh
Channel.0.Profile.3.ATCLimit=50
```

```
Channel.0.Profile.3.RTPMulticastEnable=False
Channel.0.Profile.3.RTPMulticastAddress=
Channel.0.Profile.3.RTPMulticastPort=0
Channel.0.Profile.3.RTPMulticastTTL=5
Channel.0.Profile.3.CropEncodingEnable=False
Channel.0.Profile.3.CropAreaCoordinate=320,28,1600,1052
Channel.0.Profile.3.CropRatio=Manual
Channel.0.Profile.3.Resolution=1920x1080
Channel.0.Profile.3.FrameRate=25
Channel.0.Profile.3.CompressionLevel=10
Channel.0.Profile.3.Bitrate=2048
Channel.0.Profile.3.H265.BitrateControlType=VBR
Channel.0.Profile.3.H265.PriorityType=FrameRate
Channel.0.Profile.3.H265.GOVLength=50
Channel.0.Profile.3.H265.Profile=Main
Channel.0.Profile.3.H265.EntropyCoding=CABAC
Channel.0.Profile.3.H265.SmartCodecEnable=False
Channel.0.Profile.3.H265.MaxGOVLength=200
Channel.0.Profile.3.H265.MinGOVLength=1
Channel.0.Profile.3.H265.DynamicFPSEnable=False
Channel.0.Profile.3.H265.MinDynamicFPS=1
Channel.0.Profile.3.H265.DynamicGOVEnable=False
Channel.0.Profile.3.H265.DynamicGOVLength=200
Channel.0.Profile.3.H265.MaxDynamicGOVLength=480
Channel.0.Profile.3.AudioInputEnable=False
Channel.0.Profile.3.ViewModeType=Overview
Channel.0.Profile.3.IsFixedProfile=True
Channel.0.Profile.3.ProfileToken=c848c619-4653-4483-ba3e-c1ca0030056e
Channel.0.Profile.4.Name=testtt22
Channel.0.Profile.4.AudioInputEnable=False
Channel.0.Profile.4.IsFixedProfile=False
Channel.0.Profile.4.ProfileToken=51df39de-d73f-4247-bb2f-fd3e466d413a
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoProfiles": [
```

```
        {
            "Channel": 0,
            "Profiles": [
                {
                    "Profile": 1,
                    "Name": "MJPEG",
                    "EncodingType": "MJPEG",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastAddress": "",
                    "RTPMulticastPort": 0,
                    "RTPMulticastTTL": 5,
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1920x1080",
                    "FrameRate": 2,
                    "CompressionLevel": 10,
                    "Bitrate": 6144,
                    "PriorityType": "Bitrate",
                    "AudioInputEnable": true,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": true,
                    "ProfileToken": "0bfe0cc8-6a06-4973-87dc-dc695a3224b0"
                },
                {
                    "Profile": 2,
                    "Name": "H.264",
                    "EncodingType": "H264",
                    "ATCTrigger": "Network",
                    "ATCEventType": "MotionDetection",
                    "ATCMode": "Disabled",
                    "ATCSensitivity": "VeryHigh",
                    "ATCLimit": 50,
                    "RTPMulticastEnable": false,
                    "RTPMulticastAddress": "",
                    "RTPMulticastPort": 0,
```

```
            "RTPMulticastTTL": 5,
            "CropEncodingEnable": false,
            "CropAreaCoordinate": "320,28,1600,1052",
            "CropRatio": "Manual",
            "Resolution": "1920x1080",
            "FrameRate": 30,
            "CompressionLevel": 10,
            "Bitrate": 30720,
            "H264": {
                "BitrateControlType": "VBR",
                "PriorityType": "FrameRate",
                "GOVLength": 1,
                "Profile": "BaseLine",
                "EntropyCoding": "CAVLC",
                "SmartCodecEnable": false,
                "MaxGOVLength": 240,
                "MinGOVLength": 1,
                "DynamicFPSEnable": false,
                "MinDynamicFPS": 1,
                "DynamicGOVEnable": false,
                "DynamicGOVLength": 200,
                "MaxDynamicGOVLength": 480
            },
            "AudioInputEnable": false,
            "ViewModeType": "Overview",
            "IsFixedProfile": true,
            "ProfileToken": "7c3efdcd-f995-4fd4-9c8c-766044665c23"
        },
        {
            "Profile": 3,
            "Name": "H.265",
            "EncodingType": "H265",
            "ATCTrigger": "Network",
            "ATCEventType": "MotionDetection",
            "ATCMode": "Disabled",
            "ATCSensitivity": "VeryHigh",
            "ATCLimit": 50,
            "RTPMulticastEnable": false,
            "RTPMulticastAddress": "",
            "RTPMulticastPort": 0,
            "RTPMulticastTTL": 5,
```

```
                    "CropEncodingEnable": false,
                    "CropAreaCoordinate": "320,28,1600,1052",
                    "CropRatio": "Manual",
                    "Resolution": "1920x1080",
                    "FrameRate": 25,
                    "CompressionLevel": 10,
                    "Bitrate": 2048,
                    "H265": {
                        "BitrateControlType": "VBR",
                        "PriorityType": "FrameRate",
                        "GOVLength": 50,
                        "Profile": "Main",
                        "EntropyCoding": "CABAC",
                        "SmartCodecEnable": false,
                        "MaxGOVLength": 200,
                        "MinGOVLength": 1,
                        "DynamicFPSEnable": false,
                        "MinDynamicFPS": 1,
                        "DynamicGOVEnable": false,
                        "DynamicGOVLength": 200,
                        "MaxDynamicGOVLength": 480
                    },
                    "AudioInputEnable": false,
                    "ViewModeType": "Overview",
                    "IsFixedProfile": true,
                    "ProfileToken": "c848c619-4653-4483-ba3e-c1ca0030056e"
                },
                {
                    "Profile": 4,
                    "Name": "testtt22",
                    "AudioInputEnable": false,
                    "IsFixedProfile": false,
                    "ProfileToken": "51df39de-d73f-4247-bb2f-fd3e466d413a"
                }
            ]
        }
    ]
}
```


106.8.4.4. Removing Profile 4 on Channel 0

To use the **remove** action, the **Profile** parameter should be set.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=remove&Channel=0&Profile=4
```

106.8.4.5. Adding a profile

Setting the name to Test and encoding type to MJPEG

To add a profile with the **add** action, the **Name** and **EncodingType** parameters must be set together.

The **add** action returns the profile index which should be sent for the **update** action.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=add&Name=Test&EncodingType=MJPEG
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=add&Channel=0&Name=Temp3&EncodingType=MJPEG
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Profile=4
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "Response": "Success",
    Profile=4
}
```

106.8.4.6. Updating Profile 4

Setting the resolution to 1920x1080 and encoding type to H.264

To modify the profile with the **update** action, the **Profile** parameter must be set.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=update&Profile=4&Resolution=1920x1080&EncodingType=H264
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=update&Profile=4&Resolution=1920x1080&EncodingType=H264&Channel=0
```

Setting ATC to send I-frames in normal scenario and full frames only for motion detection events

REQUEST

The following request example is for network cameras only.

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=update&Profile=4&ATCTrigger=Event&ATCEventType=MotionDetection&ATCMode=AdjustFramerate
```

Switching ATC to network mode (adjusting video bandwidth according to network traffic)

REQUEST

The following request example is for network cameras only.

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=update&Profile=4&ATCTrigger=Network&ATCMode=AdjustFramerate
```

106.9. Audio Input

106.9.1. Description

The **audioinput** submenu configures and views the audio input settings.

Access level

Action	Camera	NVR	Encoder	Decoder
view	Suser	User	Suser	User
set	Admin	User	Admin	User

NOTE

Attribute to check for Audio Input: "attributes/Media/Support/**AudioIn**"
Permission to check for User Access: "security/users/AudioInAccess"
Attribute to check for maximum Audio Inputs: "attributes/Media/Limit/**MaxAudioInput**"

106.9.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
audioinput&action=<value>[&<parameter>=<value>]
```

106.9.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the audio input settings
	Channel	REQ	<csv>	Channel ID
	SampleRate	RES	<int> 8000	Audio input sample rate (read only)
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables the audio input.
	Mode	REQ, RES	<enum> Mono, Stereo	Audio input mode (read-only for NVR)

Action	Parameters	Request/ Response	Type/ Value	Description
	Source	REQ, RES	<enum> MIC, ExternalMIC, LineIn	Audio input source • MIC: Built-in microphone in the camera • ExternalMIC: External microphone • LineIn: Audio device connected through a cable The value may vary depending on the model. Please check the device attributes using attributes.cgi. CAMERA ONLY
	PowerOnExternalMIC	REQ, RES	<bool> True, False	Enables or disables the function in which the camera supplies power to the connected external microphone when the external microphone has no separate power supply. This parameter is valid only when **Source** is set to ExternalMIC. CAMERA ONLY
	EncodingType	REQ, RES	<enum> G711, G726, AAC, OPUS	Audio input codec (read-only for NVR) The value may vary depending on the model. **Note** Attribute to check for supported AudioEncodings: "attributes/Media/Support/**AudioEncodingType**"

Action	Parameters	Request/ Response	Type/ Value	Description
	Bitrate	REQ, RES	<enum> 16000, 24000, 32000, 40000, 64000	Audio input bit rate (read-only for NVR) The value may vary depending on the model. Please check the device attributes using attributes.cgi. The values 16000, 24000, 32000, or 40000 are available if **EncodingType** is set to G726; the value 64000 is available if **EncodingType** is set to G711.
	Gain	REQ, RES	<int>	Audio input gain (read-only for NVR) If **Enable** is set to True, the value is in the range of 1 to 10. If **Enable** is set to False, the value is 0.
	NoiseReduction	REQ, RES	<bool> True, False	For enabling noise reduction feature
	NoiseReductionSensitivity	REQ, RES	<int>	Noise reduction sensitivity value

106.9.4. Examples

106.9.4.1. Getting audio input settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audioinput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.SampleRate=8000
Channel.0.Mode=Mono
Channel.0.Source=LineIn
Channel.0.PowerOnExternalMIC=True
Channel.0.EncodingType=G711
```

```
Channel.0.Bitrate=64000
Channel.0.Gain=5
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AudioInputs": [
        {
            "Channel": 0,
            "SampleRate": 8000,
            "Mode": "Mono",
            "Source": "LineIn",
            "PowerOnExternalMIC": true,
            "EncodingType": "G711",
            "Bitrate": "64000",
            "Gain": 5
        }
    ]
}
```


The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.SampleRate=0
Channel.0.Mode=Mono
Channel.0.EncodingType=
Channel.0.Bitrate=0
Channel.0.Gain=1
Channel.1.Enable=True
Channel.1.SampleRate=0
```

```
Channel.1.Mode=Mono
Channel.1.EncodingType=
Channel.1.Bitrate=0
Channel.1.Gain=1
Channel.2.Enable=True
Channel.2.SampleRate=8
Channel.2.Mode=Mono
Channel.2.EncodingType=G711
Channel.2.Bitrate=64000
Channel.2.Gain=1
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AudioInputs": [
        {
            "Channel": 0,
            "Enable": true,
            "SampleRate": 0,
            "Mode": "Mono",
            "EncodingType": "",
            "Bitrate": "0",
            "Gain": 1
        },
        {
            "Channel": 1,
            "Enable": true,
            "SampleRate": 0,
            "Mode": "Mono",
            "EncodingType": "",
            "Bitrate": "0",
            "Gain": 1
        },
        {
            "Channel": 2,
            "Enable": true,
```

```
            "SampleRate": 8,
            "Mode": "Mono",
            "EncodingType": "G711",
            "Bitrate": "64000",
            "Gain": 1
        }
        ...
    ]
}
```

106.9.4.2. Setting the audio gain to 1

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audioinput&action=set&Gain=1
```

106.10. Audio Input Options

106.10.1. Description

The **audioinputoptions** submenu views the audio input settings dependency.

Access level

Action	Camera
view	Suser
set	Suser

106.10.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
audioinputoptions&action=<value>[&<parameter>=<value>]
```

106.10.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the audio input configurable settings
	Channel	REQ	<csv>	Channel ID
	Codecs.#.SampleRate	RES	<enum>	Displays the configurable sample rates for the codec.

Action	Parameters	Request/ Response	Type/ Value	Description
	Codecs.#.Bitrate	RES	<enum>	Displays the configurable bitrate for the codec.

106.10.4. Examples

106.10.4.1. Getting audio input configurable settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audioinputoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
  "AudioInputOptions": [
    {
      "Channel": 0,
      "Codecs": {
        "G.711": {
          "SampleRate": [
            8000
          ],
          "Bitrate": [
            64000
          ]
        },
        "G.726": {
          "SampleRate": [
            8000
          ],
          "Bitrate": [
            16000,
            24000,
            32000,
            40000
          ]
```

```
      },
      "AAC": {
        "SampleRate": [
          16000
        ],
        "Bitrate": [
          48000
        ]
      },
      "OPUS": {
        "SampleRate": [
          16000
        ],
        "Bitrate": [
          64000
        ]
      }
    }
  }
 ]
}
```

106.11. Audio Output

106.11.1. Description

The **audiooutput** submenu configures and views the audio output settings.

Access level

Action	Camera	NVR	Encoder
view	Suser	User	Suser
set	Admin	(Not supported)	Admin

NOTE

Attribute to check for Audio Output: "attributes/Media/Support/Device**AudioOut**"
Permission to check for User Access: "security/users/AudioOutAccess"
Attribute to check for maximum Audio Inputs: "attributes/Media/Limit/**MaxAudioOutput**"

106.11.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
```

```
audiooutput&action=<value>[&<parameter>=<value>]
```

106.11.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the audio output settings
	Channel	REQ	<csv>	Channel ID
	SampleRate	RES	<int> 8000	Audio output sample rate
	UnitSize	RES	<int>	Unit size of audio output
set	Channel	REQ, RES	<int>	Channel ID (read-only for NVR)
	DecodingType	REQ, RES	<enum> G711, G726	Audio output decoding type (read-only for NVR) The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	Mode	REQ, RES	<enum> Mono, Stereo	Audio output mode (read-only for NVR) The value may vary depending on the model.
	Bitrate	REQ, RES	<enum> 16000, 24000, 32000, 40000, 64000	Audio output bit rate (read-only for NVR) The values 16000, 24000, 32000, and 40000 are available if **DecodingType** is set to G726; the value 64000 is available if **DecodingType** is set to G711. The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	Enable	REQ, RES	<bool> True, False	Enables or disables the audio output. (read-only for NVR)
	Gain	REQ, RES	<int>	Audio output gain (read-only for NVR) If **Enable** is set to True, the value is in the range of 1 to 10. If **Enable** is set to False, the value is 0.

Action	Parameters	Request/ Response	Type/ Value	Description
	EQPreset	REQ, RES	<enum> NEUTRAL, BASS_BOOST, MID_BOOST, TREBLE_BOOST	Audio output Equalizer presets.

106.11.4. Examples

106.11.4.1. Getting the audio output settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audiooutput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.SampleRate=8000
Channel.0.DecodingType=G711
Channel.0.Mode=Mono
Channel.0.Bitrate=64000
Channel.0.Enable=True
Channel.0.Gain=5
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AudioOutputs": [
        {
            "Channel": 0,
```

```
            "SampleRate": 8000,
            "DecodingType": "G711",
            "Mode": "Mono",
            "Bitrate": "64000",
            "Enable": true,
            "Gain": 5
        }
    ]
}
```

The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.1.Enable=True
Channel.1.UnitSize=8000
Channel.1.SampleRate=8000
Channel.1.Mode=Mono
Channel.1.DecodingType=G711
Channel.1.Bitrate=0
Channel.1.Gain=1
Channel.2.Enable=True
Channel.2.UnitSize=8000
Channel.2.SampleRate=8000
Channel.2.Mode=Mono
Channel.2.DecodingType=G711
Channel.2.Bitrate=0
Channel.2.Gain=1
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AudioOutputs": [
        {
            "Channel": 0,
            "Enable": false
        },
        {
            "Channel": 1,
            "Enable": true,
            "UnitSize": 8000,
            "SampleRate": 8000,
            "Mode": "Mono",
            "DecodingType": "G711",
            "Bitrate": "0",
            "Gain": 1
        },
        {
            "Channel": 2,
            "Enable": true,
            "UnitSize": 8000,
            "SampleRate": 8000,
            "Mode": "Mono",
            "DecodingType": "G711",
            "Bitrate": "0",
            "Gain": 1
        }
    ]
}
```

106.11.4.2. Setting the audio output to mono

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=audiooutput&action=set&Enable=True&Mode=Mono
```

106.12. Stream URI

106.12.1. Description

The **streamuri** submenu requests the streaming URI (Uniform Resource Identifier) for a given profile based on the configuration parameters.

Access level

Action	Camera	NVR	Encoder
view	Guest	User	Guest

NOTE	Attribute to check for UDP streaming: "attributes/Media/Support/Stream.UDP" Attribute to check for TCP streaming: "attributes/Media/Support/Stream.TCP" Attribute to check for Multicast streaming: "attributes/Media/Support/Stream.Multicast" Attribute to check for RTP streaming over RTSP Over TCP: "attributes/Media/Support/Stream.RTPOverRTSPOverTCP" Attribute to check for RTSP streaming over HTTP: "attributes/Media/Support/Stream.RTSPOverHTTP" Attribute to check for RTSP streaming over HTTPS: "attributes/Media/Support/Stream.RTSPOverHTTPS"

106.12.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
streamuri&action=view[&<parameter>=<value>]
```

106.12.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID
	DeviceID	REQ	<int>	Device ID NVR ONLY
	Profile	REQ	<int>	Stream URI profile **Note** **Profile**, **MediaType**, **Mode**, **StreamType**, **TransportProtocol**, and **RTSPOverHTTP** must be sent together with the **view** action.
	MediaType	REQ	<enum> Live, Search, Backup	Streaming media type
	Mode	REQ	<enum> IframeOnly, Full	Streaming mode

Action	Parameters	Request/ Response	Type/ Value	Description
	ClientType	REQ	<enum> PC, Mobile	Client type **Note** **Profile**, **MediaType**, **Mode**, **StreamType**, **TransportProtocol**, **RTSPOverHTTP** and **ClientType** must be sent together with the **view** action for NVR. NVR ONLY
	StreamType	REQ	<enum> RTPUnicast, RTPMulticast	Streaming type
	TransportProtocol	REQ	<enum> TCP, UDP	Streaming transfer protocol
	RTSPOverHTTP	REQ	<bool> True, False	Enables or disables the RTSP (Real Time Streaming Protocol) over HTTP (Hyper Transfer Protocol)
	OverlappedID	REQ	<int>	Overlapped recording ID For more information about overlapped recording, please refer to 'Chapter 5 Overlapped Recording' of the 'Recording' document. **OverlappedID** is valid only when **MediaType** is set to Backup. CAMERA ONLY
	RecordStreamType	REQ	<enum> Main, Sub	Requests a stream type that users want. This parameter is valid only when **MediaType** is set to Search or Backup. NVR ONLY
	URI	RES	<string>	URI

106.12.4. Examples

106.12.4.1. Getting the stream URI of Profile 2

To get the stream URI using the **view** action, **Profile**, **MediaType**, **Mode**, **StreamType**, **TransportProtocol**, and **RTSPOverHTTP** parameters must be sent together (For NVR, **ClientType** is also mandatory).

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&Profile=2&MediaType=L
ive&Mode=Full&StreamType=RTPUnicast&TransportProtocol=TCP&RTSPOverHTTP=False
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://<Device IP>:<Port Number>/profile2/media.smp
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "URI": "rtsp://://<Device IP>:<Port Number>//profile2/media.smp"
}
```

The following example is for NVR only.

REQUEST

```
http://Device IP/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&MediaType=Live&Mode=I
frameOnly&ClientType=Mobile&RTSPOverHttp=True&StreamType=RTPUnicast&Transpor
tProtocol=TCP
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=http://<Device IP>:<Port Number>/LiveChannel/0/media.smp/iframe
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "URI": " http://<Device IP>:<Port
Number>/LiveChannel/0/media.smp/iframe"
}
```


106.12.4.2. Getting the stream URI of POS

To get the stream URI for POS using the **view** action, DeviceID, **MediaType**, **Mode**, and **ClientType** are mandatory.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&DeviceID=0&MediaType=Live&Mode=
Full&ClientType=PC
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://<Device IP>:<Port Number>/PosLive/0/media.smp
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
```

```
    "URI": "rtsp:///<Device IP>:<Port Number>/PosLive/0/media.smp"
}
```

106.13. Session Key

106.13.1. Description

The **sessionkey** submenu requests the session key settings.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User

106.13.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
sessionkey&action=view[&<parameter>=<value>]
```

106.13.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	SessionKey	RES	<int>	Session key

106.13.4. Examples

106.13.4.1. Getting the session key information

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=sessionkey&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SessionKey=1234567
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SessionKey": "1234567"
}
```


106.14. Multicast

106.14.1. Description

The **multicast** submenu controls multicast operations.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
control	Guest

106.14.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
multicast&action=control[&<parameter>=<value>...]
```

106.14.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<Int>	Channel ID MULTI-DIRECTIONAL CAMERA ONLY
	Profile	REQ	<int>	Multicast profile **Note** **Profile** and **Mode** must be sent together with the **control** action.
	Mode	REQ	<enum> Start, Stop	Multicast mode

106.14.4. Examples

To use the **control** action, the **Profile** and **Mode** parameters must be set together.

106.14.4.1. Starting the multicast for Profile 2

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=multicast&action=control&Profile=2&Mode=Start
```

106.14.4.2. Stopping the multicast for Profile 2

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=multicast&action=control&Profile=2&Mode=Stop
```

With a Multi Directional camera, to use the **control** action, **Channel**, **Profile** and **Mode** parameters must be set together.

106.14.4.3. Starting the multicast for Channel 1, Profile 2

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=multicast&action=control&Channel=1&
Profile=2&Mode=Start
```

106.14.4.4. Stopping the multicast for Channel1, Profile 2

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=multicast&action=control&Channel=1&Profile=2&Mode=Sto
p
```

106.15. Video Codec Info

106.15.1. Description

The **videocodecinfo** submenu requests video codec information.

Access level

Action	Camera	NVR	Encoder
view	Admin	User	Guest

106.15.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videocodecinfo&action=view[&<parameter>=<value>...]
```

106.15.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID Uses from ‘0’ to max channel count-1 for channel ID. A user can request multiple channels with comma. ENCODER ONLY MULTI DIRECTION CAMERA ONLY
	ChannelIDList	REQ	<csv>	Channel ID List Uses from ‘0’ to max channel count-1 for channel ID. A user can request multiple channels with comma. NVR ONLY
	AllProfiles	REQ	<bool> True, False	When all profiles needs to be included in the response at once When user request with this parameter, encoder would send information about all profiles in requested channel. This parameter should be used with channel parameter.
	Profile	REQ	<int>	Profile number An encoder has different information per each profile. So User should request information separately.

Action	Parameters	Request/ Response	Type/ Value	Description
	EncodingType	REQ	<enum> MJPEG, MPEG4, H264, H265	Video encoding type
	ViewMode	REQ	<csv> All, SingleView, Panaoram, DoublePanorama, QuadView, LeftHalfView, RightHalfView, Overview, 360Panorama, QuadView.#, Bright, Dark, SubregionView.1, SubregionView.2	View mode
	Channel.#.EncodingType	RES	<enum> MJPEG, MPEG4, H264, H265	Video encoding type by channel ID
	<EncodingType>.IsDigitalPTZSupported	RES	<bool> True, False	Digital PTZ is supported by a particular encoding or not. CAMERA ONLY
	<EncodingType>.Compression.DefaultCompressionLevel	RES	<int>	Default value of the compression level CAMERA ONLY
	<EncodingType>.Compression.MinCompressionLevel	RES	<int>	Minimum value of the compression level CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.Compression.MaxCompressionLevel	RES	<int>	Maximum value of the compression level CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.Width	RES	<int>	Supported resolution - Width pixels
	<EncodingType>.General.<Width>X<Height>.Height	RES	<int>	Supported resolution - Height pixels
	<EncodingType>.General.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - Maximum FPS
	<EncodingType>.General.<Width>X<Height>.MinFPS	RES	<int>	Supported resolution - Minimum FPS
	<EncodingType>.General.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - Maximum CBR bitrate CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - Minimum CBR bitrate CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - Default CBR bitrate CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.MaxVBRTargetBitrate	RES	<int>	Supported resolution - Maximum VBR bitrate CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.MinVBRTargetBitrate	RES	<int>	Supported resolution - Minimum VBR bitrate CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.DefaultVBRTargetBitrate	RES	<int>	Supported resolution - Default VBR bitrate CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.General.<Width>X<Height>.MaxUnknownTargetBitrate	RES	<int>	Supported resolution - Maximum Unknown bitrate NVR ONLY
	<EncodingType>.General.<Width>X<Height>.MinUnknownTargetBitrate	RES	<int>	Supported resolution - Minimum Unknown bitrate NVR ONLY
	<EncodingType>.General.<Width>X<Height>.IsTruncated	RES	<enum> Normal, Crop	It indicates whether target resolution can show entire image captured from sensor or not. If a resolution is a cut out of the original image, this parameter presents as Crop. CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution – Default FPS CAMERA ONLY
	<EncodingType>.General.<Width>X<Height>.FrameLockMaxFPS	RES	<int>	Users can request their device not to fluctuate the framerate if the device supports the IsFixedFrameRateProfile feature. This indicates the maximum supported FPS for this situation. It would be shown only in supported models. CAMERA ONLY
	<EncodingType>.Record.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	The maximum CBR bitrate of a record profile Record profile is available only in MJPEG or H.264 CAMERA ONLY
	<EncodingType>.Record.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Record profile default CBR bitrate CAMERA ONLY
	<EncodingType>.Record.<Width>X<Height>.MaxVBRTargetBitrate	RES	<int>	Record profile maximum VBR bitrate CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.Record.<Width>X<Height>.DefaultVBRTargetBitrate	RES	<int>	Record profile default VBR bitrate CAMERA ONLY
	<EncodingType>.Record.<Width>X<Height>.DefaultFPS	RES	<int>	Record profile – Default FPS CAMERA ONLY
	<EncodingType>.Record.<Width>X<Height>.FrameLockMaxFPS	RES	<int>	User can request device not to fluctuate its framerate if a device supports IsFixedFrameRateProfile feature. This indicates about max supported FPS for this situation. It would be shown only in supported models. CAMERA ONLY
	<EncodingType>.Email.<Width>X<Height>.MaxFPS	RES	<int>	Email profile maximum FPS Email profile is available only for MJPEG. CAMERA ONLY
	<EncodingType>.Email.<Width>X<Height>.DefaultFPS	RES	<int>	Email profile default FPS Email profile is available only for MJPEG. CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.Width	RES	<int>	Supported resolution - Width in pixels CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.Height	RES	<int>	Supported resolution - Height in pixels CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - Maximum FPS CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution – Default FPS CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.DigitalPTZ.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - Minimum CBR bitrate CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - Maximum CBR bitrate CAMERA ONLY
	<EncodingType>.DigitalPTZ.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - Default CBR bitrate CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.Width	RES	<int>	Supported resolution - width of pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.Height	RES	<int>	Supported resolution - height of pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - maximum FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution - default FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - maximum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - minimum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - default CBR bitrate by encoding type and view mode CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxVBRTargetBitrate	RES	<int>	Supported resolution - maximum VBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MinVBRTargetBitrate	RES	<int>	Supported resolution - minimum VBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.DefaultVBRTargetBitrate	RES	<int>	Supported resolution - default VBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MaxUnknownTargetBitrate	RES	<int>	Supported resolution - maximum Supported resolution – maximum unknown bitrate by encoding type and view mode NVR ONLY
	<EncodingType>.General.<ViewMode>.<Width>X<Height>.MinUnknownTargetBitrate	RES	<int>	Supported resolution – minimum unknown bitrate by encoding type and view mode NVR ONLY
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.Width	RES	<int>	Supported resolution - width of pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.Height	RES	<int>	Supported resolution - height of pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - maximum FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution - default FPS by encoding type and view mode CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - minimum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.DigitalPTZ.<ViewMode>.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - maximum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - default CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.Width	RES	<int>	Supported resolution - width in pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.Height	RES	<int>	Supported resolution - height in pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - maximum FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution – default FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - minimum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - maximum CBR bitrate by encoding type and view mode CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.Record.<ViewMode>.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - default CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.Width	RES	<int>	Supported resolution - width in pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.Height	RES	<int>	Supported resolution - height in pixels by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - maximum FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution - default FPS by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - minimum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - maximum CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.Email.<ViewMode>.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - default CBR bitrate by encoding type and view mode CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.Width	RES	<int>	Supported resolution - Width in pixels CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.Height	RES	<int>	Supported resolution - Height in pixels CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - Maximum FPS CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.VoIP.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution – Default FPS CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - Minimum CBR bitrate CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - Maximum CBR bitrate CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - Default CBR bitrate CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.MinVBRTargetBitrate	RES	<int>	Supported resolution - Minimum VBR bitrate CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.MaxVBRTargetBitrate	RES	<int>	Supported resolution - Maximum VBR bitrate CAMERA ONLY
	<EncodingType>.VoIP.<Width>X<Height>.DefaultVBRTargetBitrate	RES	<int>	Supported resolution - Default VBR bitrate CAMERA ONLY
	<EncodingType>.DPM.<Width>X<Height>.Width	RES	<int>	Supported resolution - Width in pixels
	<EncodingType>.DPM.<Width>X<Height>.Height	RES	<int>	Supported resolution - Height in pixels
	<EncodingType>.DPM.<Width>X<Height>.MaxFPS	RES	<int>	Supported resolution - Maximum FPS
	<EncodingType>.DPM.<Width>X<Height>.MinFPS	RES	<int>	Supported resolution - Minimum FPS
	<EncodingType>.DPM.<Width>X<Height>.DefaultFPS	RES	<int>	Supported resolution - Default FPS
	<EncodingType>.DPM.<Width>X<Height>.MaxCBRTargetBitrate	RES	<int>	Supported resolution - Maximum CBR bitrate

Action	Parameters	Request/ Response	Type/ Value	Description
	<EncodingType>.DPM.<Width>X<Height>.MinCBRTargetBitrate	RES	<int>	Supported resolution - Minimum CBR bitrate
	<EncodingType>.DPM.<Width>X<Height>.DefaultCBRTargetBitrate	RES	<int>	Supported resolution - Default CBR bitrate
	<EncodingType>.DPM.<Width>X<Height>.MaxVBRTargetBitrate	RES	<int>	Supported resolution - Maximum VBR bitrate
	<EncodingType>.DPM.<Width>X<Height>.MinVBRTargetBitrate	RES	<int>	Supported resolution - Minimum VBR bitrate
	<EncodingType>.DPM.<Width>X<Height>.DefaultVBRTargetBitrate	RES	<int>	Supported resolution - Default VBR bitrate
	<EncodingType>.DPM.<Width>X<Height>.IsTruncated	RES	<enum> Normal, Crop	It indicates whether target resolution can show entire image captured from sensor or not. If a resolution is a cut out of the original image, this parameter presents as Crop.
	<EncodingType>.DPM.<Width>X<Height>.FrameLockMaxFPS	RES	<int>	User can request device not to fluctuate its framerate if a device supports IsFixedFrameRateProfile feature. This indicates about max supported FPS for this situation. It would be shown only in supported models.

106.15.4. Examples

106.15.4.1. Getting all resolution information based on Encoding Type

The command without **Profile** returns all available video resolution information based on Encoding Type.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videocodecinfo&action=view&EncodingType=MJPEG
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=MJPEG
Channel.0.ViewMode=Overview
MJPEG.IsDigitalPTZSupported=False
MJPEG.General.1920X1080.Width=1920
MJPEG.General.1920X1080.Height=1080
MJPEG.General.1920X1080.MaxFPS=30000
MJPEG.General.1920X1080.DefaultFPS=15000
MJPEG.General.1920X1080.MaxCBRTargetBitrate=30720
MJPEG.General.1920X1080.MinCBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultCBRTargetBitrate=10240
MJPEG.General.1920X1080.MaxVBRTargetBitrate=30720
MJPEG.General.1920X1080.MinVBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultVBRTargetBitrate=10240
MJPEG.General.1920X1080.IsTruncated=Normal
MJPEG.General.1280X1024.Width=1280
MJPEG.General.1280X1024.Height=1024
MJPEG.General.1280X1024.MaxFPS=30000
MJPEG.General.1280X1024.DefaultFPS=15000
MJPEG.General.1280X1024.MaxCBRTargetBitrate=30720
MJPEG.General.1280X1024.MinCBRTargetBitrate=1024
MJPEG.General.1280X1024.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X1024.MaxVBRTargetBitrate=30720
MJPEG.General.1280X1024.MinVBRTargetBitrate=1024
MJPEG.General.1280X1024.DefaultVBRTargetBitrate=10240
MJPEG.General.1280X1024.IsTruncated=Crop
MJPEG.General.1280X960.Width=1280
MJPEG.General.1280X960.Height=960
MJPEG.General.1280X960.MaxFPS=30000
MJPEG.General.1280X960.DefaultFPS=15000
MJPEG.General.1280X960.MaxCBRTargetBitrate=30720
MJPEG.General.1280X960.MinCBRTargetBitrate=1024
MJPEG.General.1280X960.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X960.MaxVBRTargetBitrate=30720
MJPEG.General.1280X960.MinVBRTargetBitrate=1024
MJPEG.General.1280X960.DefaultVBRTargetBitrate=10240
```

```
MJPEG.General.1280X960.IsTruncated=Crop
MJPEG.General.1280X720.Width=1280
MJPEG.General.1280X720.Height=720
MJPEG.General.1280X720.MaxFPS=30000
MJPEG.General.1280X720.DefaultFPS=15000
MJPEG.General.1280X720.MaxCBRTargetBitrate=30720
MJPEG.General.1280X720.MinCBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X720.MaxVBRTargetBitrate=30720
MJPEG.General.1280X720.MinVBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultVBRTargetBitrate=10240
MJPEG.General.1280X720.IsTruncated=Normal
MJPEG.General.1024X768.Width=1024
MJPEG.General.1024X768.Height=768
MJPEG.General.1024X768.MaxFPS=30000
MJPEG.General.1024X768.DefaultFPS=15000
MJPEG.General.1024X768.MaxCBRTargetBitrate=30720
MJPEG.General.1024X768.MinCBRTargetBitrate=1024
MJPEG.General.1024X768.DefaultCBRTargetBitrate=10240
MJPEG.General.1024X768.MaxVBRTargetBitrate=30720
MJPEG.General.1024X768.MinVBRTargetBitrate=1024
MJPEG.General.1024X768.DefaultVBRTargetBitrate=10240
MJPEG.General.1024X768.IsTruncated=Crop
MJPEG.General.800X600.Width=800
MJPEG.General.800X600.Height=600
MJPEG.General.800X600.MaxFPS=30000
MJPEG.General.800X600.DefaultFPS=15000
MJPEG.General.800X600.MaxCBRTargetBitrate=30720
MJPEG.General.800X600.MinCBRTargetBitrate=512
MJPEG.General.800X600.DefaultCBRTargetBitrate=10240
MJPEG.General.800X600.MaxVBRTargetBitrate=30720
MJPEG.General.800X600.MinVBRTargetBitrate=512
MJPEG.General.800X600.DefaultVBRTargetBitrate=10240
MJPEG.General.800X600.IsTruncated=Crop
MJPEG.General.800X448.Width=800
MJPEG.General.800X448.Height=448
MJPEG.General.800X448.MaxFPS=30000
MJPEG.General.800X448.DefaultFPS=15000
MJPEG.General.800X448.MaxCBRTargetBitrate=30720
MJPEG.General.800X448.MinCBRTargetBitrate=512
MJPEG.General.800X448.DefaultCBRTargetBitrate=10240
```

```
MJPEG.General.800X448.MaxVBRTargetBitrate=30720
MJPEG.General.800X448.MinVBRTargetBitrate=512
MJPEG.General.800X448.DefaultVBRTargetBitrate=10240
MJPEG.General.800X448.IsTruncated=Normal
MJPEG.General.720X576.Width=720
MJPEG.General.720X576.Height=576
MJPEG.General.720X576.MaxFPS=30000
MJPEG.General.720X576.DefaultFPS=15000
MJPEG.General.720X576.MaxCBRTargetBitrate=30720
MJPEG.General.720X576.MinCBRTargetBitrate=512
MJPEG.General.720X576.DefaultCBRTargetBitrate=10240
MJPEG.General.720X576.MaxVBRTargetBitrate=30720
MJPEG.General.720X576.MinVBRTargetBitrate=512
MJPEG.General.720X576.DefaultVBRTargetBitrate=10240
MJPEG.General.720X576.IsTruncated=Crop
MJPEG.General.720X480.Width=720
MJPEG.General.720X480.Height=480
MJPEG.General.720X480.MaxFPS=30000
MJPEG.General.720X480.DefaultFPS=15000
MJPEG.General.720X480.MaxCBRTargetBitrate=30720
MJPEG.General.720X480.MinCBRTargetBitrate=512
MJPEG.General.720X480.DefaultCBRTargetBitrate=10240
MJPEG.General.720X480.MaxVBRTargetBitrate=30720
MJPEG.General.720X480.MinVBRTargetBitrate=512
MJPEG.General.720X480.DefaultVBRTargetBitrate=10240
MJPEG.General.720X480.IsTruncated=Crop
MJPEG.General.640X480.Width=640
MJPEG.General.640X480.Height=480
MJPEG.General.640X480.MaxFPS=30000
MJPEG.General.640X480.DefaultFPS=15000
MJPEG.General.640X480.MaxCBRTargetBitrate=30720
MJPEG.General.640X480.MinCBRTargetBitrate=512
MJPEG.General.640X480.DefaultCBRTargetBitrate=10240
MJPEG.General.640X480.MaxVBRTargetBitrate=30720
MJPEG.General.640X480.MinVBRTargetBitrate=512
MJPEG.General.640X480.DefaultVBRTargetBitrate=10240
MJPEG.General.640X480.IsTruncated=Crop
MJPEG.General.640X360.Width=640
MJPEG.General.640X360.Height=360
MJPEG.General.640X360.MaxFPS=30000
MJPEG.General.640X360.DefaultFPS=15000
```

```
MJPEG.General.640X360.MaxCBRTargetBitrate=30720
MJPEG.General.640X360.MinCBRTargetBitrate=512
MJPEG.General.640X360.DefaultCBRTargetBitrate=10240
MJPEG.General.640X360.MaxVBRTargetBitrate=30720
MJPEG.General.640X360.MinVBRTargetBitrate=512
MJPEG.General.640X360.DefaultVBRTargetBitrate=10240
MJPEG.General.640X360.IsTruncated=Normal
MJPEG.General.320X240.Width=320
MJPEG.General.320X240.Height=240
MJPEG.General.320X240.MaxFPS=30000
MJPEG.General.320X240.DefaultFPS=15000
MJPEG.General.320X240.MaxCBRTargetBitrate=30720
MJPEG.General.320X240.MinCBRTargetBitrate=256
MJPEG.General.320X240.DefaultCBRTargetBitrate=10240
MJPEG.General.320X240.MaxVBRTargetBitrate=30720
MJPEG.General.320X240.MinVBRTargetBitrate=256
MJPEG.General.320X240.DefaultVBRTargetBitrate=10240
MJPEG.General.320X240.IsTruncated=Crop
MJPEG.Record.1920X1080.MaxFPS=2000
MJPEG.Record.1920X1080.DefaultFPS=2000
MJPEG.Record.1920X1080.MinCBRTargetBitrate=2048
MJPEG.Record.1920X1080.MaxCBRTargetBitrate=6144
MJPEG.Record.1920X1080.DefaultCBRTargetBitrate=6144
MJPEG.Record.1920X1080.MinVBRTargetBitrate=2048
MJPEG.Record.1920X1080.MaxVBRTargetBitrate=6144
MJPEG.Record.1920X1080.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X1024.MaxFPS=3000
MJPEG.Record.1280X1024.DefaultFPS=3000
MJPEG.Record.1280X1024.MinCBRTargetBitrate=1024
MJPEG.Record.1280X1024.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X1024.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X1024.MinVBRTargetBitrate=1024
MJPEG.Record.1280X1024.MaxVBRTargetBitrate=6144
MJPEG.Record.1280X1024.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X960.MaxFPS=3000
MJPEG.Record.1280X960.DefaultFPS=3000
MJPEG.Record.1280X960.MinCBRTargetBitrate=1024
MJPEG.Record.1280X960.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X960.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X960.MinVBRTargetBitrate=1024
MJPEG.Record.1280X960.MaxVBRTargetBitrate=6144
```

```
MJPEG.Record.1280X960.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X720.MaxFPS=3000
MJPEG.Record.1280X720.DefaultFPS=3000
MJPEG.Record.1280X720.MinCBRTargetBitrate=1024
MJPEG.Record.1280X720.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X720.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X720.MinVBRTargetBitrate=1024
MJPEG.Record.1280X720.MaxVBRTargetBitrate=6144
MJPEG.Record.1280X720.DefaultVBRTargetBitrate=6144
MJPEG.Record.1024X768.MaxFPS=3000
MJPEG.Record.1024X768.DefaultFPS=3000
MJPEG.Record.1024X768.MinCBRTargetBitrate=1024
MJPEG.Record.1024X768.MaxCBRTargetBitrate=6144
MJPEG.Record.1024X768.DefaultCBRTargetBitrate=6144
MJPEG.Record.1024X768.MinVBRTargetBitrate=1024
MJPEG.Record.1024X768.MaxVBRTargetBitrate=6144
MJPEG.Record.1024X768.DefaultVBRTargetBitrate=6144
MJPEG.Record.800X600.MaxFPS=5000
MJPEG.Record.800X600.DefaultFPS=5000
MJPEG.Record.800X600.MinCBRTargetBitrate=512
MJPEG.Record.800X600.MaxCBRTargetBitrate=6144
MJPEG.Record.800X600.DefaultCBRTargetBitrate=6144
MJPEG.Record.800X600.MinVBRTargetBitrate=512
MJPEG.Record.800X600.MaxVBRTargetBitrate=6144
MJPEG.Record.800X600.DefaultVBRTargetBitrate=6144
MJPEG.Record.800X448.MaxFPS=5000
MJPEG.Record.800X448.DefaultFPS=5000
MJPEG.Record.800X448.MinCBRTargetBitrate=512
MJPEG.Record.800X448.MaxCBRTargetBitrate=6144-
MJPEG.Record.800X448.DefaultCBRTargetBitrate=6144
MJPEG.Record.800X448.MinVBRTargetBitrate=512
MJPEG.Record.800X448.MaxVBRTargetBitrate=6144
MJPEG.Record.800X448.DefaultVBRTargetBitrate=6144
MJPEG.Record.720X576.MaxFPS=5000
MJPEG.Record.720X576.DefaultFPS=5000
MJPEG.Record.720X576.MinCBRTargetBitrate=512
MJPEG.Record.720X576.MaxCBRTargetBitrate=6144
MJPEG.Record.720X576.DefaultCBRTargetBitrate=6144
MJPEG.Record.720X576.MinVBRTargetBitrate=512
MJPEG.Record.720X576.MaxVBRTargetBitrate=6144
MJPEG.Record.720X576.DefaultVBRTargetBitrate=6144
```

```
MJPEG.Record.720X480.MaxFPS=5000
MJPEG.Record.720X480.DefaultFPS=5000
MJPEG.Record.720X480.MinCBRTargetBitrate=512
MJPEG.Record.720X480.MaxCBRTargetBitrate=6144
MJPEG.Record.720X480.DefaultCBRTargetBitrate=6144
MJPEG.Record.720X480.MinVBRTargetBitrate=512
MJPEG.Record.720X480.MaxVBRTargetBitrate=6144
MJPEG.Record.720X480.DefaultVBRTargetBitrate=6144
MJPEG.Record.640X480.MaxFPS=5000
MJPEG.Record.640X480.DefaultFPS=5000
MJPEG.Record.640X480.MinCBRTargetBitrate=512
MJPEG.Record.640X480.MaxCBRTargetBitrate=6144
MJPEG.Record.640X480.DefaultCBRTargetBitrate=6144
MJPEG.Record.640X480.MinVBRTargetBitrate=512
MJPEG.Record.640X480.MaxVBRTargetBitrate=6144
MJPEG.Record.640X480.DefaultVBRTargetBitrate=6144
MJPEG.Record.640X360.MaxFPS=5000
MJPEG.Record.640X360.DefaultFPS=5000
MJPEG.Record.640X360.MinCBRTargetBitrate=512
MJPEG.Record.640X360.MaxCBRTargetBitrate=6144
MJPEG.Record.640X360.DefaultCBRTargetBitrate=6144
MJPEG.Record.640X360.MinVBRTargetBitrate=512
MJPEG.Record.640X360.MaxVBRTargetBitrate=6144
MJPEG.Record.640X360.DefaultVBRTargetBitrate=6144
MJPEG.Record.320X240.MaxFPS=5000
MJPEG.Record.320X240.DefaultFPS=5000
MJPEG.Record.320X240.MinCBRTargetBitrate=256
MJPEG.Record.320X240.MaxCBRTargetBitrate=6144
MJPEG.Record.320X240.DefaultCBRTargetBitrate=6144
MJPEG.Record.320X240.MinVBRTargetBitrate=256
MJPEG.Record.320X240.MaxVBRTargetBitrate=6144
MJPEG.Record.320X240.DefaultVBRTargetBitrate=6144
MJPEG.Email.1920X1080.MaxFPS=5000
MJPEG.Email.1920X1080.DefaultFPS=5000
MJPEG.Email.1280X1024.MaxFPS=5000
MJPEG.Email.1280X1024.DefaultFPS=5000
MJPEG.Email.1280X960.MaxFPS=5000
MJPEG.Email.1280X960.DefaultFPS=5000
MJPEG.Email.1280X720.MaxFPS=5000
MJPEG.Email.1280X720.DefaultFPS=5000
MJPEG.Email.1024X768.MaxFPS=5000
```

```
MJPEG.Email.1024X768.DefaultFPS=5000
MJPEG.Email.800X600.MaxFPS=5000
MJPEG.Email.800X600.DefaultFPS=5000
MJPEG.Email.800X448.MaxFPS=5000
MJPEG.Email.800X448.DefaultFPS=5000
MJPEG.Email.720X576.MaxFPS=5000
MJPEG.Email.720X576.DefaultFPS=5000
MJPEG.Email.720X480.MaxFPS=5000
MJPEG.Email.720X480.DefaultFPS=5000
MJPEG.Email.640X480.MaxFPS=5000
MJPEG.Email.640X480.DefaultFPS=5000
MJPEG.Email.640X360.MaxFPS=5000
MJPEG.Email.640X360.DefaultFPS=5000
MJPEG.Email.320X240.MaxFPS=5000
MJPEG.Email.320X240.DefaultFPS=5000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "ViewModes": [
                {
                    "ViewMode": "Overview",
                    "Codecs": [
                        {
                            "EncodingType": "MJPEG",
                            "IsDigitalPTZSupported": false,
                            "General": [
                                {
                                    "Width": 1920,
                                    "Height": 1080,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 15000,
                                    "MaxCBRTargetBitrate": 30720,
                                    "MinCBRTargetBitrate": 2048,
```

```
                                "DefaultCBRTargetBitrate": 10240,
                                "MaxVBRTargetBitrate": 30720,
                                "MinVBRTargetBitrate": 2048,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 1280,
                                "Height": 1024,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
                                "MaxCBRTargetBitrate": 30720,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 10240,
                                "MaxVBRTargetBitrate": 30720,
                                "MinVBRTargetBitrate": 1024,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Crop"
                            },
                            {
                                "Width": 1280,
                                "Height": 960,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
                                "MaxCBRTargetBitrate": 30720,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 10240,
                                "MaxVBRTargetBitrate": 30720,
                                "MinVBRTargetBitrate": 1024,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Crop"
                            },
                            {
                                "Width": 1280,
                                "Height": 720,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
                                "MaxCBRTargetBitrate": 30720,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 10240,
                                "MaxVBRTargetBitrate": 30720,
```

```
                                "MinVBRTargetBitrate": 1024,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Normal"
                            },
                            {
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                                "Height": 768,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
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                                "MinCBRTargetBitrate": 1024,
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                                "MinVBRTargetBitrate": 1024,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Crop"
                            },
                            {
                                "Width": 800,
                                "Height": 600,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
                                "MaxCBRTargetBitrate": 30720,
                                "MinCBRTargetBitrate": 512,
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                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 10240,
                                "IsTruncated": "Crop"
                            },
                            {
                                "Width": 800,
                                "Height": 448,
                                "MaxFPS": 30000,
                                "DefaultFPS": 15000,
                                "MaxCBRTargetBitrate": 30720,
                                "MinCBRTargetBitrate": 512,
                                "DefaultCBRTargetBitrate": 10240,
                                "MaxVBRTargetBitrate": 30720,
                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 10240,
```

```
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 720,
                                    "Height": 576,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 15000,
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                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 10240,
                                    "IsTruncated": "Crop"
                                },
                                {
                                    "Width": 720,
                                    "Height": 480,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 15000,
                                    "MaxCBRTargetBitrate": 30720,
                                    "MinCBRTargetBitrate": 512,
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                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 10240,
                                    "IsTruncated": "Crop"
                                },
                                {
                                    "Width": 640,
                                    "Height": 480,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 15000,
                                    "MaxCBRTargetBitrate": 30720,
                                    "MinCBRTargetBitrate": 512,
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                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 10240,
                                    "IsTruncated": "Crop"
                                },
```

```
                {
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                    "Height": 360,
                    "MaxFPS": 30000,
                    "DefaultFPS": 15000,
                    "MaxCBRTargetBitrate": 30720,
                    "MinCBRTargetBitrate": 512,
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                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 10240,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 320,
                    "Height": 240,
                    "MaxFPS": 30000,
                    "DefaultFPS": 15000,
                    "MaxCBRTargetBitrate": 30720,
                    "MinCBRTargetBitrate": 256,
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                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 256,
                    "DefaultVBRTargetBitrate": 10240,
                    "IsTruncated": "Crop"
                }
            ],
            "Record": [
                {
                    "Width": 1920,
                    "Height": 1080,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000,
                    "MinCBRTargetBitrate": 2048,
                    "MaxCBRTargetBitrate": 6144,
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                    "MinVBRTargetBitrate": 2048,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 6144
                },
                {
```

```
                "Width": 1280,
                "Height": 1024,
                "MaxFPS": 3000,
                "DefaultFPS": 3000,
                "MinCBRTargetBitrate": 1024,
                "MaxCBRTargetBitrate": 6144,
                "DefaultCBRTargetBitrate": 6144,
                "MinVBRTargetBitrate": 1024,
                "MaxVBRTargetBitrate": 6144,
                "DefaultVBRTargetBitrate": 6144
            },
            {
                "Width": 1280,
                "Height": 960,
                "MaxFPS": 3000,
                "DefaultFPS": 3000,
                "MinCBRTargetBitrate": 1024,
                "MaxCBRTargetBitrate": 6144,
                "DefaultCBRTargetBitrate": 6144,
                "MinVBRTargetBitrate": 1024,
                "MaxVBRTargetBitrate": 6144,
                "DefaultVBRTargetBitrate": 6144
            },
            {
                "Width": 1280,
                "Height": 720,
                "MaxFPS": 3000,
                "DefaultFPS": 3000,
                "MinCBRTargetBitrate": 1024,
                "MaxCBRTargetBitrate": 6144,
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                "MaxVBRTargetBitrate": 6144,
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            },
            {
                "Width": 1024,
                "Height": 768,
                "MaxFPS": 3000,
                "DefaultFPS": 3000,
                "MinCBRTargetBitrate": 1024,
```

```
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 6144,
                                    "MinVBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 6144
                                },
                                {
                                    "Width": 800,
                                    "Height": 600,
                                    "MaxFPS": 5000,
                                    "DefaultFPS": 5000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 6144,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 6144
                                },
                                {
                                    "Width": 800,
                                    "Height": 448,
                                    "MaxFPS": 5000,
                                    "DefaultFPS": 5000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 6144,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 6144
                                },
                                {
                                    "Width": 720,
                                    "Height": 576,
                                    "MaxFPS": 5000,
                                    "DefaultFPS": 5000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 6144,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 6144
```

```
                                        },
                                        {
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                                            "Height": 480,
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                                            "DefaultFPS": 5000,
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                                            "DefaultCBRTargetBitrate": 6144,
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                                            "MaxVBRTargetBitrate": 6144,
                                            "DefaultVBRTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 640,
                                            "Height": 480,
                                            "MaxFPS": 5000,
                                            "DefaultFPS": 5000,
                                            "MinCBRTargetBitrate": 512,
                                            "MaxCBRTargetBitrate": 6144,
                                            "DefaultCBRTargetBitrate": 6144,
                                            "MinVBRTargetBitrate": 512,
                                            "MaxVBRTargetBitrate": 6144,
                                            "DefaultVBRTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 640,
                                            "Height": 360,
                                            "MaxFPS": 5000,
                                            "DefaultFPS": 5000,
                                            "MinCBRTargetBitrate": 512,
                                            "MaxCBRTargetBitrate": 6144,
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                                            "DefaultVBRTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 320,
                                            "Height": 240,
                                            "MaxFPS": 5000,
```

```
                    "DefaultFPS": 5000,
                    "MinCBRTargetBitrate": 256,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 6144,
                    "MinVBRTargetBitrate": 256,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 6144
                }
            ],
            "Email": [
                {
                    "Width": 1920,
                    "Height": 1080,
                    "MaxFPS": 5000,
                    "DefaultFPS": 5000
                },
                {
                    "Width": 1280,
                    "Height": 1024,
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                    "DefaultFPS": 5000
                },
                {
                    "Width": 1280,
                    "Height": 960,
                    "MaxFPS": 5000,
                    "DefaultFPS": 5000
                },
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 5000,
                    "DefaultFPS": 5000
                },
                {
                    "Width": 1024,
                    "Height": 768,
                    "MaxFPS": 5000,
                    "DefaultFPS": 5000
                },
                {
```

```
                "Width": 800,
                "Height": 600,
                "MaxFPS": 5000,
                "DefaultFPS": 5000
            },
            {
                "Width": 800,
                "Height": 448,
                "MaxFPS": 5000,
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                "Height": 576,
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                "DefaultFPS": 5000
            },
            {
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                "Height": 480,
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                "DefaultFPS": 5000
            },
            {
                "Width": 640,
                "Height": 360,
                "MaxFPS": 5000,
                "DefaultFPS": 5000
            },
            {
                "Width": 320,
                "Height": 240,
                "MaxFPS": 5000,
                "DefaultFPS": 5000
            }
```

```
                              ]
                          }
                      ]
                  }
              ]
          }
      ]
}
```


The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```


```
Channel.0.Profile=1
Channel.0.ViewMode=Overview
Channel.0.EncodingType=MJPEG
MJPEG.IsDigitalPTZSupported=False
MJPEG.General.4000X3000.Width=4000
MJPEG.General.4000X3000.Height=3000
MJPEG.General.4000X3000.MaxFPS=2
MJPEG.General.4000X3000.MinUnknownTargetBitrate=5120
MJPEG.General.4000X3000.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.4000X3000.Width=4000
MJPEG.General.Overview.4000X3000.Height=3000
MJPEG.General.Overview.4000X3000.MaxFPS=2
MJPEG.General.Overview.4000X3000.MinUnknownTargetBitrate=5120
MJPEG.General.Overview.4000X3000.MaxUnknownTargetBitrate=6144
MJPEG.General.3840X2160.Width=3840
MJPEG.General.3840X2160.Height=2160
MJPEG.General.3840X2160.MaxFPS=2
MJPEG.General.3840X2160.MinUnknownTargetBitrate=4096
MJPEG.General.3840X2160.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.3840X2160.Width=3840
MJPEG.General.Overview.3840X2160.Height=2160
MJPEG.General.Overview.3840X2160.MaxFPS=2
MJPEG.General.Overview.3840X2160.MinUnknownTargetBitrate=4096
MJPEG.General.Overview.3840X2160.MaxUnknownTargetBitrate=6144
```

```
MJPEG.General.2592X1944.Width=2592
MJPEG.General.2592X1944.Height=1944
MJPEG.General.2592X1944.MaxFPS=2
MJPEG.General.2592X1944.MinUnknownTargetBitrate=2048
MJPEG.General.2592X1944.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.2592X1944.Width=2592
MJPEG.General.Overview.2592X1944.Height=1944
MJPEG.General.Overview.2592X1944.MaxFPS=2
MJPEG.General.Overview.2592X1944.MinUnknownTargetBitrate=2048
MJPEG.General.Overview.2592X1944.MaxUnknownTargetBitrate=6144
MJPEG.General.2592X1464.Width=2592
MJPEG.General.2592X1464.Height=1464
MJPEG.General.2592X1464.MaxFPS=2
MJPEG.General.2592X1464.MinUnknownTargetBitrate=2048
MJPEG.General.2592X1464.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.2592X1464.Width=2592
MJPEG.General.Overview.2592X1464.Height=1464
MJPEG.General.Overview.2592X1464.MaxFPS=2
MJPEG.General.Overview.2592X1464.MinUnknownTargetBitrate=2048
MJPEG.General.Overview.2592X1464.MaxUnknownTargetBitrate=6144
MJPEG.General.1920X1080.Width=1920
MJPEG.General.1920X1080.Height=1080
MJPEG.General.1920X1080.MaxFPS=5
MJPEG.General.1920X1080.MinUnknownTargetBitrate=1024
MJPEG.General.1920X1080.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1920X1080.Width=1920
MJPEG.General.Overview.1920X1080.Height=1080
MJPEG.General.Overview.1920X1080.MaxFPS=5
MJPEG.General.Overview.1920X1080.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1920X1080.MaxUnknownTargetBitrate=6144
MJPEG.General.1600X1200.Width=1600
MJPEG.General.1600X1200.Height=1200
MJPEG.General.1600X1200.MaxFPS=5
MJPEG.General.1600X1200.MinUnknownTargetBitrate=1024
MJPEG.General.1600X1200.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1600X1200.Width=1600
MJPEG.General.Overview.1600X1200.Height=1200
MJPEG.General.Overview.1600X1200.MaxFPS=5
MJPEG.General.Overview.1600X1200.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1600X1200.MaxUnknownTargetBitrate=6144
MJPEG.General.1280X1024.Width=1280
```

```
MJPEG.General.1280X1024.Height=1024
MJPEG.General.1280X1024.MaxFPS=5
MJPEG.General.1280X1024.MinUnknownTargetBitrate=1024
MJPEG.General.1280X1024.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1280X1024.Width=1280
MJPEG.General.Overview.1280X1024.Height=1024
MJPEG.General.Overview.1280X1024.MaxFPS=5
MJPEG.General.Overview.1280X1024.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1280X1024.MaxUnknownTargetBitrate=6144
MJPEG.General.1280X960.Width=1280
MJPEG.General.1280X960.Height=960
MJPEG.General.1280X960.MaxFPS=5
MJPEG.General.1280X960.MinUnknownTargetBitrate=1024
MJPEG.General.1280X960.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1280X960.Width=1280
MJPEG.General.Overview.1280X960.Height=960
MJPEG.General.Overview.1280X960.MaxFPS=5
MJPEG.General.Overview.1280X960.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1280X960.MaxUnknownTargetBitrate=6144
MJPEG.General.1280X720.Width=1280
MJPEG.General.1280X720.Height=720
MJPEG.General.1280X720.MaxFPS=5
MJPEG.General.1280X720.MinUnknownTargetBitrate=1024
MJPEG.General.1280X720.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1280X720.Width=1280
MJPEG.General.Overview.1280X720.Height=720
MJPEG.General.Overview.1280X720.MaxFPS=5
MJPEG.General.Overview.1280X720.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1280X720.MaxUnknownTargetBitrate=6144
MJPEG.General.1024X768.Width=1024
MJPEG.General.1024X768.Height=768
MJPEG.General.1024X768.MaxFPS=5
MJPEG.General.1024X768.MinUnknownTargetBitrate=1024
MJPEG.General.1024X768.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1024X768.Width=1024
MJPEG.General.Overview.1024X768.Height=768
MJPEG.General.Overview.1024X768.MaxFPS=5
MJPEG.General.Overview.1024X768.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1024X768.MaxUnknownTargetBitrate=6144
MJPEG.General.800X600.Width=800
MJPEG.General.800X600.Height=600
```

```
MJPEG.General.800X600.MaxFPS=5
MJPEG.General.800X600.MinUnknownTargetBitrate=512
MJPEG.General.800X600.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.800X600.Width=800
MJPEG.General.Overview.800X600.Height=600
MJPEG.General.Overview.800X600.MaxFPS=5
MJPEG.General.Overview.800X600.MinUnknownTargetBitrate=512
MJPEG.General.Overview.800X600.MaxUnknownTargetBitrate=6144
MJPEG.General.800X448.Width=800
MJPEG.General.800X448.Height=448
MJPEG.General.800X448.MaxFPS=5
MJPEG.General.800X448.MinUnknownTargetBitrate=512
MJPEG.General.800X448.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.800X448.Width=800
MJPEG.General.Overview.800X448.Height=448
MJPEG.General.Overview.800X448.MaxFPS=5
MJPEG.General.Overview.800X448.MinUnknownTargetBitrate=512
MJPEG.General.Overview.800X448.MaxUnknownTargetBitrate=6144
MJPEG.General.720X576.Width=720
MJPEG.General.720X576.Height=576
MJPEG.General.720X576.MaxFPS=5
MJPEG.General.720X576.MinUnknownTargetBitrate=512
MJPEG.General.720X576.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.720X576.Width=720
MJPEG.General.Overview.720X576.Height=576
MJPEG.General.Overview.720X576.MaxFPS=5
MJPEG.General.Overview.720X576.MinUnknownTargetBitrate=512
MJPEG.General.Overview.720X576.MaxUnknownTargetBitrate=6144
MJPEG.General.720X480.Width=720
MJPEG.General.720X480.Height=480
MJPEG.General.720X480.MaxFPS=5
MJPEG.General.720X480.MinUnknownTargetBitrate=512
MJPEG.General.720X480.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.720X480.Width=720
MJPEG.General.Overview.720X480.Height=480
MJPEG.General.Overview.720X480.MaxFPS=5
MJPEG.General.Overview.720X480.MinUnknownTargetBitrate=512
MJPEG.General.Overview.720X480.MaxUnknownTargetBitrate=6144
MJPEG.General.640X480.Width=640
MJPEG.General.640X480.Height=480
MJPEG.General.640X480.MaxFPS=5
```

```
MJPEG.General.640X480.MinUnknownTargetBitrate=512
MJPEG.General.640X480.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.640X480.Width=640
MJPEG.General.Overview.640X480.Height=480
MJPEG.General.Overview.640X480.MaxFPS=5
MJPEG.General.Overview.640X480.MinUnknownTargetBitrate=512
MJPEG.General.Overview.640X480.MaxUnknownTargetBitrate=6144
MJPEG.General.640X360.Width=640
MJPEG.General.640X360.Height=360
MJPEG.General.640X360.MaxFPS=5
MJPEG.General.640X360.MinUnknownTargetBitrate=512
MJPEG.General.640X360.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.640X360.Width=640
MJPEG.General.Overview.640X360.Height=360
MJPEG.General.Overview.640X360.MaxFPS=5
MJPEG.General.Overview.640X360.MinUnknownTargetBitrate=512
MJPEG.General.Overview.640X360.MaxUnknownTargetBitrate=6144
MJPEG.General.320X240.Width=320
MJPEG.General.320X240.Height=240
MJPEG.General.320X240.MaxFPS=5
MJPEG.General.320X240.MinUnknownTargetBitrate=256
MJPEG.General.320X240.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.320X240.Width=320
MJPEG.General.Overview.320X240.Height=240
MJPEG.General.Overview.320X240.MaxFPS=5
MJPEG.General.Overview.320X240.MinUnknownTargetBitrate=256
MJPEG.General.Overview.320X240.MaxUnknownTargetBitrate=6144
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "Profiles": [
                {
```

```
                    "Profile": 1,
                    "ViewModes": [
                        {
                            "ViewMode": "Overview",
                            "Codecs": [
                                {
                                    "EncodingType": "MJPEG",
                                    "IsDigitalPTZSupported": false,
                                    "General": [
                                        {
                                            "Width": 4000,
                                            "Height": 3000,
                                            "MaxFPS": 2,
                                            "MinUnknownTargetBitrate": 5120,
                                            "MaxUnknownTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 3840,
                                            "Height": 2160,
                                            "MaxFPS": 2,
                                            "MinUnknownTargetBitrate": 4096,
                                            "MaxUnknownTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 2592,
                                            "Height": 1944,
                                            "MaxFPS": 2,
                                            "MinUnknownTargetBitrate": 2048,
                                            "MaxUnknownTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 2592,
                                            "Height": 1464,
                                            "MaxFPS": 2,
                                            "MinUnknownTargetBitrate": 2048,
                                            "MaxUnknownTargetBitrate": 6144
                                        },
                                        {
                                            "Width": 1920,
                                            "Height": 1080,
                                            "MaxFPS": 5,
```

```
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 1600,
                    "Height": 1200,
                    "MaxFPS": 5,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 1280,
                    "Height": 1024,
                    "MaxFPS": 5,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 1280,
                    "Height": 960,
                    "MaxFPS": 5,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 5,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 1024,
                    "Height": 768,
                    "MaxFPS": 5,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 6144
                },
                {
                    "Width": 800,
                    "Height": 600,
```

```
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 800,
                                                "Height": 448,
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 720,
                                                "Height": 576,
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 720,
                                                "Height": 480,
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 640,
                                                "Height": 480,
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 640,
                                                "Height": 360,
                                                "MaxFPS": 5,
                                                "MinUnknownTargetBitrate": 512,
                                                "MaxUnknownTargetBitrate": 6144
                                            },
                                            {
                                                "Width": 320,
```

```
                                            "Height": 240,
                                            "MaxFPS": 5,
                                            "MinUnknownTargetBitrate": 256,
                                            "MaxUnknownTargetBitrate": 6144
                                        }
                                    ]
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
    ...
}
```

106.15.4.2. Getting all resolution information based on view modes

The following command returns all available video resolution information based on view mode.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videocodecinfo&action=view&ViewMode=All
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=MJPEG
Channel.0.ViewMode=Overview
MJPEG.IsDigitalPTZSupported=False
MJPEG.General.3328X1872.Width=3328
MJPEG.General.3328X1872.Height=1872
MJPEG.General.3328X1872.MaxFPS=5000
MJPEG.General.3328X1872.DefaultFPS=5000
MJPEG.General.3328X1872.MaxCBRTargetBitrate=30720
MJPEG.General.3328X1872.MinCBRTargetBitrate=4096
MJPEG.General.3328X1872.DefaultCBRTargetBitrate=12288
```

```
MJPEG.General.3328X1872.MaxVBRTargetBitrate=30720
MJPEG.General.3328X1872.MinVBRTargetBitrate=4096
MJPEG.General.3328X1872.DefaultVBRTargetBitrate=12288
MJPEG.General.3328X1872.IsTruncated=Normal
MJPEG.General.3072X1728.Width=3072
MJPEG.General.3072X1728.Height=1728
MJPEG.General.3072X1728.MaxFPS=5000
MJPEG.General.3072X1728.DefaultFPS=5000
MJPEG.General.3072X1728.MaxCBRTargetBitrate=30720
MJPEG.General.3072X1728.MinCBRTargetBitrate=4096
MJPEG.General.3072X1728.DefaultCBRTargetBitrate=12288
MJPEG.General.3072X1728.MaxVBRTargetBitrate=30720
MJPEG.General.3072X1728.MinVBRTargetBitrate=4096
MJPEG.General.3072X1728.DefaultVBRTargetBitrate=12288
MJPEG.General.3072X1728.IsTruncated=Normal
MJPEG.General.2592X1944.Width=2592
MJPEG.General.2592X1944.Height=1944
MJPEG.General.2592X1944.MaxFPS=5000
MJPEG.General.2592X1944.DefaultFPS=5000
MJPEG.General.2592X1944.MaxCBRTargetBitrate=30720
MJPEG.General.2592X1944.MinCBRTargetBitrate=3072
MJPEG.General.2592X1944.DefaultCBRTargetBitrate=10240
MJPEG.General.2592X1944.MaxVBRTargetBitrate=30720
MJPEG.General.2592X1944.MinVBRTargetBitrate=3072
MJPEG.General.2592X1944.DefaultVBRTargetBitrate=10240
MJPEG.General.2592X1944.IsTruncated=Crop
MJPEG.General.2688X1520.Width=2688
MJPEG.General.2688X1520.Height=1520
MJPEG.General.2688X1520.MaxFPS=5000
MJPEG.General.2688X1520.DefaultFPS=5000
MJPEG.General.2688X1520.MaxCBRTargetBitrate=30720
MJPEG.General.2688X1520.MinCBRTargetBitrate=3072
MJPEG.General.2688X1520.DefaultCBRTargetBitrate=10240
MJPEG.General.2688X1520.MaxVBRTargetBitrate=30720
MJPEG.General.2688X1520.MinVBRTargetBitrate=3072
MJPEG.General.2688X1520.DefaultVBRTargetBitrate=10240
MJPEG.General.2688X1520.IsTruncated=Crop
MJPEG.General.1920X1080.Width=1920
MJPEG.General.1920X1080.Height=1080
MJPEG.General.1920X1080.MaxFPS=30000
MJPEG.General.1920X1080.DefaultFPS=15000
```

```
MJPEG.General.1920X1080.MaxCBRTargetBitrate=30720
MJPEG.General.1920X1080.MinCBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultCBRTargetBitrate=10240
MJPEG.General.1920X1080.MaxVBRTargetBitrate=30720
MJPEG.General.1920X1080.MinVBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultVBRTargetBitrate=10240
MJPEG.General.1920X1080.IsTruncated=Normal
MJPEG.General.1600X1200.Width=1600
MJPEG.General.1600X1200.Height=1200
MJPEG.General.1600X1200.MaxFPS=30000
MJPEG.General.1600X1200.DefaultFPS=15000
MJPEG.General.1600X1200.MaxCBRTargetBitrate=30720
MJPEG.General.1600X1200.MinCBRTargetBitrate=2048
MJPEG.General.1600X1200.DefaultCBRTargetBitrate=10240
MJPEG.General.1600X1200.MaxVBRTargetBitrate=30720
MJPEG.General.1600X1200.MinVBRTargetBitrate=2048
MJPEG.General.1600X1200.DefaultVBRTargetBitrate=10240
MJPEG.General.1600X1200.IsTruncated=Crop
MJPEG.General.1280X1024.Width=1280
MJPEG.General.1280X1024.Height=1024
MJPEG.General.1280X1024.MaxFPS=30000
MJPEG.General.1280X1024.DefaultFPS=15000
MJPEG.General.1280X1024.MaxCBRTargetBitrate=30720
MJPEG.General.1280X1024.MinCBRTargetBitrate=1024
MJPEG.General.1280X1024.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X1024.MaxVBRTargetBitrate=30720
MJPEG.General.1280X1024.MinVBRTargetBitrate=1024
MJPEG.General.1280X1024.DefaultVBRTargetBitrate=10240
MJPEG.General.1280X1024.IsTruncated=Crop
MJPEG.General.1280X960.Width=1280
MJPEG.General.1280X960.Height=960
MJPEG.General.1280X960.MaxFPS=30000
MJPEG.General.1280X960.DefaultFPS=15000
MJPEG.General.1280X960.MaxCBRTargetBitrate=30720
MJPEG.General.1280X960.MinCBRTargetBitrate=1024
MJPEG.General.1280X960.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X960.MaxVBRTargetBitrate=30720
MJPEG.General.1280X960.MinVBRTargetBitrate=1024
MJPEG.General.1280X960.DefaultVBRTargetBitrate=10240
MJPEG.General.1280X960.IsTruncated=Crop
MJPEG.General.1280X720.Width=1280
```

```
MJPEG.General.1280X720.Height=720
MJPEG.General.1280X720.MaxFPS=30000
MJPEG.General.1280X720.DefaultFPS=15000
MJPEG.General.1280X720.MaxCBRTargetBitrate=30720
MJPEG.General.1280X720.MinCBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultCBRTargetBitrate=10240
MJPEG.General.1280X720.MaxVBRTargetBitrate=30720
MJPEG.General.1280X720.MinVBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultVBRTargetBitrate=10240
MJPEG.General.1280X720.IsTruncated=Normal
MJPEG.General.1024X768.Width=1024
MJPEG.General.1024X768.Height=768
MJPEG.General.1024X768.MaxFPS=30000
MJPEG.General.1024X768.DefaultFPS=15000
MJPEG.General.1024X768.MaxCBRTargetBitrate=30720
MJPEG.General.1024X768.MinCBRTargetBitrate=1024
MJPEG.General.1024X768.DefaultCBRTargetBitrate=10240
MJPEG.General.1024X768.MaxVBRTargetBitrate=30720
MJPEG.General.1024X768.MinVBRTargetBitrate=1024
MJPEG.General.1024X768.DefaultVBRTargetBitrate=10240
MJPEG.General.1024X768.IsTruncated=Crop
MJPEG.General.800X600.Width=800
MJPEG.General.800X600.Height=600
MJPEG.General.800X600.MaxFPS=30000
MJPEG.General.800X600.DefaultFPS=15000
MJPEG.General.800X600.MaxCBRTargetBitrate=30720
MJPEG.General.800X600.MinCBRTargetBitrate=512
MJPEG.General.800X600.DefaultCBRTargetBitrate=10240
MJPEG.General.800X600.MaxVBRTargetBitrate=30720
MJPEG.General.800X600.MinVBRTargetBitrate=512
MJPEG.General.800X600.DefaultVBRTargetBitrate=10240
MJPEG.General.800X600.IsTruncated=Crop
MJPEG.General.800X448.Width=800
MJPEG.General.800X448.Height=448
MJPEG.General.800X448.MaxFPS=30000
MJPEG.General.800X448.DefaultFPS=15000
MJPEG.General.800X448.MaxCBRTargetBitrate=30720
MJPEG.General.800X448.MinCBRTargetBitrate=512
MJPEG.General.800X448.DefaultCBRTargetBitrate=10240
MJPEG.General.800X448.MaxVBRTargetBitrate=30720
MJPEG.General.800X448.MinVBRTargetBitrate=512
```

```
MJPEG.General.800X448.DefaultVBRTargetBitrate=10240
MJPEG.General.800X448.IsTruncated=Normal
MJPEG.General.720X576.Width=720
MJPEG.General.720X576.Height=576
MJPEG.General.720X576.MaxFPS=30000
MJPEG.General.720X576.DefaultFPS=15000
MJPEG.General.720X576.MaxCBRTargetBitrate=30720
MJPEG.General.720X576.MinCBRTargetBitrate=512
MJPEG.General.720X576.DefaultCBRTargetBitrate=10240
MJPEG.General.720X576.MaxVBRTargetBitrate=30720
MJPEG.General.720X576.MinVBRTargetBitrate=512
MJPEG.General.720X576.DefaultVBRTargetBitrate=10240
MJPEG.General.720X576.IsTruncated=Crop
MJPEG.General.720X480.Width=720
MJPEG.General.720X480.Height=480
MJPEG.General.720X480.MaxFPS=30000
MJPEG.General.720X480.DefaultFPS=15000
MJPEG.General.720X480.MaxCBRTargetBitrate=30720
MJPEG.General.720X480.MinCBRTargetBitrate=512
MJPEG.General.720X480.DefaultCBRTargetBitrate=10240
MJPEG.General.720X480.MaxVBRTargetBitrate=30720
MJPEG.General.720X480.MinVBRTargetBitrate=512
MJPEG.General.720X480.DefaultVBRTargetBitrate=10240
MJPEG.General.720X480.IsTruncated=Crop
MJPEG.General.640X480.Width=640
MJPEG.General.640X480.Height=480
MJPEG.General.640X480.MaxFPS=30000
MJPEG.General.640X480.DefaultFPS=15000
MJPEG.General.640X480.MaxCBRTargetBitrate=30720
MJPEG.General.640X480.MinCBRTargetBitrate=512
MJPEG.General.640X480.DefaultCBRTargetBitrate=10240
MJPEG.General.640X480.MaxVBRTargetBitrate=30720
MJPEG.General.640X480.MinVBRTargetBitrate=512
MJPEG.General.640X480.DefaultVBRTargetBitrate=10240
MJPEG.General.640X480.IsTruncated=Crop
MJPEG.General.640X360.Width=640
MJPEG.General.640X360.Height=360
MJPEG.General.640X360.MaxFPS=30000
MJPEG.General.640X360.DefaultFPS=15000
MJPEG.General.640X360.MaxCBRTargetBitrate=30720
MJPEG.General.640X360.MinCBRTargetBitrate=512
```

```
MJPEG.General.640X360.DefaultCBRTargetBitrate=10240
MJPEG.General.640X360.MaxVBRTargetBitrate=30720
MJPEG.General.640X360.MinVBRTargetBitrate=512
MJPEG.General.640X360.DefaultVBRTargetBitrate=10240
MJPEG.General.640X360.IsTruncated=Normal
MJPEG.General.320X240.Width=320
MJPEG.General.320X240.Height=240
MJPEG.General.320X240.MaxFPS=30000
MJPEG.General.320X240.DefaultFPS=15000
MJPEG.General.320X240.MaxCBRTargetBitrate=30720
MJPEG.General.320X240.MinCBRTargetBitrate=256
MJPEG.General.320X240.DefaultCBRTargetBitrate=10240
MJPEG.General.320X240.MaxVBRTargetBitrate=30720
MJPEG.General.320X240.MinVBRTargetBitrate=256
MJPEG.General.320X240.DefaultVBRTargetBitrate=10240
MJPEG.General.320X240.IsTruncated=Crop
MJPEG.Record.3328X1872.MaxFPS=1000
MJPEG.Record.3328X1872.DefaultFPS=1000
MJPEG.Record.3328X1872.MinCBRTargetBitrate=4096
MJPEG.Record.3328X1872.MaxCBRTargetBitrate=6144
MJPEG.Record.3328X1872.DefaultCBRTargetBitrate=6144
MJPEG.Record.3328X1872.MinVBRTargetBitrate=4096
MJPEG.Record.3328X1872.MaxVBRTargetBitrate=6144
MJPEG.Record.3328X1872.DefaultVBRTargetBitrate=6144
MJPEG.Record.3072X1728.MaxFPS=1000
MJPEG.Record.3072X1728.DefaultFPS=1000
MJPEG.Record.3072X1728.MinCBRTargetBitrate=4096
MJPEG.Record.3072X1728.MaxCBRTargetBitrate=6144
MJPEG.Record.3072X1728.DefaultCBRTargetBitrate=6144
MJPEG.Record.3072X1728.MinVBRTargetBitrate=4096
MJPEG.Record.3072X1728.MaxVBRTargetBitrate=6144
MJPEG.Record.3072X1728.DefaultVBRTargetBitrate=6144
MJPEG.Record.2592X1944.MaxFPS=1000
MJPEG.Record.2592X1944.DefaultFPS=1000
MJPEG.Record.2592X1944.MinCBRTargetBitrate=3072
MJPEG.Record.2592X1944.MaxCBRTargetBitrate=6144
MJPEG.Record.2592X1944.DefaultCBRTargetBitrate=6144
MJPEG.Record.2592X1944.MinVBRTargetBitrate=3072
MJPEG.Record.2592X1944.MaxVBRTargetBitrate=6144
MJPEG.Record.2592X1944.DefaultVBRTargetBitrate=6144
MJPEG.Record.2688X1520.MaxFPS=1000
```

```
MJPEG.Record.2688X1520.DefaultFPS=1000
MJPEG.Record.2688X1520.MinCBRTargetBitrate=3072
MJPEG.Record.2688X1520.MaxCBRTargetBitrate=6144
MJPEG.Record.2688X1520.DefaultCBRTargetBitrate=6144
MJPEG.Record.2688X1520.MinVBRTargetBitrate=3072
MJPEG.Record.2688X1520.MaxVBRTargetBitrate=6144
MJPEG.Record.2688X1520.DefaultVBRTargetBitrate=6144
MJPEG.Record.1920X1080.MaxFPS=2000
MJPEG.Record.1920X1080.DefaultFPS=2000
MJPEG.Record.1920X1080.MinCBRTargetBitrate=2048
MJPEG.Record.1920X1080.MaxCBRTargetBitrate=6144
MJPEG.Record.1920X1080.DefaultCBRTargetBitrate=6144
MJPEG.Record.1920X1080.MinVBRTargetBitrate=2048
MJPEG.Record.1920X1080.MaxVBRTargetBitrate=6144
MJPEG.Record.1920X1080.DefaultVBRTargetBitrate=6144
MJPEG.Record.1600X1200.MaxFPS=2000
MJPEG.Record.1600X1200.DefaultFPS=2000
MJPEG.Record.1600X1200.MinCBRTargetBitrate=2048
MJPEG.Record.1600X1200.MaxCBRTargetBitrate=6144
MJPEG.Record.1600X1200.DefaultCBRTargetBitrate=6144
MJPEG.Record.1600X1200.MinVBRTargetBitrate=2048
MJPEG.Record.1600X1200.MaxVBRTargetBitrate=6144
MJPEG.Record.1600X1200.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X1024.MaxFPS=3000
MJPEG.Record.1280X1024.DefaultFPS=3000
MJPEG.Record.1280X1024.MinCBRTargetBitrate=1024
MJPEG.Record.1280X1024.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X1024.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X1024.MinVBRTargetBitrate=1024
MJPEG.Record.1280X1024.MaxVBRTargetBitrate=6144
MJPEG.Record.1280X1024.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X960.MaxFPS=3000
MJPEG.Record.1280X960.DefaultFPS=3000
MJPEG.Record.1280X960.MinCBRTargetBitrate=1024
MJPEG.Record.1280X960.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X960.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X960.MinVBRTargetBitrate=1024
MJPEG.Record.1280X960.MaxVBRTargetBitrate=6144
MJPEG.Record.1280X960.DefaultVBRTargetBitrate=6144
MJPEG.Record.1280X720.MaxFPS=3000
MJPEG.Record.1280X720.DefaultFPS=3000
```

```
MJPEG.Record.1280X720.MinCBRTargetBitrate=1024
MJPEG.Record.1280X720.MaxCBRTargetBitrate=6144
MJPEG.Record.1280X720.DefaultCBRTargetBitrate=6144
MJPEG.Record.1280X720.MinVBRTargetBitrate=1024
MJPEG.Record.1280X720.MaxVBRTargetBitrate=6144
MJPEG.Record.1280X720.DefaultVBRTargetBitrate=6144
MJPEG.Record.1024X768.MaxFPS=3000
MJPEG.Record.1024X768.DefaultFPS=3000
MJPEG.Record.1024X768.MinCBRTargetBitrate=1024
MJPEG.Record.1024X768.MaxCBRTargetBitrate=6144
MJPEG.Record.1024X768.DefaultCBRTargetBitrate=6144
MJPEG.Record.1024X768.MinVBRTargetBitrate=1024
MJPEG.Record.1024X768.MaxVBRTargetBitrate=6144
MJPEG.Record.1024X768.DefaultVBRTargetBitrate=6144
MJPEG.Record.800X600.MaxFPS=5000
MJPEG.Record.800X600.DefaultFPS=5000
MJPEG.Record.800X600.MinCBRTargetBitrate=512
MJPEG.Record.800X600.MaxCBRTargetBitrate=6144
MJPEG.Record.800X600.DefaultCBRTargetBitrate=6144
MJPEG.Record.800X600.MinVBRTargetBitrate=512
MJPEG.Record.800X600.MaxVBRTargetBitrate=6144
MJPEG.Record.800X600.DefaultVBRTargetBitrate=6144
MJPEG.Record.800X448.MaxFPS=5000
MJPEG.Record.800X448.DefaultFPS=5000
MJPEG.Record.800X448.MinCBRTargetBitrate=512
MJPEG.Record.800X448.MaxCBRTargetBitrate=6144
MJPEG.Record.800X448.DefaultCBRTargetBitrate=6144
MJPEG.Record.800X448.MinVBRTargetBitrate=512
MJPEG.Record.800X448.MaxVBRTargetBitrate=6144
MJPEG.Record.800X448.DefaultVBRTargetBitrate=6144
MJPEG.Record.720X576.MaxFPS=5000
MJPEG.Record.720X576.DefaultFPS=5000
MJPEG.Record.720X576.MinCBRTargetBitrate=512
MJPEG.Record.720X576.MaxCBRTargetBitrate=6144
MJPEG.Record.720X576.DefaultCBRTargetBitrate=6144
MJPEG.Record.720X576.MinVBRTargetBitrate=512
MJPEG.Record.720X576.MaxVBRTargetBitrate=6144
MJPEG.Record.720X576.DefaultVBRTargetBitrate=6144
MJPEG.Record.720X480.MaxFPS=5000
MJPEG.Record.720X480.DefaultFPS=5000
MJPEG.Record.720X480.MinCBRTargetBitrate=512
```

```
MJPEG.Record.720X480.MaxCBRTargetBitrate=6144
MJPEG.Record.720X480.DefaultCBRTargetBitrate=6144
MJPEG.Record.720X480.MinVBRTargetBitrate=512
MJPEG.Record.720X480.MaxVBRTargetBitrate=6144
MJPEG.Record.720X480.DefaultVBRTargetBitrate=6144
MJPEG.Record.640X480.MaxFPS=5000
MJPEG.Record.640X480.DefaultFPS=5000
MJPEG.Record.640X480.MinCBRTargetBitrate=512
MJPEG.Record.640X480.MaxCBRTargetBitrate=6144
MJPEG.Record.640X480.DefaultCBRTargetBitrate=6144
MJPEG.Record.640X480.MinVBRTargetBitrate=512
MJPEG.Record.640X480.MaxVBRTargetBitrate=6144
MJPEG.Record.640X480.DefaultVBRTargetBitrate=6144
MJPEG.Record.640X360.MaxFPS=5000
MJPEG.Record.640X360.DefaultFPS=5000
MJPEG.Record.640X360.MinCBRTargetBitrate=512
MJPEG.Record.640X360.MaxCBRTargetBitrate=6144
MJPEG.Record.640X360.DefaultCBRTargetBitrate=6144
MJPEG.Record.640X360.MinVBRTargetBitrate=512
MJPEG.Record.640X360.MaxVBRTargetBitrate=6144
MJPEG.Record.640X360.DefaultVBRTargetBitrate=6144
MJPEG.Record.320X240.MaxFPS=5000
MJPEG.Record.320X240.DefaultFPS=5000
MJPEG.Record.320X240.MinCBRTargetBitrate=256
MJPEG.Record.320X240.MaxCBRTargetBitrate=6144
MJPEG.Record.320X240.DefaultCBRTargetBitrate=6144
MJPEG.Record.320X240.MinVBRTargetBitrate=256
MJPEG.Record.320X240.MaxVBRTargetBitrate=6144
MJPEG.Record.320X240.DefaultVBRTargetBitrate=6144
MJPEG.Email.3328X1872.MaxFPS=5000
MJPEG.Email.3328X1872.DefaultFPS=5000
MJPEG.Email.3072X1728.MaxFPS=5000
MJPEG.Email.3072X1728.DefaultFPS=5000
MJPEG.Email.2592X1944.MaxFPS=5000
MJPEG.Email.2592X1944.DefaultFPS=5000
MJPEG.Email.2688X1520.MaxFPS=5000
MJPEG.Email.2688X1520.DefaultFPS=5000
MJPEG.Email.1920X1080.MaxFPS=5000
MJPEG.Email.1920X1080.DefaultFPS=5000
MJPEG.Email.1600X1200.MaxFPS=5000
MJPEG.Email.1600X1200.DefaultFPS=5000
```

```
MJPEG.Email.1280X1024.MaxFPS=5000
MJPEG.Email.1280X1024.DefaultFPS=5000
MJPEG.Email.1280X960.MaxFPS=5000
MJPEG.Email.1280X960.DefaultFPS=5000
MJPEG.Email.1280X720.MaxFPS=5000
MJPEG.Email.1280X720.DefaultFPS=5000
MJPEG.Email.1024X768.MaxFPS=5000
MJPEG.Email.1024X768.DefaultFPS=5000
MJPEG.Email.800X600.MaxFPS=5000
MJPEG.Email.800X600.DefaultFPS=5000
MJPEG.Email.800X448.MaxFPS=5000
MJPEG.Email.800X448.DefaultFPS=5000
MJPEG.Email.720X576.MaxFPS=5000
MJPEG.Email.720X576.DefaultFPS=5000
MJPEG.Email.720X480.MaxFPS=5000
MJPEG.Email.720X480.DefaultFPS=5000
MJPEG.Email.640X480.MaxFPS=5000
MJPEG.Email.640X480.DefaultFPS=5000
MJPEG.Email.640X360.MaxFPS=5000
MJPEG.Email.640X360.DefaultFPS=5000
MJPEG.Email.320X240.MaxFPS=5000
MJPEG.Email.320X240.DefaultFPS=5000
Channel.0.EncodingType=H264
H264.IsDigitalPTZSupported=True
H264.Compression.DefaultCompressionLevel=16
H264.Compression.MinCompressionLevel=0
H264.Compression.MaxCompressionLevel=50
H264.General.3328X1872.Width=3328
H264.General.3328X1872.Height=1872
H264.General.3328X1872.MaxFPS=30000
H264.General.3328X1872.DefaultFPS=30000
H264.General.3328X1872.MaxCBRTargetBitrate=20480
H264.General.3328X1872.MinCBRTargetBitrate=3584
H264.General.3328X1872.DefaultCBRTargetBitrate=9216
H264.General.3328X1872.MaxVBRTargetBitrate=30720
H264.General.3328X1872.MinVBRTargetBitrate=2048
H264.General.3328X1872.DefaultVBRTargetBitrate=9216
H264.General.3328X1872.IsTruncated=Normal
H264.General.3328X1872.FrameLockMaxFPS=30000
H264.General.3072X1728.Width=3072
H264.General.3072X1728.Height=1728
```

```
H264.General.3072X1728.MaxFPS=30000
H264.General.3072X1728.DefaultFPS=30000
H264.General.3072X1728.MaxCBRTargetBitrate=20480
H264.General.3072X1728.MinCBRTargetBitrate=3584
H264.General.3072X1728.DefaultCBRTargetBitrate=9216
H264.General.3072X1728.MaxVBRTargetBitrate=30720
H264.General.3072X1728.MinVBRTargetBitrate=2048
H264.General.3072X1728.DefaultVBRTargetBitrate=9216
H264.General.3072X1728.IsTruncated=Normal
H264.General.3072X1728.FrameLockMaxFPS=30000
H264.General.2592X1944.Width=2592
H264.General.2592X1944.Height=1944
H264.General.2592X1944.MaxFPS=30000
H264.General.2592X1944.DefaultFPS=30000
H264.General.2592X1944.MaxCBRTargetBitrate=20480
H264.General.2592X1944.MinCBRTargetBitrate=2048
H264.General.2592X1944.DefaultCBRTargetBitrate=7680
H264.General.2592X1944.MaxVBRTargetBitrate=30720
H264.General.2592X1944.MinVBRTargetBitrate=1536
H264.General.2592X1944.DefaultVBRTargetBitrate=7680
H264.General.2592X1944.IsTruncated=Crop
H264.General.2592X1944.FrameLockMaxFPS=30000
H264.General.2688X1520.Width=2688
H264.General.2688X1520.Height=1520
H264.General.2688X1520.MaxFPS=30000
H264.General.2688X1520.DefaultFPS=30000
H264.General.2688X1520.MaxCBRTargetBitrate=20480
H264.General.2688X1520.MinCBRTargetBitrate=2048
H264.General.2688X1520.DefaultCBRTargetBitrate=7680
H264.General.2688X1520.MaxVBRTargetBitrate=30720
H264.General.2688X1520.MinVBRTargetBitrate=1536
H264.General.2688X1520.DefaultVBRTargetBitrate=7680
H264.General.2688X1520.IsTruncated=Crop
H264.General.2688X1520.FrameLockMaxFPS=30000
H264.General.1920X1080.Width=1920
H264.General.1920X1080.Height=1080
H264.General.1920X1080.MaxFPS=30000
H264.General.1920X1080.DefaultFPS=30000
H264.General.1920X1080.MaxCBRTargetBitrate=20480
H264.General.1920X1080.MinCBRTargetBitrate=1024
H264.General.1920X1080.DefaultCBRTargetBitrate=3072
```

```
H264.General.1920X1080.MaxVBRTargetBitrate=30720
H264.General.1920X1080.MinVBRTargetBitrate=1536
H264.General.1920X1080.DefaultVBRTargetBitrate=3072
H264.General.1920X1080.IsTruncated=Normal
H264.General.1920X1080.FrameLockMaxFPS=30000
H264.General.1600X1200.Width=1600
H264.General.1600X1200.Height=1200
H264.General.1600X1200.MaxFPS=30000
H264.General.1600X1200.DefaultFPS=30000
H264.General.1600X1200.MaxCBRTargetBitrate=20480
H264.General.1600X1200.MinCBRTargetBitrate=1024
H264.General.1600X1200.DefaultCBRTargetBitrate=3072
H264.General.1600X1200.MaxVBRTargetBitrate=30720
H264.General.1600X1200.MinVBRTargetBitrate=1536
H264.General.1600X1200.DefaultVBRTargetBitrate=3072
H264.General.1600X1200.IsTruncated=Crop
H264.General.1600X1200.FrameLockMaxFPS=30000
H264.General.1280X1024.Width=1280
H264.General.1280X1024.Height=1024
H264.General.1280X1024.MaxFPS=30000
H264.General.1280X1024.DefaultFPS=30000
H264.General.1280X1024.MaxCBRTargetBitrate=20480
H264.General.1280X1024.MinCBRTargetBitrate=1024
H264.General.1280X1024.DefaultCBRTargetBitrate=1536
H264.General.1280X1024.MaxVBRTargetBitrate=30720
H264.General.1280X1024.MinVBRTargetBitrate=1536
H264.General.1280X1024.DefaultVBRTargetBitrate=1536
H264.General.1280X1024.IsTruncated=Crop
H264.General.1280X1024.FrameLockMaxFPS=30000
H264.General.1280X960.Width=1280
H264.General.1280X960.Height=960
H264.General.1280X960.MaxFPS=30000
H264.General.1280X960.DefaultFPS=30000
H264.General.1280X960.MaxCBRTargetBitrate=20480
H264.General.1280X960.MinCBRTargetBitrate=1024
H264.General.1280X960.DefaultCBRTargetBitrate=1536
H264.General.1280X960.MaxVBRTargetBitrate=30720
H264.General.1280X960.MinVBRTargetBitrate=1536
H264.General.1280X960.DefaultVBRTargetBitrate=1536
H264.General.1280X960.IsTruncated=Crop
H264.General.1280X960.FrameLockMaxFPS=30000
```

```
H264.General.1280X720.Width=1280
H264.General.1280X720.Height=720
H264.General.1280X720.MaxFPS=30000
H264.General.1280X720.DefaultFPS=30000
H264.General.1280X720.MaxCBRTargetBitrate=20480
H264.General.1280X720.MinCBRTargetBitrate=1024
H264.General.1280X720.DefaultCBRTargetBitrate=1536
H264.General.1280X720.MaxVBRTargetBitrate=30720
H264.General.1280X720.MinVBRTargetBitrate=1536
H264.General.1280X720.DefaultVBRTargetBitrate=1536
H264.General.1280X720.IsTruncated=Normal
H264.General.1280X720.FrameLockMaxFPS=30000
H264.General.1024X768.Width=1024
H264.General.1024X768.Height=768
H264.General.1024X768.MaxFPS=30000
H264.General.1024X768.DefaultFPS=30000
H264.General.1024X768.MaxCBRTargetBitrate=20480
H264.General.1024X768.MinCBRTargetBitrate=1024
H264.General.1024X768.DefaultCBRTargetBitrate=1536
H264.General.1024X768.MaxVBRTargetBitrate=30720
H264.General.1024X768.MinVBRTargetBitrate=1536
H264.General.1024X768.DefaultVBRTargetBitrate=1536
H264.General.1024X768.IsTruncated=Crop
H264.General.1024X768.FrameLockMaxFPS=30000
H264.General.800X600.Width=800
H264.General.800X600.Height=600
H264.General.800X600.MaxFPS=30000
H264.General.800X600.DefaultFPS=30000
H264.General.800X600.MaxCBRTargetBitrate=20480
H264.General.800X600.MinCBRTargetBitrate=512
H264.General.800X600.DefaultCBRTargetBitrate=1024
H264.General.800X600.MaxVBRTargetBitrate=30720
H264.General.800X600.MinVBRTargetBitrate=512
H264.General.800X600.DefaultVBRTargetBitrate=1024
H264.General.800X600.IsTruncated=Crop
H264.General.800X600.FrameLockMaxFPS=30000
H264.General.800X448.Width=800
H264.General.800X448.Height=448
H264.General.800X448.MaxFPS=30000
H264.General.800X448.DefaultFPS=30000
H264.General.800X448.MaxCBRTargetBitrate=20480
```

```
H264.General.800X448.MinCBRTargetBitrate=512
H264.General.800X448.DefaultCBRTargetBitrate=1024
H264.General.800X448.MaxVBRTargetBitrate=30720
H264.General.800X448.MinVBRTargetBitrate=512
H264.General.800X448.DefaultVBRTargetBitrate=1024
H264.General.800X448.IsTruncated=Normal
H264.General.800X448.FrameLockMaxFPS=30000
H264.General.720X576.Width=720
H264.General.720X576.Height=576
H264.General.720X576.MaxFPS=30000
H264.General.720X576.DefaultFPS=30000
H264.General.720X576.MaxCBRTargetBitrate=20480
H264.General.720X576.MinCBRTargetBitrate=512
H264.General.720X576.DefaultCBRTargetBitrate=1024
H264.General.720X576.MaxVBRTargetBitrate=30720
H264.General.720X576.MinVBRTargetBitrate=512
H264.General.720X576.DefaultVBRTargetBitrate=1024
H264.General.720X576.IsTruncated=Crop
H264.General.720X576.FrameLockMaxFPS=30000
H264.General.720X480.Width=720
H264.General.720X480.Height=480
H264.General.720X480.MaxFPS=30000
H264.General.720X480.DefaultFPS=30000
H264.General.720X480.MaxCBRTargetBitrate=20480
H264.General.720X480.MinCBRTargetBitrate=512
H264.General.720X480.DefaultCBRTargetBitrate=1024
H264.General.720X480.MaxVBRTargetBitrate=30720
H264.General.720X480.MinVBRTargetBitrate=512
H264.General.720X480.DefaultVBRTargetBitrate=1024
H264.General.720X480.IsTruncated=Crop
H264.General.720X480.FrameLockMaxFPS=30000
H264.General.640X480.Width=640
H264.General.640X480.Height=480
H264.General.640X480.MaxFPS=30000
H264.General.640X480.DefaultFPS=30000
H264.General.640X480.MaxCBRTargetBitrate=20480
H264.General.640X480.MinCBRTargetBitrate=512
H264.General.640X480.DefaultCBRTargetBitrate=1024
H264.General.640X480.MaxVBRTargetBitrate=30720
H264.General.640X480.MinVBRTargetBitrate=512
H264.General.640X480.DefaultVBRTargetBitrate=1024
```

```
H264.General.640X480.IsTruncated=Crop
H264.General.640X480.FrameLockMaxFPS=30000
H264.General.640X360.Width=640
H264.General.640X360.Height=360
H264.General.640X360.MaxFPS=30000
H264.General.640X360.DefaultFPS=30000
H264.General.640X360.MaxCBRTargetBitrate=20480
H264.General.640X360.MinCBRTargetBitrate=512
H264.General.640X360.DefaultCBRTargetBitrate=1024
H264.General.640X360.MaxVBRTargetBitrate=30720
H264.General.640X360.MinVBRTargetBitrate=512
H264.General.640X360.DefaultVBRTargetBitrate=1024
H264.General.640X360.IsTruncated=Normal
H264.General.640X360.FrameLockMaxFPS=30000
H264.General.320X240.Width=320
H264.General.320X240.Height=240
H264.General.320X240.MaxFPS=30000
H264.General.320X240.DefaultFPS=30000
H264.General.320X240.MaxCBRTargetBitrate=20480
H264.General.320X240.MinCBRTargetBitrate=256
H264.General.320X240.DefaultCBRTargetBitrate=512
H264.General.320X240.MaxVBRTargetBitrate=30720
H264.General.320X240.MinVBRTargetBitrate=256
H264.General.320X240.DefaultVBRTargetBitrate=512
H264.General.320X240.IsTruncated=Crop
H264.General.320X240.FrameLockMaxFPS=30000
H264.Record.3328X1872.MaxFPS=30000
H264.Record.3328X1872.DefaultFPS=30000
H264.Record.3328X1872.MinCBRTargetBitrate=3584
H264.Record.3328X1872.MaxCBRTargetBitrate=6144
H264.Record.3328X1872.DefaultCBRTargetBitrate=5120
H264.Record.3328X1872.MinVBRTargetBitrate=2048
H264.Record.3328X1872.MaxVBRTargetBitrate=6144
H264.Record.3328X1872.DefaultVBRTargetBitrate=5120
H264.Record.3328X1872.FrameLockMaxFPS=30000
H264.Record.3072X1728.MaxFPS=30000
H264.Record.3072X1728.DefaultFPS=30000
H264.Record.3072X1728.MinCBRTargetBitrate=3584
H264.Record.3072X1728.MaxCBRTargetBitrate=6144
H264.Record.3072X1728.DefaultCBRTargetBitrate=5120
H264.Record.3072X1728.MinVBRTargetBitrate=2048
```

```
H264.Record.3072X1728.MaxVBRTargetBitrate=6144
H264.Record.3072X1728.DefaultVBRTargetBitrate=5120
H264.Record.3072X1728.FrameLockMaxFPS=30000
H264.Record.2592X1944.MaxFPS=30000
H264.Record.2592X1944.DefaultFPS=30000
H264.Record.2592X1944.MinCBRTargetBitrate=2048
H264.Record.2592X1944.MaxCBRTargetBitrate=6144
H264.Record.2592X1944.DefaultCBRTargetBitrate=5120
H264.Record.2592X1944.MinVBRTargetBitrate=1536
H264.Record.2592X1944.MaxVBRTargetBitrate=6144
H264.Record.2592X1944.DefaultVBRTargetBitrate=5120
H264.Record.2592X1944.FrameLockMaxFPS=30000
H264.Record.2688X1520.MaxFPS=30000
H264.Record.2688X1520.DefaultFPS=30000
H264.Record.2688X1520.MinCBRTargetBitrate=2048
H264.Record.2688X1520.MaxCBRTargetBitrate=6144
H264.Record.2688X1520.DefaultCBRTargetBitrate=5120
H264.Record.2688X1520.MinVBRTargetBitrate=1536
H264.Record.2688X1520.MaxVBRTargetBitrate=6144
H264.Record.2688X1520.DefaultVBRTargetBitrate=5120
H264.Record.2688X1520.FrameLockMaxFPS=30000
H264.Record.1920X1080.MaxFPS=30000
H264.Record.1920X1080.DefaultFPS=30000
H264.Record.1920X1080.MinCBRTargetBitrate=1024
H264.Record.1920X1080.MaxCBRTargetBitrate=6144
H264.Record.1920X1080.DefaultCBRTargetBitrate=5120
H264.Record.1920X1080.MinVBRTargetBitrate=1536
H264.Record.1920X1080.MaxVBRTargetBitrate=6144
H264.Record.1920X1080.DefaultVBRTargetBitrate=5120
H264.Record.1920X1080.FrameLockMaxFPS=30000
H264.Record.1600X1200.MaxFPS=30000
H264.Record.1600X1200.DefaultFPS=30000
H264.Record.1600X1200.MinCBRTargetBitrate=1024
H264.Record.1600X1200.MaxCBRTargetBitrate=6144
H264.Record.1600X1200.DefaultCBRTargetBitrate=5120
H264.Record.1600X1200.MinVBRTargetBitrate=1536
H264.Record.1600X1200.MaxVBRTargetBitrate=6144
H264.Record.1600X1200.DefaultVBRTargetBitrate=5120
H264.Record.1600X1200.FrameLockMaxFPS=30000
H264.Record.1280X1024.MaxFPS=30000
H264.Record.1280X1024.DefaultFPS=30000
```

```
H264.Record.1280X1024.MinCBRTargetBitrate=1024
H264.Record.1280X1024.MaxCBRTargetBitrate=6144
H264.Record.1280X1024.DefaultCBRTargetBitrate=5120
H264.Record.1280X1024.MinVBRTargetBitrate=1536
H264.Record.1280X1024.MaxVBRTargetBitrate=6144
H264.Record.1280X1024.DefaultVBRTargetBitrate=5120
H264.Record.1280X1024.FrameLockMaxFPS=30000
H264.Record.1280X960.MaxFPS=30000
H264.Record.1280X960.DefaultFPS=30000
H264.Record.1280X960.MinCBRTargetBitrate=1024
H264.Record.1280X960.MaxCBRTargetBitrate=6144
H264.Record.1280X960.DefaultCBRTargetBitrate=5120
H264.Record.1280X960.MinVBRTargetBitrate=1536
H264.Record.1280X960.MaxVBRTargetBitrate=6144
H264.Record.1280X960.DefaultVBRTargetBitrate=5120
H264.Record.1280X960.FrameLockMaxFPS=30000
H264.Record.1280X720.MaxFPS=30000
H264.Record.1280X720.DefaultFPS=30000
H264.Record.1280X720.MinCBRTargetBitrate=1024
H264.Record.1280X720.MaxCBRTargetBitrate=6144
H264.Record.1280X720.DefaultCBRTargetBitrate=5120
H264.Record.1280X720.MinVBRTargetBitrate=1536
H264.Record.1280X720.MaxVBRTargetBitrate=6144
H264.Record.1280X720.DefaultVBRTargetBitrate=5120
H264.Record.1280X720.FrameLockMaxFPS=30000
H264.Record.1024X768.MaxFPS=30000
H264.Record.1024X768.DefaultFPS=30000
H264.Record.1024X768.MinCBRTargetBitrate=1024
H264.Record.1024X768.MaxCBRTargetBitrate=6144
H264.Record.1024X768.DefaultCBRTargetBitrate=5120
H264.Record.1024X768.MinVBRTargetBitrate=1536
H264.Record.1024X768.MaxVBRTargetBitrate=6144
H264.Record.1024X768.DefaultVBRTargetBitrate=5120
H264.Record.1024X768.FrameLockMaxFPS=30000
H264.Record.800X600.MaxFPS=30000
H264.Record.800X600.DefaultFPS=30000
H264.Record.800X600.MinCBRTargetBitrate=512
H264.Record.800X600.MaxCBRTargetBitrate=6144
H264.Record.800X600.DefaultCBRTargetBitrate=5120
H264.Record.800X600.MinVBRTargetBitrate=512
H264.Record.800X600.MaxVBRTargetBitrate=6144
```

```
H264.Record.800X600.DefaultVBRTargetBitrate=5120
H264.Record.800X600.FrameLockMaxFPS=30000
H264.Record.800X448.MaxFPS=30000
H264.Record.800X448.DefaultFPS=30000
H264.Record.800X448.MinCBRTargetBitrate=512
H264.Record.800X448.MaxCBRTargetBitrate=6144
H264.Record.800X448.DefaultCBRTargetBitrate=5120
H264.Record.800X448.MinVBRTargetBitrate=512
H264.Record.800X448.MaxVBRTargetBitrate=6144
H264.Record.800X448.DefaultVBRTargetBitrate=5120
H264.Record.800X448.FrameLockMaxFPS=30000
H264.Record.720X576.MaxFPS=30000
H264.Record.720X576.DefaultFPS=30000
H264.Record.720X576.MinCBRTargetBitrate=512
H264.Record.720X576.MaxCBRTargetBitrate=6144
H264.Record.720X576.DefaultCBRTargetBitrate=5120
H264.Record.720X576.MinVBRTargetBitrate=512
H264.Record.720X576.MaxVBRTargetBitrate=6144
H264.Record.720X576.DefaultVBRTargetBitrate=5120
H264.Record.720X576.FrameLockMaxFPS=30000
H264.Record.720X480.MaxFPS=30000
H264.Record.720X480.DefaultFPS=30000
H264.Record.720X480.MinCBRTargetBitrate=512
H264.Record.720X480.MaxCBRTargetBitrate=6144
H264.Record.720X480.DefaultCBRTargetBitrate=5120
H264.Record.720X480.MinVBRTargetBitrate=512
H264.Record.720X480.MaxVBRTargetBitrate=6144
H264.Record.720X480.DefaultVBRTargetBitrate=5120
H264.Record.720X480.FrameLockMaxFPS=30000
H264.Record.640X480.MaxFPS=30000
H264.Record.640X480.DefaultFPS=30000
H264.Record.640X480.MinCBRTargetBitrate=512
H264.Record.640X480.MaxCBRTargetBitrate=6144
H264.Record.640X480.DefaultCBRTargetBitrate=5120
H264.Record.640X480.MinVBRTargetBitrate=512
H264.Record.640X480.MaxVBRTargetBitrate=6144
H264.Record.640X480.DefaultVBRTargetBitrate=5120
H264.Record.640X480.FrameLockMaxFPS=30000
H264.Record.640X360.MaxFPS=30000
H264.Record.640X360.DefaultFPS=30000
H264.Record.640X360.MinCBRTargetBitrate=512
```

```
H264.Record.640X360.MaxCBRTargetBitrate=6144
H264.Record.640X360.DefaultCBRTargetBitrate=5120
H264.Record.640X360.MinVBRTargetBitrate=512
H264.Record.640X360.MaxVBRTargetBitrate=6144
H264.Record.640X360.DefaultVBRTargetBitrate=5120
H264.Record.640X360.FrameLockMaxFPS=30000
H264.Record.320X240.MaxFPS=30000
H264.Record.320X240.DefaultFPS=30000
H264.Record.320X240.MinCBRTargetBitrate=256
H264.Record.320X240.MaxCBRTargetBitrate=6144
H264.Record.320X240.DefaultCBRTargetBitrate=5120
H264.Record.320X240.MinVBRTargetBitrate=256
H264.Record.320X240.MaxVBRTargetBitrate=6144
H264.Record.320X240.DefaultVBRTargetBitrate=5120
H264.Record.320X240.FrameLockMaxFPS=30000
H264.DigitalPTZ.3328X1872.Width=3328
H264.DigitalPTZ.3328X1872.Height=1872
H264.DigitalPTZ.3328X1872.MaxFPS=30000
H264.DigitalPTZ.3328X1872.DefaultFPS=30000
H264.DigitalPTZ.3328X1872.MaxCBRTargetBitrate=4000
H264.DigitalPTZ.3328X1872.MinCBRTargetBitrate=4000
H264.DigitalPTZ.3328X1872.DefaultCBRTargetBitrate=4000
H264.DigitalPTZ.3072X1728.Width=3072
H264.DigitalPTZ.3072X1728.Height=1728
H264.DigitalPTZ.3072X1728.MaxFPS=30000
H264.DigitalPTZ.3072X1728.DefaultFPS=30000
H264.DigitalPTZ.3072X1728.MaxCBRTargetBitrate=4000
H264.DigitalPTZ.3072X1728.MinCBRTargetBitrate=4000
H264.DigitalPTZ.3072X1728.DefaultCBRTargetBitrate=4000
H264.DigitalPTZ.1920X1080.Width=1920
H264.DigitalPTZ.1920X1080.Height=1080
H264.DigitalPTZ.1920X1080.MaxFPS=30000
H264.DigitalPTZ.1920X1080.DefaultFPS=30000
H264.DigitalPTZ.1920X1080.MaxCBRTargetBitrate=4000
H264.DigitalPTZ.1920X1080.MinCBRTargetBitrate=4000
H264.DigitalPTZ.1920X1080.DefaultCBRTargetBitrate=4000
H264.DigitalPTZ.1280X720.Width=1280
H264.DigitalPTZ.1280X720.Height=720
H264.DigitalPTZ.1280X720.MaxFPS=30000
H264.DigitalPTZ.1280X720.DefaultFPS=30000
H264.DigitalPTZ.1280X720.MaxCBRTargetBitrate=4000
```

```
H264.DigitalPTZ.1280X720.MinCBRTargetBitrate=4000
H264.DigitalPTZ.1280X720.DefaultCBRTargetBitrate=4000
Channel.0.EncodingType=H265
H265.IsDigitalPTZSupported=True
H265.Compression.DefaultCompressionLevel=16
H265.Compression.MinCompressionLevel=0
H265.Compression.MaxCompressionLevel=50
H265.General.3328X1872.Width=3328
H265.General.3328X1872.Height=1872
H265.General.3328X1872.MaxFPS=30000
H265.General.3328X1872.DefaultFPS=30000
H265.General.3328X1872.MaxCBRTargetBitrate=20480
H265.General.3328X1872.MinCBRTargetBitrate=2560
H265.General.3328X1872.DefaultCBRTargetBitrate=6144
H265.General.3328X1872.MaxVBRTargetBitrate=30720
H265.General.3328X1872.MinVBRTargetBitrate=2048
H265.General.3328X1872.DefaultVBRTargetBitrate=6144
H265.General.3328X1872.IsTruncated=Normal
H265.General.3328X1872.FrameLockMaxFPS=30000
H265.General.3072X1728.Width=3072
H265.General.3072X1728.Height=1728
H265.General.3072X1728.MaxFPS=30000
H265.General.3072X1728.DefaultFPS=30000
H265.General.3072X1728.MaxCBRTargetBitrate=20480
H265.General.3072X1728.MinCBRTargetBitrate=2560
H265.General.3072X1728.DefaultCBRTargetBitrate=6144
H265.General.3072X1728.MaxVBRTargetBitrate=30720
H265.General.3072X1728.MinVBRTargetBitrate=2048
H265.General.3072X1728.DefaultVBRTargetBitrate=6144
H265.General.3072X1728.IsTruncated=Normal
H265.General.3072X1728.FrameLockMaxFPS=30000
H265.General.2592X1944.Width=2592
H265.General.2592X1944.Height=1944
H265.General.2592X1944.MaxFPS=30000
H265.General.2592X1944.DefaultFPS=30000
H265.General.2592X1944.MaxCBRTargetBitrate=20480
H265.General.2592X1944.MinCBRTargetBitrate=2048
H265.General.2592X1944.DefaultCBRTargetBitrate=5120
H265.General.2592X1944.MaxVBRTargetBitrate=30720
H265.General.2592X1944.MinVBRTargetBitrate=1024
H265.General.2592X1944.DefaultVBRTargetBitrate=5120
```

```
H265.General.2592X1944.IsTruncated=Crop
H265.General.2592X1944.FrameLockMaxFPS=30000
H265.General.2688X1520.Width=2688
H265.General.2688X1520.Height=1520
H265.General.2688X1520.MaxFPS=30000
H265.General.2688X1520.DefaultFPS=30000
H265.General.2688X1520.MaxCBRTargetBitrate=20480
H265.General.2688X1520.MinCBRTargetBitrate=2048
H265.General.2688X1520.DefaultCBRTargetBitrate=5120
H265.General.2688X1520.MaxVBRTargetBitrate=30720
H265.General.2688X1520.MinVBRTargetBitrate=1024
H265.General.2688X1520.DefaultVBRTargetBitrate=5120
H265.General.2688X1520.IsTruncated=Crop
H265.General.2688X1520.FrameLockMaxFPS=30000
H265.General.1920X1080.Width=1920
H265.General.1920X1080.Height=1080
H265.General.1920X1080.MaxFPS=30000
H265.General.1920X1080.DefaultFPS=30000
H265.General.1920X1080.MaxCBRTargetBitrate=20480
H265.General.1920X1080.MinCBRTargetBitrate=1024
H265.General.1920X1080.DefaultCBRTargetBitrate=2048
H265.General.1920X1080.MaxVBRTargetBitrate=30720
H265.General.1920X1080.MinVBRTargetBitrate=1024
H265.General.1920X1080.DefaultVBRTargetBitrate=2048
H265.General.1920X1080.IsTruncated=Normal
H265.General.1920X1080.FrameLockMaxFPS=30000
H265.General.1600X1200.Width=1600
H265.General.1600X1200.Height=1200
H265.General.1600X1200.MaxFPS=30000
H265.General.1600X1200.DefaultFPS=30000
H265.General.1600X1200.MaxCBRTargetBitrate=20480
H265.General.1600X1200.MinCBRTargetBitrate=1024
H265.General.1600X1200.DefaultCBRTargetBitrate=2048
H265.General.1600X1200.MaxVBRTargetBitrate=30720
H265.General.1600X1200.MinVBRTargetBitrate=1024
H265.General.1600X1200.DefaultVBRTargetBitrate=2048
H265.General.1600X1200.IsTruncated=Crop
H265.General.1600X1200.FrameLockMaxFPS=30000
H265.General.1280X1024.Width=1280
H265.General.1280X1024.Height=1024
H265.General.1280X1024.MaxFPS=30000
```

```
H265.General.1280X1024.DefaultFPS=30000
H265.General.1280X1024.MaxCBRTargetBitrate=20480
H265.General.1280X1024.MinCBRTargetBitrate=1024
H265.General.1280X1024.DefaultCBRTargetBitrate=1536
H265.General.1280X1024.MaxVBRTargetBitrate=30720
H265.General.1280X1024.MinVBRTargetBitrate=1024
H265.General.1280X1024.DefaultVBRTargetBitrate=1536
H265.General.1280X1024.IsTruncated=Crop
H265.General.1280X1024.FrameLockMaxFPS=30000
H265.General.1280X960.Width=1280
H265.General.1280X960.Height=960
H265.General.1280X960.MaxFPS=30000
H265.General.1280X960.DefaultFPS=30000
H265.General.1280X960.MaxCBRTargetBitrate=20480
H265.General.1280X960.MinCBRTargetBitrate=1024
H265.General.1280X960.DefaultCBRTargetBitrate=1536
H265.General.1280X960.MaxVBRTargetBitrate=30720
H265.General.1280X960.MinVBRTargetBitrate=1024
H265.General.1280X960.DefaultVBRTargetBitrate=1536
H265.General.1280X960.IsTruncated=Crop
H265.General.1280X960.FrameLockMaxFPS=30000
H265.General.1280X720.Width=1280
H265.General.1280X720.Height=720
H265.General.1280X720.MaxFPS=30000
H265.General.1280X720.DefaultFPS=30000
H265.General.1280X720.MaxCBRTargetBitrate=20480
H265.General.1280X720.MinCBRTargetBitrate=1024
H265.General.1280X720.DefaultCBRTargetBitrate=1536
H265.General.1280X720.MaxVBRTargetBitrate=30720
H265.General.1280X720.MinVBRTargetBitrate=1024
H265.General.1280X720.DefaultVBRTargetBitrate=1536
H265.General.1280X720.IsTruncated=Normal
H265.General.1280X720.FrameLockMaxFPS=30000
H265.General.1024X768.Width=1024
H265.General.1024X768.Height=768
H265.General.1024X768.MaxFPS=30000
H265.General.1024X768.DefaultFPS=30000
H265.General.1024X768.MaxCBRTargetBitrate=20480
H265.General.1024X768.MinCBRTargetBitrate=1024
H265.General.1024X768.DefaultCBRTargetBitrate=1536
H265.General.1024X768.MaxVBRTargetBitrate=30720
```

```
H265.General.1024X768.MinVBRTargetBitrate=1024
H265.General.1024X768.DefaultVBRTargetBitrate=1536
H265.General.1024X768.IsTruncated=Crop
H265.General.1024X768.FrameLockMaxFPS=30000
H265.General.800X600.Width=800
H265.General.800X600.Height=600
H265.General.800X600.MaxFPS=30000
H265.General.800X600.DefaultFPS=30000
H265.General.800X600.MaxCBRTargetBitrate=20480
H265.General.800X600.MinCBRTargetBitrate=512
H265.General.800X600.DefaultCBRTargetBitrate=1024
H265.General.800X600.MaxVBRTargetBitrate=30720
H265.General.800X600.MinVBRTargetBitrate=512
H265.General.800X600.DefaultVBRTargetBitrate=1024
H265.General.800X600.IsTruncated=Crop
H265.General.800X600.FrameLockMaxFPS=30000
H265.General.800X448.Width=800
H265.General.800X448.Height=448
H265.General.800X448.MaxFPS=30000
H265.General.800X448.DefaultFPS=30000
H265.General.800X448.MaxCBRTargetBitrate=20480
H265.General.800X448.MinCBRTargetBitrate=512
H265.General.800X448.DefaultCBRTargetBitrate=1024
H265.General.800X448.MaxVBRTargetBitrate=30720
H265.General.800X448.MinVBRTargetBitrate=512
H265.General.800X448.DefaultVBRTargetBitrate=1024
H265.General.800X448.IsTruncated=Normal
H265.General.800X448.FrameLockMaxFPS=30000
H265.General.720X576.Width=720
H265.General.720X576.Height=576
H265.General.720X576.MaxFPS=30000
H265.General.720X576.DefaultFPS=30000
H265.General.720X576.MaxCBRTargetBitrate=20480
H265.General.720X576.MinCBRTargetBitrate=512
H265.General.720X576.DefaultCBRTargetBitrate=1024
H265.General.720X576.MaxVBRTargetBitrate=30720
H265.General.720X576.MinVBRTargetBitrate=512
H265.General.720X576.DefaultVBRTargetBitrate=1024
H265.General.720X576.IsTruncated=Crop
H265.General.720X576.FrameLockMaxFPS=30000
H265.General.720X480.Width=720
```

```
H265.General.720X480.Height=480
H265.General.720X480.MaxFPS=30000
H265.General.720X480.DefaultFPS=30000
H265.General.720X480.MaxCBRTargetBitrate=20480
H265.General.720X480.MinCBRTargetBitrate=512
H265.General.720X480.DefaultCBRTargetBitrate=1024
H265.General.720X480.MaxVBRTargetBitrate=30720
H265.General.720X480.MinVBRTargetBitrate=512
H265.General.720X480.DefaultVBRTargetBitrate=1024
H265.General.720X480.IsTruncated=Crop
H265.General.720X480.FrameLockMaxFPS=30000
H265.General.640X480.Width=640
H265.General.640X480.Height=480
H265.General.640X480.MaxFPS=30000
H265.General.640X480.DefaultFPS=30000
H265.General.640X480.MaxCBRTargetBitrate=20480
H265.General.640X480.MinCBRTargetBitrate=512
H265.General.640X480.DefaultCBRTargetBitrate=1024
H265.General.640X480.MaxVBRTargetBitrate=30720
H265.General.640X480.MinVBRTargetBitrate=512
H265.General.640X480.DefaultVBRTargetBitrate=1024
H265.General.640X480.IsTruncated=Crop
H265.General.640X480.FrameLockMaxFPS=30000
H265.General.640X360.Width=640
H265.General.640X360.Height=360
H265.General.640X360.MaxFPS=30000
H265.General.640X360.DefaultFPS=30000
H265.General.640X360.MaxCBRTargetBitrate=20480
H265.General.640X360.MinCBRTargetBitrate=512
H265.General.640X360.DefaultCBRTargetBitrate=1024
H265.General.640X360.MaxVBRTargetBitrate=30720
H265.General.640X360.MinVBRTargetBitrate=512
H265.General.640X360.DefaultVBRTargetBitrate=1024
H265.General.640X360.IsTruncated=Normal
H265.General.640X360.FrameLockMaxFPS=30000
H265.General.320X240.Width=320
H265.General.320X240.Height=240
H265.General.320X240.MaxFPS=30000
H265.General.320X240.DefaultFPS=30000
H265.General.320X240.MaxCBRTargetBitrate=20480
H265.General.320X240.MinCBRTargetBitrate=256
```

```
H265.General.320X240.DefaultCBRTargetBitrate=512
H265.General.320X240.MaxVBRTargetBitrate=30720
H265.General.320X240.MinVBRTargetBitrate=256
H265.General.320X240.DefaultVBRTargetBitrate=512
H265.General.320X240.IsTruncated=Crop
H265.General.320X240.FrameLockMaxFPS=30000
H265.Record.3328X1872.MaxFPS=30000
H265.Record.3328X1872.DefaultFPS=30000
H265.Record.3328X1872.MinCBRTargetBitrate=2560
H265.Record.3328X1872.MaxCBRTargetBitrate=6144
H265.Record.3328X1872.DefaultCBRTargetBitrate=5120
H265.Record.3328X1872.MinVBRTargetBitrate=2048
H265.Record.3328X1872.MaxVBRTargetBitrate=6144
H265.Record.3328X1872.DefaultVBRTargetBitrate=5120
H265.Record.3328X1872.FrameLockMaxFPS=30000
H265.Record.3072X1728.MaxFPS=30000
H265.Record.3072X1728.DefaultFPS=30000
H265.Record.3072X1728.MinCBRTargetBitrate=2560
H265.Record.3072X1728.MaxCBRTargetBitrate=6144
H265.Record.3072X1728.DefaultCBRTargetBitrate=5120
H265.Record.3072X1728.MinVBRTargetBitrate=2048
H265.Record.3072X1728.MaxVBRTargetBitrate=6144
H265.Record.3072X1728.DefaultVBRTargetBitrate=5120
H265.Record.3072X1728.FrameLockMaxFPS=30000
H265.Record.2592X1944.MaxFPS=30000
H265.Record.2592X1944.DefaultFPS=30000
H265.Record.2592X1944.MinCBRTargetBitrate=2048
H265.Record.2592X1944.MaxCBRTargetBitrate=6144
H265.Record.2592X1944.DefaultCBRTargetBitrate=5120
H265.Record.2592X1944.MinVBRTargetBitrate=1024
H265.Record.2592X1944.MaxVBRTargetBitrate=6144
H265.Record.2592X1944.DefaultVBRTargetBitrate=5120
H265.Record.2592X1944.FrameLockMaxFPS=30000
H265.Record.2688X1520.MaxFPS=30000
H265.Record.2688X1520.DefaultFPS=30000
H265.Record.2688X1520.MinCBRTargetBitrate=2048
H265.Record.2688X1520.MaxCBRTargetBitrate=6144
H265.Record.2688X1520.DefaultCBRTargetBitrate=5120
H265.Record.2688X1520.MinVBRTargetBitrate=1024
H265.Record.2688X1520.MaxVBRTargetBitrate=6144
H265.Record.2688X1520.DefaultVBRTargetBitrate=5120
```

```
H265.Record.2688X1520.FrameLockMaxFPS=30000
H265.Record.1920X1080.MaxFPS=30000
H265.Record.1920X1080.DefaultFPS=30000
H265.Record.1920X1080.MinCBRTargetBitrate=1024
H265.Record.1920X1080.MaxCBRTargetBitrate=6144
H265.Record.1920X1080.DefaultCBRTargetBitrate=5120
H265.Record.1920X1080.MinVBRTargetBitrate=1024
H265.Record.1920X1080.MaxVBRTargetBitrate=6144
H265.Record.1920X1080.DefaultVBRTargetBitrate=5120
H265.Record.1920X1080.FrameLockMaxFPS=30000
H265.Record.1600X1200.MaxFPS=30000
H265.Record.1600X1200.DefaultFPS=30000
H265.Record.1600X1200.MinCBRTargetBitrate=1024
H265.Record.1600X1200.MaxCBRTargetBitrate=6144
H265.Record.1600X1200.DefaultCBRTargetBitrate=5120
H265.Record.1600X1200.MinVBRTargetBitrate=1024
H265.Record.1600X1200.MaxVBRTargetBitrate=6144
H265.Record.1600X1200.DefaultVBRTargetBitrate=5120
H265.Record.1600X1200.FrameLockMaxFPS=30000
H265.Record.1280X1024.MaxFPS=30000
H265.Record.1280X1024.DefaultFPS=30000
H265.Record.1280X1024.MinCBRTargetBitrate=1024
H265.Record.1280X1024.MaxCBRTargetBitrate=6144
H265.Record.1280X1024.DefaultCBRTargetBitrate=5120
H265.Record.1280X1024.MinVBRTargetBitrate=1024
H265.Record.1280X1024.MaxVBRTargetBitrate=6144
H265.Record.1280X1024.DefaultVBRTargetBitrate=5120
H265.Record.1280X1024.FrameLockMaxFPS=30000
H265.Record.1280X960.MaxFPS=30000
H265.Record.1280X960.DefaultFPS=30000
H265.Record.1280X960.MinCBRTargetBitrate=1024
H265.Record.1280X960.MaxCBRTargetBitrate=6144
H265.Record.1280X960.DefaultCBRTargetBitrate=5120
H265.Record.1280X960.MinVBRTargetBitrate=1024
H265.Record.1280X960.MaxVBRTargetBitrate=6144
H265.Record.1280X960.DefaultVBRTargetBitrate=5120
H265.Record.1280X960.FrameLockMaxFPS=30000
H265.Record.1280X720.MaxFPS=30000
H265.Record.1280X720.DefaultFPS=30000
H265.Record.1280X720.MinCBRTargetBitrate=1024
H265.Record.1280X720.MaxCBRTargetBitrate=6144
```

```
H265.Record.1280X720.DefaultCBRTargetBitrate=5120
H265.Record.1280X720.MinVBRTargetBitrate=1024
H265.Record.1280X720.MaxVBRTargetBitrate=6144
H265.Record.1280X720.DefaultVBRTargetBitrate=5120
H265.Record.1280X720.FrameLockMaxFPS=30000
H265.Record.1024X768.MaxFPS=30000
H265.Record.1024X768.DefaultFPS=30000
H265.Record.1024X768.MinCBRTargetBitrate=1024
H265.Record.1024X768.MaxCBRTargetBitrate=6144
H265.Record.1024X768.DefaultCBRTargetBitrate=5120
H265.Record.1024X768.MinVBRTargetBitrate=1024
H265.Record.1024X768.MaxVBRTargetBitrate=6144
H265.Record.1024X768.DefaultVBRTargetBitrate=5120
H265.Record.1024X768.FrameLockMaxFPS=30000
H265.Record.800X600.MaxFPS=30000
H265.Record.800X600.DefaultFPS=30000
H265.Record.800X600.MinCBRTargetBitrate=512
H265.Record.800X600.MaxCBRTargetBitrate=6144
H265.Record.800X600.DefaultCBRTargetBitrate=5120
H265.Record.800X600.MinVBRTargetBitrate=512
H265.Record.800X600.MaxVBRTargetBitrate=6144
H265.Record.800X600.DefaultVBRTargetBitrate=5120
H265.Record.800X600.FrameLockMaxFPS=30000
H265.Record.800X448.MaxFPS=30000
H265.Record.800X448.DefaultFPS=30000
H265.Record.800X448.MinCBRTargetBitrate=512
H265.Record.800X448.MaxCBRTargetBitrate=6144
H265.Record.800X448.DefaultCBRTargetBitrate=5120
H265.Record.800X448.MinVBRTargetBitrate=512
H265.Record.800X448.MaxVBRTargetBitrate=6144
H265.Record.800X448.DefaultVBRTargetBitrate=5120
H265.Record.800X448.FrameLockMaxFPS=30000
H265.Record.720X576.MaxFPS=30000
H265.Record.720X576.DefaultFPS=30000
H265.Record.720X576.MinCBRTargetBitrate=512
H265.Record.720X576.MaxCBRTargetBitrate=6144
H265.Record.720X576.DefaultCBRTargetBitrate=5120
H265.Record.720X576.MinVBRTargetBitrate=512
H265.Record.720X576.MaxVBRTargetBitrate=6144
H265.Record.720X576.DefaultVBRTargetBitrate=5120
H265.Record.720X576.FrameLockMaxFPS=30000
```

```
H265.Record.720X480.MaxFPS=30000
H265.Record.720X480.DefaultFPS=30000
H265.Record.720X480.MinCBRTargetBitrate=512
H265.Record.720X480.MaxCBRTargetBitrate=6144
H265.Record.720X480.DefaultCBRTargetBitrate=5120
H265.Record.720X480.MinVBRTargetBitrate=512
H265.Record.720X480.MaxVBRTargetBitrate=6144
H265.Record.720X480.DefaultVBRTargetBitrate=5120
H265.Record.720X480.FrameLockMaxFPS=30000
H265.Record.640X480.MaxFPS=30000
H265.Record.640X480.DefaultFPS=30000
H265.Record.640X480.MinCBRTargetBitrate=512
H265.Record.640X480.MaxCBRTargetBitrate=6144
H265.Record.640X480.DefaultCBRTargetBitrate=5120
H265.Record.640X480.MinVBRTargetBitrate=512
H265.Record.640X480.MaxVBRTargetBitrate=6144
H265.Record.640X480.DefaultVBRTargetBitrate=5120
H265.Record.640X480.FrameLockMaxFPS=30000
H265.Record.640X360.MaxFPS=30000
H265.Record.640X360.DefaultFPS=30000
H265.Record.640X360.MinCBRTargetBitrate=512
H265.Record.640X360.MaxCBRTargetBitrate=6144
H265.Record.640X360.DefaultCBRTargetBitrate=5120
H265.Record.640X360.MinVBRTargetBitrate=512
H265.Record.640X360.MaxVBRTargetBitrate=6144
H265.Record.640X360.DefaultVBRTargetBitrate=5120
H265.Record.640X360.FrameLockMaxFPS=30000
H265.Record.320X240.MaxFPS=30000
H265.Record.320X240.DefaultFPS=30000
H265.Record.320X240.MinCBRTargetBitrate=256
H265.Record.320X240.MaxCBRTargetBitrate=6144
H265.Record.320X240.DefaultCBRTargetBitrate=5120
H265.Record.320X240.MinVBRTargetBitrate=256
H265.Record.320X240.MaxVBRTargetBitrate=6144
H265.Record.320X240.DefaultVBRTargetBitrate=5120
H265.Record.320X240.FrameLockMaxFPS=30000
H265.DigitalPTZ.3328X1872.Width=3328
H265.DigitalPTZ.3328X1872.Height=1872
H265.DigitalPTZ.3328X1872.MaxFPS=30000
H265.DigitalPTZ.3328X1872.DefaultFPS=30000
H265.DigitalPTZ.3328X1872.MaxCBRTargetBitrate=4000
```

```
H265.DigitalPTZ.3328X1872.MinCBRTargetBitrate=4000
H265.DigitalPTZ.3328X1872.DefaultCBRTargetBitrate=4000
H265.DigitalPTZ.3072X1728.Width=3072
H265.DigitalPTZ.3072X1728.Height=1728
H265.DigitalPTZ.3072X1728.MaxFPS=30000
H265.DigitalPTZ.3072X1728.DefaultFPS=30000
H265.DigitalPTZ.3072X1728.MaxCBRTargetBitrate=4000
H265.DigitalPTZ.3072X1728.MinCBRTargetBitrate=4000
H265.DigitalPTZ.3072X1728.DefaultCBRTargetBitrate=4000
H265.DigitalPTZ.1920X1080.Width=1920
H265.DigitalPTZ.1920X1080.Height=1080
H265.DigitalPTZ.1920X1080.MaxFPS=30000
H265.DigitalPTZ.1920X1080.DefaultFPS=30000
H265.DigitalPTZ.1920X1080.MaxCBRTargetBitrate=4000
H265.DigitalPTZ.1920X1080.MinCBRTargetBitrate=4000
H265.DigitalPTZ.1920X1080.DefaultCBRTargetBitrate=4000
H265.DigitalPTZ.1280X720.Width=1280
H265.DigitalPTZ.1280X720.Height=720
H265.DigitalPTZ.1280X720.MaxFPS=30000
H265.DigitalPTZ.1280X720.DefaultFPS=30000
H265.DigitalPTZ.1280X720.MaxCBRTargetBitrate=4000
H265.DigitalPTZ.1280X720.MinCBRTargetBitrate=4000
H265.DigitalPTZ.1280X720.DefaultCBRTargetBitrate=4000
```

JSON RESPONSE

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HTTP/1.0 200 OK
Content-type: application/json
<Body>
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{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "ViewModes": [
                {
                    "ViewMode": "Overview",
                    "Codecs": [
                        {
                            "EncodingType": "MJPEG",
                            "IsDigitalPTZSupported": false,
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                                    "General": [
                                        {
                                            "Width": 3328,
                                            "Height": 1872,
                                            "MaxFPS": 5000,
                                            "DefaultFPS": 5000,
                                            "MaxCBRTargetBitrate": 30720,
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                                            "MinVBRTargetBitrate": 4096,
                                            "DefaultVBRTargetBitrate": 12288,
                                            "IsTruncated": "Normal"
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                                            "MinVBRTargetBitrate": 4096,
                                            "DefaultVBRTargetBitrate": 12288,
                                            "IsTruncated": "Normal"
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                                            "MaxFPS": 5000,
                                            "DefaultFPS": 5000,
                                            "MaxCBRTargetBitrate": 30720,
                                            "MinCBRTargetBitrate": 3072,
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                                            "MinVBRTargetBitrate": 3072,
                                            "DefaultVBRTargetBitrate": 10240,
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                                        "DefaultFPS": 5000,
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                                        "MinCBRTargetBitrate": 3072,
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                                        "MinVBRTargetBitrate": 3072,
                                        "DefaultVBRTargetBitrate": 10240,
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                                        "Width": 1920,
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                                        "MinCBRTargetBitrate": 2048,
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                                        "MinVBRTargetBitrate": 2048,
                                        "DefaultVBRTargetBitrate": 10240,
                                        "IsTruncated": "Normal"
                                    },
                                    {
                                        "Width": 1600,
                                        "Height": 1200,
                                        "MaxFPS": 30000,
                                        "DefaultFPS": 15000,
                                        "MaxCBRTargetBitrate": 30720,
                                        "MinCBRTargetBitrate": 2048,
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                                        "DefaultVBRTargetBitrate": 10240,
                                        "IsTruncated": "Crop"
                                    },
                                    {
                                        "Width": 1280,
                                        "Height": 1024,
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                    "MaxFPS": 30000,
                    "DefaultFPS": 15000,
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                    "DefaultCBRTargetBitrate": 10240,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 1024,
                    "DefaultVBRTargetBitrate": 10240,
                    "IsTruncated": "Crop"
                },
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                    "Width": 1280,
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                    "DefaultFPS": 15000,
                    "MaxCBRTargetBitrate": 30720,
                    "MinCBRTargetBitrate": 1024,
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                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 1024,
                    "DefaultVBRTargetBitrate": 10240,
                    "IsTruncated": "Crop"
                },
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                    "Width": 1280,
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                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 1024,
                    "DefaultVBRTargetBitrate": 10240,
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                    "Width": 1024,
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```

```
                },
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                {
```

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                    "DefaultVBRTargetBitrate": 5120,
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                {
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                {
```

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                                    "MinCBRTargetBitrate": 4000,
                                    "DefaultCBRTargetBitrate": 4000
                                }
                            ]
                        }
                    ]
                }
            ]
        }
```

```
    ]
}
```

The following response example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videocodecinfo&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.Profile=1
Channel.1.ViewMode=Overview
Channel.1.EncodingType=MJPEG
MJPEG.IsDigitalPTZSupported=False
MJPEG.General.4000X3000.Width=4000
MJPEG.General.4000X3000.Height=3000
MJPEG.General.4000X3000.MaxFPS=2
MJPEG.General.4000X3000.MinUnknownTargetBitrate=5120
MJPEG.General.4000X3000.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.4000X3000.Width=4000
MJPEG.General.Overview.4000X3000.Height=3000
MJPEG.General.Overview.4000X3000.MaxFPS=2
MJPEG.General.Overview.4000X3000.MinUnknownTargetBitrate=5120
MJPEG.General.Overview.4000X3000.MaxUnknownTargetBitrate=6144
MJPEG.General.3840X2160.Width=3840
MJPEG.General.3840X2160.Height=2160
MJPEG.General.3840X2160.MaxFPS=2
MJPEG.General.3840X2160.MinUnknownTargetBitrate=4096
MJPEG.General.3840X2160.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.3840X2160.Width=3840
MJPEG.General.Overview.3840X2160.Height=2160
MJPEG.General.Overview.3840X2160.MaxFPS=2
MJPEG.General.Overview.3840X2160.MinUnknownTargetBitrate=4096
MJPEG.General.Overview.3840X2160.MaxUnknownTargetBitrate=6144
MJPEG.General.2592X1944.Width=2592
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MJPEG.General.2592X1944.Height=1944
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MJPEG.General.Overview.2592X1944.Width=2592
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MJPEG.General.Overview.2592X1944.MaxFPS=2
MJPEG.General.Overview.2592X1944.MinUnknownTargetBitrate=2048
MJPEG.General.Overview.2592X1944.MaxUnknownTargetBitrate=6144
MJPEG.General.2592X1464.Width=2592
MJPEG.General.2592X1464.Height=1464
MJPEG.General.2592X1464.MaxFPS=2
MJPEG.General.2592X1464.MinUnknownTargetBitrate=2048
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MJPEG.General.Overview.2592X1464.Height=1464
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MJPEG.General.Overview.2592X1464.MinUnknownTargetBitrate=2048
MJPEG.General.Overview.2592X1464.MaxUnknownTargetBitrate=6144
MJPEG.General.1920X1080.Width=1920
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MJPEG.General.1920X1080.MaxFPS=5
MJPEG.General.1920X1080.MinUnknownTargetBitrate=1024
MJPEG.General.1920X1080.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1920X1080.Width=1920
MJPEG.General.Overview.1920X1080.Height=1080
MJPEG.General.Overview.1920X1080.MaxFPS=5
MJPEG.General.Overview.1920X1080.MinUnknownTargetBitrate=1024
MJPEG.General.Overview.1920X1080.MaxUnknownTargetBitrate=6144
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MJPEG.General.1600X1200.Height=1200
MJPEG.General.1600X1200.MaxFPS=5
MJPEG.General.1600X1200.MinUnknownTargetBitrate=1024
MJPEG.General.1600X1200.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1600X1200.Width=1600
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MJPEG.General.Overview.1600X1200.MaxFPS=5
MJPEG.General.Overview.1600X1200.MinUnknownTargetBitrate=1024
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MJPEG.General.1280X1024.MaxFPS=5
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MJPEG.General.1280X960.Width=1280
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MJPEG.General.1280X960.MinUnknownTargetBitrate=1024
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MJPEG.General.1280X720.Width=1280
MJPEG.General.1280X720.Height=720
MJPEG.General.1280X720.MaxFPS=5
MJPEG.General.1280X720.MinUnknownTargetBitrate=1024
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MJPEG.General.Overview.1280X720.Width=1280
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MJPEG.General.1024X768.Width=1024
MJPEG.General.1024X768.Height=768
MJPEG.General.1024X768.MaxFPS=5
MJPEG.General.1024X768.MinUnknownTargetBitrate=1024
MJPEG.General.1024X768.MaxUnknownTargetBitrate=6144
MJPEG.General.Overview.1024X768.Width=1024
MJPEG.General.Overview.1024X768.Height=768
MJPEG.General.Overview.1024X768.MaxFPS=5
MJPEG.General.Overview.1024X768.MinUnknownTargetBitrate=1024
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MJPEG.General.800X448.MaxFPS=5
MJPEG.General.800X448.MinUnknownTargetBitrate=512
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MJPEG.General.720X480.MaxUnknownTargetBitrate=6144
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MJPEG.General.Overview.720X480.MaxFPS=5
MJPEG.General.Overview.720X480.MinUnknownTargetBitrate=512
MJPEG.General.Overview.720X480.MaxUnknownTargetBitrate=6144
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MJPEG.General.Overview.640X480.MaxUnknownTargetBitrate=6144
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MJPEG.General.640X360.MaxFPS=5
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MJPEG.General.640X360.MaxUnknownTargetBitrate=6144
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MJPEG.General.Overview.640X360.Height=360
MJPEG.General.Overview.640X360.MaxFPS=5
MJPEG.General.Overview.640X360.MinUnknownTargetBitrate=512
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MJPEG.General.320X240.MaxUnknownTargetBitrate=6144
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MJPEG.General.Overview.320X240.Height=240
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JSON RESPONSE

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Content-type: application/json
<Body>
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                                        "MaxFPS": 2,
                                        "MinUnknownTargetBitrate": 2048,
                                        "MaxUnknownTargetBitrate": 15360
                                    },
                                    {
                                        "Width": 2592,
                                        "Height": 1464,
                                        "MaxFPS": 2,
                                        "MinUnknownTargetBitrate": 2048,
                                        "MaxUnknownTargetBitrate": 15360
                                    },
                                    {
                                        "Width": 1920,
                                        "Height": 1080,
                                        "MaxFPS": 15,
                                        "MinUnknownTargetBitrate": 1024,
                                        "MaxUnknownTargetBitrate": 30720
                                    },
                                    {
                                        "Width": 1600,
                                        "Height": 1200,
                                        "MaxFPS": 15,
                                        "MinUnknownTargetBitrate": 1024,
                                        "MaxUnknownTargetBitrate": 30720
                                    },
                                    {
                                        "Width": 1280,
                                        "Height": 1024,
                                        "MaxFPS": 15,
                                        "MinUnknownTargetBitrate": 1024,
                                        "MaxUnknownTargetBitrate": 30720
                                    },
                                    {
                                        "Width": 1280,
                                        "Height": 960,
```

```
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 1024,
                    "Height": 768,
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 1024,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 800,
                    "Height": 600,
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 512,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 800,
                    "Height": 448,
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 512,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 720,
                    "Height": 576,
                    "MaxFPS": 15,
                    "MinUnknownTargetBitrate": 512,
                    "MaxUnknownTargetBitrate": 30720
                },
                {
                    "Width": 720,
```

```
                                            "Height": 480,
                                            "MaxFPS": 15,
                                            "MinUnknownTargetBitrate": 512,
                                            "MaxUnknownTargetBitrate": 30720
                                        },
                                        {
                                            "Width": 640,
                                            "Height": 480,
                                            "MaxFPS": 15,
                                            "MinUnknownTargetBitrate": 512,
                                            "MaxUnknownTargetBitrate": 30720
                                        },
                                        {
                                            "Width": 640,
                                            "Height": 360,
                                            "MaxFPS": 15,
                                            "MinUnknownTargetBitrate": 512,
                                            "MaxUnknownTargetBitrate": 30720
                                        },
                                        {
                                            "Width": 320,
                                            "Height": 240,
                                            "MaxFPS": 15,
                                            "MinUnknownTargetBitrate": 256,
                                            "MaxUnknownTargetBitrate": 30720
                                        }
                                    ]
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```


106.15.5. Examples

106.15.5.1. Getting all ‘general’ resolution information based on channel

The command without **Profile** returns all available channel-based general resolution information.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videocodecinfo&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=MJPEG
Channel.0.ViewMode=Overview
MJPEG.General.1920X1080.Width=1920
MJPEG.General.1920X1080.Height=1080
MJPEG.General.1920X1080.MaxFPS=2000
MJPEG.General.1920X1080.DefaultFPS=2000
MJPEG.General.1920X1080.MaxCBRTargetBitrate=10240
MJPEG.General.1920X1080.MinCBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultCBRTargetBitrate=2560
MJPEG.General.1920X1080.MaxVBRTargetBitrate=10240
MJPEG.General.1920X1080.MinVBRTargetBitrate=2048
MJPEG.General.1920X1080.DefaultVBRTargetBitrate=2560
MJPEG.General.1920X1080.IsTruncated=Normal
MJPEG.General.1280X720.Width=1280
MJPEG.General.1280X720.Height=720
MJPEG.General.1280X720.MaxFPS=2000
MJPEG.General.1280X720.DefaultFPS=2000
MJPEG.General.1280X720.MaxCBRTargetBitrate=10240
MJPEG.General.1280X720.MinCBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultCBRTargetBitrate=2048
MJPEG.General.1280X720.MaxVBRTargetBitrate=10240
MJPEG.General.1280X720.MinVBRTargetBitrate=1024
MJPEG.General.1280X720.DefaultVBRTargetBitrate=2048
MJPEG.General.1280X720.IsTruncated=Normal
MJPEG.General.928X576.Width=928
MJPEG.General.928X576.Height=576
MJPEG.General.928X576.MaxFPS=2000
MJPEG.General.928X576.DefaultFPS=2000
MJPEG.General.928X576.MaxCBRTargetBitrate=10240
MJPEG.General.928X576.MinCBRTargetBitrate=512
```

```
MJPEG.General.928X576.DefaultCBRTargetBitrate=1024
MJPEG.General.928X576.MaxVBRTargetBitrate=10240
MJPEG.General.928X576.MinVBRTargetBitrate=512
MJPEG.General.928X576.DefaultVBRTargetBitrate=1024
MJPEG.General.928X576.IsTruncated=Normal
MJPEG.General.704X576.Width=704
MJPEG.General.704X576.Height=576
MJPEG.General.704X576.MaxFPS=2000
MJPEG.General.704X576.DefaultFPS=2000
MJPEG.General.704X576.MaxCBRTargetBitrate=10240
MJPEG.General.704X576.MinCBRTargetBitrate=512
MJPEG.General.704X576.DefaultCBRTargetBitrate=1024
MJPEG.General.704X576.MaxVBRTargetBitrate=10240
MJPEG.General.704X576.MinVBRTargetBitrate=512
MJPEG.General.704X576.DefaultVBRTargetBitrate=1024
MJPEG.General.704X576.IsTruncated=Normal
MJPEG.General.928X288.Width=928
MJPEG.General.928X288.Height=288
MJPEG.General.928X288.MaxFPS=2000
MJPEG.General.928X288.DefaultFPS=2000
MJPEG.General.928X288.MaxCBRTargetBitrate=10240
MJPEG.General.928X288.MinCBRTargetBitrate=512
MJPEG.General.928X288.DefaultCBRTargetBitrate=1024
MJPEG.General.928X288.MaxVBRTargetBitrate=10240
MJPEG.General.928X288.MinVBRTargetBitrate=512
MJPEG.General.928X288.DefaultVBRTargetBitrate=1024
MJPEG.General.928X288.IsTruncated=Normal
MJPEG.General.704X288.Width=704
MJPEG.General.704X288.Height=288
MJPEG.General.704X288.MaxFPS=2000
MJPEG.General.704X288.DefaultFPS=2000
MJPEG.General.704X288.MaxCBRTargetBitrate=10240
MJPEG.General.704X288.MinCBRTargetBitrate=512
MJPEG.General.704X288.DefaultCBRTargetBitrate=1024
MJPEG.General.704X288.MaxVBRTargetBitrate=10240
MJPEG.General.704X288.MinVBRTargetBitrate=512
MJPEG.General.704X288.DefaultVBRTargetBitrate=1024
MJPEG.General.704X288.IsTruncated=Normal
MJPEG.General.352X288.Width=352
MJPEG.General.352X288.Height=288
MJPEG.General.352X288.MaxFPS=2000
```

```
MJPEG.General.352X288.DefaultFPS=2000
MJPEG.General.352X288.MaxCBRTargetBitrate=10240
MJPEG.General.352X288.MinCBRTargetBitrate=256
MJPEG.General.352X288.DefaultCBRTargetBitrate=512
MJPEG.General.352X288.MaxVBRTargetBitrate=10240
MJPEG.General.352X288.MinVBRTargetBitrate=256
MJPEG.General.352X288.DefaultVBRTargetBitrate=512
MJPEG.General.352X288.IsTruncated=Normal
MJPEG.Email.1920X1080.MaxFPS=2000
MJPEG.Email.1920X1080.DefaultFPS=2000
MJPEG.Email.1280X720.MaxFPS=2000
MJPEG.Email.1280X720.DefaultFPS=2000
MJPEG.Email.928X576.MaxFPS=2000
MJPEG.Email.928X576.DefaultFPS=2000
MJPEG.Email.704X576.MaxFPS=2000
MJPEG.Email.704X576.DefaultFPS=2000
MJPEG.Email.928X288.MaxFPS=2000
MJPEG.Email.928X288.DefaultFPS=2000
MJPEG.Email.704X288.MaxFPS=2000
MJPEG.Email.704X288.DefaultFPS=2000
MJPEG.Email.352X288.MaxFPS=2000
MJPEG.Email.352X288.DefaultFPS=2000
Channel.0.EncodingType=H264
H264.General.1920X1080.Width=1920
H264.General.1920X1080.Height=1080
H264.General.1920X1080.MaxFPS=25000
H264.General.1920X1080.DefaultFPS=25000
H264.General.1920X1080.MaxCBRTargetBitrate=10240
H264.General.1920X1080.MinCBRTargetBitrate=1024
H264.General.1920X1080.DefaultCBRTargetBitrate=2560
H264.General.1920X1080.MaxVBRTargetBitrate=15360
H264.General.1920X1080.MinVBRTargetBitrate=1536
H264.General.1920X1080.DefaultVBRTargetBitrate=2560
H264.General.1920X1080.IsTruncated=Normal
H264.General.1280X720.Width=1280
H264.General.1280X720.Height=720
H264.General.1280X720.MaxFPS=25000
H264.General.1280X720.DefaultFPS=25000
H264.General.1280X720.MaxCBRTargetBitrate=8192
H264.General.1280X720.MinCBRTargetBitrate=1024
H264.General.1280X720.DefaultCBRTargetBitrate=2048
```

```
H264.General.1280X720.MaxVBRTargetBitrate=12288
H264.General.1280X720.MinVBRTargetBitrate=1536
H264.General.1280X720.DefaultVBRTargetBitrate=2048
H264.General.1280X720.IsTruncated=Normal
H264.General.928X576.Width=928
H264.General.928X576.Height=576
H264.General.928X576.MaxFPS=25000
H264.General.928X576.DefaultFPS=25000
H264.General.928X576.MaxCBRTargetBitrate=6144
H264.General.928X576.MinCBRTargetBitrate=512
H264.General.928X576.DefaultCBRTargetBitrate=1024
H264.General.928X576.MaxVBRTargetBitrate=9216
H264.General.928X576.MinVBRTargetBitrate=512
H264.General.928X576.DefaultVBRTargetBitrate=1024
H264.General.928X576.IsTruncated=Normal
H264.General.704X576.Width=704
H264.General.704X576.Height=576
H264.General.704X576.MaxFPS=25000
H264.General.704X576.DefaultFPS=25000
H264.General.704X576.MaxCBRTargetBitrate=6144
H264.General.704X576.MinCBRTargetBitrate=512
H264.General.704X576.DefaultCBRTargetBitrate=1024
H264.General.704X576.MaxVBRTargetBitrate=9216
H264.General.704X576.MinVBRTargetBitrate=512
H264.General.704X576.DefaultVBRTargetBitrate=1024
H264.General.704X576.IsTruncated=Normal
H264.General.928X288.Width=928
H264.General.928X288.Height=288
H264.General.928X288.MaxFPS=25000
H264.General.928X288.DefaultFPS=25000
H264.General.928X288.MaxCBRTargetBitrate=6144
H264.General.928X288.MinCBRTargetBitrate=512
H264.General.928X288.DefaultCBRTargetBitrate=1024
H264.General.928X288.MaxVBRTargetBitrate=9216
H264.General.928X288.MinVBRTargetBitrate=512
H264.General.928X288.DefaultVBRTargetBitrate=1024
H264.General.928X288.IsTruncated=Normal
H264.General.704X288.Width=704
H264.General.704X288.Height=288
H264.General.704X288.MaxFPS=25000
H264.General.704X288.DefaultFPS=25000
```

```
H264.General.704X288.MaxCBRTargetBitrate=6144
H264.General.704X288.MinCBRTargetBitrate=512
H264.General.704X288.DefaultCBRTargetBitrate=1024
H264.General.704X288.MaxVBRTargetBitrate=9216
H264.General.704X288.MinVBRTargetBitrate=512
H264.General.704X288.DefaultVBRTargetBitrate=1024
H264.General.704X288.IsTruncated=Normal
H264.General.352X288.Width=352
H264.General.352X288.Height=288
H264.General.352X288.MaxFPS=25000
H264.General.352X288.DefaultFPS=25000
H264.General.352X288.MaxCBRTargetBitrate=2048
H264.General.352X288.MinCBRTargetBitrate=256
H264.General.352X288.DefaultCBRTargetBitrate=512
H264.General.352X288.MaxVBRTargetBitrate=3072
H264.General.352X288.MinVBRTargetBitrate=256
H264.General.352X288.DefaultVBRTargetBitrate=512
H264.General.352X288.IsTruncated=Normal
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "ViewModes": [
                {
                    "ViewMode": "Overview",
                    "Codecs": [
                        {
                            "EncodingType": "MJPEG",
                            "General": [
                                {
                                    "Width": 1920,
                                    "Height": 1080,
                                    "MaxFPS": 2000,
                                    "DefaultFPS": 2000,
```

```
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 2048,
                                    "DefaultCBRTargetBitrate": 2560,
                                    "MaxVBRTargetBitrate": 10240,
                                    "MinVBRTargetBitrate": 2048,
                                    "DefaultVBRTargetBitrate": 2560,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 1280,
                                    "Height": 720,
                                    "MaxFPS": 2000,
                                    "DefaultFPS": 2000,
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2048,
                                    "MaxVBRTargetBitrate": 10240,
                                    "MinVBRTargetBitrate": 1024,
                                    "DefaultVBRTargetBitrate": 2048,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 928,
                                    "Height": 576,
                                    "MaxFPS": 2000,
                                    "DefaultFPS": 2000,
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 10240,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 704,
                                    "Height": 576,
                                    "MaxFPS": 2000,
                                    "DefaultFPS": 2000,
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 512,
```

```
                                "DefaultCBRTargetBitrate": 1024,
                                "MaxVBRTargetBitrate": 10240,
                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 1024,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 928,
                                "Height": 288,
                                "MaxFPS": 2000,
                                "DefaultFPS": 2000,
                                "MaxCBRTargetBitrate": 10240,
                                "MinCBRTargetBitrate": 512,
                                "DefaultCBRTargetBitrate": 1024,
                                "MaxVBRTargetBitrate": 10240,
                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 1024,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 704,
                                "Height": 288,
                                "MaxFPS": 2000,
                                "DefaultFPS": 2000,
                                "MaxCBRTargetBitrate": 10240,
                                "MinCBRTargetBitrate": 512,
                                "DefaultCBRTargetBitrate": 1024,
                                "MaxVBRTargetBitrate": 10240,
                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 1024,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 352,
                                "Height": 288,
                                "MaxFPS": 2000,
                                "DefaultFPS": 2000,
                                "MaxCBRTargetBitrate": 10240,
                                "MinCBRTargetBitrate": 256,
                                "DefaultCBRTargetBitrate": 512,
                                "MaxVBRTargetBitrate": 10240,
```

```
                    "MinVBRTargetBitrate": 256,
                    "DefaultVBRTargetBitrate": 512,
                    "IsTruncated": "Normal"
                }
            ],
            "Email": [
                {
                    "Width": 1920,
                    "Height": 1080,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
                },
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
                },
                {
                    "Width": 928,
                    "Height": 576,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
                },
                {
                    "Width": 704,
                    "Height": 576,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
                },
                {
                    "Width": 928,
                    "Height": 288,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
                },
                {
                    "Width": 704,
                    "Height": 288,
                    "MaxFPS": 2000,
                    "DefaultFPS": 2000
```

```
                    },
                    {
                        "Width": 352,
                        "Height": 288,
                        "MaxFPS": 2000,
                        "DefaultFPS": 2000
                    }
                ]
            },
            {
                "EncodingType": "H264",
                "General": [
                    {
                        "Width": 1920,
                        "Height": 1080,
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
                        "MaxCBRTargetBitrate": 10240,
                        "MinCBRTargetBitrate": 1024,
                        "DefaultCBRTargetBitrate": 2560,
                        "MaxVBRTargetBitrate": 15360,
                        "MinVBRTargetBitrate": 1536,
                        "DefaultVBRTargetBitrate": 2560,
                        "IsTruncated": "Normal"
                    },
                    {
                        "Width": 1280,
                        "Height": 720,
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
                        "MaxCBRTargetBitrate": 8192,
                        "MinCBRTargetBitrate": 1024,
                        "DefaultCBRTargetBitrate": 2048,
                        "MaxVBRTargetBitrate": 12288,
                        "MinVBRTargetBitrate": 1536,
                        "DefaultVBRTargetBitrate": 2048,
                        "IsTruncated": "Normal"
                    },
                    {
                        "Width": 928,
                        "Height": 576,
```

```
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
                        "MaxCBRTargetBitrate": 6144,
                        "MinCBRTargetBitrate": 512,
                        "DefaultCBRTargetBitrate": 1024,
                        "MaxVBRTargetBitrate": 9216,
                        "MinVBRTargetBitrate": 512,
                        "DefaultVBRTargetBitrate": 1024,
                        "IsTruncated": "Normal"
                    },
                    {
                        "Width": 704,
                        "Height": 576,
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
                        "MaxCBRTargetBitrate": 6144,
                        "MinCBRTargetBitrate": 512,
                        "DefaultCBRTargetBitrate": 1024,
                        "MaxVBRTargetBitrate": 9216,
                        "MinVBRTargetBitrate": 512,
                        "DefaultVBRTargetBitrate": 1024,
                        "IsTruncated": "Normal"
                    },
                    {
                        "Width": 928,
                        "Height": 288,
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
                        "MaxCBRTargetBitrate": 6144,
                        "MinCBRTargetBitrate": 512,
                        "DefaultCBRTargetBitrate": 1024,
                        "MaxVBRTargetBitrate": 9216,
                        "MinVBRTargetBitrate": 512,
                        "DefaultVBRTargetBitrate": 1024,
                        "IsTruncated": "Normal"
                    },
                    {
                        "Width": 704,
                        "Height": 288,
                        "MaxFPS": 25000,
                        "DefaultFPS": 25000,
```

```
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 352,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 2048,
                                    "MinCBRTargetBitrate": 256,
                                    "DefaultCBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 3072,
                                    "MinVBRTargetBitrate": 256,
                                    "DefaultVBRTargetBitrate": 512,
                                    "IsTruncated": "Normal"
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```


106.15.5.2. Getting all ‘profiles’ resolution information based on Channel

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videocodecinfo&action=view&Channel=0&AllProfiles=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Channel.0.EncodingType=MJPEG
Channel.0.ViewMode=Overview
Profile.1.MJPEG.General.1920X1080.Width=1920
Profile.1.MJPEG.General.1920X1080.Height=1080
Profile.1.MJPEG.General.1920X1080.MaxFPS=2000
Profile.1.MJPEG.General.1920X1080.DefaultFPS=2000
Profile.1.MJPEG.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.1920X1080.MinCBRTargetBitrate=2048
Profile.1.MJPEG.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.1.MJPEG.General.1920X1080.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.1920X1080.MinVBRTargetBitrate=2048
Profile.1.MJPEG.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.1.MJPEG.General.1920X1080.IsTruncated=Normal
Profile.1.MJPEG.General.1280X720.Width=1280
Profile.1.MJPEG.General.1280X720.Height=720
Profile.1.MJPEG.General.1280X720.MaxFPS=2000
Profile.1.MJPEG.General.1280X720.DefaultFPS=2000
Profile.1.MJPEG.General.1280X720.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.1280X720.MinCBRTargetBitrate=1024
Profile.1.MJPEG.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.1.MJPEG.General.1280X720.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.1280X720.MinVBRTargetBitrate=1024
Profile.1.MJPEG.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.1.MJPEG.General.1280X720.IsTruncated=Normal
Profile.1.MJPEG.General.928X576.Width=928
Profile.1.MJPEG.General.928X576.Height=576
Profile.1.MJPEG.General.928X576.MaxFPS=2000
Profile.1.MJPEG.General.928X576.DefaultFPS=2000
Profile.1.MJPEG.General.928X576.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.928X576.MinCBRTargetBitrate=512
Profile.1.MJPEG.General.928X576.DefaultCBRTargetBitrate=1024
Profile.1.MJPEG.General.928X576.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.928X576.MinVBRTargetBitrate=512
Profile.1.MJPEG.General.928X576.DefaultVBRTargetBitrate=1024
Profile.1.MJPEG.General.928X576.IsTruncated=Normal
Profile.1.MJPEG.General.704X576.Width=704
Profile.1.MJPEG.General.704X576.Height=576
Profile.1.MJPEG.General.704X576.MaxFPS=2000
```

```
Profile.1.MJPEG.General.704X576.DefaultFPS=2000
Profile.1.MJPEG.General.704X576.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.704X576.MinCBRTargetBitrate=512
Profile.1.MJPEG.General.704X576.DefaultCBRTargetBitrate=1024
Profile.1.MJPEG.General.704X576.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.704X576.MinVBRTargetBitrate=512
Profile.1.MJPEG.General.704X576.DefaultVBRTargetBitrate=1024
Profile.1.MJPEG.General.704X576.IsTruncated=Normal
Profile.1.MJPEG.General.928X288.Width=928
Profile.1.MJPEG.General.928X288.Height=288
Profile.1.MJPEG.General.928X288.MaxFPS=2000
Profile.1.MJPEG.General.928X288.DefaultFPS=2000
Profile.1.MJPEG.General.928X288.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.928X288.MinCBRTargetBitrate=512
Profile.1.MJPEG.General.928X288.DefaultCBRTargetBitrate=1024
Profile.1.MJPEG.General.928X288.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.928X288.MinVBRTargetBitrate=512
Profile.1.MJPEG.General.928X288.DefaultVBRTargetBitrate=1024
Profile.1.MJPEG.General.928X288.IsTruncated=Normal
Profile.1.MJPEG.General.704X288.Width=704
Profile.1.MJPEG.General.704X288.Height=288
Profile.1.MJPEG.General.704X288.MaxFPS=2000
Profile.1.MJPEG.General.704X288.DefaultFPS=2000
Profile.1.MJPEG.General.704X288.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.704X288.MinCBRTargetBitrate=512
Profile.1.MJPEG.General.704X288.DefaultCBRTargetBitrate=1024
Profile.1.MJPEG.General.704X288.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.704X288.MinVBRTargetBitrate=512
Profile.1.MJPEG.General.704X288.DefaultVBRTargetBitrate=1024
Profile.1.MJPEG.General.704X288.IsTruncated=Normal
Profile.1.MJPEG.General.352X288.Width=352
Profile.1.MJPEG.General.352X288.Height=288
Profile.1.MJPEG.General.352X288.MaxFPS=2000
Profile.1.MJPEG.General.352X288.DefaultFPS=2000
Profile.1.MJPEG.General.352X288.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.352X288.MinCBRTargetBitrate=256
Profile.1.MJPEG.General.352X288.DefaultCBRTargetBitrate=512
Profile.1.MJPEG.General.352X288.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.352X288.MinVBRTargetBitrate=256
Profile.1.MJPEG.General.352X288.DefaultVBRTargetBitrate=512
Profile.1.MJPEG.General.352X288.IsTruncated=Normal
```

```
Profile.1.MJPEG.Email.1920X1080.MaxFPS=2000
Profile.1.MJPEG.Email.1920X1080.DefaultFPS=2000
Profile.1.MJPEG.Email.1280X720.MaxFPS=2000
Profile.1.MJPEG.Email.1280X720.DefaultFPS=2000
Profile.1.MJPEG.Email.928X576.MaxFPS=2000
Profile.1.MJPEG.Email.928X576.DefaultFPS=2000
Profile.1.MJPEG.Email.704X576.MaxFPS=2000
Profile.1.MJPEG.Email.704X576.DefaultFPS=2000
Profile.1.MJPEG.Email.928X288.MaxFPS=2000
Profile.1.MJPEG.Email.928X288.DefaultFPS=2000
Profile.1.MJPEG.Email.704X288.MaxFPS=2000
Profile.1.MJPEG.Email.704X288.DefaultFPS=2000
Profile.1.MJPEG.Email.352X288.MaxFPS=2000
Profile.1.MJPEG.Email.352X288.DefaultFPS=2000
Channel.0.EncodingType=H264
Profile.2.H264.General.1920X1080.Width=1920
Profile.2.H264.General.1920X1080.Height=1080
Profile.2.H264.General.1920X1080.MaxFPS=25000
Profile.2.H264.General.1920X1080.DefaultFPS=25000
Profile.2.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.2.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.2.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.2.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.2.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.2.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.2.H264.General.1920X1080.IsTruncated=Normal
Profile.2.H264.General.1280X720.Width=1280
Profile.2.H264.General.1280X720.Height=720
Profile.2.H264.General.1280X720.MaxFPS=25000
Profile.2.H264.General.1280X720.DefaultFPS=25000
Profile.2.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.2.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.2.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.2.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.2.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.2.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.2.H264.General.1280X720.IsTruncated=Normal
Profile.2.H264.General.928X576.Width=928
Profile.2.H264.General.928X576.Height=576
Profile.2.H264.General.928X576.MaxFPS=25000
Profile.2.H264.General.928X576.DefaultFPS=25000
```

```
Profile.2.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.2.H264.General.928X576.MinCBRTargetBitrate=512
Profile.2.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.2.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.2.H264.General.928X576.MinVBRTargetBitrate=512
Profile.2.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.2.H264.General.928X576.IsTruncated=Normal
Profile.2.H264.General.704X576.Width=704
Profile.2.H264.General.704X576.Height=576
Profile.2.H264.General.704X576.MaxFPS=25000
Profile.2.H264.General.704X576.DefaultFPS=25000
Profile.2.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.2.H264.General.704X576.MinCBRTargetBitrate=512
Profile.2.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.2.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.2.H264.General.704X576.MinVBRTargetBitrate=512
Profile.2.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.2.H264.General.704X576.IsTruncated=Normal
Profile.2.H264.General.928X288.Width=928
Profile.2.H264.General.928X288.Height=288
Profile.2.H264.General.928X288.MaxFPS=25000
Profile.2.H264.General.928X288.DefaultFPS=25000
Profile.2.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.2.H264.General.928X288.MinCBRTargetBitrate=512
Profile.2.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.2.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.2.H264.General.928X288.MinVBRTargetBitrate=512
Profile.2.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.2.H264.General.928X288.IsTruncated=Normal
Profile.2.H264.General.704X288.Width=704
Profile.2.H264.General.704X288.Height=288
Profile.2.H264.General.704X288.MaxFPS=25000
Profile.2.H264.General.704X288.DefaultFPS=25000
Profile.2.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.2.H264.General.704X288.MinCBRTargetBitrate=512
Profile.2.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.2.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.2.H264.General.704X288.MinVBRTargetBitrate=512
Profile.2.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.2.H264.General.704X288.IsTruncated=Normal
Profile.2.H264.General.352X288.Width=352
```

```
Profile.2.H264.General.352X288.Height=288
Profile.2.H264.General.352X288.MaxFPS=25000
Profile.2.H264.General.352X288.DefaultFPS=25000
Profile.2.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.2.H264.General.352X288.MinCBRTargetBitrate=256
Profile.2.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.2.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.2.H264.General.352X288.MinVBRTargetBitrate=256
Profile.2.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.2.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.3.H264.General.1920X1080.Width=1920
Profile.3.H264.General.1920X1080.Height=1080
Profile.3.H264.General.1920X1080.MaxFPS=25000
Profile.3.H264.General.1920X1080.DefaultFPS=25000
Profile.3.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.3.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.3.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.3.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.3.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.3.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.3.H264.General.1920X1080.IsTruncated=Normal
Profile.3.H264.General.1280X720.Width=1280
Profile.3.H264.General.1280X720.Height=720
Profile.3.H264.General.1280X720.MaxFPS=25000
Profile.3.H264.General.1280X720.DefaultFPS=25000
Profile.3.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.3.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.3.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.3.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.3.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.3.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.3.H264.General.1280X720.IsTruncated=Normal
Profile.3.H264.General.928X576.Width=928
Profile.3.H264.General.928X576.Height=576
Profile.3.H264.General.928X576.MaxFPS=25000
Profile.3.H264.General.928X576.DefaultFPS=25000
Profile.3.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.3.H264.General.928X576.MinCBRTargetBitrate=512
Profile.3.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.928X576.MaxVBRTargetBitrate=9216
```

```
Profile.3.H264.General.928X576.MinVBRTargetBitrate=512
Profile.3.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.928X576.IsTruncated=Normal
Profile.3.H264.General.704X576.Width=704
Profile.3.H264.General.704X576.Height=576
Profile.3.H264.General.704X576.MaxFPS=25000
Profile.3.H264.General.704X576.DefaultFPS=25000
Profile.3.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.3.H264.General.704X576.MinCBRTargetBitrate=512
Profile.3.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.3.H264.General.704X576.MinVBRTargetBitrate=512
Profile.3.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.704X576.IsTruncated=Normal
Profile.3.H264.General.928X288.Width=928
Profile.3.H264.General.928X288.Height=288
Profile.3.H264.General.928X288.MaxFPS=25000
Profile.3.H264.General.928X288.DefaultFPS=25000
Profile.3.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.3.H264.General.928X288.MinCBRTargetBitrate=512
Profile.3.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.3.H264.General.928X288.MinVBRTargetBitrate=512
Profile.3.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.928X288.IsTruncated=Normal
Profile.3.H264.General.704X288.Width=704
Profile.3.H264.General.704X288.Height=288
Profile.3.H264.General.704X288.MaxFPS=25000
Profile.3.H264.General.704X288.DefaultFPS=25000
Profile.3.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.3.H264.General.704X288.MinCBRTargetBitrate=512
Profile.3.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.3.H264.General.704X288.MinVBRTargetBitrate=512
Profile.3.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.704X288.IsTruncated=Normal
Profile.3.H264.General.352X288.Width=352
Profile.3.H264.General.352X288.Height=288
Profile.3.H264.General.352X288.MaxFPS=25000
Profile.3.H264.General.352X288.DefaultFPS=25000
Profile.3.H264.General.352X288.MaxCBRTargetBitrate=2048
```

```
Profile.3.H264.General.352X288.MinCBRTargetBitrate=256
Profile.3.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.3.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.3.H264.General.352X288.MinVBRTargetBitrate=256
Profile.3.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.3.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.4.H264.General.1920X1080.Width=1920
Profile.4.H264.General.1920X1080.Height=1080
Profile.4.H264.General.1920X1080.MaxFPS=25000
Profile.4.H264.General.1920X1080.DefaultFPS=25000
Profile.4.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.4.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.4.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.4.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.4.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.4.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.4.H264.General.1920X1080.IsTruncated=Normal
Profile.4.H264.General.1280X720.Width=1280
Profile.4.H264.General.1280X720.Height=720
Profile.4.H264.General.1280X720.MaxFPS=25000
Profile.4.H264.General.1280X720.DefaultFPS=25000
Profile.4.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.4.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.4.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.4.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.4.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.4.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.4.H264.General.1280X720.IsTruncated=Normal
Profile.4.H264.General.928X576.Width=928
Profile.4.H264.General.928X576.Height=576
Profile.4.H264.General.928X576.MaxFPS=25000
Profile.4.H264.General.928X576.DefaultFPS=25000
Profile.4.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.4.H264.General.928X576.MinCBRTargetBitrate=512
Profile.4.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.4.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.4.H264.General.928X576.MinVBRTargetBitrate=512
Profile.4.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.4.H264.General.928X576.IsTruncated=Normal
Profile.4.H264.General.704X576.Width=704
```

```
Profile.4.H264.General.704X576.Height=576
Profile.4.H264.General.704X576.MaxFPS=25000
Profile.4.H264.General.704X576.DefaultFPS=25000
Profile.4.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.4.H264.General.704X576.MinCBRTargetBitrate=512
Profile.4.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.4.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.4.H264.General.704X576.MinVBRTargetBitrate=512
Profile.4.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.4.H264.General.704X576.IsTruncated=Normal
Profile.4.H264.General.928X288.Width=928
Profile.4.H264.General.928X288.Height=288
Profile.4.H264.General.928X288.MaxFPS=25000
Profile.4.H264.General.928X288.DefaultFPS=25000
Profile.4.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.4.H264.General.928X288.MinCBRTargetBitrate=512
Profile.4.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.4.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.4.H264.General.928X288.MinVBRTargetBitrate=512
Profile.4.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.4.H264.General.928X288.IsTruncated=Normal
Profile.4.H264.General.704X288.Width=704
Profile.4.H264.General.704X288.Height=288
Profile.4.H264.General.704X288.MaxFPS=25000
Profile.4.H264.General.704X288.DefaultFPS=25000
Profile.4.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.4.H264.General.704X288.MinCBRTargetBitrate=512
Profile.4.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.4.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.4.H264.General.704X288.MinVBRTargetBitrate=512
Profile.4.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.4.H264.General.704X288.IsTruncated=Normal
Profile.4.H264.General.352X288.Width=352
Profile.4.H264.General.352X288.Height=288
Profile.4.H264.General.352X288.MaxFPS=25000
Profile.4.H264.General.352X288.DefaultFPS=25000
Profile.4.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.4.H264.General.352X288.MinCBRTargetBitrate=256
Profile.4.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.4.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.4.H264.General.352X288.MinVBRTargetBitrate=256
```

```
Profile.4.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.4.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.5.H264.General.1920X1080.Width=1920
Profile.5.H264.General.1920X1080.Height=1080
Profile.5.H264.General.1920X1080.MaxFPS=25000
Profile.5.H264.General.1920X1080.DefaultFPS=25000
Profile.5.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.5.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.5.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.5.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.5.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.5.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.5.H264.General.1920X1080.IsTruncated=Normal
Profile.5.H264.General.1280X720.Width=1280
Profile.5.H264.General.1280X720.Height=720
Profile.5.H264.General.1280X720.MaxFPS=25000
Profile.5.H264.General.1280X720.DefaultFPS=25000
Profile.5.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.5.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.5.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.5.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.5.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.5.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.5.H264.General.1280X720.IsTruncated=Normal
Profile.5.H264.General.928X576.Width=928
Profile.5.H264.General.928X576.Height=576
Profile.5.H264.General.928X576.MaxFPS=25000
Profile.5.H264.General.928X576.DefaultFPS=25000
Profile.5.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.5.H264.General.928X576.MinCBRTargetBitrate=512
Profile.5.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.5.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.5.H264.General.928X576.MinVBRTargetBitrate=512
Profile.5.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.5.H264.General.928X576.IsTruncated=Normal
Profile.5.H264.General.704X576.Width=704
Profile.5.H264.General.704X576.Height=576
Profile.5.H264.General.704X576.MaxFPS=25000
Profile.5.H264.General.704X576.DefaultFPS=25000
Profile.5.H264.General.704X576.MaxCBRTargetBitrate=6144
```

```
Profile.5.H264.General.704X576.MinCBRTargetBitrate=512
Profile.5.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.5.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.5.H264.General.704X576.MinVBRTargetBitrate=512
Profile.5.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.5.H264.General.704X576.IsTruncated=Normal
Profile.5.H264.General.928X288.Width=928
Profile.5.H264.General.928X288.Height=288
Profile.5.H264.General.928X288.MaxFPS=25000
Profile.5.H264.General.928X288.DefaultFPS=25000
Profile.5.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.5.H264.General.928X288.MinCBRTargetBitrate=512
Profile.5.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.5.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.5.H264.General.928X288.MinVBRTargetBitrate=512
Profile.5.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.5.H264.General.928X288.IsTruncated=Normal
Profile.5.H264.General.704X288.Width=704
Profile.5.H264.General.704X288.Height=288
Profile.5.H264.General.704X288.MaxFPS=25000
Profile.5.H264.General.704X288.DefaultFPS=25000
Profile.5.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.5.H264.General.704X288.MinCBRTargetBitrate=512
Profile.5.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.5.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.5.H264.General.704X288.MinVBRTargetBitrate=512
Profile.5.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.5.H264.General.704X288.IsTruncated=Normal
Profile.5.H264.General.352X288.Width=352
Profile.5.H264.General.352X288.Height=288
Profile.5.H264.General.352X288.MaxFPS=25000
Profile.5.H264.General.352X288.DefaultFPS=25000
Profile.5.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.5.H264.General.352X288.MinCBRTargetBitrate=256
Profile.5.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.5.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.5.H264.General.352X288.MinVBRTargetBitrate=256
Profile.5.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.5.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.6.H264.General.1920X1080.Width=1920
```

```
Profile.6.H264.General.1920X1080.Height=1080
Profile.6.H264.General.1920X1080.MaxFPS=25000
Profile.6.H264.General.1920X1080.DefaultFPS=25000
Profile.6.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.6.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.6.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.6.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.6.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.6.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.6.H264.General.1920X1080.IsTruncated=Normal
Profile.6.H264.General.1280X720.Width=1280
Profile.6.H264.General.1280X720.Height=720
Profile.6.H264.General.1280X720.MaxFPS=25000
Profile.6.H264.General.1280X720.DefaultFPS=25000
Profile.6.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.6.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.6.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.6.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.6.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.6.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.6.H264.General.1280X720.IsTruncated=Normal
Profile.6.H264.General.928X576.Width=928
Profile.6.H264.General.928X576.Height=576
Profile.6.H264.General.928X576.MaxFPS=25000
Profile.6.H264.General.928X576.DefaultFPS=25000
Profile.6.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.6.H264.General.928X576.MinCBRTargetBitrate=512
Profile.6.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.6.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.6.H264.General.928X576.MinVBRTargetBitrate=512
Profile.6.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.6.H264.General.928X576.IsTruncated=Normal
Profile.6.H264.General.704X576.Width=704
Profile.6.H264.General.704X576.Height=576
Profile.6.H264.General.704X576.MaxFPS=25000
Profile.6.H264.General.704X576.DefaultFPS=25000
Profile.6.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.6.H264.General.704X576.MinCBRTargetBitrate=512
Profile.6.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.6.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.6.H264.General.704X576.MinVBRTargetBitrate=512
```

```
Profile.6.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.6.H264.General.704X576.IsTruncated=Normal
Profile.6.H264.General.928X288.Width=928
Profile.6.H264.General.928X288.Height=288
Profile.6.H264.General.928X288.MaxFPS=25000
Profile.6.H264.General.928X288.DefaultFPS=25000
Profile.6.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.6.H264.General.928X288.MinCBRTargetBitrate=512
Profile.6.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.6.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.6.H264.General.928X288.MinVBRTargetBitrate=512
Profile.6.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.6.H264.General.928X288.IsTruncated=Normal
Profile.6.H264.General.704X288.Width=704
Profile.6.H264.General.704X288.Height=288
Profile.6.H264.General.704X288.MaxFPS=25000
Profile.6.H264.General.704X288.DefaultFPS=25000
Profile.6.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.6.H264.General.704X288.MinCBRTargetBitrate=512
Profile.6.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.6.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.6.H264.General.704X288.MinVBRTargetBitrate=512
Profile.6.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.6.H264.General.704X288.IsTruncated=Normal
Profile.6.H264.General.352X288.Width=352
Profile.6.H264.General.352X288.Height=288
Profile.6.H264.General.352X288.MaxFPS=25000
Profile.6.H264.General.352X288.DefaultFPS=25000
Profile.6.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.6.H264.General.352X288.MinCBRTargetBitrate=256
Profile.6.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.6.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.6.H264.General.352X288.MinVBRTargetBitrate=256
Profile.6.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.6.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.7.H264.General.1920X1080.Width=1920
Profile.7.H264.General.1920X1080.Height=1080
Profile.7.H264.General.1920X1080.MaxFPS=25000
Profile.7.H264.General.1920X1080.DefaultFPS=25000
Profile.7.H264.General.1920X1080.MaxCBRTargetBitrate=10240
```

```
Profile.7.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.7.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.7.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.7.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.7.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.7.H264.General.1920X1080.IsTruncated=Normal
Profile.7.H264.General.1280X720.Width=1280
Profile.7.H264.General.1280X720.Height=720
Profile.7.H264.General.1280X720.MaxFPS=25000
Profile.7.H264.General.1280X720.DefaultFPS=25000
Profile.7.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.7.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.7.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.7.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.7.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.7.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.7.H264.General.1280X720.IsTruncated=Normal
Profile.7.H264.General.928X576.Width=928
Profile.7.H264.General.928X576.Height=576
Profile.7.H264.General.928X576.MaxFPS=25000
Profile.7.H264.General.928X576.DefaultFPS=25000
Profile.7.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.7.H264.General.928X576.MinCBRTargetBitrate=512
Profile.7.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.7.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.7.H264.General.928X576.MinVBRTargetBitrate=512
Profile.7.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.7.H264.General.928X576.IsTruncated=Normal
Profile.7.H264.General.704X576.Width=704
Profile.7.H264.General.704X576.Height=576
Profile.7.H264.General.704X576.MaxFPS=25000
Profile.7.H264.General.704X576.DefaultFPS=25000
Profile.7.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.7.H264.General.704X576.MinCBRTargetBitrate=512
Profile.7.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.7.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.7.H264.General.704X576.MinVBRTargetBitrate=512
Profile.7.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.7.H264.General.704X576.IsTruncated=Normal
Profile.7.H264.General.928X288.Width=928
Profile.7.H264.General.928X288.Height=288
```

```
Profile.7.H264.General.928X288.MaxFPS=25000
Profile.7.H264.General.928X288.DefaultFPS=25000
Profile.7.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.7.H264.General.928X288.MinCBRTargetBitrate=512
Profile.7.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.7.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.7.H264.General.928X288.MinVBRTargetBitrate=512
Profile.7.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.7.H264.General.928X288.IsTruncated=Normal
Profile.7.H264.General.704X288.Width=704
Profile.7.H264.General.704X288.Height=288
Profile.7.H264.General.704X288.MaxFPS=25000
Profile.7.H264.General.704X288.DefaultFPS=25000
Profile.7.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.7.H264.General.704X288.MinCBRTargetBitrate=512
Profile.7.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.7.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.7.H264.General.704X288.MinVBRTargetBitrate=512
Profile.7.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.7.H264.General.704X288.IsTruncated=Normal
Profile.7.H264.General.352X288.Width=352
Profile.7.H264.General.352X288.Height=288
Profile.7.H264.General.352X288.MaxFPS=25000
Profile.7.H264.General.352X288.DefaultFPS=25000
Profile.7.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.7.H264.General.352X288.MinCBRTargetBitrate=256
Profile.7.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.7.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.7.H264.General.352X288.MinVBRTargetBitrate=256
Profile.7.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.7.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.8.H264.General.1920X1080.Width=1920
Profile.8.H264.General.1920X1080.Height=1080
Profile.8.H264.General.1920X1080.MaxFPS=25000
Profile.8.H264.General.1920X1080.DefaultFPS=25000
Profile.8.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.8.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.8.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.8.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.8.H264.General.1920X1080.MinVBRTargetBitrate=1536
```

```
Profile.8.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.8.H264.General.1920X1080.IsTruncated=Normal
Profile.8.H264.General.1280X720.Width=1280
Profile.8.H264.General.1280X720.Height=720
Profile.8.H264.General.1280X720.MaxFPS=25000
Profile.8.H264.General.1280X720.DefaultFPS=25000
Profile.8.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.8.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.8.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.8.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.8.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.8.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.8.H264.General.1280X720.IsTruncated=Normal
Profile.8.H264.General.928X576.Width=928
Profile.8.H264.General.928X576.Height=576
Profile.8.H264.General.928X576.MaxFPS=25000
Profile.8.H264.General.928X576.DefaultFPS=25000
Profile.8.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.8.H264.General.928X576.MinCBRTargetBitrate=512
Profile.8.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.8.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.8.H264.General.928X576.MinVBRTargetBitrate=512
Profile.8.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.8.H264.General.928X576.IsTruncated=Normal
Profile.8.H264.General.704X576.Width=704
Profile.8.H264.General.704X576.Height=576
Profile.8.H264.General.704X576.MaxFPS=25000
Profile.8.H264.General.704X576.DefaultFPS=25000
Profile.8.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.8.H264.General.704X576.MinCBRTargetBitrate=512
Profile.8.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.8.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.8.H264.General.704X576.MinVBRTargetBitrate=512
Profile.8.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.8.H264.General.704X576.IsTruncated=Normal
Profile.8.H264.General.928X288.Width=928
Profile.8.H264.General.928X288.Height=288
Profile.8.H264.General.928X288.MaxFPS=25000
Profile.8.H264.General.928X288.DefaultFPS=25000
Profile.8.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.8.H264.General.928X288.MinCBRTargetBitrate=512
```

```
Profile.8.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.8.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.8.H264.General.928X288.MinVBRTargetBitrate=512
Profile.8.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.8.H264.General.928X288.IsTruncated=Normal
Profile.8.H264.General.704X288.Width=704
Profile.8.H264.General.704X288.Height=288
Profile.8.H264.General.704X288.MaxFPS=25000
Profile.8.H264.General.704X288.DefaultFPS=25000
Profile.8.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.8.H264.General.704X288.MinCBRTargetBitrate=512
Profile.8.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.8.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.8.H264.General.704X288.MinVBRTargetBitrate=512
Profile.8.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.8.H264.General.704X288.IsTruncated=Normal
Profile.8.H264.General.352X288.Width=352
Profile.8.H264.General.352X288.Height=288
Profile.8.H264.General.352X288.MaxFPS=25000
Profile.8.H264.General.352X288.DefaultFPS=25000
Profile.8.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.8.H264.General.352X288.MinCBRTargetBitrate=256
Profile.8.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.8.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.8.H264.General.352X288.MinVBRTargetBitrate=256
Profile.8.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.8.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.9.H264.General.1920X1080.Width=1920
Profile.9.H264.General.1920X1080.Height=1080
Profile.9.H264.General.1920X1080.MaxFPS=25000
Profile.9.H264.General.1920X1080.DefaultFPS=25000
Profile.9.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.9.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.9.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.9.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.9.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.9.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.9.H264.General.1920X1080.IsTruncated=Normal
Profile.9.H264.General.1280X720.Width=1280
Profile.9.H264.General.1280X720.Height=720
```

```
Profile.9.H264.General.1280X720.MaxFPS=25000
Profile.9.H264.General.1280X720.DefaultFPS=25000
Profile.9.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.9.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.9.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.9.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.9.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.9.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.9.H264.General.1280X720.IsTruncated=Normal
Profile.9.H264.General.928X576.Width=928
Profile.9.H264.General.928X576.Height=576
Profile.9.H264.General.928X576.MaxFPS=25000
Profile.9.H264.General.928X576.DefaultFPS=25000
Profile.9.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.9.H264.General.928X576.MinCBRTargetBitrate=512
Profile.9.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.9.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.9.H264.General.928X576.MinVBRTargetBitrate=512
Profile.9.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.9.H264.General.928X576.IsTruncated=Normal
Profile.9.H264.General.704X576.Width=704
Profile.9.H264.General.704X576.Height=576
Profile.9.H264.General.704X576.MaxFPS=25000
Profile.9.H264.General.704X576.DefaultFPS=25000
Profile.9.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.9.H264.General.704X576.MinCBRTargetBitrate=512
Profile.9.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.9.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.9.H264.General.704X576.MinVBRTargetBitrate=512
Profile.9.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.9.H264.General.704X576.IsTruncated=Normal
Profile.9.H264.General.928X288.Width=928
Profile.9.H264.General.928X288.Height=288
Profile.9.H264.General.928X288.MaxFPS=25000
Profile.9.H264.General.928X288.DefaultFPS=25000
Profile.9.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.9.H264.General.928X288.MinCBRTargetBitrate=512
Profile.9.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.9.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.9.H264.General.928X288.MinVBRTargetBitrate=512
Profile.9.H264.General.928X288.DefaultVBRTargetBitrate=1024
```

```
Profile.9.H264.General.928X288.IsTruncated=Normal
Profile.9.H264.General.704X288.Width=704
Profile.9.H264.General.704X288.Height=288
Profile.9.H264.General.704X288.MaxFPS=25000
Profile.9.H264.General.704X288.DefaultFPS=25000
Profile.9.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.9.H264.General.704X288.MinCBRTargetBitrate=512
Profile.9.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.9.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.9.H264.General.704X288.MinVBRTargetBitrate=512
Profile.9.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.9.H264.General.704X288.IsTruncated=Normal
Profile.9.H264.General.352X288.Width=352
Profile.9.H264.General.352X288.Height=288
Profile.9.H264.General.352X288.MaxFPS=25000
Profile.9.H264.General.352X288.DefaultFPS=25000
Profile.9.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.9.H264.General.352X288.MinCBRTargetBitrate=256
Profile.9.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.9.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.9.H264.General.352X288.MinVBRTargetBitrate=256
Profile.9.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.9.H264.General.352X288.IsTruncated=Normal
Channel.0.EncodingType=H264
Profile.10.H264.General.1920X1080.Width=1920
Profile.10.H264.General.1920X1080.Height=1080
Profile.10.H264.General.1920X1080.MaxFPS=25000
Profile.10.H264.General.1920X1080.DefaultFPS=25000
Profile.10.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.10.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.10.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.10.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.10.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.10.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.10.H264.General.1920X1080.IsTruncated=Normal
Profile.10.H264.General.1280X720.Width=1280
Profile.10.H264.General.1280X720.Height=720
Profile.10.H264.General.1280X720.MaxFPS=25000
Profile.10.H264.General.1280X720.DefaultFPS=25000
Profile.10.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.10.H264.General.1280X720.MinCBRTargetBitrate=1024
```

```
Profile.10.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.10.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.10.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.10.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.10.H264.General.1280X720.IsTruncated=Normal
Profile.10.H264.General.928X576.Width=928
Profile.10.H264.General.928X576.Height=576
Profile.10.H264.General.928X576.MaxFPS=25000
Profile.10.H264.General.928X576.DefaultFPS=25000
Profile.10.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.10.H264.General.928X576.MinCBRTargetBitrate=512
Profile.10.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.10.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.10.H264.General.928X576.MinVBRTargetBitrate=512
Profile.10.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.10.H264.General.928X576.IsTruncated=Normal
Profile.10.H264.General.704X576.Width=704
Profile.10.H264.General.704X576.Height=576
Profile.10.H264.General.704X576.MaxFPS=25000
Profile.10.H264.General.704X576.DefaultFPS=25000
Profile.10.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.10.H264.General.704X576.MinCBRTargetBitrate=512
Profile.10.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.10.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.10.H264.General.704X576.MinVBRTargetBitrate=512
Profile.10.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.10.H264.General.704X576.IsTruncated=Normal
Profile.10.H264.General.928X288.Width=928
Profile.10.H264.General.928X288.Height=288
Profile.10.H264.General.928X288.MaxFPS=25000
Profile.10.H264.General.928X288.DefaultFPS=25000
Profile.10.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.10.H264.General.928X288.MinCBRTargetBitrate=512
Profile.10.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.10.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.10.H264.General.928X288.MinVBRTargetBitrate=512
Profile.10.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.10.H264.General.928X288.IsTruncated=Normal
Profile.10.H264.General.704X288.Width=704
Profile.10.H264.General.704X288.Height=288
Profile.10.H264.General.704X288.MaxFPS=25000
```

```
Profile.10.H264.General.704X288.DefaultFPS=25000
Profile.10.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.10.H264.General.704X288.MinCBRTargetBitrate=512
Profile.10.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.10.H264.General.704X288.MaxVBRTargetBitrate=9216
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JSON RESPONSE

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                                },
                                {
                                    "Width": 928,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 704,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 352,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 2048,
                                    "MinCBRTargetBitrate": 256,
```

```
                                "DefaultCBRTargetBitrate": 512,
                                "MaxVBRTargetBitrate": 3072,
                                "MinVBRTargetBitrate": 256,
                                "DefaultVBRTargetBitrate": 512,
                                "IsTruncated": "Normal"
                            }
                        ]
                    }
                ]
            },
            {
                "ViewMode": "Overview",
                "Profile": 8,
                "Codecs": [
                    {
                        "EncodingType": "H264",
                        "General": [
                            {
                                "Width": 1920,
                                "Height": 1080,
                                "MaxFPS": 25000,
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 10240,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 2560,
                                "MaxVBRTargetBitrate": 15360,
                                "MinVBRTargetBitrate": 1536,
                                "DefaultVBRTargetBitrate": 2560,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 1280,
                                "Height": 720,
                                "MaxFPS": 25000,
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 8192,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 2048,
                                "MaxVBRTargetBitrate": 12288,
                                "MinVBRTargetBitrate": 1536,
                                "DefaultVBRTargetBitrate": 2048,
```

```
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 928,
                    "Height": 576,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 704,
                    "Height": 576,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 928,
                    "Height": 288,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal"
                },
```

```
                        {
                            "Width": 704,
                            "Height": 288,
                            "MaxFPS": 25000,
                            "DefaultFPS": 25000,
                            "MaxCBRTargetBitrate": 6144,
                            "MinCBRTargetBitrate": 512,
                            "DefaultCBRTargetBitrate": 1024,
                            "MaxVBRTargetBitrate": 9216,
                            "MinVBRTargetBitrate": 512,
                            "DefaultVBRTargetBitrate": 1024,
                            "IsTruncated": "Normal"
                        },
                        {
                            "Width": 352,
                            "Height": 288,
                            "MaxFPS": 25000,
                            "DefaultFPS": 25000,
                            "MaxCBRTargetBitrate": 2048,
                            "MinCBRTargetBitrate": 256,
                            "DefaultCBRTargetBitrate": 512,
                            "MaxVBRTargetBitrate": 3072,
                            "MinVBRTargetBitrate": 256,
                            "DefaultVBRTargetBitrate": 512,
                            "IsTruncated": "Normal"
                        }
                    ]
                }
            ]
        },
        {
            "ViewMode": "Overview",
            "Profile": 9,
            "Codecs": [
                {
                    "EncodingType": "H264",
                    "General": [
                        {
                            "Width": 1920,
                            "Height": 1080,
                            "MaxFPS": 25000,
```

```
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 10240,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 2560,
                                "MaxVBRTargetBitrate": 15360,
                                "MinVBRTargetBitrate": 1536,
                                "DefaultVBRTargetBitrate": 2560,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 1280,
                                "Height": 720,
                                "MaxFPS": 25000,
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 8192,
                                "MinCBRTargetBitrate": 1024,
                                "DefaultCBRTargetBitrate": 2048,
                                "MaxVBRTargetBitrate": 12288,
                                "MinVBRTargetBitrate": 1536,
                                "DefaultVBRTargetBitrate": 2048,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 928,
                                "Height": 576,
                                "MaxFPS": 25000,
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 6144,
                                "MinCBRTargetBitrate": 512,
                                "DefaultCBRTargetBitrate": 1024,
                                "MaxVBRTargetBitrate": 9216,
                                "MinVBRTargetBitrate": 512,
                                "DefaultVBRTargetBitrate": 1024,
                                "IsTruncated": "Normal"
                            },
                            {
                                "Width": 704,
                                "Height": 576,
                                "MaxFPS": 25000,
                                "DefaultFPS": 25000,
                                "MaxCBRTargetBitrate": 6144,
```

```
                "MinCBRTargetBitrate": 512,
                "DefaultCBRTargetBitrate": 1024,
                "MaxVBRTargetBitrate": 9216,
                "MinVBRTargetBitrate": 512,
                "DefaultVBRTargetBitrate": 1024,
                "IsTruncated": "Normal"
            },
            {
                "Width": 928,
                "Height": 288,
                "MaxFPS": 25000,
                "DefaultFPS": 25000,
                "MaxCBRTargetBitrate": 6144,
                "MinCBRTargetBitrate": 512,
                "DefaultCBRTargetBitrate": 1024,
                "MaxVBRTargetBitrate": 9216,
                "MinVBRTargetBitrate": 512,
                "DefaultVBRTargetBitrate": 1024,
                "IsTruncated": "Normal"
            },
            {
                "Width": 704,
                "Height": 288,
                "MaxFPS": 25000,
                "DefaultFPS": 25000,
                "MaxCBRTargetBitrate": 6144,
                "MinCBRTargetBitrate": 512,
                "DefaultCBRTargetBitrate": 1024,
                "MaxVBRTargetBitrate": 9216,
                "MinVBRTargetBitrate": 512,
                "DefaultVBRTargetBitrate": 1024,
                "IsTruncated": "Normal"
            },
            {
                "Width": 352,
                "Height": 288,
                "MaxFPS": 25000,
                "DefaultFPS": 25000,
                "MaxCBRTargetBitrate": 2048,
                "MinCBRTargetBitrate": 256,
                "DefaultCBRTargetBitrate": 512,
```

```
                                    "MaxVBRTargetBitrate": 3072,
                                    "MinVBRTargetBitrate": 256,
                                    "DefaultVBRTargetBitrate": 512,
                                    "IsTruncated": "Normal"
                                }
                            ]
                        }
                    ]
                },
                {
                    "ViewMode": "Overview",
                    "Profile": 10,
                    "Codecs": [
                        {
                            "EncodingType": "H264",
                            "General": [
                                {
                                    "Width": 1920,
                                    "Height": 1080,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2560,
                                    "MaxVBRTargetBitrate": 15360,
                                    "MinVBRTargetBitrate": 1536,
                                    "DefaultVBRTargetBitrate": 2560,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 1280,
                                    "Height": 720,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 8192,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2048,
                                    "MaxVBRTargetBitrate": 12288,
                                    "MinVBRTargetBitrate": 1536,
                                    "DefaultVBRTargetBitrate": 2048,
                                    "IsTruncated": "Normal"
```

```
                                },
                                {
                                    "Width": 928,
                                    "Height": 576,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 704,
                                    "Height": 576,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 928,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
```

```
                                    "Width": 704,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 352,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 2048,
                                    "MinCBRTargetBitrate": 256,
                                    "DefaultCBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 3072,
                                    "MinVBRTargetBitrate": 256,
                                    "DefaultVBRTargetBitrate": 512,
                                    "IsTruncated": "Normal"
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```

106.15.5.3. Getting an ‘profile’ resolution information based on Profile number

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videocodecinfo&action=view&Channel=0&Profile=3
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=H264
Profile.3.H264.General.1920X1080.Width=1920
Profile.3.H264.General.1920X1080.Height=1080
Profile.3.H264.General.1920X1080.MaxFPS=25000
Profile.3.H264.General.1920X1080.DefaultFPS=25000
Profile.3.H264.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.3.H264.General.1920X1080.MinCBRTargetBitrate=1024
Profile.3.H264.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.3.H264.General.1920X1080.MaxVBRTargetBitrate=15360
Profile.3.H264.General.1920X1080.MinVBRTargetBitrate=1536
Profile.3.H264.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.3.H264.General.1920X1080.IsTruncated=Normal
Profile.3.H264.General.1280X720.Width=1280
Profile.3.H264.General.1280X720.Height=720
Profile.3.H264.General.1280X720.MaxFPS=25000
Profile.3.H264.General.1280X720.DefaultFPS=25000
Profile.3.H264.General.1280X720.MaxCBRTargetBitrate=8192
Profile.3.H264.General.1280X720.MinCBRTargetBitrate=1024
Profile.3.H264.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.3.H264.General.1280X720.MaxVBRTargetBitrate=12288
Profile.3.H264.General.1280X720.MinVBRTargetBitrate=1536
Profile.3.H264.General.1280X720.DefaultVBRTargetBitrate=2048
Profile.3.H264.General.1280X720.IsTruncated=Normal
Profile.3.H264.General.928X576.Width=928
Profile.3.H264.General.928X576.Height=576
Profile.3.H264.General.928X576.MaxFPS=25000
Profile.3.H264.General.928X576.DefaultFPS=25000
Profile.3.H264.General.928X576.MaxCBRTargetBitrate=6144
Profile.3.H264.General.928X576.MinCBRTargetBitrate=512
Profile.3.H264.General.928X576.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.928X576.MaxVBRTargetBitrate=9216
Profile.3.H264.General.928X576.MinVBRTargetBitrate=512
Profile.3.H264.General.928X576.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.928X576.IsTruncated=Normal
Profile.3.H264.General.704X576.Width=704
```

```
Profile.3.H264.General.704X576.Height=576
Profile.3.H264.General.704X576.MaxFPS=25000
Profile.3.H264.General.704X576.DefaultFPS=25000
Profile.3.H264.General.704X576.MaxCBRTargetBitrate=6144
Profile.3.H264.General.704X576.MinCBRTargetBitrate=512
Profile.3.H264.General.704X576.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.704X576.MaxVBRTargetBitrate=9216
Profile.3.H264.General.704X576.MinVBRTargetBitrate=512
Profile.3.H264.General.704X576.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.704X576.IsTruncated=Normal
Profile.3.H264.General.928X288.Width=928
Profile.3.H264.General.928X288.Height=288
Profile.3.H264.General.928X288.MaxFPS=25000
Profile.3.H264.General.928X288.DefaultFPS=25000
Profile.3.H264.General.928X288.MaxCBRTargetBitrate=6144
Profile.3.H264.General.928X288.MinCBRTargetBitrate=512
Profile.3.H264.General.928X288.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.928X288.MaxVBRTargetBitrate=9216
Profile.3.H264.General.928X288.MinVBRTargetBitrate=512
Profile.3.H264.General.928X288.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.928X288.IsTruncated=Normal
Profile.3.H264.General.704X288.Width=704
Profile.3.H264.General.704X288.Height=288
Profile.3.H264.General.704X288.MaxFPS=25000
Profile.3.H264.General.704X288.DefaultFPS=25000
Profile.3.H264.General.704X288.MaxCBRTargetBitrate=6144
Profile.3.H264.General.704X288.MinCBRTargetBitrate=512
Profile.3.H264.General.704X288.DefaultCBRTargetBitrate=1024
Profile.3.H264.General.704X288.MaxVBRTargetBitrate=9216
Profile.3.H264.General.704X288.MinVBRTargetBitrate=512
Profile.3.H264.General.704X288.DefaultVBRTargetBitrate=1024
Profile.3.H264.General.704X288.IsTruncated=Normal
Profile.3.H264.General.352X288.Width=352
Profile.3.H264.General.352X288.Height=288
Profile.3.H264.General.352X288.MaxFPS=25000
Profile.3.H264.General.352X288.DefaultFPS=25000
Profile.3.H264.General.352X288.MaxCBRTargetBitrate=2048
Profile.3.H264.General.352X288.MinCBRTargetBitrate=256
Profile.3.H264.General.352X288.DefaultCBRTargetBitrate=512
Profile.3.H264.General.352X288.MaxVBRTargetBitrate=3072
Profile.3.H264.General.352X288.MinVBRTargetBitrate=256
```

```
Profile.3.H264.General.352X288.DefaultVBRTargetBitrate=512
Profile.3.H264.General.352X288.IsTruncated=Normal
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "Profile": 3,
            "ViewModes": [
                {
                    "ViewMode": "Overview",
                    "Codecs": [
                        {
                            "EncodingType": "H264",
                            "General": [
                                {
                                    "Width": 1920,
                                    "Height": 1080,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 10240,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2560,
                                    "MaxVBRTargetBitrate": 15360,
                                    "MinVBRTargetBitrate": 1536,
                                    "DefaultVBRTargetBitrate": 2560,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 1280,
                                    "Height": 720,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 8192,
                                    "MinCBRTargetBitrate": 1024,
```

```
                    "DefaultCBRTargetBitrate": 2048,
                    "MaxVBRTargetBitrate": 12288,
                    "MinVBRTargetBitrate": 1536,
                    "DefaultVBRTargetBitrate": 2048,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 928,
                    "Height": 576,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 704,
                    "Height": 576,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal"
                },
                {
                    "Width": 928,
                    "Height": 288,
                    "MaxFPS": 25000,
                    "DefaultFPS": 25000,
                    "MaxCBRTargetBitrate": 6144,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 9216,
```

```
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 704,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 6144,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 9216,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal"
                                },
                                {
                                    "Width": 352,
                                    "Height": 288,
                                    "MaxFPS": 25000,
                                    "DefaultFPS": 25000,
                                    "MaxCBRTargetBitrate": 2048,
                                    "MinCBRTargetBitrate": 256,
                                    "DefaultCBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 3072,
                                    "MinVBRTargetBitrate": 256,
                                    "DefaultVBRTargetBitrate": 512,
                                    "IsTruncated": "Normal"
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```

106.16. WiseStream

106.16.1. Description

The **wisestream** submenu is used to configure and view the WiseStream settings.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes.cgi/attributes/Media/Support/WiseStream"

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User

106.16.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
wisestream&action=<value>[&<parameter>=<value>]
```

106.16.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video-profile policy settings
	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Mode	REQ, RES	<enum> Off, Low, Medium, High	WiseStream mode
	AISupportEnable	REQ, RES	<bool> True, False	AI is supported when wisestream is working **Note** **AISupportEnable** requires an AI engine

106.16.4. Examples

106.16.4.1. Getting WiseStream settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=wisestream&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Off
Channel.0.AISupportEnable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "WiseStream": [
        {
            "Channel": 0,
            "Mode": "Off",
            "AISupportEnable": false
        }
    ]
}
```

106.16.4.2. Setting the WiseStream mode

Setting WiseStream mode to **High**

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=wisestream&action=set&Mode=High
```

106.17. Camera Password Change

106.17.1. Description

The **camerapasswordchange** submenu requests the password change for all the connected cameras.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
control	User

106.17.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
camerapasswordchange&action=view
```

106.17.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	IsPasswordChanging	REQ	<bool>	Password change status
control	Mode	REQ	<enum> Start, Cancel	Mode to start or cancel the request
	NewPassword	REQ	<string>	Password to apply **NewPassword** is mandatory when **Mode** is Start.
	IsPasswordEncrypted	REQ	<bool> True, False	If set to True, password is encrypted using an RSA key and sent as POST content.
	Channel	RES	<int>	Channel number
	Model	RES	<string>	Model name
	Address	RES	<string>	IP address or URL

Action	Parameters	Request/ Response	Type/ Value	Description
	Status	RES	<enum> Success, Fail, Connection Fail, CameraBlock, NoResponse, AuthFail	Result status

106.17.4. Examples

106.17.4.1. Getting status of password change request

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=camerapasswordchange&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
IsPasswordChanging=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "IsPasswordChanging": false
}
```

106.17.4.2. Setting the new password

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=camerapasswordchange&action=control&Mode=Start&NewPassword={password}
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Model=QNO-6082R
Channel.0.Address=192.168.75.231
Channel.0.Status=ConnectionFail
Channel.1.Model=PNM-9322VQP
Channel.1.Address=192.168.75.221
Channel.1.Status=Success
Channel.2.Model=PNO-9080R
Channel.2.Address=192.168.75.207
Channel.2.Status=Fail
Channel.3.Model=PNO-9080RVM
Channel.3.Address=192.168.75.176
Channel.3.Status=Success
Channel.4.Model=SNP-6320RH
Channel.4.Address=192.168.75.75
Channel.4.Status=Success
Channel.5.Model=PNF-9010R
Channel.5.Address=192.168.75.40
Channel.5.Status=Success
Channel.6.Model=SNO-L6083R
Channel.6.Address=192.168.75.33
Channel.6.Status=Success
Channel.7.Model=SNB-6004
Channel.7.Address=192.168.75.2
Channel.7.Status=ConnectionFail
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "PasswordChangeStatus": [
        {
            "Channel": 0,
            "Model": "QNO-6082R",
            "Address": "192.168.75.231",
            "Status": "ConnectionFail"
        },
        {
            "Channel": 1,
            "Model": "PNM-9322VQP",
            "Address": "192.168.75.221",
            "Status": "Success"
        },
        {
            "Channel": 2,
            "Model": "PNO-9080R",
            "Address": "192.168.75.207",
            "Status": "Fail"
        },
        {
            "Channel": 3,
            "Model": "PNO-9080RVM",
            "Address": "192.168.75.176",
            "Status": "Success"
        },
        {
            "Channel": 4,
            "Model": "SNP-6320RH",
            "Address": "192.168.75.75",
            "Status": "Success"
        },
        {
            "Channel": 5,
            "Model": "PNF-9010R",
            "Address": "192.168.75.40",
            "Status": "Success"
        },
```

```
        {
            "Channel": 6,
            "Model": "SNO-L6083R",
            "Address": "192.168.75.33",
            "Status": "Success"
        },
        {
            "Channel": 7,
            "Model": "SNB-6004",
            "Address": "192.168.75.2",
            "Status": "ConnectionFail"
        }
    ]
}
```

106.17.4.3. Cancel the password change request

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=camerapasswordchange&action=control&Mode=Cancel
```

106.18. Media Options

106.18.1. Description

The **mediaoptions** submenu is used to view media options available in the device.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

106.18.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
mediaoptions&action=<value>[&<parameter>=<value>...]
```

106.18.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the media options like crop ratio, minimum and maximum crop resolution supported by the device.
	Channel	REQ, RES	<csv>	Channel
	ViewModeType	RES	<enum> Overview, 360Panorama, Panorama, DoublePanorama, QuadView, SingleView, OneOverviewAndTripleView, OneOverviewAndOctaView, CropView, LeftHalfView, RightHalfView, QuadView.# , QuadView.#	View mode type
	CropRatio	RES	<enum> Manual, 16:9, 4:3	Crop ratio
	MinCropResolution	RES	<string>	Minimum crop resolution
	MaxCropResolution	RES	<string>	Maximum crop resolution

106.18.4. Examples

106.18.4.1. Getting media options

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=mediaoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MediaOptions": [
```

```
        {
            "Channel": 0,
            "ViewModes": [
                {
                    "ViewModeType": "Overview",
                    "ResolutionsByCropRatio": [
                        {
                            "CropRatio": "16:9",
                            "MinCropResolution": "640x360",
                            "MaxCropResolution": "1280x720"
                        },
                        {
                            "CropRatio": "4:3",
                            "MinCropResolution": "320x240",
                            "MaxCropResolution": "1280x960"
                        },
                        {
                            "CropRatio": "Manual",
                            "MinCropResolution": "320x240",
                            "MaxCropResolution": "1280x1024"
                        }
                    ]
                }
            ]
        }
    ]
}
```

106.19. Synchronization Point Settings

106.19.1. Description

The **setsynchronizationpoint** submenu requests the intra frame for the given profile.

Access level

Action	Camera	NVR	Encoder
control	User	User	User

106.19.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
```

```
setsynchronizationpoint&action=control[&<parameter>=<value>...]
```

106.19.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Profile	REQ	<int>	Profile index

106.19.4. Examples

106.19.4.1. Requesting Intra frame for Channel 0, Profile 2

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=setsynchronizationpoint&action=control&Channel=0&Prof
ile=2
```

106.20. Video Source Options

106.20.1. Description

The **videosourceoptions** submenu views possible video source settings. Clients can identify each channel's video type or sensor size using this submenu.

NOTE This chapter applies to the encoder only.

Access level

Action	Camera	NVR	Encoder
view	Admin		Admin

106.20.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videosourceoptions&action=<value>[&<parameter>=<value>]
```

106.20.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the video source options settings.
	Channel	REQ, RES	<csv>	Channel ID
	VideoType	RES	<enum> NTSC, PAL, AHD, TVI, CVI, CVBS	Video type The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	VideoType.<VideoType enum>.SensorCaptureSize	RES	<csv> 2M, 4M, 5M, SD, 720P, 2M_double_shutter	Capture size The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	VideoType.<Video Type enum>.SensorCaptureSize.<Sensor Capture size>.SensorCaptureFrameRate	RES	<csv> 15, 20, 25, 30, 50, 60	Capture frame rate The value may vary depending on the model. Please check the device attributes using attributes.cgi.
	AvailableSensorCaptureFrameRateForAISupport	RES	<csv> 15, 20, 25, 30, 50, 60	Available lists of configurable SensorCaptureFrameRate if the app is using the AI engine. The value may vary depending on the model. Please check the device attributes using attributes.cgi.

106.20.4. Examples

106.20.4.1. Getting video source settings

The following example is for encoder only

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videosourceoptions&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.VideoType=AHD
Channel.0.VideoType.AHD.SensorCaptureSize=4M,2M
Channel.0.VideoType=TVI
Channel.0.VideoType.TVI.SensorCaptureSize=2M
Channel.0.VideoType=CVI
Channel.0.VideoType.CVI.SensorCaptureSize=2M
Channel.0.VideoType=CVBS
Channel.0.VideoType.CVBS.SensorCaptureSize=SD
…
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoOptions": [
        {
            "Channel": 0,
            "VideoTypes": [
                {
                    "VideoType": "AHD",
                    "SensorCaptureSize": [
                        "4M",
                        "2M"
                    ]
                },
                {
                    "VideoType": "TVI",
                    "SensorCaptureSize": [
                        "2M"
                    ]
                },
                {
                    "VideoType": "CVI",
                    "SensorCaptureSize": [
                        "2M"
                    ]
                },
```

```
                {
                    "VideoType": "CVBS",
                    "SensorCaptureSize": [
                        "SD"
                    ]
                }
            ]
        }
        ...
    ]
}
```

106.20.4.2. Getting video source settings

The following example is response of 2-channel camera

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videosourceoptions&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.VideoType=PAL
Channel.0.VideoType.PAL.SensorCaptureSize=32M,24M,15M,8M
Channel.0.VideoType.PAL.SensorCaptureSize.32M.SensorCaptureFrameRate=15
Channel.0.VideoType.PAL.SensorCaptureSize.24M.SensorCaptureFrameRate=20
Channel.0.VideoType.PAL.SensorCaptureSize.15M.SensorCaptureFrameRate=30
Channel.0.VideoType.PAL.SensorCaptureSize.8M.SensorCaptureFrameRate=60
Channel.1.VideoType=PAL
Channel.1.VideoType.PAL.SensorCaptureSize=2M
Channel.1.VideoType.PAL.SensorCaptureSize.2M.SensorCaptureFrameRate=60,30,20
,15
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "VideoOptions": [
        {
            "Channel": 0,
            "VideoTypes": [
                {
                    "VideoType": "PAL",
                    "SensorCaptureSize": [
                        "32M",
                        "24M",
                        "15M",
                        "8M"
                    ],
                    "FrameRates": [
                        {
                            "SensorCaptureSize": "32M",
                            "SensorCaptureFrameRate": [
                                "15"
                            ]
                        },
                        {
                            "SensorCaptureSize": "24M",
                            "SensorCaptureFrameRate": [
                                "20"
                            ]
                        },
                        {
                            "SensorCaptureSize": "15M",
                            "SensorCaptureFrameRate": [
                                "30"
                            ]
                        },
                        {
                            "SensorCaptureSize": "8M",
                            "SensorCaptureFrameRate": [
                                "60"
                            ]
                        }
```

```
                    ]
                }
            ],
            "AvailableSensorCaptureFrameRateForAISupport": [
                15
            ]
        },
        {
            "Channel": 1,
            "VideoTypes": [
                {
                    "VideoType": "PAL",
                    "SensorCaptureSize": [
                        "2M"
                    ],
                    "FrameRates": [
                        {
                            "SensorCaptureSize": "2M",
                            "SensorCaptureFrameRate": [
                                "60",
                                "30",
                                "20",
                                "15"
                            ]
                        }
                    ]
                }
            ],
            "AvailableSensorCaptureFrameRateForAISupport": [
                15
            ]
        }
    ]
}
```


106.21. Camera Register

106.21.1. Description

The **cameraregister** submenu can view, add, update, and remove network camera settings from NVR and encoders.

Access level

Action	Camera	NVR	Decoder
view	-	User	User
add/update		User	User
remove		User	User

106.21.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
cameraregister&action=<value>[&<parameter>=<value>]
```

106.21.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	DataRate	RES	float	In Mbps
	CPUUsage	RES	float	In percentage
	Status	RES	<enum> Success, ConnectFail, AuthFail, MaxUserLimit, InvalidVideoFormat, AccBlock, Disconnected	Registered network camera status
	Model	RES	<string>	Model name
	CamChannel	RES	<int>	Receive a request from the hta. Current set max is 16. MULTI-DIRECTION CAMERA ONLY
	IsBypassSupported	RES	<bool> True, False	NVR supports bypass CGI if this parameter is set to True.
add/update	Channel	REQ, RES		Read only
	Protocol	REQ, RES	<enum> SAMSUNG, ONVIF, RTSP	Read only

Action	Parameters	Request/ Response	Type/ Value	Description
	UserID	REQ, RES	<string>	Camera user ID
	UserPassword	REQ	<string>	Camera user password
	IsPasswordEncrypted	REQ, RES	<bool> True, False	When password encryption is used enable this.
	AddressType	REQ, RES	<enum> IPv4, IPv6, SamsungDDNS, URL, MAC	Cameras address type
	IPAddress	REQ, RES	<string>	IP address If AddressType=IPv4,IPv6
	DevicePort	REQ, RES	<int>	SVNP Port number This parameter is valid only when **Protocol** is set to SAMSUNG and **AddressType** is NOT set to SamsungDDNS.
	HTTPPort	REQ, RES	<int>	HTTP port number of camera. This parameter is valid only when **Protocol** is set to SAMSUNG and **AddressType** is NOT set to SamsungDDNS; or, **Protocol** is set to ONVIF and **AddressType** is NOT set to URL.
	AuthenticationType	REQ, RES	<enum> UserToken, Digest, CertificateBased	Authentication type This parameter is valid only when **Protocol** is set to ONVIF.
	StreamingMode	REQ, RES	<enum> TCP, UDP, HTTP, HTTPS	Streaming mode This parameter is valid only when **Protocol** is set to ONVIF, or RTSP. The value HTTPS is only available only when **Protocol** is set to RTSP.
	MACAddress	REQ, RES	<string>	MAC address This parameter is valid only when **AddressType** is set to MAC.

Action	Parameters	Request/ Response	Type/ Value	Description
	URL	REQ, RES	<string>	URL This parameter is valid only when **AddressType** is set to SamsungDDNS, or URL.
	CamChannel	REQ, RES	<int>	When registering camera is a multidirectional camera.Refers to the channel number of multidirectional camera.
	VideoState	REQ, RES	<enum> On, Off	Changes the video streaming state for a channel.
	AudioState	REQ, RES	<enum> On, Off	Changes the audio streaming state for a channel.
remove	Channel	REQ	<int>	Remove camera channel

106.21.4. Examples

106.21.4.1. Getting registerd camera info

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=cameraregister&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "RegisteredCameras": [
        {
            "Channel": 0,
            "Model": "XNB-8000",
            "UserID": "admin",
            "Protocol": "SAMSUNG",
            "AddressType": "IPv4",
            "IPAddress": "192.168.71.113",
            "DevicePort": 4520,
            "HTTPPort": 80,
```

```
            "Status": "Success",
            "DataRate": 1.204000,
            "CPUUsage": 0.200000
        },
        {
            "Channel": 1,
            "Model": "XNP-6120H",
            "UserID": "admin",
            "Protocol": "SAMSUNG",
            "AddressType": "IPv4",
            "IPAddress": "192.168.71.98",
            "DevicePort": 4520,
            "HTTPPort": 80,
            "Status": "Success",
            "DataRate": 1.982000,
            "CPUUsage": 0.300000
        }
    ]
}
```

106.21.4.2. Adding camera to channel 16

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=cameraregister&action=add&Channel=16&Protocol=SAMSUNG&UserID=admin&UserPassword={password}&IsPasswordEncrypted=False&AddressType=Ipv4&IPAddress=[CAM-IP]
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.21.4.3. Adding multi-channel camera's channel 1 to NVR channel 16

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=cameraregister&action=add&Channel=16&Protocol=SAMSUNG
&UserID=admin&UserPassword={password}&IsPasswordEncrypted=False&AddressType=
Ipv4&IPAddress=[CAM-IP]&CamChannel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.21.4.4. Updating camera IP in channel 16

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=cameraregister&action=update&Channel=16&
Protocol=SAMSUNG&UserID=admin&UserPassword={password}&IsPasswordEncrypted=Fa
lse& AddressType=Ipv4&IPAddress=[CAM-IP]
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.21.4.5. Removing camera IP from channel 16

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=cameraregister&action=remove&Channel=16
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.22. Auto Register

106.22.1. Description

The **autoregister** submenu registers a network camera to NVR sequentially.

Access level

Action	Camera	NVR	Decoder
view	—	User	User
control		User	User

106.22.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
autoregister&action=<value>[&<parameter>=<value>]
```

106.22.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	IsRegistering	RES	<bool> True, False	State
control	UserID	REQ	<string>	User ID
	UserPassword	REQ	<string>	Password

Action	Parameters	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<string>	Used when password encryption is used
	Protocol	REQ	<enum> SAMSUNG, ONVIF, RTSP	Camera registration protocol selection
	DevicePort	REQ	<int>	Device port number This parameter is valid only when **Protocol** is set to SAMSUNG.
	HTTPPort	REQ	<int>	HTTP port number This parameter is valid only when **Protocol** is set to SAMSUNG or ONVIF.
	IPAddress	REQ	<csv>	IP address

106.22.4. Examples

106.22.4.1. Getting Auto Registerd Cameras info

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=autoregister&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "IsRegistering": false
}
```

106.22.4.2. Add Camera Auto Register

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=autoregister&action=control&UserID=admin&UserPassword
={password}&IsPasswordEncrypted=False&Protocol=SAMSUNG&DevicePort=4520&HTTPP
ort=80&IPAddress=192.168.71.81,192.168.71.98,192.168.71.113
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.23. Camera Discovery

106.23.1. Description

The **cameradiscovery** submenu discovers a network camera, and sets camera ip use parameters.

Access level

Action	Camera	NVR	Decoder
view	—	User	User
set		User	User

106.23.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
cameradiscovery&action=<value>[&<parameter>=<value>]
```

106.23.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Timeout	REQ	<int>	Discovery action Time out value The values must be within the range of 30 to 60.
	IPType	REQ, RES	<enum> IPv4, IPv6	IP address type
	Protocol	REQ, RES	<enum> SAMSUNG, ONVIF, ANALOG	Discovered camera protocol

Action	Parameters	Request/ Response	Type/ Value	Description
	Model	REQ	<string>	Camera model name
	Interface	REQ	<enum> Network1, Network2, Network3, Network4	Camera default network inteface
	Status	REQ	<enum> Registered, NotRegistered	Status of discovered camera whether it's already registered or not registered.
set	Password	REQ	<string>	Password The **set** action is valid only when **IPType** is set to IPv4 and **Protocol** is Samsung.
	IsPasswordEncrypted	REQ	<csv>	When password is encrypted enable this in the request.
	MACAddress	REQ, RES	<string>	Mac address of the discovered camera
	IPAssignType	REQ, RES	<enum> DHCP, Manual	IP type that needs to be assigned
	IPAddress	REQ, RES	<string>	IP address to be assinged
	SubnetMask	REQ, RES	<string>	Subnet mask to be assigned
	Gateway	REQ, RES	<string>	Gateway to be assigned
	DevicePort	REQ, RES	<int>	Device port to be assigned
	HTTPPort	REQ, RES	<int>	http port to be assinged

106.23.4. Examples

106.23.4.1. Getting discovered camera during 30'seconds

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=cameradiscovery&action=view&Timeout=30
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "DiscoveredCameras": [
        {
            "Protocol": "SAMSUNG",
            "CamerasByIPType": [
                {
                    "IPType": "IPv4",
                    "Cameras": [
                        {
                            "Model": "SNF-7010",
                            "Interface": "Network1",
                            "Status": "NotRegistered",
                            "IPAssignType": "Manual",
                            "IPAddress": "192.168.31.244",
                            "HTTPPort": 80,
                            "MACAddress": "00:09:18:30:a8:e1",
                            "DevicePort": 4520,
                            "SubnetMask": "255.255.255.0",
                            "Gateway": "192.168.31.1"
                        },
                        {
                            "Model": "SNO-L6013R",
                            "Interface": "Network1",
                            "Status": "NotRegistered",
                            "IPAssignType": "Manual",
                            "IPAddress": "192.168.31.246",
                            "HTTPPort": 80,
                            "MACAddress": "00:16:6c:ad:a2:5e",
                            "DevicePort": 4520,
                            "SubnetMask": "255.255.255.0",
                            "Gateway": "192.168.31.1"
                        }
                    ]
                }
            ]
        },
        {
            "Protocol": "ONVIF",
```

```
            "CamerasByIPType": [
                {
                    "IPType": "IPv4",
                    "Cameras": [
                        {
                            "Model": "ENR-800S ",
                            "Interface": "Network1",
                            "Status": "NotRegistered",
                            "IPAssignType": "Manual",
                            "IPAddress": "192.168.111.230",
                            "HTTPPort": 80
                        },
                        {
                            "Model": "ENR-1600S ",
                            "Interface": "Network1",
                            "Status": "NotRegistered",
                            "IPAssignType": "Manual",
                            "IPAddress": "192.168.111.239",
                            "HTTPPort": 80
                        }
                    ]
                }
            ]
        }
    ]
}
```

106.23.4.2. Change/Update the discovered camera network setting.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=cameradiscovery&action=set&Protocol=SAMSUNG&IPType=IPv4&IsPasswordEncrypted=False&Password={password}&MACAddress=00:00:00:00:00:30&DevicePort=4553&HTTPPort=80&IPAddress=192.168.71.182&SubnetMask=255.255.255.0&Gateway=192.168.71.1&IPAssignType=Manual
```

Note: Here, the mac address is used to uniquely identify the device and change network settings.

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

106.24. Camera Connection

106.24.1. Description

The **cameraconnection** submenu is used for testing the camera connection.

Access level

Action	Camera	NVR	Decoder
view	—	User	User
test		User	User

106.24.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
cameraconnection&action=<value>[&<parameter>=<value>]
```

106.24.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	IsTestRunning	RES	<bool> True, False	Connection command running status view
test	Mode	REQ	<enum> Start, Cancel	Start test or cancel the connection test.
	Channel	REQ	<int>	Test camera channel This parameter is valid only when **Mode** is set to Start.
	CamChannel	REQ	<int>	Multi channel camera's channel number NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Protocol	REQ	<enum> SAMSUNG, ONVIF, RTSP	Protocol This parameter is valid only when **Mode** is set to Start.
	UserID	REQ	<string>	Camera user ID
	UserPassword	REQ	<string>	Camera user password
	IsPasswordEncrypted	REQ	<bool> True, False	Enable when the password is encyrpted
	Model	REQ	<string>	Camera model name <optional>
	AddressType	REQ	<enum> IPv4, IPv6, SamsungDDNS, URL, MAC	Camera address type.
	IPAddress	REQ	<string>	IP address This parameter is valid only when **Mode** is set to Start, and **AddressType** is to IPv4, or IPv6.
	DevicePort	REQ	<int>	Device port number This parameter is valid only when **Mode** is set to Start, **Protocol** is to SAMSUNG, and **AddressType** is NOT to SamsungDDNS.
	HTTPPort	REQ	<int>	HTTP port number If Mode = Start, Protocol=SAMSUNG and AddressType!=SamsungDDNS or Protocol=ONVIF and AddressType!=URL
	AuthenticationType	REQ	<enum> UserToken, Digest, CertificateBased	Authentication type If Mode = Start, Protocol=ONVIF
	StreamingMode	REQ	<enum> TCP, UDP, HTTP, HTTPS	Streaming mode If Mode = Start,Protocol=ONVIF,RTSP; HTTPS is only for Protocol=RTSP

Action	Parameters	Request/ Response	Type/ Value	Description
	MACAddress	REQ	<string>	Mac address If Mode = Start, AddressType=MAC
	URL	REQ	<string>	URL If Mode = Start, AddressType=SamsungDDNS,URL
	Status	RES	<string>	Status If Mode = Start

106.24.4. Examples

106.24.4.1. Getting camera connection test status

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=cameraconnection&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "IsTestRunning": false
}
```

106.24.4.2. Testing camera connection

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=cameraconnection&action=test&Mode=Start&Channel=1&Pro
tocol=SAMSUNG&UserID=admin&UserPassword={password}&IsPasswordEncrypted=False
&AddressType=IPv4&IPAddress=192.168.71.113
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Status": "Success"
}
```

106.25. Video Encoder Instances information

106.25.1. Description

The **videoencoderinstances** submenu requests the information about concurrent stream counts for a device. User can know about all limitations about stream count.

Access level

Action	Camera	Encoder
view	Guest	Guest

106.25.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videoencoderinstances&action=<value>[&<parameter>=<value>...]
```

106.25.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Read stream related information
	TotalConcurrentStreamCount	RES	<int>	Total count that a device can streams concurrently. This shall be different with sum of all channels' ConcorruntStreamCount.
	Channel	RES	<csv>	Channel value.
	Channel.#.ConcurrentStreamCount	RES	<int>	Total count that a channel can streams concurrently.

Action	Parameters	Request/ Response	Type/ Value	Description
	Limitation.#.NotSupportedConcurrentStreamViewModes	RES	<csv> Overview, QuadView, DoublePanorama, Panorama, QuadView.1, QuadView.2, QuadView.3, QuadView.4	Some fisheye models wouldn't support simultaneous streams among viewmodes. This parameter shows about those limitations. This parameter is optional and would be shown only in fisheye models and if that fisheye camera has some limitations for concurrent streaming among some viewmodes. FISHEYE ONLY

106.25.4. Examples

106.25.4.1. Getting information

The concurrent streaming count information would be provided like below.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videoencoderinstances&action=view
```

The following response example is for Fisheye models.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalConcurrentStreamCount=3
Channel.0.ConcurrentStreamCount=1
Channel.1.ConcurrentStreamCount=1
Channel.2.ConcurrentStreamCount=1
Limitation.1.NotSupportedConcurrentStreamViewModes=DoublePanorama,Panorama
Limitation.2.NotSupportedConcurrentViewModes=QuadView,QuadView.1,QuadView.2,
QuadView.3,QuadView.4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoEncoderInstances": [
        {
            "TotalConcurrentStreamCount": 3,
            "ConcurrentStreamCounts": [
                {
                    "Channel": 0,
                    "ConcurrentStreamCount": 1
                },
                {
                    "Channel": 1,
                    "ConcurrentStreamCount": 1
                },
                {
                    "Channel": 2,
                    "ConcurrentStreamCount": 1
                }
            ],
            "Limitations": [
                {
                    "Index": 1,
                    "Limitation": "Panorama,DoublePanorama"
                },
                {
                    "Index": 2,
                    "Limitation":
"QuadView,QuadView.1,QuadView.2,QuadView.3,QuadView.4"
                }
            ]
        }
    ]
}
```


The following response example is for common multi channel cameras.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalConcurrentStreamCount=9
Channel.0.ConcurrentStreamCount=3
Channel.1.ConcurrentStreamCount=3
Channel.2.ConcurrentStreamCount=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoEncoderInstances": [
        {
            "TotalConcurrentStreamCount": 9,
            "ConcurrentStreamCounts": [
                {
                    "Channel": 0,
                    "ConcurrentStreamCount": 3
                },
                {
                    "Channel": 1,
                    "ConcurrentStreamCount": 3
                },
                {
                    "Channel": 2,
                    "ConcurrentStreamCount": 3
                }
            ]
        }
    ]
}
```

The following response example is for common single channel cameras.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalConcurrentStreamCount=3
Channel.0.ConcurrentStreamCount=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoEncoderInstances": [
        {
            "TotalConcurrentStreamCount": 3,
            "ConcurrentStreamCounts": [
                {
                    "Channel": 0,
                    "ConcurrentStreamCount": 3
                }
            ]
        }
    ]
}
```

106.26. Video Input

106.26.1. Description

The **videoinput** submenu configures the video input format of the device.

NOTE This chapter applies to the encoder only.

Access level

Action	Encoder
view	User
set	User

106.26.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
videoinput&action=<value>[&<parameter>=<value>...]
```

106.26.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the current videoinput format
set	Format	REQ/RES	<enum> NTSC,PAL	Video input format.

106.26.4. Examples

106.26.4.1. Retrieving current videoinput format.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoinput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Format=NTSC
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "Format":"NTSC"
}
```

106.26.4.2. Setting videoinput format

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videoinput&action=set&Format=PAL
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.27. Metadata Sharing

106.27.1. Description

The **metadatashare** submenu is for configuring and viewing the metadata sharing settings. When metadata sharing is enabled, metadata is shared between all video sources.

Access level

Action	Camera
view	Admin
set	Admin

NOTE Attribute to check for Metadata Share: "attributes/Media/Support/**MetadataShare**"

106.27.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
metadatashare&action=<value>[&<parameter>=<value>]
```

106.27.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the current Metadata sharing Mode
set	Mode	REQ, RES	<enum> Off, All	Off : Metadata is not shared. All : All channels share Metadata. A new type of <Mode> can be added in the future. See an example with metadata share enabled in section Frame metadata example when Metadata sharing is turned on

106.27.4. Examples

106.27.4.1. Getting the metadata share settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=metadatashare&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Mode=Off
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```

```
{
    "Mode": "Off"
}
```

106.27.4.2. Change/Update the metadata share settings

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=metadatashare&action=set&Mode=All
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.28. Speaker Volume Control

106.28.1. Description

The **speakervolume** submenu is used to control the volume of speaker and all connection subspeakers when the SpeakerType is configured as MASTER or SERVER.
When SpeakerType is configured as INDEPENDENT, it controls the volume of that individual speaker.

NOTE	This chapter applies to IPAS only. Device ids should match with the device ids listed in the system.cgi **registeredsubdevices** submenu response. In IPAS, The availability of this submenu depends on the AudioOutAccess level.

Access level

Action	IPAS
view	Admin
set	Admin

106.28.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
speakervolume&action=<value>[&<parameter>=<value>]
```

106.28.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				
set	Volume	REQ,RES	<int>	Values 0 to 100 (100 refers to max volume) **Note** Applicable only when SpeakerType is INDEPENDENT, MASTER or SERVER.
	SpeakerDevice.<ID>.Volume	REQ, RES	<int>	Values 0 to 100 (100 refers to max volume) **Note** Applicable only when SpeakerType is MASTER or SERVER.

106.28.4. Examples

106.28.4.1. Getting the current volume (When the speakertype is MASTER or SERVER)

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=speakervolume&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Volume": 70,
    "SpeakerDevice":[
        {
            "ID": 1121,
            "Volume": 30
        },
        {
            "ID": 1123,
            "Volume": 40
        }
    ]
}
```


106.28.4.2. Setting the volume of INDEPENDENT, MASTER or SERVER Speaker

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=speakervolume&action=set&Volume=72
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


106.28.4.3. Setting the volume of Individual Speaker

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=speakervolume&action=set&SpeakerDevice.1121.Volume=40
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

106.28.4.4. Getting the current volume (When the speakertype is INDEPENDENT)

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=speakervolume&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Volume": 70
}
```

106.29. BackChannel Stream Information

106.29.1. Description

The **backchannelinfo** submenu is used to get the backchannel streaming information.

NOTE	This chapter applies to IPAS only. Device ids should match with the device ids listed in the system.cgi **registeredsubdevices** submenu response.

Access level

Action	IPAS
view	Admin

106.29.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
backchannelinfo&action=<value>[&<parameter>=<value>]
```

106.29.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	StreamType	REQ	<enum> RTSPOverTCP, RTSPOverUDP, RTSPOverHTTP	Stream type selection (If not passed by default RTSPOverTCP is used) **Note** This parameter is deprecated.
	DeviceIDs	REQ	<CSV>	Filter the response based on selected one or more deviceids Example: DeviceIDs=1121,1123 **Note** Applicable only when SpeakerType is MASTER or SERVER.
	SpeakerDevice.<ID>.BackChannelRTSPURL	RES	<string>	RTSP URL **Note** When speaker type is INDEPENDENT. Always the ID is 0
	SpeakerDevice.<ID>.AudioCodec	RES	<enum> G711	Audio Codec **Note** When speaker type is INDEPENDENT. Always the ID is 0

Action	Parameter	Request/ Response	Type/ Value	Description
	SpeakerDevice.<ID>.Bitrate	RES	<int>	Bitrate of the audio codec, 64000 for G711 **Note** When speaker type is INDEPENDENT. Always the ID is 0

106.29.4. Examples

106.29.4.1. Getting the backchannel rtsp urls

When SpeakerType is **INDEPENDENT** below response can be expected.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=backchannelinfo&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SpeakerDevice":[
        {
            "ID": 0,
            "BackChannelRTSPURL": "rtsp://192.168.66.22:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        }
    ]
}
```

When SpeakerType is **MASTER** or **SERVER** below response can be expected.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=backchannelinfo&action=view&StreamType=RTSPOverTCP
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SpeakerDevice":[
        {
            "ID": 1121,
            "BackChannelRTSPURL": "rtsp://192.168.66.22:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        },
        {
            "ID": 1123,
            "BackChannelRTSPURL": "rtsp://192.168.66.23:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        },
        {
            "ID": 1124,
            "BackChannelRTSPURL": "rtsp://192.168.66.25:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        }
    ]
}
```


106.29.4.2. Getting the backchannel rtsp urls based on device ids

NOTE

When SpeakerType is MASTER or SERVER
DeviceIDs based filtering is supported.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=backchannelinfo&action=view&StreamType=RTSPOverTCP&DeviceIDs=1121,1123
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SpeakerDevice":[
        {
            "ID": 1121,
            "BackChannelRTSPURL": "rtsp://192.168.66.22:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        },
        {
            "ID": 1123,
            "BackChannelRTSPURL": "rtsp://192.168.66.23:8554/BackChannel",
            "AudioCodec":"G711",
            "Bitrate": 64000
        }
    ]
}
```


106.30. Speaker Status

106.30.1. Description

The **speakerstatus** submenu is used check the current status of the speaker

NOTE This chapter applies to IPAS only and if the Device Type is SPEAKER or SPEAKER_WITH_AUDIO_SOURCE

Access level

Action	IPAS
view	Suser

106.30.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
speakerstatus&action=<value>[&<parameter>=<value>]
```

106.30.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Status	RES	<enum> NOT_READY, READY, PLAYING	Current status of the speaker

106.30.4. Examples

106.30.4.1. Getting the current status

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=speakerstatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "READY"
}
```

106.31. Network Audio Input

106.31.1. Description

The **networkaudioinput** submenu configures and views the network audio input settings.

NOTE Supported only in IPAS.

Access level

Action	IPAS
view	Admin
set	Admin

NOTE Attribute to check for Audio Input: "attributes/Media/Support/**NetworkAudioIn**"
Permission to check for User Access: "security/users/AudioOutAccess"

106.31.2. Syntax

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=
networkaudioinput&action=<value>[&<parameter>=<value>]
```

106.31.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the audio input settings
	Channel	REQ	<csv>	Channel ID
	EncodingType	RES	<enum> G711	Audio Codec (read only)
	SampleRate	RES	<int>	Audio input sample rate (read only) Default is 8000.
	Bitrate	RES	<enum> 64000	Audio input bit rate (read-only)
set	Channel	REQ, RES	<int>	Channel ID
	EnableUnicast	REQ, RES	<bool> True, False	Enables or disables the unicast listenting.
	UnicastPort	REQ, RES	<int> 1~65535	Valid port number to receive the audio data
	EnableMulticast	REQ, RES	<bool> True, False	Enables or disable multicast listening
	MulticastAddress	REQ, RES	<string>	The value must be in the format of an IPv4 address, and within the range of 224.0.0.1 to 239.255.255.254.
	MulticastPort	REQ, RES	<int> 1~65535	Multicat port to receive the audio data

106.31.4. Examples

106.31.4.1. Getting audio input settings

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=networkaudioinput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Channel.0.EncodingType=G711
Channel.0.SampleRate=8000
Channel.0.Bitrate=64000
Channel.0.EnableUnicast=True
Channel.0.UnicastPort=2088
Channel.0.EnableMulticast=True
Channel.0.MulticastAddress=239.0.0.11
Channel.0.MulticastPort=3044
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "NetworkAudioInputs": [
        {
            "Channel": 0,
            "EncodingType": "G711",
            "SampleRate": 8000,
            "Bitate": "64000",
            "EnableUnicast": true,
            "UnicastPort": 2088,
            "EnableMulticast": true,
            "MulticastAddress": "239.0.0.11",
            "MulticastPort": 3044
        }
    ]
}
```

106.31.4.2. Setting the networkaudioinput Unicast settings

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=networkaudioinput&action=set&EnableUnicast=True&Unica
stPort=2088
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


106.31.4.3. Setting the networkaudioinput Multicast settings

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=networkaudioinput&action=set&EnableMulticast=True&Mul
ticastPort=3044&MulticastAddress=239.0.0.11
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 107. RTP/RTSP Streaming

107.1. Live URL/Authentication

107.1.1. Live URL for camera

The URLs for RTSP (Real Time Streaming Protocol) are as follows.

```
[Type1] rtsp://<Device IP>/<encoding>/media.smp

[Type2] rtsp://<Device IP>/profile<no>/media.smp

[Type3] rtsp://<Device IP>/multicast/<encoding>/media.smp

[Type4] rtsp://<Device IP>/multicast/profile<no>/media.smp

[Type5] rtsp://<Device IP>/<profile name>/media.smp

[Type6] rtsp://<Device IP>/multicast/<profile name>/media.smp
```

When Camera supports multiple sources:

```
[Type1] rtsp://<Device IP>/<chid>/<encoding>/media.smp

[Type2] rtsp://<Device IP>/<chid>/profile<no>/media.smp

[Type3] rtsp://<Device IP>/<chid>/multicast/<encoding>/media.smp

[Type4] rtsp://<Device IP>/<chid>/multicast/profile<no>/media.smp

[Type5] rtsp://<Device IP>/<chid>/<profile name>/media.smp

[Type6] rtsp://<Device IP>/<chid>/multicast/<profile name>/media.smp
```

NOTE	Refer to '2.11 Stream URI (page 54)' for information on how to get RTSP URLs using **streamurl** submenu in **media.cgi**.

- <Device IP>: The hostname or the network camera's IP address
- <encoding>: Video encoding type (MJPEG/MPEG4/H.264/H.265)

NOTE	Check the device attributes using **attributes.cgi** for the encoding type that the device

supports.

- profile<no>: Video profile number (1 ... 10)
- <profile name> : Video profile name
- <chid>: Channel index starting from 0.

NOTE The default RTSP port number is 554. You can change the port number by using HTTP/CGI commands if required.

107.1.2. Live URL for NVR

The URLs for RTSP (Real Time Streaming Protocol) are as follows.

```
[Type1] rtsp://<Device IP>:558/LiveChannel/<chid>/media.smp

[Type2] rtsp://<Device IP>:558/LiveChannel/<chid>/media.smp/session=<sid>

[Type3] rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/multicast&session=<sid>

[Type4] rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/iframe&multicast&session=<sid>

[Type5] rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/profile=<profileNo>&session=<sid>

[Type6] rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/ProfileUsage=<profileType>&session=<sid
>
```

- <Device IP>: The hostname or the network camera’s IP address
- <chid>: Channel index starting from 0.
- <sid>: Session ID
  As all live channels with same session ID are considered as one live session, all playback sessions with same session ID are considered as one session. If session ID is not changed, each RTSP connection will be considered as one session.
- <profileNo>: Any profile number from the profilepolicy submenu, that is currently being used in NVR

  Capability to verify: "Stream.MultiProfiles" attribute under Media/Support
- <profileType>: Any of the following types can be used "Live", "Record", or "Network"

  Capability to verify:" Stream.ProfileUsageBased" attribute under Media/Support

NOTE <sid> should be unique for live, playback and backup mode.

NOTE The default RTSP port number is 558. You can change the port number by using the HTTP/CGI commands if required.

107.1.3. Authentication

RTSP complies with RFC 2326 and uses RFC 2096 (Digest method) to authorize users.

The RTSP communication procedure is as follows.

1 Application (the client) delivers the DESCRIBE method, having no Authorization header.

2 Network camera (the server) responds with the 401 (Unauthorized) status code. At this time, the WWW-Authenticate response header is included, showing the MD5 authorization information.

3 The application generates an Authorization request header using the given MD5 information, and then delivers the DESCRIBE method again, including the Authorization request header.

107.2. RTSP Methods

The followings are types of RTSP methods.

Method	Description
OPTIONS	Notifies with the methods supported by the server.
GET_PARAMETER	Used for testing the client with no entity contents or for testing (ping) to ensure the server is alive.
DESCRIBE	Search for media streams identified by the presentation description or the aggregate URL.
SETUP	Specifies how a single media stream must be transported.
PLAY	Causes one or all media streams to be played, or restarted.
PAUSE	Temporarily halts one or all media streams.
TEARDOWN	Stops media stream transfer, and terminates the session and releases all resources related to the session.

107.3. RTSP Liveness

When **Timeout** is set to 60s with network.cgi (please refer to the 'Network' document), the device regularly checks if the RTSP command has been received. If the device receives no command within 1 minute, the RTP/RTSP session is automatically disconnected.

Therefore, if **Timeout** of **rtsp** submenu with **network.cgi** is set to 60s, the RTSP client should send the GET_PARAMETER or OPTION command within every 60 seconds to maintain the RTP/RTSP session.

107.4. RTSP Playback

107.4.1. Playback URL for camera

The URLs to view the saved video through RTSP (Real Time Streaming Protocol) are as follows.

```
[Type1] rtsp://<Device IP>/recording/<Start Time>/play.smp
[Type2] rtsp://<Device IP>/recording/<Start Time>-<End Time>/play.smp
[Type3] rtsp://<Device IP>/recording/play.smp
```

When Camera supports multiple sources:

```
[Type1] rtsp://<Device IP>/<chid>/recording/<Start Time>/play.smp
[Type2] rtsp://<Device IP>/<chid>/recording/<Start Time>-<End Time>/play.smp
[Type2.a] rtsp://<Device IP>/<chid>/recording/<Start Time>-<End
Time>/OverlappedID=<overlapid>/play.smp
[Type3] rtsp://<Device IP>/<chid>/recording/play.smp
```

- <Device IP>: The hostname or the network camera's IP address
- <Start Time>: Playback start time (in YYYYMMDDHHMMSS format). 20010203040506 denotes 4: 05:06 AM in February 3, 2001.
- <End Time>: Playback end time (YYYYMMDDHHMMSS format)
- profile<no>: Video profile number (1 … 10)
- <chid>: Channel index starting from 0
- <overlapid>: Overlap id of the playback stream, default overlapid is 0

NOTE

Note 1

[Type1] continues to play the saved video from the <Start Time>.
[Type2] plays the saved video from <Start Time> to <End Time>. If the <Start Time> is set to later than the <End Time>, the video is played backward.
[Type3] plays the video in the range specified by Range header of the PLAY method. The time specified in Range header only supports the Absolute Time format defined in Ch 3.7 of RFC2326.

NOTE

Note 2

[Type1] and [Type2] use Local Time, and [Type3] uses UTC Time.

NOTE

Note 3

Multicast is not available during playback.

107.4.2. Playback URL for NVR

The URLs to view the saved video through RTSP (Real Time Streaming Protocol) are as follows:

```
[Type1] rtsp://<Device IP>:558/PlaybackChannel/<chid>/media.smp
[Type2] rtsp://<Device
IP>:558/PlaybackChannel/<chid>/media.smp/session=<sid>
[Type3] rtsp://<Device
IP>:558/PlaybackChannel/<chid>/media.smp/overlap=<id>&session=<sid>
[Type4] rtsp://<Device
IP>:558/PlaybackChannel/<chid>/media.smp/overlap=<id>&session=<sid>&iframe
[Type5] rtsp://<Device
IP>:558/PlaybackChannel/<chid>/media.smp/start=<starttime>&end=<endtime>&ove
rlap=<overlapid>&session=<sid>
[Type6] rtsp://<Device
IP>:558/PlaybackChannel/<chid>/media.smp/start=<starttime>&end=<endtime>&ove
rlap=<overlapid>&session=<sid>&substream
```

- <Device IP>: The hostname or the network camera's IP address
- <chid>: Channel index starting from 0
- <sid>: Session ID
- <Start Time>: Playback start time
  Playback start time should be in the format of <YYYYMMDDTHHMMSS> (e.g, 20141206T111500) for local time and <YYYYMMDDTHHMMSSZ> (e.g, 20141206T110000Z) for UTC time.
- <End Time>: Playback end time
  Playback end time should be in the format of <YYYYMMDDTHHMMSS> (e.g, 20141206T111500) for local time and <YYYYMMDDTHHMMSSZ> (e.g, 20141206T110000Z) for UTC time.
- substream: If the NVR supports dual track recording, this option can be used to play the substream. If not passed, mainstream would be used by default. Check attributes/Recording/Support/DualTrackRecording capability for dual track recording support.

107.4.3. Media Track

In Playback's SDP, which can be acquired through the DESCRIBE method, information of all available saving codecs is included as an individual track. Use SETUP method to connect to each track, and to receive data of the corresponding track if the video's codec changes during the playback.

107.4.3.1. Examples

Playback SDP for camera

REQUEST

```
DESCRIBE rtsp://<Device IP>/recording/play.smp RTSP/1.0
```

RESPONSE

```
RTSP/1.0 200 OK
Content-type: application/sdp
<Body>
```

```
……
m=video 40000 RTP/AVP 26
a=rtpmap:26 JPEG/90000
a=control:rtsp://<Device IP>/recording/play.smp/trackID=JPEG
m=video 40000 RTP/AVP 98
a=rtpmap:98 H264/90000
a=control:rtsp://<Device IP>/recording/play.smp/trackID=H264
m=audio 40000 RTP/AVP 0
a=rtpmap:0 PCMU/8000
a=control:rtsp://<Device IP>/recording/play.smp/trackID=PCMU
m=audio 40000 RTP/AVP 97
a=rtpmap:97 G726-32/8000
a=control:rtsp://<Device IP>/recording/play.smp/trackID=G726-32
……
```

Playback SDP for NVR

REQUEST

```
DESCRIBE rtsp://<DeviceIP>:558/PlaybackChannel/<chid>/media.smp RTSP/1.0
```

RESPONSE

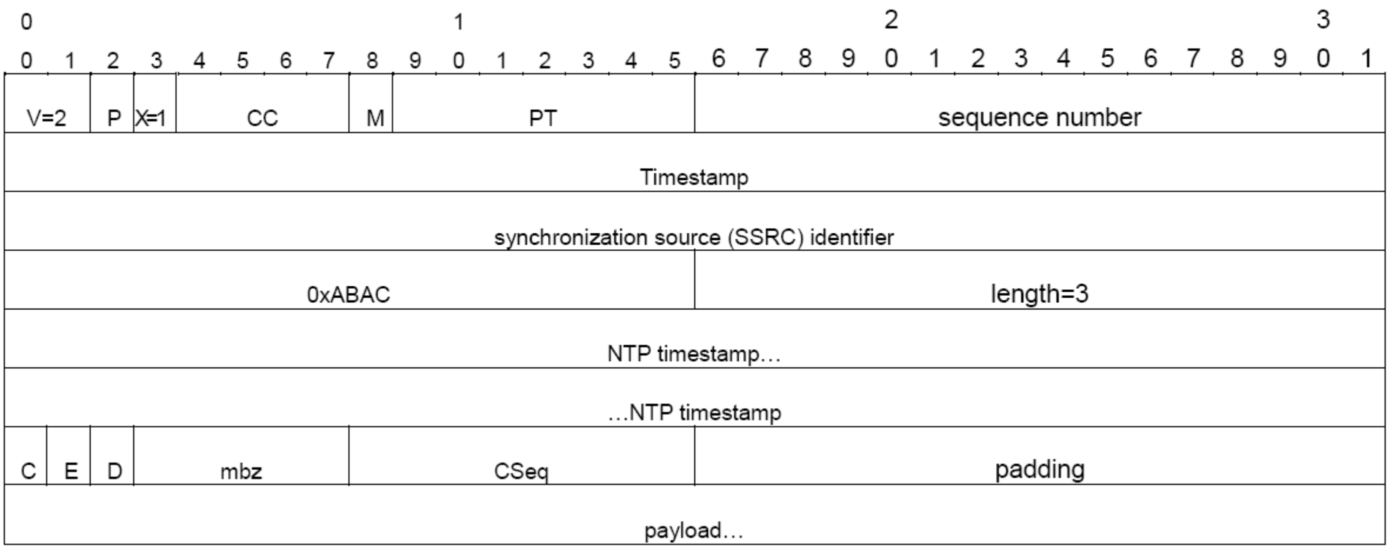
```
RTSP/1.0 200 OK
Content-type: application/sdp
<Body>
```

```
……
m=video 0 RTP/AVP 98
b=AS:500
```

```
a=rtpmap:98 H264/90000
a=control:track1
a=x-onvif-track:video1
m=video 0 RTP/AVP 96
b=AS:500
a=rtpmap:96 MP4V-ES/90000
a=control:track2
a=x-onvif-track:video2
m=video 0 RTP/AVP 26
b=AS:500
a=rtpmap:26 JPEG/90000
a=control:track3
a=x-onvif-track:video3
m=audio 0 RTP/AVP 103
b=AS:500
a=control:track4
a=rtpmap:103 G726-40/8000/1
a=x-onvif-track:audio1
……
```

107.4.4. RTP packet format

To know exactly when the currently playing video was recorded, an RTP extension header with the ID of 0xABAC included in the first packet of each frame data is also transferred. In this case, the RTP packet type is as follows.



Each item in the extension header is defined below.

Item	Description
0xABAC	Extension header ID

Item	Description
NTP timestamp	The recording time of the frame is displayed as a timestamp in NTP [RFC1305] format.
C	Indicates the "clean point", which means the frame can be decoded without having the previous frame. For example, it means an I-frame in H.264.
E	Indicates the last frame of one continuous saving (recording) section. It is set to 1 if the frame is the last one of a playable section, or if frames have no saved video within the playable section.
D	If this frame had been recorded prior to the one already transferred, it is set to 1. When playing in reversed mode, D bit of every I-frame is set to 1.
mbz	Reserved. It is set to 0 in all cases.
CSeq	It is set to the value of subordinate 1 byte of RTSP PLAY command's Cseq, which requested the video section being played.
padding	Reserved and not used, this is for sorting the extension header by 4-byte ordering.

NOTE

The above RTP extension header has the same format as the playback extension header defined by ONVIF Streaming Spec version 2.10.
The RTP extension for streaming MJPEG over 3 Megapixels is followed as defined by ONVIF Streaming Spec version 2.10 for both NVR and the camera.

107.4.5. Range Header

When requesting playback while connecting to an RTSP URL that contains no playback time information, the Range header to specify the playback range must be included in the RTSP PLAY command. In this case, the only time format of Range header is utc-range (the Absolute Time format specified by Ch 3.7 of RFC2326).

107.4.5.1. Examples

Specifying playback start time as June 15, 2012 20:33:15

REQUEST

```
PLAY rtsp://<Device IP>/recording/play.smp RTSP/1.0
Range: clock=20120615T203315Z-
```

RESPONSE

```
RTSP/1.0 200 OK
```

Specifying playback start/end time as June 15, 2012 20:33:15~40

REQUEST

```
PLAY rtsp://<Device IP>/recording/play.smp RTSP/1.0
```

```
Range: clock=20120615T203315Z-20120615T203340Z
```

RESPONSE

```
RTSP/1.0 200 OK
```

107.4.6. Scale Header

You can specify playback speed and direction by adding Scale header in RTSP PLAY command. If it is set to 1.0, the video is played at x1 speed; if it is set to 2.0, the video is played at x2 speed. If the value is lower than 0, the video is played in reverse.

If Scale header is not specified, the video is played at x1.0 speed by default.

CAUTION If the requested speed is not supported by the camera, it is automatically changed to one that is supported.

107.4.6.1. Examples

Setting playback speed

REQUEST

```
PLAY rtsp://<Device IP>/recording/play.smp RTSP/1.0
Scale: 12.3
```

RESPONSE

```
RTSP/1.0 200 OK
Scale: 8.0
```

107.4.7. Rate-Control Header

You can change how network traffic is controlled by adding Rate-Control header in RTSP PLAY command. Rate-Control header can have "yes" or "no" as its value. If it is set to "yes", the video is transmitted at the bitrate used to record it. If it is set to "no", the video is transmitted as quickly as possible. If the Rate-Control is set to "no", the Scale header can have only 1.0 or -1.0 (reverse play) for its value.

If Rate-Control header is not included, it is set to "yes" by default.

107.4.7.1. Examples

Requesting to transmit saved video as fast as possible

REQUEST

```
PLAY rtsp://<Device IP>/recording/play.smp RTSP/1.0
```

```
Rate-Control: no
```

RESPONSE

```
RTSP/1.0 200 OK
```

107.4.8. Frames Header

You can select and transmit specific kinds of frames by adding a Frames header in the RTSP PLAY command. The Frames header can have "intra", "intra/<Interval>" or "all" as its value. If a Frames header is not included, it is set to "all" by default.

Header	Description
Frames: intra	Transmit only I-frame.
Frames: intra/<Interval>	Transmit I-frames only at the minimum interval of <Interval> ms.
Frames: all	Transmit all the frames.

NOTE <interval> feature is supported only in cameras.

107.4.8.1. Examples

Transmitting only one I-frame per 3 seconds in saved video

REQUEST

```
PLAY rtsp://<Device IP>/recording/play.smp RTSP/1.0
Frames: intra/3000
```

RESPONSE

```
RTSP/1.0 200 OK
```

107.5. RTSP bi-directional audio

The audio out function can be used in the camera through RTSP. Since standard RTSP does not support two-way transmission, the Require header should be used to use this feature.

If client delivers a DESCRIBE request after adding an item of ‘www.onvif.org/ver20/backchannel’ in the Require header, the stream information for "audio out" with a=sendonly attribute is included in the SDP response. After making a session for "audio out" by using the SETUP command and delivering the PLAY command, the RTP audio data transmitted by client is output through the camera’s "audio out" port.

107.5.1. Examples

Playback SDP for camera

REQUEST

```
DESCRIBE rtsp://<Device IP>/profile1/media.smp RTSP/1.0
CSeq: 1
Accept: application/sdp
Require: www.onvif.org/ver20/backchannel
```

RESPONSE

```
RTSP/1.0 200 OK
Content-type: application/sdp
<Body>
```

```
......
m=audio 40002 RTP/AVP 0
a=rtpmap:0 PCMU/8000
a=control:rtsp://<Device IP>/profile1/media.smp/trackID=a
a=recvonly
......
m=audio 40006 RTP/AVP 0
a=rtpmap:0 PCMU/8000
a=control:rtsp://<Device IP>/profile1/media.smp/trackID=t
a=sendonly
......
```

NOTE	As shown in the above SDP, if a media including trackID=a is set up, the "audio in" function is activated. When a media including trackID=t is set up, the "audio out" function is activated.

Playback SDP for NVR

REQUEST

```
DESCRIBE rtsp://<Device IP>:558/LiveChannel/<chid>/media.smp RTSP/1.0
CSeq: 1
Accept: application/sdp
Require: www.onvif.org/ver20/backchannel
```

RESPONSE

```
RTSP/1.0 200 OK
Content-type: application/sdp
<Body>
```

```
……
m=audio 8002 RTP/AVP 0
b=AS:64
a=control:trackID=a
a=rtpmap:0 PCMU/8000/1
a=recvonly
……
m=audio 8006 RTP/AVP 0
a=control:trackID=back
a=rtpmap:0 PCMU/8000/1
a=sendonly
……
```

107.6. RTSP Backup

NOTE RTSP backup is applicable only for NVR.

The URLs for RTSP backup are as follows:

```
[Type1] rtsp://<Device IP>:558/BackupChannel/<chid>/media.smp
[Type2] rtsp://<Device IP>:558/BackupChannel/<chid>/media.smp/session=<sid>
[Type3] rtsp://<Device
IP>:558/BackupChannel/<chid>/media.smp/overlap=<id>&session=<sid>
[Type4] rtsp://<Device
IP>:558/BackupChannel/<chid>/media.smp/overlap=<id>&session=<sid>&iframe
[Type5] rtsp://<Device
IP>:558/BackupChannel/<chid>/media.smp/start=<starttime>&end=<endtime>&overl
ap=<overlapid>&session=<sid>
```

- <Device IP>: The hostname or the network camera’s IP address
- <chid>: Channel index starting from 0
- <sid>: Session ID
- <Start Time>: Playback start time
  Playback start time should be in the format of <YYYYMMDDTHHMMSS> (e.g, 20141206T111500) for

local time and <YYYYMMDDTHHMMSSZ> (e.g, 20141206T110000Z) for UTC time.

- <End Time>: Playback end time
  Playback end time should be in the format of <YYYYMMDDTHHMMSS> (e.g, 20141206T111500) for local time and <YYYYMMDDTHHMMSSZ> (e.g, 20141206T110000Z) for UTC time.

107.7. SRTP Streaming

This feature is supported only in the camera as of now. The client can check the media/support section of attributes for "Stream.SRTP.UDP" and "Stream.SRTP.Multicast" attributes.

NOTE At present Stream.SRTP.Multicast is not supported

References

- RFC 2326 – Real Time Streaming Protocol (RTSP) (https://tools.ietf.org/html/rfc3326)
- RFC 3711 – The Secure Real-time Transport Protocol (SRTP)(https://tools.ietf.org/html/rfc3711)
- RFC 5285 – A General Mechanism for RTP Header Extensions (https://tools.ietf.org/html/rfc5285)
- RFC 4567 – Key Management Extensions for Session Description Protocol (SDP) and Real Time Streaming Protocol (RTSP) (https://tools.ietf.org/html/rfc4567)
- RFC 3830 – MIKEY: Multimedia Internet KEYing (https://tools.ietf.org/html/rfc3830)

SRTP Packet format

In the following RTP data the payload is encrypted.

```
   0                   1                   2                   3
  0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
 +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+<+
 |V=2|P|X|  CC   |M|     PT      |       sequence number         | |
 +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
 |                           timestamp                           | |
 +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
 |           synchronization source (SSRC) identifier            | |
 +=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+ |
 |            contributing source (CSRC) identifiers             | |
 |                               ....                            | |
 +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
 |                   RTP extension (OPTIONAL)                    | |
+>+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
| |                          payload  ...                        | |
| |                               +-------------------------------+ |
| |                               | RTP padding   | RTP pad count | |
+>+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+<+
| ~                     SRTP MKI (OPTIONAL)                       ~ |
| +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
| :                 authentication tag (RECOMMENDED)              : |
| +-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ |
|                                                                   |
+- Encrypted Portion*                      Authenticated Portion ---+
```

The following security policies are used

- Encryption algorithm – AES-CM
- Master key length – 128 bits
- Master key salt – 112 bits
- Authentication algorithm – HMAC-SHA1
- Authentication key length – 160bits
- Authentication tag length – 80bits
- MKI length – 32 bits

Extension Header

To help the client recognize the Key frame without decrypting the RTP payload, an RTP extension is used, For [RFC5285], if a one byte extension profile 0xBEDE is used, the extension payload would have 0xE0050000 in the first packet of the key frame, and in the case of MJPEG the first packet of all frames would have this extension header.

It is possible that the first packet of a key frame already includes a header extension. Since RTP packets only allow one extension profile, this poses a problem. In this case, an empty RTP packet containing the key frame header extension shall be inserted right before the actual first packet in the frame. This empty packet is considered part of the frame so it must have the same RTP timestamp as the rest of the frame and must not be marked.

Key Exchange

There are several key exchange methods, but only the MIKEY method is supported as of now [RFC 3830].

Sample MIKEY message after base64 encoding would look like:


```
AQAFAP1td9ABAADCD1UcAAAAAAoAAdOOGc75XD0BAAAAGAABAQEBEAIBAQMBFAcBAQgBAQoBAQsB
CgAAACcAIQAe30C59UrClE0e27UP5h/Wty9UL8+dfzg+2ttmmo3uRAAAAC8A
```


For secure communication, RTSP over TLS is used and 322 would be the default port, but it can be changed using the rtspovertls submenu in network cgi. Since the Key exchange happens using the SDP or SETUP, it can be protected only using the RTSP over TLS.

In the setup request the client passes the MIKEY message and the device uses this master key provided.

```
SETUP rtsp://<RTSP_URI> RTSP/1.0
CSeq: 2
User-Agent: RTSPClient
Transport: RTP/SAVP/UDP;unicast
KeyMgmt:
prot=mikey;uri="";data="AQAFAP1td9ABAADCD1UcAAAAAAoAAdOOGc75XD0BAAAAGAABAQEB
```

```
EAIBAQ
MBFAcBAQgBAQoBAQsBCgAAACcAIQAe30C59UrClE0e27UP5h/Wty9UL8+dfzg+2ttmmo3kBAAAAC
8A"
```

Transport profile should be RTP/SAVP, if it’s different MIKEY may not be recognized.

Re-keying can be done using the SET PARAMETER command, and the device accepts the new key without resetting the ROC (Roll over Counter).

```
SET_PARAMETER rtsp://<RTSP_URI> RTSP/1.0
CSeq: 8
Session: 1570
User-Agent: RTSPClient
Content-Type: text/parameters
Content-Length: 145
mikey:
AQAFAGgCr8EBAADSvxgkAAAAAAoAAdOOK7UihqIBAAAAGAABAQEBEAIBAQMBFAcBAQgBAQoBAQsB
CgAAACcAIQAepekjs88g+Q7AU6LAvRsoVyn18ZW1JuXI9qht4g6+BAAAAAIA
```

107.8. BestShot Streaming

Any device that supports metaimagetransfer also supports Bestshot streaming using RTSP/RTP.

A Bestshot image can also be retrieved on the client end using an additional RTP session dedicated to bestshot image streaming. It would be provided only when a Describe request has "**Require: Bestshot**"

Request (Example):

```
DESCRIBE rtsp://192.168.9.100:554/profile2/media.smp RTSP/1.0
CSeq: 2
Authorization: Digest
username="admin",realm="iPOLiS",nonce="C9AB5CEC50A78C07C25704A9675B171E",uri
="rtsp://192.168.9.100:554/profile2/media.smp",cnonce="814ae708b84f4f0f9f2e5
1fe1624166d",nc=00000001,response="40899fab43a5df53c72f8b3683f023e2",qop="au
th"
User-Agent: oscar-rtsp-client
Require: Bestshot
```

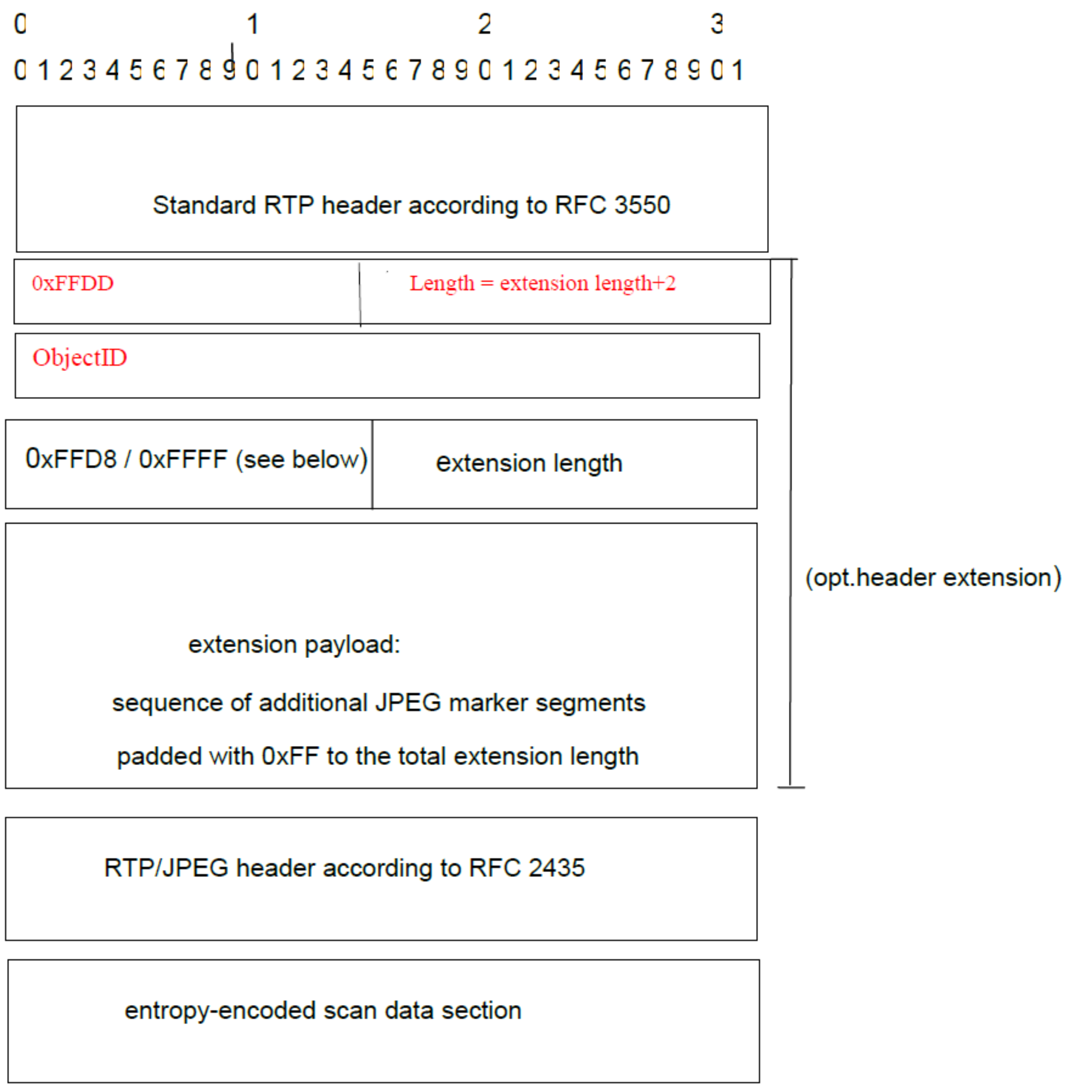
SDP RESPONSE format

```
a=control:rtsp://192.168.75.52:554/0/onvif/profile1/media.smp
a=range:npt=now-
m=video 40104 RTP/AVP 26
```

```
a=rtpmap:26 JPEG/90000
a=control:rtsp://192.168.75.52:554/0/onvif/profile1/media.smp/trackID=v
m=application 40106 RTP/AVP 107
b=AS:8
a=rtpmap:107 vnd.onvif.metadata/90000
a=control:rtsp://192.168.75.52:554/0/onvif/profile1/media.smp/trackID=m
a=recvonly
m=video 40108 RTP/AVP 26
i=MetaImageSession
a=rtpmap:26 JPEG/90000
a=control:rtsp://192.168.75.52:554/0/onvif/profile1/media.smp/trackID=MI
a=recvonly
```

RTP extension to pass the object ID

In the below extension, if the resolution of the JPEG is over 3 megapixels, ONVIF extension header can be used additionally. In that case, the Length of the extension header is extensionlength+2, if the resolution is less than 3 megapixels, Length of extension header is Just 1.



NOTE In the metaimage RTP session, RTP timestamp does not need to be monotonically in increasing order. The rtp timestamp will be based on the bestshot capture time, which could be before the current time. The current ntp time of camera can be retrieved from RTCP, and based on that, frame time can be calculated.

Filtering Objects from Playback

NOTE This section is only applicable to NVR with AI capabilities (attributes/System/Support/AIFeatures) and only while performing playback.

Playback url can be used as below:

```
rtsp://<ip>/PlaybackChannel/<chid>/media.smp/BestshotFilter=Face&session=<si
d>
rtsp://<ip>/PlaybackChannel/<chid>/media.smp/BestshotFilter=Vehicle&session=
<sid>
```

If **Require:Bestshot** is absent in the request, BestshotFilter is ignored in the URL.

If each channel in a single playback session (Playback sessions using same session id) carries a different Filter, the last RTSP request supersedes the others.

If FilterType is not present in the request url and **Require:Bestshot** is passed, Person type would be sent by default.

107.9. ReID In Metadata

NOTE This is applicable only for Cameras

If the **WiseAI** application is installed and running, and if it supports ReID, then client can receive ReID information in the metadata.

107.9.1. Capability

The **WiseAI** capability configuration indicates whether ReID is supported,

REQUEST

```
http://<device_IP>/opensdk/WiseAI/configuration/capability
```

RESPONSE (Parameters to check)

```
"reIDSupport": true,
"reIDSupportedTypes": [
"ReIDHumanBody",
```

```
"ReIDHumanFace"
]
```

107.9.2. Requesting ReID in Metadata

To enable ReID metadata, the RTSP SETUP request from the client must include a `Require` header specifying the desired ReID types. These values should match the supported types listed in the WiseAI capability configuration.

Example SETUP request for ReID metadata in RTP session

```
SETUP rtsp://<URL> RTSP/1.0
CSeq: 8
Transport: RTP/AVP/TCP;unicast;interleaved=0-1
Require: ReIDHumanBody,ReIDHumanFace
```

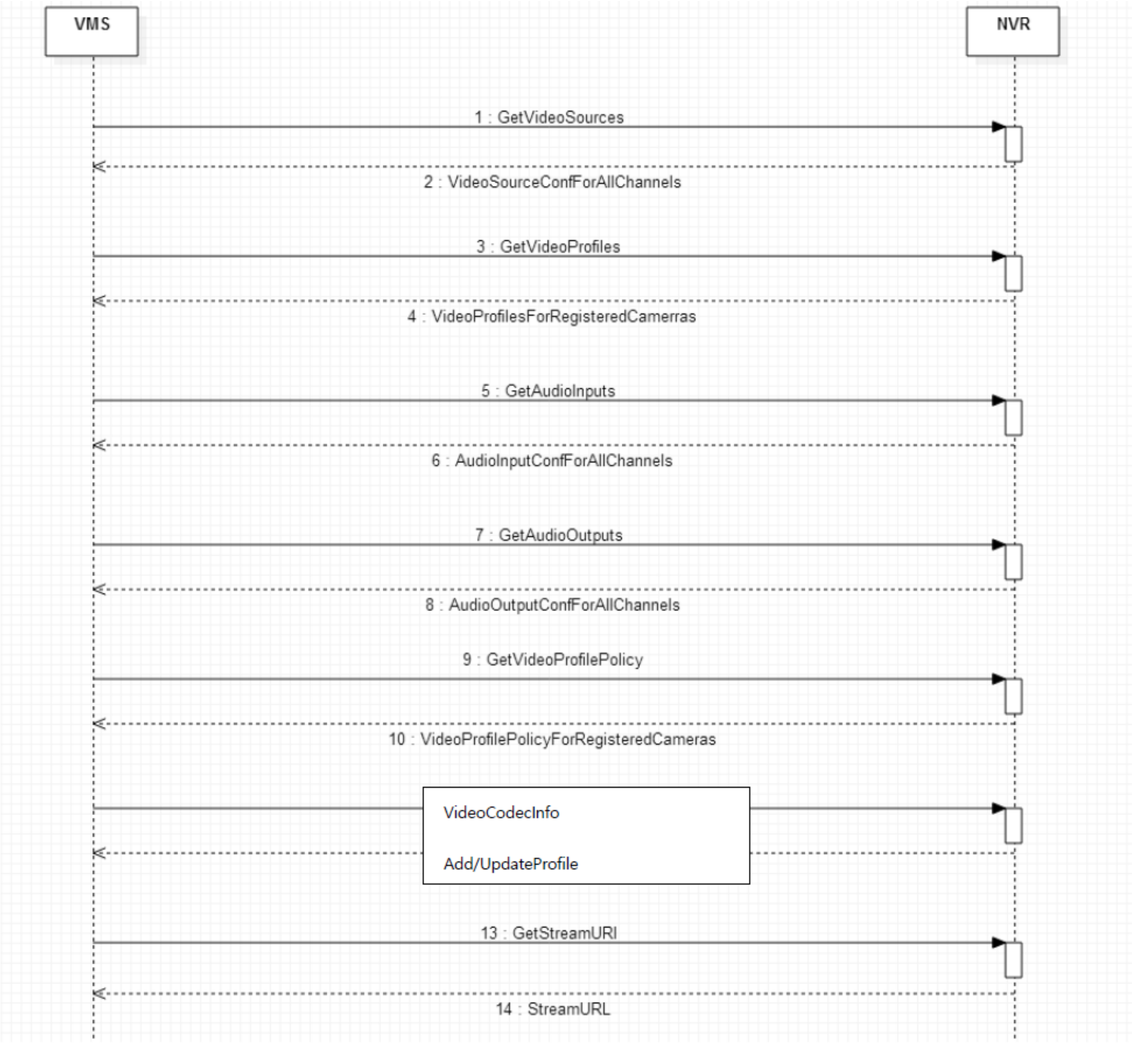
Sample Metadata Format with ReID information

```
<tt:Object ObjectId="11">
    <tt:Appearance>
        <tt:Shape>
            <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
            <tt:CenterOfGravity x="31.0" y="218.0"/>
        </tt:Shape>
        <tt:Class>
            <tt:Type Likelihood="0.8">Human</tt:Type>
        </tt:Class>
        <tt:ImageRef>/download/objectid_11_1548728056100.jpg</tt:ImageRef>
        <tt:ImageRefShape>
            <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
            <tt:CenterOfGravity x="31.0" y="218.0"/>
        </tt:ImageRefShape>
        <tt:ReID Samples="704" Type="Person" BytesPerSample="4"
Endianness="Little" Version="1.0">base64encoded binary</tt:ReID>
    </tt:Appearance>
</tt:Object>
<tt:Object ObjectId="13" Parent="11">
    <tt:Appearance>
        <tt:Shape>
            <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
```

```
            <tt:CenterOfGravity x="31.0" y="218.0"/>
        </tt:Shape>
        <tt:Class>
            <tt:Type Likelihood="0.8">HumanFace</tt:Type>
        </tt:Class>
        <tt:ImageRef>/download/objectid_13_1548728056120.jpg</tt:ImageRef>
        <tt:ImageRefShape>
            <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
            <tt:CenterOfGravity x="31.0" y="218.0"/>
        </tt:ImageRefShape>
        <tt:ReID Samples="513" Type="Face"  BytesPerSample="4"
Endianness="Little" Version="1.0">base64encoded binary</tt:ReID>
    </tt:Appearance>
</tt:Object>
```


NOTE The ReID **Type** can be, "Person" or "Face"

Chapter 108. Simple Connection Progress in Encoder



108.1. Get Video Sources

This command will return information about all video sources, whether a camera is registered or not, and whether video is enabled or not. Users can also identify which channels are connected. Users can request a specific channel's information with channel parameters.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videosource&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Type=NTSC
Channel.0.SensorCaptureSize=Unknown
Channel.0.Name=CAM 01
Channel.0.State=On
Channel.1.Type=NTSC
Channel.1.SensorCaptureSize=Unknown
Channel.1.Name=CAM 02
Channel.1.State=On
Channel.2.Type=NTSC
Channel.2.SensorCaptureSize=Unknown
Channel.2.Name=CAM 03
Channel.2.State=On
```

108.2. Get Video Profiles

This command will return information about the video profiles for all requested channels.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videoprofile&action=view&Channel=0
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.RTPMulticastEnable=False
Channel.0.Profile.1.RTPMulticastAddress=
Channel.0.Profile.1.RTPMulticastPort=0
Channel.0.Profile.1.RTPMulticastTTL=5
Channel.0.Profile.1.Resolution=1280x720
Channel.0.Profile.1.FrameRate=2
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.Bitrate=2560
Channel.0.Profile.1.MJPEG.PriorityType=Bitrate
Channel.0.Profile.1.AudioInputEnable=False
```

```
Channel.0.Profile.1.ViewModeType=Overview
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.ProfileToken=DefaultProfile-01-0
```

108.3. Get Audio Inputs

This command will return information about audio input configuration, such as enabled status, encoding type, etc. for all channels. Audio input counts may differ to channel counts, so you need check the results carefully.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audioinput&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.SampleRate=0
Channel.0.Mode=Mono
Channel.0.EncodingType=G711
Channel.0.Bitrate=0
Channel.0.Gain=1
Channel.1.Enable=True
Channel.1.SampleRate=0
Channel.1.Mode=Mono
Channel.1.EncodingType=G711
Channel.1.Bitrate=0
Channel.1.Gain=1
```

108.4. Get Audio Outputs

This command will return information about the audio talk configuration, such as enabled/disabled status, decoding type, etc. for all channels. Audio input counts may differ from channel counts, so you need check the results carefully.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audiooutput&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.SampleRate=8000
Channel.0.DecodingType=G711
Channel.0.Mode=Mono
Channel.0.Bitrate=64000
Channel.0.Enable=True
Channel.0.Gain=5
```

108.5. Get Video Profile Policy

This command will return information about the configured profile index numbers for Live, Network, and Recording for all registered channels.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofilepolicy&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.DefaultProfile=2
Channel.0.EventProfile=1
Channel.1.DefaultProfile=2
Channel.1.EventProfile=1
Channel.2.DefaultProfile=2
Channel.2.EventProfile=1
Channel.3.DefaultProfile=2
Channel.3.EventProfile=1
Channel.4.DefaultProfile=2
```

```
Channel.4.EventProfile=1
```

108.6. Get resolution information for creating or updating profile

This command will return information about profile configuration (resolution, encoding type, max framerate, etc.. An encoder may have a different resolution or codec limitation for each profile, so users should submit requests with a profile number or use the AllProfiles parameter.

REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=videocodecinfo&action=view&Channel=0&AllProfiles=True
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=MJPEG
Channel.0.ViewMode=Overview
Profile.1.MJPEG.General.1920X1080.Width=1920
Profile.1.MJPEG.General.1920X1080.Height=1080
Profile.1.MJPEG.General.1920X1080.MaxFPS=2000
Profile.1.MJPEG.General.1920X1080.DefaultFPS=2000
Profile.1.MJPEG.General.1920X1080.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.1920X1080.MinCBRTargetBitrate=2048
Profile.1.MJPEG.General.1920X1080.DefaultCBRTargetBitrate=2560
Profile.1.MJPEG.General.1920X1080.MaxVBRTargetBitrate=10240
Profile.1.MJPEG.General.1920X1080.MinVBRTargetBitrate=2048
Profile.1.MJPEG.General.1920X1080.DefaultVBRTargetBitrate=2560
Profile.1.MJPEG.General.1920X1080.IsTruncated=Normal
Profile.1.MJPEG.General.1280X720.Width=1280
Profile.1.MJPEG.General.1280X720.Height=720
Profile.1.MJPEG.General.1280X720.MaxFPS=2000
Profile.1.MJPEG.General.1280X720.DefaultFPS=2000
Profile.1.MJPEG.General.1280X720.MaxCBRTargetBitrate=10240
Profile.1.MJPEG.General.1280X720.MinCBRTargetBitrate=1024
Profile.1.MJPEG.General.1280X720.DefaultCBRTargetBitrate=2048
Profile.1.MJPEG.General.1280X720.MaxVBRTargetBitrate=10240
```

```
Profile.1.MJPEG.General.1280X720.MinVBRTargetBitrate=1024
Profile.1.MJPEG.General.1280X720.DefaultVBRTargetBitrate=2048
```

108.7. Add/update video profile

Users can add a new profile or update to a specific profile with various parameter device supports. Users can get setting parameter limitations by request using the videocodecinfo submenu.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=add&Channel=0&Name=test&EncodingType=H264
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Profile=4
OK
```

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=update&Channel=0&Profile=2&FrameRate=15
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

108.8. Get Stream URI For Live

This command will return the URL to get live streams from a requested channel and profile.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&MediaType=Live&Mode=Full&StreamType=RTPUnicast&TransportProtocol=TCP&RTSPOverHTTP=False&Profile=2
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://192.168.71.123:554/0/profile2/media.smp
```

Chapter 109. Annex A: Metadata Schema

Event Type	Metadata Format	Comments
POS	<tt:MetaDataStream xmlns:tt="http://www.onvif.org/ver10/schema"> <tt:POSData > <tt:POSFrame ID="0" UtcTime="2015-09-10T12:24:57.321"> ..... </tt:POSFrame> </tt:POSData> </tt:MetaDataStream>	Applicable device: NVR
GPS	<tt:MetaDataStream xmlns:tt="http://www.onvif.org/ver10/schema"> <tt:GPSData> <tt:Data Format="GPRMC" UtcTime="2015-09-10T12:24:57.321"> GPRMC,123519,A,4807.038,N,01131.000,E,022.4,084.4,230394,003.1,W*6A </tt:Data> </tt:GPSData> </tt:MetaDataStream>	Applicable device: NVR
Digital Auto Tracking (Area Details Event) ProfileToken: Media Profile Token which is currently used for Digital Auto Tracking	<tt:VideoAnalytics> <tt:Frame UtcTime="2016-03-21T10:54:08.448Z"> <tt:Transformation> <tt:Translate x="-1.0" y="1.0" /> <tt:Scale x="0.000521" y="-0.000926" /> </tt:Transformation> <tt:Object ObjectId="5211"> <tt:Appearance> <tt:Shape> <tt:BoundingBox left="2855.0" top="682.0" right="3242.0" bottom="937.0" /> <tt:CenterOfGravity x="3048.5" y="809.5" /> </tt:Shape> </tt:Appearance> </tt:Object> <tt:Extension> <tnssamsung:DigitalAutoTracking MediaProfileToken="cb4fbc38-e5f6-4ff0-b2e8-2e166b4414d1"/> </tt:Extension> </tt:Frame> </tt:VideoAnalytics>	Applicable device: Camera

Event Type	Metadata Format	Comments
Digital Auto Tracking (State Event) ProfileToken: Media Profile Token which is currently used for Digital Auto Tracking	<tt:MetadataStream> <tt:Event> <wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:VideoSource/tnssamsung:DigitalAutoTracking</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-03-21T11:01:26.346Z" PropertyOperation="Changed"> <tt:Source> <tt:SimpleItem Name="MediaProfileToken" Value="cb4fbc38-e5f6-4ff0-b2e8-2e166b4414d1" /> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="true" /> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage> </tt:Event> </tt:MetadataStream>	Applicable device: Camera
Motion Detection Event ROIId: ROI ID which triggers the event. Index starts from 1. It is a string parameter and it can be CSV value, to support motion happening in multiple ROIS at the same time	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:RuleEngine/MotionRegionDetector/Motion</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-03-31T00:15:58.421Z" PropertyOperation="Initialized"> <tt:Source> tt:SimpleItem Name="VideoSourceToken" Value="cb4fbc38-e5f6-4ff0-b2e8-2e166b4414d1"/> </tt:Source> <tt:Key> <tt:SimpleItemDescription Name="ROIId" Type="1,2,3,4"/> </tt:Key> <tt:Data> tt:SimpleItem Name="State" Value="false"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera

Event Type	Metadata Format	Comments
Fog Detection	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:VideoSource/ImageTooBlurry/ImagingService</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z" PropertyOperation="Initialized"> <tt:Source> <tt:SimpleItem Name="Source" Value="VideoSource1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="false"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera
Sound Classification (Audio Analytics)	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:AudioAnalytics/Audio/DetectedSound </wsnt:Topic> <wsnt:Message> <tt:Message PropertyOperation="Initialized" UtcTime="2016-12-14T15:54:44Z"> <tt:Source> <tt:SimpleItem Value=""AudioSourceConfigurationToken" Name=""AudioSourceConfigurationToken1"/> <tt:SimpleItem Value=""AudioAnalyticsConfigurationToken" Name=""AudioAnalyticsConfigurationToken1"/> <tt:SimpleItem Value=""Rule" Name=""Rule1"/> </tt:Source> <tt:Data> <tt:SimpleItem Value="true" Name="isSoundDetected"/> <tt:SimpleItem Value="2016-12-14T15:54:44Z" Name="UTCTime"/> <tt:SimpleItem Value="gun_shot" Name="AudioClassifications"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	gun_shot, scream, glass_breaking, explosion Applicable device: Camera

Event Type	Metadata Format	Comments
Video Analytics	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics xmlns:tnssamsung=http://www.samsungcctv.com/2011/event/topics">tns1:VideoAnalytics/tnssamsung:VideoAnalytics</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z"> <tt:Source> <tt:SimpleItem Value="VideoSource1" Name="VideoSourceToken"/> </tt:Source> <tt:Data> <tt:SimpleItem Value="0" Name="State"/> <tt:SimpleItem Value="None" Name="Action"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Right, Left, Enter, Exit, Appear/Disapper, Intrusion, Loitering, None Applicable device: Camera
Face Detection	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics xmlns:tnssamsung=http://www.samsungcctv.com/2011/event/topics">tns1:VideoSource/tnssamsung:FaceDetection</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z"> <tt:Source> <tt:SimpleItem Name="VideoSourceToken" Value="VideoSource1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="Face" Value="0"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera

Event Type	Metadata Format	Comments
Tampering / SceneChange	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:VideoSource/GlobalSceneChange/ImagingService</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z" PropertyOperation="Initialized"> <tt:Source> <tt:SimpleItem Name="Source" Value="VideoSource1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="false"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera, NVR
Relay	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:Device/tns1:Trigger/tns1:Relay</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z" PropertyOperation="Initialized"> <tt:Source> <tt:SimpleItem Name="RelayToken" Value="Relay-1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="LogicalState" Value="inactive"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera,NVR

Event Type	Metadata Format	Comments
Digitial Input	<wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:Device/tns1:Trigger/tns1:DigitalInput</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2016-12-14T15:54:44Z" PropertyOperation="Initialized"> <tt:Source> <tt:SimpleItem Name="InputToken" Value="DigitalInput-1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="LogicalState" Value="false"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Applicable device: Camera, NVR
ObjectDetection	<tt:Event> <wsnt:NotificationMessage> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:RuleEngine/ObjectDetection/Object</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2019-11-14T05:50:51.290Z" PropertyOperation="Changed"> <tt:Source> <tt:SimpleItem Name="VideoSource" Value="VideoSourceToken-0"/> <tt:SimpleItem Name="RuleName" Value="ObjectDetectionRule-1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="ClassTypes" Value="Person Vehicle"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage> </tt:Event>	

Event Type	Metadata Format	Comments
Mask Detection	<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:RuleEngine/Detection/Mask</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2020-07-14T15:54:44Z" xmlns:tt="https://www.onvif.org/ver10/schema/"> <tt:Source> <tt:SimpleItem Name="VideoSourceToken" Value="VideoSouceTok1"/> <tt:SimpleItem Name="RuleName" Value="MaskDetectionRule1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="true"/> <tt:SimpleItem Name="ObjectID" Value="100023"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	

Event Type	Metadata Format	Comments
BodyTemperatureDetection	<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:RuleEngine/Detection/BodyTemperature</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2020-07-14T15:54:44Z" xmlns:tt="https://www.onvif.org/ver10/schema/"> <tt:Source> <tt:SimpleItem Name="VideoSourceToken" Value="VideoSouceToken-1"/> <tt:SimpleItem Name="RuleName" Value="BodyTemperatureDetectionRule-1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="true"/> <tt:SimpleItem Name="ObjectID" Value="100023"/> <tt:SimpleItem Name="Temperature" Value="312.0"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	Temperature unit is in Kelvin

Event Type	Metadata Format	Comments
SocialDistancingViolation	<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"> <wsnt:Topic Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2 xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:RuleEngine/Detection/SocialDistancingViolation</wsnt:Topic> <wsnt:Message> <tt:Message UtcTime="2020-11-10T05:51:11Z" xmlns:tt="https://www.onvif.org/ver10/schema/"> <tt:Source> <tt:SimpleItem Name="VideoSource" Value="VideoSouceTok1"/> <tt:SimpleItem Name="RuleName" Value="SocialDistancingViolationRule-1"/> </tt:Source> <tt:Data> <tt:SimpleItem Name="State" Value="true"/> <tt:SimpleItem Name="ObjectIDs" Value="100023 100020 110221"/> </tt:Data> </tt:Message> </wsnt:Message> </wsnt:NotificationMessage>	

Chapter 110. Annex B: Frame Metadata Sample

110.1. Non-AI-supported models

Non-AI-supported models only support object information with bounding box.

Here the boundingbox follows the onvif spec and we can get the normalized bounding box with the help of translate and scale.

```
<tt:MetadataStream>
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2020-07-01T18:28:00.146Z">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0"/>
                <tt:Scale x="0.000521" y="-0.000926"/>
            </tt:Transformation>
            <tt:Object ObjectId="489">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="2990.0" top="1159.0"
right="3751.0" bottom="2096.0"/>
                        <tt:CenterOfGravity x="3370.5" y="1627.5"/>
                    </tt:Shape>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

110.2. BodyTemperatureDetection frame metadata example

```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
    xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
    xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2019-05-15T12:24:57.321">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0" />
                <tt:Scale x="0.000781" y="-0.001042" />
            </tt:Transformation>
```

```
            <tt:Object ObjectId="15" Parent="12">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0" />
                        <tt:CenterOfGravity x="31.0" y="218.0" />
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Face</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Face</tt:Type>
                    </tt:Class>
                    <tt:HumanFace>
                        <fc:Temperature>311.75</fc:Temperature>
                    </tt:HumanFace>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

110.3. AI-supported Models

Those marked in Blue are custom extensions for our camera that we defined:

- **Parent :** Describes the relationship between objects. e.g. Vehicle and Number Plate (Vehicle object is parent to Number plate), Human and Face (Human is parent to the Face)
- **ImageRef:** Temporary image stored in the camera or central server for the client to retrieve
- **Possible ClassTypes:** Human, Face, LicensePlate, Vehicle, Head, and Unknown

See the below section to get the class type information along with the confidence level.

```
<tt:Class>
    <tt:ClassCandidate>
        <tt:Type>LicensePlate</tt:Type>
        <tt:Likelihood>0.8</tt:Likelihood>
    </tt:ClassCandidate>
    <tt:Type Likelihood="0.8">LicensePlate</tt:Type>
</tt:Class>
```

The table of supported attributes

> **NOTE** The attribute fields described below in the metadata are optional, and the onvif metadata schema is as follows.

	Objects	Attributes	Supported attributes items
Attributes	Person	Gender	Female, Male
		Upper (Color)	Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple (up to 2 colors at the same time)
		Lower (Color)	
		Bag	Bag (If a Bag is detected)
	Vehicle	Type	Car (Sedan/SUV/Van...etc), Bus, Truck, Motorcycle, Bicycle
		Color	Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple (up to 2 colors at the same time)
	Face	Gender	Female, Male
		Age	Young (0-19), Adult (20-44), Middle-aged (45-64), Senior (65-)
		Mask	Wearing Mask or Not
		Glasses	Wearing Glasses or Not
	Licenseplate		

Sample Meta Frame

Onvif Metadata spec is used to notify the object-related attributes. Those marked in BLUE are custom extensions to ONVIF metadata spec.

```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2019-05-15T12:24:57.321">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0"/>
                <tt:Scale x="0.000781" y="-0.001042"/>
            </tt:Transformation>
            <tt:Object ObjectId="15" Parent="12">
                <tt:Appearance>
                    <tt:Shape>
```

```
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
                        <tt:CenterOfGravity x="31.0" y="218.0"/>
                    </tt:Shape>
                    <tt:Color>
                        <tt:ColorCluster>
                            <tt:Color X="58" Y="105" Z="212"/>
                            <tt:Covariance XX="7.2" YY="6" ZZ="3"/>
                            <tt:Weight>90</tt:Weight>
                            <tt:ColorString>WHITE</tt:ColorString>
                        </tt:ColorCluster>
                        <tt:ColorCluster>
                            <tt:Color X="165" Y="44" Z="139"/>
                            <tt:Covariance XX="4" YY="4" ZZ="4"/>
                            <tt:Weight>5</tt:Weight>
                            <tt:ColorString>BLUE</tt:ColorString>
                        </tt:ColorCluster>
                    </tt:Color>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>LicensePlate</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">LicensePlate</tt:Type>
                    </tt:Class>
                    <tt:VehicleInfo>
                        <tt:Type Likelihood="0.8">car</tt:Type>
                    </tt:VehicleInfo>
                    <tt:HumanFace>
                        <fc:Gender>Male</fc:Gender>
                        <fc:AgeType>Adult</fc:AgeType>
                        <fc:Accessory>
                            <fc:Opticals>
                                <fc:Wear>true</fc:Wear>
                            </fc:Opticals>
                            <fc:Mask>
                                <fc:Wear>true</fc:Wear>
                            </fc:Mask>
                        </fc:Accessory>
                    </tt:HumanFace>
                    <tt:HumanBody>
```

```
                        <bd:Gender>Male</bd:Gender>
                        <bd:Clothing>
                            <bd:Tops>
                                <bd:Color>
                                    <tt:ColorCluster>
                                        <tt:Color X="58" Y="105" Z="212"/>
                                        <tt:Covariance XX="7.2" YY="6"
ZZ="3"/>
                                        <tt:Weight>90</tt:Weight>

<tt:ColorString>WHITE</tt:ColorString>
                                    </tt:ColorCluster>
                                    <tt:ColorCluster>
                                        <tt:Color X="165" Y="44" Z="139"/>
                                        <tt:Covariance XX="4" YY="4"
ZZ="4"/>
                                        <tt:Weight>5</tt:Weight>

<tt:ColorString>BLUE</tt:ColorString>
                                    </tt:ColorCluster>
                                </bd:Color>
                                <bd:Length>Long</bd:Length>
                            </bd:Tops>
                            <bd:Bottoms>
                                <bd:Color>
                                    <tt:ColorCluster>
                                        <tt:Color X="58" Y="105" Z="212"/>
                                        <tt:Covariance XX="7.2" YY="6"
ZZ="3"/>
                                        <tt:Weight>90</tt:Weight>

<tt:ColorString>WHITE</tt:ColorString>
                                    </tt:ColorCluster>
                                    <tt:ColorCluster>
                                        <tt:Color X="165" Y="44" Z="139"/>
                                        <tt:Covariance XX="4" YY="4"
ZZ="4"/>
                                        <tt:Weight>5</tt:Weight>

<tt:ColorString>BLUE</tt:ColorString>
                                    </tt:ColorCluster>
```

```
                                </bd:Color>
                                <bd:Length>Long</bd:Length>
                            </bd:Bottoms>
                        </bd:Clothing>
                        <bd:Belonging>
                            <bd:Bag>
                                <bd:Category>Bag</bd:Category>
                            </bd:Bag>
                        </bd:Belonging>
                    </tt:HumanBody>

<tt:ImageRef>../download/objectid_1_1548728068_100.jpg</tt:ImageRef>
                    <tt:ImageRefShape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
                        <tt:CenterOfGravity x="31.0" y="218.0"/>
                    </tt:ImageRefShape>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

ImageRef

A URL can also have absolute address

```
<tt:ImageRef>http://192.168.75.150/download/objectid_1_1548728068_100.jpg</t
t:ImageRef>
```

Metadata Sample for SocialdistancingViolation:

In the below Meta, the **ProximateObjects** section provides the distance between objects in meters.

```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2019-05-15T12:24:57.321">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0"/>
```

```
                <tt:Scale x="0.000781" y="-0.001042"/>
            </tt:Transformation>
            <tt:Object ObjectId="15">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
                        <tt:CenterOfGravity x="31.0" y="218.0"/>
                    </tt:Shape>
                    <tt:Color>
                        <tt:ColorCluster>
                            <tt:Color X="58" Y="105" Z="212"/>
                            <tt:Covariance XX="7.2" YY="6" ZZ="3"/>
                            <tt:Weight>90</tt:Weight>
                            <tt:ColorString>WHITE</tt:ColorString>
                        </tt:ColorCluster>
                    </tt:Color>
                    <tt:ProximateObjects>
                        <tt:ProximateObject Id="11" Distance="0.5"/>
                        <tt:ProximateObject Id="16" Distance="0.5"/>
                    </tt:ProximateObjects>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Human</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Human</tt:Type>
                    </tt:Class>

<tt:ImageRef>http://192.168.75.150/download/objectid_1_1548728068_100.jpg</t
t:ImageRef>
                    <tt:ImageRefShape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0"/>
                        <tt:CenterOfGravity x="31.0" y="218.0"/>
                    </tt:ImageRefShape>
                </tt:Appearance>
            </tt:Object>

<tt:SceneImageRef>http://192.168.75.150/download/FullImage_id.jpg</tt:SceneI
mageRef>
```

```
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

SceneImageRef

```
<tt:SceneImageRef>/download/objectid_212_1637556386253.jpg</tt:SceneImageRef
>
```

"Metadata Sample for ObjectDetection"

In the below Meta, the "SceneImageRef" tag provides a URL to download the full screen image, and "SceneImageInformation" in "Extension" provides detailed information about the object detected in the full screen image (SceneImageRef).

```
<tt:MetadataStream>
  <tt:VideoAnalytics>
    <tt:Frame UtcTime="2021-11-22T04:46:16.410Z">
      <tt:Transformation>
        <tt:Translate x="-1.0" y="1.0"/>
        <tt:Scale x="0.000772" y="-0.001316"/>
      </tt:Transformation>
      <tt:Object ObjectId="212">
        <tt:Appearance>
          <tt:Shape>
            <tt:BoundingBox left="907.0" top="1202.0" right="1438.0"
bottom="1463.0"/>
            <tt:CenterOfGravity x="1172.5" y="1332.5"/>
          </tt:Shape>
          <tt:Class>
            <tt:ClassCandidate>
                <tt:Type>Vehical</tt:Type>
                <tt:Likelihood>0.8</tt:Likelihood>
            </tt:ClassCandidate>
            <tt:Type Likelihood="0.8">Vehicle</tt:Type>
          </tt:Class>
          <tt:VehicleInfo>
            <tt:Type>Bus</tt:Type>
          </tt:VehicleInfo>

<tt:ImageRef>/download/objectid_212_1637556386253.jpg</tt:ImageRef>
```

```
          <tt:ImageRefShape>
            <tt:BoundingBox left="862.0" top="1173.0" right="1458.0"
bottom="1487.0"/>
            <tt:CenterOfGravity x="1160.0" y="1330.0"/>
          </tt:ImageRefShape>
        </tt:Appearance>
      </tt:Object>

<tt:SceneImageRef>/download/objectid_212_1637556386253.jpg</tt:SceneImageRef
>
      <tt:Extension>
        <tnssamsung:SceneImageInformation>
          <tt:Object ObjectId="212">
            <tt:Appearance>
              <tt:Shape>
                <tt:BoundingBox left="862.0" top="1173.0" right="1458.0"
bottom="1487.0"/>
                <tt:CenterOfGravity x="1160.0" y="1330.0"/>
              </tt:Shape>
            </tt:Appearance>
          </tt:Object>
        </tnssamsung:SceneImageInformation>
      </tt:Extension>
    </tt:Frame>
  </tt:VideoAnalytics>
</tt:MetadataStream>
```

110.4. AI NVR Metadata Samples

Face Recognition Event

Face recognition event is notified in the ONVIF format as below:

```
<wsnt:NotificationMessage>
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Ru
leEngine/Recognition/Face</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-09T10:35:31Z"
PropertyOperation="Changed">
            <tt:Source>
```

```
                <tt:SimpleItem Name="VideoSourceConfiguration"
Value="Source_CH-3"/>
                <tt:SimpleItem Name="VideoAnalyticsConfiguration"
Value="VideoAnalyticsConf_CH-3"/>
                <tt:SimpleItem Name="RuleName" Value="FaceRecognitonRule_CH-
3"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="Likelihood" Value="0.943513"/>
                <tt:SimpleItem Name="ImageUri" Value="/stw-
cgi/ai.cgi?msubmenu=imageget&amp;action=view&amp;type=bestshotBuffer&amp;ID=
0000000000000000_37_1_2_2020-07-09T10&#58;35&#58;31Z_21"/>
                <tt:SimpleItem Name="ImageCaptureTimeUTC" Value="2020-07-
09T10:35:31Z"/>
                <tt:SimpleItem Name="EnrollmentID" Value="□□ 1/iii"/>
                <tt:SimpleItem Name="RefImageUri" Value="/stw-
cgi/ai.cgi?msubmenu=imageget&amp;action=view&amp;type=imagelibrary&amp;ID=fa
ce_1001_78"/>
                <tt:ElementItem Name="BoundingBox">
                    <tt:Rectangle left="-0.153425" top="-0.538889"
right="0.168533" bottom="0.215741"/>
                </tt:ElementItem>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```


Face Detection Event


```
<wsnt:NotificationMessage>
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Vi
deoSource/DynamicEvent</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-09T10:35:32Z">
            <tt:Source>
                <tt:SimpleItem Name="Source" Value="3"/>
                <tt:SimpleItem Name="EventName"
Value="ObjectDetection.Face"/>
            </tt:Source>
            <tt:Data>
```

```
                <tt:SimpleItem Name="State" Value="true"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

Face Metadata

Face metadata is notified in the ONVIF metadata format as below

```
<?xml version="1.0" encoding="UTF-8"?>
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:ttr="https://www.onvif.org/ver20/analytics/radiometry"
xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"
xmlns:tns1="http://www.onvif.org/ver10/topics"
xmlns:tnssamsung="http://www.samsungcctv.com/2011/event/topics"
xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2020-07-09T10:09:33Z">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0"/>
                <tt:Scale x="0.000521" y="-0.000926"/>
            </tt:Transformation>
            <tt:Object ObjectId="357701">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="1503.0" top="570.0"
right="2095.0" bottom="1298.0"/>
                        <tt:CenterOfGravity x="1799.0" y="934.0"/>
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Other</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Head</tt:Type>
                    </tt:Class>
                </tt:Appearance>
            </tt:Object>
            <tt:Object ObjectId="357702" Parent="357703">
```

```
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="1566.0" top="710.0"
right="2044.0" bottom="1270.0"/>
                        <tt:CenterOfGravity x="1805.0" y="990.0"/>
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Face</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Face</tt:Type>
                    </tt:Class>
                    <fc:HumanFace>
                        <fc:Gender>Male</fc:Gender>
                        <fc:AgeType>Adult</fc:AgeType>
                        <fc:Accessory>
                            <fc:Opticals>
                                <fc:Wear>False</fc:Wear>
                            </fc:Opticals>
                            <fc:Hat>
                                <fc:Wear>False</fc:Wear>
                            </fc:Hat>
                        </fc:Accessory>
                    </fc:HumanFace>
                    <tt:ImageRef UtcTime="2020-07-
09T10:09:33Z">0000000000000000_37_1_2_2020-07-
09T10:09:33Z_357702</tt:ImageRef>
                    <tt:ImageRefShape>
                        <tt:BoundingBox left="1447.0" top="486.0"
right="2163.0" bottom="1326.0"/>
                        <tt:CenterOfGravity x="1805.0" y="906.0"/>
                    </tt:ImageRefShape>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```


License Plate Detection Event

```
<wsnt:NotificationMessage>
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Vi
deoSource/DynamicEvent</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-09T10:34:39Z">
            <tt:Source>
                <tt:SimpleItem Name="Source" Value="3"/>
                <tt:SimpleItem Name="EventName"
Value="ObjectDetection.LicensePlate"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="State" Value="true"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

LPR Metadata

LPR Metadata follows ONVIF metadata schema

```
<?xml version="1.0" encoding="UTF-8"?>
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
xmlns:ttr="https://www.onvif.org/ver20/analytics/radiometry"
xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2"
xmlns:tns1="http://www.onvif.org/ver10/topics"
xmlns:tnssamsung="http://www.samsungcctv.com/2011/event/topics"
xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2020-07-09T10:08:00Z">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0"/>
                <tt:Scale x="0.000521" y="-0.000926"/>
            </tt:Transformation>
            <tt:Object ObjectId="357699">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="1998.0" top="628.0"
right="3323.0" bottom="1759.0"/>
```

```
                        <tt:CenterOfGravity x="2660.5" y="1193.5"/>
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Vehical</tt:Type>
                            <tt:Likelihood>0.7</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.7">Vehicle</tt:Type>
                    </tt:Class>
                </tt:Appearance>
            </tt:Object>
            <tt:Object ObjectId="357700" Parent="357699">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="0.0" top="0.0" right="0.0"
bottom="0.0"/>
                        <tt:CenterOfGravity x="0.0" y="0.0"/>
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>LicensePlate</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">LicensePlate</tt:Type>
                    </tt:Class>
                    <tt:ImageRef UtcTime="2020-07-
09T10:07:50Z">0000000000000000_37_4_2_2020-07-
09T10:07:50Z_357700</tt:ImageRef>
                    <tt:ImageRefShape>
                        <tt:BoundingBox left="1212.0" top="1229.0"
right="1881.0" bottom="1549.0"/>
                        <tt:CenterOfGravity x="1546.5" y="1389.0"/>
                    </tt:ImageRefShape>
                    <tt:LicensePlateInfo>
                        <tt:PlateNumber
Likelihood="1">19N1004</tt:PlateNumber>
                    </tt:LicensePlateInfo>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
```

```
</tt:MetadataStream>
```

110.5. Frame metadata example when Metadata sharing is turned on

When the Metadata Share function is turned on, the "VideoSourceToken" attritube is included in the tt:Frame tag to indicate which video source the metadata originated from.

```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
    xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
    xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        < tt:Frame UtcTime="2019-05-15T12:24:57.321"
VideoSourceToken="VideoSourceToken-1">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0" />
                <tt:Scale x="0.000781" y="-0.001042" />
            </tt:Transformation>
            <tt:Object ObjectId="15" Parent="12">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0" />
                        <tt:CenterOfGravity x="31.0" y="218.0" />
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Face</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Face</tt:Type>
                    </tt:Class>
                    <tt:HumanFace>
                        <fc:Temperature>311.75</fc:Temperature>
                    </tt:HumanFace>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

Image Adjustment

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 111. Overview

111.1. Description

image.cgi configures camera features such as SSDR, white balance, exposure and OSD. This document describes how to configure each function in detail.

The following submenus are used for image settings.

- **camera**: Configures various settings such as BLC, HLC, Iris, SSNR, etc.
- **whitebalance**: Configures the relative amount of red, green and blue primary colors in the image so that neutral colors are reproduced correctly.
- **imageenhancements**: Adjusts the image sharpness, brightness, saturation, etc.
- **imageenhancements2**: Image setting can be configured with imagepresetmode.
- **irled**: Configures IR LED settings.
- **flip**: Flips the image horizontally/vertically.
- **ssdr**: Configures the SSDR (Samsung Super Dynamic Range) settings.
- **focus**: Changes the focus settings according to the environment.
- **overlay**: Configures OSD (On-screen Display) on the image.
- **multilineosd**: Configures multiple lines of OSD (On-screen Display) on the image.
- **multiimageosd:** Supports overlaying images on the live stream.
- **smartcodec**: Configures smart codec settings.
- **privacy**: Specifies a certain area of the video to be protected for privacy.
- **fisheyelens**: Configures the settings for fisheye lenses.
- **fisheyesetup**: Configures the settings, mount position and view modes for fisheye lenses.
- **viewmodes**: Specifies the viewing mode.
- **imagepreset**: Configures the image preset schedule settings.
- **imagepreset2**: Configures each image preset mode.
- **imagealignment**: Configures the image alignment of multiple sensors.
- **imageoptions**: Explains the dependency between various features in image cgi.
- **ptr**: Configures the PTR settings.
- **ptrzusage**: Provides information on the number of times the PTR feature was used, to track durability.
- **autoimagealignment**: Adjusts panoramic camera’s panoramic image.
- **imagepresetschedule**: Configures image preset mode’s scheduling.
- **thermalpalettesetting**: Configures the thermal color palette, upper/lower temperature level, emissivity, and color bar overlay.
- **thermalpalettesettingoptions**: Provides temperature unit’s configurable temperature range.

- **spottemperaturereading**: Provides requested spot's temperature information.
- **directionindicator**: Configures settings about direction indicator.
- **noisereduct**: Provides features related image quality control for an analog camera connected to a Hybrid NVR.
- **ptrpreset**: Provides features that can configure settings for ptr presets.
- **stereosensorcalibration**: Calibrates the coordinates of both sensors, which is performed by superimposing a thermal image over a non-thermal image based on the scale and adjustment of the offset.
- **radiometrysettings**: Configures radiometry settings for the camera.
- **radiometrysettingsoptions**: Requests information about radiometry settings.
- **blackbodyconfig**: Configures the use of black body device for the camera.
- **blackbodyconfigoptions**: Requests information about black body settings.
- **focuspreset**: Supports how to configure presets for each focus.
- **thermalnuc**: Thermal camera non uniformity correction settings.
- **lpcapturesetup**: Supports to install camera for LPR capturing.
- **wiseautofocus**: Wiseautofocus related settings.
- **autofocuszone**: autofocus zone setup.
- **whiteled**: Configures White LED settings.
- **dynamicprivacymask**: Configures dynamic privacymask settings.

Chapter 112. Camera

112.1. Description

The **camera** submenu configures various settings, such as BLC, HLC, Iris, and SSNR.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

112.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
camera&action=<value>[&<parameter>=<value>]
```

112.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the camera settings such as BLC, HLC, SSNR, etc.
	Channel	REQ, RES	<csv>	Channel ID
	LensModel	RES	<enum> SLA-T46XX, SLA-T10XX, SLA-T10XX/SLA-T16XX	Lens models of the camera This parameter is only shown in case of model that support changing lens. **Note** It is not applied for the fisheye camera.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.SSNRMode	RES	<enum> Off, Manual, Auto, AIAuto	SSNR Mode • Off: Disables SSNR mode. • Manual: SSNR manual mode • Auto: SSNR auto mode • AIAuto: SSNR auto mode using AI engine (AI engine required) **SSNRMode** Auto Mode and **SSNREnable** Parameter true values are the same. These two parameters should not be set together. **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.
	ImagePresetMode.#.LCEMode	RES	<enum> Full, Horizontal, Sky, Ground, Center75%, Center50%, Center25%, Custom	LCEMode Mode can display the screen with a high contrast based on the set area. • Full: Full screen. • Horizontal: Rectangular area in the middle of the screen. • Sky: Rectangular area at the upper part of the screen • Ground: Rectangular area at the lower part of the screen • Center75%: 75% Area size from the center • Center50%: 50% Area size from the center • Center25%: 25% Area size from the center • Custom: If LCEMode is set to this, LCETop, LCEBottom, LCELeft, LCERight parameters can be set.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.LCETop	RES	<int>	Top position info for the LCE region It sets the top position based on the distance from the upper end of the screen LCEBottom should be greater than LCETop
	ImagePresetMode.#.LCEBottom	RES	<int>	Bottom position info for the LCE region It sets the bottom position based on the distance from the lower end of the screen LCEBottom should be greater than LCETop
	ImagePresetMode.#.LCELeft	RES	<int>	Left position info for the LCE region It sets the left position based on the distance from the left end of the screen LCERight should be greater than LCELeft
	ImagePresetMode.#.LCERight	RES	<int>	Right position info for the LCE region It sets the right position based on the distance from the right end of the screen LCERight should be greater than LCELeft
	ImagePresetMode.#.CompensationMode	RES	<enum> Off, BLC, HLC, WDR	Compensation mode • Off: Disables the compensation function • BLC: Backlight compensation • HLC: High Light compensation • WDR: Wide Dynamic Range

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.BLCLevel	RES	<enum> Low, Medium, High	BLC (Backlight Compensation) level The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC.
	ImagePresetMode.#.NormalizedBLCLevel	RES	<int>	Controls BLC (Backlight Compensation) level, but this adjusts the level in more detail than BLCLevel does.
	ImagePresetMode.#.BLCAreaTop	RES	<int>	Top position info for the BLC region It sets the top position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.BLCAreaBottom	RES	<int>	Bottom position info for the BLC region It sets the bottom position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.
	ImagePresetMode.#.BLCAreaLeft	RES	<int>	Left position info for the BLC region It sets the left position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.BLCAreaRight	RES	<int>	Right position info for the BLC region It sets the right position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.
	ImagePresetMode.#.BLCAreaCoordinates	REQ,RES	<string>	Sets BLC region as real coordinates. This is supported in certain models only, so users should check their attirbutes first to see if their model supports this feature. BLCAreaCoordinates value must be specified in the format of <x1,y1,x2,y2>.
	ImagePresetMode.#.HLCMode	RES	<enum> NightOnly, AllDay, On, Off	HLC (High Light Compensation) mode
	ImagePresetMode.#.HLCLevel	RES	<enum> Low, Medium, High	HLC (High Light Compensation) level **HLCLevel** is valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.NormalizedHLCLevel	RES	<int>	Controls HLC (High Light Compensation) level, but in more detail than HLCLevel does.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.HLC MaskTone	RES	<int>	Tone of the mask used for a high light compensation area **HLCMaskTone** is valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.HLC MaskColor	RES	<enum> Black, Blue, Red, Cyan, Magenta	Color of the mask used for a high light compensation area **HLCMaskColor** is valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.HLC Dimming	RES	<enum> "On", "Off"	Enables or disables the HLC Dimming feature. If enabled, the brightness of the image is adjusted automatically according to the surrounding environment
	ImagePresetMode.#.HLC AreaTop	RES	<int>	Top position info for the HLC region It sets the top position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.HLC AreaBottom	RES	<int>	Bottom position info for the HLC region It sets the bottom position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.HLCAreaLeft	RES	<int>	Left position info for the HLC region It sets the left position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.HLCAreaRight	RES	<int>	Right position info for the HLC region It sets the right position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	ImagePresetMode.#.HLCAreaCoordinates	REQ,RES	<string>	Sets HLC region with real coordinates. This is supported in certain models only, so users should check their attirbutes first to see if their model supports this feature. HLCAreaCoordinates value must be expressed in the form of <x1,y1,x2,y2>.
	ImagePresetMode.#.WDRLevel	RES	<enum> Low, Medium, High	WDR (Wide Dynamic Range) sensitivity level **WDRLevel** is valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRLimit	RES	<int>	WDR limit **WDRLimit** is valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRBlackLevel	RES	<enum> Low, Medium, High	WDR black level **WDRBlackLevel** is valid only when **CompensationMode** is set to WDR.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.WDRWhiteLevel	RES	<enum> Low, Medium, High	WDR white level **WDRWhiteLevel** is valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRControlMode	RES	<enum> AlwaysOn, Auto, TurnOffInLowLight, TurnOffInBW	WDRControlMode • AlwaysOn: WDRSeamlessTransition, WDRLowLight, WDRIRLEDEnable are not enabled • TurnOffInLowLight: WDRLowLight • TurnOffInBW: WDRIRLEDEnable **Note** For more details on the parameter features, please see the description of each parameter (WDRSeamlessTransition, WDRLowLight, WDRIRLEDEnable).
	ImagePresetMode.#.WDRAreaTop	RES	<int>	Top position info for WDR region It sets the top position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRAreaBottom	RES	<int>	Bottom position info for WDR region It sets the bottom position based on the distance from the center of the screen. The **HLCLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.WDRAreaLeft	RES	<int>	Left position info for WDR region It sets the left position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft** , and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRAreaRight	RES	<int>	Right position info for WDR region It sets the right position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	ImagePresetMode.#.WDRAreaCoordinates	RES	<string>	Sets WDR region with real coordinates. This is supported in certain models only, so users should check the attributes first to see if their model supports this feature. WDRAreaCoordinates value must be specified in the format of <x1,y1,x2,y2>.
	ImagePresetMode.#.WDRSeamlessTransition	RES	<enum> Off, On	Enables or disables WDR Seamless Transition. If enabled, the image is analyzed and WDR mode is activated automatically when backlight correction is necessary.
	ImagePresetMode.#.WDRLowLight	RES	<enum> Off, On	Enables or disables WDR Low Light. If enabled, the WDR mode is turned off automatically in a low-light environment.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.WDRIRLEDEnable	RES	<enum> Off, On	Enables or disables the WDR IR LED feature. If enabled, the WDR mode is turned off automatically when the B/W mode operates.
	ImagePresetMode.#.WDRType	RES	<enum> ExposureControl, GainControl	WDR operation mode. **ExposureControl** is the default value when WDR is enabled. • ExposureControl: Captures two frames, one with short exposure for bright areas and the other with a long exposure for dark areas, and composites them to correct backlighting. Default mode. When this value is enabled then **WDRLevel** can be configured. When this parameter is not shown in attributes.cgi, then only ExposureControl operation mode is supported. • GainControl: Captures a single frame at a different gain level and composites the two images, which has the advantage of reducing ghosting.
	ImagePresetMode.#.ShutterMode	RES	<enum> Auto, Manual, ESC, AFLK	Shutter mode • Auto: Adjusts the shutter speed automatically. • Manual: Shutter speed of the camera can be adjusted manually. • ESC: Electronic Shutter Control; adjusts the shutter speed automatically according to the ambient brightness. • AFLK: Anti Flicker; Adjusts the shutter speed when a screen flickers frequently due to a mismatch with the ambient lighting.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.PreferedShutterSpeed	RES	<enum>	Preferred shutter speed **PreferedShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; This parameter is read-only if **ShutterMode** is set to AFLK.
	ImagePresetMode.#.ManualShutterSpeed	RES	<enum>	Manual shutter speed **ManualShutterSpeed** is valid only when **ShutterMode** is set to Manual.
	ImagePresetMode.#.AutoLongShutterSpeed	RES	<enum>	Maximum shutter speed **AutoLongShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; this parameter is read-only if **ShutterMode** is set to AFLK.
	ImagePresetMode.#.AutoShortShutterSpeed	RES	<enum>	Minimum shutter speed **AutoShortShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; This parameter is read-only if **ShutterMode** is set to AFLK.
	ImagePresetMode.#.PreferShutterAISupportEnable	RES	<bool> True, False	Enables or disables auto PreferShutter using AI engine support **Note** **PreferShutterAISupportEnable** requires the camera to have an AI engine.
	ImagePresetMode.#.AFLKMode	RES	<enum> Off, On, 50, 60	AFLK (Anti-Flicker) mode
	ImagePresetMode.#.SimpleFocus	RES	<bool> True, False	Automatic focus adjustment after day and night **Note** Attribute to check for SimpleFocus: "attributes/Image/Support/SimpleFocus"
	ImagePresetMode.#.NegativeModeEnable	RES	<bool> True, False	Enables or disables negative video mode

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.LensModel	RES	<enum> SLA-T46XX, SLA-T10XX, SLA-T10XX/SLA-T16XX	Not applied for fisheye models Enum values of different lenses for cameras on which the lens can be replaced.
	ImagePresetMode.#.IrisMode	RES	<enum> Auto, Manual, P-Iris, P-Iris-SLAM3180PN, P-Iris-M13VP288IR, P-Iris-SLAM2890PN, ICS	Lens Iris mode • Auto: Automatically adjusts the iris. • Manual: Adjusts the iris and focus manually. • P-Iris: Controls the iris of the lens equipped with a step motor.
	ImagePresetMode.#.PIrisMode	RES	<enum> Auto, Manual	P-Iris (Precise Iris) mode • Auto: Automatically adjusts the amount of surrounding light. • Manual: Adjusts the iris manually. • **PIrisMode** is valid only when **IrisMode** is set to P-Iris. **Note** For P-Iris support better to check "attributes/Image/Support/P-Iris" and "cgis/image/camera" submenu PIrisMode parameter. In some dome models, which only supports Auto Piris, PIrisMode parameter may not be supported, eventhough "attributes/Image/Support/P-Iris" support would be true.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.PIrisPosition	RES	<int>	P-Iris (Precise Iris) position **PIrisPosition** is valid only when **IrisMode** is set to P-Iris and **PIrisMode** is set to Manual. The value should be specified in even numbers only.
	ImagePresetMode.#.IrisFno	RES	<enum>	Iris F-stop number **IrisFno** is valid only when **IrisMode** is set to P-Iris or Manual.
	ImagePresetMode.#.SensUpMode	RES	<enum> Off, Auto	Enables or disables Sens-up mode Sens-up function makes a camera more sensitive to light in order to produce clear images even in low light conditions. **SensUpMode** and **SensUpLevel** are valid only when **ShutterMode** is not set to Manual or AFLK. The **SensUpMode** and **SensUpLevel** parameters are not available if **AGCMode** is set to Off.
	ImagePresetMode.#.SensUpLevel	RES	<enum>	Sens-up Level **SensUpMode** and **SensUpLevel** are not available if **AGCMode** is set to Off. **SensUpLevel** parameter is valid only when **SensUpMode** is set to Auto.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.AGCMode	RES	<enum> Off, Low, Medium, High, Manual, MaxGain	AGC (Automatic Gain Control) mode Adjusts the gain value of the video to control the video brightness. • Off: Disables AGC. • Low/ Medium /High: As the level increases to high, the screen is brighter in low lighting. • Manual: Adjusts the AGC level manually • MaxGain: Automatically adjusts the gain level to the **MaxGainLevel** value and less. If **AGCMode** is set to Off or Manual, **DayNightMode** cannot be set to Auto; therefore, **SSNREnable**, **SSNRLevel**, **SensupModeEnable**, and **SensupLevel** are invalid. And if AGCMode is set to MaxGain and MaxGainLevel less than 30dB, DayNightMode cannot be set to Auto
	ImagePresetMode.#.AGCLevel	RES	<int>	AGC level **AGCLevel** is valid only when **AGCMode** is set to Manual.
	ImagePresetMode.#.AGCMaxGainLevel	RES	<int>	AGC max gain level **AGCMaxGainLevel** is valid only when **AGCMode** is set to MaxGain
	ImagePresetMode.#.SSNREnable	RES	<bool> True, False	Enables or disables SSNR (Samsung Super Noise Reduction). If **AGCMode** is set to Off, **SSNREnable** and **SSNRLevel** parameters are not valid.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.SSNR2DLevel	RES	<int>	SSNR 2D level If **AGCMode** is set to Off, **SSNREnable** and **SSNR2DLevel** parameters are not valid. **SSNR2DLevel** is valid only when **SSNREnable** is set to True. **SSNR2DLevel** and **SSNRLevel** cannot be set at the same time.
	ImagePresetMode.#.SSNR3DLevel	RES	<int>	SSNR 3D level If **AGCMode** is set to Off, **SSNREnable** and **SSNR3DLevel** parameters are not valid. **SSNR3DLevel** is valid only when **SSNREnable** is set to True. **SSNR3DLevel** and **SSNRLevel** cannot be set at the same time.
	ImagePresetMode.#.SSNRLevel	RES	<int>	SSNR level If **AGCMode** is set to Off, **SSNREnable** and **SSNRLevel** parameters are not valid. **SSNRLevel** is valid only when **SSNREnable** is set to True. Changing SSNRLevel applies the same value to both **SSNR2DLevel** and **SSNR3DLevel.**
	ImagePresetMode.#.ThermalColorPalette	RES	<enum> WhiteHot, BlackHot, Rainbow, Custom, Sepia, Red, Iron	For thermal cameras only Allows various color palettes.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.ExposureControlSpeed	RES	<int> 0, 1	For vehicle cameras only • 0: Normal • 1: High
	ImagePresetMode.#.SubShutterSpeedRatio	RES	<enum> x2, x3, x4, x5, x6, x7, x8	For the double shutter camera, it can control the sub shutter speed compared to base shutter speed. Ex.) If the shutter speed is 1/5, or x5, the subshutterspeed is 1/25, or 5 times faster.
	ImagePresetMode.#.AFLKShutterSpeed	RES	<enum> 1/25, 1/30, 1/50, 1/60, 1/100, 1/120	Shutter speed in AFLK mode
	ImagePresetMode.#.WDRMotionArtifactReduction	RES	<bool> True,False	Configure WDRMotion artifcat reduction setting
set	Channel	REQ, RES	<int>	Channel ID If several devices are connected, it indicates the channel ID. For a network camera, it shows ‘0’.
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have its own ImagePreset settings. Image settings in any Mode can be changed, but modes except UserPreset1, UserPreset2 have different default image settings.

Action	Parameters	Request/ Response	Type/ Value	Description
	LCEMode	REQ, RES	<enum> Off, Low, Medium, High, Manual, Full, Horizontal, Sky, Ground, Center75Percent, Center50Percent, Center25Percent, Custom	LCE (Local Contrast Enhancement) Mode LCEMode can display the screen with a high contrast based on the set area. • Full: Full screen. • Horizontal: Rectangular area in the middle of the screen • Sky: Rectangular area at the upper part of the screen • Ground: Rectangular area at the lower part of the screen • Center75Percent: 75% Area size from the center • Center50Percent: 50% Area size from the center • Center25Percent: 25% Area size from the center • Custom: If LCEMode is set to this, the LCETop, LCEBottom, LCELeft, and LCERight parameters can be set.
	LCETop	REQ, RES	<int>	Top position info for the LCE region It sets the top position based on the distance from the upper end of the screen LCEBottom should be greater than LCETop
	LCEBottom	REQ, RES	<int>	Bottom position info for the LCE region It sets the bottom position based on the distance from the bottom end of the screen LCEBottom should be greater than LCETop

Action	Parameters	Request/ Response	Type/ Value	Description
	LCELeft	REQ, RES	<int>	Left position info for the LCE region It sets the left position based on the distance from the left end of the screen LCERight should be greater than LCELeft
	LCERight	REQ, RES	<int>	Right position info for the LCE region It sets the right position based on the distance from the right end of the screen LCERight should be greater than LCELeft
	ThermalColorPalette	REQ, RES	<enum> WhiteHot, BlackHot, Rainbow, Custom, Sepia, Red, Iron, Red-WH, Iron-WH, Midrange-WH	Thermal Color Palette Mode. Each mode can show different types of thermal video.
	CompensationMode	REQ, RES	<enum> Off, BLC, HLC, WDR	Compensation mode • Off: Disables the compensation function • BLC: Backlight Compensation • HLC: High Light Compensation • WDR: Wide Dynamic Range
	BLCLevel	REQ, RES	<enum> Low, Medium, High	BLC (Backlight Compensation) level The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC.

Action	Parameters	Request/ Response	Type/ Value	Description
	NormalizedBLCLevel	RES	<int>	Controls BLC (Backlight Compensation) level, but this adjusts the level in more detail than BLCLevel does.
	BLCAreaTop	REQ, RES	<int>	Top position info for BLC region It sets the top position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.
	BLCAreaBottom	REQ, RES	<int>	Bottom position info for BLC region It sets the bottom position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn’t return the value after a request.

Action	Parameters	Request/ Response	Type/ Value	Description
	BLCAreaLeft	REQ, RES	<int>	Left position info for BLC region It sets the left position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn't return the value after a request.
	BLCAreaRight	REQ, RES	<int>	Right position info for BLC region It sets the right position based on the distance from the center of the screen. The **BLCLevel**, **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaLeft**, and **BLCAreaRight** parameters are valid only when **CompensationMode** is set to BLC. For PTZ models, the **BLCAreaTop**, **BLCAreaBottom**, **BLCAreaRight**, and **BLCAreaLeft** parameters cannot be set altogether; only one of them can be set at a time. These are request only parameters for PTZ models; the device doesn't return the value after a request.

Action	Parameters	Request/ Response	Type/ Value	Description
	BLCAreaCoordinates	REQ,RES	<string>	Sets BLC region as real coordinates. This is supported in certain models only, so users should check the attributes first to see if their model supports this feature. BLCAreaCoordinates value must be specified in the format of <x1,y1,x2,y2>.
	HLCMode	REQ, RES	<enum> NightOnly, AllDay, On, Off	HLC (High Light Compensation) mode
	HLCLevel	REQ, RES	<enum> Low, Medium, High	HLC (High Light Compensation) level **HLCLevel** is valid only when **CompensationMode** is set to HLC.
	HLCMaskTone	REQ, RES	<int>	Tone of the mask used for high light compensation area **HLCMaskTone** is valid only when **CompensationMode** is set to HLC.
	HLCMaskColor	REQ, RES	<enum> Black, Blue, Red, Cyan, Magenta	Color of the mask used for high light compensation area **HLCMaskColor** is valid only when **CompensationMode** is set to HLC.
	HLCDimming	REQ, RES	<bool> "On", "Off"	Enables or disables HLC Dimming Feature. If enabled, The brightness of the image is adjusted automatically according to the surrounding environment

Action	Parameters	Request/ Response	Type/ Value	Description
	HLCAreaTop	REQ, RES	<int>	Top position info for HLC region It sets the top position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	HLCAreaBottom	REQ, RES	<int>	Bottom position info for HLC region It sets the bottom position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	HLCAreaLeft	REQ, RES	<int>	Left position info for HLC region It sets the left position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.
	HLCAreaRight	REQ, RES	<int>	Right position info for HLC region It sets the right position based on the distance from the center of the screen. The **HLCLevel**, **HLCAreaTop**, **HLCAreaBottom**, **HLCAreaLeft**, and **HLCAreaRight** parameters are valid only when **CompensationMode** is set to HLC.

Action	Parameters	Request/ Response	Type/ Value	Description
	HLCAreaCoordinates	REQ, RES	<string>	Sets HLC region with real coordinates. This is supported in certain models only, so users should check the attributes first to see if their model supports this feature. HLCAreaCoordinates value must be expressed in the form of <x1,y1,x2,y2>.
	NormalizedHLCLevel	REQ, RES	<int>	Controls HLC (Highlight Compensation) level, but this adjusts the level in more detail than HLCLevel does.
	WDRLevel	REQ, RES	<enum> Low, Medium, High	WDR (Wide Dynamic Range) sensitivity level **WDRLevel** is valid only when **CompensationMode** is set to WDR.
	WDRLimit	REQ, RES	<int>	WDR limit **WDRLimit** is valid only when **CompensationMode** is set to WDR.
	WDRBlackLevel	REQ, RES	<enum> Low, Medium, High	WDR black level **WDRBlackLevel** is valid only when **CompensationMode** is set to WDR.
	WDRWhiteLevel	REQ, RES	<enum> Low, Medium, High	WDR white level **WDRWhiteLevel** is valid only when **CompensationMode** is set to WDR.

Action	Parameters	Request/ Response	Type/ Value	Description
	WDRControlMode	REQ, RES	<enum> AlwaysOn, Auto, TurnOffInLowLight, TurnOffInBW	WDRControlMode • AlwaysOn:WDRSeamlessTransition,WDRLowLight,WDRIRLEDEnable are not enabled. • Auto: WDRSeamlessTransition • TurnOffInLowLight: WDRLowLight • TurnOffInBW: WDRIRLEDEnable **Note** For more details on the parameter features, please refer to the description of each parameter (WDRSeamlessTransition, WDRLowLight, WDRIRLEDEnable).
	WDRAreaTop	REQ, RES	<int>	Top position info for WDR region It sets the top position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	WDRAreaBottom	REQ, RES	<int>	Bottom position info for WDR region It sets the bottom position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.

Action	Parameters	Request/ Response	Type/ Value	Description
	WDRAreaLeft	REQ, RES	<int>	Left position info for WDR region It sets the left position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	WDRAreaRight	REQ, RES	<int>	Right position info for WDR region It sets the right position based on the distance from the center of the screen. The **WDRLevel**, **WDRAreaTop**, **WDRAreaBottom**, **WDRAreaLeft**, and **WDRAreaRight** parameters are valid only when **CompensationMode** is set to WDR.
	WDRAreaCoordinates	REQ, RES	<string>	Sets WDR region with real coordinates. This is supported in certain models only, so users should check the attributes first to see if their model supports this feature. WDRAreaCoordinates value must be expressed in the form of <x1,y1,x2,y2>.

Action	Parameters	Request/ Response	Type/ Value	Description
	WDRType	REQ, RES	<enum> ExposureControl, GainControl	WDR operation mode. **ExposureControl** is the default value when WDR is enabled. • ExposureControl: Captures two frames, one with short exposure for bright areas and the other with a long exposure for dark areas, and composites them to correct backlighting. Default mode. When this value is enabled then **WDRLevel** can be configured. When this parameter is not shown in attributes.cgi, then only ExposureControl operation mode is supported. • GainControl: Captures a single frame at a different gain level and composites the two images, which has the advantage of reducing ghosting.
	ShutterMode	REQ, RES	<enum> Auto, Manual, ESC, AFLK	Shutter mode • Auto: Adjusts the shutter speed automatically • Manual: Shutter speed of the camera can be adjusted manually • ESC: Electronic Shutter Control; adjusts the shutter speed automatically according to the ambient brightness • AFLK: Anti Flicker; Adjusts the shutter speed when a screen flickers frequently due to a mismatch with the ambient lighting
	ManualShutterSpeed	REQ, RES	<enum>	Manual shutter speed **ManualShutterSpeed** is valid only when **ShutterMode** is set to Manual.

Action	Parameters	Request/ Response	Type/ Value	Description
	AutoLongShutterSpeed	REQ, RES	<enum>	Maximum shutter speed **AutoLongShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; this parameter is read-only if **ShutterMode** is set to AFLK.
	AutoShortShutterSpeed	REQ, RES	<enum>	Minimum shutter speed **AutoShortShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; This parameter is read-only if **ShutterMode** is set to AFLK.
	PreferedShutterSpeed	REQ, RES	<enum>	Prefer shutter speed **PreferedShutterSpeed** is valid only when **ShutterMode** is NOT set to AFLK; This parameter is read-only if **ShutterMode** is set to AFLK.
	PreferShutterAISupportEnable	REQ, RES	<bool> True, False	Enable or Disable for auto PreferShutter using AI engine support **Note** **PreferShutterAISupportEnable** requires the camera to have an AI engine.
	AFLKMode	REQ, RES	<enum> Off, On, 50, 60	AFLK (Anti-Flicker) mode

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightMode	REQ, RES	<enum> Color, BW, Auto, ExternalBW, Schedule	Day and night mode The controls the IR-cut filter if one is available on the device. • Color: IR-cut filter is on; always outputs in color. • BW: IR-cut filter is off; always outputs in black and white. • Auto: IR-cut filter is automatically on or off based on the light level; normally outputs in color but will output black and white under low luminance at night. • ExternalBW: Controls the color of the video when the alarm input terminal is synchronized with an external device. • Schedule: Outputs in color only in the specified time. If **AGCMode** is set to Off, **DayNightMode** cannot be set to Auto. **Note** Attribute to check for DayNight: "attributes/Image/Support/DayNight"

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightModeSchedule.<ddd>	REQ, RES	<bool> 0, 1	Enables or disables the day and night mode for a scheduled day (every Sunday, Monday, etc.) <ddd> stands for day of the week and should be specified in short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. • 0: Disabled • 1: Enabled e.g. 'DayNightModeSchedule.SUN=1' indicates that day and night mode is activated every Sunday12:00 AM to 11:59 PM. **DayNightModeSchedule.<ddd>** is valid only when **DayNightMode** is set to Schedule. **DayNightModeSchedule.<ddd>** and **DayNightModeSchedule.EveryDay** parameters cannot be set together; Everyday cannot be set together with a specific day of the week such as Sunday, Monday, Saturday, etc. for enabling or disabling the day and night mode.

<table>
<tr><th>Action</th><th>Parameters</th><th>Request/ Response</th><th>Type/ Value</th><th>Description</th></tr>
<tr><td></td><td>DayNightModeSchedule.
EveryDay</td><td>REQ, RES</td><td><bool>

0, 1</td><td>Enables or disables the day and night mode every day (always)
• 0: Disabled
• 1: Enabled

e.g. ’DayNightModeSchedule.EveryDay=1’ indicates that day and night mode is activated every day12:00 AM to 11:59 PM.

DayNightModeSchedule.EveryDay is valid only when DayNightMode is set to Schedule.

DayNightModeSchedule.<ddd> and DayNightModeSchedule.EveryDay parameters cannot be set together; Everyday cannot be set together with a specific day of the week such as Sunday, Monday, Saturday, etc. for enabling or disabling the day and night mode.</td></tr>
</table>

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightModeSchedule.<ddd>.FromTo	REQ, RES	<string>	Schedule for day and night mode (day of the week and time) **DayNightModeSchedule.<ddd>.FromTo** must be specified in the format of <hh:mm-hh:mm>. <ddd> stands for day of the week and should be specified in short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g. 'DayNightModeSchedule.SUN.FromTo=1:0-8:59' indicates that day and night mode is enabled from 1:00 AM to 8:59 AM every Sunday. **DayNightModeSchedule.<ddd>.FromTo** and **DayNightModeSchedule.EveryDay.FromTo** parameters cannot be set together; day and night mode schedule for every day and day of the week such as Sunday, Monday, Saturday, etc. cannot be set together.

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightModeSchedule.EveryDay.FromTo	REQ, RES	<string>	Schedule for day and night mode (Every day) **DayNightModeSchedule.EveryDay.FromTo** value must be specified in the format of <hh:mm-hh:mm>. e.g. 'DayNightModeSchedule.EveryDay.FromTo=1:00-8:59' denotes the day and night mode is enabled from 1:00 AM to 8:59 AM every day. **DayNightModeSchedule.<ddd>.FromTo** and **DayNightModeSchedule.EveryDay.FromTo** parameters cannot be set together; day and night mode schedule for every day and day of the week such as Sunday, Monday, Saturday, etc. cannot be set together.
	SimpleFocus	REQ, RES	<bool> True, False	Automatic focus adjustment after day and night **Note** Attribute to check for SimpleFocus: "attributes/Image/Support/SimpleFocus"
	DayNightSwitchingTime	REQ, RES	<enum>	Duration of switch between day and night mode **DayNightSwitchingTime** is valid only when **DayNightMode** is set to Auto.
	DayNightSwitchingTimeColorToBW	REQ, RES	<enum>	Duration of switch between day and night mode for switching from color to black/white **DayNightSwitchingTimeColorToBW** is valid only when **DayNightMode** is set to Auto.

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightSwitchingBrightnessColorToBW	REQ, RES	<enum> Low, Medium, High	Brightness level of day and night mode for switching from color to black/white **DayNightSwitchingBrightnessColorToBW** is valid only when **DayNightMode** is set to Auto.
	DayNightSwitchingTimeBWToColor	REQ, RES	<enum>	Duration of switch between day and night mode for switching from black/white to color **DayNightSwitchingTimeBWToColor** is valid only when **DayNightMode** is set to Auto.
	DayNightSwitchingBrightnessBWToColor	REQ, RES	<enum> Low, Medium, High	Brightness level of day and night mode for switching from black/white to color **DayNightSwitchingBrightnessBWToColor** is valid only when **DayNightMode** is set to Auto.
	DayNightSwitchingMode	REQ, RES	<enum> VeryFast, Fast, Normal, Slow, VerySlow, Custom	Interval of switch between day and night mode **DayNightSwitchingMode** is valid only when **DayNightMode** is set to Auto.
	DNSwitchColorToBWThreshold	REQ, RES	<int>	Threshold value of changing from day to night mode. **DNSwitchColorToBWThreshold** is valid only when **DayNightSwitchingMode** is set to Custom. **Note** Attribute to check for DNSwitchColorToBWThreshold: "attributes/Image/Support/ DayNightSwitchThresholdAdjust"

Action	Parameters	Request/ Response	Type/ Value	Description
	DNSwitchBWToColorThreshold	REQ, RES	<int>	Threshold value of changing from night to day mode. **DNSwitchBWToColorThreshold** is valid only when **DayNightSwitchingMode** is set to Custom. **Note** Attribute to check for DNSwitchBWToColorThreshold: "attributes/Image/Support/DayNightSwitchThresholdAdjust"
	DayNightAlarmIn	REQ, RES	<enum> SwitchToBWIfOpens, SwitchToBWIfCloses	Day and night setup depending on the alarm state. **DayNightAlarmIn** is valid only when **DayNightMode** is set to ExternalBW.
	NegativeModeEnable	REQ, RES	<bool> True, False	Enables or disables negative video mode
	IrisMode	REQ, RES	<enum> Auto, Manual, P-Iris, P-Iris-SLAM3180PN, P-Iris-M13VP288IR , P-Iris-SLAM2890PN, ICS	Lens Iris mode • Auto: Automatically adjusts the iris. • Manual: Adjusts the Iris and focus manually. • P-Iris: Controls the iris of the lens equipped with a step motor.

<table>
<tr><th>Action</th><th>Parameters</th><th>Request/ Response</th><th>Type/ Value</th><th>Description</th></tr>
<tr><td></td><td>PIrisMode</td><td>REQ, RES</td><td><enum>
Auto,
Manual</td><td>P-Iris (Precise Iris) mode
• Auto: Automatically adjusts the amount of surrounding light.
• Manual: Adjusts the Iris manually.
PIrisMode is valid only when IrisMode is set to P-Iris.
Note
For P-Iris support better to check "attributes/Image/Support/P-Iris" and "cgis/image/camera" submenu PIrisMode parameter.
In some dome models, which only supports Auto Piris, PIrisMode parameter may not be supported, eventhough "attributes/Image/Support/P-Iris" support would be true.</td></tr>
<tr><td></td><td>PIrisPosition</td><td>REQ, RES</td><td><int></td><td>P-Iris (Precise Iris) position
PIrisPosition is valid only when IrisMode is set to P-Iris, and PIrisMode is set to Manual.
The value should be specified in even numbers only.</td></tr>
<tr><td></td><td>IrisFno</td><td>REQ, RES</td><td><enum></td><td>Iris F-stop number
IrisFno is valid only when IrisMode is set to P-Iris or Manual.</td></tr>
</table>

Action	Parameters	Request/ Response	Type/ Value	Description
	SensUpMode	REQ, RES	<enum> Off, Auto	Enables or disables Sens-up mode Sens-up function makes a camera more sensitive to light in order to produce clear images even in low light conditions. **SensUpMode** and **SensUpLevel** are valid only when **ShutterMode** is not set to Manual or AFLK. The **SensUpMode** and **SensUpLevel** parameters are not available if **AGCMode** is set to Off.
	SensUpLevel	REQ, RES	<enum>	Sens-up Level **SensUpMode** and **SensUpLevel** are not available if **AGCMode** is set to Off. **SensUpLevel** parameter is valid only when **SensUpMode** is set to Auto.

Action	Parameters	Request/ Response	Type/ Value	Description
	AGCMode	REQ, RES	<enum> Off, Low, Medium, High, Manual, MaxGain	AGC (Automatic Gain Control) mode It adjusts the gain value of the video to control the video brightness. • Off: Disables AGC. • Low/ Medium /High: As the level increases to high, the screen is brighter in a low lighting condition. • Manual: Adjusts the AGC level manually • MaxGain: Automatically adjust the gain level to the **MaxGainLevel** value and less. If **AGCMode** is set to Off, **DayNightMode** cannot be set to Auto, and therefore **SSNREnable**, **SSNRLevel**, **SensupModeEnable** and **SensupLevel** are invalid. And If AGCMode is set to MaxGain and MaxGainLevel less than 30dB, DayNightMode cannot be set to Auto
	AGCLevel	REQ, RES	<int>	AGC level **AGCLevel** is valid only when **AGCMode** is set to Manual.
	AGCMaxGainLevel	REQ, RES	<int>	AGC max gain level **AGCMaxGainLevel** is valid only when **AGCMode** is set to MaxGain
	SSNREnable	REQ, RES	<bool> True, False	Enables or disables SSNR (Samsung Super Noise Reduction). If **AGCMode** is set to Off, **SSNREnable** and **SSNRLevel** parameters are not valid.

Action	Parameters	Request/ Response	Type/ Value	Description
	SSNRMode	REQ, RES	<enum> Off, Manual, Auto, AIAuto	SSNR Mode • Off: Disables SSNR Mode. • Manual: SSNR Manual Mode • Auto: SSNR Auto mode • AIAuto**:** SSNR auto mode using AI engine (AI engine required) **SSNRMode** Auto Mode and **SSNREnable** Parameter true values are the same. These two parameters should not be sent together.
	SSNRLevel	REQ, RES	<int>	SSNR level If **AGCMode** is set to Off, **SSNREnable** and **SSNRLevel** parameters are not valid. **SSNRLevel** is valid only when **SSNREnable** is set to True. Changing SSNRLevel applies the same value to both **SSNR2DLevel** and **SSNR3DLevel**
	SSNR2DLevel	REQ, RES	<int>	SSNR 2D level If **AGCMode** is set to Off, **SSNREnable** and **SSNR2DLevel** parameters are not valid. **SSNR2DLevel** is valid only when **SSNREnable** is set to True. **SSNR2DLevel** and **SSNRLevel** cannot be set at the same time.

Action	Parameters	Request/ Response	Type/ Value	Description
	SSNR3DLevel	REQ, RES	<int>	SSNR 3D level If **AGCMode** is set to Off, **SSNREnable** and **SSNR3DLevel** parameters are not valid. **SSNR3DLevel** is valid only when **SSNREnable** is set to True. **SSNR3DLevel** and **SSNRLevel** cannot be set at the same time.
	WDRSeamlessTransition	REQ, RES	<enum> Off, On	Enables or disables WDR Seamless transition. If enabled, the image is analyzed and WDR mode is activated automatically when backlight correction is necessary.
	WDRLowLight	REQ, RES	<enum> Off, On	Enables or disables WDR Low Light. If enabled, the WDR mode is turned off automatically in a low-light environment
	WDRIRLEDEnable	REQ, RES	<enum> Off, On	Enables or disables WDR IR LED feature. If enabled, the WDR mode is turned off automatically when the B/W mode operates
	LensModel	REQ, RES	<enum> SLA-T46XX, SLA-T10XX, SLA-T10XX/SLA-T16XX	Not applied for fisheye models Enum values of different lenses, for cameras on which the lens can be replaced. • TNB-9000 also support changing lens, and it uses canon lens. So enum values of TNB-9000 are different with other lens changeable models’. Please get exact values from attributes because its values might be changed in each models.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop, AWC	Image preview mode Allows viewing of the preview image of the configuration, rather than saving the image configuration to camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied • AWC: AWC mode will be started
	TemperatureUnit	REQ, RES	<enum> Celsius, Fahrenheit	For thermal cameras only
	ThermalColorPalette	REQ, RES	<enum> WhiteHot, BlackHot, Rainbow, Custom, Sepia, Red, Iron	For thermal cameras only Allows various color palettes
	ThermalVariationSensitivity	REQ, RES	<int>	Sensitivity of thermal variation
	MaximumIris	REQ, RES	<enum> F1.2, F1.4, F1.6, F1.8, F2.0, F2.2, F2.5, F2.8, F3.2, F3.5, F4.0, F4.5, F5.0, F5.6	Maximum Iris Limits auto adjustment aperture value when **PIrisMode** is Auto on a camera with the EF lens. (Limiting the opening of the aperture prevents the depth of field from shallow)

Action	Parameters	Request/ Response	Type/ Value	Description
	PreferIris	REQ, RES	<enum> F1.2, F1.4, F1.6, F1.8, F2.0, F2.2, F2.5, F2.8, F3.2, F3.5, F4.0, F4.5, F5.0, F5.6	Prefer Iris If **PIrisMode** is Auto on a camera with the EF lens and has sufficient brightness, the camera will keep the set aperture value.
	ExposureControlSpeed	REQ, RES	<int> 0, 1	For vehicle cameras only • 0: Normal • 1: High
	SubShutterSpeedRatio	REQ, RES	<enum> x2, x3, x4, x5, x6, x7, x8	For the double shutter camera, it can control sub shutter speed compared to base shutter speed. Ex) If shutter speed is 1/5, or x5, the subshutterspeed is 1/25, or 5 times faster.
	AFLKShutterSpeed	REQ, RES	<enum> 1/25, 1/30, 1/50, 1/60, 1/100, 1/120	Shutter speed in AFLK mode, can check imageoptions submenu "AFLKMode.#.ShutterSpeed" parameter for allowed values per mode.
	TemperatureRangeMode	REQ, RES	<enum> Narrow, Wide, Auto	Change temperature range mode. **Narrow** : temperature detection range is narrow, but more accurate. **Wide** : temperature detection range is Wide, but less accurate. **Auto** : temperature detection range is changed automatically based on screen temperature range.
	ManualShutterSpeed	REQ, RES	<int>	This feature allows user to set shutterspeed manually with integer value.
	WDRMotionArtifactReduction	REQ, RES	<bool> True,False	Configure WDRMotion artifcat reduction setting
	GainSequencerEnable	REQ, RES	<bool> True,False	Enables or disables Gain Sequencer feature
	GainFrameInterval	REQ, RES	<int>	Frame interval for Gain Sequencer

Action	Parameters	Request/ Response	Type/ Value	Description
	GainNumberOfSteps	REQ, RES	<int>	Number of steps for Gain Sequencer
	AGCLevelFloat	REQ, RES	<float>	AGC level (same setting as `AGCLevel`) with floating point precision. Use this for new integrations that require fractional AGC control; `AGCLevel` remains for backward-compatible integer-only control.
	GainSequencerStep.#.GainLevel	REQ, RES	<float>	Gain level for each step in Gain Sequencer

112.4. Examples

112.4.1. Getting the current camera information

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=camera&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.CompensationMode=Off
Channel.0.BLCAreaTop=30
Channel.0.BLCAreaBottom=70
Channel.0.BLCAreaLeft=30
Channel.0.BLCAreaRight=70
Channel.0.BLCAreaCoordinates=975,548,2319,1304
Channel.0.BLCLevel=Medium
Channel.0.NormalizedBLCLevel=50
Channel.0.HLCLevel=Medium
Channel.0.HLCMaskTone=100
Channel.0.HLCMode=On
Channel.0.HLCMaskColor=Black
Channel.0.HLCDimming=Off
Channel.0.HLCAreaTop=30
Channel.0.HLCAreaBottom=70
Channel.0.HLCAreaLeft=30
```

```
Channel.0.HLCAreaRight=70
Channel.0.HLCAreaCoordinates=975,548,2319,1304
Channel.0.NormalizedHLCLevel=50
Channel.0.WDRLevel=High
Channel.0.WDRAreaTop=1
Channel.0.WDRAreaBottom=100
Channel.0.WDRAreaLeft=1
Channel.0.WDRAreaRight=100
Channel.0.WDRAreaCoordinates=0,0,3327,1871
Channel.0.AutoShortShutterSpeed=1/30
Channel.0.AutoLongShutterSpeed=1/12000
Channel.0.PreferShutterSpeed=1/200
Channel.0.AFLKMode=Off
Channel.0.SSNREnable=True
Channel.0.SSNRMode=Manual
Channel.0.SSNRLevel=12
Channel.0.SSNR2DLevel=12
Channel.0.SSNR3DLevel=12
Channel.0.IrisMode=Auto
Channel.0.PIrisMode=Auto
Channel.0.PIrisPosition=100
Channel.0.AGCMode=MaxGain
Channel.0.AGCLevel=1
Channel.0.AGCLevelFloat=1.2
Channel.0.AGCMaxGainLevel=60
Channel.0.ManualShutterSpeed=200
Channel.0.DayNightMode=Auto
Channel.0.DayNightSwitchingTime=5s
Channel.0.DayNightSwitchingMode=Normal
Channel.0.DNSwitchColorToBWThreshold=70
Channel.0.DNSwitchBWToColorThreshold=30
Channel.0.SimpleFocus=False
Channel.0.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.DayNightModeSchedule.EveryDay=1
Channel.0.DayNightModeSchedule.EveryDay.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.SUN=1
Channel.0.DayNightModeSchedule.SUN.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.MON=1
Channel.0.DayNightModeSchedule.MON.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.TUE=1
Channel.0.DayNightModeSchedule.TUE.FromTo=00:00-23:59
```

```
Channel.0.DayNightModeSchedule.WED=1
Channel.0.DayNightModeSchedule.WED.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.THU=1
Channel.0.DayNightModeSchedule.THU.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.FRI=1
Channel.0.DayNightModeSchedule.FRI.FromTo=00:00-23:59
Channel.0.DayNightModeSchedule.SAT=1
Channel.0.DayNightModeSchedule.SAT.FromTo=00:00-23:59
Channel.0.WDRControlMode=AlwaysOn
Channel.0.TemperatureRangeMode=Auto
Channel.0.GainSequencerEnable=False
Channel.0.GainFrameInterval=1
Channel.0.GainNumberOfSteps=2
Channel.0.GainSequencerStep.1.GainLevel=0.0
Channel.0.GainSequencerStep.2.GainLevel=0.0
Channel.0.GainSequencerStep.3.GainLevel=0.0
Channel.0.GainSequencerStep.4.GainLevel=0.0
Channel.0.GainSequencerStep.5.GainLevel=0.0
Channel.0.GainSequencerStep.6.GainLevel=0.0
Channel.0.ImagePresetMode.UserPreset1.CompensationMode=Off
Channel.0.ImagePresetMode.UserPreset1.BLCAreaTop=30
Channel.0.ImagePresetMode.UserPreset1.BLCAreaBottom=70
Channel.0.ImagePresetMode.UserPreset1.BLCAreaLeft=30
Channel.0.ImagePresetMode.UserPreset1.BLCAreaRight=70
Channel.0.ImagePresetMode.UserPreset1.BLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.UserPreset1.BLCLevel=Medium
Channel.0.ImagePresetMode.UserPreset1.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.UserPreset1.HLCLevel=Medium
Channel.0.ImagePresetMode.UserPreset1.HLCMaskTone=100
Channel.0.ImagePresetMode.UserPreset1.HLCMode=On
Channel.0.ImagePresetMode.UserPreset1.HLCMaskColor=Black
Channel.0.ImagePresetMode.UserPreset1.HLCDimming=Off
Channel.0.ImagePresetMode.UserPreset1.HLCAreaTop=30
Channel.0.ImagePresetMode.UserPreset1.HLCAreaBottom=70
Channel.0.ImagePresetMode.UserPreset1.HLCAreaLeft=30
Channel.0.ImagePresetMode.UserPreset1.HLCAreaRight=70
Channel.0.ImagePresetMode.UserPreset1.HLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.UserPreset1.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.UserPreset1.WDRLevel=High
Channel.0.ImagePresetMode.UserPreset1.WDRAreaTop=1
Channel.0.ImagePresetMode.UserPreset1.WDRAreaBottom=100
```

```
Channel.0.ImagePresetMode.UserPreset1.WDRAreaLeft=1
Channel.0.ImagePresetMode.UserPreset1.WDRAreaRight=100
Channel.0.ImagePresetMode.UserPreset1.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.UserPreset1.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.UserPreset1.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.UserPreset1.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.UserPreset1.AFLKMode=Off
Channel.0.ImagePresetMode.UserPreset1.SSNREnable=True
Channel.0.ImagePresetMode.UserPreset1.SSNRMode=Manual
Channel.0.ImagePresetMode.UserPreset1.SSNRLevel=12
Channel.0.ImagePresetMode.UserPreset1.SSNR2DLevel=12
Channel.0.ImagePresetMode.UserPreset1.SSNR3DLevel=12
Channel.0.ImagePresetMode.UserPreset1.IrisMode=Auto
Channel.0.ImagePresetMode.UserPreset1.PIrisMode=Auto
Channel.0.ImagePresetMode.UserPreset1.PIrisPosition=100
Channel.0.ImagePresetMode.UserPreset1.AGCMode=MaxGain
Channel.0.ImagePresetMode.UserPreset1.AGCLevel=1
Channel.0.ImagePresetMode.UserPreset1.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.UserPreset1.DayNightMode=Auto
Channel.0.ImagePresetMode.UserPreset1.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.UserPreset1.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.UserPreset1.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.UserPreset1.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.UserPreset1.SimpleFocus=False
Channel.0.ImagePresetMode.UserPreset1.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.EveryDay.FromTo=0
0:00-23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.SUN.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.MON.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.TUE.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.WED.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleTHU=1
```

```
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.THU.FromTo=00:00-23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.FRI.FromTo=00:00-23:59
Channel.0.ImagePresetMode.UserPreset1.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.UserPreset1.DayNightModeSchedule.SAT.FromTo=00:00-23:59
Channel.0.ImagePresetMode.UserPreset1.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.UserPreset2.CompensationMode=Off
Channel.0.ImagePresetMode.UserPreset2.BLCAreaTop=30
Channel.0.ImagePresetMode.UserPreset2.BLCAreaBottom=70
Channel.0.ImagePresetMode.UserPreset2.BLCAreaLeft=30
Channel.0.ImagePresetMode.UserPreset2.BLCAreaRight=70
Channel.0.ImagePresetMode.UserPreset2.BLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.UserPreset2.BLCLevel=Medium
Channel.0.ImagePresetMode.UserPreset2.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.UserPreset2.HLCLevel=Medium
Channel.0.ImagePresetMode.UserPreset2.HLCMaskTone=100
Channel.0.ImagePresetMode.UserPreset2.HLCMode=On
Channel.0.ImagePresetMode.UserPreset2.HLCMaskColor=Black
Channel.0.ImagePresetMode.UserPreset2.HLCDimming=Off
Channel.0.ImagePresetMode.UserPreset2.HLCAreaTop=30
Channel.0.ImagePresetMode.UserPreset2.HLCAreaBottom=70
Channel.0.ImagePresetMode.UserPreset2.HLCAreaLeft=30
Channel.0.ImagePresetMode.UserPreset2.HLCAreaRight=70
Channel.0.ImagePresetMode.UserPreset2.HLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.UserPreset2.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.UserPreset2.WDRLevel=High
Channel.0.ImagePresetMode.UserPreset2.WDRAreaTop=1
Channel.0.ImagePresetMode.UserPreset2.WDRAreaBottom=100
Channel.0.ImagePresetMode.UserPreset2.WDRAreaLeft=1
Channel.0.ImagePresetMode.UserPreset2.WDRAreaRight=100
Channel.0.ImagePresetMode.UserPreset2.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.UserPreset2.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.UserPreset2.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.UserPreset2.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.UserPreset2.AFLKMode=Off
Channel.0.ImagePresetMode.UserPreset2.SSNREnable=True
Channel.0.ImagePresetMode.UserPreset2.SSNRMode=Manual
Channel.0.ImagePresetMode.UserPreset2.SSNRLevel=12
```

```
Channel.0.ImagePresetMode.UserPreset2.SSNR2DLevel=12
Channel.0.ImagePresetMode.UserPreset2.SSNR3DLevel=12
Channel.0.ImagePresetMode.UserPreset2.IrisMode=Auto
Channel.0.ImagePresetMode.UserPreset2.PIrisMode=Auto
Channel.0.ImagePresetMode.UserPreset2.PIrisPosition=100
Channel.0.ImagePresetMode.UserPreset2.AGCMode=MaxGain
Channel.0.ImagePresetMode.UserPreset2.AGCLevel=1
Channel.0.ImagePresetMode.UserPreset2.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.UserPreset2.DayNightMode=Auto
Channel.0.ImagePresetMode.UserPreset2.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.UserPreset2.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.UserPreset2.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.UserPreset2.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.UserPreset2.SimpleFocus=False
Channel.0.ImagePresetMode.UserPreset2.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.EveryDay.FromTo=0
0:00-23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.SUN.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.MON.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.TUE.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.WED.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.THU.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.FRI.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.UserPreset2.DayNightModeSchedule.SAT.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.UserPreset2.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.OutdoorDaytime.CompensationMode=Off
```

```
Channel.0.ImagePresetMode.OutdoorDaytime.BLCAreaTop=30
Channel.0.ImagePresetMode.OutdoorDaytime.BLCAreaBottom=70
Channel.0.ImagePresetMode.OutdoorDaytime.BLCAreaLeft=30
Channel.0.ImagePresetMode.OutdoorDaytime.BLCAreaRight=70
Channel.0.ImagePresetMode.OutdoorDaytime.BLCAreaCoordinates=975,548,2319,130
4
Channel.0.ImagePresetMode.OutdoorDaytime.BLCLevel=Medium
Channel.0.ImagePresetMode.OutdoorDaytime.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.OutdoorDaytime.HLCLevel=Medium
Channel.0.ImagePresetMode.OutdoorDaytime.HLCMaskTone=100
Channel.0.ImagePresetMode.OutdoorDaytime.HLCMode=On
Channel.0.ImagePresetMode.OutdoorDaytime.HLCMaskColor=Black
Channel.0.ImagePresetMode.OutdoorDaytime.HLCDimming=Off
Channel.0.ImagePresetMode.OutdoorDaytime.HLCAreaTop=30
Channel.0.ImagePresetMode.OutdoorDaytime.HLCAreaBottom=70
Channel.0.ImagePresetMode.OutdoorDaytime.HLCAreaLeft=30
Channel.0.ImagePresetMode.OutdoorDaytime.HLCAreaRight=70
Channel.0.ImagePresetMode.OutdoorDaytime.HLCAreaCoordinates=975,548,2319,130
4
Channel.0.ImagePresetMode.OutdoorDaytime.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.OutdoorDaytime.WDRLevel=Medium
Channel.0.ImagePresetMode.OutdoorDaytime.WDRAreaTop=1
Channel.0.ImagePresetMode.OutdoorDaytime.WDRAreaBottom=100
Channel.0.ImagePresetMode.OutdoorDaytime.WDRAreaLeft=1
Channel.0.ImagePresetMode.OutdoorDaytime.WDRAreaRight=100
Channel.0.ImagePresetMode.OutdoorDaytime.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.OutdoorDaytime.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.OutdoorDaytime.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.OutdoorDaytime.PreferShutterSpeed=1/250
Channel.0.ImagePresetMode.OutdoorDaytime.AFLKMode=Off
Channel.0.ImagePresetMode.OutdoorDaytime.SSNREnable=True
Channel.0.ImagePresetMode.OutdoorDaytime.SSNRMode=Manual
Channel.0.ImagePresetMode.OutdoorDaytime.SSNRLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.SSNR2DLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.SSNR3DLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.IrisMode=Auto
Channel.0.ImagePresetMode.OutdoorDaytime.PIrisMode=Auto
Channel.0.ImagePresetMode.OutdoorDaytime.PIrisPosition=100
Channel.0.ImagePresetMode.OutdoorDaytime.AGCMode=MaxGain
Channel.0.ImagePresetMode.OutdoorDaytime.AGCLevel=1
Channel.0.ImagePresetMode.OutdoorDaytime.AGCMaxGainLevel=60
```

```
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightMode=Auto
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.OutdoorDaytime.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.OutdoorDaytime.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.OutdoorDaytime.SimpleFocus=False
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.EveryDay.FromT
o=00:00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.SUN.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.MON.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.TUE.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.WED.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.THU.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.FRI.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.OutdoorDaytime.DayNightModeSchedule.SAT.FromTo=00:
00-23:59
Channel.0.ImagePresetMode.OutdoorDaytime.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.OutdoorNightime.CompensationMode=Off
Channel.0.ImagePresetMode.OutdoorNightime.BLCAreaTop=30
Channel.0.ImagePresetMode.OutdoorNightime.BLCAreaBottom=70
Channel.0.ImagePresetMode.OutdoorNightime.BLCAreaLeft=30
Channel.0.ImagePresetMode.OutdoorNightime.BLCAreaRight=70
Channel.0.ImagePresetMode.OutdoorNightime.BLCAreaCoordinates=975,548,2319,13
04
Channel.0.ImagePresetMode.OutdoorNightime.BLCLevel=Medium
Channel.0.ImagePresetMode.OutdoorNightime.NormalizedBLCLevel=50
```

```
Channel.0.ImagePresetMode.OutdoorNightime.HLCLevel=Medium
Channel.0.ImagePresetMode.OutdoorNightime.HLCMaskTone=100
Channel.0.ImagePresetMode.OutdoorNightime.HLCMode=On
Channel.0.ImagePresetMode.OutdoorNightime.HLCMaskColor=Black
Channel.0.ImagePresetMode.OutdoorNightime.HLCDimming=Off
Channel.0.ImagePresetMode.OutdoorNightime.HLCAreaTop=30
Channel.0.ImagePresetMode.OutdoorNightime.HLCAreaBottom=70
Channel.0.ImagePresetMode.OutdoorNightime.HLCAreaLeft=30
Channel.0.ImagePresetMode.OutdoorNightime.HLCAreaRight=70
Channel.0.ImagePresetMode.OutdoorNightime.HLCAreaCoordinates=975,548,2319,13
04
Channel.0.ImagePresetMode.OutdoorNightime.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.OutdoorNightime.WDRLevel=Medium
Channel.0.ImagePresetMode.OutdoorNightime.WDRAreaTop=1
Channel.0.ImagePresetMode.OutdoorNightime.WDRAreaBottom=100
Channel.0.ImagePresetMode.OutdoorNightime.WDRAreaLeft=1
Channel.0.ImagePresetMode.OutdoorNightime.WDRAreaRight=100
Channel.0.ImagePresetMode.OutdoorNightime.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.OutdoorNightime.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.OutdoorNightime.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.OutdoorNightime.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.OutdoorNightime.AFLKMode=Off
Channel.0.ImagePresetMode.OutdoorNightime.SSNREnable=True
Channel.0.ImagePresetMode.OutdoorNightime.SSNRMode=Manual
Channel.0.ImagePresetMode.OutdoorNightime.SSNRLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.SSNR2DLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.SSNR3DLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.IrisMode=Auto
Channel.0.ImagePresetMode.OutdoorNightime.PIrisMode=Auto
Channel.0.ImagePresetMode.OutdoorNightime.PIrisPosition=100
Channel.0.ImagePresetMode.OutdoorNightime.AGCMode=MaxGain
Channel.0.ImagePresetMode.OutdoorNightime.AGCLevel=1
Channel.0.ImagePresetMode.OutdoorNightime.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.OutdoorNightime.DayNightMode=Auto
Channel.0.ImagePresetMode.OutdoorNightime.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.OutdoorNightime.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.OutdoorNightime.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.OutdoorNightime.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.OutdoorNightime.SimpleFocus=False
Channel.0.ImagePresetMode.OutdoorNightime.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleEveryDay=1
```

```
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.EveryDay.From
To=00:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.SUN.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.MON.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.TUE.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.WED.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.THU.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.FRI.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.OutdoorNightime.DayNightModeSchedule.SAT.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.OutdoorNightime.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.IndoorBacklight.CompensationMode=WDR
Channel.0.ImagePresetMode.IndoorBacklight.BLCAreaTop=30
Channel.0.ImagePresetMode.IndoorBacklight.BLCAreaBottom=70
Channel.0.ImagePresetMode.IndoorBacklight.BLCAreaLeft=30
Channel.0.ImagePresetMode.IndoorBacklight.BLCAreaRight=70
Channel.0.ImagePresetMode.IndoorBacklight.BLCAreaCoordinates=975,548,2319,13
04
Channel.0.ImagePresetMode.IndoorBacklight.BLCLevel=Medium
Channel.0.ImagePresetMode.IndoorBacklight.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.IndoorBacklight.HLCLevel=Medium
Channel.0.ImagePresetMode.IndoorBacklight.HLCMaskTone=100
Channel.0.ImagePresetMode.IndoorBacklight.HLCMode=On
Channel.0.ImagePresetMode.IndoorBacklight.HLCMaskColor=Black
Channel.0.ImagePresetMode.IndoorBacklight.HLCDimming=Off
Channel.0.ImagePresetMode.IndoorBacklight.HLCAreaTop=30
Channel.0.ImagePresetMode.IndoorBacklight.HLCAreaBottom=70
Channel.0.ImagePresetMode.IndoorBacklight.HLCAreaLeft=30
```

```
Channel.0.ImagePresetMode.IndoorBacklight.HLCAreaRight=70
Channel.0.ImagePresetMode.IndoorBacklight.HLCAreaCoordinates=975,548,2319,13
04
Channel.0.ImagePresetMode.IndoorBacklight.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.IndoorBacklight.WDRLevel=Low
Channel.0.ImagePresetMode.IndoorBacklight.WDRAreaTop=1
Channel.0.ImagePresetMode.IndoorBacklight.WDRAreaBottom=100
Channel.0.ImagePresetMode.IndoorBacklight.WDRAreaLeft=1
Channel.0.ImagePresetMode.IndoorBacklight.WDRAreaRight=100
Channel.0.ImagePresetMode.IndoorBacklight.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.IndoorBacklight.AutoShortShutterSpeed=1/60
Channel.0.ImagePresetMode.IndoorBacklight.AutoLongShutterSpeed=1/1500
Channel.0.ImagePresetMode.IndoorBacklight.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.IndoorBacklight.AFLKMode=Off
Channel.0.ImagePresetMode.IndoorBacklight.SSNREnable=True
Channel.0.ImagePresetMode.IndoorBacklight.SSNRMode=Manual
Channel.0.ImagePresetMode.IndoorBacklight.SSNRLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.SSNR2DLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.SSNR3DLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.IrisMode=Manual
Channel.0.ImagePresetMode.IndoorBacklight.PIrisMode=Auto
Channel.0.ImagePresetMode.IndoorBacklight.PIrisPosition=100
Channel.0.ImagePresetMode.IndoorBacklight.AGCMode=MaxGain
Channel.0.ImagePresetMode.IndoorBacklight.AGCLevel=1
Channel.0.ImagePresetMode.IndoorBacklight.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.IndoorBacklight.DayNightMode=Auto
Channel.0.ImagePresetMode.IndoorBacklight.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.IndoorBacklight.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.IndoorBacklight.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.IndoorBacklight.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.IndoorBacklight.SimpleFocus=False
Channel.0.ImagePresetMode.IndoorBacklight.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.EveryDay.From
To=00:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.SUN.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.MON.FromTo=00
:00-23:59
```

```
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.TUE.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.WED.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.THU.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.FRI.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.IndoorBacklight.DayNightModeSchedule.SAT.FromTo=00
:00-23:59
Channel.0.ImagePresetMode.IndoorBacklight.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.IndoorBrightScene.CompensationMode=Off
Channel.0.ImagePresetMode.IndoorBrightScene.BLCAreaTop=30
Channel.0.ImagePresetMode.IndoorBrightScene.BLCAreaBottom=70
Channel.0.ImagePresetMode.IndoorBrightScene.BLCAreaLeft=30
Channel.0.ImagePresetMode.IndoorBrightScene.BLCAreaRight=70
Channel.0.ImagePresetMode.IndoorBrightScene.BLCAreaCoordinates=975,548,2319,
1304
Channel.0.ImagePresetMode.IndoorBrightScene.BLCLevel=Medium
Channel.0.ImagePresetMode.IndoorBrightScene.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.IndoorBrightScene.HLCLevel=Medium
Channel.0.ImagePresetMode.IndoorBrightScene.HLCMaskTone=100
Channel.0.ImagePresetMode.IndoorBrightScene.HLCMode=On
Channel.0.ImagePresetMode.IndoorBrightScene.HLCMaskColor=Black
Channel.0.ImagePresetMode.IndoorBrightScene.HLCDimming=Off
Channel.0.ImagePresetMode.IndoorBrightScene.HLCAreaTop=30
Channel.0.ImagePresetMode.IndoorBrightScene.HLCAreaBottom=70
Channel.0.ImagePresetMode.IndoorBrightScene.HLCAreaLeft=30
Channel.0.ImagePresetMode.IndoorBrightScene.HLCAreaRight=70
Channel.0.ImagePresetMode.IndoorBrightScene.HLCAreaCoordinates=975,548,2319,
1304
Channel.0.ImagePresetMode.IndoorBrightScene.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.IndoorBrightScene.WDRLevel=Medium
Channel.0.ImagePresetMode.IndoorBrightScene.WDRAreaTop=1
Channel.0.ImagePresetMode.IndoorBrightScene.WDRAreaBottom=100
Channel.0.ImagePresetMode.IndoorBrightScene.WDRAreaLeft=1
```

```
Channel.0.ImagePresetMode.IndoorBrightScene.WDRAreaRight=100
Channel.0.ImagePresetMode.IndoorBrightScene.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.IndoorBrightScene.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.IndoorBrightScene.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.IndoorBrightScene.PreferShutterSpeed=1/60
Channel.0.ImagePresetMode.IndoorBrightScene.AFLKMode=Off
Channel.0.ImagePresetMode.IndoorBrightScene.SSNREnable=True
Channel.0.ImagePresetMode.IndoorBrightScene.SSNRMode=Manual
Channel.0.ImagePresetMode.IndoorBrightScene.SSNRLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.SSNR2DLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.SSNR3DLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.IrisMode=Auto
Channel.0.ImagePresetMode.IndoorBrightScene.PIrisMode=Auto
Channel.0.ImagePresetMode.IndoorBrightScene.PIrisPosition=100
Channel.0.ImagePresetMode.IndoorBrightScene.AGCMode=MaxGain
Channel.0.ImagePresetMode.IndoorBrightScene.AGCLevel=1
Channel.0.ImagePresetMode.IndoorBrightScene.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightMode=Auto
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.IndoorBrightScene.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.IndoorBrightScene.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.IndoorBrightScene.SimpleFocus=False
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightAlarmIn=SwitchToBWIfClos
es
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.EveryDay.Fr
omTo=00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.SUN.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.MON.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.TUE.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.WED.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleTHU=1
```

```
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.THU.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.FRI.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.IndoorBrightScene.DayNightModeSchedule.SAT.FromTo=
00:00-23:59
Channel.0.ImagePresetMode.IndoorBrightScene.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.NumberPlate.CompensationMode=Off
Channel.0.ImagePresetMode.NumberPlate.BLCAreaTop=30
Channel.0.ImagePresetMode.NumberPlate.BLCAreaBottom=70
Channel.0.ImagePresetMode.NumberPlate.BLCAreaLeft=30
Channel.0.ImagePresetMode.NumberPlate.BLCAreaRight=70
Channel.0.ImagePresetMode.NumberPlate.BLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.NumberPlate.BLCLevel=Medium
Channel.0.ImagePresetMode.NumberPlate.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.NumberPlate.HLCLevel=Medium
Channel.0.ImagePresetMode.NumberPlate.HLCMaskTone=100
Channel.0.ImagePresetMode.NumberPlate.HLCMode=On
Channel.0.ImagePresetMode.NumberPlate.HLCMaskColor=Black
Channel.0.ImagePresetMode.NumberPlate.HLCDimming=Off
Channel.0.ImagePresetMode.NumberPlate.HLCAreaTop=30
Channel.0.ImagePresetMode.NumberPlate.HLCAreaBottom=70
Channel.0.ImagePresetMode.NumberPlate.HLCAreaLeft=30
Channel.0.ImagePresetMode.NumberPlate.HLCAreaRight=70
Channel.0.ImagePresetMode.NumberPlate.HLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.NumberPlate.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.NumberPlate.WDRLevel=Medium
Channel.0.ImagePresetMode.NumberPlate.WDRAreaTop=1
Channel.0.ImagePresetMode.NumberPlate.WDRAreaBottom=100
Channel.0.ImagePresetMode.NumberPlate.WDRAreaLeft=1
Channel.0.ImagePresetMode.NumberPlate.WDRAreaRight=100
Channel.0.ImagePresetMode.NumberPlate.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.NumberPlate.AutoShortShutterSpeed=1/300
Channel.0.ImagePresetMode.NumberPlate.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.NumberPlate.PreferShutterSpeed=1/500
Channel.0.ImagePresetMode.NumberPlate.AFLKMode=Off
Channel.0.ImagePresetMode.NumberPlate.SSNREnable=True
Channel.0.ImagePresetMode.NumberPlate.SSNRMode=Manual
Channel.0.ImagePresetMode.NumberPlate.SSNRLevel=4
```

```
Channel.0.ImagePresetMode.NumberPlate.SSNR2DLevel=4
Channel.0.ImagePresetMode.NumberPlate.SSNR3DLevel=4
Channel.0.ImagePresetMode.NumberPlate.IrisMode=Auto
Channel.0.ImagePresetMode.NumberPlate.PIrisMode=Auto
Channel.0.ImagePresetMode.NumberPlate.PIrisPosition=100
Channel.0.ImagePresetMode.NumberPlate.AGCMode=MaxGain
Channel.0.ImagePresetMode.NumberPlate.AGCLevel=1
Channel.0.ImagePresetMode.NumberPlate.AGCMaxGainLevel=30
Channel.0.ImagePresetMode.NumberPlate.DayNightMode=Auto
Channel.0.ImagePresetMode.NumberPlate.DayNightSwitchingTime=5s
Channel.0.ImagePresetMode.NumberPlate.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.NumberPlate.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.NumberPlate.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.NumberPlate.SimpleFocus=False
Channel.0.ImagePresetMode.NumberPlate.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.EveryDay.FromTo=0
0:00-23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.SUN.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.MON.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.TUE.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.WED.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.THU.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.FRI.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.NumberPlate.DayNightModeSchedule.SAT.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.NumberPlate.WDRControlMode=AlwaysOn
Channel.0.ImagePresetMode.Vivid.CompensationMode=Off
```

```
Channel.0.ImagePresetMode.Vivid.BLCAreaTop=30
Channel.0.ImagePresetMode.Vivid.BLCAreaBottom=70
Channel.0.ImagePresetMode.Vivid.BLCAreaLeft=30
Channel.0.ImagePresetMode.Vivid.BLCAreaRight=70
Channel.0.ImagePresetMode.Vivid.BLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.Vivid.BLCLevel=Medium
Channel.0.ImagePresetMode.Vivid.NormalizedBLCLevel=50
Channel.0.ImagePresetMode.Vivid.HLCLevel=Medium
Channel.0.ImagePresetMode.Vivid.HLCMaskTone=100
Channel.0.ImagePresetMode.Vivid.HLCMode=On
Channel.0.ImagePresetMode.Vivid.HLCMaskColor=Black
Channel.0.ImagePresetMode.Vivid.HLCDimming=Off
Channel.0.ImagePresetMode.Vivid.HLCAreaTop=30
Channel.0.ImagePresetMode.Vivid.HLCAreaBottom=70
Channel.0.ImagePresetMode.Vivid.HLCAreaLeft=30
Channel.0.ImagePresetMode.Vivid.HLCAreaRight=70
Channel.0.ImagePresetMode.Vivid.HLCAreaCoordinates=975,548,2319,1304
Channel.0.ImagePresetMode.Vivid.NormalizedHLCLevel=50
Channel.0.ImagePresetMode.Vivid.WDRLevel=Medium
Channel.0.ImagePresetMode.Vivid.WDRAreaTop=1
Channel.0.ImagePresetMode.Vivid.WDRAreaBottom=100
Channel.0.ImagePresetMode.Vivid.WDRAreaLeft=1
Channel.0.ImagePresetMode.Vivid.WDRAreaRight=100
Channel.0.ImagePresetMode.Vivid.WDRAreaCoordinates=0,0,3327,1871
Channel.0.ImagePresetMode.Vivid.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.Vivid.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.Vivid.PreferShutterSpeed=1/60
Channel.0.ImagePresetMode.Vivid.AFLKMode=Off
Channel.0.ImagePresetMode.Vivid.SSNREnable=True
Channel.0.ImagePresetMode.Vivid.SSNRMode=Manual
Channel.0.ImagePresetMode.Vivid.SSNRLevel=12
Channel.0.ImagePresetMode.Vivid.SSNR2DLevel=12
Channel.0.ImagePresetMode.Vivid.SSNR3DLevel=12
Channel.0.ImagePresetMode.Vivid.IrisMode=Auto
Channel.0.ImagePresetMode.Vivid.PIrisMode=Auto
Channel.0.ImagePresetMode.Vivid.PIrisPosition=100
Channel.0.ImagePresetMode.Vivid.AGCMode=MaxGain
Channel.0.ImagePresetMode.Vivid.AGCLevel=1
Channel.0.ImagePresetMode.Vivid.AGCMaxGainLevel=60
Channel.0.ImagePresetMode.Vivid.DayNightMode=Auto
Channel.0.ImagePresetMode.Vivid.DayNightSwitchingTime=5s
```

```
Channel.0.ImagePresetMode.Vivid.DayNightSwitchingMode=Normal
Channel.0.ImagePresetMode.Vivid.DNSwitchColorToBWThreshold=70
Channel.0.ImagePresetMode.Vivid.DNSwitchBWToColorThreshold=30
Channel.0.ImagePresetMode.Vivid.SimpleFocus=False
Channel.0.ImagePresetMode.Vivid.DayNightAlarmIn=SwitchToBWIfCloses
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleEveryDay=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.EveryDay.FromTo=00:00-
23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleSUN=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.SUN.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleMON=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.MON.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleTUE=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.TUE.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleWED=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.WED.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleTHU=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.THU.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleFRI=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.FRI.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.DayNightModeScheduleSAT=1
Channel.0.ImagePresetMode.Vivid.DayNightModeSchedule.SAT.FromTo=00:00-23:59
Channel.0.ImagePresetMode.Vivid.WDRControlMode=AlwaysOn
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Camera": [
        {
            "Channel": 0,
            "CompensationMode": "Off",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
```

```
            "BLCAreaCoordinates": [
                {
                    "x1": 1125,
                    "y1": 632
                },
                {
                    "x2": 2676,
                    "y2": 1505
                }
            ],
            "HLCLevel": "Medium",
            "NormalizedHLCLevel": 50,
            "HLCMaskTone": 100,
            "HLCMode": "On",
            "HLCMaskColor": "Black",
            "HLCDimming": "Off",
            "HLCAreaTop": 30,
            "HLCAreaBottom": 70,
            "HLCAreaLeft": 30,
            "HLCAreaRight": 70,
            "HLCAreaCoordinates": [
                {
                    "x1": 1125,
                    "y1": 632
                },
                {
                    "x2": 2676,
                    "y2": 1505
                }
            ],
            "WDRLevel": "High",
            "WDRAreaTop": 1,
            "WDRAreaBottom": 100,
            "WDRAreaLeft": 1,
            "WDRAreaRight": 100,
            "WDRAreaCoordinates": [
                {
                    "x1": 0,
                    "y1": 0
                },
                {
```

```
                "x2": 3839,
                "y2": 2159
            }
        ],
        "AutoShortShutterSpeed": "1/25",
        "AutoLongShutterSpeed": "1/12000",
        "PreferShutterSpeed": "1/200",
        "AFLKMode": "Off",
        "SSNREnable": true,
        "SSNRMode": "Manual",
        "SSNRLevel": 12,
        "SSNR2DLevel": 12,
        "SSNR3DLevel": 12,
        "IrisMode": "Auto",
        "PIrisMode": "Auto",
        "PIrisPosition": 100,
        "AGCMode": "MaxGain",
        "AGCLevel": 1,
        "AGCLevelFloat": 1.2,
        "AGCMaxGainLevel": 60,
        "ManualShutterSpeed": 200,
        "DayNightMode": "Auto",
        "DayNightSwitchingTime": "5s",
        "DayNightSwitchingMode": "Normal",
        "DNSwitchColorToBWThreshold": 70,
        "DNSwitchBWToColorThreshold": 30,
        "SimpleFocus": false,
        "TemperatureRangeMode": "Auto",
        "GainSequencerEnable": false,
        "GainFrameInterval": 1,
        "GainNumberOfSteps": 2,
        "GainSequencerStep": [
            {
                "Step": 1,
                "GainLevel": 0.0
            },
            {
                "Step": 2,
                "GainLevel": 0.0
            },
            {
```

```
                    "Step": 3,
                    "GainLevel": 0.0
                },
                {
                    "Step": 4,
                    "GainLevel": 0.0
                },
                {
                    "Step": 5,
                    "GainLevel": 0.0
                },
                {
                    "Step": 6,
                    "GainLevel": 0.0
                }
            ],
            "DayNightAlarmIn": "SwitchToBWIfCloses",
            "DayNightModeSchedules": {
                "EveryDay": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SUN": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "MON": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "TUE": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "WED": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "THU": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
```

```
                },
                "FRI": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn",
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "CompensationMode": "Off",
                    "BLCLevel": "Medium",
                    "NormalizedBLCLevel": 50,
                    "BLCAreaTop": 30,
                    "BLCAreaBottom": 70,
                    "BLCAreaLeft": 30,
                    "BLCAreaRight": 70,
                    "BLCAreaCoordinates": [
                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "HLCLevel": "Medium",
                    "NormalizedHLCLevel": 50,
                    "HLCMaskTone": 100,
                    "HLCMode": "On",
                    "HLCMaskColor": "Black",
                    "HLCDimming": "Off",
                    "HLCAreaTop": 30,
                    "HLCAreaBottom": 70,
                    "HLCAreaLeft": 30,
                    "HLCAreaRight": 70,
```

```
                    "HLCAreaCoordinates": [
                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "WDRLevel": "High",
                    "WDRAreaTop": 1,
                    "WDRAreaBottom": 100,
                    "WDRAreaLeft": 1,
                    "WDRAreaRight": 100,
                    "WDRAreaCoordinates": [
                        {
                            "x1": 0,
                            "y1": 0
                        },
                        {
                            "x2": 3839,
                            "y2": 2159
                        }
                    ],
                    "AutoShortShutterSpeed": "1/25",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/200",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 12,
                    "SSNR2DLevel": 12,
                    "SSNR3DLevel": 12,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 60,
                    "DayNightMode": "Auto",
```

```
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "MON": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "TUE": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "WED": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "THU": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "FRI": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SAT": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        }
                    },
                    "WDRControlMode": "AlwaysOn"
```

```
            },
            {
                "ImagePresetMode": "UserPreset2",
                "CompensationMode": "Off",
                "BLCLevel": "Medium",
                "NormalizedBLCLevel": 50,
                "BLCAreaTop": 30,
                "BLCAreaBottom": 70,
                "BLCAreaLeft": 30,
                "BLCAreaRight": 70,
                "BLCAreaCoordinates": [
                    {
                        "x1": 1125,
                        "y1": 632
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "HLCLevel": "Medium",
                "NormalizedHLCLevel": 50,
                "HLCMaskTone": 100,
                "HLCMode": "On",
                "HLCMaskColor": "Black",
                "HLCDimming": "Off",
                "HLCAreaTop": 30,
                "HLCAreaBottom": 70,
                "HLCAreaLeft": 30,
                "HLCAreaRight": 70,
                "HLCAreaCoordinates": [
                    {
                        "x1": 1125,
                        "y1": 632
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "WDRLevel": "High",
```

```
                    "WDRAreaTop": 1,
                    "WDRAreaBottom": 100,
                    "WDRAreaLeft": 1,
                    "WDRAreaRight": 100,
                    "WDRAreaCoordinates": [
                        {
                            "x1": 0,
                            "y1": 0
                        },
                        {
                            "x2": 3839,
                            "y2": 2159
                        }
                    ],
                    "AutoShortShutterSpeed": "1/25",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/200",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 12,
                    "SSNR2DLevel": 12,
                    "SSNR3DLevel": 12,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 60,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
```

```
                "SUN": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "MON": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "TUE": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "WED": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "THU": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "FRI": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn"
        },
        {
            "ImagePresetMode": "OutdoorDaytime",
            "CompensationMode": "Off",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
            "BLCAreaCoordinates": [
```

```
                    {
                        "x1": 1125,
                        "y1": 632
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "HLCLevel": "Medium",
                "NormalizedHLCLevel": 50,
                "HLCMaskTone": 100,
                "HLCMode": "On",
                "HLCMaskColor": "Black",
                "HLCDimming": "Off",
                "HLCAreaTop": 30,
                "HLCAreaBottom": 70,
                "HLCAreaLeft": 30,
                "HLCAreaRight": 70,
                "HLCAreaCoordinates": [
                    {
                        "x1": 1125,
                        "y1": 632
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "WDRLevel": "Medium",
                "WDRAreaTop": 1,
                "WDRAreaBottom": 100,
                "WDRAreaLeft": 1,
                "WDRAreaRight": 100,
                "WDRAreaCoordinates": [
                    {
                        "x1": 0,
                        "y1": 0
                    },
                    {
                        "x2": 3839,
```

```
                    "y2": 2159
                }
            ],
            "AutoShortShutterSpeed": "1/25",
            "AutoLongShutterSpeed": "1/12000",
            "PreferShutterSpeed": "1/250",
            "AFLKMode": "Off",
            "SSNREnable": true,
            "SSNRMode": "Manual",
            "SSNRLevel": 12,
            "SSNR2DLevel": 12,
            "SSNR3DLevel": 12,
            "IrisMode": "Auto",
            "PIrisMode": "Auto",
            "PIrisPosition": 100,
            "AGCMode": "MaxGain",
            "AGCLevel": 1,
            "AGCMaxGainLevel": 60,
            "DayNightMode": "Auto",
            "DayNightSwitchingTime": "5s",
            "DayNightSwitchingMode": "Normal",
            "DNSwitchColorToBWThreshold": 70,
            "DNSwitchBWToColorThreshold": 30,
            "SimpleFocus": false,
            "DayNightAlarmIn": "SwitchToBWIfCloses",
            "DayNightModeSchedules": {
                "EveryDay": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SUN": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "MON": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "TUE": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
```

```
                },
                "WED": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "THU": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "FRI": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn"
        },
        {
            "ImagePresetMode": "OutdoorNightime",
            "CompensationMode": "Off",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
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                {
                    "x2": 2676,
                    "y2": 1505
                }
            ],
            "HLCLevel": "Medium",
            "NormalizedHLCLevel": 50,
```

```
                    "HLCMaskTone": 100,
                    "HLCMode": "On",
                    "HLCMaskColor": "Black",
                    "HLCDimming": "Off",
                    "HLCAreaTop": 30,
                    "HLCAreaBottom": 70,
                    "HLCAreaLeft": 30,
                    "HLCAreaRight": 70,
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                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "WDRLevel": "Medium",
                    "WDRAreaTop": 1,
                    "WDRAreaBottom": 100,
                    "WDRAreaLeft": 1,
                    "WDRAreaRight": 100,
                    "WDRAreaCoordinates": [
                        {
                            "x1": 0,
                            "y1": 0
                        },
                        {
                            "x2": 3839,
                            "y2": 2159
                        }
                    ],
                    "AutoShortShutterSpeed": "1/25",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/200",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 12,
                    "SSNR2DLevel": 12,
```

```
                    "SSNR3DLevel": 12,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 60,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "MON": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "TUE": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "WED": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "THU": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "FRI": {
                            "DayNightSchedule": true,
```

```
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn"
        },
        {
            "ImagePresetMode": "IndoorBacklight",
            "CompensationMode": "WDR",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
            "BLCAreaCoordinates": [
                {
                    "x1": 1125,
                    "y1": 632
                },
                {
                    "x2": 2676,
                    "y2": 1505
                }
            ],
            "HLCLevel": "Medium",
            "NormalizedHLCLevel": 50,
            "HLCMaskTone": 100,
            "HLCMode": "On",
            "HLCMaskColor": "Black",
            "HLCDimming": "Off",
            "HLCAreaTop": 30,
            "HLCAreaBottom": 70,
            "HLCAreaLeft": 30,
            "HLCAreaRight": 70,
            "HLCAreaCoordinates": [
                {
                    "x1": 1125,
```

```
                "y1": 632
            },
            {
                "x2": 2676,
                "y2": 1505
            }
        ],
        "WDRLevel": "Low",
        "WDRAreaTop": 1,
        "WDRAreaBottom": 100,
        "WDRAreaLeft": 1,
        "WDRAreaRight": 100,
        "WDRAreaCoordinates": [
            {
                "x1": 0,
                "y1": 0
            },
            {
                "x2": 3839,
                "y2": 2159
            }
        ],
        "AutoShortShutterSpeed": "1/50",
        "AutoLongShutterSpeed": "1/1500",
        "PreferShutterSpeed": "1/200",
        "AFLKMode": "Off",
        "SSNREnable": true,
        "SSNRMode": "Manual",
        "SSNRLevel": 24,
        "SSNR2DLevel": 24,
        "SSNR3DLevel": 24,
        "IrisMode": "Manual",
        "PIrisMode": "Auto",
        "PIrisPosition": 100,
        "AGCMode": "MaxGain",
        "AGCLevel": 1,
        "AGCMaxGainLevel": 60,
        "DayNightMode": "Auto",
        "DayNightSwitchingTime": "5s",
        "DayNightSwitchingMode": "Normal",
        "DNSwitchColorToBWThreshold": 70,
```

```
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "MON": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "TUE": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "WED": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "THU": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "FRI": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SAT": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        }
                    },
                    "WDRControlMode": "AlwaysOn"
                },
                {
                    "ImagePresetMode": "IndoorBrightScene",
```

```
                    "CompensationMode": "Off",
                    "BLCLevel": "Medium",
                    "NormalizedBLCLevel": 50,
                    "BLCAreaTop": 30,
                    "BLCAreaBottom": 70,
                    "BLCAreaLeft": 30,
                    "BLCAreaRight": 70,
                    "BLCAreaCoordinates": [
                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "HLCLevel": "Medium",
                    "NormalizedHLCLevel": 50,
                    "HLCMaskTone": 100,
                    "HLCMode": "On",
                    "HLCMaskColor": "Black",
                    "HLCDimming": "Off",
                    "HLCAreaTop": 30,
                    "HLCAreaBottom": 70,
                    "HLCAreaLeft": 30,
                    "HLCAreaRight": 70,
                    "HLCAreaCoordinates": [
                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "WDRLevel": "Medium",
                    "WDRAreaTop": 1,
                    "WDRAreaBottom": 100,
                    "WDRAreaLeft": 1,
```

```
                    "WDRAreaRight": 100,
                    "WDRAreaCoordinates": [
                        {
                            "x1": 0,
                            "y1": 0
                        },
                        {
                            "x2": 3839,
                            "y2": 2159
                        }
                    ],
                    "AutoShortShutterSpeed": "1/25",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/50",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 12,
                    "SSNR2DLevel": 12,
                    "SSNR3DLevel": 12,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 60,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
```

```
                },
                "MON": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "TUE": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "WED": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "THU": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "FRI": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn"
        },
        {
            "ImagePresetMode": "NumberPlate",
            "CompensationMode": "Off",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
            "BLCAreaCoordinates": [
                {
                    "x1": 1125,
                    "y1": 632
```

```
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "HLCLevel": "Medium",
                "NormalizedHLCLevel": 50,
                "HLCMaskTone": 100,
                "HLCMode": "On",
                "HLCMaskColor": "Black",
                "HLCDimming": "Off",
                "HLCAreaTop": 30,
                "HLCAreaBottom": 70,
                "HLCAreaLeft": 30,
                "HLCAreaRight": 70,
                "HLCAreaCoordinates": [
                    {
                        "x1": 1125,
                        "y1": 632
                    },
                    {
                        "x2": 2676,
                        "y2": 1505
                    }
                ],
                "WDRLevel": "Medium",
                "WDRAreaTop": 1,
                "WDRAreaBottom": 100,
                "WDRAreaLeft": 1,
                "WDRAreaRight": 100,
                "WDRAreaCoordinates": [
                    {
                        "x1": 0,
                        "y1": 0
                    },
                    {
                        "x2": 3839,
                        "y2": 2159
                    }
                ],
```

```
                    "AutoShortShutterSpeed": "1/300",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/500",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 4,
                    "SSNR2DLevel": 4,
                    "SSNR3DLevel": 4,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 30,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "MON": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "TUE": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "WED": {
                            "DayNightSchedule": true,
```

```
                    "FromTo": "0:00-23:59"
                },
                "THU": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "FRI": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                },
                "SAT": {
                    "DayNightSchedule": true,
                    "FromTo": "0:00-23:59"
                }
            },
            "WDRControlMode": "AlwaysOn"
        },
        {
            "ImagePresetMode": "Vivid",
            "CompensationMode": "Off",
            "BLCLevel": "Medium",
            "NormalizedBLCLevel": 50,
            "BLCAreaTop": 30,
            "BLCAreaBottom": 70,
            "BLCAreaLeft": 30,
            "BLCAreaRight": 70,
            "BLCAreaCoordinates": [
                {
                    "x1": 1125,
                    "y1": 632
                },
                {
                    "x2": 2676,
                    "y2": 1505
                }
            ],
            "HLCLevel": "Medium",
            "NormalizedHLCLevel": 50,
            "HLCMaskTone": 100,
            "HLCMode": "On",
            "HLCMaskColor": "Black",
```

```
                    "HLCDimming": "Off",
                    "HLCAreaTop": 30,
                    "HLCAreaBottom": 70,
                    "HLCAreaLeft": 30,
                    "HLCAreaRight": 70,
                    "HLCAreaCoordinates": [
                        {
                            "x1": 1125,
                            "y1": 632
                        },
                        {
                            "x2": 2676,
                            "y2": 1505
                        }
                    ],
                    "WDRLevel": "Medium",
                    "WDRAreaTop": 1,
                    "WDRAreaBottom": 100,
                    "WDRAreaLeft": 1,
                    "WDRAreaRight": 100,
                    "WDRAreaCoordinates": [
                        {
                            "x1": 0,
                            "y1": 0
                        },
                        {
                            "x2": 3839,
                            "y2": 2159
                        }
                    ],
                    "AutoShortShutterSpeed": "1/25",
                    "AutoLongShutterSpeed": "1/12000",
                    "PreferShutterSpeed": "1/50",
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRMode": "Manual",
                    "SSNRLevel": 12,
                    "SSNR2DLevel": 12,
                    "SSNR3DLevel": 12,
                    "IrisMode": "Auto",
                    "PIrisMode": "Auto",
```

```
                    "PIrisPosition": 100,
                    "AGCMode": "MaxGain",
                    "AGCLevel": 1,
                    "AGCMaxGainLevel": 60,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Normal",
                    "DNSwitchColorToBWThreshold": 70,
                    "DNSwitchBWToColorThreshold": 30,
                    "SimpleFocus": false,
                    "DayNightAlarmIn": "SwitchToBWIfCloses",
                    "DayNightModeSchedules": {
                        "EveryDay": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SUN": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "MON": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "TUE": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "WED": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "THU": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "FRI": {
                            "DayNightSchedule": true,
                            "FromTo": "0:00-23:59"
                        },
                        "SAT": {
```

```
                                "DayNightSchedule": true,
                                "FromTo": "0:00-23:59"
                            }
                        },
                        "WDRControlMode": "AlwaysOn"
                    }
                ]
            }
        ]
}
```

112.4.2. Setting the AFLK mode as 50

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=camera&action=set&AFLKMode=50
```

sets image preset mode as Vivid with AFLK mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&AFLKMode=50&ImagePresetMode=Vivid
```

112.4.3. Setting day and night mode to BW

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&DayNightMode=BW
```

112.4.4. Setting the time schedule for day and night mode

Using day and night mode from 5: 32 AM to 8:32 PM on Wednesday

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&DayNightMode=BW&DayNightModeSchedul
e.WED=1&DayNightModeSchedule.WED.FromTo=5:32-20:32
```

Using day and night mode from 1:00 AM to 8:59 AM everyday

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&DayNightMode=BW&DayNightModeSchedul
e.EveryDay=1&DayNightModeSchedule.EveryDay.FromTo=1:0-8:59
```

112.4.5. Setting the compensation mode to BLC

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&CompensationMode=BLC
```

112.4.6. Setting the Iris mode to manual

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&IrisMode=Manual
```

112.4.7. Setting the P-Iris mode to manual

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=camera&action=set&IrisMode=P-
Iris&PIrisMode=Manual
```

112.4.8. Setting the day and night mode to color

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&DayNightMode=Color
```

112.4.9. Setting the switching time and mode for day and night mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&DayNightMode=Auto&DayNightSwitching
Time=10s&DayNightSwitchingMode=Slow
```

Chapter 113. White Balance

113.1. Description

The **whitebalance** submenu configures white balance settings.

NOTE This chapter applies to network cameras only.

Attribute to check for White Balance support: "attributes/Image/Support/WhiteBalance"

Access level

Action	Camera
view	User
set	User
control	User

113.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
whitebalance&action=<value>[&<parameter>=<value>]
```

113.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads white balance settings
	Channel	REQ, RES	<csv>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.WhiteBalanceMode	RES	<enum> Manual, AWC, ATW, Outdoor, Indoor, ATW1, ATW2, 3200K, 5600K, Mercury, Sodium, NarrowATW	White balance mode • Manual: Manually adjust the red and blue gains of the camera video. • AWC: Corrects the colors of the camera video to be optimized to the current lighting condition and screen mode. • ATW: Automatically corrects the colors of the camera video. (2400K~10500K) • Outdoor: Automatically corrects the video colors of the camera to be optimized to the outdoor environment. (1800K~10500K) • Indoor: Automatically corrects the video colors of the camera to be optimized to the indoor environment. (4500K~8500K) • Mercury: Automatically corrects the video colors of the camera to be optimized to a mercury lamp environment. • Sodium: Automatically corrects the video colors of the camera to be optimized to a sodium lamp environment. • NarrowATW: Automatically corrects the video colors of the camera to be optimized to current lighting condition and screen mode in a narrower range than ATW mode (2800K~9000K) **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.WhiteBalanceManualRedLevel	RES	<int>	Red level of the white balance The higher the value, the greater the red level. **WhiteBalanceManualRedLevel** is valid only when **WhiteBalanceMode** is set to Manual.
	ImagePresetMode.#.WhiteBalanceManualBlueLevel	RES	<int>	Blue level of the white balance. The higher the value, the greater the blue level. **WhiteBalanceManualBlueLevel** is valid only when **WhiteBalanceMode** is set to Manual.
set	Channel	REQ, RES	<int>	Channel ID
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped and the original image settings saved in the camera will be applied. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	WhiteBalanceMode	REQ, RES	<enum> Manual, AWC, ATW, Outdoor, Indoor, ATW1, ATW2, 3200K, 5600K, Mercury, Sodium, NarrowATW	White balance mode • Manual: Adjusts the Red and blue gains of the camera video manually • AWC: Corrects the colors of the camera video to be optimized to the current lighting condition and screen mode (2400K~10500K) • ATW: Corrects the colors of the camera video automatically • Outdoor: Automatically corrects the video colors of the camera to be optimized to the outdoor environment (1800K~10500K) • Indoor: Automatically corrects the video colors of the camera to be optimized to the indoor environment. (4500K~8500K) • Mercury: Automatically corrects the video colors of the camera to be optimized to a mercury lamp environment. • Sodium: Automatically corrects the video colors of the camera to be optimized to a sodium lamp environment. • NarrowATW: Automatically corrects the video colors of the camera to be optimized to current lighting condition and screen mode in a narrower range than ATW mode (2800K~9000K)
	WhiteBalanceManualRedLevel	REQ, RES	<int>	Red level of the white balance The higher the value, the greater the red level. **WhiteBalanceManualRedLevel** is valid only when **WhiteBalanceMode** is set to Manual.

Action	Parameters	Request/ Response	Type/ Value	Description
	WhiteBalanceManualBlueLevel	REQ, RES	<int>	Blue level of the white balance. The higher the value, the greater the blue level. **WhiteBalanceManualBlueLevel** is valid only when **WhiteBalanceMode** is set to Manual.
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNighttime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each ImagePresetMode can have ImagePreset settings. Image settings in any Mode can be changed. Default settings for UserPreset1 and UserPreset2 are the same, whereas other preset mode like OutdoorDaytime, OutdoorNighttime,etc., have specif default settings depending on its name.
control	AWCSet			AWC mode set This parameter is used for a one-time control action to obtain the optimal conditions for the current lighting. It does not require any value. This parameter is valid only when **WhiteBalanceMode** is set to AWC.

113.4. Examples

113.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.WhiteBalanceMode=ATW
Channel.0.WhiteBalanceManualRedLevel=604
Channel.0.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.UserPreset1.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.UserPreset1.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.UserPreset1.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.UserPreset2.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.UserPreset2.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.UserPreset2.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.OutdoorDaytime.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.OutdoorDaytime.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.OutdoorDaytime.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.OutdoorNightime.WhiteBalanceMode=Outdoor
Channel.0.ImagePresetMode.OutdoorNightime.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.OutdoorNightime.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.IndoorBacklight.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.IndoorBacklight.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.IndoorBacklight.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.IndoorBrightScene.WhiteBalanceMode=Indoor
Channel.0.ImagePresetMode.IndoorBrightScene.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.IndoorBrightScene.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.NumberPlate.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.NumberPlate.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.NumberPlate.WhiteBalanceManualBlueLevel=403
Channel.0.ImagePresetMode.Vivid.WhiteBalanceMode=ATW
Channel.0.ImagePresetMode.Vivid.WhiteBalanceManualRedLevel=604
Channel.0.ImagePresetMode.Vivid.WhiteBalanceManualBlueLevel=403
```

113.4.2. Setting the white balance to manual

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=set&WhiteBalanceMode=Manual
```

113.4.3. Setting the image preset mode's image

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=set&WhiteBalanceMode=Manual&Image
PresetMode=Vivid
```

113.4.4. Setting the white balance (Red level to 70, Blue level to 120)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=set&WhiteBalanceManualRedLevel=70
&WhiteBalanceManualBlueLevel=120
```

113.4.5. AWC set

For AWC mode setting, **WhiteBalanceMode** must be set to AWC.

REQUEST 1

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=set&WhiteBalanceMode=AWC
```

REQUEST 2

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whitebalance&action=control&AWCSet
```

Chapter 114. Image Enhancements

114.1. Description

The **imageenhancements** submenu configures the image enhancement settings.

CAUTION	It is recommended to use the **imageenhancements2** submenu instead. The **imageenhancements** submenu will be deprecated starting January 2027.

Access level

Action	Camera	Encoder	NVR
view	User	Admin	User
set	User	Admin	User
control	User	Admin	

114.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imageenhancements&action=<value>[&<parameter>=<value>]
```

114.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads image enhancement settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	OpticalDefogFilterEnable	REQ, RES	<bool> True, False	Enables or disables the optical defog filter.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied. CAMERA ONLY
	SharpnessEnable	REQ, RES	<bool> True, False	Enables or disables image sharpness CAMERA ONLY **Note** Attribute to check for Sharpness: "attributes/Image/Support/Sharpness"
	SharpnessLevel	REQ, RES	<int>	Sharpness level SharpnessLevel is valid only when SharpnessEnable is set to True. CAMERA ONLY
	Brightness	REQ, RES	<int>	Brightness level The higher the value, the greater the brightness. CAMERA ONLY **Note** Attribute to check for Brightness: "attributes/Image/Support/Brightness"

Action	Parameters	Request/ Response	Type/ Value	Description
	Gamma	REQ, RES	<int>	Gamma level The higher the value, the greater the gamma value. CAMERA ONLY **Note** Attribute to check for Gamma: "attributes/Image/Support/Gamma"
	GammaControl	REQ, RES	<bool>	Gamma Control Adjust the brightness of the image. When On is selected, dark areas are generally brighter. CAMERA ONLY **Note** When setting On, the screen change due to changes in brightness and gamma items may not be significant.
	Saturation	REQ, RES	<int>	Saturation level The higher the value, the greater the saturation. CAMERA ONLY **Note** Attribute to check for Saturation: "attributes/Image/Support/Saturation"

<table>
<tr><th>Action</th><th>Parameters</th><th>Request/ Response</th><th>Type/ Value</th><th>Description</th></tr>
<tr><td></td><td>Contrast</td><td>REQ, RES</td><td><int></td><td>Contrast level

The higher the value, the greater the contrast.

CAMERA ONLY

Note
Attribute to check for Saturation: "image/imageenhancements/set/Contrast", "attributes/Image/Support/Contrast",</td></tr>
<tr><td></td><td>DefogMode</td><td>REQ, RES</td><td><enum>
Off, Auto, Manual</td><td>Defogging Mode

This enhances the quality of images created in a foggy environment.
Off: Disables defogging function
Auto: Automatically compensates the image according to the fog level
Manual: Manually sets the amount of compensation
Note
Attribute to check for Defog: "attributes/Image/Support/Defog"</td></tr>
<tr><td></td><td>DefogLevel</td><td>REQ, RES</td><td><int></td><td>Defogging Level

The greater the value, the higher the defogging level.

DefogLevel is valid only when DefogMode is set to Manual or Auto.</td></tr>
</table>

Action	Parameters	Request/ Response	Type/ Value	Description
	DISEnable	REQ, RES	<bool> True, False	Enables or disables DIS (Digital Image Stabilization). This compensates the image automatically when the camera vibrates due to external factors such as wind. CAMERA ONLY **Note** Attribute to check for DIS: "attributes/Image/Support/DIS"
	DISFocalLength	REQ,RES	<int> 2800 to 50000 (micro meter)	DIS Focal length value adjustable, can check imageoptions submenu "DISFocalLengthStepSize" parameter for allowed step size. CAMERA ONLY
	CAR	REQ, RES	<enum> Off, Low, Medium, High	Color Aberration Reduction This reduces the distortion or blurring of purple in the very bright area of an image. CAMERA ONLY
	LDCEnable	REQ, RES	<bool> True, False	Enables or disables LDC (Lens Distortion Control). LDCEnable and LDCMode cannot be set at the same time. **LDCEnable** True is same as Manual **LDCMode** **LDCEnable** False is same as Off **LDCMode** CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	LDCLevel	REQ, RES	<int>	LDC level The greater the value, the higher the LDC level. **LDCLevel** is valid only when **LDCMode** is not set to Off. **LDCLevel** is valid only when **LDCEnable** is not set to False. CAMERA ONLY
	LDCMode	REQ, RES	<enum> Off, Auto, Manual, FillMode_Manual, FillMode_Auto, StretchMode_Manual, StretchMode_Auto	Provides option to control the LDC feature. • Off: Disables LDC function • Auto: Automatically adjusts the LDC • Manual: Manually sets the LDC level • FillMode: The angle of up and down view is not changed, but the angle of left and right view can be cut. FillMode_Manual mode works LDC with LDCLevel value, FillMode_Auto mode is works LDC with automatically • StretchMode: The FOV is not changed, but the aspect ratio can be changed. StretchMode_Manual mode works LDC with LDCLevel value, StretchMode_Auto mode is works LDC with automatically CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	XCEEnable	REQ RES	<bool>	XCE(eXternd Contrast Enhancement) Enabling This feature is similar to using unsharp mask filtering **Note** Attribute to check for XCEEnable: "attributes/Image/Support/XCE" CAMERA ONLY
	XCELevel	REQ RES	<int>	User can set filter level **Note** Attribute to check for XCELevel: "attributes/Image/Support/XCE" CAMERA ONLY
	HorizontalPosition	REQ RES	<int>	It can adjust horizontal analog camera video position ENCODER ONLY
	VerticalPosition	REQ RES	<int>	It can adjust vertical analog camera video position ENCODER ONLY
control	Channel	REQ	<int>	Channel ID
	Reset	REQ	<bool> True, False	The following image parameters are reset to the default value; Contrast, Brightness, SharpnessLevel, Saturation

114.4. Examples

114.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imageenhancements&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain
```

```
<Body>
```

```
Channel.0.Brightness=50
Channel.0.SharpnessEnable=True
Channel.0.SharpnessLevel=12
Channel.0.Gamma=5
Channel.0.GammaControl=On
Channel.0.Saturation=50
Channel.0.DefogMode=Off
Channel.0.DefogLevel=5
Channel.0.DISEnable=False
Channel.0.LDCEnable=False
Channel.0.LDCMode=Off
Channel.0.LDCLevel=15
Channel.0.Contrast=50
Channel.0.XCEEnable=False
Channel.0.XCELevel=50
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ImageEnhancements": [
        {
            "Channel": 0,
            "Brightness": 50,
            "SharpnessEnable": true,
            "SharpnessLevel": 12,
            "Gamma": 5,
            "GammaControl": "On",
            "Saturation": 50,
            "DefogMode": "Off",
            "DefogLevel": 5,
            "DISEnable": false,
            "LDCEnable": false,
            "LDCMode": "Off",
            "LDCLevel": 15,
```

```
            "Contrast": 50,
            "XCEEnable": false,
            "XCELevel": 50
        }
    ]
}
```

114.4.2. Setting the sharpness level to 5

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imageenhancements&action=set&SharpnessLevel=5
```

114.4.3. Setting the image enhancements

Saturation to 60, Defog mode to Manual, DIS to On, and Brightness to 60

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imageenhancements&action=set&Saturation=60&DefogMode=
Manual&DISEnable=True&Brightness=60
```

Chapter 115. IR LED

115.1. Description

The **irled** submenu configures IR LED (Infra-red Light-Emitting-Diode) settings. IR LED creates brighter and clearer images in low illumination situations.

NOTE This chapter applies to network cameras only.

Attribute to check for Feature Support: "attributes/Image/Support/IRLED"

Access level

Action	Camera
view	User
set	User

115.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
irled&action=<value>[&<parameter>=<value>]
```

115.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads IR LED settings.
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied.
	Mode	REQ, RES	<enum> Off, On, Sensor, Schedule, Manual, Auto, Auto1, Auto2	IR LED mode • Off: Turns off IR LED. • On: Turns on IR LED. • Sensor: Changes the IR according to the light flux detected by the light sensor. • Schedule: Sets the IR usage time. • Manual: Controls the IR manually. • Auto, Auto1, Auto2: Controls the IR Level automatically. See product specs for detailed information on the actual differences between the modes
	LEDOnLevel	REQ, RES	<int>	LED On level LED is turned on when the light flux is lower than the LED usage level. **LEDOnLevel** is valid only when **Mode** is set to Sensor.

Action	Parameters	Request/ Response	Type/ Value	Description
	LEDOffLevel	REQ, RES	<int>	LED Off level LED is turned off when the light flux is higher than the LED usage level. **LEDOffLevel** is valid only when **Mode** is set to Sensor.
	LEDPowerControlMode	REQ, RES	<enum> Off, Auto	LED power control • Off: Turns off the LED power control. • Auto: Controls the screen saturation caused by a nearby object.
	LEDMaxPower	REQ, RES	<enum> Low, Medium, High	Maximum power for IR LED It sets the highest brightness of IR.
	LEDMaxPowerLevel	REQ, RES	<int>	Maximum power level for IR LED It sets the highest IR brightness
	Schedule.EveryDay.FromTo	REQ, RES	<string>	Schedule for IR LED The schedule sets the time when the IR will remain turned on in the format of <hh:mm-hh:mm>. **Schedule.EveryDay.FromTo** is valid only when **Mode** is set to Sensor.
	Level	REQ, RES	<int>	IR LED level **Level** is valid only when **Mode** is set to Manual
	Zone.#.Level	REQ, RES	<int>	IR LED level by zone It can be set if **Mode** is set as Manual.

115.4. Examples

115.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=irled&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=DayNight
Channel.0.LEDOnLevel=40
Channel.0.LEDOffLevel=60
Channel.0.Schedule.EveryDay.FromTo=19:0-5:0
Channel.0.LEDMaxPower=Medium
Channel.0.LEDPowerControlMode=Auto
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "IRled": [
        {
            "Channel": 0,
            "Mode": "DayNight",
            "LEDOnLevel": 40,
            "LEDOffLevel": 60,
            "Schedule": {
                "EveryDay": {
                    "FromTo": "19:0-5:0"
                }
            },
            "LEDMaxPower": "Medium",
            "LEDPowerControlMode": "Auto"
        }
```

```
    ]
}
```

115.4.2. Setting LED On level

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&Mode=Sensor&LEDOnLevel=50
```

115.4.3. Setting LED Off level

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&Mode=Sensor&LEDOffLevel=40
```

115.4.4. Setting LED max power and power control mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&LEDMaxPower=High&LEDPowerControlMode
=Auto
```

115.4.5. Setting the schedule for IR LED

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&Mode=Sensor&Schedule.EveryDay.FromTo
=00:00-01:00
```

115.4.6. Disabling IR LED

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&Channel=0&Mode=Off
```

115.4.7. Setting IR LED mode to manual and its level

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=irled&action=set&Channel=0&Mode=Manual&Level=50
```

Chapter 116. Flip

116.1. Description

The **flip** submenu configures image flip settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

116.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
flip&action=<value>&[<parameter>=<value>]
```

116.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads image flip settings.
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	HorizontalFlipEnable	REQ, RES	<bool> True, False	Enables or disables horizontal image flipping.
	VerticalFlipEnable	REQ, RES	<bool> True, False	Enables or disables vertical image flipping.
	Rotate	REQ, RES	<enum> 0, 90, 270	Rotates a video by a specified angle

116.4. Examples

116.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=flip&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.HorizontalFlipEnable=False
Channel.0.VerticalFlipEnable=False
Channel.0.Rotate=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "Flip": [
        {
            "Channel": 0,
            "HorizontalFlipEnable": false,
            "VerticalFlipEnable": false,
            "Rotate": "0"
        }
    ]
}
```


116.4.2. Enabling horizontal flip

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=flip&action=set&HorizontalFlipEnable=True
```

116.4.3. Enabling vertical flip

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/image.cgi?msubmenu=flip&action=set&VerticalFlipEnable=True
```

Chapter 117. SSDR

117.1. Description

The **SSDR** submenu configures the SSDR (Samsung Super Dynamic Range) settings. It regulates the overall brightness by increasing the brightness of the dark area alone in a scene where the difference between bright and dark is severe.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

117.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
ssdr&action=<value>[&<parameter>=<value>]
```

117.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads SSDR (Samsung Super Dynamic Range) settings.
	Channel	REQ, RES	<csv>	Channel ID
	ImagePresetMode.#.Level	RES	<int>	SSDR level The higher the value, the greater the SSDR level. **Level** is valid only when **Enable** is set to True. **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.DynamicRange	RES	<enum> Narrow, Wide	SSDR mode **DynamicRange** is valid only when **Enable** is set to True.
	ImagePresetMode.#.Enable	RES	<bool> True, False	Enables or disables SSDR.
	ImagePresetMode.#.HighLightLevel	RES	<int>	Configure highlight level of video
set	Channel	REQ, RES	<int>	Channel ID
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have ImagePreset settings. Image settings in any Mode can be changed, but modes except UserPreset1 and UserPreset2 have different default image settings.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied.
	Enable	REQ, RES	<bool> True, False	Enables or disables SSDR.
	Level	REQ, RES	<int>	SSDR level The higher the value, the greater the SSDR level. **Level** is valid only when **Enable** is set to True.
	DynamicRange	REQ, RES	<enum> Narrow, Wide	SSDR mode **DynamicRange** is valid only when **Enable** is set to True.
	HighLightLevel	REQ, RES	<int>	Configure highlight level of video

117.4. Examples

117.4.1. Getting the current information for SSDR of channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=ssdr&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
Channel.0.Enable=True
Channel.0.Level=12
Channel.0.DynamicRange=Narrow
Channel.0.ImagePresetMode.UserPreset1.Enable=True
Channel.0.ImagePresetMode.UserPreset1.Level=12
Channel.0.ImagePresetMode.UserPreset1.DynamicRange=Narrow
Channel.0.ImagePresetMode.UserPreset2.Enable=True
Channel.0.ImagePresetMode.UserPreset2.Level=12
Channel.0.ImagePresetMode.UserPreset2.DynamicRange=Narrow
Channel.0.ImagePresetMode.OutdoorDaytime.Enable=True
Channel.0.ImagePresetMode.OutdoorDaytime.Level=12
Channel.0.ImagePresetMode.OutdoorDaytime.DynamicRange=Narrow
Channel.0.ImagePresetMode.OutdoorNightime.Enable=True
Channel.0.ImagePresetMode.OutdoorNightime.Level=12
Channel.0.ImagePresetMode.OutdoorNightime.DynamicRange=Narrow
Channel.0.ImagePresetMode.IndoorBacklight.Enable=True
Channel.0.ImagePresetMode.IndoorBacklight.Level=12
Channel.0.ImagePresetMode.IndoorBacklight.DynamicRange=Narrow
Channel.0.ImagePresetMode.IndoorBrightScene.Enable=True
Channel.0.ImagePresetMode.IndoorBrightScene.Level=12
Channel.0.ImagePresetMode.IndoorBrightScene.DynamicRange=Narrow
Channel.0.ImagePresetMode.NumberPlate.Enable=True
Channel.0.ImagePresetMode.NumberPlate.Level=12
Channel.0.ImagePresetMode.NumberPlate.DynamicRange=Narrow
Channel.0.ImagePresetMode.Vivid.Enable=True
Channel.0.ImagePresetMode.Vivid.Level=10
Channel.0.ImagePresetMode.Vivid.DynamicRange=Narrow
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "SSDR": [
        {
            "Channel": 0,
            "Enable": true,
            "Level": 12,
            "DynamicRange": "Narrow",
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "UserPreset2",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "OutdoorDaytime",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "OutdoorNightime",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "IndoorBacklight",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
```

```
                {
                    "ImagePresetMode": "IndoorBrightScene",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "Enable": true,
                    "Level": 12,
                    "DynamicRange": "Narrow"
                },
                {
                    "ImagePresetMode": "Vivid",
                    "Enable": true,
                    "Level": 10,
                    "DynamicRange": "Narrow"
                }
            ]
        }
    ]
}
```

117.4.2. Enabling SSDR

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=ssdr&action=set&Enable=True
```

117.4.3. Setting the image preset mode as IndoorBacklight

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=ssdr&action=set&Enable=True&ImagePresetMode=IndoorBac
klight
```

117.4.4. Setting the SSDR level to 10

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=ssdr&action=set&Level=10
```

Chapter 118. Focus

118.1. Description

The **focus** submenu configures the focus settings.

NOTE NVR supports only control action.

Access level

Action	Camera	NVR
view	User	
set	User	
control	User	User

118.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
focus&action=<value>[&<parameter>=<value>]
```

118.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the focus settings.
	Channel	REQ, RES	<csv>	Channel ID
	ImagePresetMode.#.FastAutoFocus	RES	<bool> True, False	Enables or disables Fast Auto Focus mode. **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.
	ImagePresetMode.#.IRShift	RES	<enum> Off, 780, 850, 940	IR wave length setting

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.TemperatureCompensationEnable	RES	<bool> True, False	Enables or disables temperature compensation. This is only available when **IrisMode** is set to ICS-EG3Z3915TCS-MPEIR.
	ImagePresetMode.#.Mode	RES	<enum> Auto, Manual, OneShotAutoFocus	Focus mode • Auto: Adjusts the focus automatically according to the zoom operation. • Manual: Manually Adjust the focus according to the zoom operation. • OneShotAutoFocus: Auto focus is performed once after the zoom operation.
	ImagePresetMode.#.ZoomTrackingMode	RES	<enum> Off, AutoTracking, Tracking	Focus sync mode with zoom operation • Off: Only the zoom lens operates. • AutoTracking: Focus is adjusted as per the change in zoom position. • Tracking: Focus is adjusted to predetermined positions as per the change in zoom position.
	ImagePresetMode.#.ZoomTrackingSpeed	RES	<enum> Slow, Medium, Fast	Focus speed in zoom tracking mode **ZoomTrackingSpeed** is valid only when **ZoomTrackingMode** is not set to Off.
	ImagePresetMode.#.LensResetSchedule	RES	<enum> Manual, 1Day, 2Days, 3Days, 4Days, 5Days, 6Days, 7Days	Lens reset schedule
	ImagePresetMode.#.SimpleFocusSchedule.Enable	RES	<bool> True, False	State of the simple focus schedule.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.SimpleFocusSchedule.DurationInDays	RES	<int>	Duration in days when the simple focus should be done.
	ImagePresetMode.#.SimpleFocusSchedule.TimeInHours	RES	<int> 0 to 23	Hour of the day when the simple focus should be done.
	ImagePresetMode.#.AutoFocusRange	RES	<enum> Wide, Narrow	Focus range setting
set	Channel	REQ, RES	<int>	Channel ID
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have ImagePreset settings. Image settings in any Mode can be changed, but modes except UserPreset1 and UserPreset2 have different default image settings.
	Mode	REQ, RES	<enum> Auto, Manual, OneShotAutoFocus	Focus mode • Auto: Adjusts the focus automatically according to the zoom operation.. • Manual: Adjusts the Focus manually according to the zoom operation. • OneShotAutoFocus: Auto Focus performed once after the zoom operation.

Action	Parameters	Request/ Response	Type/ Value	Description
	ZoomTrackingMode	REQ, RES	<enum> Off, AutoTracking, Tracking	Focus sync mode with zoom operation • Off: Only the zoom lens operates. • AutoTracking: Focus is adjusted as per the change in zoom position. • Tracking: Focus is adjusted to predetermined positions as per the change in zoom position.
	ZoomTrackingSpeed	REQ, RES	<enum> Slow, Medium, Fast	Focus speed in zoom tracking mode **ZoomTrackingSpeed** is valid only when **ZoomTrackingMode** is not set to Off.
	LensResetSchedule	REQ, RES	<enum> Manual, 1Day, 2Days, 3Days, 4Days, 5Days, 6Days, 7Days	Lens reset schedule
	SimpleFocusSchedule.Enable	REQ,RES	<bool> True, False	Enables or disables the simple focus schedule.
	SimpleFocusSchedule.DurationInDays	REQ,RES	<int>	Duration in days when the simple focus should be done.
	SimpleFocusSchedule.TimeInHours	REQ,RES	<int> 0 to 23	Hour of the day when the simple focus should be done.
	FastAutoFocus	REQ, RES	<bool> True, False	Enables or disables Fast Auto Focus mode.
	IRShift	REQ, RES	<enum> Off, 780, 850, 940	IR wave length setting
	TemperatureCompensationEnable	Req, Res	<bool> True, False	Enables or disables temperature compensation This is only available when **IrisMode** is set to ICS-EG3Z3915TCS-MPEIR.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop, AWC	Image preview mode Allows viewing of the preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started. • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied. • AWC: AWC mode will be started.
	AutoFocusRange	REQ, RES	<enum> Wide, Narrow	Focus range setting
	FocusFeedbackEnable	REQ, RES	<bool>	It can enable focus feedback feature which can provide current focus data from frame metadata. User can adjust current focus by using this data.
control	Channel	REQ, RES	<int>	Channel ID
	Focus	REQ	<enum>	Focus The higher the value, the clearer far objects become. This makes close objects blurry.
	Zoom	REQ	<enum>	Zoom in/out

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ	<enum> SimpleFocus, Reset, AutoFocus, LensInitialize	Focus mode. • SimpleFocus: Fits the focus automatically. • Reset: Restores the focus adjustment to the default flange back position. • AutoFocus: Performs auto focus operation. • LensInitialize: Same feature as Reset but it is used in PTZ model. **Note** Attribute to check for SimpleFocus: "attributes/Image/Support/SimpleFocus". Attribute to check for Reset Focus: "attributes/Image/Support/ResetFocus".
	FocusAreaCoordinate	REQ	<string>	Focus area coordinates The coordinates should be in the format of x1,y1,x2,y2.
	FocusContinuous	REQ	<enum> Near, Far, Stop	User can request continuous focus move. **Note** Attribute to check for FocusContinuous: "attributes/Image/Support/FocusContinuousAdjust"
	ZoomContinuous	REQ	<enum> In, Out, Stop	User can request continuous zoom move. **Note** Attribute to check for ZoomContinuous: "attributes/Image/Support/ZoomContinuousAdjust"
	ResetMaxFocus	REQ	<bool>	User can reset max focus data in focus feedback from frame metadata.

118.4. Examples

118.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.FastAutoFocus=True
Channel.0.ImagePresetMode.UserPreset1.FastAutoFocus=True
Channel.0.ImagePresetMode.UserPreset2.FastAutoFocus=True
Channel.0.ImagePresetMode.OutdoorDaytime.FastAutoFocus=True
Channel.0.ImagePresetMode.OutdoorNightime.FastAutoFocus=True
Channel.0.ImagePresetMode.IndoorBacklight.FastAutoFocus=True
Channel.0.ImagePresetMode.IndoorBrightScene.FastAutoFocus=True
Channel.0.ImagePresetMode.NumberPlate.FastAutoFocus=True
Channel.0.ImagePresetMode.Vivid.FastAutoFocus=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Focus": [
        {
            "Channel": 0,
            "FastAutoFocus": true,
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "FastAutoFocus": true
                },
                {
```

```
                    "ImagePresetMode": "UserPreset2",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "OutdoorDaytime",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "OutdoorNightime",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "IndoorBacklight",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "IndoorBrightScene",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "FastAutoFocus": true
                },
                {
                    "ImagePresetMode": "Vivid",
                    "FastAutoFocus": true
                }
            ]
        }
    ]
}
```


118.4.2. Setting zoom tracking speed

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=set&ZoomTrackingSpeed=Medium
```

118.4.3. Setting the image preset mode as Vivid

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=set&ZoomTrackingSpeed=Medium&ImagePreset
Mode=Vivid
```

118.4.4. Setting zoom tracking mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=set&ZoomTrackingMode=AutoTracking
```

118.4.5. Setting focus

Setting focus mode to SimpleFocus

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&Mode=SimpleFocus
```

Setting focus to 100

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=control&Focus=100
```

Resetting focus

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&Mode=Reset
```

Setting focus to 10

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=control&Focus=10
```

Adjust focus continuously

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&FocusContinuous=Near
```

Stop focus adjust

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&FocusContinuous=Stop
```

118.4.6. Setting zoom

Setting zoom to -10

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=control&Zoom=-10
```

Setting zoom to 10

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=control&Zoom=10
```

Adjust zoom continuously

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&ZoomContinuous=In
```

Stop zoom Adjust

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=control&ZoomContinuous=Stop
```

118.4.7. Setting the focus area

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/image.cgi?msubmenu=focus&action=control&Mode=SimpleFocus&FocusAreaCoordi
nate=400,400,600,600
```

118.4.8. Simple focus schedule

NOTE Only some camera models support this feature, can refer to the cgi section of attribute response.
stw-cgi/attributes.cgi/image/focus

Enable the simplefocus schedule.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focus&action=set&SimpleFocusSchedule.Enable=True&Simp
leFocusSchedule.DurationInDays=1&SimpleFocusSchedule.TimeInHours=2&Channel=0
```

View the current setting.

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=focus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Focus": [
        {
            "Channel": 0,
            "FastAutoFocus": true,
            "SimpleFocusSchedule": {
                "Enable": true,
                "DurationInDays": 1,
                "TimeInHours": 2
            },
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
```

```
                "FastAutoFocus": true,
                "SimpleFocusSchedule": {
                    "Enable": true,
                    "DurationInDays": 1,
                    "TimeInHours": 2
                }
            },
            {
                "ImagePresetMode": "UserPreset2",
                "FastAutoFocus": true,
                "SimpleFocusSchedule": {
                    "Enable": true,
                    "DurationInDays": 1,
                    "TimeInHours": 2
                }
            },
            {
                "ImagePresetMode": "OutdoorDaytime",
                "FastAutoFocus": true,
                "SimpleFocusSchedule": {
                    "Enable": true,
                    "DurationInDays": 1,
                    "TimeInHours": 2
                }
            },
            {
                "ImagePresetMode": "OutdoorNightime",
                "FastAutoFocus": true,
                "SimpleFocusSchedule": {
                    "Enable": true,
                    "DurationInDays": 1,
                    "TimeInHours": 2
                }
            },
            {
                "ImagePresetMode": "IndoorBacklight",
                "FastAutoFocus": true,
                "SimpleFocusSchedule": {
                    "Enable": true,
                    "DurationInDays": 1,
                    "TimeInHours": 2
```

```
                    }
                },
                {
                    "ImagePresetMode": "IndoorBrightScene",
                    "FastAutoFocus": true,
                    "SimpleFocusSchedule": {
                        "Enable": true,
                        "DurationInDays": 1,
                        "TimeInHours": 2
                    }
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "FastAutoFocus": true,
                    "SimpleFocusSchedule": {
                        "Enable": true,
                        "DurationInDays": 1,
                        "TimeInHours": 2
                    }
                },
                {
                    "ImagePresetMode": "Vivid",
                    "FastAutoFocus": true,
                    "SimpleFocusSchedule": {
                        "Enable": true,
                        "DurationInDays": 1,
                        "TimeInHours": 2
                    }
                }
            ]
        }
    ]
}
```

Chapter 119. Overlay

119.1. Description

The **overlay** submenu configures title and time OSD (on-screen display) settings.

> **NOTE** This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

119.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
overlay&action=<value>&[&<parameter>=<value>]
```

119.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads title and time OSD (on-screen display) settings
	Channel	REQ, RES	<csv>	Channel ID
	ImagePresetMode.#.PTZPositionEnable	RES	<bool> True, False	Enables or disables the PTZ position overlay **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.
	ImagePresetMode.#.PresetNameEnable	RES	<bool> True, False	Enables or disables the preset name overlay.
	ImagePresetMode.#.CameraIDEnable	RES	<bool> True, False	Enables or disables the camera ID overlay.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.AzimuthEnable	RES	<bool> True, False	Enables or disables the azimuth overlay. **Note** Attribute to check for Azimuth: "attributes/PTZSupport/Support/Azimuth"
	ImagePresetMode.#.QuickZoomAndFocusEnable	RES	<bool> True, False	Enables or disables the QuickZoomAndFocus overlay.
Set	Channel	REQ, RES	<int>	Channel ID
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have ImagePreset settings. Image settings in any Mode can be changed, but modes except UserPreset1 and UserPreset2 have different default image settings.
	TitleEnable	REQ, RES	<bool> True, False	Enables or disables title overlay
	Title	REQ, RES	<string>	Camera title **Title** is valid only when **TitleEnable** is set to True.
	TitlePositionX	REQ, RES	<int>	Camera title overlay position row The value may vary depending on the model. **TitlePositionX** and **TitlePositionY** are valid only when **TitleEnable** is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
	TitlePositionY	REQ, RES	<int>	Camera title overlay position column The value may vary depending on the model. **TitlePositionX** and **TitlePositionY** are valid only when **TitleEnable** is set to True.
	TimeEnable	REQ, RES	<bool> True, False	Enables or disables time overlay
	TimeFormat	REQ, RES	<enum>	Time overlay format The time can be specified in the format of <YYYY-MM-DD>, <MM-DD-YYYY>, or <DD-MM-YYYY>. **TimeFormat** is valid only when the **TimeEnable** is set to True.
	TimePositionX	REQ, RES	<int>	Time overlay position row **TimePositionX** and **TimePositionY** are valid only when **TimeEnable** is set to True.
	TimePositionY	REQ, RES	<int>	Time overlay position column **TimePositionX** and **TimePositionY** are valid only when **TimeEnable** is set to True.
	WeekdayEnable	REQ, RES	<bool> True, False	Enables or disables the day of week overlay **WeekdayEnable** is valid only when **TimeEnable** is set to True.
	PTZPositionEnable	REQ, RES	<bool> True, False	Enables or disables the PTZ position overlay
	AzimuthEnable	REQ, RES	<bool> True, False	Enables or disables the azimuth overlay **Note** Attribute to check for Azimuth: "attributes/PTZSupport/Support/Azimuth"

Action	Parameters	Request/ Response	Type/ Value	Description
	CameraIDEnable	REQ, RES	<bool> True, False	Enables or disables the camera ID overlay
	PresetNameEnable	REQ, RES	<bool> True, False	Enables or disables the preset name overlay
	PresetNamePositionX	REQ, RES	<int>	Presetname overlay position row **PresetNamePositionX** and **PresetNamePositionY** are valid only when **PresetNameEnable** is set to True.
	PresetNamePositionY	REQ, RES	<int>	Presetname overlay position column **PresetNamePositionX** and **PresetNamePositionY** are valid only when **PresetNameEnable** is set to True.
	OSDColor	REQ, RES	<enum> White, Red, Blue, Green, Yellow	Color
	FontSize	REQ, RES	<enum> Small, Medium, Large	Font Size
	MacAddressEnable	REQ, RES	<bool> True, False	Enables or disables the device's MAC address overlay
	MacAddressPositionX	REQ, RES	<int>	MAC address overlay position row The value may vary depending on the model. **MacAddressPositionX** and **MacAddressPositionY** are valid only when **MacAddressEnable** is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
	MacAddressPositionY	REQ, RES	<int>	MAC address overlay position column The value may vary depending on the model. **MacAddressPositionX** and **MacAddressPositionY** are valid only when **MacAddressEnable** is set to True.
	IPAddressEnable	REQ, RES	<bool> True, False	Enables or disables the device's IP address overlay
	IPAddressPositionX	REQ, RES	<int>	IP address overlay position row The value may vary depending on the model. **IPAddressPositionX** and **IPAddressPositionY** are valid only when **IPAddressEnable** is set to True.
	IPAddressPositionY	REQ, RES	<int>	IP address overlay position column The value may vary depending on the model. **IPAddressPositionX** and **IPAddressPositionY** are valid only when **IPAddressEnable** is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped and the original image settings saved in the camera will be applied.
	QuickZoomAndFocusEnable	REQ, RES	<bool> True, False	Enables or disables the QuickZoomAndFocus overlay.
control	DeviceTemperatureEnable	REQ	<bool> True, False	Control temperature related information to OSD. Determine whether to show the temperature of device on a screen.
	TemperatureScale	REQ	<enum> Celsius, Fahrenheit	Determine the format of temperature to be shown. Default foramt is the Celsius.

119.4. Examples

119.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=overlay&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.TitleEnable=True
Channel.0.Title=PNP-9200RH
Channel.0.TitlePositionX=1
Channel.0.TitlePositionY=1
Channel.0.TimeEnable=True
Channel.0.TimeFormat=YYYY-MM-DD
Channel.0.TimePositionX=38
Channel.0.TimePositionY=26
Channel.0.WeekdayEnable=True
Channel.0.FontSize=Small
Channel.0.PTZPositionEnable=True
Channel.0.CameraIDEnable=False
Channel.0.PresetNameEnable=True
Channel.0.AzimuthEnable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Overlay": [
        {
            "Channel": 0,
            "TitleEnable": true,
            "Title": "PNP-9200RH",
            "TitlePositionX": 1,
            "TitlePositionY": 1,
            "TimeEnable": true,
            "TimeFormat": "YYYY-MM-DD",
            "TimePositionX": 38,
            "TimePositionY": 26,
            "WeekdayEnable": true,
            "FontSize": "Small",
            "PTZPositionEnable": true,
            "CameraIDEnable": false,
            "PresetNameEnable": true,
            "AzimuthEnable": false
        }
```

```
    ]
}
```

119.4.2. Setting the title overlay position

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=overlay&action=set&TitleEnable=True&TitlePositionX=6&
TitlePositionY=6
```

119.4.3. Setting time overlay

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=overlay&action=set&TimeEnable=True&TimeFormat=YYYY-
MM-DD&TimePositionX=5&TimePositionY=5&WeekdayEnable=True
```

119.4.4. Enabling azimuth overlay (OSD)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=overlay&action=set&AzimuthEnable=True
```

119.4.5. Enabling device temperature overlay (OSD) with Farenheit format

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=overlay&action=control&DeviceTemperatureEnable=True&T
emperatureScale=Fahrenheit
```

Chapter 120. Multi Line Overlay

120.1. Description

The **multilineosd** submenu configures multiple OSD (on-screen display) title and time settings.

NOTE

This chapter applies to network cameras only.
Attribute to check for maximum value of OSD title Supported:
"attributes/Image/Limit/MaxOSDTitles"
Attribute to check for maximum value of OSD Date Supported:
"attributes/Image/Limit/MaxOSDDates"

Access level

Action	Camera
view	User
add/update	User
remove	User

120.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
multilineosd&action=<value>&[&<parameter>=<value>]
```

120.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads title and time OSD (on-screen display) settings
	Channel	REQ, RES	<csv>	Channel ID
	Index	REQ, RES	<csv>	OSD Index can be more than the sum of **MaxOSDTitles** and **MaxOSDDate**. It is a mandatory parameter.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.Index.#.OSDType	RES	<enum> Date, Title	OSD Type is a mandatory parameter **Note** Each ImagePresetMode.# can have different image settings. For more details, refer to the "ImagePresetMode" parameter in set actions.
	ImagePresetMode.#.Index.#.WeekDay	RES	<bool> True, False	Enables or disables the day of week overlay. **WeekDay** is valid only when **OSDType** is set to Date.
	ImagePresetMode.#.Index.#.DateFormat	RES	<enum> YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY	Date overlay format The time can be specified in any one of the supported formats. **DateFormat** is valid only when the **OSDType** is set to Date.
	ImagePresetMode.#.Index.#.OSD	RES	<string>	OSD text to be displayed on the image OSD is valid only when **OSDType** is set to Title.
	ImagePresetMode.#.Index.#.OSDBlink	RES	<enum> OFF, HDMI	Enables or disables OSD blinking for HDMI video output of the camera.
	ImagePresetMode.#.Index.#.Enable	RES	<bool> True, False	Enables or disables title or date overlay.
	ImagePresetMode.#.Index.#.PositionX	RES	<int>	OSD overlay position row The value may vary depending on the model. The max allowed value changes when font size is changed. Refer to the Image Options command for information on the exact ranges for this value.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.Index.#.PositionY	RES	<int>	OSD overlay position column The value may vary depending on the model. The max allowed value changes when font size is changed. Refer to the Image Options command for information on the exact ranges for this value.
	ImagePresetMode.#.Index.#.OSDColor	RES	<enum> White, Red, Blue, Green, Yellow	OSD Font Color
	ImagePresetMode.#.Index.#.FontSize	RES	<enum> Small, Medium, Large	OSD Font Size
	ImagePresetMode.#.Index.#.CustomFontSize	RES	<int>	OSD custom font size **Note** **FontSize** and **CustomFontSize** cannot be set at the same time
	ImagePresetMode.#.Index.#.Transparency	RES	<enum> Off, Low, Medium, High	To control the transparency of text
add/ update	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode	REQ	<enum> UserPreset 1, UserPreset 2, OutdoorDa ytime, OutdoorNig htime, IndoorBack light, IndoorBrig htScene, NumberPla te, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have ImagePreset settings. Image settings in any Mode can be changed, but modes except UserPreset1 and UserPreset2 have different default image settings.
	Index	REQ, RES	<csv>	OSD Index can be more than the sum of **MaxOSDTitles** and **MaxOSDDate**. It is a mandatory parameter.
	OSDType	REQ, RES	<enum> Date,Title	OSD Type is a mandatory parameter
	Enable	REQ, RES	<bool> True, False	Enables or disables title or date overlay
	OSD	REQ, RES	<string>	OSD text to be displayed on the image OSD is valid only when **OSDType** is set to Title.
	PositionX	REQ, RES	<int>	OSD overlay position row The value may vary depending on the model. The max allowed value changes when Font Size is changed. Refer to the Image Options command for information on the exact ranges for this value.

Action	Parameters	Request/ Response	Type/ Value	Description
	PositionY	REQ, RES	<int>	OSD overlay position column The value may vary depending on the model. The max allowed value changes when Font Size is changed. Refer to the Image Options command for information on the exact ranges for this value
	DateFormat	REQ, RES	<enum> YYYY-MM-DD, MM-DD-YYYY, DD-MM-YYYY	Date overlay format The time can be specified in any one of the supported formats. **DateFormat** is valid only when the **OSDType** is set to Date.
	WeekDay	REQ, RES	<bool> True, False	Enables or disables the day of week overlay **WeekDay** is valid only when **OSDType** is set to Date.
	OSDColor	REQ, RES	<enum> White, Red, Blue, Green, Yellow	OSD font color
	FontSize	REQ, RES	<enum> Small, Medium, Large	OSD font size
	CustomFontSize	REQ, RES	<int>	OSD custom font size **Note** **FontSize** and **CustomFontSize** cannot be set at the same time
	OSDBlink	REQ, RES	<enum> OFF, HDMI	Enables or disables OSD blinking for HDMI video output of camera.

Action	Parameters	Request/ Response	Type/ Value	Description
	Transparency	REQ, RES	<enum> Off, Low, Medium, High	To control the transparency of text
	AdditionalOSD	REQ, RES	<csv> None, PTZPosition, CameraID, PresetName, Azimuth	To enable additional OSD information
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped and the original image settings saved in the camera will be applied.
remove	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode	REQ	<enum> UserPreset 1, UserPreset 2, OutdoorDa ytime, OutdoorNig htime, IndoorBack light, IndoorBrig htScene, NumberPla te, Vivid, Indoor, Outdoor	Removes image preset mode’s image settings.
	Index	REQ, RES	<csv>	OSD Index can be more than the sum of **MaxOSDTitles** and **MaxOSDDate**. It is a mandatory parameter
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped and the original image settings saved in the camera will be applied.

120.4. Examples

120.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multilineosd&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Index.6.Enable=False
Channel.0.Index.6.OSDType=Date
Channel.0.Index.6.DateFormat=YYYY-MM-DD
Channel.0.Index.6.PositionX=1
Channel.0.Index.6.PositionY=2
Channel.0.Index.6.WeekDay=False
Channel.0.Index.6.FontSize=Small
Channel.0.Index.6.OSDColor=White
Channel.0.Index.6.Transparency=Off
Channel.0.ImagePresetMode.UserPreset1.Index.6.Enable=False
Channel.0.ImagePresetMode.UserPreset1.Index.6.OSDType=Date
Channel.0.ImagePresetMode.UserPreset1.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.UserPreset1.Index.6.PositionX=1
Channel.0.ImagePresetMode.UserPreset1.Index.6.PositionY=2
Channel.0.ImagePresetMode.UserPreset1.Index.6.WeekDay=False
Channel.0.ImagePresetMode.UserPreset1.Index.6.FontSize=Small
Channel.0.ImagePresetMode.UserPreset1.Index.6.OSDColor=White
Channel.0.ImagePresetMode.UserPreset1.Index.6.Transparency=Off
Channel.0.ImagePresetMode.UserPreset2.Index.6.Enable=False
Channel.0.ImagePresetMode.UserPreset2.Index.6.OSDType=Date
Channel.0.ImagePresetMode.UserPreset2.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.UserPreset2.Index.6.PositionX=1
Channel.0.ImagePresetMode.UserPreset2.Index.6.PositionY=2
Channel.0.ImagePresetMode.UserPreset2.Index.6.WeekDay=False
Channel.0.ImagePresetMode.UserPreset2.Index.6.FontSize=Small
Channel.0.ImagePresetMode.UserPreset2.Index.6.OSDColor=White
Channel.0.ImagePresetMode.UserPreset2.Index.6.Transparency=Off
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.Enable=False
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.OSDType=Date
```

```
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.PositionX=1
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.PositionY=2
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.WeekDay=False
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.FontSize=Small
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.OSDColor=White
Channel.0.ImagePresetMode.OutdoorDaytime.Index.6.Transparency=Off
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.Enable=False
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.OSDType=Date
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.PositionX=1
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.PositionY=2
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.WeekDay=False
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.FontSize=Small
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.OSDColor=White
Channel.0.ImagePresetMode.OutdoorNightime.Index.6.Transparency=Off
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.Enable=False
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.OSDType=Date
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.PositionX=1
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.PositionY=2
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.WeekDay=False
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.FontSize=Small
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.OSDColor=White
Channel.0.ImagePresetMode.IndoorBacklight.Index.6.Transparency=Off
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.Enable=False
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.OSDType=Date
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.PositionX=1
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.PositionY=2
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.WeekDay=False
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.FontSize=Small
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.OSDColor=White
Channel.0.ImagePresetMode.IndoorBrightScene.Index.6.Transparency=Off
Channel.0.ImagePresetMode.NumberPlate.Index.6.Enable=False
Channel.0.ImagePresetMode.NumberPlate.Index.6.OSDType=Date
Channel.0.ImagePresetMode.NumberPlate.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.NumberPlate.Index.6.PositionX=1
Channel.0.ImagePresetMode.NumberPlate.Index.6.PositionY=2
Channel.0.ImagePresetMode.NumberPlate.Index.6.WeekDay=False
Channel.0.ImagePresetMode.NumberPlate.Index.6.FontSize=Small
```

```
Channel.0.ImagePresetMode.NumberPlate.Index.6.OSDColor=White
Channel.0.ImagePresetMode.NumberPlate.Index.6.Transparency=Off
Channel.0.ImagePresetMode.Vivid.Index.6.Enable=False
Channel.0.ImagePresetMode.Vivid.Index.6.OSDType=Date
Channel.0.ImagePresetMode.Vivid.Index.6.DateFormat=YYYY-MM-DD
Channel.0.ImagePresetMode.Vivid.Index.6.PositionX=1
Channel.0.ImagePresetMode.Vivid.Index.6.PositionY=2
Channel.0.ImagePresetMode.Vivid.Index.6.WeekDay=False
Channel.0.ImagePresetMode.Vivid.Index.6.FontSize=Small
Channel.0.ImagePresetMode.Vivid.Index.6.OSDColor=White
Channel.0.ImagePresetMode.Vivid.Index.6.Transparency=Off
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MultiLineOSD": [
        {
            "Channel": 0,
            "OSDs": [
                {
                    "Index": 6,
                    "OSDType": "Date",
                    "Enable": false,
                    "DateFormat": "YYYY-MM-DD",
                    "PositionX": 1,
                    "PositionY": 2,
                    "WeekDay": false,
                    "FontSize": "Small",
                    "OSDColor": "White",
                    "Transparency": "Off"
                }
            ],
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "OSDs": [
                        {
```

```
                    "Index": 6,
                    "OSDType": "Date",
                    "Enable": false,
                    "DateFormat": "YYYY-MM-DD",
                    "PositionX": 1,
                    "PositionY": 2,
                    "WeekDay": false,
                    "FontSize": "Small",
                    "OSDColor": "White",
                    "Transparency": "Off"
                }
            ]
        },
        {
            "ImagePresetMode": "UserPreset2",
            "OSDs": [
                {
                    "Index": 6,
                    "OSDType": "Date",
                    "Enable": false,
                    "DateFormat": "YYYY-MM-DD",
                    "PositionX": 1,
                    "PositionY": 2,
                    "WeekDay": false,
                    "FontSize": "Small",
                    "OSDColor": "White",
                    "Transparency": "Off"
                }
            ]
        },
        {
            "ImagePresetMode": "OutdoorDaytime",
            "OSDs": [
                {
                    "Index": 6,
                    "OSDType": "Date",
                    "Enable": false,
                    "DateFormat": "YYYY-MM-DD",
                    "PositionX": 1,
                    "PositionY": 2,
                    "WeekDay": false,
```

```
                "FontSize": "Small",
                "OSDColor": "White",
                "Transparency": "Off"
            }
        ]
    },
    {
        "ImagePresetMode": "OutdoorNightime",
        "OSDs": [
            {
                "Index": 6,
                "OSDType": "Date",
                "Enable": false,
                "DateFormat": "YYYY-MM-DD",
                "PositionX": 1,
                "PositionY": 2,
                "WeekDay": false,
                "FontSize": "Small",
                "OSDColor": "White",
                "Transparency": "Off"
            }
        ]
    },
    {
        "ImagePresetMode": "IndoorBacklight",
        "OSDs": [
            {
                "Index": 6,
                "OSDType": "Date",
                "Enable": false,
                "DateFormat": "YYYY-MM-DD",
                "PositionX": 1,
                "PositionY": 2,
                "WeekDay": false,
                "FontSize": "Small",
                "OSDColor": "White",
                "Transparency": "Off"
            }
        ]
    },
    {
```

```
                    "ImagePresetMode": "IndoorBrightScene",
                    "OSDs": [
                        {
                            "Index": 6,
                            "OSDType": "Date",
                            "Enable": false,
                            "DateFormat": "YYYY-MM-DD",
                            "PositionX": 1,
                            "PositionY": 2,
                            "WeekDay": false,
                            "FontSize": "Small",
                            "OSDColor": "White",
                            "Transparency": "Off"
                        }
                    ]
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "OSDs": [
                        {
                            "Index": 6,
                            "OSDType": "Date",
                            "Enable": false,
                            "DateFormat": "YYYY-MM-DD",
                            "PositionX": 1,
                            "PositionY": 2,
                            "WeekDay": false,
                            "FontSize": "Small",
                            "OSDColor": "White",
                            "Transparency": "Off"
                        }
                    ]
                },
                {
                    "ImagePresetMode": "Vivid",
                    "OSDs": [
                        {
                            "Index": 6,
                            "OSDType": "Date",
                            "Enable": false,
                            "DateFormat": "YYYY-MM-DD",
```

```
                            "PositionX": 1,
                            "PositionY": 2,
                            "WeekDay": false,
                            "FontSize": "Small",
                            "OSDColor": "White",
                            "Transparency": "Off"
                        }
                    ]
                }
            ]
        }
    ]
}
```

120.4.2. Adding multiple text overlay positions

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multilineosd&action=add&Index=3&OSDType=Title&Enable=
True&PositionX=6&PositionY=6&OSD=TestOSD
```

120.4.3. Setting a title at the specific image preset mode

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multilineosd&action=add&Index=3&OSDType=Title&Enable=
True&PositionX=6&PositionY=6&OSD=TestOSD&ImagePresetMode=Vivid
```

120.4.4. Setting time overlay

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multilineosd&action=update&Index=6&OSDType=Date&Enabl
e=True&DateFormat=YYYY-MM-DD&PositionX=5&PositionY=5&WeekDay=True
```

120.4.5. Enabling azimuth overlay (OSD)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multilineosd&action=update&AdditionalOSD=AzimuthEnabl
```

```
e
```

Chapter 121. Multi Image Overlay

121.1. Description

The **multiimageosd** submenu supports overlaying images on the live stream.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
install	User
remove	User

121.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
multiimageosd&action=<value>&[&<parameter>=<value>]
```

121.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads Image overlay settings
	Channel	REQ, RES	<csv>	Channel ID
	Index	REQ	<csv>	List of overlay image index
	MaxResolution	RES	<string>	Maximum image resolution that can be uploaded for image overlay
	IsInstalled	RES	<bool>	Indicates if an image is already installed or not
set	Channel	REQ, RES	<int>	Channel ID
	PositionX	REQ, RES	<int>	X Coordinate of the image
	PositionY	REQ, RES	<int>	Y Coordinate of the image
install	Channel	REQ, RES	<int>	Channel ID
	Index	REQ, RES	<csv>	Overlay Image Index

Action	Parameters	Request/ Response	Type/ Value	Description
	FileType	REQ, RES	<enum> BMP, GIF, JPG, JPEG, PNG, TIF	Specifies the format of the image that should be installed
remove	Channel	REQ, RES	<int>	Channel ID
	Index	REQ, RES	<int>	Overlay Image Index

121.4. Examples

121.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multiimageosd&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Index.1.IsInstalled=True
Channel.0.MaxResolution=128x128
Channel.0.Index.1.PositionX=1086
Channel.0.Index.1.PositionY=728
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "multiimageosd": [
        {
            "Channel": 0,
            "MaxResolution": "128x128",
```

```
            "Imageosds": [
                {
                    "Index": 1,
                    "IsInstalled": true
                    "PositionX": 1086,
                    "PositionY": 728
                }
            ]
        }
    ]
}
```

121.4.2. Installing overlay image

Make sure that the image you are uploading is in a format supported by the device.

Refer to the **image/multiimageosd/install/FileType** response in attributes.cgi for information on supported file formats.

The maximum image resolution supported is provided in the **MaxResolution** parameter.

REQUEST

```
*[.underline]#Step 1:#*
```

Image file needs to be base64 encoded before sending a POST request.

```
> openssl base64 -in image.bmp -out image_base64.bin
```

Step 2:

curl -v --digest -u <userid>:<password> --data-urlencode @image_base64.bin "http://<Device IP>/stw-cgi/image.cgi?msubmenu=multiimageosd&action=install&Index=1&FileType=BMP" -H "Expect:"

The above command will produce a request to the device that looks something like:

```
 POST /stw-
cgi/image.cgi?msubmenu=multiimageosd&action=install&Index=1&FileType=BMP
HTTP/1.1
 User-Agent: curl/7.26.0
 Host: 192.168.22.47
 Accept: */*
 Content-Length: 147610
```

```
Content-Type: application/x-www-form-urlencoded
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


121.4.3. Removing an installed overlay image

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=multiimageosd&action=remove&Channel=0&Index=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 122. Smart Codec

122.1. Description

The **smartcodec** submenu configures smart codec settings.

Using Smart Codec, specific areas are compressed with high quality and other areas with low quality.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin
add	Admin
remove	Admin

122.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
smartcodec&action=<value>[&<parameter>=<value>]
```

122.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads smart codec settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ, RES	<enum> FaceDetection, Manual	Smart codec mode • FaceDetection: Detects the face and compresses it with high quality • Manual: Defines The area (rectangle type) manually for high quality compression **Note** **add** and **remove** actions are valid only when **Mode** is set to Manual.
	Sensitivity	REQ, RES	<int>	Face detection sensitivity **Sensitivity** is valid only when **Mode** is set to FaceDetection.
	QualityLevel	REQ, RES	<enum>	Quality of user-defined smart codec area or the face detection area
add	Channel	REQ	<int>	Channel ID
	Area.#.Coordinate	REQ, RES	<string>	Area coordinates for smart codec The coordinates should be in the format of <x1,y1,x2,y2,x3,y3,x4>. **Note** To use the **add** action, **Mode** must have been set to Manual, and **Area.#.Coordinate** must be sent together. Attribute to check for maximum smart codec area(s): "attributes/Image/Limit/MaxSmartCodecArea"
remove	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Area	REQ	<csv> All, #	Smart codec area number/index to be deleted This parameter is valid only if **Mode** is set to Manual. **Note** To use the **remove** action, **Mode** must have been set to Manual, and **Area** must be sent together.

122.4. Examples

122.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=smartcodec&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Manual
Channel.0.QualityLevel=Auto
Channel.0.Area.1.Coordinate=828,384,1860,840
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SmartCodec": [
        {
            "Channel": 0,
```

```
            "Mode": "Manual",
            "QualityLevel": "Auto",
            "Areas": [
                {
                    "Area": 1,
                    "Coordinate": [
                        {
                            "x": 828,
                            "y": 384
                        },
                        {
                            "x": 1860,
                            "y": 840
                        }
                    ]
                }
            ]
        }
    ]
}
```

122.4.2. Setting smart codec mode to face detection

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=smartcodec&action=set&Mode=FaceDetection
```

122.4.3. Adding smart codec area 1

To add a smart codec area for high quality compression using the **add** action, the **Mode** parameter must have already been set to Manual; If **Mode** is set to FaceDetection, an error will be returned to the following request.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=smartcodec&action=add&Area.1.Coordinate=86,154,973,96
5
```

122.4.4. Removing smart codec area 1

To remove a smart codec area using the **remove** action, the **Mode** parameter must have already been set to Manual; If **Mode** is set to FaceDetection, an error will be returned to the following request.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=smartcodec&action=remove&Area=1
```

122.4.5. Removing all smart codec areas

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=smartcodec&action=remove&Area=All
```

Chapter 123. Privacy

123.1. Description

The **privacy** submenu configures privacy mask settings.

> **NOTE**
>
> This chapter applies to network cameras only.
> Attribute to check for privacy mask zoom threshold:
> "attributes/Image/Support/PrivacyMask.ZoomThreshold"
> Attribute to check for privacy mask color:
> "attributes/Image/Support/Privacy.MaskColor.Global"
> Attribute to check for maximum privacy masks: "attributes/Image/Limit/MaxPrivacyMask"
> NVR supports only control action.

Access level

Action	Camera	NVR
view	User	
add, update	User	
set	User	
remove	User	
control		User

123.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
privacy&action=<value>[&<parameter>=<value>]
```

123.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads privacy mask settings.
	Channel	REQ	<csv>	Channel ID
	CommonMaskColor	RES	<enum> Gray, Green, Red, Blue, Black, White	This parameter is supported when the **Privacy.MaskColor.Global** parameter is True. This parameter is used to configure color for all privacy masks.

Action	Parameters	Request/ Response	Type/ Value	Description
	Position.Pan	RES	<float>	Pan value when the device supports PTZ.
	Position.Tilt	RES	<float>	Tilt value when the device supports PTZ.
add	Channel	REQ, RES	<int>	Channel ID
	MaskIndex	REQ, RES	<int>	Index of the privacy area The value may vary depending on the model. **MaskIndex** parameter is optional for **add** action. The **MaskIndex** value will be returned in response to a **view** request.
	MaskName	REQ, RES	<string>	Privacy area's name The name may consist of letters, numbers, hyphens and periods. **Note** To use the **add** action, **MaskName**, **MaskColor**, and **MaskCoordinate** parameters must be sent together.
	MaskColor	REQ, RES	<enum> Gray, Green, Red, Blue, Black, White	Color of the privacy area **Note** To use the **add** action, **MaskName**, **MaskColor**, and **MaskCoordinate** parameters must be sent together. **Note** In some cameras, changing the color of each mask is not supported. If **Privacy.MaskColor.Global** parameter is False, then this feature is available. If this feature is not available, then **CommonMaskColor** parameter is used and this parameter is used as Common MaskColor.

Action	Parameters	Request/ Response	Type/ Value	Description
	MaskCoordinate	REQ, RES	<string>	Vertices of the privacy area **Note** To use the **add** action, **MaskName**, **MaskColor**, and **MaskCoordinate** parameters must be sent together.
	ZoomThresholdEnable	REQ, RES	<bool> True, False	Enables or disables the zoom threshold
	AdjustMode	REQ	<enum> Start	When adding a privacy area with this parameter, the privacy area will be transparent until some times passes or an update command comes with **AdjustMode** as Stop. **Note** This parameter is supported when the device has a zoom module.
update	MaskIndex	REQ	<int>	Masks the index number
	ZoomThresholdEnable	REQ, RES	<bool> True, False	Enables or disables the zoom threshold
	AdjustMode	REQ	<enum> Start, Stop	To adjust a privacy area's zoom status without removing it. When this comes as Start, the area will be transparent until some times passes or an update command comes as Stop. **Note** To use this parameter, the **MaskIndex** parameter must be sent together. This is supported when the device has a zoom module.
	Channel	REQ, RES	<int>	Channel information
	MaskName	REQ, RES	<string>	Privacy area's name The name may consist of letters, numbers, hyphens and periods.

Action	Parameters	Request/ Response	Type/ Value	Description
	MaskColor	REQ, RES	<enum> Gray, Green, Red, Blue, Black, White	Color of the privacy area
	MaskCoordinate	REQ, RES	<string>	Vertices of the privacy area
set	Enable	REQ, RES	<bool> True, False	Whether to use a privacy area
	Channel	REQ	<int>	ID of the channel that will use the privacy function
	MaskPattern	REQ, RES	<enum> Solid, Mosaic1, Mosaic2, Mosaic3, Mosaic4	Masks the pattern
	CurrentIndexOrder	REQ	<csv>	The current **MaskIndex** list **Note** To change the **MaskIndex** order, the **NewIndexOrder** parameters must be sent together.
	NewIndexOrder	REQ	<csv>	New **MaskIndex** list for changing the orders **Note** To change the **MaskIndex** order, the **CurrentIndexOrder** parameters must be sent together.
	CornerMasking	REQ,RES	<bool>	Enables corner masking of video image.
	CornerMasking	REQ,RES	<bool>	Enables corner masking of video image.
remove	Channel	REQ	<int>	ID of the channel to remove the privacy area(s) from.

Action	Parameters	Request/ Response	Type/ Value	Description
	MaskIndex	REQ	<csv> All, #	Privacy area(s) to remove **Note** To use the **remove** action, **MaskIndex** parameter must be sent.
control	Channel	REQ	<int>	ID of the channel from which to control the privacy area **Note** The control action is supported only in PTZ models.
	Mode	REQ	<enum> Start, Move	Privacy mask control mode • Start: Moves the screen's center to the center of the drawn area. • Move: Moves the screen's center to the center of the MaskIndex's area.
	MaskIndex	REQ	<int>	Privacy area for which the PTZ coordinates should be adjusted This parameter should be sent only when Mode is Move
	MaskCoordinate	REQ	<string>	Co-ordinates of the privacy mask that is to be drawn. This parameter is used to request PTZ camera to move so that the Privacy area is at the center of the scene.

123.4. Examples

123.4.1. Getting the current information for Privacy of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain

<Body>
```

```
Channel.0.Enable=True
Channel.0.MaskPattern=Solid
Channel.0.MaskIndex=1
Channel.0.MaskName=privacy1
Channel.0.MaskColor=White
Channel.0.MaskCoordinate=456,432,1467,651
Channel.0.ZoomThresholdEnable=True
Channel.0.Position.Pan=3.78
Channel.0.Position.Tilt=223.58
Channel.0.MaskIndex=2
Channel.0.MaskName=privacy2
Channel.0.MaskColor=White
Channel.0.MaskCoordinate=570,456,1353,627
Channel.0.ZoomThresholdEnable=False
Channel.0.Position.Pan=10.12
Channel.0.Position.Tilt=201.11
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PrivacyMask": [
        {
            "Channel": 0,
            "Enable": true,
            "MaskPattern": "Solid",
            "Masks": [
                {
                    "MaskIndex": 1,
                    "MaskName": "privacy1",
                    "MaskColor": "White",
                    "MaskCoordinate": [
                        {
```

```
                        "x": 456,
                        "y": 432
                    },
                    {
                        "x": 1467,
                        "y": 651
                    }
                ],
                "ZoomThresholdEnable": true,
                "Position": {
                    "Pan": 3.77999999713897707,
                    "Tilt": 23.5799999923706056
                }
            },
            {
                "MaskIndex": 2,
                "MaskName": "privacy2",
                "MaskColor": "White",
                "MaskCoordinate": [
                    {
                        "x": 570,
                        "y": 456
                    },
                    {
                        "x": 1353,
                        "y": 627
                    }
                ],
                "ZoomThresholdEnable": false,
                "Position": {
                    "Pan": 10.11999998113497601,
                    "Tilt": 201.1099999926183082
                }
            }
        ]
    }
  ]
}
```

123.4.2. Enabling a privacy area

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=set&Enable=True
```

123.4.3. Disabling a privacy area

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=privacy&action=set&Enable=False
```

123.4.4. Setting the privacy protection area

To use the **add** action, **MaskName**, **MaskColor**, and **MaskCoordinate** parameters must be set together. If the mask area is a polygon type, the coordinate format is <x1,y1,x2,y2,x3,y3,x4,y4> and for a rectangle type, the coordinate format is <x1,y1,x2,y2>.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=privacy&action=add&MaskColor=White&MaskCoordinate=70,
140,817,807,1000,697,1140,420&MaskName=stwcamera
```

123.4.5. Setting a privacy mask pattern

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=privacy&action=set&MaskPattern=Solid
```

123.4.6. Changing the mask index at the user’s request

To reorder the current mask indexes, **CurrentIndexOrder** and **NewIndexOrder** must have valid values based on the **view** action. As the result of the below command, the 5th privacy mask will be returned as the 1st mask, and the 4th privacy mask will be returned as the 2nd mask, and so on.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=privacy&action=set&currentIndexOrder=1,3,4,5&NewIndex
Order=5,4,1,3
```

123.4.7. Removing the privacy protection area

To use the **remove** action, the **MaskIndex** parameter must be sent. The mask index value can be requested with the view action.

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=remove&MaskIndex=1
```

123.4.8. Adding a privacy mask with zoom threshold enabled

On enabling a privacy mask with the zoom threshold feature, privacy masks are not displayed between the zoom wide position and threshold position.

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=add&MaskName=M3&MaskColor=Black&MaskCoordinate=100,100,200,200&ZoomThresholdEnable=True
```

123.4.9. Adding a privacy mask when adjusting zoom threshold

Before SUNAPI version 2.6.1, when enabling a privacy mask with the zoom threshold feature, privacy masks were not displayed between the zoom wide position and the threshold position. With the new parameter **AdjustMode**, the mask will be displayed as transparent until an update command with **AdjustMode** as Stop is received or a specific time passes. Then the mask will change to a solid color automatically.

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=add&MaskIndex=1&MaskName=M3&MaskColor=Black&MaskCoordinate=100,100,200,200&ZoomThresholdEnable=True&AdjustMode=Start
```

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=privacy&action=update&MaskIndex=1&ZoomThresholdEnable=True&AdjustMode=Stop
```

123.4.10. Adding a privacy mask in PTZ model using control action

To move the PTZ camera, so that the privacy area being drawn is shown in the middle of the image, Mode and MaskCoordinate parameter must be sent together.

123.4.11. Setting the image preset

To set the image preset with the **set** action, **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** must be set together. (If **ScheduleEnable** had been already set to True, the **ScheduleEnable** parameter could be removed for the following request.)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepreset&action=set&ScheduleEnable=True&Schedule.2
.Mode=DefinitionFocus&Schedule.2.EveryDay.FromTo=00:00-00:01
```

Chapter 124. Fisheye Lens

124.1. Description

The **fisheyelens** submenu configures the settings for fisheye lenses.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for Feature Support: "attributes/Image/Support/FisheyeLens"

Access level

Action	Camera
view	Guest
set	Admin

124.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
fisheyelens&action=<value>[&<parameter>=<value>]
```

124.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for fisheye lenses
	Channel	REQ, RES	<csv>	Channel ID
	RPLnumber	RES	<string>	RPL number for Immervision fisheye lens
	LensModel	RES	<enum> CBC, Ambarella, BoowonOpticals, AL1	Lens model
	ViewModeType	RES	<enum> SubregionView.1, SubregionView.2	View mode
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	CameraPosition	REQ, RES	<enum> Wall, Ceiling, Ground	Camera position

124.4. Examples

124.4.1. Getting the current fisheye lens settings of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=fisheyelens&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.LensModel=CBC
Channel.0.CameraPosition=Wall
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Overlay": [
        {
            "Channel": 0,
            "LensModel": "CBC",
            "CameraPosition": "Wall"
        }
    ]
}
```

124.4.2. Setting camera position to ceiling

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=fisheyelens&action=set&CameraPosition=Ceiling
```

124.4.3. Setting camera position to wall

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=fisheyelens&action=set&CameraPosition=Wall
```

Chapter 125. View Modes

125.1. Description

The **viewmodes** submenu enables selection of the view mode.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for Fisheye lens Support: "attributes/Image/Support/FisheyeLens"
> Attribute to check for maximum view modes: "attributes/Image/Limit/ViewModes"
> Attribute to check for maximum active view modes:
> "attributes/Image/Limit/ViewModes.MaxActive"
> Attribute to check for Dewarped View: "attributes/Image/Support/ DewarpedView"

Access level

Action	Camera
view	Guest
set	Admin

125.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
viewmodes&action=<value>[&<parameter>=<value>]
```

125.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the settings for the fisheye lens settings and view mode settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ViewMode.#.Type	REQ, RES	<enum> Panorama, DoublePanorama, QuadView, QuadView.#, SingleView, CropView, 360Panorama, SubregionView.1, SubregionView.2	View mode **Note** Attribute to check for panorama view mode: "attributes/Image/Support/ViewModes.Panorama". Attribute to check for double panorama view mode: "attributes/Image/Support/ViewModes.DoublePanorama". Attribute to check for singleview view mode: "attributes/Image/Support/ViewModes.SingleView Attribute to check for cropview view mode: "attributes/Image/Support/ViewModes.CropView Attribute to check for quadview view mode: "attributes/Image/Support/ViewModes.QuadView". Attribute to check for 360Panorama view mode: "attributes/Image/Support/ViewModes.360Panorama".

125.4. Examples

125.4.1. Getting the current view mode settings of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=viewmodes&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.ViewMode.1.Type=Panorama
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ViewModes": [
        {
            "Channel": 0,
            "Views": [
                {
                    "ViewMode": 1,
                    "Type": "Panorama"
                }
            ]
        }
    ]
}
```


125.4.2. Selecting a view mode type

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=viewmodes&action=set&Channel=0&ViewMode.1.Type=Single
View
```

Chapter 126. Fisheye Setup

126.1. Description

The **fisheyesetup** submenu enables selection of both view mode and camera position at the same time.

> **NOTE**
>
> This chapter applies to network cameras only.
> Attribute to check for Fisheye lens Support: "attributes/Image/Support/FisheyeLens"
> Attribute to check for maximum view modes: "attributes/Image/Limit/ViewModes"
> Attribute to check for maximum active view modes:
> "attributes/Image/Limit/ViewModes.MaxActive"
> Attribute to check for Dewarped View: "attributes/Image/Support/ DewarpedView"

Access level

Action	Camera
view	Guest
set	User

126.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
fisheyesetup&action=<value>[&<parameter>=<value>]
```

126.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the settings for the view mode.
	Channel	REQ, RES	<csv>	Channel ID
	LensModel	RES	<enum> CBC, Ambarella, BoowonOpticals, AL1	Lens model
set	Channel	REQ, RES	<int>	Channel ID
	CameraPosition	REQ, RES	<enum> Wall, Ceiling, Ground	Camera position

Action	Parameters	Request/ Response	Type/ Value	Description
	ViewModeIndex	REQ, RES	<int>	• 0 indicates fisheye view • 1 indicated dewarp view
	ViewModeType	REQ, RES	<enum> Panorama, DoublePanorama, QuadView, Quadview.#., SingleView, CropView, 360Panorama	View mode. Each CameraPosition may support a different viewModeType. Refer to the imageoptions command to understand the dependency between the two parameters. **Note** Attribute to check for panorama view mode: "attributes/Image/Support/ViewModes.Panorama". Attribute to check for double panorama view mode: "attributes/Image/Support/ViewModes.DoublePanorama". Attribute to check for singleview view mode: "attributes/Image/Support/ViewModes.SingleView Attribute to check for cropview view mode: "attributes/Image/Support/ViewModes.CropView Attribute to check for quadview view mode: "attributes/Image/Support/ViewModes.QuadView". Attribute to check for 360Panorama view mode: "attributes/Image/Support/ViewModes.360Panorama".

126.4. Examples

126.4.1. Getting the current view mode settings of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=fisheyesetup&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.LensModel=CBC
Channel.0.CameraPosition=Ceiling
Channel.0.ViewModeIndex=1
Channel.0.ViewModeType=QuadView
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Viewmodes": [
        {
            "Channel": 0,
            "LensModel": "CBC",
            "CameraPosition": "Ceiling",
            "ViewModeIndex": "1",
            "ViewModeType": "QuadView"
        }
    ]
}
```


126.4.2. Selecting a view mode type and camera position

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=fisheyesetup&action=set&Channel=0&CameraPosition=Ceil
ing&ViewModeIndex=1&ViewModeType=QuadView
```

Chapter 127. Image Preset

127.1. Description

The **imagepreset** submenu configures the image preset.

CAUTION	It is recommended to use the **imagepreset2** submenu instead. The **imagepreset** submenu will be deprecated starting January 2027.

NOTE	This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

127.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imagepreset&action=<value>[&<parameter>=<value>]
```

127.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop, AWC	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied
	Mode	REQ, RES	<enum> DefinitionFocus, MotionFocus, ReducedNoise, BrightVideo, MotionFocus+ReducedNoise, MotionFocus+BrightVideo, VividVideo, UserPreset	Image preset mode • DefinitionFocus: Puts priority on definition • MotionFocus: Puts priority on the motion • ReducedNoise: Puts priority on noise reduction • BrightVideo: Puts priority on the brightness of the video • MotionFocus+ReducedNoise: Puts priority on the motion and noise reduction • MotionFocus+BrightVideo: Puts priority on motion and brightness of the video • VividVideo: Puts more on noise reduction and saturation

Action	Parameters	Request/ Response	Type/ Value	Description
	ScheduleEnable	REQ, RES	<bool> True, False	Enables or disables the schedule for image preset operation If **ScheduleEnable** is set to True, **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** should be sent together to set the schedule.

Action	Parameters	Request/ Response	Type/ Value	Description
	Schedule.#.Mode	REQ, RES	<enum> Off, DefinitionFocus, MotionFocus, ReducedNoise, BrightVideo, MotionFocus+ReducedNoise, MotionFocus+BrightVideo, VividVideo, UserPreset	Image preset mode in the corresponding schedule • Off: Disables the image preset mode • DefinitionFocus: Puts priority on definition • MotionFocus: Puts priority on the motion • ReducedNoise: Puts priority on noise reduction • BrightVideo: Puts priority on the brightness of the video • MotionFocus+ReducedNoise: Puts priority on the motion and noise reduction • MotionFocus+BrightVideo: Puts priority on motion and brightness of the video • VividVideo: Puts more on noise reduction and saturation **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** parameters are applicable only when **ScheduleEnable** is set to True. **Note** To use the **set** action, **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** must be sent together.

Action	Parameters	Request/ Response	Type/ Value	Description
	Schedule.#.EveryDay. FromTo	REQ, RES	<string>	Start time for the image preset The value is specified in the <hh:mm-hh:mm> format. **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** parameters are applicable only when **ScheduleEnable** is set to True. **Note** To use the **set** action, **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** must be sent together.

127.4. Examples

127.4.1. Getting the current image preset settings

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepreset&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=UserPreset
Channel.0.ScheduleEnable=True
Channel.0.Schedule.1.Mode=DefinitionFocus
Channel.0.Schedule.1.EveryDay.FromTo=00:00-06:59
Channel.0.Schedule.2.Mode=MotionFocus
Channel.0.Schedule.2.EveryDay.FromTo=07:00-08:59
Channel.0.Schedule.3.Mode=ReducedNoise
Channel.0.Schedule.3.EveryDay.FromTo=09:00-10:59
Channel.0.Schedule.4.Mode=BrightVideo
```

```
Channel.0.Schedule.4.EveryDay.FromTo=11:00-12:59
Channel.0.Schedule.5.Mode=VividVideo
Channel.0.Schedule.5.EveryDay.FromTo=14:00-23:59
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ImagePreset": [
        {
            "Channel": 0,
            "Mode": "UserPreset",
            "ScheduleEnable": true,
            "Schedules": [
                {
                    "Schedule": 1,
                    "Mode": "DefinitionFocus",
                    "EveryDay": {
                        "FromTo": "00:00-06:59"
                    }
                },
                {
                    "Schedule": 2,
                    "Mode": "MotionFocus",
                    "EveryDay": {
                        "FromTo": "07:00-08:59"
                    }
                },
                {
                    "Schedule": 3,
                    "Mode": "ReducedNoise",
                    "EveryDay": {
                        "FromTo": "09:00-10:59"
                    }
                },
                {
                    "Schedule": 4,
                    "Mode": "BrightVideo",
```

```
                    "EveryDay": {
                        "FromTo": "11:00-12:59"
                    }
                },
                {
                    "Schedule": 5,
                    "Mode": "VividVideo",
                    "EveryDay": {
                        "FromTo": "14:00-23:59"
                    }
                }
            ]
        }
    ]
}
```

127.4.2. Setting the image preset

To set the image preset with the **set** action, **Schedule.#.Mode** and **Schedule.#.EveryDay.FromTo** must be set together. (If **ScheduleEnable** had been already set to True, the **ScheduleEnable** parameter could be removed for the following request.)

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepreset&action=set&ScheduleEnable=True&Schedule.2.Mode=DefinitionFocus&Schedule.2.EveryDay.FromTo=00:00-00:01
```

Chapter 128. Image Alignment

128.1. Description

The **imagealignment** submenu configures the image alignment of multiple sensors.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for multiple sensor support: "attributes/Image/Support/MultiImager"

Access level

Action	Camera
view	Admin
set	Admin
control	Admin

128.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imagealignment&action=<value>[&<parameter>=<value>]
```

128.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	SensorID	RES	<int>	Image sensor ID
	SensorResolution	RES	<string>	Shows each sensor’s resolution
set	Channel	REQ, RES	<int>	Channel ID
	AlphaBlending	REQ, RES	<bool> True, False	Enable or disable the Alpha Blending feature of all sensors of a multi-imager camera

Action	Parameters	Request/ Response	Type/ Value	Description
	AlphaBlendingLevel	REQ, RES	<int>	Level when **AlphaBlending** is enabled. When this value is greater than 0, the overlapping image between 2 sensors will be larger. **Note** To check if this feature is supported, refer to the values of the following attributes "attributes/Image/Support/Image Alignment"
	FovAngle	REQ, RES	<enum> 180Degree, 209Degree	Field of view angle
control	Channel	REQ	<int>	Channel ID
	Mode	REQ	<enum> Move, Reset	Image preset mode • Move: Changes the sensor calibration • Reset : Resets the sensor calibration
	SensorID	REQ	<int>	Senor ID for which calibration needs to be applied **Note** For information on the Maximum number of sensors supported, refer to the following CGI section in attributes **image/imagealignment/control/ SensorID**

Action	Parameters	Request/ Response	Type/ Value	Description
	Horizontal	REQ	<int>	Distance the image should be adjusted in horizontal direction. Value is Normalized to a range of –N to N, with negative values indicating the image should be adjusted in the leftward direction. A positive value indicates the image should be adjusted in the rightward direction. **Note** For information on the actual range of the normalized value, refer to the following CGI section in attributes **image/imagealignment/control/Horizontal**
	Vertical	REQ	<int>	Distance the image should be adjusted in vertical direction. Value is Normalized to a range of –N to N; with negative values indicating the image should be adjusted in an upward direction. A positive value indicates the image should be adjusted in a downward direction. **Note** For information on the actual range of the normalized value, refer to the following CGI section in attributes image/imagealignment/control/Vertical

128.4. Examples

128.4.1. Aligning the images from different sensors

To adjust the image by 5 units in a rightward direction and 5 units in an upward direction, the following command can be used.

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagealignment&action=control&Channel=0&Mode=Move&SensorID=1&Horizontal=5&Vertical=-5&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "Response": "Success"
}
```

128.4.2. Get request to check settings

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagealignment&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain
```

```
<Body>
```

```
Channel.0.SensorID.1.Horizontal=0
Channel.0.SensorID.1.Vertical=0
Channel.0.SensorID.1.SensorResolution=1914:2240
Channel.0.SensorID.2.Horizontal=-10
Channel.0.SensorID.2.Vertical=0
Channel.0.SensorID.2.SensorResolution=1626:2240
Channel.0.SensorID.3.Horizontal=-30
Channel.0.SensorID.3.Vertical=0
Channel.0.SensorID.3.SensorResolution=1626:2240
Channel.0.SensorID.4.Horizontal=-50
Channel.0.SensorID.4.Vertical=0
Channel.0.SensorID.4.SensorResolution=1554:2240
Channel.0.FovAngle=192Degree
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    {
        "ImageAlignments": [
            {
                "Channel": 0,
                "FovAngle": "192Degree",
                "ImageAlignment": [
                    {
                        "SensorID": 1,
                        "Horizontal": 0,
                        "Vertical": 0,
                        "SensorResolution": "1914:2240"
                    },
                    {
                        "SensorID": 2,
                        "Horizontal": -10,
```

```
                        "Vertical": 0,
                        "SensorResolution": "1626:2240"
                    },
                    {
                        "SensorID": 3,
                        "Horizontal": -30,
                        "Vertical": 0,
                        "SensorResolution": "1626:2240"
                    },
                    {
                        "SensorID": 4,
                        "Horizontal": -50,
                        "Vertical": 0,
                        "SensorResolution": "1554:2240"
                    }
                ]
            }
        ]
    }
}
```

Chapter 129. Image Options

129.1. Description

The **imageoptions** submenu is used to explain the dependency between various features in image cgi.

NOTE	This chapter applies to network cameras only. Attribute to check for Fisheye lens Support: "attributes/Image/Support/FisheyeLens" Attribute to check for OSD Font Size Support: "image/multilineosd/add/FontSize"

Access level

Action	Camera
view	Admin

129.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imageoptions&action=<value>[&<parameter>=<value>]
```

129.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<csv>	Channel ID list
	CameraPosition	RES	<enum> Wal	This parameter will be supported only in Fisheye models. **CameraPosition** and **ViewModeType** parameters are dependent. When camera position is changed, the supported view modes will change.

Action	Parameters	Request/ Response	Type/ Value	Description
	ViewModeType	RES	<enum> Overview, 360Panorama, Panorama, DoublePanorama, QuadView, SingleView, OneOverviewAndTripleView,OneOverviewAndOctaView,CropView, LeftHalfView, RightHalfView, QuadView.#	This parameter will be supported only in Fisheye models. Supported View mode in each camera position
	TopLeftCoordinates	RES	<enum>	The top left co-ordinate of the view mode in whole image Format=x1,y1
	BottomRightCoordinates	RES	<enum>	The bottom left co-ordinate of the view mode in whole image Format=x1,y1
	MaxAGCSensorFrameRate	RES	<enum> 25, 30, 55, 60, 100, 120	Indicates the maximum sensor frame rate that can be set when AGC mode is enabled.
	MaxDISSensorFrameRate	RES	<enum> 25, 30, 55, 60	Indicates the maximum sensor frame rate the DIS can be set to.
	ImagePresetMode	RES	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	The predefined image preset has a defined set of values for the following parameters: SSDRLevel, SSNRLevel, AutoShortShutterSpeed, AutoLongShutterSpeed, Saturation The default value for each parameter in each of the image preset modes is listed in this command.

Action	Parameters	Request/ Response	Type/ Value	Description
	AGCMode	RES	<enum> Off, Low, Medium, High, Manual, MaxGain, AGCMaxGain	AGC (Auto Gain Control) mode adjusts the brightness by controlling the sensitivity of image gain when capturing an object in dim light.
	SSDRLevel	RES	<int>	SSDR level in each image preset mode
	SSNRLevel	RES	<int>	SSNR level in each image preset mode
	AutoShortShutterSpeed	RES	<enum> 2, 1, 1/2, 1/4, 1/5, 1/8, 1/15, 1/20, 1/25, 1/30, 1/50, 1/60, 1/100, 1/120, 1/150, 1/180, 1/200, 1/240, 1/250, 1/300, 1/360, 1/480, 1/500, 1/600, 1/700, 1/1000, 1/1500, 1/2500, 1/5000, 1/10000, 1/12000	Minimum shutter speed in each image preset mode
	AutoLongShutterSpeed	RES	<enum> 2, 1, 1/2, 1/4, 1/5, 1/8, 1/15, 1/20, 1/25, 1/30, 1/50, 1/60, 1/100, 1/120, 1/150, 1/180, 1/200, 1/240, 1/250, 1/300, 1/360, 1/480, 1/500, 1/600, 1/700, 1/1000, 1/1500, 1/2500, 1/5000, 1/10000, 1/12000	Maximum shutter speed in each image preset mode
	Saturation	RES	<int>	Saturation value in each image preset mode
	OSDType	RES	<enum> Date, Title	OSD Type
	PositionX	RES	<int>	Max position X for each OSD Type **Note** This parameter shall be used when Font Size option is not supported

Action	Parameters	Request/ Response	Type/ Value	Description
	PositionY	RES	<int>	Max position Y for each OSD Type **Note** This parameter shall be used when Font Size option is not supported
	FontSize	RES	<enum> Small, Medium, Large	Supported OSD Font sizes
	FontSize.#.OSDType	RES	<enum> Date, Title	OSD Types supported
	FontSize.#.PositionX	RES	<int>	Max position X for each OSD Type and each font type **Note** This parameter shall be used when Font Size option is not supported
	FontSize.#.PositionY	RES	<int>	Max position Y for each OSD Type and each font type **Note** This parameter shall be used when Font Size option is not supported
	CompensationMode.#.DefaultAutoShortShutterSpeed	RES	<enum>	The default minimum shutter speed for each compensation mode
	CompensationMode.#.DefaultAutoLongShutterSpeed	RES	<enum>	The default maximum shutter speed for each compensation mode
	CompensationMode.#.SensorCaptureFrameRate.#.DefaultPreferShutterSpeed	RES	<enum>	The default prefer shutter speed for each compensation mode at different sensor capture rates
	CompensationMode.#.SensorCaptureFrameRate.#.AutoShortShutterSpeed	RES	<enum>	The possible minimum shutter speed for each compensation mode at different sensor capture rates

Action	Parameters	Request/ Response	Type/ Value	Description
	CompensationMode.#.SensorCaptureFrameRate.#.AutoLongShutterSpeed	RES	<enum>	The possible maximum shutter speed for each compensation mode at different sensor capture rates
	CompensationMode.#.SensorCaptureFrameRate.#.PreferShutterSpeed	RES	<enum>	The possible prefer shutter speed for each compensation mode at different sensor capture rates
	CompensationMode.#.SensorCaptureFrameRate.#.DefaultMaxShutterSpeed	RES	<enum>	The default maximum shutter speed for each compensation mode at different sensor capture rates
	CompensationMode.#.SensorCaptureFrameRate.#.DefaultMinShutterSpeed	RES	<enum>	The default minimum shutter speed for each compensation mode at different sensor capture rates
	BLCAreaCoordinates.MinWidth	RES	<int>	User can set BLC area with real coordinates in some models. These show the valid range for that.
	BLCAreaCoordinates.MinHeight	RES	<int>	User can set BLC area with real coordinates in some models. These show the valid range for that.
	BLCAreaCoordinates.MaxWidth	RES	<int>	User can set BLC area with real coordinates in some models. These show the valid range for that.
	BLCAreaCoordinates.MaxHeight	RES	<int>	User can set BLC area with real coordinates in some models. These show the valid range for that.
	OSDType.Title.SupportedSpecialCharacters	RES	<csv>	This shows all special characters that can be written on OSD.
	OSDType.Title.SupportedLanguages	RES	<csv>	This shows all all available languages that can be written on OSD.
	BacklightType	RES	<enum> BLC, HLC, WDR	This shows all the supported backlight types.

Action	Parameters	Request/ Response	Type/ Value	Description
	BacklightType.BLC.BLCAreaTop.Min	RES	<int>	User can set BLC area with normalized values. These show the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaTop.Max	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaBottom.Min	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaBottom.Max	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaLeft.Min	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaLeft.Max	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.BLC.BLCAreaRight.Min	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.

Action	Parameters	Request/ Response	Type/ Value	Description
	BacklightType.BLC.BLCAreaRight.Max	RES	<int>	User can set BLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaTop.Min	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaTop.Max	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaBottom.Min	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaBottom.Max	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaLeft.Min	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaLeft.Max	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.

Action	Parameters	Request/ Response	Type/ Value	Description
	BacklightType.HLC.HLCAreaRight.Min	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.HLC.HLCAreaRight.Max	RES	<int>	User can set HLC area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaTop.Min	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaTop.Max	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaBottom.Min	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaBottom.Max	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaLeft.Min	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.

Action	Parameters	Request/ Response	Type/ Value	Description
	BacklightType.WDR.WDRAreaLeft.Max	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaRight.Min	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	BacklightType.WDR.WDRAreaRight.Max	RES	<int>	User can set WDR area with normalized values. These shows the valid range for that. The values must be within the range of 1 to 100.
	MaxWDRSensorFrameRate	RES	<enum> 20, 30, 50, 60	This shows the maximum framerates that can be supported in a device when WDR is enabled.
	LensModel.#.IrisFno	RES	<enum>	It shows which lensmodel support what IrisFno values.
	LensModel.#.SupportLDCMode	RES	<csv> Off, Manual, Auto, FillMode_Manual, FillMode_Auto, StretchMode_Manual, StretchMode_Auto	This shows which lensmodel supports which LDC modes.
	DISFocalLengthStepSize	RES	<int>	It shows how much DISFocalLength step size is. e.g. stepsize = 100, set value = 2150. applied set value is 2100.
	DayNightSwitchingMode.#.BWToColor	RES	<int>	"DayNightSwitchingMode" have enums that shows switch speed according to light. Each mode has default "BWToColor" & "ColorToBW" value.

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightSwitchingMode.#.ColorToBW	RES	<int>	"DayNightSwitchingMode" have enums that shows switch speed according to light. Each mode has default "BWToColor" & "ColorToBW" value.
	BacklightType.WDR.#.Max	RES	<int>	User can set WDR area with normalized values. This shows the valid range of that. The values must be within the range of 1 to 100.
	BacklightType.WDR.#.Min	RES	<int>	User can set WDR area with normalized values. This shows the valid range of that. The values must be within the range of 1 to 100.
	WDRAreaCoordinates.MinWidth	RES	<int>	User can set WDR area with real coordinates in some models. These show the valid range for that.
	WDRAreaCoordinates.MinHeight	RES	<int>	User can set WDR area with real coordinates in some models. These show the valid range for that.
	WDRAreaCoordinates.MaxWidth	RES	<int>	User can set WDR area with real coordinates in some models. These show the valid range for that.
	WDRAreaCoordinates.MaxHeight	RES	<int>	User can set WDR area with real coordinates in some models. These show the valid range for that.
	HLCAreaCoordinates.MinWidth	RES	<int>	User can set HLC area with real coordinates in some models. These show the valid range for that.
	HLCAreaCoordinates.MinHeight	RES	<int>	User can set HLC area with real coordinates in some models. These show the valid range for that.

Action	Parameters	Request/ Response	Type/ Value	Description
	HLCAreaCoordinates.MaxWidth	RES	<int>	User can set HLC area with real coordinates in some models. These show the valid range for that.
	HLCAreaCoordinates.MaxHeight	RES	<int>	User can set HLC area with real coordinates in some models. These show the valid range for that.
	LCEMode.#.Top	RES	<int>	LCE Coordinate top position for each mode. **Note** Some of the supported LCE modes are Horizontal, Sky, Full, Ground, Center75Percent, Center50Percent, Center25Percent
	LCEMode.#.Left	RES	<int>	LCE Coordinates left positon for each mode.
	LCEMode.#.Bottom	RES	<int>	LCE Coordinates bottom position for each mode.
	LCEMode.#.Right	RES	<int>	LCE Coordinates right posittion for each mode.
	AFLKMode.#.ShutterSpeed	RES	<enum> 1/25, 1/30, 1/50, 1/60, 1/100, 1/120	Supported shutter speeds for each AFLKMode
	MinManualShutterSpeed	RES	<int>	Supported minimum shutter speed value
	MaxManualShutterSpeed	RES	<int>	Supported maximum shutter speed value
	SupplementLightDurationMode.#.BWToColor	RES	<int>	SupplementLight default sensitivity (BW to Color level) for each sensitivity duration mode Here **#** refers to any of these modes, ["VeryHigh", "High", "Normal", "Low", "VeryLow"]

Action	Parameters	Request/ Response	Type/ Value	Description
	SupplementLightDurationMode.#.ColorToBW	RES	<int>	SupplementLight default sensitivity (Color to BW level) for each sensitivity duration mode Here **#** refers to any of these modes, ["VeryHigh", "High", "Normal", "Low", "VeryLow"]

129.4. Examples

129.4.1. Aligning the images from different sensors

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imageoptions&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
Channel.0.MaxAGCSensorFrameRate=30
Channel.0.MaxWDRSensorFrameRate=30
Channel.0.ImagePresetMode.UserPreset1.SSDRLevel=12
Channel.0.ImagePresetMode.UserPreset1.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.UserPreset1.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.UserPreset1.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.UserPreset1.SSNRLevel=12
Channel.0.ImagePresetMode.UserPreset1.SSNR2DLevel=12
Channel.0.ImagePresetMode.UserPreset1.SSNR3DLevel=12
Channel.0.ImagePresetMode.UserPreset1.AGCMode=High
Channel.0.ImagePresetMode.UserPreset1.Saturation=50
Channel.0.ImagePresetMode.UserPreset2.SSDRLevel=12
Channel.0.ImagePresetMode.UserPreset2.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.UserPreset2.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.UserPreset2.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.UserPreset2.SSNRLevel=12
```

```
Channel.0.ImagePresetMode.UserPreset2.SSNR2DLevel=12
Channel.0.ImagePresetMode.UserPreset2.SSNR3DLevel=12
Channel.0.ImagePresetMode.UserPreset2.AGCMode=High
Channel.0.ImagePresetMode.UserPreset2.Saturation=50
Channel.0.ImagePresetMode.OutdoorDaytime.SSDRLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.OutdoorDaytime.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.OutdoorDaytime.PreferShutterSpeed=1/250
Channel.0.ImagePresetMode.OutdoorDaytime.SSNRLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.SSNR2DLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.SSNR3DLevel=12
Channel.0.ImagePresetMode.OutdoorDaytime.AGCMode=High
Channel.0.ImagePresetMode.OutdoorDaytime.Saturation=55
Channel.0.ImagePresetMode.OutdoorNightime.SSDRLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.OutdoorNightime.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.OutdoorNightime.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.OutdoorNightime.SSNRLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.SSNR2DLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.SSNR3DLevel=12
Channel.0.ImagePresetMode.OutdoorNightime.AGCMode=High
Channel.0.ImagePresetMode.OutdoorNightime.Saturation=30
Channel.0.ImagePresetMode.IndoorBacklight.SSDRLevel=12
Channel.0.ImagePresetMode.IndoorBacklight.AutoShortShutterSpeed=1/60
Channel.0.ImagePresetMode.IndoorBacklight.AutoLongShutterSpeed=1/1500
Channel.0.ImagePresetMode.IndoorBacklight.PreferShutterSpeed=1/200
Channel.0.ImagePresetMode.IndoorBacklight.SSNRLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.SSNR2DLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.SSNR3DLevel=24
Channel.0.ImagePresetMode.IndoorBacklight.AGCMode=High
Channel.0.ImagePresetMode.IndoorBacklight.Saturation=40
Channel.0.ImagePresetMode.IndoorBrightScene.SSDRLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.IndoorBrightScene.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.IndoorBrightScene.PreferShutterSpeed=1/60
Channel.0.ImagePresetMode.IndoorBrightScene.SSNRLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.SSNR2DLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.SSNR3DLevel=12
Channel.0.ImagePresetMode.IndoorBrightScene.AGCMode=High
Channel.0.ImagePresetMode.IndoorBrightScene.Saturation=60
Channel.0.ImagePresetMode.NumberPlate.SSDRLevel=12
```

```
Channel.0.ImagePresetMode.NumberPlate.AutoShortShutterSpeed=1/300
Channel.0.ImagePresetMode.NumberPlate.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.NumberPlate.PreferShutterSpeed=1/500
Channel.0.ImagePresetMode.NumberPlate.SSNRLevel=4
Channel.0.ImagePresetMode.NumberPlate.SSNR2DLevel=4
Channel.0.ImagePresetMode.NumberPlate.SSNR3DLevel=4
Channel.0.ImagePresetMode.NumberPlate.AGCMode=Low
Channel.0.ImagePresetMode.NumberPlate.Saturation=50
Channel.0.ImagePresetMode.Vivid.SSDRLevel=10
Channel.0.ImagePresetMode.Vivid.AutoShortShutterSpeed=1/30
Channel.0.ImagePresetMode.Vivid.AutoLongShutterSpeed=1/12000
Channel.0.ImagePresetMode.Vivid.PreferShutterSpeed=1/60
Channel.0.ImagePresetMode.Vivid.SSNRLevel=12
Channel.0.ImagePresetMode.Vivid.SSNR2DLevel=12
Channel.0.ImagePresetMode.Vivid.SSNR3DLevel=12
Channel.0.ImagePresetMode.Vivid.AGCMode=High
Channel.0.ImagePresetMode.Vivid.Saturation=60
Channel.0.OSDType.Title.MaxCharacterLimit=85
Channel.0.OSDType.Title.PositionX=3328
Channel.0.OSDType.Title.PositionY=1872
Channel.0.OSDType.Title.SupportedSpecialCharacters=. -#*+/(),
Channel.0.OSDType.Date.PositionX=3328
Channel.0.OSDType.Date.PositionY=1872
Channel.0.FontSize.Small.OSDType.Title.MaxCharacterLimit=85
Channel.0.FontSize.Small.OSDType.Title.PositionX=3328
Channel.0.FontSize.Small.OSDType.Title.PositionY=1872
Channel.0.FontSize.Small.OSDType.Date.PositionX=3328
Channel.0.FontSize.Small.OSDType.Date.PositionY=1872
Channel.0.FontSize.Medium.OSDType.Title.MaxCharacterLimit=85
Channel.0.FontSize.Medium.OSDType.Title.PositionX=3328
Channel.0.FontSize.Medium.OSDType.Title.PositionY=1872
Channel.0.FontSize.Medium.OSDType.Date.PositionX=3328
Channel.0.FontSize.Medium.OSDType.Date.PositionY=1872
Channel.0.FontSize.Large.OSDType.Title.MaxCharacterLimit=85
Channel.0.FontSize.Large.OSDType.Title.PositionX=3328
Channel.0.FontSize.Large.OSDType.Title.PositionY=1872
Channel.0.FontSize.Large.OSDType.Date.PositionX=3328
Channel.0.FontSize.Large.OSDType.Date.PositionY=1872
Channel.0.CompensationMode.Off.DefaultAutoShortShutterSpeed=1/5
Channel.0.CompensationMode.Off.DefaultAutoLongShutterSpeed=1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.DefaultPreferShutte
```

```
rSpeed=1/200
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.DefaultMinShutterSp
eed=1/25
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.AutoLongShutterSpee
d=1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,
1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.25.PreferShutterSpeed=
1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/
480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.DefaultPreferShutte
rSpeed=1/200
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.DefaultMinShutterSp
eed=1/30
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.AutoLongShutterSpee
d=1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480
,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.Off.SensorCaptureFrameRate.30.PreferShutterSpeed=
1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480,1
/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.DefaultAutoShortShutterSpeed=1/5
Channel.0.CompensationMode.BLC.DefaultAutoLongShutterSpeed=1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.DefaultPreferShutte
rSpeed=1/200
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.DefaultMinShutterSp
eed=1/25
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
```

```
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.AutoLongShutterSpee
d=1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,
1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.25.PreferShutterSpeed=
1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/
480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.DefaultPreferShutte
rSpeed=1/200
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.DefaultMinShutterSp
eed=1/30
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.AutoLongShutterSpee
d=1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480
,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.BLC.SensorCaptureFrameRate.30.PreferShutterSpeed=
1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480,1
/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.DefaultAutoShortShutterSpeed=1/5
Channel.0.CompensationMode.HLC.DefaultAutoLongShutterSpeed=1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.DefaultPreferShutte
rSpeed=1/200
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.DefaultMinShutterSp
eed=1/25
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.AutoLongShutterSpee
d=1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,
1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.25.PreferShutterSpeed=
1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/
```

```
480,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.DefaultPreferShutte
rSpeed=1/200
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.DefaultMaxShutterSp
eed=1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.DefaultMinShutterSp
eed=1/30
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.AutoShortShutterSpe
ed=2,1,1/2,1/4,1/5,1/8,1/15,1/20,1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180
,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500,1/2500,
1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.AutoLongShutterSpee
d=1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480
,1/500,1/600,1/700,1/1000,1/1500,1/2500,1/5000,1/10000,1/12000
Channel.0.CompensationMode.HLC.SensorCaptureFrameRate.30.PreferShutterSpeed=
1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,1/480,1
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Channel.0.CompensationMode.WDR.DefaultAutoLongShutterSpeed=1/1500
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,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500
Channel.0.CompensationMode.WDR.SensorCaptureFrameRate.25.AutoLongShutterSpee
d=1/25,1/30,1/50,1/60,1/100,1/120,1/150,1/180,1/200,1/240,1/250,1/300,1/360,
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480,1/500,1/600,1/700,1/1000,1/1500
Channel.0.CompensationMode.WDR.SensorCaptureFrameRate.30.DefaultPreferShutte
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eed=1/1500
Channel.0.CompensationMode.WDR.SensorCaptureFrameRate.30.DefaultMinShutterSp
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Channel.0.CompensationMode.WDR.SensorCaptureFrameRate.30.AutoShortShutterSpe
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,1/200,1/240,1/250,1/300,1/360,1/480,1/500,1/600,1/700,1/1000,1/1500
Channel.0.CompensationMode.WDR.SensorCaptureFrameRate.30.AutoLongShutterSpee
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Channel.0.BacklightType.BLC.BLCAreaBottom.Max=100
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Channel.0.BacklightType.BLC.BLCAreaLeft.Max=70
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Channel.0.BacklightType.BLC.BLCAreaRight.Max=100
Channel.0.BacklightType.BLC.BLCAreaRight.Min=31
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Channel.0.BacklightType.HLC.HLCAreaBottom.Max=100
Channel.0.BacklightType.HLC.HLCAreaBottom.Min=31
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Channel.0.SupplementLightDurationMode.VeryHigh.ColorToBW=60
Channel.0.SupplementLightDurationMode.High.BWToColor=35
Channel.0.SupplementLightDurationMode.High.ColorToBW=65
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Channel.0.SupplementLightDurationMode.Normal.ColorToBW=70
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JSON RESPONSE

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HTTP/1.0 200 OK

Content-type: application/json

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```

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                    "1/300",
                    "1/360",
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                    "1/500",
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                "DefaultMinShutterSpeed": "1/30",
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                    "1/8",
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                    "1/60",
                    "1/100",
                    "1/120",
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                    "1/180",
                    "1/200",
                    "1/240",
                    "1/250",
```

```
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                                "1/200",
                                "1/240",
                                "1/250",
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```

```
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                        ]
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                "Outdoor",
                "Indoor",
                "AWC",
                "NarrowATW"
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                                "Min": 1
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                        ],
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                            {
                                "Max": 100,
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                        ],
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                                "Min": 1
```

```
                    }
                ],
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                        "Min": 31
                    }
                ]
            },
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                    }
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                        "Min": 31
                    }
                ],
                "HLCAreaLeft": [
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                        "Max": 70,
                        "Min": 1
                    }
                ],
                "HLCAreaRight": [
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            },
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```

```
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                ],
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            }
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    "BLCAreaCoordinates": {
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```

```
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                "ColorToBW": 65
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            {
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                "BWToColor": 30,
                "ColorToBW": 70
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            {
                "DayNightSwitchingMode": "Slow",
                "BWToColor": 25,
                "ColorToBW": 75
            },
            {
                "DayNightSwitchingMode": "VerySlow",
                "BWToColor": 20,
                "ColorToBW": 80
            }
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            "ColorToBW": 60
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        {
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            "BWToColor": 35,
            "ColorToBW": 65
        },
        {
            "SupplementLightDurationMode": "Normal",
```

```
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                "ColorToBW": 70
            },
            {
                "SupplementLightDurationMode": "Low",
                "BWToColor": 25,
                "ColorToBW": 75
            },
            {
                "SupplementLightDurationMode": "VeryLow",
                "BWToColor": 20,
                "ColorToBW": 80
            }
        ]

        }
    ]
}
```

Chapter 130. PTR

130.1. Description

The **ptr** submenu configures the PTR settings.

> **NOTE** This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
control	Admin

130.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
ptr&action=<value>[&<parameter>=<value>]
```

130.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the focus settings.
	Channel	REQ, RES	<csv>	Channel ID
	PanAngle	RES	<int>	Sets the Pan angle of the lens. **Note** Supported range may differ depending on model.
	TiltAngle	RES	<int>	Sets the Tilt angle of the lens. **Note** Supported range may differ depending on model.
	RotateAngle	RES	<int>	Sets the rotation angle of lens. **Note** Supported range may differ depending on model.
control	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	AutoRotate	REQ, RES	<bool> True, False	Enables or disables the auto rotation feature of a PTRZ camera.
	Pan	REQ, RES	<int>	Panning to specified angle. This is different from a normal PTZ camera’s Pan operation. The purpose of PTR differs.
	Tilt	REQ, RES	<int>	Tilt to specified angle. This is different from a normal PTZ camera’s Pan operation. The purpose of PTR differs.
	Rotate	RES	<int>	Rotate to specified angle.
	Mode	REQ	<enum> Continuous, Relative, Reset	If set to Relative, PTR moves in relative mode. If set to Reset, it returns to the initial position. If the value is not set by user, it moves in continuous mode by default.

130.4. Examples

130.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=ptr&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ptr": [
        {
            "Channel": 0,
            "PanAngle": 20,
```

```
            "TiltAngle": 12,
            "RotateAngle": 40
        }
    ]
}
```

130.4.2. Controlling Pan, Tilt and Rotate position

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=ptr&action=control&Channel=0&Pan=50&Tilt=50&Rotate=50
&Mode=Continuous
```

130.4.3. Controlling Auto Rotate

REQUEST

```
http://<Device IP>/ stw-
cgi/image.cgi?msubmenu=ptr&action=control&Channel=0&AutoRotate=True
```

Chapter 131. PTRZ Usage

131.1. Description

The **ptrzusage** submenu provides information on the number of times the PTR feature was used, to track durability.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

131.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
ptrzusage&action=<value>[&<parameter>=<value>]
```

131.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the PTR usage information
	Channel	REQ, RES	<csv>	Channel ID
	Pan	RES	<int>	Number of times Pan was used
	Tilt	RES	<int>	Number of times Tilt was used
	Rotate	RES	<int>	Number of times Rotate was used

131.4. Examples

131.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=ptrzusage&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json

<Body>
```


```
{
    "ptr": [
        {
            "Channel": 0,
            "Pan": 0,
            "Tilt": 0,
            "Rotate": 4
        }
    ]
}
```

Chapter 132. Image Enhancements 2

132.1. Description

The **imageenhancements2** submenu configures the image enhancement settings with image preset mode.

Access level

Action	Camera
view	User
set	User
control	User

132.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imageenhancements2&action=<value>[&<parameter>=<value>]
```

132.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads image enhancement settings
	Channel	REQ, RES	<csv>	Channel ID
	ImagePresetMode.#.SharpnessEnable	RES	<bool> True, False	Enables or disables image sharpness **Note** Attribute to check for Sharpness: "attributes/Image/Support/Sharpness" Each ImagePresetMode.# can have different image setting. For more details, refer the "ImagePresetMode" parameter in set action.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.SharpnessLevel	RES	<int>	Sharpness level **SharpnessLevel** is valid only when **SharpnessEnable** is set to True.
	ImagePresetMode.#.Brightness	RES	<int>	Brightness level The higher the value, the greater the brightness. **Note** Attribute to check for Brightness: "attributes/Image/Support/Brightness"
	ImagePresetMode.#.Gamma	RES	<int>	Gamma level The higher the value, the greater the gamma value. **Note** Attribute to check for Gamma: "attributes/Image/Support/Gamma"
	ImagePresetMode.#.Saturation	RES	<int>	Saturation level The higher the value, the greater the saturation. **Note** Attribute to check for Saturation: "attributes/Image/Support/Saturation"

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.DefogMode	RES	<enum> Off, Auto, Manual	Defogging Mode This enhances the quality of images created in a foggy environment. • Off: Disables defogging function • Auto: Automatically compensates the image according to the fog level • Manual: Manually sets the amount of compensation **Note** Attribute to check for Defog: "attributes/Image/Support/Defog"
	ImagePresetMode.#.DefogLevel	RES	<int>	Defogging Level The greater the value, the higher the defogging level. **DefogLevel** is valid only when **DefogMode** is set to Manual or Auto.
	ImagePresetMode.#.CAR	RES	<enum> Off, Low, Medium, High	Color Aberration Reduction This reduces the distortion or blurring of purple in the very bright area of an image.
	ImagePresetMode.#.DISEnable	RES	<bool> True, False	Enables or disables DIS (Digital Image Stabilization). This compensates the image automatically when the camera vibrates due to external factors such as wind. **Note** Attribute to check for DIS: "attributes/Image/Support/DIS"

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#. DISFocalLength	RES	<int> 2800 to 50000 (micro meter)	DIS Focal length value adjustable, can check imageoptions submenu "DISFocalLengthStepSize" parameter for allowed step size. CAMERA ONLY
	ImagePresetMode.#. OISEnable	RES	<bool> True, False	Enables or disables OIS (Optical Image Stabilization).
	ImagePresetMode.#. OISZoomThreshold	RES	<int>	OIS zoom threshold.
	ImagePresetMode.#.Contrast	RES	<int>	Contrast level The higher the value, the greater the contrast. **Note** Attribute to check for Saturation: "image/imageenhancements/set/Contrast"
	ImagePresetMode.#. OpticalDefogFilterEnable	RES	<bool> True, False	Enable or disable optical defog filter.
	ImagePresetMode.#.XCEEnable	RES	<bool>	XCE(eXternd Contrast Enhancement) Enabling This feature is similar to using unsharp mask filtering
	ImagePresetMode.#.XCELevel	RES	<int>	User can set filter level
	ImagePresetMode.#.LDCEnable	RES	<bool>	LDC status
	ImagePresetMode.#.LDCLevel	RES	<int>	User can set filter level

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.LDCMode	RES	<enum> Off, Auto, Manual, FillMode_Manual, FillMode_Auto, StretchMode_Manual, StretchMode_Auto	Provides option to control the LDC feature. • Off: Disables LDC function • Auto: Automatically adjusts the LDC • Manual: Manually sets the LDC level • FillMode: The angle of up and down view is not changed, but the angle of left and right view can be cut. FillMode_Manual mode works LDC with LDCLevel value, FillMode_Auto mode is works LDC with automatically • StretchMode: The FOV is not changed, but the aspect ratio can be changed. StretchMode_Manual mode works LDC with LDCLevel value, StretchMode_Auto mode is works LDC with automatically
	ImagePresetMode.#.GammaControl	RES	<bool>	Gamma Control Adjust the brightness of the image. When On is selected, dark areas are generally brighter. CAMERA ONLY **Note** When setting On, the screen change due to changes in brightness and gamma items may not be significant.
	ImagePresetMode.#.CACEnable	RES	<bool>	Enables chromatic aberration correction This feature is similar to using uns

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.CACLevel	RES	<int>	Adjust the chromatic aberration correction level. A higher value applies more correction. This option is enabled only when you enabled CACEnable value.
	ImagePresetMode.#.ContrastAdaptiveLevel	RES	<int>	Adjust contrast adaptive level
set	Channel	REQ, RES	<int>	Channel ID
	OpticalDefogFilterEnable	REQ, RES	<bool> True, False	Enable or disable optical defog filter.
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Each **ImagePresetMode** can have ImagePreset settings. Image settings in any Mode can be changed. But modes except UserPreset1, UserPreset2 have different default image settings.
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied.

Action	Parameters	Request/ Response	Type/ Value	Description
	SharpnessEnable	REQ, RES	<bool> True, False	Enables or disables image sharpness **Note** Attribute to check for Sharpness: "attributes/Image/Support/Sharpness"
	SharpnessLevel	REQ, RES	<int>	Sharpness level **SharpnessLevel** is valid only when **SharpnessEnable** is set to True.
	Brightness	REQ, RES	<int>	Brightness level The higher the value, the greater the brightness. **Note** Attribute to check for Brightness: "attributes/Image/Support/Brightness"
	Gamma	REQ, RES	<int>	Gamma level The higher the value, the greater the gamma value. **Note** Attribute to check for Gamma: "attributes/Image/Support/Gamma"

Action	Parameters	Request/ Response	Type/ Value	Description
	GammaControl	REQ, RES	<bool>	Gamma Control Adjust the brightness of the image. When On is selected, dark areas are generally brighter. CAMERA ONLY **Note** When setting On, the screen change due to changes in brightness and gamma items may not be significant.
	Saturation	REQ, RES	<int>	Saturation level The higher the value, the greater the saturation. **Note** Attribute to check for Saturation: "attributes/Image/Support/Saturation"
	Contrast	REQ, RES	<int>	Contrast level The higher the value, the greater the contrast. **Note** Attribute to check for Saturation: "image/imageenhancements/set/Contrast"

Action	Parameters	Request/ Response	Type/ Value	Description
	DefogMode	REQ, RES	<enum> Off, Auto, Manual	Defogging Mode This enhances the quality of images created in a foggy environment. • Off: Disables defogging function • Auto: Automatically compensates the image according to the fog level • Manual: Manually sets the amount of compensation **Note** Attribute to check for Defog: "attributes/Image/Support/Defog"
	DefogLevel	REQ, RES	<int>	Defogging Level The greater the value, the higher the defogging level. **DefogLevel** is valid only when **DefogMode** is set to Manual or Auto.
	DISEnable	REQ, RES	<bool> True, False	Enables or disables DIS (Digital Image Stabilization). This compensates the image automatically when the camera vibrates due to external factors such as wind. **Note** Attribute to check for DIS: "attributes/Image/Support/DIS"

Action	Parameters	Request/ Response	Type/ Value	Description
	DISFocalLength	REQ,RES	<int> 2800 to 50000 (micro meter)	DIS Focal length value adjustable, can check imageoptions submenu "DISFocalLengthStepSize" parameter for allowed step size. CAMERA ONLY
	OISEnable	REQ,RES	<bool> True, False	Enables or disables OIS (Optical Image Stabilization).
	OISZoomThreshold	REQ,RES	<int>	OIS zoom threshold.
	CAR	REQ, RES	<enum> Off, Low, Medium, High	Color Aberration Reduction This reduces the distortion or blurring of purple in the very bright area of an image.
	LDCEnable	REQ, RES	<bool> True, False	Enables or disables LDC (Lens Distortion Control). LDCEnable and LDCMode cannot be set at the same time. **LDCEnable** True is same as Manual **LDCMode** **LDCEnable** False is same as Off **LDCMode**
	LDCLevel	REQ, RES	<int>	LDC level The greater the value, the higher the LDC level. **LDCLevel** is valid only when **LDCMode** is not set to Off. **LDCLevel** is valid only when **LDCEnable** is not set to False.

Action	Parameters	Request/ Response	Type/ Value	Description
	LDCMode	REQ, RES	<enum> Off, Auto, Manual	Provides option to control the LDC feature. • Off: Disables LDC function • Auto: Automatically adjusts the LDC • Manual: Manually sets the LDC level • FillMode: The angle of up and down view is not changed, but the angle of left and right view can be cut. FillMode_Manual mode works LDC with LDCLevel value, FillMode_Auto mode is works LDC with automatically • StretchMode: The FOV is not changed, but the aspect ratio can be changed. StretchMode_Manual mode works LDC with LDCLevel value, StretchMode_Auto mode is works LDC with automatically
	XCEEnable	REQ, RES	<bool>	XCE(eXternd Contrast Enhancement) Enabling This feature is similar to using unsharp mask filtering
	XCELevel	REQ RES	<int>	User can set filter level
	CACEnable	REQ, RES	<bool>	Enables chromatic aberration correction This feature is similar to using uns
	CACLevel	REQ, RES	<int>	Adjust the chromatic aberration correction level. A higher value applies more correction. This option can be configured only when you enabled CACEnable.
	ContrastAdaptiveLeve l	REQ, RES	<int>	Adjust contrast adaptive level

Action	Parameters	Request/ Response	Type/ Value	Description
	HDREnable	REQ, RES	<bool>	Enable HDR function
control	Channel	REQ	<int>	Channel ID
	Reset	REQ	<bool> True, False	The following image parameters are reset to the default value; Contrast, Brightness, SharpnessLevel, Saturation

132.4. Examples

132.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imageenhancements2&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
Channel.0.Brightness=50
Channel.0.SharpnessEnable=True
Channel.0.SharpnessLevel=12
Channel.0.Gamma=5
Channel.0.Saturation=50
Channel.0.Contrast=50
Channel.0.HDREnable=True
Channel.0.ImagePresetMode.UserPreset1.Brightness=50
Channel.0.ImagePresetMode.UserPreset1.SharpnessEnable=True
Channel.0.ImagePresetMode.UserPreset1.SharpnessLevel=12
Channel.0.ImagePresetMode.UserPreset1.Gamma=5
Channel.0.ImagePresetMode.UserPreset1.Saturation=50
Channel.0.ImagePresetMode.UserPreset1.Contrast=50
Channel.0.ImagePresetMode.UserPreset2.Brightness=50
Channel.0.ImagePresetMode.UserPreset2.SharpnessEnable=True
```

```
Channel.0.ImagePresetMode.UserPreset2.SharpnessLevel=12
Channel.0.ImagePresetMode.UserPreset2.Gamma=5
Channel.0.ImagePresetMode.UserPreset2.Saturation=50
Channel.0.ImagePresetMode.UserPreset2.Contrast=50
Channel.0.ImagePresetMode.OutdoorDaytime.Brightness=50
Channel.0.ImagePresetMode.OutdoorDaytime.SharpnessEnable=True
Channel.0.ImagePresetMode.OutdoorDaytime.SharpnessLevel=16
Channel.0.ImagePresetMode.OutdoorDaytime.Gamma=5
Channel.0.ImagePresetMode.OutdoorDaytime.Saturation=55
Channel.0.ImagePresetMode.OutdoorDaytime.Contrast=55
Channel.0.ImagePresetMode.OutdoorNightime.Brightness=50
Channel.0.ImagePresetMode.OutdoorNightime.SharpnessEnable=True
Channel.0.ImagePresetMode.OutdoorNightime.SharpnessLevel=6
Channel.0.ImagePresetMode.OutdoorNightime.Gamma=5
Channel.0.ImagePresetMode.OutdoorNightime.Saturation=30
Channel.0.ImagePresetMode.OutdoorNightime.Contrast=30
Channel.0.ImagePresetMode.IndoorBacklight.Brightness=70
Channel.0.ImagePresetMode.IndoorBacklight.SharpnessEnable=True
Channel.0.ImagePresetMode.IndoorBacklight.SharpnessLevel=6
Channel.0.ImagePresetMode.IndoorBacklight.Gamma=2
Channel.0.ImagePresetMode.IndoorBacklight.Saturation=40
Channel.0.ImagePresetMode.IndoorBacklight.Contrast=40
Channel.0.ImagePresetMode.IndoorBrightScene.Brightness=50
Channel.0.ImagePresetMode.IndoorBrightScene.SharpnessEnable=True
Channel.0.ImagePresetMode.IndoorBrightScene.SharpnessLevel=16
Channel.0.ImagePresetMode.IndoorBrightScene.Gamma=5
Channel.0.ImagePresetMode.IndoorBrightScene.Saturation=60
Channel.0.ImagePresetMode.IndoorBrightScene.Contrast=50
Channel.0.ImagePresetMode.NumberPlate.Brightness=50
Channel.0.ImagePresetMode.NumberPlate.SharpnessEnable=True
Channel.0.ImagePresetMode.NumberPlate.SharpnessLevel=12
Channel.0.ImagePresetMode.NumberPlate.Gamma=5
Channel.0.ImagePresetMode.NumberPlate.Saturation=50
Channel.0.ImagePresetMode.NumberPlate.Contrast=50
Channel.0.ImagePresetMode.Vivid.Brightness=50
Channel.0.ImagePresetMode.Vivid.SharpnessEnable=True
Channel.0.ImagePresetMode.Vivid.SharpnessLevel=14
Channel.0.ImagePresetMode.Vivid.Gamma=5
Channel.0.ImagePresetMode.Vivid.Saturation=60
Channel.0.ImagePresetMode.Vivid.Contrast=60
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ImageEnhancements": [
        {
            "Channel": 0,
            "Brightness": 50,
            "SharpnessEnable": true,
            "SharpnessLevel": 12,
            "Gamma": 5,
            "Saturation": 50,
            "Contrast": 50,
            "HDREnable": true,
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "Brightness": 50,
                    "SharpnessEnable": true,
                    "SharpnessLevel": 12,
                    "Gamma": 5,
                    "Saturation": 50,
                    "Contrast": 50
                },
                {
                    "ImagePresetMode": "UserPreset2",
                    "Brightness": 50,
                    "SharpnessEnable": true,
                    "SharpnessLevel": 12,
                    "Gamma": 5,
                    "Saturation": 50,
                    "Contrast": 50
                },
                {
                    "ImagePresetMode": "OutdoorDaytime",
                    "Brightness": 50,
                    "SharpnessEnable": true,
                    "SharpnessLevel": 16,
```

```
                "Gamma": 5,
                "Saturation": 55,
                "Contrast": 55
            },
            {
                "ImagePresetMode": "OutdoorNightime",
                "Brightness": 50,
                "SharpnessEnable": true,
                "SharpnessLevel": 6,
                "Gamma": 5,
                "Saturation": 30,
                "Contrast": 30
            },
            {
                "ImagePresetMode": "IndoorBacklight",
                "Brightness": 70,
                "SharpnessEnable": true,
                "SharpnessLevel": 6,
                "Gamma": 2,
                "Saturation": 40,
                "Contrast": 40
            },
            {
                "ImagePresetMode": "IndoorBrightScene",
                "Brightness": 50,
                "SharpnessEnable": true,
                "SharpnessLevel": 16,
                "Gamma": 5,
                "Saturation": 60,
                "Contrast": 50
            },
            {
                "ImagePresetMode": "NumberPlate",
                "Brightness": 50,
                "SharpnessEnable": true,
                "SharpnessLevel": 12,
                "Gamma": 5,
                "Saturation": 50,
                "Contrast": 50
            },
            {
```

```
                    "ImagePresetMode": "Vivid",
                    "Brightness": 50,
                    "SharpnessEnable": true,
                    "SharpnessLevel": 14,
                    "Gamma": 5,
                    "Saturation": 60,
                    "Contrast": 60
                }
            ]
        }
    ]
}
```

132.4.2. Setting the sharpness level to 5

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imageenhancements2&action=set&SharpnessLevel=5
```

132.4.3. Setting the sharpness level to 5 of the image preset mode(IndoorBacklight)

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imageenhancements2&action=set&SharpnessLevel=5&ImagePresetMode=IndoorBacklight
```

132.4.4. Setting the image enhancements

Saturation to 60, Defog mode to Manual, DIS to On, and Brightness to 60

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imageenhancements2&action=set&Saturation=60&DefogMode=Manual&DISEnable=True&Brightness=60
```

Chapter 133. Auto Image Alignment

133.1. Description

The **autoimagealignment** submenu is defined for panoramic cameras that support image stitching.

> **NOTE**
> Check channel based AutoImageAlignmentSupport attribute to check if a particular channel supports it. Attribute to check for feature support: "attributes/Image/Support/[ChannelID]/AutoImageAlignmentSupport"

In case of PNM-9030V, only the first channel (panorama channel) supports this feature.

Access level

Action	Camera
view	Admin
set	Admin

133.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
autoimagealignment&action=<value>[&<parameter>=<value>]
```

133.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<int>	Channel ID of panoramic channel
set	Channel	REQ, RES	<int>	Channel ID A camera uses from ‘0’ to max fixed channel count-1 for channel ID.
	FovDistance	REQ, RES	<float> 1.5-100	Distance in meters. A camera would optimize stitching for this distance.
	FovAngle	REQ, RES	<string> 180Degree, 220Degree	Field of view angle
	HorizontalAdjustmentEnable	REQ, RES	<bool> True, False	Enable or disable image alignment based on the horizon

Action	Parameters	Request/ Response	Type/ Value	Description
	HorizontalAdjustmentPosition	REQ, RES	<int>	Y-axis position of the horizon. This value is used to adjust the horizontal alignment between multiple lenses.
check	Status	RES	<enum> Idle, InProgress, Complete	
	Channel	REQ, RES		

133.4. Examples

133.4.1. Getting auto image alignment settings of the panorama channel

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=autoimagealignment&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
Channel.0.FovDistance=5.0
Channel.0.FovAngle=180Degree
```

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=autoimagealignment&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "AutoImageAlignment": [
        {
            "Channel": 0,
            "FovDistance": 5,
            "FovAngle": "180Degree"
        }
    ]
}
```

133.4.2. Setting calibration value to channel 1 (stiched panoramic channel)

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=autoimagealignment&action=set&Channel=1&FovDistance=25.3&FovAngle=180Degree
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

Chapter 134. Image Preset 2

134.1. Description

The **imagepreset2** submenu configures the image preset mode, ImagePreview.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User
control	User

134.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imagepreset2&action=<value>[&<parameter>=<value>]
```

134.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop, AWC	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied
	Mode	REQ, RES	<enum> DefinitionFocus, MotionFocus, ReducedNoise, BrightVideo, MotionFocus+ReducedNoise, MotionFocus+BrightVideo, VividVideo, UserPreset	Image preset mode • DefinitionFocus: Puts priority on definition • MotionFocus: Puts priority on the motion • ReducedNoise: Puts priority on noise reduction • BrightVideo: Puts priority on the brightness of the video • MotionFocus+ReducedNoise: Puts priority on the motion and noise reduction • MotionFocus+BrightVideo: Puts priority on motion and brightness of the video • VividVideo: Puts more on noise reduction and saturation

Action	Parameters	Request/ Response	Type/ Value	Description
control	ResetImagePresetMode	REQ	<enum> UserPreset 1, UserPreset 2, OutdoorDaytime, OutdoorNighttime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid	Reset Image Preset mode's image settings. Each Image setting's default value will be applied. About default value, please refer the **imageoptions** submenu.
	Channel	REQ	<int>	Channel ID

134.4. Examples

134.4.1. Getting the current image preset settings

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepreset2&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ImagePreset": [
        {
            "Channel": 0,
            "Mode": "Vivid"
        }
    ]
}
```

134.4.2. Reset image preset mode’s image setting

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepreset2&action=control&ResetImagePresetMode=Vivi
d
```

Chapter 135. Imagepreset schedule

135.1. Description

The **imagepresetschedule** submenu configures the image preset mode’s scheduling.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User

135.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
imagepresetschedule&action=<value>[&<parameter>=<value>]
```

135.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	<ddd>.<hourIndex>.Enable	REQ, RES	<bool> True, False	Enables or disables Day and Hour in schedule.

Action	Parameters	Request/ Response	Type/ Value	Description
	<ddd>.<hourIndex>.Mode	REQ, RES	<csv> UserPreset1, UserPreset2, OutdoorDaytime, OutdoorNightime, IndoorBacklight, IndoorBrightScene, NumberPlate, Vivid, Indoor, Outdoor	Set image preset mode in the corresponding hour's schedule. **Note** To use the **This parameter**, **<ddd>.Enable=True, <ddd>.<hourIndex>.Enable=True** must be sent together. **<ddd>.FromTo** must be specified in the format of <hh:mm-hh:mm>. <ddd> stands for day of the week and should be specified in short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. Ex) SUN.Enable=True&SUN.0.Enable=True&SUN.0.Mode=Vivid
	<ddd>.<hourIndex>.Minute	REQ, RES	<csv>	**Note** To use the This parameter, **<ddd>.Enable=True, <ddd>.<hourIndex>.Enable=True** must be sent together. **<ddd>.<hourIndex>.Minute** value must be specified in the format of <hh-hh>. Ex) SUN.Enable=True&SUN.0.Enable=True&SUN.0.Minute=30-40

Action	Parameters	Request/ Response	Type/ Value	Description
	<ddd>.Enable	REQ, RES	<bool> True, False	Enable or disable <ddd> day's scheduling. **Note** If this parameter is used without **<ddd>.<hourIndex>.Enable** parameter, Enable <ddd> day's all hour. Ex) **SUN.Enable=True** means Sunday 24hours.
	Activate	REQ, RES	<enum> Always, Scheduled	Enable or disable scheduling. • Always: Disables scheduling. Apply current ImagePreset mode. • Scheduled: Enables scheduling.

135.4. Examples

135.4.1. Getting the current image preset schedule settings

REQUEST

```
http://<IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "ImagePresetSchedule": [
        {
            "Channel": 0,
```

```
            "Activate": "Off",
            "Schedule": {
                "SUN": [
                    {
                        "HourIndex": 0,
                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
                        "HourIndex": 1,
                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
                        "HourIndex": 2,
                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
                        "HourIndex": 3,
                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
                        "HourIndex": 4,
                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
                        "HourIndex": 5,
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                    },
                    {
                        "HourIndex": 6,
```

```
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                    "HourIndex": 7,
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                },
                {
                    "HourIndex": 11,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 12,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
```

```
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            "Minute": "00-00"
        },
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        },
        {
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        },
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        },
        {
            "HourIndex": 18,
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            "Mode": "UserPreset1",
            "Minute": "00-00"
        },
        {
            "HourIndex": 19,
            "Enable": false,
            "Mode": "UserPreset1",
            "Minute": "00-00"
        },
```

```
                    {
                        "HourIndex": 20,
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                    },
                    {
                        "HourIndex": 21,
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                    },
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                    }
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                },
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                },
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                },
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            "Minute": "00-00"
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                },
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                    },
                    {
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                },
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                },
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                },
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                },
                {
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                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 7,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 8,
                    "Enable": false,
```

```
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                    "Minute": "00-00"
                },
                {
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                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 10,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 11,
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                },
                {
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                    "Minute": "00-00"
                },
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                },
                {
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                "Minute": "00-00"
            },
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            },
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            },
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            },
            {
```

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                }
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                },
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            },
            {
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                },
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                    "Minute": "00-00"
                },
                {
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                    "Enable": false,
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                },
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                },
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                    "Minute": "00-00"
                },
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                    "Minute": "00-00"
                },
                {
                    "HourIndex": 18,
```

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            },
            {
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                "Minute": "00-00"
            },
            {
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            }
        ],
        "FRI": [
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                "Minute": "00-00"
```

```
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                },
                {
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                },
                {
                    "HourIndex": 3,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 4,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 5,
                    "Enable": false,
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                    "Minute": "00-00"
                },
                {
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                    "Minute": "00-00"
                },
                {
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                    "Mode": "UserPreset1",
```

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                },
                {
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                    "Minute": "00-00"
                },
                {
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                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
                    "HourIndex": 10,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
                {
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                    "Enable": false,
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                    "Minute": "00-00"
                },
                {
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                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                },
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                    "Minute": "00-00"
                },
                {
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                    "Enable": false,
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            },
            {
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                "Mode": "UserPreset1",
                "Minute": "00-00"
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            },
            {
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                "Minute": "00-00"
            },
            {
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            },
            {
                "HourIndex": 19,
                "Enable": false,
                "Mode": "UserPreset1",
                "Minute": "00-00"
            },
            {
                "HourIndex": 20,
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                "Minute": "00-00"
            },
            {
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```

```
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            },
            {
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            },
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            }
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                "Minute": "00-00"
            },
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            },
            {
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```
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                        "Minute": "00-00"
                    },
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                    },
                    {
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                        "Minute": "00-00"
                    },
                    {
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                        "Enable": false,
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                        "Minute": "00-00"
                    },
                    {
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                        "Enable": false,
                        "Mode": "UserPreset1",
                        "Minute": "00-00"
                    },
                    {
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                        "Enable": false,
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                },
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                },
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                },
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                    "Minute": "00-00"
                },
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```

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                },
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                    "Minute": "00-00"
                },
                {
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                },
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                    "Minute": "00-00"
                },
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                    "Minute": "00-00"
                },
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                },
                {
                    "HourIndex": 23,
                    "Enable": false,
                    "Mode": "UserPreset1",
                    "Minute": "00-00"
                }
            ]
        }
```

```
        }
    ]
}
```

135.4.2. Setting the image preset schedule

Enable Scheduler

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepresetschedule&action=set&Activate=Scheduled
```

Enable Sunday 24 hours

REQUEST

```
http://<Device IP>/stw-cgi/stw-
cgi/image.cgi?msubmenu=imagepresetschedule&action=set&SUN.Enable=True
```

Enable Sunday 0~1 hour

REQUEST

```
http://<Device IP>/stw-cgi/stw-
cgi/image.cgi?msubmenu=imagepresetschedule&action=set&SUN.Enable=True&SUN.0.
Enable=True
```

Set Sunday hour's mode or minute

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepresetschedule&action=set&SUN.Enable=True&SUN.0.
Enable=True&SUN.0.Mode=Vivid
```

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=imagepresetschedule&action=set&SUN.Enable=True&SUN.0.
Enable=True&SUN.0.Minute=30-40
```

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=set&SUN.Enable=True&SUN.0.Enable=True&SUN.0.Mode=Vivid&SUN.0.Minute=30-40
```

Enable Tuesday 0~4 hours

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=set&TUE.Enable=True&TUE.0.Enable=True&TUE.1.Enable=True&TUE.2.Enable=True&TUE.3.Enable=True
```

Enable Monday 0~4 hours & Tuesday 0~4 hours

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=set&MON.Enable=True&MON.0.Enable=True&MON.1.Enable=True&MON.2.Enable=True&MON.3.Enable=True&TUE.Enable=True&TUE.0.Enable=True&TUE.1.Enable=True&TUE.2.Enable=True&TUE.3.Enable=True
```

Enable Monday 0~2:30 as Vivid mode

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=set&MON.Enable=True&MON.0.Enable=True&MON.0.Mode=Vivid&MON.1.Enable=True&MON.1.Mode=Vivid&MON.2.Enable=True&MON.2.Mode=Vivid&MON.2.Minute=0-30
```

Enable Monday 0~2:30 as Vivid mode & Tuesday 0~2:30 as IndoorBrightScene mode

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=imagepresetschedule&action=set&MON.Enable=True&MON.0.Enable=True&MON.0.Mode=Vivid&MON.1.Enable=True&MON.1.Mode=Vivid&MON.2.Enable=True&MON.2.Mode=Vivid&MON.2.Minute=0-30&TUE.Enable=True&TUE.0.Enable=True&TUE.0.Mode=IndoorBrightScene&TUE.1.Enable=True&TUE.1.Mode=IndoorBrightScene&TUE.2.Enable=True&TUE.2.Mode=IndoorBrig
```

```
htScene&TUE.2.Minute=0-30
```

Chapter 136. Thermal Palette Setting

136.1. Description

The **thermalpalettesetting** submenu configures the thermal color palette, upper&lower temperature level, Emissivity, ColorBar overlay.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

136.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
thermalpalettesetting&action=<value>[&<parameter>=<value>]
```

136.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	ThermalColorPalette	REQ, RES	<enum> WhiteHot, BlackHot, Rainbow, Custom, Sepia, Red, Iron, Red-WH, Iron-WH, Midrange-WH	Configures thermal color palette.

Action	Parameters	Request/ Response	Type/ Value	Description
	UpperTemperatureLevel	REQ, RES	<int>	Set boundary that devides high temperature range and middle temperature range. **Note** Only if **ThermalColorPalette** is Red-WH,Iron-WH,Midrange-WH, this parameter is applicable. UpperTemperatureLevel should be greater than LowerTemperatureLevel. Refer **thermalpalettesettingoptions** submunu for setable range.
	LowerTemperatureLevel	REQ, RES	<int>	Set boundary that devides middle temperature range and low temperature range. **Note** Only if **ThermalColorPalette** is Red-WH,Iron-WH,Midrange-WH, this parameter is applicable. UpperTemperatureLevel should be greater than LowerTemperatureLevel. Refer **thermalpalettesettingoptions** submunu for setable range.
	NormalizedEmissivity	REQ, RES	<int>	Configure overall NormalizedEmissivity of material shown in video. This configuration can effect camera's accuracy to detect temperature. **Note** Emissivity: The **emissivity** of the surface of a material is its effectiveness in emitting energy as thermal radiation.
	ColorBarOverlay	REQ, RES	<bool> True, False	Enable or disable Colorbar overlay in video.

Action	Parameters	Request/ Response	Type/ Value	Description
	ThermalVariationSensitivity	REQ, RES	<int>	Sensitivity of thermal variation **Note** This parameter applies to thermal channels only Attribute to check for ThermalVariationSensitivity: "attributes/Image/Support/[ChannleID]/ ThermalVariationSensitivity"

136.4. Examples

136.4.1. Getting the current thermal palette settings

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=thermalpalettesetting&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "ThermalPaletteSetting": [
        {
            "Channel": 0,
            "ThermalColorPalette": "WhiteHot",
            "UpperTemperatureLevel": 60,
            "LowerTemperatureLevel": 20,
            "NormalizedEmissivity": 95,
            "ColorBarOverlay": false,
            "ThermalVariationSensitivity": 12
        }
    ]
```

```
}
```

136.4.2. Setting the image preset schedule

Set thermal color palette, Emissivity, Color bar overlay, Upper temperature leve, Lower temperature level

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=thermalpalettesetting&action=set&Channel=0&ThermalCol
orPalette=Red-
WH&NormalizedEmissivity=51&ColorBarOverlay=True&UpperTemperatureLevel=70&Low
erTemperatureLevel=30
```

Chapter 137. Thermal Palette Setting Options

137.1. Description

The **thermalpalettesettingoptions** submenu provides temperature unit's setable temperature range.

This submenu is applicable for temperature detection camera.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

137.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
thermalpalettesettingoptions&action=<value>[&<parameter>=<value>]
```

137.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset
	Channel	REQ, RES	<csv>	Channel ID
	MinimumTemperatureLevel.Celsius	RES	<int>	Shows minimum celsius temperature value.
	MaximumTemperatureLevel.Celsius	RES	<int>	Shows maximum celsius temperature value.
	MinimumTemperatureLevel.Fahrenheit	RES	<int>	Shows minimum fahrenheit temperature value.
	MaximumTemperatureLevel.Fahrenheit	RES	<int>	Shows maximum fahrenheit temperature value.

137.4. Examples

137.4.1. Getting setable temperature range of each units

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/image.cgi?msubmenu=thermalpalettesettingoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ThermalPaletteSettingOptions": [
        {
            "Channel": 0,
            "MinimumTemperatureLevel": {
                "Celsius": {
                    "Min": -19,
                    "Max": 128
                },
                "Fahrenheit": {
                    "Min": -2,
                    "Max": 263
                }
            },
            "MaximumTemperatureLevel": {
                "Celsius": {
                    "Min": -18,
                    "Max": 129
                },
                "Fahrenheit": {
                    "Min": -1,
                    "Max": 264
                }
            }
        }
    ]
}
```

Chapter 138. Spot Temperature Reading

138.1. Description

The **spottemperaturereading** submenu provides the requested spot’s temperature information.

This submenu is applicable for temperature detection cameras.

NOTE	This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

138.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
spottemperaturereading&action=<value>[&<parameter>=<value>]
```

138.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the image preset.
	Channel	REQ, RES	<int>	Channel ID
	ScreenCoordinates	REQ	<string>	Request screen coordinates to display temperature information.
	Temperature	RES	<float>	Shows the spot’s current temperature.
	Unit	RES	<enum> Celsius, Fahrenheit	Shows the current temperature’s unit.
	ScreenResolution	REQ	<string>	Shows the current screen’s resolution. **Note** This parameter should be set with **ScreenCoordinates**.

138.4. Examples

138.4.1. Getting the requested spot’s temperature

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=spottemperaturereading&action=view&Channel=0&ScreenRe
solution=640x480&ScreenCoordinates=334,216
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SpotTemperatureReading": [
        {
            "Channel": 0,
            "Temperature": 30.5,
            "Unit": "Celsius"
        }
    ]
}
```

Chapter 139. Direction Indicator

139.1. Description

The **directionindicator** submenu provides the feature to set and show options related to each direction.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

139.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
directionindicator&action=<value>[&<parameter>=<value>]
```

139.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads settings for the direction-related settings.
	Channel	REQ, RES	<csv>	Channel ID
	Enable	REQ, RES	<bool> True, False	Used to request to enable or disable this feature. Enables or disables the direction indicating feature
	Direction	REQ, RES	<csv> Top, TopRight, Right, BottomRight, Bottom, BottomLeft, Left, TopLeft	Users can select specific directions that they want. Multiple directions are allowed to set.
set	Channel	REQ	<int>	Channel value to set.

Action	Parameters	Request/ Response	Type/ Value	Description
	Direction.#.AzimuthEnable	REQ,RES	<bool> True, False	Device shows an azimuth value in each direction if user sets this value to True.
	Direction.#.Show	REQ,RES	<bool> True, False	Device shows a direction string in each direction if user sets this value to True.
	Direction.#.Text	REQ,RES	<string>	User can set name for each direction using this parameter.
	Direction.#.PositionX	REQ,RES	<int>	User can specify the position of the information that will appear on screen.
	Direction.#.PositionY	REQ,RES	<int>	User can specify the position of the information that will appear on screen.
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows viewing the preview image of the configuration rather than saving the image configuration to camera.

139.4. Examples

139.4.1. Getting direction information

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=directionindicator&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: plain/text_
<Body>
```

```
Channel.0.Enable=False
Channel.0.FontSize=Small
Channel.0.Direction.Top.AzimuthEnable=False
Channel.0.Direction.Top.Show=False
Channel.0.Direction.Top.Text=
```

```
Channel.0.Direction.Top.PositionX=50
Channel.0.Direction.Top.PositionY=1
Channel.0.Direction.TopRight.AzimuthEnable=False
Channel.0.Direction.TopRight.Show=False
Channel.0.Direction.TopRight.Text=
Channel.0.Direction.TopRight.PositionX=100
Channel.0.Direction.TopRight.PositionY=1
Channel.0.Direction.Right.AzimuthEnable=False
Channel.0.Direction.Right.Show=False
Channel.0.Direction.Right.Text=
Channel.0.Direction.Right.PositionX=100
Channel.0.Direction.Right.PositionY=13
Channel.0.Direction.BottomRight.AzimuthEnable=False
Channel.0.Direction.BottomRight.Show=False
Channel.0.Direction.BottomRight.Text=
Channel.0.Direction.BottomRight.PositionX=100
Channel.0.Direction.BottomRight.PositionY=26
Channel.0.Direction.Bottom.AzimuthEnable=False
Channel.0.Direction.Bottom.Show=False
Channel.0.Direction.Bottom.Text=
Channel.0.Direction.Bottom.PositionX=50
Channel.0.Direction.Bottom.PositionY=26
Channel.0.Direction.BottomLeft.AzimuthEnable=False
Channel.0.Direction.BottomLeft.Show=False
Channel.0.Direction.BottomLeft.Text=
Channel.0.Direction.BottomLeft.PositionX=1
Channel.0.Direction.BottomLeft.PositionY=26
Channel.0.Direction.Left.AzimuthEnable=False
Channel.0.Direction.Left.Show=False
Channel.0.Direction.Left.Text=
Channel.0.Direction.Left.PositionX=1
Channel.0.Direction.Left.PositionY=13
Channel.0.Direction.TopLeft.AzimuthEnable=False
Channel.0.Direction.TopLeft.Show=False
Channel.0.Direction.TopLeft.Text=
Channel.0.Direction.TopLeft.PositionX=1
Channel.0.Direction.TopLeft.PositionY=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "DirectionIndicators": [
        {
            "Channel": 0,
            "Enable": false,
            "FontSize": "Small",
            "Directions": [
                {
                    "Direction": "Top",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 50,
                    "PositionY": 1
                },
                {
                    "Direction": "TopRight",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 100,
                    "PositionY": 1
                },
                {
                    "Direction": "Right",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 100,
                    "PositionY": 13
                },
                {
                    "Direction": "BottomRight",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 100,
```

```
                    "PositionY": 26
                },
                {
                    "Direction": "Bottom",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 50,
                    "PositionY": 26
                },
                {
                    "Direction": "BottomLeft",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 1,
                    "PositionY": 26
                },
                {
                    "Direction": "Left",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 1,
                    "PositionY": 13
                },
                {
                    "Direction": "TopLeft",
                    "AzimuthEnable": false,
                    "Show": false,
                    "Text": "",
                    "PositionX": 1,
                    "PositionY": 1
                }
            ]
        }
    ]
}
```

139.4.2. Setting information for the top direction

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=directionindicator&action=set&Channel=0&AzimuthEnable
=True&Show=True&Direction=Top&Text=North&PositionX=50&PositionY=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: plain/text
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 140. Noise Reduction

140.1. Description

The **noisereduct** submenu provides features related to image quality control for analog camera connected to Hybrid NVR.

NOTE This chapter applies to Hybrid NVR Only.

Access level

Action	NVR
view	USER

140.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
noisereduct&action=<value>[&<parameter>=<value>]
```

140.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	RES		Channel ID
set	Channel	REQ		Channel ID
	NoiseLevel	REQ,RES	<int>	Noise reduction level. The higher the level value, the more the noise reduction and the higher the image quality.
	HorizontalPos	REQ,RES	<int>	Horizontal position. It’s a feature for screen adjustment, not for noise reduction.
	VerticalPos	REQ,RES	<int>	Vertical position. It’s a feature for screen adjustment, not for noise reduction.
	HorizontalScale	REQ,RES	<int>	Horizontal scale. It’s a feature for screen adjustment, not for noise reduction.
	VerticalScale	REQ,RES	<int>	Vertical scale. It’s a feature for screen adjustment, not for noise reduction.

140.4. Examples

140.4.1. Getting each channel noise reduction value

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=noisereduct&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NoiseReduct": [
        {
            "Channel": 1,
            "NoiseLevel": 0,
            "HorizontalPos": 0,
            "VerticalPos": 0,
            "HorizontalScale": 0,
            "VerticalScale": 0
        }
    ]
}
```


140.4.2. Setting each channel noise reduction value

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=noisereduct&action=set&Channel=1&NoiseLevel=2
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 141. PTR Preset

141.1. Description

The **ptrpreset** submenu provides features that can configure settings for ptr presets.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
update	Admin
control	Admin

141.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
ptrpreset&action=<value>[&<parameter>=<value>]
```

141.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Name	RES	<string> Up to 12 characters	The predefined name of presets.
	Enable	RES	<bool> True, False	If this value is set to false, the preset position has not been set yet.
	Configurable	RES	<bool> True, False	User can change the preset position when this value is set to true; other presets have predefined values.
update	Preset	REQ	<int>	The number of preset to be saved
control	Preset	REQ	<int> 1~4	The number of preset to be moved
	PresetName	REQ	<string> initialize, 360, 200, user	The name of preset to be moved. These are predefined values and match with Preset numbers in sequence.

141.4. Examples

141.4.1. Getting the current camera information for PTZ preset (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=ptrpreset&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "PtrPresets": [
        {
            "Preset": 1,
            "Name": "initialize",
            "Enable": 1,
            "Configurable": "False"
        },
        {
            "Preset": 2,
            "Name": "360",
            "Enable": 1,
            "Configurable": "False"
        },
        {
            "Preset": 3,
            "Name": "200",
            "Enable": 0,
            "Configurable": "False"
        },
        {
            "Preset": 4,
            "Name": "user",
            "Enable": 1,
            "Configurable": "True"
```

```
        }
    ]
}
```

141.4.2. Moving to predefined preset (Preset 2 means 360 preset)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=ptrpreset&action=control&Preset=2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

141.4.3. Updating the current position to match the user preset

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=ptrpreset&action=update&Preset=4
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

Chapter 142. Stereo Calibration

142.1. Description

The **stereosensorcalibration** submenu provides commands to calibrate the coordinates of both sensors, which is performed by superimposing a thermal image over a non-thermal image based on the scale and adjustment of the offset.

NOTE This chapter is applicable to dual thermal cameras.

Access level

Action	Camera
view	Admin
set	Admin

142.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
stereosensorcalibration&action=<value>[&<parameter>=<value>]
```

142.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	HeightScale	RES	<float>	Height scale Height scale between normal and thermal images
	WidthScale	RES	<float>	Width scale Width scale between normal and thermal images
	ZoomScale	RES	<float>	Zoom scale Zoom scale between normal and thermal images
	CalibrationCompleted	RES	<bool> True, False	Calibration completed Shows whether calibration settings are performed or not

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ,RES	<int>	Channel ID
	CalibrationOffset	REQ,RES	<string>	Position offset of the thermal image The CalibrationOffset is expressed in <x,y>.

142.4. Examples

142.4.1. Getting the current calibration setting values from camera

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=stereosensorcalibration&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.HeightScale=3.44
Channel.1.WidthScale=3.44
Channel.1.ZoomScale=1.22
Channel.1.CalibrationCompleted=False
Channel.1.CalibrationOffset=0,0
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "StereoSensorCalibration": [
        {
            "Channel": 1,
```

```
            "HeightScale": 3.44,
            "WidthScale": 3.44,
            "ZoomScale": 1.22,
            "CalibrationCompleted": false,
            "CalibrationOffset": [
                {
                    "x": 0,
                    "y": 0
                }
            ]
        }
    ]
}
```


142.4.2. Setting calibration position offset

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=stereosensorcalibration&action=set&CalibrationOffset=
-1,2&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
```

```
}
```

Chapter 143. Radiometry Settings

143.1. Description

The **radiometrysettings** submenu configures radiometry settings in the camera.

NOTE

This chapter is applicable to dual thermal cameras.
Attribute to check for feature support:
"attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

143.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
radiometrysettings&action=<value>[&<parameter>=<value>]
```

143.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ,RES	<int>	Channel ID
	DistanceToObject	REQ,RES	<int>	Distance to the object for which the temperature is to be measured (in centimeters)
	EnableRelativeHumidity	REQ, RES	<bool> True, False	Whether to use the **RelativeHumidity** setting
	RelativeHumidity	REQ, RES	<float>	Relative humidity (in percentage)
	EnableAtmosphericTemperature	REQ, RES	<bool> True, False	Whether to use the **AtmosphericTemperature** setting
	AtmosphericTemperature	REQ, RES	<float>	Atmospheric temperature
	EnableTemperatureOffset	REQ, RES	<bool> True, False	Whether to use the **TemperatureOffset** setting

Action	Parameters	Request/ Response	Type/ Value	Description
	TemperatureOffset	REQ, RES	<float>	Offset for measured temperature Offset value to apply to the measured temperature value.

143.4. Examples

143.4.1. Getting current radiometry settings for Channel 1

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=radiometrysettings&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.DistanceToObject=300
Channel.1.EnableRelativeHumidity=True
Channel.1.RelativeHumidity=50.0
Channel.1.EnableAtmosphericTemperature=True
Channel.1.AtmosphericTemperature=23.0
Channel.1.EnableTemperatureOffset=True
Channel.1.TemperatureOffset=0.0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "RadiometrySettings": [
        {
            "Channel": 1,
            "DistanceToObject": 300,
```

```
            "EnableRelativeHumidity": true,
            "RelativeHumidity": 50,
            "EnableAtmosphericTemperature": true,
            "AtmosphericTemperature": 23,
            "EnableTemperatureOffset": true,
            "TemperatureOffset": 0
        }
    ]
}
```

143.4.2. Setting radiometry settings to use distance to object, relative humidity, and atmospheric temperature

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=radiometrysettings&action=set&Channel=1&DistanceToObject=350&EnableRelativeHumidity=True&RelativeHumidity=50&EnableAtmosphericTemperature=True&AtmosphericTemperature=25.5
```

TEXT RESPONSE

```
HTTP/1.0 200 OK

Content-type: text/plain

<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "Response": "Success"
```

```
}
```

Chapter 144. Radiometry Settings

144.1. Description

The **radiometrysettingsoptions** submenu provides information on radiometry settings.

> **NOTE**
> This chapter is applicable to dual thermal cameras.
> Attribute to check for feature support:
> "attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

144.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
radiometrysettings&action=<value>[&<parameter>=<value>]
```

144.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<csv>	Channel ID
	DefaultDistanceToObject	RES	<int>	Default value for DistanceToObject
	DefaultRelativeHumidity	RES	<float>	Default value for Relative humidity
	DefaultAtmosphericTemperature	RES	<float>	Default value for Atmospheric temperature
	Temperature.Celsius.Min	RES	<float>	Minimum Celsius temperature
	Temperature.Celsius.Max	RES	<float>	Maximum Celsius temperature
	Temperature.Fahrenheit.Min	RES	<float>	Minimum Fahrenheit temperature
	Temperature.Fahrenheit.Max	RES	<float>	Maximum Fahrenheit temperature
	TemperatureOffset.Celsius.Min	RES	<float>	Minimum Celsius temperature for temperatureoffset parameter

Action	Parameters	Request/ Response	Type/ Value	Description
	TemperatureOffset.Celsius.Max	RES	<float>	Maximum Celsius temperature for temperatureoffset parameter
	TemperatureOffset.Fahrenheit.Min	RES	<float>	Minimum Fahrenheit temperature for temperatureoffset parameter
	TemperatureOffset.Fahrenheit.Max	RES	<float>	Maximum Fahrenheit temperature for temperatureoffset parameter

144.4. Examples

144.4.1. Getting radiometry settings for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=radiometrysettingsoptions&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "RadiometrySettingsOptions": [
        {
            "Channel": 1,
            "DefaultDistanceToObject": 300,
            "DefaultRelativeHumidity": 50,
            "DefaultAtmosphericTemperature": 23,
            "Temperature": {
                "Celsius": {
                    "Min": 10,
                    "Max": 35
                },
                "Fahrenheit": {
                    "Min": 50,
```

```
                    "Max": 95
                }
            },
            "TemperatureOffset": {
                "Celsius": {
                    "Min": -3,
                    "Max": 3
                },
                "Fahrenheit": {
                    "Min": -37.4,
                    "Max": 37.4
                }
            }
        }
    ]
}
```

Chapter 145. BlackBody Config

145.1. Description

The **blackbodyconfig** submenu is used to configure the use of a black body device with the camera.

> **NOTE**
> This chapter is applicable to dual thermal cameras.
> Attribute to check for feature support:
> "attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

145.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
blackbodyconfig&action=<value>[&<parameter>=<value>]
```

145.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	CurrentBlackBodyTemperature	RES	<float>	Current temperature of BlackBody It shows the temperature of the set blackbody. If Enable is false, it is displayed as 0
set	Channel	REQ,RES	<int>	Channel ID
	Enable	REQ,RES	<bool> True, False	Enables or disables BlackBody
	Coordinates	REQ,RES	<string> Format=x1, y1,x2,y2	Coordinates of BlackBody position The coordinates are expressed in < x1,y1,x2,y2>
	EnableBlackBodyBeepAlarm	REQ, RES	<bool> True, False	Enables or disables BlackBody alarm
	BlackBodyTemperature	REQ, RES	<float>	Temperature of BlackBody

Action	Parameters	Request/ Response	Type/ Value	Description
	BlackBodyTemperatureOffset	REQ, RES	<float>	Temperature offset of BlackBody Blackbody temperature modification
	BlackBodyEmissivity	REQ, RES	<int>	Emissivity of BlackBody
	BlackBodyDistance	REQ, RES	<int>	Distance between BlackBody and Camera
	BlackBodyHeight	REQ, RES	<int>	Height of BlackBody
	CameraHeight	REQ, RES	<int>	Height of Camera
	CameraAngle	REQ, RES	<float>	Angle of Camera

145.4. Examples

145.4.1. Getting the current blackbody config for Channel 1

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=blackbodyconfig&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.Enable=True
Channel.1.CurrentBlackBodyTemperature=37.0
Channel.1.Coordinates=10,10,20,20
Channel.1.EnableBlackBodyBeepAlarm=False
Channel.1.BlackBodyTemperature=37.0
Channel.1.BlackBodyTemperatureOffset=1.0
Channel.1.BlackBodyEmissivity=96
Channel.1.BlackBodyDistance=300
Channel.1.BlackBodyHeight=180
Channel.1.CameraHeight=180
Channel.1.CameraAngle=0.0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "BlackBodyConfig": [
        {
            "Channel": 1,
            "Enable": true,
            "CurrentBlackBodyTemperature": 37,
            "Coordinates": [
                {
                    "x": 10,
                    "y": 10
                },
                {
                    "x": 20,
                    "y": 20
                }
            ],
            "EnableBlackBodyBeepAlarm": false,
            "BlackBodyTemperature": 37,
            "BlackBodyTemperatureOffset": 1,
            "BlackBodyEmissivity": 96,
            "BlackBodyDistance": 300,
            "BlackBodyHeight": 180,
            "CameraHeight": 180,
            "CameraAngle": 0
        }
    ]
}
```


145.4.2. Setting blackbody configurations for Channel 1

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=blackbodyconfig&action=set&Channel=1&Coordinates=10,1
0,20,20&EnableBlackBodyBeepAlarm=True&BlackBodyTemperature=30&BlackBodyTempe
```

```
ratureOffset=1.0&BlackBodyEmissivity=70&BlackBodyDistance=300&BlackBodyHeigh
t=100&CameraHeight=150&CameraAngle=3
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 146. BlackBody Config Options

146.1. Description

The **blackbodyconfigoptions** submenu provides information about blackbody settings.

> **NOTE**
> This chapter is applicable to dual thermal cameras.
> Attribute to check for feature support:
> "attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

146.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
balckbodyconfigoptions&action=<value>[&<parameter>=<value>]
```

146.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<csv>	Channel ID
	Coordinates.MinWidth	RES	<int>	Minimum size of Coordinates width
	Coordinates.MinHeight	RES	<int>	Minimum size of Coordinates height
	Coordinates.MaxWidth	RES	<int>	Maximum size of Coordinates width
	Coordinates.MaxHeight	RES	<int>	Maximum size of Coordinates height
	DefaultBlackBodyTemperature	RES	<float>	Default value for BlackBodyTemperature
	DefaultBlackBodyEmissivity	RES	<int>	Default value for BlackBodyEmissivity
	DefaultBlackBodyDistance	RES	<int>	Default value for BlackBodyDistance
	DefaultBlackBodyHeight	RES	<int>	Default value for BlackBodyHeight
	DefaultCameraHeight	RES	<int>	Default value for CameraHeight
	DefaultCameraAngle	RES	<float>	Default value for CameraAngle

Action	Parameters	Request/ Response	Type/ Value	Description
	Temperature.Celsius.Min	RES	<float>	Minimum Celsius temperature for BlackBodyTemperature
	Temperature.Celsius.Max	RES	<float>	Maximum Celsius temperature for BlackBodyTemperature
	Temperature.Fahrenheit.Min	RES	<float>	Minimum Fahrenheit temperature for BlackBodyTemperature
	Temperature.Fahrenheit.Max	RES	<float>	Maximum Fahrenheit temperature for BlackBodyTemperature
	TemperatureOffset.Celsius.Min	RES	<float>	Minimum Celsius temperature for BlackBodyTemperatureOffset
	TemperatureOffset.Celsius.Max	RES	<float>	Maximum Celsius temperature for BlackBodyTemperatureOffset
	TemperatureOffset.Fahrenheit.Min	RES	<float>	Minimum Fahrenheit temperature for BlackBodyTemperatureOffset
	TemperatureOffset.Fahrenheit.Max	RES	<float>	Maximum Fahrenheit temperature for BlackBodyTemperatureOffset

146.4. Examples

146.4.1. Getting blackbody config options for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=blackbodyconfigoptions&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "BlackBodyConfigOptions": [
        {
            "Channel": 1,
```

```
            "Coordinates": {
                "MinWidth": 0,
                "MinHeight": 0,
                "MaxWidth": 319,
                "MaxHeight": 239
            },
            "DefaultBlackBodyTemperature": 37,
            "DefaultBlackBodyEmissivity": 96,
            "DefaultBlackBodyDistance": 300,
            "DefaultBlackBodyHeight": 180,
            "DefaultCameraHeight": 180,
            "DefaultCameraAngle": 0,
            "Temperature": {
                "Celsius": {
                    "Min": 30,
                    "Max": 45
                },
                "Fahrenheit": {
                    "Min": 86,
                    "Max": 113
                }
            },
            "TemperatureOffset": {
                "Celsius": {
                    "Min": -1,
                    "Max": 1
                },
                "Fahrenheit": {
                    "Min": -1.8,
                    "Max": 1.8
                }
            }
        }
    ]
}
```

Chapter 147. Focus Preset

147.1. Description

The **focuspreset** submenu supports the configuration of presets per focus.

> **NOTE** Attribute to check for feature support: "attributes/Image/Support/[ChannelID]/FocusPreset"

Access level

Action	Camera
view	User
set	User
add	User
control/remove	User

147.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
focuspreset&action=<value>[&<parameter>=<value>]
```

147.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<csv>	Channel ID
set	Channel	REQ	<int>	Channel ID
	Enable	REQ	<bool> True, False	Enables presets per channel
add	Channel	REQ	<int>	Channel ID
	Preset	REQ,RES	<int>	Index of preset
	Name	REQ,RES	<string>	Name of preset
	IRShiftEnable	REQ,RES	<bool> True, False	Enables IR-based focus
	IRShiftValue	REQ,RES	<int> 0 to 200	IR focus shift value

Action	Parameters	Request/ Response	Type/ Value	Description
control/ remove	Channel	REQ	<int>	Channel ID
	Preset	REQ	<int>	Index of preset

147.4. Examples

147.4.1. Getting the current focus preset configuration

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focuspreset&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "FocusPresets": [
        {
            "Channel": 0,
            "Enable": true,
            "Presets": [
                {
                    "Preset": 1,
                    "Name": "test",
                    "IRShiftEnable": true,
                    "IRShiftValue": 100
                }
            ]
        }
    ]
}
```

147.4.2. Add the current focus setting to the new preset

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focuspreset&action=add&Preset=1&Name=test&IRShiftEnab
le=True&IRShiftValue=100
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "Response": "Success"
}
```

147.4.3. Control to move to a saved focus preset

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focuspreset&action=control&Preset=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "Response": "Success"
}
```

147.4.4. Remove a focus preset

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=focuspreset&action=remove&Preset=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 148. Thermal NUC

148.1. Description

The **thermalnuc** submenu used in scheduling the NUC (Non Uniformity Correction).

NOTE: Attribute to check for feature support: "attributes/Image/Support/[ChannelID]/ThermalNUC"

Access level

Action	Camera
view	Admin
set	Admin
control	Admin

148.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
thermalnuc&action=<value>[&<parameter>=<value>]
```

148.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<csv>	Channel ID
set	Channel	REQ	<int>	Channel ID
	DisplayEnable	REQ,RES	<bool> True,False	Enable OSD when NUC is in progress.
	DisplayPosition	REQ,RES	<enum> LowerLeft,LowerRight, UpperLeft, UpperRight	OSD display position.
	IntervalMode	REQ,RES	<enum> Auto,5min,15min,Schedule	Correction interval **Note** When the interval mode is set as **Schedule**, the user can configure a custom day and time to apply the correction.

Action	Parameters	Request/ Response	Type/ Value	Description
	<ddd>.Enable	REQ,RES	<bool> True, False	Day for which the schedule should be enabled. **Note** Here, <ddd> refers to the three letters of the weekday; SUN, MON, TUE, etc.
	<ddd>.<hourIndex>.Enable	REQ,RES	<bool> True, False	Hour of a particular day when the schedule should be enabled. **Note** Here, <hourIndex> refers to an hour index from 0 to 23
	<ddd>.<hourIndex>.Mode	REQ,RES	<enum> Auto, 5min, 15min	Correction interval mode.
control	Channel	REQ	<int>	Channel ID for which the NUC should be applied.

148.4. Examples

148.4.1. Getting the settings related to the current NUC schedule

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=thermalnuc&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "ThermalNUCSetting": [
        {
            "Channel": 1,
            "DisplayEnable": "False",
```

```
            "DisplayPosition": "LowerLeft",
            "IntervalMode": "Auto",
            "Schedule": {
                "SUN": [
                    {
                        "HourIndex": 0,
                        "Enable": false,
                        "Mode": "Auto"
                    },
                    {
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                        "Enable": false,
                        "Mode": "Auto"
                    },
                    {
                        "HourIndex": 2,
                        "Enable": false,
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                    },
                    {
                        "HourIndex": 3,
                        "Enable": false,
                        "Mode": "Auto"
                    },
                    {
                        "HourIndex": 4,
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                    {
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                    },
                    {
                        "HourIndex": 6,
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                    },
                    {
                        "HourIndex": 7,
```

```
                    "Enable": false,
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                },
                {
                    "HourIndex": 8,
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                },
                {
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                {
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                {
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                    "Mode": "Auto"
                },
                {
                    "HourIndex": 15,
                    "Enable": false,
```

```
                "Mode": "Auto"
            },
            {
                "HourIndex": 16,
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                "Mode": "Auto"
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            {
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            },
            {
                "HourIndex": 23,
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```

```
                }
            ],
            "MON": [
                {
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                {
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                {
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                },
                {
                    "HourIndex": 7,
                    "Enable": false,
```

```
                    "Mode": "Auto"
                },
                {
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                },
                {
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                    "Enable": false,
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                {
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```
                    },
                    {
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                    {
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                    }
```

```
                ],
                "TUE": [
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```

```
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                },
                {
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            ],
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            {
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            {
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```

```
                "Mode": "Auto"
            },
            {
                "HourIndex": 17,
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            },
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            },
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                "Mode": "Auto"
            },
            {
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                "Mode": "Auto"
            },
            {
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            {
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            }
        ],
        "SAT": [
            {
                "HourIndex": 0,
```

```
                        "Enable": false,
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                    },
                    {
                        "HourIndex": 1,
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                    {
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```

```
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```
                    },
                    {
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                    {
                        "HourIndex": 22,
                        "Enable": false,
                        "Mode": "Auto"
                    },
                    {
                        "HourIndex": 23,
                        "Enable": false,
                        "Mode": "Auto"
                    }
                ]
            }
        }
    ]
}
```

148.4.2. Configure the NUC schedule

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=thermalnuc&action=set&DisplayEnable=True&DisplayPosition=LowerLeft&IntervalMode=Schedule&SUN.0.Enable=True&SUN.Enable=True&SUN.0.Mode=Auto&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 149. LP Capture Setup

149.1. Description

The **lpcapturesetup** guides you to an optimized camera installation angle when using license plate recognition on the road. Set the camera's height and road distance and move the camera to see the guided values.

NOTE This chapter is applicable for cameras which have gyrosensor and support LPR feature.

Access level

Action	Camera
view	User
set	User
check	User

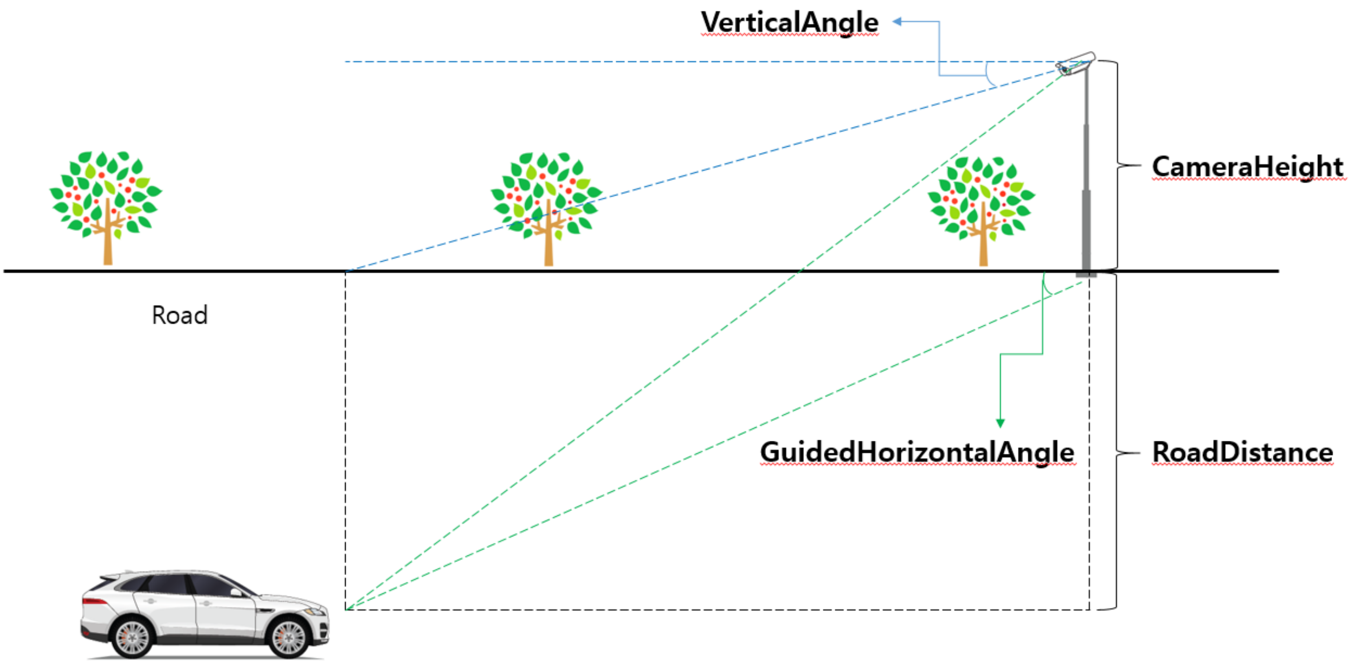
149.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
lpcapturesetup&action=<value>[&<parameter>=<value>]
```

149.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	GuidedHorizontalAngleMin	RES	<int>	Minimum recommended value of the horizontal angle at which the camera is looking at the vehicle
	GuidedHorizontalAngleMax	RES	<int>	Maximum recommended value of the horizontal angle at which the camera is looking at the vehicle
	VerticalAngleMax	RES	<int>	Maximum recommended value of camera vertical angle
	VerticalAngleMin	RES	<int>	Minimum recommended value of camera vertical angle
	RollAngleMax	RES	<int>	Maximum recommended value of camera rotation angle
	RollAngleMin	RES	<int>	Minimum recommended value of camera rotation angle

Action	Parameters	Request/ Response	Type/ Value	Description
set	CameraHeight	REQ,RES	<int>	Height(m) of camera
	RoadDistance	REQ,RES	<int>	Vertical distance(m) between car and camera. Check below picture for reference
check	VerticalAngle	RES	<int>	Vertical angle of the current camera position
	RollAngle	RES	<int>	Rotate angle of the current camera position
	GuidedHorizontalAngle	RES	<int>	The horizontal angle at which the camera should point at the vehicle. This value is calculated by CameraHeight, RoadDistance, and VerticalAngle. Check below picture for reference



149.4. Examples

149.4.1. Setting the camera height & RoadDistance

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=lpcapturesetup&action=set&CameraHeight=5&RoadDistance
=5
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


149.4.2. Getting current vertical angle, roll angle and guided horizontal angle

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=lpcapturesetup&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VerticalAngle": 20,
    "RollAngle": 5,
    "GuidedHorizontalAngle": 25,
}
```


149.4.3. Getting setting values & recommended values range

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=lpcapturesetup&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "CameraHeight": 5,
    "RoadDistance": 5,
    "GuidedHorizontalAngleMin": 0,
    "GuidedHorizontalAngleMax": 30,
    "VerticalAngleMin": 0,
    "VerticalAngleMax": 30,
    "RollAngleMin": 0,
    "RollAngleMax": 15,
}
```

Chapter 150. WiseAutoFocus settings

150.1. Description

The **wiseautofocus** submenu configures the wiseautofocus settings.

NOTE Attribute to check for feature support: "attributes/Image/Support/[ChannelID]/WiseAutoFocus"

Access level

Action	Camera
view	User
set	User

150.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
wiseautofocus&action=<value>[&<parameter>=<value>]
```

150.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel IDs
set	Channel	REQ,RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enable the wiseautofocus feature
	QuickZoom	REQ, RES	<bool> True,False	Enable te quick zoom feature
	FocusLevel	REQ,RES	<int> 1 to 100	Focus level value.

150.4. Exmaples

150.4.1. View the wiseautofocus settings

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=wiseautofocus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "WiseAutoFocus": [
        {
            "Channel": 0,
            "Enable": false,
            "QuickZoom": true,
            "FocusLevel": 50
        }
    ]
}
```


150.4.2. set the wiseautofocus settings

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=wiseautofocus&action=set&Channel=0&Enable=True&QuickZ
oom=True&FocusLevel=34
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 151. AutoFocusZone Setup

151.1. Description

The **autofocuszone** submenu used to setup the autofocuszone related settings.

NOTE Attribute to check for feature support:
"attributes/Image/Support/[ChannelID]/AutoFocusZone"
"attributes/Image/Limit/[ChannelID]/MaxAutoFocusZoneCount"

Access level

Action	Camera
view	Admin
set	Admin
add	Admin
update	Admin
control	Admin
remove	Admin
check	Admin

151.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
autofocuszone&action=<value>[&<parameter>=<value>]
```

151.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel Ids
	Channel	RES	<int>	Channel ID
	Enable	RES	<bool> True, False	If this is true, autofocuszone feature is enabled
	ZoneIndex.#.ZoneName	RES	<string>	Userconfigured zone name.
	ZoneIndex.#.GridSize.Width	RES	<int>	Width of the grid.
	ZoneIndex.#.GridSize.Height	RES	<int>	Height of the grid.

Action	Parameters	Request/ Response	Type/ Value	Description
	ZoneIndex.#.Cell.#.Enable	RES	<bool> True,False	Cell enable state.
set	Channel	REQ	<int>	Channel ID
	Enable	REQ	<bool> True,False	Enable or disable autofocus zone
	ZoneIndex.#.ZoneName	REQ	<string>	Configure zone name.
add	Channel	REQ	<int>	Channel ID
	ZoneIndex	REQ	<int>	Zone index to that needs to be added newly.
	ZoneName	REQ	<string>	Zonename string
	GridSize.Height	REQ	<int>	height of the grid
	GridSize.Width	REQ	<int>	width of the grid
	SelectedCells	REQ	<csv>	Selected cells to be added. Selected cell will be 1 and the other will be 0.
update	Channel	REQ	<int?	Channel ID
	ZoneIndex	REQ	<int>	Zone index to be updated
	FocusChangedCell	REQ	<int>	Index of cell for which refocus is needed.
	DisabledCell	REQ	<int>	Index of cell for which focus needs to be disabled.
control	Channel	REQ	<int>	Channel ID.
	Mode	REQ	<enum> Move, StopScanning, NoZoomMoveToZone, StartEdit, RenewEdit, ExitEdit	Control Mode
	ZoneIndex	REQ	<int>	Zone index
	Cell	REQ	<int>	Cell index
	EditTimeoutSecs	RES	<int>	Timeout duration for the edit session, when mode is set to StartEdit or RenewEdit.
remove	Channel	REQ	<int>	Channel ID.

Action	Parameters	Request/ Response	Type/ Value	Description
	ZoneIndex	REQ	<int>	Zone index to be removed.
check	Channel	REQ	<int>	Channel ID
	Type	REQ	<enum> GridSize,ScanStatus	Can check the gridsize or ScanStatus, based on this parameter.
	GridSize.MaxWidth	RES	<int>	Max width of the grid.
	GridSize.MaxHeight	RES	<int>	Max height of the grid.
	GridSize.MinWidth	RES	<int>	Min width of the grid.
	GridSize.MinHeight	RES	<int>	Min height of the grid.
	GridSize.ZoomRatio	RES	<int>	Current zoom ratio.
	ScanStatus.SaveStatus	RES	<enum> None, InProgress, Complete	Scan status after adding or updating zone.
	ScanStatus.TotalCellCount	RES	<int>	Shows total cells count
	ScanStatus.EstimatedTimeInSecs	RES	<int>	Shows estimated times for adding or updating zone in seconds

151.4. Examples

151.4.1. Enable the autofocuszone

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=set&Channel=0&Enable=True
```

151.4.2. Check the supported gridsize information

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=check&Channel=0&Type=GridSize
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "GridSize": {
        "MaxWidth": 20,
        "MaxHeight": 20,
        "MinWidth": 1,
        "MinHeight": 1,
        "ZoomRatio": 20
    }
}
```

151.4.3. Add zone

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=add&Channel=0&ZoneIndex=1&ZoneNa
me=test&GridSize.Width=4&GridSize.Height=4&SelectedCells=1,1,1,1,0,1,1,1,1,0
,1,1,1,1,1,1
```

151.4.4. Update Cell [Disabling]

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=update&Channel=0&ZoneIndex=1&Dis
abledCell=2
```

151.4.5. Update cell [Refocus]

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=update&Channel=0&ZoneIndex=1&Foc
usChangedCell=2
```

151.4.6. Move to a cell

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=Move&Zone
Index=1&Cell=3
```

151.4.7. Move to a zone

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=Move&Zone
Index=1
```

151.4.8. Move to Zone without Zoom

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=NoZoomMov
eToZone&ZoneIndex=1
```

151.4.9. Check the scan status

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=check&Channel=0&Type=ScanStatus
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ScanStatus": {
        "SaveStatus": "Complete",
        "TotalCellCount": 0,
        "EstimatedTimeInSecs": 0
    }
}
```


151.4.10. Stop Scanning that is in progress

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Mode=StopScanning&Channe
l=0
```

151.4.11. Enter the focus scan edit mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=StartEdit
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EditTimeoutSecs": 60
}
```

151.4.12. Renew focus scan edit mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=RenewEdit
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EditTimeoutSecs": 60
}
```

151.4.13. Exit focus scan edit mode

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=control&Channel=0&Mode=ExitEdit
```

151.4.14. Remove zone

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=autofocuszone&action=remove&ZoneIndex=1
```

Chapter 152. White LED

152.1. Description

The **whiteled** submenu configures White LED settings. White LED creates brighter and clearer images in low illumination situations.

NOTE This chapter applies to network cameras only.

Attribute to check for Feature Support: "attributes/Image/Support/WhiteLED"

Access level

Action	Camera
view	Admin
set	Admin

152.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
whiteled&action=<value>[&<parameter>=<value>]
```

152.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads White LED settings.
	Channel	REQ	<csv>	Channel ID
	ImagePresetMode.#.Mode	RES	<enum> SupplementLight, LightAlarm, Off	White LED mode • Off: disable White LED • SupplementLight: Use White LED as supplement light • LightAlarm: Use White LED as light alarm

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.SupplementLightMode	RES	<enum> Auto, Schedule	Supplement light mode • Auto: Controls the White LED automatically • Schedule: Controls the White LED according to the set schedule **ImagePresetMode.#.SupplementLightMode** is valid only when **ImagePresetMode.#.Mode** is set to **SupplementLight**.
	ImagePresetMode.#.SupplementLightDNSwitchTime	RES	<enum> 5s,7s,10s,15s,20s,30s,40s,60s	For the led light to light up or off, the ambient change must last longer than this duration, applicable only when **SupplementLightMode** is configured as **Auto**.
	ImagePresetMode.#.SupplementLightDurationMode	RES	<enum> VeryHigh, High, Normal, Low, VeryLow, Custom	Used to configure the sensitivity mode, for example if configured as **VeryHigh** even if ambient light level increases little lights will be off, applicable only when **Mode** is configured as **SupplementLight** and **SupplementLightMode** is configured as **Auto**.
	ImagePresetMode.#.SupplementLightColorToBWThreshold	RES	<int> 1~100	To manually configure the sensitiviy of ambient light, applicable only when **SupplementLightDurationMode** is configured as **Custom**, this value would be always greater than the SupplementLightBWToColorThreshold value.
	ImagePresetMode.#.SupplementLightBWToColorThreshold	RES	<int> 1~100	To manually configure the sensitiviy of ambient light, applicable only when **SupplementLightDurationMode** is configured as **Custom**, this value would be always less than the SupplementLightColorToBWThreshold value.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePresetMode.#.LightAlarmMode	RES	<enum> Constant, Flashing	Light alarm mode • Constant: An LED light constantly emits light during a time period set in [Dwell time] if an event occurs. • Flashing: An LED light flashes with an interval of 1.5 seconds during a time period set in [Dwell time] if an event occurs. **ImagePresetMode.#.LightAlarmMode** is valid only when **ImagePresetMode.#.Mode** is set to **LightAlarm**.
	ImagePresetMode.#.DwellTime	RES	<int>	Set the dwell time of an LED light **ImagePresetMode.#.DwellTime** is valid only when **ImagePresetMode.#.Mode** is set to **LightAlarm**.
	ImagePresetMode.#.Level	RES	<int>	White LED brightness level
	ImagePresetMode.#.Schedule.EveryDay.FromTo	RES	<int>	Schedule for White LED **ImagePresetMode.#.Schedule.EveryDay.FromTo** is valid only when **ImagePresetMode.#.SupplementLightMode** is set to **Schedule**.
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop	Image preview mode Allows the user to view a preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped, and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied.
	Mode	REQ	<enum> SupplementLight, LightAlarm, Off	White LED mode • Off: disable White LED • SupplementLight: Use White LED as supplement light • LightAlarm: Use White LED as light alarm
	SupplementLightMode	REQ	<enum> Auto, Schedule	Supplement light mode • Auto: Controls the White LED automatically • Schedule: Controls the White LED according to the set schedule **SupplementLightMode** is valid only when **Mode** is set to **SupplementLight**.
	SupplementLightDNSwitchTime	REQ	<enum> 5s,7s,10s,15s,20s,30s,40s,60s	For the led light to light up or off, the ambient change must last longer than this duration, applicable only when **SupplementLightMode** is configured as **Auto**.

Action	Parameters	Request/ Response	Type/ Value	Description
	SupplementLightDurationMode	REQ	<enum> VeryHigh, High, Normal, Low, VeryLow, Custom	Used to configure the sensitivity mode, for example if configured as **VeryHigh** even if ambient light level increases little lights will be off, applicable only when **Mode** is configured as **SupplementLight** and **SupplementLightMode** is configured as **Auto**.
	SupplementLightColorToBWThreshold	REQ	<int> 1~100	To manually configure the sensitiviy of ambient light, applicable only when **SupplementLightDurationMode** is configured as **Custom**, this value would be always greater than the SupplementLightBWToColorThreshold value.
	SupplementLightBWToColorThreshold	REQ	<int> 1~100	To manually configure the sensitiviy of ambient light, applicable only when **SupplementLightDurationMode** is configured as **Custom**, this value would be always less than the SupplementLightColorToBWThreshold value.
	LightAlarmMode	REQ	<enum> Constant, Flashing	Light alarm mode • Constant: An LED light constantly emits light during a time period set in [Dwell time] if an event occurs. • Flashing: An LED light flashes with an interval of 1.5 seconds during a time period set in [Dwell time] if an event occurs. **LightAlarmMode** is valid only when **Mode** is set to **LightAlarm**.
	DwellTime	REQ	<int>	Set the dwell time of an LED light **DwellTime** is valid only when **Mode** is set to **LightAlarm**.
	Level	REQ	<int>	White LED brightness level

Action	Parameters	Request/ Response	Type/ Value	Description
	Schedule.EveryDay.FromTo	REQ	<int>	Schedule for White LED **Schedule.EveryDay.FromTo** is valid only when **SupplementLightMode** is set to **Schedule**.
	ImagePresetMode	REQ	<enum> UserPreset1, UserPreset2,OutdoorDaytime,OutdoorNightime,IndoorBacklight,IndoorBrightScene,NumberPlate,Vivid	Image preset mode to which the setting is applicable.

152.4. Examples

152.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=whiteled&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=SupplementLight
Channel.0.Level=5
Channel.0.SupplementLightMode=Auto
Channel.0.SupplementLightDNSwitchTime=5s
Channel.0.SupplementLightDurationMode=Custom
Channel.0.SupplementLightColorToBWThreshold=70
Channel.0.SupplementLightBWToColorThreshold=22
Channel.0.Schedule.EveryDay.FromTo=00:00-00:00
Channel.0.LightAlarmMode=Constant
```

```
Channel.0.DwellTime=10
Channel.0.FocusFeedbackEnable=false
Channel.0.ImagePresetMode.UserPreset1.Mode=SupplementLight
Channel.0.ImagePresetMode.UserPreset1.Level=5
Channel.0.ImagePresetMode.UserPreset1.SupplementLightMode=Auto
Channel.0.ImagePresetMode.UserPreset1.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.UserPreset1.SupplementLightDurationMode=Custom
Channel.0.ImagePresetMode.UserPreset1.SupplementLightColorToBWThreshold=70
Channel.0.ImagePresetMode.UserPreset1.SupplementLightBWToColorThreshold=22
Channel.0.ImagePresetMode.UserPreset1.Schedule.EveryDay.FromTo=00:00-00:00
Channel.0.ImagePresetMode.UserPreset1.LightAlarmMode=Constant
Channel.0.ImagePresetMode.UserPreset1.DwellTime=10
Channel.0.ImagePresetMode.UserPreset2.Mode=SupplementLight
Channel.0.ImagePresetMode.UserPreset2.Level=5
Channel.0.ImagePresetMode.UserPreset2.SupplementLightMode=Auto
Channel.0.ImagePresetMode.UserPreset2.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.UserPreset2.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.UserPreset2.SupplementLightColorToBWThreshold=70
Channel.0.ImagePresetMode.UserPreset2.SupplementLightBWToColorThreshold=30
Channel.0.ImagePresetMode.UserPreset2.Schedule.EveryDay.FromTo=00:00-00:00
Channel.0.ImagePresetMode.UserPreset2.LightAlarmMode=Constant
Channel.0.ImagePresetMode.UserPreset2.DwellTime=10
Channel.0.ImagePresetMode.OutdoorDaytime.Mode=SupplementLight
Channel.0.ImagePresetMode.OutdoorDaytime.Level=5
Channel.0.ImagePresetMode.OutdoorDaytime.SupplementLightMode=Auto
Channel.0.ImagePresetMode.OutdoorDaytime.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.OutdoorDaytime.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.OutdoorDaytime.SupplementLightColorToBWThreshold=7
0
Channel.0.ImagePresetMode.OutdoorDaytime.SupplementLightBWToColorThreshold=3
0
Channel.0.ImagePresetMode.OutdoorDaytime.Schedule.EveryDay.FromTo=00:00-
00:00
Channel.0.ImagePresetMode.OutdoorDaytime.LightAlarmMode=Constant
Channel.0.ImagePresetMode.OutdoorDaytime.DwellTime=10
Channel.0.ImagePresetMode.OutdoorNightime.Mode=SupplementLight
Channel.0.ImagePresetMode.OutdoorNightime.Level=5
Channel.0.ImagePresetMode.OutdoorNightime.SupplementLightMode=Auto
Channel.0.ImagePresetMode.OutdoorNightime.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.OutdoorNightime.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.OutdoorNightime.SupplementLightColorToBWThreshold=
```

```
70
Channel.0.ImagePresetMode.OutdoorNightime.SupplementLightBWToColorThreshold=
30
Channel.0.ImagePresetMode.OutdoorNightime.Schedule.EveryDay.FromTo=00:00-
00:00
Channel.0.ImagePresetMode.OutdoorNightime.LightAlarmMode=Constant
Channel.0.ImagePresetMode.OutdoorNightime.DwellTime=10
Channel.0.ImagePresetMode.IndoorBacklight.Mode=SupplementLight
Channel.0.ImagePresetMode.IndoorBacklight.Level=5
Channel.0.ImagePresetMode.IndoorBacklight.SupplementLightMode=Auto
Channel.0.ImagePresetMode.IndoorBacklight.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.IndoorBacklight.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.IndoorBacklight.SupplementLightColorToBWThreshold=
70
Channel.0.ImagePresetMode.IndoorBacklight.SupplementLightBWToColorThreshold=
30
Channel.0.ImagePresetMode.IndoorBacklight.Schedule.EveryDay.FromTo=00:00-
00:00
Channel.0.ImagePresetMode.IndoorBacklight.LightAlarmMode=Constant
Channel.0.ImagePresetMode.IndoorBacklight.DwellTime=10
Channel.0.ImagePresetMode.IndoorBrightScene.Mode=SupplementLight
Channel.0.ImagePresetMode.IndoorBrightScene.Level=5
Channel.0.ImagePresetMode.IndoorBrightScene.SupplementLightMode=Auto
Channel.0.ImagePresetMode.IndoorBrightScene.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.IndoorBrightScene.SupplementLightDurationMode=Norm
al
Channel.0.ImagePresetMode.IndoorBrightScene.SupplementLightColorToBWThreshol
d=70
Channel.0.ImagePresetMode.IndoorBrightScene.SupplementLightBWToColorThreshol
d=30
Channel.0.ImagePresetMode.IndoorBrightScene.Schedule.EveryDay.FromTo=00:00-
00:00
Channel.0.ImagePresetMode.IndoorBrightScene.LightAlarmMode=Constant
Channel.0.ImagePresetMode.IndoorBrightScene.DwellTime=10
Channel.0.ImagePresetMode.NumberPlate.Mode=SupplementLight
Channel.0.ImagePresetMode.NumberPlate.Level=5
Channel.0.ImagePresetMode.NumberPlate.SupplementLightMode=Auto
Channel.0.ImagePresetMode.NumberPlate.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.NumberPlate.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.NumberPlate.SupplementLightColorToBWThreshold=70
Channel.0.ImagePresetMode.NumberPlate.SupplementLightBWToColorThreshold=30
```

```
Channel.0.ImagePresetMode.NumberPlate.Schedule.EveryDay.FromTo=00:00-00:00
Channel.0.ImagePresetMode.NumberPlate.LightAlarmMode=Constant
Channel.0.ImagePresetMode.NumberPlate.DwellTime=10
Channel.0.ImagePresetMode.Vivid.Mode=SupplementLight
Channel.0.ImagePresetMode.Vivid.Level=5
Channel.0.ImagePresetMode.Vivid.SupplementLightMode=Auto
Channel.0.ImagePresetMode.Vivid.SupplementLightDNSwitchTime=5s
Channel.0.ImagePresetMode.Vivid.SupplementLightDurationMode=Normal
Channel.0.ImagePresetMode.Vivid.SupplementLightColorToBWThreshold=70
Channel.0.ImagePresetMode.Vivid.SupplementLightBWToColorThreshold=30
Channel.0.ImagePresetMode.Vivid.Schedule.EveryDay.FromTo=00:00-00:00
Channel.0.ImagePresetMode.Vivid.LightAlarmMode=Constant
Channel.0.ImagePresetMode.Vivid.DwellTime=10
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "WhiteLed": [
        {
            "Channel": 0,
            "Mode": "SupplementLight",
            "Level": 5,
            "SupplementLightMode": "Auto",
            "SupplementLightDNSwitchTime": "5s",
            "SupplementLightDurationMode": "Custom",
            "SupplementLightColorToBWThreshold": 70,
            "SupplementLightBWToColorThreshold": 22,
            "Schedule": {
                "EveryDay": {
                    "FromTo": "00:00-00:00"
                }
            },
            "LightAlarmMode": "Constant",
            "DwellTime": 10,
            "FocusFeedbackEnable": false,
```

```
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Custom",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 22,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "UserPreset2",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "OutdoorDaytime",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
```

```
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "OutdoorNightime",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "IndoorBacklight",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
```

```
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "IndoorBrightScene",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "Mode": "SupplementLight",
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                },
                {
                    "ImagePresetMode": "Vivid",
                    "Mode": "SupplementLight",
```

```
                    "Level": 5,
                    "SupplementLightMode": "Auto",
                    "SupplementLightDNSwitchTime": "5s",
                    "SupplementLightDurationMode": "Normal",
                    "SupplementLightColorToBWThreshold": 70,
                    "SupplementLightBWToColorThreshold": 30,
                    "Schedule": {
                        "EveryDay": {
                            "FromTo": "00:00-00:00"
                        }
                    },
                    "LightAlarmMode": "Constant",
                    "DwellTime": 10
                }
            ]
        }
    ]
}
```

152.4.2. Setting the LED mode of UserPreset1 to supplement light and set schedule from 18:00 to 22:00

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whiteled&action=set&Mode=SupplementLight&SupplementLi
ghtMode=Schedule&Schedule.EveryDay.FromTo=18:00-
22:00&ImagePresetMode=UserPreset1
```

152.4.3. Setting the LED mode of UserPreset1 to light alarm and set dwell time to 10

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=whiteled&action=set&Mode=LightAlarm&DwellTime=10&Imag
ePresetMode=UserPreset1
```

152.4.4. Setting the LED mode of current preset to off

REQUEST

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=whiteled&action=set&Mode=Off
```

152.4.5. Configure Supplement light mode with custom sensitivity

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=dynamicprivacymask&action=set&Channel=0&Mode=Suppleme
ntLight&SupplementLightDNSwitchTime=5s&SupplementLightDurationMode=Custom&Su
pplementLightBWToColorThreshold=22&SupplementLightColorToBWThreshold=82
```

Chapter 153. Dynamic Prvacy Mask

153.1. Description

The **dynamicprivacymask** submenu configures dynamic privacy mask settings. dynamic privacy mask can do image masking on moving objects.

NOTE This chapter applies to network cameras only.

You can configure which profile to be used in masking by configuring IsDPMProfile in videoprofile submenu in media cgi.

Access level

Action	Camera
view	User
set	User

153.2. Syntax

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=
dynamicprivacymask&action=<value>[&<parameter>=<value>]
```

153.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads White LED settings.
set	Channel	REQ, RES	<int>	Channel ID
	ObjectTypes	REQ, RES	<csv> Vehicle, Person	Select the object to mask
	Enable	REQ, RES	<bool>	Enable/Disable dynamic privacy masking feature

153.4. Examples

153.4.1. Getting the current information of Channel 0

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/image.cgi?msubmenu=dynamicprivacymask&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json

<Body>
```


```
{
    "DynamicPrivacyMask": [
        {
            "Channel": 0,
            "Enable": true,
            "ObjectTypes": ["Person", "Vehicle"]
        }
    ]
}
```


153.4.2. Setting dynamic privacy mask Enable with Person, Vehicle objects

REQUEST

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=dynamicprivacymask&action=set&&Channel=0&Enable=True&
ObjectTypes=Person,Vehicle
```

PTZ

SUNAPI

v2.6.8
2026-04-09

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Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 154. Overview

154.1. Description

SUNAPI allows control and configuration of the PTZ (pan, tilt, zoom) functionality of video surveillance devices. PTZ functionality is covered by two CGIs: **ptzcontrol.cgi** and **ptzconfig.cgi**.

The following submenus are used to control PTZ functionality:

- **absolute**: Moves the camera to the absolute coordinates.
- **relative**: Moves the camera to the new position relative to the current position.
- **continuous**: Controls continuous PTZ operation.
- **query**: Requests camera’s current position information.
- **preset**: Recalls and moves to the specified preset.
- **swing**: Starts and stops a Swing operation, moving between two preset points.
- **group**: Starts and stops a Group operation, grouping multiple presets and calling them in sequence.
- **tour**: Starts and stops a Tour operation, calling groups of presets in sequence.
- **trace**: Starts and stops a Tracking operation.
- **home**: Moves the camera to Home position.
- **areazoom**: Defines the zoom area.
- **stop**: Stops a PTZ movement.
- **move**: Controls PTZ movement.
- **aux**: Controls the auxiliary PTZ operations.
- **digitalautotracking**: Starts or stops digital auto tracking if a profile is configured as a DPTZ profile.
- **rs485Command**: Sends custom RS-485 commands to devices in the format of hex.
- **osdmenu**: Controls the Onscreen Display (OSD) menu for analog cameras.
- **digitalrtz**: Moves the fisheye camera position based on the rotation angle.
- **supportedptzactions**: Provides information about supported submenus and action lists for each view mode.
- **movestatus**: Gets the current moving status of PTZ camera.

The following submenus are used to configure PTZ settings:

- **swing**: Configures the Swing settings.
- **group**: Configures the Group settings.
- **tour**: Configures the Tour settings.
- **trace**: Configures the Trace settings.
- **autorun**: Configures the Auto run settings.

- **home**: Sets the current position to the Home position.
- **preset**: Specifies the camera preset number and name.
- **presetimageconfig**: Configures the camera settings according to the selected preset.
- **presetvideoanalysis**: Configures the video analysis settings of the selected preset.
- **presetvideoanalysis2**: Same as presetvideoanalysis. However, it is possible to configure the parameters for each area/ROI.
- **ptzsettings**: Enables or disables auto flip and configures digital zoom limits.
- **ptlimits**: Enables or disables pan and tilt limits.
- **digitalautotracking**: Sets the object type to track. This is for AI cameras.
- **panzeroposition**: Configures the current pan position.
- **ptzprotocol**: Specifies the PTZ operation protocol.
- **ptzmode**: Configures the PTZ mode settings.
- **ptcorrection**: Configures the default position for Pan and Tilt.
- **exclusiveptzcontrol**: Configures the exclusive access authority for the PTZ features.
- **presetobjectdetection**: Configures the object detection based on selected preset.

Chapter 155. PTZ Control

155.1. Absolute Position Control

155.1.1. Description

The **absolute** submenu of **ptzcontrol.cgi** controls an absolute PTZ operation that moves the camera to the specified position.

> **NOTE**
>
> This chapter applies to network cameras only. To find out whether Absolute Pan/Tilt/Zoom is supported by the device or not, refer to the Attributes/PTZSupport/Support/Absolute.Pan, Absolute.Tilt and Absolute.Zoom attributes, respectively, in the device attributes section.

Access level

Action	Camera
control	Suser

155.1.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
absolute&action=control[&<parameter>=<value>]
```

155.1.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Pan	REQ	<float>	Panning, left/right movement
	Tilt	REQ	<float>	Tilting, left/right movement • Negative values: Moves downward. • Positive values: Moves upward
	Zoom	REQ	<float>	Zoom in/out **Zoom** and **ZoomPulse** cannot be sent together.

Action	Parameters	Request/ Response	Type/ Value	Description
	ZoomPulse	REQ	<int>	Controls the zoom by segmenting its phases from 1 to 9999(Normalized value). **Zoom** and **ZoomPulse** cannot be sent together.
	ActualZoomPulse	REQ	<int>	Controls the zoom by segmenting its phases; the range can be obtained from the attributes xml. (Actual hardware value) This parameter cannot be sent with **Zoom** or **ZoomPulse**.
	ViewModeIndex	REQ	<int>	View mode of index number
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	RotateInPlace	REQ	<float>	Based on center of video, tilt horizon of video.

155.1.4. Examples

155.1.4.1. Moving the camera to the right in 90 degrees

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=absolute&action=control&Pan=90
```

155.1.4.2. Setting the zoom to 30

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=absolute&action=control&Zoom=30
```

155.2. Relative Position Control

155.2.1. Description

The **relative** submenu of **ptzcontrol.cgi** controls a PTZ operation that moves the camera to the new position relative to the current position.

NOTE	This chapter applies to network cameras only. To find out whether Relative Pan/Tilt/Zoom is supported by the device or not, refer to the Attributes/PTZSupport/Support/Relative.Pan, Relative.Tilt and Relative.Zoom attributes, respectively, in the device attributes section.

Access level

Action	Camera
control	Suser

155.2.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
relative&action=control[&<parameter>=<value>]
```

155.2.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Pan	REQ	<float>	Panning, left/right movement • Negative values: Moves to the left • Positive values: Moves to the right.
	Tilt	REQ	<float>	Tilting, left/right movement • Negative values: Moves downward. Positive values: Moves upward
	Zoom	REQ	<float>	Zoom in/out **Zoom** and **ZoomPulse** cannot be sent together.
	ZoomPulse	REQ	<int>	Controls the zoom by segmenting its phases from -9999 to 9999. **Zoom** and **ZoomPulse** cannot be sent together.
	RotateInPlace	REQ	<float>	Based on center of video, tilt horizon of video.

Action	Parameters	Request/ Response	Type/ Value	Description
	ViewModeIndex	REQ	<int>	View mode of index number
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.

155.2.4. Examples

155.2.4.1. Moving camera to the left in 90 degrees on the basis of the current position

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=relative&action=control&Pan=-90
```

155.2.4.2. Moving camera to the upward in 45 degrees on the basis of the current position

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=relative&action=control&Tilt=45
```

155.3. Continuous PTZ Operation Control

155.3.1. Description

The **continuous** submenu of **ptzcontrol.cgi** controls continuous PTZ operation. Once a command is entered, the camera continues to move until requested to stop.

NOTE	To find out whether Continuous Pan/Tilt/Zoom/Focus/Iris is supported by the device or not, refer to the Attributes/PTZSupport/Support/Continuous.Pan, Continuous.Tilt, Continuous.Zoom, Continuous.Focus and

Continuous.Iris attributes, respectively, in the device attributes section.

Access level

Action	Camera	NVR	Encoder	Decoder
control	Suser	User	Suser	User

155.3.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
continuous&action=control[&<parameter>=<value>]
```

155.3.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Pan	REQ	<int>	Pans at the specified speed • Negative values: Moves to the left • Positive values: Moves to the right.
	Tilt	REQ	<int>	Tilts at the specified speed • Negative values: Moves downward. • Positive values: Moves upward
	Zoom	REQ	<int>	Zooms in/out
	NormalizedSpeed	REQ	<bool> True, False	Enables or disables the normalized speed range for Pan, Tilt, and Zoom If **NormalizedSpeed** is not sent, or set as False, the Pan, Tilt, and Zoom speed range will be device dependent values; If **NormalizedSpeed** is set as True, the speed values for Pan, Tilt, and Zoom will be in the range of -100 to 100. **NormalizedSpeed** must be sent together with **Pan**, **Tilt**, or **Zoom**. CAMERA ONLY
	RotateInPlace	REQ	<float>	Based on center of video, tilt horizon of video.
	Focus	REQ	<enum> Near, Far, Stop	Focus control This parameter cannot be sent with any other parameters.
	Iris	REQ	<enum> Open, Close, Stop	Iris control NVR ONLY
	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Duration	REQ	<int>	Time (seconds) to continue the requested operation
	ViewModeType	REQ	<enum> Panorama, DoublePanorama, QuadView, QuadView.#	Selecting particular view mode to do DPTZ All profiles with same view mode will move in sync
	SubViewIndex	REQ	<int>	Used to select the tile for PTZ in Quad view mode. This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY
	IgnoreIfBusy	REQ	<bool>	Ignores the command if the ptz controller is busy. If value is not passed, then this field is assumed to be false. CAMERA ONLY

155.3.4. Examples

155.3.4.1. Panning the camera

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Pan=5
```

155.3.4.2. Zooming in with the camera

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Zoom=3
```

155.3.4.3. Tilting the camera for 6 seconds

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Tilt=5&Duration=6
```

155.4. Requesting Camera’s Position Information

155.4.1. Description

The **query** submenu of **ptzcontrol.cgi** requests camera’s current position and zooming information.

NOTE This chapter applies to network cameras only.

To find out whether the camera can provide its Pan/Tilt/Zoom information by query submenu, refer to the Attributes/PTZSupport/Support/Query.Pan, Query.Tilt and Query.Zoom attributes, respectively, in the device attributes section.

Access level

Action	Camera
view	Suser

155.4.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
query&action=view[&<parameter>=<value>]
```

155.4.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID
	Query	REQ	<csv> Rotate, Pan, Tilt, Zoom, ActualZoomPulse, RotateInPlace, SubregionViewCoordinates	Current position of Pan, Tilt and (Zoom or ActualZoomPulse) (+ Rotate in case of Fisheye models) **Note** **Query** must be sent together with the **view** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	Rotate	RES	<float>	Current Rotate angle FISHEYE ONLY
	Pan	RES	<float>	Current Pan position
	Tilt	RES	<float>	Current Tilt position
	Zoom	RES	<float>	Lens Magnification
	ZoomPulse	RES	<int>	Controls the zoom by segmenting its phases from 1 to 9999. (Normalized values)
	SubregionViewCoordinates	RES	<string>	Shows subregions view's coordinates information
	RotateInPlace	RES	<float>	Shows Rotate degree information in Place
	ActualZoomPulse	RES	<int>	Controls the zoom by segmenting its phases; the range can be obtained from attributes xml. (Actual value)
	ViewModeIndex	REQ	<int>	View mode of index number
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.

155.4.4. Examples

155.4.4.1. Getting the position information of a camera

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=query&action=view&Channel=0&Query=Pan,Tilt,Zoom
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Pan=180.00
Tilt=25.00
```

```
Zoom=1.00
ZoomPulse=1789
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Query": [
        {
            "Channel": 0,
            "Pan": 180,
            "Tilt": 25,
            "Zoom": 1,
            "ZoomPulse": 1789
        }
    ]
}
```


155.4.4.2. Getting the zoom information

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=query&action=view&Channel=0&Query=Zoom
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Zoom=11.00
ZoomPulse=1789
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```

```
{
    "Query": [
        {
            "Channel": 0,
            "Zoom": 11,
            "ZoomPulse": 1789
        }
    ]
}
```

155.5. Moving to Preset Position

155.5.1. Description

The **preset** submenu of **ptzcontrol.cgi** controls the camera to move to the specified preset.

> **NOTE**
> To find out whether preset functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Preset attribute in the device attributes section.

Access level

Action	Camera	NVR	Encoder	Decoder
control	Suser	User	Suser	User

155.5.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
preset&action=control[&<parameter>=<value>]
```

155.5.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Preset	REQ	<int>	Preset number The number of presets supported is dependent on the device; To find out the max presets is supported by the device, please refer to the "Attributes/PTZSupport/Limit/MaxPreset" attribute in the device attributes section. **Note** Either **Preset** or **PresetName** must be sent together with the **control** action.
	PresetName	REQ	<string>	Preset name
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

155.5.4. Examples

155.5.4.1. Moving the camera to Preset 1 position

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=preset&action=control&Preset=1
```

155.5.4.2. Moving the camera to the preset named ‘PresetName1’

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=preset&action=control&PresetName=PresetName1
```

155.6. Swing Control

155.6.1. Description

The **swing** submenu of **ptzcontrol.cgi** starts and stops a Swing operation. A swing is a monitoring function that moves between two preset points.

> **NOTE** To find out whether swing functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Swing attribute in the device attributes section.

Access level

Action	Camera	NVR
control	Suser	User

155.6.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
swing&action=control[&<parameter>=<value>]
```

155.6.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Mode	REQ	<enum> Pan, Tilt, PanTilt, Stop	Swing mode • Pan: Performs the Swing monitoring only in the Pan mode • Tilt: Performs the Swing monitoring only in the Tilt mode • PanTilt: Performs the Swing monitoring using both Pan and Tilt functions • Stop: Stops the Swing monitoring **Note** **Mode** must be sent together with the **control** action.

155.6.4. Examples

155.6.4.1. Swing in the Pan mode

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/ptzcontrol.cgi?msubmenu=swing&action=control&Channel=0&Mode=Pan
```

155.6.4.2. Swing in the Pan and Tilt mode

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=swing&action=control&Channel=0&Mode=PanTilt
```

155.7. Group Control

155.7.1. Description

The **group** submenu of **ptzcontrol.cgi** starts and stops a Group operation in which various presets are grouped and called in sequence.

NOTE To find out whether group functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Group attribute in the device attributes section.

Access level

Action	Camera	NVR
control	Suser	User

155.7.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
group&action=control[&<parameter>=<value>]
```

155.7.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Group	REQ	<int>	Group number
	SubViewIndex	REQ	<int>	Index of the tile in Quad view mode.

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ	<enum> Start, Stop	Group mode • Start: Starts Group action. • Stop: Stops Group action **Note** **Mode** must be sent together with the **control** action.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

155.7.4. Examples

155.7.4.1. Starting Group 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=group&action=control&Channel=0&Group=1&Mode=Star
t
```

155.8. Tour Control

155.8.1. Description

The **tour** submenu of **ptzcontrol.cgi** starts **and stops a Tour operation, calling groups of presets in sequence.**

NOTE To find out whether tour functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Tour attribute in the device attributes section.

Access level

Action	Camera	NVR
control	Suser	User

155.8.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
tour&action=control[&<parameter>=<value>]
```

155.8.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Tour	REQ	<int>	Tour number
	Mode	REQ	<enum> Start, Stop	Tour mode • Start: Starts Tour action. • Stop: Stops Tour action **Note** **Mode** must be sent together with the **control** action.

155.8.4. Examples

155.8.4.1. Starting Tour 1

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
tour&action=control&Channel=0&Tour=1&Mode=Start
```

155.9. Trace Control

155.9.1. Description

The **trace** submenu of **ptzcontrol.cgi** starts and stops a Trace action.

NOTE To find out whether trace functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Trace attribute in the device attributes section.

Access level

Action	Camera	NVR
control	Suser	User

155.9.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
trace&action=control[&<parameter>=<value>]
```

155.9.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Trace	REQ	<int>	Corresponding Trace number **Note** **Trace** and **Mode** must be sent together with the **control** action.
	Mode	REQ	<enum> Start, Stop	Trace mode • Start: Starts Trace action. • Stop: Stops Trace action

155.9.4. Examples

155.9.4.1. Starting the Trace 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=trace&action=control&Channel=0&Trace=1&Mode=Star
t
```

155.9.4.2. Stopping Trace

Stopping a Trace action means finishing memorizing the trace of movements of the device

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=trace&action=control&Channel=0&Trace=1&Mode=Stop
```

155.10. Moving to Home Position

155.10.1. Description

The **home** submenu of **ptzcontrol.cgi** moves the camera to the Home position.

NOTE This chapter applies to network cameras only.

To find out whether Home Position is supported by the device or not, refer to the Attributes/PTZSupport/Support/Home attribute in the device attributes section.

Access level

Action	Camera
control	Suser

155.10.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
home&action=control[&<parameter>=<value>]
```

155.10.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

155.10.4. Examples

155.10.4.1. Moving camera to the Home position in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=home&action=control&Channel=0
```

155.11. Area Zoom

155.11.1. Description

The **areazoom** submenu of **ptzcontrol.cgi** defines the area for zooming in and out. It specifies which part of the area should be zoomed in or out.

NOTE	This chapter applies to network cameras only. To find out whether area zoom functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/ AreaZoom attribute in the device attributes section.

Access level

Action	Camera
control	Suser

155.11.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
areazoom&action=control[&<parameter>=<value>]
```

155.11.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
Control	Channel	REQ	<int>	Channel ID
	Type	REQ	<enum> ZoomIn, 1x	Zoom type
	Profile	REQ	<int>	Index of profile being used while performing areazoom (an optional parameter that mainly improves the accuracy of areazoom if the profile is cropped)
	X1	REQ	<int>	X-coordinate of the start point of the zoom area X1, Y1, X2, and Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set; and scaled to 10000 in horizontal resolution pixels. **X1** is valid only when **Type** is set to ZoomIn.
	Y1	REQ	<int>	Y-coordinate of the start point of the zoom area X1, Y1, X2, and Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set; and scaled to 10000 in horizontal resolution pixels. **Y1** is valid only when **Type** is set to ZoomIn.

Action	Parameters	Request/ Response	Type/ Value	Description
	X2	REQ	<int>	X-coordinate of the end point of the zoom area X1, Y1, X2, and Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set; and scaled to 10000 in horizontal resolution pixels. **X2** is valid only when **Type** is set to ZoomIn.
	Y2	REQ	<int>	Y-coordinate of the end point of the zoom area X1, Y1, X2, and Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set; and scaled to 10000 in horizontal resolution pixels. **Y2** is valid only when **Type** is set to ZoomIn.
	TileWidth	REQ	<int>	Width of the zoom area **TileWidth** and **TileHeight** must be set together with **X1**, **Y1**, **X2** and **Y2** for absolute coordinates; X1, Y1, X2, Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set together. **TileWidth** is valid only when **Type** is set to ZoomIn.
	TileHeight	REQ	<int>	Height of the zoom area **TileWidth** and **TileHeight** must be set together with **X1**, **Y1**, **X2** and **Y2** for absolute coordinates; X1, Y1, X2, Y2 are interpreted as relative if **TileWidth** and **TileHeight** are not set together. **TileHeight** is valid only when **Type** is set to ZoomIn.

155.11.4. Examples

155.11.4.1. Defining the relative coordinates of the zoom area

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=areazoom&action=control&X1=100&X2=200&Y1=100&Y2=
200
```

155.11.4.2. Defining the absolute coordinates of the zoom area

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=areazoom&action=control&X1=100&X2=200&Y1=100&Y2=
200&TileWidth=200&TileHeight=200
```

155.12. Stop Control

155.12.1. Description

The **stop** submenu of **ptzcontrol.cgi** stops a PTZ operation.

Access level

Action	Camera	NVR	Encoder	Decoder
control	Suser	User	Suser	User

155.12.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
stop&action=control[&<parameter>=<value>]
```

155.12.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	ViewModeType	REQ	<enum> Panorama, DoublePanorama, QuadView, QuadView.#	Selecting particular view mode to stop DPTZ All profiles with same view mode will stop in sync
	OperationType	REQ	<csv> All, Pan, Tilt, Zoom, RotateInPlace	Operation type **Note** **OperationType** must be sent together with the **control** action.

155.12.4. Examples

155.12.4.1. Stopping all PTZ operation

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=stop&action=control&OperationType=All
```

155.13. Movement Control

155.13.1. Description

The **move** submenu of **ptzcontrol.cgi** controls a PTZ movement.

Access level

Action	Camera	NVR
control	Suser	User

155.13.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
move&action=control[&<parameter>=<value>]
```

155.13.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	MoveSpeed	REQ	<int>	Speed **MoveSpeed** is valid only when **Direction** is not set to Stop.
	Direction	REQ	<enum> Left, Right, Up, Down, LeftUp, LeftDown, RightUp, RightDown, Stop	Direction. • Left: Moves to the left. • Right: Moves to the right. • Up: Moves up. • Down: Moves down. • LeftUp: Moves to the top left. • LeftDown: Moves to the bottom left. • RightUp: Moves to the top right. • RightDown: Moves to the bottom right. • Stop: Stops PTZ movement. **MoveSpeed** should be sent along with **Direction** except for **Direction=Stop**. **Note** **Direction** must be sent together with the **control** action.
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

155.13.4. Examples

155.13.4.1. Moving the camera left

This example moves the camera to the left and sets the speed of camera movement.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=move&action=control&Direction=Left&MoveSpeed=2
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=move&action=Control&channel=1&MoveSpeed=100&Dire
ction=Right
```

155.14. Move status

movestatus submenu is used for getting the current moving status of PTZ camera.

Access level

Action	Camera
view	Suser

155.14.1. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
movestatus&action=<value>[&<parameter>=<value>]
```

155.14.2. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<int>	Channel ID
	PanTilt	RES	<enum> Idle, Moving	PanTilt moving status.
	Zoom	RES	<enum> Idle, Moving	Zoom moving status.

155.14.3. Example

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=movestatus&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "MoveStatus": [
        {
            "Channel": 0,
            "PanTilt": "Idle",
            "Zoom": "Idle"
        }
    ]
}
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PanTilt=Idle
Channel.0.Zoom=Idle
```

155.15. Aux control

155.15.1. Description:

The **aux** submenu of **ptzcontrol.cgi** is for controlling the auxiliary PTZ operations; the supported auxiliary commands are listed in the attribute xml (refer to "AuxCommands").

Access level

Action	Camera
control	Suser

Action	Camera
check	Suser

155.15.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
aux&action=<value>[&<parameter>=<value>]
```

155.15.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel ID
	Command	REQ	<string> HeaterOn, FanOn, HeaterOff, WiperOn, VibrationOn, PanDriveHeaterOn, PanDriveHeaterOff	Refer to "AuxCommands" in attributes.cgi for supported commands.
check	Channel	REQ	<int>	Channel ID
	AuxType	REQ	<enum> Heater	Refer to "AuxType" in the attributes.cgi/cgis section for supported values.
	Activate	RES	<bool> True, False	Checks if a specific aux is activated

155.15.4. Examples

155.15.4.1. Control auxiliary camera functions

This example controls an auxiliary camera function: the heater.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=aux&action=control&Command=HeaterOn
```

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=aux&action=control&Command=HeaterOff
```

155.15.4.2. Check whether an auxiliary function is activated or not

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=aux&action=check&Channel=0&AuxType=Heater
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Channel": 0,
    "Activate": true
}
```

155.16. Digital Auto tracking

155.16.1. Description

If a profile is configured as a DPTZ profile, digital auto tracking can be started and stopped using this submenu.

NOTE This chapter applies to network cameras only. For a profile to support digitalautotracking, the IsDigitalPTZProfile parameter in profile should be true.

Access level

Action	Camera
control	Suser

155.16.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
digitalautotracking&action=<value>[&<parameter>=<value>]
```

155.16.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel number
	Profile	REQ	<int>	Profile number on which digital auto tracking should be performed.
	Mode	REQ	<enum> Start, Stop	Starts and stops the digital auto tracking

155.16.4. Examples

155.16.4.1. Enabling digital autotracking in a profile

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=digitalautotracking&action=control&Profile=2&Mode=Start
```

155.17. RS485 Command

155.17.1. Description

This submenu is used for sending custom RS485 commands directly to the serial device connected.

NOTE This chapter applies to only Encoder.

Access level

Action	Encoder
control	Suser

155.17.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
rs485Command&action=<value>[&<parameter>=<value>]
```

155.17.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel number
	Command	REQ	<string>	Hex values as string

155.17.4. Examples

155.17.4.1. Sending a custom serial command in hex string

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=rs485Command&action=control&Channel=1&Command=AA1BC02B
```

155.18. OSD Menu

155.18.1. Description

This submenu is for controlling the OSD submenu of analog camera connected.

NOTE This chapter applies to only Hybrid NVR

Access level

Action	NVR
view	Suser
control	Suser

155.18.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
osdmenu&action=<value>[&<parameter>=<value>]
```

155.18.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel number
	Channel.#.OSDMenuState	RES	<bool> True, False	The value True will be returned if OSD menu control is supported by the channel. **Note** In Hybrid NVR, only analog channels support this feature.
control	Channel	REQ	<int>	Channel number

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ	<enum> On, Off, Up, Down, Right, Left, Select, Return	Supported operations

155.18.4. Examples

155.18.4.1. Getting OSD menu state

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=osdmenu&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.OSDMenuState=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "OSDMenus": [
        {
            "Channel": 0,
            "OSDMenuState": false
        }
    ]
}
```

155.18.4.2. Sending OSD menu control command

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=osdmenu&action=control&Channel=1&Mode=Up
```

155.19. digitalrtz

155.19.1. Description

The **digitalrtz** submenu allows the user to move a selected point using Tilt and Rotate values. TNF-9010 has 5 channels, and the last 4 channels are dewapred channels. Users can calculate the Rotate and Tilt values if they know center point's coordinates and the selected point's coordinates from an overview stream. So user can directly move to a desired point using this feature. This will help user to configure initial settings quickly to observe each parking area. With the **view** action, the user can retrieve the current Tilt and Rotate values.

NOTE This chapter applies to **fisheye** cameras only.

Access level

Action	Camera
view	User
control	User

155.19.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
digitalrtz&action=<value>[&<parameter>=<value>...]
```

155.19.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<int>	Channel ID
	Rotate	RES	<float> 0~360	Rotate angle
	Tilt	RES	<float> 0~90	Tilts value
	Zoom	RES	<float>	Zoom value

Action	Parameter	Request/ Response	Type/ Value	Description
	ZoomPulse	RES	<int>	Normalized zoom value from 1 to 9999.
	ActualZoomPulse	RES	<int>	The current position of the zoom lens in hardware pulse units.
	RotateInPlace	RES	<float>	Based on center of video, tilt horizon of video.
	SubViewIndex	REQ	<int> 1~4	Used to select the tile for PTZ in Quad view mode. This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number
control	Rotate	REQ	<float> 0~360	Rotate angle
	Tilt	REQ	<float> 0~90	Tilts value
	Channel	REQ	<int>	Channel ID
	SubViewIndex	REQ	<int> 1~4	Used to select the tile for PTZ in Quad view mode. This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number
	RotateInPlace	REQ	<float>	Based on center of video, tilt horizon of video.
	Mode	REQ	<enum> Absolute, Relative	Control mode; **Absolute** mode is to move the fisheye view to the specific point defined by Rotate and Tilt values; **Relative** mode is to move the fisheye view based on current view. If **Mode** is not sent, the default value will be **Absolute**.

155.19.4. Examples

155.19.4.1. Going to a specific point

REQUEST

```
http://<Device
IP>/ptzcontrol.cgi?msubmenu=digitalrtz&action=control&Tilt=6.152&Rotate=61.6
99&SubViewIndex=3
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

155.19.4.2. Rotating the fisheye view relative to the current view

In relative mode (`Mode=Relative`), the fisheye view rotates clockwise by the specified `Rotate` value (0~360) from its current position. For example, `Rotate=5` rotates the view 5 degrees clockwise from the current view.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzcontrol.cgi?msubmenu=digitalrtz&action=control&Channel=0&Rotate=4&Mod
e=Relative
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

155.19.4.3. Getting current Rotate and Tilt values

Getting the current position

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/ptzcontrol.cgi?msubmenu=digitalrtz&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Digitalrtz": [
        {
            "Channel": 0,
            "Rotate": 90.0,
            "Tilt": 47.0,
            "Zoom": 1.6,
            "ZoomPulse": 999,
            "ActualZoomPulse": 160,
            "RotateInPlace": 180.0
        }
    ]
}
```


155.20. Supported PTZ actions

155.20.1. Description

The **supportedptzactions** submenu provides supported PTZ action lists for each view mode.

NOTE This chapter applies to **fisheye** cameras only.

Access level

Action	Camera
view	Suser

155.20.2. Syntax

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=
supportedptzactions&action=view&[&<parameter>=<value>...]
```

155.20.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ,RES	<csv>	Channel ID
	Support.ViewMode	RES	<enum> Overview, QuadView, DoublePanorama, Panorama, QuadView.1, QuadView.2, QuadView.3, QuadView.4	View mode supports in the target device
	Support.Submenus	RES	<csv>	Supported submenus support in the target device
	Support.Operations	RES	<csv>	Supported operations (Pan, Tilt, Zoom)

155.20.4. Examples

155.20.4.1. Going to a specific point

REQUEST

```
http://<Device IP>/ptzcontrol.cgi?msubmenu=supportedptzactions&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Channels": [
        {
            "Channel": 0,
            "Support": [
                {
                    "ViewMode": "Overview",
                    "Submenus": [],
                    "Operations": []
                },
                {
```

```
                "ViewMode": "QuadView",
                "Submenus": [
                    "absolute",
                    "relative",
                    "continuous",
                    "query",
                    "preset",
                    "group",
                    "move",
                    "stop",
                    "digitalrtz"
                ],
                "Operations": [
                    "Pan",
                    "Tilt",
                    "Zoom"
                ]
            },
            {
                "ViewMode": "DoublePanorama",
                "Submenus": [
                    "continuous",
                    "stop"
                ],
                "Operations": [
                    "Pan",
                    "Tilt"
                ]
            },
            {
                "ViewMode": "Panorama",
                "Submenus": [
                    "continuous",
                    "stop"
                ],
                "Operations": [
                    "Pan"
                ]
            }
        ]
    },
```

```
        {
            "Channel": 1,
            "Support": [
                {
                    "ViewMode": "QuadView.1",
                    "Submenus": [
                        "absolute",
                        "relative",
                        "continuous",
                        "query",
                        "preset",
                        "group",
                        "move",
                        "stop",
                        "digitalrtz"
                    ],
                    "Operations": [
                        "Pan",
                        "Tilt",
                        "Zoom"
                    ]
                }
            ]
        },
        {
            "Channel": 2,
            "Support": [
                {
                    "ViewMode": "QuadView.2",
                    "Submenus": [
                        "absolute",
                        "relative",
                        "continuous",
                        "query",
                        "preset",
                        "group",
                        "move",
                        "stop",
                        "digitalrtz"
                    ],
                    "Operations": [
```

```
                        "Pan",
                        "Tilt",
                        "Zoom"
                    ]
                }
            ]
        },
        {
            "Channel": 3,
            "Support": [
                {
                    "ViewMode": "QuadView.3",
                    "Submenus": [
                        "absolute",
                        "relative",
                        "continuous",
                        "query",
                        "preset",
                        "group",
                        "move",
                        "stop",
                        "digitalrtz"
                    ],
                    "Operations": [
                        "Pan",
                        "Tilt",
                        "Zoom"
                    ]
                }
            ]
        },
        {
            "Channel": 4,
            "Support": [
                {
                    "ViewMode": "QuadView.4",
                    "Submenus": [
                        "absolute",
                        "relative",
                        "continuous",
                        "query",
```

```
                        "preset",
                        "group",
                        "move",
                        "stop",
                        "digitalrtz"
                    ],
                    "Operations": [
                        "Pan",
                        "Tilt",
                        "Zoom"
                    ]
                }
            ]
        }
    ]
}
```

Chapter 156. PTZ Configuration

156.1. Swing Setup

156.1.1. Description

The **swing** submenu of **ptzconfig.cgi** configures the Swing settings.

A Swing is a monitoring function that moves between two preset points.

> **NOTE** To find out whether swing functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Swing attribute in the device attributes section.

Access level

Action	Camera	NVR
view	Suser	User
set	Suser	(Not supported)

156.1.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
swing&action=<value>[&<parameter>=<value>]
```

156.1.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the Swing settings.
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID (read-only for NVR)

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ, RES	<enum> Pan, Tilt, PanTilt	Swing mode (read-only for NVR) • Pan: Performs the Swing monitoring only by using the Pan function • Tilt: Performs the Swing monitoring only by using the Tilt function • PanTilt: Performs the Swing monitoring by using both Pan and Tilt functions **Note** **Mode**, **FromPreset**, **ToPreset**, **Speed**, and **DwellTime** must be sent together with the **set** action.
	FromPreset	REQ, RES	<int>	First preset for the Swing operation **Note** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer to the "Attributes/PTZSupport/Limit/MaxPreset" attribute in the device attributes section. CAMERA ONLY
	ToPreset	REQ, RES	<int>	Second preset for the Swing operation **Note** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer to the "Attributes/PTZSupport/Limit/MaxPreset" attribute in the device attributes section. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	Speed	REQ, RES	<int>	Moving Speed CAMERA ONLY
	DwellTime	REQ, RES	<int>	Interval between first and second presets CAMERA ONLY

156.1.4. Examples

156.1.4.1. Getting the current Swing settings in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=swing&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode.Pan.FromPreset=1
Channel.0.Mode.Pan.ToPreset=2
Channel.0.Mode.Pan.Speed=64
Channel.0.Mode.Pan.DwellTime=3
Channel.0.Mode.Tilt.FromPreset=3
Channel.0.Mode.Tilt.ToPreset=4
Channel.0.Mode.Tilt.Speed=25
Channel.0.Mode.Tilt.DwellTime=35
Channel.0.Mode.PanTilt.FromPreset=5
Channel.0.Mode.PanTilt.ToPreset=6
Channel.0.Mode.PanTilt.Speed=60
Channel.0.Mode.PanTilt.DwellTime=36
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "PTZSwing": [
        {
            "Channel": 0,
            "SwingSequence": [
                {
                    "Mode": "Pan",
                    "FromPreset": 1,
                    "ToPreset": 2,
                    "Speed": 64,
                    "DwellTime": 3
                },
                {
                    "Mode": "Tilt",
                    "FromPreset": 2,
                    "ToPreset": 1,
                    "Speed": 64,
                    "DwellTime": 3
                },
                {
                    "Mode": "PanTilt",
                    "FromPreset": 1,
                    "ToPreset": 2,
                    "Speed": 64,
                    "DwellTime": 3
                }
            ]
        }
    ]
}
```


(The following response example is for NVR only.)

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.3.Mode=PanTilt
Channel.4.Mode=PanTilt
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZSwing": [
        {
            "Channel": 3,
            "Mode": "PanTilt"
        },
        {
            "Channel": 4,
            "Mode": "PanTilt"
        }
    ]
}
```


156.1.4.2. Setting Swing moving from Preset 1 to 2 in the Pan mode only

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=swing&action=set&Channel=0&Mode=Pan&FromPreset=1&
ToPreset=2&Speed=1&DwellTime=1
```

156.1.4.3. Setting Swing moving from Preset 2 to 3 in the Tilt mode only

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=swing&action=set&Channel=0&Mode=Tilt&FromPreset=2
&ToPreset=3&Speed=3&DwellTime=3
```

156.1.4.4. Swing moving from Preset 3 to 4 in both Pan and Tilt modes

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=swing&action=set&Channel=0&Mode=PanTilt&FromPreset=3&ToPreset=4&Speed=3&DwellTime=3
```

156.2. Group Setup

156.2.1. Description

The **group** submenu of **ptzconfig.cgi** configures the Group settings. Multiple presets are grouped and called in sequence according to the Group feature.

NOTE To find out whether group functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Group attribute in the device attributes section.

Access level

Action	Camera	NVR
view	Suser	User
add, update	Suser	(Not supported)
remove	Suser	(Not supported)

156.2.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
group&action=<value>[&<parameter>=<value>]
```

156.2.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the Group settings.
	Channel	REQ, RES	<csv>	Channel ID
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY
add/ update	Channel	REQ, RES	<int>	Channel ID (read-only for NVR)

Action	Parameters	Request/ Response	Type/ Value	Description
	Group	REQ, RES	<int>	Group Number (read-only for NVR) **Note** **Group**, **Preset**, **Speed**, and **DwellTime** must be sent together with the **add** action, and **Group** and **PresetSequence** together with the **update** action, To find out whether Max Groups is supported by the device**,** refer to the **"**Attributes/PTZSupport /Limit/MaxGroupCount**"** attribute in the device attributes section.
	Name	REQ, RES	<string>	Group name CAMERA ONLY
	PresetSequence	REQ, RES	<int>	Preset sequence **Note** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer to the "Attributes/PTZSupport/Limit/Max Preset" attribute in the device attributes section. CAMERA ONLY
	Preset	REQ, RES	<int>	Preset number **Note** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer to the **"**Attributes/PTZSupport/Limit/Max Preset*"* attribute in the device attributes section. CAMERA ONLY
	Speed	REQ, RES	<int>	Speed CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	DwellTime	REQ, RES	<int>	Interval between presets It specifies a waiting time before the next preset is called. CAMERA ONLY
	ViewModeIndex	REQ	<int>	View mode index number CAMERA ONLY
remove	Channel	REQ	<int>	Channel ID CAMERA ONLY
	Group	REQ	<csv>	Group number CAMERA ONLY
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

156.2.4. Examples

156.2.4.1. Getting the current Group settings in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=group&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Group.1.PresetSequence.1.Preset=1
Channel.0.Group.1.PresetSequence.1.Speed=64
Channel.0.Group.1.PresetSequence.1.DwellTime=3
Channel.0.Group.1.PresetSequence.2.Preset=2
Channel.0.Group.1.PresetSequence.2.Speed=64
Channel.0.Group.1.PresetSequence.2.DwellTime=3
Channel.0.Group.2.PresetSequence.1.Preset=1
Channel.0.Group.2.PresetSequence.1.Speed=37
Channel.0.Group.2.PresetSequence.1.DwellTime=3
```

```
Channel.0.Group.2.PresetSequence.2.Preset=3
Channel.0.Group.2.PresetSequence.2.Speed=20
Channel.0.Group.2.PresetSequence.2.DwellTime=3
Channel.0.Group.2.PresetSequence.3.Preset=4
Channel.0.Group.2.PresetSequence.3.Speed=60
Channel.0.Group.2.PresetSequence.3.DwellTime=3
Channel.0.Group.2.PresetSequence.4.Preset=6
Channel.0.Group.2.PresetSequence.4.Speed=48
Channel.0.Group.2.PresetSequence.4.DwellTime=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZGroups": [
        {
            "Channel": 0,
            "Groups": [
                {
                    "Group": 1,
                    "PresetSequences": [
                        {
                            "PresetSequence": 1,
                            "Preset": 1,
                            "Speed": 64,
                            "DwellTime": 3
                        },
                        {
                            "PresetSequence": 2,
                            "Preset": 2,
                            "Speed": 64,
                            "DwellTime": 3
                        }
                    ]
                },
                {
                    "Group": 2,
                    "PresetSequences": [
```

```
                        {
                            "PresetSequence": 1,
                            "Preset": 1,
                            "Speed": 37,
                            "DwellTime": 3
                        },
                        {
                            "PresetSequence": 2,
                            "Preset": 3,
                            "Speed": 20,
                            "DwellTime": 3
                        },
                        {
                            "PresetSequence": 3,
                            "Preset": 4,
                            "Speed": 60,
                            "DwellTime": 3
                        },
                        {
                            "PresetSequence": 4,
                            "Preset": 6,
                            "Speed": 48,
                            "DwellTime": 3
                        }
                    ]
                }
            ]
        }
    ]
}
```

(The following example response is for NVR only.)

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.3.Group=1
Channel.3.Group=2
```

```
Channel.3.Group=3
Channel.4.Group=1
Channel.4.Group=2
Channel.4.Group=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZGroups": [
        {
            "Channel": 3,
            "Groups": [
                {
                    "Group": 1
                },
                {
                    "Group": 2
                },
                {
                    "Group": 3
                }
            ]
        },
        {
            "Channel": 4,
            "Groups": [
                {
                    "Group": 1
                },
                {
                    "Group": 2
                },
                {
                    "Group": 3
                }
            ]
        }
```

```
    ]
}
```

156.2.4.2. Adding Group 1 calling Preset 2

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=group&action=add&Channel=0&Group=1&PresetSequence
=1&Preset=2&Speed=60&DwellTime=2
```

156.2.4.3. Updating Group 1 to call Preset 3 in the second sequence

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=group&action=update&Channel=0&Group=1&PresetSeque
nce=2&Preset=3&Speed=50&DwellTime=3
```

156.2.4.4. Removing Group 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=group&action=remove&Channel=0&Group=1
```

156.3. Tour Setup

156.3.1. Description

The **tour** submenu of **ptzconfig.cgi** configures the Tour settings. Groups of presets are called in sequence according to the Tour feature.

NOTE To find out whether tour functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Tour attribute in the device attributes section.

Access level

Action	Camera	NVR
view	Suser	User
add, update	Suser	(Not supported)
remove	Suser	(Not supported)

156.3.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
tour&action=<value>[&<parameter>=<value>]
```

156.3.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the Tour settings.
	Channel	REQ, RES	<csv>	Channel ID
add/update	Channel	REQ, RES	<int>	Channel ID (read-only for NVR) **Note** **Group** and **DwellTime** must be sent together withthe **add** action, and **GroupSequence** together withthe **update** action.
	Tour	REQ, RES	<int>	Tour number (read-only for NVR) **Note** To find out whether Max Tours is supported by the device, refer to the "Attributes/PTZSupport/Limit /MaxTourCount" attribute in the device attributes section.
	TourMode	REQ, RES	<enum> EnhancedSequences,GroupSequences	Tour mode can be changed using this parameter, when GroupSequences is selected only group can be configured in tour, when EnhancedSequences is selected other sequence type selection is possible. CAMERA ONLY
	GroupSequence	REQ, RES	<int>	Group sequence CAMERA ONLY
	Group	REQ, RES	<int>	Group number CAMERA ONLY
	EnhancedSequence	REQ, RES	<int>	When TourMode is configured as EnhancedSequences, used to configure the enhanced sequence index. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	SequenceType	REQ, RES	<enum> None,Preset,Group, Swing,Trace,Home	When TourMode is configured as EnhancedSequences, this parameter is used to configure the sequence type. CAMERA ONLY
	PresetIndex	REQ, RES	<int>	When the SequenceType is configured as Preset, this parameter is used to configure the preset index. CAMERA ONLY
	GroupIndex	REQ, RES	<int>	When the SequenceType is configured as Group, this parameter is used to configure the Group index. CAMERA ONLY
	TraceIndex	REQ, RES	<int>	When the SequenceType is configured as Trace, this parameter is used to configure the Trace index. CAMERA ONLY
	SwingMode	REQ, RES	<enum> Stop, Pan, Tilt, PanTilt	When the SequenceType is configured as Swing, this parameter is used to configure the swingmode. CAMERA ONLY
	DwellTime	REQ, RES	<int>	Interval between groups (second) It specifies a waiting time before a new group is called. CAMERA ONLY
remove	Channel	REQ	<int>	Channel ID CAMERA ONLY
	Tour	REQ	<csv>	Tour number CAMERA ONLY

156.3.4. Examples

156.3.4.1. Getting the current Group settings in Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=tour&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Tour.GroupSequence.1.Group=1
Channel.0.Tour.GroupSequence.1.DwellTime=10
Channel.0.Tour.GroupSequence.2.Group=2
Channel.0.Tour.GroupSequence.2.DwellTime=25
Channel.0.Tour.GroupSequence.3.Group=3
Channel.0.Tour.GroupSequence.3.DwellTime=36
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PTZTours": [
        {
            "Channel": 0,
            "Tours": [
                {
                    "Tour": 1,
                    "GroupSequences": [
                        {
                            "GroupSequence": 1,
                            "Group": 1,
                            "DwellTime": 10
                        },
                        {
                            "GroupSequence": 2,
                            "Group": 2,
                            "DwellTime": 25
                        },
                        {
                            "GroupSequence": 3,
                            "Group": 3,
```

```
                        "DwellTime": 36
                    }
                ]
            }
        ]
    }
    ]
}
```

JSON RESPONSE [When Tour Mode is supported]

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZTours": [
        {
            "Channel": 0,
            "Tours": [
                {
                    "Tour": 1,
                    "TourMode": "EnhancedSequences",
                    "EnhancedSequences": [
                        {
                            "EnhancedSequence": 1,
                            "SequenceType": "Preset",
                            "PresetIndex": 1,
                            "DwellTime": 3
                        },
                        {
                            "EnhancedSequence": 2,
                            "SequenceType": "Home",
                            "DwellTime": 3
                        },
                        {
                            "EnhancedSequence": 3,
                            "SequenceType": "Swing",
                            "SwingMode": "Pan",
                            "DwellTime": 3
                        },
```

```
                        {
                            "EnhancedSequence": 4,
                            "SequenceType": "Group",
                            "GroupIndex": 1,
                            "DwellTime": 3
                        },
                        {
                            "EnhancedSequence": 5,
                            "SequenceType": "Trace",
                            "TraceIndex": 1,
                            "DwellTime": 3
                        }
                    ]
                }
            ]
        }
    ]
}
```

(The following response example is for NVR only.)

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.3.Tour=1
Channel.4.Tour=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PTZTours": [
        {
            "Channel": 3,
```

```
                "Tours": [
                    {
                        "Tour": 1
                    }
                ]
            },
            {
                "Channel": 4,
                "Tours": [
                    {
                        "Tour": 1
                    }
                ]
            }
        ]
}
```

156.3.4.2. Adding Tour 1 calling the Group 1

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=tour&action=add&Channel=0&Tour=1&GroupSequence=1&Group=1&DwellTime=1
```

156.3.4.3. Updating Tour 1 to call the Group 2 in the second sequence

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=tour&action=update&Channel=0&Tour=1&GroupSequence=2&Group=2&DwellTime=1
```

NOTE Update using POST is also supported in Camera

UPDATE REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=tour&action=update
```

POST BODY

```
{
    "PTZTours": [
```

```
        {
            "Channel": 0,
            "Tours": [
                {
                    "Tour": 1,
                    "TourMode": "EnhancedSequences",
                    "EnhancedSequences": [
                        {
                            "EnhancedSequence": 1,
                            "SequenceType": "Preset",
                            "PresetIndex": 1,
                            "DwellTime": 3
                        },
                        {
                            "EnhancedSequence": 2,
                            "SequenceType": "Home",
                            "DwellTime": 3
                        },
                        {
                            "EnhancedSequence": 3,
                            "SequenceType": "Swing",
                            "SwingMode": "Pan",
                            "DwellTime": 3
                        }
                    ]
                }
            ]
        }
    ]
}
```

156.3.4.4. Removing Tour 1

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=tour&action=remove&Tour=1
```

156.4. Trace Setup

156.4.1. Description

The **trace** submenu of **ptzconfig.cgi** configures settings for a Trace action. Movement of the device is remembered and reproduced when required.

NOTE	To find out whether trace functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Trace attribute in the device attributes section.

Access level

Action	Camera	NVR
view	SUser	User
memorize	Suser	(Not supported)

156.4.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
trace&action=<value>[&<parameter>=<value>]
```

156.4.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the Trace settings
	Channel	REQ, RES	<csv>	Channel ID
	Trace.#.IsMemorized	RES	<bool> True, False	Show wheterer the trace is memorized or not.
set	Channel	REQ, RES	<csv>	Channel ID
	Trace.#.Name	REQ, RES	<string>	Trace name
memorize	Channel	REQ	<int>	Channel ID
	Mode	REQ	<enum> Start, Stop	Trace mode • Start: Starts Trace action. • Stop: Stops Trace action (finishes memorizing) **Note** **Mode** and **Trace** must be sent together with the **memorize** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	Trace	REQ	<int>	Trace number **Note** To find out whether Max Traces is supported by the device, refer to the "Attributes/PTZSupport/Limit/MaxTraceCount" attribute in the device attributes section.

156.4.4. Examples

156.4.4.1. Getting current Trace settings in Channel 0

Below response if for camera

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=trace&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK

Content-type: application/json
<Body>
```


```
{
    "PTZTrace": [
        {
            "Channel": 0,
            "Traces": [
                {
                    "Trace": 1,
                    "IsMemorized": true,
                    "Name": "trac1"
                },
                {
                    "Trace": 2,
                    "IsMemorized": false,
                    "Name": "Trace2"
```

```
                },
                {
                    "Trace": 3,
                    "IsMemorized": false,
                    "Name": ""
                },
                {
                    "Trace": 4,
                    "IsMemorized": false,
                    "Name": ""
                }
            ]
        }
    ]
}
```

The following example is for NVR; only NVR supports the **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=trace&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.3.Trace=1
Channel.3.Trace=2
Channel.3.Trace=3
Channel.3.Trace=4
Channel.4.Trace=1
Channel.4.Trace=2
Channel.4.Trace=3
Channel.4.Trace=4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "PTZTraces": [
        {
            "Channel": 3,
            "Traces": [
                {
                    "Trace": 1
                },
                {
                    "Trace": 2
                },
                {
                    "Trace": 3
                },
                {
                    "Trace": 4
                }
            ]
        },
        {
            "Channel": 4,
            "Traces": [
                {
                    "Trace": 1
                },
                {
                    "Trace": 2
                },
                {
                    "Trace": 3
                },
                {
                    "Trace": 4
                }
            ]
        }
```

```
    ]
}
```

156.4.4.2. Set the trace name in camera

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=trace&action=set&Channel=0&Trace.1.Enable=True&Trace.1.Name=trace1
```

156.4.4.3. Memorizing the Trace action

To use the **memorize** action, **Mode** and **Trace** must be set together. (Only network cameras support the **memorize** action.)

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=trace&action=memorize&Channel=0&Mode=Start&Trace=1
```

156.5. Auto Run Setup

156.5.1. Description

The **autorun** submenu of **ptzconfig.cgi** configures settings for an Auto run. The Autorun makes the PTZ camera to start predefined operations such as going to a preset and starting Swing operation.

NOTE This chapter applies to network cameras only.

To find out whether auto run functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/AutoRun attribute in the device attributes section.

Access level

Action	Camera
view	Suser
set	Suser
update	Suser

156.5.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
```

```
autorun&action=<value>[&<parameter>=<value>]
```

156.5.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the Auto run settings.
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID **Note** To use the **set** action, **Mode**, **ActivationTime**, and one of **Preset**, **Group**, **Trace**, **AutoPanSpeed**, **AutoPanTiltAngle**, **Tour**, and **SwingMode** must be sent together. (The parameters to be sent differ according to the **Mode** value.) For example, if the value of **Mode** is set to Preset, **Mode**, **ActivationTime**, and **Preset** must be sent together.

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ, RES	<enum> Off, Home, Preset, Swing, Group, Tour, Trace, AutoPan, Schedule	Auto run mode • Off: Disables Auto run • Home: Automatically moves to the Home position • Preset: Automatically moves to the preset number • Swing: Automatically executes the operation in Swing mode • Group: Automatically executes the operation in Group mode • Tour: Automatically executes the operation in Tour mode • Trace: Automatically executes the operation in Trace mode • AutoPan: Automatically executes the 360 degree rotation in the pan direction • Schedule: Automatically executes the operation according to a schedule **Note** When **Mode** is not set to Off, **ActivationTime** is a mandatory parameter.
	ActivationTime	REQ, RES	<enum> 5s, 10s, 20s, 30s, 1m, 2m, 3m, 4m, 5m, Custom	Activation time (second/ minute) **ActivationTime** is valid only when **Mode** is not set to Off. When set to "Custom", it operates based on ActivationCustomTime.
	ActivationCustomTime	REQ, RES	<int>	Activation custom time (second) **ActivationCustomTime** is valid only when **ActivationTime** is set to Custom.

Action	Parameters	Request/ Response	Type/ Value	Description
	Preset	REQ, RES	<int>	Preset number The number of presets supported is dependent on the device; please refer to the device attributes. **Preset** is valid only when **Mode** is set to Preset.
	Group	REQ, RES	<int>	Group number **Group** is valid only when **Mode** is set to Group.
	Trace	REQ, RES	<int>	Trace number **Trace** is valid only when **Mode** is set to Trace.
	AutoPanSpeed	REQ, RES	<int>	Speed in camera’s automatic panning operation **AutoPanSpeed** is valid only when **Mode** is set to AutoPan.
	AutoPanTiltAngle	REQ, RES	<int>	Angle that is placed for the camera to move around during Panning and Tilting operation **AutoPanTiltAngle** is valid only when **Mode** is set to AutoPan.
	Tour	REQ, RES	<int>	Tour number **Tour** is valid only when **Mode** is set to Tour.
	SwingMode	REQ, RES	<enum> Pan, Tilt, PanTilt	Swing mode **SwingMode** is valid only when **Mode** is set to Swing.
	AutorunDuringSequence	REQ, RES	<bool> True, False	When enabled autorun will work during sequence operation.

Action	Parameters	Request/ Response	Type/ Value	Description
update	Channel	REQ, RES	<int>	Channel ID **Note** To use the **update** action, **ScheduleMode**, **FromTo**, and one of **Preset**, **Group**, **Trace**, **AutoPanSpeed**, **AutoPanTiltAngle**, **Tour**, and **SwingMode** (depending on the **ScheduleMode** value) must be sent together.
	ScheduleMode	REQ, RES	<enum> Home, Preset, Swing, Group, Tour, Trace, AutoPan	Automatically runs Schedule Mode
	FromTo	REQ, RES	<string>	Time schedule for auto run The schedule is specified in the format of <ddd hh:mm-ddd hh:mm>. Only mm=00 is allowed in FromTo (i.e, only a granular level houris allowed).
	Preset	REQ, RES	<int>	Preset number The number of the preset supported depends on the device; please refer to the device attributes. **Preset** is valid only when **ScheduleMode** is set to Preset.
	Group	REQ, RES	<int>	Group number **Group** is valid only when **ScheduleMode** is set to Group.
	Trace	REQ, RES	<int>	Trace number **Trace** is valid only when **ScheduleMode** is set to Trace.

Action	Parameters	Request/ Response	Type/ Value	Description
	AutoPanSpeed	REQ, RES	<int>	Speed of the camera's automatic panning operation **AutoPanSpeed** is valid only when **ScheduleMode** is set to AutoPan.
	AutoPanTiltAngle	REQ, RES	<int>	Angle that is set for the camera to move between during the Panning and Tilting operation **AutoPanTiltAngle** is valid only when **ScheduleMode** is set to AutoPan.
	Tour	REQ, RES	<int>	Tour number **Tour** is valid only when **ScheduleMode** is set to Tour.
	SwingMode	REQ, RES	<enum> Pan, Tilt, PanTilt	Swing mode **SwingMode** is valid only when **ScheduleMode** is set to Swing.

156.5.4. Examples

156.5.4.1. Getting the current Auto run settings in Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=autorun&action=view&Channel=0
```

If **Mode** is set to Group, the following response will be returned.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Group
Channel.0.ActivationTime=5s
Channel.0.Group=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AutoRun": [
        {
            "Channel": 0,
            "Mode": "Group",
            "ModeHome": {
                "ActivationTime": "5s"
            },
            "ModePreset": {
                "ActivationTime": "5s",
                "Preset": 1
            },
            "ModeSwing": {
                "ActivationTime": "5s",
                "SwingMode": "Pan"
            },
            "ModeGroup": {
                "ActivationTime": "5s",
                "Group": 1
            },
            "ModeTour": {
                "Tour": 1,
                "ActivationTime": "5s"
            },
            "ModeTrace": {
                "ActivationTime": "5s",
                "Trace": 1
            },
            "ModeAutoPan": {
                "ActivationTime": "5s",
                "AutoPanSpeed": 20,
                "AutoPanTiltAngle": 20
            },
            "ModeSchedule": {
                "ActivationTime": "5s",
```

```
                "Schedules": [
                    {
                        "FromTo": "SUN 00:00-SUN 14:00",
                        "ScheduleMode": "Swing",
                        "SwingMode": "Pan"
                    },
                    {
                        "FromTo": "SUN 14:00-TUE 06:00",
                        "ScheduleMode": "Home"
                    },
                    {
                        "FromTo": "TUE 06:00-TUE 16:00",
                        "ScheduleMode": "AutoPan",
                        "AutoPanSpeed": 20,
                        "AutoPanTiltAngle": 20
                    },
                    {
                        "FromTo": "TUE 16:00-FRI 04:00",
                        "ScheduleMode": "Home"
                    },
                    {
                        "FromTo": "FRI 04:00-FRI 07:00",
                        "ScheduleMode": "Tour",
                        "Tour": 1
                    },
                    {
                        "FromTo": "FRI 07:00-SAT 24:00",
                        "ScheduleMode": "Home"
                    }
                ]
            }
        }
    ]
}
```


If **Mode** is set to Schedule, the following response will be returned.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Channel.0.Mode=Schedule
Channel.0.ActivationTime=5s
Channel.0.Schedule.1.FromTo=SUN 00:00-SUN 14:00
Channel.0.Schedule.1.ScheduleMode=Swing
Channel.0.Schedule.1.SwingMode=Pan
Channel.0.Schedule.2.FromTo=SUN 14:00-TUE 06:00
Channel.0.Schedule.2.ScheduleMode=Home
Channel.0.Schedule.3.FromTo=TUE 06:00-TUE 16:00
Channel.0.Schedule.3.ScheduleMode=AutoPan
Channel.0.Schedule.3.AutoPanSpeed=20
Channel.0.Schedule.3.AutoPanTiltAngle=20
Channel.0.Schedule.4.FromTo=TUE 16:00-FRI 04:00
Channel.0.Schedule.4.ScheduleMode=Home
Channel.0.Schedule.5.FromTo=FRI 04:00-FRI 07:00
Channel.0.Schedule.5.ScheduleMode=Tour
Channel.0.Schedule.6.FromTo=FRI 07:00-SAT 24:00
Channel.0.Schedule.6.ScheduleMode=Home
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AutoRun": [
        {
            "Channel": 0,
            "Mode": "Schedule",
            "ModeHome": {
                "ActivationTime": "5s"
            },
            "ModePreset": {
                "ActivationTime": "5s",
                "Preset": 1
            },
            "ModeSwing": {
                "ActivationTime": "5s",
```

```
            "SwingMode": "Pan"
        },
        "ModeGroup": {
            "ActivationTime": "5s",
            "Group": 1
        },
        "ModeTour": {
            "Tour": 1,
            "ActivationTime": "5s"
        },
        "ModeTrace": {
            "ActivationTime": "5s",
            "Trace": 1
        },
        "ModeAutoPan": {
            "ActivationTime": "5s",
            "AutoPanSpeed": 20,
            "AutoPanTiltAngle": 20
        },
        "ModeSchedule": {
            "ActivationTime": "5s",
            "Schedules": [
                {
                    "FromTo": "SUN 00:00-SUN 14:00",
                    "ScheduleMode": "Swing",
                    "SwingMode": "Pan"
                },
                {
                    "FromTo": "SUN 14:00-TUE 06:00",
                    "ScheduleMode": "Home"
                },
                {
                    "FromTo": "TUE 06:00-TUE 16:00",
                    "ScheduleMode": "AutoPan",
                    "AutoPanSpeed": 20,
                    "AutoPanTiltAngle": 20
                },
                {
                    "FromTo": "TUE 16:00-FRI 04:00",
                    "ScheduleMode": "Home"
                },
```

```
                    {
                        "FromTo": "FRI 04:00-FRI 07:00",
                        "ScheduleMode": "Tour",
                        "Tour": 1
                    },
                    {
                        "FromTo": "FRI 07:00-SAT 24:00",
                        "ScheduleMode": "Home"
                    }
                ]
            }
        }
    ]
}
```


156.5.4.2. Disabling Auto run

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=set&Channel=0&Mode=Off
```

156.5.4.3. Setting Auto run in Swing mode to be activated in 30 seconds

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=set&Channel=0&Mode=Swing&Activatio
nTime=30s
```

156.5.4.4. Configuring Auto run schedule with the Home mode for entire week

Mode must be set to Schedule first, using the below example.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=update&FromTo=SUN 00:00-SAT
23:59&ScheduleMode=Home
```

After the above example has been applied, the view action will return the following responses.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Schedule
Channel.0.ActivationTime=5s
Channel.0.Schedule.1.FromTo=SUN 00:00-SAT 24:00
Channel.0.Schedule.1.ScheduleMode=Home
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AutoRun": [
        {
            "Channel": 0,
            "Mode": "Schedule",
            "ModeHome": {
                "ActivationTime": "5s"
            },
            "ModePreset": {
                "ActivationTime": "5s",
                "Preset": 1
            },
            "ModeSwing": {
                "ActivationTime": "30s",
                "SwingMode": "Pan"
            },
            "ModeGroup": {
                "ActivationTime": "5s",
```

```
                "Group": 1
            },
            "ModeTour": {
                "Tour": 1,
                "ActivationTime": "5s"
            },
            "ModeTrace": {
                "ActivationTime": "5s",
                "Trace": 1
            },
            "ModeAutoPan": {
                "ActivationTime": "5s",
                "AutoPanSpeed": 20,
                "AutoPanTiltAngle": 20
            },
            "ModeSchedule": {
                "ActivationTime": "5s",
                "Schedules": [
                    {
                        "FromTo": "SUN 00:00-SAT 24:00",
                        "ScheduleMode": "Home"
                    }
                ]
            }
        }
    ]
}
```

156.5.4.5. Configuring Auto run schedule for Monday 8PM to Wednesday 7PM with the Preset mode moving to preset 1

Mode must be set to Schedule first, using the below example.

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=autorun&action=update&FromTo=MON 20:00-WED 19:00&ScheduleMode=Preset&Preset=1
```

After the above example has been applied, the view action will return the following responses.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Schedule
Channel.0.ActivationTime=5s
Channel.0.Schedule.1.FromTo=SUN 00:00-MON 20:00
Channel.0.Schedule.1.ScheduleMode=Home
Channel.0.Schedule.2.FromTo=MON 20:00-WED 19:00
Channel.0.Schedule.2.ScheduleMode=Preset
Channel.0.Schedule.2.Preset=1
Channel.0.Schedule.3.FromTo=WED 19:00-SAT 24:00
Channel.0.Schedule.3.ScheduleMode=Home
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AutoRun": [
        {
            "Channel": 0,
            "Mode": "Schedule",
            "ModeHome": {
                "ActivationTime": "5s"
            },
            "ModePreset": {
                "ActivationTime": "5s",
                "Preset": 1
            },
            "ModeSwing": {
```

```
                "ActivationTime": "30s",
                "SwingMode": "Pan"
            },
            "ModeGroup": {
                "ActivationTime": "5s",
                "Group": 1
            },
            "ModeTour": {
                "Tour": 1,
                "ActivationTime": "5s"
            },
            "ModeTrace": {
                "ActivationTime": "5s",
                "Trace": 1
            },
            "ModeAutoPan": {
                "ActivationTime": "5s",
                "AutoPanSpeed": 20,
                "AutoPanTiltAngle": 20
            },
            "ModeSchedule": {
                "ActivationTime": "5s",
                "Schedules": [
                    {
                        "FromTo": "SUN 00:00-MON 20:00",
                        "ScheduleMode": "Home"
                    },
                    {
                        "FromTo": "MON 20:00-WED 19:00",
                        "ScheduleMode": "Preset",
                        "Preset": 1
                    },
                    {
                        "FromTo": "WED 19:00-SAT 24:00",
                        "ScheduleMode": "Home"
                    }
                ]
            }
        }
    ]
}
```

156.5.4.6. Configuring Auto run schedule for Tuesday 2AM to Wednesday 9AM with the Group mode

Mode must be set to Schedule first, using the below example.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=update&FromTo=TUE 02:00-WED
09:00&ScheduleMode=Group&Group=1
```

After the above example applied, the view action will return the followings;

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=autorun&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Mode=Schedule
Channel.0.ActivationTime=5s
Channel.0.Schedule.1.FromTo=SUN 00:00-MON 20:00
Channel.0.Schedule.1.ScheduleMode=Home
Channel.0.Schedule.2.FromTo=MON 20:00-TUE 02:00
Channel.0.Schedule.2.ScheduleMode=Preset
Channel.0.Schedule.2.Preset=1
Channel.0.Schedule.3.FromTo=TUE 02:00-WED 09:00
Channel.0.Schedule.3.ScheduleMode=Group
Channel.0.Schedule.3.Group=1
Channel.0.Schedule.4.FromTo=WED 09:00-WED 19:00
Channel.0.Schedule.4.ScheduleMode=Preset
Channel.0.Schedule.4.Preset=1
Channel.0.Schedule.5.FromTo=WED 19:00-SAT 24:00
Channel.0.Schedule.5.ScheduleMode=Home
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "AutoRun": [
        {
            "Channel": 0,
            "Mode": "Schedule",
            "ModeHome": {
                "ActivationTime": "5s"
            },
            "ModePreset": {
                "ActivationTime": "5s",
                "Preset": 1
            },
            "ModeSwing": {
                "ActivationTime": "30s",
                "SwingMode": "Pan"
            },
            "ModeGroup": {
                "ActivationTime": "5s",
                "Group": 1
            },
            "ModeTour": {
                "Tour": 1,
                "ActivationTime": "5s"
            },
            "ModeTrace": {
                "ActivationTime": "5s",
                "Trace": 1
            },
            "ModeAutoPan": {
                "ActivationTime": "5s",
                "AutoPanSpeed": 20,
                "AutoPanTiltAngle": 20
            },
            "ModeSchedule": {
                "ActivationTime": "5s",
                "Schedules": [
                    {
                        "FromTo": "SUN 00:00-MON 20:00",
```

```
                        "ScheduleMode": "Home"
                    },
                    {
                        "FromTo": "MON 20:00-TUE 02:00",
                        "ScheduleMode": "Preset",
                        "Preset": 1
                    },
                    {
                        "FromTo": "TUE 02:00-WED 09:00",
                        "ScheduleMode": "Group",
                        "Group": 1
                    },
                    {
                        "FromTo": "WED 09:00-WED 19:00",
                        "ScheduleMode": "Preset",
                        "Preset": 1
                    },
                    {
                        "FromTo": "WED 19:00-SAT 24:00",
                        "ScheduleMode": "Home"
                    }
                ]
            }
        }
    ]
}
```

156.6. Home Position Setup

156.6.1. Description

The **home** submenu of **ptzconfig.cgi** sets the current preset position to the Home position.

NOTE This chapter applies to network cameras only.

To find out whether home position is supported by the device or not, refer to the Attributes/PTZSupport/Support/Home attribute in the device attributes section.

Access level

Action	Camera
set	Suser

156.6.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
home&action=<value>[&<parameter>=<value>]
```

156.6.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ, RES	<int>	Channel ID
	ViewModeIndex	REQ	<int>	View mode of index number

156.6.4. Examples

156.6.4.1. Setting the current position to the Home position

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=home&action=set
```

156.7. Preset Configuration

156.7.1. Description

The **preset** submenu of **ptzconfig.cgi** configures the preset number and name.

NOTE To find out whether preset is supported by the device or not, refer to the Attributes/PTZSupport/Support/Preset attribute in the device attributes section.

Access level

Action	Camera	NVR
view	Suser	User
add, update	Suser	User
remove	Suser	User

156.7.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
preset&action=<value>[&<parameter>=<value>]
```

156.7.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the preset position settings.
	Channel	REQ, RES	<csv>	Channel ID
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY
add/update	Channel	REQ, RES	<int>	Channel ID
	Preset	REQ, RES	<int>	Preset of index number **Note** For the **add** or **update** action, **Preset** and **Name** must be sent together.
	Name	REQ, RES	<string>	Preset name
	SubViewIndex	REQ	<int>	Sub view of index number This parameter is valid only when **ViewMode.#.Type** of **image.cgi** is set to Quadview.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY
	DisablePTZMotion	REQ, RES	<bool>	Sets the PTZ not to move when control presets.
remove	Channel	REQ	<int>	Channel ID
	Preset	REQ	<csv>	Preset of index number **Note** **Preset** must be sent together with the **remove** action.
	ViewModeIndex	REQ	<int>	View mode of index number CAMERA ONLY

156.7.4. Examples

156.7.4.1. Getting the current preset information

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=preset&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Preset.1.Name=test1
Channel.0.Preset.2.Name=test2
Channel.1.Preset.1.Name=test3
Channel.1.Preset.2.Name=test4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZPresets": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "Name": "test1"
                },
                {
                    "Preset": 2,
                    "Name": "test2"
                }
            ]
        },
        {
            "Channel": 1,
            "Presets": [
                {
                    "Preset": 1,
                    "Name": "test3"
                },
                {
```

```
                    "Preset": 2,
                    "Name": "test4"
                }
            ]
        }
    ]
}
```

156.7.4.2. Getting the preset information for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=preset&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Preset.1.Name=Preset1
Channel.0.Preset.2.Name=Preset2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
{
    "PTZPresets": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "Name": "Preset1"
                },
                {
```

```
                    "Preset": 2,
                    "Name": "Preset2"
                }
            ]
        }
    ]
}
```

156.7.4.3. Adding ‘Preset 1’ with the name ‘preset001’

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=preset&action=add&Preset=1&Name=preset001
```

156.7.4.4. Adding ‘Preset 3’ with the name ‘preset003’ to Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=preset&action=add&Channel=0&Preset=3&Name=Preset0
03
```

156.7.4.5. Removing presets 1 and 3

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=preset&action=remove&Channel=0&Preset=1,3
```

156.7.4.6. Removing all presets at once

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=preset&action=remove&Channel=0&Preset=All
```

156.8. Preset Image Configuration

156.8.1. Description

The **presetimageconfig** submenu of **ptzconfig.cgi** configures the camera image settings according to the preset selected.

NOTE This chapter applies to network cameras only.

To find out whether preset functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Preset attribute in the device attributes section.

Access level

Action	Camera
view	Suser
set	Suser
check	Suser

156.8.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
presetimageconfig&action=<value>[&<parameter>=<value>]
```

156.8.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads camera image settings.
	Channel	REQ, RES	<csv>	Channel ID
	Preset	REQ, RES	<csv>	Preset number of the channel **Note** To use the **view** action, the **Channel** and **Preset** parameters must be sent together.
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Preset	REQ, RES	<int>	Preset number **Note** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer to "Attributes/PTZSupport/Limit/Max Preset" attribute in the device attributes section. **Note** **Preset** must be sent together with the **set** action.
	SSDREnable	REQ, RES	<bool> True, False	Enables or disables SSDR (Samsung Super Dynamic Range), which regulates the overall brightness by increasing the brightness of the dark area alone in a scene where the gap between brightness and darkness widens.
	SSDRLevel	REQ, RES	<int>	SSDR level **SSDRLevel** is valid only when **SSDREnable** is set to True.
	DynamicRange	REQ, RES	<enum> Narrow, Wide	SSDR dynamic range **DynamicRange** is valid only when **SSDREnable** is set to True.

<table>
<tr><th>Action</th><th>Parameters</th><th>Request/ Response</th><th>Type/ Value</th><th>Description</th></tr>
<tr><td></td><td>WhiteBalanceMode</td><td>REQ, RES</td><td><enum> ATW, Manual, Outdoor, Indoor, Mercury, Sodium</td><td>White balance mode
ATW: Automatically corrects the colors of the camera video
Manual: Manually adjusts the red and blue gains of the camera video
Outdoor: Automatically corrects the video colors of the camera to be optimized for the outdoor environment
Indoor: Automatically corrects the video colors of the camera to be optimized for the indoor environment.
Mercury: Automatically corrects the video colors of the camera to be optimized to a mercury lamp environment.
Sodium: Automatically corrects the video colors of the camera to be optimized to a sodium lamp environment.</td></tr>
<tr><td></td><td>WhiteBalanceManualRedLevel</td><td>REQ, RES</td><td><int></td><td>White balance red level

WhiteBalanceManualRedLevel is valid only when WhiteBalanceMode is set to Manual.</td></tr>
<tr><td></td><td>WhiteBalanceManualBlueLevel</td><td>REQ, RES</td><td><int></td><td>White balance blue level

WhiteBalanceManualBlueLevel is valid only when WhiteBalanceMode is set to Manual.</td></tr>
<tr><td></td><td>CompensationMode</td><td>REQ, RES</td><td><enum> WDR, HLC, BLC, Off</td><td>Compensation mode
WDR: Wide Dynamic Range
HLC: High Light Compensation
BLC: Backlight Compensation
Off: Disables the compensation function</td></tr>
</table>

Action	Parameters	Request/ Response	Type/ Value	Description
	BLCLevel	REQ, RES	<enum> Low, Medium, High	BLC (Backlight Compensation) level **BLCLevel** is valid only when **CompensationMode** is set to BLC.
	BLCAreaTop	REQ	<int>	Top position info for BLC region **BLCAreaTop** is valid only when **CompensationMode** is set to BLC. It is a request only parameter for PTZ models.
	BLCAreaBottom	REQ	<int>	Bottom position info for BLC region **BLCAreaBottom** is valid only when **CompensationMode** is set to BLC. It is a request only parameter for PTZ models.
	BLCAreaLeft	REQ	<int>	Left position info for BLC region **BLCAreaLeft** is valid only when **CompensationMode** is set to BLC. It is a request only parameter for PTZ models.
	BLCAreaRight	REQ	<int>	Right position info for BLC region **BLCAreaRight** is valid only when **CompensationMode** is set to BLC. It is a request only parameter for PTZ models.
	HLCMode	REQ, RES	<enum> NightOnly, AllDay	HLC (High Light Compensation) mode **HLCMode** is valid only when **CompensationMode** is set to HLC.
	HLCLevel	REQ, RES	<enum> Low, Medium, High	HLC level **HLCLevel** is valid only when **Compensation Mode** is set to HLC.

Action	Parameters	Request/ Response	Type/ Value	Description
	HLCMaskTone	REQ, RES	<int>	Tone of the mask used for high light compensation **HLCMaskTone** is valid only when **CompensationMode** is set to HLC.
	HLCMaskColor	REQ, RES	<enum>	Color of the mask used for high light compensation area **HLCMaskColor** is valid only when **CompensationMode** is set to HLC.
	WDRLimit	REQ, RES	<int>	WDR (Wide Dynamic Range) limit **WDRLimit** is valid only when **CompensationMode** is set to WDR.
	WDRLevel	REQ, RES	<enum> Low, Medium, High	WDR sensitivity level **WDRLevel** is valid only when **CompensationMode** is set to WDR.
	WDRBlackLevel	REQ, RES	<enum> Low, Medium, High	WDR black level **WDRBlackLevel** is valid only when **CompensationMode** is set to WDR.
	WDRWhiteLevel	REQ, RES	<enum> Low, Medium, High	WDR white level **WDRWhiteLevel** is valid only when **CompensationMode** is set to WDR.
	Brightness	REQ, RES	<int>	Brightness level
	ShutterMode	REQ, RES	<enum> ESC, Manual, AFLK	Shutter mode • ESC: Electronic shutter control. Automatically adjusts the shutter speed according to the ambient brightness • Manual: Manually adjusts the shutter speed of the camera • AFLK: Anti flicker. Adjusts the shutter speed when a screen flickers due to frequent mismatches of the ambient lighting.

Action	Parameters	Request/ Response	Type/ Value	Description
	AFLKMode	REQ, RES	<enum> Off, On	Anti-flicker mode
	ManualShutterSpeed	REQ, RES	<enum>	Shutter speed **ManualShutterSpeed** is valid only when **ShutterMode** is set to Manual, and **CompensationMode** is set to Off.
	SSNREnable	REQ, RES	<bool> True, False	Enables or disables SSNR (Samsung Super Noise Reduction)
	SSNRLevel	REQ, RES	<int>	SSNR level **SSNRLevel** is valid only when **SSNREnable** is set to True.
	SSNRMode	REQ, RES	<enum> Off, Manual, Auto	SSNRMode mode • Off: Off SSNR feature. • Manual: Manually adjusts the SSNR. • Auto: Automatically adjusts the SSNR if a moving object is detected.
	SSNR2DLevel	REQ, RES	<int>	Adjust level of 2DNR to reduce noise. **SSNR2DLevel** is valid only when SSNR is not Off.
	SSNR3DLevel	REQ, RES	<int>	Adjust level of 3DNR to reduce noise. **SSNR3DLevel** is valid only when SSNR is not Off.
	LDCEnable	REQ, RES	<bool> True, False	Enables or disables LDC (Lens Distortion Control). **LDCEnable** and **LDCMode** cannot be set at the same time. **LDCEnable** True is same as Manual LDCMode. **LDCEnable** False is same as Off LDCMode.

Action	Parameters	Request/ Response	Type/ Value	Description
	LDCMode	REQ, RES	<enum> Off, Auto, Manual	Provides option to control the LDC feature. • Off: Disables LDC function • Auto: Automatically adjusts the LDC • Manual: Manually sets the LDC level
	LDCLevel	REQ, RES	<int>	LDC Level The greater the value, the higher the LDC level. **LDCLevel** is valid only when **LDCMode** is not set to Off. **LDCLevel** is valid only when **LDCEnable** is not set to False.
	SensupMode	REQ, RES	<enum> Off, Auto	Sens-up mode • Off: Disables sens-up feature • Auto: Automatically senses the darkness level in a low contrast scene and extends the accumulation time according to the bright and sharp image.
	SensupLevel	REQ, RES	<enum>	Sens-up level **SensupLevel** is valid only when **SensUpMode** is set to Auto.
	IrisMode	REQ, RES	<enum> Auto, Manual	Lens Iris mode • Auto: Automatically adjusts the iris. • Manual: Manually adjusts the iris and focus.
	IrisFno	REQ, RES	<enum>	Iris F-stop number **IrisFno** is valid only when **IrisMode** is set to Manual.

Action	Parameters	Request/ Response	Type/ Value	Description
	AGCMode	REQ, RES	<enum> Off, Low, Medium, High, Manual	AGC (Automatic Gain Control) mode. Adjusts the gain value of the video to control the video brightness. • Off: Disables AGC. • Low/ Medium /High: As the level of the screen increases to the high and brighter level in a low lighting condition. • Manual: Manually adjusts the AGC level
	AGCLevel	REQ, RES	<int>	AGC level **AGCLevel** is valid only when **AGCMode** is set to Manual.
	DISEnable	REQ, RES	<bool> True, False	Enables or disables DIS (Digital Image Stabilization) Automatically compensates the image when the camera vibrates due to the external factors including wind
	OISEnable	REQ, RES	<bool> True, False	Enables or disables OIS (Optical Image Stabilization).
	OISZoomThreshold	REQ, RES	<int>	OIS zoom threshold.
	DefogMode	REQ, RES	<enum> Off, Auto, Manual	Defogging mode
	DefogLevel	REQ, RES	<int>	Defogging level **DefogLevel** is valid only when **DefogMode** is set to Manual.
	DayNightMode	REQ, RES	<enum> Color, BW, Auto	Day and night mode • Color: Always outputs in color • BW: Always outputs in black and white. • Auto: Normally outputs in color but black and white under low luminance at night.

Action	Parameters	Request/ Response	Type/ Value	Description
	DayNightSwitchingTime	REQ, RES	<enum> 5s, 7s, 10s, 15s, 20s, 30s, 40s, 60s	Duration of switch between day and night mode **DayNightSwitchingTime** is valid only when **DayNightMode** is set to Auto.
	DayNightSwitchingMode	REQ, RES	<enum> Slow, Fast	Interval of switch between day and night mode **DayNightSwitchingMode** is valid only when **DayNightMode** is set to Auto.
	SharpnessEnable	REQ, RES	<bool> True, False	Enable or disables image sharpness feature
	SharpnessLevel	REQ, RES	<int>	Sharpness level **SharpnessLevel** is valid only when **SharpnessEnable** is set to True.
	Saturation	REQ, RES	<int>	Saturation level
	AfterAction	REQ, RES	<enum> Off, AutoRun, Tracking, VideoAnalytics, HomeAfterTracking, AutoRunAfterTracking, Washer, GoToHome	Action after the corresponding preset e.g. when **AfterAction** is set to Tracking, the device automatically starts tracking when it moves to a certain preset.
	AfterActionTrackingTime	REQ, RES	<enum> 10s, 20s, 30s, 40s, 50s, 1m, 2m, m, 4m, 5m, 10m	Tracking time where auto tracking is executed as an after action **AfterActionTrackingTime** is valid only when **AfterAction** is set to Tracking.

Action	Parameters	Request/ Response	Type/ Value	Description
	FocusMode	REQ, RES	<enum> Manual, Auto, OneShotAutoFocus	Focus mode • Manual: Manually adjusts the focus. • Auto: Automatically adjust the focus according to the zoom factor • OneShotAutoFocus: Adjust the focus for one shot
	AutoFocusRange	REQ, RES	<enum> Wide, Narrow	Setting AutoFocus to target for Wide or Narrow view.
	DigitalZoomEnable	REQ, RES	<bool> True, False	Enables or disables the digital zoom.
	MaxDigitalZoom	REQ, RES	<enum>	Digital zoom limit **MaxDigitalZoom** is valid only when **DigitalZoomEnable** is set to True.
	OpticalDefogFilterEnable	REQ, RES	<bool>	Enables or disables the optical defog filter setting.
	XCEEnable	REQ, RES	<bool>	Enables XCE(eXternd Contrast Enhancement) This feature is similar to using unsharp mask filtering
	XCELevel	REQ, RES	<int>	Sets XCE filter level
	GammaControl	RES	<enum> Off, On	Gamma Control Adjust the brightness of the image. When On is selected, dark areas are generally brighter.
check	Channel	REQ, RES	<int>	
	CurrentType	REQ, RES	<enum> Global, Preset	
	CurrentPresetIndex	REQ, RES	<int>	Visible only when CurrentType is set to Preset

156.8.4. Examples

156.8.4.1. Getting the current preset image information of preset 1 in Channel 0

To use the **view** action, the channel number and preset number must be set together.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=view&Channel=0&Preset=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Preset.1.SSDREnable=True
Channel.0.Preset.1.SSDRLevel=8
Channel.0.Preset.1.WhiteBalanceMode=ATW
Channel.0.Preset.1.WhiteBalanceManualRedLevel=138
Channel.0.Preset.1.WhiteBalanceManualBlueLevel=140
Channel.0.Preset.1.Brightness=50
Channel.0.Preset.1.AFLKMode=Off
Channel.0.Preset.1.SSNREnable=True
Channel.0.Preset.1.SSNRLevel=2
Channel.0.Preset.1.IrisMode=Auto
Channel.0.Preset.1.IrisFno=F1.6
Channel.0.Preset.1.AGCMode=High
Channel.0.Preset.1.AGCLevel=0
Channel.0.Preset.1.DISEnable=False
Channel.0.Preset.1.DayNightMode=Auto
Channel.0.Preset.1.DayNightSwitchingTime=5s
Channel.0.Preset.1.DayNightSwitchingMode=Slow
Channel.0.Preset.1.SharpnessEnable=True
Channel.0.Preset.1.SharpnessLevel=24
Channel.0.Preset.1.Saturation=50
Channel.0.Preset.1.DefogMode=Off
Channel.0.Preset.1.DefogLevel=5
Channel.0.Preset.1.AfterAction=VideoAnalytics
Channel.0.Preset.1.AfterActionTrackingTime=10s
Channel.0.Preset.1.FocusMode=OneShotAutoFocus
Channel.0.Preset.1.DigitalZoomEnable=False
Channel.0.Preset.1.MaxDigitalZoom=2x
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PresetImageConfig": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "SSDREnable": true,
                    "SSDRLevel": 8,
                    "WhiteBalanceMode": "ATW",
                    "WhiteBalanceManualRedLevel": 138,
                    "WhiteBalanceManualBlueLevel": 140,
                    "Brightness": 50,
                    "AFLKMode": "Off",
                    "SSNREnable": true,
                    "SSNRLevel": 2,
                    "IrisMode": "Auto",
                    "IrisFno": "F1.6",
                    "AGCMode": "High",
                    "AGCLevel": 0,
                    "DISEnable": false,
                    "DayNightMode": "Auto",
                    "DayNightSwitchingTime": "5s",
                    "DayNightSwitchingMode": "Slow",
                    "SharpnessEnable": true,
                    "SharpnessLevel": 24,
                    "Saturation": 50,
                    "DefogMode": "Off",
                    "DefogLevel": 5,
                    "AfterAction": "VideoAnalytics",
                    "AfterActionTrackingTime": "10s",
                    "FocusMode": "OneShotAutoFocus",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                }
```

```
            ]
        }
    ]
}
```

156.8.4.2. Setting SSDR level in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=1&S
SDREnable=True&SSDRLevel=1
```

156.8.4.3. Setting white balance in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=1&W
hiteBalanceMode=Manual&WhiteBalanceManualRedLevel=1&WhiteBalanceManualBlueLe
vel=1
```

156.8.4.4. Setting compensation mode to BLC in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=1&C
ompensationMode=BLC
```

156.8.4.5. Setting BLC level to medium in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=1&C
ompensationMode=BLC&BLCLevel=Medium
```

156.8.4.6. Setting day and night switching time in Preset 2

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=2&D
```

```
ayNightMode=Auto&DayNightSwitchingTime=5s
```

156.8.4.7. Setting focus mode to auto in Preset 2

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetimageconfig&action=set&Channel=0&Preset=2&F
ocusMode=Auto
```

156.9. Preset Video Analysis Setup

156.9.1. Description

The **presetvideoanalysis** submenu of **ptzconfig.cgi** configures the video analysis settings of the preset selected.

NOTE This chapter applies to network cameras only.

To find out whether preset functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Preset attribute in the device attributes section.

Access level

Action	Camera
view	Suser
set	Suser
remove	Suser

156.9.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
presetvideoanalysis&action=<value>[&<parameter>=<value>]
```

156.9.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads video analysis settings.
	Channel	REQ, RES	<csv>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Preset	REQ, RES	<csv>	Preset number **Note** To use the **view** action, the **Channel** and **Preset** parameters must be sent together.
set	Channel	REQ, RES	<int>	Channel ID
	Preset	REQ, RES	<int>	Preset number **NOTE** The number of presets supported is dependent on the device; To find out whether max presets is supported by the device, please refer "Attributes/PTZSupport/Limit/MaxPreset" attribute in the device attributes section. **Note** To use the **set** action, **Preset**, **DefinedArea.#.Mode**, and **DefinedArea.#.Coordinate** must be sent together to set the user defined area; then **Preset**, **Line.#.Mode** and **Line.#.Coordinate** must be sent together to set the line.
	DetectionType	REQ, RES	<enum> MotionDetection, IntelligentVideo, Off	Detection type • MotionDetection: Detects motion within the specified area. • IntelligentVideo: Detects objects that are appearing, disappearing, entering, exiting, changing their scenes, etc within the specified area. • Off: No detection.

Action	Parameters	Request/ Response	Type/ Value	Description
	Sensitivity	REQ, RES	<enum> VeryLow, Low, Medium, High, VeryHigh	Sensitivity level **Sensitivity** is valid only when **DetectionType** is NOT set to Off.
	MinimumObjectSize	REQ, RES	<string>	Minimum size of objects that are detected by motion detection The size is specified in the format of <width, height>. The value of **MinimumObjectSize** should be less than the value of **MaximumObjectSize**. **MinimumObjectSize** is valid only when **DetectionType** is Notset to Off.
	MaximumObjectSize	REQ, RES	<string>	Maximum size of object that are detected by motion detection The size is specified in the format of <width, height>. The value of **MaximumObjectSize** should be greater than the value of **MinimumObjectSize**. **MaximumObjectSize** is valid only when **DetectionType** is NOTset to Off.
	MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum size of objects that are detectable by motion detection (in pixels) The size is specified in the format of <width, height>, where the value of **MinimumObjectSizeInPixels** should be less than the value of **MaximumObjectSizeInPixels**. **MinimumObjectSizeInPixels** is valid only when **DetectionType** is NOTset to Off.

Action	Parameters	Request/ Response	Type/ Value	Description
	MaximumObjectSizeInPixels	REQ, RES	<string>	Maximum size of object that are detectable by motion detection. The size is specified in the format of <width, height>, where the value of **MaximumObjectSizeInPixels** should be greater than the value of **MinimumObjectSizeInPixels**. **MaximumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Coordinate	REQ, RES	<string>	ROI (Region of Interest) coordinates **ROI.#.Coordinate** is valid only when **DetectionType** is NOTset to Off.
	ROIMode	REQ, RES	<enum> Inside, Outside	ROI detection mode • Inside: Detects motion within the specified ROI • Outside: Detects motion outside the specified ROI **ROIMode** is valid only when **DetectionType** is NOT set to Off.
	IVRuleType	REQ, RES	<enum> Passing, EnterExit, AppearDisappear	Rule type for IntelligentVideo **IVRuleType** is valid only when **DetectionType** is NOT set to Off.

Action	Parameters	Request/ Response	Type/ Value	Description
	DefinedArea.#.Mode	REQ, RES	<csv> Off, AppearDisappear, Entering, Exiting	Defined virtual area detection mode • Off: Disables virtual area detection mode • AppearDisappear: Detects objects that are appearing or disappearing in the specified virtual area. • Entering: Detects objects that are entering the specified virtual area. • Exiting: Detects objects exiting the specified virtual area. When **DetectionType** is set to IntelligentVideo, **DefinedArea.#.Coordinate** should be sent along with **DefinedArea.#.Mode**.
	DefinedArea.#.Coordinate	REQ, RES	<string>	Top left and bottom right verticles of the defined virtual area for motion detection When **DetectionType** is set to IntelligentVideo, **DefinedArea.#.Mode** should be sent along with **DefinedArea.#.Coordinate**.
	EntireAreaMode	REQ, RES	<enum> Off, AppearDisappear, Scenechange	Entire area detection mode • Off: Disables the entire area detection mode • AppearDisappear: Detects objects appearing or disappearing in the entire area • Scenechange: Detects scene change events, which are triggered when a large portion of the scene is changed. **EntireAreaMode** is valid only when **DetectionType** is set to IntelligentVideo.

Action	Parameters	Request/ Response	Type/ Value	Description
	DetectionResultOverlay	REQ, RES	<bool> True, False	Enables or disables detection result overlay **DetectionResultOverlay** is valid only when **DetectionType** is NOT set to Off.
	DisplayRules	REQ, RES	<bool> True, False	Enables or disables the rule display **DisplayRules** is valid only when **DetectionType** is set to IntelligentVideo.
	Line.#.Mode	REQ, RES	<csv> LeftSide, RightSide	Line detection mode • LeftSide: Detects motion to the left of the virtual line. • RightSide: Detects motion to the right of the virtual line. When **DetectionType** is set to IntelligentVideo. **Line.#.Coordinate** should be sent along with **Line.#.Mode**.
	Line.#.Coordinate	REQ, RES	<string>	X and Y coordinates of the two points which define the virtual line When **DetectionType** is set to IntelligentVideo, **Line.#.Mode** should be sent along with **Line.#.Coordinate**.
remove	Channel	REQ	<int>	Channel ID
	Preset	REQ	<int>	Preset number **Note** **Preset** must be sent together with the **remove** action.
	ROIIndex	REQ	<csv>	Index of the ROI
	LineIndex	REQ	<csv>	Index of the virtual line
	DefinedAreaIndex	REQ	<csv>	Index of the virtual area

156.9.4. Examples

156.9.4.1. Getting the video analysis setting information of Preset 1 for Channel 0

To use the **view** action, the channel number and preset number must be set together.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=view&Channel=0&Preset=
1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Preset.1.DetectionType=Off
Channel.0.Preset.1.Sensitivity=Medium
Channel.0.Preset.1.MinimumObjectSize=4,7
Channel.0.Preset.1.MaximumObjectSize=50,89
Channel.0.Preset.1.MinimumObjectSizeInPixels=96,96
Channel.0.Preset.1.MaximumObjectSizeInPixels=972,972
Channel.0.Preset.1.IVRuleType=Passing
Channel.0.Preset.1.EntireAreaMode=Off,
Channel.0.Preset.1.DetectionResultOverlay=False,
Channel.0.Preset.1.DisplayRules=False,
Channel.0.Preset.1.ROIMode=Inside
Channel.0.Preset.1.ROI.1.Coordinate=189,291,708,624
Channel.0.Preset.1.ROI.2.Coordinate=1188,180,1683,393
Channel.0.Preset.1.ROI.3.Coordinate=924,684,1581,882
Channel.0.Preset.1.ROI.4.Coordinate=90,837,639,990
Channel.0.Preset.1.Line.1.Mode=RightSide
Channel.0.Preset.1.Line.1.Coordinate=1302,141,1302,774
Channel.0.Preset.1.DefinedArea.2.Mode=AppearDisappear
Channel.0.Preset.1.DefinedArea.2.Coordinate=588,402,1785,972
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PresetVideoAnalysis": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "DetectionType": "Off",
                    "Sensitivity": "Medium",
                    "MinimumObjectSize": "4,7",
                    "MaximumObjectSize": "50,89",
                    "MinimumObjectSizeInPixels": "96,96",
                    "MaximumObjectSizeInPixels": "972,972",
                    "IVRuleType": "Passing",
                    "EntireAreaMode": "Off",
                    "DetectionResultOverlay": false,
                    "DisplayRules": false,
                    "ROIMode": "Inside",
                    "ROIs": [
                        {
                            "ROI": 1,
                            "Coordinate": [
                                {
                                    "x": 189,
                                    "y": 291
                                },
                                {
                                    "x": 708,
                                    "y": 624
                                }
                            ]
                        },
                        {
                            "ROI": 2,
                            "Coordinate": [
                                {
                                    "x": 1188,
                                    "y": 180
                                },
                                {
                                    "x": 1683,
```

```
                            "y": 393
                        }
                    ]
                },
                {
                    "ROI": 3,
                    "Coordinate": [
                        {
                            "x": 924,
                            "y": 684
                        },
                        {
                            "x": 1581,
                            "y": 882
                        }
                    ]
                },
                {
                    "ROI": 4,
                    "Coordinate": [
                        {
                            "x": 90,
                            "y": 837
                        },
                        {
                            "x": 639,
                            "y": 990
                        }
                    ]
                }
            ],
            "Lines": [
                {
                    "Line": 1,
                    "Mode": "RightSide",
                    "Coordinate": [
                        {
                            "x": 1302,
                            "y": 141
                        },
                        {
```

```
                                "x": 1302,
                                "y": 774
                            }
                        ]
                    }
                ],
                "DefinedAreas": [
                    {
                        "DefinedArea": 2,
                        "Mode": "AppearDisappear",
                        "Coordinate": [
                            {
                                "x": 588,
                                "y": 402
                            },
                            {
                                "x": 1785,
                                "y": 972
                            }
                        ]
                    }
                ]
            }
        ]
    }
  ]
}
```

156.9.4.2. Setting the sensitivity level for motion detection to high in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=set&Channel=0&Preset=1
&DetectionType=MotionDetection&Sensitivity=High
```

156.9.4.3. Setting the minimum and maximum object size in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=set&Channel=0&Preset=1
```

```
&DetectionType=MotionDetection&MinimumObjectSize=5,5&MaximumObjectSize=10,10
```

156.9.4.4. Setting the detection mode and coordinates of defined area 1 in Preset 2

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=set&Channel=0&Preset=2&DetectionType=IntelligentVideo&DefinedArea.1.Mode=AppearDisappear&DefinedArea.1.Coordinate=441,231,429,912,1200,933,1100,177
```

156.9.4.5. Setting the detection mode and coordinates of line 1 in Preset 3

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=set&Channel=0&Preset=3&DetectionType=IntelligentVideo&Line.1.Mode=RightSide&Line.1.Coordinate=650,750,622,410
```

156.9.4.6. Removing the defined area of index 1 in Preset 1

To use the **remove** action, the **Preset** parameter must be set.

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=remove&Channel=0&Preset=1&DefinedAreaIndex=1
```

156.9.4.7. Removing line index 1 in Preset 1

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis&action=remove&Channel=0&Preset=1&LineIndex=1
```

156.10. Preset Video Analysis 2 Setup

156.10.1. Description

The **presetvideoanalysis2** submenu is similar to the **presetvideoanalysis** submenu, but allows the operator to configure the parameters for each area/ROI.

NOTE This chapter applies to network cameras only.

To find out whether preset functionality is supported by the device or not, refer to the Attributes/PTZSupport/Support/Preset attribute in the device attributes section.

Access level

Action	Camera
view	Suser
set	Suser
remove	Suser

156.10.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
presetvideoanalysis2&action=<value>[&<parameter>=<value>]
```

156.10.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads video analysis settings.
	Channel	REQ, RES	<csv>	Channel ID
	Preset	REQ, RES	<csv>	Preset number **Note** To use the **view** action, the **Channel** and **Preset** parameters must be sent together.
	DetectionType	REQ	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type **DetectionType** for the **view** action requests the current preset video analysis settings and does not change the configured detection type.
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Preset	REQ, RES	<int>	Preset number **Note** The number of presets supported depends on the device. To find out whether max presets is supported by the device, please refer to the "Attributes/PTZSupport/Limit/MaxPreset" attribute in the device attributes section. **Note** To use the **set** action, **Preset**, **DefinedArea.#.Mode**, and **DefinedArea.#.Coordinate** must be sent together to set the user-defined area; then **Preset**, **Line.#.Mode** and **Line.#.Coordinate** must be sent together to set the line.
	DetectionType	REQ, RES	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type • MotionDetection: Detects motion within the specified area. • IntelligentVideo: Detects objects that are appearing, disappearing, entering, exiting, changing their scenes, etc. within the specified area. • Off: No detection.
	SensitivityLevel	REQ, RES	<int>	Sensitivity level for IntelligentVideo. **Sensitivity** is valid only when **DetectionType** is NOT set to Off.

Action	Parameters	Request/ Response	Type/ Value	Description
	DetectionType.#.MinimumObjectSize	REQ, RES	<string>	Minimum size of objects detectable by motion detection. Objects smaller than the specified minimum size are not detected. The size is specified in the format of <width, height>. The value of **MinimumObjectSize** must be less than the value of **MaximumObjectSize**. **MinimumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	DetectionType.#.MaximumObjectSize	REQ, RES	<string>	Maximum size of objects detectable by motion detection Objects bigger than the maximum size are not detected. The size is specified in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**. **MaximumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	DetectionType.#.MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum object size in pixels The size is specified in the format of <width, height>. **MinimumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.

Action	Parameters	Request/ Response	Type/ Value	Description
	DetectionType.#.MaximumObjectSizeInPixels	REQ, RES	<string>	Maximum object size in pixels The size is specified in the format of <width, height>. **MaximumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Coordinate	REQ, RES	<string>	ROI (Region of Interest) coordinates **ROI.#.Coordinate** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Mode	REQ, RES	<enum> Inside, Outside	ROI detection mode • Inside: Detects motion within the specified ROI • Outside: Detects motion outside the specified ROI **ROIMode** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Duration	REQ, RES	<int>	Event activation duration in seconds for ROI.
	ROI.#.SensitivityLevel	REQ, RES	<int>	Sensitivity level for Motion Detection
	ROI.#.ThresholdLevel	REQ, RES	<int>	Threshold level for Motion Detection
	ROI.#.HandoverIndex	REQ, RES	<int>	Handover index for designated ROI

Action	Parameters	Request/ Response	Type/ Value	Description
	DefinedArea.#.Mode	REQ, RES	<csv> Off, AppearDisappear, Entering, Exiting, Intrusion, Loitering	Defined virtual area detection mode • AppearDisappear: Detects objects appearing or disappearing in the specified virtual area. • Entering: Detects objects entering the specified virtual area. • Exiting: Detects objects exiting the specified virtual area. • Intrusion: Detects objects moving inside of the specified virtual Area • Loitering: Detects objects loitering more than threshold time in the specified virtual area. **DefinedArea.#.Mode** is valid only when **DetectionType** is set to IntelligentVideo. **Note** **DefinedArea.#.Mode**, **DefinedArea.#.Coordinate**, **Line.#.Mode**, and **Line.#.Coordinate** must be sent together with the **set** action.
	DefinedArea.#.Coordinate	REQ, RES	<string>	Top left and bottom right vertices of the defined virtual area for motion detection **DefinedArea.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. **DefinedArea.#.Mode** must be sent together with **DefinedArea.#.Coordinate**.
	DefinedArea.#.Type	REQ, RES	<enum> Inside, Outside	Define Area type • Inside: Detects video analytics within the specified area • Outside: Detects video analytics outside the specified area

Action	Parameters	Request/ Response	Type/ Value	Description
	DefinedArea.#.AppearanceDuration	REQ, RES	<int>	Appearance Duration in seconds
	DefinedArea.#.LoiteringDuration	REQ, RES	<int>	Loitering Duration in seconds
	DefinedArea.#.RuleName	REQ,RES	<string>	Rule name.
	DefinedArea.#.ObjectTypeFilter	REQ,RES	<csv> Vehicle, Person, Face	If set, only for the configured object types, rule will be triggered, this parameter is supported only if AI based IVA is supported.
	DefinedArea.#.ObjectTypeFilterDetails	REQ,RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck	If detailed filter support is needed, this parameter can be configured.
	DefinedArea.#.IntrusionDuration	REQ, RES	<int>	Intrusion duration in seconds for designated defined area
	DefinedArea.#.HandoverIndex	REQ, RES	<int>	Handover index for designated Defined Area
	EntireAreaMode	REQ, RES	<enum> Off, AppearDisappear, Scenechange	Entire area detection mode • Off: Disables the entire area detection mode • AppearDisappear: Detects objects appearing or disappearing in the entire area • Scenechange: Detects scene change events, which are triggered when a large portion of the scene is changed. **EntireAreaMode** is valid only when **DetectionType** is set to IntelligentVideo.

Action	Parameters	Request/ Response	Type/ Value	Description
	Line.#.Mode	REQ, RES	<csv> Left, Right, BothDirections ,Off	Line detection mode • Left: Detects motion to the left of the virtual line. • Right: Detects motion to the right of the virtual line. • BothDirections: Detects motion on both sides of line • Off : No event will be detected **Line.#.Mode** is valid only when **DetectionType** is set to IntelligentVideo. The **Line.#.Mode** parameter must be sent along with **Line.#.Coordinate**.
	Line.#.Coordinate	REQ, RES	<string>	X and Y coordinates of the two points which define the virtual line The coordinates are specified in the form of <x1,y1,x2,y2>; x1 and y1 are the start points and x2 and y2 are the end points. **Line.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. The **Line.#.Coordinate** parameter must be sent together with **Line.#.Mode**.
	Line.#.RuleName	REQ,RES	<string>	Rule name
	Line.#.ObjectTypeFilter	REQ, RES	<csv> Vehicle,Person	If set, only for the configured object types, rule will be triggered, this parameter is supported only if AI based IVA is supported.

Action	Parameters	Request/ Response	Type/ Value	Description
	Line.#.ObjectTypeFilterDetails	REQ, RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck	If detailed filter support is needed, this parameter can be configured.
	Line.#.HandoverIndex	REQ, RES	<int>	Handover index for designated Line
remove	Channel	REQ	<int>	Channel ID
	Preset	REQ	<int>	Preset number **Note** **Preset** must be sent together with the **remove** action.
	ROIIndex	REQ	<csv>	Index of the ROI
	LineIndex	REQ	<csv>	Index of the virtual line
	DefinedAreaIndex	REQ	<csv>	Index of the virtual area

156.10.4. Examples

156.10.4.1. Getting the video analysis setting information of Preset 1 for Channel 0

To use the **view** action, the channel number and preset number must be set together.

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=view&Channel=0&Preset=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PresetVideoAnalysis": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "DetectionType": "IntelligentVideo",
                    "SensitivityLevel": 80,
                    "ObjectSizeByDetectionTypes": [
                        {
                            "DetectionType": "MotionDetection",
                            "MinimumObjectSize": "0,0",
                            "MaximumObjectSize": "99,99",
                            "MinimumObjectSizeInPixels": "24,24",
                            "MaximumObjectSizeInPixels": "1920,1080"
                        },
                        {
                            "DetectionType": "IntelligentVideo",
                            "MinimumObjectSize": "4,7",
                            "MaximumObjectSize": "50,89",
                            "MinimumObjectSizeInPixels": "97,97",
                            "MaximumObjectSizeInPixels": "972,972"
                        }
                    ],
                    "ROIs": [
                        {
                            "ROI": 1,
                            "Mode": "Inside",
                            "SensitivityLevel": 80,
                            "ThresholdLevel": 5,
                            "Coordinates": [
                                {
                                    "x": 0,
                                    "y": 0
                                },
                                {
                                    "x": 0,
                                    "y": 1079
                                },
                                {
```

```
                        "x": 1919,
                        "y": 1079
                    },
                    {
                        "x": 1919,
                        "y": 0
                    }
                ],
                "HandoverIndex": 0,
                "Duration": 0
            }
        ],
        "DefinedAreas": [
            {
                "DefinedArea": 1,
                "Type": "Inside",
                "Mode": [
                    "Intrusion"
                ],
                "Coordinates": [
                    {
                        "x": 849,
                        "y": 218
                    },
                    {
                        "x": 849,
                        "y": 724
                    },
                    {
                        "x": 1553,
                        "y": 724
                    },
                    {
                        "x": 1553,
                        "y": 218
                    }
                ],
                "AppearanceDuration": 10,
                "LoiteringDuration": 10,
                "IntrusionDuration": 2,
                "HandoverIndex": 0
```

```
                        }
                    ],
                    "EntireAreaMode": "Off"
                }
            ]
        }
    ]
}
```

156.10.4.2. Setting the sensitivity level for IV detection to 20 in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=set&Channel=0&Preset=
1&DetectionType=IntelligentVideo&SensitivityLevel=20
```

156.10.4.3. Setting the minimum and maximum object size in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=set&Channel=0&Preset=
1&DetectionType=MotionDetection&DetectionType.MotionDetection.MinimumObjectS
ize=5,5&DetectionType.MotionDetection.MaximumObjectSize=10,10
```

156.10.4.4. Setting the detection mode and coordinates of defined area 1 in Preset 2

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=set&DefinedArea.1.Coo
rdinate=159,138,114,363,540,429,576,96&
DefinedArea.1.Mode=AppearDisappear&DefinedArea.1.AppearanceDuration=10&Defin
edArea.1.LoiteringDuration=10&Channel=0&Preset=2&DetectionType=IntelligentVi
deo
```

156.10.4.5. Setting the detection mode and coordinates of line 1 in Preset 3

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=set&Line.1.Coordinate
=129,183,246,884&Line.1.Mode=Right&Channel=0&Preset=3&DetectionType=Intellig
```

```
entVideo
```

156.10.4.6. Removing the defined area of index 1 in Preset 1

To use the **remove** action, the **Preset** parameter must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=remove&Channel=0&Pres
et=1&DefinedAreaIndex=1
```

156.10.4.7. Removing line index 1 in Preset 1

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=presetvideoanalysis2&action=remove&Channel=0&Pres
et=1&LineIndex=1
```

156.11. PTZ Settings

156.11.1. Description

The **ptzsettings** submenu of **ptzconfig.cgi** enables or disables auto flip and configures digital zoom limits.

NOTE This chapter applies to network cameras only.

To find out whether digital zoom is supported by the device or not, refer to the Attributes/PTZSupport/Support/DigitalZoom attribute in the device attributes section.

Access level

Action	Camera
view	User
set	User

156.11.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
ptzsettings&action=<value>[&<parameter>=<value>]
```

156.11.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads PTZ settings.
	Channel	REQ, RES	<int>	Channel ID
	ImagePresetMode.#.DigitalZoomEnable	RES	<bool> True, False	Enables or disables the digital zoom for each image preset mode
	ImagePresetMode.#.MaxDigitalZoom	RES	<enum> 2x, 3x, 4x, 5x, 6x...	Max digital zoom scale for each image preset mode
	CurrentMountPosition	RES	<enum> Floor,Ceiling,Canted	When MountPosition is set to Auto, it represents the actual position.
set	Channel	REQ, RES	<int>	Channel ID
	AutoFlipEnable	REQ, RES	<bool> True, False	Enables or disables auto flip
	DigitalZoomEnable	REQ, RES	<bool> True, False	Enables or disables the digital zoom.
	MaxDigitalZoom	REQ, RES	<enum>	Digital zoom limits
	ProportionalPTSpeedMode	REQ, RES	<enum> Off, Slow, Medium, Fast	Proportional PT speed mode
	ProportionalPTSpeed	REQ, RES	<int>	Proportional PT speed value (from 1 to 100)
	RememberLastPosition	REQ,RES	<bool> True, False	Whether to remember the last PTZ position for RemberLastPositionDuration
	RememberLastPostionDuration	REQ,REQ	<int>	Remember the last position for the set duration.
	DigitalPTZEnable	REQ, RES	<bool> True, False	Whether to route PTZ commands for DPTZ or External PTZ in supported models
	NorthDirection	REQ		Sets current position as a north direction. **NorthDirection** is an empty parameter with no value required, and should be sent alone in a request.

Action	Parameters	Request/ Response	Type/ Value	Description
	ImagePreview	REQ	<enum> Start, Stop, AWC	Image preview mode Allows viewing of the preview image of the configuration, rather than saving the image configuration to the camera. If this parameter is ignored, then preview mode will be stopped and the original image configuration will be applied. • Start: Image preview mode will be started • Stop: Image preview mode will be stopped, and the original image settings saved in the camera will be applied • AWC: AWC mode will be started
	SpeedType	REQ, RES	<enum> Linear, Exponential	Selects speed of PTZ movement to increase linearly or exponentially.
	ImagePresetMode	REQ, RES	<enum> UserPreset 1, UserPreset 2, OutdoorDa ytime, OutdoorNig htime, IndoorBack light, IndoorBrig htScene, NumberPla te, Vivid	Selects image preset mode.

Action	Parameters	Request/ Response	Type/ Value	Description
	MountPosition	REQ, RES	<enum> Floor,Ceiling,Auto,Canted	Based on mountposition, direction of relative PTZ changes. **Note** **PTZSupport/Support/MountPosition** value in attributes is true if this feature is supported.
	QuickZoomEnable	REQ,RES	<bool> True, False	Enables the quick zoom feature.

156.11.4. Examples

156.11.4.1. Getting the current PTZ settings of channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.AutoFlipEnable=True
Channel.0.DigitalZoomEnable=False
Channel.0.MaxDigitalZoom=2x
Channel.0.ProportionalPTSpeedMode=Medium
Channel.0.RememberLastPosition=False
Channel.0.RememberLastPositionDuration=100
Channel.0.SpeedType=Linear
Channel.0.MountPosition=Ceiling
Channel.0.ImagePresetMode.UserPreset1.DigitalZoomEnable=True
Channel.0.ImagePresetMode.UserPreset1.MaxDigitalZoom=3x
Channel.0.ImagePresetMode.UserPreset2.DigitalZoomEnable=False
Channel.0.ImagePresetMode.UserPreset2.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.OutdoorDaytime.DigitalZoomEnable=False
Channel.0.ImagePresetMode.OutdoorDaytime.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.OutdoorNightime.DigitalZoomEnable=False
```

```
Channel.0.ImagePresetMode.OutdoorNightime.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.IndoorBacklight.DigitalZoomEnable=False
Channel.0.ImagePresetMode.IndoorBacklight.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.IndoorBrightScene.DigitalZoomEnable=False
Channel.0.ImagePresetMode.IndoorBrightScene.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.NumberPlate.DigitalZoomEnable=False
Channel.0.ImagePresetMode.NumberPlate.MaxDigitalZoom=2x
Channel.0.ImagePresetMode.Vivid.DigitalZoomEnable=False
Channel.0.ImagePresetMode.Vivid.MaxDigitalZoom=2x
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZSettings": [
        {
            "Channel": 0,
            "AutoFlipEnable": true,
            "DigitalZoomEnable": false,
            "MaxDigitalZoom": "2x",
            "ProportionalPTSpeedMode": "Medium",
            "RememberLastPosition": false,
            "RememberLastPositionDuration": 100,
            "SpeedType": "Linear",
            "MountPosition": "Ceiling",
            "ImagePreset": [
                {
                    "ImagePresetMode": "UserPreset1",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "3x"
                },
                {
                    "ImagePresetMode": "UserPreset2",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "OutdoorDaytime",
```

```
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "OutdoorNightime",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "IndoorBacklight",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "IndoorBrightScene",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "NumberPlate",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                },
                {
                    "ImagePresetMode": "Vivid",
                    "DigitalZoomEnable": false,
                    "MaxDigitalZoom": "2x"
                }
            ]
        }
    ]
}
```

156.11.4.2. Setting auto flip to be enabled

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&Channel=0&AutoFlipEnable=T
rue
```

156.11.4.3. Setting maximum limit of digital zoom

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&Channel=0&MaxDigitalZoom=4
x
```

156.11.4.4. Setting proportional speed mode to Slow

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&Channel=0&ProportionalPTSp
eedMode=Slow
```

156.11.4.5. Setting proportional speed with integer

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&Channel=0&ProportionalPTSp
eed=50
```

156.11.4.6. Setting proportional speed mode to Off

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&Channel=0&ProportionalPTSp
eedMode=Off
```

156.11.4.7. Setting the current pan position as North direction

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&NorthDirectoion
```

156.11.4.8. Change the mount position of camera

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzsettings&action=set&MountPosition=Floor
```

156.12. PT Operation Limits

156.12.1. Description

The **ptlimits** submenu of **ptzconfig.cgi** enables or disables pan and tilt limits.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	User
set	User
control	User

156.12.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
ptlimits&action=<value>[&<parameter>=<value>]
```

156.12.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads limit settings for pan and tilt functions.
	Channel	REQ, RES	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	PanLimitEnable	REQ, RES	<bool> True, False	Enables or disables pan limits
	TiltLimitEnable	REQ, RES	<bool> True, False	Enables or disables tilt limits
	TiltRange	REQ, RES	<enum>	Setting the Tilt range value of camera.
	DaysAfterReboot	REQ, RES	<int>	Duration to reset the PT module. From 0 to 7 days.
	StartTime	REQ, RES	<int>	Time to initialize the PT module. From 0 to 23.
control	Channel	REQ	<int>	Channel number

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ	<enum> PanBegin, PanEnd, TiltBegin, TiltEnd, Exit	PT operation limits

156.12.4. Examples

156.12.4.1. Getting the current PTZ settings

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptlimits&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PanLimitEnable=False
Channel.0.TiltLimitEnable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PTLimits": [
        {
            "Channel": 0,
            "PanLimitEnable": false,
            "TiltLimitEnable": false
        }
    ]
}
```

156.12.4.2. Enabling Pan limits

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptlimits&action=set&PanLimitEnable=True
```

156.13. PTZ Protocol

156.13.1. Description

The **ptzprotocol** submenu of **ptzconfig.cgi** configures the protocol used for PTZ operation.

Please refer to the **serial** submenu of system.cgi for the serial port settings.

NOTE This chapter applies to network cameras, encoder and hybrid NVR.

Access level

Action	Camera	Encoder	NVR
view	Suser	Suser	Suser
set	Suser	Suser	Suser

156.13.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
ptzprotocol&action=<value>[&<parameter>=<value>]
```

156.13.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the values of all response parameters and shows the current information.
	Channel	REQ, RES	<int>	Channel ID
	Status	RES	<enum> Connect, Disconnect	Returns the connection status of the analog channel serial. This parameter is supported only for analog channels in Hybrid NVR NVR ONLY
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Protocol	REQ, RES	<enum> Samsung-T, Samsung-E, Pelco-D, Pelco-P, Panasonic, Vicon, Honeywell, AD, GE, Bosch, Sungjin	Protocol **Note** **Protocol** must be sent together with the **set** action.
	ConnectionPortType	REQ, RES	<enum> RS-485, Coaxial	Selecting connection port type The value may vary depending on the model. Please check the device attributes using attributes.cgi. ENCODER ONLY
	CameraID	REQ, RES	<int>	Camera ID for the PTZ protocol
	CoaxProtocol	REQ, RES	<enum> Auto,None	Returns the Coax protocol type This parameter is only supported for analog cameras in Hybrid NVR. NVR ONLY
	PortNum	REQ, RES	<int>	In Hybrid NVR, the analog port number This parameter is only supported for analog channels in Hybrid NVR NVR ONLY

156.13.4. Examples

156.13.4.1. Getting the current PTZ Protocol

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=ptzprotocol&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Channel.0.Protocol=Samsung-T
Channel.0.CameraID=1
Channel.0.ConnectionPortType=RS-485
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZProtocol": [
        {
            "Channel": 0,
            "Protocol": "Samsung-T",
            "CameraID": 1,
            "ConnectionPortType": "RS-485"
        }
    ]
}
```


JSON RESPONSE(Hybrid NVR)

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PTZProtocol": [
        {
            "Channel": 0,
            "PortNum" : 0,
            "CoaxProtocol":"Auto",
            "Protocol": "Samsung-T",
            "CameraID": 1,
            "ConnectionPortType": "RS-485"
        }
```

```
    ]
}
```

156.13.4.2. Setting the PTZ Protocol to 'Samsung-E' protocol

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzprotocol&action=set&Protocol=Samsung-E
```

156.13.4.3. Setting connection port Type to RS-485

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzprotocol&action=set&Channel=0&Protocol=Samsung
-T&CameraID=1&ConnectionPortType=RS-485
```

156.14. PTZ Mode

156.14.1. Description

The **ptzmode** submenu of **ptzconfig.cgi** configures the mode used for PTZ operation.

NOTE	This chapter applies to network cameras only. If "/attributes/PTZSupport/GlobalPTZMode" attribute is true, this setting is globally applied for the device.

Access level

Action	Camera
view	Suser
set	Suser

156.14.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
ptzmode&action=<value>[&<parameter>=<value>]
```

156.14.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the values of all response parameters and shows the current information.
	Channel	REQ,	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Mode	REQ, RES	<enum> DigitalPTZ, ExternalPTZ	PTZ Mode

156.14.4. Examples

156.14.4.1. Getting the current PTZ Mode

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=ptzmode&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PtzMode": [
        {
            "Channel": 0,
            "Mode": "ExternalPTZ"
        }
    ]
}
```

156.14.4.2. Setting the PTZ Mode to ‘DigitalPTZ’

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=ptzmode&action=set&Mode=DigitalPTZ
```

156.15. Pan Zero Position

156.15.1. Description

The **panzeroposition** submenu configures the current pan position for the pan zero position of the requested channel.

NOTE This chapter applies to network cameras only.

Refer to Attributes/PTZSupport/Support/PanZeroPosition in the attributes section for more information.

Access level

Action	Camera
set	User

156.15.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
panzeroposition&action=<value>[&<parameter>=<value>]
```

156.15.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ, RES	<int>	Channel ID

156.15.4. Examples

156.15.4.1. Getting the current PTZ mode

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=panzeroposition&action=set&Channel=0
```

156.16. Digital Auto Tracking

156.16.1. Description

The **digitalautotracking** submenu configures digital auto tracking settings.

NOTE This chapter applies to network cameras only.

AI Camera DPTZ Channel (Channel 1) only supports this.

Access level

Action	Camera
view	User
set	User

156.16.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
digitalautotracking&action=<value>[&<parameter>=<value>]
```

156.16.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID
set	Channel	REQ,RES	<int>	Channel ID
	ObjectTypeFilter	REQ,RES	<csv> Vehicle,Person,Face	When the filter is configured, digital auto tracking would be enabled only for configured object types.

156.16.4. Examples

156.16.4.1. Getting the current setting

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=digitalautotracking&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "digitalautotracking": [
        {
            "Channel": 1,
            "ObjectTypeFilter": [
                "Person",
```

```
                    "Vehicle"
                ]
            }
        ]
}
```

156.16.4.2. Setting digitalautotracking configuration

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=digitalautotracking&action=set&Channel=1&ObjectTy
peFilter=Person,Vehicle
```

156.17. PT Position Correction

156.17.1. Description

The **ptcorrection** submenu configures PT's default position.

NOTE This chapter applies to PT cameras only.

Access level

Action	Camera
view	User
set	User

156.17.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
ptcorrection&action=<value>[&<parameter>=<value>]
```

156.17.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ	<int>	Channel ID that sets the current PT's position to be 0. You can check PT's position in query submenu in ptzconrol cgi.

156.17.4. Examples

156.17.4.1. Set the current PT position to be 0

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=ptcorrection&action=set
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

156.18. Exclusive PTZ Control

156.18.1. Description

The **exclusiveptzcontrol** submenu configures exclusive access authority for PTZ controls, allowing execution of only one PTZ operation at a time.

NOTE This chapter applies to PTZ cameras only.

Access level

Action	Camera
view	User
set	User

156.18.2. Syntax

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=
exclusiveptzcontrol&action=<value>[&<parameter>=<value>]
```

156.18.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ	<int>	Channel ID
	Enable	REQ,RES	<bool> True,False	Sets the exclusive PTZ control authority to enabled or not.

156.18.4. Examples

156.18.4.1. Enable the exclusive PTZ control authority

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=exclusiveptzcontrol&action=set&Channel=0&Enable=T
rue
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

156.18.4.2. Request the current settings

REQUEST

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=exclusiveptzcontrol&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ExclusivePTZControl": [
        {
            "Channel": 0,
            "Enable": false
        }
    ]
}
```


156.19. Preset Object Detection

156.19.1. Description

The **presetobjectdetection** submenu enables ptzpreset location based object detection capability in the camera.

Access level

Action	Camera
view	User
set	User

156.19.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
presetobjectdetection&action=<value>[&<parameter>=<value>]
```

156.19.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	ChanelID
	Preset	REQ	<csv>	Optional preset index
set	Channel	REQ, RES	<int>	ChanelID
	Preset	REQ, RES	<int>	Valid PTZ preset index

Action	Parameter	Request/ Response	Type/ Value	Description
	Enable	REQ, RES	<bool> True, False	Enable object detection
	ObjectTypes	REQ, RES	<csv> Vehicle, Face, Person, LicensePlate	When a configured object is detected, an event would be generated.
	ObjectTypeDetails	REQ, RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck	These parameters indicate the details of the ObjectType top-level object. If none of these details are set, the top-level object wil not be detected either.
	Sensitivity	REQ, RES	<int> 1 to 100	Sensitivity level
	Duration	REQ, RES	<int> 1 to 5	Duration in secs
	MinimumObjectSize	REQ, RES	<string>	Minimum size of objects Objects smaller than the specified minimum size are not detected. The size is expressed in the form of <width, height>. The value of **MinimumObjectSize** must be smaller than the value of **MaximumObjectSize**.

Action	Parameter	Request/ Response	Type/ Value	Description
	MaximumObjectSize	REQ, RES	<string>	Maximum size of objects Objects bigger than the maximum size are not detected. The size is expressed in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**.
	MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum object size in pixel The size is specified in the form of <width, height>.
	MaximumObjectSizeInPixels	REQ, RES	<string>	Miximum object size in pixel The size is expressed in the form of <width, height>.
	EnableMetadataInExcludeArea	REQ, RES	<bool> True, False	By enabling this parameter, metadata would be generated even for exclude region.
	ExcludeArea.#.Coordinate	REQ, RES	<string>	Exclude region coordinates In this region, object detection does not work
	HandoverIndex	REQ, RES	<int>	**HandOverIndex** associated with object detection
remove	Channel	REQ	<int>	Channel id.
	Preset	REQ	<int>	Preset Index.
	ExcludeAreaIndex	REQ	<csv>	Exclude area index to be removed

156.19.4. Examples

156.19.4.1. Getting objectdetection settings

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetobjectdetection&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PresetObjectDetection": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Preset": 1,
                    "Enable": true,
                    "Duration": 2,
                    "Sensitivity": 80,
                    "MinimumObjectSize": "1,2",
                    "MaximumObjectSize": "50,89",
                    "MinimumObjectSizeInPixels": "32,32",
                    "MaximumObjectSizeInPixels": "972,972",
                    "EnableMetadataInExcludeArea": false,
                    "ObjectTypes": [
                        "Person",
                        "Vehicle",
                        "Face",
                        "LicensePlate"
                    ],
                    "ObjectTypeDetails": {
                        "Vehicle": {
                            "Types": [
                                "Bicycle",
                                "Car",
                                "Motorcycle",
                                "Bus",
                                "Truck"
                            ]
                        }
                    },
                    "HandoverIndex": 0
                },
                {
```

```
                "Preset": 2,
                "Enable": true,
                "Duration": 2,
                "Sensitivity": 80,
                "MinimumObjectSize": "1,2",
                "MaximumObjectSize": "50,89",
                "MinimumObjectSizeInPixels": "32,32",
                "MaximumObjectSizeInPixels": "972,972",
                "EnableMetadataInExcludeArea": false,
                "ObjectTypes": [
                    "Person",
                    "Vehicle",
                    "Face",
                    "LicensePlate"
                ],
                "ObjectTypeDetails": {
                    "Vehicle": {
                        "Types": [
                            "Bicycle",
                            "Car",
                            "Motorcycle",
                            "Bus",
                            "Truck"
                        ]
                    }
                },
                "HandoverIndex": 0
            },
            {
                "Preset": 3,
                "Enable": false,
                "Duration": 2,
                "Sensitivity": 80,
                "MinimumObjectSize": "1,2",
                "MaximumObjectSize": "50,89",
                "MinimumObjectSizeInPixels": "32,32",
                "MaximumObjectSizeInPixels": "972,972",
                "EnableMetadataInExcludeArea": false,
                "ObjectTypes": [
                    "Person",
                    "Vehicle",
```

```
                        "Face",
                        "LicensePlate"
                    ],
                    "ObjectTypeDetails": {
                        "Vehicle": {
                            "Types": [
                                "Bicycle",
                                "Car",
                                "Motorcycle",
                                "Bus",
                                "Truck"
                            ]
                        }
                    },
                    "HandoverIndex": 0
                }
            ]
        }
    ]
}
```

156.19.4.2. Configure object detection

REQUEST

```
http://<Device IP>/stw-cgi/ptzconfig.cgi?msubmenu=presetobjectdetection&action=set&Channel=0&Preset=3&Enable=True&ObjectTypes=Vehicle,Face,Person&Sensitivity=20&Duration=3&EnableMetadataInExcludeArea=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Event

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 157. Overview

157.1. Events

Using SUNAPI event-related APIs, users can define when and how certain actions should be performed by a video surveillance product. For example, a camera can be set to upload images to an FTP server or send notification messages when it detects motion. Events can be scheduled to run at certain times, or they can be triggered by things happening, such as motion detection or a signal from an input port.

eventsources.cgi uses the following submenus to configure events:

- **videoanalysis**: Configures settings for video analysis. Video Analytics creates a virtual area or boundary line over the site where the video surveillance device is installed, then generates actions when it detects motion in the virtual area or across the line. It can detect if there is motion inside or outside the defined area, and it can detect if objects enter, exit, appear, or disappear in the area. In addition, it can detect if an object crosses the boundary line from the left or right side.
- **videoanalysis2**: This is the same as the video analysis submenu, but it allows configuration of parameters for each ROI/area.
- **audioanalysis**: Configures the settings for audio analytics events.
- **fogdetection**: Configures the settings for fog detection. This submenu can specify the sensitivity level for fog detection and enable the auto defog.
- **facedetection**: Configures the settings for face detection. This submenu can specify the sensitivity level for face detection or set up a virtual area that causes the camera to react when a face is detected inside or outside the area.
- **tamperingdetection**: Configures the settings for scene changes, which are usually a sign of tampering with the video surveillance device. The submenu can specify various settings such as the sensitivity level for tampering detection.
- **audiodetection**: Configures the settings for audio detection. Settings include the detection input threshold level.
- **videoloss**: Configures video loss detection settings.
- **autotracking**: Configures auto tracking settings.
- **timer**: Configures the schedule for the timer, which can generate events periodically.
- **alarminput**: Configures the settings of the alarm input function.
- **firealarminput**: Configures the settings of the fire alarm input function.
- **networkalarminput**: Configures the network alarm input function.
- **networkdisconnect**: Configures whether an event should be generated when a network disconnection occurs.
- **defocusdetection**: Configures the settings for defocus detection.
- **peoplecount**: Configures the settings for people count feature.
- **vehiclecount**: Configures the settings for vehicle count feature.

- **heatmap**: Configures the settings for heat map feature.
- **sourceoptions**: Gives information about the list of event sources available and associated action triggers.
- **samples**: Gets the samples of event levels for the corresponding event source.
- **queuemanagementsetup**: Configures the settings for the queue management feature.
- **gsensor**: Configures the settings for the G sensor feature.
- **temperaturechangedetection:** Configures the settings for temperature change detection.
- **temperaturechangedetectionoptions**: Gives information about temperature change detection options.
- **shockdetection**: Configures the settings for shock detection.
- **wiperhousingdetection**: Configures the settings for wiper housing detection.
- **boxtemperaturedetection**: Configures the settings for box temperature detection.
- **boxtemperaturedetectionoptions**: Gives information about box temperature detection options.
- **objectdetection**: Configures object detection algorithm for the camera.
- **metaimagetransfer**: Configures metadata image sending capability.
- **facerecognition**: Configures face recognition on selected channels.
- **ocr**: Configures character recognition on selected channels.
- **thermaldetectionmode**: Configures thermal camera operation mode.
- **bodytemperaturedetection**: Configures human body temperature detection.
- **bodytemperaturedetectionoptions**: Provides the settings for bodytemperaturedetection options.
- **temperaturemeasurementregion**: Configures the region to measure human body temperature.
- **maskdetection**: Configures face mask detection.
- **cellmotion**: Configures cell-based motion detection.
- **parkingdetection**: Configures the settings for parking detection.
- **ledindicator**: Configures LED settings used for indicating current parking status.
- **callrequest**: Configures the settings for SIP Call request events.
- **dtmf**: Configures the settings for DTMF events.
- **tamperingswitch**: Configures the settings for tampering switch events.
- **casetampering**: Configures the settings for casetampering events.
- **proximitysensor**: Configures the settings for proximity sensor events.
- **socialdistancingviolation**: Configures the settings for social distancing violation detection.
- **overspeed**: Configures the settings for overspeed events.
- **mqttpublication**: Configures the settings for MQTT publication messages.
- **mqttsubscription**: Configures the settings for MQTT subscription messages.

- **earlyfiredetection**: Configures the settings for the early fire detection.
- **earlyfiredetectionoptions**: Provides options for the early fire detection settings.
- **temperaturedifference**: Configures the settings for the temperature difference.
- **temperaturedifferenceoptions**: Provides options for the temperature difference settings.
- **tankleveldetection**: Configures the settings for tank level detection.

eventactions.cgi uses the following submenus to configure the actions that a NVR will perform when an event occurs:

- **smtp:** Specifies the event types for sending alert messages by email, as well as the relevant email settings.
- **complexaction**: Specifies the event types for an alarm output and the duration as well as the preset index that the camera will move to when a specified event occurs.

157.2. Event rule

An event rule defines when and how to react to events generated by the camera. For example, a rule may stipulate that the video surveillance device makes alarm output for 5 seconds when a motion is detected between 2 AM and 2 PM on Saturday.

Use **eventrules.cgi** to create various rules by specifying the event type, event action, and applicable time.

- **rules**: This submenu is used to add or update event rules.
- **handover**: Configures the settings for the handover feature.
- **scheduler**: Configures the schedule settings for report generation for the people count, heat map, and queue management features.
- **handover2**: This is the same as the handover submenu, but can be used to configure the handover parameters in a generic way, such that the handover feature can be associated with different camera events.
- **schedulelist**: This submenu is used to configure schedulers in a device that can be associated with the dynamicrules submenu.
- **dynamicrules**: This submenu is used to manage event actions upon receiving an event in a device.
- **dynamicrulesoptions**: This submenu provides information about the list of event sources and associated action triggers available in dynamicrules.
- **internalhandovercalibration**: Manages the internal handover calibration settings which are used for the smart zoom feature.
- **indicationpass**: This submenu is used to manage the configuration to pass information to an opponent camera when a parking detection event occurs.
- **ioboxregister**: Provides information on how to register an IO box.
- **audiooutfiles**: This submenu is used to manage the audio files in the camera and IP Audio System (IPAS).
- **audiooutfilesschedule**: Configures the schedule settings for audio clip playback.

- **ledpreset**: This submenu is used to manage LED preset-related settings.
- **ttsfiles**: This submenu is used to manage the tts files in the IP Audio System (IPAS).

157.3. Event status

The event status can be requested using **eventstatus.cgi**.

The current status may be sent once or continuously. When the status is sent continuously, it can be configured to receive all events data during the first response, but later only event data changes are sent.

- **eventstatus**: Requests the current event status.
- **pushnotification**: Configures the push notification feature in NVR.
- **eventscheme**: Chooses whether old or new topics should be used in the ONVIF event service.
- **metadataschema**: Provides schema of metadata events generated from the camera when an event occurs.
- **eventstatusschema**: Provides event schema of all events.

157.4. Transfer method and configuration

transfer.cgi configures the FTP, and SMTP server settings so that images and messages can be transferred when an event occurs.

Chapter 158. Event Sources

158.1. Video Analytics Setup

158.1.1. Description

The **videoanalysis** submenu configures the video analysis settings.

CAUTION It is recommended to use the **videoanalysis2** submenu instead. The **videoanalysis** submenu will be deprecated starting January 2027.

NOTE This chapter applies to network cameras and encoders only.
Attribute to check for video analytics support: "attributes/Eventsource/Support/Channel.#.IVRule"
Attribute to check for maximum IV Rules: "attributes/Eventsource/Limit/MaxIVRule"
Attribute to check for motion detection support: "attributes/Eventsource/Support/VA.MotionDetection"
Camera/Encoder gives adjusted coordinates based on flip/mirror/rotate settings when the "/attributes/Eventsource/Support/AdjustMDIVRuleOnFlipMirror" parameter is set to False. Otherwise, camera provides original coordinates and client has to perform coordinate adjustment based on flip/mirror/rotate settings.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin
remove	Admin	Admin

158.1.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
videoanalysis&action=<value>[&<parameter>=<value>]
```

158.1.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the video analysis settings.
	Channel	REQ, RES	<csv>	Channel ID

Action	Parameter	Request/ Response	Type/ Value	Description
	DetectionType	REQ	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type **DetectionType** for the **view** action requests the current video analysis settings and does not change the configured detection type.
set	Channel	REQ, RES	<int>	Channel ID
	DetectionType	REQ, RES	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type • MotionDetection: Detects motion within the specified area. • IntelligentVideo: Detects object appearance, disappearance, entry, exit, scene changes, and other activities within the specified area. • Off: No detection. • MDAndIV: Both motion detection and intelligent video functions work together.
	Sensitivity	REQ, RES	<enum> VeryLow, Low, Medium, High, VeryHigh	Sensitivity level for MotionDetection or IntelligentVideo. **Sensitivity** is valid only when **DetectionType** is NOT set to Off.
	ObjectSize	REQ, RES	<enum> VerySmall, Small, Medium, Large, VeryLarge	Size of detectable objects for motion detection

Action	Parameter	Request/ Response	Type/ Value	Description
	MinimumObjectSize	REQ, RES	<string>	Minimum size of objects detectable by motion detection. Objects smaller than the specified minimum size are not detected. The size is specified in the format of <width, height>. The value of **MinimumObjectSize** must be less than the value of **MaximumObjectSize**. **MinimumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	MaximumObjectSize	REQ, RES	<string>	Maximum size of objects detectable by motion detection Objects larger than the maximum size are not detected. The size is specified in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**. **MaximumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum object size in pixels The size is specified in the format of <width, height>. **MinimumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.

Action	Parameter	Request/ Response	Type/ Value	Description
	MaximumObjectSizeInPixels	REQ, RES	<string>	Maximum object size in pixel The size is specified in the format of <width, height>. **MaximumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Coordinate	REQ, RES	<string>	ROI (Region of Interest) coordinates **ROI.#.Coordinate** is valid only when **DetectionType** is NOT set to Off.
	ROIMode	REQ, RES	<enum> Inside, Outside	ROI detection mode • Inside: Detects motion within the specified ROI • Outside: Detects motion outside the specified ROI **ROIMode** is valid only when **DetectionType** is NOT set to Off.
	DetectionResultOverlay	REQ, RES	<bool> True, False	Whether to mark detected motions on the screen in a box when an event occurs **DetectionResultOverlay** is valid only when **DetectionType** is NOT set to Off.
	DisplayRules	REQ, RES	<bool> True, False	Whether to show the video analytics on the web client monitoring page **DisplayRules** is valid only when **DetectionType** is set to IntelligentVideo.

Action	Parameter	Request/ Response	Type/ Value	Description
	DefinedArea.#.Mode	REQ, RES	<csv> AppearDisappear, Entering, Exiting	Defined virtual area detection mode • AppearDisappear: Detects objects appearing or disappearing in the specified virtual area. • Entering: Detects objects entering the specified virtual area. • Exiting: Detects objects exiting the specified virtual area. **DefinedArea.#.Mode** is valid only when **DetectionType** is set to IntelligentVideo. **DefinedArea.#.Coordinate** must be sent together with **DefinedArea.#.Mode**. **Note** **DefinedArea.#.Mode**, **DefinedArea.#.Coordinate**, **Line.#.Mode**, and **Line.#.Coordinate** must be sent together with the **set** action.
	DefinedArea.#.Coordinate	REQ, RES	<string>	Top left and bottom right vertices of the defined virtual area for motion detection **DefinedArea.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. **DefinedArea.#.Mode** must be sent together with **DefinedArea.#.Coordinate**.

<table>
<tr><th>Action</th><th>Parameter</th><th>Request/
Response</th><th>Type/
Value</th><th>Description</th></tr>
<tr><td></td><td>EntireAreaMode</td><td>REQ, RES</td><td><csv>
Off, AppearDisappear, Scenechange</td><td>Entire area detection mode
Off: Disables the entire area detection mode
AppearDisappear: Detects objects appearing or disappearing in the entire area
Scenechange: Detects scene change events, which are triggered when a large portion of the scene is changed.
EntireAreaMode is valid only when DetectionType is set to IntelligentVideo.</td></tr>
<tr><td></td><td>Line.#.Mode</td><td>REQ, RES</td><td><csv>
LeftSide, RightSide</td><td>Line detection mode
LeftSide: Detects motion to the left of the virtual line.
RightSide: Detects motion to the right of the virtual line.
Line.#.Mode is valid only when DetectionType is set to IntelligentVideo.

The Line.#.Mode parameter must be sent along with Line.#.Coordinate.</td></tr>
</table>

Action	Parameter	Request/ Response	Type/ Value	Description
	Line.#.Coordinate	REQ, RES	<string>	X and Y coordinates of the two points which define the virtual line The coordinates are specified in the form of <x1,y1,x2,y2>; x1 and y1 are the start points and x2 and y2 are the end points. **Line.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. The **Line.#.Coordinate** parameter must be sent together with **Line.#.Mode**.
remove	Channel	REQ	<int>	ID of the channel to be deleted
	LineIndex	REQ	<csv> All, #	Index of the virtual line to be deleted
	DefinedAreaIndex	REQ	<csv> All, #	Index of the virtual area to be deleted
	ROIIndex	REQ	<csv> All, #	Index of the ROI to be deleted

158.1.4. Examples

158.1.4.1. Getting the current video analytics settings of 'Channel 0'

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=videoanalysis&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.DetectionType=IntelligentVideo
Channel.0.Sensitivity=Medium
Channel.0.MinimumObjectSize=4,5
```

```
Channel.0.MaximumObjectSize=47,64
Channel.0.MinimumObjectSizeInPixels=192,192
Channel.0.MaximumObjectSizeInPixels=1944,1944
Channel.0.ROIMode=Inside
Channel.0.ROI.1.Coordinate=1725,925,1157,1613,2544,1225,2307,438
Channel.0.DefinedArea.1.Mode=Entering
Channel.0.DefinedArea.1.Coordinate=1307,782,2000,707,1763,1519,957,1613
Channel.0.Line.1.Mode=LeftSide,RightSide
Channel.0.Line.1.Coordinate=1194,682,1075,1382
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoAnalysis": [
        {
            "Channel": 0,
            "DetectionType": "IntelligentVideo",
            "Sensitivity": "Medium",
            "MinimumObjectSize": "4,6",
            "MaximumObjectSize": "48,64",
            "MinimumObjectSizeInPixels": "192,192",
            "MaximumObjectSizeInPixels": "1944,1944",
            "ROIMode": "Inside",
            "ROIs": [
                {
                    "ROI": 1,
                    "Coordinates": [
                        {
                            "x": 1725,
                            "y": 925
                        },
                        {
                            "x": 1157,
                            "y": 1613
                        },
                        {
                            "x": 2544,
```

```
                        "y": 1225
                    },
                    {
                        "x": 2307,
                        "y": 438
                    }
                ]
            }
        ],
        "Lines": [
            {
                "Line": 1,
                "Mode": [
                    "RightSide",
                    "LeftSide"
                ],
                "Coordinates": [
                    {
                        "x": 1194,
                        "y": 682
                    },
                    {
                        "x": 1075,
                        "y": 1382
                    }
                ]
            }
        ],
        "DefinedAreas": [
            {
                "DefinedArea": 1,
                "Mode": [
                    "Entering"
                ],
                "Coordinates": [
                    {
                        "x": 1307,
                        "y": 782
                    },
                    {
                        "x": 2000,
```

```
                        "y": 707
                    },
                    {
                        "x": 1763,
                        "y": 1519
                    },
                    {
                        "x": 957,
                        "y": 1613
                    }
                ]
            }
        ]
    }
  ]
}
```

158.1.4.2. Setting intelligent video analysis

Setting video analytics to detect objects appearing/disappearing within the defined virtual area

DetectionType must be set to IntelligentVideo to define the virtual area (using the **DefinedArea.#.Coordinate** parameter) and the detection mode within the area (using the **DefinedArea.#.Mode** parameter).

NOTE	Attribute to check for maximum area rules: "attributes/Eventsource/Limit/MaxIVRule.Area"

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=videoanalysis&action=set&DetectionType=IntelligentVideo&DefinedArea.1.Mode=AppearDisappear&DefinedArea.1.Coordinate=151,437,1139,184,384,639,1347,1061
```

Setting video analytics to detect objects crossing the virtual line from the right side

DetectionType must be set to IntelligentVideo to define the coordinates of the virtual line (using the **Line.#.Coordinate** parameter) and the detection mode (using the **Line.#.Mode** parameter).

NOTE	Attribute to check for Maximum Line Rules: "attributes/Eventsource/Limit/MaxIVRule.Line"

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=set&DetectionType=Intelli
gentVideo&Line.2.Mode=RightSide&Line.2.Coordinate=650,750,622,410
```

158.1.4.3. Setting motion detection with high sensitivity

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=set&DetectionType=MotionD
etection&Sensitivity=High
```

158.1.4.4. Setting the minimum/maximum size of detectable objects

The minimum object size value must be less than the value of **MaximumObjectSize**, and the maximum object size value should be greater than the value of **MinimumObjectSize**.

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=set&DetectionType=Intelli
gentVideo&MinimumObjectSize=0,0&MaximumObjectSize=20,20
```

158.1.4.5. Setting the coordinates of ROI 1

Setting the coordinates of ROI 1 and setting the detection mode to detect motion within the specified ROI

Based on the ROI type, the number of coordinates that has to be sent to create a new ROI will vary.

NOTE	Attribute to check for Maximum ROI Supported: "attributes/Eventsource/Limit/MaxROI" Attribute to check for ROI type: "attributes/Eventsource/Support/ROIType" Attribute to check for ROI Minimum X Coordinate: "attributes/Eventsource/Limit/ROICoordinate.MinX" Attribute to check for ROI Maximum X Coordinate: "attributes/Eventsource/Limit/ROICoordinate.MinY" Attribute to check for ROI Minimum Y Coordinate: "attributes/Eventsource/Limit/ROICoordinate.MaxX" Attribute to check for ROI Maximum Y Coordinate: "attributes/Eventsource/Limit/ROICoordinate. MaxY"

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=set&DetectionType=MotionD
```

```
etection&ROI.1.Coordinate=380,209,1540,858,540,297,1380,759&ROIMode=Inside
```

158.1.4.6. Removing all configured lines and ROI in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=remove&Channel=0
```

158.1.4.7. Removing configured line number 1 in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=remove&Channel=0&LineInde
x=1
```

158.1.4.8. Removing a line, area and ROI in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis&action=remove&Channel=0&LineInde
x=1&DefinedAreaIndex=1,2&ROIIndex=1
```

158.2. Video Analytics2 Setup

158.2.1. Description

The **videoanalysis2** submenu is similar to the **videoanalysis** submenu but this submenu supports the configuration of the parameters for each area/ROI.

NOTE	This chapter applies to network cameras and encoders only. Attribute to check for video analytics support: "attributes/Eventsource/Support/Channel.#.IVRule" Attribute to check for maximum IV Rules: "attributes/Eventsource/Limit/MaxIVRule" Attribute to check for motion detection support: "attributes/Eventsource/Support/VA.MotionDetection" Camera/Encoder gives adjusted coordinates based on flip/mirror/rotate settings when the "/attributes/Eventsource/Support/AdjustMDIVRuleOnFlipMirror" parameter is set to False. Otherwise, camera provides original coordinates and client has to perform coordinate adjustment based on flip/mirror/rotate settings.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin
remove	Admin	Admin

158.2.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
videoanalysis2&action=<value>[&<parameter>=<value>]
```

158.2.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads video analysis settings
	Channel	REQ, RES	<csv>	Channel ID
	DetectionType	REQ	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type **DetectionType** for the **view** action requests the current video analysis settings and does not change the configured detection type.
set	Channel	REQ, RES	<int>	Channel ID
	DetectionType	REQ, RES	<enum> MotionDetection, IntelligentVideo, Off, MDAndIV	Detection type • MotionDetection: Detects motion within the specified area. • IntelligentVideo: Detects objects' appearing, disappearing, entering, exiting, scene changes, etc within the specified area. • Off: No detection. • MDAndIV: Both motion detection and intelligent video functions work together.
	SensitivityLevel	REQ, RES	<int>	Sensitivity level for IntelligentVideo. **Sensitivity** is valid only when **DetectionType** is NOT set to Off.

Action	Parameter	Request/ Response	Type/ Value	Description
	DetectionType.#.DetectionResultOverlay	REQ, RES	<bool> True, False	Whether to mark detected motions on the screen in a box when an event occurs **DetectionResultOverlay** is valid only when **DetectionType** is NOT set to Off.
	DisplayRules	REQ, RES	<bool> True, False	Whether to show the video analytics on the web client monitoring page **DisplayRules** is valid only when **DetectionType** is set to IntelligentVideo.
	DetectionType.#.MinimumObjectSize	REQ, RES	<string>	Minimum size of objects detectable by motion detection. Objects smaller than the specified minimum size are not detected. The size is specified in the format of <width, height>. The value of **MinimumObjectSize** must be less than the value of **MaximumObjectSize**. **MinimumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	DetectionType.#.MaximumObjectSize	REQ, RES	<string>	Maximum size of objects detectable by motion detection Objects larger than the maximum size are not detected. The size is specified in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**. **MaximumObjectSize** is valid only when **DetectionType** is NOT set to Off.

Action	Parameter	Request/ Response	Type/ Value	Description
	DetectionType.#.MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum object size in pixels The size is specified in the format of <width, height>. **MinimumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	DetectionType.#.MaximumObjectSizeInPixels	REQ, RES	<string>	Maximum object size in pixel The size is specified in the format of <width, height>. **MaximumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	DetectionType.#.EnableMetadataInExcludeArea	REQ	<bool> True, False	When this value is false, metadata will not be delivered in exclude region By enabling this parameter, metadata would be generated even for exclude region.
	ROI.#.Coordinate	REQ, RES	<string>	ROI (Region of Interest) coordinates **ROI.#.Coordinate** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.Mode	REQ, RES	<enum> Inside, Outside	ROI detection mode • Inside: Detects motion within the specified ROI • Outside: Detects motion outside the specified ROI **ROIMode** is valid only when **DetectionType** is NOT set to Off.
	ROI.#.SensitivityLevel	REQ, RES	<int>	Sensitivity level for Motion Detection
	ROI.#.ThresholdLevel	REQ, RES	<int>	Threshold level for Motion Detection
	ROI.#.ObjectTypeFilter	REQ, RES	<csv> Vehicle, Person	This parameter is only supported on models supporting MotionDetection with AI; if this filter is not set, all objects in the defined area would be detected.

Action	Parameter	Request/ Response	Type/ Value	Description
	ROI.#.RuleName	REQ, RES	<string>	Name of rule This parameter can only be set on models supporting MotionDetection with AI
	DefinedArea.#.Mode	REQ, RES	<csv> AppearDisappear, Entering, Exiting, Intrusion, Loitering	Defined virtual area detection mode • AppearDisappear: Detects objects appearing or disappearing in the specified virtual area. • Entering: Detects objects entering the specified virtual area. • Exiting: Detects objects exiting the specified virtual area. • Intrusion: Detects objects intruding in the specified virtual area • Loitering: Detects objects remaining in the specified virtual area over a certain period **DefinedArea.#.Mode** is valid only when **DetectionType** is set to IntelligentVideo. **Note** **DefinedArea.#.Mode**, **DefinedArea.#.Coordinate**, **Line.#.Mode**, and **Line.#.Coordinate** must be sent together with the **set** action.
	DefinedArea.#.Coordinate	REQ, RES	<string>	Top left and bottom right vertices of the defined virtual area for motion detection **DefinedArea.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. **DefinedArea.#.Mode** must be sent together with **DefinedArea.#.Coordinate**.

Action	Parameter	Request/ Response	Type/ Value	Description
	DefinedArea.#.Type	REQ, RES	<enum> Inside, Outside	Define Area type • Inside: Detects video analytics within the specified area • Outside: Detects video analytics outside the specified area
	DefinedArea.#.AppearanceDuration	REQ, RES	<int>	Appearance Duration in seconds
	DefinedArea.#.LoiteringDuration	REQ, RES	<int>	Loitering Duration in seconds
	DefinedArea.#.IntrusionDuration	REQ, RES	<int>	Intrusion duration in seconds
	DefinedArea.#.RuleName	REQ, RES	<string>	Name of rule This parameter can only be set on models supporting IntelligentVideo with AI
	DefinedArea.#.ObjectTypeFilter	REQ, RES	<csv> Vehicle, Person	This parameter is only supported on models supporting IntelligentVideo with AI; if this filter is not set, all objects in the defined area would be detected.
	DefinedArea.#.ObjectTypeFilterDetails	REQ, RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck, Person.Color.Orange, Person.Color.Black, Person.Color.Red	These parameters indicate the details of the ObjectType top-level object. If none of these details are set, the top-level object will not be detected either.

Action	Parameter	Request/ Response	Type/ Value	Description
	EntireAreaMode	REQ, RES	<csv> Off, AppearDisappear, Scenechange	Entire area detection mode • Off: Disables the entire area detection mode • AppearDisappear: Detects objects appearing or disappearing in the entire area • Scenechange: Detects scene change events, which are triggered when a large portion of the scene is changed. **EntireAreaMode** is valid only when **DetectionType** is set to IntelligentVideo. **Note** Deprecated parameter
	Line.#.Mode	REQ, RES	<csv> Off, Right, Left, BothDirections	Line detection mode • Left: Detects motion to the left of the virtual line. • Right: Detects motion to the right of the virtual line. • BothDirections: Detects motion on both sides of line • Off: No event will be detected **Line.#.Mode** is valid only when **DetectionType** is set to IntelligentVideo. The **Line.#.Mode** parameter must be sent along with **Line.#.Coordinate**.

Action	Parameter	Request/ Response	Type/ Value	Description
	Line.#.Coordinate	REQ, RES	<string>	X and Y coordinates of the two points which define the virtual line The coordinates are specified in the form of <x1,y1,x2,y2>; x1 and y1 are the start points and x2 and y2 are the end points. **Line.#.Coordinate** is valid only when **DetectionType** is set to IntelligentVideo. The **Line.#.Coordinate** parameter must be sent together with **Line.#.Mode**.
	Line.#.RuleName	REQ, RES	<string>	Name of rule This parameter can only be set on models supporting AI
	Line.#.ObjectTypeFilter	REQ, RES	<csv> Vehicle, Person	This parameter is only supported on models supporting AI; if this filter is not set, all objects crossing the line would be detected.
	Line.#.ObjectTypeFilterDetails	REQ, RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck, Person.Color.Orange, Person.Color.Black, Person.Color.Red	These parameters indicate the details of the ObjectType top-level object. If none of these details are set, the top-level object will be not be detected either.
	ROI.#.HandoverIndex	REQ, RES	<int>	**Hand Over Index** associated with the ROI region

Action	Parameter	Request/ Response	Type/ Value	Description
	DefinedArea.#.Handover Index	REQ, RES	<int>	**HandoverIndex** associated with the defined area
	Line.#.HandoverIndex	REQ, RES	<int>	**HandoverIndex** associated with the virtual line
	ROI.#.Duration	REQ, RES	<int>	ROI duration in seconds
	DefinedArea.#.Detection ResultOverlay	REQ, RES	<bool> True, False	Whether to mark detected motions on the screen in a box when an event occurs **DetectionResultOverlay** is valid only when **DetectionType** is NOT set to Off.
remove	Channel	REQ	<int>	ID of the channel to be deleted
	LineIndex	REQ	<csv> All, #	Index of the virtual line to be deleted
	DefinedAreaIndex	REQ	<csv> All, #	Index of the virtual area to be deleted
	ROIIndex	REQ	<csv> All, #	Index of the ROI to be deleted

158.2.4. Examples

158.2.4.1. Getting the current video analytics settings of 'Channel 0'

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoanalysis2&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoAnalysis": [
        {
            "Channel": 0,
            "DetectionType": "MDAndIV",
```

```
            "SensitivityLevel": 80,
            "ObjectSizeByDetectionTypes": [
                {
                    "DetectionType": "MotionDetection",
                    "MinimumObjectSize": "0,0",
                    "MaximumObjectSize": "99,99",
                    "MinimumObjectSizeInPixels": "48,48",
                    "MaximumObjectSizeInPixels": "3840,2160",
                    "EnableMetadataInExcludeArea": false
                },
                {
                    "DetectionType": "IntelligentVideo",
                    "MinimumObjectSize": "4,7",
                    "MaximumObjectSize": "50,89",
                    "MinimumObjectSizeInPixels": "194,194",
                    "MaximumObjectSizeInPixels": "1944,1944",
                    "EnableMetadataInExcludeArea": false
                }
            ],
            "ROIs": [
                {
                    "ROI": 1,
                    "Mode": "Inside",
                    "SensitivityLevel": 80,
                    "ThresholdLevel": 5,
                    "Coordinates": [
                        {
                            "x": 0,
                            "y": 0
                        },
                        {
                            "x": 0,
                            "y": 2159
                        },
                        {
                            "x": 3839,
                            "y": 2159
                        },
                        {
                            "x": 3839,
                            "y": 0
```

```
                }
            ],
            "HandoverIndex": 0,
            "Duration": 0
        }
    ],
    "Lines": [
        {
            "Line": 1,
            "Coordinates": [
                {
                    "x": 588,
                    "y": 454
                },
                {
                    "x": 3119,
                    "y": 1227
                }
            ],
            "Mode": "Right",
            "HandoverIndex": 0,
            "RuleName": "test1",
            "ObjectTypeFilter": [
                "Person",
                "Vehicle"
            ],
            "ObjectTypeFilterDetails": {
                "Vehicle": {
                    "Types": [
                        "Bicycle",
                        "Car",
                        "Motorcycle",
                        "Bus",
                        "Truck"
                    ]
                }
            }
        }
    ],
    "DefinedAreas": [
        {
```

```
                    "DefinedArea": 1,
                    "Type": "Inside",
                    "Mode": [
                        "AppearDisappear",
                        "Entering",
                        "Exiting",
                        "Intrusion"
                    ],
                    "Coordinates": [
                        {
                            "x": 996,
                            "y": 412
                        },
                        {
                            "x": 3443,
                            "y": 406
                        },
                        {
                            "x": 3347,
                            "y": 1749
                        },
                        {
                            "x": 1008,
                            "y": 1785
                        }
                    ],
                    "AppearanceDuration": 10,
                    "LoiteringDuration": 10,
                    "HandoverIndex": 0,
                    "IntrusionDuration": 2,
                    "RuleName": "area1",
                    "ObjectTypeFilter": [
                        "Person",
                        "Vehicle"
                    ],
                    "ObjectTypeFilterDetails": {
                        "Vehicle": {
                            "Types": [
                                "Bicycle",
                                "Car",
                                "Motorcycle",
```

```
                        "Bus",
                        "Truck"
                    ]
                }
            }
        },
        {
            "DefinedArea": 9,
            "Type": "Outside",
            "Mode": [],
            "Coordinates": [
                {
                    "x": 714,
                    "y": 426
                },
                {
                    "x": 714,
                    "y": 1769
                },
                {
                    "x": 2615,
                    "y": 1769
                },
                {
                    "x": 2615,
                    "y": 426
                }
            ],
            "AppearanceDuration": 1,
            "LoiteringDuration": 1,
            "IntrusionDuration": 0,
            "RuleName": "",
            "ObjectTypeFilter": [],
            "ObjectTypeFilterDetails": {}
        },
        {
            "DefinedArea": 10,
            "Type": "Outside",
            "Mode": [],
            "Coordinates": [
                {
```

```
                            "x": 2999,
                            "y": 208
                        },
                        {
                            "x": 2999,
                            "y": 1623
                        },
                        {
                            "x": 3425,
                            "y": 1623
                        },
                        {
                            "x": 3425,
                            "y": 208
                        }
                    ],
                    "AppearanceDuration": 1,
                    "LoiteringDuration": 1,
                    "IntrusionDuration": 0,
                    "RuleName": "",
                    "ObjectTypeFilter": [],
                    "ObjectTypeFilterDetails": {}
                }
            ]
        }
    ]
}
```


158.3. Audio Analytics Setup

158.3.1. Description

The **audioanalysis** submenu configures the audio analysis settings.

> **NOTE**
> This chapter applies to the network cameras only.
> Attribute to check for audio analytics support:
> "attributes/Eventsource/Support/AudioAnalysis"

Access level

Action	Camera
view	Admin

Action	Camera
set	Admin

158.3.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
audioanalysis&action=<value>[&<parameter>=<value>]
```

158.3.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads audio analysis settings
	Channel	REQ, RES	<csv>	Channel ID
	ConfigurationToken	RES	<string>	Audio source configuration token
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables audio analysis
	SensitivityLevel	REQ, RES	<int>	Sensitivity level for audio analysis **Note** Deprecated
	ThresholdLevel	REQ, RES	<int>	Threshold level for audio analysis
	NoiseReduction	REQ, RES	<bool> True, False	Enables or disables noise reduction
	SoundType	REQ, RES	<csv> Scream, Gunshot, Explosion, GlassBreak	Types of sounds supported
	HandoverIndex	REQ, RES	<int>	Handoverindex associated with audio analysis.

158.3.4. Examples

158.3.4.1. Getting the current audio analytics settings of 'Channel 0'

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=audioanalysis&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AudioAnalysis": [
        {
            "Channel": 0,
            "Enable": false,
            "NoiseReduction": false,
            "ThresholdLevel": 50,
            "SoundType": [],
            "HandoverIndex": 2,
            "ConfigurationToken": "AudioSourceConfig1"
        }
    ]
}
```


158.3.4.2. Changing audio analysis configuration

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=audioanalysis&action=set&Channel=0&Enable=True
&NoiseReduction=True&ThresholdLevel=25&SoundType=Scream,Gunshot
```

158.4. Fog Detection Setup

158.4.1. Description

The **fogdetection** submenu configures the fog detection settings.

NOTE This chapter applies to the network cameras only.
Attribute to check for feature support: "attributes/Eventsource/Support/FogDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.4.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
fogdetection&action=<value>[&<parameter>=<value>]
```

158.4.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads fog detection settings
	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables fog detection
	AutoDefog	REQ, RES	<bool> True, False	Enables or disables auto defog
	Duration	REQ, RES	<int>	Fog detection duration in seconds
	SensitivityLevel	REQ, RES	<int>	Sensitivity level for fog detection
	ThresholdLevel	REQ, RES	<int>	Threshold level for fog detection

158.4.4. Examples

158.4.4.1. Getting the settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=fogdetection&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "FogDetection": [
        {
            "Channel": 0,
            "Enable": false,
            "SensitivityLevel": 35,
```

```
            "ThresholdLevel": 50,
            "AutoDefog": false,
            "Duration": 30
        }
    ]
}
```

158.4.4.2. Changing fog detection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=fogdetection&action=set&Channel=0&Enable=True&
SensitivityLevel=10&ThresholdLevel=12&AutoDefog=True&Duration=50
```

158.5. Face Detection Setup

158.5.1. Description

The **facedetection** submenu configures human face detection settings.

> **NOTE**
>
> This chapter applies to the network cameras only.
> Attribute to check for feature support:
> "attributes/Eventsource/Support/Channel.#.FaceDetection"
> Attribute to check for max face detection areas:
> "attributes/Eventsource/Limit/MaxFaceDetectionArea"
>
> Camera gives adjusted coordinates based on flip/mirror/rotate settings when "/attributes/Eventsource/Support/AdjustMDIVRuleOnFlipMirror" parameter is set to False. Otherwise, camera provides original coordinates and client has to perform coordinate adjustment based on the flip/mirror/rotate settings.

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin

158.5.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
facedetection&action=<value>[&<parameter>=<value>]
```

158.5.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads face detection settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use face detection
	Sensitivity	REQ, RES	<int>	Face detection sensitivity level
	OverlayColor	REQ, RES	<enum> Red, Orange, Yellow, Green, Blue, Navy, Violet, Black, White	Color of face detected area line
	MarkDetectedFaces	REQ, RES	<bool> True, False	Whether to mark detected faces on the screen
	DetectionAreaMode	REQ, RES	<enum> Inside, Outside	Face detection area mode • Inside: Detects faces within the specified area • Outside: Exclude region. In this region, face detection does not work
	DetectionArea.#.Mode	REQ, RES	<enum> Inside, Outside	Face detection area mode • Inside: Detects faces within the specified area • Outside: Exclude region. In this region, face detection does not work
	DetectionArea.#.Coordinate	REQ, RES	<string>	Face detection area coordinates **Note** Please refer to "MaxFaceDetectionArea" in attributes page for settable area count.

Action	Parameter	Request/ Response	Type/ Value	Description
	DynamicArea	REQ, RES	<bool>	Enable or disable motion based dynamic area for face detection
remove	Channel	REQ	<int>	Channel ID
	DetectionAreaIndex	REQ	<csv> All, #	Index of the detection area to be deleted **Note** **DetectionAreaIndex** must be sent together with the **remove** action.

158.5.4. Examples

158.5.4.1. Getting the settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.0.Sensitivity=5
Channel.0.DynamicArea=True
Channel.0.OverlayColor=Red
Channel.0.MarkDetectedFaces=False
Channel.0.DetectionAreaMode=Inside
Channel.0.DetectionArea.1.Mode=Inside
Channel.0.DetectionArea.1.Coordinate=591,126,276,129,255,372,663,372
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "FaceDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Sensitivity": 5,
            "DynamicArea": true,
            "OverlayColor": "Red",
            "MarkDetectedFaces": false,
            "DetectionAreaMode": "Inside",
            "DetectionAreas": [
                {
                    "DetectionArea": 1,
                    "Mode": "Inside",
                    "Coordinates": [
                        {
                            "x": 591,
                            "y": 126
                        },
                        {
                            "x": 276,
                            "y": 129
                        },
                        {
                            "x": 255,
                            "y": 372
                        },
                        {
                            "x": 663,
                            "y": 372
                        }
                    ]
                }
            ]
        }
    ]
}
```


158.5.4.2. Enabling face detection and setting the sensitivity level to 10

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=set&Enable=True&Sensitivi
ty=10
```

158.5.4.3. Enabling motion based dynamic area setting

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=set&DynamicArea=True
```

158.5.4.4. Setting the coordinates of detection area index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=set&Enable=True&Detection
Area.1.Coordinate=86,154,973,965,513,725,32,992
```

158.5.4.5. Removing all detection area indexes in all channels

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=remove&DetectionAreaIndex
=All
```

158.5.4.6. Removing all detection area indexes in Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=facedetection&action=remove&Channel=0&Detectio
nAreaIndex=All
```

158.6. Tampering Detection

158.6.1. Description

The **tamperingdetection** submenu configures event detection for scene changes (tampering attempts). It can detect tampering attempts such as a sudden change in the camera’s viewing direction, a blocked lens, and other overall changes to the scenes in the video.

NOTE	This chapter applies to network cameras and encoders only. Attribute to check for feature support: "attributes/Eventsource/Support/TamperingDetection"

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

158.6.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
tamperingdetection&action=<value>[&<parameter>=<value>]
```

158.6.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads tampering detection settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use tampering detection
	Sensitivity	REQ, RES	<enum> VeryLow, Low, Medium, High, VeryHigh	Scene change detection sensitivity level
	DarknessDetection	REQ, RES	<bool> True, False	Enables or disables darkness detection
	Duration	REQ, RES	<int>	Tampering detection duration in seconds
	SensitivityLevel	REQ, RES	<int>	Sensitivity level for tampering
	ThresholdLevel	REQ, RES	<int>	Threshold level for tampering
	HandoverIndex	REQ, RES	<int>	**HandOverIndex** associated with tampering detection
	RuleName	REQ, RES	<string>	Name of rule

Action	Parameter	Request/ Response	Type/ Value	Description
	ChannelIDList	REQ	<csv>	List of channels to be configured NVR ONLY

158.6.4. Examples

158.6.4.1. Getting tampering detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tamperingdetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.0.Sensitivity=Low
Channel.0.SensitivityLevel=80
Channel.0.ThresholdLevel=9
Channel.0.Duration=22
Channel.0.DarknessDetection=False
Channel.0.HandoverIndex=1
Channel.0.RuleName=TamperingRule-0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TamperingDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Sensitivity": "High",
```

```
            "SensitivityLevel": 80,
            "ThresholdLevel": 9,
            "Duration": 22,
            "DarknessDetection": false,
            "HandoverIndex": 1,
            "RuleName": "TamperingRule-0"
        }
    ]
}
```

158.6.4.2. Setting tampering detection sensitivity level to high

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tamperingdetection&action=set&Enable=True&Sens
itivity=High
```

158.7. Audio Detection

158.7.1. Description

The **audiodetection** submenu configures audio detection settings.

NOTE	This chapter applies to the network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/AudioDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.7.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
audiodetection&action=<value>[&<parameter>=<value>]
```

158.7.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads audio detection settings
	Channel	REQ, RES	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use audio detection
	InputThresholdLevel	REQ, RES	<int>	Audio detection input threshold level
check	Channel	REQ, RES	<int>	Channel ID
	MaxSamples	REQ	<int>	Maximum Samples for audio detection
	SequenceID	RES	<int>	Sequence ID
	Level	RES	<int>	Audio level

158.7.4. Examples

158.7.4.1. Getting audio detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=audiodetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.InputThresholdLevel=50
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "AudioDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "InputThresholdLevel": 50
        }
    ]
}
```

158.7.4.2. Enabling audio detection with an input threshold level of 50

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=audiodetection&action=set&Enable=True&InputThr
esholdLevel=50
```

158.7.4.3. Checking audio detection level for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=audiodetection&action=check&MaxSamples=5
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.SequenceID.18738854.Level=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AudioDetection": [
        {
```

```
                "Channel": 0,
                "Samples": [
                    {
                        "SequenceID": 18738854,
                        "Level": 0
                    }
                ]
            }
        ]
}
```

158.8. Video Loss

158.8.1. Description

The **videoloss** submenu configures video loss detection settings. The event is triggered when the camera is disconnected.

> **NOTE**
> This feature is available for encoder products.
> Attribute to check for feature support: "attributes/Eventsource/Support/VideoLoss"

Access level

Action	Encoder
view	Admin
set	Admin

158.8.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
videoloss&action=<value>[&<parameter>=<value>]
```

158.8.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads video loss detection settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use video loss detection

158.8.4. Examples

158.8.4.1. Getting video loss settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoloss&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "VideoLoss": [
        {
            "Channel": 0,
            "Enable": true
        }
    ]
}
```

158.8.4.2. Enabling video loss detection

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=videoloss&action=set&Enable=True
```

158.9. Auto Tracking

158.9.1. Description

The **autotracking** submenu configures auto tracking settings. If the camera supports AI engine, then this submenu supports AI-based tracking, in which case users can choose a specific object type to track.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for feature support: "attributes/Eventsource/Support/Tracking"
> Attribute to check for maximum tracking areas: "attributes/Eventsource/Limit/MaxTrackingArea"
> Attribute to check for supporting AI-based tracking: "attributes/PTZSupport/Support/AIAutoTracking"

Access level

Action	Camera
view	Admin
set	Admin
control	Admin
add	Admin
remove	Admin

158.9.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
autotracking&action=<value>[&<parameter>=<value>]
```

158.9.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the auto tracking settings
	Channel	REQ, RES	<csv>	Channel ID
	TrackingAreas	RES	<int>	The number of configured tracking areas
control	Channel	REQ	<int>	Channel ID
	TargetLockOn	REQ	<bool> True, False	Whether to lock the target

Action	Parameter	Request/ Response	Type/ Value	Description
	TargetLockCoordinate	REQ	<string>	The coordinates to lock the target The coordinates are specified in the form of <x,y>; x and y are scaled from 1 to 10000. **Note** **TargetLockCoordinate** must be sent together with the **control** action.
	TrackingArea.#	REQ	<enum> Start, End	Tracking area
	Mode	REQ	<enum> Start, Move	Tracking Area Mode **Note** When **Mode** is set to Move, TrackingAreaID parameter should be sent along with Mode. When **Mode** is set to Start, TrackingCoordinate parameter should be sent along with Mode.
	TrackingAreaID	REQ	<string>	Tracking area ID
	TrackingCoordinate	REQ, RES	<string> <format=x1,y1,x2,y2>	Tracking Coordinate
set	Channel	REQ, RES	<Int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables the auto tracking
	CameraHeight	REQ, RES	<enum>	Camera height
	ObjectSize	REQ, RES	<enum> Small, Medium, Large	Size of the object to be tracked
	Sensitivity	REQ, RES	<enum> Low, Medium, High	Auto tracking sensitivity

Action	Parameter	Request/ Response	Type/ Value	Description
	ZoomControl	REQ, RES	<enum> Off, OneShot, Continuous	Zoom mode • Off: Disables zoom • OneShot: Operates zoom one time when detecting an object • Continuous: Operates zoom continuously
	LostMode	REQ, RES	<enum> Stop, Research, ZoomOut	Action for the remaining tracking duration defined in Auto Release after the tracking object escapes
	AutoReturn	REQ, RES	<enum> Off, 1s, 2s, 3s, 4s, 5s, 6s, 7s, 8s, 9s, 10s, 20s, 30s, 40s, 50s, 1m, 2m, 3m, 4m, 5m	Time interval to return the camera to the initial position after the Auto Release duration
	AutoRelease	REQ, RES	<enum> Off, 10s, 20s, 30s, 40s, 50s, 1m, 2m, 3m, 4m, 5m	Tracking duration, to continue tracking the object
	DisplayIndicator	REQ, RES	<enum> Off, Rectangle, Pointer, Target	Indicator to surround the object when auto-tracking
	TrackingArea.#.Enable	REQ, RES	<bool> True, False	Enables or disables the corresponding tracking area
	DisplayTrackingArea	REQ, RES	<bool> True, False	Whether to display tracking area
	TrackingAreaEnable	REQ, RES	<bool> True, False	Enables or disables the global setting for tracking areas
	ObjectFilterEnable	REQ, RES	<bool> True, False	Enables or disables the specific object type tracking

Action	Parameter	Request/ Response	Type/ Value	Description
	ObjectTypeFilter	REQ, RES	<csv> Person, Vehicle	Object Type to track
	LostModeTimer	REQ, RES	<int> 1 to 60	Lost mode timeout, when the LostMode is configured as Stop.
	Redection	REQ, RES	<bool> True, False	Enable or disable redection.
	RedetectionTimer	REQ, RES	<int> 5 to 60	Time window for re-detection, applicable when redection is enabled.
	AfterAction	REQ, RES	<enum> None, GoToHome, LastPosition, StartAutoRun	Action to perform after autotracking.
add	Channel	REQ, RES	<int>	Channel ID
	TrackingAreaID	REQ, RES	<string>	Tracking area ID **Note** **TrackingAreaID** and **Coordinate** must be sent together with the **add** action.
	Coordinate	REQ, RES	<string>	Coordinates The coordinates are specified in the form of <x1,y1,x2,y2>; x and y are points based on real pixels. Please refer **ROICoordinate Min and Max X, Y** in attributes to get the actual pixel range that a device supports. **Note** **TrackingAreaID** and **Coordinate** must be sent together with the **add** action.
remove	Channel	REQ	<Int>	Channel ID

Action	Parameter	Request/ Response	Type/ Value	Description
	TrackingAreaID	REQ	<csv>	Tracking area ID **Note** **TrackingAreaID** must be sent together with the **remove** action.

158.9.4. Examples

158.9.4.1. Getting auto tracking settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.CameraHeight=250cm
Channel.0.ObjectSize=Medium
Channel.0.ZoomControl=On
Channel.0.LostMode=Research
Channel.0.DisplayIndicator=Off
Channel.0.TrackingAreaEnable=True
Channel.0.TrackingArea.Tracking04.Coordinate=1,1,3840,2160
Channel.0.TrackingArea.Tracking03.Coordinate=1,1,3839,2159
Channel.0.TrackingArea.Tracking02.Coordinate=1000,1500,1500,2000
Channel.0.TrackingAreas=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AutoTracking": [
        {
            "Channel": 0,
            "Enable": true,
            "CameraHeight": "250cm",
            "ObjectSize": "Medium",
            "ZoomControl": "On",
            "LostMode": "Research",
            "DisplayIndicator": "Off",
            "TrackingAreaEnable": true,
            "TrackingAreas": [
                {
                    "TrackingArea": "Tracking04",
                    "Coordinates": [
                        {
                            "x": 1,
                            "y": 1
                        },
                        {
                            "x": 3840,
                            "y": 2160
                        }
                    ]
                },
                {
                    "TrackingArea": "Tracking03",
                    "Coordinates": [
                        {
                            "x": 1,
                            "y": 1
                        },
                        {
                            "x": 3839,
                            "y": 2159
                        }
                    ]
                },
                {
                    "TrackingArea": "Tracking02",
                    "Coordinates": [
```

```
                        {
                            "x": 1000,
                            "y": 1500
                        },
                        {
                            "x": 1500,
                            "y": 2000
                        }
                    ]
                }
            ]
        }
    ]
}
```

158.9.4.2. Responses from PTZ models supporting AI Autotracking

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AutoTracking": [
        {
            "Channel": 0,
            "Enable": false,
            "CameraHeight": "250cm",
            "ObjectSize": "Medium",
            "ZoomControl": "On",
            "LostMode": "Research",
            "DisplayIndicator": "Off",
            "TrackingAreaEnable": true,
            "ObjectFilterEnable": true,
            "ObjectTypeFilter": [
                "Person",
                "Vehicle"
            ]
        }
    ]
```

```
}
```

158.9.4.3. Specifying auto tracking settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=set&Enable=True&CameraHeig
ht=300cm&LostMode=Research
```

158.9.4.4. Locking the target

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=control&Channel=0&TargetLo
ckOn=True
```

158.9.4.5. Tracking area 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=control&Channel=0&Tracking
Area.1=Start
```

158.9.4.6. Configuring tracking areas

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=add&Channel=0&TrackingArea
ID=TA1&Coordinate=1,1,5000,5000
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.9.4.7. Removing tracking areas

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=remove&Channel=0&TrackingA
reaID=TA1
```

158.9.4.8. Moving to a tracking area

REQUEST

```
http://<Device IP>/ stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=control&Mode=Move&Tracking
AreaID=TA1
```

158.9.4.9. Starting a tracking area

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=autotracking&action=control&Mode=Start&Trackin
gCoordinate=1884,690,2958,1536
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.TrackingCoordinate=1383,657,2457,1503
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TrackingArea": [
        {
            "Channel": 0,
            "TrackingCoordinate": [
                {
                    "x": 1383,
                    "y": 657
                },
                {
                    "x": 2457,
                    "y": 1503
                }
            ]
        }
    ]
}
```


158.10. Scheduled Events

158.10.1. Description

The **timer** submenu configures the schedule for the timer, which can generate events periodically.

NOTE This chapter applies to network cameras and encoders only.
Attribute to check for feature support: "attributes/Eventsource/Support/Timer"

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

158.10.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
timer&action=<value>[&<parameter>=<value>]
```

158.10.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the scheduled events settings.
set	Enable	REQ, RES	<bool> True, False	Whether to use automatic scheduled events
	ScheduleInterval	REQ, RES	<enum> 5, 10, 15, 30, 45, 60	Schedule interval in seconds/minutes
	ScheduleIntervalUnit	REQ, RES	<enum> Seconds, Minutes	Schedule interval units

158.10.4. Examples

158.10.4.1. Getting the event schedule settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=timer&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
ScheduleInterval=60
ScheduleIntervalUnit=Seconds
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false,
    "ScheduleInterval": "60",
    "ScheduleIntervalUnit": "Seconds"
}
```

158.10.4.2. Setting the schedule to record video every 5 minutes

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=timer&action=set&ScheduleInterval=5&ScheduleIn
tervalUnit=Minutes
```

158.11. Alarm Input

158.11.1. Description

The **alarminput** submenu sets the alarm input (activated/ deactivated) and the alarm input state.

NOTE

Attribute to check for feature support: "attributes/Eventsource/Support/AlarmInput"
Attribute to check for maximum alarm inputs: "attributes/Eventsource/Limit/MaxAlarmInput"
Attribute to check for supervised feature support: "attributes/Eventsource/Support/AlarmInput.Supervised"

Access level

Action	Camera	Encoder	NVR
view	Admin	Admin	User
set	Admin	Admin	User

158.11.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
alarminput&action=<value>[&<parameter>=<value>]
```

158.11.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the alarm input settings.

Action	Parameter	Request/ Response	Type/ Value	Description
set	AlarmInput.#.Enable	REQ, RES	<bool> True, False	Enables or disables alarm input.
	AlarmInput.#.State	REQ, RES	<enum> NormallyOpen, NormallyClose	Alarm triggering condition • NormallyOpen: The alarm input sensor’s "open circuit" state is considered normal, and the alarm is triggered if it goes into to a "closed circuit" state. • NormallyClose: The alarm input sensor’s "closed circuit" state is considered normal, and the alarm is triggered if it goes into to an "open circuit" state.
	AlarmInput.#.IOPortIndex	RES	<int>	Physical IO Port Index
	AlarmInput.#.SupervisedCheck	RES	<int>	Optional, Only exposed to Indexes that support Supervised

NOTE represents the index number of the alarm input.

158.11.4. Examples

158.11.4.1. Getting the current alarm input setting information

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=alarminput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1.Enable=True
AlarmInput.1.State=NormallyOpen
AlarmInput.1.IOPortIndex=1
AlarmInput.2.Enable=True
AlarmInput.2.State=NormallyOpen
AlarmInput.2.IOPortIndex=2
```

```
AlarmInput.3.Enable=True
AlarmInput.3.State=NormallyOpen
AlarmInput.4.IOPortIndex=3
AlarmInput.4.Enable=True
AlarmInput.4.State=NormallyOpen
AlarmInput.4.IOPortIndex=4
AlarmInput.4.SupervisedCheck=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AlarmInputs": [
        {
            "AlarmInput": 1,
            "Enable": true,
            "State": "NormallyOpen",
            "IOPortIndex": 1
        },
        {
            "AlarmInput": 2,
            "Enable": true,
            "State": "NormallyOpen",
            "IOPortIndex": 2
        },
        {
            "AlarmInput": 3,
            "Enable": true,
            "State": "NormallyOpen",
            "IOPortIndex": 3
        },
        {
            "AlarmInput": 4,
            "Enable": true,
            "State": "NormallyOpen",
            "IOPortIndex": 4,
            "SupervisedCheck": false
        }
```

```
    ]
}
```

158.11.4.2. Setting Alarm Input 1 to 'Enabled'

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=alarminput&action=set&Alarminput.1.Enable=True
```

158.11.4.3. Setting the state of Alarm Input 1 to 'NormallyOpen'

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=alarminput&action=set&AlarmInput.1.State=Norma
llyOpen
```

158.12. Fire Alarm Input

158.12.1. Description

The **firealarminput** submenu sets the fire alarm input (activated/ deactivated).

NOTE Attribute to check for feature support: "attributes/Eventsource/Support/FireAlarmInput"

Access level

Action	Camera	Encoder	NVR
view	Admin	Admin	User
set	Admin	Admin	User

158.12.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
firealarminput&action=<value>[&<parameter>=<value>]
```

158.12.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the firealarm input settings.

Action	Parameter	Request/ Response	Type/ Value	Description
set	FireAlarmInput.#.Enable	REQ, RES	<bool> True, False	Enables or disables fire alarm input.

NOTE represents the index number of the alarm input.

158.12.4. Examples

158.12.4.1. Getting the current alarm input setting information

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=firealarminput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "FireAlarmInputs": [
        {
            "FireAlarmInput": 1,
            "Enable": true
        }
    ]
}
```

158.12.4.2. Setting Fire Alarm Input 1 to 'Enabled'

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=firealarminput&action=set&FireAlarminput.1.Ena
ble=True
```

158.13. Network Alarm Input

158.13.1. Description

The **networkalarminput** submenu configures the network alarm input function.

NOTE	This chapter applies to NVR only. Attribute to check for feature support: "attributes/Eventsource/Support/NetworkAlarmInput" Attribute to check for max network alarm inputs: "attributes/Eventsource/Limit/MaxNetworkAlarmInput"

Access level

Action	NVR
view	User
set	User

158.13.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
networkalarminput&action=<value>[&<parameter>=<value>]
```

158.13.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the network alarm input settings.
	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use the network alarm input event

158.13.4. Examples

158.13.4.1. Getting network alarm input events on Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=networkalarminput&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NetworkAlarmInputs": [
        {
            "Channel": 0,
            "Enable": true
        }
    ]
}
```


158.13.4.2. Enabling network alarm input

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=networkalarminput&action=set&Channel=0&Enable=
True
```

158.14. Network Disconnection

158.14.1. Description

The **networkdisconnect** submenu sets the detection of network disconnections (activated/deactivated).

NOTE

This chapter applies to the network cameras only.
Attribute to check for feature support:
"attributes/Eventsource/Support/NetworkDisconnect"

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

158.14.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
networkdisconnect&action=<value>[&<parameter>=<value>]
```

158.14.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the network disconnection settings.
	Channel	REQ, RES	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use the Network Disconnection event

158.14.4. Examples

158.14.4.1. Getting network disconnection events on Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=networkdisconnect&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NetworkDisconnect": [
        {
            "Channel": 0,
            "Enable": false
        }
    ]
}
```


158.14.4.2. Enabling network disconnection event

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=networkdisconnect&action=set&Enable=True
```

158.15. Defocus Detection

158.15.1. Description

The **defocusdetection** submenu configures event detection when the camera is out of focus.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/DefocusDetection"

Access level

Action	Camera
view	Admin

Action	Camera
set	Admin

158.15.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
defocusdetection&action=<value>[&<parameter>=<value>]
```

158.15.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads defocus detection settings
	Channel	REQ, RES	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use defocus detection
	Sensitivity	REQ, RES	<int>	Defocus detection sensitivity level
	Duration	REQ, RES	<int>	Defocus detection duration in seconds
	ThresholdLevel	REQ, RES	<int>	Threshold level for defocus detection
	AutoSimpleFocus	REQ, RES	<bool> True, False	To enable/disable auto simple focus **Note** Only supported in simple focus supported models
	OneShotAF	REQ, RES	<bool> True, False	To enable/disable One shot Auto focus **Note** Only supported in OneShotAutoFocus supported models. In attributes.cgi response, refer to below capability in Eventsources/Support/Channel **DefocusDetection.OneShotAF**
	RuleName	REQ, RES	<string>	Name of rule

158.15.4. Examples

158.15.4.1. Getting defocus detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=defocusdetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.Sensitivity=22
Channel.0.ThresholdLevel=71
Channel.0.Duration=10
Channel.0.AutoSimpleFocus=False
Channel.0.RuleName=DefocusRule-0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "DefocusDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Sensitivity": 22,
            "ThresholdLevel": 71,
            "Duration": 10,
            "AutoSimpleFocus": false,
            "RuleName": "DefocusRule-0"
        }
    ]
}
```

158.15.4.2. Changing defocus detection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=defocusdetection&action=set&Channel=0&Enable=T
rue&Sensitivity=75&ThresholdLevel=88&Duration=6&AutoSimpleFocus=True&RuleNam
e=Defocus1
```

158.16. People Count

158.16.1. Description

The **peoplecount** submenu configures people count settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin
check	Admin

158.16.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
peoplecount&action=<value>[&<parameter>=<value>]
```

158.16.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads people count settings
	Channel	REQ, RES	<int>	Channel ID
	MasterName	RES	<string>	Master camera name **Note** Current camera is master camera always.
set	Channel	REQ, RES	<int>	Channel ID

Action	Parameter	Request/ Response	Type/ Value	Description
	Enable	REQ, RES	<bool> True, False	Whether to use people count feature
	CalibrationMode	REQ, RES	<enum> CameraHeight, ObjectSize	Calibration mode
	CameraHeight	REQ, RES	<int>	Height at which camera is mounted **Note** This parameter is valid only when **CalibrationMode** is set to CameraHeight.
	ObjectSizeCoordinates	REQ, RES	<string> Format=x1,y1,x2,y2	Coordinates of detectable Object size **Note** This parameter is valid only when **CalibrationMode** is set to ObjectSize.
	Line.#.Name	REQ, RES	<string>	Name of the line
	Line.#.Enable	REQ, RES	<bool> True, False	Whether to use the line or not
	Line.#.Mode	REQ, RES	<enum> LeftToRightIn, RightToLeftIn	Line detection mode: • LeftToRightIn: Detects when person crosses the line from left to right. • RightToLeftIn: Detects when person crosses the line from right to left.
	Line.#.Coordinates	REQ, RES	<string> Format=x1,y1,x2,y2	X and Y coordinates of the two points which define the line The coordinates are specified in the form of <x1,y1,x2,y2>; x1 and y1 are the start points and x2 and y2 are the end points.
	ReportEnable	REQ, RES	<bool> True, False	Whether to use report or not
	ReportFilename	REQ, RES	<string>	File name of the report

Action	Parameter	Request/ Response	Type/ Value	Description
	ReportFileType	REQ, RES	<enum> XLSX, TXT	File type of the report
	Area.#.Type	REQ, RES	<enum> Outside	Specifying the area to exclude from analyzing peoplecount
	Area.#.Coordinates	REQ, RES	<string>	Coordinates of area to set The coordinates are specified in the form of < x1,y1,x2,y2>.
remove	Channel	REQ	<int>	Channel ID
	AreaIndex	REQ	<csv>	Area Index
check	Channel	REQ	<int>	Channel ID
	LineIndex	REQ	<int>	Index of the line
	Name	RES	<string>	Line name
	InCount	RES	<int>	Number of people who entered the line
	OutCount	RES	<int>	Number of people who exited the line

158.16.4. Examples

158.16.4.1. Getting people count settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.MasterName=PeopleCount-Master
Channel.0.Enable=True
Channel.0.ReportEnable=True
Channel.0.ReportFilename=peoplecountreport
Channel.0.ReportFileType=XLSX
Channel.0.CalibrationMode=CameraHeight
Channel.0.CameraHeight=300
```

```
Channel.0.ObjectSizeCoordinate=1316,1316,1675,1675
Channel.0.Line.1.Name=testline1
Channel.0.Line.1.Enable=True
Channel.0.Line.1.Mode=LeftToRightIn
Channel.0.Line.1.Coordinate=1043,1875,2875,1943
Channel.0.Line.2.Name=testline2
Channel.0.Line.2.Enable=True
Channel.0.Line.2.Mode=LeftToRightIn
Channel.0.Line.2.Coordinate=2912,893,1206,706
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PeopleCount": [
        {
            "Channel": 0,
            "MasterName": "PeopleCount-Master",
            "Enable": true,
            "ReportEnable": true,
            "ReportFilename": "peoplecountreport",
            "ReportFileType": "XLSX",
            "CalibrationMode": "CameraHeight",
            "CameraHeight": 300,
            "ObjectSizeCoordinate": [
                {
                    "x": 1316,
                    "y": 1316
                },
                {
                    "x": 1675,
                    "y": 1675
                }
            ],
            "Lines": [
                {
                    "Line": 1,
                    "Mode": "LeftToRightIn",
```

```
                    "Name": "testline1",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 1043,
                            "y": 1875
                        },
                        {
                            "x": 2875,
                            "y": 1943
                        }
                    ]
                },
                {
                    "Line": 2,
                    "Mode": "LeftToRightIn",
                    "Name": "testline2",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 2912,
                            "y": 893
                        },
                        {
                            "x": 1206,
                            "y": 706
                        }
                    ]
                }
            ]
        }
    ]
}
```


158.16.4.2. Setting people count data

Setting up a people count line rule to detect people crossing the line from right to left

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=set&Enable=True&Line.1.Name
=testrule1&Line.1.Coordinate=11,12,600,400&Line.1.Mode=RightToLeftIn&ObjectS
```

```
izeInPixels=100,200
```

Setting up a people count report

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=set&ReportEnable=True&Repor
tFilename=testreport&ReportFileType=TXT
```

158.16.4.3. Removing exclude area

To remove an exclude region based on area index

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=remove&Channel=0&AreaIndex=
1
```

158.16.4.4. Reset Peoplecounting DB

Refer to system.cgi **databasereset** submenu.

158.16.4.5. Getting people count live data

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.LineIndex=1
Channel.0.LineIndex.1.Name=testline1
Channel.0.LineIndex.1.InCount=0
Channel.0.LineIndex.1.OutCount=0
Channel.0.LineIndex=2
Channel.0.LineIndex.2.Name=testline2
Channel.0.LineIndex.2.InCount=9
```

```
Channel.0.LineIndex.2.OutCount=21
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PeopleCount": [
        {
            "Lines": [
                {
                    "LineIndex": 1,
                    "Name": "testline1",
                    "InCount": 0,
                    "OutCount": 0
                },
                {
                    "LineIndex": 2,
                    "Name": "testline2",
                    "InCount": 9,
                    "OutCount": 21
                }
            ]
        }
    ]
}
```


158.17. Vehicle Count

158.17.1. Description

The **vehiclecount** submenu configures vehicle count settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

Action	Camera
set	Admin
check	Admin

158.17.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
vehiclecount&action=<value>[&<parameter>=<value>]
```

158.17.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads vehicle count settings
	Channel	REQ, RES	<int>	Channel ID
	MasterName	RES	<string>	Master camera name **Note** Current camera is master camera always.
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use vehicle count feature
	CalibrationMode	REQ, RES	<enum> CameraHeight, ObjectSize	Calibration mode
	CameraHeight	REQ, RES	<int>	Height at which camera is mounted **Note** This parameter is valid only when **CalibrationMode** is set to CameraHeight.
	ObjectSizeCoordinates	REQ, RES	<string> Format=x1,y1,x2,y2	Coordinates of detectable Object size **Note** This parameter is valid only when **CalibrationMode** is set to ObjectSize.
	Line.#.Name	REQ, RES	<string>	Name of the line

Action	Parameter	Request/ Response	Type/ Value	Description
	Line.#.Enable	REQ, RES	<bool> True, False	Whether to use the line or not
	Line.#.Mode	REQ, RES	<enum> LeftToRightIn, RightToLeftIn	Line detection mode: • LeftToRightIn: Detects when person crosses the line from left to right. • RightToLeftIn: Detects when person crosses the line from right to left.
	Line.#.Coordinates	REQ, RES	<string> Format=x1,y1,x2,y2	X and Y coordinates of the two points which define the line The coordinates are specified in the form of <x1,y1,x2,y2>; x1 and y1 are the start points and x2 and y2 are the end points.
	ReportEnable	REQ, RES	<bool> True, False	Whether to use report or not
	ReportFilename	REQ, RES	<string>	File name of the report
	ReportFileType	REQ, RES	<enum> XLSX, TXT	File type of the report
	Area.#.Type	REQ, RES	<enum> Outside	Specifying the area to exclude from analyzing vehiclecount
	Area.#.Coordinates	REQ, RES	<string>	Coordinates of area to set The coordinates are specified in the form of < x1,y1,x2,y2>.
remove	Channel	REQ	<int>	Channel ID
	AreaIndex	REQ	<csv>	Area Index
check	Channel	REQ	<int>	Channel ID
	ShowAIStats	REQ	<bool> True,False	Whether to show AI Stats
	LineIndex	REQ	<int>	Index of the line
	Name	RES	<string>	Line name
	InCount	RES	<int>	Number of all vehicles who entered the line

Action	Parameter	Request/ Response	Type/ Value	Description
	In.Car	RES	<int>	Number of the Car who entered the line This parameter is shown when **ShowAIStats** is True
	In.Bus	RES	<int>	Number of the Bus who entered the line This parameter is shown when **ShowAIStats** is True
	In.Truck	RES	<int>	Number of the Truck who entered the line This parameter is shown when **ShowAIStats** is True
	In.Motorcycle	RES	<int>	Number of the Motorcycle who entered the line This parameter is shown when **ShowAIStats** is True
	In.Bicycle	RES	<int>	Number of the Bicycle who entered the line This parameter is shown when **ShowAIStats** is True
	OutCount	RES	<int>	Number of all vehicles who exited the line
	Out.Car	RES	<int>	Number of the Car who exited the line This parameter is shown when **ShowAIStats** is True
	Out.Bus	RES	<int>	Number of the Bus who exited the line This parameter is shown when **ShowAIStats** is True
	Out.Truck	RES	<int>	Number of the Truck who exited the line This parameter is shown when **ShowAIStats** is True

Action	Parameter	Request/ Response	Type/ Value	Description
	Out.Motorcycle	RES	<int>	Number of the Motorcycle who exited the line This parameter is shown when **ShowAIStats** is True
	Out.Bicycle	RES	<int>	Number of the Bicycle who exited the line This parameter is shown when **ShowAIStats** is True

158.17.4. Examples

158.17.4.1. Getting vehicle count settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=vehiclecount&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.MasterName=VehicleCount-Master
Channel.0.Enable=True
Channel.0.ReportEnable=False
Channel.0.ReportFilename=
Channel.0.ReportFileType=XLSX
Channel.0.CalibrationMode=ObjectSize
Channel.0.ObjectSizeCoordinate=1790,950,2049,1209
Channel.0.Line.1.Name=name1
Channel.0.Line.1.Enable=True
Channel.0.Line.1.Mode=LeftToRightIn
Channel.0.Line.1.Coordinate=1,1180,3839,1180
Channel.0.Line.2.Name=name2
Channel.0.Line.2.Enable=True
Channel.0.Line.2.Mode=LeftToRightIn
Channel.0.Line.2.Coordinate=3839,980,1,980
```

```
Channel.0.Area.1.Type=Outside
Channel.0.Area.1.Coordinates=1136,468,1136,1715,1670,1715,1670,468
Channel.0.Area.2.Type=Outside
Channel.0.Area.2.Coordinates=
Channel.0.Area.3.Type=Outside
Channel.0.Area.3.Coordinates=
Channel.0.Area.4.Type=Outside
Channel.0.Area.4.Coordinates=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VehicleCount": [
        {
            "Channel": 0,
            "MasterName": "VehicleCount-Master",
            "Enable": true,
            "ReportEnable": false,
            "ReportFilename": "",
            "ReportFileType": "XLSX",
            "CalibrationMode": "ObjectSize",
            "ObjectSizeCoordinate": [
                {
                    "x": 1790,
                    "y": 950
                },
                {
                    "x": 2049,
                    "y": 1209
                }
            ],
            "Lines": [
                {
                    "Line": 1,
                    "Mode": "LeftToRightIn",
                    "Name": "name1",
                    "Enable": true,
```

```
                    "Coordinates": [
                        {
                            "x": 1,
                            "y": 1180
                        },
                        {
                            "x": 3839,
                            "y": 1180
                        }
                    ]
                },
                {
                    "Line": 2,
                    "Mode": "LeftToRightIn",
                    "Name": "name2",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 3839,
                            "y": 980
                        },
                        {
                            "x": 1,
                            "y": 980
                        }
                    ]
                }
            ],
            "Areas": [
                {
                    "Area": 1,
                    "Type": "Outside",
                    "Coordinates": [
                        {
                            "x": 1136,
                            "y": 468
                        },
                        {
                            "x": 1136,
                            "y": 1715
                        },
```

```
                        {
                            "x": 1670,
                            "y": 1715
                        },
                        {
                            "x": 1670,
                            "y": 468
                        }
                    ]
                },
                {
                    "Area": 2,
                    "Type": "Outside",
                    "Coordinates": []
                },
                {
                    "Area": 3,
                    "Type": "Outside",
                    "Coordinates": []
                },
                {
                    "Area": 4,
                    "Type": "Outside",
                    "Coordinates": []
                }
            ]
        }
    ]
}
```


158.17.4.2. Setting vehicle count data

Setting up a vehicle count line rule to detect vehicle crossing the line from right to left

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=vehiclecount&action=set&Enable=True&Line.1.Nam
e=testrule1&Line.1.Coordinate=11,12,600,400&Line.1.Mode=RightToLeftIn&Object
SizeInPixels=100,200
```

Setting up a vehicle count report

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=vehiclecount&action=set&ReportEnable=True&Repo
rtFilename=testreport&ReportFileType=TXT
```

158.17.4.3. Removing exclude area

To remove an exclude region based on area index

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=vehiclecount&action=remove&Channel=0&AreaIndex
=1
```

158.17.4.4. Reset Vehiclecounting DB

Refer to system.cgi **databasereset** submenu.

158.17.4.5. Getting vehicle count live data with AI Stats

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=vehiclecount&action=check&ShowAIStats=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.LineIndex=1
Channel.0.LineIndex.1.Name=name1
Channel.0.LineIndex.1.InCount=0
Channel.0.LineIndex.1.In.Car=0
Channel.0.LineIndex.1.In.Bus=0
Channel.0.LineIndex.1.In.Truck=0
Channel.0.LineIndex.1.In.Motorcycle=0
Channel.0.LineIndex.1.In.Bicycle=0
Channel.0.LineIndex.1.OutCount=0
Channel.0.LineIndex.1.Out.Car=0
Channel.0.LineIndex.1.Out.Bus=0
```

```
Channel.0.LineIndex.1.Out.Truck=0
Channel.0.LineIndex.1.Out.Motorcycle=0
Channel.0.LineIndex.1.Out.Bicycle=0
Channel.0.LineIndex=2
Channel.0.LineIndex.2.Name=name2
Channel.0.LineIndex.2.InCount=0
Channel.0.LineIndex.2.In.Car=0
Channel.0.LineIndex.2.In.Bus=0
Channel.0.LineIndex.2.In.Truck=0
Channel.0.LineIndex.2.In.Motorcycle=0
Channel.0.LineIndex.2.In.Bicycle=0
Channel.0.LineIndex.2.OutCount=0
Channel.0.LineIndex.2.Out.Car=0
Channel.0.LineIndex.2.Out.Bus=0
Channel.0.LineIndex.2.Out.Truck=0
Channel.0.LineIndex.2.Out.Motorcycle=0
Channel.0.LineIndex.2.Out.Bicycle=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VehicleCount": [
        {
            "Channel": 0,
            "Lines": [
                {
                    "LineIndex": 1,
                    "Name": "name1",
                    "InCount": 0,
                    "In": {
                        "Car": 0,
                        "Bus": 0,
                        "Truck": 0,
                        "Motorcycle": 0,
                        "Bicycle": 0
                    },
                    "OutCount": 0,
```

```
                    "Out": {
                        "Car": 0,
                        "Bus": 0,
                        "Truck": 0,
                        "Motorcycle": 0,
                        "Bicycle": 0
                    }
                },
                {
                    "LineIndex": 2,
                    "Name": "name2",
                    "InCount": 0,
                    "In": {
                        "Car": 0,
                        "Bus": 0,
                        "Truck": 0,
                        "Motorcycle": 0,
                        "Bicycle": 0
                    },
                    "OutCount": 0,
                    "Out": {
                        "Car": 0,
                        "Bus": 0,
                        "Truck": 0,
                        "Motorcycle": 0,
                        "Bicycle": 0
                    }
                }
            ]
        }
    ]
}
```


158.18. Heat Map

158.18.1. Description

The **heatmap** submenu configures heat map settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin
check	Admin

158.18.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
heatmap&action=<value>[&<parameter>=<value>]
```

158.18.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads heat map settings
	Channel	REQ, RES	<int>	Channel ID
	AutoReference	RES	<int>	Automatically detected heatmap reference values
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use heat map feature
	ReportEnable	REQ, RES	<bool> True, False	Whether to use report or not
	ReportFilename	REQ, RES	<string>	File name of the report
	ReportFileType	REQ, RES	<enum> PNG	File type of the report
	BackgroundColourLevel	REQ, RES	<int>	Background color adjustment The values should be in the range of 0 to 100; • 0: Black and White • 100: Color Image
	Area.#.Type	REQ, RES	<enum> Outside	Specifying the area to exclude from analyzing heatmap
	Area.#.Coordinates	REQ, RES	<string>	Coordinates of Area to set The coordinates are specified in the form of < x1,y1,x2,y2...>.

Action	Parameter	Request/ Response	Type/ Value	Description
	ManualModeEnable	REQ, RES	<bool> True, False	Whether to use heatmap in manual reference mode
	ManualReference	REQ, RES	<int>	Manual heatmap reference values
remove	Channel	REQ	<int>	Channel ID
	AreaIndex	REQ	<csv>	Area index
check	Channel	REQ, RES	<int>	Channel ID
	Descriptor	RES	<int>	Heat map level descriptor
	Level	RES	<csv>	Heat map Level
	Resolution	RES	<string>	Heat map image resolution as <width>x<Height>

158.18.4. Examples

158.18.4.1. Getting heat map settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=heatmap&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.ReportEnable=True
Channel.0.ReportFilename=heatmapreport
Channel.0.ReportFileType=PNG
Channel.0.ManualModeEnable=False
Channel.0.ManualReference=0
Channel.0.AutoReference=213485
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "HeatMap": [
        {
            "Channel": 0,
            "Enable": true,
            "ReportEnable": true,
            "ReportFilename": "heatmapreport",
            "ReportFileType": "PNG",
            "ManualModeEnable": false,
            "ManualReference": 0,
            "AutoReference": 213485
        }
    ]
}
```


158.18.4.2. Setting heat map data

Enabling the heatmap feature

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=heatmap&action=set&Enable=True
```

Setting up the heatmap report

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=heatmap&action=set&ReportEnable=True&ReportFilename=testreport&ReportFileType=PNG
```

158.18.4.3. Setting up heatmap in manual reference mode

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=heatmap&action=set&ManualModeEnable=True&ManualReference=1000
```

158.18.4.4. Removing heatmap data

Setting up a people count line rule to detect people crossing the line from right to left

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=heatmap&action=remove&Channel=0&AreaIndex=1
```

158.18.4.5. Check the heat map levels

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=heatmap&action=check&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```


```
Channel.0.Level=0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
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JSON RESPONSE

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}
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158.19. Source Options

158.19.1. Description

The **sourceoptions** submenu gives information about the list of event sources available in the device and corresponding action triggers.

NOTE This chapter applies to network cameras and encoders only.

Access level

Action	Camera	Encoder
view	Admin	Admin

158.19.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
```

```
sourceoptions&action=<value>[&<parameter>=<value>]
```

158.19.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	EventSource	RES	<enum> AlarmInput.#, MotionDetection, VideoLoss, NetworkEvent, FaceDetection, TamperingDetection, AudioDetection, Tracking, Timer, OpenSDK, UserInput, DefocusDetection, QueueManagement, PeopleCount, HeatMap, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, ObjectDetection, BodyTemperatureDetection, MaskDetection, SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor, ParkingDetection, VehicleCount, CaseTampering, SIPCall, AudioClipPlayback, TwoWayTalk	Source of the event
	EventAction	RES	<csv> AlarmOutput, SMTP, FTP, Record, HTTP, GoToPreset, AudioClip, LightAlarm	Action trigger when an event is raised

Action	Parameter	Request/ Response	Type/ Value	Description
	MinimumObjectSize	RES	<string> Format=w,h	Minimum size of objects detectable by motion detection. Objects smaller than the specified minimum size are not detected. The size is specified in the format of <width, height>. The value of MinimumObjectSize must be less than the value of MaximumObjectSize. **MinimumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	MaximumObjectSize	RES	<string> Format=w,h	Maximum size of objects detectable by motion detection Objects larger than the maximum size are not detected. The size is specified in the format of <width, height>. The value of MaximumObjectSize must be greater than the value of MinimumObjectSize. **MaximumObjectSize** is valid only when **DetectionType** is NOT set to Off.
	MinimumObjectSizeIn Pixels	RES	<string> Format=w,h	Minimum object size in pixels The size is specified in the format of <width, height>. **MinimumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.

Action	Parameter	Request/ Response	Type/ Value	Description
	MaximumObjectSizeInPixels	RES	<string> Format=w,h	Maximum object size in pixels The size is specified in the format of <width, height>. **MaximumObjectSizeInPixels** is valid only when **DetectionType** is NOT set to Off.
	ROIIncludeMinIndex	RES	<int>	ROI Include area minimum index number
	ROIIncludeMaxIndex	RES	<int>	ROI Include area maximum index number
	ROIExcludeMinIndex	RES	<int>	ROI Exclude area minimum index number
	ROIExcludeMaxIndex	RES	<int>	ROI Exclude area maximum index number
	ExcludeAreaMinIndex	RES	<int>	Exclude area minimum index
	ExcludeAreaMaxIndex	RES	<int>	Exclude area maximum index
	DefinedAreaIncludeMinIndex	RES	<int>	Defined Include Area minimum index number
	DefinedAreaIncludeMaxIndex	RES	<int>	Defined Include Area maximum index number
	DefinedAreaExcludeMinIndex	RES	<int>	Defined Exclude Area minimum index number
	DefinedAreaExcludeMaxIndex	RES	<int>	Defined Exclude Area maximum index number
	DetectionAreaIncludeMinIndex	RES	<int>	Minimum index number of include detection area
	DetectionAreaIncludeMaxIndex	RES	<int>	Maximum index number of include detection area
	DetectionAreaExcludeMinIndex	RES	<int>	Minimum index number of exclude detection area
	DetectionAreaExcludeMaxIndex	RES	<int>	Maximum index number of exclude detection area

Action	Parameter	Request/ Response	Type/ Value	Description
	MinimumAreaSizeInPixels	RES	<string>	Minimum area size in pixels When **EventSource** is FaceDetection. Format=w,h
	MinimumAllowedDistanceStepSize	RES	<float>	In SocialDistancingViolation EventSource, it shows the minimum distance step size setting.
	ObjectTypeFilter	RES	<csv> Person, Vehicle	List of object types available in IVA among all object types
	ObjectTypeFilterDetails	RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck, Person.Color.Orange, Person.Color.Black, Person.Color.Red	List of detailed object types available in IVA
	ObjectTypes	RES	<csv> Person, Vehicle, Face, LicensePlate	List of object types available in ObjectDetection among all object types
	ObjectTypeDetails	RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck, Person.Color.Orange, Person.Color.Black, Person.Color.Red	List of detailed object types available in ObjectDetection
	VideoOutOverlayObjectTypes	RES	<csv> Person, Vehicle, Face, LicensePlate	List of detailed object types available in VideoOutOverlay

158.19.4. Examples

158.19.4.1. Getting source options

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=sourceoptions&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
EventSource.AlarmInput.1.EventAction=FTP,SMTP,Record,AlarmOutput,AudioClip
EventSource.MotionDetection.EventAction=FTP,SMTP,Record,AlarmOutput,AudioCli
p
EventSource.MotionDetection.MinimumObjectSize=0,0
EventSource.MotionDetection.MaximumObjectSize=99,99
EventSource.MotionDetection.MinimumObjectSizeInPixels=41,41
EventSource.MotionDetection.MaximumObjectSizeInPixels=3328,1872
EventSource.MotionDetection.ROIIncludeMinIndex=1
EventSource.MotionDetection.ROIIncludeMaxIndex=8
EventSource.MotionDetection.ROIExcludeMinIndex=9
EventSource.MotionDetection.ROIExcludeMaxIndex=16
EventSource.VideoAnalysis.EventAction=FTP,SMTP,Record,AlarmOutput,AudioClip
EventSource.VideoAnalysis.MinimumObjectSize=0,0
EventSource.VideoAnalysis.MaximumObjectSize=99,99
EventSource.VideoAnalysis.MinimumObjectSizeInPixels=41,41
EventSource.VideoAnalysis.MaximumObjectSizeInPixels=3328,1872
EventSource.VideoAnalysis.DefinedAreaIncludeMinIndex=1
EventSource.VideoAnalysis.DefinedAreaIncludeMaxIndex=8
EventSource.VideoAnalysis.DefinedAreaExcludeMinIndex=9
EventSource.VideoAnalysis.DefinedAreaExcludeMaxIndex=16
EventSource.VideoAnalysis.ObjectTypeFilter=Person,Vehicle
EventSource.VideoAnalysis.ObjectTypeFilterDetails=Vehicle.Types.Bicycle,Vehi
cle.Types.Car,Vehicle.Types.Motorcycle,Vehicle.Types.Bus,Vehicle.Types.Truck
EventSource.NetworkEvent.EventAction=Record,AlarmOutput
EventSource.FaceDetection.EventAction=FTP,SMTP,Record,AlarmOutput,AudioClip
EventSource.FaceDetection.MinimumAreaSizeInPixels=960,540
EventSource.FaceDetection.DetectionAreaIncludeMinIndex=1
EventSource.FaceDetection.DetectionAreaIncludeMaxIndex=1
EventSource.FaceDetection.DetectionAreaExcludeMinIndex=2
EventSource.FaceDetection.DetectionAreaExcludeMaxIndex=9
```

```
EventSource.TamperingDetection.EventAction=FTP,SMTP,Record,AlarmOutput,Audio
Clip
EventSource.DefocusDetection.EventAction=FTP,SMTP,Record,AlarmOutput,AudioCl
ip
EventSource.FogDetection.EventAction=FTP,SMTP,Record,AlarmOutput,AudioClip
EventSource.AudioDetection.EventAction=FTP,SMTP,Record,AlarmOutput
EventSource.AudioAnalysis.EventAction=FTP,SMTP,Record,AlarmOutput
EventSource.OpenSDK.EventAction=FTP,SMTP
EventSource.Timer.EventAction=FTP
EventSource.QueueManagement.EventAction=FTP,SMTP,AlarmOutput,AudioClip
EventSource.ShockDetection.EventAction=FTP,SMTP,Record,AlarmOutput,AudioClip
EventSource.PeopleCount.EventAction=
EventSource.SocialDistancingViolation.EventAction=FTP,SMTP,Record,AlarmOutpu
t,GoToPreset,AudioClip
EventSource.SocialDistancingViolation.MinimumAllowedDistanceStepSize=0.5
EventSource.SocialDistancingViolation.MinimumObjectSize=0,0
EventSource.SocialDistancingViolation.MaximumObjectSize=99,99
EventSource.SocialDistancingViolation.MinimumObjectSizeInPixels=12,12
EventSource.SocialDistancingViolation.MaximumObjectSizeInPixels=3840,2160
EventSource.SocialDistancingViolation.ExcludeAreaMinIndex=1
EventSource.SocialDistancingViolation.ExcludeAreaMaxIndex=
EventSource.ObjectDetection.EventAction=FTP,SMTP,Record,AlarmOutput
EventSource.ObjectDetection.MinimumObjectSize=0,0
EventSource.ObjectDetection.MaximumObjectSize=99,99
EventSource.ObjectDetection.MinimumObjectSizeInPixels=12,12
EventSource.ObjectDetection.MaximumObjectSizeInPixels=3840,2160
EventSource.ObjectDetection.ExcludeAreaMinIndex=1
EventSource.ObjectDetection.ExcludeAreaMaxIndex=8
EventSource.ObjectDetection.ObjectTypes=Person,Vehicle,Face,LicensePlate
EventSource.ObjectDetection.ObjectTypeDetails=Vehicle.Types.Bicycle,Vehicle.
Types.Car,Vehicle.Types.Motorcycle,Vehicle.Types.Bus,Vehicle.Types.Truck
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EventSources": [
        {
```

```
            "EventSource": "AlarmInput.1",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "MotionDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ],
            "MinimumObjectSize": {
                "Width": 0,
                "Height": 0
            },
            "MaximumObjectSize": {
                "Width": 99,
                "Height": 99
            },
            "MinimumObjectSizeInPixels": {
                "Width": 41,
                "Height": 41
            },
            "MaximumObjectSizeInPixels": {
                "Width": 3328,
                "Height": 1872
            },
            "ROIIncludeMinIndex": 1,
            "ROIIncludeMaxIndex": 8,
            "ROIExcludeMinIndex": 9,
            "ROIExcludeMaxIndex": 16
        },
        {
            "EventSource": "VideoAnalysis",
```

```
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ],
            "MinimumObjectSize": {
                "Width": 0,
                "Height": 0
            },
            "MaximumObjectSize": {
                "Width": 99,
                "Height": 99
            },
            "MinimumObjectSizeInPixels": {
                "Width": 41,
                "Height": 41
            },
            "MaximumObjectSizeInPixels": {
                "Width": 3328,
                "Height": 1872
            },
            "DefinedAreaIncludeMinIndex": 1,
            "DefinedAreaIncludeMaxIndex": 8,
            "DefinedAreaExcludeMinIndex": 9,
            "DefinedAreaExcludeMaxIndex": 16,
            "ObjectTypeFilter": [
                "Person",
                "Vehicle"
            ],
            "ObjectTypeFilterDetails": {
                "Vehicle": {
                    "Types": [
                        "Bicycle",
                        "Car",
                        "Motorcycle",
                        "Bus",
                        "Truck"
                    ]
                }
```

```
            }
        },
        {
            "EventSource": "NetworkEvent",
            "EventAction": [
                "Record",
                "AlarmOutput"
            ]
        },
        {
            "EventSource": "FaceDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ],
            "MinimumAreaSizeInPixels": {
                "Width": 960,
                "Height": 540
            },
            "DetectionAreaIncludeMinIndex": 1,
            "DetectionAreaIncludeMaxIndex": 1,
            "DetectionAreaExcludeMaxIndex": 9
        },
        {
            "EventSource": "TamperingDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "DefocusDetection",
            "EventAction": [
                "FTP",
                "SMTP",
```

```
                "Record",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "FogDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "AudioDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput"
            ]
        },
        {
            "EventSource": "AudioAnalysis",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput"
            ]
        },
        {
            "EventSource": "OpenSDK",
            "EventAction": [
                "FTP",
                "SMTP"
            ]
        },
        {
```

```
            "EventSource": "Timer",
            "EventAction": [
                "FTP"
            ]
        },
        {
            "EventSource": "QueueManagement",
            "EventAction": [
                "FTP",
                "SMTP",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "ShockDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "AudioClip"
            ]
        },
        {
            "EventSource": "PeopleCount",
            "EventAction": []
        },
        {
            "EventSource": "SocialDistancingViolation",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput",
                "GoToPreset",
                "AudioClip"
            ],
            "MinimumAllowedDistanceStepSize": 0.5,
            "MinimumObjectSize": {
                "Width": 0,
```

```
                "Height": 0
            },
            "MaximumObjectSize": {
                "Width": 99,
                "Height": 99
            },
            "MinimumObjectSizeInPixels": {
                "Width": 12,
                "Height": 12
            },
            "MaximumObjectSizeInPixels": {
                "Width": 3840,
                "Height": 2160
            },
            "ExcludeAreaMinIndex": 1,
            "ExcludeAreaMaxIndex": 8
        },
        {
            "EventSource": "ObjectDetection",
            "EventAction": [
                "FTP",
                "SMTP",
                "Record",
                "AlarmOutput"
            ],
            "MinimumObjectSize": {
                "Width": 0,
                "Height": 0
            },
            "MaximumObjectSize": {
                "Width": 99,
                "Height": 99
            },
            "MinimumObjectSizeInPixels": {
                "Width": 12,
                "Height": 12
            },
            "MaximumObjectSizeInPixels": {
                "Width": 3840,
                "Height": 2160
            },
```

```
            "ExcludeAreaMinIndex": 1,
            "ExcludeAreaMaxIndex": 8,
            "ObjectTypes": [
                "Person",
                "Vehicle",
                "Face",
                "LicensePlate"
            ],
            "ObjectTypeDetails": {
                "Vehicle": {
                    "Types": [
                        "Bicycle",
                        "Car",
                        "Motorcycle",
                        "Bus",
                        "Truck"
                    ]
                }
            }
        }
    ]
}
```

158.20. Samples

158.20.1. Description

The **samples** submenu gives the current level of the event for the requested event source.

NOTE This chapter applies to network cameras & encoder only.

Access level

Action	Camera	Encoder
check	Admin	Admin

158.20.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
samples&action=<value>[&<parameter>=<value>]
```

158.20.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
check	Channel	REQ, RES	<int>	Channel ID
	EventSourceType	REQ, RES	<enum> MotionDetection, TamperingDetection, AudioDetection, FogDetection, DefocusDetection, AudioAnalysis, ShockDetection	Source of the event
	Index	REQ	<int>	Area/ROI index
	MaxSamples	REQ	<int>	Maximum samples required
	SequenceID	RES	<int>	Sequence ID for the sample
	Level	RES	<int>	Event level

158.20.4. Examples

158.20.4.1. Getting samples for MotionDetection

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=samples&action=check&EventSourceType=MotionDet
ection&MaxSamples=5&Channel=0&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "MotionDetection": [
        {
            "Channel": 0,
            "Samples": [
                {
                    "SequenceID": 600718837,
                    "Level": 0
```

```
                    },
                    {
                        "SequenceID": 600718738,
                        "Level": 0
                    },
                    {
                        "SequenceID": 600718636,
                        "Level": 0
                    },
                    {
                        "SequenceID": 600718531,
                        "Level": 0
                    },
                    {
                        "SequenceID": 600718427,
                        "Level": 0
                    }
                ]
            }
        ]
}
```


158.21. Queue Management Setup

158.21.1. Description

The **queuemanagementsetup** submenu configures Queue Management settings.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin
check	Admin

158.21.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
queuemanagementsetup&action=<value>[&<parameter>=<value>]
```

158.21.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads queue management settings
	Channel	REQ, RES	<int>	Channel ID
	Queue.#.Level	REQ, RES	<enum> High, Medium	Level of the queue
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use queue management feature
	ReportEnable	REQ, RES	<bool> True, False	Whether to use report
	ReportFilename	REQ, RES	<string>	Filename of the report
	ReportFileType	REQ, RES	<enum> XLSX,TXT	File type of the report
	Queue.#.Name	REQ, RES	<string>	Name of the queue
	Queue.#.Enable	REQ, RES	<bool> True, False	Whether to use the queue
	Queue.#.Coordinates	REQ, RES	<string> Format=x1,y1,x2,y2,x3,y3,x4,y4...x8,y8	X and Y coordinates of the points which define the two lines of the queue. There should be even number of points eg: x1,y1,x2,y2,x3,y3,x4,y4 Where first two points represents the first line and the remaining two points represents the second line of a queue.
	Queue.#.MaxPeople	REQ, RES	<int>	Setting the max number of people for the queue
	Queue.#.Level.High.Count	REQ, RES	<int>	Setting "High boundary value". It should be less than Max value. And Queue.#.Level.Medium.Count value is automatically set as half of "high value".
	Queue.#.Level.#.AlarmEnable	REQ, RES	<bool> True, False	Enabling or disabling alarm if queue reaches a certain threshold
	Queue.#.Level.#.Threshold	REQ, RES	<int>	Threshold to trigger queue’s level

Action	Parameter	Request/ Response	Type/ Value	Description
check	Channel	REQ	<int>	Channel ID
	QueueIndex	REQ	<csv>	QueueIndex list comma separated
	Queue.#.Count	RES	<int>	Current count in the queue.

158.21.4. Examples

158.21.4.1. Getting queue management settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "QueueManagementSetup": [
        {
            "Channel": 0,
            "Enable": false,
            "ReportEnable": false,
            "ReportFilename": "",
            "ReportFileType": "XLSX",
            "Queues": [
                {
                    "Queue": 1,
                    "MaxPeople": 32,
                    "Name": "",
                    "Enable": false,
                    "Coordinates": [
                        {
                            "x": 468,
                            "y": 258
                        },
                        {
```

```
                        "x": 468,
                        "y": 798
                    },
                    {
                        "x": 1428,
                        "y": 798
                    },
                    {
                        "x": 1428,
                        "y": 258
                    }
                ],
                "QueueLevels": [
                    {
                        "Level": "High",
                        "Count": 16,
                        "AlarmEnable": false,
                        "Threshold": 60
                    },
                    {
                        "Level": "Medium",
                        "Count": 8,
                        "AlarmEnable": false,
                        "Threshold": 60
                    }
                ]
            },
            {
                "Queue": 2,
                "MaxPeople": 32,
                "Name": "",
                "Enable": false,
                "Coordinates": [
                    {
                        "x": 480,
                        "y": 270
                    },
                    {
                        "x": 480,
                        "y": 810
                    },
```

```
                {
                    "x": 1440,
                    "y": 810
                },
                {
                    "x": 1440,
                    "y": 270
                }
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "Count": 16,
                    "AlarmEnable": false,
                    "Threshold": 60
                },
                {
                    "Level": "Medium",
                    "Count": 8,
                    "AlarmEnable": false,
                    "Threshold": 60
                }
            ]
        },
        {
            "Queue": 3,
            "MaxPeople": 32,
            "Name": "",
            "Enable": false,
            "Coordinates": [
                {
                    "x": 492,
                    "y": 282
                },
                {
                    "x": 492,
                    "y": 822
                },
                {
                    "x": 1452,
                    "y": 822
```

```
                            },
                            {
                                "x": 1452,
                                "y": 282
                            }
                        ],
                        "QueueLevels": [
                            {
                                "Level": "High",
                                "Count": 16,
                                "AlarmEnable": false,
                                "Threshold": 60
                            },
                            {
                                "Level": "Medium",
                                "Count": 8,
                                "AlarmEnable": false,
                                "Threshold": 60
                            }
                        ]
                    }
                ]
            }
        ]
}
```

158.21.4.2. Setting queue management data

Setting up a queue management rule to detect the number of people in the queue

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=set&Channel=0&Enable=True&Queue.1.Enable=True&Queue.1.MaxPeople=32&Queue.1.Name=queue1&Queue.1.Level.High.AlarmEnable=False&Queue.1.Level.High.Count=16&Queue.1.Level.High.Threshold=60&Queue.1.Level.Medium.AlarmEnable=False&Queue.1.Level.Medium.Threshold=60&Queue.1.Coordinates=468,258,468,797,1424,866,1427,258
```

Setting up a queue management report

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=set&ReportEnable=T
rue&ReportFilename=queue1&ReportFileType=XLSX
```

158.21.4.3. Getting queue management live count data

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=check&Channel=0&Qu
eueIndex=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "QueueCount": [
        {
            "Channel": 0,
            "Queues": [
                {
                    "Queue": 0,
                    "Count": 4
                }
            ]
        }
    ]
}
```


158.22. G Sensor Setup

158.22.1. Description

The **gsensor** submenu configures G Sensor settings.

NOTE	This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

158.22.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
gsensor&action=<value>[&<parameter>=<value>]
```

158.22.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads g sensor settings
	SensitivityType	RES	<enum> +X, +Y, +Z, -X, -Y, -Z	Type of sensitivity
set	Enable	REQ, RES	<bool> True, False	Enable or disable gsensor
	SesitivityType.#.Sensitivity	REQ, RES	<enum> Low, Medium, High	Type of sensitivity
	VehicleDirection	REQ, RES	<enum> +X, +Y, +Z, -X, -Y, -Z, None	Direction of vehicle moving
	AlarmOutput	REQ, RES	<csv> 1, 2, 3, 4, 5, 6, Beep, None	Type of alarm output
	Duration	REQ, RES	<enum> None, Always, 5s, 10s, 20s, 30s	Alarm duration

158.22.4. Examples

158.22.4.1. Getting g sensor settings for Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=gsensor&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
VehicleDirection=None
AlarmOutput=None
Duration=None
SensitivityType.+X.Sensitivity=Medium
SensitivityType.+Y.Sensitivity=Medium
SensitivityType.+Z.Sensitivity=Medium
SensitivityType.-X.Sensitivity=Medium
SensitivityType.-Y.Sensitivity=Medium
SensitivityType.-Z.Sensitivity=Medium
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false,
    "VehicleDirection": "None",
    "AlarmOutput": [
        "None"
    ],
    "Duration": "None",
    "SensitivityList": [
        {
            "SensitivityType": "+X",
            "Sensitivity": "Medium"
        },
```

```
        {
            "SensitivityType": "+Y",
            "Sensitivity": "Medium"
        },
        {
            "SensitivityType": "+Z",
            "Sensitivity": "Medium"
        },
        {
            "SensitivityType": "-X",
            "Sensitivity": "Medium"
        },
        {
            "SensitivityType": "-Y",
            "Sensitivity": "Medium"
        },
        {
            "SensitivityType": "-Z",
            "Sensitivity": "Medium"
        }
    ]
}
```

158.22.4.2. Setting g sensor

Setting up g sensor's Sensitivity Type

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=gsensor&action=set&Enable=False&SensitivityType.+X.Sensitivity=Low&SensitivityType.-X.Sensitivity=Low&SensitivityType.+Y.Sensitivity=Low&SensitivityType.-Y.Sensitivity=Low&SensitivityType.+Z.Sensitivity=Low&SensitivityType.-Z.Sensitivity=Low&VehicleDirection=-Z&AlarmOutput=1,2,3,4&Duration=5s
```

Setting up g sensor's sensitivity

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=gsensor&action=set&Enable=False&SensitivityType.+X.Sensitivity=High&SensitivityType.-X.Sensitivity=High&VehicleDirection=-
```

```
Z&AlarmOutput=1,2,3,4&Duration=5s
```

158.23. Temperature Change Detection

158.23.1. Description

The **temperaturechangedetection** submenu configures temperature change detection settings.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/TemperatureChangeDetection"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin

158.23.2. Syntax

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=<value>[&<pa
rameter>=<value>]
```

158.23.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads temperature change detection settings
	Channel	REQ, RES	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use temperature change detection
	TemperatureChange.ROI .#.Mode	REQ, RES	<enum> Minimum, Maximum, Average	Tempareture change mode for the ROI

Action	Parameter	Request/ Response	Type/ Value	Description
	TemperatureChange.ROI.#.Gap	REQ, RES	<enum> Celsius: 20, 40, 60, 80, 100 Fahrenheit: 40, 80, 120, 160, 200	Tempareture change gap for the ROI in celsius or fahrenheit
	TemperatureChange.ROI.#.DetectionPeriod	REQ, RES	<int>	Tempareture change detection period in seconds for the ROI
	TemperatureChange.ROI.#.Coordinates	REQ, RES	<string> Format=x1,y1,x2,y2	Coordinates of the ROI region **Note** This parameter is for backward compatibility and is deprecated Use TemperatureChange.ROI.#.PolygonCoordinates instead
	TemperatureChange.ROI.#.PolygonCoordinates	REQ, RES	<string> Format=x1,y1,x2,y2,x3,y3,x4,y4,x5,y5,x6,y6,x7,y7,x8,y8	Polygon Coordinates of the ROI region
	TemperatureChange.ROI.#.HandoverIndex	REQ, RES	<int>	Handover index associated with the ROI
	TemperatureChange.ROI.#.DetectionType	REQ, RES	<enum> Change, Increase, Decrease	Set the conditions for detecting temperature changes • Change: Observe an increase or decrease in the temperature within the set area • Increase: Observe an increase in the temperature within the set area. • Decrease: Observe a decrease in the temperature within the set area.
remove	Channel	REQ	<int>	Channel ID
	ROIIndex	REQ	<int>	ROI index of the region to be removed

158.23.4. Examples

158.23.4.1. Getting temperature chnage detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=view&Channel
=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "TemperatureChangeDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "TemperatureChange": [
                {
                    "ROI": 1,
                    "Mode": "Maximum",
                    "Gap": 20,
                    "DetectionPeriod": 1,
                    "HandoverIndex": 1,
                    "Coordinates": [
                        {
                            "x": 31,
                            "y": 38
                        },
                        {
                            "x": 376,
                            "y": 256
                        }
                    ]
                }
            ]
        }
    ]
```

```
}
```

158.23.4.2. Changing temperature change detection settings

REQUEST

```
stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=set&Channel=
0&Enable=True&TemperatureChange.ROI.1.Mode=Average&TemperatureChange.ROI.1.G
ap=20& TemperatureChange.ROI.1.DetectionPeriod=10&
TemperatureChange.ROI.1.Coordinates=100,100,350,420
```

158.23.4.3. Remvoing temperature change detection ROI Region 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=remove&Chann
el=0&ROIIdex=1
```

158.24. Temperature Change Detection Options

158.24.1. Description

The **temperaturechangedetectionoptions** submenu provides information about temperature change detection options.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/TemperatureChangeDetection"

Access level

Action	Camera
view	Admin

158.24.2. Syntax

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetectionoptions&action=<valu
e>[&<parameter>=<value>]
```

158.24.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads temperature change detection options
	Channel	REQ, RES	<int>	Channel ID
	Celsius.SupportedGap	RES	<csv> 20, 40, 60, 80, 100	Temperature change gap supported values in celsisus
	Fahrenheit.SupportedGap	RES	<csv> 40, 80, 120, 160, 200	Temperature change gap supported values in fahrenheit

158.24.4. Examples

158.24.4.1. Getting temperature chnage detection options for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetectionoptions&action=view&
Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TemperatureChangeDetectionOption": [
        {
            "Channel": 0,
            "SupportedGap": {
                "Celsius": "20,40,60,80,100",
                "Fahrenheit": "40,80,120,160,200"
            }
        }
    ]
}
```

158.25. Shock Detection Setup

158.25.1. Description

The **shockdetection** submenu configures the shock detection settings.

NOTE	This chapter applies to the network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/ShockDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.25.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
shockdetection&action=<value>[&<parameter>=<value>]
```

158.25.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads shock detection settings
	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables shock detection
	Sensitivity	REQ, RES	<int>	Shock detection sensitivity level
	ThresholdLevel	REQ, RES	<int>	Shock detection threshold level

158.25.4. Examples

158.25.4.1. Getting shock detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=shockdetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.0.Sensitivity=80
Channel.0.ThresholdLevel=50
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ShockDetection": [
        {
            "Channel": 0,
            "Enable": false,
            "Sensitivity": 80,
            "ThresholdLevel": 50
        }
    ]
}
```

158.25.4.2. Changing shock detection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=shockdetection&action=set&Channel=0&Enable=Tru
e&Sensitivity=80&ThresholdLevel=15
```

158.26. Wiper Housing Detection Setup

158.26.1. Description

The **wiperhousingdetection** submenu configures the wiper housing detection settings.

NOTE

This chapter applies to the network cameras only.
Attribute to check for feature support:
"attributes/Eventsource/Support/HousingTampering"

Access level

Action	Camera
view	Admin
set	Admin

158.26.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
wiperhousingdetection&action=<value>[&<parameter>=<value>]
```

158.26.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads wiper housing detection settings
	Channel	REQ	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	HousingTamperingEnable	REQ, RES	<bool> True, False	Enables or disables Housing Tampering
	WaterLevelWarningEnable	REQ, RES	<bool> True, False	Enables or disables Water Level Warning
	WiperOperationDuration	REQ, RES	<int>	Duration for wiper operation
	WasherOperationDuration	REQ, RES	<int>	Duration for washer operation
	OperationDelayDuration	REQ, RES	<int>	Operation delay duration

158.26.4. Examples

158.26.4.1. Getting wiper housing detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=wiperhousingdetection&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.HousingTamperingEnable=False
Channel.0.WaterLevelWarningEnable=True
Channel.0.WiperOperationDuration=30
Channel.0.WasherOperationDuration=20
Channel.0.OperationDelayDuration=5
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "WiperHousingDetection": [
        {
            "Channel": 0,
            "HousingTamperingEnable": false,
            "WaterLevelWarningEnable": true,
            "WiperOperationDuration": 30,
            "WasherOperationDuration": 20,
            "OperationDelayDuration": 5
        }
    ]
}
```


158.26.4.2. Changing wiper housing detection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=wiperhousingdetection&action=set&Channel=0&Wat
erLevelWarningEnable=True&WiperOperationDuration=10&OperationDelayDuration=1
5
```

158.27. Box Temperature Detection

158.27.1. Description

The **boxtemperaturedetection** submenu configures box temperature detection settings.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/BoxTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin
check	Admin

158.27.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
boxtemperaturedetection&action=<value>[&<parameter>=<value>]
```

158.27.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads box temperature detection settings
	Channel	REQ, RES	<int>	Channel ID
remove	Channel	REQ	<int>	Channel ID
	ROIIndex	REQ	<int>	ROI index of the region to be removed
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables box temperature detection
	ROI.#.TemperatureType	REQ, RES	<enum> Minimum, Maximum, Average	Temperature type for the ROI

Action	Parameter	Request/ Response	Type/ Value	Description
	ROI.#.DetectionType	REQ, RES	<enum> Above, Below, Increase, Decrease	Temperature detection type for the ROI
	ROI.#.ThresholdTemperature	REQ, RES	<int>	Threshold Temperature for the ROI
	ROI.#.Coordinate	REQ, RES	<string> Format=x1, y1,x2,y2	Coordinates of the ROI region
	ROI.#.Duration	REQ, RES	<int>	ROI Duration in seconds
	ROI.#.NormalizedEmissivity	REQ, RES	<int>	ROI Normalized Emissivity
	ROI.#.AreaOverlay	REQ, RES	<bool> True, False	Enables or disables ROI Area Overlay
	ROI.#.AvgTemperatureOverlay	REQ, RES	<bool> True, False	Enables or disables ROI Average Temperature Overlay
	ROI.#.MinTemperatureOverlay	REQ, RES	<bool> True, False	Enables or disables ROI Minimum Temperature Overlay
	ROI.#.MaxTemperatureOverlay	REQ, RES	<bool> True, False	Enables or disables ROI Maximum Temperature Overlay
	ROI.#.HandoverIndex	REQ, RES	<int>	Handover index to be attached to ROI. If index is 0, it means "Off"
	ROI.#.AreaName	REQ, RES	<string>	set AreaName for this ROI.
	ROI.#.PolygonCoordinates	REQ, RES	<string>	ROI coordinates can be polygon by using this parameter. But "Coordinates" parameter can set only rectangle parallel to axis.
	ROI.#.AreaColor	REQ, RES	<enum> Gray, Green, Red, Blue, Black, White	set area display color
	ROI.#.Condition.#.Duration	REQ, RES	<int>	Duration of the ROI’s #th condition to trigger the event

Action	Parameter	Request/ Response	Type/ Value	Description
	ROI.#.Condition.#.DetectionType	REQ, RES	<enum> Above, Below, Increase, Decrease	Temperature detection type for the ROI's #th condition
	ROI.#.Condition.#.ThresholdTemperature	REQ, RES	<int>	Threshold Temperature for the ROI's #th condition
	ROI.#.Condition.#.TemperatureType	REQ, RES	<enum> Minimum, Maximum, Average	Temperature type for the ROI's #th condition
	ROI.#.Condition.#.Enable	REQ, RES	<bool>	Enable ROI's #th condition
	EntireAreaAvgTemperatureOverlay	REQ, RES	<bool>	Enable average temperature overlay of the whole screen overlay
	EntireAreaMinTemperatureOverlay	REQ, RES	<bool>	Enable minimum temperature overlay of the whole screen overlay
	EntireAreaMaxTemperatureOverlay	REQ, RES	<bool>	Enable maximum temperature overlay of the whole screen overlay
check	Channel	REQ, RES	<int>	channel for check
	ROIIndex	RES	<int>	ROI index
	AreaName	RES	<string>	ROI Name
	Unit	RES	<enum>	Temperature unit
	AvgTemperature	RES	<float>	Average temperature for each ROI & whole screen
	MinTemperature	RES	<float>	Maximum temperature for each ROI & whole screen
	MaxTemperature	RES	<float>	Minimum temperature for each ROI & whole screen
	MinTemperatureCoordinates	RES	<string>	Minimum temperature point coordinates for each ROI & whole screen
	MaxTemperatureCoordinates	RES	<string>	Maximum temperature point coordinates for each ROI & whole screen

158.27.4. Examples

158.27.4.1. Getting box temperature detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "BoxTemperatureDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "EntireAreaAvgTemperatureOverlay": false,
            "EntireAreaMinTemperatureOverlay": false,
            "EntireAreaMaxTemperatureOverlay": false,
            "ROIs": [
                {
                    "ROI": 1,
                    "AreaName": "A",
                    "TemperatureType": "Maximum",
                    "DetectionType": "Above",
                    "ThresholdTemperature": 34,
                    "Coordinates": [
                        {
                            "x": 24,
                            "y": 50
                        },
                        {
                            "x": 620,
                            "y": 442
                        }
                    ],
                    "PolygonCoordinates": [
                        {
                            "x": 24,
                            "y": 67
```

```
                },
                {
                    "x": 37,
                    "y": 431
                },
                {
                    "x": 318,
                    "y": 442
                },
                {
                    "x": 620,
                    "y": 50
                }
            ],
            "Duration": 1,
            "Conditions": [
                {
                    "condition": 1,
                    "Enable": true,
                    "TemperatureType": "Maximum",
                    "DetectionType": "Above",
                    "ThresholdTemperature": 34,
                    "Duration": 1
                },
                {
                    "condition": 2,
                    "Enable": false,
                    "TemperatureType": "Minimum",
                    "DetectionType": "Below",
                    "ThresholdTemperature": 100,
                    "Duration": 5
                }
            ],
            "NormalizedEmissivity": 98,
            "AreaOverlay": false,
            "AvgTemperatureOverlay": false,
            "MinTemperatureOverlay": false,
            "MaxTemperatureOverlay": false,
            "HandoverIndex": 0
        },
        {
```

```
                    "ROI": 2,
                    "AreaName": "B",
                    "TemperatureType": "Maximum",
                    "DetectionType": "Above",
                    "ThresholdTemperature": 68,
                    "Coordinates": [
                        {
                            "x": 644,
                            "y": 156
                        },
                        {
                            "x": 735,
                            "y": 508
                        }
                    ],
                    "PolygonCoordinates": [
                        {
                            "x": 651,
                            "y": 330
                        },
                        {
                            "x": 644,
                            "y": 487
                        },
                        {
                            "x": 721,
                            "y": 508
                        },
                        {
                            "x": 735,
                            "y": 156
                        }
                    ],
                    "Duration": 5,
                    "Conditions": [
                        {
                            "condition": 1,
                            "Enable": true,
                            "TemperatureType": "Maximum",
                            "DetectionType": "Above",
                            "ThresholdTemperature": 68,
```

```
                        "Duration": 5
                    },
                    {
                        "condition": 2,
                        "Enable": false,
                        "TemperatureType": "Minimum",
                        "DetectionType": "Below",
                        "ThresholdTemperature": 100,
                        "Duration": 5
                    }
                ],
                "NormalizedEmissivity": 98,
                "AreaOverlay": false,
                "AvgTemperatureOverlay": false,
                "MinTemperatureOverlay": false,
                "MaxTemperatureOverlay": false,
                "HandoverIndex": 0
            }
        ]
    }
  ]
}
```

158.27.4.2. Changing box temperature detection settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=set&Channel=0&ROI.1.Coordinate=63,37,346,205&ROI.1.TemperatureType=Maximum&ROI.1.DetectionType=Above&ROI.1.ThresholdTemperature=10&ROI.1.Duration=26&ROI.1.NormalizedEmissivity=33&ROI.1.AreaOverlay=True&ROI.1.AvgTemperatureOverlay=True&ROI.1.MinTemperatureOverlay=True&ROI.1.MaxTemperatureOverlay=True
```

158.27.4.3. Changing box temperature detection settings with polygon coordinates, 1,2 conditions

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=set&Channel=0&ROI.1.AreaName=A&ROI.1.PolygonCoordinate=209,210,79,459,360,470,662,78&ROI.1.Condition.1.Enable=True&ROI.1.Condition.1.DetectionType=Above&ROI.1.Condition.1.Duration=1&ROI.1.Condition.1.ThresholdTemperature=50&ROI.1.Condition.1.T
```

```
emperatureType=Maximum&ROI.1.Condition.2.Enable=True&ROI.1.Condition.2.Detec
tionType=Below&ROI.1.Condition.2.Duration=5&ROI.1.Condition.2.ThresholdTempe
rature=1&ROI.1.Condition.2.TemperatureType=Minimum&ROI.1.NormalizedEmissivit
y=98&ROI.1.AreaOverlay=False&ROI.1.AvgTemperatureOverlay=False&ROI.1.MinTemp
eratureOverlay=False&ROI.1.MaxTemperatureOverlay=False&ROI.1.HandoverIndex=0
```

158.27.4.4. Removing box temperature detection ROI Region 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=remove&ROIIndex
=1&Channel=0
```

158.27.4.5. Getting box temperature detection ROI & whole screen’s min/max/avg temperature reading information

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "BoxTemperatureDetection": [
        {
            "Channel": 0,
            "Unit": "Celsius",
            "FullScreenTemperature": {
                "AvgTemperature": 20.8,
                "MinTemperature": 17.6,
                "MaxTemperature": 33.8,
                "MinTemperatureCoordinates": {
                    "x": 276,
                    "y": 11
                },
                "MaxTemperatureCoordinates": {
```

```
                    "x": 35,
                    "y": 119
                }
            },
            "ROIsTemperature": [
                {
                    "ROIIndex": 1,
                    "AreaName": "A",
                    "AvgTemperature": 20.7,
                    "MinTemperature": 20,
                    "MaxTemperature": 21.1,
                    "MinTemperatureCoordinates": {
                        "x": 81,
                        "y": 204
                    },
                    "MaxTemperatureCoordinates": {
                        "x": 85,
                        "y": 95
                    }
                },
                {
                    "ROIIndex": 2,
                    "AreaName": "B",
                    "AvgTemperature": 21.2,
                    "MinTemperature": 20.4,
                    "MaxTemperature": 29.9,
                    "MinTemperatureCoordinates": {
                        "x": 250,
                        "y": 198
                    },
                    "MaxTemperatureCoordinates": {
                        "x": 318,
                        "y": 95
                    }
                }
            ]
        }
    ]
}
```

158.28. Box Temperature Detection Options

158.28.1. Description

The **boxtemperaturedetectionoptions** submenu provides information about box temperature detection options.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/BoxTemperatureDetection"

Access level

Action	Camera
view	Admin

158.28.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=boxtemperaturedetectionoptions&action=<value>[&<parameter>=<value>]
```

158.28.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads box temperature detection options
	Channel	REQ	<int>	Channel ID
	TemperatureType	RES	<enum> Above, Below Increase, Decrease	Condition for temperature detection
	TemperatureType.#.Celsius.Min	RES	<int>	Minimum Celsius temperature
	TemperatureType.#.Celsius.Max	RES	<int>	Maximum Celsius temperature
	TemperatureType.#.Fahrenheit.Min	RES	<int>	Minimum Fahrenheit temperature
	TemperatureType.#.Fahrenheit.Max	RES	<int>	Maximum Fahrenheit temperature

Action	Parameter	Request/ Response	Type/ Value	Description
	TemperatureRangeMode	REQ	<enum> Narrow, Wide, Auto	Maximum Fahrenheit temperature

158.28.4. Examples

158.28.4.1. Getting box temperature detection options for Channel 0

REQUEST

```
http://<Device IP>/ stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetectionoptions&action=view&Cha
nnel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "BoxTemperatureDetectionOptions": [
        {
            "Channel": 0,
            "ThresholdTemperature": [
                {
                    "TemperatureType": "Above",
                    "Celsius": {
                        "Min": -20,
                        "Max": 130
                    },
                    "Fahrenheit": {
                        "Min": -4,
                        "Max": 266
                    }
                },
                {
                    "TemperatureType": "Below",
                    "Celsius": {
                        "Min": -20,
```

```
                    "Max": 130
                },
                "Fahrenheit": {
                    "Min": -4,
                    "Max": 266
                }
            },
            {
                "TemperatureType": "Increase",
                "Celsius": {
                    "Min": 10,
                    "Max": 100
                },
                "Fahrenheit": {
                    "Min": 50,
                    "Max": 212
                }
            },
            {
                "TemperatureType": "Decrease",
                "Celsius": {
                    "Min": 10,
                    "Max": 100
                },
                "Fahrenheit": {
                    "Min": 50,
                    "Max": 212
                }
            }
        ]
    }
  ]
}
```

158.28.4.2. Getting box temperature detection options for Channel 0 & Narrow range mode

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=boxtemperaturedetectionoptions&action=view&Channel=0&TemperatureRangeMode=Narrow
```

If "TemperatureRangeMode" is set, show response of TemperatureRangeMode. If it is not set, show current TemperatureRangeMode setting's(imagecgi camera submenu) response.

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "BoxTemperatureDetectionOptions": [
        {
            "Channel": 0,
            "ThresholdTemperature": [
                {
                    "TemperatureType": "Above",
                    "Celsius": {
                        "Min": -40,
                        "Max": 150
                    },
                    "Fahrenheit": {
                        "Min": -40,
                        "Max": 302
                    }
                },
                {
                    "TemperatureType": "Below",
                    "Celsius": {
                        "Min": -40,
                        "Max": 150
                    },
                    "Fahrenheit": {
                        "Min": -40,
                        "Max": 302
                    }
                },
                {
                    "TemperatureType": "Increase",
                    "Celsius": {
                        "Min": 10,
                        "Max": 100
```

```
                    },
                    "Fahrenheit": {
                        "Min": 50,
                        "Max": 212
                    }
                },
                {
                    "TemperatureType": "Decrease",
                    "Celsius": {
                        "Min": 10,
                        "Max": 100
                    },
                    "Fahrenheit": {
                        "Min": 50,
                        "Max": 212
                    }
                }
            ]
        }
    ]
}
```


158.29. Overspeed

158.29.1. Description

The **overspeed** submenu provides information and set overspeed event for NVR(GPS)

Access level

Action	NVR
view	User
set	User

158.29.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
overspeed&action=<value>[&<parameter>=<value>]
```

158.29.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads overspeed settings
set	Enable	REQ, RES	<bool> True, False	Enable or disable overspeed check
	Speed	REQ, RES	<int>	Type of limit speed value
	AlarmOutput	REQ, RES	<csv> 1, 2, 3, 4, 5, 6, Beep, None	Type of alarm output
	Duration	REQ, RES	<enum> None, Always, 5s, 10s, 20s, 30s	Alarm duration

158.29.4. Examples

158.29.4.1. Getting overspeed settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=overspeed&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
Speed=0
AlarmOutput=None
Duration=None
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false,
    "Speed": 0,
    "AlarmOutput": [
        "None"
    ],
    "Duration": "None"
}
```

158.29.4.2. Applying overspeed settings

REQUEST

```
http:// <Device IP>/stw-cgi/eventsources.cgi?msubmenu=overspeed&action=set&Enable=True&Speed=50&AlarmOutput=1&Duration=Always
```

158.30. Object Detection

158.30.1. Description

The **objectdetection** submenu enables the object detection capability in the camera.

NOTE Only when Object detection or IVA is enabled in the camera, object metadata would be generated. If ObjectDetection submenu is enabled without any object types, only metadata without objectdetection event would be generated.

Access level

Action	NVR	Camera
view	User	ADMIN
set	User	ADMIN

158.30.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
objectdetection&action=<value>[&<parameter>=<value>]
```

158.30.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	ChanelID
set	Channel	REQ, RES	<int>	ChanelID
	Enable	REQ, RES	<bool> True, False	Enable object detection
	ObjectTypes	REQ, RES	<csv> Vehicle, Face, Person, LicensePlate	When a configured object is detected, an event would be generated.
	ObjectTypeDetails	REQ, RES	<csv> Vehicle.Types.Bicycle, Vehicle.Types.Car, Vehicle.Types.Motorcycle, Vehicle.Types.Bus, Vehicle.Types.Truck	These parameters indicate the details of the ObjectType top-level object. If none of these details are set, the top-level object wil not be detected either.
	Sensitivity	REQ, RES	<int> 1 to 100	Sensitivity level
	Duration	REQ, RES	<int> 1 to 5	Duration in secs
	MinimumObjectSize	REQ, RES	<string>	Minimum size of objects Objects smaller than the specified minimum size are not detected. The size is expressed in the form of <width, height>. The value of **MinimumObjectSize** must be smaller than the value of **MaximumObjectSize**.

Action	Parameter	Request/ Response	Type/ Value	Description
	MaximumObjectSize	REQ, RES	<string>	Maximum size of objects Objects larger than the maximum size are not detected. The size is expressed in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**.
	MinimumObjectSizeInPixels	REQ, RES	<string>	Minimum object size in pixels The size is specified in the form of <width, height>.
	MaximumObjectSizeInPixels	REQ, RES	<string>	Maximum object size in pixel The size is expressed in the form of <width, height>.
	EnableMetadataInExcludeArea	REQ, RES	<bool> True, False	By enabling this parameter, metadata would be generated even for exclude region.
	ExcludeArea.#.Coordinate	REQ, RES	<string>	Exclude region coordinates In this region, object detection does not work
	HandoverIndex	REQ, RES	<int>	**HandOverIndex** associated with object detection
	VideoOutOverlay.<ObjectType>.LineColor	REQ, RES	<enum> None, Red, Orange, Yellow, Green, Blue, Navy, Violet, Black, White	Color of <ObjectTypes> detected area line **Note** <ObjectType> available list can get VideoOutOverlayObjectTypes of sourceoptions submenu
remove	Channel	REQ	<int>	
	ExcludeAreaIndex	REQ	<csv>	Exclude area index to be removed

158.30.4. Examples

158.30.4.1. Getting objectdetection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=objectdetection&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ObjectDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Duration": 1,
            "Sensitivity": 80,
            "MinimumObjectSize": "4,7",
            "MaximumObjectSize": "50,89",
            "MinimumObjectSizeInPixels": "194,194",
            "MaximumObjectSizeInPixels": "1944,1944",
            "ObjectTypes": [
                "Person",
                "Vehicle",
                "Face",
                "LicensePlate"
            ],
            "ObjectTypeDetails": {
                "Vehicle": {
                    "Types": [
                        "Bicycle",
                        "Car",
                        "Motorcycle",
                        "Bus",
                        "Truck"
                    ]
                }
            },
            "ExcludeAreas": [
```

```
                {
                    "ExcludeArea": 1,
                    "Coordinates": [
                        {
                            "x": 1248,
                            "y": 502
                        },
                        {
                            "x": 3173,
                            "y": 502
                        },
                        {
                            "x": 3317,
                            "y": 1743
                        },
                        {
                            "x": 972,
                            "y": 1701
                        }
                    ]
                }
            ],
            "HandoverIndex": 0
        }
    ]
}
```


158.30.4.2. Applying objectdetection settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=objectdetection&action=set&Channel=0&ObjectTypes=Vehicle,Person,Face,LicensePlate&Sensitivity=50&Enable=True&ExcludeArea.1.Coordinate=672,1002,1044,254,2291,326,2275,1662
```

158.31. Meta Image Transfer

158.31.1. Description

The **metaimagetransfer** submenu enables the best shot image delivery in metatadata for the selected object types.

Access level

Action	NVR	Camera
view	User	Admin
set	User	Admin

158.31.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
metaimagetransfer&action=<value>[&<parameter>=<value>]
```

158.31.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	ObjectTypes	REQ, RES	<csv> Vehicle, Person, Face, LicensePlate	Bestshot of objects required in metadata
	ImageQuality	REQ, RES	<int>	Quality of the delivered images
	Base64ObjectImageEnable	REQ, RES	<bool> True, False	Whether to include base64 object image

158.31.4. Examples

158.31.4.1. Get the metaimagetransfer settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=metaimagetransfer&action=view&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.ObjectTypes=Person,Vehicle,Face,LicensePlate
Channel.0.ImageQuality=75
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MetaImageTransfer": [
        {
            "Channel": 0,
            "ObjectTypes": [
                "Person",
                "Vehicle",
                "Face",
                "LicensePlate"
            ],
            "ImageQuality": 75,
            "Base64ObjectImageEnable": false
        }
    ]
}
```


158.31.4.2. Setting metaimagetransfer

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=objectdetection&action=set&Channel=0&ObjectTyp
es=Vehicle,Person,Face,LicensePlate&Sensitivity=50&Enable=True
```

158.32. Face Recognition

158.32.1. Description

The **facerecognition** submenu can be used to enable facerecognition feature on selected channels.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

158.32.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
facerecognition&action=<value>[&<parameter>=<value>]
```

158.32.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads facerecognition settings
set	Channel	REQ, RES	<int>	
	Enable	REQ, RES	<bool> True, False	
	GroupIDList	REQ, RES	<csv>	Detection group id list
	Similarity	REQ, RES	<int>	Similarity threshold value

158.32.4. Examples

158.32.4.1. Set face recognition settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?&msubmenu=facerecognition&action=set&Channel=1&Enable=T
rue&GroupIDList=1001&Similarity=12
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.32.4.2. View face recognition settings

REQUEST

```
http:// <Device IP>/stw-
cgi/eventsources.cgi?&msubmenu=facerecognition&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.Enable=True
Channel.1.Similarity=12
Channel.1.GroupIDList=1001
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "facerecognition": [
        {
            "Channel": 1,
            "Enable": true,
            "Similarity": 12,
            "GroupIDList": [
```

```
                "1001"
            ]
        }
    ]
}
```

158.33. OCR

158.33.1. Description

The **ocr** submenu can be used to enable/disable OCR (optical character recognition) feature on channels.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
view	User
set	User

158.33.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
ocr&action=<value>[&<parameter>=<value>]
```

158.33.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads ocr settings
set	Channel	REQ, RES	<int>	
	Enable	REQ, RES	<bool> True, False	

158.33.4. Examples

158.33.4.1. Set ocr settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=ocr&action=set&Channel=1&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.33.4.2. View ocr settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=ocr&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

Channel.1.Enable=True

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
```

```
    "ocr": [{
        "Channel": 1,
        "Enable": true
    }]
}
```

158.34. Thermal Detection Mode

158.34.1. Description

The **thermaldetectionmode** submenu provides setting for switching the operating mode of the camera between bodytemperature detection mode and normal mode.

NOTE This chapter is applicable to dual thermal cameras.

Access level

Action	Camera
view	Admin
set	Admin

158.34.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
thermaldetectionmode&action=<value>[&<parameter>=<value>]
```

158.34.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	ThermalMode	REQ, RES	<enum> BodyTemperature, Normal	Thermal mode Operating mode of dual thermal camera

Action	Parameters	Request/ Response	Type/ Value	Description
	FaceDetectionSource	REQ, RES	<enum> Visible, Thermal	Face detection source Source channel to use for face detection **Note** It can use only WhiteHot thermal color palette if **FaceDetectionSource** is set to Thermal channel

158.34.4. Examples

158.34.4.1. Getting current thermal detection mode settings (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-cgi/
eventsources.cgi?msubmenu=thermaldetectionmode&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ThermalDetectionMode": [
        {
            "Channel": 1,
            "ThermalMode": "Normal",
            "FaceDetectionSource": "Visible"
        }
    ]
}
```

158.34.4.2. Setting thermal detection mode to 'Normal' and face detection source to 'Visible'

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/eventsources.cgi?msubmenu=thermaldetectionmode&action=set&Channel=1&ThermalMode=Normal&FaceDetectionSource=Visible
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.35. Body Temperature Detection

158.35.1. Description

The **bodytemperaturedetection** submenu configures body temperature detection settings.

NOTE This chapter is applicable to dual thermal cameras.
Attribute to check for feature support:
"attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.35.2. Syntax

```
http://<Device IP>/stw-cgi/
```

```
eventsources.cgi?msubmenu=bodytemperaturedetection&action=<value>[&<paramete
r>=<value>]
```

158.35.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel Id
set	Channel	REQ, RES	<int>	Channel Id
	Enable	REQ, RES	<bool> True, False	Whether to use Body Temperature Detection event
	EnableBeepAlarm	REQ, RES	<bool> True, False	Whether to beep alarm when temperature exceeds ThresholdTemperature
	ThersholdTemperature	REQ, RES	<float>	Threshold temperature for detection
	EnableTemperatureMaxLimit	REQ, RES	<bool> Tr True, False	Whether to use TemperatureMaxLimit setting
	EnableTemperatureMinLimit	REQ, RES	<bool> True, False	Whether to use TemperatureMinLimit setting
	TemperatureMaxLimit	REQ, RES	<float>	Excludes detection above the temperature
	TemperatureMinLimit	REQ, RES	<float>	Excludes detection below the temperature
	Coordinates	REQ, RES	<string> Format=x1,y1,x2,y2	Coordinates of detection area The coordinates are expressed in < x1,y1,x2,y2>
	Duration	REQ, RES	<int>	Bodytemperature detection duration in seconds
	Sensitivity	REQ, RES	<int>	Sensitivity level
	EnableOverlay	REQ, RES	<bool> True, False	Whether to use detected area box overlay

Action	Parameters	Request/ Response	Type/ Value	Description
	MinimumObjectSize	RES,REQ	<string> Format=w,h	Minimum size of objects Objects smaller than the specified minimum size are not detected. The size is expressed in <width, height>. The value of **MinimumObjectSize** must be less than the value of **MaximumObjectSize**.
	MaximumObjectSize	RES,REQ	<string> Format=w,h	Maximum size of objects Objects larger than the maximum size are not detected. The size is expressed in <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**.
	MinimumObjectSizeInPixels	RES,REQ	<string> Format=w,h	Minimum object size in pixels The size is expressed in <width, height>.
	MaximumObjectSizeInPixels	RES,REQ	<string> Format=w,h	Maximum object size in pixel The size is expressed in <width, height>.

158.35.4. Examples

158.35.4.1. Getting current body temperature detection settings for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=bodytemperaturedetection&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "BodyTemperatureDetection": [
        {
            "Channel": 1,
            "Enable": true,
            "MinimumObjectSize": "50,50",
            "MaximumObjectSize": "80,80",
            "MinimumObjectSizeInPixels": "164,124",
            "MaximumObjectSizeInPixels": "260,195",
            "EnableBeepAlarm": true,
            "ThresholdTemperature": 37.5,
            "EnableTemperatureMaxLimit": true,
            "EnableTemperatureMinLimit": true,
            "TemperatureMaxLimit": 40.0,
            "TemperatureMinLimit": 35.0,
            "Coordinates": [
                {
                    "x": 0,
                    "y": 0
                },
                {
                    "x": 319,
                    "y": 121
                }
            ],
            "Duration": 2,
            "Sensitivity": 2,
            "EnableOverlay": false
        }
    ]
}
```


158.35.4.2. Setting body temperature detection configurations for Channel 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=bodytemperaturedetection&action=set&Channel=1&
Enable=True&EnableBeepAlarm=True&ThresholdTemperature=37.5&EnableTemperature
```

```
MaxLimit=True&TemperatureMaxLimit=39&EnableTemperatureMinLimit=True&Temperat
ureMinLimit=35&Coordinates=0,0,100,100&Duration=3&Sensitivity=3&EnableOverla
y=True&MinimumObjectSize=30,30&MaximumObjectSize=50,50
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.36. Body Temperature Detection Options

158.36.1. Description

The **bodytemperaturedetectionoptions** submenu provides information on body temperature detection settings

NOTE

This chapter is applicable to dual thermal cameras.
Attribute to check for feature support:
"attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.36.2. Syntax

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=bodytemperaturedetectionoptions&action=<value>
[&<parameter>=<value>]
```

158.36.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel Id
set	Channel	REQ, RES	<int>	Channel Id
	Temperature.#.Celsius.Min	RES	<float>	Minimum Celsius temperature
	Temperature.#.Celsius.Max	RES	<float>	Maximum Celsius temperature
	Temperature.#.Fahrenheit.Min	RES	<float>	Minimum Fahrenheit temperature
	Temperature.#.Fahrenheit.Max	RES	<float>	Maximum Fahrenheit temperature

158.36.4. Examples

158.36.4.1. Getting body temperature detection options for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=bodytemperaturedetectionoptions&action=view&Ch
annel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "BodyTemperatureDetectionOptions": [
        {
            "Channel": 1,
```

```
            "Temperature": {
                "Celsius": {
                    "Min": 30.0,
                    "Max": 45.0
                },
                "Fahrenheit": {
                    "Min": 86.0,
                    "Max": 113.0
                }
            }
        }
    ]
}
```

158.37. Temperature Measurement Region Settings

158.37.1. Description

The **temperaturemeasurementregion** submenu configures adjustment of the face area to measure the maximum temperature value.

NOTE

This chapter is applicable to dual thermal cameras.
Attribute to check for feature support:
"attributes/Eventsource/Support/[ChannelID]/BodyTemperatureDetection"

Access level

Action	Camera
view	Admin
set	Admin

158.37.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturemeasurementregion&action=<value>[&<parameter>=<value>]
```

158.37.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel Id

Action	Parameters	Request/ Response	Type/ Value	Description
set	Channel	REQ, RES	<int>	Channel Id
	HorizontalRatio	REQ, RES	<int>	Horizontal ratio of detected face marking
	VerticalRatio	REQ, RES	<int>	Vertical ratio of detected face marking
	CenterOffset	REQ, RES	<string> format=x,y	Position offset of detection area

158.37.4. Examples

158.37.4.1. Getting current temperature measurement region settings for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturemeasurementregion&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TemperatureMeasurementRegion": [
        {
            "Channel": 1,
            "HorizontalRatio": 0,
            "VerticalRatio": 0,
            "CenterOffset": [
                {
                    "x": 0,
                    "y": 0
                }
            ]
        }
    ]
```

```
}
```

158.37.4.2. Setting ratio and position of the temperature measurement region for Channel 1 (this submenu supports only JSON response)

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturemeasurementregion&action=set&Channe
l=1&HorizontalRatio=50&VerticalRatio=50&CenterOffset=0,-30
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.38. Mask Detection Setup

158.38.1. Description

The **maskdetection** submenu provides how to configure mask detection setup and enables it.

NOTE	Attribute to check for feature support: "attributes/Eventsource/Support/[ChannelID]/MaskDetection"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin

158.38.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
maskdetection&action=<value>[&<parameter>=<value>]
```

158.38.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel Id
set	Channel	REQ, RES	<int>	Channel Id
	Enable	REQ, RES	<bool> True, False	Enables or disables the mask detection
	DetectionMode	REQ, RES	<enum> MASK, NO_MASK	Detection mode
	Sensitivity	REQ, RES	<int>	Mask detection sensitivity level
	Duration	REQ, RES	<int>	Mask detection duration in seconds
	MinimumObjectSize	REQ, RES	<string> w,h	Minimum size of objects Objects smaller than the specified minimum size are not detected. The size is expressed in <width, height>. The value of **MinimumObjectSize** must be smaller than the value of **MaximumObjectSize**.
	MaximumObjectSize	REQ, RES	<string> w,h	Maximum size of objects Objects larger than the maximum size are not detected. The size is expressed in the format of <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**.
	MinimumObjectSizeInPixels	REQ, RES	<string> w,h	Minimum object size in pixels The size is expressed in <width, height>.
	MaximumObjectSizeInPixels	REQ, RES	<string> w,h	Maximum object size in pixel The size is expressed in <width, height>.

Action	Parameters	Request/ Response	Type/ Value	Description
	EnableMetadataInExcludeArea	REQ, RES	<bool> True, False	Enabling this parameter generates metadata even for exclude region.
	ExcludeArea.#.Coordinate	REQ, RES	<string>	Exclude region coordinates In this region, object detection does not work
remove	Channel	REQ	<int>	Channel Id
	ExcludeAreaIndex	REQ	<csv> x1,y1,x2,y2	Exclude area index to be removed

158.38.4. Examples

158.38.4.1. Getting current mask detection setup of Channel 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=maskdetection&action=view&Channel=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MaskDetection": [
        {
            "Channel": 1,
            "Enable": true,
            "DetectionMode": "MASK",
            "Duration": 2,
            "Sensitivity": 80,
            "MinimumObjectSize": "5,8",
            "MaximumObjectSize": "50,89",
            "MinimumObjectSizeInPixels": "194,194",
            "MaximumObjectSizeInPixels": "1944,1944",
            "EnableMetadataInExcludeArea": false,
            "ExcludeAreas": [
                {
```

```
                    "ExcludeArea": 1,
                    "Coordinates": [
                        {
                            "x": 714,
                            "y": 424
                        },
                        {
                            "x": 714,
                            "y": 1767
                        },
                        {
                            "x": 2615,
                            "y": 1767
                        },
                        {
                            "x": 2615,
                            "y": 424
                        }
                    ]
                }
            ]
        }
    ]
}
```


158.38.4.2. Setting exclude area for Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=maskdetection&action=set&Channel=0&Enable=True&ExcludeArea.1.Coordinate=714,426,714,1769,2615,1769,2615,426&Sensitivity=50
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
```

```
}
```

158.39. Cell motion

158.39.1. Description

The **cellmotion** submenu configures the cell motion detection in DVR for the connected analog camera.

> **NOTE**
> This chapter is applicable to DVR only.
> This feature works in analog camera only.

Access level

Action	DVR
view	User
set	User

158.39.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
cellmotion&action=<value>[&<parameter>=<value>]
```

158.39.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	CellMotionWidth	RES	<int>	Shows maximum width of area of interest
	CellMotionHeight	RES	<int>	Shows maximum height of area of interest
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool>	Sets whether to execute the feature
	Sensitivity	REQ, RES	<int> 1~10	Sets sensitivity level of detection
	Coordinates	REQ, RES	<string>	Information on cells allowed to detect motion

158.39.4. Examples

158.39.4.1. Getting current information of Channel 1

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=cellmotion&action=view&Channel=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel=1
Channel.1.CellMotionWidth= 32
Channel.1.CellMotionHeight= 15
Channel.1.Enable= False
Channel.1.Sensitivity= 3
Channel.1.Coordinates=
Y00: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y01: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y02: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y03: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y04: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y05: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y06: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y07: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y08: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y09: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y10: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y11: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y12: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y13: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Y14: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "CellMotionInfos": [{
        "Channel": 1,
        "CellMotionWidth": "32",
        "CellMotionHeight": "15",
        "Enable": false,
        "Sensitivity": "3",
        "Coordinates": {
            "Y00": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y01": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y02": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y03": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y04": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y05": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y06": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y07": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y08": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y09": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y10": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
```

```
            "Y11": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y12": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y13": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"],
            "Y14": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0"]
        }
    }]
}
```


158.39.4.2. Setting interest cells to enable motion detection

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=cellmotion&action=set
```

content

This submenu only supports POST for configuration. Users should send data in the HTTP body using POST. Please refer to the figure below:

```
<Body>
```


```
{"CellMotionInfos":[{"Channel":1,"CellMotionWidth":"32","CellMotionHeight":"
15","Enable":"true","Sensitivity":"3","Coordinates":{"Y00":["0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0"],"Y01":["0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0"],"Y02":["0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0"],"Y03":["0","0","0","1","1","1","1","1","1","1","1","1","1","1","1","1",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0"],"Y04":["0",
"0","0","1","1","1","1","1","1","1","1","1","1","1","1","1","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0"],"Y05":["0","0","0","1","1",
"1","1","1","1","1","1","1","1","1","1","1","0","0","0","0","0","0","0","0",
```

```
"0","0","0","0","0","0","0","0"],"Y06":["0","0","0","1","1","1","1","1","1",
"1","1","1","1","1","1","1","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0"],"Y07":["0","0","0","1","1","1","1","1","1","1","1","1","1",
"1","1","1","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0"]
,"Y08":["0","0","0","1","1","1","1","1","1","1","1","1","1","1","1","1","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0"],"Y09":["0","0",
"0","1","1","1","1","1","1","1","1","1","1","1","1","1","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0"],"Y10":["0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0"],"Y11":["0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0"],"Y12":["0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0"],"Y1
3":["0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0"],"Y14":["0","0","0",
"0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0","0",
"0","0","0","0","0","0","0","0","0","0"]}}]}
```


JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.40. Parking detection

158.40.1. Description

The **parkingdetection** submenu configures settings related to parking management for each channel. User can enable or disable a parking detection feature and set the interest area for detecting parking events.

NOTE This chapter is applicable to camera only.

Access level

Action	Camera
view	Guest
set	Admin
remove	Admin

158.40.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
parkingdetection&action=<value>[&<parameter>=<value>]
```

158.40.3. Parameters

Action	Parameters	Request/ Response	Type/	Description
view	Channel	REQ/RES	<csv>	Channel ID A camera uses from 0 to max fixed channel count-1 for channel ID. A user can requests multiple channels with a comma.
set	Channel	REQ	<int> 1 to 4	Channel ID Set the target channel that you want to configure.
	Enable	REQ/RES	<enum>	Configures whether the parking detection feature is active.
	Sensitivity	REQ/RES	<int> 1 to 10	Sensitivity level
	MinimumObjectSize	REQ/RES	<string> W, h	Minimum size of objects Objects smaller than the specified minimum size are not detected. The size is expressed in the form of <width, height>. The value of **MinimumObjectSize** must be smaller than the value of **MaximumObjectSize**.

Action	Parameters	Request/ Response	Type/	Description
	MaximumObjectSize	REQ/RES	<string> W, h	Maximum size of objects Objects larger than the maximum size are not detected. The size is expressed as <width, height>. The value of **MaximumObjectSize** must be greater than the value of **MinimumObjectSize**.
	MinimumObjectSizeInPixels	REQ/RES	<string> W, h	Minimum object size in pixels The size is expressed as <width, height>.
	MaximumObjectSizeInPixels	REQ/RES	<string> W, h	Maximum object size in pixels The size is expressed as <width, height>.
	Area.#.MaxDetectionCount	REQ/RES	<int>	Configures how many vehicles can be detected in each interest region. A parking detection event will occur only when the MaxDetectionCount of vehicles are parking in that area.
	Area.#.Coordinate	REQ/RES	<string>	Interest region coordinates Parking detection only works in this region.
	Area.#.DetectionArea.#.ZoneNumber	REQ/RES	<string>	Zone number for the detection area This parameter specifies the zone identification number for the parking detection area.
	Area.#.DetectionArea.#.ParkingLotNumber	REQ/RES	<string>	Parking lot number for the detection area This parameter specifies the parking lot identification number within the detection area.

Action	Parameters	Request/ Response	Type/	Description
	ExcludeArea.#.Coordinate	REQ/RES	<string>	Exclude region coordinates In this region, parking detection does not work.
remove	Channel	REQ	<int>	
	AreaIndex	REQ	<csv> #, All	Interest area index to be removed.
	ExcludeAreaIndex	REQ	<csv> #, All	Exclude area index to be removed.

158.40.4. Examples

158.40.4.1. Getting parking detection settings for all channel

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=parkingdetection&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.1.Enable=False
Channel.1.Sensitivity=10
Channel.1.MinimumObjectSize=21,21
Channel.1.MaximumObjectSize=99,99
Channel.1.MinimumObjectSizeInPixels=271,271
Channel.1.MaximumObjectSizeInPixels=950,950
Channel.1.Area.1.Coordinate=353,195,353,1182,1511,1182,1511,195
Channel.1.Area.1.MaxDetectionCount=3
Channel.2.Enable=False
Channel.2.Sensitivity=80
Channel.2.MinimumObjectSize=21,21
Channel.2.MaximumObjectSize=99,99
Channel.2.MinimumObjectSizeInPixels=271,271
Channel.2.MaximumObjectSizeInPixels=950,950
Channel.2.Area.1.Coordinate=414,181,414,1113,1550,1113,1550,181
```

```
Channel.2.Area.1.MaxDetectionCount=4
Channel.3.Enable=False
Channel.3.Sensitivity=80
Channel.3.MinimumObjectSize=21,21
Channel.3.MaximumObjectSize=99,99
Channel.3.MinimumObjectSizeInPixels=271,271
Channel.3.MaximumObjectSizeInPixels=950,950
Channel.3.Area.1.Coordinate=350,310,350,802,666,802,666,310
Channel.3.Area.1.MaxDetectionCount=4
Channel.4.Enable=False
Channel.4.Sensitivity=80
Channel.4.MinimumObjectSize=21,21
Channel.4.MaximumObjectSize=301,301
Channel.4.MinimumObjectSizeInPixels=271,271
Channel.4.MaximumObjectSizeInPixels=2707,2707
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ParkingDetection": [
        {
            "Channel": 1,
            "Enable": true,
            "Sensitivity": 3,
            "MinimumObjectSize": "4,5",
            "MaximumObjectSize": "65,88",
            "MinimumObjectSizeInPixels": "160,160",
            "MaximumObjectSizeInPixels": "1210,1210",
            "Areas": [
                {
                    "Area": 1,
                    "Coordinates": [
                        {
                            "x": 246,
                            "y": 230
                        },
                        {
```

```
                        "x": 448,
                        "y": 854
                    },
                    {
                        "x": 1136,
                        "y": 890
                    },
                    {
                        "x": 1324,
                        "y": 294
                    }
                ],
                "MaxDetectionCount": 4
            }
        ]
    },
    {
        "Channel": 2,
        "Enable": true,
        "Sensitivity": 5,
        "MinimumObjectSize": "4,5",
        "MaximumObjectSize": "65,88",
        "MinimumObjectSizeInPixels": "160,160",
        "MaximumObjectSizeInPixels": "1210,1210",
        "Areas": [
            {
                "Area": 1,
                "Coordinates": [
                    {
                        "x": 515,
                        "y": 392
                    },
                    {
                        "x": 1525,
                        "y": 451
                    },
                    {
                        "x": 1452,
                        "y": 873
                    },
                    {
```

```
                    "x": 193,
                    "y": 1122
                }
            ],
            "MaxDetectionCount": 1
        }
    ]
},
{
    "Channel": 3,
    "Enable": true,
    "Sensitivity": 5,
    "MinimumObjectSize": "4,5",
    "MaximumObjectSize": "65,88",
    "MinimumObjectSizeInPixels": "160,160",
    "MaximumObjectSizeInPixels": "1210,1210",
    "Areas": [
        {
            "Area": 1,
            "Coordinates": [
                {
                    "x": 1005,
                    "y": 577
                },
                {
                    "x": 308,
                    "y": 596
                },
                {
                    "x": 353,
                    "y": 982
                },
                {
                    "x": 1436,
                    "y": 1223
                }
            ],
            "MaxDetectionCount": 3
        }
    ]
},
```

```
        {
            "Channel": 4,
            "Enable": false,
            "Sensitivity": 5,
            "MinimumObjectSize": "4,5",
            "MaximumObjectSize": "65,88",
            "MinimumObjectSizeInPixels": "160,160",
            "MaximumObjectSizeInPixels": "1210,1210",
            "Areas": [
                {
                    "Area": 1,
                    "Coordinates": [
                        {
                            "x": 246,
                            "y": 507
                        },
                        {
                            "x": 1469,
                            "y": 104
                        },
                        {
                            "x": 1514,
                            "y": 484
                        },
                        {
                            "x": 339,
                            "y": 1010
                        }
                    ],
                    "MaxDetectionCount": 2
                }
            ]
        }
    ]
}
```

158.40.4.2. Setting parking detection area and max vehicle count to be detected

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=parkingdetection&action=set&Channel=1&Sensitiv
```

```
ity=10&MinimumObjectSizeInPixels=271,271&MaximumObjectSizeInPixels=950,950&A
rea.1.Coordinate=526,363,473,975,1175,1048,1282,438&Area.1.MaxDetectionCount
=3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Response": "Success"
}
```


158.40.4.3. Setting parking detection area with zone and parking lot numbers

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=parkingdetection&action=set&Channel=1&Area.1.C
oordinate=526,363,473,975,1175,1048,1282,438&Area.1.DetectionArea.1.ZoneNumb
er=A1&Area.1.DetectionArea.1.ParkingLotNumber=001
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Response": "Success"
}
```


158.41. ledindicator

158.41.1. Description

The **ledindicator** submenu configures LED settings used for indicating the current parking status to clients. The main feature for this submenu is that it determines how to use 2 LEDs. Users can use those 2 LEDs in an integrated or separate way. Users can choose this based on their parking lot's circumstance. And users can select LED colors when the parking area is full or vacant.

Access level

Action	Camera
view	Admin
set	Admin

158.41.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
ledindicator&action=<value>[&<parameter>=<value>]
```

158.41.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<csv>	Channel ID
set	LEDUsage	REQ, RES	<int> 1, 2	Define how to use 2 LEDs.
	IndicationPassEnable	REQ, RES	<bool> True, False	Determine whether to use indication pass feature. This feature will work only when the user sets LEDUsage to 1. If this works, a parking detection event will be sent to the previously configured camera and that will generate an event instead of the original camera. Please refer to the Appendix if you are interested with this feature.
	EnableFlickeringAlarm	REQ, RES	<boolean> True,False	Configuring the LED flickering setting. LED will flicker when the alarm event occurs if it is set as True.
	LED.#.EventOn.Color	REQ, RES	<enum> None, Green, Blue, Red, Yellow, Pink1, Pink2, AquaBlue, Purple	Configure the LED color when the parking area is fully occupied.

Action	Parameters	Request/ Response	Type/ Value	Description
	LED.#.EventOn.Text	REQ, RES	<string>	Configure the text when the parking area is fully occupied. This value is only for internal memos and not to be shared.
	LED.#.EventOff.Color	REQ, RES	<enum> None, Green, Blue, Red, Yellow, Pink1, Pink2, AquaBlue, Purple	Configure the LED color when the parking area is empty.
	LED.#.EventOff.Text	REQ, RES	<string>	Configure the text when the parking area is empty. This value is only for internal memos and not to be shared.
	SourceChannel.#.LEDUsageIndex	REQ, RES	<int> 1~2	Determine which LED displays the current status when an event occurs.

158.41.4. Examples

158.41.4.1. Getting LED settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=ledindicator&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.LEDUsage=2
Channel.0.EnableFlickeringAlarm=True
Channel.0.IndicationPassEnable=False
Channel.0.LED.1.EventOn.Color=Red
```

```
Channel.0.LED.1.EventOn.Text=Occupied
Channel.0.LED.1.EventOff.Color=Yellow
Channel.0.LED.1.EventOff.Text=Vacant
Channel.0.LED.2.EventOn.Color=Red
Channel.0.LED.2.EventOn.Text=Occupied
Channel.0.LED.2.EventOff.Color=Pink1
Channel.0.LED.2.EventOff.Text=Vacant
SourceChannel.1.LEDUsageIndex=1
SourceChannel.2.LEDUsageIndex=1
SourceChannel.3.LEDUsageIndex=2
SourceChannel.4.LEDUsageIndex=2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: json/application
<Body>
```


```
{
    "LEDIndicator": [
        {
            "Channel": 0,
            "LEDUsage": 1,
            "EnableFlickeringAlarm ”: false,
            "IndicationPassEnable": false,
            "LEDs": [
                {
                    "LED": 1,
                    "Type": "EventOn",
                    "Color": "Red",
                    "Text": "Occupied"
                },
                {
                    "LED": 1,
                    "Type": "EventOff",
                    "Color": "Pink1",
                    "Text": "Vacant"
                }
            ],
            "SourceChannels": [
                {
```

```
                    "SourceChannel": 1,
                    "LEDUsageIndex": 1
                },
                {
                    "SourceChannel": 2,
                    "LEDUsageIndex": 1
                },
                {
                    "SourceChannel": 3,
                    "LEDUsageIndex": 1
                },
                {
                    "SourceChannel": 4,
                    "LEDUsageIndex": 1
                }
            ]
        }
    ]
}
```

158.41.4.2. Setting to use 2 LEDs separately and to show the event status with LED 2 when a parking detection event occurs in Channel 2

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=ledindicator&action=set&LEDUsage=2&SourceChann
el.2.LEDUsageIndex=2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.42. Call Request Event Settings

158.42.1. Description

The **callrequest** submenu provides settings for CallRequest events.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Eventsource/Support/CallRequest"

Access level

Action	Camera
view	Admin
set	Admin

158.42.2. Syntax

```
http://<Device IP>/stw-cgi/ eventsources.cgi?msubmenu=
callrequest&action=<value>[&<parameter>=<value>]
```

158.42.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Ringtone	REQ, RES	<enum> Default, Ringtone1, Ringtone2, Ringtone3, Ringtone4, Ringtone5	Ringtone played on call request
	CallingTimeout	REQ, RES	<int>	Calling timeout in seconds
	TouchlessCallEnable	REQ, RES	<bool> True, False	Whether to use a touch-free call
	StatusLEDEnable	REQ, RES	<bool> True, False	Whether to use the status LED
	AllowWelcomeLED	REQ, RES	<bool> True, False	Whether to allow the welcome LED It is applied if the **StatusLEDEnable** parameter is enabled.

Action	Parameters	Request/ Response	Type/ Value	Description
	MicrophoneActivationMode	REQ, RES	<enum> AlwaysOn, DuringCalls	Microphone activation mode • AlwaysOn: A microphone to always operate even when not on SIP or VMS calls • DuringCalls: A microphone mounted on a camera to operate only on SIP or VMS calls

158.42.4. Examples

158.42.4.1. Getting the callrequest event settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=callrequest&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Ringtone=Default
Channel.0.CallingTimeout=60
Channel.0.TouchlessCallEnable=False
Channel.0.StatusLEDEnable=True
Channel.0.AllowWelcomeLED=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "CallRequest": [
        {
            "Channel": 0,
            "Ringtone": "Default",
```

```
            "CallingTimeout": 60,
            "TouchlessCallEnable": false,
            "StatusLEDEnable": true,
            "AllowWelcomeLED": false
        }
    ]
}
```

158.42.4.2. Setting the callrequest event settings

To set a ringtone to something other than the default setting, an audio out file must be added. You can check the audio out file at the URL below.

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=audiooutfiles&action=view
```

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=callrequest&action=set&Ringtone=Ringtone1&Touc
hlessCallEnable=True&StatusLEDEnable=False&AllowWelcomeLED=False&CallingTime
out=20
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.43. DTMF Event Settings

158.43.1. Description

The **DTMF** submenu configures DTMF event settings.

NOTE

This chapter applies to network cameras only.
Attribute to check for feature support: "attributes/Eventsource/Support/DTMF"

Access level

Action	Camera
view	Admin
set	Admin
add	Admin
update	Admin
remove	Admin

158.43.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
dtmf&action=<value>[&<parameter>=<value>]
```

158.43.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	Index	REQ, RES	<int>	DTMF Index
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use a DTMF event
add	Channel	REQ	<int>	Channel ID
	Code	REQ	<string>	DTMF code
	HandoverIndex	REQ	<int>	HandoverIndex associated with DTMF event
update	Channel	REQ	<int>	Channel ID
	DTMF.#.Code	REQ, RES	<string>	DTMF code
	DTMF.#.HandoverIndex	REQ	<int>	HandoverIndex associated with DTMF event

Action	Parameters	Request/ Response	Type/ Value	Description
remove	Channel	REQ	<int>	Channel ID
	Index	REQ	<csv> All, #	DTMF Index

158.43.4. Examples

158.43.4.1. Getting current DTMF event settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=dtmf&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
DTMF.1.Code=1234
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "DTMF": [
        {
            "Channel": 0,
            "Enable": true,
            "DTMF": [
                {
                    "Index": 1,
                    "Code": "1234"
                }
            ]
        }
```

```
    ]
}
```

158.43.4.2. Enabling a DTMF event

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=dtmf&action=set&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.43.4.3. Adding a new DTMF code

The number of DTMF codes supported by the camera can be checked in the attribute below.

```
http://<Device IP>/stw-
cgi/attributes.cgi/attributes/Eventsource/Limit/MaxDTMFCodeCount
```

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=dtmf&action=add&Code=1234
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.43.4.4. Updating DTMF code for index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=dtmf&action=update&DTMF.1.Code=4321
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.43.4.5. Remove DTMF code for index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=dtmf&action=remove&Index=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.44. Tampering Switch Event Settings

158.44.1. Description

The **tamperingswitch** submenu configures the TamperingSwitch event settings

NOTE	This chapter applies to the network cameras only. Attribute to check for feature support: "Attributes/Eventsource/Support/TamperingSwitch"

Access level

Action	Camera
view	Admin
set	Admin

158.44.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
tamperingswitch&action=<value>[&<parameter>=<value>]
```

158.44.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use a TamperingSwitch event

158.44.4. Examples

158.44.4.1. Getting tampering switch event settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tamperingswitch&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TamperingSwitch": [
        {
            "Channel": 0,
            "Enable": false
        }
    ]
}
```

158.44.4.2. Enabling a tampering switch event

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tamperingswitch&action=set&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.45. Proximity Sensor Event Settings

158.45.1. Description

The **proximitysensor** submenu configures the ProximitySensor event settings

NOTE This chapter applies to the network cameras only.
Attribute to check for feature support: "Attributes/Eventsource/Support/ProximitySensor"

Access level

Action	Camera
view	Admin
set	Admin

158.45.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
proximitysensor&action=<value>[&<parameter>=<value>]
```

158.45.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Whether to use a ProximitySensor event

158.45.4. Examples

158.45.4.1. Getting proximity sensor event settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=proximitysensor&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ProximitySensor": [
        {
            "Channel": 0,
            "Enable": false
        }
    ]
}
```


158.45.4.2. Enabling a proximity sensor event

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=proximitysensor&action=set&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

158.46. Social Distancing Violation Detection

158.46.1. Description

The **socialdistancingviolation** submenu configures the socialdistancingviolation event settings.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "Attributes/Eventsource/Support/SocialDistancingViolation"

Access level

Action	Camera
view	Admin
set	Admin

158.46.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=socialdistancingviolation&action=<value>[&<parameter>=<value>]
```

158.46.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables socialdistancingviolation detection.
	CameraHeight	REQ, RES	<float>	Camera height from the floor in meters
	LensFocal	REQ, RES	<float>	For finetuning the focal length of the lens
	Tilt	REQ, RES	<float>	For adjusting the tilt angle used in the algorithm
	Rotation	REQ, RES	<float>	Rotation angle setting
	MinimumAllowedDistance	REQ, RES	<float>	When the distance between people is less than the set value, an event will be generated.
	Sensitivity	REQ, RES	<int>	Sensitivity level of detection

Action	Parameter	Request/ Response	Type/ Value	Description
	Duration	REQ, RES	<int>	When the configured duration of two people within the minimum allowed distance is exceed, an event will be generated.
	MinimumObjectSize	REQ, RES	<string> Format=w,h	Normalized minimum object size (0-99)
	MaximumObjectSize	REQ, RES	<string> Format=w,h	Normalized maximum object size (0-99)
	MinimumObjectSizeInPixels	REQ, RES	<string> Format=w,h	Minimum object size in pixels (based on source resolution)
	MaximumObjectSizeInPixels	REQ, RES	<string> Format=w,h	Maximum object size in pixels (based on source resolution)
	EnableMetadataInExcludeArea	REQ, RES	<string>	To allow metadata in the excluded region
	ExcludeArea.#.Coordinate	REQ, RES	<string> Format=x1, y1,x2,y2...	Exclude area. The algorithm will not be active in this region.
check	Channel	REQ, RES	<int>	
	RealLensFocal	RES	<float>	Focal length of the lens
remove	Channel	REQ	<int>	
	ExcludeAreaIndex	REQ	<csv>	List of excluded areas to be removed

158.46.4. Examples

158.46.4.1. Getting social distancing violation settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=socialdistancingviolation&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.0.CameraHeight=3.0
Channel.0.LensFocal=4.5
Channel.0.Tilt=45.0
Channel.0.Rotation=0.0
Channel.0.MinimumAllowedDistance=2.0
Channel.0.Duration=2
Channel.0.Sensitivity=80
Channel.0.MinimumObjectSize=3,4
Channel.0.MaximumObjectSize=50,88
Channel.0.MinimumObjectSizeInPixels=128,99
Channel.0.MaximumObjectSizeInPixels=1945,1921
Channel.0.EnableMetadataInExcludeArea=False
Channel.0.ExcludeArea.1.Coordinate=
Channel.0.ExcludeArea.2.Coordinate=
Channel.0.ExcludeArea.3.Coordinate=
Channel.0.ExcludeArea.4.Coordinate=
Channel.0.ExcludeArea.5.Coordinate=
Channel.0.ExcludeArea.6.Coordinate=
Channel.0.ExcludeArea.7.Coordinate=
Channel.0.ExcludeArea.8.Coordinate=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SocialDistancingViolation": [
        {
            "Channel": 0,
            "Enable": false,
            "CameraHeight": 3,
            "LensFocal": 4.5,
            "Tilt": 45,
            "Rotation": 0,
            "MinimumAllowedDistance": 2,
            "Duration": 2,
            "Sensitivity": 80,
```

```
            "MinimumObjectSize": "3,4",
            "MaximumObjectSize": "50,88",
            "MinimumObjectSizeInPixels": "128,99",
            "MaximumObjectSizeInPixels": "1945,1921",
            "EnableMetadataInExcludeArea": false
        }
    ]
}
```

158.46.4.2. Checking the focal length of the lens

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=socialdistancingviolation&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.RealLensFocal=4.7
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SocialDistancingViolation": [
        {
            "Channel": 0,
            "RealLensFocal": 4.7
        }
    ]
}
```

158.47. MQTT Publication Settings

158.47.1. Description

The **mqttpublication** submenu configures messages for MQTT publication.

NOTE	This chapter applies to network cameras only. Attribute to check for feature support: "attributes/Network/Support/MQTT" Attribute to check for the maximum number of messages supported: "attributes/Eventsource/Limit/MaxMQTTMessageCount"

Access level

Action	Camera
view	Admin
add/update	Admin
remove	Admin

158.47.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
mqttpublication&action=<value>[&<parameter>=<value>]
```

158.47.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
add/update	Index	REQ, RES	<int>	Message Index
	Name	REQ, RES	<string>	Message Name
	UseDefaultTopicPrefix	REQ, RES	<bool> True, False	Whether to use the default topic prefix
	Topic	REQ, RES	<string>	Message Topic
	QoS	REQ, RES	<enum> 0, 1, 2	Message QoS
	Retain	REQ, RES	<bool> True, False	Whether to use the retain flag
	Payload	REQ, RES	<string>	Message Payload
remove	Index	REQ	<csv> #	Message Index list

158.47.4. Examples (This submenu supports only JSON responses.)

158.47.4.1. Getting current DTMF event settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttpublication&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MQTTPublication": [
        {
            "Index": 1,
            "Name": "test",
            "UseDefaultTopicPrefix": false,
            "Topic": "test",
            "QoS": 0,
            "Retain": false,
            "Payload": "payload"
        }
    ]
}
```


158.47.4.2. Adding a new MQTT message

The number of MQTT messages supported by the camera can be checked in the attribute below.

```
http://<Device IP>/stw-
cgi/attributes.cgi/attributes/Eventsource/Limit/MaxMQTTMessageCount
```

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttpublication&action=add&Name=test&UseDefaul
tTopicPrefix=True&Topic=topic&QoS=0&Retain=False&Payload=testpayload
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.47.4.3. Updating MQTT message for index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttpublication&action=update&Index=1&Name=tes
t1&UseDefaultTopicPrefix=True&Topic=topic1&QoS=0&Retain=False&Payload=testpa
yload1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.47.4.4. Remove MQTT message for index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttpublication&action=remove&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.48. MQTT Subsciption Settings

158.48.1. Description

The **mqttsubsciption** submenu configures messages for MQTT Subsciption.

NOTE

This chapter applies to network cameras only.
Attribute to check for feature support: "attributes/Network/Support/MQTT"
Attribute to check for the maximum number of subscription supported: "attributes/Eventsource/Limit/MaxMQTTSubscriptionCount"

Access level

Action	Camera
view	Admin
add/update	Admin
remove	Admin

158.48.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
mqttsubsciption&action=<value>[&<parameter>=<value>]
```

158.48.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
add/update	Index	REQ, RES	<int>	Subscription Index
	Name	REQ, RES	<string>	Subscription Name
	Topic	REQ, RES	<string>	Subscription Topic
	QoS	REQ, RES	<enum> 0, 1, 2	Subscription QoS

Action	Parameters	Request/ Response	Type/ Value	Description
	Type	REQ, RES	<enum> OneShot, Property	Subscription Type • OneShot: An event occurs whenever a message is received from a topic that is subscribed to and the event occurrence status will be released immediately (Stateless) • Property: When a message is received from a topic that has been subscribed to, an event will occur, and the event occurrence status will be maintained afterward. The status will be disabled if a message different from the subscribing message is received (Stateful)
	Payload	REQ, RES	<string>	Subscription Payload
remove	Index	REQ	<csv> #	Subscription Index list

158.48.4. Examples (This submenu supports only JSON responses.)

158.48.4.1. Getting current DTMF event settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttsubscription&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "MQTTSubscription": [
        {
            "Index": 1,
            "Name": "name",
```

```
                "Type": "OneShot",
                "Topic": "topic",
                "QoS": 0,
                "Payload": "payload"
            }
        ]
}
```

158.48.4.2. Adding a new MQTT subscription

The number of MQTT subscription supported by the camera can be checked in the attribute below.

```
http://<Device IP>/stw-cgi/attributes.cgi/attributes/Eventsource/Limit/MaxMQTTSubscriptionCount
```

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=mqttsubscription&action=add&Name=test&Topic=topic&QoS=0&Type=OneShot&Payload=testpayload
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

158.48.4.3. Updating MQTT subscription for index 1

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=mqttsubscription&action=update&Index=1&Name=test1&Topic=topic1&QoS=0&Type=OneShot&Payload=testpayload1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.48.4.4. Remove MQTT subscription for index 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=mqttsubscription&action=remove&Index=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


158.49. Early Fire Detection

158.49.1. Description

The **earlyfiredetection** submenu provides settings for the Early fire detection. The Early fire detection feature triggers an early-fire-detection event when it detects flame burning or smoke rising within the specified area.

NOTE	Attribute to check for feature support: "attributes/Eventsource/Support/EarlyFireDetection" Attribute to check how many areas are supported: "attributes/Eventsource/Limit/MaxEarlyFireDetectionArea" Attribute to check the supported type of area: "attributes/Eventsource/Support/EarlyFireDetection.AreaType"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin
check	Admin

158.49.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
earlyfiredetection&action=<value>[&<parameter>=<value>...]
```

158.49.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	Rule.#.RuleIndex	RES	<int>	Index of the rule
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables early fire detection
	Sensitivity	REQ, RES	<int>	Sensitivity level
	Rule.#.RuleName	REQ, RES	<string>	Name of the rule
	Rule.#.Duration	REQ, RES	<int>	Set the minimum duration for flame or smoke detection to trigger early fire detection
	Rule.#.Coordinates	REQ, RES	<string>	Coordinates of area to set
	Rule.#.DetectionDistance	REQ, RES	<int>	Specify the distance of the target to detect
	Rule.#.TemperatureOffsetEnable	REQ, RES	<bool> True, False	Enable or disable temperature offset

Action	Parameters	Request/ Response	Type/ Value	Description
	Rule.#.TemperatureOffset	REQ, RES	<int>	Offset value to be adjusted if **TemperatureOffsetEnable** is enabled **Note** You can see the TemperatureOffset range of settings in the **earlyfiredetectionoptions** submenu
	Rule.#.ThresholdTemperature	REQ, RES	<int>	Threshold temperature value to be adjusted **Note** You can see the Temperature range of settings in the **earlyfiredetectionoptions** submenu
remove	Channel	REQ	<int>	
	RuleIndex	REQ	<csv> #, All	Interest area index to be removed
check	Channel	REQ, RES	<int>	channel for check
	TemperatureUnit	RES	<enum>	Temperature unit
	Rule.#.RuleIndex	RES	<int>	Rule index
	Rule.#.RuleName	RES	<string>	Rule Name
	Rule.#.AvgTemperature	RES	<float>	Average temperature for each ROI
	Rule.#.MinTemperature	RES	<float>	Maximum temperature for each ROI
	Rule.#.MaxTemperature	RES	<float>	Minimum temperature for each ROI
	Rule.#.MinTemperatureCoordinates	RES	<string>	Minimum temperature point coordinates for each ROI
	Rule.#.MaxTemperatureCoordinates	RES	<string>	Maximum temperature point coordinates for each ROI

158.49.4. Examples (this submenu supports only JSON response)

158.49.4.1. Getting earlyfiredetection settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetection&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "EarlyFireDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Sensitivity": 80,
            "Rule": [
                {
                    "RuleIndex": 1,
                    "RuleName": "Name1",
                    "Duration": 1,
                    "DetectionDistance": 0,
                    "TemperatureOffsetEnable": false,
                    "TemperatureOffset": 0,
                    "ThresholdTemperature": 120,
                    "Coordinates": [
                        {
                            "x": 726,
                            "y": 428
                        },
                        {
                            "x": 1905,
                            "y": 428
                        },
                        {
                            "x": 1905,
                            "y": 1157
                        },
                        {
                            "x": 726,
                            "y": 1157
                        }
                    ]
                }
            ]
```

```
        }
    ]
}
```

158.49.4.2. Adding a rule

Adding early fire detection area 'Name1'. When adding a new event rule, **Rule.#.RuleName**, **Rule.#.Duration**, **Rule.#.Coordinate** must be set.

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetection&action=set&Channel=0&Enable=True&Sensitivity=1&Rule.1.RuleName=Name1&Rule.1.Duration=1&Rule.1.DetectionDistance=0&Rule.1.ThresholdTemperature=120&Rule.1.TemperatureOffsetEnable=False&Rule.1.TemperatureOffset=120&Rule.1.Coordinates=755,622,1784,622,1784,1343,755,1343
```

158.49.4.3. Enabling early fire detection and setting the sensitivity level to 5

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetection&action=set&Enable=True&Sensitivity=5
```

158.49.4.4. Getting current min/max/avg temperature reading information of the early fire detection

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetection&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EarlyFireDetection": [
        {
```

```
            "Channel": 0,
            "TemperatureUnit": "Celsius",
            "Rule": [
                {
                    "RuleIndex": 1,
                    "RuleName": "Name1",
                    "AvgTemperature": 55.2,
                    "MinTemperature": 47.3,
                    "MaxTemperature": 66.4
                }
            ]
        }
    ]
}
```

158.50. Early Fire Detection Options

158.50.1. Description

The **earlyfiredetectionoptions** submenu provides settings options for the Early fire detection.

NOTE Attribute to check for feature support: "attributes/Eventsource/Support/EarlyFireDetection"

Access level

Action	Camera
view	Admin

158.50.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetectionoptions&action=<value>[&<parameter>=<value>...]
```

158.50.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	TemperatureUnit	RES	<enum> Celsius, Fahrenheit	Currently set temperature units
	Temperature.#.Celsius.Min	RES	<int>	Minimum Celsius temperature
	Temperature.#.Celsius.Max	RES	<int>	Maximum Celsius temperature
	Temperature.#.Fahrenheit.Min	RES	<int>	Minimum Fahrenheit temperature
	Temperature.#.Fahrenheit.Max	RES	<int>	Maximum Fahrenheit temperature
	TemperatureOffset.#.Celsius.Min	RES	<int>	Minimum Celsius value for temperature offset
	TemperatureOffset.#.Celsius.Max	RES	<int>	Maximum Celsius value for temperature offset
	TemperatureOffset.#.Fahrenheit.Min	RES	<int>	Minimum Fahrenheit value for temperature offset
	TemperatureOffset.#.Fahrenheit.Max	RES	<int>	Maximum Fahrenheit value for temperature offset

158.50.3.1. Getting Earlyfire detection options

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=earlyfiredetectionoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "EarlyFireDetectionOptions": [
        {
            "Channel": 0,
            "TemperatureUnit": "Celsius",
            "Temperature": {
```

```
                    "Celsius": {
                        "Min": 60,
                        "Max": 450
                    },
                    "Fahrenheit": {
                        "Min": 140,
                        "Max": 842
                    }
                },
                "TemperatureOffset": {
                    "Celsius": {
                        "Min": -10,
                        "Max": 450
                    },
                    "Fahrenheit": {
                        "Min": -10,
                        "Max": 450
                    }
                }
            }
        ]
}
```


158.51. Temperature Difference

158.51.1. Description

The **temperaturedifference** submenu provides settings for the Temperature Difference. The Temperature Difference feature compares the difference between two temperature values in areas set in BoxTemperatureDetection and fires an event when the difference is greater than a certain value

NOTE	Attribute to check for feature support: "attributes/Eventsource/Support/TemperatureDifference" Attribute to check how many rules are supported: "attributes/Eventsource/Limit/MaxTemperatureDifferenceRule"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin

158.51.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
temperaturedifference&action=<value>[&<parameter>=<value>...]
```

158.51.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	Rule.#.RuleIndex	RES	<int>	Index of the rule
	Rule.#.Source.#.SourceIndex	RES	<int>	Source index of the rule
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables temperature difference
	Rule.#.RuleName	REQ, RES	<string>	Name of the rule
	Rule.#.Duration	REQ, RES	<int>	Set the minimum duration to trigger temperature difference
	Rule.#.ThresholdTemperature	REQ, RES	<int>	Threshold Temperature for the rule
	Rule.#.Source.#.SourceType	REQ, RES	<enum>	Specify the distance of the target to detect **Note** You can see the list of sources in the **temperaturedifferenceoptions** submenu
	Rule.#.Source.#.DataType	REQ, RES	<enum> Min, Max, Avg	Set the type of values to compare
	Rule.#.Source.#.RuleIndex	REQ, RES	<int>	The rule index for event source **Note** You can see the list of sources in the **temperaturedifferenceoptions** submenu
remove	Channel	REQ	<int>	

Action	Parameters	Request/ Response	Type/ Value	Description
	RuleIndex	REQ	<csv> #, All	Interest rule index to be removed

158.51.4. Examples (this submenu supports only JSON response)

158.51.4.1. Getting temperaturedifference settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturedifference&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TemperatureDifference": [
        {
            "Channel": 1,
            "Enable": true,
            "Rule": [
                {
                    "RuleIndex": 1,
                    "RuleName": "Rule1",
                    "Duration": 1,
                    "ThresholdTemperature": 20,
                    "Source": [
                        {
                            "SourceIndex": 1,
                            "SourceType": "BoxTemperatureDetection",
                            "DataType": "Min",
                            "RuleIndex": 1
                        },
                        {
                            "SourceIndex": 2,
                            "SourceType": "BoxTemperatureDetection",
                            "DataType": "Min",
```

```
                                "RuleIndex": 2
                            }
                        ]
                    }
                ]
            }
        ]
}
```

158.51.4.2. Adding a rule

Add a 'testrule' that compares the Min value in area 1 of the BoxTemperatureDetection to the Max value in area 2 When adding a new event rule, **Rule.#.RuleName**, **Rule.#.Duration**, **Rule.#.ThresholdTemperature** must be set.

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturedifference&action=set&Channel=1&Enable=True&Rule.1.RuleName=testrule&Rule.1.Duration=1&Rule.1.ThresholdTemperature=20&Rule.1.Source.1.SourceType=BoxTemperatureDetection&Rule.1.Source.1.DataType=Min&Rule.1.Source.1.RuleIndex=1&Rule.1.Source.2.SourceType=BoxTemperatureDetection&Rule.1.Source.2.DataType=Max&Rule.1.Source.2.RuleIndex=2
```

158.51.4.3. For rule 1, set Duration to 3 seconds and ThresholdTemperature to 30

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturedifference&action=set&Channel=1&Rule.1.Duration=3&Rule.1.ThresholdTemperature=30
```

158.52. Temperature Difference Options

158.52.1. Description

The **temperaturedifferenceoptions** submenu provides settings options for the Temperature Difference. When a rule for a TemperatureDifference supported event is added, such as BoxTemperatureDetection, it appears in the TemperatureDifferenceOptions list.

NOTE Attribute to check for feature support: "attributes/Eventsource/Support/TemperatureDifference"

Access level

Action	Camera
view	Admin

158.52.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturedifferenceoptions&action=<value>[&<parameter>=<value>...]
```

158.52.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel	REQ	<int>	Channel ID
	Source.#.SourceType	RES	<enum> BoxTemperatureDetection	
	Source.#.DataType	RES	<csv> Min, Max, Avg	
	Source.#.RuleName	RES	<string>	
	Source.#.RuleIndex	RES	<int>	
	Temperature.#.Celsius.Min	RES	<int>	Minimum Celsius temperature
	Temperature.#.Celsius.Max	RES	<int>	Maximum Celsius temperature
	Temperature.#.Fahrenheit.Min	RES	<int>	Minimum Fahrenheit temperature
	Temperature.#.Fahrenheit.Max	RES	<int>	Maximum Fahrenheit temperature

158.52.3.1. Getting temperature difference options

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=temperaturedifferenceoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TemperatureDifferenceOptions": [
        {
            "Channel": 1,
            "Source": [
                {
                    "SourceType": "BoxTemperatureDetection",
                    "DataType": [
                        "Min",
                        "Max",
                        "Avg"
                    ],
                    "RuleIndex": 1,
                    "RuleName": "A"
                },
                {
                    "SourceType": "BoxTemperatureDetection",
                    "DataType": [
                        "Min",
                        "Max",
                        "Avg"
                    ],
                    "RuleIndex": 2,
                    "RuleName": "B"
                }
            ],
            "Temperature": {
                "Celsius": {
                    "Min": 1,
                    "Max": 100
                },
                "Fahrenheit": {
                    "Min": 1,
                    "Max": 180
                }
```

```
            }
        }
    ]
}
```

158.53. Tank Level Detection

158.53.1. Description

The **tankleveldetection** submenu configures tank level detection settings.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for feature support:
> "attributes/Eventsource/Support/TankLevelDetection"

Access level

Action	Camera
view	Admin
set	Admin
remove	Admin
check	Admin

158.53.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
tankleveldetection&action=<value>[&<parameter>=<value>]
```

158.53.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads tank level detection settings
	Channel	REQ	<int>	Channel ID
	ROI.#.ROI	RES	<int>	ROI index of the region
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables tank level detection

Action	Parameter	Request/ Response	Type/ Value	Description
	ROI.#.Coordinate	REQ, RES	<string> Format=x1, y1,x2,y2,x3, y3,x4,y4	Coordinates of the ROI region
	ROI.#.DetectionType	REQ, RES	<enum> Above, Below	Detection type for the ROI
	ROI.#.ThresholdLevel	REQ, RES	<int>	Threshold Level for the ROI
	ROI.#.Duration	REQ, RES	<int>	ROI Duration in seconds
	ROI.#.RuleName	REQ, RES	<string>	set rule name for this ROI.
remove	Channel	REQ	<int>	Channel ID
	ROIIndex	REQ	<csv> All, #	ROI index of the region to be removed
check	Channel	REQ, RES	<int>	channel for check
	RuleName	RES, REQ	<string>	ROI Name
	ROIIndex	RES	<int>	ROI index
	TankLevel	RES	<int>	Current tank level

158.53.4. Examples

158.53.4.1. Getting tank level detection settings for Channel 0

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tankleveldetection&action=view&Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TankLevelDetection": [
        {
            "Channel": 0,
            "Enable": true,
```

```
                "ROIs": [
                    {
                        "ROI": 1,
                        "RuleName": "A",
                        "DetectionType": "Above",
                        "ThresholdLevel": 10,
                        "Coordinates": [
                            {
                                "x": 102,
                                "y": 100
                            },
                            {
                                "x": 505,
                                "y": 315
                            }
                        ],
                        "Duration": 5
                    }
                ]
            }
        ]
}
```


158.53.4.2. Changing tank level detection settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tankleveldetection&action=set&Channel=0&Enable
=True&ROI.1.Coordinate=102,100,505,315&ROI.1.DetectionType=Below&ROI.1.Thres
holdLevel=20&ROI.1.Duration=10&ROI.1.RuleName=A
```

158.53.4.3. Removing tank level detection ROI Region 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tankleveldetection&action=remove&ROIIndex=1&Ch
annel=0
```

158.53.4.4. Getting tank level reading information from tank level detection ROI

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=tankleveldetection&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TankLevelDetection": [
        {
            "Channel": 0,
            "ROIsLevel": [
                {
                    "ROIIndex": 1,
                    "RuleName": "A",
                    "TankLevel": 10
                }
            ]
        }
    ]
}
```


158.54. Case Tampering Detection

158.54.1. Description

The **casetampering** submenu used to configure case tampering detection.

Access level

Action	Camera	ACS
view	Admin	Admin
set	Admin	Admin

158.54.2. Syntax

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=
```

```
casetampering&action=<value>[&<parameter>=<value>...]
```

158.54.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the case tampering configuration.
set	Enable	REQ,RES	<bool> True,False	Enables or disables case tampering detection.

158.54.4. Examples

158.54.4.1. View case tampering configuration

REQUEST

```
http://<Device IP/stw-
cgi/eventsources.cgi?msubmenu=casetampering&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "CaseTampering": {
        "Enable": true
    }
}
```


158.54.4.2. Enable case tampering

REQUEST

```
http://<Device IP/stw-
cgi/eventsources.cgi?msubmenu=casetampering&action=set&Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

158.54.5. Event Format

158.54.5.1. Sunapi Schema Based

Schema Based Json Event format

```
{
    "EventName": "SystemEvent.CaseTampering",
    "Time": "2024-09-27T06:46:26.804+00:00",
    "Source": {},
    "Data": {
        "State": true
    }
}
```

Schema Based Text Response format

```
SystemEvent.CaseTampering=True
```

ONVIF Event Format (Property Event)

```
<wsnt:NotificationMessage>
          <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:De
vice/Hardware/CaseTampered</wsnt:Topic>
          <wsnt:Message>
                    <tt:Message UtcTime="2024-08-05T02:39:39.142Z"
PropertyOperation="Changed">
                              <tt:Data>
                                        <tt:SimpleItem Name="State"
Value="true"/>
                              </tt:Data>
                    </tt:Message>
          </wsnt:Message>
</wsnt:NotificationMessage>
```

Chapter 159. Event Actions

159.1. Email Sending

159.1.1. Description

The **smtp** submenu configures the settings for sending the alert message through email when the specified event occurs.

NOTE This chapter applies to the NVR only.

Access level

Action	NVR
view	User
set	User

159.1.2. Syntax

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=
smtp&action=<value>[&<parameter>=<value>]
```

159.1.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the smtp event action settings
set	MailingPeriod	REQ, RES	<enum> 5s, 10s, 30s, 45s, 5m, 10m, 15m, 20m, 25m, 30m	Mailing period
	EventDetection	REQ, RES	<bool> True, False	Whether to send an email when a specific event is detected
	AlarmInput	REQ, RES	<bool> True, False	Whether to send an email when the alarm input event or network alarm input event occurs

Action	Parameter	Request/ Response	Type/ Value	Description
	SystemEvent	REQ, RES	<csv> Gsensor, EmergencyTrigger, ChangePassword, HDDInformation, PowerOnOff, ManualRecord, Videoloss, None	System event to perform the smtp event action
	RecipientGroupID.#.EventDetection	REQ, RES	<bool> True, False	Whether to send an email to the specified group when a specific event is detected
	RecipientGroupID.#.AlarmInput	REQ, RES	<bool> True, False	Whether to send an email to the specified group when the alarm input event or network alarm input event occurs
	RecipientGroupID.#.SystemEvent	REQ, RES	<csv> Gsensor, EmergencyTrigger, ChangePassword, HDDInformation, PowerOnOff, ManualRecord None	System event to perform the smtp event action with corresponding email recipient group

159.1.4. Examples

159.1.4.1. Getting SMTP event action settings

REQUEST

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=smtp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
MailingPeriod=15m
EventDetection=True
AlarmInput=True
SystemEvent=ChangePassword,HDDInformation,PowerOnOff,ManualRecord
RecipientGroupID.Group1.EventDetection=True
RecipientGroupID.Group1.AlarmInput=True
RecipientGroupID.Group1.SystemEvent=ChangePassword,HDDInformation,PowerOnOff
,ManualRecord
RecipientGroupID.Group2.EventDetection=False
RecipientGroupID.Group2.AlarmInput=False
RecipientGroupID.Group2.SystemEvent=None
RecipientGroupID.Group3.EventDetection=False
RecipientGroupID.Group3.AlarmInput=False
RecipientGroupID.Group3.SystemEvent=None
RecipientGroupID.Group4.EventDetection=False
RecipientGroupID.Group4.AlarmInput=False
RecipientGroupID.Group4.SystemEvent=None
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MailingPeriod": "10m",
    "EventDetection": true,
    "AlarmInput": true,
    "SystemEvent": [
        "ChangePassword",
        "HDDInformation",
        "PowerOnOff",
        "ManualRecord"
    ],
```

```
    "RecipientGroups": [
        {
            "RecipientGroupID": "Group 1",
            "EventDetection": true,
            "AlarmInput": true,
            "SystemEvent": [
                "ChangePassword",
                "HDDInformation",
                "PowerOnOff",
                "ManualRecord"
            ]
        },
        {
            "RecipientGroupID": "Group 2",
            "EventDetection": false,
            "AlarmInput": false,
            "SystemEvent": [
                "None"
            ]
        }
    ]
}
```

159.1.4.2. Setting the mailing period

REQUEST

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=smtp&action=set&MailingPeriod=15m
```

159.1.4.3. Sending email when event detection occurs

REQUEST

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=smtp&action=set&EventDetection=True
```

159.1.4.4. Setting a system event for email sending action

REQUEST

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=smtp&action=set&RecipientGroupID.group1.System
```

```
Event=ChangePassword,HDDInformation,PowerOnOff,ManualRecord
```

159.2. Complex Action

159.2.1. Description

The **complexaction** submenu is used to create event action rule, particularly for devices supporting one or more channels.

> **CAUTION**
> It is recommended to use the **dynamicrules** and **dynamicruleoptions** submenus instead. The **complexaction** submenu will be deprecated starting January 2027.

Access level

Action	Camera	Encoder	NVR
view	Admin	Admin	User
set	Admin	Admin	User

159.2.2. Syntax

```
http://<Device IP>/stw-cgi/eventactions.cgi?msubmenu=
complexaction&action=<value>[&<parameter>=<value>]
```

159.2.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	EventType	REQ	<enum> Timer, NetworkDisconnect, Channel.#.FaceDetection, Channel.#.FogDetection, Channel.#.DefocusDetection, Channel.#.VideoAnalysis,AlarmInput.#, Channel.#.NetworkAlarmInput, Channel.#.TamperingDetection, Channel.#.MotionDetection, Channel.#.VideoLoss, Channel.#.CameraEvent, Channel.#.ShockDetection, Channel.#.ObjectDetection, Channel.#.MaskDetection, Channel.#.SocialDistancingViolation, Channel.#.BodyTemperatureDetection	Event type

Action	Parameter	Request/ Response	Type/ Value	Description
set	EventType	REQ, RES	<enum> Timer, NetworkDisconnect, Channel.#.FaceDetection, Channel.#.FogDetection, Channel.#.DefocusDetection, Channel.#.VideoAnalysis, AlarmInput.#, Channel.#.NetworkAlarmInput, Channel.#.TamperingDetection, Channel.#.MotionDetection, Channel.#.VideoLoss, Channel.#.CameraEvent, Channel.#.ShockDetection, Channel.#.ObjectDetection,Channel.#.MaskDetection, Channel.#.SocialDistancingViolation, Channel.#.BodyTemperatureDetection	Event type
	Channel.#.PresetIndex	REQ, RES	<int>	Preset index number as per Channel ID
	AlarmOutput	REQ, RES	<csv> 1, 2, 3, 4, Beep, None	Alarm output
	Duration	REQ, RES	<enum> None, Always, 5s, 10s, 20s, 30s	Alarm output duration

Action	Parameter	Request/ Response	Type/ Value	Description
	Enable	REQ, RES	<bool> True, False	Whether to activate or deactivate the rule • True: Activated • False: Deactivated CAMERA ONLY ENCODER ONLY
	ScheduleType	REQ, RES	<enum> Always, Scheduled	Time schedule for event operation • Always: All the time • Scheduled: Only when scheduled CAMERA ONLY ENCODER ONLY
	<ddd>	REQ, RES	<bool> 0, 1	Day of week selected for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <ddd> represents day of the week, and should be specified in the short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g.) 'SUN=1' indicates recording is activated every Sunday 12:00 AM to 11:59 PM unless the specific time is not set using the <dddh> parameter such like SUN1=1, SUN2=1, etc. CAMERA ONLY ENCODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EveryDay	REQ, RES	<bool> 0, 1	Whether to activate or deactivate the event operation every day • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. 'EveryDay=1', denoting that the recording is activated every day, is the same as when the **ScheduleType** parameter is set to Always. CAMERA ONLY ENCODER ONLY
	<dddh>	REQ, RES	<bool> 0, 1	Time of day selected for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <dddh> stands for the day of the week and time in hours. e.g. SUN1 means 1:00 AM on Sunday. MON2 means 2:00 AM on Monday. This parameter is valid only when <corresponding weekday> is set to 1; e.g. **SUN**=1 is required for SUN0 ... SUN23. 'SUN=1&SUN18=1' means that the event operation is activated every Sunday 6:00 PM to 6:59 PM. CAMERA ONLY ENCODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EveryDay<h>	REQ, RES	<bool> 0, 1	Daily time for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. This parameter is valid only when **EveryDay** is set to 1; e.g. EveryDay=1 is required for EveryDay0 ... EveryDay23. 'EveryDay=1&EveryDay18=1' means the event operation is activated every day 6:00 PM to 6:59 PM. CAMERA ONLY ENCODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>.FromTo	REQ, RES	<string>	The time of week selected for event operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when **Activate** is set to Scheduled. This parameter is also valid only when <corresponding weekday><hour> is set to 1; e.g. SUN0=1 is required for SUN0.FromTo. 'SUN=1&SUN18=1&SUN18.FromTo=12-20' means that the event operation is activated every Sunday from 6:12 PM to 6:20 PM. CAMERA ONLY ENCODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EventAction	REQ, RES	<csv> AlarmOutput.#, SMTP, FTP, Record, HTTP, GoToPreset, LightAlarm	Event action • AlarmOutput.#: Activates alarm output when the configured event occurs. • SMTP: Sends notification and image as attachment via SMTP when the configured event occurs. • FTP: Uploads image via FTP when the configured event occurs. • Record: Saves event video on the storage device when the configured event occurs. (Refer to 'Recording' document (of **recording.cgi**) for information on configuring recording settings such as pre and post event buffers) • GoToPreset: Moves to the specified preset position when the configured event occurs. • LightAlarm: Triggers the LED light when a configured event occurs. **Note** **transfer.cgi** is used to configure FTP/SMTP server settings. CAMERA ONLY ENCODER ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	AlarmOutput.#.Duration	REQ, RES	<enum> None, Always, 5s, 10s, 15s	Alarm output duration when the event occurs **AlarmOutput.#.Duration** is valid only when **EventAction** is set to AlarmOutput.# . CAMERA ONLY ENCODER ONLY

159.2.4. Examples

159.2.4.1. Getting Complexation event action settings from NVR

REQUEST

```
http://<Device IP>/stw-
cgi/eventactions.cgi?msubmenu=complexaction&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1.Channel.0.PresetIndex=0
AlarmInput.1.AlarmOutput=1,2,3,4
AlarmInput.1.Duration=10s
AlarmInput.2.Channel.1.PresetIndex=0
AlarmInput.2.AlarmOutput=1,2,3,4
AlarmInput.2.Duration=10s
AlarmInput.3.Channel.2.PresetIndex=0
AlarmInput.3.AlarmOutput=1,2,3,4
AlarmInput.3.Duration=10s
...
Channel.0.NetworkAlarmInput.Channel.0.PresetIndex=0
Channel.0.NetworkAlarmInput.AlarmOutput=1,2,3,4
Channel.0.NetworkAlarmInput.Duration=10s
Channel.1.NetworkAlarmInput.Channel.1.PresetIndex=0
Channel.1.NetworkAlarmInput.AlarmOutput=1,2,3,4
Channel.1.NetworkAlarmInput.Duration=10s
```

```
Channel.2.NetworkAlarmInput.Channel.2.PresetIndex=0
Channel.2.NetworkAlarmInput.AlarmOutput=1,2,3,4
Channel.2.NetworkAlarmInput.Duration=10s
Channel.3.NetworkAlarmInput.Channel.3.PresetIndex=0
Channel.3.NetworkAlarmInput.AlarmOutput=1,2,3,4
Channel.3.NetworkAlarmInput.Duration=10s
…
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
{
    "ComplexActions": [
        {
            "EventType": "AlarmInput.#",
            "Actions": [
                {
                    "AlarmInput": 1,
                    "Presets": [
                        {
                            "Channel": 0,
                            "PresetIndex": 0
                        }
                    ],
                    "AlarmOutput": [
                        "None"
                    ],
                    "Duration": "10s"
                }
            ]
        },
        {
            "EventType": "Channel.#.NetworkAlarmInput",
            "Actions": [
                {
                    "Channel": 0,
                    "Presets": [
                        {
```

```
                    "Channel": 0,
                    "PresetIndex": 0
                }
            ],
            "AlarmOutput": [
                "None"
            ],
            "Duration": "10s"
        }
    ]
},
{
    "EventType": "Channel.#.CameraEvent",
    "Actions": [
        {
            "Channel": 0,
            "Presets": [
                {
                    "Channel": 0,
                    "PresetIndex": 0
                }
            ],
            "AlarmOutput": [
                "None"
            ],
            "Duration": "10s"
        }
    ]
},
{
    "EventType": "Channel.#.VideoLoss",
    "Actions": [
        {
            "Channel": 0,
            "AlarmOutput": [
                "None"
            ],
            "Duration": "10s"
        }
    ]
}
```

```
    ]
}
```

159.2.4.2. Getting Complexation event action settings from camera

REQUEST

```
http://<Device IP>/stw-
cgi/eventactions.cgi?msubmenu=complexaction&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Timer.Enable=False
Timer.ScheduleType=Always
Timer.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Timer.AlarmOutput.1.Duration=None
Timer.EventAction=None
AlarmInput.1.Enable=True
AlarmInput.1.ScheduleType=Always
AlarmInput.1.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
AlarmInput.1.AlarmOutput.1.Duration=None
AlarmInput.1.EventAction=None
NetworkDisconnect.Enable=False
NetworkDisconnect.ScheduleType=Always
NetworkDisconnect.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
NetworkDisconnect.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
NetworkDisconnect.AlarmOutput.1.Duration=None
NetworkDisconnect.EventAction=None
Channel.0.MotionDetection.Enable=False
Channel.0.MotionDetection.ScheduleType=Always
Channel.0.MotionDetection.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.MotionDetection.AlarmOutput.1.Duration=None
Channel.0.MotionDetection.EventAction=None
Channel.0.VideoAnalysis.Enable=False
Channel.0.VideoAnalysis.ScheduleType=Always
Channel.0.VideoAnalysis.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.VideoAnalysis.AlarmOutput.1.Duration=None
Channel.0.VideoAnalysis.EventAction=None
Channel.0.FaceDetection.Enable=False
Channel.0.FaceDetection.ScheduleType=Always
Channel.0.FaceDetection.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
Channel.0.FaceDetection.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FaceDetection.AlarmOutput.1.Duration=None
Channel.0.FaceDetection.EventAction=None
Channel.0.TamperingDetection.Enable=False
Channel.0.TamperingDetection.ScheduleType=Always
Channel.0.TamperingDetection.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0
Channel.0.TamperingDetection.AlarmOutput.1.Duration=None
Channel.0.TamperingDetection.EventAction=None
Channel.0.DefocusDetection.Enable=False
Channel.0.DefocusDetection.ScheduleType=Always
Channel.0.DefocusDetection.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
Channel.0.DefocusDetection.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0
```

```
Channel.0.DefocusDetection.AlarmOutput.1.Duration=None
Channel.0.DefocusDetection.EventAction=None
Channel.0.FogDetection.Enable=False
Channel.0.FogDetection.ScheduleType=Always
Channel.0.FogDetection.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Channel.0.FogDetection.AlarmOutput.1.Duration=None
Channel.0.FogDetection.EventAction=None
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
{
    "ComplexActions": [
        {
            "EventType": "Timer",
            "Actions": [
                {
                    "Enable": false,
                    "ScheduleType": "Always",
                    "Schedule": {
                        "SUN": [
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
```

```
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0"
    ],
    "MON": [
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
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        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0",
        "0"
    ],
    "TUE": [
```

```
                    "0",
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                    "0",
                    "0",
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                    "0",
                    "0",
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0"
                ],
                "WED": [
                    "0",
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
```

```
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0"
            ],
            "THU": [
                "0",
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                "0",
                "0",
                "0",
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                "0",
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                "0",
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                "0",
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                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0"
            ],
            "FRI": [
                "0",
                "0",
                "0",
                "0",
```

```
                            "0",
                            "0",
                            "0",
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                            "0",
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                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0"
                        ],
                        "SAT": [
                            "0",
                            "0",
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                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
```

```
                        "0",
                        "0",
                        "0",
                        "0",
                        "0"
                    ]
                },
                "AlarmOutputs": [
                    {
                        "AlarmOutput": 1,
                        "Duration": "None"
                    }
                ],
                "EventActions": [
                    "None"
                ]
            }
        ]
    },
    {
        "EventType": "AlarmInput.#",
        "Actions": [
            {
                "AlarmInput": 1,
                "Enable": true,
                "ScheduleType": "Always",
                "Schedule": {
                    "SUN": [
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
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```

```
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0"
                        ],
                        "MON": [
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0"
                        ],
                        "TUE": [
                            "0",
                            "0",
```

```
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
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            ],
            "WED": [
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                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
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                "0",
                "0",
```

```
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0"
                        ],
                        "THU": [
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
                            "0"
                        ],
                        "FRI": [
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
```

```
                            "0",
                            "0",
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                        "SAT": [
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                            "0",
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                            "0",
                            "0",
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```

```
                            "0",
                            "0",
                            "0"
                        ]
                    },
                    "AlarmOutputs": [
                        {
                            "AlarmOutput": 1,
                            "Duration": "None"
                        }
                    ],
                    "EventActions": [
                        "None"
                    ]
                }
            ]
        },
        {
            "EventType": "NetworkDisconnect",
            "Actions": [
                {
                    "Enable": false,
                    "ScheduleType": "Always",
                    "Schedule": {
                        "SUN": [
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
                            "0",
```

```
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0"
                ],
                "MON": [
                    "0",
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
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                    "0",
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                    "0",
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                    "0",
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                    "0"
                ],
                "TUE": [
                    "0",
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```

```
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                            "0"
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                        "WED": [
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```

```
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                            "0"
                        ],
                        "THU": [
                            "0",
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                            "0",
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0"
                        ],
                        "FRI": [
                            "0",
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                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
                            "0",
```

```
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0"
                ],
                "SAT": [
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
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                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0"
```

```
                        ]
                    },
                    "AlarmOutputs": [
                        {
                            "AlarmOutput": 1,
                            "Duration": "None"
                        }
                    ],
                    "EventActions": [
                        "None"
                    ]
                }
            ]
        },
        {
            "EventType": "Channel.#.MotionDetection",
            "Actions": [
                {
                    "Channel": 0,
                    "Enable": false,
                    "ScheduleType": "Always",
                    "Schedule": {
                        "SUN": [
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
                            "0",
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                            "0",
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                            "0",
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                            "0",
                            "0",
                            "0",
```

```
                        "0",
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                        "0",
                        "0",
                        "0"
                    ],
                    "MON": [
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
                        "0",
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                    ],
                    "EventActions": [
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                    ]
                }
```

```
            ]
        }
    ]
}
```

159.2.4.3. Moving to preset 1 for the alarm input

REQUEST

```
http://<Device IP>/stw-
cgi/eventactions.cgi?msubmenu=complexaction&action=set&EventType=AlarmInput.
1&Channel.0.PresetIndex=1
```

159.2.4.4. Setting the alarm always out for the video loss event

REQUEST

```
http://<Device IP>/stw-
cgi/eventactions.cgi?msubmenu=complexaction&action=set&EventType=Channel.0.V
ideoLoss&AlarmDuration=Always
```

Chapter 160. Event Rules

eventrules.cgi is used to add or update event rules.

160.1. Rules

160.1.1. Description

rules submenu is used to add or update event rules. An event rule defines that which action will be performed when a certain specific event occurs at a specified day and time.

CAUTION	It is recommended to use the **dynamicrules** and **dynamicruleoptions** submenus instead. The **rules** submenu will be deprecated starting January 2027.

NOTE	This chapter is exclusively applicable to network cameras.

Access level

Action	Camera
view	Admin
add, update	Admin
remove	Admin

160.1.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
rules&action=<value>[&<parameter>=<value>]
```

160.1.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the event rules settings.

Action	Parameter	Request/ Response	Type/ Value	Description
add, update	RuleIndex	REQ, RES	<int>	Index number of the rule **Note** **RuleIndex** is a response parameter when it is used with the **add** action. And it is a request parameter for the **update** action. **Note** **RuleIndex** must be sent together with the **update** action.
	RuleName	REQ, RES	<string>	Rule name The rule name should be unique. **Note** **RuleName** and **EventSource** must be sent together with the **add** action.

Action	Parameter	Request/ Response	Type/ Value	Description
	EventSource	REQ, RES	<enum> AlarmInput.#, MotionDetection, VideoLoss, NetworkEvent, FaceDetection, TamperingDetection, AudioDetection, Tracking, Timer, OpenSDK, UserInput, DefocusDetection, PeopleCount, HeatMap, FogDetection, AudioAnalysis, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, ObjectDetection, SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor, VehicleCount	Event source **Note** **RuleName** and **EventSource** must be sent together with the **add** action. **Caution** Once any event source is selected, the same event source cannot be added to a new rule.
	Enable	REQ, RES	<bool> True, False	Whether to activate or deactivate the rule • True: Activated • False: Deactivated
	ScheduleType	REQ, RES	<enum> Always, Scheduled	Time schedule for event operation • Always: All the time • Scheduled: Only when scheduled

Action	Parameter	Request/ Response	Type/ Value	Description
	<ddd>	REQ, RES	<bool> 0, 1	Day of week selected for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <ddd> stands for week of the day and should be specified in the short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g.) 'SUN=1' indicates recording is activated every Sunday 12:00 AM to 11:59 PM unless the specific time is not set using the <dddh> parameter such like SUN1=1, SUN2=1, etc.
	EveryDay	REQ, RES	<bool> 0, 1	Whether to activate or deactivate the event operation every day • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. 'EveryDay=1', denoting that the recording is activated every day, is same as the **ScheduleType** parameter is set to Always.

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>	REQ, RES	<bool> 0, 1	Time of day selected for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <dddh> stands for the day of the week and time in hour. e.g. SUN1 means 1:00 AM on Sunday. MON2 means 2:00 AM on Monday. This parameter is valid only when <corresponding weekday> is set to 1; e.g. **SUN**=1 is required for SUN0 ... SUN23. 'SUN=1&SUN18=1' means that the event operation is activated every Sunday 6:00 PM to 6:59 PM.
	EveryDay<h>	REQ, RES	<bool> 0, 1	Time of everyday for event operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. This parameter is valid only when **EveryDay** is set to 1; e.g. EveryDay=1 is required for EveryDay0 ... EveryDay23. 'EveryDay=1&EveryDay18=1' means the event operation is activated every day 6:00 PM to 6:59 PM.

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>.FromTo	REQ, RES	<string>	The time of week selected for event operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when **Activate** is set to Scheduled. This parameter is also valid only when <corresponding weekday><hour> is set to 1; e.g. SUN0=1 is required for SUN0.FromTo. 'SUN=1&SUN18=1&SUN18.FromTo=12-20' means that the event operation is activated every Sunday 6:12 PM to 6:20 PM.
	EveryDay<h>.FromTo	REQ, RES	<string>	The time for everyday event action recording The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when **Activate** is set to Scheduled. This parameter is valid only when EveryDay<hour> is set to 1; e.g. 'EveryDay0=1' is required for EveryDay0.FromTo. 'EveryDay=1&EveryDay18=1&EveryDay18.FromTo=12-20' means that the event operation is activated every day 6:12 PM to 6:20 PM.

Action	Parameter	Request/ Response	Type/ Value	Description
	EventAction	REQ, RES	<csv> AlarmOutput.#, SMTP, FTP, Record, HTTP, GoToPreset, AudioClip, LightAlarm	Event action • AlarmOutput.#: Activates alarm output when the configured event occurs. • SMTP: Sends notification and image as attachment via SMTP when the configured event occurs. • FTP: Uploads image via FTP when the configured event occurs. • Record: Saves event video on the storage device when the configured event occurs. (Refer to 'Recording' document (of **recording.cgi**) for configuring recording settings such as pre and post event buffers) • GoToPreset: Moves to the specified preset position when the configured event occurs. • AudioClip: Plays a specific audio file in the device when the configured event occurs. **Note** **transfer.cgi** is used to configure FTP/SMTP server settings.
	AlarmOutput.#.Duration	REQ, RES	<enum> Always, 5s, 10s, 15s, 30s, 60s, 90s, 120s, 150s, 200s	Alarm output duration when the event occurs **AlarmOutput.#.Duration** is valid only when **EventAction** is set to AlarmOutput.# .
	PresetNumber	REQ, RES	<int>	Preset number **PresetNumber** is valid only when **EventAction** is set to GoToPreset.

Action	Parameter	Request/ Response	Type/ Value	Description
	AudioClipIndex	REQ, RES	<int>	Audio clip index number **AudioClipIndex** is valid only when **EventAction** is set to AudioClip.
remove	RuleIndex	REQ	<int>	Index of the rule **Note** **RuleIndex** must be sent together with the **remove** action.

160.1.4. Examples

160.1.4.1. Getting the current rules

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=rules&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Rule.1.RuleName=AlarmInput.1
Rule.1.EventSource=AlarmInput.1
Rule.1.Enable=True
Rule.1.ScheduleType=Scheduled
Rule.1.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.1.MON=0 0 0 0 0 1 1 1 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0
Rule.1.TUE=0 0 0 0 0 0 0 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0 0 0
Rule.1.WED=0 0 0 0 0 0 0 1 1 1 1 1 0 0 0 1 1 1 0 1 0 0 0 0
Rule.1.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.1.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.1.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.1.EventAction=AlarmOutput.1,SMTP,FTP,Record
Rule.1.AlarmOutput.1.Duration=Always
Rule.2.RuleName=MotionDetection
Rule.2.EventSource=MotionDetection
```

```
Rule.2.Enable=True
Rule.2.ScheduleType=Always
Rule.2.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.2.EventAction=AlarmOutput.1,SMTP,FTP
Rule.2.AlarmOutput.1.Duration=15s
Rule.3.RuleName=NetworkEvent
Rule.3.EventSource=NetworkEvent
Rule.3.Enable=True
Rule.3.ScheduleType=Always
Rule.3.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.3.EventAction=AlarmOutput.1,Record
Rule.3.AlarmOutput.1.Duration=10s
Rule.4.RuleName=TamperingDetection
Rule.4.EventSource=TamperingDetection
Rule.4.Enable=True
Rule.4.ScheduleType=Scheduled
Rule.4.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.4.MON=0 0 1 1 1 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.4.TUE=1 1 0 0 1 0 0 0 0 1 1 0 0 0 0 0 1 1 0 0 0 0 0 0
Rule.4.WED=1 0 0 0 1 0 0 0 0 0 1 0 0 0 1 1 0 1 0 1 0 0 0 0
Rule.4.THU=1 1 0 0 1 1 0 0 0 0 1 1 1 1 0 0 0 1 1 0 0 0 0 0
Rule.4.FRI=0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0
Rule.4.SAT=0 0 0 0 0 0 1 1 0 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0
Rule.4.EventAction=AlarmOutput.1,SMTP,FTP,Record
Rule.4.AlarmOutput.1.Duration=5s
Rule.5.RuleName=AudioDetection
Rule.5.EventSource=AudioDetection
Rule.5.Enable=True
Rule.5.ScheduleType=Always
```

```
Rule.5.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.5.EventAction=AlarmOutput.1,SMTP,FTP,AudioClip
Rule.5.AlarmOutput.1.Duration=5s
Rule.6.RuleName=Timer
Rule.6.EventSource=Timer
Rule.6.Enable=True
Rule.6.ScheduleType=Scheduled
Rule.6.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.6.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.6.TUE=0 0 0 0 0 0 1 1 1 0 0 0 0 1 1 0 0 0 1 1 0 1 0 0
Rule.6.WED=0 0 0 0 0 0 0 0 1 0 0 0 1 1 1 1 0 1 1 1 1 1 0 0
Rule.6.THU=0 0 0 0 0 0 0 0 1 1 0 0 1 0 0 1 1 1 0 0 0 0 0 0
Rule.6.FRI=0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0
Rule.6.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.6.EventAction=FTP
Rule.6.AlarmOutput.1.Duration=
Rule.7.RuleName=OpenSDK
Rule.7.EventSource=OpenSDK
Rule.7.Enable=True
Rule.7.ScheduleType=Always
Rule.7.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.7.EventAction=SMTP,FTP
Rule.7.AlarmOutput.1.Duration=
Rule.8.RuleName=DefocusDetection
Rule.8.EventSource=DefocusDetection
Rule.8.Enable=True
Rule.8.ScheduleType=Scheduled
Rule.8.SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Rule.8.MON=0 1 1 1 0 0 0 1 1 1 0 0 0 0 0 0 0 1 1 0 0 1 1 0
```

```
Rule.8.TUE=0 0 0 1 0 0 0 1 0 0 1 0 0 1 1 1 0 1 1 1 0 1 1 0
Rule.8.WED=0 0 0 1 0 0 0 1 0 0 1 0 0 1 0 1 1 1 0 1 1 1 1 0
Rule.8.THU=0 1 0 1 1 1 1 0 0 0 1 1 1 1 0 0 0 0 0 0 0 1 0 0
Rule.8.FRI=0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
Rule.8.SAT=0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 0 0
Rule.8.EventAction=AlarmOutput.1,SMTP,FTP,Record
Rule.8.AlarmOutput.1.Duration=10s.
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "EventRules": [
        {
            "RuleIndex": 1,
            "RuleName": "AlarmInput.1",
            "EventSource": "AlarmInput.1",
            "Enable": true,
            "ScheduleType": "Scheduled",
            "Schedule": {
                "SUN": [
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
```

```
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0"
            ],
            "MON": [
                "0",
                "0",
                "0",
                "0",
                "0",
                "1",
                "1",
                "1",
                "0",
                "0",
                "0",
                "0",
                "1",
                "1",
                "1",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
                "0"
            ],
            "TUE": [
                "0",
                "0",
                "0",
                "0",
                "0",
                "0",
```

```
            "0",
            "1",
            "0",
            "0",
            "0",
            "1",
            "1",
            "0",
            "1",
            "1",
            "0",
            "0",
            "1",
            "0",
            "0",
            "0",
            "0",
            "0"
        ],
        "WED": [
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "1",
            "1",
            "1",
            "1",
            "1",
            "0",
            "0",
            "0",
            "1",
            "1",
            "1",
            "0",
            "1",
            "0",
```

```
                    "0",
                    "0",
                    "0"
                ],
                "THU": [
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0"
                ],
                "FRI": [
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
                    "0",
```

```
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0"
        ],
        "SAT": [
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0",
            "0"
        ]
```

```
            },
            "EventAction": [
                "AlarmOutput.1",
                "SMTP",
                "FTP",
                "Record"
            ],
            "AlarmOutputs": [
                {
                    "AlarmOutput": 1,
                    "Duration": "Always"
                }
            ]
        }
    ]
}
```

160.1.4.2. Adding a rule

Adding 'TestRule01', which detects the alarm input at all times (without a time schedule) and saves the event video to the storage device

When adding a new event rule, **RuleName** and **EventSource** must be set. In some cameras event rules are added by default, and thus there is no need to create an event rule. Update action can be used to modify default rules as needed.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&RuleName=TestRule01&EventSource
=AlarmInput.1&Enable=True&ScheduleType=Always&EventAction=Record
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
RuleIndex=1
```

Adding 'TestRule02', which detects faces every Saturday and Sunday

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=FaceDetection&RuleN
ame=TestRule02&Enable=True&ScheduleType=Scheduled&SUN=1&SAT=1&MON=0&TUE=0&WE
D=0&THU=0&FRI=0
```

Adding 'TestRule03', which detects audio events from 1:00 AM to 4:00 AM everyday and saves the event video to the storage device

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=AudioDetection&Rule
Name=TestRule03&Enable=True&ScheduleType=Scheduled&EveryDay=1&EveryDay1=1&Ev
eryDay2=1&EveryDay3=1&EveryDay4=1&EventAction=Record
```

Adding 'TestRule04', which detects and records audio events every Sunday except at 2 AM and 3AM

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=AudioDetection&Rule
Name=TestRule04&Enable=True&ScheduleType=Scheduled&SUN=1&SUN0=1&SUN1=1&SUN2=
0&SUN3=0&SUN4=1&SUN5=1&SUN6=1&SUN7=1&SUN8=1&SUN9=1&SUN10=1&SUN11=1&SUN12=1&S
UN13=1&SUN14=1&SUN15=1&SUN16=1&SUN17=1&SUN18=1&SUN19=1&SUN20=1&SUN21=1&SUN22
=1&SUN23=1&EventAction=Record
```

Adding 'TestRule05', which detects audio events and records from 3:58 AM to 3:59 AM on Sundays

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=AudioDetection&Rule
Name=TestRule05&Enable=True&ScheduleType=Scheduled&SUN=1&SUN3=1&SUN3.FromTo=
58-59&EventAction=Record
```

Adding 'TestRule001', which produces an alarm for 5 seconds when it detects tampering

To set the alarm output duration, **EventAction** must be set to AlarmOutput.\#.

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=TamperingDetection&
RuleName=TestRule001&Enable=True&ScheduleType=Always&EventAction=AlarmOutput
.1&AlarmOutput.1.Duration=5s
```

Adding 'TestRule002', which moves the network camera to preset number 16 when camera detects face

You can set the preset number only when **EventAction** is set to GoToPreset.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&EventSource=FaceDetection&RuleN
ame=TestRule002&Enable=True&ScheduleType=Always&EventAction=GoToPreset&Prese
tNumber=16
```

Adding a rule with the OpenSDK event source

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=add&RuleName=ANPRRule&EventSource=O
penSDK&Enable=True&ScheduleType=Always&EventAction=FTP
```

160.1.4.3. Updating Rule 1

To update an existing event rule, you must indicate the **RuleIndex**.

RuleIndex is mandatory parameter for the update action. The Enable, EventSource, ScheduleType, and EventAction parameters can be updated independently when combined with RuleIndex parameter for the given rule.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=update&RuleIndex=1&EventSource=Moti
onDetection&EventAction=AudioClip&AudioClipIndex=1
```

160.1.4.4. Removing Rule 1

To remove a rule with the **remove** action, the **RuleIndex** parameter must be set.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=rules&action=remove&RuleIndex=1
```

NOTE	Some camera models may have predefined rules which cannot be removed or added but can be only updated.

160.2. Dynamic Rules

160.2.1. Description

The **dynamicrules** submenu is used to configure rules regarding what actions to take on what channels when an event is notified.

NOTE	Attributes to check **dynamicrules** feature support: "attributes/Eventsource/Support/DynamicRule" Attribute to check for the maximum number of rules supported: "attributes/Eventsource/Limit/MaxDynamicRule" Attribute to check for the maximum number of events supported by the rule: "attributes/Eventsource/Support/MaxDynamicRule.EventSource" Attribute to check for the maximum number of schedules supported: "attributes/Eventsource/Limit/MaxScheduleCount"

Access level

Action	Camera	NVR
view	Admin	User
add, update	Admin	User
remove	Admin	User

160.2.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
dynamicrules&action=<value>[&<parameter>=<value>]
```

160.2.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Rule.#.RuleName	RES	<string>	Rulename used for uniquely identifying a rule.
	Rule.#.Duration	RES	<int>	Duration in seconds
	Rule.#.ScheduleName	RES	<string>	Determines which schedule is associated with the rule.
	Rule.#.Enable	RES	<bool> True, False	To enable or disable rule.

Action	Parameter	Request/ Response	Type/ Value	Description
	Rule.#.Status	RES	<enum> Unavailable, Available	Indicates the operable state of the rule. CAMERA ONLY
	Rule.#.EventSource.#. Type	RES	<enum> MotionDetection, VideoAnalytics, Tampering, DefocusDetection, FogDetection, AudioDetection, AudioAnalytics, NetworkAlarmInput, PasswordChange, HDDStatus, FANError, PowerOnOff, Recording, AppEvent, MQTTSubscription, CaseTampering, EarlyFireDetection, TemperatureDifference, TemperatureChangeDetection, AccessGranted, AccessDenied	In a single rule, there can be several eventsources configured.
	Rule.#.EventSource.#. EventName_<Language>	RES	<string>	Interface language data for the language selected in the Language parameter Displayed only when the **Language** parameter is specified CAMERA ONLY
	Rule.#.EventSource.#. AppName	RES	<string>	The name of the installed app **Rule.#.EventSource.#.AppName** is valid only when **Rule.#.EventSource.#.Type** is set to **AppEvent**. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	Rule.#.EventSource.#.AppEventType	RES	<string>	The event source type of the installed app **Rule.#.EventSource.#.AppEventType** is valid only when **Rule.#.EventSource.#.Type** is set to **AppEvent**. CAMERA ONLY
	Rule.#.EventSource.#.RuleIndexType	RES	<enum> Any, Specific	Rule index of the event source type: • Any – A trigger with one or more of the event's rule indices • Specific - A specific rule index of an event as a trigger **Note** If RuleIndexType is **Specific**, it should be specified with the **Rule.#.EventSource.#.RuleIndex** parameter CAMERA ONLY
	Rule.#.EventSource.#.RuleIndex	RES	<int>	A rule index of an event as a trigger. **Rule.#.EventSource.#.RuleIndex** is valid only when **Rule.#.EventSource.#.RuleIndexType** is set to **Specific**. CAMERA ONLY
	Rule.#.EventSource.#.Channel	RES	<int>	Determines from which channel Event source type needs to be handled. CAMERA ONLY
	Rule.#.EventSource.#.ChannelIDList	RES	<csv>	Determines from which channels Event source type needs to be handled. NVR ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	Rule.#.EventSource.#. DynamicEventName	RES	<string>	Dynamic event name received from the camera NVR ONLY
	Rule.#.EventSource.#. State	RES	<bool> True, Flase	State of the event source set as the trigger condition CAMERA ONLY
	Rule.#.EventSource.#. CredentialHolderTokens	RES	<csv>	List of credential holder tokens to filter the event trigger **Rule.#.EventSource.#.CredentialHolderTokens** is valid only when **Rule.#.EventSource.#.Type** is set to **AccessGranted** or **AccessDenied**. If not specified, events from all credential holders will trigger the action. ACS ONLY
	Rule.#.EventSource.#. AccessGroupTokens	RES	<csv>	List of access group tokens to filter the event trigger **Rule.#.EventSource.#.AccessGroupTokens** is valid only when **Rule.#.EventSource.#.Type** is set to **AccessGranted** or **AccessDenied**. If not specified, events from all access groups will trigger the action. ACS ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	Rule.#.EventAction.#. Type	RES	<enum> GoToPreset, AlarmOutput.#, SMTP, EventPush, EventSpot, FTP, AudioClip, Record, Handover, MQTTPublication, LightAlarm, MonitorAlert, LEDPreset, FakeMetadata	Any of the following event actions are possible; multiple event actions can be configured.
	Rule.#.EventAction.#. Channel.#.PresetNumber	RES	<int>	Used when the event action type is GoToPreset
	Rule.#.EventAction.#. AlarmOutput.Mask	RES	<csv>	Used when the event action type is AlarmOutput NVR ONLY
	Rule.#.EventAction.#. AlarmOutput.Duration	RES	<enum> Off, 5s, 10s, 20s, 30s, Always	Duration of alarmout
	Rule.#.EventAction.#. SMTP.GroupIndex	RES	<int>	Used when the event action type is SMTP Recipient group index NVR ONLY
	Rule.#.EventAction.#. SMTP.UserIndex	RES	<int>	Used when the event action type is SMTP Recipient user index NVR ONLY
	Rule.#.EventAction.#. SMTP.Duration	RES	<enum> Off, 5s, 10s, 20s, 30s, Always	Duration NVR ONLY
	Rule.#.EventAction.#. EventSpot.Enable	RES	<bool> True, False	Used when the event action type is EventSpot Enabled or Disabled NVR ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	Rule.#.EventAction.#.EventSpot.Duration	RES	<int>	Used when the event action type is EventSpot Duration NVR ONLY
	Rule.#.EventAction.#.EventPush.Enable	RES	<bool> True, False	Used when the event action type is EventPush Enable or Disabled NVR ONLY
	Rule.#.EventAction.#.LEDPresetIndex	RES	<int>	LEDPreset Index, used when the event action type is LED Preset
	Rule.#.EventAction.#.AudioClipIndex	RES	<int>	Used when the event action type is AudioClip Audio clip index CAMERA ONLY
	Rule.#.EventAction.#.HandoverIndex	RES	<int>	Used when the event action type is Handover Handover index CAMERA ONLY
	Rule.#.EventAction.#.MQTTMessageIndex	RES	<int>	Used when the event action type is MQTT publication message MQTT publication index CAMERA ONLY
	Rule.#.EventAction.#.MonitorAlertMode	RES	<enum> Off, LED, Sound, LEDAndSound	PVM device can make sound or LED
	Rule.#.EventAction.#.FakeMetadataType	RES	<enum> MotionDetection, AlarmInput.#	Generates virtual Event Metadata, currently FakeMetadata is supported only when EventSource is PTZMotion. CAMERA ONLY
add/update	RuleName	REQ	<string>	Rulename used for uniquely identifying a rule.

Action	Parameter	Request/ Response	Type/ Value	Description
	RuleNewName	REQ	<string>	The Rulename to change. This parameter is used for the **update** action.
	Duration	REQ	<int>	Duration in seconds
	ScheduleName	REQ	<string>	Name of schedule to be associated with this rule.
	Enable	REQ	<bool> True, False	To enable or disable rule.
	Overwrite	REQ	<bool> True, False	Whether to overwrite. This parameter is used for the **update** action. **Note** If Overwrite is **True**, all other parameters must be entered.
	EventSource.#.Type	REQ	<enum> MotionDetection, VideoAnalytics, Tampering, DefocusDetection, FogDetection, AudioDetection, AudioAnalytics, NetworkAlarmInput, PasswordChange, HDDStatus, FANError, PowerOnOff, Recording, AppEvent, MQTTSubscription, CaseTampering, EarlyFireDetection, TemperatureDifference, TemperatureChangeDetection, AccessGranted, AccessDenied	For a single rule, multiple eventsources can be configured.

Action	Parameter	Request/ Response	Type/ Value	Description
	EventSource.#.AppName	REQ	<string>	The name of the installed app **EventSource.#.AppName** is valid only when **EventSource.#.Type** is set to **AppEvent**. CAMERA ONLY
	EventSource.#.AppEventType	REQ	<string>	The event source type of the installed app **EventSource.#.AppEventType** is valid only when **EventSource.#.Type** is set to **AppEvent**. CAMERA ONLY
	EventSource.#.RuleIndexType	REQ	<enum> Any, Specific	Rule index of the event source type: • Any – A trigger with one or more of the event's rule indices • Specific - A specific rule index of an event as a trigger **Note** If RuleIndexType is **Specific**, it should be specified with the **EventSource.#.RuleIndex** parameter CAMERA ONLY
	EventSource.#.RuleIndex	REQ	<int>	A rule index of an event as a trigger. **EventSource.#.RuleIndex** is valid only when **EventSource.#.RuleIndexType** is set to **Specific**. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EventSource.#.Channel	REQ	<int>	Determines from which channel Event source type needs to be handled. CAMERA ONLY
	EventSource.#.ChannelIDList	REQ	<csv>	Determines from which channels Event source type needs to be handled. NVR ONLY
	EventSource.#.DynamicEventName	REQ	<string>	Dynamic event name received from Camera
	EventSource.#.State	REQ	<bool> True, Flase	Set which state of the event source to set as the trigger condition CAMERA ONLY
	EventSource.#.CredentialHolderTokens	REQ	<csv>	List of credential holder tokens to filter the event trigger **EventSource..CredentialHolderTokens is valid only when EventSource..Type** is set to **AccessGranted** or **AccessDenied**. If not specified, events from all credential holders will trigger the action. ACS ONLY
	EventSource.#.AccessGroupTokens	REQ	<csv>	List of access group tokens to filter the event trigger **EventSource..AccessGroupTokens is valid only when EventSource..Type** is set to **AccessGranted** or **AccessDenied**. If not specified, events from all access groups will trigger the action. ACS ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EventAction.#.Type	REQ	<enum> GoToPreset, AlarmOutput.#, SMTP, EventPush, EventSpot, FTP, AudioClip, Record, Handover, MQTTPublication, LightAlarm, MonitorAlert, LEDPreset, FakeMetadata	Any of the following event actions are possible, multiple event actions can be configured.
	EventAction.#.Channel.#.PresetNumber	REQ	<int>	Used when the event action type is GoToPreset
	EventAction.#.AlarmOutput.Mask	REQ	<csv>	Used when the event action type is AlarmOutput NVR ONLY
	EventAction.#.AlarmOutput.Duration	REQ	<enum> Off, 5s, 10s, 20s, 30s, Always	Duration of alarmout
	EventAction.#.SMTP.GroupIndex	REQ	<int>	Used when the event action type is SMTP Recipient group index NVR ONLY
	EventAction.#.SMTP.UserIndex	REQ	<int>	Used when the event action type is SMTP Recipient user index NVR ONLY
	EventAction.#.SMTP.Duration	REQ	<enum> Off, 5s, 10s, 20s, 30s, Always	Duration NVR ONLY
	EventAction.#.EventSpot.Enable	REQ	<bool> True, False	Used when the event action type is EventSpot Enabled or Disabled NVR ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EventAction.#.EventSpot.Duration	REQ	<int>	Used when the event action type is EventSpot Duration NVR ONLY
	EventAction.#.EventPush.Enable	REQ	<bool> True, False	Used when the event action type is EventPush Enable or Disabled NVR ONLY
	EventAction.#.LEDPresetIndex	REQ	<int>	LEDPreset Index, used when the event action type is LED Preset
	EventAction.#.AudioClipIndex	REQ	<int>	Used when the event action type is AudioClip Audio clip index CAMERA ONLY
	EventAction.#.HandoverIndex	REQ	<int>	Used when the event action type is Handover Handover index CAMERA ONLY
	EventAction.#.MQTTMessageIndex	REQ	<int>	Used when the event action type is MQTT publication message MQTT publication index CAMERA ONLY **Note** EventAction.#.Type should be **MQTTPublication**
	EventAction.#.MonitorAlertMode	REQ	<enum> Off, LED, Sound, LEDAndSound	PVM device can make sound or LED
	EventAction.#.FakeMetadataType	REQ	<enum> MotionDetection, AlarmInput.#	Generates virtual Event Metadata, currently FakeMetadata is supported only when EventSource is PTZMotion. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
remove	RuleName	REQ	<string>	Rule name to be deleted

160.2.4. Examples (for NVR)

160.2.4.1. Getting the current dynamic rules

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=dynamicrules&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Rule.1.RuleName=AlarmInput.1
Rule.1.EventSource=AlarmInput.1
Rule.1.Enable=True
Rule.0.RuleName=HDD Status
Rule.0.ScheduleName=Schedule1
Rule.0.Duration=3
Rule.0.Enable=True
Rule.0.EventSource.0.Type=HDDStatus
Rule.0.EventSource.0.DynamicEventName=HDD status
Rule.0.EventAction.0.Type=AlarmOutput
Rule.0.EventAction.0.AlarmOutput.Mask=Beep
Rule.0.EventAction.0.AlarmOutput.Duration=Always
Rule.1.RuleName=FAN Error
Rule.1.ScheduleName=Schedule1
Rule.1.Duration=3
Rule.1.Enable=True
Rule.1.EventSource.0.Type=FANError
Rule.1.EventSource.0.DynamicEventName=Fan failure
Rule.1.EventAction.0.Type=AlarmOutput
Rule.1.EventAction.0.AlarmOutput.Mask=Beep
Rule.1.EventAction.0.AlarmOutput.Duration=Always
Rule.2.RuleName=Motion
Rule.2.ScheduleName=Schedule1
Rule.2.Duration=3
```

```
Rule.2.Enable=False
Rule.2.EventSource.0.Type=MotionDetection
Rule.2.EventSource.0.ChannelIDList=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,
17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,4
2,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63
Rule.2.EventSource.0.DynamicEventName=Motion detection
Rule.3.RuleName=ttttt
Rule.3.ScheduleName=Schedule1
Rule.3.Duration=3
Rule.3.Enable=True
Rule.3.EventSource.0.Type=VideoAnalytics
Rule.3.EventSource.0.DynamicEventName=IVA
Rule.3.EventSource.1.Type=Tampering
Rule.3.EventSource.1.DynamicEventName=Tampering
Rule.3.EventSource.2.Type=FogDetection
Rule.3.EventSource.2.DynamicEventName=Fog detection
Rule.3.EventAction.0.Type=AlarmOutput
Rule.3.EventAction.0.AlarmOutput.Mask=1,2,3,4
Rule.3.EventAction.0.AlarmOutput.Duration=None
Rule.3.EventAction.1.Type=EventPush
Rule.3.EventAction.1.EventPush.Enable=True
Rule.3.EventAction.2.Type=EventSpot
Rule.3.EventAction.2.EventSpot.Enable=True
Rule.3.EventAction.2.EventSpot.Duration=5s
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Rules": [
        {
            "Rule": 0,
            "RuleName": "HDD Status",
            "ScheduleName": "Schedule1",
            "Duration": 3,
            "Enable": true,
            "EventSources": [
                {
```

```
                "EventSource": 0,
                "Type": "HDDStatus",
                "ChannelIDList": [],
                "DynamicEventName": "HDD status"
            }
        ],
        "EventActions": [
            {
                "EventAction": 0,
                "Type": "AlarmOutput",
                "AlarmOutput": {
                    "Mask": [
                        "Beep"
                    ],
                    "Duration": "Always"
                }
            }
        ]
    },
    {
        "Rule": 1,
        "RuleName": "FAN Error",
        "ScheduleName": "Schedule1",
        "Duration": 3,
        "Enable": true,
        "EventSources": [
            {
                "EventSource": 0,
                "Type": "FANError",
                "ChannelIDList": [],
                "DynamicEventName": "Fan failure"
            }
        ],
        "EventActions": [
            {
                "EventAction": 0,
                "Type": "AlarmOutput",
                "AlarmOutput": {
                    "Mask": [
                        "Beep"
                    ],
```

```
                    "Duration": "Always"
                }
            }
        ]
    },
    {
        "Rule": 2,
        "RuleName": "Motion",
        "ScheduleName": "Schedule1",
        "Duration": 3,
        "Enable": false,
        "EventSources": [
            {
                "EventSource": 0,
                "Type": "MotionDetection",
                "ChannelIDList": [
                    "0",
                    "1",
                    "2",
                    "3",
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
                    "23",
                    "24",
```

```
                    "25",
                    "26",
                    "27",
                    "28",
                    "29",
                    "30",
                    "31",
                    "32",
                    "33",
                    "34",
                    "35",
                    "36",
                    "37",
                    "38",
                    "39",
                    "40",
                    "41",
                    "42",
                    "43",
                    "44",
                    "45",
                    "46",
                    "47",
                    "48",
                    "49",
                    "50",
                    "51",
                    "52",
                    "53",
                    "54",
                    "55",
                    "56",
                    "57",
                    "58",
                    "59",
                    "60",
                    "61",
                    "62",
                    "63"
                ],
                "DynamicEventName": "Motion detection"
```

```
                }
            ],
            "EventActions": []
        },
        {
            "Rule": 3,
            "RuleName": "ttttt",
            "ScheduleName": "Schedule1",
            "Duration": 3,
            "Enable": true,
            "EventSources": [
                {
                    "EventSource": 0,
                    "Type": "VideoAnalytics",
                    "ChannelIDList": [],
                    "DynamicEventName": "IVA"
                },
                {
                    "EventSource": 1,
                    "Type": "Tampering",
                    "ChannelIDList": [],
                    "DynamicEventName": "Tampering"
                },
                {
                    "EventSource": 2,
                    "Type": "FogDetection",
                    "ChannelIDList": [],
                    "DynamicEventName": "Fog detection"
                }
            ],
            "EventActions": [
                {
                    "EventAction": 0,
                    "Type": "AlarmOutput",
                    "AlarmOutput": {
                        "Mask": [
                            "1",
                            "2",
                            "3",
                            "4"
                        ],
```

```
                            "Duration": "None"
                        }
                    },
                    {
                        "EventAction": 1,
                        "Type": "EventPush",
                        "EventPush": {
                            "Enable": "True"
                        }
                    },
                    {
                        "EventAction": 2,
                        "Type": "EventSpot",
                        "EventSpot": {
                            "Enable": "True",
                            "Duration": "5s"
                        }
                    }
                ]
            }
        ]
}
```

160.2.4.2. Adding a dynamic rule

Adding a new dynamic rule with Rule name ABCD and several event sources; PasswordChange, CamAllEvent, PowerOnOff.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrules&action=add&RuleName=ABCD&Enable=Fal
se&ScheduleName=Schedule1&Duration=45&EventSource.0.Type=PasswordChange&Even
tSource.1.Type=CamAllEvent&EventSource.2.Type=PowerOnOff&EventSource.1.Chann
elIDList=40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,6
2,63&EventAction.0.Type=AlarmOutput&EventAction.0.AlarmOutput.Mask=1,2,3,Bee
p&EventAction.0.AlarmOutput.Duration=Always&EventAction.2.Type=GoToPreset&Ev
entAction.2.Channel.1.PresetNumber=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.2.4.3. Updating Dynamic Rule

To update an existing event rule, you must indicate the RuleName.

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrules&action=update&RuleName=ABCD&Enable=
True&EventAction.2.Type=EventPush&EventAction.2.EventPush.Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.2.4.4. Removing Dynamic Rule

To remove a rule with the **remove** action and by passing the RuleName

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrules&action=remove&RuleName=ABCD
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.2.5. Examples (for Camera)

160.2.5.1. Getting the current dynamic rules

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=dynamicrules&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Rules": [
        {
            "Rule": 0,
            "RuleName": "Test",
            "ScheduleName": "Always",
            "Duration": 5,
            "Enable": true,
            "Status": "Unavailable",
            "EventSources": [
                {
                    "EventSource": 0,
                    "Type": "MotionDetection",
                    "RuleIndexType": "Any",
                    "Channel": 1,
                    "State": true
                },
                {
                    "EventSource": 1,
                    "Type": "AppEvent",
                    "AppName": "WiseAI",
                    "AppEventType": "ObjectDetection",
                    "RuleIndexType": "Any",
                    "Channel": 0,
                    "State": true
                },
                {
                    "EventSource": 2,
                    "Type": "AppEvent",
                    "AppName": "WiseAI",
                    "AppEventType": "IvaArea",
                    "RuleIndexType": "Specific",
                    "RuleIndex": 1,
                    "Channel": 0,
                    "State": true
```

```
                }
            ],
            "EventActions": [
                {
                    "EventAction": 0,
                    "Type": "SMTP"
                }
            ]
        },
        {
            "Rule": 1,
            "RuleName": "test2",
            "ScheduleName": "Always",
            "Duration": 60,
            "Enable": true,
            "Status": "Unavailable",
            "EventSources": [
                {
                    "EventSource": 0,
                    "Type": "AlarmInput.1",
                    "RuleIndexType": "Any",
                    "Channel": 0,
                    "State": true
                },
                {
                    "EventSource": 1,
                    "Type": "TamperingDetection",
                    "RuleIndexType": "Any",
                    "Channel": 0,
                    "State": true
                },
                {
                    "EventSource": 2,
                    "Type": "DefocusDetection",
                    "RuleIndexType": "Any",
                    "Channel": 0,
                    "State": true
                }
            ],
            "EventActions": []
        }
```

```
    ]
}
```

160.2.5.2. Adding a dynamic rule

Adding a new dynamic rule with Rule name 'Test' and several event sources; MotionDetection, IvaArea and ObjectDetection of WiseAI app's event.

NOTE The camera should see a list of supported events and actions via the **dynamicrulesoptions** submenu.
The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=dynamicrules&action=add&RuleName=Test&ScheduleName=Always&Enable=True&Duration=5&EventSource.0.Type=MotionDetection&EventSource.0.RuleIndexType=Specific&EventSource.0.RuleIndex=1&EventSource.0.Channel=1&EventSource.0.State=True&EventSource.1.Type=AppEvent&EventSource.1.AppName=WiseAI&EventSource.1.AppEventType=IvaArea&EventSource.1.RuleIndexType=Any&EventSource.1.Channel=0&EventSource.1.State=False&EventSource.2.Type=AppEvent&EventSource.2.AppName=WiseAI&EventSource.2.AppEventType=ObjectDetection&EventSource.2.RuleIndexType=Any&EventSource.2.Channel=0&EventSource.2.State=True&EventAction.0.Type=SMTP&EventAction.1.Type=Handover&EventAction.1.HandoverIndex=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.2.5.3. Updating Dynamic Rule

To update an existing event rule, you must indicate the RuleName.

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrules&action=update&RuleName=Test&RuleNew
Name=Test2&Enable=True&EventAction.0.Type=FTP
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.2.5.4. Removing Dynamic Rule

To remove a rule with the **remove** action and by passing the RuleName

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrules&action=remove&RuleName=Test
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.3. Dynamic Rules Options

160.3.1. Description

The **dynamicrulesoptions** submenu provides a list of available event sources and information about their action triggers, that can be used in the **dynamicrules** submenu. Event sources and event actions that are not activated do not appear in the list and cannot be added to rules in the **dynamicrules** submenu

NOTE	This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin

160.3.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
dynamicrulesoptions&action=<value>[&<parameter>=<value>]
```

160.3.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<csv>	Channel ID
	Language	REQ	<enum>	Language of the interface to the event type
	EventSource.#.Type	RES	<string>	Event types provided by the device
	EventSource.#.Type_<Language>	RES	<string>	Interface language data for the language selected in the Language parameter Displayed only when the **Language** parameter is specified
	EventSource.#.Status	RES	<enum> Inactive, Active	Indicates whether the event is currently active
	EventSource.#.Policy	RES	<enum> OneShot, Property, None	Indicates the event policy

Action	Parameter	Request/ Response	Type/ Value	Description
	EventSource.#.ActionTypes	RES	<csv> GoToPreset, AlarmOutput.#, SMTP, EventPush, EventSpot, FTP, AudioClip, Record, Handover, MQTTPublication, LightAlarm, MonitorAlert, LEDPreset, FakeMetadata	Event action types provided by the device, currently FakeMetadata is supported only when EventSource is PTZMotion.
	EventSource.#.Rule.#.Name	RES	<string>	The name of the rule in the event
	AppEventSource.#.AppName	RES	<string>	The name of the app installed through the device's opensdk
	AppEventSource.#.Type	RES	<string>	Event types provided the opensdk app of the device
	AppEventSource.#.Type_<Language>	RES	<string>	Interface language data for the language selected in the Language parameter Displayed only when the **Language** parameter is specified
	AppEventSource.#.Status	RES	<enum> Inactive, Active	Indicates whether the event of the app is currently active
	AppEventSource.#.Policy	RES	<enum> OneShot, Property, None	'Oneshot' means this event doesn't support "EventSource.#.State" in the dynamicrules submenu. On the other side, 'Property' means the opposite.
	AppEventSource.#.ActionTypes	RES	<csv> GoToPreset, AlarmOutput.#, SMTP, EventPush, EventSpot, FTP, AudioClip, Record, Handover, MQTTPublication, LightAlarm, MonitorAlert, LEDPreset	Event types provided by the device for the event source of the app

Action	Parameter	Request/ Response	Type/ Value	Description
	AppEventSource.#.Rule.#.Name	RES	<string>	The name of the rule in the event of the app
	AppEventSource.#.Rule.#.Policy	RES	<enum> OneShot,Property, None	Information about the app event whether its an property event or oneshot event.

160.3.4. Examples

160.3.4.1. Getting the current dynamic rules options (this submenu supports only JSON responses)

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=dynamicrulesoptions&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "DynamicRulesOptions": [
        {
            "Channel": 0,
            "EventSources": [
                {
                    "Type": "AlarmInput.1",
                    "Status": "Active",
                    "Policy": "Property",
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
                        "Record",
                        "GoToPreset",
                        "AudioClip",
                        "Handover",
                        "MQTTPublication"
                    ]
```

```
                },
                {
                    "Type": "AudioAnalysis",
                    "Status": "Active",
                    "Policy": "OneShot",
                    "Rule": [
                        {
                            "Rule": 1,
                            "Name": "Scream",
                            "Policy": "OneShot"
                        },
                        {
                            "Rule": 2,
                            "Name": "Explosion",
                            "Policy": "OneShot"
                        },
                        {
                            "Rule": 3,
                            "Name": "CrashingGlass",
                            "Policy": "OneShot"
                        }
                    ],
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
                        "Record",
                        "GoToPreset",
                        "AudioClip",
                        "Handover",
                        "MQTTPublication"
                    ]
                },
                {
                    "Type": "AudioDetection",
                    "Status": "Inactive",
                    "Policy": "Property",
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
```

```
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "DayNight",
                "Status": "Active",
                "Policy": "OneShot",
                "Rule": [
                    {
                        "Rule": 1,
                        "Name": "Night",
                        "Policy": "OneShot"
                    },
                    {
                        "Rule": 2,
                        "Name": "Day",
                        "Policy": "OneShot"
                    }
                ],
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "DefocusDetection",
                "Status": "Active",
                "Policy": "Property",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
```

```
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "FogDetection",
                "Status": "Inactive",
                "Policy": "Property",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "MotionDetection",
                "Status": "Inactive",
                "Policy": "Property",
                "Rule": [
                    {
                        "Rule": 1,
                        "Name": "MotionRule-1",
                        "Policy": "Property"
                    }
                ],
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
```

```
                "Handover",
                "MQTTPublication"
            ]
        },
        {
            "Type": "NetworkDisconnect",
            "Status": "Inactive",
            "Policy": "Property",
            "ActionTypes": [
                "AlarmOutput.1",
                "SMTP",
                "FTP",
                "Record",
                "GoToPreset",
                "AudioClip",
                "Handover",
                "MQTTPublication"
            ]
        },
        {
            "Type": "ObjectDetection",
            "Status": "Active",
            "Policy": "Property",
            "ActionTypes": [
                "AlarmOutput.1",
                "SMTP",
                "FTP",
                "Record",
                "GoToPreset",
                "AudioClip",
                "Handover",
                "MQTTPublication"
            ]
        },
        {
            "Type": "ShockDetection",
            "Status": "Inactive",
            "Policy": "OneShot",
            "ActionTypes": [
                "AlarmOutput.1",
                "SMTP",
```

```
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "TamperingDetection",
                "Status": "Active",
                "Policy": "OneShot",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "Timer",
                "Status": "Active",
                "Policy": "OneShot",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "VideoAnalysis",
                "Status": "Inactive",
```

```
                "Policy": "OneShot",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "GoToPreset",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            }
        ],
        "AppEventSources": [
            {
                "Type": "CrowdCountChanged",
                "Status": "Inactive",
                "Policy": "None",
                "AppName": "WiseAI",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "IvaArea",
                "Status": "Active",
                "Policy": "OneShot",
                "AppName": "WiseAI",
                "Rule": [
                    {
                        "Rule": 1,
                        "Name": "dddd",
                        "Policy": "OneShot"
                    },
                    {
```

```
                "Rule": 2,
                "Name": "sssss",
                "Policy": "OneShot"
            }
        ],
        "ActionTypes": [
            "AlarmOutput.1",
            "SMTP",
            "FTP",
            "Record",
            "AudioClip",
            "Handover",
            "MQTTPublication"
        ]
    },
    {
        "Type": "LineCrossing",
        "Status": "Active",
        "Policy": "OneShot",
        "AppName": "WiseAI",
        "Rule": [
            {
                "Rule": 1,
                "Name": "ffff",
                "Policy": "OneShot"
            },
            {
                "Rule": 2,
                "Name": "lin2",
                "Policy": "OneShot"
            }
        ],
        "ActionTypes": [
            "AlarmOutput.1",
            "SMTP",
            "FTP",
            "Record",
            "AudioClip",
            "Handover",
            "MQTTPublication"
        ]
```

```
                },
                {
                    "Type": "MaskDetection",
                    "Status": "Active",
                    "Policy": "OneShot",
                    "AppName": "WiseAI",
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
                        "Record",
                        "AudioClip",
                        "Handover",
                        "MQTTPublication"
                    ]
                },
                {
                    "Type": "ObjectDetection",
                    "Status": "Active",
                    "Policy": "Property",
                    "AppName": "WiseAI",
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
                        "Record",
                        "AudioClip",
                        "Handover",
                        "MQTTPublication"
                    ]
                },
                {
                    "Type": "QueueHigh",
                    "Status": "Inactive",
                    "Policy": "OneShot",
                    "AppName": "WiseAI",
                    "ActionTypes": [
                        "AlarmOutput.1",
                        "SMTP",
                        "FTP",
                        "Record",
```

```
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "QueueMedium",
                "Status": "Inactive",
                "Policy": "OneShot",
                "AppName": "WiseAI",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "SlipAndFallDetection",
                "Status": "Inactive",
                "Policy": "OneShot",
                "AppName": "WiseAI",
                "ActionTypes": [
                    "AlarmOutput.1",
                    "SMTP",
                    "FTP",
                    "Record",
                    "AudioClip",
                    "Handover",
                    "MQTTPublication"
                ]
            },
            {
                "Type": "SocialDistancingViolation",
                "Status": "Inactive",
                "Policy": "Property",
                "AppName": "WiseAI",
                "ActionTypes": [
```

```
                                "AlarmOutput.1",
                                "SMTP",
                                "FTP",
                                "Record",
                                "AudioClip",
                                "Handover",
                                "MQTTPublication"
                            ]
                        }
                    ]
                }
            ]
        }
```

160.4. Handover

160.4.1. Description

The **handover** feature allows the user to redirect to the preset for the inter-operation of the PTZ camera(s) when motion or a video analytics, tampering, or audio analysis event occurs. The **handover2** submenu configures the receiver (PTZ) cameras to which the user will be redirected when motion or a video analytics, tampering, or audio analysis event occurs.

CAUTION It is recommended to use the **handover2** submenu instead. The **handover** submenu will be deprecated starting January 2027.

NOTE This chapter applies to network cameras and encoders only.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin
add/update	Admin	Admin
remove	Admin	Admin

160.4.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
handover&action=<value>[&<parameter>=<value>]
```

160.4.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads handover settings
	Channel	REQ, RES	<int>	Channel ID
	ROIIndex	REQ, RES	<int>	Index of ROI(Region of Interest)
	PresetIndex	REQ	<csv> **All**, #	Index of the preset: • All – Get handover information for all presets • # - Get handover information for specific preset **Note** **PresetIndex** parameter is valid only for PTZ cameras. For PTZ cameras, **PresetIndex** is not passed in view action; camera provides global handover information.
set	Channel	REQ, RES	<int>	Channel ID
	PresetIndex	REQ, RES	<int>	Index of the preset **Note** **PresetIndex** parameter is valid only for PTZ cameras.
	ROIIndex	REQ, RES	<int>	Index of ROI(Region of Interest)
	Enable	REQ, RES	<bool> True, False	Enable or disable handover for the specified ROI **Note** **ROIIndex** should be sent along with **Enable** parameter. For PTZ cameras, **PresetIndex** and **ROIIndex** should be sent along with **Enable** parameter.
add/update	Channel	REQ, RES	<int>	Channel ID
	PresetIndex	REQ, RES	<int>	Index of the preset **Note** **PresetIndex** parameter is valid only for PTZ cameras.

Action	Parameter	Request/ Response	Type/ Value	Description
	ROIIndex	REQ, RES	<int>	Index of ROI(Region of Interest)
	HandoverIndex	REQ, RES	<int>	Index of receiver camera **Note** **ROIIndex** should be sent along with **HandoverIndex** parameter. For PTZ cameras, **PresetIndex** and **ROIIndex** should be sent along with **HandoverIndex** parameter.
	IPType	REQ, RES	<enum> IPV4, IPV6	IP Type of receiver camera
	IPAddress	REQ, RES	<string> <formatInfo="IPv4Address or IPv6Address>	IP Address of receiver camera
	Port	REQ, RES	<int>	Port of receiver camera
	Username	REQ, RES	<string>	User name of receiver camera
	Password	REQ, RES	<string>	Password of receiver camera
	PresetNumber	REQ, RES	<int>	Preset number of receiver camera
	IsPasswordEncrypted	REQ	<bool> True, False	Enables or disables password encryption for receiver camera
remove	Channel	REQ	<int>	Channel ID
	PresetIndex	REQ	<int>	Index of the preset
	ROIIndex	REQ	<int>	Index of ROI(Region of Interest) to be deleted
	HandoverIndex	REQ	<csv>	Index of receiver camera to be deleted

160.4.4. Examples

160.4.4.1. Getting handover settings for Channel 0

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.ROIIndex.1.Enable=False
Channel.0.ROIIndex.1.HandoverIndex.1.IPType=IPv4
Channel.0.ROIIndex.1.HandoverIndex.1.IPAddress=1.1.1.10
Channel.0.ROIIndex.1.HandoverIndex.1.Port=80
Channel.0.ROIIndex.1.HandoverIndex.1.Username=admin12
Channel.0.ROIIndex.1.HandoverIndex.1.Password=432134
Channel.0.ROIIndex.1.HandoverIndex.1.PresetNumber=1
Channel.0.ROIIndex.2.Enable=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Handover": [
        {
            "Channel": 0,
            "HandoverList": [
                {
                    "ROIIndex": 1,
                    "Enable": false,
                    "UserList": [
                        {
                            "HandoverIndex": 1,
                            "IPType": "IPv4",
                            "IPAddress": "1.1.1.10",
                            "Port": 80,
                            "Username": "admin12",
                            "Password": "432134",
                            "PresetNumber": 1
                        }
                    ]
```

```
                },
                {
                    "ROIIndex": 2,
                    "Enable": false,
                    "UserList": []
                }
            ]
        }
    ]
}
```

NOTE PTZ camera supports both global handover and preset-based handover. To get the preset-based handover list, PresetIndex should be passed in view action.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover&action=view&PresetIndex=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PresetIndex.1.ROIIndex.1.Enable=False
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.IPType=IPv4
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.IPAddress=1.1.1.1
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Port=80
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Username=admin
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Password=4321
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.PresetNumber=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "Handover": [
        {
            "Channel": 0,
            "PresetList": [
                {
                    "PresetIndex": 1,
                    "HandoverList": [
                        {
                            "ROIIndex": 1,
                            "Enable": false,
                            "UserList": [
                                {
                                    "HandoverIndex": 1,
                                    "IPType": "IPv4",
                                    "IPAddress": "1.1.1.1",
                                    "Port": 80,
                                    "Username": "admin",
                                    "Password": "4321",
                                    "PresetNumber": 1
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```

160.4.4.2. Setting handover

Enabling MD handover for region 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=handover&action=set&ROIIndex=1&Enable=True
```

For PTZ cameras, PresetIndex should be passed along with ROIIndex to enable preset-based MD handover.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover&action=set&
&PresetIndex=1&ROIIndex=1&Enable=True
```

NOTE PTZ camera supports both Global MD handover & Preset-based MD handover.

160.4.4.3. Configuring receiver camera(s)

A receiver camera can be added to the MD ROI by providing the IPType, IPAddress, Port, Username, Password, and PresetNumber of the receiver camera.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=handover&action=add&PresetIndex=1&ROIIndex=1&IPT
ype=IPv4&IPAddress=1.1.1.1&Port=80&Username=admin&Password=4321&PresetNumber
=1
```

For PTZ cameras, **PresetIndex** should be passed along with ROIIndex while adding receiver camera for preset based MD handover.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=handover&action=add&PresetIndex=1&ROIIndex=1&IPT
ype=IPv4&IPAddress=1.1.1.1&Port=80&Username=admin&Password=4321&PresetNumber
=1
```

HandoVerIndex is used to update receiver camera details to specific MD ROI.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=handover&action=update&PresetIndex=1&ROIIndex=1&
HandoverIndex=1&IPType=IPv4&IPAddress=1.1.1.2&Port=80&Username=admin&Passwor
d=4321&PresetNumber=1
```

160.4.4.4. Removing receiver camera(s)

HandoverIndex is used to remove Receiver camera from the specific MD ROI.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover&action=remove&
```

```
&ROIIndex=1
```

For PTZ cameras, **PresetIndex** should be passed along with ROIIndex while removing receiver camera from preset-based MD handover.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover&action=remove&
&ROIIndex=1&PresetIndex=1
```

160.5. Handover 2

160.5.1. Description

The 'handover2' feature allows the user to redirect to the preset for the inter-operation of PTZ camera(s) when motion or video analytics or a tampering or audio analysis event occurs. The **handover2** submenu configures the receiver (PTZ) cameras to which the user will be redirected when a motion or video analytics or a tampering or audio analysis event occurs.

NOTE	This chapter applies to network cameras and encoders only. Attributes to check handover2 feature support: stw-cgi/attributes.cgi/eventrules/handover2

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin
add/update	Admin	Admin
remove	Admin	Admin

160.5.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
handover2&action=<value>[&<parameter>=<value>]
```

160.5.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads handover settings
	Channel	REQ, RES	<int>	Channel ID

Action	Parameter	Request/ Response	Type/ Value	Description
	CACertificateInstalled	RES	<bool> True, False	Indicates whether a CA certificate is installed for the target connection.
	MaxCACertificateInUse	RES	<int>	Maximum number of CA certificates that can be used
add/update	Channel	REQ, RES	<int>	Channel ID
	HandoverIndex	REQ, RES	<int>	Index of receiver camera
	IPType	REQ, RES	<enum> IPV4, IPV6, DomainName	IP Type of receiver camera
	IPAddress	REQ, RES	<string> <formatInfo="IPv4Address or IPv6Address>	IP Address of receiver camera
	Port	REQ, RES	<int>	Port of receiver camera
	ConnectionMode	REQ, RES	<enum> HTTP, HTTPS, TCP	Type of handover target connection
	Username	REQ, RES	<string>	User name of receiver camera
	Password	REQ, RES	<string>	Password of receiver camera
	Action	REQ, RES	<enum> Preset, Custom	Action type of handover • Preset : Request to move to the preset • Custom: In case of HTTP/HTTPS, request the set URL + Query. In case of TCP, send a message.
	PresetNumber	REQ, RES	<int>	Preset number of receiver camera
	Query	REQ, RES	<string>	Available only for HTTP/HTTPS. Query string to request in URL
	Message	REQ, RES	<string>	Available only for TCP. Message to be transmitted over TCP
	IsPasswordEncrypted	REQ	<bool> True, False	Enables or disables password encryption for receiver camera

Action	Parameter	Request/ Response	Type/ Value	Description
	ConnectionMode	REQ, RES	<enum> HTTP, HTTPS	Connection mode to connect receiver camera
	CACertificateInUse	REQ, RES	<csv>	CA certificates in use for target connection verification
	VerifyServerCertification	REQ, RES	<bool>	Enables the feature to verify the target connection certificate
install		REQ		Install the certificate of server
remove	Channel	REQ	<int>	Channel ID
	HandoverIndex	REQ	<csv>	Index of receiver camera to be deleted
	CertificateType	REQ	<enum> CACertificate	Certificate type to remove **Note** **CertificateType** must be sent together with the **remove** action

160.5.4. Examples

160.5.4.1. Getting handover2 settings for channel 0

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover2&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Handover": [
        {
            "Channel": 0,
            "UserList": [
                {
                    "HandoverIndex": 1,
                    "IPType": "IPv4",
                    "IPAddress": "1.1.1.1",
```

```
                "Port": 80,
                "ConnectionMode": "HTTP",
                "Username": "admin",
                "Password": "",
                "HandoverAction": "Preset",
                "PresetNumber": 1
            },
            {
                "HandoverIndex": 2,
                "IPType": "IPv4",
                "IPAddress": "1.1.1.1",
                "Port": 80,
                "ConnectionMode": "HTTP",
                "Username": "admin",
                "Password": "",
                "HandoverAction": "Custom",
                "Query": "test"
            },
            {
                "HandoverIndex": 3,
                "IPType": "IPv4",
                "IPAddress": "1.1.1.1",
                "Port": 80,
                "ConnectionMode": "HTTPS",
                "Username": "admin",
                "Password": "",
                "HandoverAction": "Preset",
                "PresetNumber": 2
            },
            {
                "HandoverIndex": 4,
                "IPType": "IPv4",
                "IPAddress": "1.1.1.1",
                "Port": 80,
                "ConnectionMode": "HTTPS",
                "Username": "admin",
                "Password": "",
                "HandoverAction": "Custom",
                "Query": "test123"
            },
            {
```

```
                    "HandoverIndex": 5,
                    "IPType": "IPv4",
                    "IPAddress": "192.168.111.200",
                    "Port": 80,
                    "ConnectionMode": "TCP",
                    "HandoverAction": "Custom",
                    "Message": "test!"
                }
            ]
        }
    ]
}
```

160.5.4.2. Configuring receiver camera(s)

A receiver camera can be added by providing the IPType, IPAddress, Port, Username, Password, and PresetNumber of the receiver camera along with **HandoverIndex**, **IsPasswordEncrypted** parameters.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover2&action=add&Channel=0&HandoverIndex=5&IPType=IPv4&IPAddress=192.168.75.79&Port=80&Username=admin&PresetNumber=120&IsPasswordEncrypted=False
```

160.5.4.3. Configuring TCP receiver camera(s)

ConnectionMode must be set to TCP and a **Message** string is required.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover2&action=add&Channel=0&HandoverIndex=7&ConnectionMode=TCP&IPType=IPv4&IPAddress=1.1.1.1&Port=80&Message=handoverViaTCP
```

160.5.4.4. Removing receiver camera(s)

HandoverIndex is used to remove Receiver Camera at specified index.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover2&action=remove&HandoverIndex=1
```

160.6. Scheduler

160.6.1. Description

The **scheduler** submenu configures the schedule settings for report generation for people count and heat map.

> **NOTE** This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

160.6.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
scheduler&action=<value>[&<parameter>=<value>]
```

160.6.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads scheduler settings
	Channel	REQ, RES	<int>	Channel ID
	Type	REQ, RES	<enum> PeopleCount, HeatMap, QueueManagement, VehicleCount	Feature Type
set	Channel	REQ, RES	<int>	Channel ID
	Type	REQ, RES	<enum> PeopleCount, HeatMap, QueueManagement, VehicleCount	Feature Type

Action	Parameter	Request/ Response	Type/ Value	Description
	ScheduleType	REQ, RES	<enum> Daily, Weekly	Type of schedule.
	WeekDay	REQ, RES	<enum> SUN, MON, TUE, WED, THU, FRI, SAT	Day of the week **Note** This parameter is valid only when **ScheduleType** is set to Weekly.
	Hour	REQ, RES	<int>	Hour of the day
	Minute	REQ, RES	<int>	Minute at which report will be generated
	EventAction	REQ, RES	<csv> AlarmOutput.#, SMTP, FTP, Record, HTTP, GoToPreset	Event action • AlarmOutput.#: Activates alarm output when the configured event occurs. • SMTP: Sends notification and image as an attachment via SMTP when the configured event occurs. • FTP: Uploads image via FTP when the configured event occurs. • Record: Saves event video on the storage device when the configured event occurs. (Refer to 'Recording' document (of **recording.cgi**) for information on configuring recording settings such as pre and post event buffers) • GoToPreset: Moves to the specified preset position when the configured event occurs. **Note** **transfer.cgi** is used to configure FTP/SMTP server settings.

Action	Parameter	Request/ Response	Type/ Value	Description
	AlarmOutput.#.Duration	REQ, RES	<enum> Always, 5s, 10s, 15s	Alarm output duration when the event occurs **AlarmOutput.#.Duration** is valid only when **EventAction** is set to AlarmOutput.

160.6.4. Examples

160.6.4.1. Getting scheduler settings

For people count

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=view&Type=PeopleCount
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PeopleCount.ScheduleType=Daily
Channel.0.PeopleCount.Hour=11
Channel.0.PeopleCount.Minute=00
Channel.0.PeopleCount.WeekDay=SUN
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "PeopleCount": [
        {
            "Channel": 0,
            "ScheduleType": "Daily",
```

```
            "Hour": 11,
            "Minute": 0,
            "WeekDay": "SUN"
        }
    ]
}
```

For heat map

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=view&Type=HeatMap
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.HeatMap.ScheduleType=Daily
Channel.0.HeatMap.Hour=06
Channel.0.HeatMap.Minute=30
Channel.0.HeatMap.WeekDay=SUN
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "HeatMap": [
        {
            "Channel": 0,
            "ScheduleType": "Daily",
            "Hour": 6,
            "Minute": 30,
            "WeekDay": "SUN"
        }
```

```
    ]
}
```

For queue management

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=view&Type=QueueManagement
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.QueueManagement.ScheduleType=Daily
Channel.0.QueueManagement.Hour=00
Channel.0.QueueManagement.Minute=00
Channel.0.QueueManagement.WeekDay=SUN
Channel.0.EventAction=AlarmOutput.1,SMTP,FTP
Channel.0.AlarmOutput.1.Duration=10s
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "QueueManagement": [
        {
            "Channel": 0,
            "ScheduleType": "Daily",
            "Hour": 0,
            "Minute": 0,
            "WeekDay": "SUN",
            "EventAction": [
                "AlarmOutput.1",
                "SMTP",
```

```
                "FTP"
            ],
            "AlarmOutputs": [
                {
                    "AlarmOutput": 1,
                    "Duration": "10s"
                }
            ]
        }
    ]
}
```

PTZ camera supports both global handover and preset-based handover. To get the preset-based handover list, PresetIndex should be passed in the view action.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=handover&action=view&PresetIndex=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PresetIndex.1.ROIIndex.1.Enable=False
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.IPType=IPv4
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.IPAddress=1.1.1.1
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Port=80
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Username=admin
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.Password=4321
Channel.0.PresetIndex.1.ROIIndex.1.HandoverIndex.1.PresetNumber=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Handover": [
        {
            "Channel": 0,
            "PresetList": [
                {
                    "PresetIndex": 1,
                    "HandoverList": [
                        {
                            "ROIIndex": 1,
                            "Enable": false,
                            "UserList": [
                                {
                                    "HandoverIndex": 1,
                                    "IPType": "IPv4",
                                    "IPAddress": "1.1.1.1",
                                    "Port": 80,
                                    "Username": "admin",
                                    "Password": "4321",
                                    "PresetNumber": 1
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```

160.6.4.2. Setting schedule configuration

Setting up a daily schedule **for report generation for people count**

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=set&ScheduleType=Daily&Hour=10&
Minute=35&Type=PeopleCount
```

Setting up a weekly schedule **for report generation for people count**

REQUEST

```
http://<Device IP>/ stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=set&ScheduleType=Weekly&Hour=10
&Minute=35&WeekDay=MON&Type=PeopleCount
```

Setting up a daily schedule **for report generation for heat map**

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=set&ScheduleType=Daily&Hour=10&
Minute=35&Type=HeatMap
```

Setting up a weekly schedule **for report generation for heat map**

REQUEST

```
http://<Device IP>/ stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=set&ScheduleType=Weekly&Hour=10
&Minute=35&WeekDay=MON&Type=HeatMap
```

160.7. Schedulelist

160.7.1. Description

The **schedulelist** submenu used to manage several schedulers.

Access level

Action	NVR
view	User
add	User
update	User
remove	User

160.7.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
schedulelist&action=<value>[&<parameter>=<value>]
```

160.7.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Schedule.#.ScheduleName	RES	<string>	Name of schedule
	Schedule.#.SUN	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.MON	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.TUE	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.WED	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.THU	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.FRI	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.SAT	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour

Action	Parameter	Request/ Response	Type/ Value	Description
	Schedule.#.HOL	RES	<string>	Array of space-separated 1 and 0 for each hour of the day. • 1: Enabled for that hour • 0: Disabled for that hour
	Schedule.#.IsFixed	RES	<bool> True, False	Array of space-separated 1 and 0 for each hour of the day. CAMERA ONLY
add	ScheduleName	REQ	<string>	Name of the schedule
	<ddd>	REQ	<bool> 0: Disable 1: Enable	Request parameter <ddd> can take any of the following parameters SUN,MON,TUE,WED,THU,FRI,SAT,HOL E.g. MON=1
	<dddh>	REQ	<bool> 0: Disable 1: Enable	Request parameter <dddh> represents the hour of the day. <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. E.g. SUN1=1 refers to SUNDAY 1 AM
	EveryDay	REQ	<bool> 0, 1	Whether to activate or deactivate the event operation every day • 0: Disabled • 1: Enabled 'EveryDay=1', denoting that the operation is activated every day. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EveryDay<h>	REQ	<bool> 0, 1	Daily time for event operation • 0: Disabled • 1: Enabled <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. This parameter is valid only when **EveryDay** is set to 1; e.g. EveryDay=1 is required for EveryDay0 ... EveryDay23. 'EveryDay=1&EveryDay18=1' means the event operation is activated every day 6:00 PM to 6:59 PM. CAMERA ONLY
	<dddh>.FromTo	REQ, RES	<string>	The time of week selected for event operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is also valid only when <corresponding weekday><hour> is set to 1; e.g. SUN0=1 is required for SUN0.FromTo. 'SUN=1&SUN18=1&SUN18.FromTo=12-20' means that the event operation is activated every Sunday from 6:12 PM to 6:20 PM. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	EveryDay<h>.FromTo	REQ, RES	<string>	The time for everyday event action operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when EveryDay<hour> is set to 1; e.g. 'EveryDay0=1' is required for EveryDay0.FromTo. 'EveryDay=1&EveryDay18=1&EveryDay18.FromTo=12-20' means that the event operation is activated every day 6:12 PM to 6:20 PM. CAMERA ONLY
update	ScheduleName	REQ	<string>	Schedule name
	ScheduleNewName	REQ	<string>	New schedule name if it needs to be changed
	<ddd>	REQ	<string> 0: Disable 1: Enable	Request parameter <ddd> can take any of the following parameters SUN,MON,TUE,WED,THU,FRI,SAT,HOL E.g. MON=1
	<dddh>	REQ	<string> 0: Disable 1: Enable	Request parameter <dddh> represents the hour of that day. <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. E.g. SUN1=1 refers to SUNDAY 1AM

Action	Parameter	Request/ Response	Type/ Value	Description
	EveryDay	REQ	<bool> 0, 1	Whether to activate or deactivate the event operation every day • 0: Disabled • 1: Enabled 'EveryDay=1', denoting that the schedule is activated every day, is the same as when the **ScheduleType** parameter is set to Always. CAMERA ONLY
	EveryDay<h>	REQ	<bool> 0, 1	Daily time for event operation • 0: Disabled • 1: Enabled <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. This parameter is valid only when **EveryDay** is set to 1; e.g. EveryDay=1 is required for EveryDay0 ... EveryDay23. 'EveryDay=1&EveryDay18=1' means the event operation is activated every day 6:00 PM to 6:59 PM. CAMERA ONLY

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>.FromTo	REQ, RES	<string>	The time of week selected for event operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is also valid only when <corresponding weekday><hour> is set to 1; e.g. SUN0=1 is required for SUN0.FromTo. 'SUN=1&SUN18=1&SUN18.FromTo=12-20' means that the event operation is activated every Sunday from 6:12 PM to 6:20 PM. CAMERA ONLY
	EveryDay<h>.FromTo	REQ, RES	<string>	The time for everyday event action operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when EveryDay<hour> is set to 1; e.g. 'EveryDay0=1' is required for EveryDay0.FromTo. 'EveryDay=1&EveryDay18=1&EveryDay18.FromTo=12-20' means that the event operation is activated every day 6:12 PM to 6:20 PM. CAMERA ONLY
remove	ScheduleName	REQ	<string>	Name of schedule to be removed

160.7.4. Examples

160.7.4.1. Getting schedulelist

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=schedulelist&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Schedule.0.ScheduleName=Schedule1
Schedule.0.SUN=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.MON=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.TUE=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.WED=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.THU=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.FRI=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.SAT=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
Schedule.0.HOL=1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Schedules": [{
        "ScheduleName": "Schedule1",
        "Schedule": {
            "SUN": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "MON": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "TUE": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "WED": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "THU": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
```

```
            "FRI": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "SAT": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"],
            "HOL": ["1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1"]
        }
    }]
}
```

160.7.4.2. Adding schedulelist

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=schedulelist&action=add&ScheduleName=schedule2&M
on=1&Mon12=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.7.4.3. Updating schedulelist

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=schedulelist&action=update&ScheduleName=schedule
2&ScheduleNewName=schedule3&SUN=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.7.4.4. Removing schedulelist

NOTE The camera only supports JSON responses.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=schedulelist&action=remove&ScheduleName=schedule
2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.8. Audio Clip

160.8.1. Description

The **audiooutfiles** submenu manages audio clip files in the device.

NOTE	This chapter applies to network cameras and IPAS. For IPAS, the install and remove operations are not supported. The availability of this submenu depends on the AudioOutAccess level and only when the DeviceType is AudioBridge, the submenu's availability is determined by the AudioInAccess level.

Access level

Action	Camera	IPAS
view	Admin	User
set	Admin	
install	Admin	
control	Admin	Admin
remove	Admin	

160.8.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
audiooutfiles&action=<value>[&<parameter>=<value>]
```

160.8.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	TotalSpace	RES	<string>	Total space of unit as KB
	FreeSpace	RES	<string>	Free space of unit as KB
	AudioClips.#.Index	RES	<int>	Audio clips index
	AudioClips.#.Name	RES	<string>	Audio clips name
set	Index	REQ, RES	<int>	Index of audio clip
	Gain	REQ, RES	<int>	Audio output gain The value is in the range of 1 to 5.
	Name	REQ, RES	<string>	File name of audio clip
	LoopCount	REQ, RES	<int> 1 ~ 5	Number of times to loop the audio clip, default is 1.
install	Index	REQ, RES	<int>	Index of audio clip
	Name	REQ, RES	<string>	File name of audio clip

Action	Parameter	Request/ Response	Type/ Value	Description
	Type	REQ, RES	<enum> WAV, MP3 **Note** MP3 files are supported only on select models. Please refer to the cgi section of attributes.cgi response to identify the supported formats.	File type of audio clip
	DeviceID	REQ	<int>	Device ID where the audio file should be stored. **Note** Applicable only when SpeakerType is MASTER or SERVER. refer **registeredsubdevices** response for the device ids with AudioSourceType **BGM** support. IPAS ONLY
	Gain	REQ, RES	<int>	Audio output gain The value is in the range of 1 to 5.

Action	Parameter	Request/ Response	Type/ Value	Description
control	Type	REQ	<enum> Donwload, Play, Stop	Control type • Download: Downloads the audio clip to the client • Play: Plays the audio clip • Stop: Stops playing of the audio clip **Note** In IPAS, when sending **stop** command, same audio filename and SpeakerID or GroupID needs to be passed that was used in **play** command. Download not supported in IPAS.
	Index	REQ	<int>	Index of audio clip CAMERA ONLY
	Name	REQ	<string>	Unique name of the audio file **Note** In IPAS, instead of Index use this parameter to refer a file uniquely. IPAS ONLY
	SpeakerID	REQ	<int>	Speaker ID where the audio should be played IPAS ONLY **Note** Applicable only when SpeakerType is MASTER or SERVER. refer **registeredsubdevices** response for the device ids with device type SPEAKER or SPEAKER_WITH_AUDIO_SOURCE support.

Action	Parameter	Request/ Response	Type/ Value	Description
	GroupID	REQ	<int>	Speaker Group ID where the audio should be played IPAS ONLY **Note** Applicable only when SpeakerType is MASTER or SERVER. refer **speakergroups** response for the supported group ids.
	Volume	REQ	<int> 1~100	Volume in which the audio file needs to be played. IPAS ONLY
	LoopCount	REQ	<int> 1~99	Audio repeat count, if not passed LoopCount is considered as 1. IPAS ONLY
remove	Index	REQ, RES	<int>	Index of audio clip
	Name	REQ	<string>	Unique name of the audio file **Note** In IPAS, instead of Index use this parameter to refer a file uniquely. IPAS ONLY

160.8.4. Examples

NOTE In IPAS, instead of **Index** parameter **Name** parameter needs to be used.

160.8.4.1. Getting basic information

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=audiooutfiles&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "AudioOutFiles": {
        "FreeSpace": 3070,
        "TotalSpace": 3072,
        "AudioClips": []
    }
}
```

160.8.4.2. Install Audio File

NOTE

When SpeakerType is MASTER or SERVER. DeviceID parameter needs to be passed.

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=audiooutfiles&action=install&Name=testaudio&Type
=WAV&DeviceID=1224
```

POST BODY [Content-Type as application/octet-stream]

```
RIFF....WAVEfmt ......
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.8.4.3. Downloads an audio clip to the client

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=audiooutfiles&action=control&Type=Download&Index
=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.8.4.4. Play Audio file

NOTE

In IPAS, When SpeakerType is MASTER or SERVER. GroupID or SpeakerID parameter needs to be passed.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=audiooutfiles&action=control&Type=Play&Index=1
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.8.4.5. Stop play operation

NOTE

In IPAS, when SpeakerType is MASTER or SERVER
Name, GroupID or SpeakerID parameter needs to be passed.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=audiooutfiles&action=control&Type=Stop&Index=1
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.8.4.6. Remove audio file

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=audiooutfiles&action=remove&Index=1
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.8.4.7. Error Responses

In case if the file is not a available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "File Name Not Found"
    }
}
```

In case if the Group ID is not available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "GroupID Not Found"
    }
}
```

In case if the SpeakerID is not available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "SpeakerID Not Found"
```

```
    }
}
```

160.9. TTS (Text to speech) Files

160.9.1. Description

The **ttsfiles** submenu manages tts files in the device.

NOTE

This chapter applies to only IPAS.
When SpeakerType is configured as INDEPENDENT or SLAVE, this submenu is not supported. For IPAS, this submenu depends on the AudioOutAccess level.

Access level

Action	IPAS
view	User
control	Admin

160.9.2. Syntax

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ttsfiles&action=<value>[&<parameter>=<value>]
```

160.9.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	TTSList.#.Name	RES	<string>	File name of the tts file
control	Type	REQ	<enum> Play, Stop	Control type • Play: Plays the tts file • Stop: Stops playing of the tts file **Note** When sending **stop** command, same tts filename and SpeakerID or GroupID needs to be passed that was used in **play** command.
	Name	REQ	<string>	Unique name of the tts file

Action	Parameter	Request/ Response	Type/ Value	Description
	SpeakerID	REQ	<int>	Speaker ID where the audio should be played **Note** Applicable only when SpeakerType is MASTER or SERVER.
	GroupID	REQ	<int>	Speaker Group ID where the audio should be played **Note** Applicable only when SpeakerType is MASTER or SERVER.
	Volume	REQ	<int> 1~100	Volume in which the tts file needs to be played.

160.9.4. Examples

160.9.4.1. Getting list of tts files

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ttsfiles&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TTSFiles": {
        "TTSList": [
            {
                "Name": "Src1_EmergencyExit.txt"
            },
            {
                "Name": "Src1_MotionWaring.txt"
            }
        ]
    }
```

```
}
```

160.9.4.2. Play Audio file

NOTE

When SpeakerType is MASTER or SERVER
GroupID or SpeakerID parameter needs to be passed.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ttsfiles&action=control&Type=Play&Name=Src1_EmergencyExit.txt&Volume=50
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.9.4.3. Stop Playing

NOTE

When SpeakerType is MASTER or SERVER
Name, GroupID or SpeakerID parameter needs to be passed.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ttsfiles&action=control&Type=Stop&Name=Src1_EmergencyExit.txt
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.9.4.4. Remove File

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ttsfiles&action=remove&Name=Src1_EmergencyExit.t
```

```
xt
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.9.4.5. Error Responses

In case if the file is not a available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "File Name Not Found"
    }
}
```

In case if the Group ID is not available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "GroupID Not Found"
    }
}
```

In case if the RequestID is not available,

RESPONSE

```
{
    "Response": "Fail",
    "Error": {
```

```
            "Code": 604,
            "Details": "Invalid Value",
            "Reason" : "RequestID Not Found"
        }
}
```

In case if the SpeakerID is not available,

RESPONSE


```
{
    "Response": "Fail",
    "Error": {
        "Code": 604,
        "Details": "Invalid Value",
        "Reason" : "SpeakerID Not Found"
    }
}
```


160.10. LED Preset

160.10.1. Description

The **ledpreset** submenu manages to make presets for LED controling.

Access level

Action	Camera	LEDBox
view	Admin	Admin
add	Admin	
update	Admin	Admin
control	Admin	Admin
remove	Admin	

160.10.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
ledpreset&action=<value>[&<parameter>=<value>]
```

160.10.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	LEDPresetIndex	REQ, RES	<int>	Index of LED preset
add/update	LEDPresetIndex	REQ, RES	<int>	Index of LED preset
	LEDUsageIndex	RES	<csv> 1,2	It means LED hardware index. Specified index LED will be affected by led preset configuration. **NOTE** Only applicable to LED Box.
	Name	REQ, RES	<string>	Name of this LED preset
	Brightness	REQ, RES	<int>	Controls LED brightness
	LightMode	REQ, RES	<enum> On, Off, Blink, Rotating, FadeIn, FadeOut, FadeInAndOut, Strobe	• On : Turn on the LEDUsageIndex LED • Off : Turn off the LEDUsageIndex LED • Blink : Turn LED on and off with a 1 second interval. • Rotating : Rotate the LED pattern around the device. • FadeIn : Increase LED brightness gradually from off to full brightness. • FadeOut : Decrease LED brightness gradually from full brightness to off. • FadeInAndOut : Repeat a full fade cycle from off to bright and back to off. • Strobe : Turn LED on and off rapidly with a 50 ms interval.
	Color	REQ, RES	<enum> Green, Blue, Red, Pink, SkyBlue, Purple, Custom	Color to apply

Action	Parameter	Request/ Response	Type/ Value	Description
	HexColorCode	REQ, RES	<string> #RRGGBB (#000000 - #FFFFFF)	HexColorCode is applied only when Color is set to Custom; otherwise, this parameter is ignored.
	Duration	REQ, RES	<int> 10 ~ 600	Duration in seconds
	ColorAfterDuration	REQ, RES	<enum> Green, Blue, Red, Pink, SkyBlue, Purple, Custom	Color to apply after duration
	HexColorCodeAfterDuration	REQ, RES	<string> #RRGGBB (#000000 - #FFFFFF)	HexColorCodeAfterDuration is applied only when ColorAfterDuration is set to Custom; otherwise, this parameter is ignored.
	LightModeAfterDuration	REQ, RES	<enum> On, Off, Blink, Rotating, FadeIn, FadeOut, FadeInAndOut, Strobe	Light mode after the configured duration * On : Turn on the LEDUsageIndex LED * Off : Turn off the LEDUsageIndex LED * Blink : Turn LED on and off with a 1 second interval. * Rotating : Rotate the LED pattern around the device. * FadeIn : Increase LED brightness gradually from off to full brightness. * FadeOut : Decrease LED brightness gradually from full brightness to off. * FadeInAndOut : Repeat a full fade cycle from off to bright and back to off. * Strobe : Turn LED on and off rapidly with a 50 ms interval.
	BrightnessAfterDuration	REQ	<int>	Controls LED brightness after duration
	EventActivePeriod	REQ, RES	<bool>	If EventActive is true, post-duration actions are ignored.

Action	Parameter	Request/ Response	Type/ Value	Description
	AntiFlashing	REQ	<bool>	Enable antiflashing feature **NOTE** Only applicable to BCR camera
	PrechargeTime	REQ	<int>	Control precharging time(before & after LED margin of shutter open period) **NOTE** Only applicable to BCR camera
remove	LEDPresetIndex	REQ	<int>	Index of LED preset
control	LEDPresetIndex	REQ	<int>	Index of LED preset

160.10.4. Examples

160.10.4.1. Getting list of led preset

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ledpreset&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "LEDPresetList": [
        {
            "LEDPresetIndex": 1,
            "LEDUsageIndex": [
                1
            ],
            "LightMode": "On",
            "Color": "Green",
            "Name": "Vacant",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
```

```
        },
        {
            "LEDPresetIndex": 2,
            "LEDUsageIndex": [
                1
            ],
            "LightMode": "On",
            "Color": "Red",
            "Name": "Occupied",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
        },
        {
            "LEDPresetIndex": 3,
            "LEDUsageIndex": [
                2
            ],
            "LightMode": "On",
            "Color": "Green",
            "Name": "Vacant",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
        },
        {
            "LEDPresetIndex": 4,
            "LEDUsageIndex": [
                2
            ],
            "LightMode": "On",
            "Color": "Red",
            "Name": "Occupied",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
        },
        {
            "LEDPresetIndex": 5,
            "LEDUsageIndex": [
                1,
```

```
                2
            ],
            "LightMode": "On",
            "Color": "Green",
            "Name": "Vacant",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
        },
        {
            "LEDPresetIndex": 6,
            "LEDUsageIndex": [
                1,
                2
            ],
            "LightMode": "On",
            "Color": "Red",
            "Name": "Occupied",
            "Brightness": 15,
            "AntiFlashing": true,
            "PrechargeTime": 15
        }
    ]
}
```

160.10.4.2. Getting list of led preset (in case of Speaker with Led Indicator)

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ledpreset&action=view
```

RESPONSE

```
{
    "LEDPresetList": [
        {
            "LEDPresetIndex": 1,
            "LightMode": "On",
            "Color": "Pink",
            "Name": "testpresetee",
            "Duration": 30,
            "EventActivePeriod": false,
```

```
            "LightModeAfterDuration": "On",
            "ColorAfterDuration": "SkyBlue"
        },
        {
            "LEDPresetIndex": 2,
            "LightMode": "On",
            "Color": "Red",
            "Name": "gggg",
            "Duration": 20,
            "EventActivePeriod": false,
            "LightModeAfterDuration": "On",
            "ColorAfterDuration": "Purple"
        },
        {
            "LEDPresetIndex": 3,
            "LightMode": "On",
            "Color": "Blue",
            "Name": "testtt",
            "Duration": 200,
            "EventActivePeriod": false,
            "LightModeAfterDuration": "On",
            "ColorAfterDuration": "Pink"
        },
        {
            "LEDPresetIndex": 4,
            "LightMode": "On",
            "Color": "Blue",
            "Name": "",
            "Duration": 200,
            "EventActivePeriod": false,
            "LightModeAfterDuration": "On",
            "ColorAfterDuration": "Pink"
        },
        {
            "LEDPresetIndex": 5,
            "LightMode": "On",
            "Color": "Red",
            "Name": "last",
            "Duration": 40,
            "EventActivePeriod": false,
            "LightModeAfterDuration": "On",
```

```
            "ColorAfterDuration": "SkyBlue"
        }
    ]
}
```

160.10.4.3. ADD PRESET

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ledpreset&action=add&LightMode=On&Color=Blue&Dur
ation=40&EventActivePeriod=False&LightModeAfterDuration=On&ColorAfterDuratio
n=Pink&Name=testtt
```

160.10.4.4. Activate PRESET

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ledpreset&action=control&LEDPresetIndex=1
```

160.10.4.5. REMOVE PRESET

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ledpreset&action=remove&LEDPresetIndex=1
```

160.10.4.6. Change ledpreset 1’s color to blue

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ledpreset&action=update&LEDPresetIndex=1&LightMo
de=On&Color=Blue
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.10.4.7. Change ledpreset 2's LightMode to Off

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ledpreset&action=update&LEDPresetIndex=2&LightMode=Off
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.10.4.8. Apply LEDPresetIndex 1 which will affect to LED hardware index 1 because LEDUsageIndex setting is 1.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ledpreset&action=control&LEDPresetIndex=1
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.10.4.9. Apply LEDPresetIndex 2 which will turn off LED hardware 1 because LightMode is Off and LEDUsageIndex is 1.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ledpreset&action=control&LEDPresetIndex=2
```

RESPONSE

```
{
    "Response": "Success"
}
```

160.11. Audio Clip Schedule

160.11.1. Description

The **audiooutfilesschedule** submenu configures the schedule settings for audio clip playback.

NOTE This chapter applies to network cameras only.

Access level

Action	Camera
view	Admin
set	Admin

160.11.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
audiooutfilesschedule&action=<value>[&<parameter>=<value>]
```

160.11.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disables playback activation • True: Activated • False: Deactivated
	AudioClipeIndex	REQ, RES	<int>	Index of audio clip
	Hour	REQ, RES	<int>	Hour of the day
	Minute	REQ, RES	<int>	Minute at which report will be generated
	WeekDay	REQ, RES	<enum> SUN, MON, TUE, WED, THU, FRI, SAT	Day of the week **Note** This parameter is valid only when **ScheduleType** is set to Weekly.

Action	Parameter	Request/ Response	Type/ Value	Description
	ScheduleType	REQ, RES	<enum> Daily, Weekly	Time schedule to play audio clip • Daily: At scheduled time every day • Weekly: At scheduled day and time

160.11.4. Examples

160.11.4.1. Getting scheduler settings of audio clip playback

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=audiooutfilesschedule&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AudioOutFilesSchedule": [
        {
            "Channel": 0,
            "AudioClip": [
                {
                    "AudioClipIndex": 1,
                    "Enable": true,
                    "ScheduleType": "Daily",
                    "Hour": 0,
                    "Minute": 0,
                    "WeekDay": "SUN"
                }
            ]
        }
    ]
}
```

160.11.4.2. Setting scheduler settings of audio clip playback

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=audiooutfilesschedule&action=set&AudioClipIndex=
1&Enable=True&ScheduleType=Weekly&Hour=10&Minute=15&WeekDay=MON
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

160.12. Internal Handover Calibration

160.12.1. Description

The **internalhandovercalibration** submenu manages the internal handover calibration settings which are used for the smart zoom feature.

NOTE This chapter is applicable for multi-directional cameras that have a PTZ channel.

Access level

Action	Camera
view	Admin
set	Admin
control	Admin
check	Admin
remove	Admin

160.12.2. Syntax

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=internalhandovercalibration&action=<value>[&<par
ameter>=<value>]
```

160.12.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	PTZChannel.#.Mapping.# .LocalScreenCoordinate	REQ, RES	<string>	Target Channel’s coordinates that PTZ Channel would head for. (X, Y)
	PTZChannel.#.Mapping.# .PTCoordinate	REQ, RES	<string>	Absolute coordinates of Pan and Tilt of PTZ channels to calibrate with target channel. (P, T)
control	Channel	REQ	<int>	Channel ID
	PTZChannel	REQ	<int>	PTZ Channel ID
	PreviewLocalScreenCoordinate	REQ	<string>	Target Channel’s coordinates that PTZ Channel would head for. (X, Y)
check	Channel	REQ	<int>	Channel ID
	CalibrationCompleted	RES	<bool> True, False	Shows whether each channel’s calibration settings are complete or not
remove	Channel	RES	<int>	Channel ID
	PTZChannels	RES	<csv> #, All	PTZ Channel ID

160.12.4. Examples

160.12.4.1. Getting internal handover calibration settings for all channels

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=internalhandovercalibration&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.PTZChannel.4.Mapping.1.LocalScreenCoordinate=555,300
Channel.0.PTZChannel.4.Mapping.1.PTCoordinate=0.00,25.00
Channel.0.PTZChannel.4.Mapping.2.LocalScreenCoordinate=570,803
```

```
Channel.0.PTZChannel.4.Mapping.2.PTCoordinate=0.70,48.61
Channel.0.PTZChannel.4.Mapping.3.LocalScreenCoordinate=1367,764
Channel.0.PTZChannel.4.Mapping.3.PTCoordinate=38.87,46.77
Channel.0.PTZChannel.4.Mapping.4.LocalScreenCoordinate=1298,294
Channel.0.PTZChannel.4.Mapping.4.PTCoordinate=35.55,24.69
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "InternalHandoverCalibrations": [
        {
            "Channel": 0,
            "PTZChannels": [
                {
                    "PTZChannel": 4,
                    "Mappings": [
                        {
                            "Index": 1,
                            "LocalScreenCoordinate": "555,300",
                            "PTCoordinate": "0.00,25.00"
                        },
                        {
                            "Index": 2,
                            "LocalScreenCoordinate": "570,803",
                            "PTCoordinate": "0.70,48.61"
                        },
                        {
                            "Index": 3,
                            "LocalScreenCoordinate": "1367,764",
                            "PTCoordinate": "38.87,46.77"
                        },
                        {
                            "Index": 4,
                            "LocalScreenCoordinate": "1298,294",
                            "PTCoordinate": "35.55,24.69"
                        }
                    ]
```

```
                }
            ]
        }
    ]
}
```

160.12.4.2. Setting calibration coordinates to a specific channel

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=internalhandovercalibration&action=set&Channel=1&PTZChannel=4&Mapping.1.LocalScreenCoordinate=306,162&Mapping.1.PTCoordinate=35.55999755859375,24.69999885559082
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

160.12.4.3. Controls PTZ channel to move requested local coordinates of a specific channel

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=internalhandovercalibration&action=control&Channel=1&PTZChannel=4&PreviewLocalScreenCoordinate=300,740
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

160.13. IO Box registration

160.13.1. Description

The **ioboxregister** submenu provides how to register an io box.

NOTE This chapter is only applicable to Camera

Access level

Action	Camera
view	Admin
add/update	Admin
remove	Admin
check	Admin

160.13.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
ioboxregister&action=<value>[&<parameter>=<value>]
```

160.13.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Channel	REQ	<int>	Channel ID
	IsRebootRequired	RES	<bool>	• True : When you change Audio source selection or IO source selection, device will reboot. • False : not reboot.
add/update	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool> True, False	Enables or disable IO box
	IOBoxIndex	REQ, RES	<int>	
	IPType	REQ, RES	<enum> IPV4, IPV6	
	IPAddress	REQ, RES	<string>	IP address of IO box
	Port	REQ, RES	<int>	Port number of IO box
	Username	REQ, RES	<string>	
	Password	REQ	<string>	

Action	Parameter	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<bool> True, False	
	ConnectionMode	REQ, RES	<enum> HTTP, HTTPS	
	AudioSourceSelection	REQ, RES	<enum> Internal, External	Select audiosource whether it is internal audiosource or external audiosource(ex.IOBox)
	IOSourceSelection	REQ, RES	<enum> Internal, External	Select IO source whether it is internal IO source or external IO source(ex.IOBox)
remove	Channel	REQ	<int>	Channel ID
	IPBoxIndex	REQ	<int>	
check	ConnectionStatus	REQ	<int>	Current status of IO box
	ModelName	RES	<string>	Shows connected iobox's model name

160.13.4. Examples

160.13.4.1. Getting the current status of io box

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ioboxregister&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ioboxregister": [
        {
            "Channel": 0,
            "IOBoxList": [
                {
                    "IOBoxIndex": 1,
                    "Enable": false,
                    "IPType": "IPv4",
```

```
                    "IPAddress": "192.168.1.100",
                    "Port": 80,
                    "ConnectionMode": "HTTP",
                    "Username": "admin",
                    "Password": ""
                }
            ]
        }
    ]
}
```


160.13.4.2. Adding a new iobox information

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ioboxregister&action=add&Channel=0&IPAddress=192
.168.1.100&IPType=IPv4&Port=80&ConnectionMode=HTTP&Username=admin&IsPassword
Encrypted=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

160.13.4.3. Connecting Io box with a camera

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=ioboxregister&action=update&Channel=0&IOBoxIndex
=1&Enable=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

160.13.4.4. Removing registed io box information

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ioboxregister&action=remove&Channel=0&IOBoxIndex=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

160.13.4.5. Checking connection status of io box

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=ioboxregister&action=check&Channel=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "ioboxregister": [
        {
            "Channel": 0,
            "ConnectionStatus": "Connected"
        }
    ]
}
```

160.14. indicationpass

160.14.1. Description

The **indicationpass** submenu allows the user to pass event information to a predefined camera when a parking detection event occurs. A fisheye lens is not good for watching a parking area just below the camera, so the user can set it to observe the opposite side of parking lot to detect parked vehicles and send analyzed data to the opposite camera. This feature will work only when the user sets the LEDUsage value in ledindicator submenu to 1.

Access level

Action	Camera	Encoder
view	Admin	Admin
add/update	Admin	Admin
remove	Admin	Admin

160.14.2. Syntax

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=
indicationpass&action=view&[&<parameter>=<value>...]
```

160.14.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads indication pass settings
	Channel	REQ, RES	<int>	Channel ID
add/update	Channel	REQ, RES	<int>	Channel ID
	IndicationPassIndex	REQ, RES	<int>	Index of receiver camera
	IPType	REQ, RES	<enum> IPV4, IPV6	IP Type of receiver camera
	IPAddress	REQ, RES	<string> <formatInfo="IPv4Address or IPv6Address>	IP Address of receiver camera
	Port	REQ, RES	<int>	Port of receiver camera
	Username	REQ, RES	<string>	User name of receiver camera
	Password	REQ, RES	<string>	Password of receiver camera

Action	Parameter	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<bool> True, False	Enables or disables password encryption for receiver camera
	ConnectionMode	REQ, RES	<enum> HTTP, HTTPS	Connection mode to connect receiver camera
remove	Channel	REQ	<int>	Channel ID
	IndicationPassIndex	REQ	<csv>	Index of receiver camera to be deleted

160.14.4. Examples

160.14.4.1. Getting the indication pass settings

REQUEST

```
http://<Device IP>/eventrules.cgi?msubmenu=indicationpass&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: json/application
<Body>
```


```
{
    "IndicationPass": [
        {
            "Channel": 0,
            "UserList": [
                {
                    "IndicationPassIndex": 1,
                    "IPType": "IPv4",
                    "IPAddress": "192.168.1.1",
                    "Port": 80,
                    "ConnectionMode": "HTTP",
                    "Username": "admin",
                    "Password": ""
                }
            ]
        }
    ]
```

```
}
```

160.14.4.2. Adding information of receiver camera

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=indicationpass&action=add&IPType=IPv4&ConnectionMode=HTTP&IPAddress=192.168.1.1&Port=80&Username=admin&IsPasswordEncrypted=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


160.14.4.3. Removing receiver camera(s)

IndicationPassIndex is used to remove the Receiver Camera at specified index.

REQUEST

```
http://<Device IP>/stw-cgi/eventrules.cgi?msubmenu=handover2&action=remove&IndicationPassIndex=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 161. Event Status

161.1. Event Status

161.1.1. Description

eventstatus is used to request the current event status (activated or deactivated).

The **eventstatus** submenu facilitates three kinds of actions: **check**, **monitor**, and **monitordiff**. The **check** action solicits the current status once. The **monitor** action periodically requests the current status, regardless of state changes, and also delivers the event status if an event takes place. The **monitordiff** action initially requests the status of all events and subsequently provides updates whenever a change from the previous state is detected.

NOTE It is suggested to use **SchemaBased** event notification support starting from SUNAPI version 2.5.8 and beyond.

Access level

Action	Camera	Encoder	NVR
check	Guest	Guest	User
monitor	Guest	Guest	User
monitordiff	Guest	Guest	User

161.1.2. Syntax

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=
eventstatus&action=<value>[&<parameter>=<value>]
```

161.1.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
check	Channel.#.EventType	REQ, RES	<csv> See Supported Events List table.	Event type If the status of VideoAnalytics is requested, the following values are returned in response: • VideoAnalytics.Passing • VideoAnalytics.Entering • VideoAnalytics.Exiting • VideoAnalytics.Appearing • VideoAnalytics.Disappering **Note** OpenSDK" Type’s meta event response. (supports only "event" type) "check" action doesn’t support "
	FireAlarmInput	REQ, RES	<csv>	Fire Alarm input number
	AlarmInput	REQ, RES	<csv>	Alarm input number
	AlarmOutput	REQ, RES	<csv>	Alarm output number
	Aux	REQ, RES	<csv>	Auxiliary devices
	SystemEvent	REQ, RES	<csv> see Supported Events List table.	System event
	AccessControl	REQ, RES	<csv> see Supported Events List table.	AccessControl event
	IncludeTimestamp	REQ	<bool> True, False	Whether a time stamp is included or not
	Timestamp	RES	<string>	Time stamp
	POS.#.EventType	REQ, RES	<csv> KeywordMatch	POS event type

Action	Parameter	Request/ Response	Type/ Value	Description
	SchemaBased	REQ	<bool> True, False	Shows schema based response. To know schema, refer to the eventstatusschema submenu.
	MQTTSubscription	REQ, RES	<csv>	MQTTSubscription index number
	OpenSDKAppStatus	REQ, RES	<bool>	Whether to show OpenSDKAppStatus event. Status can be "Installing","Inactive","Active","Uninstalling","Removed","InstallationFailed".
monitor	Channel.#.EventType	REQ, RES	<csv> see Supported Events List table.	Event type If the status of VideoAnalytics is requested, the following values are returned in response: • VideoAnalytics.Passing • VideoAnalytics.Entering • VideoAnalytics.Exiting • VideoAnalytics.Appearing • VideoAnalytics.Disappering
	FireAlarmInput	REQ, RES	<csv>	Fire Alarm input number
	AlarmInput	REQ, RES	<csv>	Alarm input number
	AlarmOutput	REQ, RES	<csv>	Alarm output number
	Aux	REQ, RES	<csv>	Auxiliary device
	SystemEvent	REQ, RES	<csv> see Supported Events List table.	System event
	AccessControl	REQ, RES	<csv> see Supported Events List table.	AccessControl event
	Periodicity	REQ	<int>	Interval in seconds

Action	Parameter	Request/ Response	Type/ Value	Description
	ChangedConfigURI	RES	<string>	Configuration change in camera The value is returned in the format of <CGI file name>?<msubmenu name>. e.g. if the motion detection related setting is changed with eventsources.cgi, the returned data will be ChangedConfigURI=eventsources.cgi?msubmenu=videoanalysis.
	IncludeTimestamp	REQ	<bool> True, False	Whether a time stamp is included or not
	Timestamp	RES	<string>	Time stamp
	POS.#.EventType	REQ, RES	<csv> KeywordMatch	POS Event Type
	EventDescription	RES	<string>	Event Description
	SchemaBased	REQ	<bool> True, False	Shows schema based response. To know schema, refer to the eventstatusschema submenu.
	AttributeUpdate	RES	<bool> True, False	Checks whether the property has been changed
	MQTTSubscription	REQ, RES	<csv>	MQTTSubscription index number
	OpenSDKAppStatus	REQ, RES	<bool>	Whether to show OpenSDKAppStatus event. Status can be "Installing","Inactive","Active","Uninstalling","Removed","InstallationFailed".
monitordiff	Channel.#.EventType	REQ, RES	<csv> see Supported Events List table.	Event type If the status of VideoAnalytics is requested, the following values are returned in response: • VideoAnalytics.Passing • VideoAnalytics.Entering • VideoAnalytics.Exiting • VideoAnalytics.Appearing • VideoAnalytics.Disappering

Action	Parameter	Request/ Response	Type/ Value	Description
	FireAlarmInput	REQ, RES	<csv>	Fire Alarm input number
	AlarmInput	REQ, RES	<csv>	Alarm input number Values may vary depending on the supported alarm input. Please check the device attributes using attributes.cgi.
	AlarmOutput	REQ, RES	<csv>	Alarm output number
	Aux	REQ, RES	<csv>	Auxiliary device
	SystemEvent	REQ, RES	<csv> see Supported Events List table.	System event
	AccessControl	REQ, RES	<csv> see Supported Events List table.	AccessControl event
	ChangedConfigURI	RES	<string>	Configuration change in camera
	IncludeTimestamp	REQ	<bool> True, False	Whether a time stamp is included or not
	Timestamp	RES	<string>	Time stamp
	POS.#.EventType	REQ, RES	<csv> KeywordMatch	POS event type
	EventDescription	RES	<string>	Event description
	SchemaBased	REQ	<bool> True, False	Shows schema based response. To know schema, refer to the eventstatusschema submenu.
	EventFilter	REQ, RES	<enum> SystemEvent, POSEvent, ChannelEvent, None	Filters by type to search an event
	AttributeUpdate	RES	<bool> True, False	Checks whether the property has been changed
	MQTTSubscription	REQ, RES	<csv>	MQTTSubscription index number

Action	Parameter	Request/ Response	Type/ Value	Description
	OpenSDKAppStatus	REQ, RES	<bool>	Whether to show OpenSDKAppStatus event. Status can be "Installing","Inactive","Active","Uninstalling","Removed","InstallationFailed".

Supported Events List

EventType	Event List
SystemEvent	PowerOn, PowerOff, PowerReboot, ConfigChange, Backup, FWUpdate, FactoryReset, HDDFull, HDDFail, HDDNone, FanError, SDFormat, SDFail, SDFull, SDInsert, SDRemove, Network, TimeChange, Record, ConfigurationBackup, ConfigurationRestore, NASFormat, NASFail, NASFull, NASConnect, NASDisconnect, CPUFanError, FrameFanError, LeftFanError, RightFanError, Recording, TimeChange, Network, InternalHDDErase, AdminLogin, RecordFiltering, NetCamTrafficOverFlow, RecordingError, OverwriteDecoding, EndofFrame, RAIDEnable, RAIDSetup, RAIDBuilding, RAIDBuildCancel, RAIDBuildFail, RAIDDegrade, RAIDRebuildStart, RAIDFail, RAIDRebuildFail, iSCSIDisconnect, BeingUpdate, InternalHDDConnect, BatteryFail, RecordFrameDrop, DualSMPSFail, USBHDDConnect, DSPDisplayStart, DSPVASystemStart, AMDLoadFail, RAIDDeviceAdd, RAIDRecordRestriction, AlarmReset, NewFWAvailable, PasswordChange, ConfigRestore, EmergencyTrigger, InternalHDDWarmup, GSensorEvent, GPSDisconnect, WiFiSignalChanged, OverSpeed, VPUError, MemoryError, CpuOverload, NetTxTrafficOverflow, CaseTampering
Channel.#.EventType	MotionDetection, FaceDetection, Videoloss, Tampering, AudioDetection, VideoAnalytics, NetworkAlarmInput, Tracking, RecordingStatus, PriorityRecordingStatus, PTZMotion, UserInput, NetworkCameraConnect, NetworkAlarmInput, AMDStart, LowFps, DefocusDetection, Profile.#.DigitalAutoTracking, FogDetection, SDFormat, SDFail, SDFull, SDInsert, AudioAnalytics, USBWIFIConnect, QueueEvent, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, ObjectDetection, OpenSDK, BodyTemperatureDetection, MaskDetection, SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor, ParkingDetection, ParkingVehicleCountChanged, AllParkingSpotOccupied, ParkingSpotOccupied, SIPCall, AudioClipPlayback, TwoWayTalk
AccessControl	AccessGranted, AccessDenied, AccessTaken, AccessNotTaken, ConfigChanged, AntiPassbackViolation, AccessPointEnabled, AccessRuleEnabled, DoorMode, DoorAlarmState, DoorFaultState, DoorPhysicalState, DoorTamperState, LockPhysicalState, DoubleLockPhysicalState

161.1.4. Examples

161.1.4.1. Checking status

Checking the current status of all event and alarm inputs/outputs

The **check** submenu requests the current status once.

The status is represented with the value True (for activated) or False (for deactivated).

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
POS.0.KeywordMatch=False
Channel.0.Videoloss=True
Channel.0.Connected=True
Channel.0.AudioDetection=False
Channel.0.NetworkCameraConnect=False
Channel.0.NetworkAlarmInput=False
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.1.Level=0
Channel.0.FaceDetection=False
Channel.0.DefocusDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappearing=False
Channel.0.ShockDetection=False
Channel.0.TemperatureChangeDetection.RegionID.1=False
Channel.0.CallRequest=False
Channel.0.TamperingSwitch=False
Channel.0.DTMFReceived=False
Channel.0.ProximitySensor=False
Channel.0.ObjectDetection=False
Channel.0.ObjectDetection.Person=False
Channel.0.ObjectDetection.Vehicle=False
```

```
Channel.0.ObjectDetection.Face=False
Channel.0.ObjectDetection.LicensePlate=False
Channel.0.ObjectDetection.Detail.Vehicle.Types.Bicycle=False
Channel.0.ObjectDetection.Detail.Vehicle.Types.Car=False
Channel.0.ObjectDetection.Detail.Vehicle.Types.Motorcycle=False
Channel.0.ObjectDetection.Detail.Vehicle.Types.Bus=False
Channel.0.ObjectDetection.Detail.Vehicle.Types.Truck=False
Channel.0.SocialDistancingViolation=False
Channel.0.AMDStart=False
Channel.0.LowFps=False
Channel.0.Tampering=False
Channel.0.SDFail=False
Channel.0.SDFull=False
Channel.0.Tracking=False
FireAlarmInput.1=False
AlarmInput.1=False
AlarmOutput.1=False
AudioOutput.1=False
SystemEvent.CPUFanError=False
SystemEvent.LeftFanError=False
SystemEvent.RightFanError=False
SystemEvent.PowerOn=True
SystemEvent.PowerReboot=False
SystemEvent.ConfigChange=False
SystemEvent.Backup=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.HDDFull=False
SystemEvent.HDDFail=False
SystemEvent.HDDNone=False
SystemEvent.Recording=True
SystemEvent.TimeChange=False
SystemEvent.Network.1=True
SystemEvent.Network.2=True
SystemEvent.Network.3=False
SystemEvent.Network.4=False
SystemEvent.InternalHDDErase=False
SystemEvent.AdminLogin=True
SystemEvent.RecordFiltering=False
SystemEvent.NetCamTrafficOverFlow=False
SystemEvent.RecordingError=False
```

```
SystemEvent.OverwriteDecoding=False
SystemEvent.EndofFrame=False
SystemEvent.RAIDEnable.1=False
SystemEvent.RAIDSetup.1=False
SystemEvent.RAIDBuilding.1=False
SystemEvent.RAIDBuildCancel.1=False
SystemEvent.RAIDBuildFail.1=False
SystemEvent.RAIDDegrade.1=False
SystemEvent.RAIDRebuildStart.1=False
SystemEvent.RAIDFail.1=False
SystemEvent.RAIDRebuildFail.1=False
SystemEvent.RAIDEnable.2=False
SystemEvent.RAIDSetup.2=False
SystemEvent.RAIDBuilding.2=False
SystemEvent.RAIDBuildCancel.2=False
SystemEvent.RAIDBuildFail.2=False
SystemEvent.RAIDDegrade.2=False
SystemEvent.RAIDRebuildStart.2=False
SystemEvent.RAIDFail.2=False
SystemEvent.RAIDRebuildFail.2=False
SystemEvent.RAIDDeviceAdd=False
SystemEvent.RAIDRecordRestriction=False
SystemEvent.iSCSIDisconnect=False
SystemEvent.BeingUpdate=False
SystemEvent.InternalHDDConnect.1=False
SystemEvent.BatteryFail=False
SystemEvent.RecordFrameDrop=False
SystemEvent.DualSMPSFail=True
SystemEvent.USBHDDConnect=False
SystemEvent.DSPDisplayStart=True
SystemEvent.DSPVASystemStart=False
SystemEvent.AMDLoadFail=False
SystemEvent.AlarmReset=False
SystemEvent.NewFWAvailable=False
SystemEvent.PasswordChange=False
SystemEvent.ConfigRestore=False
SystemEvent.CaseTampering=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "POSEvent": [
        {
            "POS": 0,
            "KeywordMatch": false
        }
    ],
    "ChannelEvent": [
        {
            "Channel": 0,
            "Videoloss": true,
            "AudioDetection": false,
            "NetworkCameraConnect": false,
            "NetworkAlarmInput": false,
            "MotionDetection": false,
            "MotionDetectionRegions": {
                "1": false
            },
            "MotionDetectionRegionsLevel": {
                "1": 0
            },
            "FaceDetection": false,
            "DefocusDetection": false,
            "ShockDetection": false,
            "CallRequest": false,
            "TamperingSwitch": false,
            "DTMFReceived": false,
            "ProximitySensor": false,
            "VideoAnalytics": {
                "Passing": false,
                "Entering": false,
                "Exiting": false,
                "Appearing": false,
                "Disappearing": false
            },
            "AMDStart": false,
            "LowFps": false,
```

```
            "Tampering": false,
            "SDFail": false,
            "SDFull": false,
            "Tracking": false,
            "TemperatureChangeDetection": false,
            "TemperatureChangeDetectionRegions": {
                "1": false
            },
            "SocialDistancingViolation": {
                "SocialDistancingViolation": false,
                "ObjectIDs": []
            },
            "ObjectDetection": {
                "ObjectDetection": false,
                "ObjectTypes": {
                    "Person": false,
                    "Vehicle": false,
                    "Face": false,
                    "LicensePlate": false
                },
                "ObjectTypeDetails": {
                    "Vehicle.Types.Bicycle": false,
                    "Vehicle.Types.Car": false,
                    "Vehicle.Types.Motorcycle": false,
                    "Vehicle.Types.Bus": false,
                    "Vehicle.Types.Truck": false
                }
            }
        }
    ],
    "FireAlarmInput": {
        "1": false
    },
    "AlarmInput": {
        "1": false
    },
    "AlarmOutput": {
        "1": false
    },
    "AudioOutput": {
        "1": false
```

```
    },
    "SystemEvent": {
        "CPUFanError": false,
        "LeftFanError": false,
        "RightFanError": false,
        "PowerOn": true,
        "PowerReboot": false,
        "ConfigChange": false,
        "Backup": false,
        "FWUpdate": false,
        "FactoryReset": false,
        "HDDFull": false,
        "HDDFail": false,
        "HDDNone": false,
        "Recording": true,
        "TimeChange": false,
        "Network": {
            "1": true,
            "2": true,
            "3": false,
            "4": false
        },
        "InternalHDDErase": false,
        "AdminLogin": true,
        "RecordFiltering": false,
        "NetCamTrafficOverFlow": false,
        "RecordingError": false,
        "OverwriteDecoding": false,
        "EndofFrame": false,
        "RAIDEvents": [
            {
                "RAID": 1,
                "RAIDEnable": false,
                "RAIDSetup": false,
                "RAIDBuilding": false,
                "RAIDBuildCancel": false,
                "RAIDBuildFail": false,
                "RAIDDegrade": false,
                "RAIDRebuildStart": false,
                "RAIDFail": false,
                "RAIDRebuildFail": false
```

```
            }
        ],
        "RAIDDeviceAdd": false,
        "RAIDRecordRestriction": false,
        "iSCSIDisconnect": false,
        "BeingUpdate": false,
        "InternalHDDConnect": {
            "1": false
        },
        "BatteryFail": false,
        "RecordFrameDrop": false,
        "DualSMPSFail": true,
        "USBHDDConnect": false,
        "DSPDisplayStart": true,
        "DSPVASystemStart": false,
        "AMDLoadFail": false,
        "AlarmReset": false,
        "NewFWAvailable": false,
        "PasswordChange": false,
        "ConfigRestore": false,
        "CaseTampering": false
    }
}
```


Checking the status of Alarm Input 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&AlarmInput=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AlarmInput": {
        "1": false
    }
}
```


Checking whether motion detection on Channel 0 is activated or deactivated

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&Channel.0.EventType=Mo
tionDetection
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.MotionDetection=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```


```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": false
```

```
        }
    ]
}
```

Checking whether video analytics on Channel 0 is activated or deactivated

Unlike other event types, VideoAnalytics produces the status of related events.

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&Channel.0.EventType=Vi
deoAnalytics
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappering=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "VideoAnalytics": {
                "Passing": false,
                "Entering": false,
                "Exiting": false,
```

```
                "Appearing": false,
                "Disappering": false
            }
        }
    ]
}
```

161.1.4.2. Monitoring status

The **monitor** submenu requests the current event status on a certain time interval (specified with the **Periodicity** parameter), irrespective of whether the event occurred or not (state changed or not).

The status is represented with the value True (for activated) or False (for deactivated).

Continuously monitoring the status of face detection events on Channel 0 and the alarm input

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitor&Channel.0.EventType=FaceDetection&AlarmInput=1
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```

```
--<MultiPartBoundary>
Content-Type: text/plain

Channel.0.FaceDetection=False
AlarmInput.1=False

--<MultiPartBoundary>
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```

```
--<MultiPartBoundary>

Content-Type: application/json
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "FaceDetection": false
        }
    ],
    "AlarmInput": {
        "1": false
    }
}
--<MultiPartBoundary>
```

Checking for any settings changed in the camera

If there have been any changes in settings, ChangedConfigURI shows the CGI URL of the changed settings. If AttributeUpdate shows as True, it indicates that the device’s attribute has changed.

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitor
```

TEXT RESPONSE

```
HTTP/1.1 200 OK

Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>

<Body>
```

```
--<MultiPartBoundary>

Content-Type: text/plain
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.1.Level=0
Channel.0.Tampering=False
```

```
Channel.0.DefocusDetection=False
Channel.0.Profile.1.DigitalAutoTracking=False
Channel.0.Profile.2.DigitalAutoTracking=False
Channel.0.Profile.3.DigitalAutoTracking=False
Channel.0.Profile.4.DigitalAutoTracking=False
Channel.0.Profile.5.DigitalAutoTracking=False
Channel.0.Profile.6.DigitalAutoTracking=False
Channel.0.Profile.7.DigitalAutoTracking=False
Channel.0.Profile.8.DigitalAutoTracking=False
Channel.0.Profile.9.DigitalAutoTracking=False
Channel.0.Profile.10.DigitalAutoTracking=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappering=False
Channel.0.ShockDetection=False
Channel.0.TemperatureChangeDetection.RegionID.1=False
FireAlarmInput.1=False
AlarmInput.1=False
AlarmOutput.1=False
SystemEvent.PowerReboot=False
SystemEvent.TimeChange=False
SystemEvent.ConfigChange=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=False
SystemEvent.NASFormat=False
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.CaseTampering=True
--<MultiPartBoundary>
```

```
Content-Type: text/plain
ChangedConfigURI=eventsources.cgi?msubmenu=thermaldetectionmode
SystemEvent.ConfigChange=True
AttributeUpdate=True

--<MultiPartBoundary>

Content-Type: text/plain
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```

```
--<MultiPartBoundary>

Content-Type: application/json
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": false,
            "MotionDetectionRegions": {
                "1": false
            },
            "MotionDetectionRegionsLevel": {
                "1": 0
            },
            "Tampering": false,
            "DefocusDetection": false,
            "DigitalAutoTracking": {
                "Profiles": {
                    "1": false,
                    "2": false,
                    "3": false,
                    "4": false,
                    "5": false,
                    "6": false,
                    "7": false,
                    "8": false,
```

```
                "9": false,
                "10": false
            }
        },
        "AudioDetection": false,
        "ShockDetection": false,
        "VideoAnalytics": {
            "Passing": false,
            "Entering": false,
            "Exiting": false,
            "Appearing": false,
            "Disappering": false
        },
        "TemperatureChangeDetection": false,
        "TemperatureChangeDetectionRegions": {
            "1": false
        }
    }
],
"FireAlarmInput": {
    "1": false
},
"AlarmInput": {
    "1": false
},
"AlarmOutput": {
    "1": false
},
"SystemEvent": {
    "PowerReboot": false,
    "TimeChange": false,
    "ConfigChange": false,
    "FWUpdate": false,
    "FactoryReset": false,
    "ConfigurationBackup": false,
    "ConfigurationRestore": false,
    "SDFormat": false,
    "SDFail": false,
    "SDFull": false,
    "SDInsert": false,
    "SDRemove": false,
```

```
        "NASFormat": false,
        "NASFail": false,
        "NASFull": false,
        "NASConnect": false,
        "NASDisconnect": true,
        "CaseTampering": true
    }
}
--<MultiPartBoundary>

Content-Type: application/json

{
    "ChangedConfigURI": "eventsources.cgi?msubmenu=thermaldetectionmode",
    "SystemEvent": {
        "ConfigChange": true,
        "AttributeUpdate": true
    }
}

--<MultiPartBoundary>

Content-Type: application/json
```

161.1.4.3. Requesting changed events

The **monitordiff** submenu initially requests the status of all events and later makes requests only when an event occurs (change from previous state), which is different from the **monitor** submenu requesting the event status periodically.

The status is represented with the value True (for activated) or False (for deactivated).

Requesting changed status motion detection events for Channel 0 only

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff&Channel.0.EventType=MotionDetection
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
```

```
<Body>
```

```
--<MultiPartBoundary>

Content-Type: text/plain
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.1.Level=0

--<MultiPartBoundary>
```

JSON RESPONSE

```
HTTP/1.1 200 OK

Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>

<Body>
```

```
--<MultiPartBoundary>

Content-Type: application/json
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": false,
            "MotionDetectionRegions": {
                "1": false
            },
            "MotionDetectionRegionsLevel": {
                "1": 0
            }
        }
    ]
}
--<MultiPartBoundary>
```

Requesting all changed events

If there have been any changes in settings, ChangedConfigURI shows the CGI URL of the changed settings.

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff
```

TEXT RESPONSE

```
HTTP/1.1 200 OK

Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>

<Body>
```

```
--<MultiPartBoundary>

Content-Type: text/plain

Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.1.Level=0
Channel.0.Tampering=False
Channel.0.DefocusDetection=False
Channel.0.Profile.1.DigitalAutoTracking=False
Channel.0.Profile.2.DigitalAutoTracking=False
Channel.0.Profile.3.DigitalAutoTracking=False
Channel.0.Profile.4.DigitalAutoTracking=False
Channel.0.Profile.5.DigitalAutoTracking=False
Channel.0.Profile.6.DigitalAutoTracking=False
Channel.0.Profile.7.DigitalAutoTracking=False
Channel.0.Profile.8.DigitalAutoTracking=False
Channel.0.Profile.9.DigitalAutoTracking=False
Channel.0.Profile.10.DigitalAutoTracking=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappering=False
```

```
Channel.0.ShockDetection=False
Channel.0.TemperatureChangeDetection.RegionID.1=False
FireAlarmInput.1=False
AlarmInput.1=False
AlarmOutput.1=False
SystemEvent.PowerReboot=False
SystemEvent.TimeChange=False
SystemEvent.ConfigChange=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=False
SystemEvent.NASFormat=False
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.CaseTampering=True
--<MultiPartBoundary>
Content-Type: text/plain
ChangedConfigURI=media.cgi?msubmenu=videoprofile
SystemEvent.ConfigChange=True

--<MultiPartBoundary>
```

JSON RESPONSE

```
HTTP/1.1 200 OK

Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>

<Body>
```

```
--<MultiPartBoundary>

Content-Type: application/json
```

```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": false,
            "MotionDetectionRegions": {
                "1": false
            },
            "MotionDetectionRegionsLevel": {
                "1": 0
            },
            "Tampering": false,
            "DefocusDetection": false,
            "DigitalAutoTracking": {
                "Profiles": {
                    "1": false,
                    "2": false,
                    "3": false,
                    "4": false,
                    "5": false,
                    "6": false,
                    "7": false,
                    "8": false,
                    "9": false,
                    "10": false
                }
            },
            "AudioDetection": false,
            "ShockDetection": false,
            "VideoAnalytics": {
                "Passing": false,
                "Entering": false,
                "Exiting": false,
                "Appearing": false,
                "Disappering": false
            },
            "TemperatureChangeDetection": false,
            "TemperatureChangeDetectionRegions": {
                "1": false
            }
        }
```

```
    ],
    "FireAlarmInput": {
        "1": false
    },
    "AlarmInput": {
        "1": false
    },
    "AlarmOutput": {
        "1": false
    },
    "SystemEvent": {
        "PowerReboot": false,
        "TimeChange": false,
        "ConfigChange": false,
        "FWUpdate": false,
        "FactoryReset": false,
        "ConfigurationBackup": false,
        "ConfigurationRestore": false,
        "SDFormat": false,
        "SDFail": false,
        "SDFull": false,
        "SDInsert": false,
        "SDRemove": false,
        "NASFormat": false,
        "NASFail": false,
        "NASFull": false,
        "NASConnect": false,
        "NASDisconnect": true,
        "CaseTampering": true
    }
}
--<MultiPartBoundary>

Content-Type: application/json

{
    "ChangedConfigURI": "media.cgi?msubmenu=videoprofile",
    "SystemEvent": {
        "ConfigChange": true
    }
}
```

```
--<MultiPartBoundary>
```

Requesting changed status for motion detection events with SchemaBased

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff&SchemaBased=True
```

If there is an OpenSDK event, the event type & meta type response is as follows:

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```


```
--<MultiPartBoundary>
{
    "EventStatus": [
        {
            "EventName": "OpenSDK",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "Channel": 0,
                "AppName": "test_dynamicEvent",
                "AppID": "test_dynamicEvent",
                "AppEvent": "LicensePlateNumber",
                "Type": "Event"
            },
            "Data": {
                "State": true
            }
        },
        {
            "EventName": "ObjectDetection",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "Channel": 0
            },
```

```
            "Data": {
                "State": true,
                "ObjectTypes": "Face, Vehicle",
                "ObjectTypeDetails":
"Vehicle.Types.Bicycle,Vehicle.Types.Truck"
            }
        },
        {
            "EventName": "SocialDistancingViolation",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false,
                "ObjectIDs": []
            }
        }
    ]
}

--<MultiPartBoundary>

Content-type:application/json; charset=utf-8

{
    "EventStatus": [
        {
            "EventName": "OpenSDK",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "Channel": 0,
                "AppName": "test_dynamicEvent",
                "AppID": "test_dynamicEvent",
                "AppEvent": "LicensePlateNumber",
                "Type": "Meta"
            },
            "Data": {
                "Info": "<tt:Message><tt:Source><tt:SimpleItem
Name=\"VideoSourceToken\" Value=\"0\"/></tt:Source><tt:Data><tt:SimpleItem
Name=\"LicensePlateNumber\" Value=\"ABC-1234\"/></tt:Data></tt:Message>"
```

```
            }
        }
    ]
}
--<MultiPartBoundary>
```

161.1.4.4. Requesting schema based events response

All events have schema and for this schema based request SUNAPI can show the schema based response.

If you want to know which schema it is, please refer to "5.5 EventStatusSchema".

Requesting Schema based response

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&SchemaBased=True
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```

```
FireAlarmInput.1=False
AlarmInput.1=False
AlarmOutput.1=False
AlarmOutput.2=False
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.1.Level=0
Channel.0.Tampering=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Intrusion=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Loitering=False
Channel.0.AudioAnalytics.Scream=False
Channel.0.AudioAnalytics.Gunshot=False
```

```
Channel.0.AudioAnalytics.Explosion=False
Channel.0.AudioAnalytics.GlassBreak=False
Channel.0.BoxTemperatureDetection=False
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=True
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
SystemEvent.CaseTampering=False
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```

```
{
    "EventStatus": [
        {
            "EventName": "FireAlarmInput",
            "Time": "2019-01-22T22:57:41.870+00:00",
            "Source": {
                "SourceID": 0
            },
            "Data": {
                "State": false
            }
        },
        {
```

```
            "EventName": "AlarmInput",
            "Time": "2019-01-22T22:57:41.870+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false,
                "SupervisedStatus": Normal
            }
        },
        {
            "EventName": "AlarmOutput",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AlarmOutput",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "MotionDetection",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0,
                "ROIID": 1
            },
            "Data": {
                "State": false,
                "Level": 0
            }
```

```
        },
        {
            "EventName": "Tampering",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioDetection",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.Scream",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.Gunshot",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
```

```
        {
            "EventName": "AudioAnalytics.Explosion",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.GlassBreak",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "ShockDetection",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "BoxTemperatureDetection",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0,
                "ROIID": 0
            },
            "Data": {
                "State": false
            }
        },
```

```
        {
            "EventName": "SocialDistancingViolation",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": true,
                "ObjectIDs": [
                    223,
                    222,
                    333
                ]
            }
        },
        {
            "EventName": "OpenSDKAppStatus",
            "Time": "2020-04-21T14:20:59.336+00:00",
            "Source": {
                "AppID": "WiseAI"
            },
            "Data": {
                "Status": "Active",
                "Description": "OPENSDK_APP_RUNNING"
            }
        },
        {
            "EventName": "OpenSDK",
            "Time": "2000-01-01T01:35:35.254+00:00",
            "Source": {
                "Channel": 0,
                "AppName": "test_dynamicEvent",
                "AppID": "test_dynamicEvent",
                "AppEvent": "LicensePlateNumber",
                "Type": "Event"
            },
            "Data": {
                "State": true
            }
        },
        {
```

```
            "EventName": "SystemEvent.TimeChange",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.PowerReboot",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.FWUpdate",
            "Time": "2019-01-22T22:57:41.871+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.FactoryReset",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.ConfigurationBackup",
```

```
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.ConfigurationRestore",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.ConfigChange",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.SDFormat",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.SDFail",
            "Time": "2019-01-22T22:57:41.872+00:00",
```

```
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.SDFull",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.SDInsert",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.SDRemove",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": true
            }
        },
        {
            "EventName": "SystemEvent.NASConnect",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
```

```
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.NASDisconnect",
        "Time": "2019-01-22T22:57:41.872+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": true
        }
    },
    {
        "EventName": "SystemEvent.NASFail",
        "Time": "2019-01-22T22:57:41.872+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.NASFull",
        "Time": "2019-01-22T22:57:41.872+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.NASFormat",
        "Time": "2019-01-22T22:57:41.872+00:00",
        "Source": {
            "Channel": 0
```

```
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.CaseTampering",
            "Time": "2019-01-22T22:57:41.872+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        }
    ]
}
```

161.2. Push Notification

161.2.1. Description

The **pushnotification** submenu is used to configure push notification feature in NVR.

NOTE This chapter applies to the NVR only.

Access level

Action	NVR
view	User
set	User

161.2.2. Syntax

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=
pushnotification&action=<value>[&<parameter>=<value>]
```

161.2.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads current pushnotification settings
set	Enable	REQ, RES	<bool> True, False	Enables or disables push notifications

161.2.4. Examples

161.2.4.1. Getting the current pushnotification settings

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=pushnotification&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{

"Enable":true

}
```

161.2.4.2. Setting the current pushnotification settings

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/eventstatus.cgi?msubmenu=pushnotification&action=set&Enable=False
```

161.3. ONVIF Event Topic

161.3.1. Description

This **eventscheme** submenu is used to choose whether old or new topics should be used in the ONVIF event service.

> **NOTE** This submenu is applicable for network camera & encoder only. Based on this setting, the onvif event service **GetEventProperties** response will change.

Access level

Action	Camera	Encoder
view	Admin	Admin
set	Admin	Admin

161.3.2. Syntax

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=
eventscheme&action=<value>[&<parameter>=<value>]
```

161.3.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads current ONVIF event topic settings
set	Type	REQ, RES	<enum> Proprietary, ONVIF	When set to Proprietary it will follow the old topic set; when set to ONVIF the new standard topics set will be used.

161.3.4. Examples

161.3.4.1. Getting the current eventscheme

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventscheme&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{

"Type": "Proprietary"

}
```

161.3.4.2. Setting the eventscheme

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventscheme&action=set&Type=ONVIF
```

161.4. Metadataschema

161.4.1. Description

This **metadataschema** submenu provides schema of metadata events generated from the camera when an event occurs.

NOTE This submenu is applicable for network camera.

Access level

Action	Camera
view	Admin

161.4.2. Syntax

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=
metadataschema&action=<value>[&<parameter>=<value>]
```

161.4.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads current metadata schema settings
	EventName	REQ, RES	<enum> AlarmInput, AlarmOutput, MotionDetection, VideoAnalytics, VideoLoss, FaceDetection, AudioDetection, AudioAnalytics, Tampering, DefocusDetection, ShockDetection, TemperatureChangeDetection, FogDetection, Tracking, BoxTemperatureDetection, ObjectDetection, MaskDetection, BodyTemperatureDetection, ParkingDetection, SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor, AllParkingSpotOccupied, ParkingSpotOccupied, SIPCall, AudioClipPlayback, TwoWayTalk	Shows the event name’s metadata schema.
	EventTopic	RES	<string>	ONVIF event topic associated with the Event
	EventSchema	RES	<string>	ONVIF event schema

161.4.4. Examples

161.4.4.1. Getting the ONVIF metadata eventschema

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=metadataschema&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MetaDataSchema": [
        {
            "EventName": "FireAlarmInput",
            "EventTopic": "tns1:Device/tns1:Trigger/FireAlarmInput",
            "EventSchema": "<tns1:Device><tns1:Trigger><FireAlarmInput
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"SourceID\"
Type=\"xsd:int\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\"
Type=\"xsd:boolean\"/></tt:Data></tt:MessageDescription></FireAlarmInput></t
ns1:Trigger></tns1:Device>"
        },
        {
            "EventName": "AlarmInput",
            "EventTopic":
"tns1:Device/tns1:Trigger/tnssamsung:DigitalInput",
            "EventSchema":
"<tns1:Device><tns1:Trigger><tnssamsung:DigitalInput
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"Index\"
Type=\"xsd:int\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"Level\"
Type=\"xsd:int\"/></tt:Data></tt:MessageDescription></tnssamsung:DigitalInpu
t></tns1:Trigger></tns1:Device>"
        },
        {
            "EventName": "AlarmOutput",
            "EventTopic": "tns1:Device/tns1:Trigger/tns1:Relay",
```

```
            "EventSchema": "<tns1:Device><tns1:Trigger><tns1:Relay
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"RelayToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"RelayLogicalState\"
Type=\"xsd:int\"/></tt:Data></tt:MessageDescription></tns1:Relay></tns1:Trig
ger></tns1:Device>"
        },
        {
            "EventName": "MotionDetection",
            "EventTopic": "tns1:VideoAnalytics/tnssamsung:MotionDetection",
            "EventSchema": "<tns1:VideoAnalytics><tnssamsung:MotionDetection
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"Window\"
Type=\"xsd:int\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"Motion\"
Type=\"xsd:int\"/></tt:Data></tt:MessageDescription></tnssamsung:MotionDetec
tion></tns1:VideoAnalytics>"
        },
        {
            "EventName": "VideoAnalytics",
            "EventTopic": "tns1:VideoAnalytics/tnssamsung:VideoAnalytics",
            "EventSchema": "<tns1:VideoAnalytics><tnssamsung:VideoAnalytics
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSourceToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\" Type=\"xsd:int\"/><tt:SimpleItemDescription Name=\"Action\"
Type=\"xsd:string\"/></tt:Data></tt:MessageDescription></tnssamsung:VideoAna
lytics></tns1:VideoAnalytics>"
        },
        {
            "EventName": "AudioDetection",
            "EventTopic": "tns1:AudioSource/tnssamsung:AudioDetection",
            "EventSchema": "<tns1:AudioSource><tnssamsung:AudioDetection
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"AudioSourceToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\"
Type=\"xsd:int\"/></tt:Data></tt:MessageDescription></tnssamsung:AudioDetect
ion></tns1:AudioSource>"
        },
```

```
        {
            "EventName": "AudioAnalytics",
            "EventTopic": "tns1:AudioAnalytics/Audio/DetectedSound",
            "EventSchema": "<tns1:AudioAnalytics><Audio><DetectedSound
wstop:topic=\"true\"><tt:MessageDescription
IsProperty=\"false\"><tt:Source><tt:SimpleItemDescription
Name=\"AudioSourceConfigurationToken\"
Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription
Name=\"AudioAnalyticsConfigurationToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"isSoundDetected\" Type=\"xsd:boolean\"/><tt:SimpleItemDescription
Name=\"UTCTime\" Type=\"xsd:dateTime\"/><tt:ElementItemDescription
Name=\"AudioClassifications\"
Type=\"tt:AudioClassDescriptor\"/></tt:Data></tt:MessageDescription></Detect
edSound></Audio></tns1:AudioAnalytics>"
        },
        {
            "EventName": "Tampering",
            "EventTopic": "tns1:VideoSource/tnssamsung:TamperingDetection",
            "EventSchema": "<tns1:VideoSource><tnssamsung:TamperingDetection
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSourceToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"Tampering\"
Type=\"xsd:int\"/></tt:Data></tt:MessageDescription></tnssamsung:TamperingDe
tection></tns1:VideoSource>"
        },
        {
            "EventName": "ShockDetection",
            "EventTopic": "tns1:VideoSource/tnssamsung:ShockDetection",
            "EventSchema": "<tns1:VideoSource><tnssamsung:ShockDetection
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSource\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\"
Type=\"xsd:boolean\"/></tt:Data></tt:MessageDescription></tnssamsung:ShockDe
tection></tns1:VideoSource>"
        },
        {
            "EventName": "ObjectDetection",
            "EventTopic": "tns1:RuleEngine/ObjectDetection/Object",
```

```
            "EventSchema": "<tns1:RuleEngine><ObjectDetection><Object
wstop:topic=\"true\"><tt:MessageDescription
IsProperty=\"true\"><tt:Source><tt:SimpleItemDescription
Name=\"VideoSource\" Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription
Name=\"RuleName\"
Type=\"xsd:string\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"ClassTypes\"
Type=\"xsd:string\"/></tt:Data></tt:MessageDescription></Object></ObjectDete
ction></tns1:RuleEngine>"
        },
        {
            "EventName": "SocialDistancingViolation",
            "EventTopic":
"tns1:RuleEngine/Detection/SocialDistancingViolation",
            "EventSchema":
"<tns1:RuleEngine><Detection><SocialDistancingViolation
wstop:topic=\"true\"><tt:MessageDescription
><tt:Source><tt:SimpleItemDescription Name=\"VideoSource\"
Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription Name=\"RuleName\"
Type=\"xsd:string\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\" Type=\"xsd:boolean\"/> <tt:SimpleItemDescription
Name=\"ObjectIDs\"
Type=\"tt:IntAttrList\"/></tt:Data></tt:MessageDescription></SocialDistancin
gViolation></Detection></tns1:RuleEngine>"
        },
        {
            "EventName": "BoxTemperatureDetection",
            "EventTopic":
"tns1:VideoAnalytics/Radiometry/BoxTemperatureReading",
            "EventSchema":
"<tns1:VideoAnalytics><Radiometry><BoxTemperatureReading
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSourceToken\"
Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription
Name=\"VideoAnalyticsConfigurationToken\"
Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription
Name=\"AnalyticsModuleName\"
Type=\"xsd:string\"/></tt:Source><tt:Data><tt:ElementItemDescription
Name=\"Reading\"
Type=\"ttr:BoxTemperatureReading\"/><tt:SimpleItemDescription
Name=\"TimeStamp\"
```

```
Type=\"xsd:dateTime\"/></tt:Data></tt:MessageDescription></BoxTemperatureRea
ding></Radiometry></tns1:VideoAnalytics>"
        },
        {
            "EventName": "MotionDetection",
            "EventTopic": "tns1:VideoSource/MotionAlarm",
            "EventSchema": "<tns1:VideoSource><MotionAlarm
wstop:topic=\"true\"><tt:MessageDescription
IsProperty=\"true\"><tt:Source><tt:SimpleItemDescription Name=\"Source\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"State\"
Type=\"xsd:boolean\"/></tt:Data></tt:MessageDescription></MotionAlarm></tns1
:VideoSource>"
        },
        {
            "EventName": "BoxTemperatureDetection",
            "EventTopic": "tns1:RuleEngine/Radiometry/BoxTemperatureAlarm",
            "EventSchema":
"<tns1:RuleEngine><Radiometry><BoxTemperatureAlarm
wstop:topic=\"true\"><tt:MessageDescription
IsProperty=\"true\"><tt:Source><tt:SimpleItemDescription
Name=\"VideoSourceConfigurationToken\"
Type=\"tt:ReferenceToken\"/><tt:SimpleItemDescription Name=\"RuleName\"
Type=\"xsd:string\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"AlarmActive\"
Type=\"xsd:boolean\"/></tt:Data></tt:MessageDescription></BoxTemperatureAlar
m></Radiometry></tns1:RuleEngine>"
        }
    ]
}
```

161.5. Event Status Schema

161.5.1. Description

The **eventstatusschema** submenu provides the schema definitions for all available events. These schemas are utilized in the **eventstatus** submenu when using `SchemaBased` parameter.

Events supported by the installed OpenApp will also be included in the response. These events follow a specific naming convention:

```
OpenSDK.{AppID}.{AppEventName}
```

- AppID is the unique identifier for the App instance. It can be retrieved from the opensdk.cgi app submenu response. In most models, AppID matches the AppName, but in some cases, they may differ.
- AppEventName refers to the event name as defined by the App developer.

NOTE This submenu is applicable for network cameras and NVR only.

Access level

Action	Camera	NVR
view	Admin	User

161.5.2. Syntax

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=
eventstatusschema&action=<value>[&<parameter>=<value>]
```

161.5.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads current event status schema settings

Action	Parameter	Request/ Response	Type/ Value	Description
	EventName	REQ, RES	<enum> AlarmInput, AlarmOutput, MotionDetection, FaceDetection, VideoLoss, Tampering, AudioDetection, DefocusDetection, FogDetection, ShockDetection, Tracking, PTZMotion, DigitalAutoTracking, VideoAnalytics, AudioAnalytics, Queue, SystemEvent, ConfigChange, TemperatureChangeDetection, BoxTemperatureDetection, ObjectDetection, OpenSDK, BodyTemperatureDetection, ParkingDetection, SocialDistancingViolation, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor, MQTTSubscription, AllParkingSpotOccupied, ParkingSpotOccupied, Connected, ParkedVehicleCountChanged, OpenSDKAppStatus, MaskDetection, TwoWayTalk, SIPCall, AudioClipPlayback	This parameter can only show the requested value’s response.

Action	Parameter	Request/ Response	Type/ Value	Description
	Language	REQ	<enum>	Requests the translated value for event topics. **Note** This can be applicable only when **MultiLanguageEventSchema** is True in attributes. CAMERA ONLY

161.5.4. Examples

161.5.4.1. Getting the eventstatusschema

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatusschema&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
EventStatus.1.Name=AlarmInput
EventStatus.1.Schema.1.Name=AlarmInput.<int>
EventStatus.1.Schema.1.Value=<boolean>
EventStatus.2.Name=AlarmOutput
EventStatus.2.Schema.1.Name=AlarmOutput.<int>
EventStatus.2.Schema.1.Value=<boolean>
EventStatus.3.Name=MotionDetection
EventStatus.3.Schema.1.Name=Channel.<int>.MotionDetection
EventStatus.3.Schema.1.Value=<boolean>
EventStatus.3.Schema.2.Name=Channel.<int>.MotionDetection.RegionID.<int>
EventStatus.3.Schema.2.Value=<boolean>
EventStatus.3.Schema.3.Name=Channel.<int>.MotionDetection.RegionID.<int>.Level
EventStatus.3.Schema.3.Value=<int>
EventStatus.4.Name=Tampering
```

```
EventStatus.4.Schema.1.Name=Channel.<int>.Tampering
EventStatus.4.Schema.1.Value=<boolean>
EventStatus.5.Name=AudioDetection
EventStatus.5.Schema.1.Name=Channel.<int>.AudioDetection
EventStatus.5.Schema.1.Value=<boolean>
EventStatus.6.Name=VideoAnalytics
EventStatus.6.Schema.1.Name=Channel.<int>.VideoAnalytics.Passing
EventStatus.6.Schema.1.Value=<boolean>
EventStatus.6.Schema.2.Name=Channel.<int>.VideoAnalytics.Passing.LineID.<int
>
EventStatus.6.Schema.2.Value=<boolean>
EventStatus.6.Schema.3.Name=Channel.<int>.VideoAnalytics.Intrusion
EventStatus.6.Schema.3.Value=<boolean>
EventStatus.6.Schema.4.Name=Channel.<int>.VideoAnalytics.Intrusion.DefinedAr
eaID.<int>
EventStatus.6.Schema.4.Value=<boolean>
EventStatus.6.Schema.5.Name=Channel.<int>.VideoAnalytics.Entering
EventStatus.6.Schema.5.Value=<boolean>
EventStatus.6.Schema.6.Name=Channel.<int>.VideoAnalytics.Entering.DefinedAre
aID.<int>
EventStatus.6.Schema.6.Value=<boolean>
EventStatus.6.Schema.7.Name=Channel.<int>.VideoAnalytics.Exiting
EventStatus.6.Schema.7.Value=<boolean>
EventStatus.6.Schema.8.Name=Channel.<int>.VideoAnalytics.Exiting.DefinedArea
ID.<int>
EventStatus.6.Schema.8.Value=<boolean>
EventStatus.6.Schema.9.Name=Channel.<int>.VideoAnalytics.Appearing
EventStatus.6.Schema.9.Value=<boolean>
EventStatus.6.Schema.10.Name=Channel.<int>.VideoAnalytics.Appearing.DefinedA
reaID.<int>
EventStatus.6.Schema.10.Value=<boolean>
EventStatus.6.Schema.11.Name=Channel.<int>.VideoAnalytics.Loitering
EventStatus.6.Schema.11.Value=<boolean>
EventStatus.6.Schema.12.Name=Channel.<int>.VideoAnalytics.Loitering.DefinedA
reaID.<int>
EventStatus.6.Schema.12.Value=<boolean>
EventStatus.7.Name=AudioAnalytics
EventStatus.7.Schema.1.Name=Channel.<int>.AudioAnalytics.Scream
EventStatus.7.Schema.1.Value=<boolean>
EventStatus.7.Schema.2.Name=Channel.<int>.AudioAnalytics.Gunshot
EventStatus.7.Schema.2.Value=<boolean>
```

```
EventStatus.7.Schema.3.Name=Channel.<int>.AudioAnalytics.Explosion
EventStatus.7.Schema.3.Value=<boolean>
EventStatus.7.Schema.4.Name=Channel.<int>.AudioAnalytics.GlassBreak
EventStatus.7.Schema.4.Value=<boolean>
EventStatus.8.Name=SystemEvent
EventStatus.8.Schema.1.Name=SystemEvent.TimeChange
EventStatus.8.Schema.1.Value=<boolean>
EventStatus.8.Schema.2.Name=SystemEvent.PowerReboot
EventStatus.8.Schema.2.Value=<boolean>
EventStatus.8.Schema.3.Name=SystemEvent.FWUpdate
EventStatus.8.Schema.3.Value=<boolean>
EventStatus.8.Schema.4.Name=SystemEvent.FactoryReset
EventStatus.8.Schema.4.Value=<boolean>
EventStatus.8.Schema.5.Name=SystemEvent.ConfigurationBackup
EventStatus.8.Schema.5.Value=<boolean>
EventStatus.8.Schema.6.Name=SystemEvent.ConfigurationRestore
EventStatus.8.Schema.6.Value=<boolean>
EventStatus.8.Schema.7.Name=SystemEvent.ConfigChange
EventStatus.8.Schema.7.Value=<boolean>
EventStatus.8.Schema.8.Name=ChangedConfigURI
EventStatus.8.Schema.8.Value=<string>
EventStatus.8.Schema.9.Name=SystemEvent.SDFormat
EventStatus.8.Schema.9.Value=<boolean>
EventStatus.8.Schema.10.Name=SystemEvent.SDFail
EventStatus.8.Schema.10.Value=<boolean>
EventStatus.8.Schema.11.Name=SystemEvent.SDFull
EventStatus.8.Schema.11.Value=<boolean>
EventStatus.8.Schema.12.Name=SystemEvent.SDInsert
EventStatus.8.Schema.12.Value=<boolean>
EventStatus.8.Schema.13.Name=SystemEvent.SDRemove
EventStatus.8.Schema.13.Value=<boolean>
EventStatus.8.Schema.14.Name=SystemEvent.NASConnect
EventStatus.8.Schema.14.Value=<boolean>
EventStatus.8.Schema.15.Name=SystemEvent.NASDisconnect
EventStatus.8.Schema.15.Value=<boolean>
EventStatus.8.Schema.16.Name=SystemEvent.NASFail
EventStatus.8.Schema.16.Value=<boolean>
EventStatus.8.Schema.17.Name=SystemEvent.NASFull
EventStatus.8.Schema.17.Value=<boolean>
EventStatus.8.Schema.18.Name=SystemEvent.NASFormat
EventStatus.8.Schema.18.Value=<boolean>
```

```
EventStatus.8.Schema.19.Name=SystemEvent.CaseTampering
EventStatus.8.Schema.19.Value=<boolean>
EventStatus.9.Name=BoxTemperatureDetection
EventStatus.9.Schema.1.Name=Channel.<int>.BoxTemperatureDetection
EventStatus.9.Schema.1.Value=<boolean>
EventStatus.9.Schema.2.Name=Channel.<int>.BoxTemperatureDetection.RegionID.<
int>
EventStatus.9.Schema.2.Value=<boolean>
EventStatus.10.Name=ObjectDetection
EventStatus.10.Schema.1.Name=Channel.<int>.ObjectDetection
EventStatus.10.Schema.1.Value=<boolean>
EventStatus.10.Schema.2.Name=Channel.<int>.ObjectDetection.Person
EventStatus.10.Schema.2.Value=<boolean>
EventStatus.10.Schema.3.Name=Channel.<int>.ObjectDetection.Vehicle
EventStatus.10.Schema.3.Value=<boolean>
EventStatus.10.Schema.4.Name=Channel.<int>.ObjectDetection.Face
EventStatus.10.Schema.4.Value=<boolean>
EventStatus.10.Schema.5.Name=Channel.<int>.ObjectDetection.LicensePlate
EventStatus.10.Schema.5.Value=<boolean>
EventStatus.10.Schema.6.Name=Channel.<int>.ObjectDetection.Detail.Vehicle.Ty
pes.Bicycle
EventStatus.10.Schema.6.Value=<boolean>
EventStatus.10.Schema.7.Name=Channel.<int>.ObjectDetection.Detail.Vehicle.Ty
pes.Car
EventStatus.10.Schema.7.Value=<boolean>
EventStatus.10.Schema.8.Name=Channel.<int>.ObjectDetection.Detail.Vehicle.Ty
pes.Motorcycle
EventStatus.10.Schema.8.Value=<boolean>
EventStatus.10.Schema.9.Name=Channel.<int>.ObjectDetection.Detail.Vehicle.Ty
pes.Bus
EventStatus.10.Schema.9.Value=<boolean>
EventStatus.10.Schema.10.Name=Channel.<int>.ObjectDetection.Detail.Vehicle.T
ypes.Truck
EventStatus.10.Schema.10.Value=<boolean>
EventStatus.11.Schema.18.Name= Channel.<int>.BodyTemperatureDetection
EventStatus.11.Schema.18.Value=<boolean>
EventStatus.12.Name=SocialDistancingViolation
EventStatus.12.Schema.1.Name=Channel.<int>.SocialDistancingViolation
EventStatus.12.Schema.1.Value=<boolean>
EventStatus.12.Schema.2.Name=Channel.<int>.SocialDistancingViolation.ObjectI
Ds
```

```
EventStatus.12.Schema.2.Value=<csv>
EventStatus.13.Name=ParkingDetection
EventStatus.13.Schema.1.Name=Channel.<int>.ParkingDetection
EventStatus.13.Schema.1.Value=<boolean>
EventStatus.14.Name=FireAlarmInput
EventStatus.14.Schema.1.Name=FireAlarmInput.<int>
EventStatus.14.Schema.1.Value=<boolean>
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: multipart/x-mixed-replace; boundary=<MultiPartBoundary>
<Body>
```


```
{
    "type": "array",
    "items": [
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "AlarmInput",
                        "AlarmOutput"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        },
                        "SourceID": {
                            "type": "number"
                        }
                    }
                },
                "Data": {
```

```
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        },
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "Tampering",
                        "AudioDetection",
                        "BodyTemperatureDetection"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        },
        {
```

```
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "AudioAnalytics.Scream",
                        "AudioAnalytics.Gunshot",
                        "AudioAnalytics.Explosion",
                        "AudioAnalytics.GlassBreak"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        },
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "MotionDetection"
                    ]
```

```
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
                    },
                    "ROIID": {
                        "type": "number"
                    }
                }
            },
            "Data": {
                "type": "object",
                "properties": {
                    "State": {
                        "type": "boolean"
                    },
                    "Level": {
                        "type": "number"
                    }
                }
            }
        }
    },
    {
        "type": "object",
        "properties": {
            "Time": {
                "type": "string"
            },
            "EventName": {
                "enum": [
                    "BoxTemperatureDetection"
                ]
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
```

```
                        },
                        "ROIID": {
                            "type": "number"
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        },
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "ObjectDetection"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        },
```

```
                    "ObjectTypes": {
                        "type": "string"
                    },
                    "ObjectTypeDetails": {
                        "type": "string"
                    }
                }
            }
        }
    },
    {
        "type": "object",
        "properties": {
            "Time": {
                "type": "string"
            },
            "EventName": {
                "enum": [
                    "VideoAnalytics.Intrusion",
                    "VideoAnalytics.Entering",
                    "VideoAnalytics.Exiting",
                    "VideoAnalytics.Appearing",
                    "VideoAnalytics.Loitering"
                ]
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
                    },
                    "DefinedAreaID": {
                        "type": "number"
                    }
                }
            },
            "Data": {
                "type": "object",
                "properties": {
                    "State": {
                        "type": "boolean"
```

```
                    }
                }
            }
        }
    },
    {
        "type": "object",
        "properties": {
            "Time": {
                "type": "string"
            },
            "EventName": {
                "enum": [
                    "VideoAnalytics.Passing"
                ]
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
                    },
                    "LineID": {
                        "type": "number"
                    }
                }
            },
            "Data": {
                "type": "object",
                "properties": {
                    "State": {
                        "type": "boolean"
                    }
                }
            }
        }
    },
    {
        "type": "object",
        "properties": {
            "Time": {
```

```
                "type": "string"
            },
            "EventName": {
                "enum": [
                    "SystemEvent.TimeChange",
                    "SystemEvent.PowerReboot",
                    "SystemEvent.FWUpdate",
                    "SystemEvent.FactoryReset",
                    "SystemEvent.ConfigurationBackup",
                    "SystemEvent.ConfigurationRestore",
                    "SystemEvent.ConfigChange",
                    "SystemEvent.SDFormat",
                    "SystemEvent.SDFail",
                    "SystemEvent.SDFull",
                    "SystemEvent.SDInsert",
                    "SystemEvent.SDRemove",
                    "SystemEvent.NASConnect",
                    "SystemEvent.NASDisconnect",
                    "SystemEvent.NASFail",
                    "SystemEvent.NASFull",
                    "SystemEvent.NASFormat"
                ]
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
                    }
                }
            },
            "Data": {
                "type": "object",
                "properties": {
                    "State": {
                        "type": "boolean"
                    }
                }
            }
        }
    },
```

```
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "ConfigChange"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        },
                        "ChangedConfigURI": {
                            "type": "string"
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        },
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "OpenSDK"
```

```
                ]
            },
            "Source": {
                "type": "object",
                "properties": {
                    "Channel": {
                        "type": "number"
                    },
                    "AppName": {
                        "type": "string"
                    },
                    "AppEvent": {
                        "type": "string"
                    },
                    "AppID": {
                        "type": "string"
                    },
                    "Type": {
                        "enum": [
                            "Event"
                        ]
                    }
                }
            },
            "Data": {
                "type": "object",
                "properties": {
                    "State": {
                        "type": "boolean"
                    }
                }
            }
        }
    },
    {
        "type": "object",
        "properties": {
            "Time": {
                "type": "string"
            },
            "EventName": {
```

```
                    "enum": [
                        "OpenSDK"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        },
                        "AppName": {
                            "type": "string"
                        },
                        "AppEvent": {
                            "type": "string"
                        },
                        "Type": {
                            "enum": [
                                "Meta"
                            ]
                        }
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "Info": {
                            "type": "string"
                        }
                    }
                }
            }
        }
    ]
}
```

Recording

SUNAPI

v2.6.8
2026-04-09

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Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 162. Overview

162.1. Description

This document explains recording.cgi.

recording.cgi configures video recording settings to enable video recording at scheduled times or upon event occurrence. It also provides submenus for configuring storage and searching recorded videos.

The following submenus of recording.cgi are used:

- **storage**: Sets the storage configuration, including whether to overwrite or auto-delete.
- **general**: Sets the recording mode and the duration of pre/post-event recording.
- **recordingschedule**: Sets the specific schedule for recording.
- **overlapped**: Requests overlapped recordings within the given time range.
- **manualrecording**: Starts or stops the manual recording.
- **calendarsearch**: Requests recording information for a given month.
- **timeline**: Requests for recording information for a given time period.
- **searchrecordingperiod**: Requests the recording period.
- **heatmapsearch**: Requests recording information using the heat map search function and controls its settings.
- **peoplecountsearch**: Requests recording information using the people count search function and controls its settings.
- **posconf**: Used to configure the POS device.
- **poseventconf**: Used to configure POS events in the device.
- **posdata**: Used to receive live POS data from the device.
- **poscalendar**: Used to get the availability of POS data in a given month.
- **metadata**: Used to search POS data for some given search criteria.
- **smartsearch**: Used to search video analytics information for given search criteria.
- **queuesearch**: Requests statistical analysis and measurement of average dwell time and number of people in queues based on given search criteria.
- **diskutility**: Requests information on the NVR’s disk array and smart attributes of HDD.
- **bookmark**: Used to manage bookmarks in NVR.
- **eventsearch**: Used to search event data for some given search criteria.
- **vehiclecountsearch**: Requests recording information using the vehicle count search function, and controls its settings.

Chapter 163. Storage

163.1. Description

The **storage** submenu requests and configures storage settings.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User

163.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
storage&action=<value>[&<parameter>=<value>]
```

163.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the storage settings.
set	Channel	REQ,RES	<int>	Channel ID NVR ONLY
	Enable	REQ, RES	<bool> True, False	Enables or disables video storage
	OverWrite	REQ, RES	<bool> True, False	Enables or disables overwriting
	DiskEndBeep	REQ, RES	<bool> True, False	Whether to beep when the disk ends NVR ONLY
	AutoDeleteEnable	REQ, RES	<bool> True, False	Whether to delete videos older than the specified number of days
	AutoDeleteDays	REQ, RES	<int>	Days before auto deletion This parameter is valid only when **AutoDeleteEnable** is set to True.

Action	Parameters	Request/ Response	Type/ Value	Description
	FileGenerationTimeOut	REQ, RES	<int>	Defines the timeout duration (in minutes) for saving barcode recognition results. If no result is detected within the set time, the current file is saved and a new file is created upon the next recognition.
	FileGenerationTimeOutEnable	REQ, RES	<bool> True, False	Enable FileGenerationTimeOut function

163.4. Examples

163.4.1. Getting the current information

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
OverWrite=False
AutoDeleteEnable=False
AutoDeleteDays=180
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": true,
    "OverWrite": false,
    "AutoDeleteEnable": false,
    "AutoDeleteDays": 180
```

```
}
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
OverWrite=True
DiskEndBeep=False
AutoDeleteEnable=True
Channel.0.AutoDeleteDays=400
Channel.1.AutoDeleteDays=400
Channel.2.AutoDeleteDays=400
Channel.3.AutoDeleteDays=400
Channel.4.AutoDeleteDays=400
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": true,
    "OverWrite": true,
    "DiskEndBeep": false,
    "AutoDeleteEnable": true,
    "ChannelwiseAutoDeleteDays": [
        {
            "Channel": 0,
            "AutoDeleteDays": 400
```

```
        },
        {
            "Channel": 1,
            "AutoDeleteDays": 400
        },
        {
            "Channel": 2,
            "AutoDeleteDays": 400
        },
        {
            "Channel": 3,
            "AutoDeleteDays": 400
        },
        {
            "Channel": 4,
            "AutoDeleteDays": 400
        }
    ]
}
```

163.4.2. Enabling video storage

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=set&Enable=True
```

163.4.3. Enabling overwriting

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=set&OverWrite=True
```

163.4.4. Configuring FileGenerationTimeOut with Enable function

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=set&FileGenerationTimeOutEnable=True&FileGenerationTimeOut=15
```

Chapter 164. General

164.1. Description

The **general** submenu requests and configures general recording settings.

NOTE	Attribute to check for Recording Support: "attributes/Recording/Support/Recording"

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User

164.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
general&action=<value>[&<parameter>=<value>]
```

164.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the general recording settings.
	Channel	REQ, RES	<csv>	Channel ID
	FullFrameBandwidth	RES	<float>	Full frame bandwidth NVR ONLY
	FullFrameRate	RES	<float>	Full frame rate NVR ONLY
	KeyFrameBandWidth	RES	<float>	Key frame bandwidth NVR ONLY
	KeyFrameRate	RES	<float>	Key frame rate NVR ONLY
	Codec	RES	<enum> MJPEG, MPEG4, H264, H265	Codec NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	RecordOverlap	RES	<csv> AudioDetection, VideoAnalysis, AlarmInput, Normal, MotionDetection, Manual, DefocusDetection, Tracking, FogDetection, AudioAnalysis, EmergencyTrigger, GSensorEvent, MaskDetection	Record overlap NVR ONLY
	Resolution	RES	<string>	Resolution NVR ONLY
	FrameRate	RES	<int>	Frame rate NVR ONLY
	CompressionLevel	RES	<int>	Compression level NVR ONLY
	SubStreamCodec	RES	<enum> MJPEG, MPEG4, H264, H265	Substream codec NVR ONLY
	SubStreamResolution	RES	<string>	Substream resolution NVR ONLY
	SubStreamFrameRate	RES	<int>	SubStream framerate NVR ONLY
set	Channel	REQ, RES	<int>	Channel ID
	SourceProfile	REQ, RES	<string>	Source profile NVR ONLY
	NormalMode	REQ, RES	<enum> I-Frame, Full, Off	Recording type for normal mode (continuous recording)
	EventMode	REQ, RES	<enum> I-Frame, Full, Off	Recording type for the occurrence of an event
	PreEventDuration	REQ, RES	<enum>	Duration of pre-event recording
	PostEventDuration	REQ, RES	<enum>	Duration of post-event recording

Action	Parameters	Request/ Response	Type/ Value	Description
	RecordedVideoFileType	REQ, RES	<enum> STW, AVI	Recording file format • STW: Hanwha Vision’s proprietary file format played with Web viewer and SD memory player • AVI: General AVI video file format played with Web viewer and Windows Media Player. CAMERA ONLY
	AudioEnable	REQ, RES	<bool> True, False	Enables or disables audio NVR ONLY
	BitrateLimit	REQ, RES	<float>	Bitrate limit NVR ONLY
	SubStreamEnable	REQ, RES	<bool> True, False	Enables or disables substream recording on a channel. NVR ONLY
	SubStreamSourceProfile	REQ, RES	<string>	Substream sourceprofile NVR ONLY

164.4. Examples

164.4.1. Getting the current information

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=general&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel=0
NormalMode=Off
EventMode=Full
```

```
PreEventDuration=3s
PostEventDuration=5s
RecordedVideoFileType=AVI
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "RecordSetup": [
        {
            "Channel": 0,
            "NormalMode": "Off",
            "EventMode": "Full",
            "PreEventDuration": "3s",
            "PostEventDuration": "5s",
            "RecordedVideoFileType": "AVI"
        }
    ]
}
```


The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=general&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.FullFrameBandWidth=3.965550
Channel.0.FullFrameRate=23.500000
Channel.0.KeyFrameBandWidth=0.850044
Channel.0.KeyFrameRate=0.500000
Channel.0.Codec=H264
```

```
Channel.0.RecordOverlap=Normal
Channel.0.SourceProfile=H.264
Channel.0.NormalMode=Full
Channel.0.EventMode=Full
Channel.0.PreEventDuration=5s
Channel.0.PostEventDuration=30s
Channel.0.Resolution=2560x1920
Channel.0.FrameRate=30
Channel.0.CompressionLevel=10
Channel.0.AudioEnable=False
Channel.0.BitrateLimit=2.300000
Channel.0.SubStreamEnable=True
Channel.0.SubStreamSourceProfile=Live4NVR
Channel.0.SubStreamCodec=H264
Channel.0.SubStreamResolution=800x600
Channel.0.SubStreamFrameRate=30
Channel.1.FullFrameBandWidth=1.157300
Channel.1.FullFrameRate=19.960000
Channel.1.KeyFrameBandWidth=0.359871
Channel.1.KeyFrameRate=0.500000
Channel.1.Codec=H264
Channel.1.RecordOverlap=Normal
Channel.1.SourceProfile=Rec4NVR1
Channel.1.NormalMode=Full
Channel.1.EventMode=Full
Channel.1.PreEventDuration=5s
Channel.1.PostEventDuration=30s
Channel.1.Resolution=1920x1080
Channel.1.FrameRate=20
Channel.1.CompressionLevel=10
Channel.1.AudioEnable=False
Channel.1.BitrateLimit=2.300000
Channel.1.SubStreamEnable=True
Channel.1.SubStreamSourceProfile=Live4NVR1
Channel.1.SubStreamCodec=H264
Channel.1.SubStreamResolution=800x600
Channel.1.SubStreamFrameRate=20
Channel.2.FullFrameBandWidth=0.780083
Channel.2.FullFrameRate=25.000000
Channel.2.KeyFrameBandWidth=0.671478
Channel.2.KeyFrameRate=1.910000
```

```
Channel.2.Codec=H264
Channel.2.RecordOverlap=Normal
Channel.2.SourceProfile=H.264
Channel.2.NormalMode=Full
Channel.2.EventMode=Full
Channel.2.PreEventDuration=5s
Channel.2.PostEventDuration=30s
Channel.2.Resolution=320x240
Channel.2.FrameRate=25
Channel.2.CompressionLevel=16
Channel.2.AudioEnable=False
Channel.2.BitrateLimit=2.300000
Channel.2.SubStreamEnable=True
Channel.2.SubStreamSourceProfile=VideoProfile5
Channel.2.SubStreamCodec=H265
Channel.2.SubStreamResolution=3840x2160
Channel.2.SubStreamFrameRate=25
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "RecordSetup": [
        {
            "Channel": 0,
            "FullFrameBandWidth": 3.416850,
            "FullFrameRate": 17.530000,
            "KeyFrameBandWidth": 1.058580,
            "KeyFrameRate": 0.350000,
            "Codec": "H264",
            "RecordOverlap": [
                "Normal"
            ],
            "SourceProfile": "H.264",
            "NormalMode": "Full",
            "EventMode": "Full",
            "PreEventDuration": "5s",
```

```
            "PostEventDuration": "30s",
            "Resolution": "2560x1920",
            "FrameRate": 30,
            "CompressionLevel": 10,
            "AudioEnable": false,
            "BitrateLimit": 2.300000,
            "SubStreamEnable": true,
            "SubStreamSourceProfile": "Live4NVR",
            "SubStreamCodec": "H264",
            "SubStreamResolution": "800x600",
            "SubStreamFrameRate": "30"
        },
        {
            "Channel": 1,
            "FullFrameBandWidth": 1.157300,
            "FullFrameRate": 19.960000,
            "KeyFrameBandWidth": 0.359871,
            "KeyFrameRate": 0.500000,
            "Codec": "H264",
            "RecordOverlap": [
                "Normal"
            ],
            "SourceProfile": "Rec4NVR1",
            "NormalMode": "Full",
            "EventMode": "Full",
            "PreEventDuration": "5s",
            "PostEventDuration": "30s",
            "Resolution": "1920x1080",
            "FrameRate": 20,
            "CompressionLevel": 10,
            "AudioEnable": false,
            "BitrateLimit": 2.300000,
            "SubStreamEnable": true,
            "SubStreamSourceProfile": "Live4NVR1",
            "SubStreamCodec": "H264",
            "SubStreamResolution": "800x600",
            "SubStreamFrameRate": "20"
        },
        ...
    ]
}
```

164.4.2. Setting the normal record mode to Full

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&NormalMode=Full
```

164.4.3. Setting the event record mode to I-Frame

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&EventMode=I-Frame
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&Channel=1&EventMode=I-Frame
```

164.4.4. Setting the duration of pre-event recording to 3 seconds

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&PreEventDuration=3s
```

164.4.5. Setting the recording video file format to AVI

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&RecordedVideoFileType=AVI
```

164.4.6. Setting the source profile

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=general&action=set&Channel=7&SourceProfile=Profil
e2
```

Chapter 165. Recording Schedule

165.1. Description

The **recordingschedule** submenu requests and configures recording schedule settings.

This submenu is only for normal recording, not for event recording. To set the recording schedule for events, please refer to the 'Event' document.

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User

165.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
recordingschedule&action=<value>[&<parameter>=<value>]
```

165.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the recording schedule settings.
	Channel	REQ	<csv>	Channel ID
set	Channel	REQ, RES	<int>	Channel ID
	Activate	REQ, RES	<enum> Always, Scheduled	Recording type. • Always: Always records the video • Scheduled: Records only at a specific time on a specific day of the week **Note** **Activate** must be sent together with the **set** action. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	<ddd>	REQ, RES	(For Cameras) <bool> 0, 1 (For NVR) <enum> 0, 1, E, B	Enables or disables recording for the selected day of week. • 0: Disabled • 1: Enabled • E: Events • B: Both <ddd> stands for week of the day and should be specified in the short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g.) '**SUN**=1' indicates that recording is activated every Sunday from 12:00 AM to 11:59 PM, unless the specific time is set using the <dddh> parameter such like SUN1=1, SUN2=1, etc. This parameter is valid only when **Activate** is set to Scheduled.
	EveryDay	REQ, RES	(For Cameras) (For Cameras) <bool> 0, 1 (For NVR) <enum> 0, 1, E, B	Enables or disables recording for every day • 0: Disabled • 1: Enabled • E: Events • B: Both '**EveryDay**=1', indicating that the recording is activated every day, has the same effect as setting the **ScheduleType** parameter to Always. **EveryDay** is valid only when **Activate** is set to Scheduled.

Action	Parameters	Request/ Response	Type/ Value	Description
	<dddh>	REQ, RES	(For Cameras) <bool> 0, 1 (For NVR) <enum> 0, 1, E, B	Enables or disables recording for the selected hour and day. • 0: Disabled • 1: Enabled • E: Events • B: Both <dddh> stands for the week of day and time in hour. e.g. SUN1 means 1:00 AM on Sunday. MON2 means 2:00 AM on Monday. This parameter is available if <corresponding weekday> = 1. ‘ **SUN**=1’ is required for SUN0 ... SUN23. e.g.) ‘SUN=1&SUN18=1’ indicates that recording is enabled from 6:00 PM to 6:59 PM on every Sunday This parameter is valid only when **Activate** is set to Scheduled.
	EveryDay<h>	REQ, RES	(For Cameras) <bool> 0, 1 (For NVR) <enum> 0, 1, E, B	Enables or disables recording for every day and hour • 0: Disabled • 1: Enabled • E: Events • B: Both This parameter is available if **EveryDay** =1. ‘EveryDay=1’ is required for EveryDay0 ... EveryDay23. e.g.) ‘EveryDay=1&EveryDay18=1’ indicates that recording is enabled from 6:00 PM to 6:59 PM every day. This parameter is valid only when **Activate** is set to Scheduled.

Action	Parameters	Request/ Response	Type/ Value	Description
	<dddh>.FromTo	REQ, RES	<string>	Start time for recording for the specific time and day of week This parameter should be specified in the format of <mm-mm> and the first 'mm' must be smaller than or equal to the second 'mm'. This is available if <corresponding weekday><hour>=1. '**SUN0**=1' is required for SUN0.FromTo. e.g.) 'SUN=1&SUN18=1&SUN18.FromTo=12-20' indicates that recording is activated from 6:12 PM to 6:20 PM on every Sunday. This parameter is valid only when **Activate** is set to Scheduled. CAMERA ONLY
	EveryDay<h>.FromTo	REQ, RES	<string>	Start time for recording every day This parameter should be specified in the format of <mm-mm> and the first 'mm' must be smaller than or equal to the second 'mm'. This parameter is available if EveryDay<hour>=1. 'EveryDay0=1' is required for EveryDay0.FromTo. e.g.) 'EveryDay=1&EveryDay18=1& EveryDay18.FromTo=12-20' indicates the recording is activated 6:12 PM to 6:20 PM every day. This parameter is valid only when **Activate** is set to Scheduled. CAMERA ONLY

165.4. Examples

165.4.1. Getting the current information

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=view
```

When Activate is set to Always

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel=0
Activate=Always
SUN:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MON:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
TUE:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
WED:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
THU:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
FRI:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
SAT:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "RecordSchedule": [
        {
            "Channel": 0,
            "Activate": "Always",
            "Schedule": {
                "SUN": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
```

```
"0"
                ],
                "MON": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "TUE": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "WED": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "THU": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "FRI": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "SAT": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ]
            }
        }
    ]
```

```
}
```

When Activate is set to Scheduled

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel=0
Activate=Scheduled
SUN:1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
MON:1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
TUE:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
WED:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
THU:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
FRI:0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
SAT:1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "RecordSchedule": [
        {
            "Channel": 0,
            "Activate": "Scheduled",
            "Schedule": {
                "SUN": [
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1",
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1"
                ],
                "MON": [
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
```

```
"1",
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1"
                ],
                "TUE": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "WED": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "THU": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "FRI": [
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",
                    "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0"
                ],
                "SAT": [
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1",
                    "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1",
"1"
                ]
            }
        }
    ]
}
```

165.4.2. Setting the device to always record

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Always
```

165.4.3. Setting the recording schedule

To set the schedule (day and time), the **Activate** parameter must be set to Scheduled.

To record every Saturday and Sunday

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Scheduled&S
AT=1&SUN=1
```

To not record every Saturday and Sunday

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Scheduled&S
AT=0&SUN=0
```

To record at 1:00 AM every day and disable the recording at 4:00 AM every day

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Scheduled&E
veryDay=1&EveryDay1=1&EveryDay4=0
```

To record from 3:58 AM to 3:59 AM on Sunday

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Scheduled&S
UN=1&SUN3=1&SUN3.FromTo=58-59
```

To record every day from 3:58 AM to 3:59 AM

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=recordingschedule&action=set&Activate=Scheduled&EveryDay=1&EveryDay3=1&EveryDay3.FromTo=58-59
```

Chapter 166. Overlapped Recording

166.1. Description

The **overlapped** submenu requests that recordings are overlapped within a given time range.

In a case where the system time settings change, or DST is applied while recording the video, the recording is overlapped for a certain period of time. A single digit number is assigned to the ID of the overlapped recording. When it returns ‘OverlappedIDList=0,1’ this means that there are two overlapped recordings.

NOTE Attribute to check for Feature Support: "attributes/Recording/Support/Overlapped"

Access level

Action	Camera	NVR
view	User	User

166.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
overlapped&action=<value>[&<parameter>=<value>]
```

166.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	OverlappedIDList	RES	<csv>	Overlapped recording ID list
	Channel.#.OverlappedID List	RES	<csv>	Overlapped recording ID list for the given Channel MULTI DIRECTIONAL CAMERA ONLY
	ChannelIDList	REQ	<csv>	Channel ID list
	FromDate	REQ	<string>	The start date and time for when the recording occurred The date must be specified in the format of <YYYY-MM-DD hh:mm:ss>. **Note** **FromDate** and **ToDate** must be sent together for the **view** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	ToDate	REQ	<string>	The end date and time for when the recording occurred The date must be specified in the format of <YYYY-MM-DD hh:mm:ss>. **Note** **FromDate** and **ToDate** must be sent together for the **view** action.

166.4. Examples

166.4.1. Getting the overlapped recording between 12:00 AM and 12:00 PM on Aug. 1, 2014

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=overlapped&action=view&FromDate=2014-08-01 00:00:00&ToDate=2014-08-01 23:59:59
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OverlappedIDList=2,1,0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "OverlappedIDList": [
        2,
        1,
        0
```

```
    ]
}
```

Chapter 167. Manual Recording

167.1. Description

The **manualrecording** submenu controls the manual recording status (start/stop).

> **NOTE**
> This chapter applies to NVR only.
> Attribute to check for Manual Recording start:
> "attributes/Recording/Support/ManualRecordingStart"
> Attribute to check for Manual Recording stop:
> "attributes/Recording/Support/ManualRecordingStop"

Access level

Action	NVR
view	User
control	User

167.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
manualrecording&action=<value>[&<parameter>=<value>]
```

167.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads manual recording status.
control	Mode	REQ	<enum> Start, Stop	Manual recording mode **Note** **Mode** must be sent together with the **control** action.

167.4. Examples

167.4.1. Getting the current manual recording status

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=manualrecording&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Mode=Start
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Mode": "Start"
}
```

167.4.2. Starting the manual recording

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=manualrecording&action=control&Mode=Start
```

167.4.3. Stopping the manual recording

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=manualrecording&action=control&Mode=Stop
```

Chapter 168. Calendar Search

168.1. Description

The **calendarsearch** submenu requests the recording information from a given month.

> **NOTE**
> Attribute to check for Feature Support: "attributes/Recording/Support/SearchCalendar"
> Attribute to check for Failover feature support: "attributes/Recording/Support/FailOverRecording"

Access level

Action	Camera	NVR
view	User	User

168.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
calendarsearch&action=<value>[&<parameter>=<value>]
```

168.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Month	REQ	<string>	Target month for searching Month must be specified in the <YYYY-MM> format. **Note** **Month** must be sent together with the **view** action.
	ChannelIDList	REQ	<csv>	List of channels in which recordings to be searched.
	Channel.#.Result	RES	<string>	Search results
	PrimaryDeviceIPAddress	REQ	<string> FormatInfo =IPv4Address or IPv6Address	IP address of primary device to which recording to be searched Applicable only if FailOver feature is supported. NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	IgnoreChannelBasedResults	REQ	<bool> True, False	If true, consolidated results will be given for all channels NVR ONLY
	Result	RES	<string>	Search results **Note** This parameter provides a response only when **IgnoreChannelBasedResults** is set to **True** in the request. NVR ONLY

NOTE represents the channel ID.

168.4. Examples

168.4.1. Searching for recordings from February 2013

The response will be a string of 31 digits, each representing a day of the month. If a digit is '0', it indicates that there are no recordings for that day. If the digit is '1', it indicates that a recording exists for that day.

The response code below means that there are recordings for the 6th, 7th and 13th of February 2013.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=calendarsearch&action=view&Month=2013-02&ChannelIDList=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Result=0000011000001000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "CalenderSearchResults": [
        {
            "Channel": 0,
            "Result": "0000011000001000000000000000000"
        }
    ]
}
```


REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=calendarsearch&action=view&Month=2013-
02&ChannelIDList=0,1,2,3& &IgnoreChannelBasedResults=true
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Result=0000011000001000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Result": "0000011000001000000000000000000"
}
```

Chapter 169. Timeline

169.1. Description

The **timeline** submenu requests recording information from a given time period.

NOTE	Attribute to check for Feature Support: "attributes/Recording/Support/SearchTimeline"

Access level

Action	Camera	NVR
view	User	User

169.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
timeline&action=<value>[&<parameter>=<value>]
```

169.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Refer to **Recording Types** for supported values	Recording type to search Common Types • All: All video recordings including normal and event recordings. • Normal: Continuous video recordings. • Event: Video recording for all events. **Note** **Type**, **FromDate**, and **ToDate** must be sent together with the **view** action for the camera.

Action	Parameters	Request/ Response	Type/ Value	Description
	FromDate	REQ	<string>	The start date and time for when the recording occurred. The time is specified in the format of <YYYY-MM-DD hh:mm:ss>. **Note** **Type**, **FromDate**, and **ToDate** must be sent together with the **view** action.
	ToDate	REQ	<string>	The end date and time for when the recording occurred. The time is specified in the format of <YYYY-MM-DD hh:mm:ss>. **Note** **Type**, **FromDate**, and **ToDate** must be sent together with the **view** action.
	OverlappedID	REQ	<int>	Overlapped recording ID For more information about the overlapped recording, please refer to '5 Overlapped Recording'(page 22).
	ChannelIDList	REQ	<csv>	Channel ID list **Note** **ChannelIDList**, **FromDate**, and **ToDate** must be sent together with the **view** action for NVR.
	TotalCount	RES	<int>	Total number of results
	Channel.#.Result.#.StartTime	RES	<string>	Requested start date and time The time is specified in the format of <YYYY-MM-DD hh:mm:ss DST> (DST displays when supported).
	Channel.#.Result.#.EndTime	RES	<string>	Requested end date and time The time is specified in the format of <YYYY-MM-DD hh:mm:ss DST> (DST displays when supported).

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel.#.Result.#.Type	RES	<enum> Refer to **Recording Types** for supported values	Recording type for the requested period
	Channel.#.Result.#.BkID	RES	<string>	Bookmark ID NVR ONLY
	PrimaryDeviceIPAddress	REQ	<string> FormatInfo =IPv4Address or IPv6Address	IP address of primary device in which recording data to be searched. Applicable only if FailOver feature is supported NVR ONLY

Recording Types

Type	All, Normal, Event, AlarmInput, VideoAnalysis, MotionDetection, NetworkDisconnect, FaceDetection, TamperingDetection, AudioDetection, Tracking, Manual, UserInput, DefocusDetection, FogDetection, AudioAnalysis, QueueEvent, videoloss, EmergencyTrigger, InternalHDDWarmup, GSensorEvent, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, BodyTemperatureDetection, MaskDetection, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor
Channel.#.Result.#.Type	Normal, AlarmInput, VideoAnalysis, MotionDetection, NetworkDisconnect, FaceDetection, TamperingDetection, AudioDetection, Tracking, ManualRecording, UserInput, DefocusDetection, FogDetection, AudioAnalysis, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, BodyTemperatureDetection, MaskDetection, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor

Recording Types in dynamicrules supported models

Type	All, Normal, Event, Rule1, Rule2,.., Rule32
Channel.#.Result.#.Type	Normal, Rule1, Rule2,.., Rule32

NOTE

Refer to attributes cgi, cgi section
stw-cgi/attributes.cgi/recording/timeline
For supported Type list.

169.4. Examples

169.4.1. Getting the timeline

Search for normal recording between 9:00 AM and 10:00 AM on Mar. 3, 2013

Type, **FromDate**, and **ToDate** must be sent together with the **view** action for the camera.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=timeline&action=view&Type=Normal&FromDate=2013-
03-03 09:00:00&ToDate=2013-03-03 10:00:00
```

The following request example is for NVR only. **ChannelIDList**, **FromDate**, and **ToDate** must be sent together with the **view** action for NVR

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=timeline&action=view&ChannelIDList=0&FromDate=201
3-03-03 09:00:00&ToDate=2013-03-03 10:00:00
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalCount=3
Channel.0.Result.0.StartTime=2013-03-03 09:15:52
Channel.0.Result.0.EndTime=2013-03-03 09:18:19
Channel.0.Result.0.Type=Normal
Channel.0.Result.1.StartTime=2013-03-03 09:10:56
Channel.0.Result.1.EndTime=2013-03-03 09:15:51
Channel.0.Result.1.Type=Normal
Channel.0.Result.2.StartTime=2013-03-03 00:10:52
Channel.0.Result.2.EndTime=2013-03-03 00:10:56
Channel.0.Result.2.Type=Normal
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "TimeLineSearchResults": [
        {
            "Channel": 0,
            "Results": [
                {
                    "Result": 0,
                    "StartTime": "2013-03-03 09:15:52",
                    "EndTime": "2013-03-03 09:18:19",
                    "Type": "Normal"
                },
                {
                    "Result": 1,
                    "StartTime": "2013-03-03 09:10:56",
                    "EndTime": "2013-03-03 09:15:51",
                    "Type": "Normal"
                },
                {
                    "Result": 2,
                    "StartTime": "2013-03-03 00:10:52",
                    "EndTime": "2013-03-03 00:10:56",
                    "Type": "FaceDetection"
                }
            ]
        }
    ]
}
```


Search for motion detection events between 9:00 AM and 10:00 AM on Mar. 3, 2013

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=timeline&action=view&Type=MotionDetection&FromDat
e=2013-03-03 09:00:00&ToDate=2013-03-03 10:00:00
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=timeline&action=view&ChannelIDList=0&FromDate=201
4-02-10 00:00:01&ToDate=2014-02-10 23:59:59&Type=MotionDetection
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalCount=2
Channel.0.Result.0.StartTime=2013-03-03 18:16:33
Channel.0.Result.0.EndTime=2013-03-03 18:16:44
Channel.0.Result.0.Type=MotionDetection
Channel.0.Result.1.StartTime=2013-03-03 18:08:36
Channel.0.Result.1.EndTime=2013-03-03 18:11:36
Channel.0.Result.1.Type=MotionDetection
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TimeLineSearchResults": [
        {
            "Channel": 0,
            "Results": [
                {
                    "Result": 0,
                    "StartTime": "2013-03-03 18:16:33",
                    "EndTime": "2013-03-03 18:16:44",
                    "Type": "MotionDetection"
                },
                {
                    "Result": 1,
                    "StartTime": "2013-03-03 18:08:36",
```

```
                    "EndTime": "2013-03-03 18:11:36",
                    "Type": "MotionDetection"
                }
            ]
        }
    ]
}
```

NOTE Please check if the video storage is enabled (recording.cgi > storage > Enable=True) and physically connected, if you receive the error code (STATUS_UNKNOWN_ERROR) in the response message.

Chapter 170. Recording Period

170.1. Description

The **searchrecordingperiod** submenu requests the recording period.

> **NOTE**
> This chapter applies to NVR only.
> Attribute to check for Feature Support: "attributes/Recording/Support/SearchPeriod"

Access level

Action	NVR
view	User

170.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
searchrecordingperiod&action=<value>[&<parameter>=<value>]
```

170.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads recording period
	StartTime	RES	<string>	First recording start time
	EndTime	RES	<string>	Last recording end time
	ResultsInUTC	REQ	<bool> True, False	Enable or disable search result in UTC

170.4. Examples

170.4.1. Searching the recording period

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=searchrecordingperiod&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
StartTime=2014-09-22 16:05:34
EndTime=2014-10-02 11:47:11
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "StartTime": "2014-09-22 16:05:34",
    "EndTime": "2014-10-02 11:47:11"
}
```

Chapter 171. Heat Map Search

171.1. Description

The **heatmapsearch** submenu requests recording information using the heat map search function and controls the heat map search settings.

> **NOTE**
> Attribute to check for Feature Support in NVR: "attributes/Recording/Support/SearchHeatMap"
> Attribute to check for Feature Support in Camera: "recording/heatmapsearch".

Access level

Action	Camera	NVR
view	Admin	User
control	Admin	User

171.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
heatmapsearch&action=<value>[&<parameter>=<value>]
```

171.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> GridColor, Results, Status	Type If **Type** is set to GridColor, the **view** action must be sent together with **ChannelIDList**, **FromDate**, **ToDate**, **OverlappedID**, **MotionType**, and **GridRegion**; and if **Type** is set to Results or Status, the **view** action must be sent together with **SearchToken**. **Note** GridColor Type is supported only for NVR.

Action	Parameters	Request/ Response	Type/ Value	Description
	ChannelIDList	REQ	<csv>	List of channels in which recordings will be searched **ChannelIDList** is valid only when **Type** is set to Gridcolor. NVR ONLY
	FromDate	REQ	<string>	The start date and time for when the recording occurred. **FromDate** is valid only when **Type** is set to Gridcolor. NVR ONLY
	ToDate	REQ	<string>	The end date and time for when the recording occurred. **ToDate** is valid only when **Type** is set to Gridcolor. NVR ONLY
	OverlappedID	REQ	<int>	Overlapped recording ID For more information about the overlapped recording, please refer to '5 Overlapped Recording'(page 22). **OverlappedID** is valid only when **Type** is set to Gridcolor. NVR ONLY
	MotionType	REQ	<enum> Person, Vehicle, Anything	Motion type **MotionType** is valid only when **Type** is set to Gridcolor. NVR ONLY
	GridRegion	REQ	<string>	Grid area **GridRegion** is valid only when **Type** is set to Gridcolor. NVR ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	SearchToken	REQ	<string>	Search Session token **SearchToken** is valid only when **Type** is set to Results or Status.
	Channel.#.ResultGridColorLevel	RES	<string>	Grid color level result **Channel.#.ResultGridColorLevel** is valid only when **Type** is set to Gridcolor. NVR ONLY
	TotalCount	RES	<int>	Total count **TotalCount** is valid only when **Type** is set to Results. NVR ONLY
	Channel.#.Result.#.StartTime	RES	<string>	Start time for a search in the corresponding results and channel **Channel.#.Result.#.StartTime** is valid only when **Type** is set to Results. DST is shown when supported. NVR ONLY
	Channel.#.Result.#.EndTime	RES	<string>	End time for search in the corresponding results and channel **Channel.#.Result.#.EndTime** is valid only when **Type** is set to Results. DST is shown when supported. NVR ONLY
	Channel.#.Result.#.Type	RES	<enum> Person, Vehicle, Anything	Search type in the corresponding results and channel **Channel.#.Result.#.Type** is valid only when **Type** is set to Results. NVR ONLY
	Status	RES	<enum> Completed, NotCompleted	Search status **Status** is valid only when **Type** is set to Status.

Action	Parameters	Request/ Response	Type/ Value	Description
control	Mode	REQ	<enum> Start, Cancel	Mode
	ChannelIDList	REQ	<csv>	Channel ID list
	FromDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The start date and time for search
	ToDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The end date and time for search
	OverlappedID	REQ	<int>	Overlapped recording ID NVR ONLY
	MotionType	REQ	<enum> Person, Vehicle, Anything	Type of motion to search NVR ONLY
	GridRegion	REQ	<string>	Grid region for search NVR ONLY
	SearchToken	REQ, RES	<string>	Search token **SearchToken** is a request-only parameter when **Mode** is set to Cancel, but it will return data when **Mode** is set to Start.
	ResultImageType	REQ	<enum> WithBackground, WithoutBackground	Type of Heat Map Result Image **Note** This parameter is valid only when ResultAsImage is set to True. CAMERA ONLY

Action	Parameters	Request/ Response	Type/ Value	Description
	ResultAsImage	REQ	<bool> True, False	HeatMap result as image CAMERA ONLY **Note** Currently, the camera supports heat map results only in image format. For this reason, the **ResultAsImage** parameter should be set to True for camera, and this parameter should send along with the **Mode** parameter when **Mode** is set to Start.

171.4. Examples

171.4.1. Heat map search color

Type must be set to GridColor, and **ChannelIDList**, **FromDate**, **ToDate**, **OverlappedID**, **MotionType** must be sent together with **view** action.

Heat map search color is supported only for NVR. The following example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=heatmapsearch&action=view&Type=GridColor&ChannelI
DList=0,1&FromDate=2013-12-01 10:11:12&ToDate=2014-06-04
08:09:10&OverlappedId=52&MotionType=Person&GridRegion=0000011111100000000001
11111000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
000000
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.ResultGridColorLevel=000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
```

```
0000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000
Channel.1.ResultGridColorLevel=000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "HeatmapGridColors": [
        {
            "Channel": 0,
            "ResultGridColorLevel":
"000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000"
        },
        {
            "Channel": 1,
            "ResultGridColorLevel":
"000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000"
        }
    ]
}
```

171.4.2. Heat map search

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=heatmapsearch&action=control&ChannelIDList=0&Mode
=Start&FromDate=2016-09-24T00:00:00Z&ToDate=2016-09-
```

```
24T23:59:59Z&ResultAsImage=True&ResultImageType=WithBackground
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=HeatMap-2016-09-24T04:32:17-824
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "HeatMap-2016-09-24T04:32:17-824"
}
```

The following example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=heatmapsearch&action=control&Mode=Start&ChannelID
List=5&FromDate=2013-12-01T10:11:12Z&ToDate=2014-12-
20T08:09:10Z&OverlappedId=10&MotionType=Person&GridRegion=111111111111111111
1111111111111111111111111111111111111111111111111111111111111111111111111111
1111111111111111111111111111111111111111111111111111111111111111111111111111
1111111111111111111111111111111111111111111111111111111111111111111111111111
1111111111
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=0926800821074333
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SearchToken": "0104700429062444"
}
```


171.4.3. Getting the status of searching

Type must be set to Status, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=heatmapsearch&action=view&Type=Status&SearchToken
=0926800821074333
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Completed
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Status": "Completed"
```

```
}
```

171.4.4. Getting the result

Type must be set to Results, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=heatmapsearch&action=view&Type=Results&SearchToke
n=0926800821074333
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
<PNG Image>
```

The following response example is for NVR only.

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalCount=2
Channel.0.Result.1.StartTime=2014-11-02T05:16:12Z
Channel.0.Result.1.EndTime=2014-11-02T05:16:42Z
Channel.0.Result.1.Type=Person
Channel.0.Result.2.StartTime=2014-11-02T06:25:53Z
Channel.0.Result.2.EndTime=2014-11-02T06:26:23Z
Channel.0.Result.2.Type=Person
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "TotalCount": 2,
    "HeatmapSearchResults": [
        {
            "Channel": 0,
            "Results": [
                {
                    "Result": 1,
                    "StartTime": "2014-11-02T05:16:12Z",
                    "EndTime": "2014-11-02T05:16:42Z",
                    "Type": "Person"
                },
                {
                    "Result": 2,
                    "StartTime": "2014-11-02T06:25:53Z",
                    "EndTime": "2014-11-02T06:26:23Z",
                    "Type": "Person"
                }
            ]
        }
    ]
}
```


171.4.5. Cancelling the search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=heatmapsearch&action=control&Mode=Cancel&SearchToken=0926800821074333
```

Chapter 172. People Count Search

172.1. Description

The **peoplecountsearch** submenu requests recording information using the people count search function, and controls the people count search settings.

NOTE | This chapter applies to Camera only.
Attribute to check for Feature Support: "recording/peoplecountsearch"

Access level

Action	Camera
view	Admin
control	Admin

172.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
peoplecountsearch&action=<value>[&<parameter>=<value>]
```

172.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	Type If **Type** is set to Results or Status, the **view** action must be sent together with **SearchToken**.
	SearchToken	REQ	<string>	Search token
	ResultInterval	RES	<enum> Hourly, Daily, Weekly, Monthly	Search result interval

Action	Parameters	Request/ Response	Type/ Value	Description
	Camera.#.Line.#.Direction.#.Result	RES	<csv>	People Count Search Results. If **ResultInterval** is Hourly, search results are in terms of hours and the number of results is fixed, i.e. 24. Here, the first result in the array represents the 0th hour of the day, and the last result in the array represents the 23rd hour of the day. If **ResultInterval** is Daily, search results are in terms of days and the first result in the array represents the first day of the month while the last result in the array represents the last day of the month. If **ResultInterval** is Weekly, search results are in terms of weeks. If **ResultInterval** is Monthly, search results are in terms of months and the first result in the array represents the first month of the year, while the last result in the array represents the last month of the year. **Note** **ResultInterval** is fixed by the camera based on **FromDate** and **ToDate** of the search
	Status	RES	<enum> Completed, NotCompleted	Search status **Status** is valid only when **Type** is set to Status.
control	Mode	REQ	<enum> Start, Cancel	Mode
	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	FromDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The start date and time for search
	ToDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The end date and time for search
	SearchToken	REQ, RES	<string>	Search token **SearchToken** is a request-only parameter when **Mode** is set to Cancel, but it will return data when **Mode** is set to Start.
	Camera.#.Line.#.Direction	REQ	<csv> In,Out	People Count Line Direction to search

172.4. Examples

172.4.1. People Count Search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=peoplecountsearch&action=control&Channel=0&Mode=Start&FromDate=2016-09-24T00:00:00Z&ToDate=2016-09-24T23:59:59Z&Camera.PeopleCount-Master.Line.Gate1.Direction=In,Out&Camera.PeopleCount-Master.Line.Gate2.Direction=In,Out
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=PeopleCount-2016-09-24T02:32:51-614
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": " PeopleCount-2016-09-24T02:32:51-614"
}
```

172.4.2. Getting the status of searching

Type must be set to Status, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=peoplecountsearch&action=view&Type=Status&SearchT
oken=PeopleCount-2016-09-24T02:32:51-614
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Completed
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "Completed"
```

```
}
```

172.4.3. Getting the result

Type must be set to Results, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=peoplecountsearch&action=view&Type=Results&Search
Token=PeopleCount-2016-09-24T02:32:51-614
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ResultInterval = Hourly
Camera.PeopleCount-
Master.Line.Gate1.Direction.In.Result=0,0,0,0,0,0,2,0,0,0,0,0,0,0,0,6,0,0,0,0,
0,3,0,2,2
Camera.PeopleCount-
Master.Line.Gate1.Direction.Out.Result=0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0
,0,2,0,5,3
Camera.PeopleCount-
Master.Line.Gate1.Direction.In.Result=0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,
0,11,0,0,0
Camera.PeopleCount-
Master.Line.Gate2.Direction.Out.Result=0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0
,0,11,0,3,2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ResultInterval": "Hourly",
    "PeopleCountSearchResults": [
```

```
        {
            "Camera": "PeopleCount-Master",
            "LineResults": [
                {
                    "Line": "Gate1",
                    "DirectionResults": [
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,2,0,0,0,0,0,0,0,6,0,0,0,0,0,3,0,2,2"
                        },
                        {
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0,0,2,0,5,3"
                        }
                    ]
                },
                {
                    "Line": "Gate2",
                    "DirectionResults": [
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,11,0,0,0"
                        },
                        {
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0,0,11,0,3,2"
                        }
                    ]
                }
            ]
        }
    ]
}
```

172.4.4. Cancelling the search

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=peoplecountsearch&action=control&Mode=Cancel&Sear
chToken=PeopleCount-2016-09-24T02:32:51-614
```

Chapter 173. Vehicle Count Search

173.1. Description

The **vehiclecountsearch** submenu requests recording information using the vehicle count search function, and controls the vehicle count search settings.

> **NOTE**
> This chapter applies to Camera only.
> Attribute to check for Feature Support: "recording/vehiclecountsearch"

Access level

Action	Camera
view	Admin
control	Admin

173.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
vehiclecountsearch&action=<value>[&<parameter>=<value>]
```

173.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	Type If **Type** is set to Results or Status, the **view** action must be sent together with **SearchToken**.
	SearchToken	REQ	<string>	Search token
	ShowAIStats	REQ	<bool> True,False	Whether to show AI Stats
	ResultInterval	RES	<enum> Hourly, Daily, Weekly, Monthly	Search result interval

Action	Parameters	Request/ Response	Type/ Value	Description
	Camera.#.Line.#.Direction.#.Result	RES	<csv>	Vehicle Count Search Results. If **ResultInterval** is Hourly, search results are in terms of hours and the number of results is fixed, i.e. 24. Here, the first result in the array represents the 0^{th} hour of the day, and the last result in the array represents the 23^{rd} hour of the day. If **ResultInterval** is Daily, search results are in terms of days and the first result in the array represents the first day of the month while the last result in the array represents the last day of the month. If **ResultInterval** is Weekly, search results are in terms of weeks. If **ResultInterval** is Monthly, search results are in terms of months and the first result in the array represents the first month of the year, while the last result in the array represents the last month of the year.
	Camera.#.Line.#.Direction.#.Car	RES	<csv>	Car Count Search Results. This parameter is shown when **ShowAIStats** is True
	Camera.#.Line.#.Direction.#.Bus	RES	<csv>	Bus Count Search Results. This parameter is shown when **ShowAIStats** is True
	Camera.#.Line.#.Direction.#.Truck	RES	<csv>	Truck Count Search Results. This parameter is shown when **ShowAIStats** is True
	Camera.#.Line.#.Direction.#.Motorcycle	RES	<csv>	Motorcycle Count Search Results. This parameter is shown when **ShowAIStats** is True

Action	Parameters	Request/ Response	Type/ Value	Description
	Camera.#.Line.#.Direction.#.Bicycle	RES	<csv>	Bicycle Count Search Results. This parameter is shown when **ShowAIStats** is True **Note** **ResultInterval** is fixed by the camera based on **FromDate** and **ToDate** of the search
	Status	RES	<enum> Completed, NotCompleted	Search status **Status** is valid only when **Type** is set to Status.
control	Mode	REQ	<enum> Start, Cancel	Mode
	Channel	REQ	<int>	Channel ID
	FromDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The start date and time for search
	ToDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The end date and time for search
	SearchToken	REQ, RES	<string>	Search token **SearchToken** is a request-only parameter when **Mode** is set to Cancel, but it will return data when **Mode** is set to Start.
	Camera.#.Line.#.Direction	REQ	<csv> In,Out	Vehicle Count Line Direction to search

173.4. Examples

173.4.1. Vehicle Count Search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=vehiclecountsearch&action=control&Channel=0&Mode=Start&FromDate=2016-09-24T00:00:00Z&ToDate=2016-09-24T23:59:59Z&Camera.VehicleCount-Master.Line.Gate1.Direction=In,Out&Camera.VehicleCount-Master.Line.Gate2.Direction=In,Out
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=VehicleCount-2016-09-24T02:32:51-614
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": " VehicleCount-2016-09-24T02:32:51-614"
}
```

173.4.2. Getting the status of searching

Type must be set to Status, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=vehiclecountsearch&action=view&Type=Status&SearchToken=VehicleCount-2016-09-24T02:32:51-614
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
Status=Completed
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "Completed"
}
```

173.4.3. Getting the result

Type must be set to Results, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=vehiclecountsearch&action=view&Type=Results&Searc
hToken=VehicleCount-2016-09-24T02:32:51-614
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ResultInterval = Hourly
Camera.VehicleCount-
Master.Line.Gate1.Direction.In.Result=0,0,0,0,0,0,2,0,0,0,0,0,0,0,6,0,0,0,0,0,
0,3,0,2,2
Camera.VehicleCount-
Master.Line.Gate1.Direction.Out.Result=0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0,0
,0,2,0,5,3
Camera.VehicleCount-
Master.Line.Gate1.Direction.In.Result=0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,
```

```
0,11,0,0,0
Camera.VehicleCount-
Master.Line.Gate2.Direction.Out.Result=0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0,0
,0,11,0,3,2
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ResultInterval": "Hourly",
    "VehicleCountSearchResults": [
        {
            "Camera": "VehicleCount-Master",
            "LineResults": [
                {
                    "Line": "Gate1",
                    "DirectionResults": [
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,2,0,0,0,0,0,0,0,6,0,0,0,0,0,0,3,0,2,2"
                        },
                        {
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0,0,0,2,0,5,3"
                        }
                    ]
                },
                {
                    "Line": "Gate2",
                    "DirectionResults": [
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,0,11,0,0,0"
                        },
                        {
```

```
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0,0,11,0,3,2"
                        }
                    ]
                }
            ]
        }
    ]
}
```


173.4.4. Cancelling the search

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=vehiclecountsearch&action=control&Mode=Cancel&Sea
rchToken=VehicleCount-2016-09-24T02:32:51-614
```

Chapter 174. POS Configuration

174.1. Description

The **posconf** submenu is used for configuring a POS device.

NOTE	This chapter applies to NVR only. Attribute to check for Feature Support: "attributes/System/Limit/MaxPOS"

Access level

Action	NVR
view	User
set	User

174.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
posconf&action=<value>[&<parameter>=<value>]
```

174.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	DeviceIDList	REQ	<csv>	Optional request parameter for sending a specific device ID. **Note** DeviceID starts from 0.
set	DeviceID	REQ, RES	<int> Start, Cancel	Mode
	DeviceName	REQ, RES	<string>	Name of the device
	Port	REQ, RES	<int>	Port number to which POS is configured
	ChannelIDList	REQ, RES	<csv>	Channels to which POs is mapped

Action	Parameters	Request/ Response	Type/ Value	Description
	EncodingType	REQ, RES	<enum> US-ASCII, UTF-8, UTF-16, EUC-KR, ISO-2022-KR, EUC-JP, SHIFT-JIS, ISO-2022-JP, EUC-CN, ISO-2022-CN, BIG5, GB2312, ISO-8859-1, ISO-8859-2, ISO-8859-3, WINDOWS-1250, WINDOWS-1251, WINDOWS-1252, WINDOWS-1253, WINDOWS-1254, CP850, CP866, CP932, CP949, CP950, CP1250, CP1251, CP1252, CP1253, CP1254, CP1257	POS encoding type setting
	ReceiptStart	REQ, RES	<string>	Receipt start identifier
	ReceiptStartType	REQ, RES	<enum> Text,HexCode,RegExpression	Receipt start identifier type
	ReceiptEnd	REQ, RES	<string>	Receipt end identifier

Action	Parameters	Request/ Response	Type/ Value	Description
	ReceiptEndType	REQ, RES	<enum> Text,HexCode,RegExpression	Receipt end identifier type
	EventPlaybackStartTime	REQ, RES	<int>	Playback start time
	EventPlaybackStartTimeUnits	REQ, RES	<enum> Seconds	Time units
	Enable	REQ, RES	<bool> True, False	Enabling and Disabling a device
	DeviceType	REQ, RES	<enum> User Defined, EPSON,WINCOR,AXIHON,RADIANTSYSTEM,IBM,ANPR	Pos Device Types

174.4. Examples

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=posconf&action=view&DeviceIDList=0,1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
DeviceID.0.DeviceName=TEXT 01
DeviceID.0.Enable=True
DeviceID.0.Port=7001
DeviceID.0.EventPlaybackStartTime=0
DeviceID.0.EventPlaybackStartTimeUnits=Seconds
DeviceID.0.ReceiptStart=(1)
DeviceID.0.ReceiptEnd=(2)
DeviceID.0.EncodingType=US-ASCII
```

```
DeviceID.0.ChannelIDList=0,1,2,3,4,5,6,7,16,17,18,19,20,21,22,23,32,33,34,35
,36,37,38,39,48,49,50,51,52,53,54,55
DeviceID.1.DeviceName=TEXT 02
DeviceID.1.Enable=True
DeviceID.1.Port=7002
DeviceID.1.EventPlaybackStartTime=0
DeviceID.1.EventPlaybackStartTimeUnits=Seconds
DeviceID.1.ReceiptStart=(1)
DeviceID.1.ReceiptEnd=(2)
DeviceID.1.EncodingType=US-ASCII
DeviceID.1.ChannelIDList=8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31,40,41
,42,43,44,45,46,47,56,57,58,59,60,61,62,63
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "POSDevices": [
        {
            "DeviceID": 0,
            "DeviceName": "TEXT 01",
            "Enable": true,
            "Port": 7001,
            "EventPlaybackStartTime": 0,
            "EventPlaybackStartTimeUnits": "Seconds",
            "ReceiptStart": "(1)",
            "ReceiptEnd": "(2)",
            "EncodingType": "US-ASCII",
            "ChannelIDList": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "16",
```

```
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23",
                "32",
                "33",
                "34",
                "35",
                "36",
                "37",
                "38",
                "39",
                "48",
                "49",
                "50",
                "51",
                "52",
                "53",
                "54",
                "55"
            ]
        },
        {
            "DeviceID": 1,
            "DeviceName": "TEXT 02",
            "Enable": true,
            "Port": 7002,
            "EventPlaybackStartTime": 0,
            "EventPlaybackStartTimeUnits": "Seconds",
            "ReceiptStart": "(1)",
            "ReceiptEnd": "(2)",
            "EncodingType": "US-ASCII",
            "ChannelIDList": [
                "8",
                "9",
                "10",
                "11",
                "12",
```

```
                "13",
                "14",
                "15",
                "24",
                "25",
                "26",
                "27",
                "28",
                "29",
                "30",
                "31",
                "40",
                "41",
                "42",
                "43",
                "44",
                "45",
                "46",
                "47",
                "56",
                "57",
                "58",
                "59",
                "60",
                "61",
                "62",
                "63"
            ]
        }
    ]
}
```

Chapter 175. POS Event Configuration

175.1. Description

The **poseventconf** submenu is used to configure a POS event in the device.

NOTE This chapter applies to NVR only.
Attribute to check for Feature Support: "attributes/System/Limit/MaxPOS"

Access level

Action	NVR
view	User
set	User
add/update	User
remove	User

175.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
poseventconf&action=<value>[&<parameter>=<value>...]
```

175.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	AmountEventEnable	REQ, RES	<bool> True, False	Enables or disables the event based on the amount
	TotalType	REQ, RES	<enum> Equal, Above, Below	Total amount condition
	TotalAmount	REQ, RES	<float>	Total amount
add/update	KeywordIndex	REQ, RES	<int>	Index of keyword to be added/updated
	KeywordCondition	REQ, RES	<string>	Keyword string
remove	KeywordIndex	REQ	<int>	Valid keyword index that needs to be removed.

175.4. Examples

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=poseventconf&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AmountEventEnable=True
TotalAmount=100.000000
TotalType=Above
KeywordIndex.1.KeywordCondition=Apple
KeywordIndex.2.KeywordCondition=banana
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AmountEventEnable": true,
    "TotalAmount": 100,
    "TotalType": "Above",
    "Keywords": [
        {
            "KeywordIndex": 1,
            "KeywordCondition": "Apple"
        },
        {
            "KeywordIndex": 2,
            "KeywordCondition": "banana"
        }
    ]
}
```

Chapter 176. POS Data

176.1. Description

The **posdata** submenu is used to receive the live POS data from the device.

> **NOTE**
> This chapter applies to NVR only.
> Attribute to check for Feature Support: "attributes/System/Limit/MaxPOS"

Access level

Action	NVR
monitordiff	User

176.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
posdata&action=<value>[&<parameter>=<value>...]
```

176.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
monitordiff	DeviceIDList	REQ, RES	<csv>	Device ID list for which POS data needs to be monitored
	ReceivedDate	RES	<string>	Date and time
	DeviceID	RES	<int>	Device ID **Note** DeviceID starts from 0.
	Receipt	RES	<string>	Receipt information

176.4. Examples

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=posdata&action=monitordiff
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
--<MultiPartBoundary>
Content-type:text/plain

ReceivedDate=2016-07-28T05:06:55Z
DeviceID=1
Receipt=
03-06-16 2:43P
<keyword>APPLE</keyword> 9.00
BERRY 3.50
MELON 10.50
PLUM 3.00

SUBTOTAL 26.00
TAX 03.00
TOTAL 29.00
CASH 30.00
CHANGE 01.00


--<MultiPartBoundary>
Content-type:text/plain

ReceivedDate=2016-07-28T05:06:55Z
DeviceID=0
Receipt=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
--<MultiPartBoundary>
Content-type:application/json

{
    "ReceivedDate": "2016-07-28T05:06:55Z",
    "DeviceID": 1,
    "Receipt": "\r\n03-06-16 2:43P\r\n<keyword>APPLE</keyword>
\t\t9.00\r\nBERRY\t \t3.50\r\nMELON \t\t10.50\r\nPLUM\t
\t3.00\r\n\r\nSUBTOTAL \t26.00\r\nTAX \t\t03.00\r\nTOTAL \t\t29.00\r\nCASH
\t\t30.00\r\nCHANGE \t\t01.00\r\n"
}

--<MultiPartBoundary>
Content-type:application/json

{
    "ReceivedDate": "2016-07-28T05:06:55Z",
    "DeviceID": 0,
    "Receipt": "\r\n02-06-16 2:43P\r\nOKRA \t\t5.00\r\nOIL\t \t9.50\r\nLEMON
\t\t2.50\r\nGREEN BANANNAS\t3.00\r\nYELLOW
BANANNAS\t3.00\r\n\r\n\r\nSUBTOTAL \t23.00\r\nTAX \t\t02.70\r\nTOTAL
\t\t25.70\r\nCASH \t\t30.00\r\nCHANGE \t\t04.30\r\n"
}
```

Chapter 177. POS Calendar

177.1. Description

The **poscalendar** submenu is used to get the availability of POS data in a month.

> **NOTE**
> This chapter applies to NVR only.
> Attribute to check for Feature Support: "attributes/System/Limit/MaxPOS"

Access level

Action	NVR
view	User

177.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
poscalendar&action=<value>[&<parameter>=<value>...]
```

177.3. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view	Month	REQ	<string>	Target month for searching Month must be specified in <YYYY-MM> format.
	Calendar	RES	<string>	String of 31 characters consisting of 0s and 1s to represent each day of the month; if data is available it is set to 1 and set to 0 otherwise

177.4. Examples

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=poscalendar&action=view&Month=2016-09
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Calendar=00000111100000000001100101010101
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Calendar": "00000111100000000001100101010101"
}
```

Chapter 178. Meta Data Search

178.1. Description

The **metadata** submenu searches POS, GPS, video, audio, and image information for a given search criteria.

NOTE	This chapter applies to NVR only. Attribute to check for Feature Support in NVR: "attributes/Recording/Support/SearchMetadata" Attribute to check for MaxResults supported in NVR: "attributes/Recording/metadata/view/MaxResults" Attribute to check for Maximum Allowed time gap between search from date and search to date: "attributes/Recording/Limit/MaxMetadataSearchDays"

Access level

Action	NVR
view	User
control	User

178.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
metadata&action=<value>[&<parameter>=<value>]
```

178.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	Search Type
	SearchToken	REQ	<sring>	Search Session token
	ResultFromIndex	REQ	<int>	Index from which search results to be fetched
	ResultFromTime	REQ	<string>	Time from which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	ResultToTime	REQ	<string>	Time to which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ
	MaxResults	REQ	<int>	Maximum number of search results to return
	Status	RES	<enum> Results, Status	Search status
	TotalResultsFound	RES	<int>	Total results
	TotalCount	RES	<int>	Total result count
	TimedOut	RES	<bool> True, False	
	SearchTokenExpiryTime	RES	<string> UTCFormat =YYYY-MM-DDTHH:MM:SSZ	Time at which search token expires
	Result.#.Date	RES	<string> UTCFormat =YYYY-MM-DDTHH:MM:SSZ	Result date
	Result.#.PlayTime	RES	<string> UTCFormat =YYYY-MM-DDTHH:MM:SSZ	Result play time
	Result.#.DeviceID	RES	<int>	POS ID
	Result.#.TextData	RES	<string>	POS receipt
	Result.#.KeywordsMatched	RES	<csv>	Keywords found in POS receipt
	Result.#.ChannelIDList	RES	<csv>	Result channel ID List
	Result.#.BkID	RES	<string>	Bookmark ID

Action	Parameters	Request/ Response	Type/ Value	Description
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Search Mode
	MetadataType	REQ	<enum> POS	Type of metadata to be searched
	DeviceIDList	REQ	<csv>	POS ID list
	OverlappedID	REQ	<int>	Overlapped number
	Keyword	REQ	<string>	Search keyword
	IsWholeWord	REQ	<bool> True, False	To match whole word or not for search
	IsCaseSensitive	REQ	<bool> True, False	Whether search is case-sensitive or not
	FromDate	REQ	<string> UTCFormat =YYYY-MM-DDTHH:MM:SSZ	From date of the search
	ToDate	REQ	<string> UTCFormat =YYYY-MM-DDTHH:MM:SSZ	To date of search
	SearchToken	REQ,RES	<string>	Metadata search token **SearchToken** is a request parameter when **Mode** is Cancel, and a response parameter when **Mode** is Start. **Note** Search Token will expire in 60 seconds. Client has to send Renew command periodically to increase the expiry time by 60 seconds.

178.4. Examples

178.4.1. Start search for POS data

Request without any filters

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-13T00:00:00Z&ToDate=2016-07-16T23:59:59Z
```

Request with Overlapped ID

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11
```

Request with Overlapped ID and Single Keyword

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=Apple
```

Request with Overlapped ID and Keyword Green or Apple

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&IsWholeWord=false&Keyword=Green%20Apple
```

Request with Overlapped ID and Keyword "Green,Apple"sss

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=Green,Apple
```

Request with Overlapped ID, Keyword and IsCaseSensitive

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=APPLE&IsCaseSensitive=true
```

Request with Overlapped ID, Keyword, IsCaseSensitive and Single DeviceID

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=OKRA&IsCaseSensitive=true&DeviceIDList=
0
```

Request with Overlapped ID, Keyword, IsCaseSensitive and Multiple DeviceIDs

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=OKRA&IsCaseSensitive=true&DeviceIDList=
1,2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=7475
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "7475"
}
```

178.4.2. Cancel Search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Cancel&SearchToken=7475
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

178.4.3. To get search status

REQUEST

```
http://<Device-IP>/stw-cgi/recording.cgi?msubmenu=metadata&action=view&Type=Status&SearchToken=7475
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
Status=Completed
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "Completed"
}
```

178.4.4. To renew search token (For 1 minute)

REQUEST

```
http://<Device-IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Renew&SearchToken=74
75
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "Response": "Success"
}
```

178.4.5. To get the results of search (First 100 results)

REQUEST

```
http://<Device-IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=view&Type=Results&ResultFromIndex
=1&MaxResults=100&SearchToken=6619
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2016-07-19T07:22:47Z
TotalResultsFound=399
TotalCount=100

Result.1.DeviceID=1
Result.1.Date=2016-07-18T07:28:01Z
Result.1.ChannelIDList=0,1,2,3,4,5,6,7
Result.1.KeywordsMatched=
Result.1.TextData=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00


SUBTOTAL 23.00
TAX 02.70
TOTAL 25.70
CASH 30.00
CHANGE 04.30

Result.2.DeviceID=2
```

```
Result.2.Date=2016-07-18T07:28:00Z
Result.2.ChannelIDList=8,9,10,11,12,13,14,15
Result.2.KeywordsMatched=
Result.2.TextData=
03-06-16 2:43P
APPLE 9.00
BERRY 3.50
MELON 10.50
PLUM 3.00


SUBTOTAL 26.00
TAX 03.00
TOTAL 29.00
CASH 30.00
CHANGE 01.00

Result.3.DeviceID=1
Result.3.Date=2016-07-18T07:27:56Z
Result.3.ChannelIDList=0,1,2,3,4,5,6,7
Result.3.KeywordsMatched=
Result.3.TextData=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SearchTokenExpiryTime": "2016-07-19T07:22:47Z",
    "ToralResultsFound": 399,
    "TotalCount": 100,
    "MetaDataSearchResults": [
        {
```

```
            "Result": 1,
            "DeviceID": 2,
            "Date": "2016-07-18T07:22:47Z",
            "ChannelIDList": [
                8,
                9,
                10,
                11,
                12,
                13,
                14,
                15
            ],
            "KeywordsMatched": [],
            "TextData": "\r\n03-06-16 2:43P\r\nAPPLE \t\t9.00\r\nBERRY\t
\t3.50\r\nMELON \t\t10.50\r\nPLUM\t \t3.00\r\n\r\nSUBTOTAL \t26.00\r\nTAX
\t\t03.00\r\nTOTAL \t\t29.00\r\nCASH \t\t30.00\r\nCHANGE \t\t01.00\r\n"
        },
        {
            "Result": 2,
            "DeviceID": 1,
            "Date": "2016-07-18T07:22:47Z",
            "ChannelIDList": [
                0,
                1,
                2,
                3,
                4,
                5,
                6,
                7
            ],
            "KeywordsMatched": [],
            "TextData": "\r\n02-06-16 2:43P\r\nOKRA \t\t5.00\r\nOIL\t
\t9.50\r\nLEMON \t\t2.50\r\nGREEN BANANNAS\t3.00\r\nYELLOW
BANANNAS\t3.00\r\n\r\n\r\nSUBTOTAL \t23.00\r\nTAX \t\t02.70\r\nTOTAL
\t\t25.70\r\nCASH \t\t30.00\r\nCHANGE \t\t04.30\r\n"
        },
        {
            "Result": 3,
            "DeviceID": 2,
```

```
            "Date": "2016-07-18T07:22:43Z",
            "ChannelIDList": [
                8,
                9,
                10,
                11,
                12,
                13,
                14,
                15
            ],
            "KeywordsMatched": [],
            "TextData": "\r\n03-06-16 2:43P\r\nAPPLE \t\t9.00\r\nBERRY\t
\t3.50\r\nMELON \t\t10.50\r\nPLUM\t \t3.00\r\n\r\nSUBTOTAL \t26.00\r\nTAX
\t\t03.00\r\nTOTAL \t\t29.00\r\nCASH \t\t30.00\r\nCHANGE \t\t01.00\r\n"
        }
    ]
}
```

178.4.6. To get the results of search (Next 100 results)

REQUEST

```
http://<Device-IP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=view&Type=Results&ResultFromIndex
=101&MaxResults=100&SearchToken=6619
```

Chapter 179. Smart Search

179.1. Description

The **smartsearch** submenu is used to search video analytics information for a given search criteria.

> **NOTE**
>
> This chapter applies to NVR only.
> Attribute to check for Feature Support in NVR for each channel:
> "attributes.cgi/attributes/recording/Support/1/SmartSearch"
> Attribute to check for maximum number of include areas supported in NVR:
> "attributes/recording/Limit/MaxSmartSearchIncludeAreas"
> Attribute to check for maximum number of exclude areas supported in NVR:
> "attributes/recording/Limit/MaxSmartSearchExcludeAreas"
> Attribute to check for maximum number of lines supported in NVR:
> "attributes/recording/Limit/MaxSmartSearchlines"

Access level

Action	NVR
view	User
control	User

179.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
smartsearch&action=<value>[&<parameter>=<value>]
```

179.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	Search Type
	SearchToken	REQ	<sring>	Search session token
	TotalCount	RES	<int>	Result count
	TimedOut	RES	<bool> True, False	Asynchronous search timeout.
	Channel.#.Result.#.Event Time	RES	<string>	Time at which event happened

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel.#.Result.#.EventType	RES	<enum> Motion, Enter, Exit, Pass	Type of event
	Status	RES	<enum> Completed, NotCompleted	Search status This parameter is returned only when Type is set to Status in the request.
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Search mode
	OverlappedID	REQ	<int>	Overlapped ID number
	Channel	REQ	<int>	Channel ID
	FromDate	REQ	<string>	From date of the search
	ToDate	REQ	<string>	To date of search
	Area.#.EventType	REQ	<csv> Motion, Enter, Exit	Type of area event
	Line.#.EventType	REQ	<enum> BothDirections, Right, Left	Type of line event
	Area.#.Type	REQ	<enum> Inside, Outside	Area type
	Area.#.Coordinates	REQ	<string> Format=x1,y1,x2,y2	Coordinates of the area
	Area.#.Filter	REQ	<enum>+ Person, Vehicle, Unknown	Object filter
	Line.#.Coordinates	REQ	<string> Format=x1,y1,x2,y2	Coordinates of the line

Action	Parameters	Request/ Response	Type/ Value	Description
	SearchToken	REQ,RES	<string>	Smart search token **SearchToken** is a request-only parameter when **Mode** is set to Cancel. It is a response parameter when **Mode** is set to Start.

179.4. Examples

179.4.1. Start smart search for selected area

Request without any filters

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=smartsearch&action=control&Mode=Start&channel=5&F
romDate=2016-06-15T00:00:00Z&ToDate=2016-06-
15T23:59:59Z&Area.1.EventType=Motion,Enter,Exit&Area.1.Type=Inside&Area.1.Co
ordinates=-0.903226,0.870504,-0.903226,-0.877698,0.903226,-
0.877698,0.903226,0.870504
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=4174
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "4174"
```

```
}
```

179.4.2. Check the status of smart search

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=smartsearch&action=view&Type=Status&SearchToken=4
174
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Completed
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Status": "Completed"
}
```


179.4.3. To get results of smart search

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=smartsearch&action=view&Type=Results&SearchToken=
4174
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
TotalCount=5
Channel.5.Result.1.EventTime=2016-06-15T00:08:35Z
Channel.5.Result.1.EventType=Motion
Channel.5.Result.2.EventTime=2016-06-15T00:32:34Z
Channel.5.Result.2.EventType=Motion
Channel.5.Result.3.EventTime=2016-06-15T00:32:35Z
Channel.5.Result.3.EventType=Motion
Channel.5.Result.4.EventTime=2016-06-15T01:32:02Z
Channel.5.Result.4.EventType=Motion
Channel.5.Result.5.EventTime=2016-06-15T01:32:03Z
Channel.5.Result.5.EventType=Motion
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TotalCount": 5,
    "SmartSearchResults": [
        {
            "Channel": 5,
            "Results": [
                {
                    "Result": 1,
                    "EventTime": "2016-06-15T00:08:35Z",
                    "EventType": "Motion"
                },
                {
                    "Result": 2,
                    "EventTime": "2016-06-15T00:32:34Z",
                    "EventType": "Motion"
                },
                {
                    "Result": 3,
                    "EventTime": "2016-06-15T00:32:35Z",
                    "EventType": "Motion"
```

```
                },
                {
                    "Result": 4,
                    "EventTime": "2016-06-15T01:32:02Z",
                    "EventType": "Motion"
                },
                {
                    "Result": 5,
                    "EventTime": "2016-06-15T01:32:03Z",
                    "EventType": "Motion"
                }
            ]
        }
    ]
}
```


179.4.4. Start Smart search for the lines

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=smartsearch&action=control&Mode=Start&channel=5&F
romDate=2016-06-15T00:00:00Z&ToDate=2016-06-
15T23:59:59Z&Line.1.EventType=Right&Line.1.Coordinates=-
0.903226,0.870504,0.903226,0.870504
```

Chapter 180. Queue Search

180.1. Description

The **queuesearch** submenu provides statistical analysis and measurement of average dwell time and number of people in queues based on a given search criteria.

> **NOTE**
> This chapter applies to camera only.
> Attribute to check for Feature Support: "Recording/Support/QueueManagement"
> Attribute to check for Max Queues Supported: "Eventsource/Limit/MaxQueues"

Access level

Action	Camera
view	Admin
control	Admin

180.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
queuesearch&action=<value>[&<parameter>=<value>]
```

180.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	Type If **Type** is set to Results or Status, the **view** action must be sent together with **SearchToken**.
	SearchToken	REQ	<string>	Search token
	ResultInterval	RES	<enum> Hourly, Daily, Weekly, Monthly	Search result interval

Action	Parameters	Request/ Response	Type/ Value	Description
	Queue.#.AveragePeople Result	RES	<csv>	Queue Search Average People Results. If **ResultInterval** is Hourly, search results are returned in terms of hours and the number of results is fixed, i.e. 24. Here, the first result in the array represents the 0^{th} hour of the day, and the last result in the array represents the 23^{rd} hour of the day. If **ResultInterval** is Daily, search results are returned in terms of days and the first result in the array represents the first day of the month while the last result in the array represents the last day of the month. If **ResultInterval** is Weekly, search results are returned in terms of weeks. If **ResultInterval** is Monthly, search results are returned in terms of months and the first result in the array represents the first month of the year, while the last result in the array represents the last month of the year. **Note** **ResultInterval** is fixed by the camera based on **FromDate** and **ToDate** of the search

Action	Parameters	Request/ Response	Type/ Value	Description
	Queue.#.Level.#.CumulativeTimeResult	RES	<csv>	Queue Search Cumulative Time Result If **ResultInterval** is Hourly, search results are returned in terms of hours and the number of results is fixed, i.e. 24. Here, the first result in the array represents the 0th hour of the day, and the last result in the array represents the 23rd hour of the day. If **ResultInterval** is Daily, search results are returned in terms of days and the first result in the array represents the first day of the month while the last result in the array represents the last day of the month. If **ResultInterval** is Weekly, search results are returned in terms of weeks. If **ResultInterval** is Monthly, search results are returned in terms of months and the first result in the array represents the first month of the year, while the last result in the array represents the last month of the year. **Note** **ResultInterval** is fixed by the camera based on **FromDate** and **ToDate** of the search
	Status	RES	<enum> Completed, NotCompleted	Search status **Status** is valid only when **Type** is set to Status.
control	Mode	REQ	<enum> Start, Cancel	Mode
	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	FromDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The start date and time for search
	ToDate	REQ	<string> <format=YYYY-MM-DDTHH:MM:SSZ>	The end date and time for search
	SearchToken	REQ, RES	<string>	Search token **SearchToken** is a request-only parameter when **Mode** is set to Cancel, but it will return data when **Mode** is set to Start.
	Queue.#.AveragePeople	REQ	<bool> True, False	Enables or disables Queue Average People search
	Queue.#.Level.#.CumulativeTime	REQ	<bool> True, False	Enables or disables Queue Cumulative Time search

180.4. Examples

180.4.1. Queue Search

Queue search for average people result

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.AveragePeople=True&Queue.2.AveragePeople=True&Queue.3.AveragePeople=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "QueueManagement-2017-09-15T04:3"
}
```

Queue search for cumulative time result

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.Level.High.CumulativeTime=True&Queue.1.Level.Medium.CumulativeTime=True&Queue.2.Level.High.CumulativeTime=True&Queue.2.Level.Medium.CumulativeTime=True&Queue.3.Level.High.CumulativeTime=True&Queue.3.Level.Medium.CumulativeTime=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "QueueManagement-2017-09-15T04:4"
}
```

Queue search for average people and cumulative time result

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.AveragePeople=True&Queue.2.AveragePeople=True&Queue.3.AveragePeople=True&Queue.1.Level.High.CumulativeTime=True&Queue.1.Level.Medium.CumulativeTime=True&Queue.2.Level.High.CumulativeTime=True&Queue.2.Level.Medium.CumulativeTime=True&Queue.3.Level.High.CumulativeTime=True&Queue.3.Level.Medium.CumulativeTime=True
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SearchToken": "QueueManagement-2017-09-15T04:5"
}
```


180.4.2. Getting the status of Queue search

Type must be set to Status, and **SearchToken** must be sent with **view** action.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Status&SearchToken=Q
ueueManagement-2017-09-15T04:4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Status": "Completed"
}
```


180.4.3. Getting Queue search result

Type must be set to Results, and **SearchToken** must be sent with **view** action.

Getting Average People search result

REQUEST

```
http://<Device IP>/ stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Results&SearchToken=
QueueManagement-2017-09-15T04:3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
        {
            "Queue": 1,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        },
        {
            "Queue": 2,
            "AveragePeopleResult": [
```

```
            "0",
            "1",
            "2",
            "3",
            "4",
            "5",
            "6",
            "7",
            "8",
            "9",
            "10",
            "11",
            "12",
            "13",
            "14",
            "15",
            "16",
            "17",
            "18",
            "19",
            "20",
            "21",
            "22",
            "23"
        ]
    },
    {
        "Queue": 3,
        "AveragePeopleResult": [
            "0",
            "1",
            "2",
            "3",
            "4",
            "5",
            "6",
            "7",
            "8",
            "9",
            "10",
            "11",
```

```
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        }
    ]
}
```

Getting Cumulative Time search result

REQUEST

```
http://<Device IP>/ stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Results&SearchToken=
QueueManagement-2017-09-15T04:4
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
        {
            "Queue": 1,
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
```

```
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                }
            ]
        },
        {
            "Queue": 2,
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                }
            ]
        },
        {
            "Queue": 3,
```

```
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                }
            ]
        }
    ]
}
```

Getting Average People and Cumulative Time search result

REQUEST

```
http://<Device IP>/ stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Results&SearchToken=
QueueManagement-2017-09-15T04:5
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
```

```
        {
            "Queue": 1,
            "AveragePeopleResult": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23"
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                }
            ]
        },
        {
            "Queue": 2,
            "AveragePeopleResult": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23"
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
```

```
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                }
            ]
        },
        {
            "Queue": 3,
            "AveragePeopleResult": [
                "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10",
"11", "12",
                "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23"
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
"21", "22", "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0", "1", "2", "3", "4", "5", "6", "7", "8", "9",
"10", "11", "12",
                        "13", "14", "15", "16", "17", "18", "19", "20",
```

```
"21", "22", "23"
                    ]
                }
            ]
        }
    ]
}
```

180.4.4. Cancelling Queue search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=queuesearch
&action=control&Mode=Cancel&SearchToken=QueueManagement-2017-09-15T04:3
```

Chapter 181. Disk Utility

181.1. Description

The **diskutility** submenu gets the details of HDD array from NVR

NOTE This submenu is only available for NVR

Access level

Action	NVR
view	User

181.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
diskutility&action=<value>[&<parameter>=<value>]
```

181.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	REQ	<int>	Disk index
	Disk.#.Index	RES	<int>	Disk index
	Disk.#.Name	RES	<string>	Disk model name
	Disk.#.SMART	RES	<string>	Smart HDD details

181.4. Examples

181.4.1. Getting the disk information

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=diskutility&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Disk.0.Index=14
Disk.0.Name=ST1000VM002-1CT162
Disk.1.Index=15
Disk.1.Name=WDC WD60PURX-64T0ZY1
Disk.2.Index=16
Disk.2.Name=WDC WD40PURX-64N96Y0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Disks": [
        {
            "Index": 14,
            "Name": "ST1000VM002-1CT162 "
        },
        {
            "Index": 15,
            "Name": "WDC WD60PURX-64T0ZY1 "
        },
        {
            "Index": 16,
            "Name": "WDC WD40PURX-64N96Y0 "
        }
    ]
}
```


REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=diskutility&action=view&Index=14
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
```

```
<Body>
```

```
Enable=True
Disk.0.Index=14
Disk.0.Name=ST1000VM002-1CT162
Disk.0.SMART=<html><body><h3 style="color:rgb(60,179,133);"> Status :
GOOD</h3>
<pre>Model Name : ST1000VM002-1CT162
Serial : W1G0AJYW
Firmware Version : SC23
Capacity : 1 TB
Temperature : 35&#8451; / 95&#8457;

</pre><pre>ID Attribute Name Current Worst Threshhold RawValue Status
001 Read Error Rate 117 099 006 000000057888 GOOD
003 Spin-Up Time 098 097 000 000000000000 GOOD
004 Start/Stop Count 096 096 020 000000004413 GOOD
005 Reallocated Sectors 100 100 036 000000000000 GOOD
007 Seek Error Rate 080 060 030 000000024466 GOOD
009 Power-On Hours Count 076 076 000 000000021334 GOOD
010 Spin Retry Count 100 100 097 000000000000 GOOD
012 Power Cycle Count 099 099 020 000000001131 GOOD
184 End-to-End error 100 100 099 000000000000 GOOD
187 Reported Uncorrectable 088 088 000 000000000012 GOOD
188 Command Timeout 100 099 000 000000000002 GOOD
189 High Fly Writes 001 001 000 000000000363 GOOD
190 Temperature Diff 065 051 045 000000000035 GOOD
191 G-sense error rate 100 100 000 000000000000 GOOD
192 Power-off retract 100 100 000 000000001125 GOOD
193 Load/Unload cycle 098 098 000 000000004413 GOOD
194 HDA temperature 035 049 000 000000000035 GOOD
197 Current pending 100 100 000 000000000000 GOOD
198 Offline scan wrong 100 100 000 000000000000 GOOD
199 UDMA CRC error rate 200 200 000 000000000000 GOOD
</pre></body></html>
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```


```
{
    "Disks": [
        {
            "Index": 14,
            "Name": "ST1000VM002-1CT162 ",
            "SMART": "<html><body><h3 style=\"color:rgb(60,179,133);\">
Status : GOOD</h3>\n<pre>Model Name : ST1000VM002-1CT162\nSerial :
W1G0AJYW\nFirmware Version : SC23\nCapacity : 1 TB\nTemperature : 35&#8451;
/ 95&#8457; \n\n</pre><pre>ID Attribute Name Current Worst Threshhold
RawValue Status\n001 Read Error Rate 117 099 006 000000057888 GOOD \n003
Spin-Up Time 098 097 000 000000000000 GOOD \n004 Start/Stop Count 096 096
020 000000004413 GOOD \n005 Reallocated Sectors 100 100 036 000000000000
GOOD \n007 Seek Error Rate 080 060 030 000000024466 GOOD \n009 Power-On
Hours Count 076 076 000 000000021334 GOOD \n010 Spin Retry Count 100 100 097
000000000000 GOOD \n012 Power Cycle Count 099 099 020 000000001131 GOOD
\n184 End-to-End error 100 100 099 000000000000 GOOD \n187 Reported
Uncorrectable 088 088 000 000000000012 GOOD \n188 Command Timeout 100 099
000 000000000002 GOOD \n189 High Fly Writes 001 001 000 000000000363 GOOD
\n190 Temperature Diff 065 051 045 000000000035 GOOD \n191 G-sense error
rate 100 100 000 000000000000 GOOD \n192 Power-off retract 100 100 000
000000001125 GOOD \n193 Load/Unload cycle 098 098 000 000000004413 GOOD
\n194 HDA temperature 035 049 000 000000000035 GOOD \n197 Current pending
100 100 000 000000000000 GOOD \n198 Offline scan wrong 100 100 000
000000000000 GOOD \n199 UDMA CRC error rate 200 200 000 000000000000 GOOD
\n</pre></body></html>"
        }
    ]
}
```

Chapter 182. Bookmark

182.1. Description

The **bookmark** submenu can be used to bookmark a video clip in NVR.

NOTE	This submenu is available only for NVR that supports System/Support/AIFeatures

Access level

Action	NVR
view	User
add	User
update	User
remove	User

182.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
bookmark&action=<value>[&<parameter>=<value>]
```

182.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ChannelIDList	REQ	<csv>	Channel id list
	FromDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	OverlappedID	REQ	<int>	Overlapped id
	TotalCount	RES	<int>	Total result count
	Result.#.FromDate	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.ToDate	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	Result.#.ChannelIDList	RES	<csv>	Channel id list
	Result.#.Name	RES	<string>	Name of bookmark
	Result.#.Category	RES	<enum> TIME, EVENT, SMART, TEXT, AI_PERSON, AI_FACE, AI_FACE_RECOGNITION, AI_VEHICLE	Bookmark category
	Result.#.SubCategory	RES	<enum> AlarmInput, MotionDetection,Videoloss, Passing, Entering, Exiting, Appearing, Disappering, Tampering, FaceDetection, Loitering, Tracking, DefocusDetection, FogDetection, AudioDetection, Scream, Gunshot, Explosion, GlassBreak, GSensorEvent, EmergencyTrigger, Intrusion	Subcategory of bookmark
	Result.#.OverlappedID	RES	<int>	Recording overlapped id
	Result.#.ObjectID	RES	<int>	Object ID
	Result.#.BkID	RES	<string>	Unique bookmark ID (UUID)
add	FromDate	REQ	<sting>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ChannelIDList	REQ	<csv>	Channel id list
	ObjectID	REQ	<int>	Object ID
	Name	REQ	<string>	Bookmark Name

Action	Parameters	Request/ Response	Type/ Value	Description
	Category	REQ	<enum> TIME, EVENT, SMART, TEXT, AI_PERSON, AI_FACE, AI_FACE_RECOGNITION, AI_VEHICLE	Bookmark category
	OverlappedID	REQ	<int>	Recording overlapped id
update	BkID	REQ	<string>	Bookmark unique id (UUID)
	Name	REQ	<string>	Bookmark name
remove	BkID	REQ	<string>	Unique bookmark id to be removed

182.4. Examples

182.4.1. Adding a bookmark

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=bookmark&action=add&FromDate=2020-04-12T09:00:00Z&ToDate=2020-04-13T09:03:00Z&ChannelIDList=0&Category=TIME&Name=TestBookMark&OverlappedID=-1&ObjectID=214
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
BkID=3757c60d27b8484cab29468d0c800a23
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "BkID": "3757c60d27b8484cab29468d0c800a23"
}
```

182.4.2. Removing a bookmark

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=bookmark&action=remove&BkID=3757c60d27b8484cab29468d0c800a23
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

182.4.3. Updating a bookmark

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=bookmark&action=update&BkID=3757c60d27b8484cab29468d0c800a23&Name=TestBookMark2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


182.4.4. Viewing a bookmark

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=bookmark&action=view&Category=TEXT
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalResultsFound=1
TotalCount=1

Result.0.FromDate=2020-03-31T09:00:00Z
Result.0.ToDate=2020-03-31T09:03:00Z
Result.0.ChannelIDList=0
Result.0.Name=TestBookMark
Result.0.Category=TIME
```

```
Result.0.SubCategory=
Result.0.OverlappedID=100
Result.0.ObjectID=0
Result.0.BkID=3757c60d27b8484cab29468d0c800a23
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TotalResultsFound": 1,
    "TotalCount": 1,
    "BookmarkResults": [
        {
            "Result": 0,
            "FromDate": "2020-03-31T09:00:00Z",
            "ToDate": "2020-03-31T09:03:00Z",
            "ChannelIDList": [
                "0"
            ],
            "Name": "TestBookMark",
            "Category": "TIME",
            "SubCategory": "",
            "OverlappedID": 100,
            "ObjectID": 0,
            "BkID": "3757c60d27b8484cab29468d0c800a23"
        }
    ]
}
```

Chapter 183. Event Search

183.1. Description

The **eventsearch** submenu is used to search event information for a given time period.

> **NOTE**
> This submenu is supported only in NVR.
> Attribute to check for Feature Support: "attributes/Recording/Support/SearchEvent"

Access level

Action	NVR
view	User
control	User

183.2. Syntax

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=
eventsearch&action=<value>[&<parameter>=<value>]
```

183.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	If Type is passed as **Status**, search status is informed. If Type is passed as **Result**, search result is provided.
	SearchToken	REQ	<string>	Search session token
	ResultFromIndex	REQ	<int>	Index from which search results are fetched
	ResultFromTime	REQ	<string>	Time from which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	ResultToTime	REQ	<string>	Time to which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ
	MaxResults	REQ	<int>	Maximum number of search results to return
	Status	RES	<enum> Completed, NotComple ted	Search status
	TotalResultsFound	RES	<int>	Total results
	TotalCount	RES	<int>	Total count of result
	TimedOut	RES	<bool> True, False	Search timeout.
	IntervalFrom	RES	<string>	Start time of search result. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ
	IntervalTo	RES	<string>	End time of search result. Time in UTC format. YYYY-MM-DDTHH:MM::SSZ
	SearchTokenExpiryTime	RES	<string>	Time when the search token expires Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.StartDateTime	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.EndDateTime	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.Channel	RES	<int>	Result channel ID
	Result.#.OverlapId	RES	<int>	Recording overlapped id

Action	Parameters	Request/ Response	Type/ Value	Description
	Result.#.EventType	RES	<enum> Refer to **Recording Types** for supported values	EventType
	Result.#.BkID	RES	<string>	Bookmark ID
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Used to start, cancel, renew or stop search
	FromDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ChannelIDList	REQ	<csv>	On which channel search has to be performed
	OverlappedID	REQ	<int>	Recording overlapped id
	Type	REQ	<enum> Refer to **Recording Types** for supported values	Recording type to search Common Types • All: All video recordings including normal and event recordings. • Normal: Continuous video recordings. • Event: Video recording for all events.
	WaitTime	REQ	<int>	Timeout second.(Default:60 sec.)

Recording Types

Type	All, Normal, Event, AlarmInput, VideoAnalysis, MotionDetection, NetworkDisconnect, FaceDetection, TamperingDetection, AudioDetection, Tracking, Manual, UserInput, DefocusDetection, FogDetection, AudioAnalysis, QueueEvent, videoloss, EmergencyTrigger, InternalHDDWarmup, GSensorEvent, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, BodyTemperatureDetection, MaskDetection, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor

Channel.#.Result.#.Type	Normal, AlarmInput, VideoAnalysis, MotionDetection, NetworkDisconnect, FaceDetection, TamperingDetection, AudioDetection, Tracking, ManualRecording, UserInput, DefocusDetection, FogDetection, AudioAnalysis, ShockDetection, TemperatureChangeDetection, BoxTemperatureDetection, BodyTemperatureDetection, MaskDetection, CallRequest, TamperingSwitch, DTMFReceived, ProximitySensor

183.4. Examples

183.4.1. Event search

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=eventSearch&action=control&Mode=Start&OverlappedID=100&FromDate=2023-04-04T00:00:00Z&ToDate=2023-04-04T01:00:00Z&ChannelIDList=0
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=22775
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken":  "22775"
}
```

183.4.2. Viewing search result status

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=eventsearch&action=view&Type=Status&SearchToken=2
```

```
2775
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2023-04-04T01:36:35Z
Status=Completed
TotalResultsFound=0
TotalCount=4
TimedOut=False
IntervalFrom=2023-04-04T00:01:54Z
IntervalTo=2023-04-04T00:25:15Z
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchTokenExpiryTime":    "2023-04-04T01:36:35Z",
    "Status":   "Completed",
    "TotalResultsFound":    0,
    "TotalCount":   4,
    "TimedOut": "False",
    "IntervalFrom": "2023-04-04T00:01:54Z",
    "IntervalTo":   "2023-04-04T00:25:15Z",
    "Results":  []
}
```

183.4.3. Viewing search result

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=eventsearch&action=view&Type=Results&SearchToken=
```

```
22775&ResultFromIndex=1&MaxResults=100
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2023-04-04T01:38:05Z
Status=Completed
TotalResultsFound=4
TotalCount=4
TimedOut=False
Result.0.StartDateTime=2023-04-04T00:24:50Z
Result.0.EndDateTime=2023-04-04T00:25:15Z
Result.0.Channel=0
Result.0.OverlapId=100
Result.0.EventType=Normal
Result.0.BkID=00000000000000000000000000000000
...
Result.3.StartDateTime=2023-04-04T00:01:54Z
Result.3.EndDateTime=2023-04-04T00:23:38Z
Result.3.Channel=0
Result.3.OverlapId=100
Result.3.EventType=Normal
Result.3.BkID=00000000000000000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchTokenExpiryTime":    "2023-04-04T01:38:05Z",
    "Status":   "Completed",
    "TotalResultsFound":    4,
    "TotalCount":   4,
    "TimedOut": "False",
    "Results":  [{
```

```
            "Result":   0,
            "StartDateTime":    "2023-04-04T00:24:50Z",
            "EndDateTime":  "2023-04-04T00:25:15Z",
            "Channel":  0,
            "OverlapId":    100,
            "EventType":    "Normal",
            "BkID": "00000000000000000000000000000000"
        },
        ...
        {
            "Result":   3,
            "StartDateTime":    "2023-04-04T00:01:54Z",
            "EndDateTime":  "2023-04-04T00:23:38Z",
            "Channel":  0,
            "OverlapId":    100,
            "EventType":    "Normal",
            "BkID": "00000000000000000000000000000000"
        }]
}
```

I/O

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 184. Overview

184.1. Description

io.cgi configures the alarm output settings. It provides manual control to the end user to perform control operations on alarm outputs at any time.

The following submenus are used for I/O functionalities:

- **alarmoutput**: Sets and controls the alarm output, state and duration.
- **aux**: Controls the auxiliary equipment state.
- **userinput**: Controls the on/off state of the manual trigger event.
- **alarmreset**: Resets the alarm.
- **ioport**: Sets the configurable alarm IO modes.
- **ledcontrol**: It can check current LED’s status or turn off LED.

Chapter 185. Alarm Output

185.1. Description

The **alarmoutput** submenu configures the alarm output settings.

NOTE	Attribute to check for alarm outputs support: "attributes/IO/Support/AlarmOutput" Attribute to check for maximum alarm outputs: "attributes/IO/Limit/MaxAlarmOutput"

Access level

Action	Camera	NVR
view	Suser	User
control	Suser	User
set	Suser	User

185.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
alarmoutput&action=<value>[&<parameter>=<value>...]
```

185.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the alarm output values.
	AlarmOutput.#.Type	RES	<enum> Alarmout, Beep	Alarm output type (read-only)
	AlarmOutput.#.IOPortIndex	RES	<int>	IO port index It shows the real index of the IO port. Users can match logical and real indexes with this parameter. CAMERA ONLY
control	AlarmOutput.#.State	REQ	<enum> On, Off	Turns the alarm on or off. **Note** **AlarmOutput.#.State** must be sent together with the **control** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	AlarmOutput.#.ManualDuration	REQ	<enum> Always, 1s, 2s, 3s, 4s, 5s, 6s, 7s, 8s, 9s, 10s, 11s, 12s, 13s, 14s, 15s	Alarm output duration **AlarmOutput.#.ManualDuration** is only valid if **AlarmOutput.#.State** is set to On. CAMERA ONLY
set	AlarmOutput.#.IdleState	REQ, RES	<enum> NormallyOpen, NormallyClose	Alarm output state (read-only for NVR)
	AlarmOutput.#.<dddh>	REQ, RES	<enum> Off, On, EventSync	Alarm output time <dddh> stands for the day of the week and time in hours (e.g., SUN1 means 1:00 AM on Sunday. and MON2 means 2:00 AM on Monday). NVR ONLY
	AlarmOutput.#.ManualDuration	REQ, RES	<enum> Always, 1s, 2s, 3s, 4s, 5s, 6s, 7s, 8s, 9s, 10s, 11s, 12s, 13s, 14s, 15s	Alarm output default duration

NOTE # represents the index number of the alarm output.

185.4. Examples

185.4.1. Getting the current alarm output settings

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=alarmoutput&action=view
```

CAMERA TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
AlarmOutput.1.Type=Alarmout
AlarmOutput.1.IdleState=NormallyOpen
AlarmOutput.1.ManualDuration=Always
AlarmOutput.1.IOPortIndex=2
```

CAMERA JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "AlarmOutputs": [
        {
            "AlarmOutput": 1,
            "Type": "Alarmout",
            "IdleState": "NormallyOpen",
            "ManualDuration": "Always",
            "IOPortIndex": 2
        }
    ]
}
```

NVR TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmOutput.1.Type=Alarmout
AlarmOutput.1.IdleState=NormallyOpen
AlarmOutput.1.SUN0=EventSync
AlarmOutput.1.SUN1=EventSync
AlarmOutput.1.SUN2=EventSync
AlarmOutput.1.SUN3=EventSync
```

```
...
```

NVR JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AlarmOutputs": [
        {
            "AlarmOutput": 1,
            "Type": "Alarmout",
            "IdleState": "NormallyOpen",
            "SUN": [
                "Off",
                "Off",
                "Off",
                "Off",
                "Off",
                "Off",
                "Off",
                "Off",
                "Off",
                "On",
                "On",
                "On",
                "On",
                "On",
                "On",
                "On",
                "On",
                "EventSync",
                "EventSync",
                "EventSync",
                "EventSync",
                "EventSync",
                "EventSync",
                "EventSync"
            ],
            "MON": [
```

```
                ...
            ],
            ...
            "FRI": [...]
        }
    ]
}
```

185.4.2. Turning on Alarm Output 1 with continuous output

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=alarmoutput&action=control&AlarmOutput.1.State=On&AlarmO
utput.1.ManualDuration=Always
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=alarmoutput&action=control&AlarmOutput.1.State=On
```

185.4.3. Setting the duration of Alarm Output 1 to 10 seconds

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=alarmoutput&action=set&AlarmOutput.1.IdleState=NormallyO
pen&AlarmOutput.1.ManualDuration=10s
```

Chapter 186. Auxiliary Devices

186.1. Description

The **aux** submenu controls the auxiliary device’s on/off state.

NOTE

This chapter applies to network cameras only.
Attribute to check for auxiliary device support: "attributes/IO/Support/Aux"
Attribute to check for maximum auxiliary devices: "attributes/IO/Limit/MaxAux"

Access level

Action	Camera
view	Suser
control	Suser

186.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
aux&action=<value>[&<parameter>=<value>...]
```

186.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the auxiliary device settings.
	Aux.#.State	RES	<enum> On, Off	Auxiliary device state
control	Aux.#.State	REQ	<enum> On, Off	Auxiliary device state **Note** **Aux.#.State** must be sent together with the **control** action.

NOTE

represents the index number of the auxiliary device.

186.4. Examples

186.4.1. Getting the current auxiliary device settings

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=aux&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Aux.1.State=On
Aux.2.State=Off
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "AuxDevices": [
        {
            "Aux": 1,
            "State": "On"
        },
        {
            "Aux": 2,
            "State": "Off"
        }
    ]
}
```


186.4.2. Deactivating auxiliary device 1

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=aux&action=control&Aux.1.State=Off
```

Chapter 187. User Input State

187.1. Description

The **userinput** submenu controls the user input on/off state. When user input is ‘on’, an event can be manually triggered by a user.

NOTE: This chapter applies to network cameras only.
Attribute to check for user input support: "attributes/Eventsource/Support/UserInput"

Access level

Action	Camera
view	Admin
control	Admin

187.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
userinput&action=<value>[&<parameter>=<value>...]
```

187.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the user input state
	State	RES	<enum> On, Off	User input state
control	State	REQ	<enum> On, Off	User input state **Note** **State** must be sent together with the **control** action.

187.4. Examples

187.4.1. Getting the current user input state

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=userinput&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
State=On
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "State": "On"
}
```


187.4.2. Deactivating the user input

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=userinput&action=control&State=Off
```

Chapter 188. Alarm Reset

188.1. Description

The **alarmreset** submenu resets the alarm.

NOTE This chapter applies to NVR only.

Access level

Action	NVR
control	User

188.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
alarmreset&action=<value>[&<parameter>=<value>...]
```

188.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Reset	REQ		Alarm reset No value is required for this parameter.

188.4. Examples

188.4.1. Resetting the alarm

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=alarmreset&action=control
```

Chapter 189. IO Ports Configuration

189.1. Description

The **ioport** submenu configures the configurable alarm IO port.

> **NOTE**
>
> This chapter applies to network cameras only.
> Attribute to check for configurable alarm IO support:
> "attributes/IO/Support/ConfigurableIO"
> Attribute to check for maximum configurable alarm IO:
> "attributes/IO/Limit/MaxConfigurableIO"

Access level

Action	Camera
view	Admin
set	Admin

189.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
ioport&action=<value>[&<parameter>=<value>...]
```

189.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the configurable alarm IO settings
	Port	REQ	<csv>	Physical port number
	Type	RES	<enum> Internal, External	Indicates whether it is built-in IO or external IO
set	Port.#.Mode	REQ, RES	<enum> Input, Output	Port operation mode

189.4. Examples

189.4.1. Getting the configurable alarm IO settings

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=ioport&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Port.1.Mode=Input
Port.1.Type=Internal
Port.2.Mode=Output
Port.2.Type=Internal
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Ports": [
        {
            "Port": 1,
            "Mode": "Input",
            "Type": "Internal"
        },
        {
            "Port": 2,
            "Mode": "Output",
            "Type": "Internal"
        }
    ]
}
```

189.4.2. Setting the operation mode of the configurable alarm IO port 1 to "Output"

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=ioport&action=set&Port.1.Mode=Output
```

Chapter 190. LED Control

190.1. Description

The **ledcontrol** submenu controls LED mode (turning off) and checks LED status.

Access level

Action	Camera	LEDBox
check	Admin	Admin
control	Admin	Admin
set	Admin	

190.2. Syntax

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=
ledcontrol&action=<value>[&<parameter>=<value>...]
```

190.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
check				Reads the LED status.
	LED.#.Color	RES	<enum> Green, Blue, Red, Pink, SkyBlue, Purple	User input state
	LED.#.LightMode	RES	<enum> On, Off	
	LED.#.LEDPresetIndex	RES	<int>	By using "eventrules"cgi "ledpreset" submenu "control" action, you can apply LEDPresetIndex to the LED. It shows which index is being used. * 0: Not set, * #: Preset index being used
control	LightMode	REQ	<enum> Off	Control Light Mode * Off: Turn off

Action	Parameters	Request/ Response	Type/ Value	Description
	LEDIndex	REQ	<csv>	"Select which LEDUsageIndex (eventrules.cgi ledpreset submenu) to apply. If the hardware has 1 LED, only 1 is available.
set	ExternalLEDEnable	REQ, RES	<bool> True, False	Turn on external LED
	ExternalLEDBrightness	REQ, RES	<int>	Control external LED brightness

190.4. Examples

190.4.1. Getting the current LED Status

REQUEST

```
http://<Device IP>/stw-cgi/io.cgi?msubmenu=ledcontrol&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ledcontrol": [
        {
            "LED": 1,
            "LightMode": "Off",
            "LEDPresetIndex": 0
        },
        {
            "LED": 2,
            "LightMode": "Off",
            "LEDPresetIndex": 0
        }
    ]
}
```

190.4.2. Turn off LEDUsageIndex 1

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=ledcontrol&action=control&LightMode=Off&LEDIndex=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

190.4.3. Turn on external LED and adjust brightness

REQUEST

```
http://<Device IP>/stw-
cgi/io.cgi?msubmenu=ledcontrol&action=set&ExternalLEDEnable=True&ExternalLED
Brightness=1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Display

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 191. Overview

191.1. Description

display.cgi is used to change the monitor layout configuration.

The following submenus are used to provide monitor layout functionalities:

- **decoderboardinfo**: Retrieves information about connected boards and configures the maximum allowed board count.
- **wall**: Adds, updates, controls, and removes monitor layouts for each connected board.
- **videooutlayout**: Configures the layout modes for encoder models.
- **spotout**: Configures the layout of the analog video output.
- **pvmsetup**: Configures the public view monitor (PVM) settings.
- **pvmpowerschedule**: Configures the schedule settings for the public view monitor (PVM).
- **bannerimage**: Configures the banner for video output.

Chapter 192. Decoder Board Info

192.1. Description

The **decoderboardinfo** submenu retrieves information about connected decoder boards and configures the maximum allowed board count.

NOTE This chapter applies to decoders (NVR) only.

Access level

Action	Camera	NVR	Decoder
view	-	-	User
set	-	-	User

192.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
decoderboardinfo&action=<value>[&<parameter>=<value>]
```

192.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index.#.Inserted	RES	<int>	Inserted board number
	Index.#.IsReady	RES	<bool>	Board enable status
set	AllowedBoardsCount	RES, REQ	<int>	Max. allowed board count

192.4. Examples

192.4.1. Getting connected board information

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=decoderboardinfo&action=view
```

TEXT RESPONSE

```
AllowedBoardsCount=8
Index.1.Inserted=True
Index.1.IsReady=True
Index.2.Inserted=True
```

```
Index.2.IsReady=True
Index.3.Inserted=False
Index.3.IsReady=False
Index.4.Inserted=False
Index.4.IsReady=False
Index.5.Inserted=False
Index.5.IsReady=False
Index.6.Inserted=False
Index.6.IsReady=False
Index.7.Inserted=False
Index.7.IsReady=False
Index.8.Inserted=False
Index.8.IsReady=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AllowedBoardsCount": 8,
    "DecoderBoards": [
        {
            "Index": 1,
            "Inserted": true,
            "IsReady": true
        },
        {
            "Index": 2,
            "Inserted": true,
            "IsReady": true
        },
        {
            "Index": 3,
            "Inserted": false,
            "IsReady": false
        },
        {
            "Index": 4,
            "Inserted": false,
```

```
                "IsReady": false
            },
            {
                "Index": 5,
                "Inserted": false,
                "IsReady": false
            },
            {
                "Index": 6,
                "Inserted": false,
                "IsReady": false
            },
            {
                "Index": 7,
                "Inserted": false,
                "IsReady": false
            },
            {
                "Index": 8,
                "Inserted": false,
                "IsReady": false
            }
        ]
}
```

192.4.2. Setting max allowed board counts

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
decoderboardinfo&action=set&AllowedBoardsCount=2
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 193. Wall

193.1. Description

The **wall** submenu adds, updates, controls, and removes monitor layouts for each connected board.

NOTE	In SPD-150/151 decoder models, this submenu is not supported when DeviceMode is set to Stand Alone mode. Please refer to the system.cgi deviceinfo submenu for the DeviceMode. This chapter applies to decoders only.

Access level

Action	Camera	NVR	Decoder
view	-	User	User
add/update	-	User	User
control	-	User	User
remove	-	User	User

193.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
wall&action=<value>[&<parameter>=<value>]
```

193.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	REQ	<csv>	Wall index
add/update	Index.#.Name	REQ,RES	<string>	Inserted board number
	Index.#.SplitMode	REQ,RES	<string>	Format=NoOfRowsxNoOfColumns 1X1...
	Index.#.MonitorOut	REQ,RES	<csv>	Display monitor index
	Index.#.Coordinates	REQ,RES	<string>	Format=x1,y1,x2,y2
	Index.#.EnableSequence	REQ,RES	<bool> True, False	Enable wall sequence

Action	Parameters	Request/ Response	Type/ Value	Description
	Index.#.Layout.#.Name	REQ,RES	<string>	Layout name **Note** Layout Name **Update** action doesn't work, if that layout is in use.
	Index.#.Layout.#.IsCurrentLayout	REQ,RES	<bool> True, False	Current layout
	Index.#.Layout.#.EnableSequence	REQ,RES	<bool>	Monitor display sequence enabled
	Index.#.Layout.#.SequenceTime	REQ,RES	<int>	Display sequence time
	Index.#.Layout.#.MonitorOut.#.SplitMode	REQ,RES	<enum> 1x1, 2x2, 3x3, 4x4	Each monitor split mode
	Index.#.Layout.#.MonitorOut.#.Tile.#.SourceType	REQ,RES	<enum> Stream, MonitorIn, None	Display source type
	Index.#.Layout.#.MonitorOut.#.Tile.#.MonitorIn	RES	<int>	If SourceType=MonitorIn
	Index.#.Layout.#.MonitorOut.#.Tile.#.Channel	REQ,RES	<int>	Channel number This parameter is only valid if **SourceType** is set to Stream.
	Index.#.Layout.#.MonitorOut.#.Tile.#.Profile	REQ,RES	<Int>	If SourceType=Stream
	Index.#.Layout.#.MonitorOut.#.Tile.#.Coordinates	REQ,RES	<string>	Format=x1,y1,x2,y2
	Index.#.Layout.#.MonitorOut.#.Tile.#.Location	REQ,RES	<string>	Format=RowxColumn
	Index.#.Layout.#.MonitorOut.#.Tile.#.MergeID	REQ,RES	<int>	Tile merge id
	Index.#.Layout.#.MonitorOut.#.Tile.#.Merge	REQ,RES	<string>	Format=NoOfRowsxNoOfColumns
	Index.#.Layout.#.MonitorOut.#.Tile.#.ImageLocation	REQ,RES	<string>	Format=RowxColumn

Action	Parameters	Request/ Response	Type/ Value	Description
update	Index.#.MonitorOut.\#.CurrentLayoutIndex=\#	REQ	<int>	The index of the layout depends on the current wall's index. Index.# : Wall index MonitorOut.# : Monitor index(number) CurrentLayoutIndex : LayoutIndex **NOTE** CurrentLayoutIndex: main monitor layout index (the index is smaller than the number of layouts).
control	Mode	REQ	<enum> Register, Show	Control action is possible only when the decoder is in vms mode.
	IsStreamServerPasswordEncrypted	REQ	<bool> True, False	If SourceType=Stream
	CommonUserID	REQ	<string>	Each camera (channel) RTSP account
	CommonPassword	REQ	<string>	Rtsp password
	MonitorOut.#.SplitMode	REQ	<enum> 1x1, 2x1, 2x2, 3x1, 3x3, 4x4, 1+5, 1+7, 1+12	Each monitor split mode
	MonitorOut.#.Tile.#.SourceType	REQ	<enum> Stream, MonitorIn	Display source type
	MonitorOut.#.Tile.#.ActionType	REQ	<enum> MediaOpen, MediaClose	Stream control
	MonitorOut.#.Tile.#.HighProfileRTSPURL	REQ	<string>	If SourceType=Stream
	MonitorOut.#.Tile.#.LowProfileRTSPURL	REQ	<string>	If SourceType=Stream
	MonitorOut.#.Tile.#.StreamServerUserID	REQ	<string>	If SourceType=Stream
	MonitorOut.#.Tile.#.StreamServerPassword	REQ	<string>	If SourceType=Stream

Action	Parameters	Request/ Response	Type/ Value	Description
	MonitorOut.#.Tile.#.CameraIP	REQ	<string>	If SourceType=Stream
	MonitorOut.#.Tile.#.CameraName	REQ	<string>	If SourceType=Stream
	MonitorOut.#.Tile.#.Coordinates	REQ	<string>	Format=x1,y1,x2,y2
	MonitorOut.#.Tile.#.Merge	REQ	<string>	Format=NoOfRowsxNoOfColumns
	MonitorOut.#.Tile.#.ImageCoordinates	REQ	<string>	Format=x1,y1,x2,y2
remove	Index	REQ	<csv>	Wall index number
	Index.#.Layout	REQ	<csv>	Wall attached layout number

193.4. Examples

193.4.1. Getting wall configuration information

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=wall&action=view&index=2
```

TEXT RESPONSE

```
Index.2.Name=Wall 02
Index.2.SplitMode=1x1
Index.2.MonitorOut=2
Index.2.Coordinates=13.4348,0,11,11
Index.2.EnableSequence=False
Index.2.Layout.1.Name=Layout 01
Index.2.Layout.1.IsCurrentLayout=True
Index.2.Layout.1.EnableSequence=True
Index.2.Layout.1.SequenceTime=10
Index.2.Layout.1.MonitorOut.2.SplitMode=2x2
Index.2.Layout.1.MonitorOut.2.Tile.1.SourceType=Stream
Index.2.Layout.1.MonitorOut.2.Tile.1.MonitorIn=1
Index.2.Layout.1.MonitorOut.2.Tile.1.Channel=0
Index.2.Layout.1.MonitorOut.2.Tile.1.Profile=2
Index.2.Layout.1.MonitorOut.2.Tile.1.Location=1x1
Index.2.Layout.1.MonitorOut.2.Tile.1.MergeID=
```

```
Index.2.Layout.1.MonitorOut.2.Tile.1.Merge=
Index.2.Layout.1.MonitorOut.2.Tile.1.ImageLocation=
Index.2.Layout.1.MonitorOut.2.Tile.2.SourceType=Stream
Index.2.Layout.1.MonitorOut.2.Tile.2.MonitorIn=1
Index.2.Layout.1.MonitorOut.2.Tile.2.Channel=2
Index.2.Layout.1.MonitorOut.2.Tile.2.Profile=2
Index.2.Layout.1.MonitorOut.2.Tile.2.Location=1x2
Index.2.Layout.1.MonitorOut.2.Tile.2.MergeID=
Index.2.Layout.1.MonitorOut.2.Tile.2.Merge=
Index.2.Layout.1.MonitorOut.2.Tile.2.ImageLocation=
Index.2.Layout.1.MonitorOut.2.Tile.3.SourceType=Stream
Index.2.Layout.1.MonitorOut.2.Tile.3.MonitorIn=1
Index.2.Layout.1.MonitorOut.2.Tile.3.Channel=1
Index.2.Layout.1.MonitorOut.2.Tile.3.Profile=2
Index.2.Layout.1.MonitorOut.2.Tile.3.Location=2x1
Index.2.Layout.1.MonitorOut.2.Tile.3.MergeID=
Index.2.Layout.1.MonitorOut.2.Tile.3.Merge=
Index.2.Layout.1.MonitorOut.2.Tile.3.ImageLocation=
Index.2.Layout.1.MonitorOut.2.Tile.4.SourceType=Stream
Index.2.Layout.1.MonitorOut.2.Tile.4.MonitorIn=1
Index.2.Layout.1.MonitorOut.2.Tile.4.Channel=3
Index.2.Layout.1.MonitorOut.2.Tile.4.Profile=2
Index.2.Layout.1.MonitorOut.2.Tile.4.Location=2x2
Index.2.Layout.1.MonitorOut.2.Tile.4.MergeID=
Index.2.Layout.1.MonitorOut.2.Tile.4.Merge=
Index.2.Layout.1.MonitorOut.2.Tile.4.ImageLocation=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Wall": [
        {
            "Index": 2,
            "Name": "Wall 02",
            "SplitMode": "1x1",
            "MonitorOut": [
                "2"
```

```
            ],
            "Coordinates": [
                {
                    "x": 13.434783,
                    "y": 0
                },
                {
                    "x": 11,
                    "y": 11.0
                }
            ],
            "EnableSequence": false,
            "Layout": [
                {
                    "Index": 1,
                    "Name": "Layout 01",
                    "IsCurrentLayout": true,
                    "EnableSequence": true,
                    "SequenceTime": 10,
                    "MonitorOut": [
                        {
                            "Index": 2,
                            "SplitMode": "2x2",
                            "Tile": [
                                {
                                    "Index": 1,
                                    "SourceType": "Stream",
                                    "MonitorIn": 1,
                                    "Channel": 0,
                                    "Profile": 2,
                                    "Location": "1x1",
                                    "MergeID": null,
                                    "Merge": "",
                                    "ImageLocation": ""
                                },
                                {
                                    "Index": 2,
                                    "SourceType": "Stream",
                                    "MonitorIn": 1,
                                    "Channel": 2,
                                    "Profile": 2,
```

```
                                    "Location": "1x2",
                                    "MergeID": null,
                                    "Merge": "",
                                    "ImageLocation": ""
                                },
                                {
                                    "Index": 3,
                                    "SourceType": "Stream",
                                    "MonitorIn": 1,
                                    "Channel": 1,
                                    "Profile": 2,
                                    "Location": "2x1",
                                    "MergeID": null,
                                    "Merge": "",
                                    "ImageLocation": ""
                                },
                                {
                                    "Index": 4,
                                    "SourceType": "Stream",
                                    "MonitorIn": 1,
                                    "Channel": 3,
                                    "Profile": 2,
                                    "Location": "2x2",
                                    "MergeID": null,
                                    "Merge": "",
                                    "ImageLocation": ""
                                }
                            ]
                        }
                    ]
                }
            ]
        }
    ]
}
```

193.4.2. Adding wall (1 monitor, 1 layout) configuration information

REQUEST

```
http://<Device IP>/stw-
cgi/display.cgi?msubmenu=wall&action=add&Index.1.Name=Wall01&Index.1.SplitMo
```

```
de=1x1&Index.1.MonitorOut=1&Index.1.Coordinates=0,0,11,11&Index.1.EnableSequ
ence=False&Index.1.Layout.1.Name=Layout01&Index.1.Layout.1.IsCurrentLayout=T
rue&Index.1.Layout.1.EnableSequence=True&Index.1.Layout.1.SequenceTime=10&In
dex.1.Layout.1.MonitorOut.1.SplitMode=1x1&Index.1.Layout.1.MonitorOut.1.Tile
.1.SourceType=Stream&Index.1.Layout.1.MonitorOut.1.Tile.1.MonitorIn=1&Index.
1.Layout.1.MonitorOut.1.Tile.1.Channel=0&Index.1.Layout.1.MonitorOut.1.Tile.
1.Profile=2&Index.1.Layout.1.MonitorOut.1.Tile.1.Location=1x1&Index.1.Layout
.1.MonitorOut.1.Tile.1.MergeID=1&Index.1.Layout.1.MonitorOut.1.Tile.1.Merge=
False&Index.1.Layout.1.MonitorOut.1.Tile.1.ImageLocation=1x1
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


193.4.3. Updating (1 monitor, 1 layout) configuration information

REQUEST

```
http:// <Device IP>/stw-
cgi/display.cgi?msubmenu=wall&action=update&Index.1.Name=Wall02
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

193.4.4. Updating (1 monitor, 1 layout) current layout index

REQUEST

```
http:// <Device IP>/stw-
cgi/display.cgi?msubmenu=wall&action=update&Index.1.MonitorOut.1.CurrentLayo
utIndex=1
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

193.4.5. Removing wall configuration information

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=wall&action=remove&index=2
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

193.4.6. Control wall mode register

NOTE Control is supported only in VMS mode (Please refer to system.cgi deviceinfo submenu for mode selection).

REQUEST

```
http://<device-ip>/stw-cgi/display.cgi?msubmenu=wall&action=control&Mode=Register&IsStreamServerPasswordEncrypted=False&CommonUserID=admin&CommonPassword=password!&MonitorOut.1.SplitMode=1x1&MonitorOut.1.Tile.1.SourceType=Stream&MonitorOut.1.Tile.1.ActionType=MediaOpen&MonitorOut.1.Tile.1.HighProfileRTSPURL=rtsp://192.168.71.144/profile1/media.smp&MonitorOut.1.Tile.1.LowProfileRTSPURL=rtsp://192.168.71.144/profile2/media.smp&MonitorOut.1.Tile.1.StreamServerUserID=admin&MonitorOut.1.Tile.1.StreamServerPassword=password!&MonitorOut.1.Tile.1.CameraIP=192.168.71.144&MonitorOut.1.Tile.1.CameraName=TestCamera&MonitorOut.1.Tile.1.Coordinates=0,0,1920,1080&MonitorOut.1.Tile.1.Merge=1x1
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
  "Response": "Success"
}
```

193.4.7. Control wall mode show

REQUEST

```
http://192.168.71.48/stw-
cgi/display.cgi?msubmenu=wall&action=control&Mode=show
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 194. Encoder video out layout

194.1. Description

The **videooutlayout** submenu configures the layout modes for encoder models.

NOTE This chapter applies to 16 channel encoders only.

Access level

Action	Camera	NVR	Encoder
view	-	-	Admin
set	-	-	Admin

194.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
videooutlayout&action=<value>[&<parameter>=<value>]
```

194.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	LayoutMode	REQ, RES	<enum> 1x1, 2x2, 3x3, 4x4	Layout modes

194.4. Examples

194.4.1. Getting the current layout mode

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=videooutlayout&action=view
```

TEXT RESPONSE

```
LayoutMode=3x3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "LayoutMode": "3x3"
}
```


194.4.2. Setting the layout mode to 4x4

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
videolayoutmode&action=set&LayoutMode=4x4
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 195. SpotOut

195.1. Description

The **spotout** submenu configures the layout of analog video output.

NOTE This chapter applies to NVR only.
Attribute to check for **spotout** support: "attributes/System/Limit/MaxAnalogSpotCount"

Access level

Action	Camera	NVR	Decoder
view	-	User	-
set	-	User	-

195.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
spotout&action=<value>[&<parameter>=<value>]
```

195.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	Enable	REQ, RES	<bool> True, False	Enables or disables analog monitor spotout
	LayoutMode	REQ, RES	<enum> 1, 2, 4, 9, 16	Layout mode Check whether spotout is supported or not on System/Limit/MaxAnalogSpotCount under attributes.cgi
	SequenceMode	REQ, RES	<bool> True, False	Enables or disables sequence mode
	ChannelList	REQ, RES	<csv>	Monitoring channels

195.4. Examples

195.4.1. Getting spotout configuration information

REQUEST

```
http://<Device IP> /stw-cgi/display.cgi?msubmenu=spotout&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Enable": true,
    "LayoutMode": 9,
    "SequenceMode": true,
    "ChannelList": ["0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0"]
}
```


195.4.2. Setting spotout configuration information

REQUEST

```
http://<Device IP>/stw-
cgi/display.cgi?msubmenu=spotout&action=set&Enable=True&LayoutMode=16&Sequen
ceMode=True&ChannelList=1,1,1,1,1,1,1,1,1,1,1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```

Chapter 196. PVM Setup

196.1. Description

The **pvmsetup** submenu configures the public view monitor (PVM) settings.

Access level

Action	Camera	NVR	Decoder
view	Admin	-	-
set	Admin	-	-
control	Admin	-	-

196.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
pvmsetup&action=<value>[&<parameter>=<value>]
```

196.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the pvmsetup settings.
set	IRRemoteControlEnable	RES, REQ	<bool> True, False	Enables or disables the IR remote control
	ScreenMode	RES, REQ	<enum> Single, PIP, PBP	Screen mode • Single: Single video mode • PIP: Picture-in-Picture mode • PBP: Picture-by-Picture mode
	PIPMainScreen	RES, REQ	<enum> Camera, HDMI	Main screen in PIP mode **PIPMainScreen** is valid only when **ScreenMode** is set to **PIP**.
	PIPSubScreenPosition	RES, REQ	<enum> LeftTop, RightTop, LeftBottom, RightBottom	Subscreen position of PIP mode **PIPSubScreenPosition** is valid only when **ScreenMode** is set to **PIP**.

Action	Parameters	Request/ Response	Type/ Value	Description
	PIPSubScreenSize	RES, REQ	<enum> Small, Middle, Large	Subscreen size in PIP mode **PIPSubScreenSize** is valid only when **ScreenMode** is set to **PIP**.
	PBPHDMIScreenPosition	RES, REQ	<enum> Left, Right	HDMI Screen position in PBP mode **PBPHDMIScreenPosition** is valid only when **ScreenMode** is set to **PBP**.
	MonitorPower	REQ	<enum> On, Off	Power on or off the monitor

196.4. Examples

196.4.1. Getting the current PVM monitor settings

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=pvmsetup&action=view
```

TEXT RESPONSE

```
IRRemoteControlEnable=False
ScreenMode=Single
PIPMainScreen=Camera
PIPSubScreenPosition=LeftTop
PIPSubScreenSize=Small
PBPHDMIScreenPosition=Left
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PvmSetup": {
        "IRRemoteControlEnable": false,
        "ScreenMode": "Single",
        "PIPMainScreen": "Camera",
        "PIPSubScreenPosition": "LeftTop",
```

```
        "PIPSubScreenSize": "Small",
        "PBPHDMIScreenPosition": "Left"
    }
}
```

196.4.2. Setting the PVM monitor to PIP, the main screen to HDMI, the position of the subscreen to RightTop, and the subscreen size to Large

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
pvmsetup&action=set&ScreenMode=PIP&PIPMainScreen=HDMI&PIPSubScreenPosition=R
ightTop&PIPSubScreenSize=Large
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

196.4.3. Control the PVM monitor power to Off

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
pvmsetup&action=control&MonitorPower=Off
```

TEXT RESPONSE

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```



```
{
    "Response": "Success"
}
```

Chapter 197. PVM power schedule

197.1. Description

The **pvmpowerschedule** submenu configures the schedule settings for the public view monitor (PVM).

Access level

Action	Camera	NVR	Decoder
view	Admin	-	-
set	Admin	-	-

197.2. Syntax

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=
pvmpowerschedule&action=<value>[&<parameter>=<value>]
```

197.2.1. Parameters

Action	Parameter	Request/ Response	Type/ Value	Description
view				Reads the pvmpowerschedule settings.
set	ScheduleType	REQ, RES	<enum> Always, Scheduled	Time schedule for operation • Always: All the time • Scheduled: Only when scheduled

Action	Parameter	Request/ Response	Type/ Value	Description
	<ddd>	REQ, RES	<bool> 0, 1	Day of week selected for the operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <ddd> stands for week of the day and should be specified in the short form such as SUN, MON, TUE, WED, THU, FRI, and SAT in uppercase. e.g.) 'SUN=1' indicates every Sunday 12:00 AM to 11:59 PM unless the specific time is not set using the <dddh> parameter such like SUN1=1, SUN2=1, etc.
	EveryDay	REQ, RES	<bool> 0, 1	Whether to activate or deactivate the operation for every day • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. 'EveryDay=1', denoting that the every day, is same as the **ScheduleType** parameter is set to Always.

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>	REQ, RES	<bool> 0, 1	Time of day selected for the operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <dddh> stands for the day of the week and time in hour. e.g. SUN1 means 1:00 AM on Sunday. MON2 means 2:00 AM on Monday. This parameter is valid only when <corresponding weekday> is set to 1; e.g. **SUN**=1 is required for SUN0 ... SUN23. 'SUN=1&SUN18=1' means that the operation is activated every Sunday 6:00 PM to 6:59 PM.
	EveryDay<h>	REQ, RES	<bool> 0, 1	Time of everyday for operation • 0: Disabled • 1: Enabled This parameter is valid only when **Activate** is set to Scheduled. <h> stands for the hours such as 0,1,2,3,4..10,11,12,...,23. This parameter is valid only when **EveryDay** is set to 1; e.g. EveryDay=1 is required for EveryDay0 ... EveryDay23. 'EveryDay=1&EveryDay18=1' means the operation is activated every day 6:00 PM to 6:59 PM.

Action	Parameter	Request/ Response	Type/ Value	Description
	<dddh>.FromTo	REQ, RES	<string>	The time of week selected for the operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when **Activate** is set to Scheduled. This parameter is also valid only when <corresponding weekday><hour> is set to 1; e.g. SUN0=1 is required for SUN0.FromTo. 'SUN=1&SUN18=1&SUN18.FromTo=12-20' means that the operation is activated every Sunday 6:12 PM to 6:20 PM.
	EveryDay<h>.FromTo	REQ, RES	<string>	The time for everyday for the operation The time is specified in the format of <mm-mm>. The first 'mm' must be smaller than or equal to the second 'mm'. This parameter is valid only when **Activate** is set to Scheduled. This parameter is valid only when EveryDay<hour> is set to 1; e.g. 'EveryDay0=1' is required for EveryDay0.FromTo. 'EveryDay=1&EveryDay18=1&EveryDay18.FromTo=12-20' means that the operation is activated every day 6:12 PM to 6:20 PM.

197.3. Examples

197.3.1. Getting the current PVM monitor settings

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=pvmpowerschedule&action=view
```

TEXT RESPONSE

```
ScheduleType=Always
SUN=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MON=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
TUE=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
WED=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
THU=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
FRI=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
SAT=0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "PvmPowerSchedule": {
        "ScheduleType": "Always",
        "Schedule": {
            "SUN": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "MON": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "TUE": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "WED": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "THU": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "FRI": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
            "SAT": [ "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
```

```
"0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0" ],
        }
    }
}
```

197.3.2. Updating pvm power schedule

Setting the schedule from 3:58 AM to 3:59 AM on Sundays

REQUEST

```
http://<Device IP>/stw-
cgi/eventrules.cgi?msubmenu=schedulelist&action=set&ScheduleType=Scheduled&S
UN=1&SUN3=1&SUN3.FromTo=58-59
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

197.4. Banner Image

197.4.1. Description

The **bannerimage** submenu is used to configure the banners for video output.

NOTE This chapter applies to cameras only.

Access level

Action	Camera
view	Admin
set	Admin
install	Admin
remove	Admin

197.4.2. Syntax

```
http://<Device IP>/stw-cgi/video.cgi?msubmenu=
bannerimage&action=<value>[&<parameter>=<value>]
```

197.4.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ	<csv>	Channel ID
	Status	RES	<enum> Idle, Uploading	If camera is uploading (installing) images to SD card, Status is "Uploading"; otherwise, it is "Idle".
	BannerIndex.#.BannerIndex	RES	<int>	Index of the banner
	BannerIndex.#.ImageName	RES	<int>	Image name of the banner
set	Channel	REQ, RES	<int>	Channel ID
	Enable	REQ, RES	<bool>	Enables banner feature. **Note** At least one image must exist to be enabled.
	BannerIndex.#.PositionY	REQ, RES	<int> 1 ~ 100	Vertical position of the banner image
install	Channel	REQ	<int>	Channel ID

Action	Parameters	Request/ Response	Type/ Value	Description
	ImageName	REQ	<string> Ex)01_banner.png	This command should have a post message with a image file included. It will be uploaded to SD card. **Note** See "attributes/Media/Support/BannerImage.ImageFormat" for supported image file formats Images can only be installed if **Enable** is false
remove	Channel	REQ	<int>	Channel ID
	Mode	REQ	<enum> All, Selected	• All: remove all saved images. • Selected: remove a particular image. "BannerIndex" should be used together.
	BannerIndex	REQ	<csv>	Index of image to be deleted. **Note** Images can only be removed if **Enable** is false

197.4.4. Examples

197.4.4.1. Requesting view action (this submenu supports only JSON response)

Getting a configuration for bannerimage

REQUEST

```
http://<Device IP>/stw-cgi/display.cgi?msubmenu=bannerimage&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "BannerImage": [
        {
```

```
            "Channel": 0,
            "Enable": false,
            "Status": "Idle",
            "Banners": [
                {
                    "BannerIndex": 1,
                    "ImageName": "bannerTest1.bmp",
                    "PositionY": 1
                }
            ]
        }
    ]
}
```

Transfer

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 198. Overview

198.1. Description

transfer.cgi configures the FTP/SMTP server settings to transfer images or event notifications when an event occurs.

The actual transfer through FTP/SMTP is controlled by the EventAction parameter in **eventrules.cgi** for camera and SMTP submenu **eventactions.cgi** for NVR. This occurs when specific events occur, such as Alarm Input, Video Analytics, Video Loss, Network Event, Face Detection, Tampering Detection, Audio Detection, Tracking, or Timer.

The following submenus are used to control the transfer API:

- **ftp**: Sets the File Transfer Protocol (FTP) configuration for sending images or videos.
- **jetsonflashingserver**: Sets the FTP configuration where the flashing file is uploaded. (Use FTP to prepare for large files).
- **smtp**: Sets the SMTP (Simple Mail Transfer Protocol) configuration for sending notification messages. Images can be attached as email attachments.
- **smtpusers**: Sets the SMTP mail users.
- **smtpgroups**: Sets the SMTP mail user groups.
- **dataserver**: Sets the data server.

Chapter 199. FTP

199.1. Description

The **ftp** submenu configures the FTP (File Transfer Protocol) server settings.

> **NOTE**
> This chapter applies to network cameras only.
> Attribute to check for Feature Support: "**attributes/transfer/Support/FTP**"

Access level

Action	Camera
view	Admin
set	Admin
test	Admin

199.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
ftp&action=<value>[&<parameter>=<value>]
```

199.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the FTP settings
	SFTP.KeyName	RES	<string>	Name of the keyfile, if keyfile is already installed.
	FileNameFormatOptions	RES	<csv>	Shows file name formats.
	DirectoryDepth	RES	<int>	Indicates how many directory level can be used.
set	Host	REQ, RES	<string>	Domain name of the FTP server (required) The host is specified in the format of <IP Address> or <Host name> e.g. ftp.samsung.com or 192.168.100.1
	Port	REQ, RES	<int>	FTP port number
	Username	REQ, RES	<string>	FTP user name
	Password	REQ, RES	<string>	FTP user password

Action	Parameters	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<bool>	Returns true if password sent is encrypted. Encrypted password should be sent as a post message.
	Mode	REQ, RES	<enum> Active, Passive	FTP transmission mode
	Path	REQ, RES	<string>	The path on the FTP server where alarm images are saved. Value must be a relative path.
	ReportFileType	REQ, RES	<enum> Image, VideoClip	The type of file which is sent to the FTP server. When this parameter is not supported, only images are provided if a configured event is triggered.
	PreEventDurationInSecs	REQ, RES	<enum> 1s, 2s, 3s	Duration of pre-event recording. The unit is second. This parameter is valid when "ReportFileType" is "VideoClip"
	PostEventDurationInSecs	REQ, RES	<enum> 10s, 15s, 20s	Duration of post-event recording. The unit is second. This parameter is valid when "ReportFileType" is "VideoClip"
	FileNameFormatEnable	REQ, RES	<bool> True, False	Enables or disables file name format feature.
	FileNameFormat	REQ, RES	<string>	Up to 127 characters allowed.
	Encryption	REQ, RES	<enum> None, SFTP	Option to select Secure FTP or normal FTP
	SFTP.AuthMode	REQ, RES	<enum> Password, KeyFile	When Encryption is set to SFTP, auth mode can be configured. It can be either password based or keyfile based.
install	IsPassPhraseEncrypted	REQ	<bool> True, False	When installing the key file, if the passphrase is encrypted need to pass this parameter as true.
remove	SFTP.KeyName	REQ	<string>	To remove the key file, need to pass the key filename
test	Status	RES	<string>	Tries to connect to the server with the configured settings and returns the result in the form of a string

199.4. Examples

199.4.1. Getting FTP server information

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=ftp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Host=192.168.100.1
Mode=Active
Port=22
Path=/home/test
Username=test
Password=
```

```
Host=192.168.100.1
Mode=Active
Port=22
Path=/home/test
Username=test
Password=
ReportFileType=Image
PreEventDuration=3s
PostEventDuration=10s
Encryption=None
SFTP.AuthMode=Password
SFTP.KeyName=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Host": "192.168.100.1",
    "Mode": "Active",
    "Port": 22,
    "Path": "/home/test",
    "Username": "test",
    "Password": ""
}
```

```
{
   "Host": "192.168.100.1",
    "Mode": "Active",
    "Port": 22,
    "Path": "/home/test",
    "Username": "test",
    "Password": "",
    "Encryption": "None",
    "SFTP": {
        "AuthMode": "Password",
        "KeyName": ""
    }
}
```

199.4.2. Setting the FTP server

The FTP host, user name, password, mode, and path can be configured as shown in the command below.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=ftp&action=set&Password={password}&Host=223.255.25
4.254&Port=4&Username=tes&Mode=Active&Path=test1234
```

199.4.3. Setting the FTP server with an encrypted password

The FTP host, user name, mode, and path can be configured as shown in the command below. Password should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi for information on obtaining the RSA key).

Base 64 Encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=ftp&action=set&Host=223.255.254.254&Port=4&Usernam
e=tes&Mode=Active&Path=test1234&IsPasswordEncrypted=True
```

199.4.4. Checking the connection status of the FTP Server

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=ftp&action=test
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Success
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Status": "Success"
}
```

199.4.5. INSTALL SFTP Key (with encrypted keypassphrase)

Key Passphrase can be encrypted using the RSA public key and sent in below json. (Refer to security.cgi and Application programming guide for information on obtaining the RSA key and encrypting the keypassphrase).

REQUEST

```
http://<DeviceIP>/stw-
cgi/transfer.cgi?msubmenu=ftp&action=install&IsPassPhraseEncrypted=True
```

POST BODY


```
{
    "SFTPKey": {
        "KeyName": "ssss.key",
        "KeyData": "c2RmZHNkZGRkZGRkZGRk..........MTIxMjEyMTIxMjExMg==",
        "KeyPassPhrase": "RdkHuyI65lDT4s5SVy......lhaeXQObvRipEA==""
    }
}
```


199.4.6. Remove SFTP Key File

REQUEST

```
http://<DeviceIP>/stw-
cgi/transfer.cgi?msubmenu=ftp&action=remove&SFTP.KeyName=ssss.key
```

199.4.7. SFTP set operation

Setting the encryption mode and sftp authentication mode.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=ftp&action=set&Password={password}&Host=223.255.25
4.254&Port=4&Username=tes&Mode=Active&Path=test1234&Encryption=SFTP&SFTP.Aut
hMode=KeyFile
```

Chapter 200. jetsonflashingserver

200.1. Description

The **jetsonflashingserver** submenu configures the FTP (File Transfer Protocol) server settings(where the flashing file is uploaded).

> **NOTE**
> This chapter applies to network cameras only.
> CGIs to check for Feature Support: "**cgis/transfer/jetsonflashingserver**"

Access level

Action	Camera
view	Admin
set	Admin
test	Admin
check	Admin

200.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
jetsonflashingserver&action=<value>[&<parameter>=<value>]
```

200.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the current settings
	SFTP.KeyName	RES	<string>	Name of the keyfile, if keyfile is already installed.
set	Host	REQ, RES	<string>	Domain name of the FTP server (required) The host is specified in the format of <IP Address> or <Host name> e.g. jetsonflashingserver.hanwhavision.com or 192.168.100.1
	Port	REQ, RES	<int>	FTP port number
	Username	REQ, RES	<string>	FTP user name
	Password	REQ, RES	<string>	FTP user password

Action	Parameters	Request/ Response	Type/ Value	Description
	IsPasswordEncrypted	REQ	<bool>	Returns true if password sent is encrypted. Encrypted password should be sent as a post message.
	Mode	REQ, RES	<enum> Active, Passive	FTP transmission mode
	Path	REQ, RES	<string>	The path to the FTP server where the flashing file was uploaded (including file name and extension). Value must be the relative path. It must be url encoded.
	Encryption	REQ, RES	<enum> None, SFTP	Option to select Secure FTP or normal FTP
	SFTP.AuthMode	REQ, RES	<enum> Password, KeyFile	When Encryption is set to SFTP, auth mode can be configured. It can be either password based or keyfile based.
test	Status	RES	<string>	Tries to connect to the server with the configured settings and returns the result in the form of a string
check				Check the current jetsonflashing steps

Action	Parameters	Request/ Response	Type/ Value	Description
	CurrentStep	RES	<enum> DOWNLOAD, UNZIP, MOVE, DOCKER_CONTAINER_RUN, DOCKER_RUN, DOCKER_CHECK, USB_MODE_CHECK, DOCKER_START, FLASHING, DOCKER_STOP, DOCKER_KILL, DOCKER_CLEANUP, USB_OFF, CLEANUP, END	
	Progress	RES	<int>	

200.4. Examples

200.4.1. Getting jetsonflashingserver information

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=jetsonflashingserver&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Host=192.168.100.1
Mode=Active
Port=22
Path=/test/NX_test.tar.gz
Username=test
Password=
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Host": "192.168.100.1",
    "Mode": "Active",
    "Port": 22,
    "Path": "/test/NX_test.tar.gz",
    "Username": "test",
    "Password": ""
}
```


200.4.2. Setting the jetsonflashingserver

The FTP host, user name, password, mode, and path can be configured as shown in the command below.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=jetsonflashingserver&action=set&Password={password
}&Host=223.255.254.254&Port=4&Username=tes&Mode=Active&Path=%2Ftest%2FNX_tes
t.tar.gz
```

200.4.3. Setting the jetsonflashingserver with an encrypted password

The FTP host, user name, mode, and path can be configured as shown in the command below. Password should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi for information on obtaining the RSA key).

Base 64 Encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=jetsonflashingserver&action=set&Host=223.255.254.254&Port=4&Username=tes&Mode=Active&Path=%2Ftest%2FNX_test.tar.gz&IsPasswordEncrypted=True
```

200.4.4. Checking the connection status of the jetsonflashingserver

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=jetsonflashingserver&action=test
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Success
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Status": "Success"
}
```

200.4.5. Check the current steps of jetsonflashing

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=jetsonflashingserver&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
  "CurrentStep": "END",
  "Progress": 100
}
```

Chapter 201. SMTP

201.1. Description

The **smtp** submenu configures the SMTP (Simple Mail Transfer Protocol) server settings.

> **NOTE** Attribute to check for Feature Support: "**attributes/transfer/Support/SMTP**"

Access level

Action	Camera	NVR
view	Admin	User
set	Admin	User
test	Admin	User

201.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
smtp&action=<value>[&<parameter>=<value>]
```

201.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the SMTP settings.
	CACertificateInstalled	RES	<bool> True, False	Indicates whether a CA certificate is installed for the SMTP server.
	MaxCACertificateInUse	RES	<int>	Maximum number of CA certificates that can be used
set	Authentication	REQ, RES	<enum> None, SMTP, POPBefore SMTP	SMTP authentication
	Host	REQ, RES	<string>	Domain name of the SMTP server (Required). The host is specified in the format of <IP Address> or <Host name> e.g. smtp.samsung.com or 192.168.100.1, and the max length is 63.

Action	Parameters	Request/ Response	Type/ Value	Description
	Port	REQ, RES	<int>	SMTP port number
	Encryption	REQ, RES	<enum> None, SSL	SMTP encryption type
	Username	REQ, RES	<string>	SMTP user name **Username** is only valid if **Authentication** is set as SMTP or POPBeforeSMTP.
	Password	REQ, RES	<string>	SMTP user password **Password** is only valid if **Authentication** is set as SMTP or POPBeforeSMTP.
	IsPasswordEncrypted	REQ	<bool>	Returns true if password sent is encrypted. Encrypted password should be sent as a post message.
	Sender	REQ, RES	<string>	Email address of the sender The sender's email address is specified in the format of <Email Address>.
	Recipient	REQ, RES	<string>	Email address of the recipient The recipient's email address is specified in the format of <Email Address>. Attribute to check max recipients: "**attributes/transfer/Limit/SMTP.MaxRecipients**" CAMERA ONLY
	Subject	REQ, RES	<string>	Email title CAMERA ONLY
	Message	REQ, RES	<string>	Email content CAMERA ONLY
	CACertificateInUse	REQ, RES	<csv>	CA certificates in use for SMTP server verification

Action	Parameters	Request/ Response	Type/ Value	Description
	VerifyServerCertification	REQ, RES	<bool>	Enables the feature to verify the SMTP server certificate
install		REQ		Install the certificate of SMTP server
remove	CertificateType	REQ	<enum> CACertificate	Certificate type to remove **Note** **CertificateType** must be sent together with the **remove** action
test	Status	RES	<string>	Tries to connect to the server with the configured settings and returns the result in the form of a string.

201.4. Examples

201.4.1. Getting SMTP server information

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtp&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Host=192.168.100.1
Port=25
Username=test
Password=
Authentication=SMTP
Sender=test.a@hanwhasecurity.com
Recipient=test1.a@hanwhasecurity.com
Subject=hello
Message=Test message
Encryption=None
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Host": "192.168.100.1",
    "Port": 25,
    "Username": "test",
    "Password": "",
    "Authentication": "SMTP",
    "Sender": "test.a@hanwhasecurity.com",
    "Recipient": "test1.a@hanwhasecurity.com",
    "Subject": "hello",
    "Message": "Test message",
    "Encryption": "None"
}
```


201.4.2. Setting the SMTP server

The SMTP host, port, user name, password, sender/recipient email addresses, email subject and message can be configured as shown in the command below.

The **Username** and **Password** parameters can be specified only when SMTP server authentication is set as SMTP or POPBeforeSMTP.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtp&action=set&Authentication=SMTP&Host=www.hanwh
asecurity.com&Port=25&Encryption=None&Username=test5678&Password={password}&
Sender=test@test.com&Recipient=stw@hanwhasecurity.com&Subject=test&Message=t
est
```

The following request example is for NVR only.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtp&action=set&Authentication=SMTP&Host=192.168.1
00.1&Port=25&Encryption=None&Username=aaa&Password={password}&Sender=test@te
```

```
st.com
```

201.4.3. Setting the SMTP server with encrypted password

The SMTP host, port, user name, sender/recipient email addresses, email subject and message can be configured as shown in the command below. Password should be encrypted with RSA and RSA_PKCS1_PADDING (Refer to security.cgi for information on obtaining the RSA key).

Base 64 Encoded data should be sent as a POST message.

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtp&action=set&Authentication=SMTP&Host=www.hanwhasecurity.com&Port=25&Encryption=None&Username=test5678&Sender=test@test.com&Recipient=stw@hanwhasecurity.com&Subject=test&Message=test&IsPasswordEncrypted=True
```

201.4.4. Checking the connection status of the SMTP Server

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtp&action=test
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Success
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
"Status": "Success"
}
```

Chapter 202. SMTP Users

202.1. Description

The **smtpusers** submenu configures the SMTP mail users.

> **NOTE**
>
> This chapter applies to NVR only.
> Attribute to check for maximum users: "**attributes/security/Limit/MaxSMTPUser**"
> Attribute to check for maximum users in group:
> "**attributes/security/Limit/MaxSMTPUserPerGroup**"

Access level

Action	NVR
view	User
add, update	User
remove	User

202.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
smtpusers&action=<value>[&<parameter>=<value>]
```

202.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	UserIndex	REQ	<int>	User index number
	GroupID	REQ	<string>	Group ID
add, update	UserIndex	REQ, RES	<int>	User index number
	GroupID	REQ, RES	<string>	Group ID
	UserName	REQ, RES	<string>	User name
	Recipient	REQ, RES	<string>	Email address of the recipient. The recipient's email address is specified in the format of <Email Address>.
remove	UserIndex	REQ	<int>	User index number

202.4. Examples

202.4.1. Getting the current SMTP user settings

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtpusers&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GroupID=Group 1
UserIndex=1
UserName=Recipient 1
Recipient=t1@hanwhasecurity.com
GroupID=Group 1
UserIndex=2
UserName=s12
Recipient=s2@hanwhasecurity.com
GroupID=Group 1
UserIndex=3
UserName=s3
Recipient=s12@s12.com
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SMTPUsers": [
        {
            "GroupID": "Group 1",
            "UserIndex": 1,
            "UserName": "Recipient 1",
            "Recipient": "t1@hanwhasecurity.com"
        },
```

```
        {
            "GroupID": "Group 1",
            "UserIndex": 2,
            "UserName": "s12",
            "Recipient": "s2@hanwhasecurity.com"
        },
        {
            "GroupID": "Group 1",
            "UserIndex": 3,
            "UserName": "s3",
            "Recipient": "s12@hanwhasecurity.com"
        }
    ]
}
```

202.4.2. Getting the ‘User Index 1’ settings

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpusers&action=view&UserIndex=1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GroupID=Group 1
UserIndex=1
UserName=Recipient 1
Recipient=t1@hanwhasecurity.com
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "SMTPUsers": [
        {
            "GroupID": "Group 1",
            "UserIndex": 1,
            "UserName": "Recipient 1",
            "Recipient": "t1@hanwhasecurity.com"
        }
    ]
}
```

202.4.3. Adding a new SMTP user

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtpusers&action=add&GroupID=Group2&UserName=s12&Recipient=s12@s12
```

CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
UserIndex=13
```

The following response example is for NVR only.

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "Response": "Success"
}
```

202.4.4. Updating the user name

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpusers&action=update&UserIndex=13&UserName=t13
```

202.4.5. Removing the user

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpusers&action=remove&UserIndex=7
```

Chapter 203. SMTP User Group

203.1. Description

The **smtpgroups** submenu configures the SMTP mail user groups.

NOTE	This chapter applies to NVR only. Attribute to check for maximum groups: "**attributes/security/Limit/MaxSMTPGroup**"

Access level

Action	NVR
view	User
add, update	User
remove	User

203.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
smtpgroups&action=<value>[&<parameter>=<value>]
```

203.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	GroupID	REQ	<string>	User group ID
add, update	GroupID	REQ, RES	<string>	User group ID
	GroupName	REQ, RES	<string>	Group name
remove	GroupID	REQ	<string>	User group ID

203.4. Examples

203.4.1. Getting the current SMTP user group settings

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=smtpgroups&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
GroupID=Group 1
GroupName=Group 1
GroupID=Group2
GroupName=Group2
GroupID=Group1
GroupName=Group1
GroupID=Group 3
GroupName=Group 3
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SMTPGroups": [
        {
            "GroupID": "Group 1",
            "GroupName": "Group 01"
        },
        {
            "GroupID": "Group2",
            "GroupName": "Group2"
        },
        {
            "GroupID": "Group 3",
            "GroupName": "Group 3"
        }
    ]
}
```


203.4.2. Getting the settings of ‘Group 1’

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/transfer.cgi?msubmenu=smtpgroups&action=view&GroupID=Group 1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GroupID=Group 1
GroupName=Group 1
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SMTPGroups": [
        {
            "GroupID": "Group 1",
            "GroupName": "Group 1"
        }
    ]
}
```


203.4.3. Adding an SMTP user group

When adding a group with only **GroupName**, NVR creates a group whose **GroupID** is the same as **GroupName**. If you want to create it separately, send a command with both **GroupID** and **GroupName**.

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpgroups&action=add&GroupName=Group3
```

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpgroups&action=add&GroupID=TestGroup&GroupName=
```

```
Group3
```

203.4.4. Updating the user group name

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpgroups&action=update&GroupID=Group3&GroupName=
Group3
```

203.4.5. Removing the user group

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=smtpgroups&action=remove&GroupID=Group3
```

Chapter 204. Data Server

204.1. Description

The **dataserver** submenu configures the data server.

NOTE This chapter applies to the camera only.

Access level

Action	Camera
view	Admin
set	Admin
test	Admin

204.2. Syntax

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=
dataserver&action=<value>[&<parameter>=<value>]
```

204.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads data server settings
set	Enable	REQ, RES	<bool>	Value that shows if the data server is in active status or not
	IPv4Address	REQ, RES	<string>	IPv4 address of the data server
	Port	REQ, RES	<int>	Data server port number
	Username	REQ, RES	<string>	Data server user name
	Password	REQ, RES	<string>	Data server user password
	IsPasswordEncrypted	REQ, RES	<bool>	Returns true if password sent is encrypted. Encrypted password should be sent as a post message.
	StoreName	REQ, RES	<string>	Description of data server Up to 20 characters allowed.

Action	Parameters	Request/ Response	Type/ Value	Description
test	ConnectionState	RES	<enum> Failed, Connecting, Success	Representing value that shows if the data server is connected or not

204.4. Examples

204.4.1. Getting the current data server settings

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=dataserver&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=False
IPv4Address=192.168.1.1
Port=3434
UserName=admin
Password=
StoreName=testServer
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Enable": false,
    "IPv4Address": "192.168.1.1",
    "Port": 3434,
    "UserName": "admin",
    "Password": "",
    "StoreName": "testServer"
```

```
}
```

204.4.2. Setting the data server

REQUEST

```
http://<Device IP>/stw-
cgi/transfer.cgi?msubmenu=dataserver&action=set&Enable=True&IPv4Address=192.
168.1.1&Port=3434&UserName=admin&Password={password}&StoreName=testServer
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "Success"
}
```

204.4.3. Testing the data server’s settings

REQUEST

```
http://<Device IP>/stw-cgi/transfer.cgi?msubmenu=dataserver&action=test
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
ConnectionState=Connecting
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "ConnectionState": "Connecting"
}
```

AI

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 205. Overview

205.1. Description

ai.cgi is used to configure AI-related features on the device.

The following submenus are used to provide AI functionalities:

- **metaattributesearch**: Used to search recordings with metadata attributes.
- **ocrsearch**: Used to search recordings for OCR text.
- **objectdetectfromimage**: Used to detect the requested object type from images.
- **imagelibrary**: Used to manage images on the device that is used for recognition/training.
- **facerecognitionsearch**: Used to search recordings for the provided face image.
- **aiengine**: Used to configure AI engine on the device
- **aitimeline**: Used to view AI timeline for the requested time duration

Chapter 206. Metaattributesearch

206.1. Description

The **metaattributesearch** submenu is used to search recordings with metadata attributes.

NOTE	This submenu is applicable to NVR only

Access level

Action	NVR
view	User
control	User

206.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
metaattributesearch&action=<value>[&<parameter>=<value>...]
```

206.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	If Type is passed as Status, search status is informed. If Type is passed as Result, search result is provided.
	SearchToken	REQ	<string>	Search session token
	ResultFromIndex	REQ	<int>	Index from which search results are fetched
	ResultFromTime	REQ	<string>	Time from which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	ResultToTime	REQ	<string>	Time to which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	MaxResults	REQ	<int>	Maximum number of search results to return
	Status	RES	<enum> Completed, NotCompleted	Search status
	TotalResultsFound	RES	<int>	Total results
	TotalCount	RES	<int>	Total count of result
	TimedOut	RES	<bool> True, False	Asynchronous search timeout.
	ResultFromDate	RES	<string>	Start time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	End time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	SearchTokenExpiryTime	RES	<string>	Time when the search token expires Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.DateTime	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.Channel	RES	<int>	Result channel ID
	Result.#.Attributes	RES	<string>	Attribute text
	Result.#.ImageURL	RES	<string>	Image URL
	Result.#.Resolution	RES	<string>	Image resolution format=width x height
	Result.#.BkID	RES	<string>	Bookmark ID
	Result.#.ObjectID	RES	<int>	Object ID
	Result.#.BoundingBox	RES	<csv>	Bounding box information of the object in the following format: left, top, right, bottom

Action	Parameters	Request/ Response	Type/ Value	Description
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Search mode
	ClassType	REQ	<enum> Person, Vehicle, Face	Search type
	ChannelIDList	REQ	<csv>	Search channel ID list
	OverlappedID	REQ	<int>	Overlapped number
	SearchAttributes.Person.Gender	REQ	<csv> Any, Male, Female	Gender type
	SearchAttributes.Person.Clothing.Tops.ColorString	REQ	<csv> Any, Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple	Color of top (clothing)
	SearchAttributes.Person.Clothing.Tops.Length	REQ	<csv> Any, Long, Short	Length of top (clothing)
	SearchAttributes.Person.Clothing.Bottoms.ColorString	REQ	<csv> Any, Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple	Color of bottom (clothing)
	SearchAttributes.Person.Clothing.Bottoms.Length	REQ	<csv> Any, Long, Short	Length of bottom (clothing)
	SearchAttributes.Person.Clothing.Hat	REQ	<csv> Any, Wear, No	Wearing a hat or not

Action	Parameters	Request/ Response	Type/ Value	Description
	SearchAttributes.Person.Belonging.Bag	REQ	<csv> Any, Wear, No	Carrying a bag or not
	SearchAttributes.Face.Gender	REQ	<csv> Any, Male, Female	Gender
	SearchAttributes.Face.AgeType	REQ	<csv> Any, Young, Adult, Middle, Senior	Age group
	SearchAttributes.Face.Hat	REQ	<csv> Any, Wear, No	Wearing a hat or not
	SearchAttributes.Face.Opticals	REQ	<csv> Any, Wear, No	Wearing glasses or not
	SearchAttributes.Vehicle.Type	REQ	<csv> Any, Car, Bus, Truck, Motorcycle, Bicycle, Train	Vehicle type
	SearchAttributes.Vehicle.ColorString	REQ	<csv> Any, Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple	Vehicle color
	FromDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Async	REQ	<bool>+ True, False	Asynchronous search option
	WaitTime	REQ	<int>	Timeout second (Default: 60 sec.)

Action	Parameters	Request/ Response	Type/ Value	Description
	SearchToken	RES	<string>	Search token given as response, which can be used for view operation.
	TotalCount	RES	<int>	Total count
	ResultFromDate	RES	<string>	Result of 'from' date Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	Result of 'to' date Time in UTC format YYYY-MM-DDTHH:MM:SSZ

206.4. Examples

206.4.1. Meta attribute search

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=metaattributesearch&action=control&Mode=Start&Async=True&ClassType=Person&ChannelIDList=0,1,2,3,63&OverlappedID=-1&FromDate=1970-01-01T01:02:03Z&ToDate=2021-01-26T01:02:03Z&SearchAttributes.Person.Gender=Male&SearchAttributes.Person.Clothing.Tops.ColorString=Black,Red&SearchAttributes.Person.Clothing.Bottoms.ColorString=Gray,White&SearchAttributes.Person.Belonging.Bag=Wear
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=35833
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
```

```
<Body>
```

```
{
    "SearchToken": "35833"
}
```

206.4.2. Viewing search result status

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=metaattributesearch&action=view&Type=Status&SearchToken=35833
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2019-06-16T00:28:29Z
Status=Completed
TotalResultsFound=0
TotalCount=1547
TimedOut=False
ResultFromDate=2019-06-15T05:38:11Z
ResultToDate=2019-06-16T00:26:49Z
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchTokenExpiryTime": "2019-06-16T00:28:29Z",
    "Status": "Completed",
    "TotalResultsFound": 0,
    "TotalCount": 1547,
```

```
    "TimedOut": "False",
    "ResultFromDate": "2019-06-15T05:38:11Z",
    "ResultToDate": "2019-06-16T00:26:49Z"
}
```

206.4.3. Viewing search result

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=metaattributesearch&action=view&Type=Results&ResultFromI
ndex=1&MaxResults=100&SearchToken=35833
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2019-06-15T23:14:45Z
Status=Completed
TotalResultsFound=100
TotalCount=1414
TimedOut=False
Result.0.DateTime=2019-06-15T23:13:32Z
Result.0.Channel=2
Result.0.Attributes.Person.Gender=Male
Result.0.Attributes.Person.Clothing.Tops.ColorString=Black
Result.0.Attributes.Person.Clothing.Bottoms.ColorString=Gray
Result.0.Attributes.Person.Belonging.Bag=Wear
Result.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=metaattributesearch&ID=0000000
000000000_10_0_2_1560640412_298530
Result.0.Resolution=208x336
Result.0.Channel=2
Result.0.ObjectID=298530
Result.0.Coordinate_Del=1372,160,1579,497
Result.0.BoundingBox=-0.285231,-0.851852,-0.17739,-0.539815
Result.0.BkID=00000000000000000000000000000000
...
Result.99.DateTime=2019-06-15T22:32:27Z
```

```
Result.99.Channel=2
Result.99.Attributes.Person.Gender=Male
Result.99.Attributes.Person.Clothing.Tops.ColorString=Red
Result.99.Attributes.Person.Clothing.Bottoms.ColorString=Gray
Result.99.Attributes.Person.Belonging.Bag=Wear
Result.99.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=metaattributesearch&ID=0000000
000000000_10_0_2_1560637947_286642
Result.99.Resolution=160x256
Result.99.Channel=2
Result.99.ObjectID=286642
Result.99.Coordinate_Del=1221,0,1382,262
Result.99.BoundingBox=-0.363897,-1,-0.280021,-0.757407
Result.99.BkID=00000000000000000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchTokenExpiryTime": "2019-06-15T23:14:45Z",
    "Status": "Completed",
    "TotalResultsFound": 100,
    "TotalCount": 1414,
    "TimedOut": "False",
    "Results": [
        {
            "Result": 0,
            "DateTime": "2019-06-15T23:13:32Z",
            "Channel": 2,
            "Attributes": {
                "Person": {
                    "Gender": ["Male"],
                    "Clothing": {
                        "Tops": {
                            "ColorString": ["Black"]
                        },
                        "Bottoms": {
                            "ColorString": ["Gray"]
```

```
                    }
                },
                "Belonging": {
                    "Bag": ["Wear"]
                }
            }
        },
        "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=metaattributesearch&ID=0000000
000000000_10_0_2_1560640412_298530",
        "Resolution": "208x336",
        "ObjectID": 298530,
        "Coordinate_Del": "1372,160,1579,497",
        "BoundingBox": [
            {
                "left": -0.285231,
                "top": -0.851852,
                "right": -0.177390,
                "bottom": -0.539815
            }],
        "BkID": "00000000000000000000000000000000"
    },
...
    {
        "Result": 99,
        "DateTime": "2019-06-15T22:32:27Z",
        "Channel": 2,
        "Attributes": {
            "Person": {
                "Gender": ["Male"],
                "Clothing": {
                    "Tops": {
                        "ColorString": ["Red"]
                    },
                    "Bottoms": {
                        "ColorString": ["Gray"]
                    }
                },
                "Belonging": {
                    "Bag": ["Wear"]
                }
```

```
                }
            },
            "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=metaattributesearch&ID=0000000
000000000_10_0_2_1560637947_286642",
            "Resolution": "160x256",
            "ObjectID": 286642,
            "Coordinate_Del": "1221,0,1382,262",
            "BoundingBox": [
                {
                    "left": -0.363897,
                    "top": -1,
                    "right": -0.280021,
                    "bottom": -0.757407
                }
            ],
            "BkID": "00000000000000000000000000000000"
        }
    ]
}
```

Chapter 207. OCR (Optical Character Recognition) search

207.1. Description

The **ocrsearch** submenu is used to search a video that matches an input string.

> **NOTE** This submenu is applicable to NVR only

Access level

Action	NVR
view	User
control	User

207.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
ocrsearch&action=<value>[&<parameter>=<value>...]
```

207.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> Results, Status	If Type is passed as Status, search status is informed. If Type is passed as Result, search result is provided.
	SearchToken	REQ	<string>	Search session token
	ResultFromIndex	REQ	<int>	Index from which search results are fetched
	ResultFromTime	REQ	<string>	Time from which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	ResultToTime	REQ	<string>	Time to which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	MaxResults	REQ	<int>	Maximum number of search results to return
	Status	RES	<enum>	Search status
	TotalResultsFound	RES	<int>	Total results
	TotalCount	RES	<int>	Total result count
	TimedOut	RES	<bool> True, False	Asynchronous search timeout.
	ResultFromDate	RES	<string>	Start time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	End time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	SearchTokenExpiryTime	RES	<string>	Time when the search token expires Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.DateTime	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.Channel	RES	<int>	Result channel ID
	Result.#.Text	RES	<string>	Text found matching the search keyword
	Result.#.ImageURL	RES	<string>	Image URL
	Result.#.Resolution	RES	<string>	Image resolution
	Result.#.BkID	RES	<string>	Bookmark ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Result.#.BoundingBox	RES	<csv>	Bounding box information of the object in the following format: left, top, right, bottom
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Used to start, cancel, or renew search
	ChannelIDList	REQ	<csv>	On which channel search has to be performed
	OverlappedID	REQ	<int>	Recording overlapped id
	SearchText	REQ	<string>	Text to search in the recording
	FromDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Async	REQ	<bool>+ True, False	Asynchronous search option
	WaitTime	REQ	<int>	Timeout second (Default: 60 sec.)
	SearchToken	RES	<string>	Search token given as response, which can be used for view operation.
	TotalCount	RES	<int>	Total result count
	ResultFromDate	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	Time in UTC format YYYY-MM-DDTHH:MM:SSZ

207.4. Examples

207.4.1. OCR search

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=ocrsearch&action=control&Mode=Start&Async=True&ChannelID
List=0,1,2,3,4&OverlappedID=-1&FromDate=1970-01-01T01-02-03&ToDate=2021-01-
```

```
01T01-02-03&SearchText=*nu*
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=48928
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchToken": "48928",
}
```

207.4.2. Viewing search result status

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=ocrsearch&action=view&Type=Status&SearchToken=48928
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=
Status=Completed
TotalResultsFound=0
TotalCount=68
TimedOut=False
ResultFromDate=2019-06-15T05:43:29Z
```

```
ResultToDate=2019-06-16T00:27:06Z
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "SearchTokenExpiryTime": "",
    "Status": "Completed",
    "TotalResultsFound": 0,
    "TotalCount": 68,
    "TimedOut": "False",
    "ResultFromDate": "2019-06-15T05:43:29Z",
    "ResultToDate": "2019-06-16T00:27:06Z"
}
```


207.4.3. Viewing search result

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=ocrsearch&action=view&Type=Results&ResultFromIndex=1&Max
Results=5&SearchToken=48928
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchTokenExpiryTime=2019-06-15T23:21:12Z
Status=Completed
TotalResultsFound=51
TotalCount=55
TimedOut=False
Result.0.DateTime=2019-06-15 23:17:07
Result.0.Channel=0
Result.0.Text=ulnu8
```

```
Result.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=ocrsearch&ID=0000000000000000_
10_4_0_1560640627_3078
Result.0.Resolution=96x40
Result.0.BoundingBox=-0.534375,-0.0296296,-0.429167,0.0518519
Result.0.BkID=00000000000000000000000000000000
...
Result.50.DateTime=2019-06-15 06:01:17
Result.50.Channel=0
Result.50.Text=ulnu8
Result.50.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=ocrsearch&ID=0000000000000000_
10_4_0_1560578477_194
Result.50.Resolution=72x24
Result.50.BoundingBox=-0.322917,0.172222,-0.246875,0.22963
Result.50.BkID=00000000000000000000000000000000
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "SearchTokenExpiryTime": "2019-06-15T23:21:12Z",
    "Status": "Completed",
    "TotalResultsFound": 51,
    "TotalCount": 55,
    "TimedOut": "False",
    "Results": [
        {
            "Result": 0,
            "DateTime": "2019-06-15 23:17:07",
            "Channel": 0,
            "Text": "ulnu8",
            "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=ocrsearch&ID=0000000000000000_
10_4_0_1560640627_3078",
            "Resolution": "96x40",
            "BoundingBox": [
                {
```

```
                    "left": -0.534375,
                    "top": -0.029630,
                    "right": -0.429167,
                    "bottom": 0.051852
                }
            ],
            "BkID": "00000000000000000000000000000000"
        },
    ...
        {
            "Result": 50,
            "DateTime": "2019-06-15 06:01:17",
            "Channel": 0,
            "Text": "ulnu8",
            "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=ocrsearch&ID=0000000000000000_
10_4_0_1560578477_194",
            "Resolution": "72x24",
            "BoundingBox": [
                {
                    "left": -0.322917,
                    "top": 0.172222,
                    "right": -0.246875,
                    "bottom": 0.229630
                }
            ],
            "BkID": "00000000000000000000000000000000"
        }
    ]
}
```

Chapter 208. Object Detection from Image

208.1. Description

The **objectdetectfromimage** submenu is used to detect the requested object type from images.

NOTE This submenu is applicable to NVR only

Access level

Action	NVR
control	User

208.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
objectdetectfromimage&action=<value>[&<parameter>=<value>...]
```

208.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	ObjectType	REQ	<enum> Face	Object type to be detected in the past image
	Result.#.TempGroupID	RES	<string>	Temporary group ID
	Result.#.TempImageID	RES	<int>	Temporary image ID
	Result.#.ImageURL	RES	<int>	Image URL of the detected image
	Result.#.Resolution	RES	<string> widthxheig ht	Resolution of the detected object image
	Result.#.Coordinates	RES	<string>	Coordinates of the detected object in the original image

Action	Parameters	Request/ Response	Type/ Value	Description
	Response	RES	<enum> Success, Fail (No Face Detected), Fail (Too Many Face Detected), Fail (Unknown)	Response code

208.4. Examples

208.4.1. Object detected from image

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=objectdetectfromimage&action=control&ObjectType=Face
```

```
Image sent as POST Content
```

```
Base64(image)
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success",
    "Results": [
        {
            "Result": 0,
            "TempGroupID": 1000,
            "TempImageID": 168,
            "ImageURL": "/stw-
```

```
cgi/ai.cgi?msubmenu=imageget&action=view&type=objectdetetfromimage&ID=168",
            "Resolution": "64x72",
            "Coordinates": [
                {
                    "x": 538,
                    "y": 349
                },
                {
                    "x": 602,
                    "y": 349
                },
                {
                    "x": 602,
                    "y": 421
                },
                {
                    "x": 538,
                    "y": 421
                }
            ]
        }
    ]
}
```

Chapter 209. Image library management

209.1. Description

The **imagelibrary** submenu is used to manage a collection of images in a device.

NOTE	This submenu is applicable to NVR only

Access level

Action	NVR
view	User
add	User
update	User
remove	User
control	User

209.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
imagelibrary&action=<value>[&<parameter>=<value>...]
```

209.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	GroupID	REQ,RES	<int>	Group ID
	GroupID.#.Type	RES	<enum> Face	Group type
	GroupID.#.GroupName	RES	<string>	Group name
	GroupID.#.ImageID.#.ImageURL	RES	<string>	Image URL
	GroupID.#.ImageID.#.Resolution	RES	<string>	Resolution in format "widthxheight"
	GroupID.#.ImageID.#.Name	RES	<string>	Name of image
add	GroupID	REQ	<int>	Group ID (if a group ID already exists, a new image will be added to it)

Action	Parameters	Request/ Response	Type/ Value	Description
	Type	REQ	<enum> Face	Type of image
	GroupName	REQ	<string>	Group name
	Name	REQ	<string>	Image name
	ImageRef	REQ	<string>	Image reference to add in the following format: {MetaType}_{Channel}_{UTC}_{ObjectID} e.g., 1_0_45545454_1234
	TempGroupID	REQ	<int>	Temporary group ID from the object detection submenu
	TempImageID	REQ	<int>	Temporary image ID from object detection submenu
	GroupID	RES	<int>	Response group ID
	ImageID	RES	<int>	Response image ID
	Response	RES	<enum> Success, Fail (No Face Detected), Fail (Too Many Face Detected), Fail (Unknown)	Response message
update	GroupID	REQ	<int>	Group ID which needs to be updated
	Type	REQ	<enum> Face	Type of image
	GroupName	REQ	<string>	New group name
	ImageID	REQ	<int>	Image ID
	Name	REQ	<string>	New name to be updated
remove	GroupID	REQ	<int>	Remove image from group ID
	ImageIDList	REQ	<csv>	List of image IDs to be removed
	ImageID	REQ	<int>	Image ID to be removed

Action	Parameters	Request/ Response	Type/ Value	Description
control	Function	REQ	<enum> Backup, Restore	Back up or restore images
	GroupID	REQ	<int>	Group ID to back up or restore
	GroupName	REQ	<string>	Group name to back up or restore
	Password	REQ	<string>	Password for the backup file
	IsPasswordEncrypted	REQ	<bool> True, False	When password is sent as encrypted text

209.4. Examples

209.4.1. Viewing image library

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=imagelibrary&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Group.0.GroupID=1001
Group.0.GroupName=tete
Group.0.Image.0.ImageID=127
Group.0.Image.0.Name=hgg
Group.0.Image.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1001_127
Group.0.Image.0.Resolution=24x24
Group.1.GroupID=1002
Group.1.GroupName=i
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Groups": [
        {
            "GroupID": 1001,
            "GroupName": "tete",
            "Images": [
                {
                    "ImageID": 127,
                    "Name": "hgg",
                    "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1001_127"
,
                    "Resolution": "24x24"
                }
            ]
        },
        {
            "GroupID": 1002,
            "GroupName": "i",
            "Images": []
        }
    ]
}
```

209.4.2. Adding a new group

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=imagelibrary&action=add&GroupName=sample
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GroupID=1003
ImageID=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "GroupID": 1003,
    "ImageID": 0
}
```


209.4.3. Adding a new image

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=imagelibrary&action=add&GroupID=1003&TempImageID=180&Tem
pGroupID=1000&Name=sampleImage
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
GroupID=1003
ImageID=186
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "GroupID": 1003,
    "ImageID": 186
}
```

209.4.4. Updating an action

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=imagelibrary&action=update&GroupID=1003&GroupName=sample2
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Response": "Success"
}
```


209.4.5. Removing an action

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=imagelibrary&action=remove&GroupID=1003
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

209.4.6. Making a backup and restoring

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=imagelibrary&action=control&Function=Backup&GroupID=1004
&IsPasswordEncrypted=True&Password=Shd/DVG1Y+vLVpTnSTbj8aOW3xd1e7eGSlM/lGjY/
ju5JB67OTET+YvdNxzNog3ZtUS/ssxAPVm24O2vCQt4CJFz8MfpzfPf7ucQy3QIaivgvm4pwRM1b
PksdUe4Ec3KpyhV+sJWvmWUP+3Hd0hsztOVEXI7Fd5CtZERilV0vtTRc5DZ6nuY1Vhe+KSDIzJc4
TOSoWNEA9Sv3yUSmZjWprJ4EnlWuZzs3Fdi7l7Xpruq7StRG50myAsu9v6bpoUVaE/ZdM+V1JfDR
xD0ooxVuQDhFY9N0ek0j/JaV1OJq8rb/A4dBhwr59JvdwKgJwoMmQRp/Rt0+4+hGz4fSihbPg==
```

TEXT/ JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
EncryptFileFormat
```

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=imagelibrary&action=control&Function=Restore&IsPasswordE
ncrypted=True&Password=P182w74Fk394fSftxjHvBR4V3zcMkqXzcRUW0zrLpOMkcoMSeDduS
bt3NP1TkCUfOJbP/olPmu42ha84Azj52rJdF3P8jNPCFuHHsiQUAsk0S/5UlSwcuc2lCQkQmMwQz
M4mLgHxS1+ZgUtBFy9FA/PKvgqvOeOm6hLywxT/nHT/Aj+ObeDcr5t17rB2n1dWlP5v1cipBbJaA
mmLk1SFwZnhilHJ0gE8mDuX8VmQ4WouCuuvsKjRFYZESLhzYCrjoQFX+xZObxrwqrdLuWaoJiWH6
JYr1+o75nsQoWJW7RpiJqbfgl0aJDvjvuOl8/YKtb0+J5m4sH5pdSzQk4e3+Q==&GroupName=te
st
```

```
File sent in POST Content
<Backup File>
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
<MultiPartBoundary>
Content-type:application/json
{
    "Status": "DownloadAck"
}
--<MultiPartBoundary>
Content-type:application/json
{
    "Progress": 0
}
--<MultiPartBoundary>
Content-type:application/json
{
    "Progress": 100
}
--<MultiPartBoundary>
Content-type:application/json
{
    "Status": "DownloadOK"
}
HTTP/1.1 200 OK
Content-type:application/json;charset=utf-8
```

Chapter 210. AI Timeline

210.1. Description

The **aitimeline** submenu is used to view the AI details for the specified duration.

NOTE	This submenu is applicable to NVR only

Access level

Action	NVR
view	User

210.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
aitimeline&action=<value>[&<parameter>=<value>...]
```

210.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ChannelIDList	REQ	<csv>	Requested channel list
	ClassType	REQ	<enum> Person, Vehicle, Face, FaceRecognition, LicensePlate	Class type to search
	FromDate	REQ	<string>	'From' date in UTC format: YYYY-MM-DDTHH:MM:SSZ
	ToDate	REQ	<string>	'To' date in UTC format: YYYY-MM-DDTHH:MM:SSZ
	OverlappedID	REQ	<int>	Recording overlapped id
	TotalCount	RES	<int>	Total result count
	Result.#.Channel	RES	<int>	Result channel number

Action	Parameters	Request/ Response	Type/ Value	Description
	Result.#.Type	RES	<enum> Person, Vehicle, Face, FaceRecognition, LicensePlate	Result classification type
	Result.#.FromDate	RES	<string>	UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.ToDate	RES	<string>	UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.BkID	RES	<string>	Bookmark ID

210.4. Examples

210.4.1. AI timeline search for a duration

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?&msubmenu=aitimeline&action=view&ClassType=Face&FromDate=1970-01-01T01:02:03Z&ToDate=2021-01-26T01:02:03Z&OverlappedID=-1
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalCount=5000
Result.0.Channel=12
Result.0.Type=Face
Result.0.FromDate=2020-06-12T15:24:43Z
Result.0.ToDate=2020-06-12T15:24:43Z
Result.0.BkID=00000000000000000000000000000000
Result.1.Channel=12
Result.1.Type=Face
```

```
Result.1.FromDate=2020-06-12T15:24:43Z
Result.1.ToDate=2020-06-12T15:24:43Z
Result.1.BkID=00000000000000000000000000000000
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "TotalCount": 5000,
    "Results": [
        {
            "Channel": 12,
            "Type": "Face",
            "FromDate": "2020-06-12T15:24:43Z",
            "ToDate": "2020-06-12T15:24:43Z",
            "BkID": "00000000000000000000000000000000"
        },
        {
            "Channel": 12,
            "Type": "Face",
            "FromDate": "2020-06-12T15:24:43Z",
            "ToDate": "2020-06-12T15:24:43Z",
            "BkID": "00000000000000000000000000000000"
        },
        ...
    ]
}
```

Chapter 211. AI Engine

211.1. Description

The **aiengine** submenu is used to manage and view AI engine status.

NOTE This submenu is applicable to NVR only

Access level

Action	NVR
view	User
set	User

211.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
aiengine&action=<value>[&<parameter>=<value>...]
```

211.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	ChannelIDList	REQ	<csv>	Requested channel list
	TotalEngineUsage.ObjectEngineUsage	RES	<int>	Total engine usage rate of object detection
	TotalEngineUsage.RecognitionEngineUsage	RES	<int>	Total engine usage rate of recognition
	EngineStatus.Channel.#.CamType	RES	<enum> Unknown, MetaDataCam, AIMetaDataCam, NoneMetaCam	Used to notify if the camera is connected to the channel; determines whether to send metadata or not.
	EngineStatus.Channel.#.ObjectEngine	RES	<bool> True, False	Enable or disable state of the object engine
	EngineStatus.Channel.#.RecognitionEngine	RES	<bool> True, False	Enable or disable state of the recognition engine

Action	Parameters	Request/ Response	Type/ Value	Description
	EngineStatus.Channel.#. ObjectEngineUsage	RES	<int>	Each usage rate of object detection
	EngineStatus.Channel.#. RecognitionEngineUsage	RES	<int>	Each usage rate of recognition
	FaceRecognitionAgreementTime	RES	<string>	Face recognition agreement time in UTC format. UTCFormat=YYYY-MM-DDTHH:MM:SSZ
set	Channel.#.ObjectEngine	REQ	<bool> True, False	Enable or disable object detection engine
	Channel.#.RecognitionEngine	REQ	<bool> True, False	Enable or disable recognition engine
	AgreementStatus.FaceRecognition	REQ	<bool> True, False	Agreement to run face recognition algorithm.

211.4. Examples

211.4.1. Viewing AI engine stats

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?&msubmenu=aiengine&action=view
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalEngineUsage.ObjectEngineUsage=46.875
TotalEngineUsage.RecognitionEngineUsage=6.25
EngineStatus.Channel.0.CamType=Unknown
EngineStatus.Channel.0.ObjectEngine=False
EngineStatus.Channel.0.RecognitionEngine=False
EngineStatus.Channel.0.ObjectEngineUsage=0
EngineStatus.Channel.0.RecognitionEngineUsage=0
EngineStatus.Channel.1.CamType=Unknown
EngineStatus.Channel.1.ObjectEngine=False
```

```
EngineStatus.Channel.1.RecognitionEngine=False
EngineStatus.Channel.1.ObjectEngineUsage=0
EngineStatus.Channel.1.RecognitionEngineUsage=0
...
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "TotalEngineUsage": {
        "ObjectEngineUsage": 46.875000,
        "RecognitionEngineUsage": 6.250000
    },
    "EngineStatus": [
        {
            "Channel": 0,
            "CamType": "Unknown",
            "ObjectEngine": false,
            "RecognitionEngine": false,
            "ObjectEngineUsage": 0,
            "RecognitionEngineUsage": 0
        },
        {
            "Channel": 1,
            "CamType": "Unknown",
            "ObjectEngine": false,
            "RecognitionEngine": false,
            "ObjectEngineUsage": 0,
            "RecognitionEngineUsage": 0
        },
        ...
    ]
}
```


211.4.2. Enabling AI engine

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?&msubmenu=aiengine&action=set&Channel.0.ObjectEngine=True&Channel.0.RecognitionEngine=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Response": "Success"
}
```

Chapter 212. Face Recognition Search

212.1. Description

The **facerecognitionsearch** submenu is used to search for face recognition information in recordings on channels with face recognition enabled.

> **NOTE** This submenu is applicable to NVR only

Access level

Action	NVR
view	User
control	User
remove	User

212.2. Syntax

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=
facerecognitionsearch&action=<value>[&<parameter>=<value>...]
```

212.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Type	REQ	<enum> SearchImages, Results, Status	Changes according to the type information provided
	SearchToken	REQ	<string>	Search token received in control operation
	ResultFromIndex	REQ	<int>	The index from which results are displayed..
	ResultFromTime	REQ	<string>	Time from which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ

Action	Parameters	Request/ Response	Type/ Value	Description
	ResultToTime	REQ	<string>	Time to which search results are fetched. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	MaxResults	REQ	<int>	Maximum results needed in view
	Status	RES	<enum> Completed, NotCompleted	When Type is set to Status, search status is provided
	TotalResultsFound	RES	<int>	Total results
	TotalCount	RES	<int>	Total count
	TimedOut	RES	<bool> True, False	Asynchronous search timeout.
	ResultFromDate	RES	<string>	Start time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	End time of search result. Time in UTC format. YYYY-MM-DDTHH:MM:SSZ
	SearchTokenExpiryTime	RES	<string>	Search token expiry date in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.DateTime	RES	<string>	Date time in UTC format YYYY-MM-DDTHH:MM:SSZ
	Result.#.Channel	RES	<int>	Result channel id
	Result.#.ObjectID	RES	<int>	Object id
	Result.#.ImageURL	RES	<string>	Image URL can be used to download image
	Result.#.Resolution	RES	<string> widthxheight	Result image resolution
	Result.#.BkID	RES	<string>	Bookmark ID

Action	Parameters	Request/ Response	Type/ Value	Description
	Result.#.BoundingBox	RES	<csv>	Bounding box information in the following format: left, top, right, bottom
	Result.#.SearchImage.#.Type	RES	<enum> Library, File	When Type is set to SearchImages
	Result.#.SearchImage.#.GroupName	RES	<int>	Group name
	Result.#.SearchImage.#.ImageName	RES	<int>	Image name
	Result.#.SearchImage.#.ImageURL	RES	<string>	Image URL
	Result.#.SearchImage.#.Similarity	RES	<int>	Similarity score 1-100
control	Mode	REQ	<enum> Start, Cancel, Renew, Stop	Search mode
	ChannelIDList	REQ	<csv>	Channels on which search is performed
	OverlappedID	REQ	<int>	Recording overlapped id
	FromDate	REQ	<string>	Time in UTC format "YYYY-MM-DDTHH:MM:SSZ"
	ToDate	REQ	<string>	Time in UTC format "YYYY-MM-DDTHH:MM:SSZ"
	Async	REQ	<bool>+ True, False	Asynchronous search option
	WaitTime	REQ	<int>	Timeout second (Default: 60 sec.)
	SearchImage.#.Type	REQ	<enum> Library	Search image type
	SearchImage.#.GroupID	REQ	<int>	Group ID from image library
	SearchImage.#.ImageID	REQ	<int>	Image ID from image library
	Similarity	REQ	<int>	Similarity threshold for filtering results above this value
	SearchToken	RES	<string>	Seach token for the requested search query

Action	Parameters	Request/ Response	Type/ Value	Description
	TotalCount	RES	<int>	Total result count
	ResultFromDate	RES	<string>	Result of ‘from’ date in UTC format YYYY-MM-DDTHH:MM:SSZ
	ResultToDate	RES	<string>	Result of ‘to’ date in UTC format YYYY-MM-DDTHH:MM:SSZ
remove	SearchImage	REQ	<int>	Search image index
	SearchToken	REQ	<string>	Search token

212.4. Examples

212.4.1. Starting search

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=facerecognitionsearch&action=control&Mode=Start&Async=Tr
ue&ChannelIDList=0,1,2,3,4,5,6,7,8&OverlappedID=-1&FromDate=2000-03-
01T10:59:23Z&ToDate=2021-03-
11T11:59:23Z&SearchImage.0.Type=Library&SearchImage.0.GroupID=1001&SearchIma
ge.0.ImageID=3
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SearchToken=97923
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "SearchToken": "97923"
}
```

212.4.2. Viewing the search result

REQUEST

```
http://<Device IP>/stw-cgi/ai.cgi?msubmenu=facerecognitionsearch&action=view&Type=Status&SearchToken=97923
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Completed
TotalResultsFound=0
TotalCount=48
TimedOut=False
SearchTokenExpiryTime=2020-06-15T19:30:13Z
ResultFromDate=2020-06-12T11:34:41Z
ResultToDate=2020-06-12T15:24:43Z
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Status": "Completed",
    "TotalResultsFound": 0,
    "TotalCount": 48,
    "TimedOut": "False",
    "SearchTokenExpiryTime": "2020-06-15T19:26:12Z"
    "ResultFromDate": "2020-06-12T11:34:41Z"
    "ResultToDate": "2020-06-12T15:24:43Z"
```

```
}
```

REQUEST

```
http://<Device IP>/stw-
cgi/ai.cgi?msubmenu=facerecognitionsearch&action=view&Type=Results&ResultFro
mIndex=1&MaxResults=100&SearchToken=97923
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Status=Completed
TotalResultsFound=48
TotalCount=48
TimedOut=False
SearchTokenExpiryTime=2020-06-16T08:13:57Z
Result.0.DateTime=2020-06-12T15:24:43Z
Result.0.Channel=2
Result.0.ObjectID=3745
Result.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=frsearch&ID=000918E1A14D0000_1
0_1_2_1591975483_3745
Result.0.Resolution=288x392
Result.0.BoundingBox=-1,-0.261111,-0.848398,0.103704
Result.0.BkID=00000000000000000000000000000000
Result.0.SearchImage.0.Type=Library
Result.0.SearchImage.0.GroupName=sample2
Result.0.SearchImage.0.ImageName=sampleImage
Result.0.SearchImage.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1003_186
Result.0.SearchImage.0.Similarity=54
...
Result.47.DateTime=2020-06-12T11:34:41Z
Result.47.Channel=3
Result.47.ObjectID=63
Result.47.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=frsearch&ID=000918E1A14D0000_1
0_1_3_1591961681_63
```

```
.Result.47.Resolution=80x72
Result.47.BoundingBox=-0.952083,-0.131481,-0.861458,0.0111111
Result.47.BkID=00000000000000000000000000000000
Result.47.SearchImage.0.Type=Library
Result.47.SearchImage.0.GroupName=sample2
Result.47.SearchImage.0.ImageName=sampleImage
Result.47.SearchImage.0.ImageURL=/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1003_186
Result.47.SearchImage.0.Similarity=54
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "Status": "Completed",
    "TotalResultsFound": 48,
    "TotalCount": 48,
    "TimedOut": "False",
    "SearchTokenExpiryTime": "2020-06-16T08:14:07Z",
    "Results": [
        {
            "DateTime": "2020-06-12T15:24:43Z",
            "Channel": 2,
            "ObjectID": 3745,
            "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=frsearch&ID=000918E1A14D0000_1
0_1_2_1591975483_3745",
            "Resolution": "288x392",
            "BoundingBox": [
                {
                    "left": -1,
                    "top": -0.261111,
                    "right": -0.848398,
                    "bottom": 0.103704
                }
            ],
            "BkID": "00000000000000000000000000000000",
            "SearchImages": [
```

```
                {
                    "Type": "Library",
                    "GroupName": "sample2",
                    "ImageName": "sampleImage",
                    "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1003_186"
,
                    "Similarity": 54
                }
            ]
        },
        ...
        {
            "DateTime": "2020-06-12T11:34:41Z",
            "Channel": 3,
            "ObjectID": 63,
            "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=frsearch&ID=000918E1A14D0000_1
0_1_3_1591961681_63",
            "Resolution": "80x72",
            "BoundingBox": [
                {
                    "left": -0.952083,
                    "top": -0.131481,
                    "right": -0.861458,
                    "bottom": 0.011111
                }
            ],
            "BkID": "00000000000000000000000000000000",
            "SearchImages": [
                {
                    "Type": "Library",
                    "GroupName": "sample2",
                    "ImageName": "sampleImage",
                    "ImageURL": "/stw-
cgi/ai.cgi?msubmenu=imageget&action=view&type=imagelibrary&ID=face_1003_186"
,
                    "Similarity": 54
                }
            ]
        }
```

```
    ]
}
```

Access Control

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

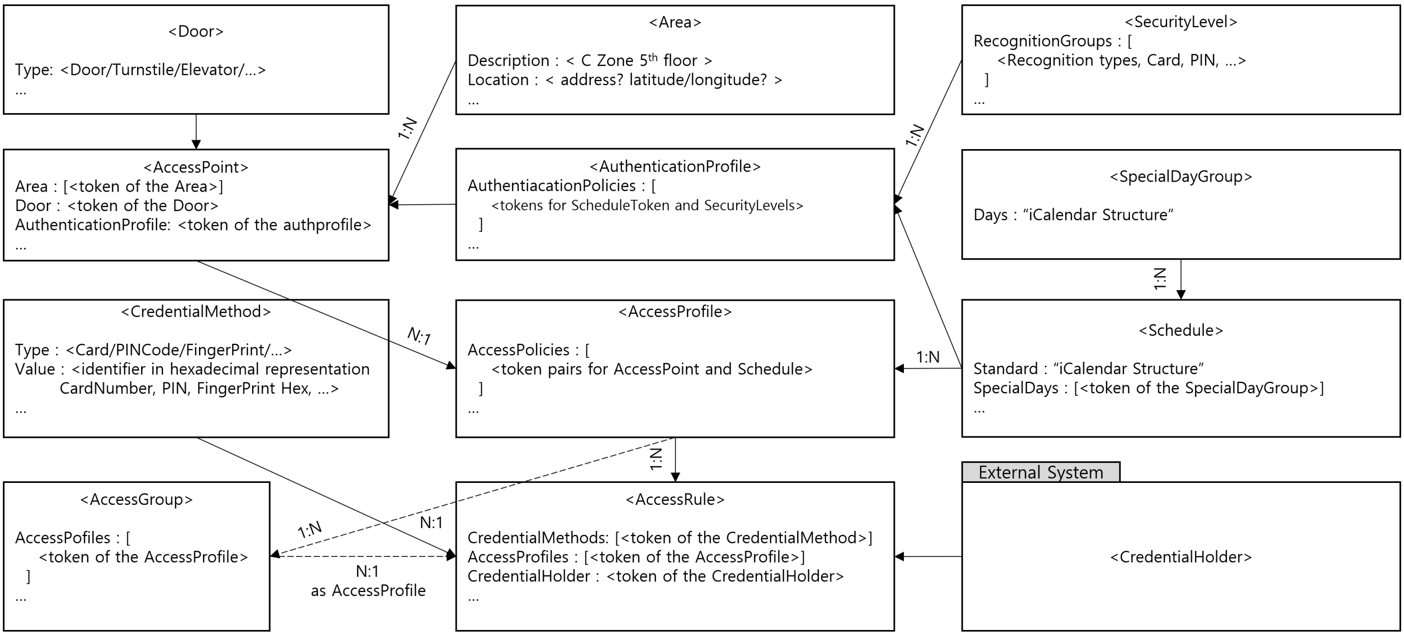
Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 213. Introduction

In SUNAPI, accesscontrol.cgi is newly added to manage access control systems. This guide provides an overview of SUNAPI and describes how to manage an access control system.

Basic Components and its relationship in an Access Control System



Chapter 214. Overview

The following submenus are defined in accesscontrol.cgi.

- **capabilities**: Checks the ACS capabilities provided by the device.
- **configuration**: Supports ACS config backup, restore, and reset.
- **door**: Manages actual physical doors such as door, turnstile, elevator, etc.
- **credentialreader**: Manages reader information to read authentication information such as card, PIN, etc.
- **area**: Manages user-created conceptual areas.
- **securitylevel**: Defines security levels using individual or combined recognition methods (logical AND/OR), or no recognition method (open).
- **authenticationprofile**: Specifies how credential holders are granted access by defining when different security levels are required.
- **accesspoint**: Manages AccessPoint, which represents the relationship between N areas accessible through one door.
- **specialdaygroup**: Manages exceptions to regular schedules, such as public holidays, election days, and foundation days.
- **accessschedule**: Access schedule management (DateFrom-To, TimeFrom-To, SpecialDay, SpecialTime)
- **accessprofile**: AccessProfile consisting of N Access Policies, Access Policy consists of N AccessSchedules for 1 AccessPoint
- **accessgroup**: Group consisting of AccessProfiles
- **credentialholder**: Manages users who can possess credentials.
- **credentialholderimage**: Registers and manages credentialholder’s image.
- **credentialmethod**: Registers and manages credential methods such as card, PINCode, and FingerPrint.
- **accessrule**: Complete Credential consisting of CredentialMethod, CredentialHolder, and AccessProfile.
- **clientfilter**: Restricts access to access control functions by filtering IP (Peer information used for filtering can be expanded).

Chapter 215. Capability

215.1. Description

The **capabilities** submenu of **accesscontrol.cgi** is used to retrieve the capabilities of the accesscontrol system.

NOTE: This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser

215.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
capabilities&action=view
```

215.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Door.MaxCount	RES	<int>	Indicates the maximum number of doors supported by the device.
	CredentialReader.MaxCount	RES	<int>	Indicates the maximum number of readers supported by the device.
	AuthenticationProfile.MaxAuthenticationProfiles	RES	<int>	Indicates the maximum number of authentication profiles.
	AuthenticationProfile.MaxPoliciesPerAuthenticationProfile	RES	<int>	Indicates the maximum number of authentication policies per authentication profile.
	SecurityLevel.MaxSecurityLevels	RES	<int>	Indicates the maximum number of security levels.
	SecurityLevel.MaxRecognitionGroupsPerSecurityLevel	RES	<int>	Indicates the maximum number of recognition groups per security level.
	SecurityLevel.MaxRecognitionMethodsPerRecognitionGroup	RES	<int>	Indicates the maximum number of recognition methods per recognition group.

Action	Parameters	Request/ Response	Type/ Value	Description
	SecurityLevel.SupportedAuthenticationModes	RES	<enum> SingleCredential DualCredential	A list of supported authentication modes
	Schedule.MaxCount	RES	<int>	Indicates the maximum number of schedules the device supports.
	Schedule.MaxTimePeriodsPerDay	RES	<int>	Indicates the maximum number of time periods per day the device supports in a schedule including special days schedule.
	Schedule.MaxSpecialDayGroups	RES	<int>	Indicates the maximum number of special day group entities the device supports.
	Schedule.MaxDaysInSpecialDayGroup	RES	<int>	Indicates the maximum number of days per SpecialDayGroup entity the device supports.
	Schedule.MaxSpecialDaysPerSchedules	RES	<int>	Indicates the maximum number of SpecialDaysSchedule entities referred by a schedule that the device supports.
	Schedule.ExtendedRecurrenceSupported	RES	<bool>	Indicates that the device supports extended iCalendar recurrence format
	Schedule.SpecialDaysSupported	RES	<bool>	Indicates that the device supports special days
	Schedule.StateReportingSupported	RES	<bool>	Indicates that the device supports state reporting
	AccessPoint.MaxCount	RES	<int>	Indicates the maximum number of access points supported by the device.
	AccessPoint.MaxAreas	RES	<int>	Indicates the maximum number of areas supported by the device.
	AccessProfile.MaxCount	RES	<int>	Indicates the maximum number of access profiles supported by the device.
	AccessProfile.MaxAccessPoliciesPerAccessProfile	RES	<int>	Indicates the maximum number of access policies per access profile supported by the device.

Action	Parameters	Request/ Response	Type/ Value	Description
	AccessProfile.MaxSchedulesPerAccessPolicy	RES	<int>	Indicates the maximum number of schedules per access policy supported by the device.
	AccessProfile.MultipleSchedulesPerAccessPointSupported	RES	<bool>	Indicates whether or not several access policies can refer to the same access point in an access profile.
	AccessGroup.MaxCount	RES	<int>	The maximum number of access profiles per access group supported by the device.
	CredentialMethod.MaxCount	RES	<int>	The maximum number of credentialmethod supported by the device.
	Credential.CredentialValiditySupported	RES	<bool>	Indicates that the device supports credential validity.
	Credential.CredentialAccessProfileValiditySupported	RES	<bool>	Indicates that the device supports validity on the association between a credential and an access profile.
	Credential.ValiditySupportsTimeValue	RES	<bool>	Indicates that the device supports both date and time value for validity. If set to false, then the time value is ignored.
	Credential.MaxCount	RES	<int>	The maximum number of credential supported by the device.
	Credential.MaxAccessProfilesPerCredential	RES	<int>	The maximum number of access profiles for a credential.
	Credential.MaxCredentialMethodsPerCredential	RES	<int>	The maximum number of credential methods for a credential.
	Credential.MaxGroupsPerCredential	RES	<int>	The maximum number of groups for a credential.
	Credential.ResetAntipassbackSupported	RES	<bool>	Indicates the device supports resetting of anti-passback violations and notifying on anti-passback violations.

Action	Parameters	Request/ Response	Type/ Value	Description
	Credential.DefaultCredentialSuspensionDuration	RES	<string>	The default time period that the credential will temporary be suspended (e.g. by using the wrong PIN a predetermined number of times). The time period is defined as an [ISO 8601] duration string (e.g. "PT5M")
	Credential.SupportedIdentifierType	RES	<csv>	Card, PIN, Fingerprint, Face, Iris, Vein, Palm, Retina, LicensePlate For custom defined identifier types, free text can be used.

215.4. Examples

215.4.1. Get device capabilities

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=capabilities&action=view
```

RESPONSE


```
{
    "Capabilities": {
        "Door": {
            "MaxCount": 15
        },
        "CredentialReader": {
            "MaxCount": 4
        },
        "AuthenticationProfile": {
            "MaxAuthenticationProfiles": 10,
            "MaxPoliciesPerAuthenticationProfile": 10
        },
        "SecurityLevel": {
            "MaxSecurityLevels": 10,
            "MaxRecognitionGroupsPerSecurityLevel": 10,
            "MaxRecognitionMethodsPerRecognitionGroup": 10,
            "SupportedAuthenticationModes": [
                "SingleCredential",
```

```
                "DualCredential"
            ]
        },
        "Schedule": {
            "MaxCount": 15,
            "MaxTimePeriodsPerDay": 10,
            "MaxSpecialDayGroups": 10,
            "MaxDaysInSpecialDayGroup": 10,
            "MaxSpecialDaysPerSchedules": 10,
            "ExtendedRecurrenceSupported": false,
            "SpecialDaysSupported": true,
            "StateReportingSupported": false
        },
        "AccessPoint": {
            "MaxCount": 15,
            "MaxAreas": 10
        },
        "AccessProfile": {
            "MaxCount": 15,
            "MaxAccessPoliciesPerAccessProfile": 15,
            "MultipleSchedulesPerAccessPointSupported": true
        },
        "CredentialMethod": {
            "MaxCount": 10
        },
        "Credential": {
            "CredentialValiditySupported": true,
            "CredentialAccessProfileValiditySupported": true,
            "ValiditySupportsTimeValue": true,
            "MaxCount": 10,
            "MaxAccessProfilesPerCredential": 10,
            "ResetAntipassbackSupported": true,
            "DefaultCredentialSuspensionDuration": "PT1M",
            "SupportedIdentifierType": [
                "Card",
                "PIN",
                "Fingerprint",
                "Face",
                "Iris",
                "Vein",
                "Palm",
```

```
                "Retina",
                "LicensePlate"
            ]
        }
    }
}
```

Chapter 216. configuration

216.1. Description

The **configuration** submenu of **accesscontrol.cgi** is used to manage the configuration.

> **NOTE** This chapter applies to ACS system only.

Access level

Action	ACS
backup	Suser
restore	Suser
reset	Suser
add	Suser

216.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
configuration&action=backup
```

216.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
backup				
restore				
reset				
add	Type	REQ	<enum> Credential CredentialHolder	

216.4. Examples

216.4.1. Config Backup

backup makes a copy of all accesscontrol settings for backup.
The format of the configuration is device-dependent.
The device can restart after making a copy of the accesscontrol settings for backup.

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=configuration&action=backup
```

RESPONSE

```
Content-Type: application/octet-stream
Content-Disposition: attachment; filename=acsconfigbackup.bin

<config file content>
```

216.4.2. Config Restore

restore the accesscontrol configuration by using the backup.
The format of the configuration is device-dependent.
The device can restart after a configuration restore.

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=configuration&action=restore

There are two types of Content-Type that can be used in the Request Header.
 - application/x-www-form-urlencoded
 - application/octet-stream

Content-Length must indicate the entire size of the file content.
When using urlencoded type, the size after being encoded in base64 format
must be indicated.

POST /stw-cgi/accesscontrol.cgi?msubmenu=configuration&action=restore
HTTP/1.1
Content-type: application/x-www-form-urlencoded; charset=utf-8
Content-Length: <content length>

<config file content>
```

RESPONSE


```
{
    "Response": "Success"
```

```
}
```

216.4.3. Reset Config

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=configuration&action=reset
```

RESPONSE


```
{
    "Response": "Success"
}
```

Chapter 217. door

217.1. Description

The **door** submenu of **accesscontrol.cgi** is used to manage doors.

> **NOTE** This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser
control	Suser
check	Suser

217.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
door&action=view[&<parameter>=<value>]
```

217.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
update	Token	REQ,RES	<string>	add: The token of the created door will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	Description	REQ	<string>	
	Type	REQ	<enum> Door, Elevator	

Action	Parameters	Request/ Response	Type/ Value	Description
	Mode	REQ	<enum> Lock, Unlock	
	ExitSignalUnlock	REQ	<bool>	Indicates whether unlocking the door on request-to-exit is supported.
	ExitSignalGrantWithoutUnlock	REQ	<bool>	When this value is set to true, the ExitSignalUnlock value is ignored, and when a REX signal occurs, unlock is not performed and only a Grant event is generated.
	NotifyOnREXSignal	REQ	<bool>	Configures whether a change in the state of the IOPort connected to the REX is notified as a REXActivated event.
	DoorHeldMonitoring	REQ	<bool>	Indicates whether monitoring for DoorOpenTooLong is supported. (DoorStatus > DoorAlarmState > DoorOpenTooLong)
	DoorHeldMonitoringWhenUnlocked	REQ	<bool>	Indicates whether monitoring DoorOpenTooLong event when unlocked schedule is supported.
	DoorHeldAlertSignal	REQ	<bool>	Indicates whether the buzzer is triggered when DoorOpenTooLong is detected.
	DoorForcedEntryMonitoring	REQ	<bool>	Indicates whether monitoring DoorForcedOpen is supported. (DoorStatus > DoorAlarmState > DoorForcedOpen)
	DoorForcedEntryAlertSignal	REQ	<bool>	Indicates whether triggering buzzer when DoorForcedOpen is detected is supported.
	RelockMode	REQ	<enum> OnGrantTime OnClose OnOpen	Condition for relocking.
	Capabilities.Access	REQ	<bool>	Indicates whether performing momentary access is supported.
	Capabilities.AccessTimingOverride	REQ	<bool>	Indicates whether overriding configured timing is supported.

Action	Parameters	Request/ Response	Type/ Value	Description
	Capabilities.Lock	REQ	<bool>	Supports LockDoor
	Capabilities.Unlock	REQ	<bool>	Supports UnlockDoor
	Capabilities.Block	REQ	<bool>	Supports BlockDoor
	Capabilities.DoubleLock	REQ	<bool>	Supports DoubleLockDoor
	Capabilities.LockDown	REQ	<bool>	Supports LockDown and put it in LockedDown mode
	Capabilities.LockOpen	REQ	<bool>	Supports LockOpen and put it in LockedOpen mode
	Capabilities.DoorMonitor	REQ	<bool>	Supports DoorPhysicalState event
	Capabilities.LockMonitor	REQ	<bool>	Supports LockPhysicalState event
	Capabilities.DoubleLockMonitor	REQ	<bool>	Supports DoubleLockPhysicalState event
	Capabilities.Alarm	REQ	<bool>	Supports door alarm and the DoorAlarm event
	Capabilities.Tamper	REQ	<bool>	This Door instance has a Tamper detector and supports the DoorTamper event
	Capabilities.Fault	REQ	<bool>	Supports door fault and the DoorFault event
	IOPort.ReaderIn	REQ	<int>	IO port number
	IOPort.ReaderOut	REQ	<int>	IO port number
	IOPort.Lock	REQ	<int>	IO port number
	IOPort.DoorPositionSwitch	REQ	<int>	IO port number
	IOPort.RexIn	REQ	<int>	IO port number
	IOPort.RexOut	REQ	<int>	IO port number
	Timings.ReleaseTime	REQ	<int>	The time from when the latch is unlocked until it is relocked again
	Timings.MaxOpenTime	REQ	<int>	The time from when the door is physically opened until the door is set in the DoorOpenTooLong alarm state
	Timings.ExtendedReleaseTime	REQ	<int>	Some individuals need extra time to open the door before the latch relocks. If supported, ExtendedReleaseTime shall be added to ReleaseTime

Action	Parameters	Request/ Response	Type/ Value	Description
	Timings.DelayTimeBeforeRelock	REQ	<int>	If the door is physically opened after access is granted, then DelayTimeBeforeRelock is the time from when the door is physically opened until the latch goes back to locked state
	Timings.ExtendedOpenTime	REQ	<int>	Some individuals need extra time to pass through the door. If supported, ExtendedOpenTime shall be added to MaxOpenTime
control	Command	REQ	<enum> Access, Lock, Unlock	Access: temporarily allow access Lock: lock door Unlock : unlock door LockDown : the device shall only allow the LockDownRelease request. LockDownRelease : releasing the LockedDown state of a door.
check	Token	REQ, RES	<string>	
	Name	RES	<string>	
	Enable	RES	<bool>	
	Description	RES	<string>	
	Type	RES	<enum> Door, Elevator	
	State.DoorPhysicalState	RES	<enum> Unknown, Open, Closed, Fault	
	State.LockPhysicalState	RES	<enum> Unknown, Locked, Unlocked, Fault	
	State.DoubleLockPhysicalState	RES	<enum> Unknown, Locked, Unlocked, Fault	

Action	Parameters	Request/ Response	Type/ Value	Description
	State.Alarm	RES	<enum> Normal, DoorForcedOpen, DoorOpenTooLong	
	State.Tamper.Reason	RES	<string>	
	State.Tamper.State	RES	<enum> Unknown, NotInTamper, TamperDetected	
	State.Fault.Reason	RES	<string>	
	State.Fault.State	RES	<enum> Unknown, NotInFault, FaultDetected	
	State.DoorMode	RES	<enum> Unknown, Locked, Unlocked, Accessed, LockedDown, DoubleLocked	

217.4. Examples

217.4.1. Get doors

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=door&action=view
```

RESPONSE

```
{
  "Doors": [
    {
```

```
        "Token": "doortoken1",
        "Name": "door1",
        "Enable": false,
        "Description": "Access Point 1",
        "Type" : "Door",
        "Mode" : "Lock",
        "OverrideSchedule": "",
        "ExitSignalUnlock": true,
        "DoorHeldMonitoring": true,
        "DoorHeldMonitoringWhenUnlocked": false,
        "DoorHeldAlertSignal": true,
        "DoorForcedEntryMonitor": true,
        "DoorForcedEntryAlertSignal": true,
        "Capabilities": {
          "Access": true,
          "AccessTimingOverride": true,
          "Lock": true,
          "Unlock": true,
          "Block": false,
          "DoubleLock": false,
          "LockDown": false,
          "LockOpen": false,
          "DoorMonitor": false,
          "LockMonitor": false,
          "DoubleLockMonitor": false,
          "Alarm": true,
          "Tamper": false,
          "Fault": true
        },
        "IOPort": {
          "ReaderIn": 1,
          "ReaderOut": 2,
          "Lock": 3,
          "DoorPositionSwitch": 4,
          "RexIn": 5,
          "RexOut": 6
        },
        "Timings": {
          "ReleaseTime": 1,
          "MaxOpenTime": 1,
          "ExtendedReleaseTime": 0,
```

```
        "DelayTimeBeforeRelock": 5,
        "ExtendedOpenTime ": 0
      }
    }
  ]
}
```

217.4.2. Update door

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=door&action=update&Token=doortoken1&Type=Elevator
```

RESPONSE

```
{
    "Response": "Success"
}
```

217.4.3. Check door state

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=door&action=check&Token=doortoken1
```

RESPONSE

```
{
  "Doors": [
    {
      "Token": "doortoken1",
      "Name": "door1",
      "Enable": true,
      "Description": "Access Point 1",
      "Type" : "Door",
      "State": {
        "DoorPhysicalState": "Closed",
        "LockPhysicalState": "Locked",
        "DoubleLockPhysicalState": "Unlocked",
```

```
          "Alarm": "Normal",
          "Tamper": {
            "Reason": "",
            "State": "NotInTamper"
          },
          "Fault": {
            "Reason": "",
            "State": "NotInFault"
          },
          "DoorMode": "Locked"
        }
      }
    ]
}
```

217.4.4. Remove door

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=door&action=remove&Token=doortoken1,doortoken
2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 218. credentialreader

218.1. Description

The **credentialreader** submenu of **accesscontrol.cgi** is for managing the readers.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
set	Suser
control	Suser

218.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
credentialreader&action=view
```

218.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Index	REQ,RES	<int>	
set	Index	REQ,RES	<int>	
	Mode	REQ,RES	<enum> OSDP Wiegand	
	BaudRate	REQ,RES	<enum> 9600	Not fixed yet Only for OSDP
	EnableSecureCommunication	REQ,RES	<bool>	Only for OSDP
	Reader.#.Address	REQ	<int>	
	Reader.#.Enable	REQ	<bool>	
	Reader.#.Silence	REQ	<bool>	
check	Index	REQ,RES	<int>	
	Reader.Index	REQ,RES	<int>	
	Reader.#.Address	RES	<string>	

Action	Parameters	Request/ Response	Type/ Value	Description
	Reader.#.LastRead.Identifier	RES	<string>	
	Reader.#.LastRead.Time	RES	<string>	
control	Index	REQ	<int>	
	Reader.Index	REQ	<int>	
	Mode	REQ	<enum> ResetOSDP SecureKey	

218.4. Examples

218.4.1. Get credentialreaders

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialreader&action=view
```

RESPONSE

```
{
  "CredentialReaders": [
    {
      "Index": 1,
      "Mode": "OSDP",
      "BaudRate": 9600,
      "EnableSecureCommunication": false,
      "Reader": [
        {
          "Index": 1,
          "Address": 0
        },
        {
          "Index": 2,
          "Address": 0
        }
      ]
    }
  ]
```

```
}
```

218.4.2. Set credentialreader

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialreader&action=set&Index=1&Mode=OSDP
&BaudRate=9600&EnableSecureCommunication=True&Reader.1.Address=1&Reader.2.Ad
dress=10
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 219. area

219.1. Description

The **area** submenu of **accesscontrol.cgi** is for managing the area.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

219.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=area&action=view
```

219.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ,RES	<string>	add: The token of the created area will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Description	REQ	<string>	
remove	Token	REQ	<string>	

219.4. Examples

219.4.1. Get areas

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=area&action=view
```

RESPONSE

```
{
  "Areas": [
    {
      "Token": "areatoken1",
      "Name": "area1",
      "Description": "C Zone"
    }
  ]
}
```

219.4.2. Add area

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=area&action=add&Name=test_area&Description=ad
d_area_test
```

RESPONSE

```
{
    "Token": "areatoken1"
}
```

219.4.3. Update area

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=area&action=update&Token=areatoken1&Descripti
on=DZone
```

RESPONSE

```
{
    "Response": "Success"
}
```

219.4.4. Remove areas

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=area&action=remove&Token=areatoken1,areatoken2
```

RESPONSE


```
{
    "Response": "Success"
}
```

Chapter 220. securitylevel

220.1. Description

The **securitylevel** submenu of **accesscontrol.cgi** is for managing the securitylevel.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser

220.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
securitylevel&action=view
```

220.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
	Name	REQ	<string>	
	Priority	REQ	<int>	A higher number indicates that the security level is considered more secure than security levels with lower priorities. The priority is used when an authentication profile have overlapping schedules with different security levels. When an access point is accessed, the authentication policies are walked through in priority order (highest priority first). When a schedule is found covering the time of access, the associated security level is used and processing stops. Two security levels cannot have the same priority
	Description	REQ	<string>	
	RecognitionGroups.Index	REQ	<int>	

Action	Parameters	Request/ Response	Type/ Value	Description
	RecognitionGroups.#.Methods.Index	REQ	<int>	
	RecognitionGroups.#.Methods.#.Type	REQ	<enum> Card PIN	The recognition groups are used to define a logical OR between the groups. No recognition groups mean that the access point is open. No recognition methods mean that the access point is closed. Note that when a recognition group is updated, then any previous recognition methods are replaced with the new list.
	RecognitionGroups.#.Methods.#.Order	REQ	<int>	

220.4. Examples

220.4.1. Get securitylevels

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=securitylevel&action=view
```

RESPONSE


```
{
  "SecurityLevels": [
    {
      "Token": "securityleveltoken1",
      "Name": "securitylevel1",
      "Priority": 1,
      "Description": "card first and pin",
      "RecognitionGroups" : [
        {
          "Index": 1,
          "Methods": [
            {
              "Index": 1,
              "Type": "Card",
              "Order": 1
            },
```

```
            {
              "Index": 2,
              "Type": "PIN",
              "Order": 2
            }
          ]
        }
      ]
    }
  ]
}
```

Chapter 221. authenticationprofile

221.1. Description

The **authenticationprofile** submenu of **accesscontrol.cgi** is for managing the authenticationprofile.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

221.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
authenticationprofile&action=view
```

221.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ,RES	<string>	add: The token of the created authenticationprofile will be the response. update: Used only when requesting
	Name	REQ	<string>	
	DefaultSecurityLevelToken	REQ	<string>	The default security level is used if none of the authentication policies has a schedule covering the time of access (or if no authentication policies are defined).
	Description	REQ	<string>	

Action	Parameters	Request/ Response	Type/ Value	Description
	AuthenticationPolicies.#. Direction	REQ	<enum> InOut In Out	default is InOut
	AuthenticationPolicies.#. ScheduleToken	REQ	<string>	Each authentication policy associates a security level with a schedule (during which the specified security level will be required at the access point) Note that when an authentication profile is updated, then any previous authentication policies are replaced with the new list.
	AuthenticationPolicies.#. SecurityLevels.#.ActiveRegularSchedule	REQ	<bool>	
	AuthenticationPolicies.#. SecurityLevels.#.ActiveSpecialDaySchedule	REQ	<bool>	
	AuthenticationPolicies.#. SecurityLevels.#.SecurityLevelToken	REQ	<string>	
	AuthenticationPolicies.#. SecurityLevels.#.AuthenticationMode	REQ	<enum> Single,Dual	Single: Normal mode where only one credential holder is required to be granted access Dual: Two credential holders are required to be granted access
remove	Token	REQ	<string>	

221.4. Examples

221.4.1. Get authenticationprofiles

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=authenticationprofile&action=view
```

RESPONSE

```
{
  "AuthenticationProfiles": [
    {
      "Token": "authenticationprofiletoken1",
      "Name": "authenticationprofile1",
      "DefaultSecurityLevelToken": "securityleveltoken1",
      "Description": "test",
```

```
      "AuthenticationPolicies" : [
        {
          "Index": 1,
          "ScheduleToken": "scheduletoken1",
          "SecurityLevels": [
            {
              "Index": 1,
              "ActiveRegularSchedule": false,
              "ActiveSpecialDaySchedule": false,
              "AuthenticationMode": "Single",
              "SecurityLevelToken": "securityleveltoken2"
            }
          ]
        }
      ]
    }
  ]
}
```

221.4.2. Add authenticationprofile

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=authenticationprofile&action=add&Name=authenticationprofile1&Description=test&DefaultSecurityLevelToken=securityleveltoken1&AuthenticationPolicies.1.ScheduleToken=scheduletoken1&AuthenticationPolicies.1.SecurityLevels.1.ActiveRegularSchedule=False&AuthenticationPolicies.1.SecurityLevels.1.ActiveSpecialDaySchedule=False&AuthenticationPolicies.1.SecurityLevels.1.AuthenticationMode=Single&AuthenticationPolicies.1.SecurityLevels.1.SecurityLevelToken=securityleveltoken1
```

RESPONSE

```
{
    "Token": "authenticationprofiletoken1"
}
```

221.4.3. Update authenticationprofile

Note that when an authentication profile is updated, then any previous authentication policies are replaced with the new list.

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=authenticationprofile&action=update&Token=authenticationprofiletoken1&AuthenticationPolicies.1.ScheduleToken=scheduletoken1&AuthenticationPolicies.1.SecurityLevels.1.ActiveRegularSchedule=True&AuthenticationPolicies.1.SecurityLevels.1.ActiveSpecialDaySchedule=True&AuthenticationPolicies.1.SecurityLevels.1.AuthenticationMode=Dual&AuthenticationPolicies.1.SecurityLevels.1.SecurityLevelToken=securityleveltoken1
```

RESPONSE

```
{
    "Response": "Success"
}
```

221.4.4. Remove authenticationprofiles

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=authenticationprofile&action=remove&Token=authenticationprofiletoken1,authenticationprofiletoken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 222. accesspoint

222.1. Description

The **accesspoint** submenu of **accesscontrol.cgi** is for managing the accesspoint.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

222.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
accesspoint&action=view
```

222.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ,RES	<string>	add: The token of the created accesspoint will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	Description	REQ	<string>	
	Areas	REQ	<csv>	token list of areas
	DoorType	REQ	<enum> Door, Elevator	
	DoorToken	REQ	<string>	
	Capabilities.DisableAccessPoint	REQ	<bool>	whether or not supports Enable/Disable

Action	Parameters	Request/ Response	Type/ Value	Description
	Capabilities.Duress	REQ	<bool>	whether or not supports duress events
	Capabilities.Anonymous Access	REQ	<bool>	whether or not support REX(RequestToExit) switch or other input that allows anonymous access
	Capabilities.SupportedRecognitionTypes	REQ	<csv> Card, PIN, …	
	AuthenticationProfileToken	REQ	<string>	
remove	Token	REQ	<string>	

222.4. Examples

222.4.1. Get accesspoints

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accesspoint&action=view
```

RESPONSE

```
{
  "AccessPoints": [
    {
      "Token": "accesspointtoken1",
      "Name": "accesspoint1",
      "Enable": true,
      "Description": "",
      "Areas": [
        "areatoken1"
      ],
      "DoorType" : "Door",
      "DoorToken": "doortoken1",
      "Capabilities": {
        "DisableAccessPoint": true,
        "Duress": false,
        "AnonymousAccess": false,anonymous access
        "SupportedRecognitionTypes": ["Card", "PIN"]
```

```
      },
      "AuthenticationProfileToken": "authenticationprofiletoken1"
    }
  ]
}
```

222.4.2. Add accesspoint

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accesspoint&action=add&Name=accesspointname1&
DoorToken=doortoken1&DisableAccessPoint=True&Duress=False&AnonymousAccess=Fa
lse&SupportedRecognitionTypes=Card&AuthenticationProfileToken=authentication
profiletoken1
```

RESPONSE

```
{
    "Token": "accesspointtoken1"
}
```

222.4.3. Update accesspoint

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accesspoint&action=update&Token=accesspointto
ken1&SupportedRecognitionTypes=Card,PIN&Areas=areatoken1,areatoken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

222.4.4. Remove accesspoints

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accesspoint&action=remove&Token=accesspointto
```

```
ken1,accesspointtoken2
```

RESPONSE


```
{
    "Response": "Success"
}
```

Chapter 223. accessschedule

223.1. Description

The **accessschedule** submenu of **accesscontrol.cgi** is for managing the accessschedule.

> **NOTE** This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

223.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
accessschedule&action=view
```

223.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ,RES	<string>	add: The token of the created accessschedule will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	Description	REQ	<string>	
	Standard	REQ	<string>	iCalendar format * Must be transmitted as url-encoded
remove	Token	REQ	<string>	

223.4. Examples

223.4.1. Get accessschedules

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessschedule&action=view
```

RESPONSE


```
{
  "Schedules": [
    {
      "Token": "scheduletoken1",
      "Name": "weekdayschedule",
      "Enable": true,
      "Description": "Access on Weekdays from 09:00:00 to 18:00:00",
      "Standard": "BEGIN:VCALENDAR\r\nBEGIN:VEVENT\r\nSUMMARY:Access on
weekdays from 9 AM to 6 PM for
employees\r\nDTSTART:19700101T090000\r\nDTEND:
19700101T180000\r\nRRULE:FREQ=WEEKLY;BYDAY=MO,TU,WE,TH,FR\r\nEND:VEVENT\r\nE
ND:VCALENDAR"
    }
  ]
}
```


223.4.2. Add accessschedule

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessschedule&action=add&Name=accessschedule
name2&Standard=BEGIN%3AVCALENDAR%5Cr%5CnBEGIN%3AVEVENT%5Cr%5CnSUMMARY%3AAcce
ss%20on%20weekdays%20from%208%20AM%20to%205%20PM%20for%20employees%5Cr%5CnDT
START%3A19700101T080000%5Cr%5CnDTEND%3A%2019700101T170000%5Cr%5CnRRULE%3AFRE
Q%3DWEEKLY%3BBYDAY%3DMO%2CTU%2CWE%2CTH%2CFR%5Cr%5CnEND%3AVEVENT%5Cr%5CnEND%3
AVCALENDAR
```

RESPONSE


```
{
    "Token": "accessscheduletoken2"
}
```

223.4.3. Update accessschedule(remove specialdays group)

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessschedule&action=update&Token=accesssche
duletoken1&<parameter>=<value>
```

RESPONSE

```
{
    "Response": "Success"
}
```

223.4.4. Remove accessschedules

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessschedule&action=remove&Token=accesssche
duletoken1,accessscheduletoken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 224. accessprofile

224.1. Description

The **accessprofile** submenu of **accesscontrol.cgi** is for managing the accessprofile.

NOTE	This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

224.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
accessprofile&action=view
```

224.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ, RES	<string>	add: The token of the created accessprofile will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	Type	REQ	<enum> Grant Deny	
	Description	REQ	<string>	
	AccessPolicies.Index	REQ	<int>	
	AccessPolicies.#.ScheduleToken	REQ	<string>	

Action	Parameters	Request/ Response	Type/ Value	Description
	AccessPolicies.#.EntityToken	REQ	<string>	
	AccessPolicies.#.EntityType	REQ	<enum> AccessPoint	enum of the EntityTypes AccessPoint or other for future
remove	Token	REQ	<string>	

224.4. Examples

224.4.1. Get accessprofiles

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessprofile&action=view
```

RESPONSE


```
{
  "AccessProfiles": [
    {
      "Token": "accessprofiletoken1",
      "Name": "accessprofile1",
      "Enable": false,
      "Type": "Grant",
      "Description": "",
      "AccessPolicies": [
        {
          "Index": 1,
          "ScheduleToken": "scheduletoken1",
          "EntityToken": "accesspointtoken1",
          "EntityType": "AccessPoint"
        }
      ]
    }
  ]
}
```


224.4.2. Add accessprofile

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessprofile&action=add&Name=weekdayprofile&
AccessPolicies.1.ScheduleToken=weekdayschedule&AccessPolicies.1.EntityToken=
accesspointtoken1
```

RESPONSE

```
{
    "Token": "accessprofiletoken2"
}
```

224.4.3. Update accessprofile

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessprofile&action=update&Token=weekdayprof
ile&Description=for%20employees
```

RESPONSE

```
{
    "Response": "Success"
}
```

224.4.4. Remove accessprofiles

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessprofile&action=remove&Token=weekdayprof
ile
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 225. accessgroup

225.1. Description

The **accessgroup** submenu of **accesscontrol.cgi** is for managing the accessgroup.

NOTE	This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

225.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
accessgroup&action=view
```

225.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
	AccessRules.Token	RES	<string>	
	AccessRules.Name	RES	<string>	
	AccessRules.ImagePath	RES	<string>	
	AccessRules.AccessLevel	RES	<string>	
	AccessRules.Enable	RES	<bool>	
add/update	Token	REQ,RES	<string>	add: The token of the created accessgroup will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	Description	REQ	<string>	
	AccessProfiles	REQ	<csv>	csv list of the AccessProfiles

Action	Parameters	Request/ Response	Type/ Value	Description
remove	Token	REQ	<string>	

225.4. Examples

225.4.1. Get accessgroups

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessgroup&action=view
```

RESPONSE

```
{
  "AccessGroups": [
    {
      "Token": "accessgrouptoken1",
      "Name": "accessgroup1",
      "Enable": false,
      "Description": "",
      "AccessProfiles": [
          "profiletoken1",
          "profiletoken2"
      ]
    }
  ]
}
```

225.4.2. Add accessgroup

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessgroup&action=add&Name=accessgroup2&Acce
ssProfiles=profiletoken1,profiletoken2
```

RESPONSE

```
{
    "Token": "accessgrouptoken2"
```

```
}
```

225.4.3. Update accessgroup

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessgroup&action=update&Token=accessgroupto
ken2&AccessProfiles=profiletoken1,profiletoken2,profiletoken3,profiletoken4,
profiletoken5
```

RESPONSE

```
{
    "Response": "Success"
}
```

225.4.4. Remove accessgroups

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessgroup&action=remove&Token=accessgroupto
ken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 226. credentialholder

226.1. Description

The **credentialholder** submenu of **accesscontrol.cgi** is for managing the credentialholder.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

226.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
credentialholder&action=view
```

226.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
add/update	Token	REQ,RES	<string>	add: The token of the created credentialholder will be the response. update: Used only when requesting
	FirstName	REQ	<string>	
	MiddleName	REQ	<string>	
	LastName	REQ	<string>	
	Description	REQ	<string>	
	EmployeeID	REQ	<string>	
	ImagePath	RES	<string>	Readonly, Path to access registered images via credentialholderimage, blank if no image is registered
remove	Token	REQ	<string>	

226.4. Examples

226.4.1. Get credentialholders

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholder&action=view
```

RESPONSE


```
{
  "CredentialHolders": [
    {
      "Token": "credentialholdertoken1",
      "FirstName": "FirstName",
      "MiddleName": "MiddleName",
      "LastName": "LastName",
      "Description": "",
      "ImagePath": "/files/acs_image/credentialholdertoken1.png"
    }
  ]
}
```


226.4.2. Add credentialholder

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholder&action=add&FirstName=Name1&M
iddleName=Name2&LastName=Name3&Description=addcredentialholdertest
```

RESPONSE


```
{
    "Token": "credentialholdertoken2"
}
```


226.4.3. Update credentialholder

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/accesscontrol.cgi?msubmenu=credentialholder&action=update&Token=credenti
alholdertoken2&Description=updatecredentialholdertest
```

RESPONSE

```
{
    "Response": "Success"
}
```

226.4.4. Remove credentialholders

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholder&action=remove&Token=credenti
alholdertoken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 227. credentialholderimage

227.1. Description

The **credentialholderimage** submenu of **accesscontrol.cgi** is for managing the credentialholderimage.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
remove	Suser

227.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
credentialholderimage&action=view
```

227.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	CredentialHolderToken	REQ, RES	<string>	<CredentialHolderToken>: <CredentialHolderImagePath>
add	CredentialHolderToken	REQ,RES	<string>	Upload the credentialholder's image
remove	CredentialHolderToken	REQ	<string>	

227.4. Examples

227.4.1. Get credentialholderimages

You can check the image by accessing the path confirmed by Response.
Example. 192.168.0.10/files/acs_image/testholder1.png

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholderimage&action=view
```

RESPONSE

```
{
  "CredentialHolderImages": {
    "credentialholderimagetoken1":
"/files/acs_image/credentialholderimagetoken1.png",
    "credentialholderimagetoken2":
"/files/acs_image/credentialholderimagetoken2.png",
    "credentialholderimagetoken3":
"/files/acs_image/credentialholderimagetoken3.png"
  }
}
```

227.4.2. Add credentialholderimage

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholderimage&action=add&CredentialHo
lderToken=credentialholdertoken1

POST /stw-
cgi/accesscontrol.cgi?msubmenu=credentialholderimage&action=add&CredentialHo
lderToken=credentialholdertoken1 HTTP/1.1
Content-Length: <content length>

<image file content>
```

RESPONSE

```
{
    "Response": "Success"
}
```

227.4.3. Remove credentialholderimages

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialholderimage&action=remove&Credentia
lHolderToken=credentialholdertoken1
```

RESPONSE


```
{
    "Response": "Success"
}
```

Chapter 228. credentialmethod

228.1. Description

The **credentialmethod** submenu of **accesscontrol.cgi** is for managing the credentialmethod.

NOTE This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

228.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
credentialmethod&action=view
```

228.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
	Limit	REQ	<string>	Maximum number of entries to return. If not specified, less than one or higher than what the ACS Panel supports, the number of items is determined by the ACS Panel(Capability.CredentialMethod.MaxCount).
	StartToken	REQ	<string>	Start returning entries from this start token. If not specified, entries shall start from the beginning of the dataset.
	NextStartToken	RES	<string>	StartToken to use in next call to get the following items. If absent, no more items to get.

Action	Parameters	Request/ Response	Type/ Value	Description
add/update	Token	REQ,RES	<string>	add: The token of the created credentialmethod will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Enable	REQ	<bool>	
	NeverExpire	REQ	<bool>	
	ValidFrom	REQ	<string>	The start dateTime validity of the credentialmethod. format : "yyyy-MM-dd’T’HH:mm:ss"
	ValidTo	REQ	<string>	The expiration date/time validity of the credentialmethod. format : "yyyy-MM-dd’T’HH:mm:ss"
	FacilityCode	REQ	<string>	
	CredentialHolderToken	REQ	<string>	
	Format	REQ	<enum> 26BIT 37BIT_WITH_FAC 37BIT_WITHOUT_FAC 32BIT_MIFARE_CSN 34BIT_HID_FORMA 40BIT_W40_2	
	Type	REQ	<enum> Card, PIN, ...	
	IsValueEncrypted	REQ	<bool>	
	Value	REQ	<string>	The value of the identifier in hexadecimal representation

Action	Parameters	Request/ Response	Type/ Value	Description
	ExemptedFromAuthentication	REQ	<bool>	If set to true, this credential identifier is not considered for authentication. For example if the access point requests Card plus PIN, and the credential identifier of type PIN is exempted from authentication, then the access point will not prompt for the PIN
remove	Token	REQ	<string>	

228.4. Examples

228.4.1. Get credentialmethods

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialmethod&action=view
```

RESPONSE


```
{
  "CredentialMethods": [
    {
      "Token": "credentialmethodtoken1",
      "Name": "PIN1",
      "Enable": true,
      "NeverExpire": true,
      "ValidFrom": "2024-06-18T00:00:00",
      "ValidTo": "9999-12-31T23:59:59",
      "FacilityCode": "facility_code_1",
      "CredentialHolderToken": "",
      "Type": "PIN",
      "Format": "26BIT",
      "ExemptedFromAuthentication": false,
      "Value": "ecb489bec489bec48948a"
    }
  ]
}
```

228.4.2. Add credentialmethod

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialmethod&action=add&Name=credential_c
ard&Enable=True&Type=Card&Value=baef48eabf48aebf4
```

RESPONSE

```
{
    "Token": "credentialmethodtoken2"
}
```

228.4.3. Update credentialmethod

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialmethod&action=update&Token=credenti
almethodtoken2&Enable=False
```

RESPONSE

```
{
    "Response": "Success"
}
```

228.4.4. Remove credentialmethods

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=credentialmethod&action=remove&Token=credenti
almethodtoken2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 229. accessrule

229.1. Description

The **accessrule** submenu of **accesscontrol.cgi** is for managing the accessrule.

> **NOTE** This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
add	Suser
update	Suser
remove	Suser

229.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=accessrule&action=view
```

229.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Token	REQ, RES	<string>	
	Limit	REQ	<string>	Maximum number of entries to return. If not specified, less than one or higher than what the ACS Panel supports, the number of items is determined by the ACS Panel(Capability.Credential.MaxCount) .
	StartToken	REQ	<string>	Start returning entries from this start token. If not specified, entries shall start from the beginning of the dataset.
	NextStartToken	RES	<string>	StartToken to use in next call to get the following items. If absent, no more items to get.

Action	Parameters	Request/ Response	Type/ Value	Description
add/update	Token	REQ,RES	<string>	add: The token of the created accessrule will be the response. update: Used only when requesting
	Name	REQ	<string>	
	Description	REQ	<string>	
	CredentialHolderToken	REQ	<string>	CredentialHolder token from external system
	ValidFrom	REQ	<string>	The start dateTime validity of the accessrule format : "yyyy-MM-dd'T'HH:mm:ss"
	ValidTo	REQ	<string>	The expiration date/time validity of the accessrule format : "yyyy-MM-dd'T'HH:mm:ss"
	CredentialMethods	REQ	<csv>	csv list of the CredentialMethods
	NeverExpire	REQ	<bool>	
	Pin	REQ	<string>	token of the PIN type CredentialMethod
	AccessProfileTokens	REQ	<csv>	csv list of the AccessProfiles
	AccessGroupTokens	REQ	<csv>	csv list of the AccessGroups, can be empty
	ExtendedGrantTime	REQ	<bool>	indicating that the credential holder needs extra time to get through the door. ExtendedReleaseTime will be added to ReleaseTime, and ExtendedOpenTime will be added to OpenTime
	State.Enabled	REQ	<bool>	
	State.Reason	REQ	<string>	
	State.AntipassbackViolated	REQ	<bool>	Indicates if anti-passback is violated for the accessrule
remove	Token	REQ	<string>	

229.4. Examples

229.4.1. Get accessrules

REQUEST

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=accessrule&action=view
```

RESPONSE


```
{
  "AccessRules": [
    {
      "Token": "accressruletoken1",
      "Name": "AccessRule_1",
      "Description": "",
      "ValidFrom": "2009-10-10T12:00:00",
      "ValidTo": "2009-10-10T12:00:00",
      "CredentialHolderToken": "<token_from_external>",
      "CredentialMethods": [
        "Card1"
      ],
      "AccessProfileTokens": [
        "accessprofiletoken1"
      ],
      "AccessGroupTokens": [
        "accessgrouptoken1"
      ],
      "ExtendedGrantTime": false,
      "State": {
        "Enabled": true,
        "Reason": "",
        "AntipassbackViolated": false
      }
    }
  ]
}
```


229.4.2. Add accessrule

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessrule&action=add&Name=AccessRule_2&Crede
ntialHolderToken=holderfromexternalsystem&CredentialMethods=credentialmethod
token1,credentialmethodtoken2&AccessProfileTokens=accessprofiletoken1,access
```

```
profiletoken2,accessprofiletoken3,accessprofiletoken4
```

RESPONSE

```
{
    "Token": "accessruletoken2"
}
```

229.4.3. Update accessrule

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessrule&action=update&Token=accessruletoke
n2&ValidFrom=2024-01-01T00%3A00%3A00&ValidTo=2024-12-31T23%3A59%3A59
```

RESPONSE

```
{
    "Response": "Success"
}
```

229.4.4. Remove accessrules

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=accessrule&action=remove&Token=accessruletoke
n2
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 230. clientfilter

230.1. Description

The **clientfilter** submenu of **accesscontrol.cgi** is for managing the clientfilter.

> **NOTE** This chapter applies to ACS system only.

Access level

Action	ACS
view	Suser
set	Suser
add	Suser
update	Suser
remove	Suser

230.2. Syntax

```
http://<Device IP>/stw-cgi/accesscontrol.cgi?msubmenu=
clientfilter&action=view
```

230.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				
set	AccessType	REQ,RES	<enum> Allow	
	PartnerType	REQ,RES	<enum> WACS ONVIF	
	Enable	REQ,RES	<bool>	
add/update	Index	REQ,RES	<int>	
	IPType	REQ,RES	<enum> IPv4 IPv6	
	FilterEnable	REQ,RES	<bool>	
	Address	REQ,RES	<string>	IPv4Address or IPv6Address

Action	Parameters	Request/ Response	Type/ Value	Description
remove	Index	REQ,RES	<int>	

230.4. Examples

230.4.1. Get accessrules

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=clientfilter&action=view
```

RESPONSE

```
{
  "AccessType": "Allow",
  "PartnerType": "WACS",
  "Enable": true,
  "IPFilters": [
    {
      "Index": 1,
      "IPType": "IPv4",
      "FilterEnable": true,
      "Address": "123.123.123.123"
    }
  ]
}
```

230.4.2. Set clientfilter

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=clientfilter&action=set&AccessType=Allow&Part
nerType=WACS&Enable=True
```

RESPONSE

```
{
    "Response": "Success"
}
```

230.4.3. Add clientfilter

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=clientfilter&action=add&IPType=IPv4&Address=1
.1.1.1&FilterEnable=True
```

RESPONSE

```
{
    "Index": 1
}
```

230.4.4. Update clientfilter

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=clientfilter&action=update&Index=1&FilterEnab
le=False
```

RESPONSE

```
{
    "Response": "Success"
}
```

230.4.5. Remove accessrules

REQUEST

```
http://<Device IP>/stw-
cgi/accesscontrol.cgi?msubmenu=clientfilter&action=remove&Index=1
```

RESPONSE

```
{
    "Response": "Success"
}
```

Chapter 231. Annex A: Schedule Standard example

231.1. Access 24*7 for admin staff


```
{
  "Schedules": [
    {
      "Token": "scheduletoken1",
      "Name": "accessforadmin",
      "Enable": true,
      "Description": "Access 24*7 for admin staff",
      "Standard": "BEGIN:VCALENDAR
                   BEGIN:VEVENT
                   SUMMARY:Access 24*7
                   DTSTART:19700101T000000
                   DTEND:19700102T000000
                   RRULE:FREQ=WEEKLY;BYDAY=MO,TU,WE,TH,FR,SA,SU
                   END:VEVENT
                   END:VCALENDAR",
      "SpecialDays": []
    }
  ]
}
```


231.2. Access on Monday and Wednesday from 6 AM to 8 PM for cleaning staff


```
{
  "Schedules": [
    {
      "Token": "scheduletoken2",
      "Name": "accessforcleaningstaff",
      "Enable": true,
      "Description": "Access on Monday and Wednesday from 6 AM to 8 PM for
cleaning staff",
      "Standard": "BEGIN:VCALENDAR
                   BEGIN:VEVENT
                   SUMMARY:Access on Monday and Wednesday from 6 AM to 8 PM
```

```
                    DTSTART:19700105T060000
                    DTEND:19700105T200000
                    RRULE:FREQ=WEEKLY;BYDAY=MO,WE
                    END:VEVENT
                    END:VCALENDAR",
      "SpecialDays": []
    }
  ]
}
```

231.3. Access from Friday 6 PM to Monday 7 AM for maintenance staff


```
{
  "Schedules": [
    {
      "Token": "scheduletoken3",
      "Name": "accessformaintenancestaff",
      "Enable": true,
      "Description": "Access from Friday 6 PM to Monday 7 AM for maintenance
staff",
      "Standard": "BEGIN:VCALENDAR
                    BEGIN:VEVENT
                    SUMMARY:Access from Friday 6 PM to Monday 7 AM for
maintenance staff
                    DTSTART:20140523T180000
                    DTEND:20140526T070000
                    RRULE:FREQ=WEEKLY;BYDAY=FR
                    END:VEVENT
                    END:VCALENDAR",
      "SpecialDays": []
    }
  ]
}
```


231.4. Access on Weekdays from 8 AM to 5 PM for employees

```
{
```

```
  "Schedules": [
    {
      "Token": "scheduletoken4",
      "Name": "accessforemployees",
      "Enable": true,
      "Description": "Access on Weekdays from 8 AM to 5 PM for employees",
      "Standard": "BEGIN:VCALENDAR
                    BEGIN:VEVENT
                    SUMMARY:Access on weekdays from 8 AM to 5 PM for
employees
                    DTSTART:19700101T080000
                    DTEND: 19700101T170000
                    RRULE:FREQ=WEEKLY;BYDAY=MO,TU,WE,TH,FR
                    END:VEVENT
                    END:VCALENDAR",
      "SpecialDays": []
    }
  ]
}
```

231.5. Access from January 15, 2014, to January 14, 2015, from 9 AM to 6 PM

```
{
  "Schedules": [
    {
      "Token": "scheduletoken5",
      "Name": "accessforoneyear",
      "Enable": true,
      "Description": "Access from January 15, 2014, to January 14, 2015,
from 9 AM to 6 PM",
      "Standard": "BEGIN:VCALENDAR
                    BEGIN:VEVENT
                    SUMMARY:Access from Jan 15, 2014, to Jan 14, 2015, from
9 AM to 6 PM
                    DTSTART:20140115T090000
                    DTEND:20150114T180000
                    RRULE:FREQ=DAILY
                    END:VEVENT
                    END:VCALENDAR",
```

```
        "SpecialDays": []
      }
    ]
}
```

Chapter 232. Annex B: Extended iCalendar recurrence format

```
recur-rule-part =
        ( ( "YEARLY" )
        / ( "YEARLY" ";" "BYMONTH" "=" bymolist )
        / ( "MONTHLY" )
        / ( "MONTHLY" ";" "BYDAY" "=" bywdaylist )
        / ( "MONTHLY" ";" "BYMONTHDAY" "=" bymodaylist )
        / ( "WEEKLY" )
        / ( "WEEKLY" ";" "BYDAY" "=" (weekday *("," weekday)) )
        / ( "DAILY" )
        / ( "HOURLY" )
        / ( "MINUTELY" )
        / ( "SECONDLY" ) )
        [ ";" "INTERVAL" "=" 1*DIGIT ]
        [ ";" "COUNT" "=" 1*DIGIT / ";" "UNTIL" "=" enddate ]
```

Chapter 233. Annex C: Event Status

> **NOTE** Since these Event items are currently under development, detailed specifications may change.
> This chapter applies to ACS system only.

Supported Type

Type	Support
Text	false
Text Schema	true
Json	false
Json Schema	true

233.1. AccessGranted

Notification that Access has been approved with a valid credential or anonymous user.
This is not property event, will only be sent when state change occurs.

233.1.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
	IOPort	int	O	-
	ReaderIndex	int	O	Only for credential
Data	Type	<enum> Credential Anonymous	M	-
	External	bool	O	-
	AccessRuleToken	string	M	Only for credential
	CredentialHolderName	string	M	
	CredentialMethods	string	O	
	SecurityLevelToken	string	O	
	ExemptedAccess	bool	O	

233.1.2. Eventstatus sample

233.1.2.1. Json Schema

Credential

```
{
  "EventName": "AccessGranted",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1,
    "ReaderIndex": 1
  },
  "Data": {
    "Type": "Credential",
    "External": false,
    "AccessRuleToken": "accessruletoken1",
    "CredentialHolderName": "abcd",
    "CredentialMethods": "credentialmethod1 credentialmethod2",
    "SecurityLevelToken": "securityleveltoken1",
    "ExemptedAccess": false
  }
}
```


Anonymous


```
{
  "EventName": "AccessGranted",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1
  },
  "Data": {
    "Type": "Anonymous",
    "External": false
  }
}
```


233.1.2.2. Text Schema

Credential

```
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.Type=Credential
```

```
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.External=False
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.AccessRuleToken=accessruletoken1
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.CredentialHolderName=abcd
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.CredentialMethods=credentialmethod1 credentialmethod2
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.SecurityLevelToken=securityleveltoken1
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1
.ExemptedAccess=False
```

Anonymous

```
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.Type=Anonymou
s
AccessControl.AccessGranted.AccessPointToken.aptoken1.IOPort.1.External=Fals
e
```

233.2. AccessDenied

Notification that Access has been denied with a invalid credential or anonymous user.
This is not property event, will only be sent when state change occurs.

233.2.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
	IOPort	int	O	-
	ReaderIndex	int	O	Only for credential and CredentialNotFound

	Item	Type	M/O	Description
Data	Type	<enum> Credential Anonymous CredentialNotFoun d	M	-
	External	bool	O	Only for Credential and Anonymous
	AccessRuleToken	string	O	Only for Credential
	CredentialHolderNa me	string	O	
	SecurityLevelToken	string	O	
	Reason	<enum> CredentialNotEnabl ed CredentialNotActive CredentialExpired InvalidPIN NotPermittedAtThis Time Unauthorized Other	M	-
	IdentifierType	<enum> Card PIN	M	Only for Identifier
	FormatType	<enum> Not fixed yet	M	
	IdentifierValue	string	M	

233.2.2. Eventstatus sample

233.2.2.1. Json Schema

Credential

```
{
  "EventName": "AccessDenied",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1,
    "ReaderIndex": 1
```

```
  },
  "Data": {
    "Type": "Credential",
    "Reason": "CredentialExpired",
    "External": false,
    "AccessRuleToken": "accessruletoken1",
    "CredentialHolderName": "abcdewf",
    "SecurityLevelToken": "securityleveltoken1"
  }
}
```

Anonymous


```
{
  "EventName": "AccessDenied",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1
  },
  "Data": {
    "Type": "Anonymous",
    "Reason": "Other",
    "External": false,
  }
}
```


CredentialNotFound


```
{
  "EventName": "AccessDenied",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1,
    "ReaderIndex": 1
  },
  "Data": {
    "Type": "CredentialNotFound",
    "IdentifierType": "Card",
```

```
    "FormatType": "WIEGAND26",
    "IdentifierValue": "cab489acb48bc48"
  }
}
```

233.2.2.2. Text Schema

Credential

```
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
Type=Credential
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
Reason=CredentialExpired
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
External=False
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
AccessRuleToken=accessruletoken1
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
CredentialHolderName=abcdewf
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
SecurityLevelToken=securityleveltoken1
```

Anonymous

```
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.Type=Anonymous
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.Reason=Other
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.External=False
```

CredentialNotFound

```
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
Type=CredentialNotFound
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
IdentifierType=Card
AccessControl.AccessDenied.AccessPointToken.aptoken1.IOPort.1.ReaderIndex.1.
IdentifierValue=cab489acb48bc48
```

233.3. AccessTaken

Notification that Access is taken.
This is not property event, will only be sent when state change occurs.

233.3.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
Data	Type	<enum> Credential Anonymous Identifier	M	-
	AccessRuleToken	string	M	Only for Credential
	CredentialHolderName	string	O	Only for Credential
	IdentifierType	<enum> Card PIN	M	Only for Identifier
	FormatType	<enum> Not fixed yet	M	
	IdentifierValue	string	M	

233.3.2. Eventstatus sample

233.3.2.1. Json Schema

Credential

```
{
  "EventName": "AccessTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "Type": "Credential",
    "AccessRuleToken": "accessruletoken1",
    "CredentialHolderName": "credentialholdername1"
  }
}
```

Anonymous

```
{
  "EventName": "AccessTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
```

```
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "Type": "Anonymous"
  }
}
```


Identifier


```
{
  "EventName": "AccessTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "Type": "Identifier",
    "IdentifierType": "Card",
    "FormatType": "WIEGAND26",
    "IdentifierValue": "cab489acb48bc48"
  }
}
```


233.3.2.2. Text Schema

Credential

```
AccessControl.AccessTaken.AccessPointToken.aptoken1.Type=Credential
AccessControl.AccessTaken.AccessPointToken.aptoken1.AccessRuleToken=accessru
letoken1
AccessControl.AccessTaken.AccessPointToken.aptoken1.CredentialHolderName=cre
dentialholdername1
```

Anonymous

```
AccessControl.AccessTaken.AccessPointToken.aptoken1.Type=Anonymous
```

Identifier

```
AccessControl.AccessTaken.AccessPointToken.aptoken1.Type=Identifier
AccessControl.AccessTaken.AccessPointToken.aptoken1.IdentifierType=Card
AccessControl.AccessTaken.AccessPointToken.aptoken1.FormatType=WIEGAND26
AccessControl.AccessTaken.AccessPointToken.aptoken1.IdentifierValue=cab489ac
b48bc48
```

233.4. AccessNotTaken

Notification that Access is not taken in time.
This is not property event, will only be sent when state change occurs.

233.4.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
Data	Type	<enum> Credential Anonymous Identifier	M	-
	AccessRuleToken	string	M	Only for Credential
	CredentialHolderName	string	O	Only for Credential
	IdentifierType	<enum> Card PIN	M	Only for Identifier
	FormatType	<enum> Not fixed yet	M	
	IdentifierValue	string	M	

233.4.2. Eventstatus sample

233.4.2.1. Json Schema

Credential

```
{
  "EventName": "AccessNotTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
```

```
    "Type": "Credential",
    "AccessRuleToken": "accessruletoken1",
    "CredentialHolderName": "credentialholdername1"
  }
}
```

Anonymous

```
{
  "EventName": "AccessNotTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "Type": "Anonymous"
  }
}
```

Identifier

```
{
  "EventName": "AccessNotTaken",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "Type": "Identifier",
    "IdentifierType": "Card",
    "FormatType": "WIEGAND26",
    "IdentifierValue": "cab489acb48bc48"
  }
}
```

233.4.2.2. Text Schema

Credential

```
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.Type=Credential
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.AccessRuleToken=acces
```

```
sruletoken1
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.CredentialHolderName=
credentialholdername1
```

Anonymous

```
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.Type=Anonymous
```

Identifier

```
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.Type=Identifier
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.IdentifierType=Card
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.FormatType=WIEGAND26
AccessControl.AccessNotTaken.AccessPointToken.aptoken1.IdentifierValue=cab48
9acb48bc48
```

233.5. ConfigChanged

Notification that the configuration has changed (including removals).
This is not a property event; it will only be sent when a state change occurs.

233.5.1. Source & Data

	Item	Type	M/O	Description
Source	Type	<enum> AccessPoint Door Area AccessProfile AccessRule Schedule SpecialDays	M	-
	Token	string	M	-
Data	Action	<enum> Changed Removed	M	-

233.5.2. Eventstatus sample

233.5.2.1. Json Schema

Changed

```
{
  "EventName": "ConfigChanged",
```

```
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "Type": "AccessProfile",
    "Token": "accessprofiletoken1"
 },
  "Data": {
    "Action": "Changed"
  }
}
```

Removed

```
{
  "EventName": "ConfigChanged",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "Type": "Schedule",
    "Token": "scheduletoken1"
  },
  "Data": {
    "Action": "Removed"
  }
}
```

233.5.2.2. Text Schema

Changed

```
AccessControl.ConfigChanged.Type.AccessRule.Token.accessruletoken1.Action=Changed
```

Removed

```
AccessControl.ConfigChanged.Type.Door.Token.doortoken1.Action=Removed
```

233.6. AccessPointEnabled

Notification that the AccessPoint state has changed (enabled or disabled).
This is not a property event; it will only be sent when a state change occurs.

233.6.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
Data	State	bool	M	-

233.6.2. Eventstatus sample

233.6.2.1. Json Schema


```
{
  "EventName": "AccessPointEnabled",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1"
  },
  "Data": {
    "State": true
  }
}
```


233.6.2.2. Text Schema

```
AccessControl.AccessPointEnabled.Token.aptoken1.State=True
```

233.7. AccessRuleEnabled

Notification that AccessRule state has been changed(enabled or disabled).
This is not a property event; it will only be sent when a state change occurs.

233.7.1. Source & Data

	Item	Type	M/O	Description
Source	AccessRuleToken	string	M	-

	Item	Type	M/O	Description
Data	State	bool	M	-
	Reason	<enum> CredentialLockedOut CredentialBlocked CredentialLost CredentialStolen CredentialDamaged CredentialDestroyed CredentialInactivity CredentialExpired CredentialRenewalNeeded	M	-
	ClientUpdated	bool	M	-

233.7.2. Eventstatus sample

233.7.2.1. Json Schema

```
{
  "EventName": "AccessRuleEnabled",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessRuleToken": "accessruletoken1"
  },
  "Data": {
    "State": true,
    "Reason": "",
    "ClientUpdated": false
  }
}
```

233.7.2.2. Text Schema

```
AccessControl.AccessRuleEnabled.Token.accessruletoken1.State=False
AccessControl.AccessRuleEnabled.Token.accessruletoken1.Reason=CredentialBloc
ked
AccessControl.AccessRuleEnabled.Token.accessruletoken1.ClientUpdated=False
```

233.8. DoorMode

Monitors the mode of the door.

233.8.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	Mode	<enum> Unknown Locked Unlocked Accessed LockedDown DoubleLocked	M	Indicates the logical operating mode of the door.

233.8.2. Eventstatus sample

233.8.2.1. Json Schema


```
{
  "EventName": "AccessControl.DoorMode",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "Mode": "Locked"
  }
}
```


233.8.2.2. Text Schema

```
AccessControl.DoorMode.DoorToken.doortoken1.Mode=Locked
```

233.9. DoorAlarmState

Monitor the AlarmState of the door.

233.9.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-

	Item	Type	M/O	Description
Data	Status	<enum> Normal DoorForcedOpen DoorOpenTooLong	M	-
	Reason	string	O	-

233.9.2. Eventstatus sample

233.9.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoorAlarmState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "Status": "Normal"
  }
}
```

233.9.2.2. Text Schema

```
AccessControl.DoorAlarmState.DoorToken.doortoken1.Status=Locked
```

233.10. DoorFaultState

Monitor the FaultState of the door.

233.10.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	Status	<enum> Unknown NotInFault FaultDetected	M	-
	Reason	string	O	-

233.10.2. Eventstatus sample

233.10.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoorFaultState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "Status": "NotInFault",
    "Reason": ""
  }
}
```

233.10.2.2. Text Schema

```
AccessControl.DoorFaultState.DoorToken.doortoken1.Status=NotInFault
AccessControl.DoorFaultState.DoorToken.doortoken1.Reason=
```

233.11. DoorPhysicalState

Monitor the PhysicalState of the door.

233.11.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	DoorPhysicalState	<enum> Unknown Open Closed Fault	M	-

233.11.2. Eventstatus sample

233.11.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoorPhysicalState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
```

```
    "DoorToken": "doortoken1"
 },
  "Data": {
    "DoorPhysicalState": "Closed"
  }
}
```

233.11.2.2. Text Schema

```
AccessControl.DoorPhysicalState.DoorToken.doortoken1.DoorPhysicalState=Close
d
```

233.12. DoorTamperState

Monitor the TamperState of the door.

233.12.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	Status	<enum> Unknown NotInTamper TamperDetected	M	-

233.12.2. Eventstatus sample

233.12.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoorTamperState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "Status": "NotInTamper"
  }
}
```

233.12.2.2. Text Schema

```
AccessControl.DoorTamperState.DoorToken.doortoken1.Status=NotInTamper
```

233.13. LockPhysicalState

Monitor the LockPhysicalState of the door.

233.13.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	LockPhysicalState	<enum> Unknown Locked Unlocked Fault	M	-

233.13.2. Eventstatus sample

233.13.2.1. Json Schema


```
{
  "EventName": "AccessControl.LockPhysicalState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "LockPhysicalState": "Locked"
  }
}
```


233.13.2.2. Text Schema

```
AccessControl.LockPhysicalState.DoorToken.doortoken1.LockPhysicalState=Locke
d
```

233.14. DoubleLockPhysicalState

Monitor the DoubleLockPhysicalState of the door.

233.14.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	DoubleLockPhysical State	<enum> Unknown Locked Unlocked Fault	M	-

233.14.2. Eventstatus sample

233.14.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoubleLockPhysicalState",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "DoorToken": "doortoken1"
 },
  "Data": {
    "DoubleLockPhysicalState": "Unlocked",
  }
}
```

233.14.2.2. Text Schema

```
AccessControl.DoubleLockPhysicalState.DoorToken.doortoken1.DoubleLockPhysica
lState=Unlocked
```

233.15. CredentialReaderState

Monitor the CredentialReaderState of the CredentialReader.

233.15.1. Source & Data

	Item	Type	M/O	Description
Source	IOPort	int	M	-
	ReaderIndex	int	M	-

	Item	Type	M/O	Description
Data	State	<enum> Active Trouble Offline Disable	M	-

233.15.2. Eventstatus sample

233.15.2.1. Json Schema


```
{
  "EventName": "AccessControl.CredentialReaderState",
  "Time": "2025-02-26T05:15:07.196+00:00",
  "Source": {
      "IOPort": 1,
      "ReaderIndex": 0
  },
  "Data": {
      "State": "Offline"
  }
}
```


233.15.2.2. Text Schema

```
AccessControl.CredentialReaderState.IOPort.1.ReaderIndex.0.State=Offline
```

233.16. DoorSecurityLevel

Monitor the DoorSecurityLevel of the CredentialReader.

233.16.1. Source & Data

	Item	Type	M/O	Description
Source	DoorToken	string	M	-
Data	EnabledSecurityLevelTokenIn	string	M	-
	EnabledSecurityLevelTokenOut	string	M	-

233.16.2. Eventstatus sample

233.16.2.1. Json Schema

```
{
  "EventName": "AccessControl.DoorSecurityLevel",
  "Time": "2025-02-26T05:35:31.155+00:00",
  "Source": {
      "DoorToken": "1"
  },
  "Data": {
      "EnabledSecurityLevelTokenIn": "2",
      "EnabledSecurityLevelTokenOut": "2"
  }
}
```

233.16.2.2. Text Schema

```
AccessControl.DoorSecurityLevel.DoorToken.1.EnabledSecurityLevelTokenIn=2
AccessControl.DoorSecurityLevel.DoorToken.1.EnabledSecurityLevelTokenOut=2
```

233.17. REXActivated

Notifies changes in the active/normal state of the REX assigned to each door.
This is not property event, will only be sent when state change occurs.

233.17.1. Source & Data

	Item	Type	M/O	Description
Source	AccessPointToken	string	M	-
	IOPort	int	M	-
Data	State	bool	M	-

233.17.2. Eventstatus sample

233.17.2.1. Json Schema

```
{
  "EventName": "REXActivated",
  "Time": "2024-03-27T08:39:13.298+09:00",
  "Source": {
    "AccessPointToken": "aptoken1",
    "IOPort": 1
  },
```

```
  "Data": {
    "State": false
  }
}
```

Bypass

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Chapter 234. Overview

234.1. Description

bypass.cgi is used to send SUNAPI commands to the connected cameras (registered using the SUNAPI protocol only) via the NVR, and to receive the corresponding responses.

NOTE	This chapter applies to NVR only.

Chapter 235. bypass

235.1. Description

The **bypass** submenu in NVR bypasses the commands sent by SUNAPI client applications directly to the camera and returns the response from the camera.

> **NOTE**
> This submenu supports all SUNAPI commands except for multi-part response commands, such as intermediate events during firmware updates, and the monitor and monitordiff actions in eventstatus.cgi.
> For the Firmware update command, only the final response will be sent to the requesting client.
> For other commands, the response sent by the camera will be returned as the response to the bypass command.

Access level

Action	NVR
control	User

235.2. Syntax

```
http://<Device IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action==<value>[&<parameter>=<value>]
```

235.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
control	Channel	REQ	<int>	Channel number
	HeadersToSend	REQ	<csv>	Extra headers to be sent to camera such as cookies, response format etc.
	Method	REQ	<enum> GET, POST, PUT, DELETE	Parameter that specifies the HTTP method of the request. It is mandatory for PUT and DELETE methods, as these methods are blocked in the recorder. When using PUT or DELETE, the client must use POST to send the request body content.
	BypassURI	REQ	<string>	SUNAPI command to be sent to camera

235.4. Examples

235.4.1. Getting camera attributes

REQUEST

```
http://<Device IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&BypassURI=/stw-
cgi/attributes.cgi/eventsources/videoanalysis/view/DetectionType
```

TEXT RESPONSE

```
<parameter response="false" request="true" name="DetectionType">
    <dataType>
        <enum>
            <entry value="Off"/>
            <entry value="MotionDetection"/>
            <entry value="IntelligentVideo"/>
            <entry value="MDAndIV"/>
        </enum>
    </dataType>
</parameter>
```

235.4.2. Getting camera events in text format

REQUEST

```
http://<Device IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&BypassURI=/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.MotionDetection=True
Channel.0.MotionDetection.RegionID.1=True
Channel.0.MotionDetection.RegionID.2=False
Channel.0.MotionDetection.RegionID.3=False
Channel.0.MotionDetection.RegionID.4=False
```

```
Channel.0.Tampering=False
Channel.0.DefocusDetection=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappering=False
AlarmInput.1=False
AlarmOutput.1=False
```

235.4.3. Getting camera events in JSON format

REQUEST

```
http://<Device IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&
HeadersToSend=Accept:application/json&BypassURI=/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": true,
            "MotionDetectionRegions": {
                "1": true,
                "2": false,
                "3": false,
                "4": false
            },
            "Tampering": false,
            "DefocusDetection": false,
            "AudioDetection": false,
            "VideoAnalytics": {
```

```
                "Passing": false,
                "Entering": false,
                "Exiting": false,
                "Appearing": false,
                "Disappering": false
            }
        }
    ],
    "AlarmInput": {
        "1": false
    },
    "AlarmOutput": {
        "1": false
    }
}
```

NOTE HeadersToSend is optional; it is used to send special headers to the camera. "Accept:application/json" header can also be sent to the NVR to receive the response from the camera in JSON format.

235.4.4. Configuration backup

REQUEST

```
curl --digest -u <userid>:<password> "http://<Device IP>/stw-cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&BypassURI=/stw-cgi/system.cgi?msubmenu=configbackup&action=control" > config.bin
```

235.4.5. Configuration restore

Encode to base64

```
openssl base64 -in config.bin -out encoded.bin
```

REQUEST

```
curl --digest -u <userid>:<password> "http://<Device IP>/stw-cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&BypassURI=/stw-cgi/system.cgi?msubmenu=configrestore&action=control&ExcludeSettings=Network,Camera" -H "Expect:" --data-urlencode @encoded.bin
```

235.4.6. Firmware update

REQUEST

```
curl --digest -u <userid>:<password> "http://<Device IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=0&BypassURI=/stw-
cgi/system.cgi?msubmenu=firmwareupdate&action=control&Type=Normal" -H
"Expect:" -F uploadFile=@srn-4000-pkg_v2.00_150114103354.img
```

Open SDK

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 236. Overview

236.1. Description

opensdk.cgi is used to install and manage applications.

The following submenus are used for Open SDK functionalities:

- **apps**: Requests and configures general application settings. The **apps** submenu is also used to install and remove applications.
- **appstatus**: Requests the application status such as memory and CPU usage, either once or periodically.
- **manifest**: Requests the application manifest file.
- **debug**: Requests to debug an Open SDK application using ‘RemoteDebugViewer’.
- **opensdkeventinfo**: Reads the event schema from the camera’s third party application.
- **metaframeschema**: Notifies the metaframe schema supported by an app.
- **metaframecapability**: Provides all supported values and ranges of metadata parameters.

NOTE

This document applies to network cameras only.
Attribute to check for feature support: "attributes/System/Support/OpenSDK"
For multi-directional cameras, please refer to the value "attributes/System/Support/OneOpenAppPerChannel". When this value is set to true, the application can be installed on any one channel.

Chapter 237. Applications

237.1. Description

The **apps** submenu requests, configures, and controls application settings. It is also used to install and remove applications.

NOTE Attribute to check maximum applications: " **attributes/system/Limit/OpenSDK.MaxApps**"

NOTE Some models might support app installing on each channel separately. In this case, the device manages each channel's app by assigning it a unique AppID. Users can find the original name of the installed app by referencing the newly added **AppName** parameter.

Access level

Action	Camera
view	Suser
set	Admin
control	Admin
install	Admin
remove	Admin

237.2. Syntax

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=<value>[&<parameter>=<value>...]
```

237.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the application settings
	InstalledApps	RES	<int>	The number of currently installed applications

Action	Parameters	Request/ Response	Type/ Value	Description
	<AppID>.Status	RES	<enum> UnInstalling,Installed,Installing,StartedNotRunning,Running,Stopped,Starting,Stopping	Status of the application
	<AppID>.InstalledDate	RES	<string>	Installation date of the application The date is specified in the format of <ddd MM DD hh:mm:ss YYYY> (Local time).
	<AppID>.Version	RES	<string>	Application version
	<AppID>.Permission	RES	<csv> Device, PTZ, Network, SDCard, None	Application permission is a read-only parameter. This information is retrieved from the manifest XML of the application.
	<AppID>.AutoStart	RES	<bool> True, False	Enables or disables automatic startup of the application
	<AppID>.Priority	RES	<enum> Low, Medium, High	Priority
	<AppID>.Channel	RES	<csv>	The channel index where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel

Action	Parameters	Request/ Response	Type/ Value	Description
	<AppID>.IsDefault	RES	<bool> True, False	Shows whether this app is the Default app **Note** Default apps will not be uninstalled when a factory reset is performed, and the app's Priority setting will automatically be set to High and frozen
	ControlForbidden	RES	<csv> StartStop,Priority,AutoStart	• StartStop: This means app will not be controled by "control" action "Mode" parameter Start/Stop. Always auto start. • Priority: This means Priority setting is forbidden. • AutoStart: This means AutoStart setting is forbidden.
	<AppID>.AppName	RES	<string>	Original Name of an app installed. If a device supports installing same app for each channel separately, AppID will vary for each channel but AppName will be same for all channels. **Note** This parameter is meaningful only when attributes/system/support/ChannelBasedOpenApp is True
	<AppID>.ModelName	RES	<string>	Model names for features supported by the app.
	<AppID>.ChannelType	RES	<enum> Single, Multiple	If a device supports distinguising whether a particular app works on all channels or only on one channel, it would inform users through this parameter.
set	AppID	REQ	<string>	Application ID **Note** **AppID** must be sent together with the **set** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	AutoStart	REQ	<bool> True, False	Enables or disables the application to automatically start
	Priority	REQ	<enum> Low, Medium, High	Priority
	Channel	REQ	<int>	Channel ID where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel
	IsDefault	REQ	<bool> True, False	Whether this app is the default app **Note** Default apps will not be uninstalled when a factory reset is performed, and the app's Priority setting will automatically be set to High and frozen.
	ActivatedChannel	REQ	<csv>	Users can set the channel what they want to activate for the app. User can select only one channel in case of single channel app, but choose multiple channels in case of multiple type.
control	AppID	REQ	<string>	Application ID **Note** **AppID** must be sent together with the **control** action.
	Mode	REQ	<enum> Start, Stop	Mode

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel	REQ	<int>	Channel ID where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel
install	AppID	REQ	<string>	Application ID **Note** **AppID** must be sent together with the **install** action.
	Permission	RES	<csv>	Permission
	InstallType	RES	<enum> New, Upgrade	Installation type
	IgnoreCookie	REQ	<bool> True, False	Ignore Application Session ID as a cookie
	ApplicationSessionId	REQ, RES	<string>	Application Session ID if cookie is not used
	KeepOldSettings	REQ	<bool> True, False	Keeps the old settings **Note** **AppID** and **KeepOldSettings** must be sent together for the **install** action if **InstallType**parameter is NOT set to New.
	Channel	REQ	<int>	Channel ID where the application will be installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel
remove	AppID	REQ	<string>	Application ID **Note** **AppID** must be sent together with the **remove** action.

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel	REQ	<int>	Channel ID where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel
update	AppID	REQ	<string>	It can upload any files to openapp. Ex.) AI binary data, application config json file, jpg, ...

237.4. Examples

We offer two types of devices: one where a single app manages multiple channels, and another where each channel has a dedicated app.
* **AppID**: This is unique to each instance of the app.
* **AppName**: Unlike AppID, it does not need to be unique. For models that do not support AppName, the AppID can be used as a substitute for AppName.

237.4.1. Getting the currently installed apps

237.4.1.1. Single App Managing Multiple Channels Case

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=apps&action=view
```

TEXT RESPONSE

```
InstalledApps=1
WiseAI.Status=Running
WiseAI.InstalledDate=Thu Mar 23 07:07:15 2023
WiseAI.Version=1.02.00
WiseAI.Permission=Device
WiseAI.AutoStart=True
WiseAI.Priority=High
WiseAI.ControlForbidden=
WiseAI.IsDefault=True
```

JSON RESPONSE

```
{
```

```
    "InstalledApps": 1,
    "Apps": [
        {
            "AppID": "WiseAI",
            "Status": "Running",
            "InstalledDate": "Thu Mar 23 07:07:15 2023",
            "Version": "1.02.00",
            "Permission": [
                "Device"
            ],
            "AutoStart": true,
            "Priority": "High",
            "IsDefault": true,
            "ControlForbidden": []
        }
    ]
}
```

237.4.1.2. Channel-Specific Apps Case

In this case you can notice that the AppName is same, whereas the AppID is different for each channel.

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=apps&action=view
```

TEXT RESPONSE

```
InstalledApps=1
WiseAI_C0.InstalledDate=Thu Mar 23 07:07:15 2023
WiseAI_C0.Verion=1.0
WiseAI_C0.Permission=SD,Network
WiseAI_C0.Status=Stopped
WiseAI_C0.AutoStart=False
WiseAI_C0.Priority=High
WiseAI_C0.Channel=0
WiseAI_C0.AppName=WiseAI
WiseAI_C1.InstalledDate=Thu Mar 23 07:07:15 2023
WiseAI_C1.Verion=1.0
WiseAI_C1.Permission=SD,Network
WiseAI_C1.Status=Stopped
WiseAI_C1.AutoStart=False
WiseAI_C1.Priority=High
```

```
WiseAI_C1.Channel=0
WiseAI_C1.AppName=WiseAI
```

JSON RESPONSE


```
{
    "InstalledApps": 1,
    "Apps": [
        {
            "AppID": "WiseAI_C0",
            "Status": "Stopped",
            "InstalledDate": "Thu Mar 23 07:07:15 2023",
            "Version": "1.0",
            "Permission": [
                "Sd",
                "Network"
            ],
            "AutoStart": false,
            "Priority": "High",
            "Channel": "0",
            "AppName": "WiseAI"
        },
        {
            "AppID": "WiseAI_C1",
            "Status": "Stopped",
            "InstalledDate": "Thu Mar 23 07:07:15 2023",
            "Version": "1.0",
            "Permission": [
                "Sd",
                "Network"
            ],
            "AutoStart": false,
            "Priority": "High",
            "Channel": "1",
            "AppName": "WiseAI"
        }
    ]
}
```

237.4.2. Installing a new application with CURL

The OpenSDK application installation is a two-step process. First, the application file needs to be sent to the camera via HTTP POST. The camera sends a session id cookie, installation type and the permissions required by the application. Then, based on the installation type and required permissions, a user can decide whether to install the application or not by sending the install command via HTTP GET.

AppID should be same as the application cap file name (without extension).

NOTE To get JSON response add the -H "Accept: application/json" header to the request.

Without using cookies:

Step 1: CURL command for sending the application to the camera

REQUEST

```
curl -v --digest -u <userid>:<password> -F UploadedFile=@ServerPushMJPEG.cap
"http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
&IgnoreCookie=True" -H "Expect:"
```

The above command will produce a request to the device as below:

```
POST /stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
HTTP/1.1
User-Agent: curl/7.26.0
Host: 111.111.11.11
Accept: */*
Content-Length: 119523
Content-Type: multipart/form-data; boundary=----------------------------
fb674236d482
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Content-Type: text/plain
Content-Length: 32
Date: Thu, 20 Mar 2014 01:32:12 GMT
Server: lighttpd/1.4.31
<Body>
```

```
ApplicationSessionId=ServerPushMJPEG-111.111.11.111
Permission=SD,Network
InstallType=New
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Content-Type: application/json
Content-Length: 32
Date: Thu, 20 Mar 2014 01:32:12 GMT
Server: lighttpd/1.4.31
<Body>
```


```
{
"ApplicationSessionId": " ServerPushMJPEG-111.111.11.111",
"Permission": ["SD","Network"],
"InstallType": "New",
}
```


Step 2:CURL command for installing the application

REQUEST

```
curl -v --digest -u <userid>:<password> "http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
&IgnoreCookie=True&ApplicationSessionId=ServerPushMJPEG-111.111.11.111"
```

The above command will produce a request to the device as below:

```
GET /stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
HTTP/1.1
User-Agent: curl/7.26.0
Host: 111.111.11.11
Accept: */*
TEXT RESPONSE
HTTP/1.1 200 OK
Content-Type: text/plain
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
```

```
Server: lighttpd/1.4.31
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: application/json
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd/1.4.31
<Body>
```

```
{
    "Response": "Success"
}
```

Using Cookies:

Step 1: CURL command for sending the application to the camera

REQUEST

```
curl -v --digest -u <userid>:<password> -F UploadedFile=@ServerPushMJPEG.cap
"http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
" -H "Expect:"
```

The above command will produce a request to the device as follows:

```
POST /stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
HTTP/1.1
User-Agent: curl/7.26.0
Host: 111.111.11.11
Accept: */*
Content-Length: 119523
```

```
Content-Type: multipart/form-data; boundary=---------------------------
fb674236d482
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=ServerPushMJPEG-111.111.11.111
Content-Type: text/plain
Content-Length: 32
Date: Thu, 20 Mar 2014 01:32:12 GMT
Server: lighttpd/1.4.31
<Body>
```

```
Permission=SD,Network
InstallType=New
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=ServerPushMJPEG-111.111.11.111
Content-Type: application/json
Content-Length: 32
Date: Thu, 20 Mar 2014 01:32:12 GMT
Server: lighttpd/1.4.31
<Body>
```

```
{
    "Response": "Success"
}
```

Step 2:CURL command for installing the application

REQUEST

```
curl -v --digest -u <userid>:<password> --cookie
AppInstallSessionID=ServerPushMJPEG-<Device IP> "http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
"
```

The above command will produce a request to the device as below:

```
GET /stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
HTTP/1.1
User-Agent: curl/7.26.0
Host: 111.111.11.11
Accept: */*
Cookie: AppInstallSessionID=ServerPushMJPEG-111.111.11.111
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: text/plain
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd/1.4.31
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: application/json
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd/1.4.31
<Body>
```

```
{
    "Response": "Success"
}
```

237.4.3. Updating the existing application with CURL

Step 1: CURL command for sending the application to the camera

REQUEST

```
curl -v --digest -u <userid>:<password> -F UploadedFile=@ServerPushMJPEG.cap
"http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
" -H "Expect:"
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=ServerPushMJPEG-111.111.11.111
Content-Type: text/plain
Content-Length: 32
Date: Thu, 20 Mar 2014 01:32:12 GMT
Server: lighttpd/1.4.31
<Body>
```

```
Permission=SD,Network
InstallType=Upgrade
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: application/json
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd
<Body>
```

```
{
    "Response": "Success"
}
```

Step 2: CURL command for updating the application

REQUEST

```
curl -v --digest -u <userid>:<password> --cookie AppInstallSessionID=<value>
```

```
"http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerPushMJPEG&Channel=0
&KeepOldSettings=True"
```

TEXT RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: text/plain
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd
<Body>
```

```
OK
```

JSON RESPONSE

```
HTTP/1.1 200 OK
Set-Cookie: AppInstallSessionID=deleted; expires=Wed, 20-Mar-2013 01:46:03
GMT
Content-Type: application/json
Content-Length: 2
Date: Thu, 20 Mar 2014 01:46:04 GMT
Server: lighttpd

<Body>
```

```
{
    "Response": "Success"
}
```

237.4.4. Installing license

REQUEST

```
curl -v --digest -u admin:<password> -F LicenseFile=@filename
"http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=install&AppID=ServerTest&Channel=0" -H
```

```
"Expect:"
```

237.4.5. Removing the installed application

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=remove&AppID=ServerPushMJPEG&Channel=0
```

237.4.6. Starting the application

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=control&AppID=ServerPushMJPEG&Mode=Star
t&Channel=0
```

237.4.7. Setting the application priority and enabling AutoStart

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=set&AppID=ServerPushMJPEG&Priority=Medi
um&AutoStart=True&Channel=0
```

237.4.8. Updating (Uploading) a datafile to openapp

REQUEST

Binary data

```
curl -v --digest -u admin:<password> -F DataFile=@{datafilename}
"http://<IP>/stw-cgi/opensdk.cgi?msubmenu=apps&action=update&AppID=SNTest"
-H "Expect:"
```

In case of not binary data, we need to put "octet-stream type" in data.

REQUEST

```
curl -v --digest -u admin:<password> -F
DataFile=@test.txt;type=application/octet-stream
"http://<IP>/stw-
cgi/opensdk.cgi?msubmenu=apps&action=update&AppID=test_Upload_File" -H
"Expect:"
```

Chapter 238. Application Status

238.1. Description

The **appstatus** submenu requests the status of the application.

Access level

Action	Camera
view	User

238.2. Syntax

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=appstatus&action=<value>[&<parameter>=<value>...]
```

238.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the application status
	AppID	REQ	<csv> <AppID>	Application ID If the specific App ID is not sent, the status of all applications is returned.
	Channel	REQ	<int>	Channel ID Used when the application status for a specific channel is needed. If **Channel** is not sent, all channels' application status is returned. **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel
	Check	REQ	<enum> Once, Periodically	Whether to checks the application status only once or periodically. If **Check** is not sent, the status of the application is returned to Once.

Action	Parameters	Request/ Response	Type/ Value	Description
	Periodicity	REQ	<int>	Interval for checking the application status If **Periodicity** is not sent, the default periodicity is applicable. The values must be within the range of 1 to 9 and the unit is a second.
	TotalCPUUsage	RES	<int>	Total CPU used The values must be within the range of 1 to 100.
	TotalMemoryUsage	RES	<int>	Memory totally used The values must be in the range of 1 to 100.
	<AppID>.CPUUsage	RES	<int>	CPU used in the corresponding application The values must be within the range of 1 to 100.
	<AppID>.MemoryUsage	RES	<int>	Memory used in the corresponding application The values must be within the range of 1 to 100.
	<AppID>.ThreadsCount	RES	<int>	Thread count of the corresponding application
	<AppID>.Duration	RES	<string>	Duration of the corresponding application Durations are represented by the format <P[n]Y[n]M[n]DT[n]H[n]M[n]S>, following the ISO 8601 duration format.

Action	Parameters	Request/ Response	Type/ Value	Description
	<AppID>.Channel	RES	<csv>	Channel ID where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel

238.4. Examples

238.4.1. Checking the application status once

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=appstatus&action=view&AppID=ANPR
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
TotalCPUUsage=40
TotalMemoryUsage=30
ANPR.CPUUsage=30
ANPR.MemoryUsage=10
ANPR.ThreadsCount=11
ANPR.Duration=P0Y0M0DT1H0M0S
ANPR.Channel=0
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
```

```
    "TotalCPUUsage": 40,
    "TotalMemoryUsage": 30,
    "Apps": [
        {
            "AppID": "ANPR",
            "CPUUsage": 30,
            "MemoryUsage": 10,
            "ThreadsCount": 11,
            "Duration": "P0Y0M0DT1H0M0S",
            "Channel": "0"
        }
    ]
}
```

238.4.2. Monitoring the application status of Channel 0 every 5 seconds

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=appstatus&action=view&Check=Periodically&Periodicity=5&Channel=0
```

TEXT RESPONSE

```
<Body>
```

```
--<MultiPartBoundary>
Content-Type: text/plain
TotalCPUUsage=40
TotalMemoryUsage=30
ANPR.CPUUsage=30
ANPR.MemoryUsage=10
ANPR.ThreadsCount=11
ANPR.Duration=P0Y0M0DT1H0M0S
ServerPushMJPEG.CPUUsage=10
ServerPushMJPEG.MemoryUsage=5
ServerPushMJPEG.ThreadsCount=9
ServerPushMJPEG.Duration=P0Y0M0DT2H10M0S

--<MultiPartBoundary>
Content-Type: text/plain
```

```
TotalCPUUsage=40
TotalMemoryUsage=30
ANPR.CPUUsage=20
ANPR.MemoryUsage=10
ANPR.ThreadsCount=11
ANPR.Duration=P0Y0M0DT1H10M0S
ServerPushMJPEG.CPUUsage=20
ServerPushMJPEG.MemoryUsage=5
ServerPushMJPEG.ThreadsCount=9
ServerPushMJPEG.Duration=P0Y0M0DT2H15M0S
```

JSON RESPONSE

```
<Body>
```

```
--<MultiPartBoundary>
Content-Type: application/json

{
    "TotalCPUUsage": 40,
    "TotalMemoryUsage": 30,
    "Apps": [
        {
            "AppID": "ANPR",
            "CPUUsage": 30,
            "MemoryUsage": 10,
            "ThreadsCount": 11,
            "Duration": "P0Y0M0DT1H0M0S"
        },
        {
            "AppID": "ServerPushMJPEG",
            "CPUUsage": 10,
            "MemoryUsage": 5,
            "ThreadsCount": 9,
            "Duration": "P0Y0M0DT2H10M0S"
        }
    ]
}

--<MultiPartBoundary>
```

```
Content-Type: application/json

{
    "TotalCPUUsage": 40,
    "TotalMemoryUsage": 30,
    "Apps": [
        {
            "AppID": "ANPR",
            "CPUUsage": 20,
            "MemoryUsage": 10,
            "ThreadsCount": 11,
            "Duration": "P0Y0M0DT1H10M0S"
        },
        {
            "AppID": " ServerPushMJPEG",
            "CPUUsage": 20,
            "MemoryUsage": 5,
            "ThreadsCount": 9,
            "Duration": "P0Y0M0DT2H15M0S"
        }
    ]
}
```

Chapter 239. Application Manifest

239.1. Description

The **manifest** submenu requests the application manifest XML file which contains detailed information of the application such as the application name, location, version, etc.

Access level

Action	Camera
view	User

239.2. Syntax

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=manifest&action=<value>[&<parameter>=<value>...]
```

239.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view				Reads the application manifest in the xml file format.
	AppID	REQ	<string>	Application ID **Note** **AppID** must be sent together with the **view** action.
	Channel	REQ	<int>	Channel ID where the application is installed **Note** To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel

239.4. Examples

239.4.1. Getting the application manifest file

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=manifest&action=view&AppID=ServerPushMJPEG&Channel=
0
```

RESPONSE

```
<Body>
```


```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<manifest>
    <appName>ServerPushMJPEG</appName>
    <appLocation>/home/user/workspace/ServerPushMJPEG</appLocation>
    <appVersion>1.0</appVersion>
    <minSDK>2.0</minSDK>
    <targetSDK>2.0</targetSDK>
    <maxSDK>2.0</maxSDK>
    <debug>false</debug>
    <vendor>Hanwha</vendor>
    <description/>
    <platform>
        <model>SNP6320</model>
        <videoEncoding>
            <codec>MJPEG</codec>
            <resolution>640 X 480</resolution>
            <frameRate>10</frameRate>
            <compression>2</compression>
            <bitRate>10240</bitRate>
            <audio>false</audio>
        </videoEncoding>
        <rawVideo>
            <format>YUV 400</format>
            <resolution>1920 X 1080</resolution>
            <frameRate>3</frameRate>
        </rawVideo>
    </platform>
    <permissions/>
    <appConfigData>
        <portNo>8080</portNo>
    </appConfigData>
```

```
</manifest>
```

Chapter 240. Application Debug

240.1. Description

The **debug** submenu requests to debug the application using the ‘RemoteDebugViewer’ program. Only one application can be debugged at a time. All applications can be debugged.

Access level

Action	Camera
set	Admin

240.2. Syntax

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=debug&action=<value>[&<parameter>=<value>...]
```

240.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
set	AppID	REQ, RES	<string>	Application ID **Note** **AppID**, **Port**, and **Enable** must be sent together.
	Port	REQ, RES	<int>	Camera’s port number **Note** **AppID**, **Port**, and **Enable** must be sent together.
	Enable	REQ, RES	<bool>	Enabling debugging **Note** **AppID**, **Port**, and **Enable** must be sent together.

Action	Parameters	Request/ Response	Type/ Value	Description
	Channel	REQ	<int>	Channel ID where the application is installed **Note** An optional parameter. If **Channel** is not sent, the first channel's application will be debugged. To use the Channel parameter, check attributes/system/support/OneOpenAppPerChannel

240.4. Examples

240.4.1. Setting the application to debug

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=debug&action=set&AppID=abc&Port=8080&Enable=True&Channel=0
```

Chapter 241. Application Event Information

241.1. Description

The **opensdkeventinfo** submenu provides the Open SDK application's event schema. Users can get event results from the Open SDK application using eventstatus.cgi. Please refer to the document regarding this event.

Access level

Action	Camera
view	Admin

241.2. Syntax

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=opensdkeventinfo&action=<value>[&<parameter>=<value>...]
```

241.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	AppName	RES	<string>	Application name
	AppEvent	RES	<string>	Application event name
	EventTopic	RES	<string>	Event topic name
	Type	RES	<enum> Event, Meta	It can be either an Event or Metadata When type is event, schema follows the onvif event schema format. When type is Metadata, metadata xml schema is provided. e.g. Like licenplate information etc.,
	EventSchema	REQ	<string>	Event schema **Note** This schema information is set with an open application.

241.4. Examples

241.4.1. Getting the event result format from installed opensdk applications

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=opensdkeventinfo&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "OpenSDKEventInfo": [
        {
            "AppName": "LicensePlateDetection",
            "AppEvent": "LicensePlateNumber",
            "Type": "Event",
            "EventTopic":
"tns1:OpenApp/LicensePlateDetection/LicensePlateNumber",
            "EventSchema":
"<tns1:OpenApp><LicensePlateDetection><LicensePlateNumber
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSourceToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"LicensePlateNumber\"
Type=\"xsd:string\"/></tt:Data></tt:MessageDescription></LicensePlateNumber>
</LicensePlateDetection></tns1:OpenApp>"
        },
        {
            "AppName": "VehicleDetection",
            "AppEvent": "VehicleDetected",
            "Type": "Event",
            "EventTopic": "tns1:OpenApp/VehicleDetection/VehicleDetected",
            "EventSchema": "<tns1:OpenApp><VehicleDetection><VehicleDetected
wstop:topic=\"true\"><tt:MessageDescription><tt:Source><tt:SimpleItemDescrip
tion Name=\"VideoSourceToken\"
Type=\"tt:ReferenceToken\"/></tt:Source><tt:Data><tt:SimpleItemDescription
Name=\"VehicleDetected\"
Type=\"xsd:boolean\"/></tt:Data></tt:MessageDescription></VehicleDetected></
```

```
VehicleDetection></tns1:OpenApp>"
        }
    ]
}
```

Chapter 242. Metaframe Schema

242.1. Description

The **metaframeschema** submenu, used to provide the schema of the frame metadata supported by an installed app.

Access level

Action	Camera
view	User

242.2. Syntax

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=metaframeschema&action=<value>[&<parameter>=<value>...]
```

242.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<int>	ChannelID, optional parameter in the request. If passed, the result will be filtered only for that channel.
	AppID	REQ, RES	<string>	Application ID
	Schema	RES	<string>	Frame metadata schema as base64 encoded string.
	Encoding	RES	<enum> base64	Used to notify the encoding format of schema.

242.4. Examples

242.4.1. Getting the schema of frame metadata supported by an app.

REQUEST

```
http://<Device IP>/stw-cgi/opensdk.cgi?msubmenu=metaframeschema&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: application/json
<Body>
```


```
{
    "MetaFrameSchema": [
        {
            "Channel": 0,
            "AppFrameSchema": [
                {
                    "AppID": "SampleAIAPP",
                    "Schema":
"PHhzOnNjaGVtYSBhdHRyaWJ1dGVGb3JtRGVmYXVsdD0idW5xdWFsaWZpZWQiIGVsZW1lbnRGb3J
tRGVmYXVsdD0icXVhbGlmaWVkIiB0YXJnZXROYW1lc3BhY2U9Imh0dHA6Ly93d3cub252aWYub3J
nL3ZlcjEwL3NjaGVtYSIgeG1sbnM6eHM9Imh0dHA6Ly93d3cudzMub3JnLzIwMDEvWE1MU2NoZW1
hIj48eHM6ZWxlbWVudCBuYW1lPSJNZXRhZGF0YVN0cmVhbSI+PHhzOmNvbXBsZXhUeXBlPjx4czp
zZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJWaWRlb0FuYWx5dGljcyI+PHhzOmNvbXBsZXhUeXB
lPjx4czpzZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJGcmFtZSI+PHhzOmNvbXBsZXhUeXBlPjx
4czpzZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJUcmFuc2Zvcm1hdGlvbiI+PHhzOmNvbXBsZXh
UeXBlPjx4czpzZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJUcmFuc2xhdGUiPjx4czpjb21wbGV
4VHlwZT48eHM6c2ltcGxlQ29udGVudD48eHM6ZXh0ZW5zaW9uIGJhc2U9InhzOnN0cmluZyI+PHh
zOmF0dHJpYnV0ZSB0eXBlPSJ4czpmbG9hdCIgbmFtZT0ieCIvPjx4czphdHRyaWJ1dGUgdHlwZT0
ieHM6ZmxvYXQiIG5hbWU9InkiLz48L3hzOmV4dGVuc2lvbj48L3hzOnNpbXBsZUNvbnRlbnQ+PC9
4czpjb21wbGV4VHlwZT48L3hzOmVsZW1lbnQ+PHhzOmVsZW1lbnQgbmFtZT0iU2NhbGUiPjx4czp
jb21wbGV4VHlwZT48eHM6c2ltcGxlQ29udGVudD48eHM6ZXh0ZW5zaW9uIGJhc2U9InhzOnN0cml
uZyI+PHhzOmF0dHJpYnV0ZSB0eXBlPSJ4czpmbG9hdCIgbmFtZT0ieCIvPjx4czphdHRyaWJ1dGU
gdHlwZT0ieHM6ZmxvYXQiIG5hbWU9InkiLz48L3hzOmV4dGVuc2lvbj48L3hzOnNpbXBsZUNvbnR
lbnQ+PC94czpjb21wbGV4VHlwZT48L3hzOmVsZW1lbnQ+PC94czpzZXF1ZW5jZT48L3hzOmNvbXB
sZXhUeXBlPjwveHM6ZWxlbWVudD48eHM6ZWxlbWVudCBuYW1lPSJPYmplY3QiPjx4czpjb21wbGV
4VHlwZT48eHM6c2VxdWVuY2U+PHhzOmVsZW1lbnQgbmFtZT0iQXBwZWFyYW5jZSI+PHhzOmNvbXB
sZXhUeXBlPjx4czpzZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJTaGFwZSI+PHhzOmNvbXBsZXh
UeXBlPjx4czpzZXF1ZW5jZT48eHM6ZWxlbWVudCBuYW1lPSJCb3VuZGluZ0JveCI+PHhzOmNvbXB
sZXhUeXBlPjx4czpzaW1wbGVDb250ZW50Pjx4czpleHRlbnNpb24gYmFzZT0ieHM6c3RyaW5nIj4
8eHM6YXR0cmlidXRlIHR5cGU9InhzOmZsb2F0IiBuYW1lPSJsZWZ0Ii8+PHhzOmF0dHJpYnV0ZSB
0eXBlPSJ4czpmbG9hdCIgbmFtZT0idG9wIi8+PHhzOmF0dHJpYnV0ZSB0eXBlPSJ4czpmbG9hdCI
gbmFtZT0icmlnaHQiLz48eHM6YXR0cmlidXRlIHR5cGU9InhzOmZsb2F0IiBuYW1lPSJib3R0b20
iLz48L3hzOmV4dGVuc2lvbj48L3hzOnNpbXBsZUNvbnRlbnQ+PC94czpjb21wbGV4VHlwZT48L3h
zOmVsZW1lbnQ+PHhzOmVsZW1lbnQgbmFtZT0iQ2VudGVyT2ZHcmF2aXR5Ij48eHM6Y29tcGxleFR
5cGU+PHhzOnNpbXBsZUNvbnRlbnQ+PHhzOmV4dGVuc2lvbiBiYXNlPSJ4czpzdHJpbmciPjx4czp
hdHRyaWJ1dGUgdHlwZT0ieHM6ZmxvYXQiIG5hbWU9IngiLz48eHM6YXR0cmlidXRlIHR5cGU9Inh
zOmZsb2F0IiBuYW1lPSJ5Ii8+PC94czpleHRlbnNpb24+PC94czpzaW1wbGVDb250ZW50PjwveHM
```

```
6Y29tcGxleFR5cGU+PC94czplbGVtZW50PjwveHM6c2VxdWVuY2U+PC94czpjb21wbGV4VHlwZT4
8L3hzOmVsZW1lbnQ+PHhzOmVsZW1lbnQgbmFtZT0iQ29sb3IiPjx4czpjb21wbGV4VHlwZT48eHM
6c2VxdWVuY2U+PHhzOmVsZW1lbnQgbmFtZT0iQ29sb3JDbHVzdGVyIj48eHM6Y29tcGxleFR5cGU
+PHhzOnNlcXVlbmNlPjx4czplbGVtZW50IG5hbWU9IkNvbG9yIj48eHM6Y29tcGxleFR5cGU+PHh
zOnNpbXBsZUNvbnRlbnQ+PHhzOmV4dGVuc2lvbiBiYXNlPSJ4czpzdHJpbmciPjx4czphdHRyaWJ
1dGUgdHlwZT0ieHM6Ynl0ZSIgbmFtZT0iWCIvPjx4czphdHRyaWJ1dGUgdHlwZT0ieHM6Ynl0ZSI
gbmFtZT0iWSIvPjx4czphdHRyaWJ1dGUgdHlwZT0ieHM6c2hvcnQiIG5hbWU9IloiLz48L3hzOmV
4dGVuc2lvbj48L3hzOnNpbXBsZUNvbnRlbnQ+PC94czpjb21wbGV4VHlwZT48L3hzOmVsZW1lbnQ
+PHhzOmVsZW1lbnQgbmFtZT0iQ292YXJpYW5jZSI+PHhzOmNvbXBsZXhUeXBlPjx4czpzaW1wbGV
Db250ZW50Pjx4czpleHRlbnNpb24gYmFzZT0ieHM6c3RyaW5nIj48eHM6YXR0cmlidXRlIHR5cGU
9InhzOmZsb2F0IiBuYW1lPSJYWCIvPjx4czphdHRyaWJ1dGUgdHlwZT0ieHM6Ynl0ZSIgbmFtZT0
iWVkiLz48eHM6YXR0cmlidXRlIHR5cGU9InhzOmJ5dGUiIG5hbWU9IlpaIi8+PC94czpleHRlbnN
pb24+PC94czpzaW1wbGVDb250ZW50PjwveHM6Y29tcGxleFR5cGU+PC94czplbGVtZW50Pjx4czp
lbGVtZW50IHR5cGU9InhzOmZsb2F0IiBuYW1lPSJXZWlnaHQiLz48L3hzOnNlcXVlbmNlPjwveHM
6Y29tcGxleFR5cGU+PC94czplbGVtZW50PjwveHM6c2VxdWVuY2U+PC94czpjb21wbGV4VHlwZT4
8L3hzOmVsZW1lbnQ+PHhzOmVsZW1lbnQgbmFtZT0iQ2xhc3MiPjx4czpjb21wbGV4VHlwZT48eHM
6c2VxdWVuY2U+PHhzOmVsZW1lbnQgbmFtZT0iVHlwZSI+PHhzOmNvbXBsZXhUeXBlPjx4czpzaW1
wbGVDb250ZW50Pjx4czpleHRlbnNpb24gYmFzZT0ieHM6c3RyaW5nIj48eHM6YXR0cmlidXRlIHR
5cGU9InhzOmZsb2F0IiBuYW1lPSJMaWtlbGlob29kIi8+PC94czpleHRlbnNpb24+PC94czpzaW1
wbGVDb250ZW50PjwveHM6Y29tcGxleFR5cGU+PC94czplbGVtZW50PjwveHM6c2VxdWVuY2U+PC9
4czpjb21wbGV4VHlwZT48L3hzOmVsZW1lbnQ+PC94czpzZXF1ZW5jZT48L3hzOmNvbXBsZXhUeXB
lPjwveHM6ZWxlbWVudD48L3hzOnNlcXVlbmNlPjx4czphdHRyaWJ1dGUgdHlwZT0ieHM6Ynl0ZSI
gbmFtZT0iT2JqZWN0SWQiLz48eHM6YXR0cmlidXRlIHR5cGU9InhzOmJ5dGUiIG5hbWU9IlBhcmV
udCIvPjwveHM6Y29tcGxleFR5cGU+PC94czplbGVtZW50PjwveHM6c2VxdWVuY2U+PHhzOmF0dHJ
pYnV0ZSB0eXBlPSJ4czpkYXRlVGltZSIgbmFtZT0iVXRjVGltZSIvPjwveHM6Y29tcGxleFR5cGU
+PC94czplbGVtZW50PjwveHM6c2VxdWVuY2U+PC94czpjb21wbGV4VHlwZT48L3hzOmVsZW1lbnQ
+PC94czpzZXF1ZW5jZT48L3hzOmNvbXBsZXhUeXBlPjwveHM6ZWxlbWVudD48L3hzOnNjaGVtYT4
=",
                    "Encoding": "base64"
                }
            ]
        }
    ]
}
```

Chapter 243. Metaframe Capability

243.1. Description

The **metaframecapability** submenu, used to provide all supported values of a metadata field. Therefore, the client can know what to expect based on this capability.

XPATH based capability notification mechanism is being used.

Providing the XPATH of the parameter and its data type and supported values or range.

If there are dependencies with another parameter, the dependency section is provided, for which XPATH affects the supported values of this parameter.

Please check the example section.

Access level

Action	Camera
view	User

243.2. Syntax

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=metaframecapability&action=<value>[&<parameter>=<va
lue>...]
```

243.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
view	Channel	REQ, RES	<int>	ChannelID, optional parameter in request. If passed, result will be filtered only for that channel.
	AppID	REQ, RES	<string>	Application ID
	Capabilities	RES	<string>	Capability Response

243.4. Examples

243.4.1. Getting the metaframe capability of the installed apps.

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/opensdk.cgi?msubmenu=metaframecapability&action=view
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "MetaFrameCapability": [
        {
            "Channel": 0,
            "AppCapabilities": [
                {
                    "AppID": "SampleAIApp",
                    "Capabilities": [
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:LicensePlateInfo/tt
:CountryCode",
                            "type": "xs:string",
                            "enum": [
                                "KR",
                                "US",
                                "CN",
                                "FN",
                                "IN"
                            ]
                        },
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:LicensePlateInfo/tt
:PlateType",
                            "type": "xs:string",
                            "enum": [
                                "Normal",
                                "Police",
                                "Diplomat",
                                "Temporary"
                            ]
                        },
```

```
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:Class/tt:Type",
                            "type": "xs:string",
                            "enum": [
                                "Vehicle",
                                "Bicycle"
                            ]
                        },
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Type
",
                            "type": "xs:string",
                            "enum": [
                                "Car",
                                "Bus"
                            ]
                        },
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Bran
d",
                            "type": "xs:string",
                            "enum": [
                                "Kia",
                                "Hyundai",
                                "Volvo"
                            ],
                            "Dependency": [
                                {
                                    "Condition":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Type
[text()='Car']",
                                    "enum": [
                                        "Kia",
                                        "Hyundai"
                                    ]
                                },
                                {
                                    "Condition":
```

```
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Type
[text()='Bus']",
                                    "enum": [
                                        "Volvo"
                                    ]
                                }
                            ]
                        },
                        {
                            "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Mode
l",
                            "type": "xs:string",
                            "enum": [
                                "K7",
                                "Sonata",
                                "Accent",
                                "Soul"
                            ],
                            "Dependency": [
                                {
                                    "Condition":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Type
[text()='Car'] and
//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Brand
[text()='Kia']",
                                    "enum": [
                                        "K7",
                                        "Soul"
                                    ]
                                },
                                {
                                    "Condition":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Type
[text()='Car'] and
//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:VehicleInfo/tt:Brand
[text()='Hyundai']",
                                    "enum": [
                                        "Sonata",
                                        "Accent"
                                    ]
```

```
                            }
                        ]
                    },
                    {
                        "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:Class/tt:Type/@Like
lihood",
                        "type": "xs:float",
                        "minimum": 0,
                        "maximum": 1
                    },
                    {
                        "xpath":
"//tt:VideoAnalytics/tt:Frame/tt:Object/tt:Appearance/tt:Color/tt:ColorClust
er/tt:ColorString",
                        "type": "xs:string",
                        "enum": [
                            "RED",
                            "BLUE",
                            "GREEN",
                            "BLACK",
                            "WHITE"
                        ]
                    }
                ]
            }
        ]
    }
    ]
}
```

Chapter 244. App metadata Count

244.1. Description

The **appmetadatacount** submenu provides real-time visibility into how much metadata and bestshots a particular app is generating. Please check the example section.

Access level

Action	Camera
check	Suser

244.2. Syntax

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=appmetadatacount&action=<value>[&<parameter>=<value
>...]
```

244.3. Parameters

Action	Parameters	Request/ Response	Type/ Value	Description
check	AppsMetadataCountingList.Channel	RES	<int>	
	AppsMetadataCountingList.Channel.#.AppsMetadataCountingInfo.#.AppId	RES	<string>	Application ID
	AppsMetadataCountingList.Channel.#.AppsMetadataCountingInfo.#.MetadataCountingList.#.Timestamp	RES	<int>	Timestamp of when the data was created
	AppsMetadataCountingList.Channel.#.AppsMetadataCountingInfo.#.MetadataCountingList.#.Metadata	RES	<int>	metadata count of when the data was created
	AppsMetadataCountingList.Channel.#.AppsMetadataCountingInfo.#.MetadataCountingList.#.Bestshot	RES	<int>	bestshot count of when the data was created

244.4. Examples

244.4.1. Getting the metaframe capability of the installed apps.

REQUEST

```
http://<Device IP>/stw-
cgi/opensdk.cgi?msubmenu=appmetadatacount&action=check
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
  "AppsMetadataCountingList": {
    "Channel": 0,
    "AppsMetadataCountingInfo": [
      {
        "AppId": "WiseAI",
        "MetadataCountingList": [
          {
            "Timestamp": 1744164665366,
            "Metadata": 927,
            "Bestshot": 0
          }
        ]
      }
    ]
  }
}
```

IP Installer

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 245. Preface

Purpose
This protocol enables clients to discover cameras and configure IP address, port, and initial password.

Audience
This document is intended for clients who need to discover and set up Hanwha devices.

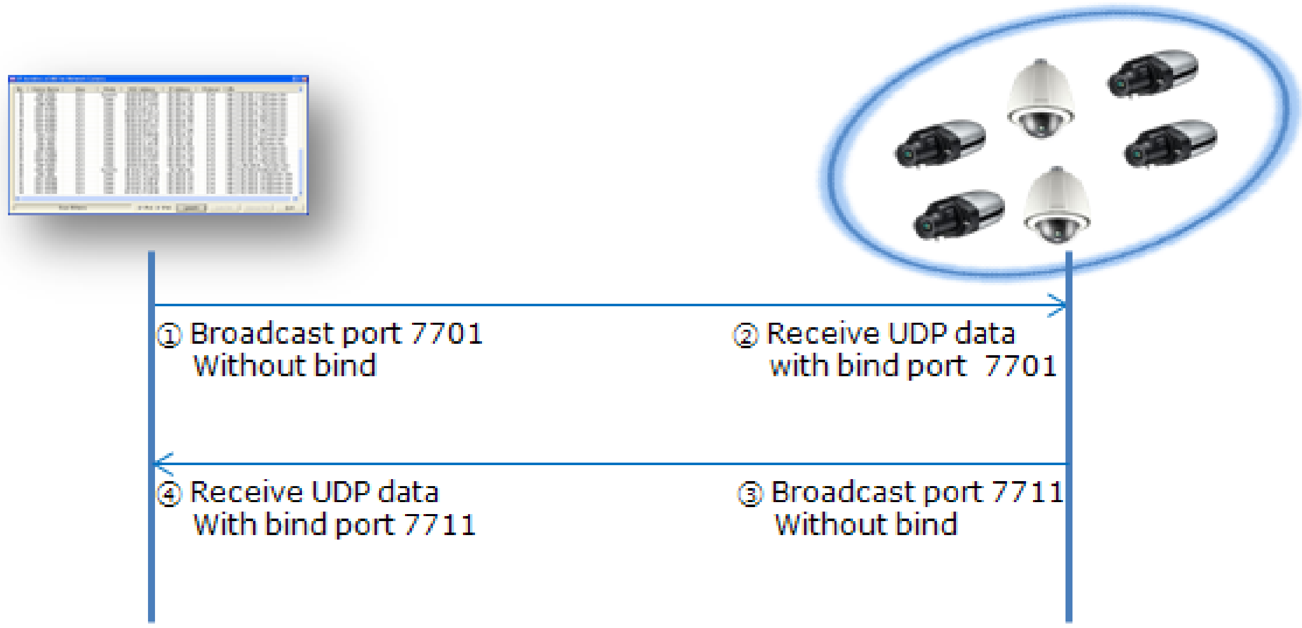
Scope
This document describes the IP installer protocol used for camera discovery and initial configuration.

Chapter 246. Introduction

This document describes the IP Installer (IP Scanner) protocol guide.
Some of Hanwha Vision’s network devices support both IPv4 and IPv6; others support only IPv4. This document covers both protocols.

Chapter 247. IPv4

247.1. Communicate Port



247.2. SendData Format (DEPRECATED)

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chPassword	char[20]	20	Password
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, it only is 10 character in the whole model name)

Name	Type	Byte size	Description
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V4_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chPassword[20];
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
} DATAPACKET_V4_OLD;
```

247.3. RecvData Format(DEPRECATED)

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chip	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chPassword	char[20]	20	Password
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address
chAlias	char[64]	64	Alias name for NVR or Encoder only • Network camera does not include this field in response packet.

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V4
{
```

```
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chPassword[20];
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[64];
} DATAPACKET_V4;
```

<Structure Field description>

Table 2. nMode

Mode Definition	VALUE
DEF_REQ_SCAN	1
DEF_REQ_APPLY	2
DEF_REQ_PORTMAPPING	4
DEF_RES_SCAN	11
DEF_RES_APPLY	22
DEF_RES_PASSWORD_ERR	33
DEF_RES_PORT_MAPPING	44
DEF_RES_PORT_MAPPING_ERR	55
DEF_RES_ROUTER_CONN_ERR	66
DEF_RES_APPLY_ERR	77

chPacketID

This value is used to identify Client. Hanwha use this value as unique ID derived from MAC address of PC and random value.

247.4. IP Scan(DEPRECATED)

247.4.1. Request

Field Name	Value
nMode	1 (DEF_REQ_SCAN)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chPassword	Unused
nPort	Unused
nStatus/nVersion	Unused, 0x08 : when client passes this version and if device don’t support SVNP ignore ipscan response.
chDeviceName	Unused
nHttpPort	Unused
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Unused
chDDNS	Unused

247.4.2. Response(DEPRECATED)

Field Name	Value
nMode	11 (DEF_RES_SCAN)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Unused
nPort	HTTP port for web-connection

Field Name	Value
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL[*]
chAlias	Alias[2] of NVR or Encoder (In case of NW camera, unused)

- **DDNS URL**[1] has the DDNS URL. If camera registered successfully the URL to DDNS server, this field is filled by the registered URL. If camera fails to register, this field is filled by the URL that is made based on the IP. (ex : http://192.168.1.200:8080)
- **Alias**[2] is used to distinguish each device of NVR/Encoder. NVR/Encoder But Network camera does not include ‘chAlias' field in response packet.
- **nDevicePort** : Port number to connect using SVNP, VNP or SSNP protocol.
- **nTcpPort** : Port number to get stream via tcp. This port is valid only if Client uses VNP.
- **nUDPPort** : Port number to get stream via udp. This port is valid only if Client uses VNP.
- **nUploadPort** : Port number to upload camera’s f/w (TCP). This port is valid only if Client uses VNP.
- **nMulticastPort** : Port number to get stream via multicast. This port is valid only if Client uses VNP.

247.5. IP Setting(DEPRECATED)

247.5.1. Request

Field Name	Value
nMode	2 (DEF_REQ_APPLY)
chPacketID	Unique value of client
chMAC	MAC address of Camera
chIP	IP address of Camera. Valid only if (nNetworkMode == 0)

Field Name	Value
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Password for connecting to camera with admin privilege
nPort	The same value of nHttpPort
nStatus	Unused
chDeviceName	Unused
nHttpPort	Port number to change
nDevicePort	Port number to connect using SVNP
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused

- Using this command, client can change the IP type, address and port number.

247.5.2. Response

Field Name	Value
nMode	22 (DEF_RES_APPLY)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document

Field Name	Value
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)

- Changed values are filled.

247.6. SendData Format for SUNAPI

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chPassword	char[20]	20	Password
is_only_support_sunapi	char	1	If device supports only SUNAPI and not SVNP
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (limited to 10 characters in new models)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port

Name	Type	Byte size	Description
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address
chAlias	Char[32]	32	Camera name
chNewModelName	Char[32]	32	New model Name(since April 2016)
nModelType	char	1	Model Type
nVersion	Unsigned short	2	version
nHttpMode	Char	1	Http Mode
nHttpsPort	Unsigned short	2	HTTPS Port
nSupportedProtocol	Char	1	Supported Protocol
nPasswordStatus	Char	1	Password Status

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V4_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chPassword[20];
    char is_only_support_sunapi;
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[32];
```

```
    char chNewModelName[32];
    char nModelType;
    unsigned short nVersion;
    char nHttpMode;
    unsigned short nHttpsPort;
    char nSupportedProtocol;
    char nPasswordStatus;
} DATAPACKET_V4_EXT;
```

247.7. RecvData Format for SUNAPI

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chip	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chPassword	char[20]	20	Password
is_only_support_sunapi	char	1	If only sunapi is supported by device
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port

Name	Type	Byte size	Description
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address
chAlias	char[32]	32	Alias name for NVR or Encoder only • Network camera does not include this field in response packet.
chNewModelName	Char[32]	32	New model Name(since April 2016)
nModelType	char	1	Model Type
nVersion	Unsigned short	2	version
nHttpMode	Char	1	Http Mode
nHttpsPort	Unsigned short	2	HTTPS Port
nSupportedProtocol	Char	1	Supported Protocol
nPasswordStatus	Char	1	Password Status

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V4
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chPassword[20];
    char is_only_support_sunapi;
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
```

```
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[32];
    char chNewModelName[32];
    char nModelType;
    unsigned short nVersion;
    char nHttpMode;
    unsigned short nHttpsPort;
    char nSupportedProtocol;
    char nPasswordStatus;
} DATAPACKET_V4_EXT;

typedef struct tagRsaScanResponse
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char MaxPasswordLen;
    char Reserved[10];
    char PayloadSize[2];
    char Payload[payloadSize];
} DATAPACKET_RSA_RESPONSE;

typedef struct tagApplyPasswordRequest
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char Reserved[10];
    char PayloadSize[2];
    char Payload[payloadSize];
} DATAPACKET_APPLY_PASSWORD_REQUEST;
```

<Structure Field description>

Table 3. nMode

Mode Definition	VALUE
DEF_REQ_SCAN_EXT	6
DEF_REQ_APPLY_EXT	7
DEF_REQ_SCAN_RSA	8
DEF_REQ_APPLY_PASSWORD	9
DEF_RES_SCAN_EXT	12
DEF_RES_SCAN_RSA	13
DEF_RES_APPLY_EXT	23
DEF_RES_APPLY_PASSWORD_ERR	24
DEF_RES_APPLY_PASSWORD	25
DEF_RES_PASSWORD_ERR	33
DEF_RES_ROUTER_CONN_ERR	66
DEF_RES_APPLY_ERR	77

chPacketID

This value is used to identify Client. Hanwha use this value as unique ID derived from MAC address of PC and random value.

247.8. IP Scan for SUNAPI

247.8.1. Request

Field Name	Value
nMode	6(DEF_REQ_SCAN_EX)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chPassword	Unused
is_only_support_sunapi	Unused
nPort	Unused
nStatus/nVersion	Unused, 0x08 : when client passes this version and if device don’t support SNVP ignore ipscan response.
chDeviceName	Unused
nHttpPort	Unused

Field Name	Value
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Unused
chDDNS	Unused
chAlias	Unused
nModelType	Unused
nVersion	Unused
nHttpMode	Unused
nHttpsPort	Unused
nSupportedProtocol	Unused
nPasswordStatus	Unused

247.8.2. Response

Field Name	Value
nMode	12(DEF_RES_SCAN_EX)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
Nonce	When nVersion 0x08 is supported will have the nonce value, that can be used for password digest.
is_only_support_sunapi	If device supports only SUNAPI
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document

Field Name	Value
nTcpPort	Unused
nUdpPort	Unused
nUploadPort/SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
MaxChannel	When nVersion 0x08 is supported will have max channel number **Note** For models that support virtual channel addition/deletion, it represents the current channel count.
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL[1]
chAlias	Alias[2] of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New model Name(since April 2016)
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox,0x05: NetworkSpeaker, 0x06: NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can’t change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)

Field Name	Value
nPasswordStatus	Password status of camera (0x00: The camera has a password, 0x01: The camera hasn’t a password) When nVersion 0x08 is supported this field can be used to check if device is initialized.

- **DDNS URL**[1] has the DDNS URL. If camera registered successfully the URL to DDNS server, this field is filled by the registered URL. If camera fails to register, this field is filled by the URL that is made based on the IP. (ex : http://192.168.1.200:8080)
- **Alias**[2] is used to distinguish each device of NVR/Encoder. NVR/Encoder But Network camera does not include ‘chAlias' field in response packet.
- **nDevicePort** : Port number to connect using SVNP, VNP or SSNP protocol.
- **nTcpPort** : Port number to get stream via tcp. This port is valid only if Client uses VNP.
- **nUDPPort** : Port number to get stream via udp. This port is valid only if Client uses VNP.
- **nUploadPort** : Port number to upload camera’s f/w (TCP). This port is valid only if Client uses VNP.
- **nMulticastPort** : Port number to get stream via multicast. This port is valid only if Client uses VNP.

247.9. SCAN Uninitialized Devices for RSA Key

247.9.1. Request

Field Name	Value
nMode	8(DEF_REQ_SCAN_RSA)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chPassword	Unused
nPort	Unused
nStatus/nVersion	Unused
chDeviceName	Unused
nHttpPort	Unused
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused

Field Name	Value
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Unused
chDDNS	Unused
chAlias	Unused
nModelType	Unused
nVersion	Unused
nHttpMode	Unused
nHttpsPort	Unused
nSupportedProtocol	Unused
nPasswordStatus	Unused

247.9.2. Response

Field Name	Value
nMode	13(DEF_RES_SCAN_RSA)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
maxPasswordLen	Maximum password length supported. If this field is 0, the maximum password length supported is 15; if it is larger than 0, the value supplied is the maximum password length.
Reserved	
PacketSize	Size of Payload
RSAKeyPayload	RSA Public Key

247.10. IP Setting for SUNAPI

247.10.1. Request

Field Name	Value
nMode	7 (DEF_REQ_APPLY_EX)
chPacketID	Unique value of client

Field Name	Value
chMAC	MAC address of Camera
chIP	IP address of Camera. Valid only if (nNetworkMode == 0)
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Password for connecting to camera with admin privilege(When version 0x08 is supported, pass sha256(Username:password:nonce) truncated to first 20 bytes of total 32 bytes)
nPort	The same value of nHttpPort
nStatus	Unused
chDeviceName	Unused
nHttpPort	Port number to change
nDevicePort	Port number to connect using SVNP
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
UserName	Pass the username in this field (When version 0x08)
chNewModelName	Unused
nModelType	Unused
nVersion	Unused,when digest password is used, this should be set to 0x08
nHttpMode	Unused
nHttpPort	Port number to change (if version value is 2)
nSupportedProtocol	Unused
nPasswordStatus	Unused

- Using this command, client can change the IP type, address and port number.

247.10.2. Response

Field Name	Value
nMode	23(DEF_RES_APPLY_EXT)

Field Name	Value
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort/SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New model Name(since April 2016)
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox, 0x05: NetworkSpeaker, 0x06 NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can’t change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest

Field Name	Value
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpsPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)
nPasswordStatus	Password status of camera (0x00: The camera has a password, 0x01: The camera hasn't a password)

- Changed values are filled.

247.11. Set Password in factory default state(SUNAPI)

247.11.1. Request

Field Name	Value
nMode	9(DEF_REQ_APPLY_PASSWORD)
chPacketID	Unique value of client
chMAC	MAC address of Camera
Reserved	
Payload size	Size of payload
Payload	RSA encrypted password

- Using this command, the client can configure the initial password of the device. Please refer to Annex A for the structure used.

247.11.2. Response

Field Name	Value
nMode	25(DEF_RES_APPLY_PASSWORD)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Unused
nPort	HTTP port for web-connection

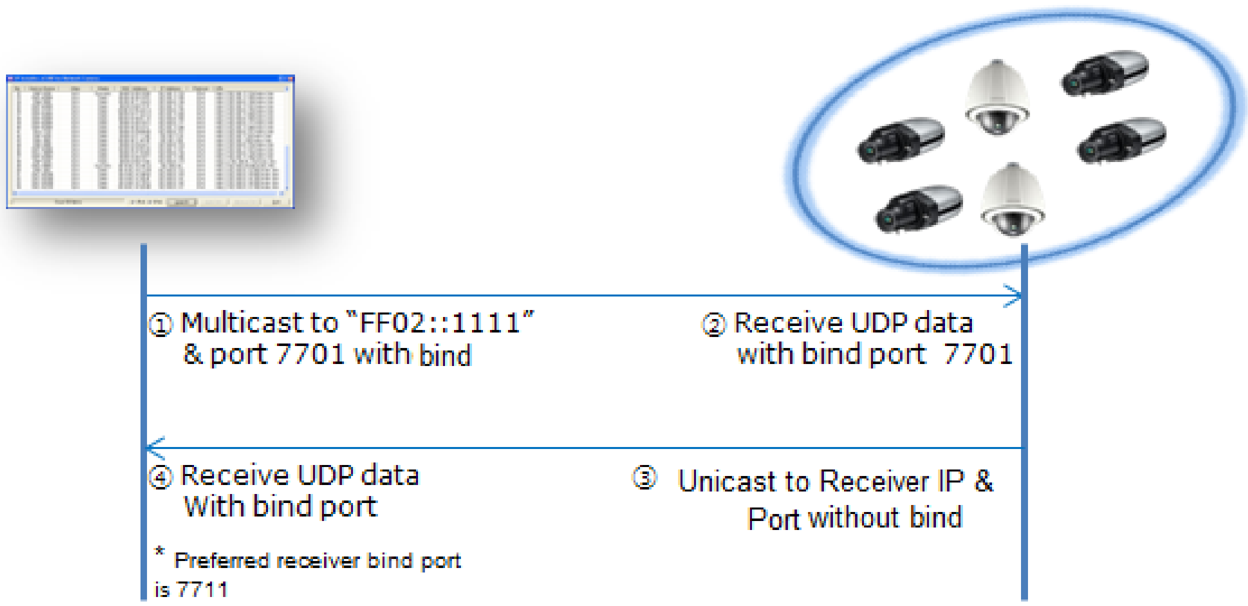
Field Name	Value
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort/SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New model Name(since April 2016)
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox, x05: NetworkSpeaker, 0x06 NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can’t change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpsPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)

Field Name	Value
nPasswordStatus	Password status of camera (0x00: The camera has a password, 0x01: The camera hasn't a password)

- Changed values are filled.

Chapter 248. IPv6

248.1. Communicate Port



248.2. SendData Format(DEPRECATED)

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chIpv6addr	char[40]	40	IPv6 address
chPassword	char[20]	20	Password
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value

Name	Type	Byte size	Description
chDeviceName	char[10]	10	Model name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V6_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chIPv6addr[40];
    char chPassword[20];
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
```

```
} DATAPACKET_V6_OLD;
```

248.3. RecvData Format(DEPRECATED)

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway
chIpv6addr	char[40]	40	IPv6 address
chPassword	char[20]	20	Password
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address

Name	Type	Byte size	Description
chAlias	char[64]	64	Alias name for NVR or Encoder only • Network camera does not include this field in response packet.

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V6_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chIPv6addr[40];
    char chPassword[20];
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[64];
} DATAPACKET_V6_OLD;
```

<Structure Field description>

Table 4. nMode

Mode Definition	VALUE
DEF_REQ_SCAN	1
DEF_REQ_APPLY	2

Mode Definition	VALUE
DEF_REQ_PORTMAPPING	4
DEF_RES_SCAN	11
DEF_RES_APPLY	22
DEF_RES_PASSWORD_ERR	33
DEF_RES_PORT_MAPPING	44
DEF_RES_PORT_MAPPING_ERR	55
DEF_RES_ROUTER_CONN_ERR	66
DEF_RES_APPLY_ERR	77

chPacketID

This value is used to identify Client. Hanwha use this value as unique ID derived from MAC address of PC and random value.

248.4. IP Scan(DEPRECATED)

248.4.1. Request

Field Name	Value
nMode	1 (DEF_REQ_SCAN)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chIpv6addr	IPv6 address
chPassword	Unused
nPort	Unused
nStatus	Unused
chDeviceName	Unused
nHttpPort	Unused
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused

Field Name	Value
nNetworkMode	Unused
chDDNS	Unused

248.4.2. Response

Field Name	Value
nMode	11 (DEF_RES_SCAN)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
chPassword	Unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL[1]
chAlias	Alias[2] of NVR or Encoder (In case of NW camera, unused)

- **DDNS URL[1]** has the DDNS URL. If camera registered successfully the URL to DDNS server, this field is filled by the registered URL. If camera fails to register, this field is filled by the URL that is made based on the IP. (ex : http://192.168.1.200:8080)
- **Alias[2]** is used to distinguish each device of NVR/Encoder. NVR/Encoder But Network camera does not include ‘chAlias' field in response packet.
- **nDevicePort** : Port number to connect using SVNP, VNP or SSNP protocol.

- **nTcpPort** : Port number to get stream via tcp. This port is valid only if Client uses VNP.
- **nUDPPort** : Port number to get stream via udp. This port is valid only if Client uses VNP.
- **nUploadPort** : Port number to upload camera's f/w (TCP). This port is valid only if Client uses VNP.
- **nMulticastPort** : Port number to get stream via multicast. This port is valid only if Client uses VNP.

248.5. IP Setting(DEPRECATED)

248.5.1. Request

Field Name	Value
nMode	2 (DEF_REQ_APPLY)
chPacketID	Unique value of client
chMAC	MAC address of Camera
chIP	IP address of Camera. Valid only if (nNetworkMode == 0)
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
chPassword	Password for connecting to camera with admin privilege
nPort	The same value of nHttpPort
nStatus	Unused
chDeviceName	Unused
nHttpPort	Port number to change
nDevicePort	Port number to connect using SVNP
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused

- Using this command, client can change the IP type, address and port number.

248.5.2. Response

Field Name	Value
nMode	22 (DEF_RES_APPLY)

Field Name	Value
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
chPassword	Unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL*
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)

- Changed values are filled.

248.6. SendData Format for SUNAPI

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask
chGateway	char[16]	16	Gateway

Name	Type	Byte size	Description
chIpv6addr	char[40]	40	IPv6 address
chPassword	char[20]	20	Password
is_only_support_sunapi	char	1	If only sunapi is supported
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, only up to 10 in the whole character)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address
chAlias	char[64]	64	Alias name for NVR or Encoder only • Network camera does not include this field in response packet.
nModelType	char	1	Model Type
nVersion	Unsigned short	2	Structure version
nHttpMode	Char	1	Http Mode
nHttpsPort	Unsigned short	2	HTTPS Port
nSupportedProtocol	Char	1	Supported Protocol
nPasswordStatus	Char	1	Password Status

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V6_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chIPv6addr[40];
    char chPassword[20];
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[64];
    char nModelType;
    unsigned short nVersion;
    char nHttpMode;
    unsigned short nHttpsPort;
    char nSupportedProtocol;
    char nPasswordStatus;
} DATAPACKET_V6_EXT;
```

248.7. RecvData Format for SUNAPI

Name	Type	Byte size	Description
nMode	Unsigned char	1	Mode, refer to below table 1
chPacketID	unsigned char[18]	18	Packet value
chMAC	char[18]	18	Camera MAC address
chIP	char[16]	16	IPv4 address
chSubnetMask	char[16]	16	Subnet mask

Name	Type	Byte size	Description
chGateway	char[16]	16	Gateway
chIpv6addr	char[40]	40	IPv6 address
chPassword	char[20]	20	Password
Is_only_sunapi	char	1	If only SUNAPI is supported
nPort	unsigned short	2	Port
nStatus	unsigned char	1	Port mapping return value
chDeviceName	char[10]	10	Model name (In new model name case, only up to 10 in the whole character)
nHttpPort	unsigned short	2	HTTP port
nDevicePort	unsigned short	2	Device port
nTcpPort	unsigned short	2	TCP port
nUdpPort	unsigned short	2	UDP port
nUploadPort	unsigned short	2	Upload port
nMulticastPort	unsigned short	2	Multicast port
nNetworkMode	unsigned char	1	Mode status (Static or DHCP)
chDDNS	char[128]	128	DDNS address
chAlias	char[32]	32	Alias name for NVR or Encoder only • Network camera does not include this field in response packet.
chNewModelName	Char[32]	32	New Model Name(since April 2016)
nModelType	char	1	Model Type
nVersion	Unsigned short	2	Structure version
nHttpMode	Char	1	Http Mode
nHttpsPort	Unsigned short	2	HTTPS Port
nSupportedProtocol	Char	1	Supported Protocol
nPasswordStatus	Char	1	Password Status

(On windows app) Struct define in C++ (Not packed)

```
typedef struct tagDataPaket_V6_OLD
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
    char chIPv6addr[40];
    char chPassword[20];
    char is_only_support_sunapi;
    unsigned short nPort;
    unsigned char nStatus;
    char chDeviceName[10];
    unsigned short nHttpPort;
    unsigned short nDevicePort;
    unsigned short nTcpPort;
    unsigned short nUdpPort;
    unsigned short nUploadPort;
    unsigned short nMulticastPort;
    unsigned char nNetworkMode;
    char chDDNS[128];
    char chAlias[64];
    char nModelType;
    unsigned short nVersion;
    char nHttpMode;
    unsigned short nHttpsPort;
    char nSupportedProtocol;
    char nPasswordStatus;
} DATAPACKET_V6_EXT;

typedef struct tagRsaScanResponse
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char chIP[16];
    char chSubnetMask[16];
    char chGateway[16];
```

```
    char MaxPasswordLen;
    char Reserved[10];
    char PayloadSize[2];
    char Payload[payloadSize];
} DATAPACKET_RSA_RESPONSE;

typedef struct tagApplyPasswordRequest
{
    unsigned char nMode;
    unsigned char chPacketID[18];
    char chMAC[18];
    char Reserved[10];
    char PayloadSize[2];
    char Payload[payloadSize];
}DATAPACKET_APPLY_PASSWORD_REQUEST;
```

<Structure Field description>

Table 5. nMode

Mode Definition	VALUE
DEF_REQ_SCAN_EXT	6
DEF_REQ_APPLY_EXT	7
DEF_REQ_SCAN_RSA	8
DEF_REQ_APPLY_PASSWORD	9
DEF_RES_SCAN_EXT	12
DEF_RES_SCAN_RSA	13
DEF_RES_APPLY_EXT	23
DEF_RES_APPLY_PASSWORD_ERR	24
DEF_RES_APPLY_PASSWORD	25
DEF_RES_PASSWORD_ERR	33
DEF_RES_ROUTER_CONN_ERR	66
DEF_RES_APPLY_ERR	77

chPacketID

This value is used to identify Client. Hanwha use this value as unique ID derived from MAC address of PC and random value.

248.8. IP Scan for SUNAPI

248.8.1. Request

Field Name	Value
nMode	6(DEF_REQ_SCAN_EX)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chIpv6addr	IPv6 address
chPassword	Unused
Is_only_sunapi	Unused
nPort	Unused
nStatus	Unused
chDeviceName	Unused
nHttpPort	Unused
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Unused
chDDNS	Unused
chAlias	Unused
nModelType	Unused
nVersion	Unused
nHttpMode	Unused
nHttpsPort	Unused
nSupportedProtocol	Unused
nPasswordStatus	Unused

248.8.2. Response

Field Name	Value
nMode	12 (DEF_RES_SCAN_EX)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
Nonce	When nVersion 0x08 is supported will have the nonce value, that can be used for password digest.
isOnlySunapi	This field is 1 in case if the device only supports SUNAPI and not SVNP
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort / SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
MaxChannel	When nVersion 0x08 is supported will have max channel number **Note** For models that support virtual channel addition/deletion, it represents the current channel count.
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL[1]
chAlias	Alias[2] of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New model name (since April 2016)

Field Name	Value
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox, 0x05: NetworkSpeaker, 0x06 NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can't change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpsPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)
nPasswordStatus	Password status of camera (if Version 0x08 need to notify the actual state of the device if password is initialized or not) (0x00: The camera has a password, 0x01: The camera hasn't a password)

- **DDNS URL**[1] has the DDNS URL. If camera registered successfully the URL to DDNS server, this field is filled by the registered URL. If camera fails to register, this field is filled by the URL that is made based on the IP. (ex : http://192.168.1.200:8080)
- **Alias**[2] is used to distinguish each device of NVR/Encoder. NVR/Encoder But Network camera does not include 'chAlias' field in response packet.
- **nDevicePort** : Port number to connect using SVNP, VNP or SSNP protocol.
- **nTcpPort** : Port number to get stream via tcp. This port is valid only if Client uses VNP.
- **nUDPPort** : Port number to get stream via udp. This port is valid only if Client uses VNP.
- **nUploadPort** : Port number to upload camera's f/w (TCP). This port is valid only if Client uses VNP.
- **nMulticastPort** : Port number to get stream via multicast. This port is valid only if Client uses VNP.

248.9. SCAN Uninitialized Devices for RSA Key

248.9.1. Request

Field Name	Value
nMode	8(DEF_REQ_SCAN_RSA)
chPacketID	Unique value of client
chMAC	Unused
chIP	Unused
chSubnetMask	Unused
chGateway	Unused
chPassword	Unused
nPort	Unused
nStatus/nVersion	Unused
chDeviceName	Unused
nHttpPort	Unused
nDevicePort	Unused
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused
nNetworkMode	Unused
chDDNS	Unused
chAlias	Unused
nModelType	Unused
nVersion	Unused
nHttpMode	Unused
nHttpsPort	Unused
nSupportedProtocol	Unused
nPasswordStatus	Unused

248.9.2. Response

Field Name	Value
nMode	13(DEF_RES_SCAN_RSA)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera

Field Name	Value
chGateway	Gateway of Camera
maxPasswordLen	Maximum password length supported. If this field is 0, the maximum password length supported is 15; if it is larger than 0, the value supplied is the maximum password length.
Reserved	
PacketSize	Size of Payload
RSAKeyPayload	RSA Public Key

248.10. IP Setting for SUNAPI

248.10.1. Request

Field Name	Value
nMode	7 (DEF_REQ_APPLY_EX)
chPacketID	Unique value of client
chMAC	MAC address of Camera
chIP	IP address of Camera. Valid only if (nNetworkMode == 0)
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
chPassword	Password for connecting to camera with admin privilege (When version 0x08 is supported, pass sha256(username:password:nonce) truncated to first 20 bytes of total 32 bytes)
Is_only_sunapi	unused
nPort	The same value of nHttpPort
nStatus	Unused
chDeviceName	Unused
nHttpPort	Port number to change
nDevicePort	Port number to connect using SVNP
nTcpPort	Unused
nUdpPort	Unused
nUploadPort	Unused
nMulticastPort	Unused

Field Name	Value
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
UserName	Pass the username in this field (When version 0x08)
nModelType	Unused
nVersion	Unused (When digest password is used should be set to 0x08)
nHttpMode	Unused
nHttpsPort	Port number to change (if version value is 2)
nSupportedProtocol	Unused
nPasswordStatus	Unused

- Using this command, client can change the IP type, address and port number.

248.10.2. Response

Field Name	Value
nMode	23 (DEF_RES_APPLY_EX)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chIpv6addr	IPv6 address
chPassword	Unused
Is_only_sunapi	unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused

Field Name	Value
nUploadPort/SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, this SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	DDNS URL*
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New Model Name(since April 2016)
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox, 0x05: NetworkSpeaker, 0x06 NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can’t change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpsPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)
nPasswordStatus	Password status of camera (0x00: The camera has a password, 0x01: The camera hasn’t a password)

- Changed values are filled.

248.11. Set password in factory default state (SUNAPI).

248.11.1. Request

Field Name	Value
nMode	9(DEF_REQ_APPLY_PASSWORD)
chPacketID	Unique value of client

Field Name	Value
chMAC	MAC address of Camera
Reserved	
Payload size	Size of payload
Payload	RSA encrypted password

- Using this command, the client can configure the initial password of the device. Please refer to Annex A for the structure used.

248.11.2. Response

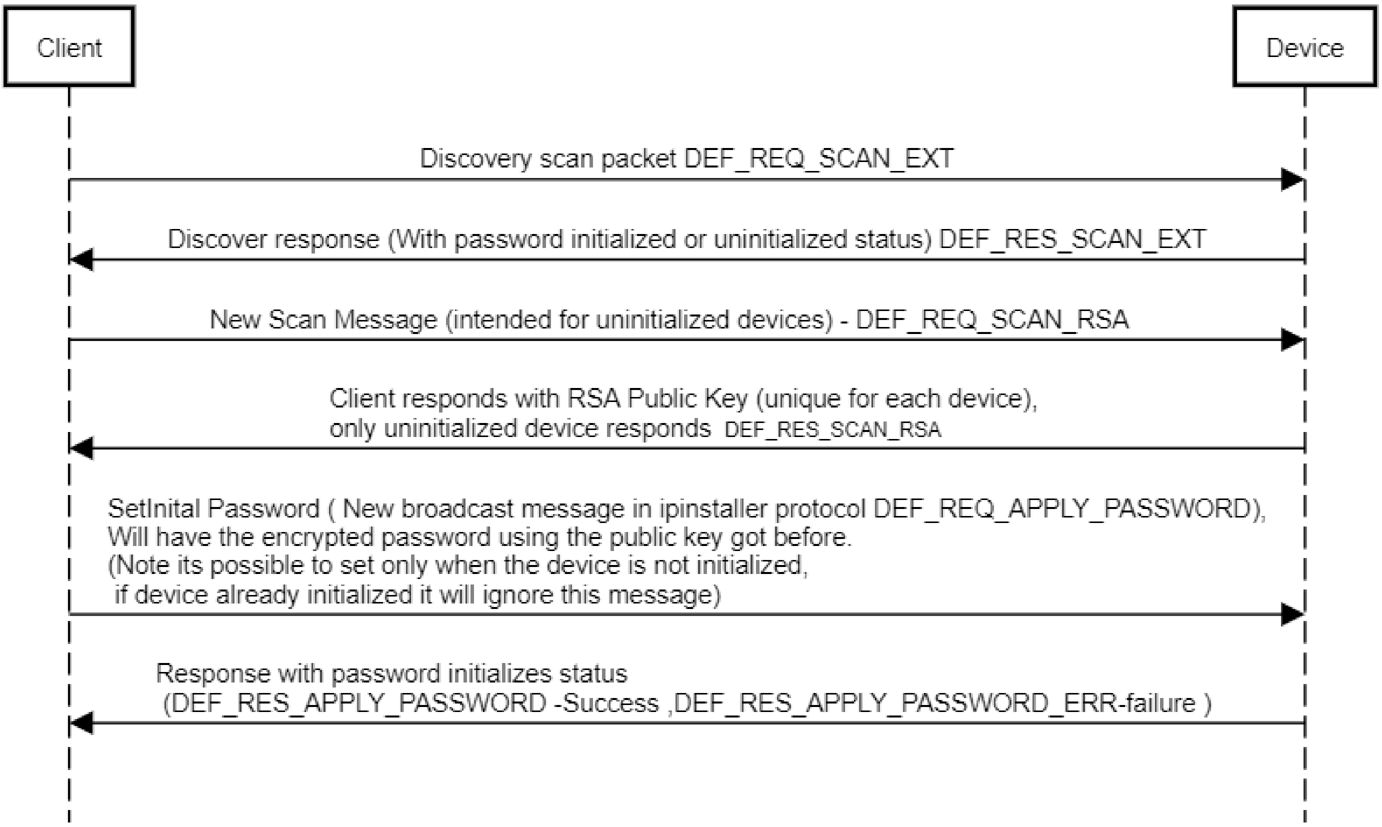
Field Name	Value
nMode	25(DEF_RES_APPLY_PASSWORD)
chPacketID	Unique value of camera
chMAC	MAC address of Camera
chIP	IP address of Camera
chSubnetMask	Subnet Mask of Camera
chGateway	Gateway of Camera
chPassword	Unused
nPort	HTTP port for web-connection
nStatus	Success or not (of UPNP port mapping)
chDeviceName	Model Name (In new model name case, it only is 10 character in the whole model name)
nHttpPort	HTTP port of web-connection
nDevicePort	Port number to connect using protocol document
nTcpPort	Unused
nUdpPort	Unused
nUploadPort/SpeakerType	Unused / When Modeltype is set to 0x05 network speaker, SpeakerType can be 0: Slave, 1: Master, 2:Server, 3: Module
nMulticastPort	Unused
nNetworkMode	Network IP type (0: Static , 1: DHCP, 2: PPPoE)
chDDNS	Unused
chAlias	Alias of NVR or Encoder (In case of NW camera, unused)
chNewModelName	New model Name(since April 2016)

Field Name	Value
nModelType	Device Type (0x00: Camera, 0x01: Encoder, 0x02: Decoder, 0x03: Recorder, 0x04: IOBox,0x05: NetworkSpeaker, 0x06 NetworkMic, 0x07: LEDBox, 0x08: EmergencyBell, 0x09 AccessController)
nVersion	version : This parameter can be a combination of the following values. • 0x01: Can’t change the HTTPS port in web page • 0x02: Can change the HTTPS port in web page • 0x04: Support New Model Name variable • 0x08: SupportPasswordVerification using digest
nHttpMode	HTTP Mode of camera (0x00: HTTP, 0x01: HTTPS)
nHttpsPort	HTTPS port for web-connection
nSupportedProtocol	Supported Protocol of camera (0x01: SVNP, 0x02: SUNAPI1.0, 0x04: SUNAPI2.0, 0x08 SUNAPI2.3.1above,0x10:SVP)
nPasswordStatus	Password status of camera (0x00: The camera has a password, 0x01: The camera hasn’t a password)

- Changed values are filled.

Password setting Sequence Diagram

IP installer password setting (RSA)



Chapter 249. Annex A

249.1. Updated Structure Used

```
static const int MAX_STRLEN_MAC = 18;
static const int MAX_STRLEN_IP = 16;
static const int MAX_STRLEN_IPv6 = 40;
static const int MAX_STRLEN_DEVICE_NAME = 10;
static const int MAX_STRLEN_DEVICE_NAME_NEW = 32;
static const int MAX_STRLEN_PASSWORD = 20;
static const int MAX_STRLEN_DDNS_URL = 128;
static const int MAX_STRLEN_ALIAS = 32;
static const int PACKETID_LEN = 18;
static const int MAX_STRLEN_RSA_PAYLOAD_SIZE = 512;

typedef struct _IPSET_T
{
    char ip_addr[MAX_STRLEN_IP];
    char subnetmask[MAX_STRLEN_IP];
    char gateway[MAX_STRLEN_IP];
} __attribute__ ((__packed__)) _ipset_t;

typedef struct _PORTSET_T
{
    unsigned short http_port;
    unsigned short device_port;
    unsigned short tcp_port;
    unsigned short udp_port;
    unsigned short upload_port;
    unsigned short multicast_port;
} __attribute__ ((__packed__)) _portset_t;

typedef struct DATAPACKET_IPv4_T
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    _ipset_t ipset;
    char password[MAX_STRLEN_PASSWORD];
    // char reserved1;
    char is_only_support_sunapi;
```

```
    unsigned short port;
    unsigned char status;
    char device_name[MAX_STRLEN_DEVICE_NAME];
    char reserved2;
    _portset_t portset;
    unsigned char network_mode;
    char ddns_url[MAX_STRLEN_DDNS_URL];
    char reserved3;
} __attribute__ ((__packed__)) datapacket_v4_t;

typedef struct DATAPACKET_IPv6_T
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    _ipset_t ipset;
    char ipv6_addr[MAX_STRLEN_IPv6];
    char password[MAX_STRLEN_PASSWORD];
    // char reserved1;
    char is_only_support_sunapi;
    unsigned short port;
    unsigned char status;
    char device_name[MAX_STRLEN_DEVICE_NAME];
    char reserved2;
    _portset_t portset;
    unsigned char network_mode;
    char ddns_url[MAX_STRLEN_DDNS_URL];
    char reserved3;
} __attribute__ ((__packed__)) datapacket_v6_t;

typedef struct DATAPACKET_EXT_IPv4_T
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    _ipset_t ipset;
    char password[MAX_STRLEN_PASSWORD];
    // char reserved1;
    char is_only_support_sunapi;
    unsigned short port;
    unsigned char status;
```

```
    char device_name[MAX_STRLEN_DEVICE_NAME];
    char reserved2;
    _portset_t portset;
    unsigned char network_mode;
    char ddns_url[MAX_STRLEN_DDNS_URL];
    char alias[MAX_STRLEN_ALIAS];
    char device_name_new[MAX_STRLEN_DEVICE_NAME_NEW]; // New Model Name (ex:
PNR-6320RH-001)
    unsigned char model_type;
    // version : This parameter can be a combination of the following values
    // 0x01: Can't change the HTTPS port in web page
    // 0x02: Can change the HTTPS port in web page
    // 0x04: Support New Model Name variable
    unsigned short version;
    unsigned char https_mode;
    unsigned char reserved3;
    unsigned short https_port;
    unsigned char supported_protocol;
    unsigned char no_password;
} __attribute__ ((__packed__)) datapacket_ext_v4_t;

typedef struct DATAPACKET_EXT_IPv6_T
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    _ipset_t ipset;
    char ipv6_addr[MAX_STRLEN_IPv6];
    char password[MAX_STRLEN_PASSWORD];
    // char reserved1;
    char is_only_support_sunapi;
    unsigned short port;
    unsigned char status;
    char device_name[MAX_STRLEN_DEVICE_NAME];
    char reserved2;
    _portset_t portset;
    unsigned char network_mode;
    char ddns_url[MAX_STRLEN_DDNS_URL];
    char alias[MAX_STRLEN_ALIAS];
    char device_name_new[MAX_STRLEN_DEVICE_NAME_NEW];
    unsigned char model_type;
```

```
    // version : This parameter can be a combination of the following values
    // 0x01: Can't change the HTTPS port in web page
    // 0x02: Can change the HTTPS port in web page
    // 0x04: Support New Model Name variable
    unsigned short version;
    unsigned char https_mode;
    unsigned char reserved3;
    unsigned short https_port;
    unsigned char supported_protocol;
    unsigned char no_password;
} __attribute__ ((__packed__)) datapacket_ext_v6_t;

typedef struct tagRsaScanResponse
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    _ipset_t ipset;
    char MaxPasswordLen;
    char reserved[10];
    unsigned short payloadSize;
    char payload[MAX_STRLEN_RSA_PAYLOAD_SIZE];
}__attribute__ ((__packed__)) datapacket_rsa_response_t;

typedef struct tagApplyPasswordRequest
{
    unsigned char mode;
    unsigned char packet_id[PACKETID_LEN];
    char mac_addr[MAX_STRLEN_MAC];
    char reserved[10];
    char reserved2;
    unsigned short payloadSize;
    char payload[MAX_STRLEN_RSA_PAYLOAD_SIZE];
}__attribute__ ((__packed__)) datapacket_apply_password_request_t;
```

Chapter 250. Revision History

Version	Description	Release Date
1.23.0	Edited according to the standard document format	30th, JULY, 2010
1.23.1	Edited the data structure according to source code	11th AUG, 2010
1.23.2	Added unpack structure for windows client & changed some sentences	20th FEB, 2012
1.23.3	Added new scan and apply command with extended structure for the SUNAPI	24th MAY, 2013
1.23.4	Support new model name (only SUNAPI)	29TH MAR,2016
1.24.0	Factory default state handling and password digest verification (nversion 0x08)	28th MAR, 2019
1.25.0	Password Initialization method added	27th JUNE 2019
1.26.0	Device Type 0x04: IOBox, 0x05: NetworkSpeaker/Mic added newly	19th AUG 2020
1.27.0	nSupportedProtocol list updated based on actual implementation	20th OCT 2020
1.28.0	Device Type modified ***0x05: NetworkSpeaker, 0x06 NetworkMic and SpeakerType added 0:Slave, 1:Master, 2:Server, 3:Module MaxChannel: definition updated***	***14th SEP 2021***
2.6.0	Added to SUNAPI Documentation Release	1st MAR 2022
2.6.1	RSA Key Response updated to include maximum password length supported Device Type 0x07: LEDBox newly added	1st AUG 2022
2.6.4	Device Type 0x08: EmergencyBell newly added	9th JAN 2024
2.6.6	Device Type 0x09 AccessController newly added	19th NOV 2024

Application Programmer’s Guide

SUNAPI

v2.6.8
2026-04-09

Contact Information
Hanwha Vision Co., Ltd.
Hanwha Vision 6, Pangyo-ro 319beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, 13488, KOREA
www.hanwhavision.com

Hanwha Vision America
500 Frank W. Burr Blvd. Suite 43 Teaneck, NJ 07666
hanwhavisionamerica.com

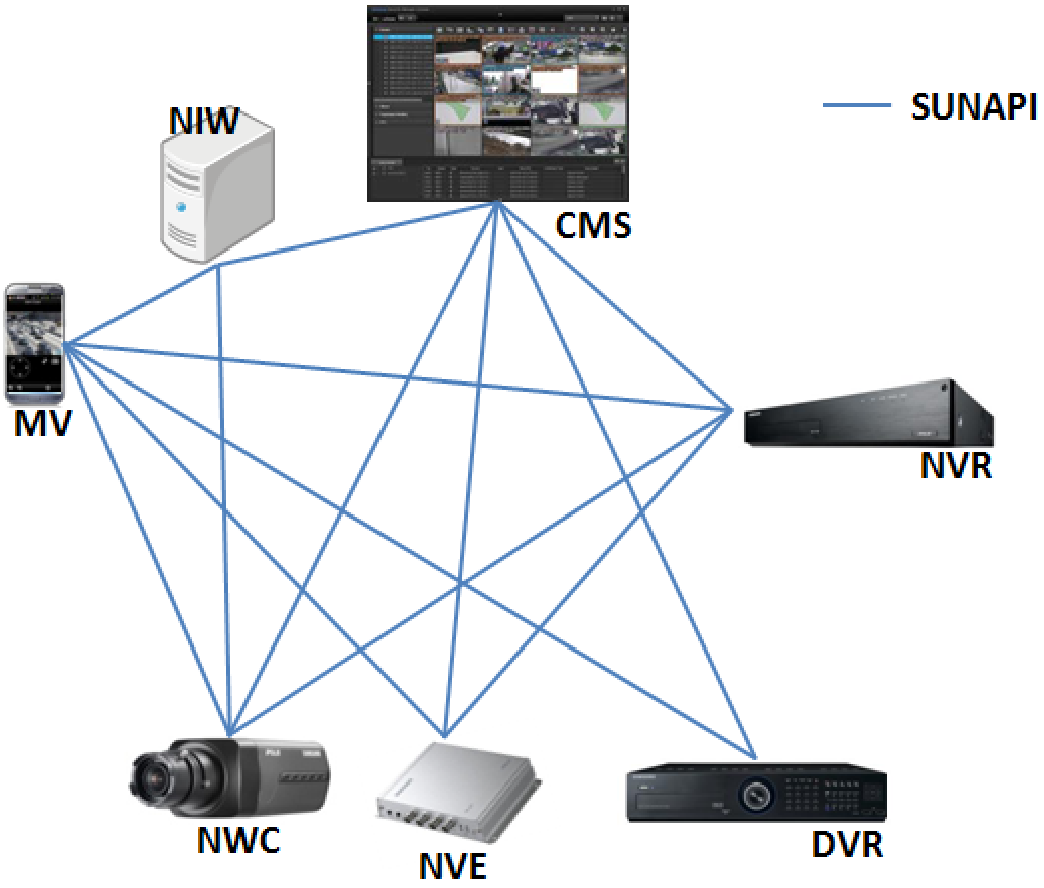
Hanwha Vision Europe
Heriot House, Heriot Road, Chertsey, Surrey, KT16 9DT, United Kingdom
hanwhavision.eu

Hanwha Vision Middle East FZE
Jafza View 18, Office 2001-2003, Po Box 263572, Jebel Ali Free Zone, Dubai, United Arab Emirates
www.hanwhavision.com/ar

Hanwha Vision

Chapter 251. Introduction

SUNAPI (Smart Unified API) is a common protocol used by Central Monitoring Software (CMS), Video Management Software (VMS), and mobile clients to communicate with Hanwha security devices, such as network cameras, DVRs, and NVRs.

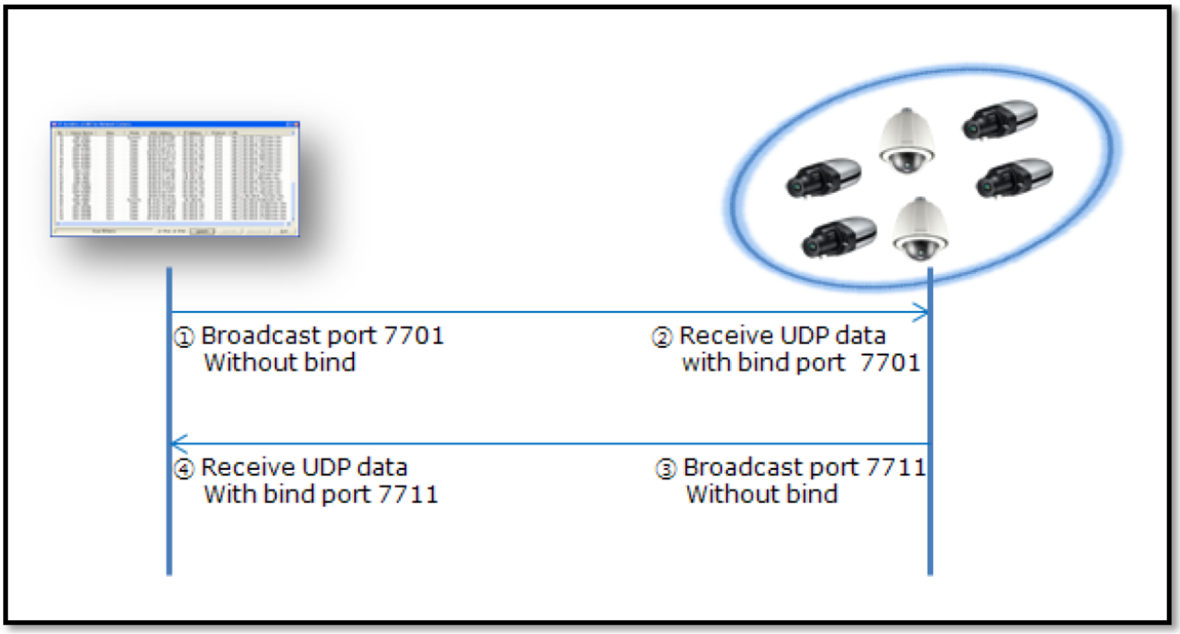


SUNAPI allows you to access product features simply by entering standard HTTP URLs. The URLs pass variables to SUNAPI's CGI, which interfaces with the specific product. This simplified interface system makes it possible for CMS to access the features of a diverse set of products in a standardized way. This makes SUNAPI a valuable tool for developers of CMS and other network video applications.

This document describes how the SUNAPI protocol can be used from a programmer's perspective. It is intended as a complementary document to the SUNAPI specification document; as such, this document does not cover all of the features described in the specification.

Chapter 252. Discovery

The Discovery protocol used by SUNAPI is a binary protocol, in which the client sends a broadcast message to a particular port and waits for a response message on a specific port number.



DISCOVERY REQUEST

```
SunapiDiscoveryRequest()
{
    IPScanRequest.nMode = DEF_REQ_SCAN;
    IPScanRequest.chPacketID = getUniqueID();// An unique 18 byte value
derived from MAC.
    Res = Send_BroadcastMessage(IPScanRequest, 255.255.255.255,7701);
}
```

DISCOVERY RESPONSE

```
SunapiDiscoveryResponse()
{
    Response = ReadBroadcastResponse(7711);
    If(Response.chPacketID != getUniqueID())
    {
        return -1;
    }

    Result.MacAddress = Response.chMac;
    Result.IPAddress = Response.IpAddress;
    Result.HttpPort = Response.nPort;
```

```
    Result.DeviceName = Response.chDeviceName;
    Result.HTTPSMode = Response.nHttpMode;
    If(Result.HTTPSMode)
        Result.HTTPsPort = Response.HttpsPort;
    Result.SunapiVersion = Response.nsupportedProtocol; //1-SVNP,2-sunapi 
1.0, 4 -sunapi 2.0
    return Result;
}
```

Chapter 253. Password Encryption

This feature was added to protect passwords sent in a URL as plain text. The client can use the following procedure to send an encrypted password to the device.

NOTE This is applicable for all submenus where password is a parameter.

Step1

Download the public key from the submenu below:

REQUEST

```
http://<ip>/stw-cgi/security.cgi?msubmenu=rsa&action=view
```

TEXT RESPONSE


```
PublicKey=-----BEGIN RSA PUBLIC KEY-----
MIIBCgKCAQEA6UfAclvda/DANJqOoWN3u292M+xLpVWgCNUEhhXeuPdgOIlYIWTh
cABwVhimgngXbn1isEwuIKZ5Q4g366/JgpSkRRCwdXZ4Xz6jObr544Dp9nCKU/UJ
3D3bQ9FJbAkBcFN7UCe6UISCcfUMrmn4PFOPSupqiCjDJ/oZgENIG8Ugtt392/QT
KX9l108IDHSj+ziL2FlJ3VW8xX7KNismZg5h8xPnwb90qQJawxyW7p5Z+ngOnJ0X
pA6X35Z0qOBsEw0L3x6QDrvKcGXA1pR6odfQlExj2uNT+Xg8NNeGiCGvFwBHooqh
yMDY1EATgAtROSeTjgnO4aCz3uB2GjAw/QIDAQAB
-----END RSA PUBLIC KEY-----
```


JSON RESPONSE


```
{
    "PublicKey": "-----BEGIN RSA PUBLIC KEY-----
\nMIIBCgKCAQEA6UfAclvda/DANJqOoWN3u292M+xLpVWgCNUEhhXeuPdgOIlYIWTh\ncABwVhim
gngXbn1isEwuIKZ5Q4g366/JgpSkRRCwdXZ4Xz6jObr544Dp9nCKU/UJ\n3D3bQ9FJbAkBcFN7UC
e6UISCcfUMrmn4PFOPSupqiCjDJ/oZgENIG8Ugtt392/QT\nKX9l108IDHSj+ziL2FlJ3VW8xX7K
NismZg5h8xPnwb90qQJawxyW7p5Z+ngOnJ0X\npA6X35Z0qOBsEw0L3x6QDrvKcGXA1pR6odfQlE
xj2uNT+Xg8NNeGiCGvFwBHooqh\nyMDY1EATgAtROSeTjgnO4aCz3uB2GjAw/QIDAQAB\n-----
END RSA PUBLIC KEY-----\n"
}
```


Step 2

Client encrypts the password using the RSA Public Key and RSA_PKCS1_PADDING padding scheme.

Step 3

Base64 encodes the binary data and sends the password in the post message. The **IsPasswordEncrypted** parameter should be set to true in the request.

Example:

```
http://<Device IP>/stw-cgi/security.cgi?msubmenu=users&action=update&Index=1&UserID=user1&Enable=True&IsPasswordEncrypted=True
```

Body


```
Vvg2Ku93HlReI+3RseQgfYQoxUFkh9P2L5RvjZ+bLovLTGMQ23OFT+BDaIwMcgvszTwCugk0TuKHENsczardZMQLSosu8RBcqKUMqDq2M1x8fO6Y4S0qlklAtaKl3d9vG0fBdV7BXRqgvK6VIGEc/6GdPyp4wzY31dalmfwObsdtN//eOU/cVP8MQSCiPod0b5fIWWwekHfDMbQhW2J8eY1KIeOo2O9+vo0gql5vBLFEeFvASZI8UEguxAbOJk4F7iaSr8IFmQhNBXsVYevcAuMgAPvGk3LbXv1DlJrUqUhj9U2r2peMGAGl4vaLPt3M2V9UUKlEn7JR/CUI8Pk1Qg==
```

Chapter 254. Setting the password in factory default state

Starting from SUNAPI version 2.5.5, the device supports the initial password setting using pw_init.cgi. Password init cgi is a hidden CGI and it requires no authentication to be used only in the factory default state.

NOTE The initial password can also be configured using the IP Installer Protocol [Refer: Ipinstaller protocol document].

254.1. To check if the camera password is initialized or not initialized

REQUEST

```
http://<ip>/init-cgi/pw_init.cgi?msubmenu=statuscheck&action=view
```

If the camera is already initialized, the response would be:

RESPONSE

```
{
    "Initialized": true,
    "Language": "English",
    "MaxChannel": 1,
    "SpecialType": "none",
    "NewPasswordPolicy": true,
    "MaxPasswordLength": 64,
    "Manufacturer": "Hanwha Vision"
}
```

If the camera is not initialized, the response would be (The RSA public below can be used to encrypt the password as explained in chapter 12.):

RESPONSE

```
{
    "Initialized": false,
    "Language": "English",
    "MaxChannel": 2,
    "SpecialType": "none",
    "NewPasswordPolicy": true,
    "MaxPasswordLength" : 64,
```

```
    "SupportedPublicKeyFormats": [
        "PKCS1",
        "X509"
    ],
    "PublicKey": "-----BEGIN RSA PUBLIC KEY-----
\nMIIBCgKCAQEAxQMliqqm+N+smTfckt4t9Ab8/PaRLTWa1OfgtnYaimcNtP9O5xJx\nS7rRu1Md
MSTbXf6MSRNUjUJYK57HpGOIUlc8jPeqo7bEO27nhRYu/zWOsgkT4VS5\nALDLh85U87Z7OsTTpn
UhjBzGJltOTDToA4T51hd0BNbFNQWWIia2dHukxzdiTiGo\n8gPkEEare7HVHBqdA6ET58jkk7dM
UbmgARpOvFjm1sgYQRdRRa0JvfZz0A6hEQA0\nqG18pqTAahyQ19k4N2AHROV7UcsblmIotJ+Lql
WhSJzcy5BWSy8bm6s4r5zJoDL1\ncyYUuvvaswaMobVTk5afysS7rRu2UW/9LwIDAQAB\n-----
END RSA PUBLIC KEY-----\n",
    "Manufacturer": "Hanwha Vision"
}
```

When **SupportedPublicKeyFormats** are specified in response, it's possible to get the rsa public key in different formats, default key format is PKCS1.

NOTE If in response **MaxPasswordLength** is not present, the maximum password length supported is 15. If present, it defines the maximum allowed password length.

Example request to get the certificate in X509 format.

REQUEST

```
http://<DeviceIP>/init-
cgi/pw_init.cgi?msubmenu=statuscheck&action=view&PublicKeyFormat=X509
```

RESPONSE

```
{
    "Initialized": false,
    "Language": "English",
    "MaxChannel": 2,
    "SpecialType": "none",
    "NewPasswordPolicy": true,
    "MaxPasswordLength" : 64,
    "SupportedPublicKeyFormats": [
        "PKCS1",
        "X509"
    ],
    "PublicKey": "-----BEGIN PUBLIC KEY-----
\nMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAxQMliqqm+N+smTfckt4t\n9Ab8/PaR
LTWa1OfgtnYaimcNtP9O5xJxS7rRu1MdMSTbXf6MSRNUjUJYK57HpGOI\nUlc8jPeqo7bEO27nhR
```

```
Yu/zWOsgkT4VS5ALDLh85U87Z7OsTTpnUhjBzGJltOTDTo\nA4T51hd0BNbFNQWWIia2dHukxzdi
TiGo8gPkEEare7HVHBqdA6ET58jkk7dMUbmg\nARpOvFjm1sgYQRdRRa0JvfZz0A6hEQA0qG18pq
TAahyQ19k4N2AHROV7UcsblmIo\ntJ+LqlWhSJzcy5BWSy8bm6s4r5zJoDL1cyYUuvvaswaMobVT
k5afysS7rRu2UW/9\nLwIDAQAB\n-----END PUBLIC KEY-----\n",
    "Manufacturer": "Hanwha Vision"
}
```


254.2. Checking the Install Wizard state in NVR

REQUEST

```
http://<ip>/init-
cgi/pw_init.cgi?msubmenu=statuscheck&action=view&ShowStage=True
```

If the NVR is already initialized, the response would be:

Stage field can take any of the following values "factoryreset, "installwizard", "installwizard_done

RESPONSE


```
{
    "Initialized": true,
    "Stage": "installwizard_done",
    "Language": "English",
    "MaxChannel": 1,
    "SpecialType": "none",
    "Manufacturer": "Hanwha Vision"
}
```


254.3. To set the initial password

Setting the initial password will work only once. If the password is already set, it will fail.

To set the password without password encryption:

REQUEST

```
http://<DeviceIP>/init-
cgi/pw_init.cgi?msubmenu=setinitpassword&action=set&Password=password!
```

NOTE The plain text initial password setting will soon be deprecated.

To set the password with password encryption, use the RSA key and follow the instructions in Chapter

Password Encryption

REQUEST (POST)

```
http://<DeviceIP>/init-
cgi/pw_init.cgi?msubmenu=setinitpassword&action=set&IsPasswordEncrypted=True
```

POST Payload

```
<base64 encoded encypted password as post content>
```

NOTE Until initial password is set, all the cgis (SUNAPI and ONVIF) will be disabled. They will be enabled immediately after setting the initial password.

Chapter 255. ONVIF Setup

After the initial password setup, ONVIF services would be enabled by default along with SUNAPI services. However, ONVIF has two kinds of operation mode in Hanwha Cameras, **Proprietary** and **ONVIF**. In **Proprietary** Mode, some of the ONVIF events will have custom namespace **tnssamsung**, when the operation mode is set to **ONVIF**, all the standard events defined in ONVIF will not have custom namespace.

To change the event scheme to **ONVIF** mode, client can send below request.

```
http://<DeviceIP>/stw-
cgi/eventstatus.cgi?msubmenu=eventscheme&action=set&Type=ONVIF
```

To change the event scheme to **Proprietary** mode, client can send below request.

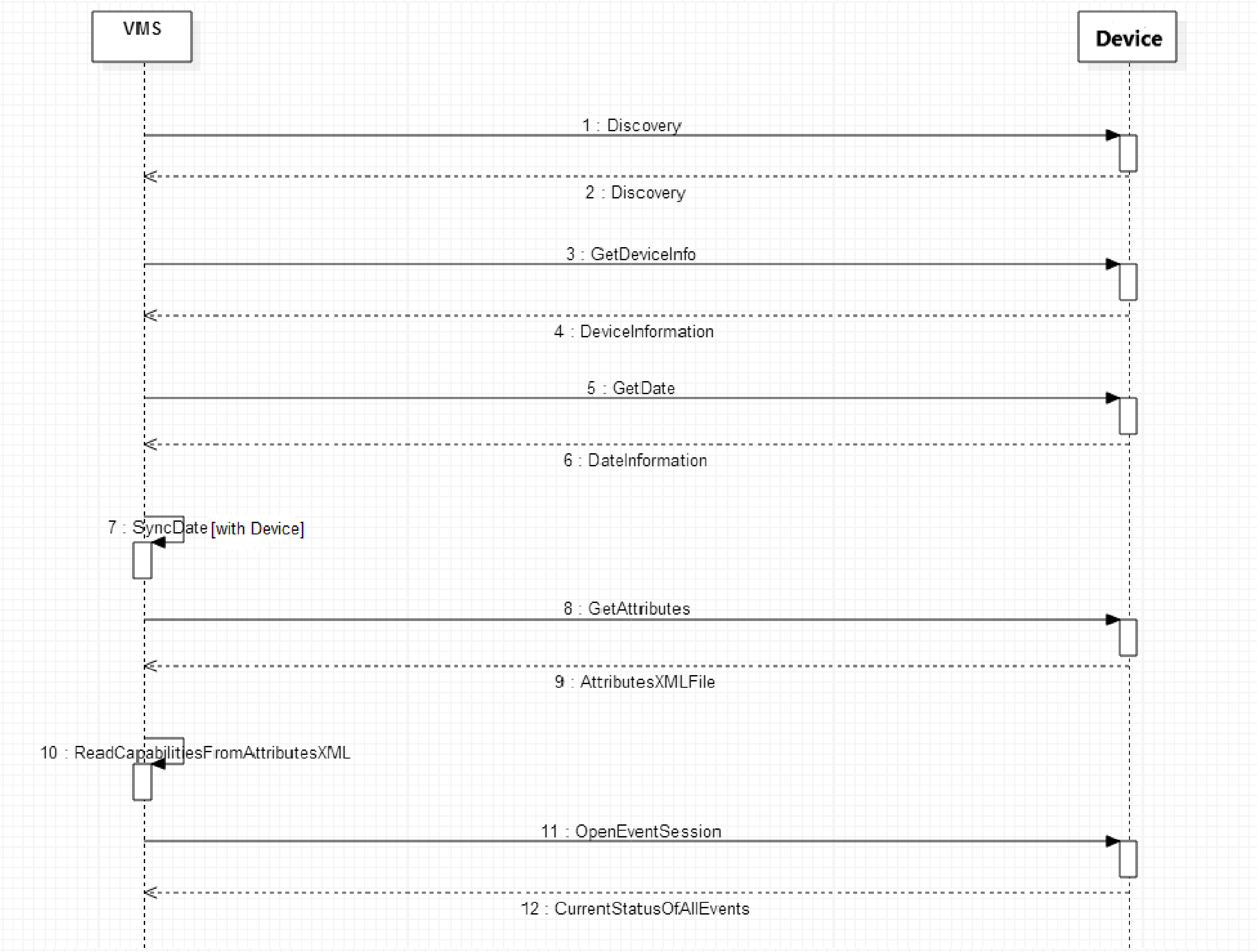
```
http://<DeviceIP>/stw-
cgi/eventstatus.cgi?msubmenu=eventscheme&action=set&Type=Proprietary
```

To check the current mode,

```
http://<DeviceIP>/stw-cgi/eventstatus.cgi?msubmenu=eventscheme&action=view
```

NOTE	As of now the **Proprietary** is the default mode. Clients using ONVIF protocols are recommended to change this mode to **ONVIF**.

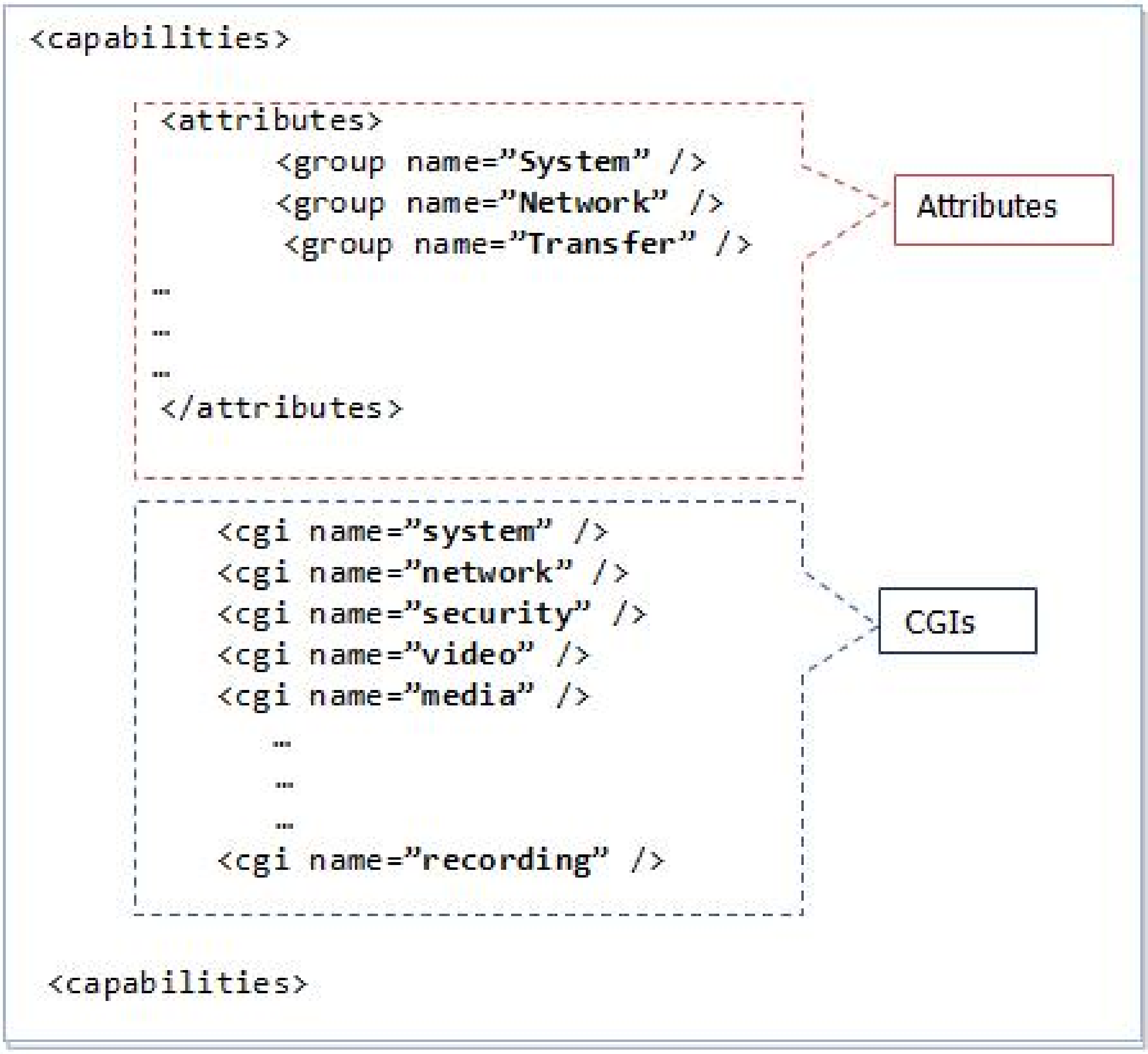
Chapter 256. Basic Setup



256.1. Attributes

Attributes XML contains two sections.

- **attributes**: Gives Information about the capabilities of the device.
  Ex: Max Channels, Max Alarm Inputs, Max Alarm Outputs etc.
- **cgis**: Gives Information about each submenu, action and parameters in SUNAPI commands.



Attributes will be changed dynamically based on the camera connection.

For more information on the attributes, please refer to [8] SUNAPI_attributes_2.6.8 in the References section.

256.2. Device Information

This command will return information about the device, such as model name, firmware version, language etc.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view
```

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Model": "PRN-4011",
    "FirmwareVersion": "v2.10_180329015157",
    "BuildDate": "2018.03.29",
    "WebURL": "http://www.hanwhasecurity.com",
    "DeviceType": "NVR",
    "ConnectedMACAddress": "00:09:18:30:97:01",
    "RequestedClientIPAddress": "192.168.71.43",
    "CGIVersion": "2.5.6",
    "MicomVersion": "36",
    "DeviceName": "PRN-4011",
    "Language": "English"
}
```

CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "Model": "PNM-C7083RVD",
    "SerialNumber": "SEP70GRC0000SY",
    "FirmwareVersion": "2.21.01_20220517_R276",
    "BuildDate": "2022.05.17",
    "WebURL": "http://www.hanwhavision.com/",
    "DeviceType": "NWC",
    "ConnectedMACAddress": "00:09:18:6E:12:B0",
    "ISPVersion": "1.00_220504",
    "BootloaderVersion": "       ",
    "CGIVersion": "2.6.1",
    "ONVIFVersion": "20.12",
    "DeviceName": "Camera",
    "DeviceLocation": "Location",
    "DeviceDescription": "Description",
    "Memo": "Memo",
    "Language": "English",
    "PasswordStrength": "Strong",
    "OpenSDKVersion": "4.02_220405",
    "FirmwareGroup": "PNM-C7083RVD"
```

```
}
```

256.3. Date Information

This command will return information about the device's Date settings, such as the Time zone, DST settings, etc. Please refer to [3] SUNAPI_system_2.6.8 in the References section.

NOTE	VMS Application has to sync the date and time with the Device.

REQUEST

```
http://<Device IP>/stw-cgi/system.cgi?msubmenu=date&action=view
```

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NTPLastUpdatedTime":   "2022-02-13 02:10:43",
    "LocalTime":    "2022-02-13 02:10:43",
    "UTCTime":  "2022-02-13 02:10:43",
    "SyncType": "Manual",
    "NTPURLList":   "203.248.240.140",
    "NTPStatus":    "Fail",
    "DSTEnable":    true,
    "POSIXTimeZone":    "STWT0STWST,M3.5.0/1:00:00,M10.5.0/1:00:00",
    "DateFormat":   "YYYY-MM-DD",
    "TimeFormat":   "HMS24"
}
```


CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "NTPURLList": [
```

```
        "pool.ntp.org",
        "asia.pool.ntp.org",
        "europe.pool.ntp.org",
        "north-america.pool.ntp.org",
        "time.nist.gov"
    ],
    "LocalTime": "2022-05-25 03:32:52",
    "UTCTime": "2022-05-25 03:32:52",
    "SyncType": "Manual",
    "DSTEnable": false,
    "TimeZoneIndex": 33,
    "POSIXTimeZone": "STWT0STWST,M3.5.0/1,M10.5.0"
}
```

256.4. Event Session

VMS application has to open an event session to receive the events from the Device.

NVR will send all events by channels, system events, alarm events, configuration change events, etc., to all connected VMS applications. Please refer to [7] SUNAPI_event_2.6.8 in the References section.

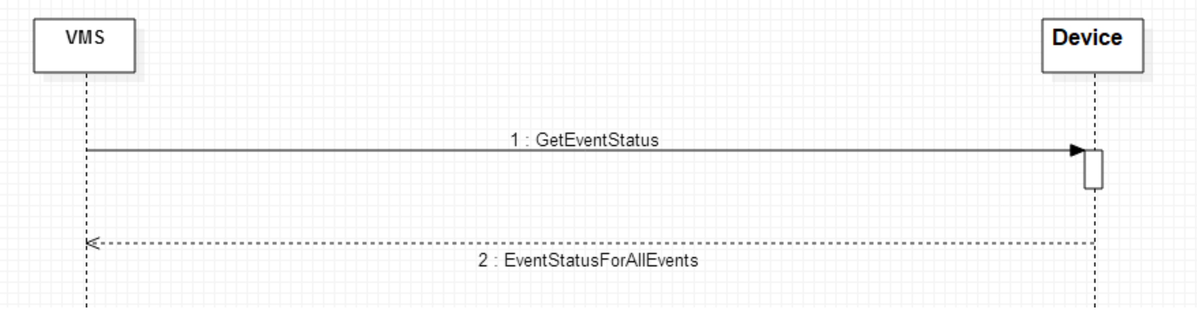
NOTE If the event is "ChangedConfigURI", VMS application has to update the corresponding configuration.

Ex: SNMP Configuration change

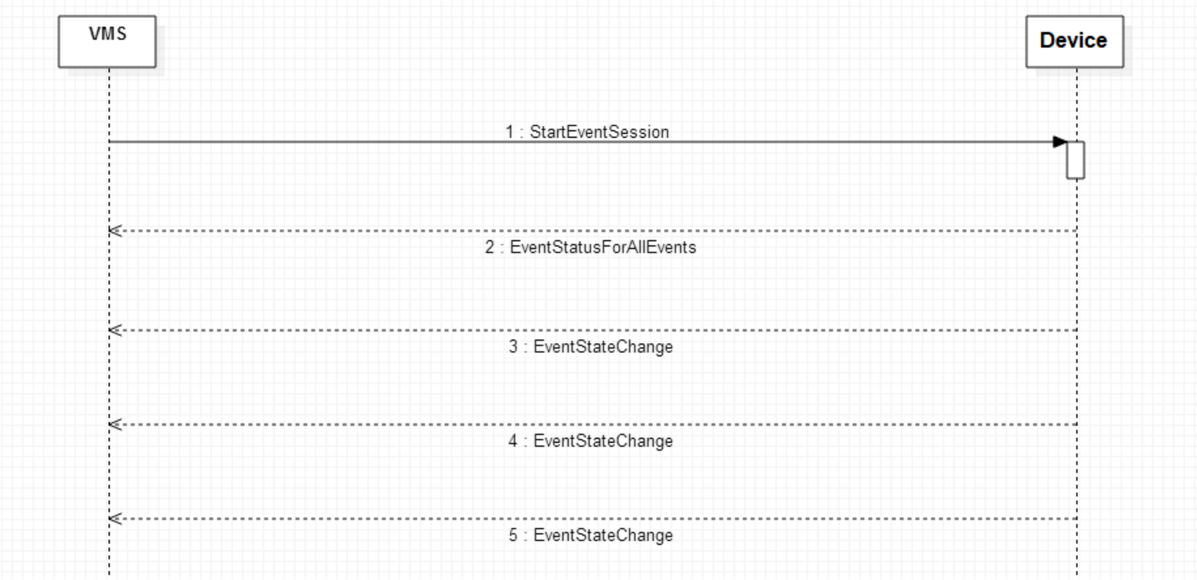
```
Timestamp=2015-05-08T02:18:59Z
SystemEvent.ConfigChange=True
ChangedConfigURI=network.cgi?msubmenu=snmp
```

Event Status has three actions –

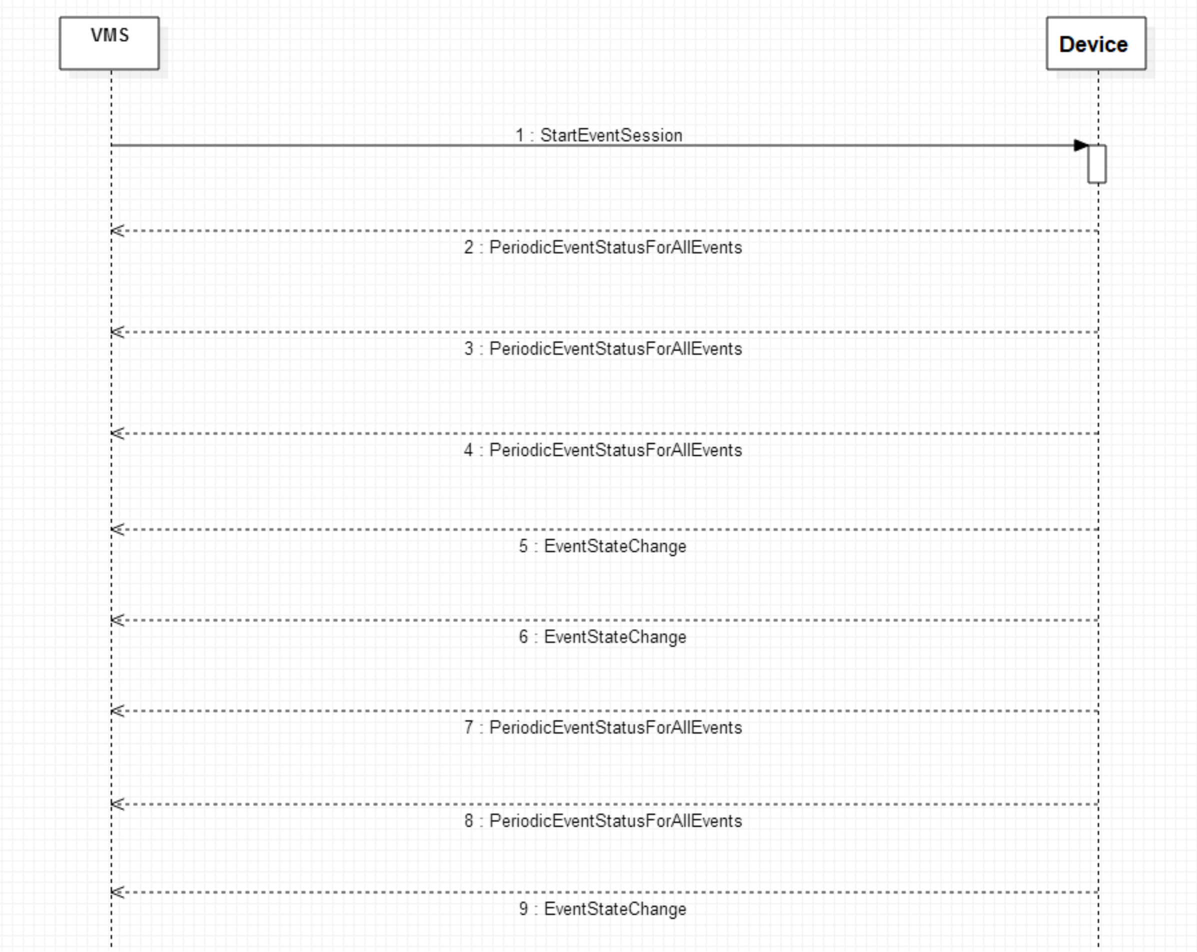
- Check : Gets the current status of all events



- MonitorDiff : Gets the event status whenever the state of the event changes



- Monitor: Gets the event status of all events periodically and when the state of the event changes



NOTE

For **monitor** and **monitordiff** actions, the connection is maintained, and there is a notification whenever an event occurs.
Please refer to Event document, on using **SchemaBased** event notification.
It is suggested to use **SchemaBased** event notification support starting from SUNAPI version 2.5.8 and beyond.

REQUEST

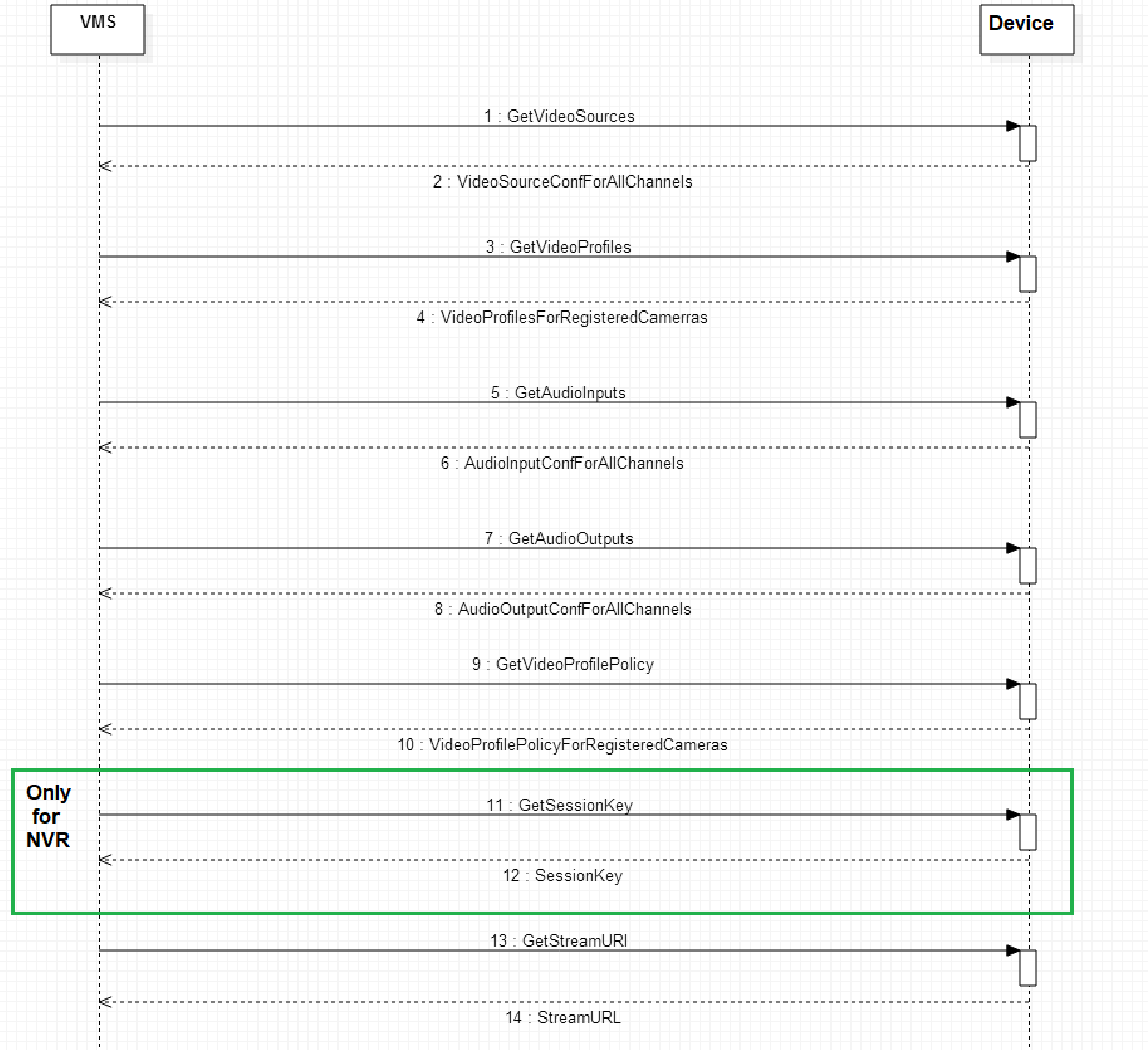
```
http://<DeviceIP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.NetworkCameraConnect=True
Channel.0.AMDStart=False
Channel.0.LowFps=False
Channel.0.Tampering=False
Channel.0.Videoloss=False
Channel.0.AudioDetection=False
Channel.0.NetworkAlarmInput=False
Channel.0.MotionDetection=False
Channel.0.FaceDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappearing=False
Channel.0.AudioAnalytics.Scream=False
Channel.0.AudioAnalytics.Gunshot=False
Channel.0.AudioAnalytics.Explosion=False
Channel.0.AudioAnalytics.GlassBreak=False
Channel.0.DefocusDetection=False
Channel.0.FogDetection=False
Channel.0.SDFail=False
Channel.0.SDFull=False
Channel.0.Tracking=False
```

Chapter 257. Live Stream Setup



257.1. Get Video Sources

This command will return information about all video sources.

For an NVR, it will also provide information on whether the camera is registered or not, whether video is enabled or not, etc. Refer to [4] in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videosource&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Type=NTSC
Channel.0.SensorCaptureSize=Unknown
Channel.0.Name=CAM 01
Channel.0.State=On
Channel.1.Type=NTSC
Channel.1.SensorCaptureSize=Unknown
Channel.1.Name=CAM 02
Channel.1.State=On
Channel.2.Type=NTSC
Channel.2.SensorCaptureSize=Unknown
Channel.2.Name=CAM 03
Channel.2.State=On
```

257.2. Get Video Profiles

This command will return information about all the video profiles.

For an NVR, we can get the video profile information of all registered cameras. Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofile&action=view
```

CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.RTPMulticastEnable=False
Channel.0.Profile.1.RTPMulticastType=IPV4
Channel.0.Profile.1.RTPMulticastAddress=
```

```
Channel.0.Profile.1.RTPMulticastAddressIPv6=
Channel.0.Profile.1.RTPMulticastPort=0
Channel.0.Profile.1.RTPMulticastTTL=5
Channel.0.Profile.1.CropEncodingEnable=False
Channel.0.Profile.1.CropAreaCoordinate=480,360,2080,1560
Channel.0.Profile.1.CropRatio=Manual
Channel.0.Profile.1.Resolution=2560x1920
Channel.0.Profile.1.FrameRate=1
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.Bitrate=6144
Channel.0.Profile.1.MJPEG.PriorityType=Bitrate
Channel.0.Profile.1.AudioInputEnable=False
Channel.0.Profile.1.ViewModeType=Overview
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.ProfileToken=DefaultProfile-01-0

Channel.0.Profile.2.Name=H.264
Channel.0.Profile.2.EncodingType=H264
Channel.0.Profile.2.RTPMulticastEnable=False
Channel.0.Profile.2.RTPMulticastType=IPV4
Channel.0.Profile.2.RTPMulticastAddress=
Channel.0.Profile.2.RTPMulticastAddressIPv6=
Channel.0.Profile.2.RTPMulticastPort=0
Channel.0.Profile.2.RTPMulticastTTL=5
Channel.0.Profile.2.CropEncodingEnable=False
Channel.0.Profile.2.CropAreaCoordinate=480,360,2080,1560
Channel.0.Profile.2.CropRatio=Manual
Channel.0.Profile.2.Resolution=2560x1920
Channel.0.Profile.2.FrameRate=30
Channel.0.Profile.2.CompressionLevel=10
Channel.0.Profile.2.Bitrate=7168
Channel.0.Profile.2.H264.BitrateControlType=VBR
Channel.0.Profile.2.H264.PriorityType=FrameRate
Channel.0.Profile.2.H264.GOVLength=60
Channel.0.Profile.2.H264.Profile=High
Channel.0.Profile.2.H264.EntropyCoding=CABAC
Channel.0.Profile.2.H264.SmartCodecEnable=False
Channel.0.Profile.2.H264.MaxGOVLength=240
Channel.0.Profile.2.H264.MinGOVLength=1
Channel.0.Profile.2.H264.DynamicFPSEnable=False
Channel.0.Profile.2.H264.MinDynamicFPS=1
```

```
Channel.0.Profile.2.H264.DynamicGOVEnable=False
Channel.0.Profile.2.H264.DynamicGOVLength=240
Channel.0.Profile.2.H264.MaxDynamicGOVLength=480
Channel.0.Profile.2.AudioInputEnable=False
Channel.0.Profile.2.ViewModeType=Overview
Channel.0.Profile.2.IsFixedProfile=True
Channel.0.Profile.2.IsDigitalPTZProfile=False
Channel.0.Profile.2.ProfileToken=DefaultProfile-02-0
Channel.0.Profile.2.IsFixedFrameRateProfile=False

Channel.0.Profile.3.Name=H.265
Channel.0.Profile.3.EncodingType=H265
Channel.0.Profile.3.RTPMulticastEnable=False
Channel.0.Profile.3.RTPMulticastType=IPV4
Channel.0.Profile.3.RTPMulticastAddress=
Channel.0.Profile.3.RTPMulticastAddressIPv6=
Channel.0.Profile.3.RTPMulticastPort=0
Channel.0.Profile.3.RTPMulticastTTL=5
Channel.0.Profile.3.CropEncodingEnable=False
Channel.0.Profile.3.CropAreaCoordinate=480,360,2080,1560
Channel.0.Profile.3.CropRatio=Manual
Channel.0.Profile.3.Resolution=2560x1920
Channel.0.Profile.3.FrameRate=30
Channel.0.Profile.3.CompressionLevel=10
Channel.0.Profile.3.Bitrate=4608
Channel.0.Profile.3.H265.BitrateControlType=VBR
Channel.0.Profile.3.H265.PriorityType=FrameRate
Channel.0.Profile.3.H265.GOVLength=60
Channel.0.Profile.3.H265.Profile=Main
Channel.0.Profile.3.H265.EntropyCoding=CABAC
Channel.0.Profile.3.H265.SmartCodecEnable=False
Channel.0.Profile.3.H265.MaxGOVLength=240
Channel.0.Profile.3.H265.MinGOVLength=1
Channel.0.Profile.3.H265.DynamicFPSEnable=False
Channel.0.Profile.3.H265.MinDynamicFPS=1
Channel.0.Profile.3.H265.DynamicGOVEnable=False
Channel.0.Profile.3.H265.DynamicGOVLength=240
Channel.0.Profile.3.H265.MaxDynamicGOVLength=480
Channel.0.Profile.3.AudioInputEnable=False
Channel.0.Profile.3.ViewModeType=Overview
Channel.0.Profile.3.IsFixedProfile=True
```

```
Channel.0.Profile.3.IsDigitalPTZProfile=False
Channel.0.Profile.3.ProfileToken=DefaultProfile-03-0
Channel.0.Profile.3.IsFixedFrameRateProfile=False

Channel.0.Profile.10.Name=MOBILE
Channel.0.Profile.10.EncodingType=H264
Channel.0.Profile.10.RTPMulticastEnable=False
Channel.0.Profile.10.RTPMulticastType=IPV4
Channel.0.Profile.10.RTPMulticastAddress=
Channel.0.Profile.10.RTPMulticastAddressIPv6=
Channel.0.Profile.10.RTPMulticastPort=0
Channel.0.Profile.10.RTPMulticastTTL=5
Channel.0.Profile.10.CropEncodingEnable=False
Channel.0.Profile.10.CropAreaCoordinate=480,360,2080,1560
Channel.0.Profile.10.CropRatio=Manual
Channel.0.Profile.10.Resolution=320x240
Channel.0.Profile.10.FrameRate=10
Channel.0.Profile.10.CompressionLevel=10
Channel.0.Profile.10.Bitrate=2048
Channel.0.Profile.10.H264.BitrateControlType=VBR
Channel.0.Profile.10.H264.PriorityType=FrameRate
Channel.0.Profile.10.H264.GOVLength=20
Channel.0.Profile.10.H264.Profile=High
Channel.0.Profile.10.H264.EntropyCoding=CABAC
Channel.0.Profile.10.H264.SmartCodecEnable=False
Channel.0.Profile.10.H264.MaxGOVLength=80
Channel.0.Profile.10.H264.MinGOVLength=1
Channel.0.Profile.10.H264.DynamicFPSEnable=False
Channel.0.Profile.10.H264.MinDynamicFPS=1
Channel.0.Profile.10.H264.DynamicGOVEnable=False
Channel.0.Profile.10.H264.DynamicGOVLength=80
Channel.0.Profile.10.H264.MaxDynamicGOVLength=160
Channel.0.Profile.10.AudioInputEnable=False
Channel.0.Profile.10.ViewModeType=Overview
Channel.0.Profile.10.IsFixedProfile=False
Channel.0.Profile.10.IsDigitalPTZProfile=False
Channel.0.Profile.10.ProfileToken=DefaultProfile-04-0
Channel.0.Profile.10.IsFixedFrameRateProfile=False
```

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Profile.1.IsFixedProfile=True
Channel.0.Profile.1.IsDigitalPTZProfile=False
Channel.0.Profile.1.Name=MJPEG
Channel.0.Profile.1.ProfileToken=Profile1
Channel.0.Profile.1.ViewModeIndex=0
Channel.0.Profile.1.ViewModeType=Overview
Channel.0.Profile.1.EncodingType=MJPEG
Channel.0.Profile.1.Bitrate=6144
Channel.0.Profile.1.Resolution=640x480
Channel.0.Profile.1.FrameRate=1
Channel.0.Profile.1.CompressionLevel=10
Channel.0.Profile.1.AudioInputEnable=True
Channel.0.Profile.2.IsFixedProfile=True
Channel.0.Profile.2.IsDigitalPTZProfile=False
Channel.0.Profile.2.Name=FisheyeView
Channel.0.Profile.2.ProfileToken=Profile2
Channel.0.Profile.2.ViewModeIndex=0
Channel.0.Profile.2.ViewModeType=Overview
Channel.0.Profile.2.EncodingType=H264
Channel.0.Profile.2.Bitrate=7280
Channel.0.Profile.2.H264.Profile=High
Channel.0.Profile.2.H264.BitrateControlType=VBR
Channel.0.Profile.2.Resolution=4000x3000
Channel.0.Profile.2.FrameRate=20
Channel.0.Profile.2.CompressionLevel=10
Channel.0.Profile.2.AudioInputEnable=True
Channel.0.Profile.3.IsFixedProfile=False
Channel.0.Profile.3.IsDigitalPTZProfile=True
Channel.0.Profile.3.Name=Dewarp1
Channel.0.Profile.3.ProfileToken=Profile3
Channel.0.Profile.3.ViewModeIndex=1
Channel.0.Profile.3.ViewModeType=QuadView
Channel.0.Profile.3.EncodingType=H264
Channel.0.Profile.3.Bitrate=5120
Channel.0.Profile.3.H264.Profile=High
```

```
Channel.0.Profile.3.H264.BitrateControlType=VBR
Channel.0.Profile.3.Resolution=2944x2208
Channel.0.Profile.3.FrameRate=20
Channel.0.Profile.3.CompressionLevel=10
Channel.0.Profile.3.AudioInputEnable=True
```

257.3. Get Audio Inputs

This command will return information about the audio input configuration, such as enabled statuses, encoding types, etc., for all the channels. Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audioinput&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=True
Channel.0.SampleRate=8000
Channel.0.Mode=Mono
Channel.0.EncodingType=G711
Channel.0.Bitrate=0
Channel.0.Gain=1
Channel.1.Enable=True
Channel.1.SampleRate=8000
Channel.1.Mode=Mono
Channel.1.EncodingType=G711
Channel.1.Bitrate=0
Channel.1.Gain=1
```

257.4. Get Audio Outputs

This command will return information about the audio talk configuration, such as the enabled/disabled statuses, decoding types, etc., for all the channels. Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=audiooutput&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Enable=False
Channel.1.Enable=False
Channel.2.Enable=True
Channel.2.UnitSize=8000
Channel.2.SampleRate=8000
Channel.2.Mode=Mono
Channel.2.DecodingType=G711
Channel.2.Bitrate=0
Channel.2.Gain=1
Channel.3.Enable=True
Channel.3.UnitSize=8000
Channel.3.SampleRate=8000
Channel.3.Mode=Mono
Channel.3.DecodingType=G711
Channel.3.Bitrate=0
Channel.3.Gain=1
```

257.5. Get Video Profile Policy

This command will return information about which profile is configured for what purpose. For a camera, we can check which profile is used as Default, Event Profile and Recording Profile. For an NVR, we can check which profile is used for Live, Recording and Network.

Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videoprofilepolicy&action=view
```

NVR RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
Channel.0.NetworkProfile=1
Channel.0.LiveProfile=5
Channel.0.RecordProfile=2
Channel.0.LiveMode=Auto
Channel.1.NetworkProfile=0
Channel.1.LiveProfile=8
Channel.1.RecordProfile=2
Channel.1.LiveMode=Auto
Channel.2.NetworkProfile=4
Channel.2.LiveProfile=3
Channel.2.RecordProfile=2
Channel.2.LiveMode=Auto
Channel.3.NetworkProfile=3
Channel.3.LiveProfile=4
Channel.3.RecordProfile=2
Channel.3.LiveMode=Auto
```

CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.DefaultProfile=2
Channel.0.EventProfile=1
Channel.0.RecordProfile=1
```

257.6. Get Session Key

This command will return the unique session key for Live, Playback, and Backup from NVR.

Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

NOTE This functionality is NVR specific

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=sessionkey&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
SessionKey=1519123
```

257.7. Get Stream URI For Live

This command will return the URL for getting live streams from the device.

Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

NVR REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&MediaType=Live&Mode=F
ull&ClientType=PC
```

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://<Device IP>:<RTSP Port>/LiveChannel/0/media.smp
```

CAMERA REQUEST

```
http://192.168.75.194/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&Profile=1&MediaType=L
ive&Mode=Full&StreamType=RTPUnicast&TransportProtocol=TCP&RTSPOverHTTP=False
```

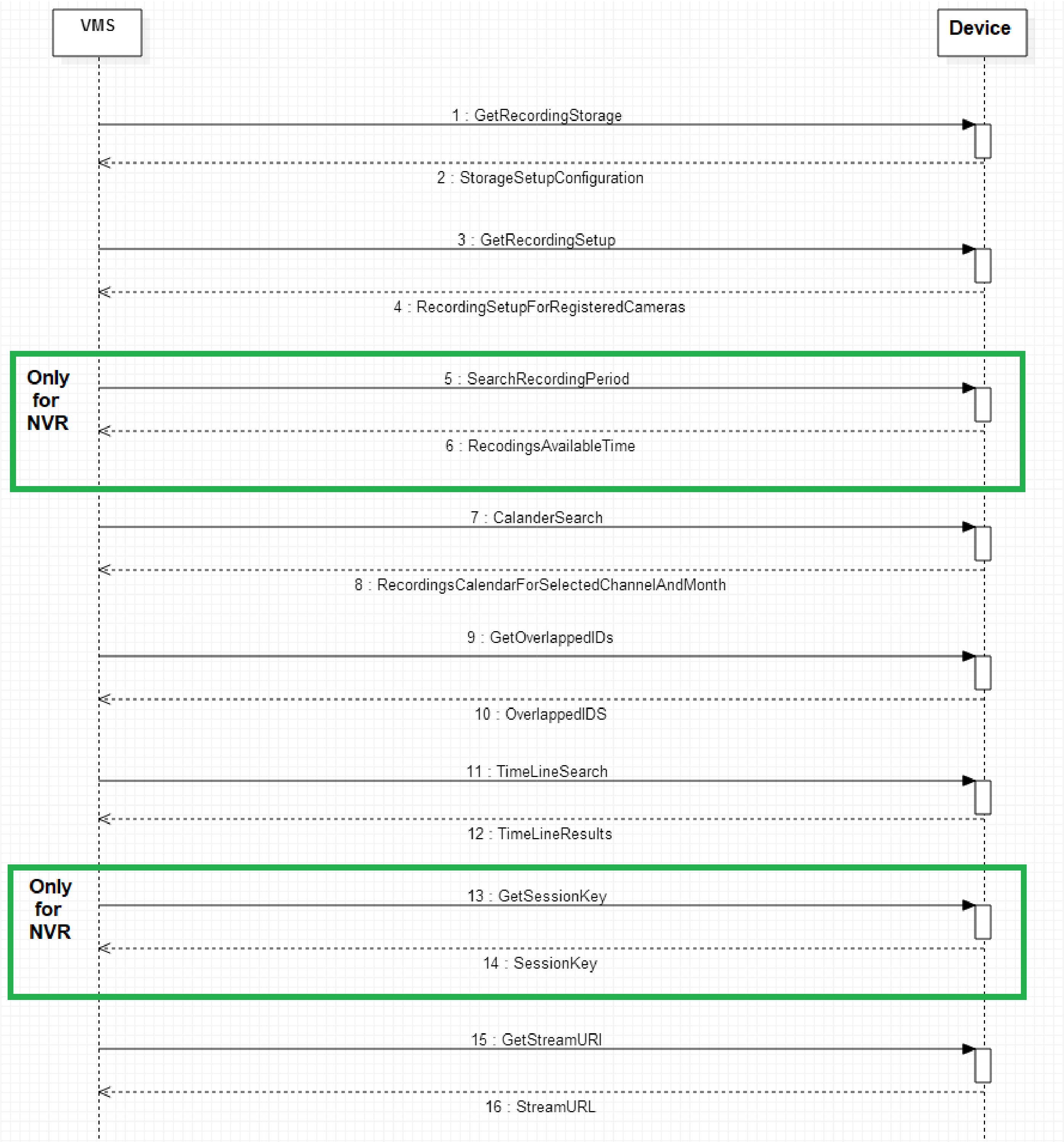
CAMERA RESPONSE

```
HTTP/1.0 200 OK
```

```
Content-type: text/plain
<Body>
```

```
URI=rtsp://192.168.75.194:554/0/profile1/media.smp
```

Chapter 258. Playback Setup



258.1. Get Storage Information

This command is used to get the current storage settings of the device.

Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=storage&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Enable=True
OverWrite=True
DiskEndBeep=False
AutoDeleteEnable=False
AutoDeleteDays=400
```

258.2. Get Recording Setup

This method will return the device's current recording configuration for all the channels. Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

Ex: current recording frame rate and recording bandwidth and codec information

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=general&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.FullFrameBandWidth=1.217640
Channel.0.FullFrameRate=19.980000
Channel.0.KeyFrameBandWidth=0.406219
Channel.0.KeyFrameRate=1.000000
Channel.0.Codec=H264
Channel.0.RecordOverlap=Normal,AlarmInput
Channel.0.SourceProfile=FisheyeView
Channel.0.NormalMode=Full
```

```
Channel.0.EventMode=I-Frame
Channel.0.PreEventDuration=Off
Channel.0.PostEventDuration=1m
Channel.0.Resolution=4000x3000
Channel.0.FrameRate=20
Channel.0.CompressionLevel=10
Channel.0.AudioEnable=False
Channel.0.BitrateLimit=148.000000
```

258.3. Search Recording Period

This command will return the overall recording duration in NVR. It retrieves the recording start and end time that are available from the storage.

NOTE This only applies to NVR

Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=searchrecordingperiod&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
StartTime=2018-03-19 11:32:13
EndTime=2018-04-19 13:03:45 DST
```

258.4. Calendar Search

This command is used to retrieve information on the availability of recordings for the selected month and channels. Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

The response is a 31-digit string, with a digit to represent each day of the month; if the digit is 0 then there is no recording for that channel on that day, and if it is 1 then recording is available for that day.

REQUEST

```
http://<Device IP>/stw-
```

```
cgi/recording.cgi?msubmenu=calendarsearch&action=view&Month=2015-05-
01T00:00:00Z&ChannelIdList=0,5
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Result=0001111000000000000000000000000
Channel.5.Result=0000011100000000000000000000000
```

258.5. Get Overlapped IDs

This command is used to get the recordings of overlapped information for the given time range.

If the system time settings changes or DST is applied while recording the video, recordings will be overlapped for certain period of time.

Eg: When the current recording time is 14:00:00 and the time in the set was changed to 10:00:00, the recording will have a 4-hour duration and two media tracks. To access these media individually we would need the overlapped ID information. This will be passed along with the playback RTSP URL and timeline search. Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

REQUEST

```
http://<Device IP>/stw-
cgi/recording.cgi?msubmenu=overlapped&action=view&FromDate=2018-03-
01T00:00:00Z&ToDate=2018-03-31T23:59:59Z
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
OverlappedIDList=36, 37
```

258.5.1. OverlapID - Behaviour of Camera

During the camera’s local recording, the local time is taken as a reference. A new Overlap ID is created when the time zone changes. Even when the time has changed multiple times, only one Overlap ID will be created. An Overlap ID is only created on a daily basis and will not be created after the current day. The latest Overlap ID is determined by the highest value.

258.5.2. OverlapID - Behaviour of NVR

In NVR UTC, time is taken as a reference for recording, therefore no overlap ID will be created when the time zone changes or DST is applied. NVR can create an overlap ID when time is changed backwards, either manually or through NTP sync. If a time shift backwards is over 5 secs, NVR creates a new overlap ID. Overlap ID is incremented each time a new overlap recording is created, and is maintained throughout the recording period and not on a daily basis.

258.6. Timeline Search

This command is used to get the recording timeline information for the specific period of time and for the specific channel. Please refer to [6] SUNAPI_recording_2.6.8 in the References section for more information.

If "SearchByUTCTime" **is set as true** in the attributes response, **then UTC time can be used for timeline search.**

If the request is sent with time in YYYY-MM-DDTHH:MM:SSZ format, then UTC time is used for search; if the time is in YYYY-MM-DDTHH:MM:SS format then local time is used.

REQUEST

```
http://<Device IP>/stw-cgi/recording.cgi?msubmenu=timeline&action=view&ChannelIDList=0&FromDate=2018-03-07T00:00:01Z&ToDate=2018-03-08T23:59:59Z
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Result.1.StartTime=2018-03-07T00:00:01Z
Channel.0.Result.1.EndTime=2018-03-07T01:56:03Z
Channel.0.Result.1.Type=Manual
Channel.0.Result.2.StartTime=2018-03-07T00:00:01Z
Channel.0.Result.2.EndTime=2018-03-07T01:56:03Z
Channel.0.Result.2.Type=Normal
Channel.0.Result.3.StartTime=2018-03-07T01:59:01Z
Channel.0.Result.3.EndTime=2018-03-07T02:07:30Z
Channel.0.Result.3.Type=Manual
```

```
Channel.0.Result.4.StartTime=2018-03-07T01:59:01Z
Channel.0.Result.4.EndTime=2018-03-07T02:07:30Z
Channel.0.Result.4.Type=Normal
Channel.0.Result.5.StartTime=2018-03-07T02:13:59Z
Channel.0.Result.5.EndTime=2018-03-07T02:15:13Z
Channel.0.Result.5.Type=Manual
Channel.0.Result.6.StartTime=2018-03-07T02:13:59Z
Channel.0.Result.6.EndTime=2018-03-07T02:15:13Z
Channel.0.Result.6.Type=Normal
Channel.0.Result.7.StartTime=2018-03-07T02:15:17Z
Channel.0.Result.7.EndTime=2018-03-07T04:52:53Z
Channel.0.Result.7.Type=Manual
Channel.0.Result.8.StartTime=2018-03-07T02:15:17Z
Channel.0.Result.8.EndTime=2018-03-07T04:52:53Z
Channel.0.Result.8.Type=Normal
TotalCount=8
```

258.7. Get Stream URI for Playback

This command will give the RTSP streaming URL in playback mode.

NOTE For a camera, the Playback and Backup URL are the same.

Please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

NVR REQUEST

```
http://<Device IP>/stw-
cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&MediaType=Search&Mode
=Full&ClientType=PC
```

NVR RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://<Device IP>:<RTSP Port>/PlaybackChannel/0/media.smp
```

CAMERA REQUEST

```
http://192.168.75.194/stw-cgi/media.cgi?msubmenu=streamuri&action=view&Channel=0&MediaType=Backup&Mode=Full&StreamType=RTPUnicast&TransportProtocol=TCP&RTSPOverHTTP=False&OverlappedID=0&Profile=1
```

CAMERA RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
URI=rtsp://192.168.75.194:554/0/recording/backup.smp
```

Chapter 259. PTZ Operation

PTZ operation can be performed using the PTZ CGI service. In this document we will discuss only basic PTZ functionality. Please refer to [5] SUNAPI_ptz_2.6.8 in the References section for more information.

259.1. Continuous Move

Pan operation can be performed as follows; in continuous move, the particular operation will continue until the stop command is sent.

REQUEST

```
http://<Device IP>/stw-cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Pan=5&Channel=5
```

Tilt Operation

REQUEST

```
http://<DeviceIP>/stw-cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Tilt=5&Channel=1
```

Zoom operation

REQUEST

```
http://<DeviceIP>/stw-cgi/ptzcontrol.cgi?msubmenu=continuous&action=control&Zoom=3&Channel=1
```

259.2. Stop

To stop all PTZ operation

REQUEST

```
http://<DeviceIP>/stw-cgi/ptzcontrol.cgi?msubmenu=stop&action=control&OperationType=All&Channel=0
```

259.3. Preset

To get preset information

REQUEST

```
http://<DeviceIP>/stw-
```

```
cgi/ptzcontrol.cgi?msubmenu=preset&action=view&Channel=0
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.Preset.1.Name=Preset1
Channel.0.Preset.2.Name=Preset2
```

To go to a particular preset

REQUEST

```
http://<DeviceIP>/stw-
cgi/ptzcontrol.cgi?msubmenu=preset&action=control&Channel=0&Preset=1
```

259.4. Identifying Capability

PTZ capability of a device can be identified using the attributes cgi.

259.4.1. Real PTZ

```
PTZSupport/Support/Absolute.Pan = true
PTZSupport/Support/Absolute.Tilt = true
PTZSupport/Support/Absolute.Zoom = true
PTZSupport/Support/DigitalPTZ = false
```

259.4.2. Zoom Only

```
PTZSupport/Support/Absolute.Pan = false
PTZSupport/Support/Absolute.Tilt = false
PTZSupport/Support/Absolute.Zoom = true
```

259.4.3. PTRZ

CGI section:

```
image/ptr/Pan/int = true
```

```
image/ptr/Tilt/int = true
image/ptr/Rotate/int = true
```

259.4.4. DPTZ

```
PTZSupport/Support/DigitalPTZ = true
PTZSupport/Limit/MaxGroupCount > 0
```

Profile-based DPTZ support:

If any of the below parameters is listed, DPTZ is only based on the selected profile. Otherwise, DPTZ is global and applicable to all profiles in the channel.

CGI section,

```
media/videoprofile/IsDigitalPTZProfile
```

259.4.5. External PTZ

```
PTZSupport/Support/Absolute.Pan = false
PTZSupport/Support/Absolute.Tilt = false
PTZSupport/Support/Absolute.Zoom = false
IO/Support/RS485 = true
PTZSupport/Limit/MaxGroupCount = 0
```

259.4.6. From SUNAPI 2.5.4

Explicit capability added to attribute section

```
PTZSupport/Support/ExternalPTZ=True
PTZSupport/Support/RealPTZ=True
PTZSupport/Support/ZoomOnly=True
Image/Support/PTRZ=true
```

Chapter 260. GPS Information

Mobile NVR supports getting the current GPS location using SUNAPI.

REQUEST

```
http://<DeviceIP>/stw-cgi/system.cgi?msubmenu=gps&action=view
```

RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Check=Periodically
Periodicity=1
GPSData=$GPRMC,hhmmss.ss,A,llll.ll,a,yyyyy.yy,a,x.x,x.x,ddmmyy,x.x,a*hh
```

Chapter 261. RTSP

261.1. RTSP Live Session

In the RTSP URL, channel information and session ID are important for NVR, while for camera channel information, the profile name or profile number is important.

Generally for NVR, after creating a session ID for live and getting the stream URI, we can establish a LIVE RTSP session.

Camera URL Format

[Type1]

```
rtsp://<Device IP>/<encoding>/media.smp
```

[Type2]

```
rtsp://<Device IP>/profile<no>/media.smp
```

[Type3]

```
rtsp://<Device IP>/multicast/<encoding>/media.smp
```

[Type4]

```
rtsp://<Device IP>/multicast/profile<no>/media.smp
```

[Type5]

```
rtsp://<Device IP>/<profile name>/media.smp
```

[Type6]

```
rtsp://<Device IP>/multicast/<profile name>/media.smp
```

Camera URL Format (multi source device)

[Type1]

```
rtsp://<Device IP>/<chid>/<encoding>/media.smp
```

[Type2]

```
rtsp://<Device IP>/<chid>/profile<no>/media.smp
```

[Type3]

```
rtsp://<Device IP>/<chid>/multicast/<encoding>/media.smp
```

[Type4]

```
rtsp://<Device IP>/<chid>/multicast/profile<no>/media.smp
```

[Type5]

```
rtsp://<Device IP>/<chid>/<profile name>/media.smp
```

[Type6]

```
rtsp://<Device IP>/<chid>/multicast/<profile name>/media.smp
```

NVR URL Format

[Type1]

```
rtsp://<Device IP>:558/LiveChannel/<chid>/media.smp
```

[Type2]

```
rtsp://<DeviceIP>:558/LiveChannel/<chid>/media.smp/session=<sid>
```

[Type3]

```
rtsp://<Device IP>:558/LiveChannel/<chid>/media.smp/multicast&session=<sid>
```

[Type4]

```
rtsp://<DeviceIP>:558/LiveChannel/<chid>/media.smp/iframe&multicast&session=
<sid>
```

[Type5]

```
rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/profile=<profileNo>&session=<sid>
```

[Type6]

```
rtsp://<Device
IP>:558/LiveChannel/<chid>/media.smp/ProfileUsage=<profileType>&session=<sid
>
```

NOTE	For NVR the default RTSP Port is 558.

In general, the following types of sessions are supported:

- Audio
- Video
- Metadata
- BackChannel

NOTE	In an NVR, all RTSP connections with the same session ID are considered to be a single session. (Eg: In 16-view mode, all of the 16 RTSP connections will have the same session ID). SessionId should be different for Live, Playback and Backup Sessions.

If the Audio talk feature is supported by a channel, client can open a new RTSP connection only with backchannel RTP session and send the audio data. This can be done dynamically, only when audio talk is required, because at one point of time only one client can access Audio talk for a channel.

Backchannel audio RTP session will be mentioned in the SDP only when the DESCRIBE request has

Require: www.onvif.org/ver20/backchannel as defined in the ONVIF streaming specifications. Please refer to [12] ONVIF Streaming Spec in the References section.

When Audio or Video configuration such as codec/resolution changes, the RTSP connection will be disconnected from the NVR and an event will be sent to the client regarding configuration change. At this point, it is the client’s responsibility to reconnect the RTSP session.

The device supports media transport through the following protocols

- TCP
- UDP
- HTTP

• HTTPS (When SSL is enabled in NVR)

• Multicast

For RTSP over HTTP and RTSP over HTTPS, port 80 and port 443 are used by default.

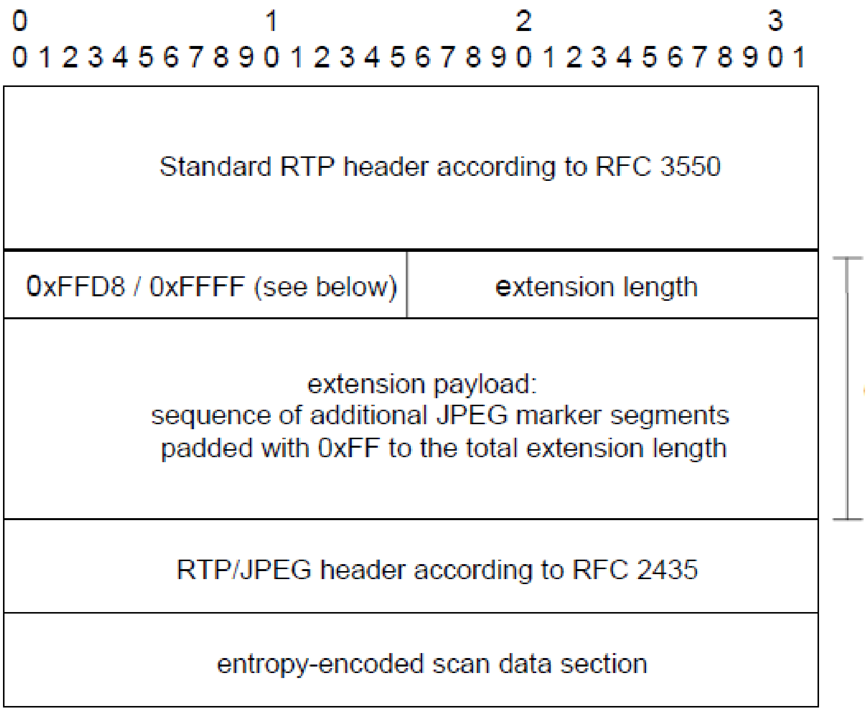
MJPEG Streaming Over 3 MP:

Since RFC 2435 does not support resolutions over 2040, we use a combination of the RTP extension and RFC 2435 for streaming MJPEG videos as described in the ONVIF streaming spec. Please refer to [12] ONVIF Streaming Spec in the References section.

<RTP HEADER> with extension flag set

<RTP Extension> → FFD8 start code, FFCO (SOF will have the height and width info)

<RFC 2435>



261.2. RTSP Playback Session

To initiate playback streaming and playback RTSP, the URL is required.

For NVR, session is also required. For NVR, when performing multichannel playback, the same session id can be used. Separate RTSP sessions are used to connect to each channel, and after sending the first play command, one RTSP session can be used to send commands like, PAUSE, PLAY etc.

NOTE Even though playback streaming can work with VLC player, we do not recommend using VLC player for testing playback streaming.

Camera URL format:

[Type1]

```
rtsp://<Device IP>/recording/<Start Time>/play.smp
```

[Type2]

```
rtsp://<Device IP>/recording/<Start Time>-<End Time>/play.smp
```

[Type3]

```
rtsp://<Device IP>/recording/play.smp
```

Camera URL format (multi source device)

[Type1]

```
rtsp://<Device IP>/<chid>/recording/<Start Time>/play.smp
```

[Type2]

```
rtsp://<Device IP>/<chid>/recording/<Start Time>-<End Time>/play.smp
```

[Type3]

```
rtsp://<Device IP>/<chid>/recording/<Start Time>-<End
Time>/OverlappedID=<overlapid>/play.smp
```

[Type4]

```
rtsp://<Device IP>/<chid>/recording/play.smp
```

NVR URL format

[Type1]

```
rtsp://<Device IP>:558/PlaybackChannel/<chid>/media.smp
```

[Type2]

```
rtsp://<Device IP>:558/PlaybackChannel/<chid>/media.smp/session=<sid>
```

[Type3]

```
rtsp://<DeviceIP>:558/PlaybackChannel/<chid>/media.smp/overlap=<id>&session=
<sid>
```

[Type4]

```
rtsp://<DeviceIP>:558/PlaybackChannel/<chid>/media.smp/overlap=<id>&session=
<sid>&iframe
```

[Type5]

```
rtsp://<DeviceIP>:558/PlaybackChannel/<chid>/media.smp/start=<starttime>&end
=<endtime>&overlap=<overlapid>&session=<sid>
```

In general all of the supported video formats (h264, MJPEG, MPEG4) and up to 5 audio formats are supplied in the RTSP DESCRIBE response as RTP sessions. Therefore, when the video/audio format changes in between recording, the playback session can still continue.

Ex: If the recording has h264 and mjpeg, initially h264 video will be sent over h264 rtp session; when the format changes to mjpeg, mjpeg rtp session will be used to send the media.

The date and time to play can be sent in two ways. It can be sent in the URL, or in the PLAY command with Range: clock field as defined in the ONVIF streaming specifications. Please refer to [12] ONVIF Streaming Spec in the References section.

The time should be specified in the following format:

- <YYYYMMDDTHHMMSS> (e.g., 20141206T111500) for local time and <YYYYMMDDTHHMMSSZ> (e.g., 20141206T110000Z) for UTC time on the NVR.
- <YYYYMMDDHHMMSS> (e.g., 20141206111500) for local time on the camera.

In playback mode, actual playback time of video can be received in two ways: one is through the RTCP, and another is using RTP Playback Extension header defined in ONVIF specifications. RTP Playback extension header will be sent only when client sends "Require: ONVIF-replay" in the setup and play commands.

Rate control can be sent in the play command, to notify whether video should be time-controlled on the NVR. If it is set to no as below, then receiver/client should control the timing.

The RTP Extension header in playback will follow the ONVIF streaming spec format. Please refer to [12] ONVIF Streaming Spec in the References section.

Table 3: RTP packet layout

<table>
<tr><td>V=
2</td><td>P</td><td>X=
1</td><td>CC</td><td>M</td><td>PT</td><td>sequence number</td></tr>
<tr><td colspan="7">timestamp</td></tr>
<tr><td colspan="7">synchronization source (SSRC) identifier</td></tr>
<tr><td colspan="6">0xABAC</td><td>length=3</td></tr>
<tr><td colspan="7">NTP timestamp…</td></tr>
<tr><td colspan="7">...NTP timestamp</td></tr>
<tr><td>C</td><td>E</td><td>D</td><td>mbz</td><td colspan="2">Cseq</td><td>padding</td></tr>
<tr><td colspan="7">payload…</td></tr>
</table>

When MJPEG over 3MP needs to be streamed, we can use the following format:

Table 4: RTP packet with JPEG header layout

<table>
<tr><td>V=
2</td><td>P</td><td>X=
1</td><td>CC</td><td>M</td><td>PT</td><td>sequence number</td></tr>
<tr><td colspan="7">timestamp</td></tr>
<tr><td colspan="7">synchronization source (SSRC) identifier</td></tr>
<tr><td colspan="6">0xABAC</td><td>length=N+4</td></tr>
<tr><td colspan="7">NTP timestamp…</td></tr>
<tr><td colspan="7">...NTP timestamp</td></tr>
<tr><td>C</td><td>E</td><td>D</td><td>mbz</td><td colspan="2">Cseq</td><td>padding</td></tr>
<tr><td colspan="6">0xFFD8</td><td>jpeglength=N</td></tr>
<tr><td colspan="7">extension payload: sequence of additional JPEG marker segments padded with 0xFF to the total extension length</td></tr>
<tr><td colspan="7">payload…</td></tr>
</table>

JPEG extension will have the same SOF information as described in the live case. We can use this SOF information in the final image we construct.

Rate-Control:

By default, rate-control is set to yes in playback mode. It is also defined in the ONVIF streaming spec.

Immediate:

Immediate field can be sent, along with play command as defined in the ONVIF streaming spec, to go to a particular time instantaneously.

261.2.1. Rewind/Fast-Forward

Rewind and Fast forward operation can be performed using the scale header defined in the RTSP specification.

```
PLAY rtsp://<Device IP>/PlaybackChannel/0/media.smp RTSP/1.0
Scale: 8
```

In the above example, the video will play at 8x speed in forward direction.

NOTE NVR supports -64x to 64x. Camera supports -8x to 8x.

When negative scale value is supplied, the video will play in reverse direction.

261.2.2. Slow Play

Slow play can be performed by specifying a scale header value between 0.1 and 0.9.

For example, if we specify the scale value as 0.5, then the video will be played at half the normal playback speed.

261.3. Backup Session

In SUNAPI, video backup is achieved using a backup RTSP session and is as follows:

NOTE Backup URL is only applicable to NVR

[Type1]

```
rtsp://<Device IP>:558/BackupChannel/<chid>/media.smp
```

[Type2]

```
rtsp://<Device IP>:558/BackupChannel/<chid>/media.smp/session=<sid>
```

[Type3]

```
rtsp://<DeviceIP>:558/BackupChannel/<chid>/media.smp/overlap=<id>&session=<s
id>
```

[Type4]

```
rtsp://<DeviceIP>:558/BackupChannel/<chid>/media.smp/overlap=<id>&session=<s
id>&iframe
```

[Type5]

```
rtsp://<DeviceIP>:558/BackupChannel/<chid>/media.smp/start=<starttime>&end=<
endtime>&overlap=<overlapid>&session=<sid>
```

A backup RTSP session is very similar to a playback session, but in backup mode the rate control is disabled by default, and therefore the media is sent rapidly.

Chapter 262. POS

This section explains how to achieve POS (Point of Sale) integration.

NOTE This section is applicable only to NVR

262.1. Capabilities

Get Max POS devices supported

REQUEST

```
http://<DeviceIP>/stw-cgi/attributes.cgi/attributes/System/Limit/MaxPOS
```

RESPONSE

```
<attribute accesslevel="*user*" value="*64*" type="*int*" name="*MaxPOS*"/>
```

Check whether device supports POS streaming or not

REQUEST

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Media/Limit/StreamingMetadata
```

RESPONSE

```
<attribute name="*StreamingMetadata*" accesslevel="*user*" value="*POS*"
type="*csv*"/>
```

Check whether channel supports Metadata streaming or not

REQUEST

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Media/Support/0/Stream.Metadata
```

RESPONSE

```
<attribute name="*Stream.Metadata*" accesslevel="*user*" value="*True*"
type="*bool*"/>
```

262.2. Configuration Setup

To get the POS configuration

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=posconf&action=view
```

REQUEST

```
http://<DeviceIP/stw-
cgi/recording.cgi?msubmenu=posconf&action=view&DeviceIDList=1,2
```

RESPONSE

```
DeviceID.1.DeviceName=TEXT 01
DeviceID.1.Enable=True
DeviceID.1.Port=7001
DeviceID.1.EventPlaybackStartTime=0
DeviceID.1.EventPlaybackStartTimeUnits=Seconds
DeviceID.1.ReceiptStart=(1)
DeviceID.1.ReceiptEnd=(2)
DeviceID.1.EncodingType=US-ASCII
DeviceID.1.ChannelIDList=0,1,2,3,4,5,6,7,16,17,18,19,20,21,22,23,32,33,34,35
,36,37,38,39,48,49,50,51,52,53,54,55
DeviceID.2.DeviceName=TEXT 02
DeviceID.2.Enable=True
DeviceID.2.Port=7002
DeviceID.2.EventPlaybackStartTime=0
DeviceID.2.EventPlaybackStartTimeUnits=Seconds
DeviceID.2.ReceiptStart=(1)
DeviceID.2.ReceiptEnd=(2)
DeviceID.2.EncodingType=US-ASCII
DeviceID.2.ChannelIDList=8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31,40,41
,42,43,44,45,46,47,56,57,58,59,60,61,62,63
```

To set the POS configuration

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=posconf&action=set&DeviceID=1&DeviceName=POS1&Ena
ble=True&Port=8001&EventPlaybackStartTime=10&ReceiptStart=Start&ReceiptEnd=E
```

```
nd&EncodingType=UTF-8&ChannelIDList=7,8,9,10
```

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=posconf&action=set&DeviceID=1&ChannelIDList=None
```

262.3. Event Setup

To get the POS events configuration

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=poseventconf&action=view
```

RESPONSE

```
AmountEventEnable=True
TotalAmount=100.000000
TotalType=Above
KeywordIndex.1.KeywordCondition=Apple
KeywordIndex.2.KeywordCondition=banana
```

To set POS event configuration

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=poseventconf&action=set&AmountEventEnable=False&T
otalAmount=9999999999.9988&TotalType=Below
```

To add event keywords

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=poseventconf&action=add&KeywordCondition=melon
```

To update the event keyword

REQUEST

```
http://<DeviceIP>/stw-
```

```
cgi/recording.cgi?msubmenu=poseventconf&action=update&KeywordIndex=2&Keyword
Condition=apple
```

To remove all event keywords

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=poseventconf&action=remove
```

To remove all particular event keywords

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=poseventconf&action=remove&KeywordIndex=2
```

262.4. Live POS Data

Similar to events, live POS data will be sent in a multi-part session. Client has to open a keep live session to receive the POS receipts. If any of the configured keywords are found in the receipt, it will be highlighted in the following format:

Ex: <keyword>APPLE</keyword>

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=posdata&action=monitordiff
```

RESPONSE

```
--<MultiPartBoundary>
Content-type:text/plain

ReceivedDate=2016-07-28T05:06:55Z
DeviceID=1
Receipt=
03-06-16 2:43P
<keyword>APPLE</keyword> 9.00
BERRY 3.50
MELON 10.50
PLUM 3.00

SUBTOTAL 26.00
```

```
TAX 03.00
TOTAL 29.00
CASH 30.00
CHANGE 01.00


--<MultiPartBoundary>
Content-type:text/plain
ReceivedDate=2016-07-28T05:06:55Z
DeviceID=0
Receipt=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00
```

Chapter 263. Metadata Search

NOTE NVR Only

263.1. Capabilities

Check whether the Metadata Search feature is supported or not

REQUEST

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Recording/Support/SearchMetadata
```

RESPONSE

```
<attribute name="SearchMetadata" accesslevel="admin" value="True"
type="bool"/>
```

Get the maximum allowed time gap between from date and to date

REQUEST

```
http://<DeviceIP>/stw-cgi/
attributes.cgi/attributes/Recording/Limit/MaxMetadataSearchDays
```

RESPONSE

```
<attribute accesslevel="admin" value="7" type="int"
name="MaxMetadataSearchDays"/>
```

Get the maximum supported value for MaxResults

REQUEST

```
http://<DeviceIP>/stw-cgi/attributes.cgi/recording/metadata/view/MaxResults
```

RESPONSE

```
http://55.101.54.147/stw-
cgi/attributes.cgi/recording/storage/set/Channel[<parameter
name="MaxResults" response="true" request="true"><dataType>]<int max="1000"
min="1"/></dataType></parameter>
```

263.2. Start Search

Request without any filters

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-13T00:00:00Z&ToDate=2016-07-16T23:59:59Z
```

Request with Overlapped ID

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11
```

Request with Overlapped ID and Single Keyword

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&Keyword=Apple
```

Request with Overlapped ID and Keyword Green or Apple

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&IsWholeWord=false&Keyword=Green%20Apple
```

Request with Overlapped ID and Keyword "Green Apple"

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=P
OS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-
16T23:59:59Z&OverlappedID=11&IsWholeWord=true&Keyword=Green%20Apple
```

Request with Overlapped ID and Keyword Green,Apple

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=POS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11&Keyword=Green,Apple
```

Request with Overlapped ID, Keyword and IsCaseSensitive

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=POS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11&Keyword=APPLE&IsCaseSensitive=true
```

Request with Overlapped ID, Keyword, IsCaseSensitive and Single DeviceID

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=POS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11&Keyword=OKRA&IsCaseSensitive=true&DeviceIDList=0
```

Request with Overlapped ID, Keyword, IsCaseSensitive and Multiple DeviceIDs

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Start&MetadataType=POS&FromDate=2016-07-15T00:00:00Z&ToDate=2016-07-16T23:59:59Z&OverlappedID=11&Keyword=OKRA&IsCaseSensitive=true&DeviceIDList=1,2
```

If search request is successful, Device will return a search token.

RESPONSE

```
SearchToken=7475
```

NOTE It is not possible to search for multiple keywords.
Ex: Search for Keyword1 and Keyword2 is not supported
Ex: Search for Keyword1 or Keyword2 is supported (By using space as delimiter)

263.3. Cancel Search

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Cancel&SearchToken=7475
```

263.4. Get Search Status

To get search status:

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=view&Type=Status&SearchToken=7475
```

263.5. Renew Search Token

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=control&Mode=Renew&SearchToken=7475
```

TEXT RESPONSE

```
OK
```

263.6. Get Search Results

To get the results of a search (Max 1000 results by default):

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=view&Type=Results&SearchToken=7475
```

To get the results of a search (First 100 results):

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=view&Type=Results&ResultFromIndex
```

```
=1&MaxResults=100&SearchToken=6619
```

TEXT RESPONSE

```
SearchTokenExpiryTime=2016-07-19T07:22:47Z
TotalResultsFound=399
TotalCount=100

Result.1.DeviceID=1
Result.1.Date=2016-07-18T07:28:01Z
Result.1.ChannelIDList=0,1,2,3,4,5,6,7
Result.1.KeywordsMatched=
Result.1.TextData=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00


SUBTOTAL 23.00
TAX 02.70
TOTAL 25.70
CASH 30.00
CHANGE 04.30

Result.2.DeviceID=2
Result.2.Date=2016-07-18T07:28:00Z
Result.2.ChannelIDList=8,9,10,11,12,13,14,15
Result.2.KeywordsMatched=
Result.2.TextData=
03-06-16 2:43P
APPLE 9.00
BERRY 3.50
MELON 10.50
PLUM 3.00


SUBTOTAL 26.00
TAX 03.00
```

```
TOTAL 29.00
CASH 30.00
CHANGE 01.00

Result.3.DeviceID=1
Result.3.Date=2016-07-18T07:27:56Z
Result.3.ChannelIDList=0,1,2,3,4,5,6,7
Result.3.KeywordsMatched=
Result.3.TextData=
02-06-16 2:43P
OKRA 5.00
OIL 9.50
LEMON 2.50
GREEN BANANNAS 3.00
YELLOW BANANNAS 3.00
```

To get the results of a search (Next 100 results):

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=metadata&action=view&Type=Results&ResultFromIndex=101&MaxResults=100&SearchToken=6619
```

NOTE

Search Token will expire in 60 seconds.
Client has to send Renew command periodically to increase the expiry time to 60 seconds more.

Chapter 264. Bypass

This section explains how the bypass feature can be used in NVR to send commands directly to a camera registered in an NVR.

NOTE NVR Only

Check whether channel is registered with SUNAPI

```
http://<NVR-IP>/stw-
cgi/attributes.cgi/attributes/Media/Support/0/Protocol.SUNAPI
```

```
<attribute accesslevel="user" value="True" type="bool"
name="Protocol.SUNAPI"/>
```

Normal get-set commands

REQUEST

```
http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=<ID>&BypassURI=<URI>
```

Configuration backup

REQUEST

```
curl --digest -u admin:7i8o9p0[ "http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/system.cgi?msubmenu=configbackup&action=control" > config.bin
```

RESPONSE

```
Downloaded File from Camera
```

Snapshot

```
http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/video.cgi?msubmenu=snapshot&action=view&Channel=0
```

POST

Configuration restore

openssl base64 -in config.bin -out encoded.bin

REQUEST

```
curl --digest -u admin:7i8o9p0[ "http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/system.cgi?msubmenu=configrestore&action=control&ExcludeSettings=Network
,Camera" -H "Expect:" --data-urlencode @encoded.bin
```

Firmware update

REQUEST

```
curl --digest -u admin:7i8o9p0[ "http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/system.cgi?msubmenu=firmwareupdate&action=control&Type=Normal" -H
"Expect:" -F uploadFile=@pkg_v2.00_150114103354.img
```

Sample requests and responses

REQUEST

```
http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

RESPONSE

```
Channel.0.Videoloss=False
Channel.0.AudioDetection=False
Channel.0.NetworkCameraConnect=True
Channel.0.NetworkAlarmInput=False
Channel.0.MotionDetection=False
Channel.0.FaceDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Disappearing=False
Channel.0.AMDStart=False
Channel.0.LowFps=False
```

```
Channel.0.Tampering=False
```

REQUEST

```
http://<NVR-IP>/stw-
cgi/bypass.cgi?msubmenu=bypass&action=control&Channel=2&BypassURI=/stw-
cgi/system.cgi?msubmenu=deviceinfo&action=view
```

RESPONSE

```
Model=XXXXXXXX
FirmwareVersion=XXXXXXXXXXX
BuildDate=XXXXXXXXXX
WebURL=XXXXXXXXXXX
DeviceType=XXXXXXXX
ConnectedMACAddress=XXXXXXXX
CGIVersion=XXXXX
MicomVersion=XXXXXXXXX
DeviceName=XXXXXXXXXXX
Language=XXXXXXXX
```

Chapter 265. Queue management

Check whether or not Queue Management feature is supported by device

REQUEST

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Recording/Support/QueueManagement
```

RESPONSE

```
<attribute accesslevel="admin" value="True" type="bool" name="QueueManagemen
t"/>
```

Get maximum Queues supported by device

REQUEST

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Eventsource/Limit/MaxQueues
```

RESPONSE

```
<attribute accesslevel="guest" value="3" type="int" name="MaxQueues"/>
```

Get Queue Management setup

REQUEST

```
http://<DeviceIP>/stw-
cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=view
```

JSON RESPONSE


```
{
    "QueueManagementSetup": [
        {
            "Channel": 0,
            "Enable": true,
            "ReportEnable": false,
            "ReportFilename": "",
            "ReportFileType": "XLS",
            "CalibrationMode": "CameraHeight",
```

```
            "CameraHeight": 300,
            "ObjectSizeCoordinates": [
                {
                    "x": 1316,
                    "y": 1316
                },
                {
                    "x": 1675,
                    "y": 1675
                }
            ],
            "Queues": [
                {
                    "Queue": 1,
                    "MaxPeople": 8,
                    "Name": "Queue1",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 1316,
                            "y": 1596
                        },
                        {
                            "x": 2991,
                            "y": 1596
                        }
                    ],
                    "QueueLevels": [
                        {
                            "Level": "High",
                            "Count": 6,
                            "AlarmEnable": true,
                            "Threshold": 180
                        },
                        {
                            "Level": "Medium",
                            "Count": 3,
                            "AlarmEnable": true,
                            "Threshold": 180
                        }
                    ]
```

```
                },
                {
                    "Queue": 2,
                    "MaxPeople": 8,
                    "Name": "Queue2",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 2316,
                            "y": 2596
                        },
                        {
                            "x": 3991,
                            "y": 2596
                        }
                    ],
                    "QueueLevels": [
                        {
                            "Level": "High",
                            "Count": 6,
                            "AlarmEnable": true,
                            "Threshold": 180
                        },
                        {
                            "Level": "Medium",
                            "Count": 3,
                            "AlarmEnable": true,
                            "Threshold": 180
                        }
                    ]
                }
            ]
        }
    ]
}
```

To change the Queue Management setup

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup
&action=set&Channel=0&Enable=True&CalibrationMode=CameraHeight&CameraHeight=
```

```
250
```

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=set&Channel=0&Enable=True&CalibrationMode=ObjectSize&ObjectSizeCoordinates=2992,1390,2,1390
```

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=set&Channel=0&ReportEnable=True&ReportFileName=QueueReport&ReportFileType=XLS
```

To change the Queue configuration

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=set&Channel=0&Queue.1.Name=Queue1&Queue.1.Enable=True&Queue.1.Coordinates=1316,1596,2991,1596&Queue.1.Level.High.Count=6&Queue.1.Level.High.AlarmEnable=True&Queue.1.Level.High.Threshold=180&Queue.1.Level.Medium.AlarmEnable=True&Queue.1.Level.Medium.Threshold=180&Queue.2.Name=Queue2&Queue.2.Enable=True&Queue.2.Coordinates=2316,2596,3991,2596&Queue.2.Level.High.Count=5&Queue.2.Level.High.AlarmEnable=True&Queue.2.Level.High.Threshold=150&Queue.2.Level.Medium.AlarmEnable=True&Queue.2.Level.Medium.Threshold=150
```

To get the current Queue levels of all Queues

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=check&Channel=0
```

JSON RESPONSE


```
{
    "QueueCount": [
        {
            "Channel": 0,
            "Queues": [
```

```
                {
                    "Queue": 1,
                    "Count": 8
                },
                {
                    "Queue": 2,
                    "Count": 15
                },
                {
                    "Queue": 3,
                    "Count": 25
                }
            ]
        }
    ]
}
```

To get the current Queue levels of selected Queues

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=queuemanagementsetup&action=check&Channel=0&QueueIndex=1,2
```

JSON RESPONSE


```
{
    "QueueCount": [
        {
            "Channel": 0,
            "Queues": [
                {
                    "Queue": 1,
                    "Count": 8
                },
                {
                    "Queue": 2,
                    "Count": 15
                }
            ]
        }
```

```
    ]
}
```

To get the scheduler/event action for Queue Management

REQUEST

```
http://<DeviceIP>/stw-
cgi/eventrules.cgi?msubmenu=scheduler&action=view&Type=QueueManagement
```

TEXT RESPONSE

```
Channel.0.QueueManagement.ScheduleType=Daily
Channel.0.QueueManagement.Hour=00
Channel.0.QueueManagement.Minute=00
Channel.0.QueueManagement.WeekDay=SUN
Channel.0.QueueManagement.EventAction= AlarmOutput.1,SMTP,FTP,
Channel.0.QueueManagement.AlarmOutput.1.Duration=5s
```

JSON RESPONSE

```
{
    "QueueManagement": [
        {
            "Channel": 0,
            "ScheduleType": "Daily",
            "Hour": 0,
            "Minute": 0,
            "WeekDay": "SUN",
            "EventAction": [
                "AlarmOutput.1",
                "SMTP",
                "FTP"
            ],
            "AlarmOutputs": [
                {
                    "AlarmOutput": 1,
                    "Duration": "5s"
                }
            ]
        }
    ]
```

```
}
```

To update the scheduler/event action for Queue Management

REQUEST

```
http://<DeviceIP>/stw-cgi/eventrules.cgi?msubmenu=scheduler&action=set&Type=QueueManagement&ScheduleType=Weekly&WeekDay=MON
```

REQUEST

```
http://<DeviceIP>/stw-cgi/eventrules.cgi?msubmenu=scheduler&action=set&Type=QueueManagement&EventAction=AlarmOutput.1&AlarmOutput.1.Duration=20s
```

REQUEST

```
http://<DeviceIP>/stw-cgi/eventrules.cgi?msubmenu=scheduler&action=set&Type=QueueManagement&EventAction=FTP,SMTP
```

To get the supported event actions for Queue Management

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=sourceoptions&action=view
```

TEXT RESPONSE

```
EventSource.QueueManagement.EventAction=FTP,SMTP,AlarmOutput
```

JSON RESPONSE


```
{
    "EventSources": [
        {
            "EventSource": "QueueManagement",
            "EventAction": [
                "FTP",
                "SMTP",
```

```
                "AlarmOutput"
            ]
        }
    ]
}
```

To check the current Queue event status

REQUEST

```
http://<DeviceIP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&Channel.0.EventType=QueueEvent
```

TEXT RESPONSE (All events)

```
Channel.0.Queue.1.Level.High=True
Channel.0.Queue.1.Level.Medium=False
Channel.0.Queue.2.Level.High=False
Channel.0.Queue.2.Level.Medium=False
```

JSON RESPONSE (All events)

```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "QueueEvents": {
                "Queues": [
                    {
                        "Queue": 1,
                        "QueueLevels": [
                            {
                                "High": true
                            },
                            {
                                "Medium": false
                            }
                        ]
                    },
                    {
                        "Queue": 2,
```

```
                        "QueueLevels": [
                            {
                                "High": false
                            },
                            {
                                "Medium": false
                            }
                        ]
                    }
                ]
            }
        }
    ]
}
```


To monitor the status of Queue events

REQUEST

```
http://<DeviceIP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitor&Channel.0.EventType=QueueEvent
```

REQUEST

```
http://<DeviceIP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff&Channel.0.EventType=QueueEvent
```

TEXT RESPONSE (Single event)

```
Channel.0.Queue.1.Level.High=True
```

JSON RESPONSE (Single event)


```
{
    "ChannelEvent": [
        {
            "Channel": 0,
            "QueueEvents": {
                "Queues": [
                    {
```

```
                        "Queue": 1,
                        "QueueLevels": [
                            {
                                "High": true
                            }
                        ]
                    }
                ]
            }
        }
    ]
}
```

To start a Queue search

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.AveragePeople=True&Queue.2.AveragePeople=True&Queue.3.AveragePeople=True
```

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.Type.High.CumulativeTime=True&Queue.1.Type.Medium.CumulativeTime=True&Queue.2.Type.High.CumulativeTime=True&Queue.2.Type.Medium.CumulativeTime=True&Queue.3.Type.High.CumulativeTime=True&Queue.3.Type.Medium.CumulativeTime=True
```

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Start&FromDate=2017-01-17T00:00:00Z&ToDate=2017-01-17T23:59:59Z&Queue.1.AveragePeople=True&Queue.2.AveragePeople=True&Queue.3.AveragePeople=True&Queue.1.Type.High.CumulativeTime=True&Queue.1.Type.Medium.CumulativeTime=True&Queue.2.Type.High.CumulativeTime=True&Queue.2.Type.Medium.CumulativeTime=True&Queue.3.Type.High.CumulativeTime=True&Queue.3.Type.Med
```

```
ium.CumulativeTime=True
```

JSON RESPONSE


```
{
    "SearchToken": "123456"
}
```


To cancel a Queue search

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=queuesearch&action=control&Channel=0&Mode=Cancel
```

JSON RESPONSE


```
{
    "Response": "Success"
}
```


To get status of a Queue search

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Status&SearchToken=1
23456
```

JSON RESPONSE


```
{
    "Status": "Completed"
}
```


To get the results of a Queue search for average People

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Results&SearchToken=
123456
```

JSON RESPONSE


```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
        {
            "Queue": 1,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        },
        {
            "Queue": 2,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
```

```
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        },
        {
            "Queue": 3,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
```

```
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        }
    ]
}
```

To get the results of a Queue search for Cumulative Time

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=queuesearch&action=view&Type=Results&SearchToken=123456
```

JSON RESPONSE


```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
        {
            "Queue": 1,
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0",
                        "1",
                        "2",
                        "3",
                        "4",
                        "5",
                        "6",
                        "7",
                        "8",
                        "9",
                        "10",
```

```
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
                    "23"
                ]
            },
            {
                "Level": "Medium",
                "CumulativeTimeResult": [
                    "0",
                    "1",
                    "2",
                    "3",
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
```

```
                    "23"
                ]
            }
        ]
    },
    {
        "Queue": 2,
        "QueueLevels": [
            {
                "Level": "High",
                "CumulativeTimeResult": [
                    "0",
                    "1",
                    "2",
                    "3",
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
                    "23"
                ]
            },
            {
                "Level": "Medium",
                "CumulativeTimeResult": [
                    "0",
```

```
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        }
    ]
},
{
    "Queue": 3,
    "QueueLevels": [
        {
            "Level": "High",
            "CumulativeTimeResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
```

```
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ]
        },
        {
            "Level": "Medium",
            "CumulativeTimeResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
```

```
                        "20",
                        "21",
                        "22",
                        "23"
                    ]
                }
            ]
        }
    ]
}
```

To get the results of a Queue search for Cumulative Time and Average People

JSON RESPONSE

```
{
    "ResultInterval": "Hourly",
    "QueueResults": [
        {
            "Queue": 1,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
```

```
                "21",
                "22",
                "23"
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0",
                        "1",
                        "2",
                        "3",
                        "4",
                        "5",
                        "6",
                        "7",
                        "8",
                        "9",
                        "10",
                        "11",
                        "12",
                        "13",
                        "14",
                        "15",
                        "16",
                        "17",
                        "18",
                        "19",
                        "20",
                        "21",
                        "22",
                        "23"
                    ]
                },
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0",
                        "1",
                        "2",
                        "3",
```

```
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
                    "23"
                ]
            }
        ]
    },
    {
        "Queue": 2,
        "AveragePeopleResult": [
            "0",
            "1",
            "2",
            "3",
            "4",
            "5",
            "6",
            "7",
            "8",
            "9",
            "10",
            "11",
            "12",
            "13",
```

```
                "14",
                "15",
                "16",
                "17",
                "18",
                "19",
                "20",
                "21",
                "22",
                "23"
            ],
            "QueueLevels": [
                {
                    "Level": "High",
                    "CumulativeTimeResult": [
                        "0",
                        "1",
                        "2",
                        "3",
                        "4",
                        "5",
                        "6",
                        "7",
                        "8",
                        "9",
                        "10",
                        "11",
                        "12",
                        "13",
                        "14",
                        "15",
                        "16",
                        "17",
                        "18",
                        "19",
                        "20",
                        "21",
                        "22",
                        "23"
                    ]
                },
```

```
                {
                    "Level": "Medium",
                    "CumulativeTimeResult": [
                        "0",
                        "1",
                        "2",
                        "3",
                        "4",
                        "5",
                        "6",
                        "7",
                        "8",
                        "9",
                        "10",
                        "11",
                        "12",
                        "13",
                        "14",
                        "15",
                        "16",
                        "17",
                        "18",
                        "19",
                        "20",
                        "21",
                        "22",
                        "23"
                    ]
                }
            ]
        },
        {
            "Queue": 3,
            "AveragePeopleResult": [
                "0",
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
```

```
            "7",
            "8",
            "9",
            "10",
            "11",
            "12",
            "13",
            "14",
            "15",
            "16",
            "17",
            "18",
            "19",
            "20",
            "21",
            "22",
            "23"
        ],
        "QueueLevels": [
            {
                "Level": "High",
                "CumulativeTimeResult": [
                    "0",
                    "1",
                    "2",
                    "3",
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
```

```
                    "19",
                    "20",
                    "21",
                    "22",
                    "23"
                ]
            },
            {
                "Level": "Medium",
                "CumulativeTimeResult": [
                    "0",
                    "1",
                    "2",
                    "3",
                    "4",
                    "5",
                    "6",
                    "7",
                    "8",
                    "9",
                    "10",
                    "11",
                    "12",
                    "13",
                    "14",
                    "15",
                    "16",
                    "17",
                    "18",
                    "19",
                    "20",
                    "21",
                    "22",
                    "23"
                ]
            }
        ]
    }
  ]
}
```

Chapter 266. People Count

Capabilities

```
http://<DeviceIP>/stw-
cgi/attributes.cgi/attributes/Recording/Support/PeopleCountSearch
```

```
<attribute name="PeopleCountSearch" accesslevel="user" value="True"
type="bool"/>
```

Get People Count configuration

REQUEST

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=view
```

TEXT RESPONSE

```
Channel.0.MasterName=PeopleCount-Master
Channel.0.Enable=True
Channel.0.ReportEnable=True
Channel.0.ReportFilename=peoplecountreport
Channel.0.ReportFileType=XLSX
Channel.0.CalibrationMode=CameraHeight
Channel.0.CameraHeight=300
Channel.0.ObjectSizeCoordinate=1316,1316,1675,1675
Channel.0.Line.1.Name=Gate1
Channel.0.Line.1.Enable=True
Channel.0.Line.1.Mode=LeftToRightIn
Channel.0.Line.1.Coordinate=1043,1875,2875,1943
Channel.0.Line.2.Name=Gate2
Channel.0.Line.2.Enable=True
Channel.0.Line.2.Mode=LeftToRightIn
Channel.0.Line.2.Coordinate=2912,893,1206,706
```

JSON RESPONSE

```
{
    "PeopleCount": [
        {
            "Channel": 0,
```

```
            "MasterName": "PeopleCount-Master",
            "Enable": true,
            "ReportEnable": true,
            "ReportFilename": "peoplecountreport",
            "ReportFileType": "XLSX",
            "CalibrationMode": "CameraHeight",
            "CameraHeight": 300,
            "ObjectSizeCoordinate": [
                {
                    "x": 1316,
                    "y": 1316
                },
                {
                    "x": 1675,
                    "y": 1675
                }
            ],
            "Lines": [
                {
                    "Line": 1,
                    "Mode": "LeftToRightIn",
                    "Name": "Gate1",
                    "Enable": true,
                    "Coordinates": [
                        {
                            "x": 1043,
                            "y": 1875
                        },
                        {
                            "x": 2875,
                            "y": 1943
                        }
                    ]
                },
                {
                    "Line": 2,
                    "Mode": "LeftToRightIn",
                    "Name": "Gate2",
                    "Enable": true,
                    "Coordinates": [
                        {
```

```
                            "x": 2912,
                            "y": 893
                        },
                        {
                            "x": 1206,
                            "y": 706
                        }
                    ]
                }
            ]
        }
    ]
}
```

To update the configuration

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=set&Channel=0&Enable=True&CalibrationMode=CameraHeight&CameraHeight=250
```

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=set&Channel=0&Enable=True&CalibrationMode=ObjectSize&ObjectSizeCoordinates=2992,1390,2,1390
```

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=set&Channel=0&Line.1.Name=FrontGate&Line.1.Enable=True&Line.1.Mode=LeftToRightIn&Line.1.Coordinates=1,2,3,4&Line.2.Name=BackGate&Line.2.Enable=True&Line.2.Mode=RightToLeftIn&Line.2.Coordinates=5,6,7,8
```

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=set&Channel=0&ReportEnable=True&ReportFileName=PeopleCountReport&ReportFileType=TXT
```

To remove exclude region

```
http://<DeviceIP>/stw-cgi/eventsources.cgi?msubmenu=peoplecount&action=remove&Channel=0&AreaIndex=
```

```
1
```

To check the live people count

```
http://<DeviceIP>/stw-
cgi/eventsources.cgi?msubmenu=peoplecount&action=check&Channel=0
```

JSON RESPONSE


```
{
    "PeopleCount": [
        {
            "Lines": [
                {
                    "LineIndex": 1,
                    "Name": "Gate1",
                    "InCount": 20,
                    "OutCount": 15
                },
                {
                    "LineIndex": 2,
                    "Name": "Gate2",
                    "InCount": 56,
                    "OutCount": 52
                }
            ]
        }
    ]
}
```


TEXT RESPONSE


```
Channel.0.LineIndex=1
Channel.0.LineIndex.1.Name=Gate1
Channel.0.LineIndex.1.InCount=20
Channel.0.LineIndex.1.OutCount=15
Channel.0.LineIndex=2
Channel.0.LineIndex.2.Name=Gate2
Channel.0.LineIndex.2.InCount=56
Channel.0.LineIndex.2.OutCount=52
```

To start a People Count search

People count search is an asynchronous search, initially search session is started as below, resulting search token is used later to check search status and to get the results.

> **NOTE**
> Here, PeopleCount-Master is **fixed** name for the camera whereas, Gate1 or Gate2 is actual name of the line in configuration.

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=peoplecountsearch&action=control&Channel=0&Mode=Start&FromDate=2016-07-01T00:00:00Z&ToDate=2016-07-01T23:59:59Z&Camera.PeopleCount-Master.Line.Gate1.Direction=In,Out&Camera.MasterCamera.Line.Gate2.Direction=In,Out
```

RESPONSE

```
{
    "SearchToken": "PeopleCount-2016-07-24T02:32:51-614"
}
```

To cancel the People Count search

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=peoplecountsearch&action=control&Mode=Cancel&SearchToken=PeopleCount-2016-07-24T02:32:51-614
```

RESPONSE

```
{
    "Response": "Success"
}
```

To get the status of a People Count search

REQUEST

```
http://<DeviceIP>/stw-cgi/recording.cgi?msubmenu=peoplecountsearch&action=view&Type=Status&SearchT
```

```
oken=PeopleCount-2016-07-24T02:32:51-614
```

RESPONSE


```
{
    "Status": "Completed"
}
```


To get the results of a People Count search

REQUEST

```
http://<DeviceIP>/stw-
cgi/recording.cgi?msubmenu=peoplecountsearch&action=view&Type=Results&Search
Token=PeopleCount-2016-07-24T02:32:51-614
```

JSON RESPONSE


```
{
    "ResultInterval": "Hourly",
    "PeopleCountSearchResults": [
        {
            "Camera": "PeopleCount-Master",
            "LineResults": [
                {
                    "Line": "Gate1",
                    "DirectionResults": [
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,2,0,0,0,0,0,0,0,6,0,0,0,0,0,3,0,2,2"
                        },
                        {
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0,0,2,0,5,3"
                        }
                    ]
                },
                {
                    "Line": "Gate2",
                    "DirectionResults": [
```

```
                        {
                            "Direction": "In",
                            "Result":
"0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,11,0,0,0"
                        },
                        {
                            "Direction": "Out",
                            "Result":
"0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0,0,11,0,3,2"
                        }
                    ]
                }
            ]
        }
    ]
}
```


TEXT RESPONSE

```
ResultInterval = Hourly
Camera.PeopleCount-Master.Line.Gate1.Direction.In.Result =
0,0,0,0,0,0,2,0,0,0,0,0,0,0,6,0,0,0,0,0,3,0,2,2
Camera.PeopleCount-Master.Line.Gate1.Direction.Out.Result =
0,0,0,0,0,0,1,0,0,0,0,0,0,0,2,0,0,0,0,0,2,0,5,3
Camera.PeopleCount-Master.Line.Gate1.Direction.In.Result =
0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,0,0,0,0,0,11,0,0,0
Camera.PeopleCount-Master.Line.Gate2.Direction.Out.Result =
0,0,0,0,0,0,2,0,0,1,1,0,0,1,6,0,0,0,0,0,11,0,3,2
```

Chapter 267. Thermal Camera Integration

NOTE The purpose of this section is to help quick integration; however, for detailed explanation of parameters, it is recommended to refer to the corresponding cgi documents

267.1. Attributes

In attributes cgi response, under Image and Support sections, you can check the below attributes for the thermal feature support:

```
<attribute accesslevel="guest" value="True" type="bool" name="ThermalFeature
s"/>
```

267.2. Color Palette Selection & Temperature Unit Selection

Supported Color Palettes:

```
WhiteHot, BlackHot, Rainbow, Custom, Sepia, Red, Iron
```

Supported Temperature Units:

```
Celsius, Fahrenheit
```

267.2.1. View

```
http://<Device IP>/stw-cgi/image.cgi?msubmenu=camera&action=view
```


```
{
    "Camera": [
        {
            "Channel": 0,
            "CompensationMode": "Off",
            "SSNREnable": true,
            "SSNRMode": "Manual",
            "SSNRLevel": 12,
            "SSNR2DLevel": 12,
            "SSNR3DLevel": 12,
            "ThermalColorPalette": "Rainbow",
```

```
            "TemperatureUnit": "Celsius",
            "DayNightAlarmIn": "SwitchToBWIfCloses",
            "WDRSeamlessTransition": "Off",
            "WDRLowLight": "Off",
            "WDRIRLEDEnable": "Off"
        }
    ]
}
```

267.2.2. Set Operation

To change the color palette

```
http://<Device IP>/stw-
cgi/image.cgi?msubmenu=camera&action=set&ThermalColorPalette=Sepia
```

267.3. Temperature Change Detection

267.3.1. Attributes

In the Eventsource Support section, check the following:

```
<attribute accesslevel="guest" value="True" type="bool" name="TemperatureCha
ngeDetection"/>
```

To get MAX ROI support, under the Eventsource Limit section, check the following:

```
<attribute accesslevel="guest" value="3" type="int" name="MaxTemperatureChan
geDetectionArea"/>
```

267.4. Configuring Temperature Change Detection

267.4.1. Options Command

This gives the supported gap both in Celsius and Fahrenheit.

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetectionoptions&action=view
```

```
{
```

```
    "TemperatureChangeDetectionOption": [
        {
            "Channel": 0,
            "SupportedGap": {
                "Celsius": "20,40,60,80,100",
                "Fahrenheit": "40,80,120,160,200"
            }
        }
    ]
}
```

267.4.2. Enable

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=set&Channel=
0&Enable=True
```

267.4.3. Set

Can set the reference temperature to Average, Maximum, or Minimum temperature in the ROI.

Example:

If Average temperature in the ROI changes more than 60 degrees over 11 secs, it will trigger an event.

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=set&Channel=
0&TemperatureChange.ROI.1.Mode=Average&TemperatureChange.ROI.1.Gap=60&Temper
atureChange.ROI.1.DetectionPeriod=11&TemperatureChange.ROI.1.Coordinates=142
,176,477,386
```

267.4.4. View

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=temperaturechangedetection&action=view
```

```
{
    "TemperatureChangeDetection": [
        {
            "Channel": 0,
```

```
            "Enable": true,
            "TemperatureChange": [
                {
                    "ROI": 1,
                    "Mode": "Average",
                    "Gap": 60,
                    "DetectionPeriod": 11,
                    "Coordinates": [
                        {
                            "x": 142,
                            "y": 176
                        },
                        {
                            "x": 477,
                            "y": 386
                        }
                    ]
                }
            ]
        }
    ]
}
```

267.5. TemperatureChange Detection Event Format

```
<wsnt:NotificationMessage>
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Vi
deoSource/tnssamsung:TemperatureChangeDetection</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2018-03-29T11:01:40.857Z">
            <tt:Source>
                <tt:SimpleItem Name="VideoSource" Value="VideoSourceToken-
01"/>
                <tt:SimpleItem Name="RuleName" Value="TemperatureChange-1"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="State" Value="true"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
```

```
</wsnt:NotificationMessage>
```

In radiometry-supported models like TNO-4030TR, the following additional submenus are supported.

267.6. Spot Temperature Reading

For reading the temperature of the screen coordinates.

```
http://<IP>/stw-
cgi/image.cgi?msubmenu=spottemperaturereading&action=view&Channel=0&ScreenRe
solution=640x480&ScreenCoordinates=334,216
```

Sample response


```
{
    "SpotTemperatureReading": [
        {
            "Channel": 0,
            "Temperature": 30,
            "Unit": "Celsius"
        }
    ]
}
```


267.7. BoxTemperatureDetection

Can configure a region to monitor avg, min, and max temperature within that region.

The **boxtemperaturedetection** submenu configures box temperature detection settings.

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=view&Channel=0
```


```
{
    "BoxTemperatureDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "ROIs": [
                {
                    "ROI": 1,
```

```
            "TemperatureType": "Average",
            "DetectionType": "Above",
            "ThresholdTemperature": 39,
            "Coordinates": [
                {
                    "x": 43,
                    "y": 23
                },
                {
                    "x": 274,
                    "y": 243
                }
            ],
            "Duration": 40,
            "NormalizedEmissivity": 27,
            "AreaOverlay": false,
            "AvgTemperatureOverlay": true,
            "MinTemperatureOverlay": true,
            "MaxTemperatureOverlay": true
        },
        {
            "ROI": 2,
            "TemperatureType": "Maximum",
            "DetectionType": "Increase",
            "ThresholdTemperature": 20,
            "Coordinates": [
                {
                    "x": 364,
                    "y": 42
                },
                {
                    "x": 556,
                    "y": 236
                }
            ],
            "Duration": 48,
            "NormalizedEmissivity": 40,
            "AreaOverlay": true,
            "AvgTemperatureOverlay": true,
            "MinTemperatureOverlay": true,
            "MaxTemperatureOverlay": false
```

```
                },
                {
                    "ROI": 3,
                    "TemperatureType": "Minimum",
                    "DetectionType": "Below",
                    "ThresholdTemperature": 5,
                    "Coordinates": [
                        {
                            "x": 319,
                            "y": 307
                        },
                        {
                            "x": 562,
                            "y": 451
                        }
                    ],
                    "Duration": 39,
                    "NormalizedEmissivity": 41,
                    "AreaOverlay": true,
                    "AvgTemperatureOverlay": false,
                    "MinTemperatureOverlay": true,
                    "MaxTemperatureOverlay": true
                }
            ]
        }
    ]
}
```

267.7.1. Changing Box Temperature Detection Settings

REQUEST

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=set&Channel=0&ROI.1.Coordinate=63,37,346,205&ROI.1.TemperatureType=Maximum&ROI.1.DetectionType=Above&ROI.1.ThresholdTemperature=10&ROI.1.Duration=26&ROI.1.NormalizedEmissivity=33&ROI.1.AreaOverlay=True&ROI.1.AvgTemperatureOverlay=True&ROI.1.MinTemperatureOverlay=True&ROI.1.MaxTemperatureOverlay=True
```

267.7.2. Removing Box Temperature Detection ROI Region 1

REQUEST

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetection&action=remove&ROIIndex
=1&Channel=0
```

267.7.3. BoxTemperatureDetectionOptions

```
http://<Device IP>/ stw-
cgi/eventsources.cgi?msubmenu=boxtemperaturedetectionoptions&action=view&Cha
nnel=0
```


```
{
    "BoxTemperatureDetectionOptions": [
        {
            "Channel": 0,
            "ThresholdTemperature": [
                {
                    "TemperatureType": "Above",
                    "Celsius": {
                        "Min": -20,
                        "Max": 130
                    },
                    "Fahrenheit": {
                        "Min": -4,
                        "Max": 266
                    }
                },
                {
                    "TemperatureType": "Below",
                    "Celsius": {
                        "Min": -20,
                        "Max": 130
                    },
                    "Fahrenheit": {
                        "Min": -4,
                        "Max": 266
                    }
                },
                {
```

```
                    "TemperatureType": "Increase",
                    "Celsius": {
                        "Min": 10,
                        "Max": 100
                    },
                    "Fahrenheit": {
                        "Min": 50,
                        "Max": 212
                    }
                },
                {
                    "TemperatureType": "Decrease",
                    "Celsius": {
                        "Min": 10,
                        "Max": 100
                    },
                    "Fahrenheit": {
                        "Min": 50,
                        "Max": 212
                    }
                }
            ]
        }
    ]
}
```

267.7.4. Box Temperature Metadata Reading (Available only as Metadata)

```
&lt;tt:MetadataStream&gt;
    &lt;tt:Event&gt;
        &lt;wsnt:NotificationMessage&gt;
            <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Vi
deoAnalytics/Radiometry/BoxTemperatureReading&lt;/wsnt:Topic&gt;
            &lt;wsnt:Message&gt;
                &lt;tt:Message UtcTime="2024-02-16T01:35:04.564Z"&gt;
                    &lt;tt:Source&gt;
                        <tt:SimpleItem Name="VideoSourceToken"
Value="VideoSourceToken-1"/>
                        <tt:SimpleItem
Name="VideoAnalyticsConfigurationToken" Value="VideoAnalyticsConfigToken-
```

```
1"/>
                        <tt:SimpleItem Name="AnalyticsModuleName"
Value="TemparetureDetectionModule-01"/>
                    </tt:Source>
                    <tt:Data>
                        <tt:ElementItem Name="Reading">
                            <ttr:BoxTemperatureReading ItemID="Z"
AreaName="name" MaxTemperature="324.5" MaxTemperatureCoordinatesX="0"
MaxTemperatureCoordinatesY="0" MinTemperature="303.1"
MinTemperatureCoordinatesX="0" MinTemperatureCoordinatesY="0"
AverageTemperature="305.4"/>
                        </tt:ElementItem>
                        <tt:SimpleItem Name="TimeStamp" Value="2024-02-
16T01:35:04.564Z"/>
                    </tt:Data>
                </tt:Message>
            </wsnt:Message>
        </wsnt:NotificationMessage>
    </tt:Event>
</tt:MetadataStream>
```

267.7.5. Box temperature Event

```
<wsnt:NotificationMessage>
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet
xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2
xmlns:tns1=http://www.onvif.org/ver10/topics
xmlns:tnssamsung=http://www.samsungcctv.com/2011/event/topics">tns1:RuleEngi
ne/Radiometry/TemperatureAlarm</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2016-03-31T00:15:58.421Z"
PropertyOperation="Initialized">
            <tt:Source>
                <tt:SimpleItem Name="VideoSourceToken"
Value="VideoSourceConfigToken-01-0"/>
                <tt:SimpleItem Name="VideoSourceConfigurationToken"
Value="cb4fbc38-e5f6-4ff0-b2e8-2e166b4414d1"/>
                <tt:SimpleItem Name="RuleName" Value="TemperatureDetection-
1"/>
                <tt:SimpleItem Name="AreaName" Value="Area-1"/>
```

```
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="AlarmActive" Value="false"/>
                <tt:SimpleItem Name="TimeStamp" Value="2022-01-
01T01:19:13.832Z"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

267.7.6. SUNAPI Event Status

Check

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

Monitor

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitor
```

Monitor diff

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff
```

The event would be delivered as below:

```
Channel.0.BoxTemperatureDetection=True
Channel.0.BoxTemperatureDetection.RegionID.1=True
```

Chapter 268. Dual Channel Thermal Camera Integration

268.1. Overview

268.1.1. Dual Channel

TNM-3620TDY has two channels, each looking in the same direction and filming the same scene, consisting of one visible channel and one thermal channel. The first channel is a general image channel, and the second channel is a thermal imaging channel.

268.1.2. Thermal Image Position Calibration

You can start a calibration process to compensate for the difference in image resolution and the minute position error between the two channels. Refer to image cgi stereosensorcalibrarion submenu.

268.1.3. Thermal Detection Mode

TNM-3620TDY has two different thermal detection modes: Body Temperature Detection mode and Normal mode. Other events are restricted while the camera is running in the Body Temperature Detection mode. The Normal mode provides the same way of representing events as before, such as Box Temperature Detection. If the thermal detection mode changes, the attributes of the camera will also change, and it can detect through the AttributeUpdate event. Refer to eventsources cgi thermaldetectionmode submenu.

Note that changing the thermal detection mode means changing to a completely different camera (because supported events and image settings change depending on the mode), so it is recommended to re-register your surveillance system.

268.2. Estimated Body Temperature Detection

268.2.1. Body Temperature Detection

The Body Temperature Detection mode is a mode in which the temperature in a specific area is measured by recognizing a person’s face. This event setting can only be set for the second channel, "thermal channel," but the event is triggered identically on both channels. Refer to eventsources cgi bodytemperaturedetection submenu for configuring the body temperature detection settings

268.2.2. Temperature Measurement Region Setting

The body temperature value is measured within the detected face area square. With the temperature measurement region setting, the user can adjust the size and position of the detected face area square to measure the body temperature value. Refer to eventsources cgi temperaturemeasurementregion submenu for changing the temperature measurement region settings

268.2.3. Improve Temperature Measurement Accuracy using blackbody device

TNM-3620TDY provides blackbody and radiometry settings. It is used to improve the accuracy of body temperature measurement. These settings work only when the camera is running in the Body

Temperature Detection mode. Refer to imagge cgi blackbodyconfig submenu for changing the settings.

268.2.4. Supported Events difference-based thermal detection mode

Supported events vary depending on the thermal detection mode set for the camera, and supported events are different for each channel; you can check image information for each channel using the following command:

http://<Device IP>/stw-cgi/attributes.cgi/attributes/Eventsource/Support

You can check whether the newly added BodyTemperatureDetection feature is supported with the following command:

```
http://<Device IP>/stw-
cgi/attributes.cgi/attributes/Eventsource/Support/[ChannelID]/
BodyTemperatureDetection
```

Supported Events	Normal Mode	Body Temperature Mode
Temperature detection	O	X
Motion detection	O	X
Tampering detection	O	X
IVA	O	X
Audio detection	O	X
Estimated body temperature detection	X	O

268.3. Sample ONVIF Event for Body temperature detection

NOTE Temperature measured in Kelvin

```
<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2">
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet
xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2
xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:RuleEngine/Detection/Body
Temperature</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-14T15:54:44Z"
xmlns:tt="https://www.onvif.org/ver10/schema/">
            <tt:Source>
```

```
                <tt:SimpleItem Name="VideoSource" Value="VideoSouceToken-
1"/>
                <tt:SimpleItem Name="RuleName"
Value="BodyTemperatureDetectionRule-1"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="State" Value="true"/>
                <tt:SimpleItem Name="ObjectID" Value="100023"/>
                <tt:SimpleItem Name="Temperature" Value="312.0"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

268.4. BodyTemperatureDetection SUNAPI event status example

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1=False
AlarmOutput.1=False
AlarmOutput.2=False
Channel.0.BodyTemperatureDetection=True
Channel.1.BodyTemperatureDetection=True
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
```

```
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=True
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AlarmInput": {
        "1": false
    },
    "AlarmOutput": {
        "1": false,
        "2": false
    },
    "ChannelEvent": [
        {
            "Channel": 0,
            "BodyTemperatureDetection": true
        },
        {
            "Channel": 1,
            "BodyTemperatureDetection": true
        }
    ],
    "SystemEvent": {
        "TimeChange": false,
        "PowerReboot": false,
        "FWUpdate": false,
        "FactoryReset": false,
        "ConfigurationBackup": false,
        "ConfigurationRestore": false,
```

```
        "ConfigChange": false,
        "SDFormat": false,
        "SDFail": false,
        "SDFull": false,
        "SDInsert": false,
        "SDRemove": true,
        "NASConnect": false,
        "NASDisconnect": true,
        "NASFail": false,
        "NASFull": false,
        "NASFormat": false
    }
}
```

268.5. BodyTemperatureDetection SUNAPI schema-based event status example

268.5.1. Getting event status schema of body temperature detection

REQUEST

```
http://<Device IP>/stw-cgi/ stw-
cgi/eventstatus.cgi?msubmenu=eventstatusschema&action=view&EventName=BodyTem
peratureDetection
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
EventStatus.1.Name=BodyTemperatureDetection
EventStatus.1.Schema.1.Name=Channel.<int>.BodyTemperatureDetection
EventStatus.1.Schema.1.Value=<boolean>
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "type": "array",
    "items": [
        {
            "type": "object",
            "properties": {
                "Time": {
                    "type": "string"
                },
                "EventName": {
                    "enum": [
                        "BodyTemperatureDetection"
                    ]
                },
                "Source": {
                    "type": "object",
                    "properties": {
                        "Channel": {
                            "type": "number"
                        },
                    }
                },
                "Data": {
                    "type": "object",
                    "properties": {
                        "State": {
                            "type": "boolean"
                        }
                    }
                }
            }
        }
    ]
}
```

268.5.2. Getting scheme-based event status

REQUEST

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&SchemaBased=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1=False
AlarmOutput.1=False
AlarmOutput.2=False
Channel.0.BodyTemperatureDetection=True
Channel.1.BodyTemperatureDetection=True
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=True
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```

```
{
    "EventStatus": [
        {
            "EventName": "AlarmOutput",
            "Time": "2020-08-20T12:03:08.454+00:00",
```

```
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "BodyTemperatureDetection",
            "Time": "2020-08-20T12:03:08.454+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "BodyTemperatureDetection",
            "Time": "2020-08-20T12:03:08.454+00:00",
            "Source": {
                "Channel": 1
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.TimeChange",
            "Time": "2020-08-20T12:03:08.454+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
    ……
    ]
}
```

268.6. Metadata format for body temperature detection

Follows ONVIF metadata format and temperature unit is in Kelvin.


```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
    xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
    xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2019-05-15T12:24:57.321">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0" />
                <tt:Scale x="0.000781" y="-0.001042" />
            </tt:Transformation>
            <tt:Object ObjectId="15" Parent="12">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0" />
                        <tt:CenterOfGravity x="31.0" y="218.0" />
                    </tt:Shape>
                    <tt:Class>
                        <tt:ClassCandidate>
                            <tt:Type>Face</tt:Type>
                            <tt:Likelihood>0.8</tt:Likelihood>
                        </tt:ClassCandidate>
                        <tt:Type Likelihood="0.8">Face</tt:Type>
                    </tt:Class>
                    <tt:HumanFace>
                        <fc:Temperature>311.75</fc:Temperature>
                    </tt:HumanFace>
                </tt:Appearance>
            </tt:Object>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

Chapter 269. AI Camera Integration

NOTE The purpose of this section is to help quick integration; however, for detailed explanation of parameters, it is recommended to refer to the corresponding cgi documents.

269.1. IVA Object Type Filter

NOTE If the filter values are not delivered, the filter would work as before. If the filter is set, only when the specified object type crosses the line or enters the area, an event will be triggered.

269.2. Line Rule

269.2.1. Set operation

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=videoanalysis2&action=set&Channel=0&Line.1.Coordinate=612,334,1815,1434&Line.1.Mode=Right&DetectionType=MDAndIV&Line.1.ObjectTypeFilter=Vehicle,Person&Line.1.RuleName=boundaryrule1
```

269.2.2. View


```
{
    "VideoAnalysis": [
        {
            "Channel": 0,
            "DetectionType": "MDAndIV",
            "SensitivityLevel": 100,
            "ObjectSizeByDetectionTypes": [
                {
                    "DetectionType": "MotionDetection",
                    "MinimumObjectSize": "0,0",
                    "MaximumObjectSize": "99,99",
                    "MinimumObjectSizeInPixels": "42,42",
                    "MaximumObjectSizeInPixels": "2560,1920",
                    "DetectionResultOverlay": false
                },
                {
                    "DetectionType": "IntelligentVideo",
                    "MinimumObjectSize": "5,7",
                    "MaximumObjectSize": "66,89",
```

```
                "MinimumObjectSizeInPixels": "173,173",
                "MaximumObjectSizeInPixels": "1728,1728",
                "DetectionResultOverlay": false
            }
        ],
        "ROIs": [
            {
                "ROI": 1,
                "Mode": "Inside",
                "SensitivityLevel": 1,
                "ThresholdLevel": 5,
                "Coordinates": [
                    {
                        "x": 0,
                        "y": 0
                    },
                    {
                        "x": 0,
                        "y": 1919
                    },
                    {
                        "x": 2559,
                        "y": 1919
                    },
                    {
                        "x": 2559,
                        "y": 0
                    }
                ],
                "HandoverIndex": 0,
                "Duration": 0
            }
        ],
        "Lines": [
            {
                "Line": 1,
                "Coordinates": [
                    {
                        "x": 612,
                        "y": 334
                    },
```

```
                    {
                        "x": 1815,
                        "y": 1434
                    }
                ],
                "Mode": "Right",
                "HandoverIndex": 0,
                "RuleName": "boundaryrule1",
                "ObjectTypeFilter": [
                    " Vehicle ",
                    " Person "
                ]
            }
        ],
        "DefinedAreas": [
            {
                "DefinedArea": 1,
                "Type": "Inside",
                "Mode": [],
                "Coordinates": [
                    {
                        "x": 1343,
                        "y": 548
                    },
                    {
                        "x": 1176,
                        "y": 932
                    },
                    {
                        "x": 1667,
                        "y": 1468
                    },
                    {
                        "x": 1843,
                        "y": 448
                    }
                ],
                "AppearanceDuration": 10,
                "LoiteringDuration": 10,
                "HandoverIndex": 0,
                "IntrusionDuration": 0
```

```
                }
            ]
        }
    ]
}
```

269.3. Area Rule

269.3.1. Set operation

```
http://<Device IP>/stw-cgi/eventsources.cgi?msubmenu=videoanalysis2&action=set&Channel=0&DefinedArea.1.Coordinate=488,638,1971,282,2335,998,1839,1618&DefinedArea.1.Type=Inside&DefinedArea.1.Mode=AppearDisappear,Entering,Exiting,Intrusion,Loitering&DefinedArea.1.AppearanceDuration=10&DefinedArea.1.LoiteringDuration=10&DefinedArea.1.IntrusionDuration=0&DefinedArea.1.ObjectTypeFilter=Vehicle,Person&DetectionType=MDAndIV&DefinedArea.1.RuleName=boundbox1
```

269.3.2. View

```
{
    "VideoAnalysis": [
        {
            "Channel": 0,
            "DetectionType": "MDAndIV",
            "SensitivityLevel": 100,
            "ObjectSizeByDetectionTypes": [
                {
                    "DetectionType": "MotionDetection",
                    "MinimumObjectSize": "0,0",
                    "MaximumObjectSize": "99,99",
                    "MinimumObjectSizeInPixels": "42,42",
                    "MaximumObjectSizeInPixels": "2560,1920",
                    "DetectionResultOverlay": false
                },
                {
                    "DetectionType": "IntelligentVideo",
                    "MinimumObjectSize": "5,7",
                    "MaximumObjectSize": "66,89",
                    "MinimumObjectSizeInPixels": "173,173",
                    "MaximumObjectSizeInPixels": "1728,1728",
```

```
                "DetectionResultOverlay": false
            }
        ],
        "ROIs": [
            {
                "ROI": 1,
                "Mode": "Inside",
                "SensitivityLevel": 1,
                "ThresholdLevel": 5,
                "Coordinates": [
                    {
                        "x": 0,
                        "y": 0
                    },
                    {
                        "x": 0,
                        "y": 1919
                    },
                    {
                        "x": 2559,
                        "y": 1919
                    },
                    {
                        "x": 2559,
                        "y": 0
                    }
                ],
                "HandoverIndex": 0,
                "Duration": 0
            }
        ],
        "Lines": [
            {
                "Line": 1,
                "Coordinates": [
                    {
                        "x": 612,
                        "y": 334
                    },
                    {
                        "x": 1815,
```

```
                        "y": 1434
                    }
                ],
                "Mode": "Right",
                "HandoverIndex": 0
            }
        ],
        "DefinedAreas": [
            {
                "DefinedArea": 1,
                "Type": "Inside",
                "Mode": [
                    "AppearDisappear",
                    "Entering",
                    "Exiting",
                    "Intrusion",
                    "Loitering"
                ],
                "Coordinates": [
                    {
                        "x": 488,
                        "y": 638
                    },
                    {
                        "x": 1971,
                        "y": 282
                    },
                    {
                        "x": 2335,
                        "y": 998
                    },
                    {
                        "x": 1839,
                        "y": 1618
                    }
                ],
                "AppearanceDuration": 10,
                "LoiteringDuration": 10,
                "HandoverIndex": 0,
                "IntrusionDuration": 0,
                "RuleName": "boundbox1",
```

```
                    "ObjectTypeFilter": [
                        " Vehicle ",
                        " Person "
                    ]
                }
            ]
        }
    ]
}
```


269.4. Object Detection Submenu

NOTE Only when Object detection or IVA is enabled, object metadata would be generated.

In **ObjectDetection** submenu, if no object types are selected, no event would be triggered and only metadata would be generated.

269.4.1. Set operation

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=objectdetection&action=set&Channel=0&ObjectTyp
es=Vehicle,Person,Face,LicensePlate&Sensitivity=50&Enable=True&ExcludeArea.1
.Coordinate=672,1002,1044,254,2291,326,2275,1662
```

269.4.2. View operation

```
http://<Device IP>/stw-
cgi/eventsources.cgi?msubmenu=objectdetection&action=view
```


```
{
    "ObjectDetection": [
        {
            "Channel": 0,
            "Enable": true,
            "Duration": 1,
            "Sensitivity": 80,
            "MinimumObjectSize": "4,7",
            "MaximumObjectSize": "50,89",
            "MinimumObjectSizeInPixels": "194,194",
            "MaximumObjectSizeInPixels": "1944,1944",
```

```
            "ObjectTypes": [
                "Person",
                "Vehicle",
                "Face",
                "LicensePlate"
            ],
            "ExcludeAreas": [
                {
                    "ExcludeArea": 1,
                    "Coordinates": [
                        {
                            "x": 1248,
                            "y": 502
                        },
                        {
                            "x": 3173,
                            "y": 502
                        },
                        {
                            "x": 3317,
                            "y": 1743
                        },
                        {
                            "x": 972,
                            "y": 1701
                        }
                    ]
                }
            ]
        }
    ]
}
```

269.5. Metaimagetransfer Submenu (BestShot Feature)

Used to enable the image sending feature in metadata

NOTE Object detection should be enabled for this functionality to work

269.5.1. View the current settings

```
http://IP/eventsources.cgi?msubmenu=metaimagetransfer&action=view
```


```
{
    " MetaImageTransfer ": [
        {
            "Channel": 0,
            "ObjectTypes": [
                "Vehicle",
                "Person",
                "Face",
                "LicensePlate"
            ],
            "ImageQuality": 75
        }
    ]
}
```


269.5.2. Set operation

```
http://IP/eventsources.cgi?msubmenu=metaimagetransfer&action=set&Channel=0&O
bjectTypes=Face,LicensePlate
```

269.6. Digital Auto Tracking

For setting the digital autotracking filter setting based on object types

NOTE Only Channel 1 supports this feature (Which is a DPTZ channel)

269.6.1. View

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=digitalautotracking&action=view
```


```
{
    "digitalautotracking": [
        {
            "Channel": 1,
            "ObjectTypeFilter": [
                "Person",
```

```
                "Vehicle"
            ]
        }
    ]
}
```

269.6.2. Set

```
http://<Device IP>/stw-
cgi/ptzconfig.cgi?msubmenu=digitalautotracking&action=set&Channel=1&ObjectTy
peFilter=Person,Vehicle
```

269.7. EventStatus Check

269.7.1. Object detection events

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

```
AlarmInput.1=False
AlarmOutput.1=False
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.FaceDetection=False
Channel.0.Tampering=False
Channel.0.AudioDetection=False
Channel.0.DefocusDetection=False
Channel.0.FogDetection=False
Channel.0.Profile.1.DigitalAutoTracking=False
Channel.0.Profile.2.DigitalAutoTracking=False
Channel.0.Profile.3.DigitalAutoTracking=False
Channel.0.Profile.4.DigitalAutoTracking=False
Channel.0.Profile.5.DigitalAutoTracking=False
Channel.0.Profile.6.DigitalAutoTracking=False
Channel.0.Profile.7.DigitalAutoTracking=False
Channel.0.Profile.8.DigitalAutoTracking=False
Channel.0.Profile.9.DigitalAutoTracking=False
Channel.0.Profile.10.DigitalAutoTracking=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Intrusion=False
Channel.0.VideoAnalytics.Entering=False
```

```
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=True
Channel.0.VideoAnalytics.Loitering=False
Channel.0.AudioAnalytics.Scream=False
Channel.0.AudioAnalytics.Gunshot=False
Channel.0.AudioAnalytics.Explosion=False
Channel.0.AudioAnalytics.GlassBreak=False
Channel.0.ObjectDetection=False
Channel.0.ObjectDetection.Person=False
Channel.0.ObjectDetection.Vehicle=False
Channel.0.ObjectDetection.Face=False
Channel.0.ObjectDetection.LicensePlate=False
Channel.0.Connected=True
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=False
SystemEvent.SDRemove=True
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
```

269.8. SchemaBased Dynamic Event format

269.8.1. Check

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&SchemaBased=True
```

269.8.2. Monitor

```
http://<Device IP>/stw-
```

```
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitor&SchemaBased=True
```

269.8.3. Monitor diff

```
http://<Device IP>/stw-
cgi/eventstatus.cgi?msubmenu=eventstatus&action=monitordiff&SchemaBased=True
```


```
{
    "EventName": "ObjectDetection",
    "Time": "2019-06-16T00:22:46.802+00:00",
    "Source": {
        "Channel": 0
    },
    "Data": {
        "State": true,
        "ObjectTypes": "Face, Vehicle"
    }
}
```


269.9. ONVIF/MetaEvent Notification (Based on ONVIF Draft)

Sample object detection event in metadata is shown below:


```
<tt:Event>
    <wsnt:NotificationMessage>
        <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet">tns1:Ru
leEngine/ObjectDetection/Object</wsnt:Topic>
        <wsnt:Message>
            <tt:Message UtcTime="2019-11-14T05:50:51.290Z"
PropertyOperation="Changed">
                <tt:Source>
                    <tt:SimpleItem Name="VideoSource"
Value="VideoSourceToken-0"/>
                    <tt:SimpleItem Name="RuleName"
Value="ObjectDetectionRule-1"/>
                </tt:Source>
                <tt:Data>
```

```
                    <tt:SimpleItem Name="ClassTypes" Value="Person
Vehicle"/>
                </tt:Data>
            </tt:Message>
        </wsnt:Message>
    </wsnt:NotificationMessage>
</tt:Event>
```

NOTE Whenever there is a change in detection types, ClassTypes field will be updated; if nothing is detected, an empty ClassType will be sent.

269.10. BestShot RTP Stream

To receive the bestshot image in RTP, please refer to [4] SUNAPI_video.audio_2.6.8 in the References section for more information.

269.11. Metadata Format

The supported attributes are shown in the table shown below:

NOTE Those marked in RED are not supported in the current release and have fixed values as marked in the table below.

	Objects	Attributes	Supported attributes items
Attributes	Person	Gender	Female, Male
		Upper(Color)	Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple (up to 2 colors at the same time)
		Lower(Color)	
		Upper(Clothing)	Long, Short **(always Long)**
		Lower(Clothing)	Long, Short **(always Long)**
		Hat	Wear Hat or Not **(always False)**
		Bag	Bag (If Bag is detected)
	Vehicle	Type	Car (Sedan/SUV/Van...etc), Bus, Truck, Motorcycle, Bicycle

	Objects	Attributes	Supported attributes items
		Color	Black, Gray, White, Red, Orange, Yellow, Green, Blue, Purple (up to 2 colors at the same time) **Note** Color information under VehicleInfo is deprecated, instead Appearance/Color information should be referred for vehicle color.
	Face	Gender	Female, Male
		Age	Young (0~19), Adult (20~44), Middle (45~64), Senior (65~)
		Mask	Wearing a mask or not
		Glasses	Wearing glasses or not
	Licenseplate		

269.11.1. Sample Meta Frame with all fields (Only for reference)

```
<tt:MetadataStream xmlns:tt="http://www.onvif.org/ver10/schema"
    xmlns:fc="http://www.onvif.org/ver20/analytics/humanface"
    xmlns:bd="http://www.onvif.org/ver20/analytics/humanbody">
    <tt:VideoAnalytics>
        <tt:Frame UtcTime="2019-05-15T12:24:57.321">
            <tt:Transformation>
                <tt:Translate x="-1.0" y="1.0" />
                <tt:Scale x="0.000781" y="-0.001042" />
            </tt:Transformation>
            <tt:Object ObjectId="15" Parent="12">
                <tt:Appearance>
                    <tt:Shape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0" />
                        <tt:CenterOfGravity x="31.0" y="218.0" />
                    </tt:Shape>
                    <tt:Color>
                        <tt:ColorCluster>
                            <tt:Color X="58" Y="105" Z="212" />
                            <tt:Covariance XX="7.2" YY="6" ZZ="3" />
                            <tt:Weight>90</tt:Weight>
                            <tt:ColorString>WHITE</tt:ColorString>
```

```
                    </tt:ColorCluster>
                    <tt:ColorCluster>
                        <tt:Color X="165" Y="44" Z="139" />
                        <tt:Covariance XX="4" YY="4" ZZ="4" />
                        <tt:Weight>5</tt:Weight>
                        <tt:ColorString>BLUE</tt:ColorString>
                    </tt:ColorCluster>
                </tt:Color>
                <tt:Class>
                    <tt:ClassCandidate>
                        <tt:Type>LicensePlate</tt:Type>
                        <tt:Likelihood>0.8</tt:Likelihood>
                    </tt:ClassCandidate>
                    <tt:Type Likelihood="0.8">LicensePlate</tt:Type>
                </tt:Class>
                <tt:GeoLocation lon="127.109806" lat="37.404676"
elevation="56.29"/>
                <tt:VehicleInfo>
                    <tt:Type Likelihood="0.8">Bus</tt:Type>
                    <tt:BodyType Likelihood="0.6">Unknwon</tt:BodyType>
                </tt:VehicleInfo>
                <tt:HumanFace>
                    <fc:Gender> Male </fc:Gender>
                    <fc:AgeType>Adult</fc:AgeType>
                    <fc:Accessory>
                        <fc:Opticals>
                            <fc:Wear>true</fc:Wear>
                        </fc:Opticals>
                        <fc:Mask>
                            <fc:Wear>true</fc:Wear>
                        </fc:Mask>
                        <fc:Hat>
                            <fc:Wear>false</fc:Wear>
                        </fc:Hat>
                    </fc:Accessory>
                </tt:HumanFace>
                <tt:HumanBody>
                    <bd:Gender> Male </bd:Gender>
                    <bd:Clothing>
                        <bd:Hat>
                            <bd:Wear>false</bd:Wear>
```

```
                            </bd:Hat>
                            <bd:Tops>
                                <bd:Color>
                                    <tt:ColorCluster>
                                        <tt:Color X="58" Y="105" Z="212" />
                                        <tt:Covariance XX="7.2" YY="6"
ZZ="3" />
                                        <tt:Weight>90</tt:Weight>

<tt:ColorString>WHITE</tt:ColorString>
                                    </tt:ColorCluster>
                                    <tt:ColorCluster>
                                        <tt:Color X="165" Y="44" Z="139" />
                                        <tt:Covariance XX="4" YY="4" ZZ="4"
/>
                                        <tt:Weight>5</tt:Weight>

<tt:ColorString>BLUE</tt:ColorString>
                                    </tt:ColorCluster>
                                </bd:Color>
                                <bd:Length>Long</bd:Length>
                            </bd:Tops>
                            <bd:Bottoms>
                                <bd:Color>
                                    <tt:ColorCluster>
                                        <tt:Color X="58" Y="105" Z="212" />
                                        <tt:Covariance XX="7.2" YY="6"
ZZ="3" />
                                        <tt:Weight>90</tt:Weight>

<tt:ColorString>WHITE</tt:ColorString>
                                    </tt:ColorCluster>
                                    <tt:ColorCluster>
                                        <tt:Color X="165" Y="44" Z="139" />
                                        <tt:Covariance XX="4" YY="4" ZZ="4"
/>
                                        <tt:Weight>5</tt:Weight>

<tt:ColorString>BLUE</tt:ColorString>
                                    </tt:ColorCluster>
                                </bd:Color>
```

```
                                <bd:Length>Long</bd:Length>
                            </bd:Bottoms>
                        </bd:Clothing>
                        <bd:Belonging>
                            <bd:Bag>
                                <bd:Category>Bag</bd:Category>
                            </bd:Bag>
                        </bd:Belonging>
                    </tt:HumanBody >

<tt:ImageRef>/download/objectid_1_1548728068_100.jpg</tt:ImageRef>
                    <tt:ImageRefShape>
                        <tt:BoundingBox left="15.0" top="141.0" right="51.0"
bottom="291.0" />
                        <tt:CenterOfGravity x="31.0" y="218.0" />
                    </tt:ImageRefShape>
                    <tt:ProximateObjects>
                        <tt:ProximateObject Id="0" Distance="0.0"/>
                    </tt:ProximateObjects>
                </tt:Appearance>
                <tt:Behaviour ObjectDirection="Right"/>
                <tt:Extension>
                    <DwellTime>
                        <RuleType>QueueManagement</RuleType>
                        <RuleName>QueueRule-1</RuleName>
                        <Duration>PT10S</Duration>
                    </DwellTime>
                </tt:Extension>
            </tt:Object>

<tt:SceneImageRef>/download/objectid_1_1548728068_100.jpg</tt:SceneImageRef>
            <tt:Extension>
                <tnssamsung:SceneImageInformation>
                    <tt:Object ObjectId="15" Parent="12">
                        <tt:Appearance>
                            <tt:Shape>
                                <tt:BoundingBox left="15.0" top="141.0"
right="51.0" bottom="291.0" />
                                <tt:CenterOfGravity x="31.0" y="218.0" />
                            </tt:Shape>
                        </tt:Appearance>
```

```
                    </tt:Object>
                </tnssamsung:SceneImageInformation>
                <tnssamsung:DigitalAutoTracking
MediaProfileToken="ProfileToken-01"/>
                <tnssamsung:PTZAutoTracking
VideoSourceToken="VideoSourceToken-0"/>
            </tt:Extension>
        </tt:Frame>
    </tt:VideoAnalytics>
</tt:MetadataStream>
```

ImageRef

A URL can also have a relative address.

../download/objected_1_23323333_100.jpg

Chapter 270. Self-signed Certificate Creation and Use

270.1. Attributes

Client can check the following attributes to check if their device supports creation of a self-signed certificate:

Request

```
http://<IP>/stw-
cgi/attributes.cgi/attributes/Security/Limit/MaxSelfSignedCertificates
```

Response

```
<attribute name="MaxSelfSignedCertificates" type="int" value="1" accesslevel
="guest"/>
```

270.2. Getting the List of Certificates

Request (HTTP-GET)

```
http://<IP>/stw-cgi/security.cgi?msubmenu=ssl&action=view
```

TEXT Response

```
Policy=HTTP
PublicCertificateInstalled=False
SelfSignedCertificateInstalled=True
PublicCertificateName= Certificate3
CertificateInUse=
Certificate.1.CertificateName =Certificate1
Certificate.1.Type=Unique
Certificate.1.Issuer=Hanwha
Certificate.1.Subject=/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddr
ess=test@hanwha.com
Certificate.1.SubjectAlternativeName=192.168.77.11
Certificate.1.IssueDate=2017-05-01
Certificate.1.ExpiryDate=2018-05-01
Certificate.1.Removable=False
Certificate.2.CertificateName =Certificate2
Certificate.2.Type=SelfSigned
```

```
Certificate.2.Issuer=CA
Certificate.2.Subject=/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddr
ess=test@hanwha.com
Certificate.2.SubjectAlternativeName=192.168.77.11
Certificate.2.IssueDate=2017-06-01
Certificate.2.ExpiryDate=2018-06-01
Certificate.2.Removable=True
Certificate.3.CertificateName =Certificate3
Certificate.3.Issuer=Hanwha
Certificate.3.Subject=/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddr
ess=test@hanwha.com
Certificate.3.SubjectAlternativeName=192.168.77.11
Certificate.3.IssueDate=2017-07-01
Certificate.3.ExpiryDate=2018-07-01
Certificate.3.Removable=True
```

JSON Response


```
{
    "Policy": "HTTP",
    "PublicCertificateInstalled": false,
    "SelfSignedCertificateInstalled": true,
    "PublicCertificateName": "Certificate3",
    "CertificateInUse": "",
    "Certificate": [
        {
            "Index": 1,
            "CertificateName": "Certificate1",
            "Type": "Unique",
            "Issuer": "Hanwha",
            "Subject":
"/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddress=test@hanwha.com",
            "SubjectAlternativeName": "192.168.77.11",
            "IssueDate": "2017-05-01",
            "ExpiryDate": "2018-05-01",
            "Removable": false
        },
        {
            "Index": 2,
            "CertificateName": "Certificate2",
            "Type": "SelfSigned",
```

```
            "Issuer": "CA",
            "Subject":
"/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddress=test@hanwha.com",
            "SubjectAlternativeName": "192.168.77.11",
            "IssueDate": "2017-06-01",
            "ExpiryDate": "2018-06-01",
            "Removable": true
        },
        {
            "Index": 3,
            "CertificateName": "Certificate3",
            "Type": "Public",
            "Issuer": "Hanwha",
            "Subject":
"/C=KR/ST=LL/L=LL/O=LL/OU=hw/CN=192.168.77.11/emailAddress=test@hanwha.com",
            "SubjectAlternativeName": "192.168.77.11",
            "IssueDate": "2017-07-01",
            "ExpiryDate": "2018-07-01",
            "Removable": true
        }
    ]
}
```

270.3. Creating a Self-signed Certificate

To create a new self-signed certificate for the device, the following cgi can be used:

Request (HTTP-GET)

```
http://<IP>/stw-
cgi/security.cgi?msubmenu=ssl&action=add&CertificateName=newCert&Type=SelfSi
gned&CommonName=192.168.75.123&SubjectAlternativeName=domain.com,testdom.com
&ExpiryDate=2020-09-
09&Country=KR&Province=Gyeonggi&Location=Bundang&Organization=Hanwha&Divisio
n=SS&EmailID=test@hanwha.com,test2@hanwha.com
```

TEXT Response

```
OK
```

JSON Response

```
{
    "Response": "Success"
}
```

270.4. Selecting a Certificate

To select a certificate to use on the camera's web server, the following method can be used:

Request (HTTP-GET)

```
http://<IP>/stw-cgi/security.cgi?msubmenu=ssl&action=set&Policy=HTTPSProprietary&CertificateInUse=newCert
```

270.5. Removing a Certificate

Any certificates with Removable status true can be removed.

Request (HTTP-GET)

```
http://<IP>/stw-cgi/security.cgi?msubmenu=ssl&action=remove&CertificateName=Certificate2
```

TEXT Response:

```
OK
```

JSON Response:

```
{
    "Response": "Success"
}
```

Chapter 271. Intercom Camera Integration

271.1. Overview

271.1.1. Supports the SIP (Session Initation Protocol)

The TID-600R provides audio and video communications using standard SIP. It can integrate with external SIP compatible systems through a SIP server.

Refer to **sipsetup**, **sipaccount**, **siprecipients** submenus of **network.cgi** for more details to change the SIP settings.

271.1.2. NAT Traversal

The TID-600R provides NAT Traversal for seamless communication between devices located on the private network and the external internet. Refer to nattraversal submenu of network.cgi for change the NAT Traversal settings.

271.2. Difference of other cameras

271.2.1. Profile for VoIP

Video delivery of SIP requires creating a profile for VoIP. The camera uses a VoIP-only profile when SIP is connected; if no profile exists, only audio is sent. VoIP profile is activated if the camera supports SIP. Whether or not SIP is supported can be checked in attribute below.

```
http://<Device IP>/stw-cgi/attributes.cgi/attributes/Network/Support/SIP
```

VoIP profile is limited in supported codecs, resolution, and bitrates, so you need to check and set the video codec information. Refer to videocodecinfo submenu of media.cgi to get video codec information.

271.2.2. Power relay output

The TID-600R model provides a power relay output. Unlike typical Open Collector type IC output, external device control is possible through power connections without requiring additional circuit configuration. The output ports that provide power relay can be found by the attributes below.

```
http://<Device IP>/stw-
cgi/attributes.cgi/attributes/IO/Support/PowerRelayIndices
```

271.3. Events

271.3.1. Call Request

The TID-600R supports SIP calls for audio and video communication. And you can also integrate with video surveillance systems for call features. When a button is pressed or a touchless sensor is detected, a

CallRequest event will be triggered and an SIP call will be requested. The event will persist until the recipient accepts the call, reaches the CallingTimeout, or sends a stop request command. If you want to check the CallRequest event on the video surveillance system and stop a SIP call, Refer to sipcall submenu of network.cgi, and refer to callrequest submenu of eventsources.cgi for change the CallRequest event settings.

Note that the CallRequest event is the fixed event for intercom model cameras and cannot be disabled.

271.3.2. DTMF Received

It can perform defined actions by receiving DTMF(Dual Tone Multi Frequency) signals through SIP. User-defined actions such as recording, door opening, and alarm triggering can be performed through a specific DTMF signal. This model supports DTMF reception over RTP payload (RFC2833) and SIP INFO Method (RFC2976). Refer to dtmf submenu of eventsources.cgi for change the DTMF event settings.

271.3.3. Tampering Switch

The tampering switch is an event that can detect security threats caused by physical damage or disassembly of the product by an external intruder. Refer to tamperingswitch submenu of eventsources.cgi for change the tampering switch event settings.

271.4. Video codec information for VoIP-only profile

271.4.1. Getting all resolution information based on Encoding Type

REQUEST

```
http://<Device IP>/stw-cgi/media.cgi?msubmenu=videocodecinfo&action=view&EncodingType=H264
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
Channel.0.EncodingType=H264
H264.General.1920X1080.Width=1920
H264.General.1920X1080.Height=1080
H264.General.1920X1080.MaxFPS=30000
H264.General.1920X1080.DefaultFPS=30000
H264.General.1920X1080.MaxCBRTargetBitrate=20480
H264.General.1920X1080.MinCBRTargetBitrate=1024
H264.General.1920X1080.DefaultCBRTargetBitrate=2560
H264.General.1920X1080.MaxVBRTargetBitrate=30720
```

```
H264.General.1920X1080.MinVBRTargetBitrate=1536
H264.General.1920X1080.DefaultVBRTargetBitrate=2560
H264.General.1920X1080.IsTruncated=Normal
H264.General.1920X1080.FrameLockMaxFPS=30000
H264.General.1280X1024.Width=1280
H264.General.1280X1024.Height=1024
H264.General.1280X1024.MaxFPS=30000
H264.General.1280X1024.DefaultFPS=30000
H264.General.1280X1024.MaxCBRTargetBitrate=20480
H264.General.1280X1024.MinCBRTargetBitrate=1024
H264.General.1280X1024.DefaultCBRTargetBitrate=2048
H264.General.1280X1024.MaxVBRTargetBitrate=30720
H264.General.1280X1024.MinVBRTargetBitrate=1536
H264.General.1280X1024.DefaultVBRTargetBitrate=2048
H264.General.1280X1024.IsTruncated=Crop
H264.General.1280X1024.FrameLockMaxFPS=30000
H264.General.1280X960.Width=1280
H264.General.1280X960.Height=960
H264.General.1280X960.MaxFPS=30000
H264.General.1280X960.DefaultFPS=30000
H264.General.1280X960.MaxCBRTargetBitrate=20480
H264.General.1280X960.MinCBRTargetBitrate=1024
H264.General.1280X960.DefaultCBRTargetBitrate=2048
H264.General.1280X960.MaxVBRTargetBitrate=30720
H264.General.1280X960.MinVBRTargetBitrate=1536
H264.General.1280X960.DefaultVBRTargetBitrate=2048
H264.General.1280X960.IsTruncated=Crop
H264.General.1280X960.FrameLockMaxFPS=30000
H264.General.1280X720.Width=1280
H264.General.1280X720.Height=720
H264.General.1280X720.MaxFPS=30000
H264.General.1280X720.DefaultFPS=30000
H264.General.1280X720.MaxCBRTargetBitrate=20480
H264.General.1280X720.MinCBRTargetBitrate=1024
H264.General.1280X720.DefaultCBRTargetBitrate=2048
H264.General.1280X720.MaxVBRTargetBitrate=30720
H264.General.1280X720.MinVBRTargetBitrate=1536
H264.General.1280X720.DefaultVBRTargetBitrate=2048
H264.General.1280X720.IsTruncated=Normal
H264.General.1280X720.FrameLockMaxFPS=30000
H264.General.1024X768.Width=1024
```

```
H264.General.1024X768.Height=768
H264.General.1024X768.MaxFPS=30000
H264.General.1024X768.DefaultFPS=30000
H264.General.1024X768.MaxCBRTargetBitrate=20480
H264.General.1024X768.MinCBRTargetBitrate=1024
H264.General.1024X768.DefaultCBRTargetBitrate=2048
H264.General.1024X768.MaxVBRTargetBitrate=30720
H264.General.1024X768.MinVBRTargetBitrate=1536
H264.General.1024X768.DefaultVBRTargetBitrate=2048
H264.General.1024X768.IsTruncated=Crop
H264.General.1024X768.FrameLockMaxFPS=30000
H264.General.800X600.Width=800
H264.General.800X600.Height=600
H264.General.800X600.MaxFPS=30000
H264.General.800X600.DefaultFPS=30000
H264.General.800X600.MaxCBRTargetBitrate=20480
H264.General.800X600.MinCBRTargetBitrate=512
H264.General.800X600.DefaultCBRTargetBitrate=1024
H264.General.800X600.MaxVBRTargetBitrate=30720
H264.General.800X600.MinVBRTargetBitrate=512
H264.General.800X600.DefaultVBRTargetBitrate=1024
H264.General.800X600.IsTruncated=Crop
H264.General.800X600.FrameLockMaxFPS=30000
H264.General.800X448.Width=800
H264.General.800X448.Height=448
H264.General.800X448.MaxFPS=30000
H264.General.800X448.DefaultFPS=30000
H264.General.800X448.MaxCBRTargetBitrate=20480
H264.General.800X448.MinCBRTargetBitrate=512
H264.General.800X448.DefaultCBRTargetBitrate=1024
H264.General.800X448.MaxVBRTargetBitrate=30720
H264.General.800X448.MinVBRTargetBitrate=512
H264.General.800X448.DefaultVBRTargetBitrate=1024
H264.General.800X448.IsTruncated=Normal
H264.General.800X448.FrameLockMaxFPS=30000
H264.General.720X576.Width=720
H264.General.720X576.Height=576
H264.General.720X576.MaxFPS=30000
H264.General.720X576.DefaultFPS=30000
H264.General.720X576.MaxCBRTargetBitrate=20480
H264.General.720X576.MinCBRTargetBitrate=512
```

```
H264.General.720X576.DefaultCBRTargetBitrate=1024
H264.General.720X576.MaxVBRTargetBitrate=30720
H264.General.720X576.MinVBRTargetBitrate=512
H264.General.720X576.DefaultVBRTargetBitrate=1024
H264.General.720X576.IsTruncated=Crop
H264.General.720X576.FrameLockMaxFPS=30000
H264.General.720X480.Width=720
H264.General.720X480.Height=480
H264.General.720X480.MaxFPS=30000
H264.General.720X480.DefaultFPS=30000
H264.General.720X480.MaxCBRTargetBitrate=20480
H264.General.720X480.MinCBRTargetBitrate=512
H264.General.720X480.DefaultCBRTargetBitrate=1024
H264.General.720X480.MaxVBRTargetBitrate=30720
H264.General.720X480.MinVBRTargetBitrate=512
H264.General.720X480.DefaultVBRTargetBitrate=1024
H264.General.720X480.IsTruncated=Crop
H264.General.720X480.FrameLockMaxFPS=30000
H264.General.640X480.Width=640
H264.General.640X480.Height=480
H264.General.640X480.MaxFPS=30000
H264.General.640X480.DefaultFPS=30000
H264.General.640X480.MaxCBRTargetBitrate=20480
H264.General.640X480.MinCBRTargetBitrate=512
H264.General.640X480.DefaultCBRTargetBitrate=1024
H264.General.640X480.MaxVBRTargetBitrate=30720
H264.General.640X480.MinVBRTargetBitrate=512
H264.General.640X480.DefaultVBRTargetBitrate=1024
H264.General.640X480.IsTruncated=Crop
H264.General.640X480.FrameLockMaxFPS=30000
H264.General.640X360.Width=640
H264.General.640X360.Height=360
H264.General.640X360.MaxFPS=30000
H264.General.640X360.DefaultFPS=30000
H264.General.640X360.MaxCBRTargetBitrate=20480
H264.General.640X360.MinCBRTargetBitrate=512
H264.General.640X360.DefaultCBRTargetBitrate=1024
H264.General.640X360.MaxVBRTargetBitrate=30720
H264.General.640X360.MinVBRTargetBitrate=512
H264.General.640X360.DefaultVBRTargetBitrate=1024
H264.General.640X360.IsTruncated=Normal
```

```
H264.General.640X360.FrameLockMaxFPS=30000
H264.General.320X240.Width=320
H264.General.320X240.Height=240
H264.General.320X240.MaxFPS=30000
H264.General.320X240.DefaultFPS=30000
H264.General.320X240.MaxCBRTargetBitrate=20480
H264.General.320X240.MinCBRTargetBitrate=256
H264.General.320X240.DefaultCBRTargetBitrate=512
H264.General.320X240.MaxVBRTargetBitrate=30720
H264.General.320X240.MinVBRTargetBitrate=256
H264.General.320X240.DefaultVBRTargetBitrate=512
H264.General.320X240.IsTruncated=Crop
H264.General.320X240.FrameLockMaxFPS=30000
H264.Record.1920X1080.MaxFPS=30000
H264.Record.1920X1080.DefaultFPS=30000
H264.Record.1920X1080.MinCBRTargetBitrate=1024
H264.Record.1920X1080.MaxCBRTargetBitrate=6144
H264.Record.1920X1080.DefaultCBRTargetBitrate=5120
H264.Record.1920X1080.MinVBRTargetBitrate=1536
H264.Record.1920X1080.MaxVBRTargetBitrate=6144
H264.Record.1920X1080.DefaultVBRTargetBitrate=5120
H264.Record.1920X1080.FrameLockMaxFPS=30000
H264.Record.1280X1024.MaxFPS=30000
H264.Record.1280X1024.DefaultFPS=30000
H264.Record.1280X1024.MinCBRTargetBitrate=1024
H264.Record.1280X1024.MaxCBRTargetBitrate=6144
H264.Record.1280X1024.DefaultCBRTargetBitrate=5120
H264.Record.1280X1024.MinVBRTargetBitrate=1536
H264.Record.1280X1024.MaxVBRTargetBitrate=6144
H264.Record.1280X1024.DefaultVBRTargetBitrate=5120
H264.Record.1280X1024.FrameLockMaxFPS=30000
H264.Record.1280X960.MaxFPS=30000
H264.Record.1280X960.DefaultFPS=30000
H264.Record.1280X960.MinCBRTargetBitrate=1024
H264.Record.1280X960.MaxCBRTargetBitrate=6144
H264.Record.1280X960.DefaultCBRTargetBitrate=5120
H264.Record.1280X960.MinVBRTargetBitrate=1536
H264.Record.1280X960.MaxVBRTargetBitrate=6144
H264.Record.1280X960.DefaultVBRTargetBitrate=5120
H264.Record.1280X960.FrameLockMaxFPS=30000
H264.Record.1280X720.MaxFPS=30000
```

```
H264.Record.1280X720.DefaultFPS=30000
H264.Record.1280X720.MinCBRTargetBitrate=1024
H264.Record.1280X720.MaxCBRTargetBitrate=6144
H264.Record.1280X720.DefaultCBRTargetBitrate=5120
H264.Record.1280X720.MinVBRTargetBitrate=1536
H264.Record.1280X720.MaxVBRTargetBitrate=6144
H264.Record.1280X720.DefaultVBRTargetBitrate=5120
H264.Record.1280X720.FrameLockMaxFPS=30000
H264.Record.1024X768.MaxFPS=30000
H264.Record.1024X768.DefaultFPS=30000
H264.Record.1024X768.MinCBRTargetBitrate=1024
H264.Record.1024X768.MaxCBRTargetBitrate=6144
H264.Record.1024X768.DefaultCBRTargetBitrate=5120
H264.Record.1024X768.MinVBRTargetBitrate=1536
H264.Record.1024X768.MaxVBRTargetBitrate=6144
H264.Record.1024X768.DefaultVBRTargetBitrate=5120
H264.Record.1024X768.FrameLockMaxFPS=30000
H264.Record.800X600.MaxFPS=30000
H264.Record.800X600.DefaultFPS=30000
H264.Record.800X600.MinCBRTargetBitrate=512
H264.Record.800X600.MaxCBRTargetBitrate=6144
H264.Record.800X600.DefaultCBRTargetBitrate=5120
H264.Record.800X600.MinVBRTargetBitrate=512
H264.Record.800X600.MaxVBRTargetBitrate=6144
H264.Record.800X600.DefaultVBRTargetBitrate=5120
H264.Record.800X600.FrameLockMaxFPS=30000
H264.Record.800X448.MaxFPS=30000
H264.Record.800X448.DefaultFPS=30000
H264.Record.800X448.MinCBRTargetBitrate=512
H264.Record.800X448.MaxCBRTargetBitrate=6144
H264.Record.800X448.DefaultCBRTargetBitrate=5120
H264.Record.800X448.MinVBRTargetBitrate=512
H264.Record.800X448.MaxVBRTargetBitrate=6144
H264.Record.800X448.DefaultVBRTargetBitrate=5120
H264.Record.800X448.FrameLockMaxFPS=30000
H264.Record.720X576.MaxFPS=30000
H264.Record.720X576.DefaultFPS=30000
H264.Record.720X576.MinCBRTargetBitrate=512
H264.Record.720X576.MaxCBRTargetBitrate=6144
H264.Record.720X576.DefaultCBRTargetBitrate=5120
H264.Record.720X576.MinVBRTargetBitrate=512
```

```
H264.Record.720X576.MaxVBRTargetBitrate=6144
H264.Record.720X576.DefaultVBRTargetBitrate=5120
H264.Record.720X576.FrameLockMaxFPS=30000
H264.Record.720X480.MaxFPS=30000
H264.Record.720X480.DefaultFPS=30000
H264.Record.720X480.MinCBRTargetBitrate=512
H264.Record.720X480.MaxCBRTargetBitrate=6144
H264.Record.720X480.DefaultCBRTargetBitrate=5120
H264.Record.720X480.MinVBRTargetBitrate=512
H264.Record.720X480.MaxVBRTargetBitrate=6144
H264.Record.720X480.DefaultVBRTargetBitrate=5120
H264.Record.720X480.FrameLockMaxFPS=30000
H264.Record.640X480.MaxFPS=30000
H264.Record.640X480.DefaultFPS=30000
H264.Record.640X480.MinCBRTargetBitrate=512
H264.Record.640X480.MaxCBRTargetBitrate=6144
H264.Record.640X480.DefaultCBRTargetBitrate=5120
H264.Record.640X480.MinVBRTargetBitrate=512
H264.Record.640X480.MaxVBRTargetBitrate=6144
H264.Record.640X480.DefaultVBRTargetBitrate=5120
H264.Record.640X480.FrameLockMaxFPS=30000
H264.Record.640X360.MaxFPS=30000
H264.Record.640X360.DefaultFPS=30000
H264.Record.640X360.MinCBRTargetBitrate=512
H264.Record.640X360.MaxCBRTargetBitrate=6144
H264.Record.640X360.DefaultCBRTargetBitrate=5120
H264.Record.640X360.MinVBRTargetBitrate=512
H264.Record.640X360.MaxVBRTargetBitrate=6144
H264.Record.640X360.DefaultVBRTargetBitrate=5120
H264.Record.640X360.FrameLockMaxFPS=30000
H264.Record.320X240.MaxFPS=30000
H264.Record.320X240.DefaultFPS=30000
H264.Record.320X240.MinCBRTargetBitrate=256
H264.Record.320X240.MaxCBRTargetBitrate=6144
H264.Record.320X240.DefaultCBRTargetBitrate=5120
H264.Record.320X240.MinVBRTargetBitrate=256
H264.Record.320X240.MaxVBRTargetBitrate=6144
H264.Record.320X240.DefaultVBRTargetBitrate=5120
H264.Record.320X240.FrameLockMaxFPS=30000
H264.VoIP.1280X720.Width=1280
H264.VoIP.1280X720.Height=720
```

```
H264.VoIP.1280X720.MaxFPS=30000
H264.VoIP.1280X720.DefaultFPS=30000
H264.VoIP.1280X720.MaxCBRTargetBitrate=2048
H264.VoIP.1280X720.MinCBRTargetBitrate=1024
H264.VoIP.1280X720.DefaultCBRTargetBitrate=1024
H264.VoIP.1280X720.MinVBRTargetBitrate=1536
H264.VoIP.1280X720.MaxVBRTargetBitrate=2048
H264.VoIP.1280X720.DefaultVBRTargetBitrate=1536
H264.VoIP.1024X768.Width=1024
H264.VoIP.1024X768.Height=768
H264.VoIP.1024X768.MaxFPS=30000
H264.VoIP.1024X768.DefaultFPS=30000
H264.VoIP.1024X768.MaxCBRTargetBitrate=2048
H264.VoIP.1024X768.MinCBRTargetBitrate=1024
H264.VoIP.1024X768.DefaultCBRTargetBitrate=1024
H264.VoIP.1024X768.MinVBRTargetBitrate=1536
H264.VoIP.1024X768.MaxVBRTargetBitrate=2048
H264.VoIP.1024X768.DefaultVBRTargetBitrate=1536
H264.VoIP.800X600.Width=800
H264.VoIP.800X600.Height=600
H264.VoIP.800X600.MaxFPS=30000
H264.VoIP.800X600.DefaultFPS=30000
H264.VoIP.800X600.MaxCBRTargetBitrate=1024
H264.VoIP.800X600.MinCBRTargetBitrate=512
H264.VoIP.800X600.DefaultCBRTargetBitrate=512
H264.VoIP.800X600.MinVBRTargetBitrate=512
H264.VoIP.800X600.MaxVBRTargetBitrate=1024
H264.VoIP.800X600.DefaultVBRTargetBitrate=512
H264.VoIP.800X448.Width=800
H264.VoIP.800X448.Height=448
H264.VoIP.800X448.MaxFPS=30000
H264.VoIP.800X448.DefaultFPS=30000
H264.VoIP.800X448.MaxCBRTargetBitrate=1024
H264.VoIP.800X448.MinCBRTargetBitrate=512
H264.VoIP.800X448.DefaultCBRTargetBitrate=512
H264.VoIP.800X448.MinVBRTargetBitrate=512
H264.VoIP.800X448.MaxVBRTargetBitrate=1024
H264.VoIP.800X448.DefaultVBRTargetBitrate=512
H264.VoIP.720X576.Width=720
H264.VoIP.720X576.Height=576
H264.VoIP.720X576.MaxFPS=30000
```

```
H264.VoIP.720X576.DefaultFPS=30000
H264.VoIP.720X576.MaxCBRTargetBitrate=1024
H264.VoIP.720X576.MinCBRTargetBitrate=512
H264.VoIP.720X576.DefaultCBRTargetBitrate=512
H264.VoIP.720X576.MinVBRTargetBitrate=512
H264.VoIP.720X576.MaxVBRTargetBitrate=1024
H264.VoIP.720X576.DefaultVBRTargetBitrate=512
H264.VoIP.720X480.Width=720
H264.VoIP.720X480.Height=480
H264.VoIP.720X480.MaxFPS=30000
H264.VoIP.720X480.DefaultFPS=30000
H264.VoIP.720X480.MaxCBRTargetBitrate=1024
H264.VoIP.720X480.MinCBRTargetBitrate=512
H264.VoIP.720X480.DefaultCBRTargetBitrate=512
H264.VoIP.720X480.MinVBRTargetBitrate=512
H264.VoIP.720X480.MaxVBRTargetBitrate=1024
H264.VoIP.720X480.DefaultVBRTargetBitrate=512
H264.VoIP.640X480.Width=640
H264.VoIP.640X480.Height=480
H264.VoIP.640X480.MaxFPS=30000
H264.VoIP.640X480.DefaultFPS=30000
H264.VoIP.640X480.MaxCBRTargetBitrate=1024
H264.VoIP.640X480.MinCBRTargetBitrate=512
H264.VoIP.640X480.DefaultCBRTargetBitrate=512
H264.VoIP.640X480.MinVBRTargetBitrate=512
H264.VoIP.640X480.MaxVBRTargetBitrate=1024
H264.VoIP.640X480.DefaultVBRTargetBitrate=512
H264.VoIP.640X360.Width=640
H264.VoIP.640X360.Height=360
H264.VoIP.640X360.MaxFPS=30000
H264.VoIP.640X360.DefaultFPS=30000
H264.VoIP.640X360.MaxCBRTargetBitrate=1024
H264.VoIP.640X360.MinCBRTargetBitrate=512
H264.VoIP.640X360.DefaultCBRTargetBitrate=512
H264.VoIP.640X360.MinVBRTargetBitrate=512
H264.VoIP.640X360.MaxVBRTargetBitrate=1024
H264.VoIP.640X360.DefaultVBRTargetBitrate=512
H264.VoIP.320X240.Width=320
H264.VoIP.320X240.Height=240
H264.VoIP.320X240.MaxFPS=30000
H264.VoIP.320X240.DefaultFPS=30000
```

```
H264.VoIP.320X240.MaxCBRTargetBitrate=512
H264.VoIP.320X240.MinCBRTargetBitrate=256
H264.VoIP.320X240.DefaultCBRTargetBitrate=256
H264.VoIP.320X240.MinVBRTargetBitrate=256
H264.VoIP.320X240.MaxVBRTargetBitrate=512
H264.VoIP.320X240.DefaultVBRTargetBitrate=256
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "VideoCodecInfo": [
        {
            "Channel": 0,
            "ViewModes": [
                {
                    "ViewMode": "Overview",
                    "Codecs": [
                        {
                            "EncodingType": "H264",
                            "General": [
                                {
                                    "Width": 1920,
                                    "Height": 1080,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MaxCBRTargetBitrate": 20480,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2560,
                                    "MaxVBRTargetBitrate": 30720,
                                    "MinVBRTargetBitrate": 1536,
                                    "DefaultVBRTargetBitrate": 2560,
                                    "IsTruncated": "Normal",
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 1280,
                                    "Height": 1024,
```

```
                "MaxFPS": 30000,
                "DefaultFPS": 30000,
                "MaxCBRTargetBitrate": 20480,
                "MinCBRTargetBitrate": 1024,
                "DefaultCBRTargetBitrate": 2048,
                "MaxVBRTargetBitrate": 30720,
                "MinVBRTargetBitrate": 1536,
                "DefaultVBRTargetBitrate": 2048,
                "IsTruncated": "Crop",
                "FrameLockMaxFPS": 30000
            },
            {
                "Width": 1280,
                "Height": 960,
                "MaxFPS": 30000,
                "DefaultFPS": 30000,
                "MaxCBRTargetBitrate": 20480,
                "MinCBRTargetBitrate": 1024,
                "DefaultCBRTargetBitrate": 2048,
                "MaxVBRTargetBitrate": 30720,
                "MinVBRTargetBitrate": 1536,
                "DefaultVBRTargetBitrate": 2048,
                "IsTruncated": "Crop",
                "FrameLockMaxFPS": 30000
            },
            {
                "Width": 1280,
                "Height": 720,
                "MaxFPS": 30000,
                "DefaultFPS": 30000,
                "MaxCBRTargetBitrate": 20480,
                "MinCBRTargetBitrate": 1024,
                "DefaultCBRTargetBitrate": 2048,
                "MaxVBRTargetBitrate": 30720,
                "MinVBRTargetBitrate": 1536,
                "DefaultVBRTargetBitrate": 2048,
                "IsTruncated": "Normal",
                "FrameLockMaxFPS": 30000
            },
            {
                "Width": 1024,
```

```
                                    "Height": 768,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MaxCBRTargetBitrate": 20480,
                                    "MinCBRTargetBitrate": 1024,
                                    "DefaultCBRTargetBitrate": 2048,
                                    "MaxVBRTargetBitrate": 30720,
                                    "MinVBRTargetBitrate": 1536,
                                    "DefaultVBRTargetBitrate": 2048,
                                    "IsTruncated": "Crop",
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 800,
                                    "Height": 600,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MaxCBRTargetBitrate": 20480,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 30720,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Crop",
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 800,
                                    "Height": 448,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MaxCBRTargetBitrate": 20480,
                                    "MinCBRTargetBitrate": 512,
                                    "DefaultCBRTargetBitrate": 1024,
                                    "MaxVBRTargetBitrate": 30720,
                                    "MinVBRTargetBitrate": 512,
                                    "DefaultVBRTargetBitrate": 1024,
                                    "IsTruncated": "Normal",
                                    "FrameLockMaxFPS": 30000
                                },
                                {
```

```
                    "Width": 720,
                    "Height": 576,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 20480,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Crop",
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 720,
                    "Height": 480,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 20480,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Crop",
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 640,
                    "Height": 480,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 20480,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Crop",
                    "FrameLockMaxFPS": 30000
                },
```

```
                {
                    "Width": 640,
                    "Height": 360,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 20480,
                    "MinCBRTargetBitrate": 512,
                    "DefaultCBRTargetBitrate": 1024,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 512,
                    "DefaultVBRTargetBitrate": 1024,
                    "IsTruncated": "Normal",
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 320,
                    "Height": 240,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 20480,
                    "MinCBRTargetBitrate": 256,
                    "DefaultCBRTargetBitrate": 512,
                    "MaxVBRTargetBitrate": 30720,
                    "MinVBRTargetBitrate": 256,
                    "DefaultVBRTargetBitrate": 512,
                    "IsTruncated": "Crop",
                    "FrameLockMaxFPS": 30000
                }
            ],
            "Record": [
                {
                    "Width": 1920,
                    "Height": 1080,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 1024,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 1536,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
```

```
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 1280,
                    "Height": 1024,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 1024,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 1536,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 1280,
                    "Height": 960,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 1024,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 1536,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 1024,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 1536,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
                    "FrameLockMaxFPS": 30000
                },
```

```
                                {
                                    "Width": 1024,
                                    "Height": 768,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 1024,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 1536,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 800,
                                    "Height": 600,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 800,
                                    "Height": 448,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 720,
```

```
                                    "Height": 576,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 720,
                                    "Height": 480,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 640,
                                    "Height": 480,
                                    "MaxFPS": 30000,
                                    "DefaultFPS": 30000,
                                    "MinCBRTargetBitrate": 512,
                                    "MaxCBRTargetBitrate": 6144,
                                    "DefaultCBRTargetBitrate": 5120,
                                    "MinVBRTargetBitrate": 512,
                                    "MaxVBRTargetBitrate": 6144,
                                    "DefaultVBRTargetBitrate": 5120,
                                    "FrameLockMaxFPS": 30000
                                },
                                {
                                    "Width": 640,
                                    "Height": 360,
                                    "MaxFPS": 30000,
```

```
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 512,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 512,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
                    "FrameLockMaxFPS": 30000
                },
                {
                    "Width": 320,
                    "Height": 240,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MinCBRTargetBitrate": 256,
                    "MaxCBRTargetBitrate": 6144,
                    "DefaultCBRTargetBitrate": 5120,
                    "MinVBRTargetBitrate": 256,
                    "MaxVBRTargetBitrate": 6144,
                    "DefaultVBRTargetBitrate": 5120,
                    "FrameLockMaxFPS": 30000
                }
            ],
            "VoIP": [
                {
                    "Width": 1280,
                    "Height": 720,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
                    "MaxCBRTargetBitrate": 2048,
                    "MinCBRTargetBitrate": 1024,
                    "DefaultCBRTargetBitrate": 1024,
                    "MinVBRTargetBitrate": 1536,
                    "MaxVBRTargetBitrate": 2048,
                    "DefaultVBRTargetBitrate": 1536
                },
                {
                    "Width": 1024,
                    "Height": 768,
                    "MaxFPS": 30000,
                    "DefaultFPS": 30000,
```

```
                                        "MaxCBRTargetBitrate": 2048,
                                        "MinCBRTargetBitrate": 1024,
                                        "DefaultCBRTargetBitrate": 1024,
                                        "MinVBRTargetBitrate": 1536,
                                        "MaxVBRTargetBitrate": 2048,
                                        "DefaultVBRTargetBitrate": 1536
                                    },
                                    {
                                        "Width": 800,
                                        "Height": 600,
                                        "MaxFPS": 30000,
                                        "DefaultFPS": 30000,
                                        "MaxCBRTargetBitrate": 1024,
                                        "MinCBRTargetBitrate": 512,
                                        "DefaultCBRTargetBitrate": 512,
                                        "MinVBRTargetBitrate": 512,
                                        "MaxVBRTargetBitrate": 1024,
                                        "DefaultVBRTargetBitrate": 512
                                    },
                                    {
                                        "Width": 800,
                                        "Height": 448,
                                        "MaxFPS": 30000,
                                        "DefaultFPS": 30000,
                                        "MaxCBRTargetBitrate": 1024,
                                        "MinCBRTargetBitrate": 512,
                                        "DefaultCBRTargetBitrate": 512,
                                        "MinVBRTargetBitrate": 512,
                                        "MaxVBRTargetBitrate": 1024,
                                        "DefaultVBRTargetBitrate": 512
                                    },
                                    {
                                        "Width": 720,
                                        "Height": 576,
                                        "MaxFPS": 30000,
                                        "DefaultFPS": 30000,
                                        "MaxCBRTargetBitrate": 1024,
                                        "MinCBRTargetBitrate": 512,
                                        "DefaultCBRTargetBitrate": 512,
                                        "MinVBRTargetBitrate": 512,
                                        "MaxVBRTargetBitrate": 1024,
```

```
                                                "DefaultVBRTargetBitrate": 512
                                            },
                                            {
                                                "Width": 720,
                                                "Height": 480,
                                                "MaxFPS": 30000,
                                                "DefaultFPS": 30000,
                                                "MaxCBRTargetBitrate": 1024,
                                                "MinCBRTargetBitrate": 512,
                                                "DefaultCBRTargetBitrate": 512,
                                                "MinVBRTargetBitrate": 512,
                                                "MaxVBRTargetBitrate": 1024,
                                                "DefaultVBRTargetBitrate": 512
                                            },
                                            {
                                                "Width": 640,
                                                "Height": 480,
                                                "MaxFPS": 30000,
                                                "DefaultFPS": 30000,
                                                "MaxCBRTargetBitrate": 1024,
                                                "MinCBRTargetBitrate": 512,
                                                "DefaultCBRTargetBitrate": 512,
                                                "MinVBRTargetBitrate": 512,
                                                "MaxVBRTargetBitrate": 1024,
                                                "DefaultVBRTargetBitrate": 512
                                            },
                                            {
                                                "Width": 640,
                                                "Height": 360,
                                                "MaxFPS": 30000,
                                                "DefaultFPS": 30000,
                                                "MaxCBRTargetBitrate": 1024,
                                                "MinCBRTargetBitrate": 512,
                                                "DefaultCBRTargetBitrate": 512,
                                                "MinVBRTargetBitrate": 512,
                                                "MaxVBRTargetBitrate": 1024,
                                                "DefaultVBRTargetBitrate": 512
                                            },
                                            {
                                                "Width": 320,
                                                "Height": 240,
```

```
                                        "MaxFPS": 30000,
                                        "DefaultFPS": 30000,
                                        "MaxCBRTargetBitrate": 512,
                                        "MinCBRTargetBitrate": 256,
                                        "DefaultCBRTargetBitrate": 256,
                                        "MinVBRTargetBitrate": 256,
                                        "MaxVBRTargetBitrate": 512,
                                        "DefaultVBRTargetBitrate": 256
                                    }
                                ]
                            }
                        ]
                    }
                ]
            }
        ]
    }
```

271.5. Usage Scenarios

271.5.1. VMS Usage (When SIP not supported)

When using with VMS that does not support SIP. Based on the callrequest submenu settings, callrequest event would be notified in eventstatus and metadata when call button is pressed. On receiving this callrequest event, VMS can establish RTSP Connection to Doorcam with Video, Audio and AudioBackChannel. On successful verification of the person at door, either power relay connected to door or emergency alarm can be activated.

Additionally VMS can also check for any ongoing SIP call using the sipcall submenu and if VMS accepts the CallRequest, it must request a start VMS call command to terminate the SIP call. And if the VMS call is terminated, you must request a stop VMS call command.

271.5.2. SIP Call Usage

On camera Sipsetup, Nat traversal settings, Sipaccount settings and sip recipients needs to be configured properly before usage. If operator has to do some action after receiving the call, DTMF event settings needs to be configured appropriately (Example: on receving a DTMF code camera can operate the powerrelay) When call button is pressed, a SIP call would be initiated with the registered recipients. After accepting the call on the recipient end, operator can send specific DTMF code either to raise alarm or operate the relay / open the door.

271.6. SUNAPI event status example

271.6.1. Getting event status

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1=False
AlarmInput.2=False
AlarmOutput.1=False
AlarmOutput.2=False
Channel.0.MotionDetection=False
Channel.0.MotionDetection.RegionID.1=False
Channel.0.Tampering=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Intrusion=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Loitering=False
Channel.0.AudioAnalytics.Scream=False
Channel.0.AudioAnalytics.Gunshot=False
Channel.0.AudioAnalytics.Explosion=False
Channel.0.AudioAnalytics.GlassBreak=False
Channel.0.ShockDetection=False
Channel.0.CallRequest=False
Channel.0.TamperingSwitch=False
Channel.0.DTMFReceived=False
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
```

```
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=True
SystemEvent.SDRemove=False
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "AlarmInput": {
        "1": false,
        "2": false
    },
    "AlarmOutput": {
        "1": false,
        "2": false
    },
    "ChannelEvent": [
        {
            "Channel": 0,
            "MotionDetection": false,
            "MotionDetectionRegions": {
                "1": false
            },
            "Tampering": false,
            "AudioDetection": false,
            "VideoAnalytics": {
                "Passing": false,
                "Intrusion": false,
                "Entering": false,
                "Exiting": false,
                "Appearing": false,
                "Loitering": false
```

```
            },
            "AudioAnalytics": {
                "Scream": false,
                "Gunshot": false,
                "Explosion": false,
                "GlassBreak": false
            },
            "ShockDetection": false,
            "CallRequest": false,
            "TamperingSwitch": false,
            "DTMFReceived": false
        }
    ],
    "SystemEvent": {
        "TimeChange": false,
        "PowerReboot": false,
        "FWUpdate": false,
        "FactoryReset": false,
        "ConfigurationBackup": false,
        "ConfigurationRestore": false,
        "ConfigChange": false,
        "SDFormat": false,
        "SDFail": false,
        "SDFull": false,
        "SDInsert": true,
        "SDRemove": false,
        "NASConnect": false,
        "NASDisconnect": true,
        "NASFail": false,
        "NASFull": false,
        "NASFormat": false
    }
}
```

271.6.2. Getting scheme-based event status

REQUEST

```
http://<Device IP>/stw-cgi/eventstatus.cgi?msubmenu=eventstatus&action=check&SchemaBased=True
```

TEXT RESPONSE

```
HTTP/1.0 200 OK
Content-type: text/plain
<Body>
```

```
AlarmInput.1=False
AlarmInput.2=False
AlarmOutput.1=False
AlarmOutput.2=False
Channel.0.MotionDetection=True
Channel.0.MotionDetection.RegionID.1=False
Channel.0.MotionDetection.RegionID.2=False
Channel.0.Tampering=False
Channel.0.AudioDetection=False
Channel.0.VideoAnalytics.Passing=False
Channel.0.VideoAnalytics.Intrusion=False
Channel.0.VideoAnalytics.Entering=False
Channel.0.VideoAnalytics.Exiting=False
Channel.0.VideoAnalytics.Appearing=False
Channel.0.VideoAnalytics.Loitering=False
Channel.0.AudioAnalytics.Scream=False
Channel.0.AudioAnalytics.Gunshot=False
Channel.0.AudioAnalytics.Explosion=False
Channel.0.AudioAnalytics.GlassBreak=False
Channel.0.ShockDetection=False
Channel.0.CallRequest=False
Channel.0.TamperingSwitch=False
Channel.0.DTMFReceived=False
Channel.0.DTMFReceived.Index.1=False
SystemEvent.TimeChange=False
SystemEvent.PowerReboot=False
SystemEvent.FWUpdate=False
SystemEvent.FactoryReset=False
SystemEvent.ConfigurationBackup=False
SystemEvent.ConfigurationRestore=False
SystemEvent.ConfigChange=False
SystemEvent.SDFormat=False
SystemEvent.SDFail=False
SystemEvent.SDFull=False
SystemEvent.SDInsert=True
```

```
SystemEvent.SDRemove=False
SystemEvent.NASConnect=False
SystemEvent.NASDisconnect=True
SystemEvent.NASFail=False
SystemEvent.NASFull=False
SystemEvent.NASFormat=False
```

JSON RESPONSE

```
HTTP/1.0 200 OK
Content-type: application/json
<Body>
```


```
{
    "EventStatus": [
        {
            "EventName": "AlarmInput",
            "Time": "2021-05-03T09:59:00.862+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AlarmInput",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AlarmOutput",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
```

```
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AlarmOutput",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "MotionDetection",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0,
                "ROIID": 1
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "MotionDetection",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0,
                "ROIID": 2
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "Tampering",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
```

```
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioDetection",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.Scream",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.Gunshot",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.Explosion",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
```

```
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "AudioAnalytics.GlassBreak",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "ShockDetection",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "CallRequest",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "TamperingSwitch",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
```

```
                "State": false
            }
        },
        {
            "EventName": "DTMFReceived",
            "Time": "2021-05-03T09:59:00.863+00:00",
            "Source": {
                "Channel": 0,
                "Index": 1
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.TimeChange",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.PowerReboot",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.FWUpdate",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
```

```
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.FactoryReset",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.ConfigurationBackup",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.ConfigurationRestore",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.ConfigChange",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
```

```
        }
    },
    {
        "EventName": "SystemEvent.SDFormat",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.SDFail",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.SDFull",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": false
        }
    },
    {
        "EventName": "SystemEvent.SDInsert",
        "Time": "2021-05-03T09:59:00.864+00:00",
        "Source": {
            "Channel": 0
        },
        "Data": {
            "State": true
        }
```

```
        },
        {
            "EventName": "SystemEvent.SDRemove",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.NASConnect",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.NASDisconnect",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": true
            }
        },
        {
            "EventName": "SystemEvent.NASFail",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
```

```
        {
            "EventName": "SystemEvent.NASFull",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        },
        {
            "EventName": "SystemEvent.NASFormat",
            "Time": "2021-05-03T09:59:00.864+00:00",
            "Source": {
                "Channel": 0
            },
            "Data": {
                "State": false
            }
        }
    ]
}
```

271.7. Metadata format

271.7.1. CallRequest event

```
<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2">
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet
xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2
xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:Device/Trigger/CallReques
t</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-14T15:54:44Z"
            xmlns:tt="https://www.onvif.org/ver10/schema/">
            <tt:Source>
                <tt:SimpleItem Name="SourceToken" Value="CallRequest-1"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="State" Value="true"/>
```

```
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

271.7.2. TamperingSwitch event

```
<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2">
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet
xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2
xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:Device/Trigger/TamperingS
witch</wsnt:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-14T15:54:44Z"
            xmlns:tt="https://www.onvif.org/ver10/schema/">
            <tt:Source>
                <tt:SimpleItem Name="SourceToken"
Value="TamperingSwitchToken-1"/>
            </tt:Source>
            <tt:Data>
                <tt:SimpleItem Name="State" Value="true"/>
            </tt:Data>
        </tt:Message>
    </wsnt:Message>
</wsnt:NotificationMessage>
```

271.7.3. DTMF event

```
<wsnt:NotificationMessage xmlns:wsnt="http://docs.oasis-open.org/wsn/b-2">
    <wsnt:Topic
Dialect="http://www.onvif.org/ver10/tev/topicExpression/ConcreteSet
xmlns:wsnt=http://docs.oasis-open.org/wsn/b-2
xmlns:tns1=http://www.onvif.org/ver10/topics">tns1:Device/Trigger/DTMF</wsnt
:Topic>
    <wsnt:Message>
        <tt:Message UtcTime="2020-07-14T15:54:44Z"
            xmlns:tt="https://www.onvif.org/ver10/schema/">
            <tt:Source>
                <tt:SimpleItem Name="RuleName" Value="DTMF-1"/>
```

```
            &lt;/tt:Source&gt;
            &lt;tt:Data&gt;
                &lt;tt:SimpleItem Name="State" Value="true"/&gt;
            &lt;/tt:Data&gt;
        &lt;/tt:Message&gt;
    &lt;/wsnt:Message&gt;
&lt;/wsnt:NotificationMessage&gt;
```

Chapter 272. Sample Application to get Device Information

Simple client example using cURL library.

```
#include <string.h>
#include <iostream>
#include <sys/stat.h>
#include <fcntl.h>
#include <curl/curl.h>
using namespace std;

class CurlObject
{
public:
    CurlObject(string &, string &, string &); //URL, Username, Password
    virtual ~CurlObject();
    bool Get();                 //Process the request
    string GetLastError();      // To get Error Message
    string GetResponseBody();   // To get Response Body
    string GetResponseHeader(); // To get Response Header
private:
    void SetDefaultCurlOptions();
    static int StringWriter(char *, size_t, size_t, string *); //Callback
Function
private:
    CURL *mpCurl;
    char mErrorStr[CURL_ERROR_SIZE];
    string mUrl;
    string mAuth;
    string mResponseBody;
    string mResponseHeader;
};

CurlObject::CurlObject(string &sUri, string &sUser, string &sPassword)
{
    mUrl = sUri;
    mAuth = sUser + ":" + sPassword;
    memset(mErrorStr, 0, sizeof(mErrorStr));
    mpCurl = curl_easy_init();
    SetDefaultCurlOptions();
```

```
    cout << mUrl << endl;
}

CurlObject::~CurlObject()
{
    curl_easy_cleanup(mpCurl);
}

void CurlObject::SetDefaultCurlOptions()
{
    if (mpCurl)
    {
        curl_easy_setopt(mpCurl, CURLOPT_NOSIGNAL, 1);
        curl_easy_setopt(mpCurl, CURLOPT_TIMEOUT, 60);        //Request
Timeout
        curl_easy_setopt(mpCurl, CURLOPT_CONNECTTIMEOUT, 10); //Connection
Timeout
        curl_easy_setopt(mpCurl, CURLOPT_ERRORBUFFER, mErrorStr);
        curl_easy_setopt(mpCurl, CURLOPT_URL, mUrl.c_str());
        curl_easy_setopt(mpCurl, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
//Digest Authentication
        curl_easy_setopt(mpCurl, CURLOPT_USERPWD, mAuth.c_str());
        curl_easy_setopt(mpCurl, CURLOPT_HEADER, 0);
        curl_easy_setopt(mpCurl, CURLOPT_FOLLOWLOCATION, 1);
        curl_easy_setopt(mpCurl, CURLOPT_SSL_VERIFYHOST, 2);
//SSL
        curl_easy_setopt(mpCurl, CURLOPT_SSL_VERIFYPEER, 0);
//SSL
        curl_easy_setopt(mpCurl, CURLOPT_HEADERFUNCTION, StringWriter);
//Callback Function
        curl_easy_setopt(mpCurl, CURLOPT_WRITEHEADER, &mResponseHeader);
//Response Header
    }
}

string CurlObject::GetLastError()
{
    return mErrorStr;
}

string CurlObject::GetResponseBody()
```

```
{
    return mResponseBody;
}

string CurlObject::GetResponseHeader()
{
    return mResponseHeader;
}

bool CurlObject::Get()
{
    bool retVal = true;
    if (mpCurl)
    {
        curl_easy_setopt(mpCurl, CURLOPT_HTTPGET, 1);
        curl_easy_setopt(mpCurl, CURLOPT_WRITEFUNCTION, StringWriter);
        curl_easy_setopt(mpCurl, CURLOPT_WRITEDATA, &mResponseBody);
        if (curl_easy_perform(mpCurl) != CURLE_OK)
        {
            cout << mErrorStr << endl;
            retVal = false;
        }
    }
    return retVal;
}

int CurlObject::StringWriter(char *pData, size_t size, size_t nmem, string
*sBuffer)
{
    int result = 0;
    if (sBuffer)
    {
        sBuffer->append(pData, size * nmem);
        result = size * nmem;
    }
    return result;
}

int main(int argc, char *argv[])
{
    string sIp = argv[1];
```

```
//Device IP
    string sUser = argv[2];
//Username
    string sPwd = argv[3];
//Password
    string sCommand = "/stw-cgi/system.cgi?msubmenu=deviceinfo&action=view";
//SUNAPI Command
    string sUrl = sIp + sCommand;
    CurlObject *pCurl = new CurlObject(sUrl, sUser, sPwd);
    if (pCurl)
    {
        if (pCurl->Get())
            cout << pCurl->GetResponseBody() << endl; //Response Body
        else
            cout << pCurl->GetLastError() << endl; //Error Message
        delete pCurl;
    }
    return 0;
}
```

Chapter 273. References

[1] SUNAPI_ipinstaller_2.6.8.pdf

[2] SUNAPI_network_2.6.8.pdf

[3] SUNAPI_system_2.6.8.pdf

[4] SUNAPI_video_audio_2.6.8.pdf

[5] SUNAPI_ptz_2.6.8.pdf

[6] SUNAPI_recording_2.6.8.pdf

[7] SUNAPI_event_2.6.8.pdf

[8] SUNAPI_attributes_2.6.8.pdf

[9] SUNAPI_image_2.6.8.pdf

[10] SUNAPI_io_2.6.8.pdf

[11] SUNAPI_security_2.6.8.pdf

[12] ONVIF Streaming Spec

[13] SUNAPI_bypass_2.6.8.pdf

[14] SUNAPI_ai_2.6.8.pdf

[15] SUNAPI_display_2.6.8.pdf

[16] SUNAPI_transfer_2.6.8.pdf